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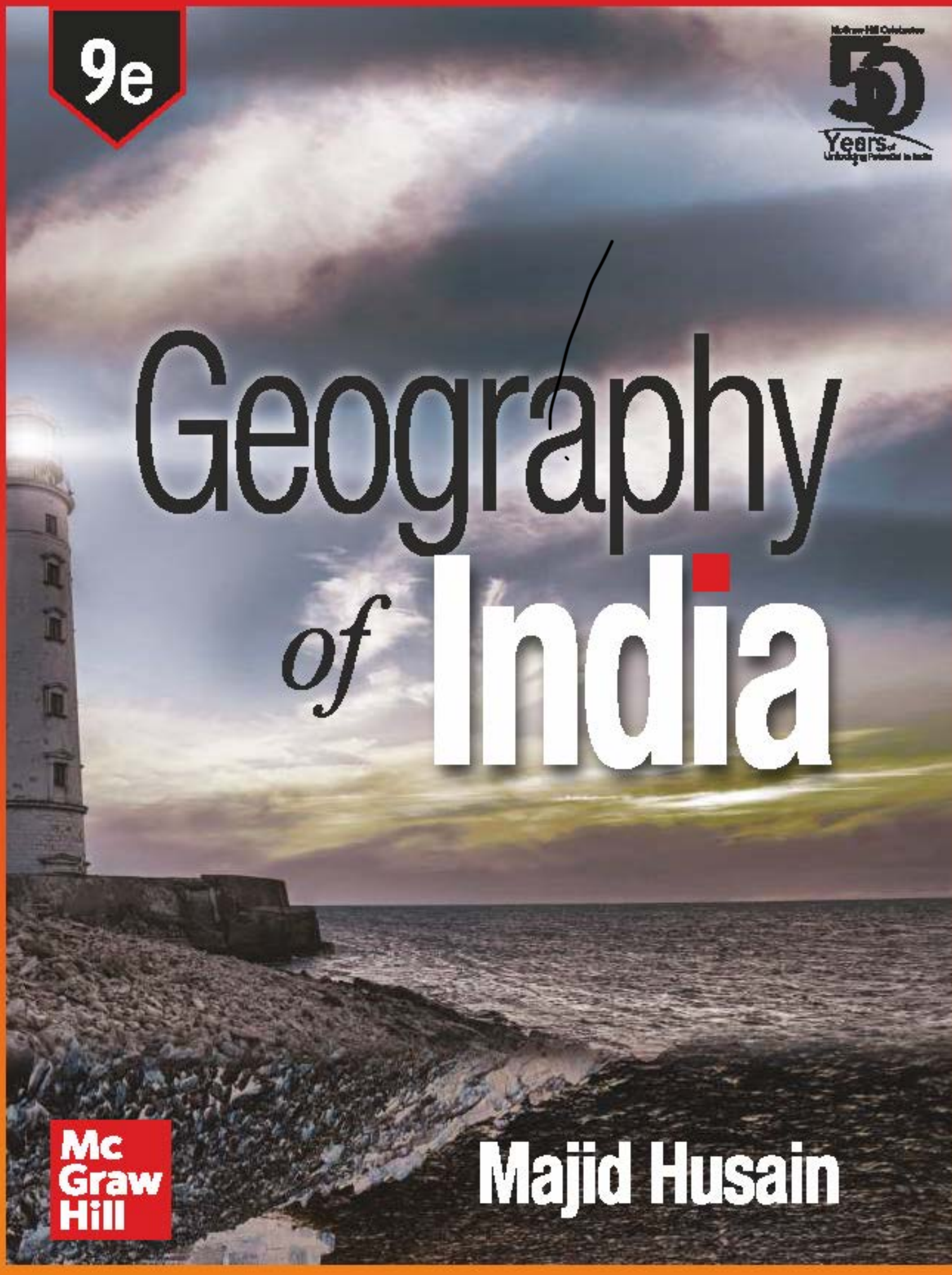

Geography of India

Majid Husain

Mc Graw Hill

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Geography *of* **India**

Majid Husain

About the Author

Born on July 2, 1938, in the village Banhera Tanda (Haridwar district) Prof. Majid Husain completed his early education from Nehru Inter College Manglaur. He had obtained his M.A. (Gold Medalist), LL.B. and Ph.D. from Swansea University, Aligarh, and pursued his higher education in Swansea (Wales) and School of Oriental and African Studies, London.

Prof. Husain had retired in 2003 as Professor and Head, Department of Geography, Jamia Millia Islamia, New Delhi. Having an excellent capability to simplify the difficult concepts in Geography in lucid and cogent style, he had authored and edited over four dozen of books. Majid Husain was an Indian Geographer of International fame, and was well known for his immense contribution to Geography through numerous publications, research works and teaching in many universities.

His unique teaching style helped students to assimilate even the most convoluted topics. He taught complete module of Geography for optional, Geography for Prelims, Environment and Ecology and mentored his students for personality test as well. Many students taught by him are working as IAS, IPS, IFS and in other elite services. His accolades were not just limited to research and teaching, he had numerous publications and useful books for UPSC and other competitive examinations to his credit.

Books written by him are also taught as textbook in University of Pennsylvania. Some of the bestselling books he has authored books

include Geography of India, World Geography, Fundamentals of Physical Geography, Geography : 3000 Terms and Concepts, Indian and World Geography, Environment and Ecology - Biodiversity, Climate Change and Disaster Management, Human Geography, Evolution of Geographical Thought, Population Geography (Perspectives in Human Geography), Models in Geography, Systematic Agricultural Geography, Bio-geography (Perspectives in Physical Geography).

The places he served include Jamia Milia Islamia, New Delhi; North East Hill University, Shillong; and University of Kashmir, Srinagar. In his 43 years long in-service experience he also served as Ex-President-Indian National Cartographic Association, Ex-Vice President-International Association of Landscape Ecology with two terms. Prof. Majid Husain was one of the most senior faculties of Geography in India, who had been honored with several awards and distinctions for his invaluable research in the field of Geography.

He was bestowed with “Man of the Year“ award by American Biographical Society, US-1997,

Delhi Govt. Award for the “Best Geography Teacher” in 1997 and “Bhugol Bhushan Award” in 2008 by Deccan Geographical Society India, “Shiksha Ratna Award” in 2011.

Prof. Majid Husain will always be remembered as an iconic academician of India.

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Geography *of* **India**

Majid Husain

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To
My students and grandchildren,
My hope for the future

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Preface to the Ninth Edition

Published first in 2008, the book Geography of India, written mainly for the candidates who prepare for competitive examinations and has become increasingly popular with each revised edition. In fact, it has become a must for the aspirants of the Indian Administrative Services and the National Education Test (NET). The book has achieved the status of the 'flag-ship' title of the McGraw-Hill as one of their best sellers. It covers all the prescribed topics of the IAS Geography optional paper II (Geography of India). It is equally helpful for the compulsory General Study Paper II (Geography of the World), and helps to solve the Geography and Ecology questions of the Preliminary Examination of the UPSC. Moreover, it also covers topics of the General Studies Paper IV like, resources of India, different types of irrigation, cropping patterns of India, transport, and public distribution system and food security. It is a text book for the Post-Graduate and Under-Graduate students in almost all the

universities in India and in many of the developing countries where Geography is taught through English medium.

The author received a number of constructive suggestions from the serious and committed students and teachers of Geography to revise the previous editions. It is in this background that the book has been thoroughly revised and updated in information and data.

The book is updated with latest change in status of Jammu & Kashmir and Ladhak. All updates are inserted in the form of Maps, Data and statistics. Updated current events, especially to mention the Amphan Cyclone, COVID- 19 situation in India along with NRC, CAA, Vizag gas leak, Indo-China Issues, Locust menace and much more forms a part of this edition. Varied topics covered are Resources of India, different types of irrigation, cropping patterns of India, transport and public distribution system and food security, disasters and Government initiatives.

The salient features of the revised edition are:

- Coverage on COVID-19, locust invasion, Vizag gas leak, Amphan cyclone and CAA & NRC
- Based on UPSC syllabus, strong in theoretical concepts, supplemented with original maps and diagrams
- Topic on geodiveristy, geotourism, fossil parks and geological heritage sites/national geological monuments
- Cratonisation and detailed heading on Indian deserts has been included
- Analysis on soil degradation in India along with the schemes and projects by the government to control land degradation and soil improvement have been included
- National New Mineral Policy 2019 and government policies in dairy enhancement added
- Governmental initiatives in renewable energy and oil industry, power sector, updates on crude oil production
- India–General land use updated, wastelands and degraded lands updated, agricultural policy, land reforms updated, government schemes in agriculture added, innovations in

agriculture, production share of various horticulture crops in India updated

- Governmental initiatives for textile development, National Steel Policy 2017, updates on fertilizer industry and drugs.
- Agreements and projects on roadways, ship recycling, aviation, traffic accidents
- Integrated Watershed Management Programme (IWMP) and updates on Metro extensions in Project Phase III
- Addition has been made on Urban Disasters, Social Media and Cyber crime
- More tables have been created and charts and pie charts provide an insight to the data

Dispute, Kolkata Kunming Corridor, the Pakistan-China Economic Corridor, and the New Silk Route - One Belt One Road (OBOR) have been incorporated in the present volume.

In addition to the above, the cartography of many of the maps have been improved and new maps have been added. The new maps in the present volume are: (i) the Panch Prayag (Five Confluences), Uttarakhand, (ii) the Kaveri Catchment Area, (iii) the Kolkata-Kunming Corridor, (iv) the China-Pakistan Economic Corridor, (v) the New Silk Route-One Belt One Road (OBOR).

Written in a simple and lucid style, it is hoped that the thorough revision of this book will be
of immense help to those interested in the subject and will also encourage them to search further
and learn more. Suggestions and constructive criticism are gratefully invited for further improvement of the book.

Majid Husain

Preface to the First Edition

Geography of India is a compulsory paper at the postgraduate and undergraduate levels of geography in all the Indian universities and autonomous colleges. In the geography optional paper of the civil services examination of the UPSC, it carries 50% weightage, equivalent to 300 marks. A sound performance followed by a high score in the General Studies Paper of the preliminary and main examinations is, therefore, significantly dependent on a clear understanding of the subject of geography, especially geography of India. In order to get high scores in the exam, a comprehensive book written in a simple and informative style supported by maps, figures, tables, etc., is a must. The book must also cover all topics and sub-topics of the syllabi prescribed as course structure by the UPSC and universities.

There are a number of books available on the subject of Indian geography by foreign authors and many Indian authors as well. While some of them contain data that is not up-to-date, others do not fully cover the syllabi of universities.

It is based on the above observations and my decades of teaching and research in geography that this book has been conceived. My goal is to provide the students of geography with a strong foundational book on Indian geography that would equip them to deal with any type of questions both in their graduate and postgraduate courses and also for the civil services examinations.

A unique feature of this book is that it has included all the topics and sub-topics prescribed by the UPSC in the new syllabus (Revised in December 2007) of Geography of India.

The following is the list of new topics as per the newly announced syllabus which has been adequately covered and documented in this book:

- Rainfall patterns.
- Forest and wildlife resources and their conservation.
- Land holdings, land tenure, and land reforms.
- Aqua-culture, sericulture, apiculture, and poultry.
- Agro-ecological regions, agrobased and aluminium industries, industrial houses and
  complexes including public sector undertakings, multinationals and liberalisation, Special Economic Zones, and Tourism including Eco-Tourism.
- Growing importance of ports on national and foreign trade, trade balance, trade policy, export processing zones.
- Indian Space Programme.
- Historical perspective of Indian Society, linguistic and ethnic diversities, religious minorities, cultural regions.
- Population problems and policies, Health Indicators, Urbanisation and remedies.
- Problems of Indian urbanisation and remedies.
- Regional Planning and Development of Island Territories.
- Cross-border terrorism, India’s role in world affairs, Geopolitics of South Asia.
- Tsunamis, Environmental awareness, Linkage of Rivers, Globalisation and Indian Economy.

Moreover there is inclusion of important entries on mountains, peaks, series, passes, rivers, lakes, national parks, tiger reserves, mining and industrial centres, places of tourist interest, and sea beaches on the map of India with a 40-word geographical write-up on each of the entries. This unique feature, I am sure would enable candidates in successfully attempting the compulsory question on the map of India in the UPSC examination.

Divided into seventeen chapters, and supported by neatly drawn maps and diagrams, the organisation of chapters and topics covered is as follows:

Chapters 1–6 discusses the physical personality of India while explaining the spatial aspect of physical elements and processes that make up the environment. This portion of the book also deals with the structure, physiography, landforms, climate, energy, natural vegetation, soils, energy, and biotic and abiotic resources of India.

Chapters 7–12 analyses the spatial distribution of agriculture and allied activities, agroclimatic and agro-ecological regions, industries, transport and communication and services and their impact on landscape.

The cultural milieu including the historical perspective of Indian Society, racial and ethnic diversity of India, major tribes, role of language, religion and traditions in the formation of cultural regions, the demographic attributes of Indian population, its problems and policies and health indicators have been examined in the thirteenth chapter.

The fourteenth chapter gives an evocative account of the rural and urban settlements while a discussion on planning and regional development constitutes the fifteenth chapter.

The spatial analysis of political aspects of India, including the geographical basis of Indian federalism, states reorganisation, regionalism, and national integration, boundary disputes, disputes on water sharing, cross border terrorism, India’s role in world affairs, Geopolitics of South Asia and Indian Ocean have been dealt with in chapter sixteen.

The last chapter presents a brief account of contemporary issues like environment hazards, earthquakes, tsunamis, landslides, population explosion and food security, epidemics, changes in land use, environmental impact assessment, environmental awareness, linkage of rivers, Globalisation and Indian economy and sustainable development.

It is hoped that the scholarly and comprehensive approach of the book will be of great help to the students and teachers alike and all those genuinely interested in knowing in detail about the geographical features and aspects of India.

Suggestions and comments for further improvement of the book are welcome and shall be gratefully considered.

—Majid Husain

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My sincere thanks to all writers whose work I have referred in this book, and also to all those who have given permission for the reproduction of copyrighted material.

I am thankful to my teacher and supervisor, Prof. Mohammad Shafi, a doyen and a pioneer geographer in the field of land use studies, whose commitment to teaching and research inspired me constantly from my student days at the Aligarh Muslim University, Aligarh. I am indebted to Prof. S.M. Tahir Rizvi, Prof. S.M. Ali, Prof. Enayat Ahmad, Prof. Moonis Raza, Prof. Mohammad Anas, Prof. S.M. Rafiullah, Janab Mukhtar Ahmad sb., Prof. S. Manzoor Alam, Prof. C.D. Deshpande, Prof. G.S. Gosal, Prof. Fakhruddin Ahmad, Prof. Nasiruddin Ahmad, Prof. Niaz.A. Siddiqui, Prof. Mehdi Raza, Prof. A. Aziz Tonki, Prof. Z.U. Alvi, Prof. Qazi Mohd. Ahmad, Prof. Aijazuddin Ahmad, Prof. R.C. Sharma, Prof. M. Farooq Siddiqui, and Prof. K.Z. Amani, for their encouragement in my academic pursuits.

The plan of the book was conceived about thirty years back when I started teaching post-graduate classes at the North Eastern Hill University, Shillong in 1977. The writing of the manuscript was,

however, in pieces and patches. It was at the persuasion of Dr. Ramesh Singh, Director, Civils India, Delhi that the dream changed into reality. I am thankful to him and the staff members of Civils India, who assisted me in more than one way in the preparation and revision of this book.

I am indebted to McGraw-Hill especially Ms. Shukti Mukherjee (Senior Manager, Portfolio), Ms. Jyoti Nagpal (Product Developer), Ms. Shreya Soni (Content Developer) and Mr. Gagan Gusai (Project Manager, Production Services) for their dedication and commitment to produce the book meticulously a short period of time.

For her sufferings and forebearance, I thank my wife who remained a perennial source of strength. Thanks are due to my daughter Dr. Shagufta Bano Husain, my sons Javed Alam and Mansoor Majid, my son-in-law Abdul Rahman Mehdi Asad, my daughters-in-law Shahida Naz and Saifi Zaka, and my grand-children (Majid Abdul Rahman, Madeeha Javed, Jibraan Mansoor, Rashad Abdul Rahman, Fareeha Javed, Areeb Mansoor, Hiba Abdul Rahman and Uzair Alam) who provided a happy environment for writing of the book. May God bless them all.

Thanks are due to Dr. Krishnanand from Shaheed Bhagat Singh College, University of Delhi for providing valuable reviews that helped in further updating the book. Dr. Shehnaz, Dr. Haseena Hashia, Dr. Mary Tahir, Dr. Tasawwur, Mr Saini from Jamia Millia Islamia, also supported the author at all times during the revision of the book while he was ailing.

—Majid Husain

About the Civil Services Examination

The Civil Services examination comprises two successive stages:

(i) Civil Services (Preliminary) Examination (Objective Type) for the selection of candidates for Main Examination; and

(ii) Civil Services (Main) Examination (Written and Interview) for the selection of candidates for the various services and posts.

Scheme and subjects for the Preliminary and Main Examination.

A. Preliminary Examination

The Examination shall comprise two compulsory Papers of 200 marks each.

Note:

(i) Both the question papers will be of the objective type (multiple choice questions).

(ii) The question papers will be set both in Hindi and English. However, questions relating to English Language

Comprehension Skills of Class X level will be tested through passages from English language only without providing Hindi translation thereof in the question paper.

B. Main Examination

The written examination will consist of the following papers:

Qualifying Papers:

Paper A: (One of the Indian Language to be selected by the candidate from the Languages included in the Eighth Schedule to the Constitution). 300 Marks

Paper B: English

300 Marks

The papers on Indian Languages and English (Paper A and Paper B) will be of Matriculation or equivalent standard and will be of qualifying nature. The marks obtained in these papers will not be counted for ranking.

Papers to be counted for merit

Paper I: Essay	200 Marks
Paper II: General Studies–I (Indian Heritage and Culture, History and Geography of the World and Society)	200 Marks
Paper III: General Studies –II (Governance, Constitution, Polity, Social Justice and International Relations)	250 Marks
Paper IV: General Studies –III (Technology, Economic Development, Bio-diversity, Environment, Security and Disaster Management)	250 Marks
Paper V: General Studies –IV (Ethics, Integrity and Aptitude)	250 Marks
Paper VI: Optional Subject – Paper 1	250 Marks
Paper VII: Optional Subject – Paper 2	250 Marks
Sub Total (Written test):	1750 Marks
Personality Test:	275 Marks
Grand Total:	2025 Marks

Candidates may choose any one of the optional subjects from amongst the list of subjects given below:

List of optional subjects for Main Examination:

(i) Agriculture
(ii) Animal Husbandry and Veterinary Science
(iii) Anthropology
(iv) Botany
(v) Chemistry
(vi) Civil Engineering
(vii) Commerce and Accountancy
(viii) Economics
(ix) Electrical Engineering
(x) Geography
(xi) Geology
(xii) History
(xiii) Law
(xiv) Management
(xv) Mathematics
(xvi) Mechanical Engineering
(xvii) Medical Science
(xviii) Philosophy
(xix) Physics
(xx) Political Science and International Relations
(xxi) Psychology
(xxii) Public Administration
(xxiii) Sociology
(xxiv) Statistics
(xxv) Zoology
(xxvi) Literature of any one of the following

Assamese, Bengali, Bodo, Dogri, Gujarati, Hindi, Kannada, Kashmiri, Konkani, Maithili, Malayalam, Manipuri, Marathi, Nepali,

Oriya, Punjabi, Sanskrit, Santhali, Sindhi, Tamil, Telugu, Urdu and English.

Geography Syllabus for Main Examination

Paper-I
Principles of Geography
Physical Geography

(i) *Geomorphology:* Factors controlling landform development; endogenetic and exogenetic forces; Origin and evolution of the earth's crust; Fundamentals of geomagnetism; Physical conditions of the earth's interior; Geosynclines; Continental drift; Isostasy; Plate tectonics; Recent views on mountain building; Vulcanicity; Earthquakes and Tsunamis; Concepts of geomorphic cycles and Landscape development; Denudation chronology; Channel morphology; Erosion surfaces; Slope development ; Applied Geomorphology: Geohydrology, economic geology and environment

(ii) *Climatology:* Temperature and pressure belts of the world; Heat budget of the earth; Atmospheric circulation; atmospheric stability and instability. Planetary and local winds; Monsoons and jet streams; Air masses and fronto

genesis, Temperate and tropical cyclones; Types and distribution of precipitation; Weather and Climate; Koppen's, Thornthwaite's and Trewartha's classification of world climates; Hydrological cycle; Global climatic change and role and response of man in climatic changes, Applied climatology and Urban climate.

(iii) *Oceanography:* Bottom topography of the Atlantic, Indian and Pacific Oceans; Temperature and salinity of the oceans; Heat and salt budgets, Ocean deposits; Waves, currents and tides; Marine resources: biotic, mineral and energy resources; Coral reefs, coral bleaching; sealevel changes; law of the sea and marine pollution.

(iv) *Biogeography:* Genesis of soils; Classification and distribution of soils; Soil profile; Soil erosion, Degradation and conservation; Factors influencing world distribution of plants and animals; Problems of deforestation and conservation measures; Social forestry; agro-forestry; Wild life; Major gene pool centres.

(v) *Environmental Geography:* Principle of ecology; Human ecological adaptations; Influence of man on ecology and environment; Global and regional ecological changes and imbalances; Ecosystem their management and conservation; Environmental degradat ion, management and conservation; Biodiversity and sustainable development; Environmental policy; Environmental hazards and remedial measures; Environmental education and legislation.

Human Geography

(i) *Perspectives in Human Geography:* Areal differentiation; regional synthesis; Dichotomy and dualism; Environmentalism; Quantitative revolution and locational analysis; radical, behavioural, human and welfare approaches; Languages, religions and secularisation; Cultural regions of the world; Human development index.

(ii) *Economic Geography:* World economic development: measurement and problems; World resources and their

distribution; Energy crisis; the limits to growth; World agriculture: typology of agricultural regions; agricultural inputs and productivity; Food and nutrition problems; Food security; famine: causes, effects and remedies; World industries: locational patterns and problems; patterns of world trade.

(iii) *Population and Settlement Geography:* Growth and distribution of world population; demographic attributes; Causes and consequences of migration; concepts of over - under-and optimum population; Population theories, world population problems and policies, Social well-being and quality of life; Population as social capital. Types and patterns of rural settlements; Environmental issues in rural settlements; Hierarchy of urban settlements; Urban morphology: Concepts of primate city and rank-size rule; Functional classification of towns; Sphere of urban influence; Rural - urban fringe; Satellite towns; Problems and remedies of urbanization; Sustainable development of cities.

(iv) *Regional Planning:* Concept of a region; types of regions and methods of regionalisation; growth centres and growth poles; regional imbalances; environmental issues in regional planning; planning for sustainable development.

(v) *Models, Theories and Laws in Human Geography:* System analysis in Human Geography; Malthusian, Marxian and Demographic Transition models; Central Place theories of Christaller and Losch; Von Thunen's model of agricultural location; Weber's model of industrial location; Rostov's model of stages of growth. Heart-land and Rimland theories; laws of international boundaries and frontiers.

PART II

Geography of India

1. *Physical Setting:* Space relationship of India with neighboring countries; Structure and relief; Drainage system and watersheds; Physiographic regions; Mechanism of Indian monsoons and rainfall patterns, Tropical cyclones and western

distur - bances; Floods and droughts; Climatic regions; Natural vegetation; Soil types and their distributions.

2. *Resources:* Land, surface and ground water, energy, minerals, biotic and marine resources; Forest and wild life resources and their conservation; Energy crisis.
3. *Agriculture:* Infrastructure: irrigation, seeds, fertilizers, power; Institutional factors: land holdings, land tenure and land reforms; Cropping pattern, agricultural productivity, agricultural intensity, crop combination, land capability; Agro and socialforestry; Green revolution and its socioeconomic and ecological implications; Significance of dry farming; Livestock resources and white revolution; aqua - culture; sericulture, apiculture and poultry; agricultural regional isation; agro-climatic zones; agro-ecological regions.
4. *Industry:* Evolution of industries; Locational factors of cotton, jute, textile, iron and steel, aluminium, fertilizer, paper, chemical and pharmaceutical, automobile, cottage and agro-based industries; Industrial houses and complexes including public sector undertakings; Industrial regionalisation; New industrial policies; Multinationals and liberalization; Special Economic Zones; Tourism including eco-tourism.
5. *Transport, Communication and Trade:* Road, railway, waterway, airway and pipeline networks and their complementary roles in regional development; Growing importance of ports on national and foreign trade; Trade balance; Trade Policy; Export processing zones; Developments in communication and information technology and their impacts on economy and society; Indian space programme.
6. *Cultural Setting:* Historical Perspective of Indian Society; Racial, linguistic and ethnic diversities; religious minorities; major tribes, tribal areas and their problems; cultural regions; Growth, distribution and density of population; Demographic attributes:
   sex-ratio, age structure, literacy rate, work-force, dependency ratio, longevity; migration (inter-regional, intra-regional and

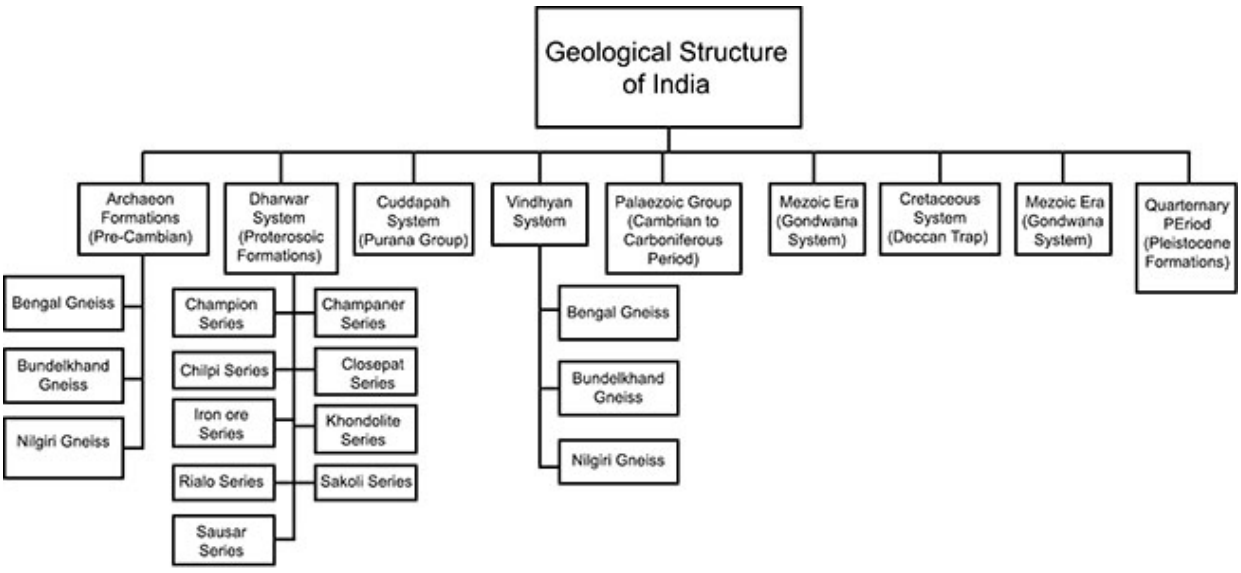
international) and associated problems; Population problems and policies; Health indicators.

7. *Settlements:* Types, patterns and morphology of rural settlements; Urban developments; Morphology of Indian cities; Functional cla ssification of Indian cities; Conurbations and metropolitan regions; urban sprawl; Slums and associated problems; town planning; Problems of urbanization and remedies.
8. *Regional Development and Planning:* Experience of regional planning in India; Five Year Plans; Integrated rural development programmes; Panchayati Raj and decentralised planning; Command area development; Watershed management; Planning for backward area,desert, drought prone, hill, tribal area development; multi-level planning; Regional planning and development of island territories.
9. *Political Aspects:* Geographical basis of Indian federalism; State reorganisation; Emergence of new states; Regional consciousness and inter state issues; international boundary of India and related issues; Cross border terrorism; India's role in world affairs; Geopolitics of South Asia and Indian Ocean realm.
10. *Contemporary Issues: Ecological issues:* Environmental hazards: landslides earth quakes, Tsunamis, floods and droughts, epidemics; Issues relating to environmental pollution; Changes in patterns of land use; Principles of environmental impact assessment and environmental management; Population explosion and food security; Environmental degradation; Deforestation, desertification and soil erosion; Problems of agrarian and industrial unrest; Regional disparities in economic development; Concept of sustainable growth and development; Environmental awareness; Linkage of rivers; Globalisation and Indian economy.

Note: Candidates will be required to answer one compulsory map question pertinent to subjects covered by this paper.

1

Geological Structure and formation of India



INTRODUCTION

The geological structure and formations of a region helps in understanding the types and character of rocks and slopes, the physical and chemical properties of soils, the availability of minerals, and the surface and underground water resources. All these resources have a direct impact on the socio-economic development of the people of a country, or region.

Geological Time Scale

The time scale is used by the earth scientists, geologists and researchers to date certain events of history on the earth.

The age of the earth as estimated by radiometric dating is approximately 4.6 billion years. Earth's past is split into sections based on the events that occurred during this time. The geologic time units are based on *stratigraphy* (correlation and classification of the rocks) and classified as *Eon* (Period greater than half a billion years); *Eras* (sub division of an Eon and they last for millions of years); *Periods* and *Epochs* **(Fig. 1.1)**

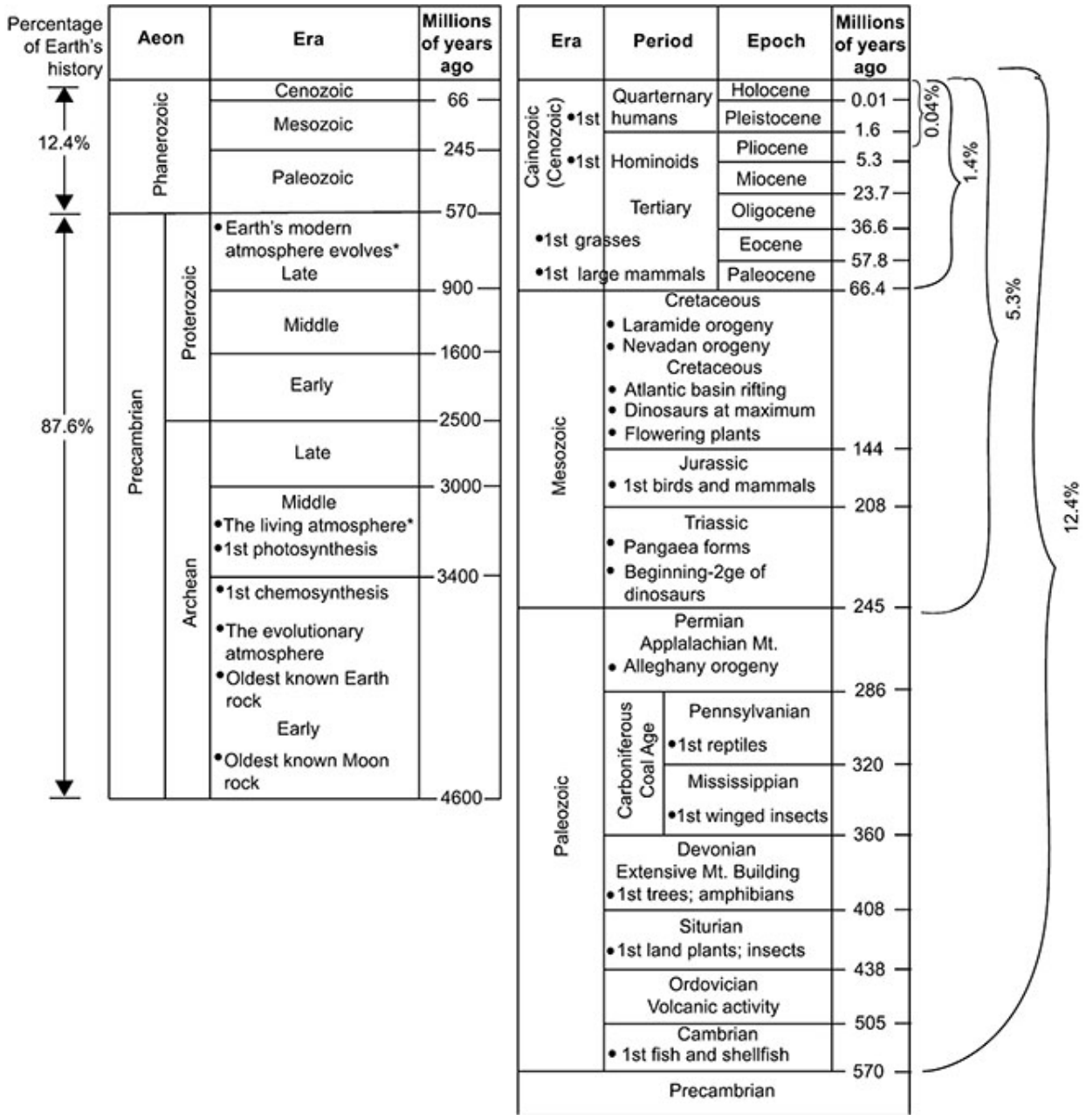


Fig. 1.1 The geological time scale is organised using absolute and relative dating methods. Absolute dates are determined through modern technological means, especially radioactive dating. In the column at left note that 87.6% of geologic time occured during the Precambrian. Data from 1983 Geologic Time Scale, Geological Society of America (after Robert W. Christopherson)

Geologically, about 200 million years ago the subcontinent of India was a part of the Gondwanaland (the Southern Continent). The geological history of India is unique, as Peninsular India was a part of the old landmass since the formation of the Earth's crust, which grew in complexity as a succession of Alpine-orogeny resulting in the upheaval of the Himalayas in the Tertiary Period and the aggradational formation of the Indo-Gangetic plain during the Pleistocene Period. The latter continues till today, through sedimentation in the flood plains of the rivers and the lower part of the Gangetic plain, namely the *Hugli basin*. The geological history of India is complex as well as varied. It begins with the first formation of the Earth's crust, first deposited sedimentary rocks, first orogeny, and extends up to the recent laying down of alluvial deposits. Many of these rock formations occur in superimposed positions and have been subjected to intense folding and faulting. The geological structure of India has been described briefly in the following sections**.**

THE ARCHAEAN FORMATIONS (PRE-CAMBRIAN)

The Archaean Era is also known as the Precambrian Period. This is the division of geologic time scale from the formation of the Earth (about 4.6 billion years ago) to the beginning of the Cambrian Period of the Paleozoic Era (about 570 million years ago).

The Precambrian time constitutes about 86.7% of the Earth's history**.** The term "Archaean", introduced by J.D. Dana in 1782, refers to the oldest plutonic rocks of the Earth's crust. The oldest known rocks of the Earth, the evolutionary atmosphere, the first chemosynthesis, the first photosynthesis, the life-supporting atmosphere and the Earth's modern atmosphere, were developed during the Precambrian Era (Archaean and Protozoic). Rocks of the Archaean system are devoid of any form of life. In other words, the Archaean rocks are all azoic or unfossiliferous. They are thoroughly crystalline, extremely contorted, faulted, and practically devoid of any sediment. They are largely intruded by plutonic intrusions and generally have a well-defined foliated structure. These rocks are

known as the basement complex **or** fundamental gneisses. Thus, all over the world, the Archaean rocks are the foundation of all the great ancient plateaus, and they form the core of all the great folded mountain ranges of the world.

In the Indian Geological Time Scale, advocated by T.S. Holland, the Precambrian Era is known as the *Purana*. The Archean system includes the Aravalli, Dharwar, Cuddapah, Vindhyan, Meghalaya Plateau, and Mikir Hills. These are also called the Archean gneiss. The mineral composition of Archaen gneiss varies from granite to gabbro. The constituent minerals are: orthoclase, oligoclase, quartz, muscovite, biotite, and hornblends. The Archaen rocks cover two-thirds of Peninsular India. They also occur in the roots of the mountain peaks all along the Greater Himalayas from the western most part of Kashmir to the eastern-most part of Arunachal Pradesh as well as in the Trans-Himalayan ranges of Zaskar (Zanskar), Ladakh, and the Karakoram **(Fig. 1.2 and Fig. 1.3).**

The Archaean rocks cover two-thirds of Peninsular India. In the Peninsular region, the Archaean rocks are known to be of three well-defined types:

(i) The Bengal Gneiss

- *The Bengal gneiss* is highly foliated, which is distributed in the Eastern Ghats, Odisha (known as Khodolites after Khond tribes in Koraput and Bolangir districts), stretching over Manbhum and Hazaribagh districts of Jharkhand, Nellore district of Andhra Pradesh, and Salem district of Tamil Nadu.
- They also occur in the Son Valley, Meghalaya Plateau and Mikir Hills.These formations are very thinly foliated. These rocks were identified for the first time in the Midnapur district of West Bengal.

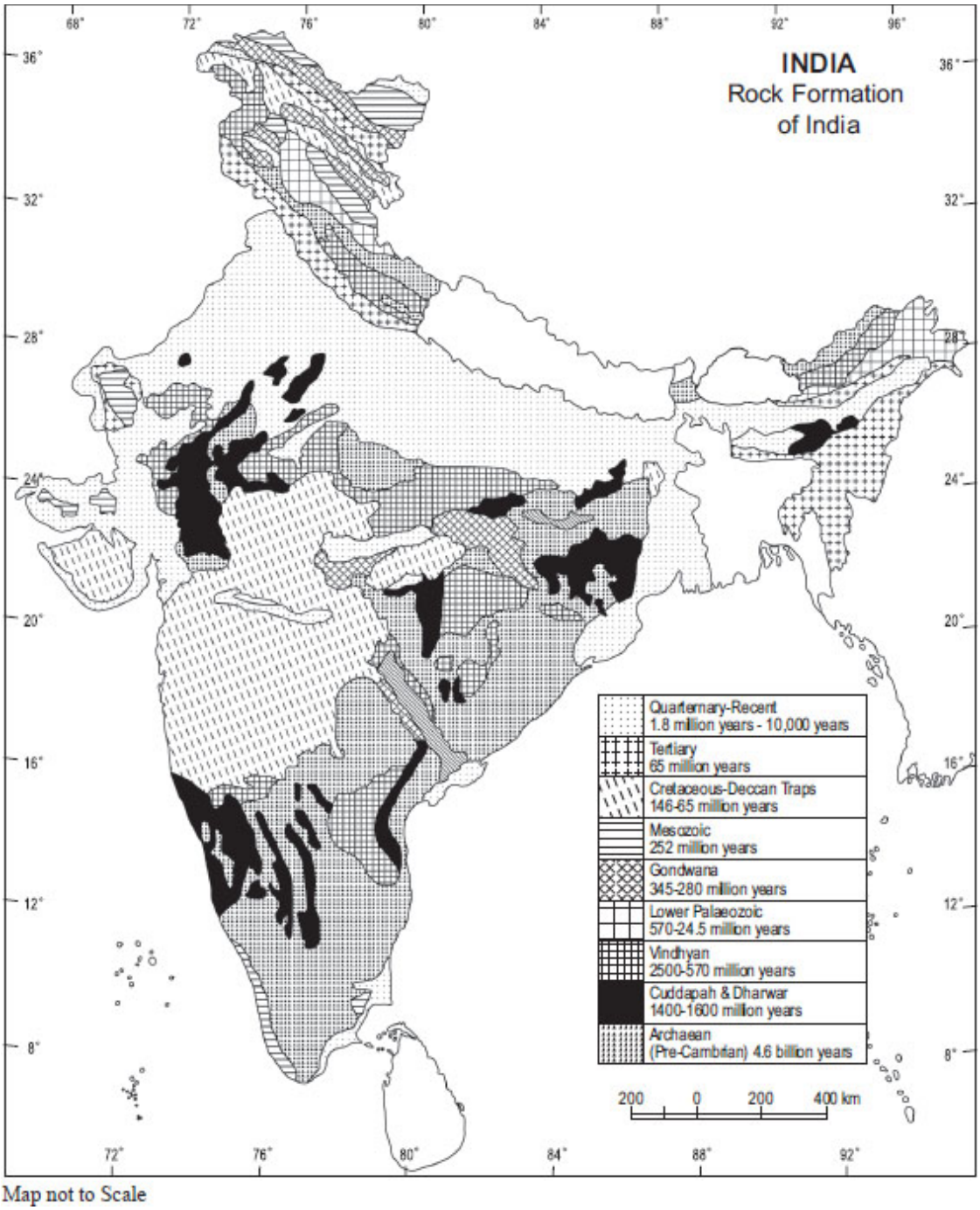


Fig. 1.2 Geological Systems

(ii) The Bundelkhand Gneiss

- *The Bundelkhand* gneiss are massive granitoid which form the second group of fundamental gneiss of the Archaean age.
- It is geographically distributed in Bundelkhand (UP),

Baghelkhand (MP), Maharashtra, Rajasthan, Andhra Pradesh, and Tamil Nadu.

- It is a coarse grained gneiss which looks like granite.
- The Bundelkhand gneiss is conspicuously criss-crossed and characterised by quartz veins.

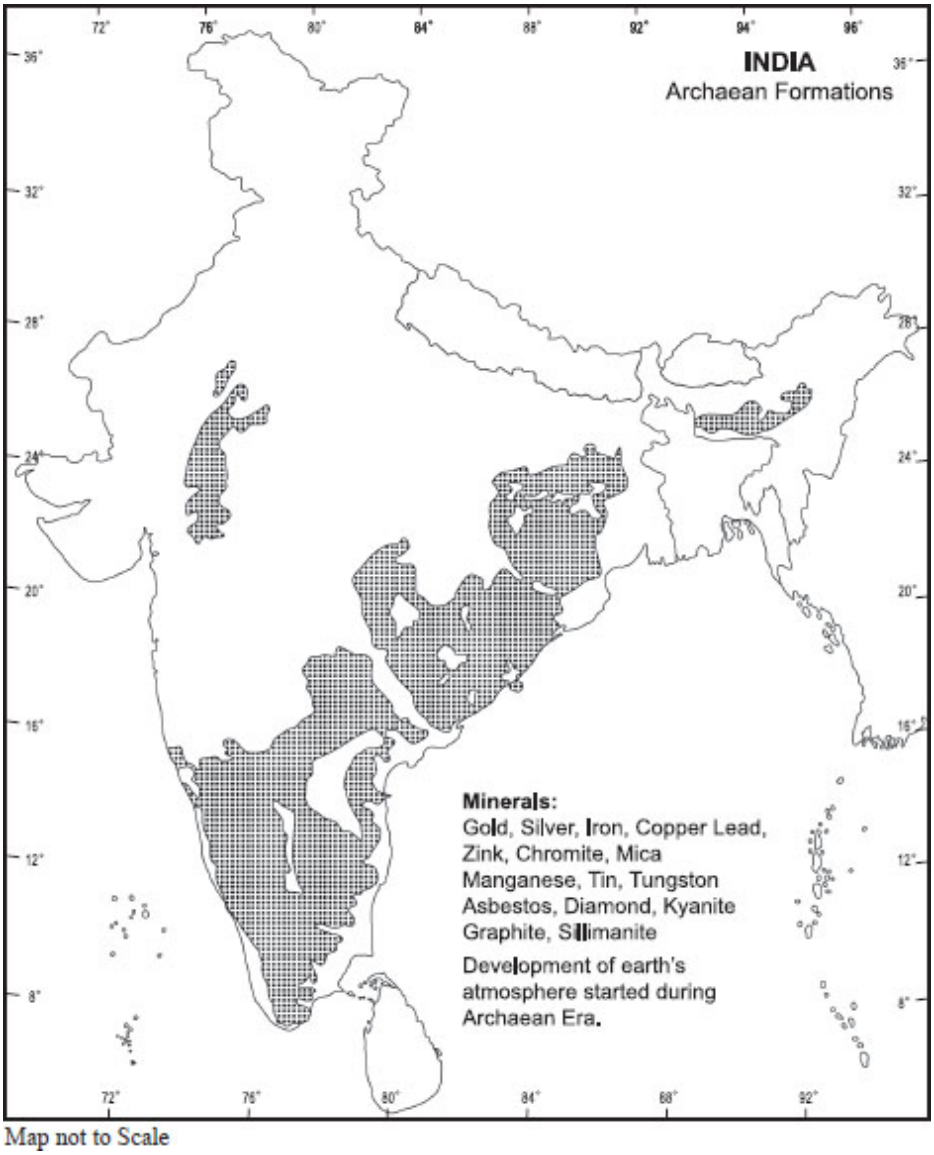


Fig. 1.3 Archaean Formations (Pre–2500 Million Years)

(iii) The Nilgiri Gneiss

- These are massive, eruptive dark-coloured gneiss. The name

being given in honour of *Job Charnock* whose tombstone in Kolkata was made of this rock.

- The Nilgiri gneiss is bluish-grey to dark coloured rock, medium to coarse grained in texture.
- This is plutonic gneiss intrudes into the other Archaean rock masses.
- Nilgiri gneiss is popularly recognised as belonging to the Charnockite series.
- It is widely found in South Arcot, Palni Hills, Shevaroy Hills and Nilgiri in Tamil Nadu, Nellore in Andhra Pradesh, Balasore in Odisha, Karnataka, Kerala, Malabar, Jharkhand, Chhattisgarh and Aravallis (Rajasthan).
- The Archaean rocks are the repositories of the mineral wealth of India. These rocks are rich in ferrous and non-ferrous minerals like iron ore, copper, manganese, mica, dolomite, lead, zinc, silver and gold.

DHARWAR SYSTEM (PROTEROZOIC FORMATIONS)

This geologic time extends from 2500 million years ago to 1800 million years ago**.** These are the first metamorphosed sedimentary rock systems known as the Dharwar System in the Indian Geological Time Scale, as they were studied for the first time in the Dharwar district of Karnataka. They are composed largely of igneous debris, schists and gneisses. The Dharwar rocks occur in scattered patches in (i) Dharwar and Bellary districts of Karnataka and extend up to the Nilgiris and Madurai districts of Tamil Nadu, (ii) Central and eastern parts of the Chotanagpur Plateau, Meghalaya Plateau and Mikir Hills, and (iii) the Aravallis, Rialo (Delhi series), from Delhi to the south of Alwar and the Himalayan region **(Fig. 1.4).**

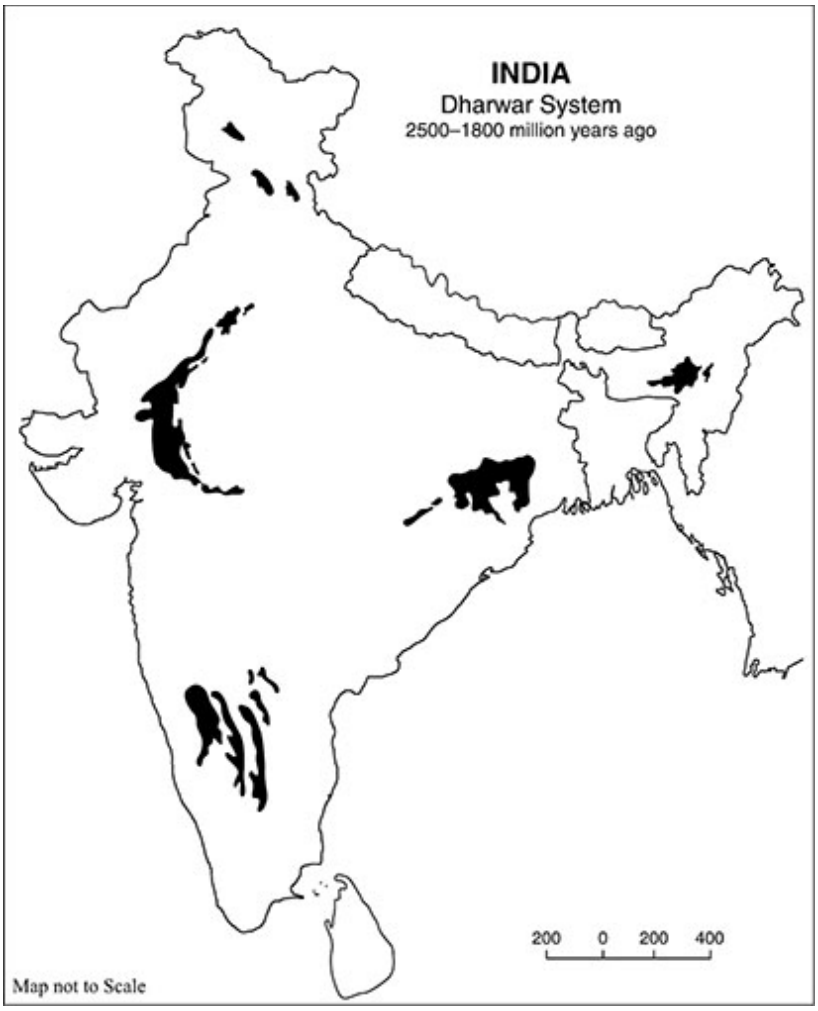


Fig. 1.4 Dharwar System (2500–1800 million years ago)

The Dharwar rocks are rich in minerals like iron ore, manganese, lead, zinc, gold, silver, dolomite, mica, copper, tungsten, nickel, precious stones and building materials.

Some of the important series of the Dharwar System are:

Name of the series of Dharwar System	Distribution in India	Significant features

Name of the series of Dharwar System	Distribution in India	Significant features
Champion Series	• North east of Mysore city to the east of Bangalore • It stretches in Kolar and Raichur districts of Karnataka	• One of the deepest gold mines in the world • Depth more than 3.5 km • Gold content is 5.5 per tonne of the ore • Headquarters of National Institute of Miner's Health is located here
Champaner Series	Lies in the outlier of Aravalli system in the vicinity of Vadodara	• Quartzites, conglomerates, phyllites, slates, limestones and marbles • Popular for green variety of marble
Chilpi Series	Occupies parts of Balaghat, Jabalpur and Chhindwara districts of Madhya Pradesh	• Minerals found: grit, phyllite, quartzites, green stones, magniferous rocks

Name of the series of Dharwar System	Distribution in India	Significant features
Closepet Series	Distributed in Balaghat and Chhindwara districts of Madhya Pradesh	• It is a Dharwarian formation • Minerals found: quartzite, copper pyrite, magniferous rocks • Ore supplied to the Malanjkhand Copper plant
Iron Ore Series	Distributed in Singhbhum, Bonai, Mayurbhanj, Keonjhar in the form of a range	• It is about 65 km in lengthIt has reserves of about 3000 million tonnes of iron ore • Iron ore is supplied to Jamshedpur, Durgapur, Rourkela and Bokaro Steel plants
Khondolite Series	Occupies large area in Eastern Ghats from the Northern extremity to Krishna Valley	• Principal rock types: khondolites, kodurites, charnockites, gneisses

Name of the series of Dharwar System	Distribution in India	Significant features
Rialo Series (Delhi Series)	Extends from Delhi (*Majnu Ka Tila*) to Alwar (Rajasthan in Northeast to Southwest direction)	• Rich in limestone and Makrana marbles • Makrana and Bhagwanpur are known for their high quality of marbles
Sakoli Series	Stretches over Jabalpur and Rewa districts	• Series belongs to the Dharwarian formation • Rich in mica, dolomite, schist and marble • (*Superiormost quality*)
Sausar Series	Series spreads over Nagpur and Bhandara districts of Maharashtra and Chhindwara district of Madhya Pradesh	It belongs to Dharwarian GroupMinerals found: mica schist, quartzite, marble and magniferous rocks

THE CUDDAPAH SYSTEM (THE PURANA GROUP)

The Cuddapah system is made of shales, slates, limestone and quartzite. The rocks are generally without fossils because there was no origin of species during their formation. The Cuddapah formations, named after the district of Cuddapah in Andhra Pradesh, are sedimentary-metamorphic formations. The Cuddapah system occurs in the (i) Cuddapah and Kurnool districts of Andhra Pradesh, (ii) Chhattisgarh, (iii) Rajasthan-Delhi to the south of Alwar, and (iv) the Lesser Himalayas in the extra-Peninsular region **(Fig. 1.5).**

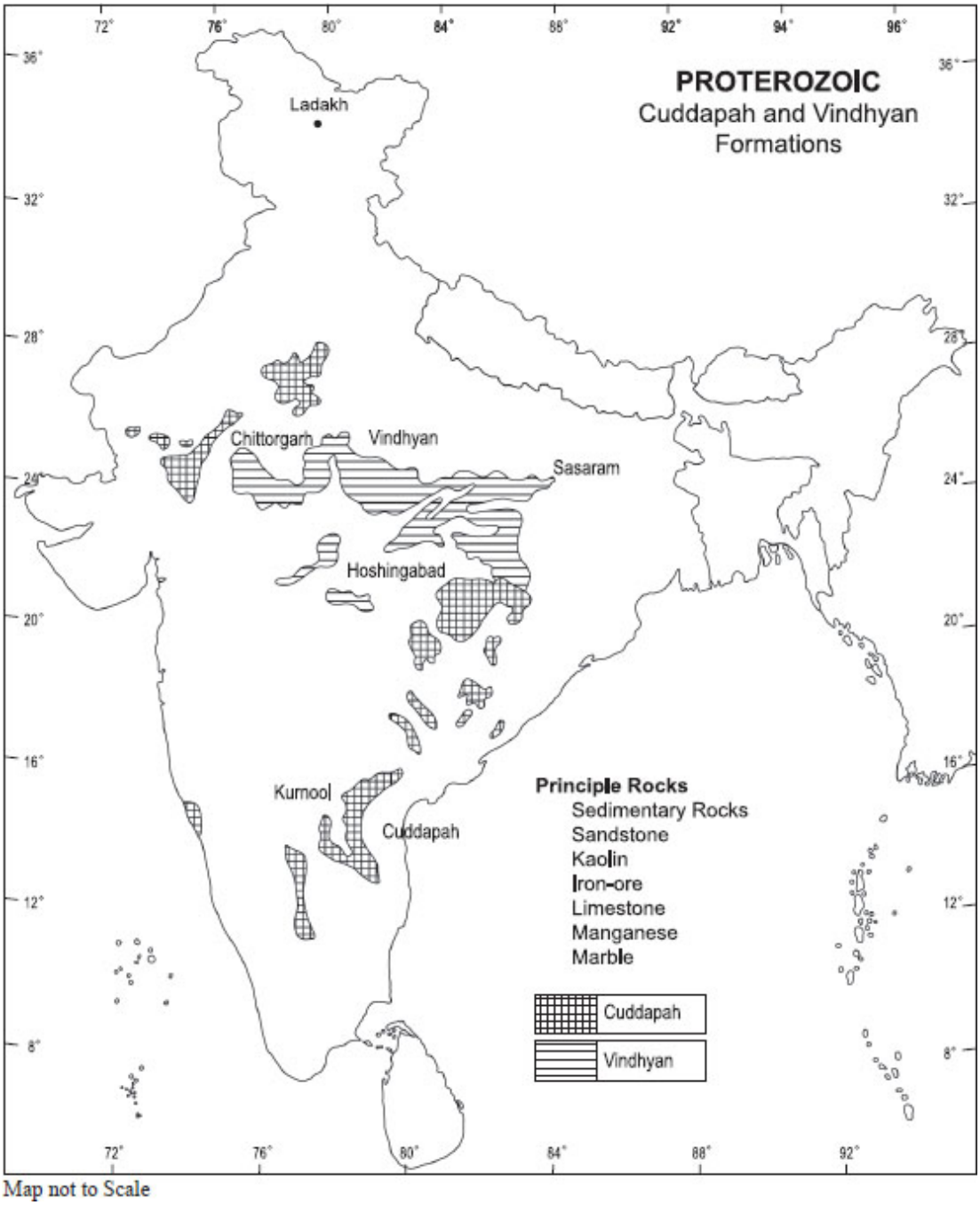


Fig. 1.5 Proterozoic Formations (2500–570 Million years ago)

At places the Cuddapah formations are 6,000 m in thickness. The enormous thickness of these rocks indicates the sinking of beds of the basin with growing sedimentation.

THE VINDHYAN SYSTEM

The Vindhyan system derives its name from the Vindhyan Mountain. This mountain forms a dividing line between the Ganga Plain and the Deccan Plateau. The system covers an extensive area of 103,600 sq km from Chittorgarh in Rajasthan to Sasaram in Bihar. It has enormous sedimentary deposits and at places their depth is more than 4000 m. In some tracts, the Vindhyan rocks are buried under Deccan lava. The Great Boundary Fault (GBF) separates the Vindhyan System from the Aravallis for a distance of about eight hundred km **(Fig. 1.5).**

The Vindhyan system is well known for red-sandstone, sandstone, building material, ornamental stone, conglomerates, diamondiferous and raw materials for cement, lime, glass and chemical industries. In certain places these rocks yield inferior quality of iron ore and manganese. The well known diamond mines of Panna and Golconda lie in the Vindhyan system. The historical buildings of Qutab Minar, Humayun's Tomb, Fatehpur Sikri, Agra Fort, Red Fort, Jama-Masjid, Birla Mandir, the Buddhist Stupa of Sanchi, etc., have been constructed from the red sandstone obtained from the Vindhyan Ranges. Coarser sandstones have been used as grindstones and millstones.

Name of the series of Vindhyan System	Distribution in India	Significant features
Bhander Series	Western parts of Vindhyan formation	• Main Rocks: Sandstones, shales, limestone • It provides good building material

Name of the series of Vindhyan System	Distribution in India	Significant features
Bijwar Series	Stretches over districts of Chhatarpur and Panna in Madhya Pradesh	• Main Rocks: Sandstone, red sandstone, quartzite • Basaltic intrusions found whose dykes are rich in diamonds
Kaimur Series	Stretches over Bundelkhand (UP), Baghelkhand (MP)	• Main Rocks: sandstone, conglomerate, shale • Red sandstone is used was used in historical monuments like Red fort, Hawa Mahal, Qutub Minar, Humayun's Tomb

The principle rocks of the Cuddapah system are sandstones, shales, limestone, quartzites slates, inferior quality of iron-ore, manganese ore, asbestos, copper, nickel, cobalt (Delhi System), marble, jasper, building material and stones for interior decoration. The metallic contents in the ores of Cuddapah rocks are, however, low and at places uneconomical for extraction.

Papaghani Series

The series has been named after the Papaghani river (Andhra Pradesh), in the valley of which these rocks have been exposed. It consists of quartzites, sandstones, shales, slates, limestones and marbles. The series is intruded by magma in the form of dykes and sills which have metamorphosed limestone into marble, talc, slate, and serpentine.

THE PALAEOZOIC GROUP (CAMBRIAN TO CARBONIFEROUS PERIOD)

The Palaeozoic Era includes the Ordovician, Silurian, Devonian, Carboniferous, and the Permian periods of the Standard Geological Time Scale. This is known as the *Dravidian Era* in the Indian Geological Time Scale.

The Palaeozoic Era extends from 570 million years ago to 24.5 million years ago. It marks the beginning of life on the Earth's surface. The formations of this period are almost absent in the Peninsular India except near Umaria in Rewa. These formations exist in the Salt Range, Pir-Panjal, Handwara, Lidder-Valley, Anantnag of Kashmir (Jammu & Kashmir), Spiti, Kangra, Shimla region (Himachal Pradesh), and Garhwal and Kumaun (Uttarakhand). It was during this period that the Pangaea was broken and the Tethys Sea came into existence. The Cambrian rocks include shales, sandstones, clays, quartzites slates, salts, marble, etc.

Palaeozoic System in the Indian Geologic Time Scale

The Gondwana formations are fluviatile and lacustrine in character. They were deposited in the river basins and lakes during the Upper Carboniferous Period. These basins later subsided along the trough faults amidst ancient rocks of the great southern continent called the Gondwanaland. These rocks were formed during the Upper Carboniferous and the Jurassic Periods (Mesozoic Era)**.**

THE MESOZOIC ERA (THE GONDWANA SYSTEM)

"Mesozoic" means middle life. The term is used for a period of geologic time in which the presence of fossil invertebrates dominated the rocks. The Mesozoic Era includes three periods: Triassic, Jurassic, and Cretaceous. In the Indian Geological Time Scale, these periods extend from the Upper Carboniferous up to the beginning of the Cenozoic Era or the *Aryan Era*.

The term 'Gondwana' was covred by H. B. Medicott in 1872. It was derived from the ancient tribe of Gonds that inhabits central provinces

of Madhya Pradesh.

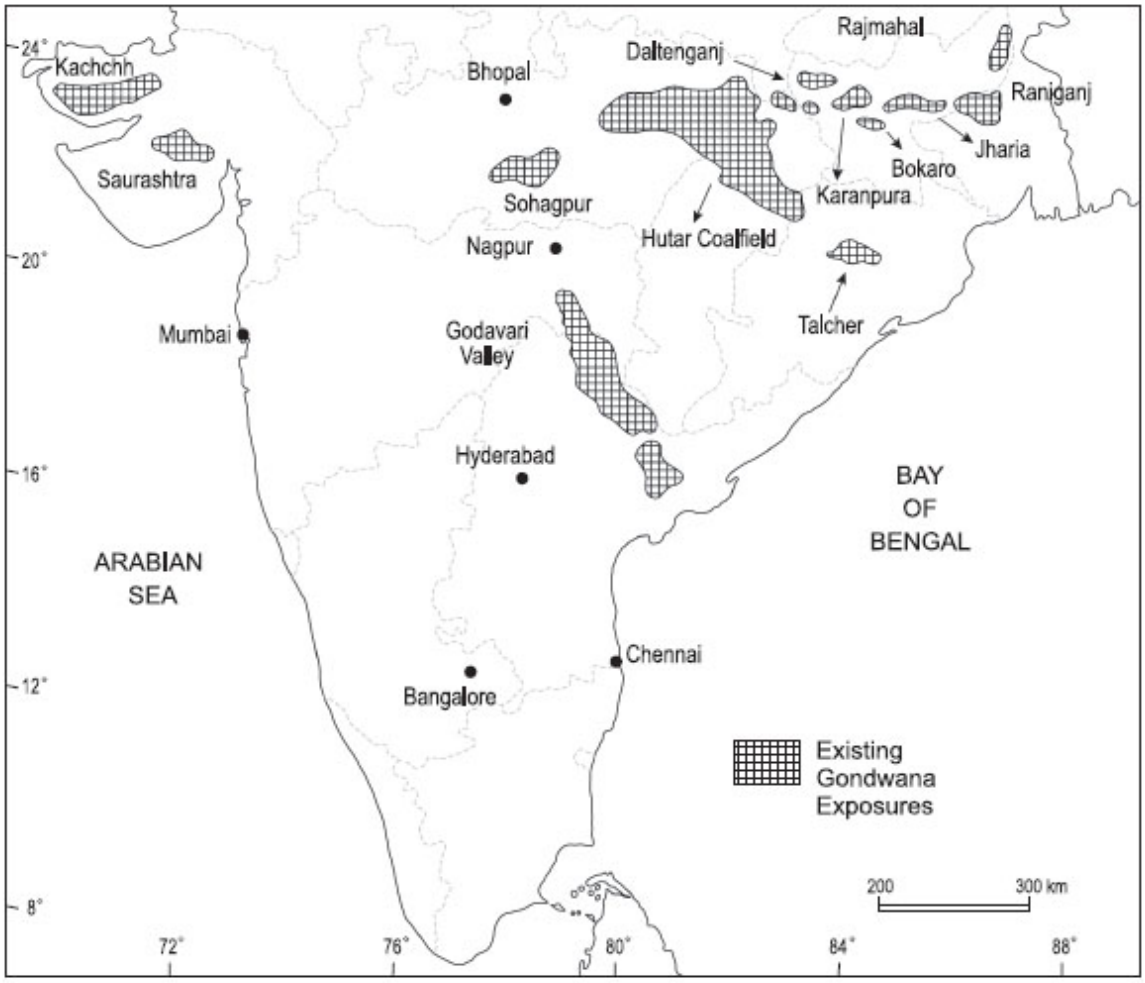
The Gondwana group begins with the Permo-Carboniferous period which, in the Standard Geologic Time Scale, is known as a period of coal formation **(Fig. 1.6).** The Lower Gondwana rocks are found in the Talcher, Panchet and Damuda series. Most of the good quality coal deposits (bituminous and anthracite) of India are found in Gondwana formations. Moreover, iron ore occurs in the iron-stone shales of Raniganj coal fields. In addition to coal and iron, kaolin, fire-clay, sandstone and grits are also found in the Gondwana formations.

Talcher Series

It is the series of the Gondwana system named after Talcher in Dhankenal district of Odisha. It is rich in good quality coal used for smelting and in thermal power plants.

Talcher coalfield has the highest coal reserves in India of 38.65 billion tonnes. This coalfield is divided into five production areas namely Talcher, Jagannath, Kalinga, Lingaraj and Hingula.

The industries in Talcher are located along the river Brahmani which flows from north west to south east.



Map not to Scale

Fig. 1.6 Gondwana System (345–280 million years before present)

The Damuda Series

The Damuda series belongs to the Middle Gondwana Period which contains enormous deposits of coal seams.The coal seams are thicker and more elongated in the eastern coal fields than in the west. The important coal bearing areas of this period are Raniganj, Jharaia, Karanpura, and Bokaro of the Damodar basin, Singrauli, Korba, and Pench valley in Chhattigarh and Madhya Pradesh, Talcher in Mahanadi Basin in Odisha, and Singareni of Satpura Basin in Madhya Pradesh. The *Jhingurda Coal Seam* with a thickness of about 131 m is the thickest coal seam in India. The Gondwana rocks are also found in Himalayas from Kashmir to Arunachal Pradesh and

Poorvanchal. The coal seams of these areas are metamorphosed. They are also found in Saurashtra, Kachchh, western Rajasthan, Coromandal Coast, and Rajmahal Hills **(Fig. 1.6).**

Panchet Series

It is the youngest series of the Lower Gondwana System, which derives its name from the hill of that name, south of Raniganj. The series consists of greenish-sandstone and shales. It is, however, devoid of coal-seams.

The iron-ore shales of the Lower Gondwana System are particularly well developed in the Raniganj coalfield of West Bengal. However, they contain inferior quality iron ore, i.e., siderite and limonite. Being inferior in quality, they are generally not mined for iron. The Gondwana System of rocks provides over 95% of the coal of India. Moreover, it provides iron-ore, limestone, sandstone, and raw material for ceramic industry.

India's best and largest coal deposits are found in the Gondwana System—mainly in the Damodar Valley of West Bengal, Jharkhand, the Mahanadi valley of Odisha and Chhattisgarh, the Godavari valley of Andhra Pradesh, and the Satpura basin of Madhya Pradesh **(Fig. 1.6).**

As stated above, the beginning of the Upper Carboniferous Period is known as the Aryan period. The salient features of the Aryan formations are:

(i) During the Upper Carboniferous Period, the Himalayan region was occupied by a vast geosyncline which was connected to the Pacific Ocean in the east through China and the Atlantic Ocean in the west through Afghanistan, Iran, Asia Minor, and the present Mediterranean Sea. This was called the *Tethys Sea*.

(ii) The area of the Kashmir Himalayas (from Pir Panjal to Hazara in the north-west and Ladakh in the north-east) witnessed violent volcanic activity.

(iii) The upper continent of Gondwanaland developed fissures and its broken parts started drifting away from each other.

The Subcontinent of India drifted towards north and north-east to collide with the Asian land mass (Eurasian Plate).

(iv) There was large scale eruption of lava in the Deccan Trap.

(v) The development and expansion of the Arabian Sea and the Bay of Bengal.

(vi) The Tertiary mountain building gave birth to Himalayas.

(vii) The Subcontinent of India assumed its present shape.

(viii) The beginning of Ice Age, belonging to the Pleistocene Period, covering large parts of the earth under ice-sheet.

(ix) Evolution and spread of man in different parts of the world.

THE CRETACEOUS SYSTEM (THE DECCAN TRAP)

The Cretaceous Period extends from about 146 million years ago to 65 million years ago. The term “Cretaceous” has been obtained from the Latin *creta*, meaning “chalk”. This is a very widely distributed system in the country which has divergent facies of deposits in different parts of India. This period is marked by the transgression of the sea (Coromandal coast, Narmada valley) and outpouring of huge quantity of lava (basalt) so as to form the Deccan Trap and intrusion of plutonic rocks such as gabbro and granite.

Towards the end of the Cretaceous period the Peninsula was affected by intense volcanic activity. During this period, enormous quantity of basaltic lava was poured out to the surface assuming a great thickness of over 3,000 m. The Lava Plateau (the Deccan Trap) is the result of that lava eruption. The Deccan lava covers about 5 lakh sq km of area in Gujarat (Katch, Kathiawad), Maharashtra, Madhya Pradesh (Malwa Plateau), Chhattisgarh, Jharkhand, northern Andhra Pradesh and north-western Karnataka **(Fig. 1.7).**

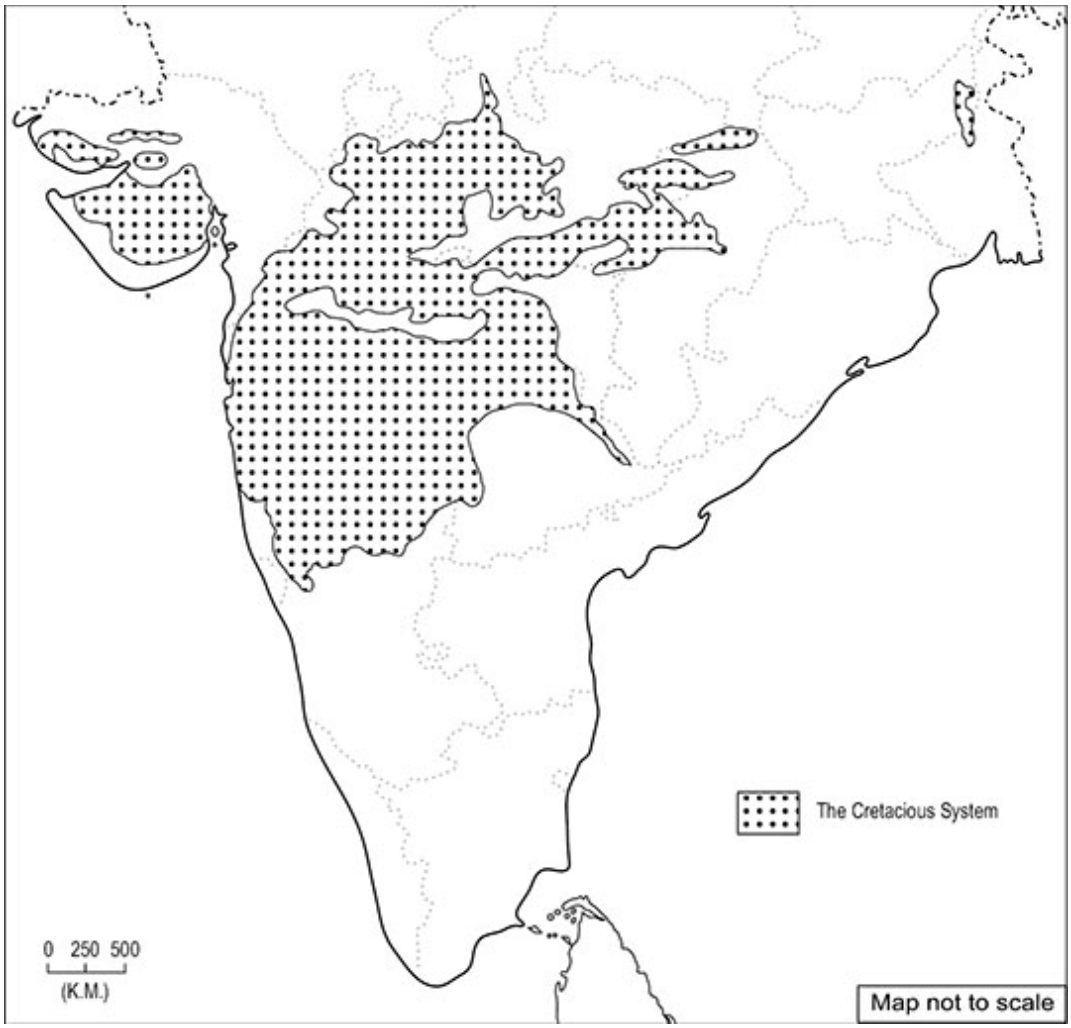


Fig. 1.7 The Cretaceous System (The Deccan Trap)—146–65 million years ago

The lava plateau of India (Deccan Trap) has a maximum thickness of about 3000 m along the coast of Mumbai from where it decreases towards south and east. It is about 800 m in Kachchh, 150 m at Amarkantak and 60 m at Belgaum (Karnataka). The individual lava flows, on an average, have a thickness of about 5 m to 29 m. Such flows have been identified in a boring near Bhusawal (Maharashtra). These are inter-bedded with sedimentary beds called “inter-trappean beds”.

The basalt of the Deccan Trap is used for the construction of roads and buildings. Moreover, quartz, bauxite, magnetite, agate, and semi-

precious stones are also found in the trap. It is also rich in magnesium, carbonate, potash, and phosphates.

THE TERTIARY SYSTEM (THE CENOZOIC ERA)

Cenozoic means recent life. The beginning of the Tertiary Period is about 66 million to 2.6 million years ago. Fossils in these rocks include many types, closely related to modern forms, including mammals, plants and invertebrates. The Cenozoic Era has two periods: The Tertiary and the Quaternary. The term ‘tertiary’ was used by Giovanni Arduino during the mid eighteenth century.

The two great events that occurred during the Tertiary Period include: (i) the final beaking-up of the old Gondwana continent, and (ii) the uplift of the Tethys geosyncline in the form of the Himalayas. During the early Tertiary Period, as the Indian plate collided with Eurasian plate, the sediments which had been accumulating in the Tethys basin had begun to rise by a slow rise of ocean bottom. The upheaval of the Himalayas altered the old topography of the subcontinent **(Fig. 1.8, 1.9).**

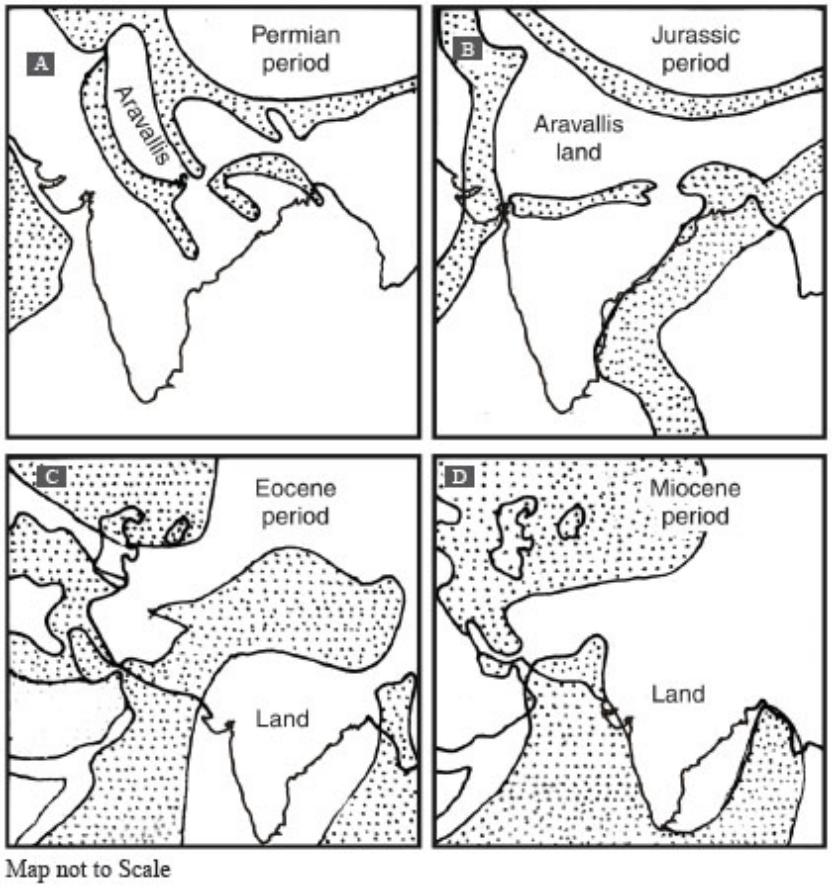


Fig. 1.8 Change in Topography of Subcontinent

Three phases of the upheaval of the Himalayas have been distinguished:

(i) During the first upheaval (Eocene—about 65 million years ago), which culminated in the Oligocene, and resulted in the upheaval of the Greater Himalayas.

(ii) It was followed by a more intense movement during the mid-Miocene period about 45 million years ago, which resulted in the folding of Lesser Himalayas.

(iii) The third upheaval took place during the Post-Pliocene period, about 1.4 million years ago which resulted in the folding of Shiwaliks or the Outer Himalayas**.** There is enough evidence to prove that the Himalayas are still rising**.**

In the Peninsular region, the Tertiary System occurred on the coast of Kachchh, Kathiawar, Konkan, Malabar, Nilgiris, and the Eastern Ghats.

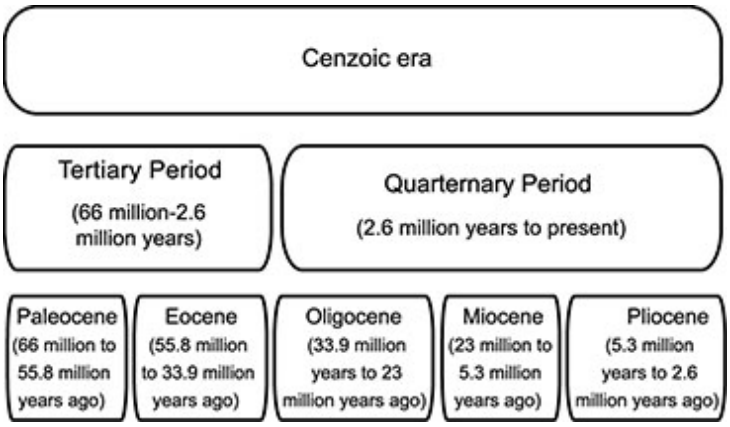


Fig. 1.9 Geological time scale of the Tertiary System

THE QUATERNARY PERIOD (THE PLEISTOCENE AND RECENT FORMATIONS)

Quaternary is the name proposed for very recent deposits, which contain fossils of species with living representatives. The northern plains of India came into existence during the Pleistocene Period **(Fig. 1.10).** During the Quaternary Period, the ice-sheets descended to as low as 1500 m in altitude. The third physical division of India which is the Great Indo-Gangetic-Brahmaputra Plain had not figured at all till the Quaternary Period. The bottom configuration of this plain occupies largely a synclinal basin, called foredeep, which is a downwarp of the Himalayan foreland of variable depth, formed concommitantly with the rise of the Himalayas to the north. The Pleistocene period is marked by Ice Age and glaciation on a large scale in the Northern Hemisphere. The moraine deposits and the Karewa formations of Kashmir Valley and the Bhadarwa (Doda District of Jammu Division) are of the Pleistocene period. It forms the terraces of the Jhelum, on the flanks of the Pir-Panjal. The thickness of the karewas at places is up to 1400 m. The river terraces of the

Narmada, Tapi, Godavari, Krishna, and Kaveri, etc., are also of the Pleistocene Period.

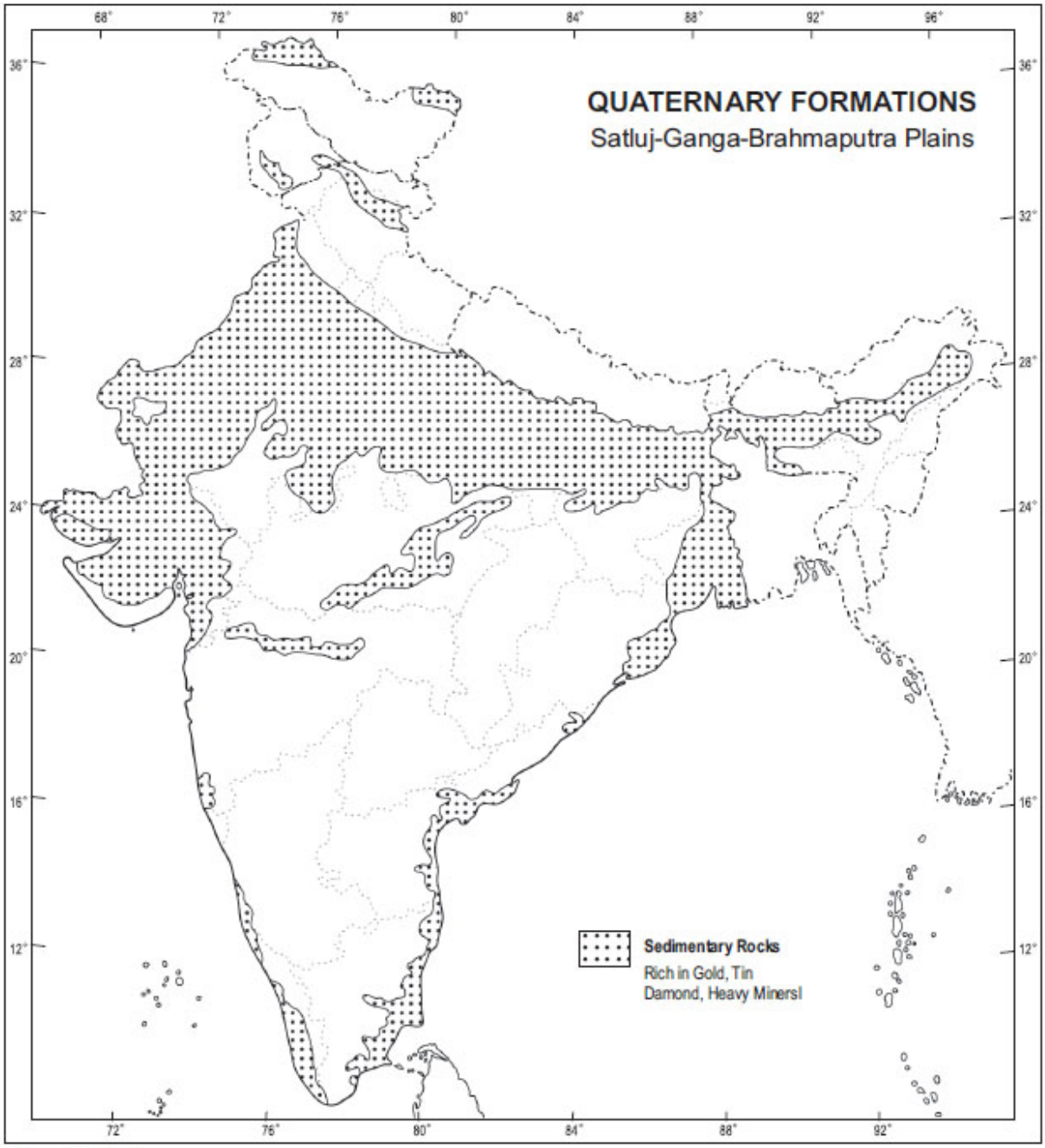
Karewas

The karewas are the lacustrine deposits of the Pleistocene period. They consist of sands, clays, loams, silt, and boulders. The karewas of Kashmir are generally found along the lower slopes of Pir-Panjal with a dip towards the Kashmir Valley. The Pampore and Pulwama karewas are well known for the cultivation of saffron, almond, and walnut.

GEODIVERSITY AND GEOTOURISM

The richness of diversity is determined by the development of biotic and abiotic resources. Geodiversity is defined as the variety of geological features like rocks, minerals, fossils, soils which are a part of the Earth's evolution. The geodiversity of the country is responsible for the wide range of geosites. The Geological Survey of India has enlisted certain areas in the subcontinent of India as National Geological Monuments. These monuments play an important role in providing an insight into the past–about its formation, the orogeny, the palaeo-environment and the exotic collection of the flora and the fauna

'*Geoconservation*' is the identification and preservation of sites that help us to understand the processes of formation of the earth.



Map not to Scale

Fig. 1.10 Quaternary Formations (1.8 million years to 10,000 years ago)

'*Geotourism*' is defined as the tourism that sustains the geographical character of a place-its environment, heritage, aesthetics, culture and the wellbeing of its residents. India is a

country with diverse physical attributes, rich cultural heritage and ancient history full of events. Hence tourism is of importance in projecting the Indian heritage to the rest of the world.

A '*Geoheritage*' is concerned with the preservation of features like landforms, natural and artificial exposure of rocks and sites where these features can be studied and examined for their preservation puroposes.

A '*geosite*' is a locality that constitutes a part of geoheritage. A geopark has been described as a territory encompassing one or more sites of scientific importance by UNESCO in its Global Geoparks Network. A geopark is of importance in archaeological, ecological and cultural values. Some of the natural geosites have been described under :

1. Natural springs and glacier lakes

R.D. Oldham in the 19^{th} century published the work of his father T.R.D. Oldham (1882) in the form of an inventory of 300 thermal springs in India. The Ministry of Power and Irrigation constituted a committee on "Hot Springs" in 1963 to study and explore the commercial utilization of the potential of the hot springs. There are about 340 hot springs. The Northwest Himalaya region of Jammu and Kashmir, Himachal Pradesh, Uttarakhand has about 340 hot springs. They are mostly concentrated along 30–50 km wide thermal band occurring along the river valleys. Naga-Lusai and West Coast provinces manifest a series of hot springs.

The thermal springs in India have classified as:

i) Himalayan Province-Territory Orogenic belt with territory magmatism
ii) Areas of Faulted blocks-Aravali belt
iii) Volcanic arc-Andaman and Nicobar arc
iv) Deep Sedimentary basin-Cambay Basin in Gujarat
v) Radioactive basin-Surajkund, Hazaribagh, Jharkhand
vi) Cratonic province-Peninsular India

The hot springs in the Peninsular region of India are related to faults in which meteoric water is in deep circulation .The circulating water acquires heat from the normal gradient of heat in the area (*Aravali range, Son-Narmada-Tapti region, Godavari and Mahanadi and South Cratonic Belts*) and emerges out at suitable localities.

Glacial lakes originate from the glaciers. They are formed when a glacier erodes the land and melts. It then fills the space created by its movement. There are about 2026 glacial lakes and water bodies in the Indian basin of the Himalayas out of which 503 are glacial lakes. The highest number of these lakes is found in the Brahmaputra basin.

A sudden burst of glacial lake produces violent flow of water and the associated debris is known as 'Glacial Lake Outburst flood (GLOF)'. A lake outburst and debris flow disaster occurred in Kedarnath, Uttarakhand in June 2013.

FOSSIL PARKS

Fossil Parks are important as they provide information about the diversity and the evolution of life through time. They help in preserving the fossils and spread awareness to the public about the values of fossils. The protected areas are maintained by the Geological Survey of India, in order to preserve the rocks, minerals, fossils and meteorites.

1. Shivalik Fossil Park (Saketi, Sirmaur district, Himachal Pradesh)

It has life size models of vertebrates that might have inhabitated Shivalik Hills, about 1.5 to 2.5 million years ago. It covers an area of about 1.5 km2 in the Markanda valley of Sirmaur. This park was built to put a check on the destruction of fossil bones. This site has been developed to a panorama of Plio-Pleistocene period (ca. 2.5 million years) through large afforestation.

2. Mandla Plant Fossils National Park (Dindori, Madhya Pradesh)

This national park preserves the fossil remains of primordial forest about 40–150 million years ago. The total area of the park is 0.27

km2 which is spread over in seven villages of Dindori district namely Ghuguwa, Umaria, Deorakhurd, Barbaspur, Chanti Hills, Chargaon and Deori Kohani.

3. Ghughua Fossil Park (Madhya Pradesh)

This national park has a trove of plant fossils belonging to 31 genera of 18 plant families. The fossils indicate to life in the area about 65 million years ago. It has a reservoir of woody plants, climbers, leaves, flowers and seeds. The dicotyledons and palm fossil woods are in abundance. Some fossils of shell bearing animals are also found indicating that the region was humid and received plenty of rainfall.

Fig: 1.11 The Ghughul National Park [*Source: Government of India*]

4. Marine Gondwana Fossil Park, Manendragarh, Sarguja district, Chattisgarh

It has fossiliferous marine Permian rocks of Talchir formation dated to 280–240 million years. It has a spread of 1 km on the right bank of Hasdeo river and Hashia nala

Some other Fossil parks managed by the Geological Survey of India include:

5. National Fossil Wood Park, Tiruvakkarai, Tamil Nadu

It has 200 fossil trees ranging in length from 3–15 m and upto 5 m in girth which are seen as lying horizontally embedded in Cuddalore Sandstone of the Mio-Pliocene age about 20 million years ago. It covers an area of 1 km2. The longest specimen measures 13.4 m×0.9 m. Scientists believe that these trees did not grow at that site but were transported there before they were petrified.

6. National Fossil Wood Park in Sathanur, Tamil Nadu

This wood park contains large trunks of petrified trees (conifers) of Upper Cretaceous age (100 million years ago). These trees were non-flowering and dominated the land vegetation. The fossilized tree trunk at Sathanur measures over 18 m in length.

7. Akal Wood Fossil Park, Jaisalmer district, Rajasthan

The fossils of the wood found in this park reflect the warm and humid climate cordering the sea some 180 million years ago. It is spread over an area of 21 hectares. It contains fossils of wooden logs (petrophyllum, ptyllophyllum, equisetitis species and dicotyledonous wood and gastropod shells of Lower Jurassic period) lying in random horizontal orientation.

8. Stromatolite Park, Bhojunda, Chittaurgarh district, Rajasthan

Stromalites are structures produced by blue green algae. They are stratiform ,columnar and nodular structures in carbonate rocks .They are the result of the combination of life activity and sediment trapping ability of algae and the preying bacteria. Their formation occurs in shallow water where the tides bring the floating sedimentary material and make it flow through carbonate particles. These are the impressions of the earliest life forms.

9. Stromatolite Park, Jharkhand, Udaipur district, Rajasthan

It is the largest and rich in the deposits of phosphorite. This site preserves and provide evidences of early life on the earth. They stretch over a length of 15 km in the rock phosphate within the Precambrian Aravalli Supergroup of rocks. The rock phosphate occurs in dolomite limestone associated with stromatolites which have bluish grey appearance.

Statewise details of the geological heritage sites /national geological monuments declared by the Geological Survey of India are given in the following table

State	Geological heritage sites/National geological monuments
Andhra Pradesh	Volcanogenic bedded barytes, Mangampeta, Cuddapah district
	Eparchaean Unconformity, Chittoor district

State	Geological heritage sites/National geological monuments
	National Geological Arch, Tirumala Hills, Chittoor district
	Erra Matti Dibbalu-Stable coastal red sediments, Vishakapatnam, Bheemunipatnam
Kerala	Laterite near Angadipuram, Malappuram district
	Varkala Cliff Section, Thiruvananthapuram district
Tamil Nadu	Fossil wood near Tiruvakkarai, South Arcot district
	National fossil wood park, Sathanur, Perambalur district
	Charnockite, St. Thomas Mount, Madras
	Badlands of Karai formation with cretaceous fossils along Karai, Perambalur district
Gujarat	Sedimentary structures–Eddy Markings, Kadam Dam, Panchmahal district
Rajasthan	Sendra Granite, Pali district
	Stromatolite Park, Bhojunda, Chittaurgarh district, Rajasthan
	Stromatolite Park, Jharkhand, Udaipur district, Rajasthan
	Akal Wood Fossil Park, Jaisalmer district
	Kishangarh Nepheline Syenite, Ajmer district
	Welded Tuff, Jodhpur district
	Malani Igneous Suite Contact, Jodhpur district
	Great Boundary Fault at Satur, Bundi district
Maharashtra	Lonar Lake, Buldhana district
Chattisgarh	Lower Permian Marine Bed, Manendragarh, Sarguja district
Karnataka	Columnar lava, St. Mary's Island, Udupi district
	Peninsular Gneiss, Lalbagh, Bangalore
	Pyroclastics and pillow lavas, Kolar Gold fields, Kolar district
Himachal Pradesh	Siwalik Fossil Park, Saketi, Sirmur
Odisha	Pillow lava in iron ore belt at Nomira, Keonjhar district
Jharkhand	Plant fossil bearing inter-trappean beds of Rajmahal Formation, upper Gondwana sequence around Mandro, Sahibganj district
Nagaland	Nagahill Ophiolite Site near Pungro

State	Geological heritage sites/National geological monuments
Sikkim	Stromatolite bearing Dolomite/Limestone of Buxa Formation at Mamley near Namchi, South district

Source: Ministry of Mines, Government of India, 2016.

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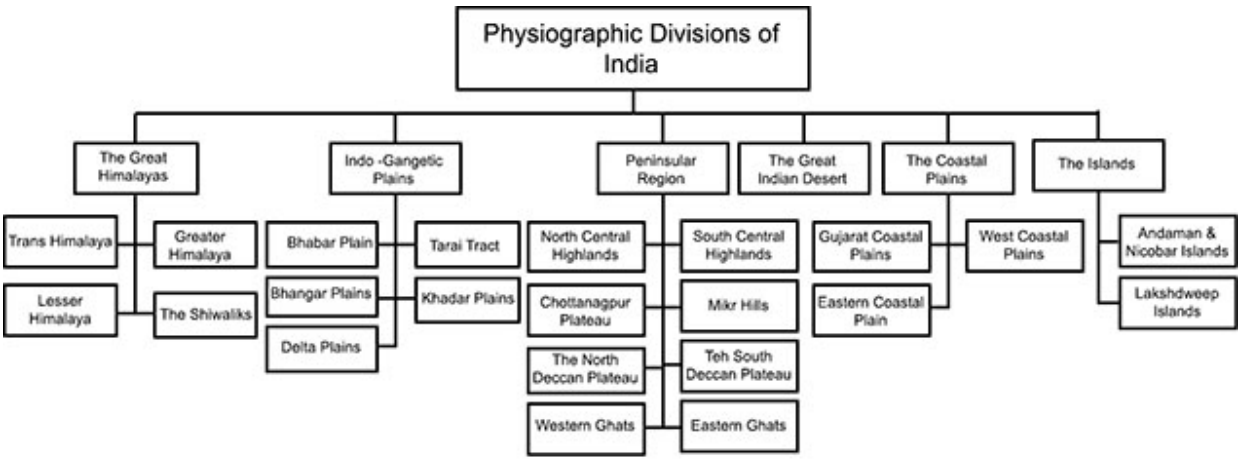
*Approximate stages in the development of earth’s atmosphere

2

Physiography



PHYSIOGRAPHIC DIVISIONS OF INDIA

Physiography deals with the study of the surface features and landforms of the Earth. On the basis of tectonic history, stratigraphy, and physiography, India may be divided into the following four physiographic divisions as shown on the map **(Fig. 2.1)**:

1. The elevated Peninsular region;
2. The mighty Himalaya and their associated young folded mountains;
3. The Indo-Gangetic-Brahmaputra Plains;
4. The Deserts.
5. The Coastal Plains and Islands.

THE HIMALAYAN REGION

The Himalaya consist of four lithotectonic mountain ranges, namely (i) the Trans-Himalaya or the Tethys Himalaya, (ii) the Greater Himalaya, (iii) the Lesser Himalaya, and (iv) the Shiwalik or the Outer Himalaya. The Indian Himalaya extend from the eastern boundary of Pakistan to the border of Myanmar for about 2500 km with a varying width of about 500 km in the west and about 320 km in the east. They lies to the north of the Ganga-Brahmaputra Plains and are separated from the plains by the Himalayan Front Fault (HFF). They include parts of Jammu and Kashmir, Himachal Pradesh, Uttarakhand, Nepal, Sikkim, Bhutan and Arunachal Pradesh. Their offshoots run in a north-south direction along the India-Myanmar boundary through Nagaland, Manipur, and Mizoram.

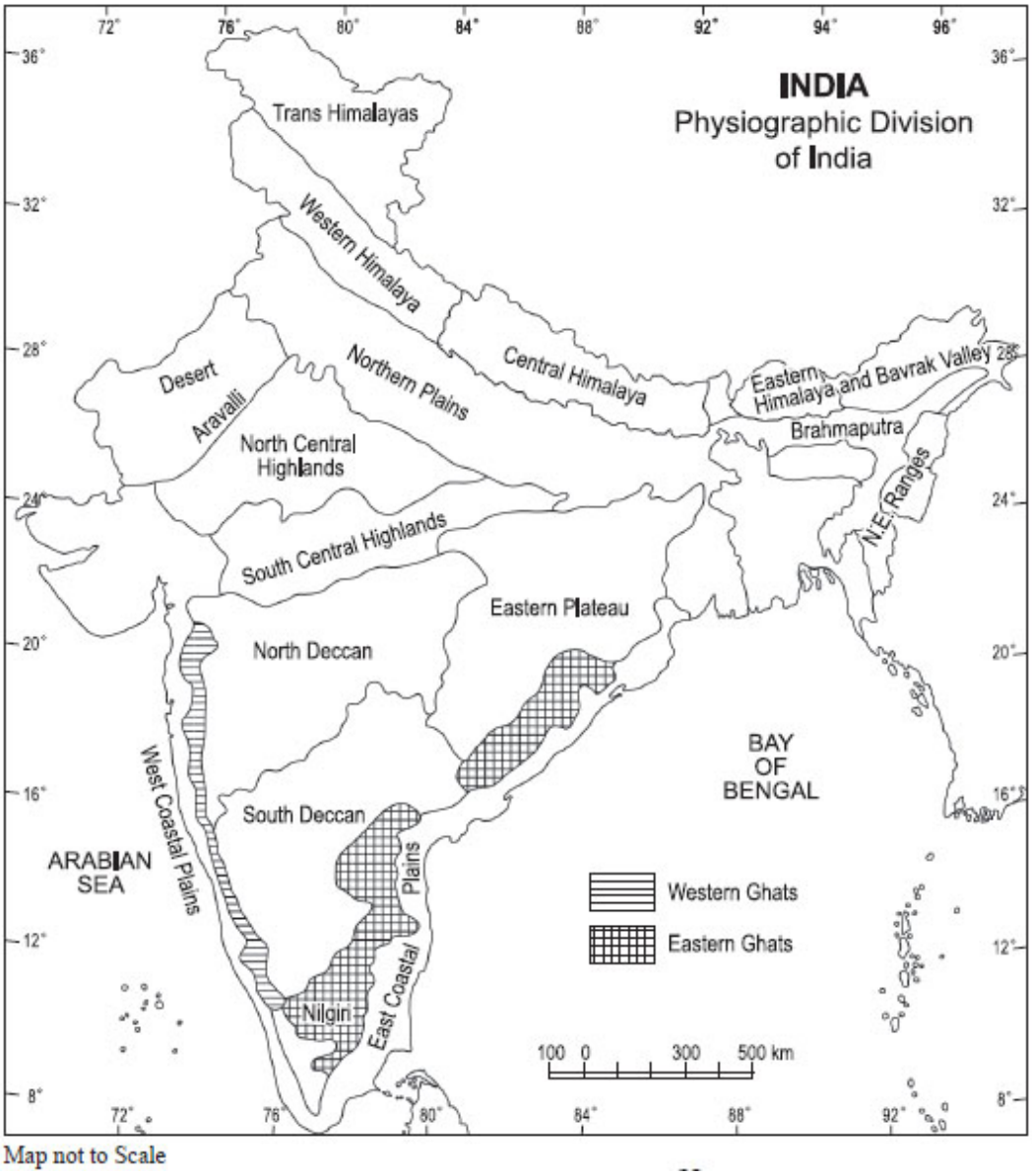


Fig. 2.1 Physiographic Divisions

Origin of the Himalaya

The origin of the Himalaya has been a point of contention among the geologists and geomorphologists. It is a complex mountain system having rocks from the Precambrian and Eocene periods. Mostly formed of sedimentary and metamorphic rocks, it has been subjected to intense folding and faulting. The main theories about the origin of the Himalays are as under:

(i) The Geosynclinal Origin

The main supporters of the geosynclinal origin of the Himalaya are Argand, Kober, and Suess. According to these geologists, the disintegration of Pangaea, about 200 million years back, led to the formation of a long Tethys Sea between the Lauratian Shield (Angaraland) of the north and the Gondwanaland of the south. This sea was occupying the region of Himalaya during the Mesozoic Era (180 million years ago). At the end of the Palaeozoic and beginning of the Mesozoic Eras, the Tethys almost girdled the whole Earth running from Europe in the west to China in the east. Eroded material from the two land masses (Eurasian Shield—Angaraland and Gondwanaland) was deposited in the Tethys Sea and assumed considerable thickness due to the sinking nature of the sea bed **(Fig. 2.2 and Fig. 2.3).** During the Cretaceous Period, the bed of the sea started rising which led to the folding of three successive ranges of the Himalaya. The first upheaval led to the formation of the Greater Himalaya during the Eocene Period (about 65 million years back). Similarly, the second upheaval took place during the Miocene Period (about 45 million years back) resulting in the formation of the Lesser Himalaya, and the third upheaval started in the Pliocene period (about 1.4 million years back) resulting in the formation of the Shiwaliks or the Outer Himalaya**.**

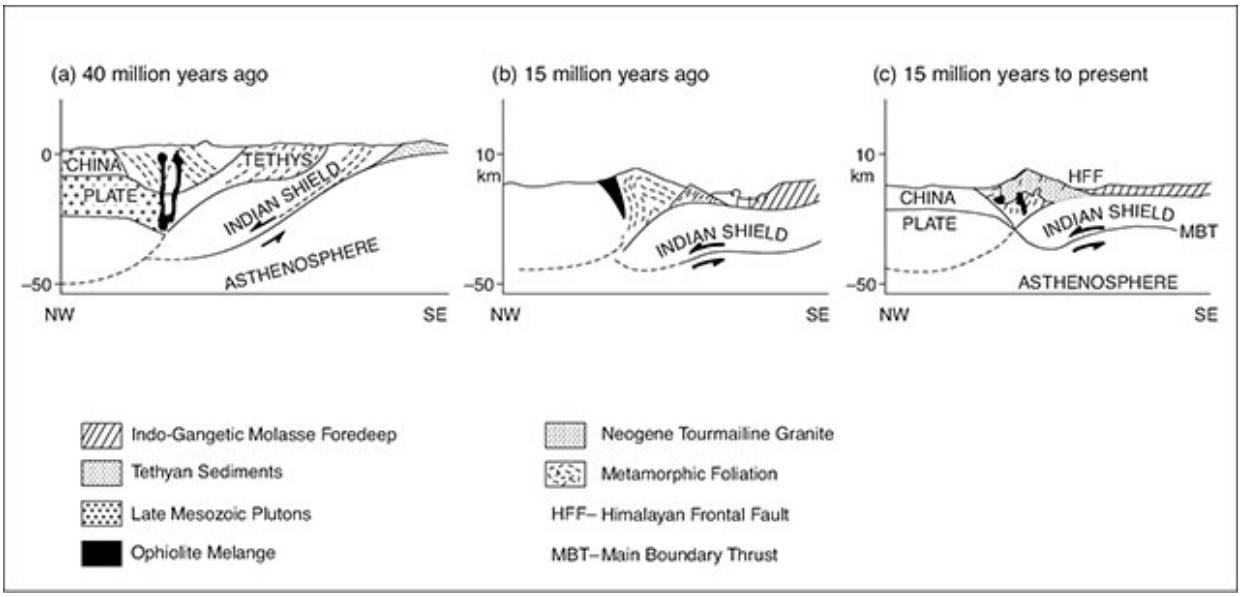


Fig. 2.2 Origin of the Himalaya

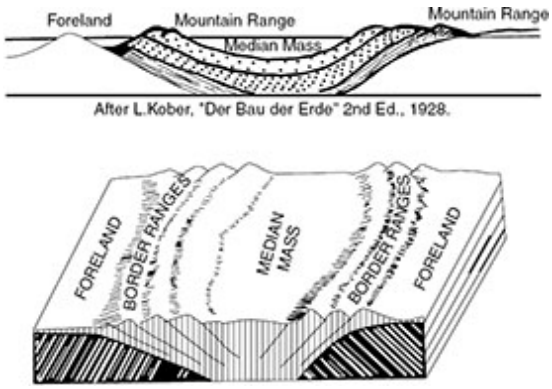


Fig. 2.3 Formation of the Himalaya

(ii) The Plate Tectonic Origin of the Himalaya

The theory of Plate Tectonics was put forward by W.J. Morgan of Princeton University in 1967. This theory is based on the concept of "Sea-Floor Spreading" advocated by H.H. Hess. According to this theory, about 70 or 65 million years ago there was an extensive geosyncline, called the Tethys, in place of the Himalaya. About 65–30 million years ago the Indian plate came very close to the Asian plate and started subducting under the Asian plate **(Fig. 2.4).** This caused lateral compression due to which the sediments of the Tethys were

squeezed and folded into three parallel ranges of the Himalaya. It has been estimated that this convergence has caused a crustal shortening of about 500km
in the Himalayan region and is compensated by sea floor spreading along the oceanic ridge in the Indian ocean region. Since the northward movement of the Indian plate is still continuing, the height of the Himalayan peaks is increasing**.** The Indian Plate is moving northward and the center of rotation is constantly changing. The northward drift of the Indian Plate and the subcontinent of India have been shown in **(Fig. 2.5 and Fig. 2.6).**

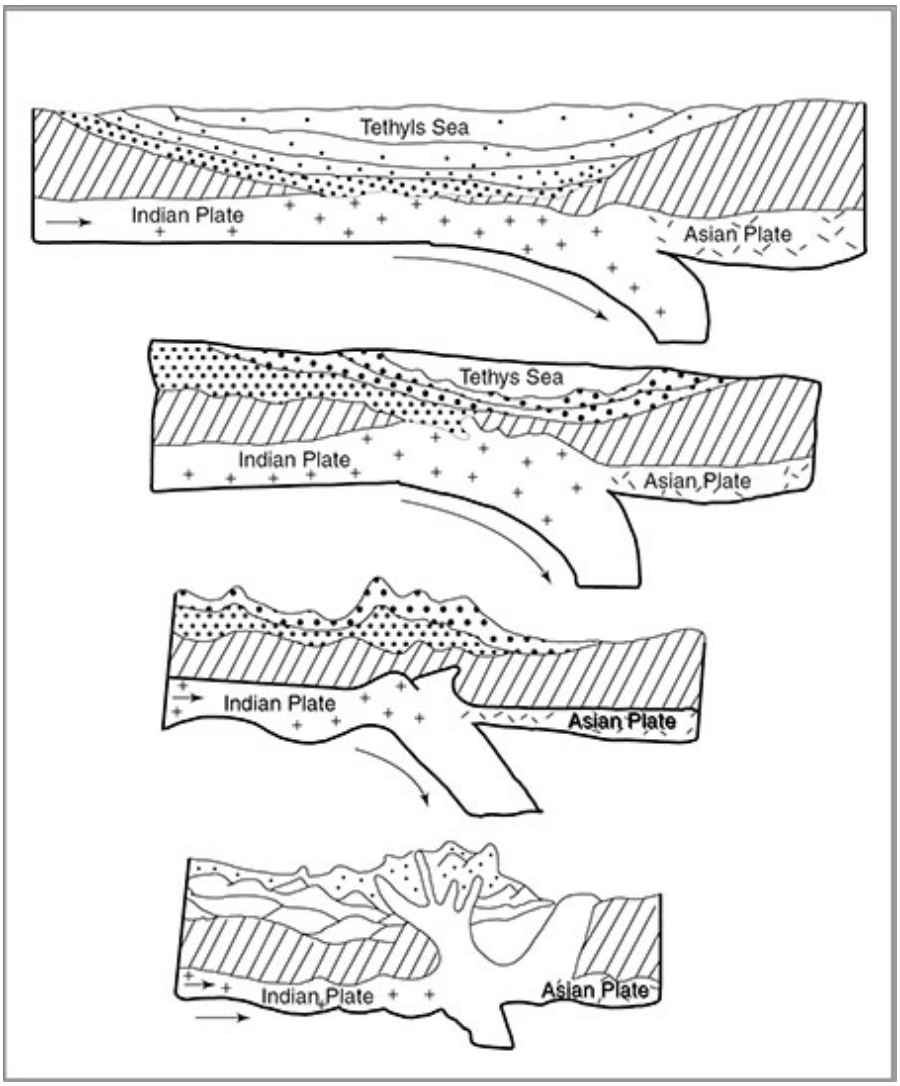


Fig. 2.4 Plate Tectonics and Origin of the Himalaya

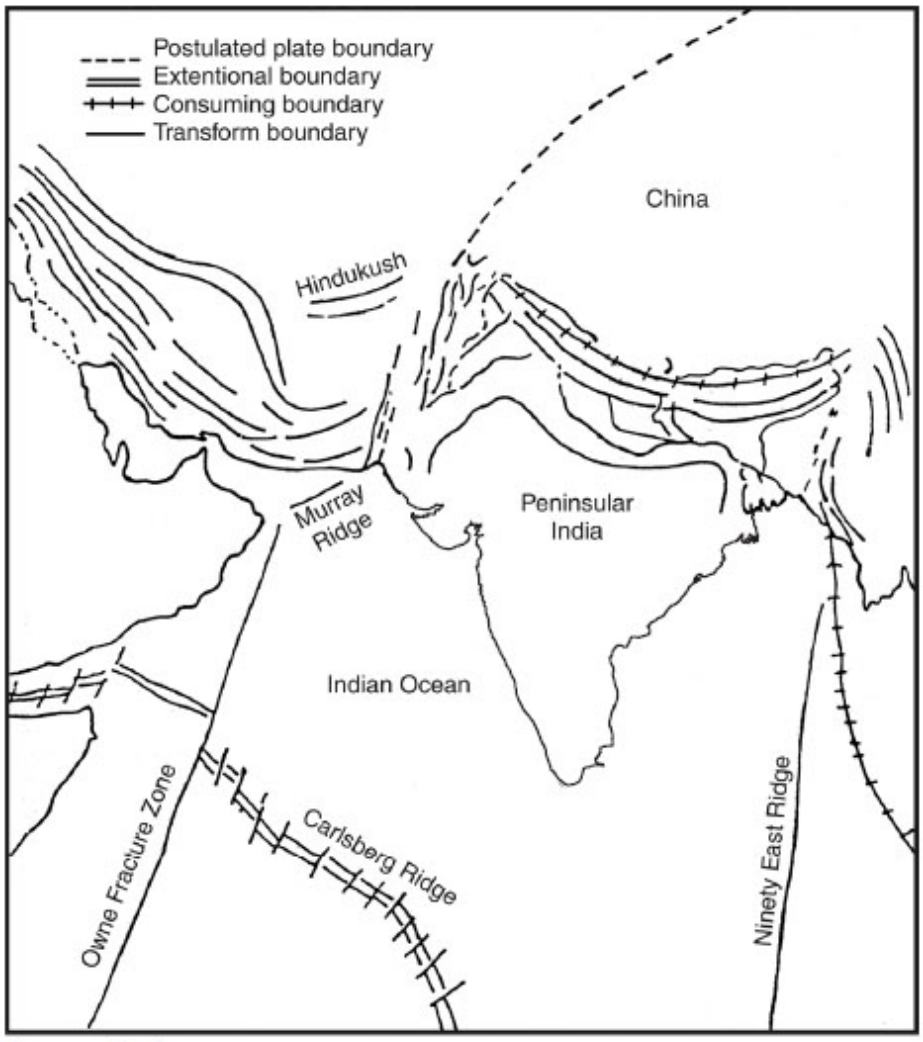


Fig. 2.5 Northward Drift of the Indian Plate

The continent-to-continent collision between the Indian and the Asiatic plates started around 65 million years ago and caused the Himalaya to rise from the Tethys geosyncline. Thus, the first major phase of uplift in the Himalaya occurred around 65 million years ago. This orogenic movement elevated the central axis of ancient crystalline and meta-sedimentary rocks which have been intruded by large masses of granite. It is believed that the first major phase of uplift initially produced the Ladakh and Zaskar ranges of the Trans-Himalaya before the formation of the Great Himalaya. Hence, it is to be realized that except the Kashmir part of the Himalaya, the

Himalayan ranges have not developed from a geosyncline and are made up of elements formerly connected to the marginal parts of the Indian shield. During the main Himalayan orogeny, this continuous geosynclinal sedimentation led to the underthrusting of the Indian shield against the Tibetan Massif which buckled down the geosynclinal deposits, resulting in the outflow of a large amount of ultrabasic rocks known as ophiolites. These ophiolites are seen as exotic blocks on the Ladakh and Zaskar Ranges of the Trans-Himalaya. The end effect of the buckling of the geosyncline was not only the crustaline thrust effect on Ladakh and Zaskar leading to their rise as ranges, but also the creation of the sharp tectonic line of the Indus suture along which large geosynclinal areas disappeared.

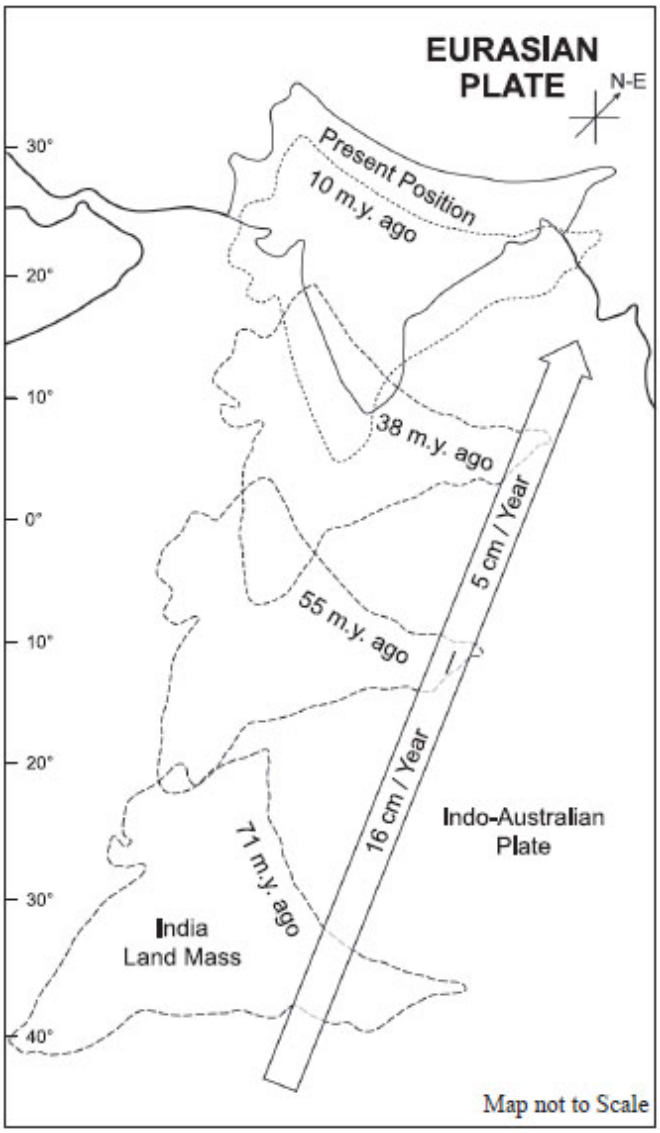


Fig. 2.6 Northward Drift of the Indian Subcontinent [*Source: Himalaya Formation www.usgs.org (US Government)*]

The intermontane basins in the Indus suture zone of Ladakh continued to receive molasses sedimentation in this period. The second major uplift which took place around 45 million years ago, caused the rapid uplift of the southern mountain front of the Lesser Himalaya, giving rise to the extremely rugged and youthful Pir-Panjal, Dhauladhar, Karol, and Mahabharat Ranges abruptly and steeply. The Greater Himalaya and the Lesser Himalaya are separated by the Main Central Thrust (MCT). These spurs of the Lesser Himalaya

again formed, in their turn, the intermontane basins of Kashmir, the Karol-basin, Dun Valley (Uttarakhand), and the Kathmandu Valley of Nepal. The foredeep which was formed further away received the thick sequence of terrestrial sediments called Shiwaliks from the middle-Miocene to the middle-Pleistocene periods, covering a span of about 1.4 million years. The Lesser Himalaya and the Shiwaliks are separated from each other by the Main Boundary Thrust (MBT). The 5000 m thick Shiwaliks dominated by boulder and conglomerate, reflect the progressive uplift of the Himalaya from which they have been derived as a result of the third major phase of uplift. The Shiwaliks are separated from the Northern Plains of India by the Himalayan Front Fault or HFF **(Fig. 2.7 and Fig. 2.8).**

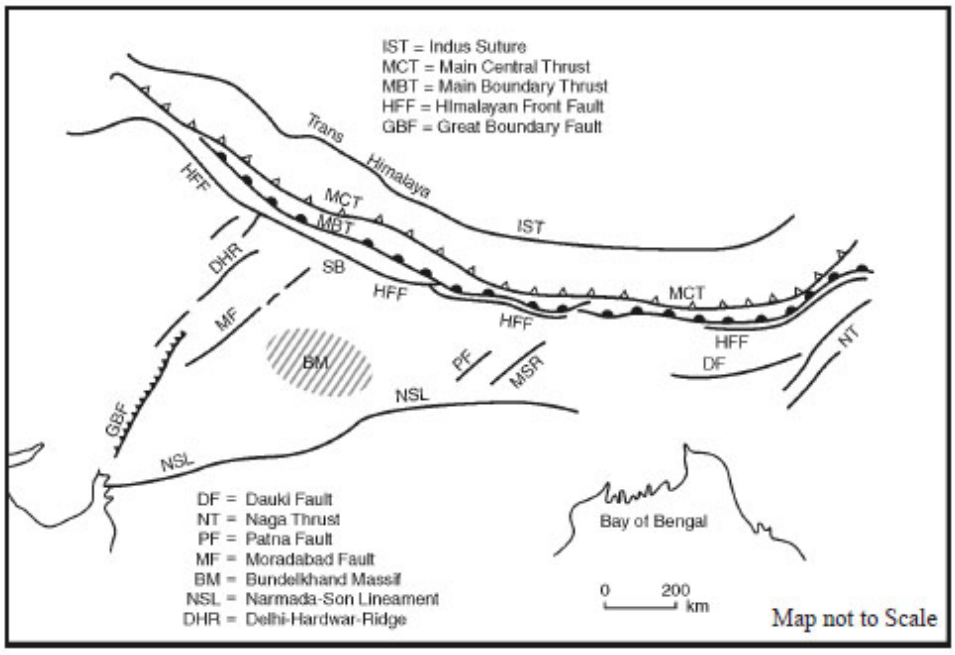


Fig. 2.7 Major Faults of the Himalaya

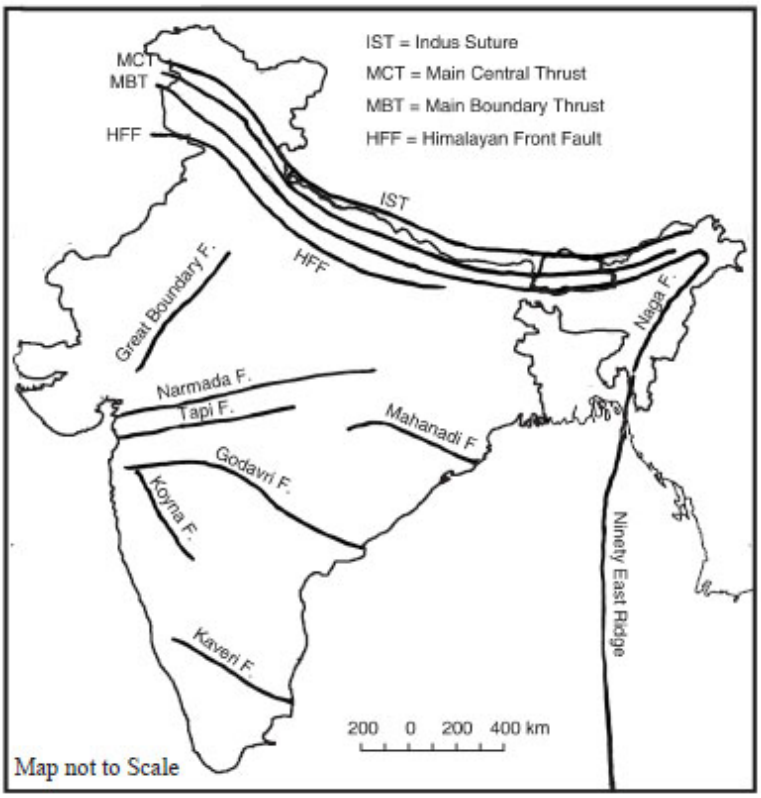


Fig. 2.8 Major Faults of India

The Shiwaliks form the normal Jura type of structures with wider basin-like synclines alternating with steep, often faulted, asymmetric anticlines. At present, the Himalayan Front Fault (HFF) is quite active recording frequent tremors and earthquakes**.**

Physiographic Divisions of the Himalaya

For a systematic study of the physiography and relief, the Himalaya may be divided into the following four divisions from north to south:

1. The Trans-Himalaya
2. The Greater Himalaya
3. The Lesser Himalaya and
4. The Shiwaliks or the Outer Himalaya.

1. The Trans-Himalaya

The Trans-Himalaya are about 40 km wide. They contain the Tethys sediments. The rocks of this region contain fossils bearing marine

sediments which are underlain by “Tertiary granite”. It has partly metamorphosed sediments and constitutes the core of the Himalayan axis. It has a great accumulation of debris in the valleys of defeated streams which could not maintain their southerly course across the rising barrier of the Himalaya **(Fig. 2.9).**

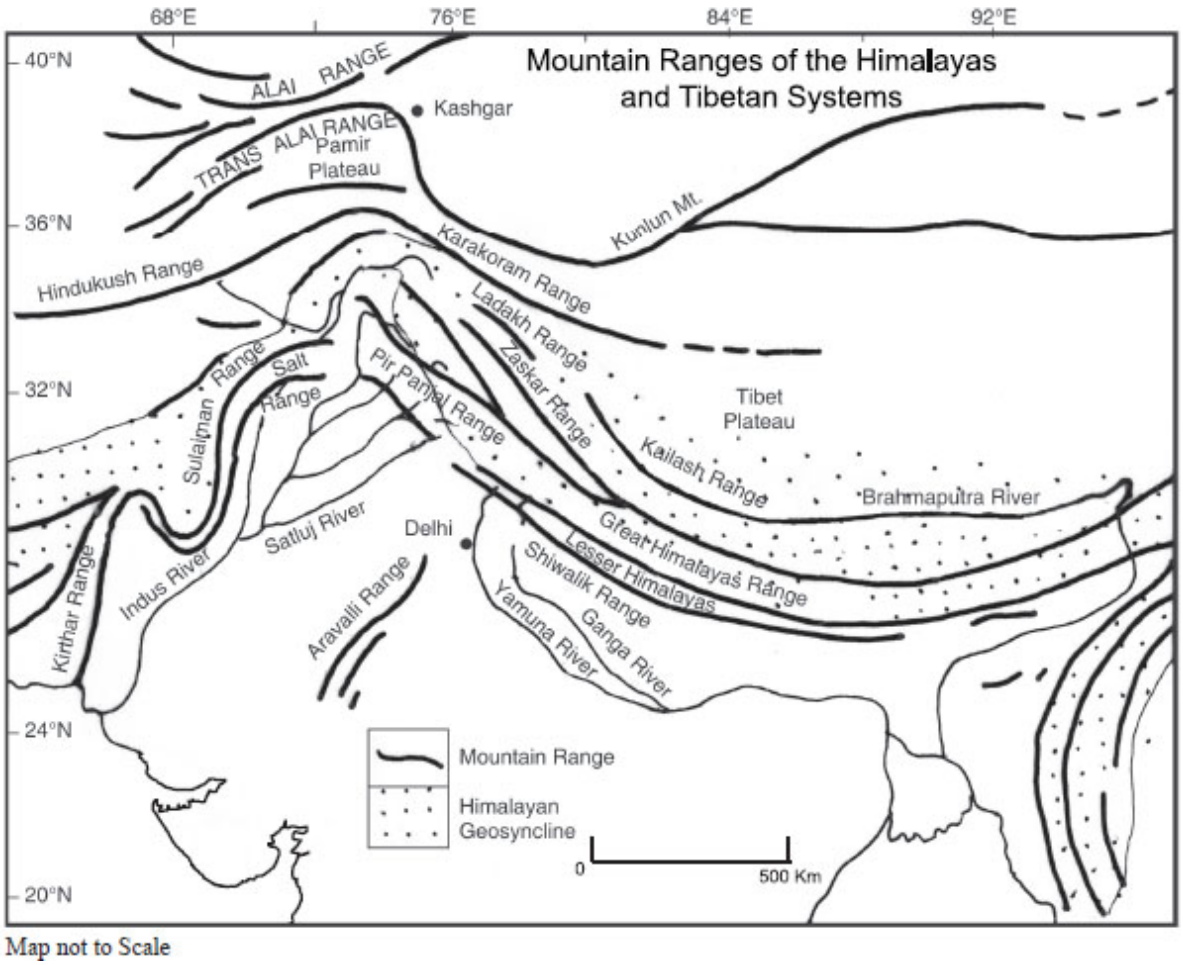


Fig. 2.9 Physiographic Divisions of the Himalayan Region [*Source: Concise Atlas of the world, DK, Penguin Company*]

2. The Greater Himalaya

The Greater Himalaya rise abruptly like a wall north of the Lesser Himalaya. The MCT separates the Greater Himalaya from the Lesser Himalaya. The Greater Himalaya are about 25 km wide with an average height above 6100 m (Wadia, D.N.). Almost all the lofty peaks of the Himalaya, Mt. Everest, Kanchanjunga, Nanga-Parbat,

Gasherbrum, Manaslu, Dhaulagiri, Annapurna, Gosainthan, Cho-Cyu, Nanda-Devi, Kamet, Badrinath, Nanda Devi, etc., lie in this zone. The Greater Himalaya are composed of crystalline, igneous or metamorphic rocks (granite, schists, and geneiss). The basal complex of the Himalaya is Archaean. At places, due to heavy thrust, older rocks are found overlying the newer rocks. The Greater Himalaya are almost a contiguous range. The range has very few gaps mainly provided by the antecedent rivers. The Greater Himalaya receive less rainfall as compared to the Lesser Himalaya and the Shiwaliks. Physical weathering is pronounced. Erosion is, however, less effective over the Greater Himalaya as compared to the Lesser Himalaya. Being lofty, they have very little forest area.

3. The Lesser Himalaya

The width of the Lesser Himalaya is about 80 km with an average height of 1300–4600 m. It consists, generally, of unfossiliferous sediments or metamorphosed crystalline. The main rocks are slate, limestone and quartzites. Along the southern margin of the Lesser Himalaya lies the autochthonous belt of highly compressed Upper Palaeozoic to Eocene rocks, often containing volcanic material. Examples of autochthonous belts are found between Murree and Panjal thrust in Kashmir, Giri thrusts in the Shimla region and Krol and MBT in Garhwal region. This region is subjected to extensive erosion due to heavy rainfall, deforestation, and urbanisation.

4. The Shiwaliks or Outer Himalaya/Sub-Himalaya

The Shiwaliks extend from Jammu Division of Jammu and Kashmir territory to Assam. In width, Shiwaliks vary from 8 km in the east to 45 km in the west with an average elevation of about 900–1500 m above sea level. It is not a continuous range. It is broader in the west and narrows down in the east. Between the Shiwaliks and the Lesser Himalaya are longitudinal valleys called Doons/Duns. Some of the important Duns are Dehra Dun, Potli, Kothri, Kathmandu, Chumbi, and Kyarda. The Shiwaliks are mainly composed of sandstones, sand-rocks, clay, conglomerates and limestones, mostly belonging to the Upper Tertiary Period.

Longitudinal Divisions of the Himalaya

The Himalaya have also been divided by Sir S. Burrard into four divisions, namely (i) The Western Himalaya, (ii) The Kumaun Himalaya, (iii) The Nepal Himalaya, (iv) The Assam Himalaya. Prof. S.P. Chatterjee (1973), divided the Himalaya into the following six transverse divisions from west to east (**Fig. 2.9, Fig. 2.10(a)** and **Fig. 2.10(b)):**

1. The Kashmir Himalaya
2. The Himachal Himalaya
3. The Kumaun Himalaya
4. The Central Himalaya
5. The Eastern Himalaya

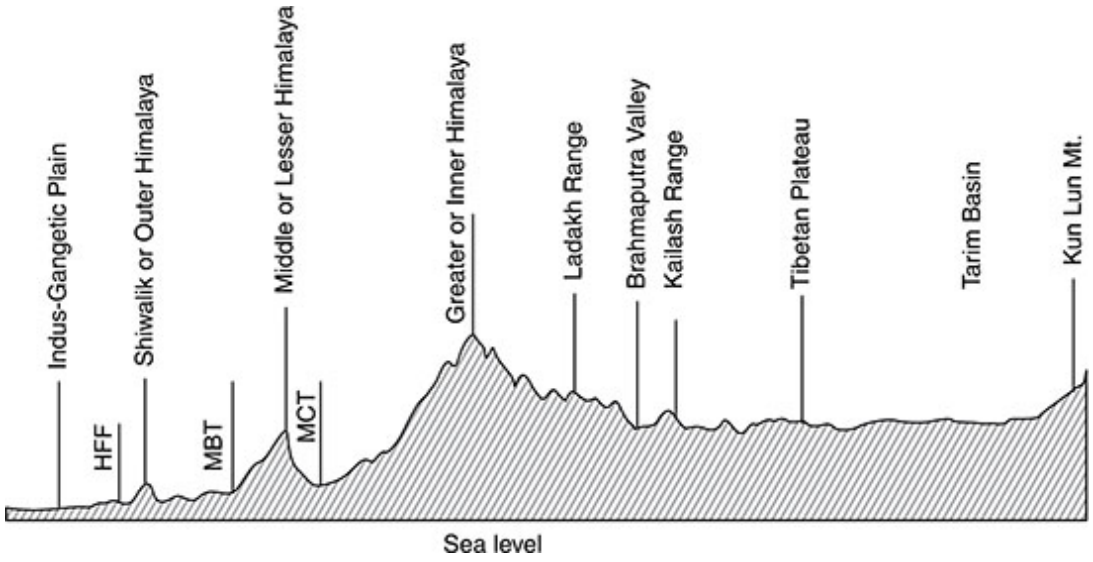
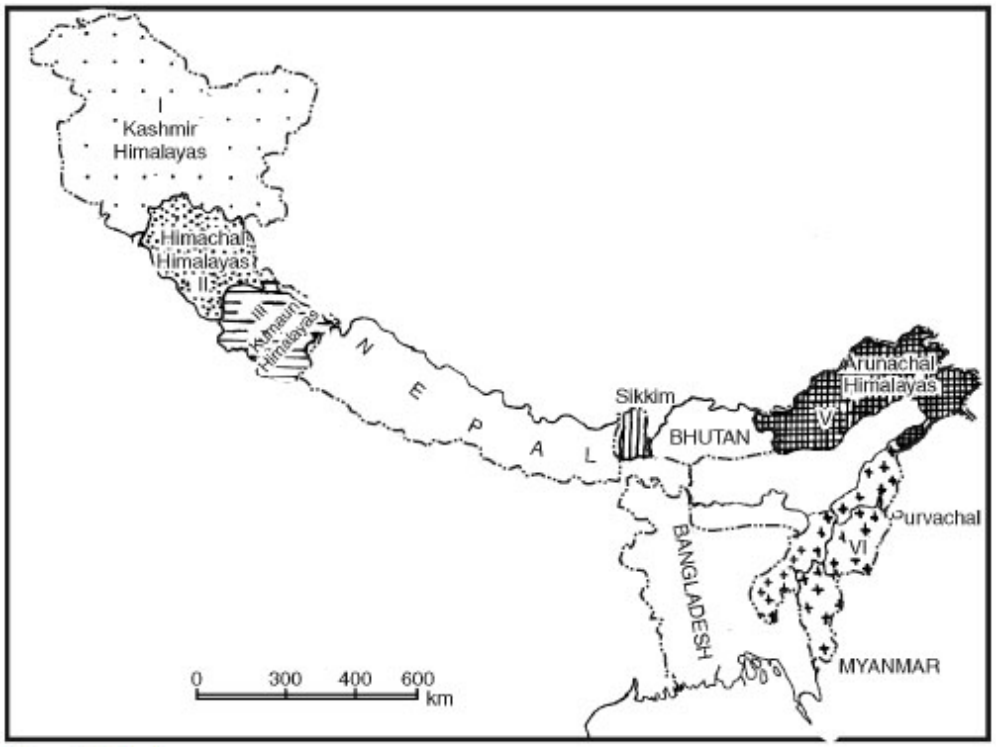


Fig. 2.10(a) Himalayan Complex



Map not to Scale

Fig. 2.10(b) Himalayan—Longitudinal Divisions

1. The Kashmir Himalaya

Sprawling over an area of about 350,000 sq km in the territory of Jammu and Kashmir, the range stretches about 700 km in length and 500 km in width. With an average height of 3000 m, it has the largest number of glaciers in India. The Ladakh region of the Kashmir Himalaya is characterised by cold desert conditions. Ladakh is one of the loftiest inhabited regions of the world (3600–4600 m). The gorge of Gilgit is 5200 m in height above the sea level of the water at its bed. Surrounded by the Greater Himalaya and the Lesser Himalaya is the Kashmir Valley. Having a height of 1585 m above the sea-level, the total area of the Kashmir Valley is about 4920 sq km. It is a structural longitudinal ”Dun” (D.N. Wadia). A special feature of the the Vale of Kashmir is the *Karewa* (lacustrine) deposits consisting of silt, sand and clay. These karewas are mainly devoted to the cultivation of saffron and have orchards of apple, peach, almond, walnut and

apricot. The Himalaya in Kashmir by high snow covered peaks, deep valleys, interlocked spurs and high mountain passes. Pir-Panjal, Banihal (Jawahar Tunnel), Zoji-La, Burzil, Khardungla, Pensi-La, Saser-La, Lanak-La, Jara-La, Taska-La, Chang-La, Umasi-La, and Qara-Tagh-La (Karakoram) are the important mountain passes of the Kashmir Himalaya**.**

The Himadri: Known as the abode of gods, this section of the Himalaya has many snow capped peaks, such knows as Nanga-Parbat (8119 m), Nanda Devi 7817 m), Trisul (7140 m), Nunkun (7135 m), Kamath (7756 m), etc.

2. The Himachal Himalaya

Stretching over Himachal Pradesh, it occupies an area of about 45,000 sq km. All the three ranges (the Greater, the Lesser, and the Outer Himalaya) are well represented in this region. The northern slopes of the Himachal Himalaya are clothed with thick forests and show plains and lakes, while the southern slopes are rugged and forest clad. Rohtang, Bara-Lacha, and Shipki-La are the important passes which join Himachal Pradesh with Tibet (China). The beautiful and highly productive valleys of Kangra, Kullu, Manali, Lahul, and Spiti lie in Himachal Pradesh. These valleys are well known for orchards and scenic beauty. Shimla, Dalhousie, Chamba, Dharamshala, Kullu-Manali are the important hill stations of this region.

3. The Kumaun Himalaya

The Kumaun Himalaya lie between the Satluj and the Kali rivers, stretching to a length of 320 km
and occupying an area of about 38,000 sq km. Its highest peak is Nanda Devi (7817 m). Among the other peaks Kamet (7756 m), Trisul (7140 m), Badrinath (7138 m), Kedarnath (6940 m), Dunagiri (7066 m), Jaonli or Shivling (6638 m), Gangotri (6615 m), and Bandarpunch (6320 m) are important. Gangotri, Milam, and Pindar are the main glaciers of Uttarakhand. The important hill stations include Mussorrie, Nainital, Ranikhet, Almora, and Bageshwar. The Kumaun Himalaya are connected to Tibet by a number of passes namely, Thaga-La

Muling-La (5669 m), Mana Pass, Niti Pass, (5068 m), Tun-Jun-La, Shalsal Pass, Balcha Dhura, Kungrinbingri Pass, Lampiya Dhura, Mangsha Dhura and Lipu Lekh.

4. The Central Himalaya

This range stretches from river Kali to river Tista for about 800 km occupying an area of about 116,800 sq km. A major part of it lies in Nepal except the extreme eastern part called Sikkim Himalaya and in the Darjeeling District of West Bengal. All the three ranges of the Himalaya are represented here. The highest peaks of the world like Mt. Everest (8850 m), Kanchenjunga (8598 m), Makalu (8481 m), Dhaulagiri (8172 m), Annapurna (8078 m), Manaslu (8154 m) and Gosainath (8014 m) are situated in this part of the Himalaya. It has very few passes. The passes of Nathu-La and Jelep-La (4538 m in Sikkim) connect Gangtok (Sikkim) with Lhasa (Tibet, China).

Kanchenjunga: Situated on the border of Sikkim and Tibet, it is the third highest mountain peak in the world. It is 8,598 m above sea level and remains snow covered throughout the year. Some of the important rivers of India like Kosi and Tista have their origin in this mountain.

5. The Eastern Himalaya

These lie between the Tista and the Brahmaputra rivers, covering a distance of about 720 km with an area of 67,500 sq km. The Eastern Himalaya occupy the state of Arunachal Pradesh (India) and Bhutan. In this part, the Himalaya rise very rapidly from the plains of Assam, and the foothills of Shiwaliks are very narrow. The Eastern Himalaya include the Aka Hills, the Dafla Hills, Miri Hills, Abor Hills, Mishmi Hills, and Namcha Barwa (7756 m). It has a number of mountain passes among which Bomdi-La, Bom La, Tunga, Yonggyap, Diphu, Pangsau, Tse-La, Dihang, Debang (Arunachal Pradesh) are the most important. In the Eastern Himalaya, due to heavy rainfall, fluvial erosion is quite pronounced.

On the southern border of Arunachal Pradesh, the Himalaya take a southerly turn and the ranges are arranged in a north-south direction. Passing through the states of Arunachal Pradesh (Tirap Division)

Nagaland, Manipur, Tripura, and Mizoram, the Himalaya are locally known as Purvanchal. The main hills of the Eastern Himalaya are Patkai-Bum (Arunachal Pradesh), Naga-Hills (Nagaland), Manipur Hills, Blue Mountains (Mizoram), Tripura Range, and Brail range. On the border of Nagaland and Myanmar lies the Arakanyoma. These hills are heavily forested. Northern Myanmar is connected through Diphu, Hpungan, Chaukan, Pangsau, and Likhapani (Arunachal Pradesh). Southwards, a pass joins Imphal (Manipur) with Mandalay (Myanmar). The Purvanchal is joined by the Meghalaya Plateau in the west. The extension of the Myanmar mountain chain continues southward up to Andaman and Nicobar Islands and even up to the Archipelago of Indonesia.

The Syntaxial Bends of the Himalaya

The general east-west trend of the Himalaya terminates suddenly at its western and eastern extremities and the ranges are sharply bent southward in deep knee-bend flextures which are called syntaxial bends. The western syntaxial bend is near Nanga Prabat where the Indus has cut a deep gorge. The geological formations here take sharp hairpin bends as if they were bent round pivotal points obstructing them. There is a similar hair-pin bend in Arunachal Pradesh where the mountains take a sharp bend from the eastern to southern direction after crossing the Brahmaputra river. The tectonic strike also undergoes a deep knee-bend from an easterly to southerly trend **(Fig. 2.11).**

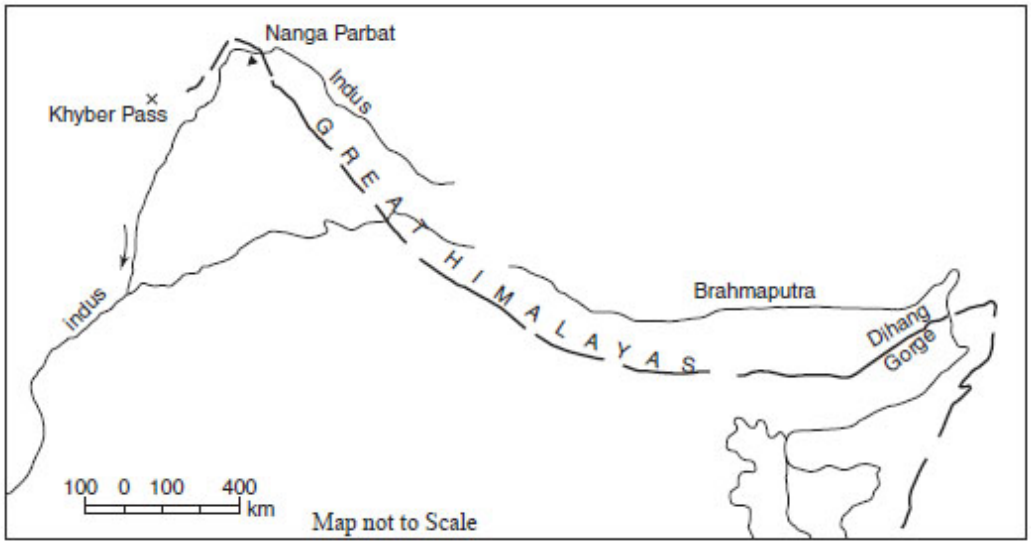


Fig. 2.11 The Syntaxial Bends of Himalaya [*Source: Fundamentals of Physical Geography; Majid Husain*]

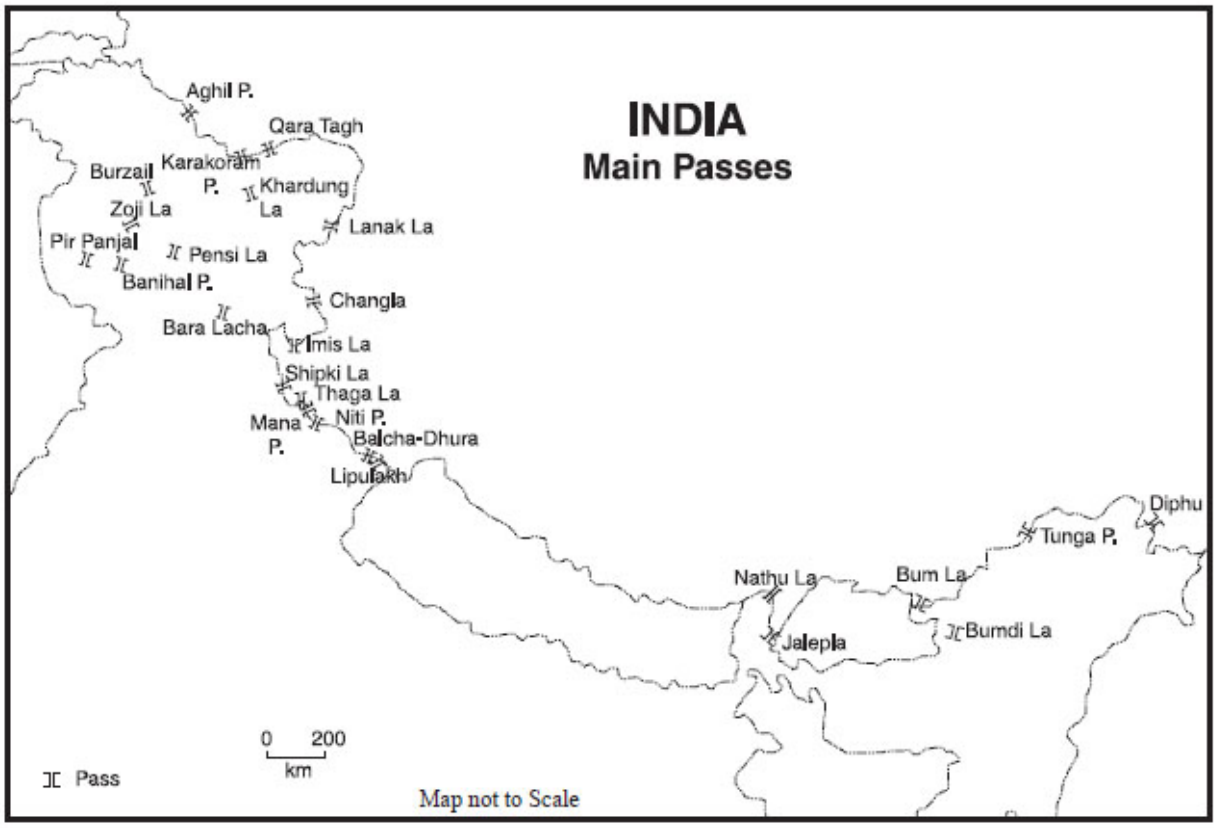


Fig. 2.12 Main Passes of the Himalaya

Main Mountain Passes of Himalayas

A mountain pass is a route through a mountain range or over a ridge. A pass plays a significant role in trade, war or migration.

Fig. 2.13 Zoji La Pass (Kashmir) *(Source: The Tribune; May 2019)*

Table 2.1

Main Mountain Passes

Name of the Pass	Altitude Above the Sea Level	Location	Significance
Aghil Pass	5000 m	Situated to the north of K2 in Karakoram	It joins Ladakh with Xinjiang (Sinkiang) province of China It remains closed during winters from November to the first week of May
Banihal Pass	2835 m	It is located in the Pir Panjal range on NH1A that has been numbered as NH44	It joins Jammu with Srinagar and it has Jawahar Tunnel (2.85 km long). The tunnel was inaugurated in December 1956
Bara Lacha	4843 m	Himachal Pradesh	It connects Manali and Leh It is snow covered from November to mid-May
Bomdi La	4331 m	Arunachal Pradesh. It is situated to the east of Bhutan in Greater Himalayas	It connects Arunachal Pradesh with Lhasa (capital of Tibet)

Name of the Pass	Altitude Above the Sea Level	Location	Significance
Burzail Pass	>5000 m	Srinagar with Kishan-Ganga Valley	It connects Kashmir Valley with the Deosai Plains of Ladakh It remains closed for trade and transport during winters
Chang-La	>5270 m	In Greater Himalayas Ladakh with Tibet	Road leads to town of Tangste Pass has a temple dedicated to Chang La Baba
Debsa Pass	5270 m	High Mountain Pass in Greater Himalayas between the Kullu and Spiti districts of Himachal Pradesh	Pass provides easier and shorter alternative to the traditional Pin Parbati Pass route between Kullu and Spiti
Dihang Pass	4000 m	Situated in Himachal Pradesh	It connects Arunachal Pradesh with Mandalay (Myanmar)
Diphu Pass		Situated in the eastern part of Arunachal Pradesh	Connects Arunachal Pradesh to Mandalay (Myanmar) It remains open throughout the year
Imis La	>4500 m	Ladakh Union Territory and Tibet in China	It connects Ladakh to Tibet (China) It has difficult terrain and steep slopes
Khardung La	>6000 m	Near Leh in Ladakh region	It connects Leh with Siachen Glacier Highest motorable pass
Khunjerab Pass	>5000 m	Karakoram Mountains in Ladakh	Connects Ladakh to Sinkiang Province of China Snow covered from November to mid-May
Likhapani	>4000 m	Arunachal Pradesh	Joins Arunachal Pradesh with Myanmar Open for trade throughout the year
Lipu Lekh		Pithoragarh district, Uttarakhand	Connects Uttarakhand to Tibet Facilitates the pilgrims for Manasarovar lake India’s important border post for trade with China

Name of the Pass	Altitude Above the Sea Level	Location	Significance
Mana Pass	5611 m	Greater Himalayas	Connects Uttarakhand to Tibet It remains snow covered for 6 months during the winters
Mangsha Dhura Pass	5000 m	Pithoragarh district of Uttarakhand	It connects Uttarakhand to Tibet Pilgrims for Manasarovar cross this pass Landslides are major problem
Muling La		Situated North to Gangotri, Uttarakhand	It connects Uttarakhand to Tibet
Nathu La	4310 m	Situated at Indo-China border, Sikkim	It forms an offshoot of the ancient Silk Road It is one of the three trading border posts between India and China It was reopened in 2006 after the 1962 war
Niti Pass	5068 m	Uttarakhand	It joins Uttarakhand to Tibet
Pangsau Pass	>4000 m	Arunachal Pradesh	It connects Arunachal Pradesh to Mandalay (Myanmar)
Pensi La	>5000 m	Greater Himalayas	It connects Kashmir Valley to Kargil (Ladakh)
Pir Panjal Pass	3494 m	Mughal Road	Traditional Pass between Jammu and Kashmir It provided the easiest shortest metalled road access but was closed down after partition of the subcontinent
Qara Tagh Pass	5540 m	Located in Karakoram Mountains	Highest Pass on ancient caravan route between Leh and Yarkand in Tarim Basin It was an offshoot of the Great Silk Road
Rohtang Pass	3979 m	Eastern side of Pir Panjal Range, 51 km from Manali	It connects Kullu, the Lahaul and Spiti valleys of Himachal Pradesh The road access is very good and was built by Border Road Organization (BRO)

Name of the Pass	Altitude Above the Sea Level	Location	Significance
Shencottah-Gap	>4300 m	Located through the Satluj Gorge	It joins Himachal Pradesh with Tibet River Satluj enters India through this Pass It is India's third border post for trade with China
Shipki La	3930 m	Himachal Pradesh	It connects Himachal Pradesh to Tibet (National Highway 5) Important border post at Indo-China border River Satluj flows through it
Thang La	5359 m	Ladakh	It is the second highest motorable pass in India after Khardung La
Traill's Pass	5212 m	Uttarakhand. Situated at the end of the Pindari Glacier in the Pithoragarh and Bageshwar districts of Uttarakhand	It links Pindari Valley to Milam Valley Very steep and rugged terrain makes it a difficult pass to be crossed
Zoji La	3528 m	Kashmir (100 km from Srinagar)	It joins Srinagar with Kargil and Leh Faces heavy snowfall Closed from December to mid-May Beacon force of BRO maintains the road Recently declared as National Highway (NH-1D), the centre has approved Rs. 6809 crores to build Asia's longest bi-directional Zoji La Pass Tunnel which will reduce the travel time between Srinagar and Leh to 15 min from 3.5 h

Glaciers and Snowline

A glacier is a body of dense ice that drifts under its own weight. A glacier is formed by accumulation of snow exceeding its ablation point over many years. It is the largest reservoir of fresh water on the earth (about 75%).

The lower limit of perpetual snow is known as “snowline”. The snowline in the Himalaya has different heights in different parts, depending on latitude, altitude, amount of precipitation, moisture, slope and local topography. The snowline in the Eastern Himalayas and Kumaon Himalayas is about 3500 km in altitude and is 2500 km in the western Himalayas. The difference in the snowline is attributed to the increase in latitude from 28° N in Kanchenjunga to 36° N in the Karakoram. In the Himalaya, glaciers (snowfields) cover about 40,000 sq km of area between Karakoram (J & K) and Arunachal Pradesh. There are about 15,000 glaciers in the Himalaya lying between the two syntaxial bends in the east and the west. In the Arunachal Himalaya, the snowline is about 4400 m, whereas in the Kashmir Himalaya it varies between 5200 to 5800 m. In the Kumaun Himalaya the snowline is about 5100 m and about 5500 m in the Karakoram **(Table 2.2)**. On the Tibetan side, the altitude of the snowline is about 900 m higher owing to great desiccation of the region and scarcity of moisture. Thus, there is a direct relationship between the presence of moisture and the altitude of the snowline. In general, more the moisture in the atmosphere, lower the altitude of the snowline and vice versa.

Table 2.2

Altitude of Snowline in the Himalaya

S. No	Himalayan Region	Altitude of Snowline
1.	North Eastern Himalaya (Arunachal Pradesh)	4400 m
2.	Kashmir Himalaya	5200 m to 5800 m
3.	Kumaun Himalaya	5100 m to 5500 m
4.	Karakoram	5500 m and above

The main glaciers in the northern mountains are found in the Greater Himalaya and the Trans-Himalayan mountains (Karakoram, Ladakh and Zaskar). The Lesser Himalaya have small glaciers, though traces of large glaciers are found in the Pir-Panjal and

Dhauladhar ranges. Some of the important glaciers of the Karakoram and the Himalaya are given in **(Table 2.3). and (Fig. 2.14)**

Table 2.3

Main Glaciers of the Himalayan Region

Name of The Glacier	Location	Length (Km)	Altitude (M)	Important Features
Siachen	Karakoram	76	5400	Largest in Nubra Valley of the Himalaya and second longest glacier outside the Polar region
Biafo	Karakoram (Baltistan region of Ladakh)	60	5128	Flows into Shigar River
Chogo Lungma	Karakoram	50	5016	Located in Gilgit Baltistan
Gangotri	Uttarakhand	28	3415	Originates below the Chaukhamba Peak and it is also known as 'Gomukh'
Rimo	Siachen Area of Ladakh	40	7385	Drains in the Shyok River
Hispar	Gilgit-Baltistan	63	5128	World's longest glacial system
Zemu	Sikkim/Nepal	26	300	Largest glacier in Eastern Himalaya and it feeds River Teesta
Drang Drung	Himalayan Range	23	4780	It lies in the Zanskar range in Kargil district of Jammu and Kashmir
Shafat	Eastern Himalayan Range	14	4400	Gives rise to the peaks of Nun and Kun The melted water flows into Suru River
Sonapani	Chandra Valley of Lahaul and Spiti region	15	4000	Longest glacier of the Pir Panjal range near the Rohtang Pass

Name of The Glacier	Location	Length (Km)	Altitude (M)	Important Features
Milam	Uttarakhand	15	2290	Major source of River Gori Ganga Biggest glacier in Kumaon Himalaya
Rupal	Kashmir	15	6326	Located in Greater Himalaya it flows northeastward
Gangri	Nun Kun Mountains; Himalayan Range	13	7135	Largest glacier in the Nun Kun peak
Bara Shigri	Chandra Valley of Lahaul in Himachal Pradesh	11	~4570	Second longest glacier in Himalaya Source of water for River Chenab It has deposits of antimony
Diamir (Nanga Parbat)	Kashmir (Pakistan Occupied)	11	4203	Diamir mountains are known as ‘King of Mountains’ and are the World’s tallest mountains
Pindari	Kumaon Region of Uttarakhand	3	3353	Gives rise to River Pindari
Chong Kumdan	Karakoram	21	7071	Ladakh; they feed and are a threat to Shyok River and might block it
Hispar	Karakoram	60	5128	Feeds Shigar River

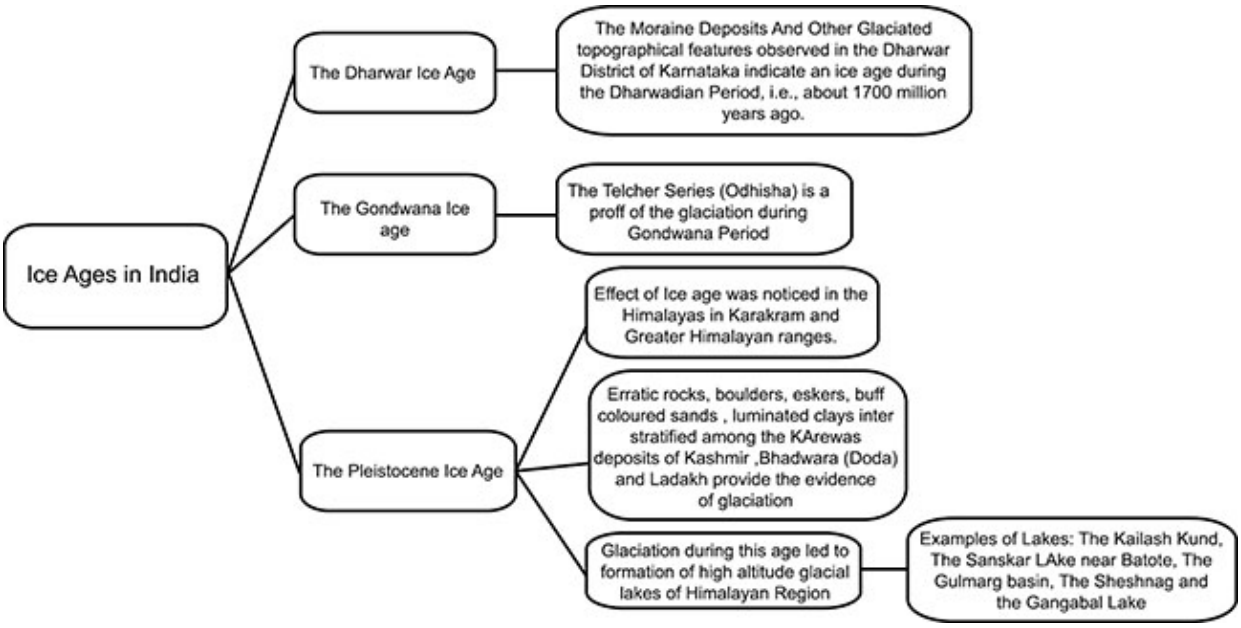
Fig. 2.14 North-West Himalaya—Major Glaciers

The glaciers of Karakoram are the remnants of the Pleistocene Age. The diurnal rate of movement of these glaciers is between 8 to 15 cm at the side and 20 to 30 cm in the middle. The glaciers of the Pir-Panjal are less numerous and smaller in size as compared to those of the Karakoram and the Greater Himalayan ranges. The glaciers of the Nanga Parbat Massif are small in size and are moving fast due to a steep slope. The glaciers are not only the source of Himalayan rivers, but also maintain a regular supply of water in these

rivers during off-monsoon period. The Himalayan glaciers are, however, receding. According to new Isro satellite study, 75% of Himalayan glaciers are melting, 8% advancing and 17% showing no change. Commissioned by the environment ministry, the study is different from an IPCC report, claiming, without sufficient evidence, that the glaciers would vanish by 2035 (Times of India 16.5.2011, p.1). The average ratreat of Gangotri-Glacier is about 12m per year. It has retreated more than 1500 m during the last 70 years.

Ice Ages in India

The subcontinent of India has recorded several ice ages. A brief description of the Indian Ice Ages has been given in the following section:



1. The Dharwar Ice Age

The moraine deposits and other glaciated topographical features observed in the Dharwar District of Karnataka indicate an ice age during the Dharwadian Period, i.e., about 1700 million years ago.

2. The Gondwana Ice Age

The Telcher Series (Odisha) of the Gondwana System provides a good proof of the glaciation during the Gondwana Period.

3. The Pleistocene Ice Age

During the Pleistocene Period the effect of ice age was noticed in the Himalaya, especially in the Karakoram and the Greater Himalayan ranges. The erratic rocks, boulders, cirques, eskers, rock polishing, buff-coloured sands, and luminated clays inter-stratified among the *karewas* deposits of Kashmir, Bhadarwa (Doda), and Ladakh give enough proof of the Pleistocene glaciation. The Pleistocene glaciation also led to the formation of a number of high altitude glacial lakes of the Himalaya. The Kailash-Kund, The Sanasar Lake near Batote, the Gulmarg-basin, the Sheshnag, and the Gangabal Lake are some of the examples of this type of lakes. The Peninsular part of India has no evidence of Pleistocene glaciation.

THE IMPORTANCE OF THE HIMALAYA

The mighty Himalaya are the most pronounced and dominating physiographic feature of the subcontinent of India. It has often been said that the Himalaya are the body and soul of India. The importance of the Himalaya has been given briefly in the following lines:

1. Climatic Influence

The impact of the Himalaya on the climate, especially on the distribution of precipitation and temperature, is quite significant. The altitude of the Himalaya, their sprawl and extension intercept the summer monsoon coming from the Bay of Bengal and the Arabian Sea. They also prevent the cold Siberian air masses from entering into India. Had there been no Himalaya, the whole of northern India would have been a desert. According to the latest meteorological studies, the Himalaya are responsible for the splitting of the jet streams into two branches, and these in turn, play an important role in the arrival, success and failure of the monsoons in India.

2. Defence

Throughout history, the foreign invaders never entered India from the northern side. Despite modern technology of warfare, the Himalaya have great defence value. At present, a network of highways has been developed up to China, Tibet, Nepal, and Bhutan borders.

3. Source of Perennial Rivers

Most of the perennial rivers of northern India have their origin in the glaciers, lakes, and springs of the Himalaya. These rivers sustain the teeming millions of the India population.

4. Source of Fertile Soils

The perennial rivers and their tributaries carry enormous quantities of alluvial soils. In fact, the Great Plains of India are covered by the fertile alluvial soils deposited by the rivers coming down from the Himalaya.

5. Generation of Hydroelectricity

The Bhakra-Nangal Dam, Giri-Bata, Chamera, Pong, Bassi, Silal, Dulhasti Baghliar Projects, Tehri Dam, Koteshwar etc., are some of the important hydel-power generating multi-purpose projects located in the Himalaya.

6. Forest Wealth

The Himalayan ranges are very rich in forest resources. There is a horizontal zonation of vegetation in the Himalaya. The natural vegetation in the Himalaya varies from the humid tropical to the conifers and alpine pastures. These forests provide fuelwood, timber, gum, resins, lac, medicinal herbs, and a variety of materials for the industries. At the higher altitudes are the alpine pastures (margs) used by the tribals for grazing cattle during the summer season.

7. Orchards

The Himalaya are known for the apple, peach, cherry, pear, mulberry, walnut, almond, and apricot orchards.

8. Minerals

The Himalaya are rich in many metallic and non-metallic minerals. Coal is found in Jammu Division of Jammu and Kashmir. Copper, lead, zinc, nickel, cobalt, gold, silver, antimony, tungsten, magnesite, limestone, semi-precious, and precious stones are found in the states of Jammu and Kashmir, Himachal Pradesh, Uttarakhand, Sikkim and Arunachal Pradesh. The poor accessibility is, however, a barrier in the exploitation of the mineral wealth of the Himalaya.

9. Tourism

The Himalaya are known for their scenic and aesthetic beauty all over the world. The Himalaya offer cool, invigorating climate when the neighbouring plains are in the grip of scorching heat of the summer season. Millions of national and international tourists visit the hill stations in the Himalaya. The famous tourist centres in the Himalaya are Srinagar, Pahalgam, Gulmarg, Sonmarg, Yushmarg, Wular-round, Chamba, Dalhousie, Dharamshala, Shimla, Solan, Kangra, Kullu, Manali, Mussoorie, Nainital, Ranikhet, Almora, and Darjeeling.

10. Pilgrimage

Apart from places of tourist interest, the Himalaya have numerous shrines and pilgrimage centres. Some of the important shrines in the Himalaya are the Amarnath, Hazratbal (Srinagar), Kailash, Vaishno Devi, Kedarnath, Badrinath, Gangotri, Yamunotri, Jwalaji, Hemkund, etc.

THE GREAT PLAINS OF INDIA

The Great Plains of India lies to the south of the Shiwalik separated by the Himalayan Front Fault (HFF). It is a transitional zone between the Himalaya of the north and Peninsular India of the south. It is an aggradational plain formed by the alluvial deposits of the Indus, Ganga, Brahmaputra and their tributaries. The plain stretches for about 2400 km from west to east. It has varying width; 90–100 km in Assam, 160 km near Rajmahal (Jharkhand), 200 km in Bihar, 280 km near Allahabad and 500 km in Punjab. In general, the width of the plain increases from east to west **(Fig. 2.15).**

The Great Plains of India consist largely of alluvial deposits brought down by the rivers originating in the Himalayan and the Peninsular region. The exact depth of alluvium has not yet been fully determined. According to recent estimates, the average depth of alluvium in the southern side of the plain (north of Bundelkhand) varies between 1300 to 1400 m, while towards the Shiwaliks, the depth of alluvium increases. The maximum depth of over 8000 m has been reached near Ambala, Yamunanagar, and Jagadhri (Haryana).

The Great Plains are remarkably homogeneous with little variation in relief features for hundreds of kilometers. The monotony of the physical landscape is broken at micro-level by the river bluffs, *Bhurs,* levees, dead-arms of river channels, the ravines and khols. Changing river courses in the areas of frequent floods is a unique geomorphic process in the plains. The frequent floods, although a cause of immense damage to life and property, lay down fresh layer of silts in the flood-plains every year, providing rich fertile soils.

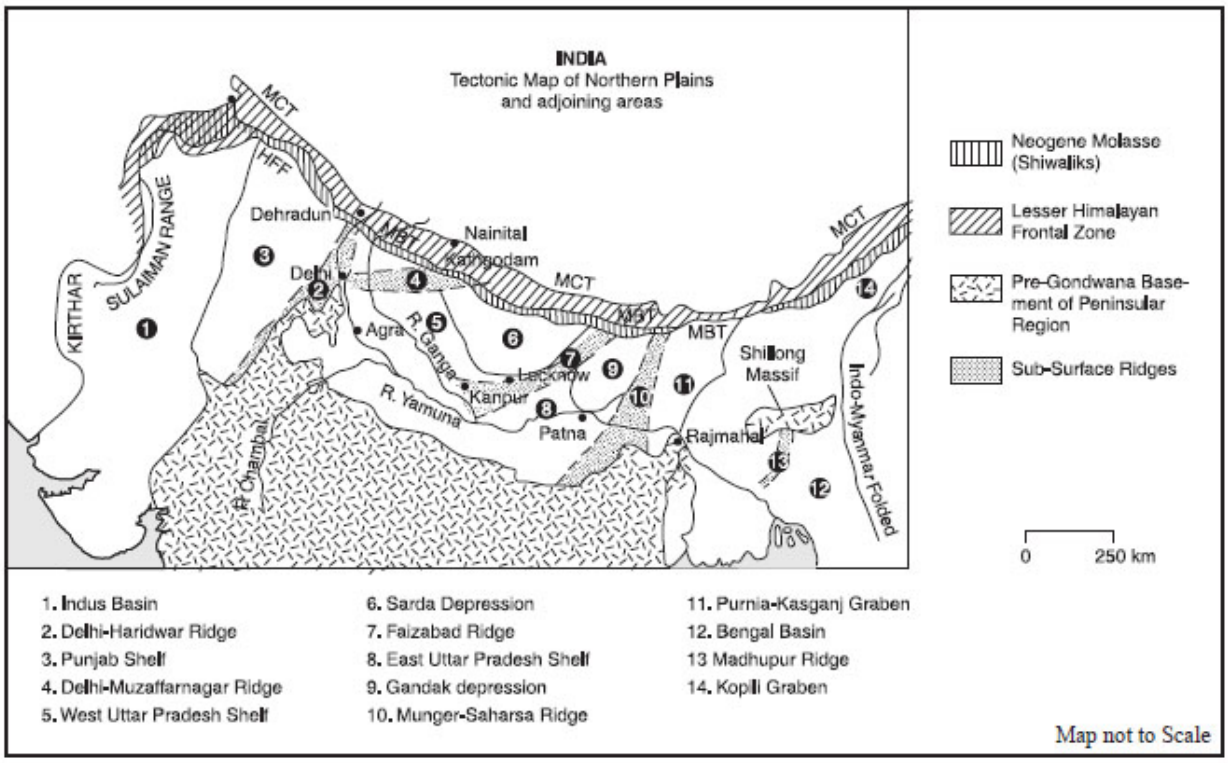


Fig. 2.15 The Great Plains of India and Adjoining Areas

Origin of the Great Plains of India

There is no unanimity amongst the geologists about the origin of the Great Plains of India. The puzzling questions are related to the enormous thickness of the alluvium, nature of the depression, mode of its formation, subterranean rock-beds and the underlying geological structure. Some of the important views about the origin of

the Northern Plains of India have been presented briefly in the following section:

1. Alluviation of the Foredeep

According to Edward Suess, an eminent Austrian geologist, a "foredeep" was formed in front of the high crust-waves of the Himalaya as they were checked in their southward advance by the more rigid landmass of the Peninsula of India. This foredeep was like a large synclinorium (a large syncline with a number of small anticlines and synclines) owing to the unevenness of its bottom. According to Suess, the bed of this foredeep had a gentle slope towards north whereas the Peninsular side depicted a steep gradient. This bed rests on the basement of hard crystalline Peninsular rocks through which the region is connected to the Himalayan and the Peninsular blocks **(Fig. 2.16).** The alluviation of this foredeep led to the formation of the Great Plains by the rivers, descending from the Himalayan region. One of the serious gaps in Suess' theory is related to his inability to give suitable explanation to the uneven slope of the bed of the plain. Moreover, no conclusive evidence is available through which it could be inferred that there exists crustal connection between the Himalayan region and the Peninsular block through the Great Plains.

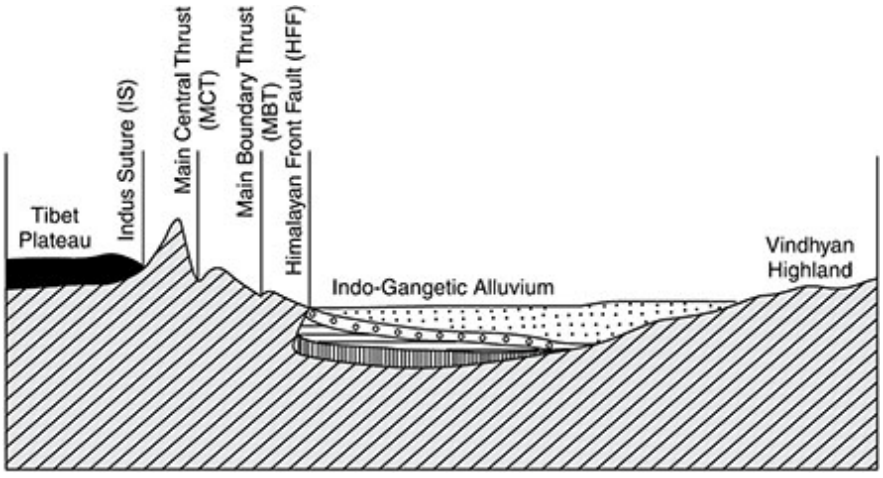


Fig. 2.16 Basement Complex—Hard Crystalline Rocks

2. Infilling of a Rift Valley

In the opinion of Sir G. Burrard, the Northern Plains of India are the result of infilling of a rift valley. Burrard opined that during the formation of the Himalaya a rift valley was created between the two parallel faults (one along the southern boundary of the Shiwaliks and the other along the northern boundary of the Peninsula). This rift valley was filled up by the detritus brought down by the Himalayan rivers. In order to support his argument Burrard has cited examples of the formation of similar rift valleys in the Himalaya and the Peninsular region especially the Narmada and Tapi rift valleys.

This concept, however, does not find approval of the modern geologists. The main criticism raised against this concept is that nowhere in the world has such a giant rift valley with a length of 2400 km and a width of over 500 km ever been formed by crustal downwarping. Also, there is no geological evidence of the formation of such a rift valley in the northern part of the Peninsular foreland. The uneven bottom and the northward slope of the bottom of the Great Plains of India have also not been explained by this concept.

3. Recession of the Sea

In the opinion of Blandford, during the Eocene Period, Peninsular India was joined together with Africa. During that period, there was one sea extending from Assam Valley to the Irrawaddy river (Myanmar) in the east and another from Iran and Baluchistan to Ladakh (Indus Valley) in the west. During the last part of the Eocene Period, arms of the Western Sea extended up to Punjab. Due to the rise of the Himalaya during the Miocene Period, these seas started receding by gradual deposits of sediments from the Himalayan rivers. After a prolonged period of sedimentation and subsidence, these gulfs (Gulf of Sind in the west and the Eastern Gulf up to the Shillong Plateau) were filled up, resulting in the formation of the Northern Plains of India.

The evidences cited in favour of the recession of the sea include: (i) the occurrence of limestone rocks in Kumaun-Garhwal region of Uttarakhand, (ii) the presence of saline water lakes in Rajasthan

(Sambar, Rajsazmand, Pushkar, Pachpadra, Nakki, Fatehsagar, Dheberr, Balasmand, etc.), (iii) the joining of the islands of the Gulf of Kachchh with the mainland, (iv) the seaward extension of the Sundarban Delta, (v) the emergence of new islands near Bangladesh coast, and (vi) the presence of marine fossils in the sediments of the Northern Plains of India. The theory, however, fails to give convincing arguments so far as the region of the central portion of the plain is concerned.

4. Remnant of the Tethys

Some of the geologists and geomorphologists opine that the Great Plains of India are a remnant of the Tethys Sea. According to them, after the upheaval of the Shiwaliks, the remaining part of the Tethys was left as a large trough which was joined to the Bay of Bengal in the east and the Arabian Sea in the west. Rivers from the Himalaya deposited their load in the trough. Because the Himalaya were rising during that period, rivers experienced rejuvenation and greater quantity of eroded material which increased the thickness of the alluviums. Due to infilling of the central part of the trough the seas located in the east and the west started receding, and the Great Plains of India came into existence.

5. Recent Views

According to the recent views, the Northern Plains of India represent a sag in the crust formed between the northward drifting of the Indian Subcontinent and the comparatively soft sediments accumulated in the Tethyan basin when the latter were crumpled and lifted up into a mountain system. Subsequently, it was filled up by the river deposits.

Physiographic Divisions of the Great Plains of India

The Great Plains of India are a remarkably homogeneous surface with an imperceptible slope. In fact, they are a featureless alluvial fertile plains formed mostly by the depositional process of the Himalayan and Vindhyan rivers. These rivers deposit enormous quantity of sediments along the foothills. Beyond the foothills, the

rivers deposit the alluvium in their flood plains**.** The Northern Plains of India may be divided into the following sub-regions:

1. The Bhabar Plain

- Location-lies to the south of the Shiwalik from west to east (Jammu Division to Assam)
- Its width is greater in the Western plains than in the eastern plains of Assam
- Its width measures 8 to 15 km
- Composition-Gravel and unasserted sediments deposited by the rivers descending from the Himalayan region and the Shiwaliks
- Due to high porosity, small streams (chos and raos) disappear in this region and only big rivers flow over the surface in this tract
- Unsuitable for crop cultivation
- Big trees with large roots thrive in this region
- Inhabitants are largely cattle keeping Gujjars

2. The Tarai Tract

- Location-Lies south to the Bhabar tract and it is 15–30 km wide
- It is wider in the eastern parts of the Great Plains in Brahmaputra region due to high rainfall
- It is a zone of excessive dampness, thick forests, rich wild life and infested with mosquitoes
- Once a marshy zone of jungle and wild grass along the southern edge of the Shiwaliks, Tarai has been almost reclaimed for agriculture
- In Uttarakhand, Uttar Pradesh, Haryana, Punjab, and Jammu Divisions (J&K) the Tarai forests have been cleared for cultivation of crops
- Cultivated crops include sugarcane, rice, wheat, maize, oilseeds, pulses, fodder

Physiographic divisions of the Great Indian Plains [*Source: NCERT Book of Geography*]

3 The Bhangar Plains

- They represent the upland alluvial tracts of the Great Plains of India, formed by the older alluviums.
- The Bhangar formations were deposited during the middle Pleistocene Period and it lies above the flood limits of the rivers.
- Its soil is dark in colour, rich in humus content and productive. It contains concretions and nodules of impure calcium carbonate or ‘Kankar’.
- In relatively drier areas, the Bhangar also exhibits small tracts of saline and alkaline efflorescences known as ‘Reh’, ‘Kallar’ or ‘Thur’.
- Bhangar is generally a well-drained and most productive land of the Great Plains of India.
- The Bhangar deposits have the fossils of elephants, horses, man, rhinoceros, hippopotamus, etc.

4. The Khadar Plains

- The new alluvium tracts along the courses of the rivers are known as the ‘Khadar’ or ‘Bet’ lands.
- Tracts are enriched by fresh deposits of silt every year during the rainy season.
- Khadar land consists of sand, silt, clay and mud. Post-independence, this region has been brought under cultivation and devoted to sugarcane, rice, wheat, maize, oilseeds, legumes, and fodder crops.
- The Khadar deposits have the fossils of living species like

man, deer, oxen, buffaloes, horses, elephants, rhino, etc.

5. Delta Plains

- The deltaic plain is an extension of the Khadar land.
- It covers about 1.9 lakh sq km of area in the lower reaches of the Ganga River.
- It is an area of deposition as the river flows in this tract sluggishly.
- The deltaic plain consists mainly of old mud, new mud and marsh.
- In the delta region, the uplands are called ‘Chars’ while marshy areas are known as ‘Bils’.
- The delta of Ganga being an active one is extending towards the Bay of Bengal.

Meso-regions of the Northern Plains of India

On the basis of geo-climatic and topographical characteristics, the Northern Plains of India may be divided into the following four meso-regions **(Fig. 2.17):**

1. The Plains of Rajasthan;
2. The Punjab Haryana Plains;
3. The Ganga Plains; and
4. The Brahmaputra Plains.

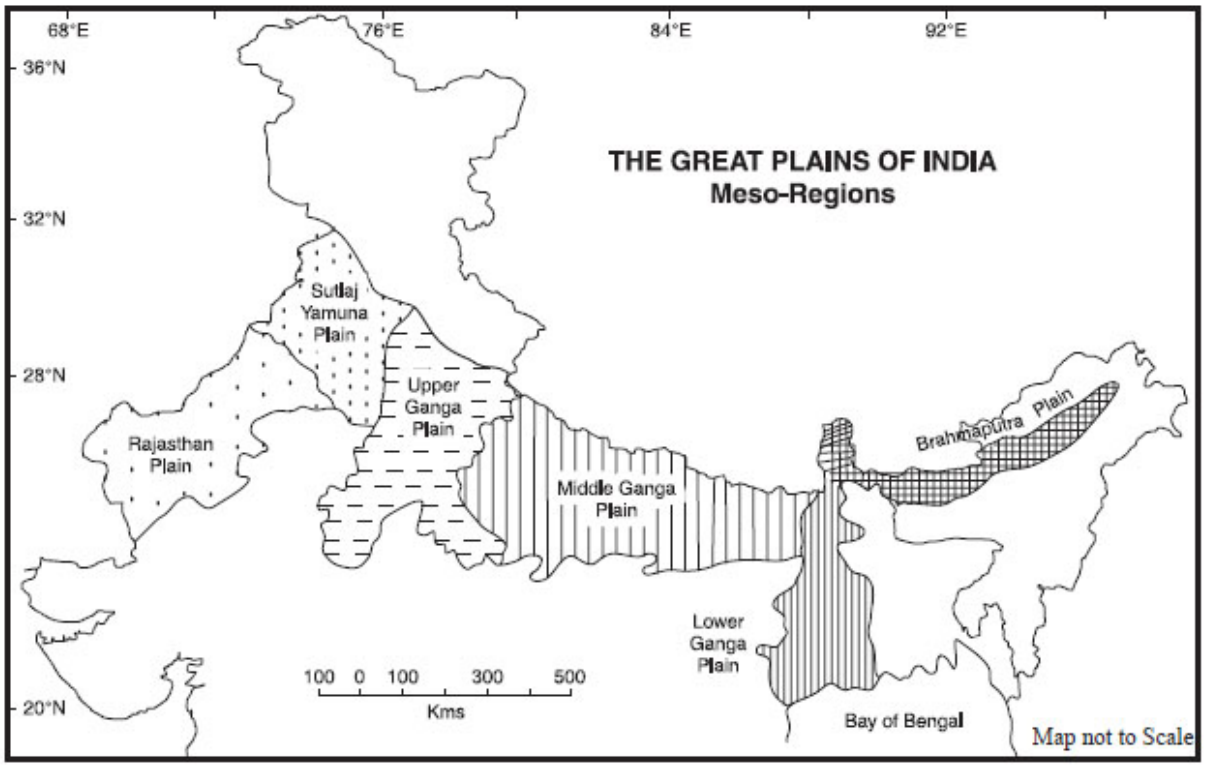


Fig. 2.17 The Great Plains of India

1. The Plains of Rajasthan

They lie to the west of the Aravallis and include the **Marusthali** and the **Bagar** of Rajasthan. The Rajasthan plains cover a total area of about 175,000 sq km. This plain has a general slope from north-east to south-west. In the lower reaches of the Luni river (Gujarat), this plain is only 20 m above sea level. A substantial part of this plain has been formed by the recession of the sea as is evidenced by the presence of salt water lakes (Sambhar, Degana, Didwana Kuchaman, Lunkaransar-Tal, and Pachpadra). The Sambhar Lake occupying an area of about 300 sq km during the rainy season lies about 65 km to the north-west of Jaipur city.

During the Permo-carboniferous Period, the greater part of the Rajasthan plain was under the sea. It has several dry beds of rivers, like Saraswati and Drisdavati which indicate that the area earlier was fertile. At present, Luni is the only flowing river which reaches the Arabian Sea through the marshes of the Rann of Kachchh. Its water

is sweet in the upper reaches but turns brackish in the lower parts. North of the Luni, there is a large area of inland drainage.

At present, the greater part of the Rajasthan Plains are a desert covered with longitudinal and transverse sand-dunes and barchans (Barkhans). A large number of playa lakes occur in the basins. In the south-western parts of the Rajasthan plain, there are the alluvial tracts known as *Rohi* (fertile plains). In the north-eastern part they consist of dry beds of the Ghaggar known as the Ghaggar Plains.

2. The Punjab Haryana Plains

Stretching over an area of about 650 km from north-east to south-west and 300 km from west to east, the Punjab-Haryana Plain is an aggradational plain, deposited by the Satluj, Beas, and Ravi rivers. The height of the plains varies from 300 m in the north-near Jammu and Kathua to 200 m in the south-east. In the east the Delhi Ridge separates it from the Gangetic Plain. The general direction of slope is from north-east to south-west and south. The main topographical features of the Punjab-Haryana Plains are bluffs, locally called as *Dhaya*, as high as three metres or more, and the *Khadar* belts known as *Bet.* The undulating topography south of the Shiwaliks is adversely affected by erosion, caused by the seasonal streams locally called as *Chos.* The south western parts, especially Hissar District is sandy, characterised by shifting sand-dunes. Satluj, Beas, and Ravi are the only perinneal rivers. Between the Satluj and the Yamuna, the Ghaggar (the ancient Saraswati) is a seasonal stream which passes through Ambala Cantt. Its course is about ten km wide and contains water only during the rainy season.

The Punjab-Haryana Plains may be divided into: (i) the Bari-Doab between the Beas and Ravi, (ii) the Bist Doab, between the Beas and Satluj, (iii) the Malwa Plain, the central part of the region, and (iv) the Haryana-Bhiwani Bagar in the southern and south-eastern parts of the region.

3. The Ganga Plains

The Ganga Plains lie between the Yamuna catchment in the west to the Bangladesh border in the east

(Fig. 2.20). It is about 1400 km from west to east and has an average width of 300 km from north to south. The general gradient of the plain is about 15 cm per km from north-west to south-east. The maximum height of this plain is found to the north of Saharanpur (276 m) followed by Roorkee
(274 m), Agra (169 m), Kanpur (125 m), Allahabad (98 m), Patna (53 m), Kolkata (6 m), and Sagar Island only 3 m above sea level.

The main topographical variations in the plains include, *Bhabar, Tarai, Bhangar, Khadar*, river bluffs (levees), abandoned courses*, Khols*, dead-channels, *Bills, Tals,* and badlands. The Ganga Plains can be subdivided into the following sub-regions:

(a) The Upper Ganga Plain The Upper Ganga Plain includes the Ganga-Yamuna Doab, Rohilkhand Division, and parts of the Agra Division. The catchment area of the Yamuna river makes its western boundary, Shiwaliks in the north and 125 m contour in the south. The elevation of the Upper Ganga Plain varies between 100 m to about 300 m. In addition to Ganga and the Yamuna, it is traversed by the Kali and Sharda rivers. A unique feature of the Upper Ganga Plain is the presence of Bhur (undulating, aeolian sandy deposits). Devoted to sugarcane, rice, wheat, maize, pulses, mustard, fodder, vegetables and orchards, it is one of the most productive plains of India in which the Green Revolution is a big success **(Fig. 2.18).**

(b) The Middle Ganga Plain Sprawling over an area of about 144,400 sq km, the Middle Ganga Plain includes central and eastern Uttar Pradesh, and the Bihar Plains up to Muzaffarpur and Patna. In the north, it is bounded by the Shiwaliks along the Indo-Nepal border. It has thick alluvial deposits with less Kankar formation. The region is homogeneous and featureless where monotony is broken by river levees, bluffs, ox-bow lakes, *Dhus*, *Tals, Jala* and *Chaurs* (marshy lands). Being a low gradient plain, the rivers often change their courses in this region. Gandak and Kosi are the main left hand tributaries, while the Son is an important right hand tributary of the Ganga in the Middle Ganga Plain.

(c) The Lower Ganga Plain (Area 80,970 sq km) The Lower Ganga

Plain extends from Patna in the west, the foot of Darjeeling Himalaya in the north to the Bay of Bengal in the south. It is bordered by Assam and Bangladesh in the east and the Chotanagpur Plateau in the west. In the lower part of the Lower Ganga Plain is Sundarban Delta. The plain has a monotonous surface.

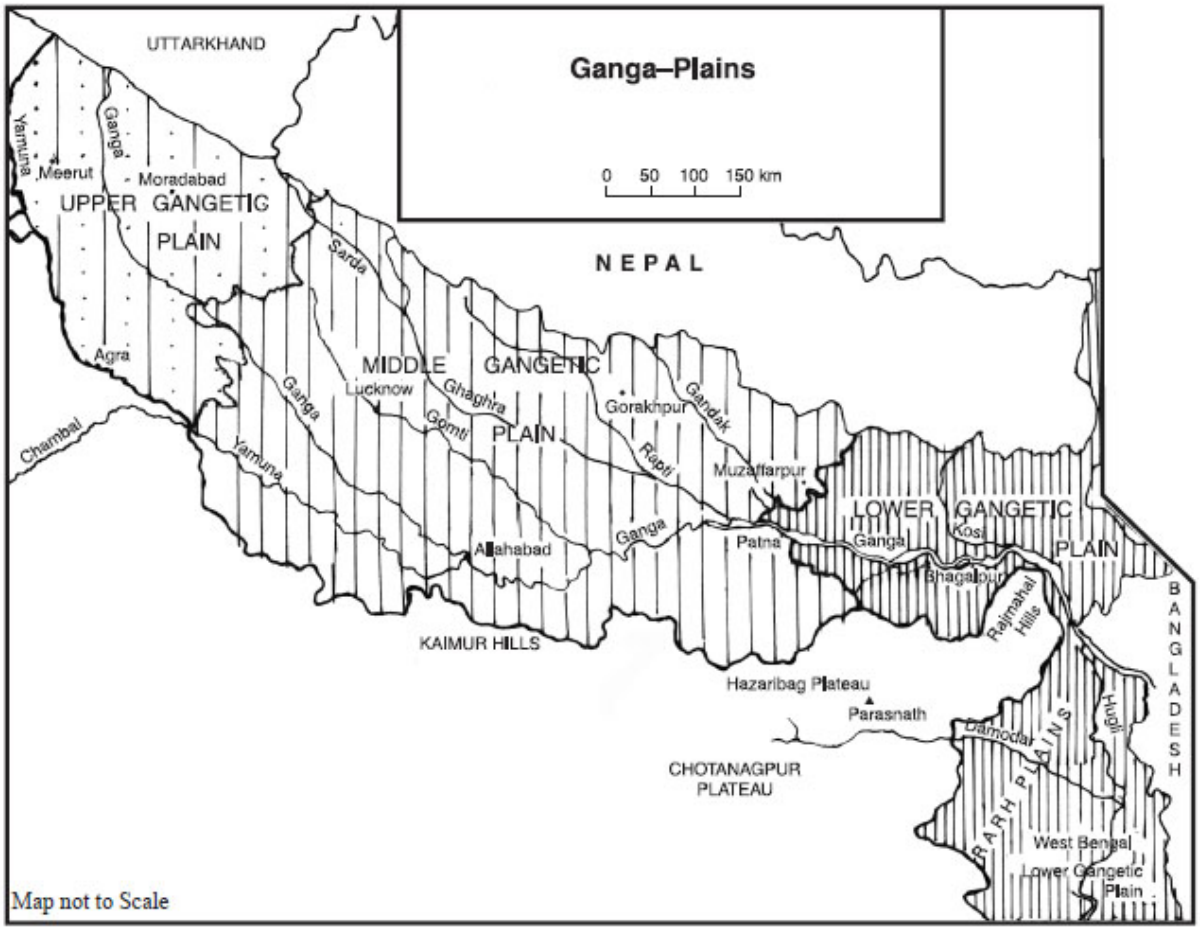


Fig. 2.18 Physiographic Divisions of Gangetic Plain

The eastern part of the plain is drained by the Tista, Jaldhakia, Sankosh joining the Brahmaputra, and the western part by the Mahananda, Ajay, and Damodar. In the extreme south-west, Kasai and Subarnarekha are the main rivers. The general slope of this plain is towards south and south-east.

The Lower Ganga Plain has been formed by the downwarping of a part of the Peninsular India between Rajmahal Hills and the

Meghalaya Plateau and subsequent sedimentation by the Ganga and the Brahmaputra rivers. The plain has a monotonous surface dissected frequently by the channels of the main streams and their tributaries.

Rahr Plain: Lying to the east of the Chotanagpur Plateau, it is a part of the Lower Gangetic Plains. Drained by the Damodar and Subarnarekha, it is covered by the lateritic-alluvium soils. Soil erosion is the main problem of the Rahr plain. Rice, maize and pulses are the main crops of the Rahr plain.

Sundarbans: The largest mangrove swamp in the world, the Sundarbans, or the beautiful forest, gets its name from the Sundari tree which grows well in marshland. It is home to the Royal Tiger and crocodiles.

4. The Brahmaputra Plain

Stretching over an area of about 56,275 sq km, it is the eastern part of the Great Plains of India. It is about 720 km long and about 80 km wide. The region is surrounded by high mountains on all sides, except on the west. It is a depositional plain. The general altitude of the Brahmaputra Plain varies between 130 m in the east to only 30 m in the west **(Fig. 2.19).** The Assam Valley is characterised by a steep slope along its northern margin but the southern side has a gradual fall from the Meghalaya Plateau. The whole length of the plain is traversed by the Brahmaputra. Due to the low gradient, the Brahmaputra is a highly braided river having numerous islands. Majuli (area 930 sq km) is the largest river island of India and the second largest in the world after the Marajo Island of the Amazon River. The valley of Brahmaputra also has a number of isolated hillocks on both flanks of the Brahmaputra River. It is one of the most productive plains of India in which rice and jute are the main crops.

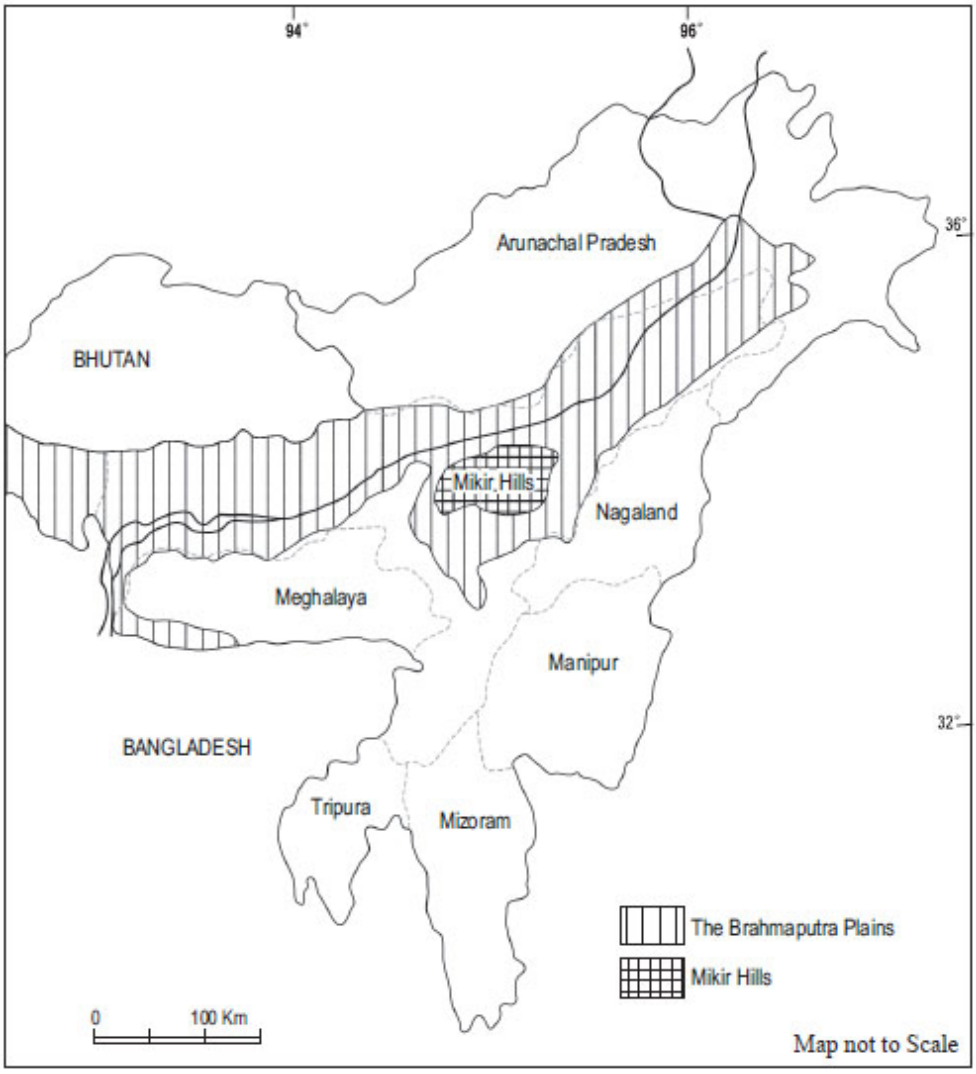


Fig. 2.19 Brahmaputra Valley

There is a marked difference between the physiography of the north and the south banks of the Brahmaputra river. The northern tributaries descending from the Arunachal and Assam Hills form a series of alluvial fans which coalesce and obstruct the courses of the tributaries forcing them to form meanders and adopt parallel course along the main stream, Brahmaputra. Consequently, there are numerous levees along the north bank. This has led to the formation of Bils, ox-bow lakes, marshy tracts, and Tarai lands with dense forest cover. The southern bank of the Brahmaputra is less uneven and less wide. Moreover, the tributaries in the southern part are considerably

larger. Here, Dhansiri and Kapili, through their headward erosion have almost isolated the Mikir and Rengma hills from the Meghalaya Plateau.

The valley of Assam may be divided into (i) the Upper Assam and (ii) the Lower Assam. These are demarcated along 94^{o} East longitude. The Upper Assam Valley includes the districts of Lakhimpur, Dibrugarh, Jorhat, and Sibsagar, and the Tezpur Tehsil of Darrang District. It is a monotonous plain except for the low hill ranges along the southern and south-eastern border. The Lower Assam Valley consists of Nagaon, Dhubri, Goalpara, Barpeta, Kamrup, Nalbari, Kokrajhar, and parts of the Darrang District. This region does not posses monotonous physiographic characteristics since its landscape is interspersed with the spurs of Meghalaya Plateau. Here, the right bank tributaries form a trellis pattern of drainage, while the left bank tributaries exhibit the dendritic pattern. Swamps and marshes are numerous in the northern region of the Lower Brahmaputra Valley.

The Brahmaputra: It is one of the great rivers of the world. Flowing eastward to the north of the Himalaya in Tibet (China), it turns sharply south and passes through Assam before entering Bangladesh. The river valley has fertile alluvial plain which is conducive to growing rice and jute. It is also famous for its tea and the two national parks at Kaziranga and Manas.

IMPORTANCE OF THE GREAT PLAINS OF INDIA

The Great Plains of India have been the repository of the Indian culture. This is covered with one of the most productive soils of the world. Its soils have the capacity to grow any crop of the tropical and temperate regions. The main points of significance of the Great Plains of India are as under:

(i) The soils of the plains are agriculturally most fertile. They are being devoted to the cereal and non-cereal crops. The plains are often termed as the “Granary of India”.

(ii) Most of the rivers traversing the Northern Plains of India are perennial in nature. A number of canals have been carved out of these rivers which make agriculture more remunerative and sustainable.

(iii) The northern plains have a rich underground water-table which is being utilised through tube-wells and pumping sets for irrigation, domestic and industrial purposes.

(iv) The rivers of the plain have very gentle gradients which makes them navigable over long distances.

(v) Development of infrastructure like roads and railways could become easy in the plains.

(vi) The sedimentary rocks of plains have petroleum and natural gas deposits.

(vii) The plains constitute less than one-third of the total area of the country, but support over 40 % of the total population of the country.

(viii) The plains have witnessed several religious, political, cultural and social movements since the dawn of history. Some of the great religions of the world, like Hinduism, Buddhism, Jainism, and Sikhism have their origin in the Great Plains of India. Several sacred places and centres of pilgrimage (Amritsar, Hardwar, Allahabad, Varanasi, Kushinagar, Bodh-Gaya, etc.)
are situated in these plains.

THE INDIAN DESERT

A desert is an arid land where the water is lost through evaporation than is gained from precipitation. It is the region where the rate of evaporation is higher so that of the rate of precipitation.

The Indian subcontinent has the following deserts:

The Thar Desert

It is a low latitude desert and is also known as tropical desert. The name Thar is derived from '*thul*' a term used for the region's sand

ridges. The Thar Desert is a large arid region that covers an area of 200,000 sq km. It is the ninth largest subtropical desert in the world. It is bordered by plains of the Indus River to the west, Punjab plain to the north and north-east, Aravalli range to the south east and Rann of Kutch to the south. More than 60% of the desert lies in Rajasthan.

The land surface of the Thar Desert is wind deposited (Aeolian) accumulation of sand over the past 1.8 million years. Its surface has high and low sand dunes separated by sandy plains and low barren hills (bhakars).The dunes are in continual motion and keep varying in their shapes and sizes. Some of the older sand dunes are of 150 m height. The region has playas (saline lake beds), locally known as dhands, scattered throughout the region. Examples: The Sambhar, Kuchaman, Didwana, Pachpadra, Phalodi (Rajasthan), Kharagoda (Gujarat), Lunkaransar are major sources of the common salt.

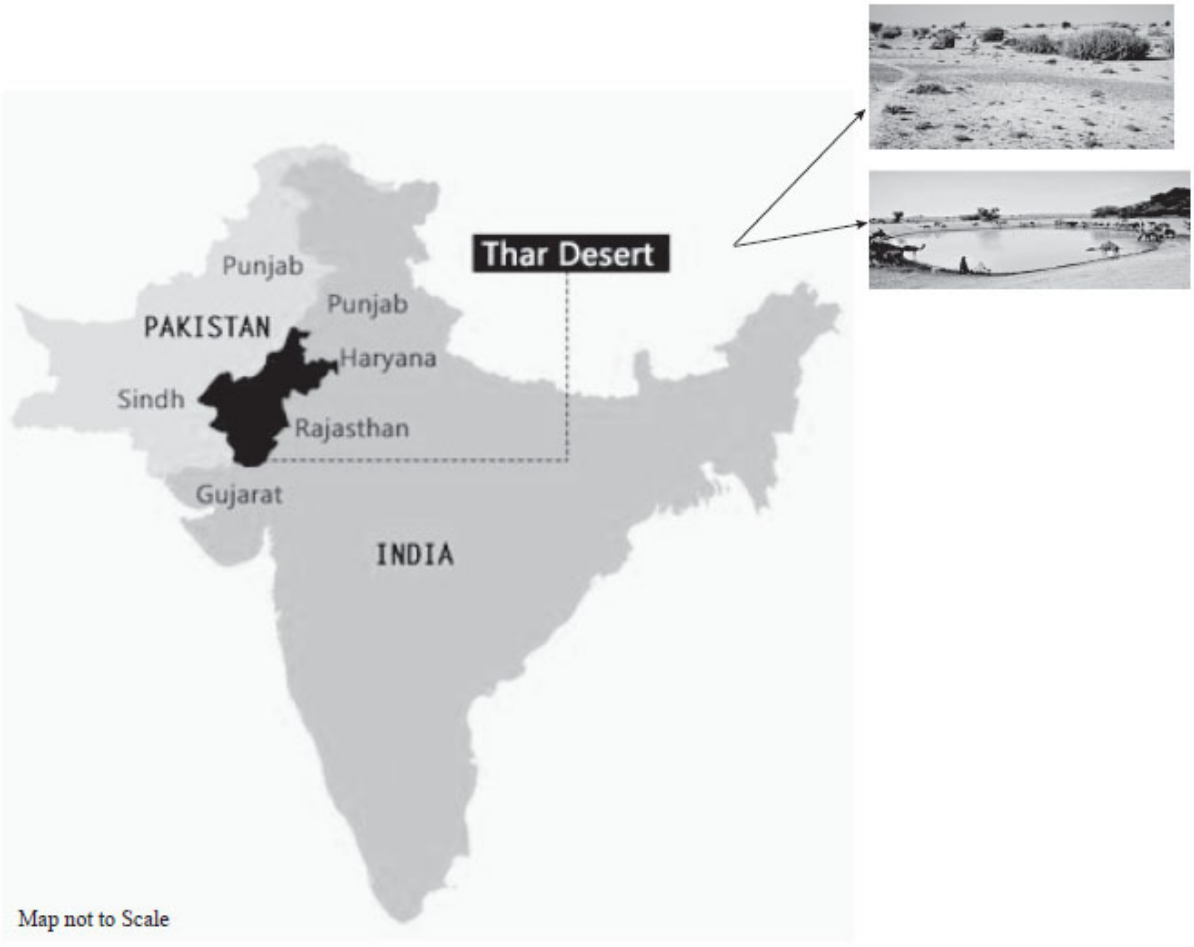


Fig. 2.20 The Great Indian Desert in Western Rajasthan [*Source: Rafal Caweda / Fotolia*]

The Thar Desert receives low annual rainfall, of about 4 inches (in the west) to 20 inches (in the east). The winds are dry north east monsoon and the deserts record temperatures up to 50° C in the months of May and June. The coldest month is January with the minimum temperature of 5–10° C.

The winters are short for two months only in December and January. Dust storms blow with velocities of up to 150 km/h during the peak summer season.

The flora includes stunted scrub, drought resistant trees, gum Arabica, acacia, jojoba, *Khejri* tree and *Euphorbia. Khejri* is an indigenous tree which plays a vital role in stabilizing the sand dunes. It can also withstand periodic burial under the sand dunes. The fauna of the Thar Desert are the falcon, kestrel, blackbucks, chinkara, Indian wild ass, foxes, partridges, quail, vultures and reptiles. Endangered species of bustard is found in the desert.

The river Luni, originating from the Pushkar valley of the Aravalli range is the only natural water source that reaches the Arabian Sea through the Rann of Kutch. The major irrigation system in the region is the Rajasthan Canal System. The Indira Gandhi Canal is the source of fresh water and irrigates the vast expanse of Indian portion of the Thar Desert. The Desert National Park is an important ecosystem. The Tal Chhapar Sanctuary is an important bird reserve area located in Churu district and also an abode of large population of blackbuck, fox, caracal and sandgrouse. The Sundha Mata Conservation reserve is located in Jalore district. The Keoladeo Ghana National Park (Desert National Park in Jaisalmer) has fossils of tree trunks and sea shells.

The Thar Desert has a population density of 83 people per sq km and it is the most densely populated desert in the world. It has become the largest wool producing area in India. Animal husbandry has increased as the harsh climatic conditions and the land terrain not favouring farming. However, kharif crops are the main agricultural

production. Bajra is the main crop. The solar and wind energy of the region is being exploited to generate electricity.

The Cold Desert of India (Ladakh)

Ladakh is the cold desert of India that lies in the Greater Himalaya, on the eastern side of Jammu and Kashmir. It is bounded by the Karakoram range in the North and by Zanskar mountains in the South. River Indus flows through Ladakh. The altitude varies from 3000 m in Kargil to more than 8000 m in Karakoram. Ladakh has very thin air and hence the sun's heat intensity is strongly felt. During the summers the day temperature is just above 0°C and at night it is as low as –40°C.The region experiences low rainfall (10 cm) annually since it lies in the rain shadow area of the Himalaya mountains.

Teh region has sparse vegetation (scanty patches of shrubs and grasses). Walnuts, apples, apricots are the main fruits of the summer season.The fauna of the region includes Yak, Wild goat and sheep, special kinds of dogs.

ORIGIN AND PHYSIOGRAPHY OF THE PENINSULAR INDIA

The origin of rocks of Peninsular India is more than 3600 million years old. Before the Carboniferous period, it was a part of the Gondwanaland. In the opinion of geologists, during the Archaean Period, the Indian Peninsula never subsided under the sea permanently. It was more rigid, stable, and had remained almost unaffected by the mountain building forces. However, it experienced block faulting and displacement during the subsequent periods as evidenced by the Dharwar and Gondwana formations and the fault valleys of the Narmada, Tapi and Son rivers.

It was during the Carboniferous Period that coal was formed in the Damodar, Son, Mahanadi, and Godavari basins. During the Cretaceous Period, large scale vulcanicity produced the Deccan Trap (the Lava Plateau of India), comprising lava sheets of several thousand metres in depth. The Deccan Trap originated about 146

million years back when the magma flowed from the depth of about 40 km below the crust.

Major Geological Formations of the Peninsular India (about 3600 million years ago)

The plateau of Peninsular India exhibits a complex system of geological structures. It has some of the oldest rocks of the world from the Precambrian period (Archaean) and the youngest rocks of the Holocene epoch (Quaternary/recent period). The major rock systems found in the Peninsular India have been described briefly in the following section:

Cratonisation

It is the process by which the formation of cratons takes place from the early rocks. A craton is an old but stable part of the continental crust that survived during the merging and the splitting of the continents. The first cratonic landmass was formed during the Archaean eon. They may be up to 2 million years old

During this period, the Earth's heat flow was about three times greater than the present time and had increased tectonic and volcanic activities. The mantle was much more liquid and the crust was thinner. As a result the oceanic ridges at the crust and hot spots were rapidly formed. It is believed that the earth's surface got broken into small plates with volcanic islands and arcs. Small cratons were melted and remelted by the hotspots and recycled in the subduction zones. The early Archaean period did not have large continents. The formation of cratons took place at the hotspots

Cratons are composed of igneous rocks like granite and have a thick crust with deep roots that extend into the mantle to the depth of 200 km.

1. The Archaean Group

Ancient crystalline and highly metamorphosed gneisses of the Archaean System are found in the plateaux of Tamil Nadu, Nilgiris, Karnataka, Andhra Pradesh, Maharashtra, Chhotanagpur, West Bengal, Odisha, Jharkhand, Chhattisgarh, Madhya Pradesh,

Meghalaya, Mikir, Bundelkhand (UP), and the Aravallis (Rajasthan). The Bengal gneiss known as '*Khondolite*' is found in the Eastern Ghats. These rocks are rich in metallic and non-metallic minerals, precious stones, and building materials, and covered more than 2/3 part of peninsula.

2. The Dharwar System

These are the oldest metamorphosed-sedimentary rocks found in narrow geosynclines flanking the Archaean gneiss (Pre-Cambrian Age). They occur mainly in (i) Dharwar, Bellary and Hospet districts of Karnataka, (ii) the Chhotanagpur Plateau, (iii) the upper reaches of Godavari (Durg, Bastar, Dantewala, Chandrapur, etc.), and (iv) the Aravallis (Delhi, Rajasthan, and Gujarat).

It is presumed that the majority of the Dharwar rocks had escaped folding completely and had deposited into the hollows and the corrugations of landmasses or were only mildly folded. These rocks are rich in iron ore, manganese, mica, copper, zinc, lead, silver, gold, slate, asbestos, marble, and limestone.

3. The Cuddapah System

The Cuddapah formations (Andhra Pradesh) occupy the deep basins of: (i) the lower valleys of Penganga and Godavari, (ii) the Talcher Series between Mahanadi and Brahmani (Odisha), the upper courses of the Narmada and Son rivers, and (iii) the west of Aravallis near Jodhpur. These rocks are rich in building material, shales, limestone, and sandstone. Some inferior quality of iron ore, manganese, copper, and asbestos are also found in these formations.

4. The Vindhyan System

The Central Indian Highlands known as the Vindhyan Mountains occupy a large basin extending from Chittorgarh (Rajasthan) in the west to Sasaram and Dehri-on-Son (Bihar) in the east. One branch of it extends from Sasaram to Hoshingabad (Madhya Pradesh). It occupies a large contiguous area stretching over one lakh sq km from the Chambal to Son rivers. Several isolated exposures of sedimentary rocks occur in the Bastar area of Chhattisgarh. In some of the exposures of the Vindhyan System are found the diamond

bearing conglomerates. The Panna District of Madhya Pradesh and the Kurnool District of Andhra Pradesh are well known for diamond production. Elsewhere in the south, the upper Vindhyans are covered by the Deccan Traps. The Vindhyans are known for the good quality of building materials. They are rich in ornamental stones, precious stones, diamonds and materials used in ceremics. The historical monuments of the Medieval Period and majestic religious places like Stupa of Sanchi, Agra Fort, Fatehpur Sikri, Red Fort, Jama-Masjid, Birla Mandir, etc., have been constructed with the red-sandstones obtained from the Vindhyan Ranges.

5. Gondwana System

The present coal belts of Peninsular India were developed during the Gondwana (Carboniferous) period. The Talcher Series, the Damuda Series and the Panchet Series are the products of this period. The rocks of the Upper Carboniferous Period, Permian, Triassic, Jurassic, Cretaceous, Tertiary, etc., are preserved in different parts of the Damodar, Mahanadi, Godavari, and Krishna river basins.

6. The Deccan Trap

The Cretaceous system is a very widely distributed system in the country. The Gondwanaland developed fissures and its broken parts started drifting from each other. There was large scale upheaval of lava (basalt) from the interior of the Earth to form the Deccan Trap. The eruption of lava was of the Hawaiian or fissure type.This period is marked by the transgression of the sea (Narmada valley and Coromandal coast), and outpouring of huge quantity of basalt so as to form the Deccan Trap. There had been intrusions of the plutonic rocks such as gabbro and granite. The basalt of the Deccan Trap is used for the construction of roads and buildings. Moreover, there are quartzites, agates, and carnelians in the lava formations of the Deccan Plateau.

7. The Tertiary System

The final fragmentation of the Gondwana took place during the Tertiary Period. There occurred faulting of the Peninsula alongwith the subsidence of the broken blocks beneath the Arabian Sea and

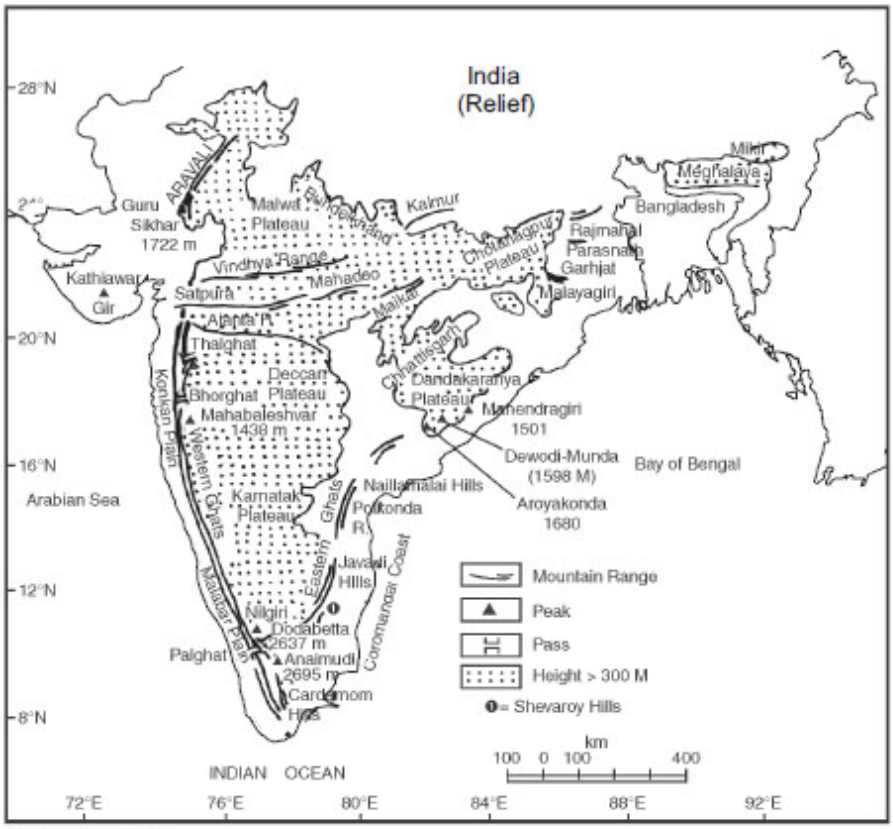
the Bay of Bengal. The Tertiary rocks are found in Kathiawar, Kachchh (Gujarat), Laki Series (Rajasthan), and along the Coromandal and Malabar coasts. In north-east, they are found in the Meghalaya Plateau; the Jaintia Series.

8. The Pleistocene Period

The Pleistocene deposits are found in the lower reaches and deltas of Mahanadi, Godavari, Krishna, and Kaveri and the western coastal plains of Gujarat, Konkan, and Malabar. These deposits are, however, more pronounced along the eastern coast of India.

Physiography and Relief Features of Peninsular India

Covering an area of about 16 lakh sq km, the peninsular upland forms the largest physiographic division of India. With a general elevation between 600–900 m, the region constitutes an irregular triangle with its base lying between the Delhi Ridge and the Rajmahal Hills and the apex formed by Kanyakumari. It is bounded by the Aravallis in the north-west, Maikal Range in the north, Hazaribagh and Rajmahal Hills in the north-east, the Western Ghats (Sahayadri Mountains.) in the west and the Eastern Ghats in the east **(Fig. 2.21).** The highest peak of Peninsular India-Anai-Mudi (Nilgiris), is 2695 metres above sea level. According to Prof. S. P Chatterji (1964), the Peninsular Uplands can be divided into the following eight macro-physiographic units (**Table 2.4**). Meso-region can be identified with a division of a state.



Map not to Scale

Fig. 2.21 Peninsular India: Relief

Table 2.4

The Physiographic Regions of Peninsular India

	Meso-Regions
1.	The North Central Highlands
2.	The South Central Highlands
3.	The Eastern Plateau
4.	The Meghalaya-Mikir Uplands
5.	The North Deccan
6.	The South Deccan
7.	The Western Ghats or Sahayadri
8.	The Eastern Ghats

Source: S.P. Chatterji, 1964, National Atlas Organisation, Kolkata.

1. The North Central Highlands

The central highlands of peninsular India include the Aravallis, the Malwa Plateau, and the Vindhyan Range **(Fig. 2.21).**

(i) **The Aravallis**: It is a range that runs from north-east to south-west for about 800 km between Delhi to Palanpur (Gujarat). It is one of the oldest folded mountains of the world. Being highly denuded, its highest peak—Guru-Sikhar—is only 1722 m in height. The Aravillis are mainly composed of quartzites, gneisses and schists of the Precambrian period. North-west of Udaipur, the Aravallis are called Jarga Hills (1431 m). The Goranghat Pass separates Gurushikar from Mount Abu. The Great Boundary Fault (GBF) separates the Aravallis from the Vindhyan Mountains.

(ii) **The Malwa Plateau:** It is bordered by the Aravallis in the north, the Vindhyan Range in the south and the Bundelkhand Plateau in the east. The Malwa Plateau has two drainage systems, one towards the Arabian Sea (Narmada and Mahi), and another towards the Bay of Bengal (Chambal, Sind, Betwa, and Ken) joining the Yamuna river.

2. The South Central Highlands

The Vindhyan Range extends from Jobat (Gujarat) and Chittorgarh (Rajasthan) to Sasaram in Bihar. It extends for about 1050 km with general elevation between 450 to 600 m. Apart from the Kaimur Hills in the east, the Maikal Range forms a connecting link between the Vindhyans and the Satpura mountains.

(i) **The Bundelkhand (Vindhyachal Plateau):** It is bounded by the Yamuna river in the north, the Vindhyans in the south, the Chambal in the north-west, and Panna-Ajaigarh Range in the south-east. The Bundelkhand upland stretches over the districts of Banda, Hamirpur, Jalaun, Jhansi, and Lalitpur (UP), and Datia, Tikamgarh, Chhatarpur and Panna (MP). The region is characterised by senile topography. The rivers like Betwa, Dhasan, and Ken have carved out steep gorges, rapids, cataracts, and waterfalls.

(ii) **The Vindhyachal-Baghelkhandor Vindhyachal Plateau**: It

includes the plateaus of Satna, Rewa (MP) and Mirzapur (UP). Its elevation varies between 150 to 1200 m with uneven relief. To the south of this lies the Narmada-Son trough (rift valley) characterised by the Archaeans and Bijwar series. South of this trough is the eastward extension of the Satpura which is an area of radial drainage. Among the basins, Singrauli and Dudhi (150–300 m) are Upper Gondwana basins, which are rich in coal deposits. Besides the Narmada and Son, this region is drained by the Karmanasa, Tons, Ken, and Belandare rivers.

Parallel to the Vindhyas between the Narmada and the Tapi rivers is the Satpura Range. Satpura consists of Rajpipla Hills, Mahadev Hills and the Maikal Range. Dhupgarh (1350 m, near Pachmarhi) is the highest peak of Satpura. Amarkantak (1064 m) is another important peak of the Satpura mountains.

3. The Chotanagpur Plateau

The Chotanagpur Plateau sprawls over parts of West Bengal, Jharkhand, Chhattisgarh, Odisha, and north-eastern part of Andhra Pradesh. This plateau has a series of the meso and micro plateaus (Ranchi, Hazaribagh, Singhbhum, Dhanbad, Palamu, Santhal, Parganas, and Purulia districts of West Bengal). It is composed of Archaean granite and gneiss rocks with patches of Dharwar (mica-schists), the Damuda series of the Gondwana Period, and the lava flow of the Cretaceous Period.

Moreover, the Chhotanagpur Plateau consists of plateaux at different levels of elevation, the highest general elevation of about 1100 m in the mid-western part is known as pat lands. The rivers which drain the Chhotanagpur Plateau are Barakar, Damodar, Subarnarekha, and Koels. These rivers have carved out deep gorges, rapids, cataracts, and waterfalls in the plateau region.

4. The Meghalaya Plateau and Mikir Hills

Consisting of the Garo, Khasi, Jaintia hills and the outlying Mikir and Rengma hills, it is a plateau which has been detached from the Indian

Peninsula by the Malda Gap. The Meghalaya Plateau has a chequered evolutionary history of emergence, submergence, planation surface with several phsases of erosion, sedimentation, diastrophism and intrusions. The Shillong Peak is the highest elevation (1823 m) in the Meghalaya Plateau, while Norkek (1515 m) is the highest peak of the Garo Hills. Mawsynram (25°15¢N, 91°44¢E) about 16 km west of Cherrapunji records the highest rainfall in the world.

The Mikir Hills are detached from the Meghalaya Plateau and are surrounded by plains from three sides. The southern range of the Mikir Hills is known as the Rengma Hills (900 m). The Mikir Hills are characterised by radial drainage with Dhansiri and Jamuna being the main rivers.

5. The North Deccan (Maharashtra Plateau)

The plateau of Maharashtra includes the entire state of Maharashtra, except the Konkan coast and the Sahyadris. It is mainly covered by the basalt of the Cretaceous Period. The basaltic sheet has a thickness of about 3 km in the western parts which diminishes towards the east and south-east. The most striking feature of the Maharashtra Plateau is the fault (1000 m), giving rise to the present shoreline of the Arabian Sea.

Through the northern part of the Maharashtra Plateau flows the Tapi River from east to west. It has a gentle slope in the south and steep gradient in the north (towards the Satpura Hills).

(i) **The Mahanadi Basin**: Sprawling over the districts of Raipur, Bilaspur, Durg, and Rajgarh, the Mahanadi basin is also known as the Chhattisgarh Plain. The region is largely dominated by the Archaean and Cuddapah formations. The Mahanadi river and its tributaries like Seonath, Hasdeo, Mand, Ib, Idra and Tel drain this plain.

(ii) **The Chhattisgarh Plain**: It is bordered by a series of hills and plateaux. The northern boundary is formed by the Lomari Plateau, Pendra Plateau, the Chhuri, and the Raigarh Hills.

The Korba coalfields of Chhattisgarh lie in this basin. The Gondwana formations are rich in bituminous coal which is supplied to the Bhilai Steel Plant. The western rimland includes the Maikal Range with crest line of 700–900 m. The southern rimland includes the Dhalli-Rajhara Hills in southern Durg district and the Raipur uplands in the south-eastern Raipur district. The Rajhara Hill contains Dharwarian rocks in which iron ore of haematite type is found. The iron ore from the Dhalli-Rajahara mines is supplied to the Bhilai Steel Plant.

(iii) **Garhjat Hills**: The Garhjat Hills are also known as the Odisha Highlands. It is bordered by the Chotanagpur Plateau in the north, Mahanadi basin in the west, Eastern Ghats in the south, and Utkal plains in the east. The region is mainly composed of Archaean rocks like granite, gneisses, and magmatic rocks. The Gondwana, Talcher, Barakar and Kamathi series are also located in this region.

(iv) **Dandakaranya**: Sprawling over the Koraput and Kalahandi districts of Odisha, Bastar District of Chhattisgarh and East Godavari, Vishakhapatnam and Srikakulam districts of Andhra Pradesh, Dandakaranya is an undulating plateau. Its Abujhmar Hills provide one of the richest iron-ore deposits at Bailadila Range. It is drained by the Tel and Udanti; tributaries of Mahanadi, and the Sabari and Sileru; tributaries of Godavari rivers.

6. The South Deccan

The South Deccan consists of several plateaus:

(i) **Karnataka Plateau:** This plateau spans in the S.W. parts state of Karnataka and the Kannur and Kozhikode districts of Kerala. It shows dominance of Archaean and Dharwar formations. This plateau has an average elevation of 600–900 m. Mulangiri (1913 m) is the highest peak in Baba-Budan Hills, followed by the Kudermukh (1892 m) peak.

The northern upland of the Karnataka plateau is known as

Malnad, while the southern part is called a Maidan. It is drained by the Kaveri and the Tungbhadra rivers. The Nandi valley is a summer resort in this region.

(ii) **The Telengana Plateau**: The plateau of Telengana consists of Dharwar and Cuddapah formations. Hyderabad, the capital and cultural city of the state lies in Telengana.

(iii) **The Tamil Nadu Uplands**: This upland lies between the South Sahyadri and Tamil Nadu coastal plains. It is largely covered by the Archaean rocks. The charnockites are found in Javadi and Shevaroy hills. Moreover, there are Cuddapah and alluvial formations. Between Coimbatore and Anaimalais, there is a broad gap, known as Palakkad Gap (Palghat), about 25 km wide, through which flows the Gayitri river from east to west joining Tamil Nadu with the coast of Kerala.

7. The Western Ghats

The Western Ghats in Sanskrit Sahyadris run parallel to the western coast for about 1600 km in the north south direction from the mouth of the Tapi river to Kanyakumari (Cape Camorin). The western slope of Sahyadri is steep while the eastern slope is gentle. These are block mountains formed due to the downwarping of a part of land into the Arabian Sea. The Sahyadris form a watershed of the peninsula. All the important rivers of Peninsular India, like the Godavari, Krishna, and Kaveri rise from the Western Ghats. The western rivers merging into the Arabian Sea are swift. The Gersoppa (Jog Falls) on Sharvati is the highest waterfall in India. The average elevation of the Western Ghats varies between 1000 to 1300 m **(Fig. 2.22).**

The important peaks of the Western Ghats are Kudermukh (1892 m), Pushpagiri (1714 m), Kalsubai (1646 m) and Salher (1567 m), Mahabaleshwar (1438 m) and Harishchandra (1424 m). In the Nilgiris the Eastern Ghat joins the Western Ghat to form a mountain knot (Nilgiri) whose highest point is Anaimudi (2695 m). South of Nilgiri lies the Palghat (Palakkad Gap). South of Palghat, the Western Ghat is known as Anaimalai Hills. Anaimudi is the highest peak of Sahayadris

Mt. is 2695 m above the sea level. The other important passes of the western Ghat are Thal Ghat and Bhor Ghat.

Bhor Ghat: (Joins Mumbai with Pune): Having an elevation of about 1,000 m above sea level, the Bhorghat joins Mumbai with Pune. It is one of the most busy passes in the Western Ghats. The frequency of trains and commercial vehicles is enormous.

Goran Ghat: Lying to the south of Mount Abu, it connects the city of Udaipur with Sirohi and Jalore in Rajasthan. It is about 1200 m above sea level. The surrounding rocks are desolate, characterised by thorny bushes and cacti.

Haldighat: It is a mountain pass in the Aravalli range of Rajasthan. Situated about 40 km from Udaipur, it connects Rajsamand and Pali district. The name is believed to have come from the turmeric-coloured yellow soil. The mountain pass is historically important as the location of the historic battle of Haldighat which took place in 1576 between Rana Pratap Singh of Mewar and Raja Mansingh of Amber, General of the Mughal Emperor—Akbar the Great.

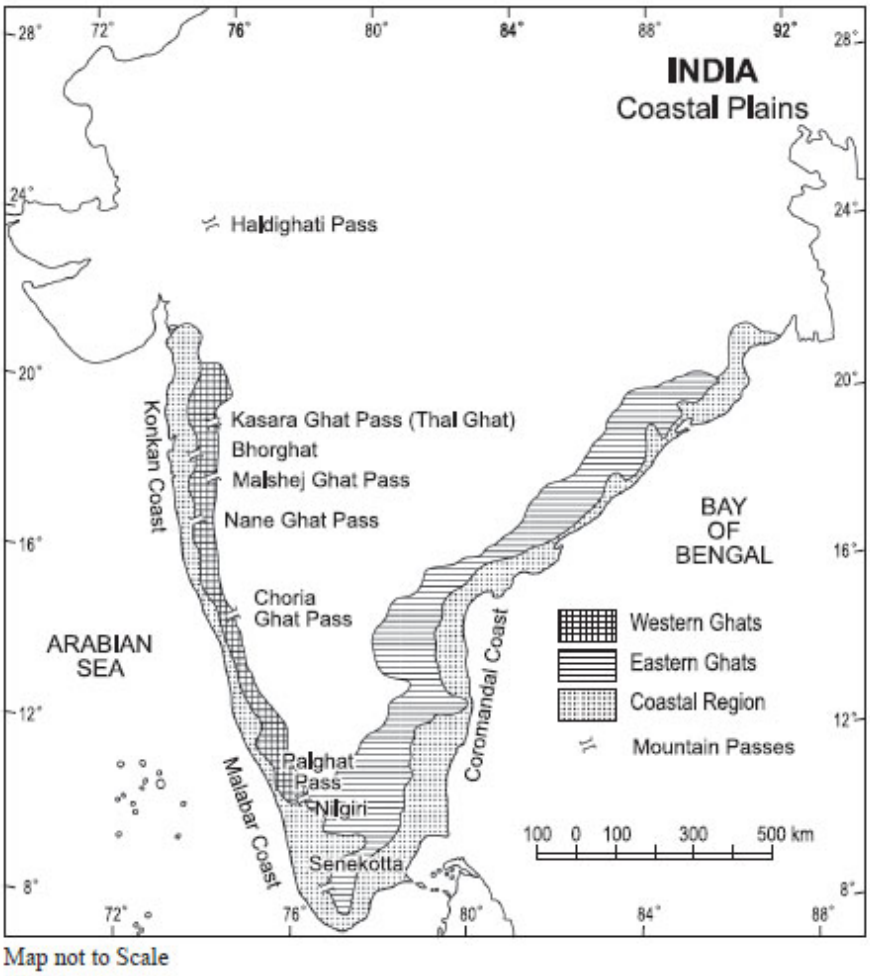


Fig. 2.22 The Eastern and Western Ghats and the Coastal Plains

Harishchandra: The Harishchandra mountain ranges in the southern parts of Maharashtra from north-west to south-east. It stretches in the districts of Pune and Osmanabad. It is covered by degraded forests.

Jog Falls: The short westward flowing Sharavati river pours down the Western Ghats, forming one of the highest waterfalls in the world at 250 m.

Kalsubai: Situated in the state of Maharashtra, it is one of the highest peaks of Western Ghats. It is 1646 metres above sea level. Inhabited by tribal people, its forest wealth has diminished due to deforestation.

Kudarmukh: Situated in the state of Karnataka, the Kudarmukh range (1892 m) is rich in iron ore. The iron-ore is of haematite and

magnetite type. Iron ore from here is exported to Iran through the port at Mangalore.

Mahabaleshwar: Having an elevation of 1438 m, Mahabaleshwar is one of the important peaks of the Western Ghats. It is a religious and cultural tourist attraction for domestic and international tourists.

The Nilgiri Hills: The Nilgiri Hills in the Western Ghats cover an area of about 2500 sq km and rise over 2500 m. Udhagamandalam, one of southern India's most famous hill resorts, is located here.

Palghat (Western Ghats; joins Coimbatore with Kochi and Kozhikode): Also known as the Palakkad Gap, it lies to the south of Nilgiri Hills. It has an elevation ranging from 75 to 300 m above the sea level. The width of this gap is about 25 km. It joins the state of Tamil Nadu with the seaports of Kerala. The river Gayitri flows through it from east to west.

Pushpagiri: This is one of the highest peaks of the Western Ghats. Its elevation is 1714 m above the sea level. It is the abode of Dravidian tribes. The forests are, however, degraded and soil erosion is the main problem.

Salher: Having an elevation of 1567 m above sea level, the Salher peak lies between Malegaon and Nashik. It is inhabited by tribal people. Heavy deforestation has reduced its aesthetic beauty and created numerous ecological problems.

Thal Ghat (Western Ghats; joins Nashik with Mumbai): Located in the Sahyadri Ranges, Thal Ghat is over 1,000 m above sea level. The National Highway No. 3 and the Bhopal-Indore Railway Line pass through the Thal Ghat.

8. The Eastern Ghats

The Eastetrn Ghats form the eastern boundary of the Deccan Plateau. It is a massive outlying block of hills. The average height of the Eastern Ghats is about 600 m. The Eastern Ghat is a series of detached hills of heterogeneous composition which are called by various local names. Between Mahanadi and Godavari, the average elevation of the Eastern Ghats is about 1100 m**.** The peak of Aroya-Konda (Andhra Pradesh) with an elevation of 1680 m is the highest

peak of the Eastern Ghats. Among other peaks Dewodi-Munda (1598 m), Singa-Raju (1516 m) and Nimalgiri (1515 m)
in the Koraput District and Mahendragiri (1501 m) in Ganjam District are the other important peaks. The predominant rocks of the Eastern Ghats are khondalites, metamorphosed-sedimentary, and charnokites (intrusive rocks being granite). Between the Krishna river and Chinnai are the Kondavidu, Nallamalai, Velikonds, Palkonda, and Erramala Ranges. Their continuation can be seen in the Seshachalam (Cuddapah and Anantapur districts), Javadi, Shevaroy, Panchaimalai, Sirumalai, and Varushnad Hills south west of Madurai (Tamil Nadu).

Significance of the Peninsular Plateau

Richly endowed with natural resources, Peninsular India has an important role in the economic development of the country. The importance of Peninsular India is mainly because of the following benefits from its location and rock formations:

(i) The Peninsular region of India is rich in both the metallic and non-metallic minerals. Mineral ores like iron, manganese, copper, bauxite, chromium, mica, gold, silver, zinc, lead, mercury, coal, diamond, precious stones, marble, building materials and decorative stones are found in abundance in this physiographic region. About 98 % of the Gondwana coal deposits of India are also found in the Peninsular region.

(ii) A substantial part of the Peninsular India is covered by black earth (Regur soil). The regur soil is conducive for the successful cultivation of cotton, millets, maize, pulses, oranges and citrus fruits. Some areas of south Peninsular India are suitable for the cultivation of tea, coffee, rubber, cashew, spices, tobacco, groundnut and oilseeds.

(iii) On the southern and eastern parts of Peninsular India are large stretches of Archaean, Dharwar, Cuddapah, and Vindhyan formations in which red, brown, and laterite soils have developed over time. These soils are the bases of rural economy.

(iv) The Western Ghats, Nilgiris and the Eastern Ghats are

covered by thick tropical moist deciduous and semi-evergreen forests. These forests provide teak, sal, sandalwood, ebony, mahogany, bamboo, cane, rosewood, iron-wood, and logwood as well as a large variety of forest products.

(v) The rivers flowing eastward into the Bay of Bengal make several gorges, waterfalls, rapids and cataracts, which have been harnessed for the generation of hydro-electricity. The rivers originating from the Western Ghats offer great opportunity for the generation of hydel power and irrigation of agricultural crops and orchards.

(vi) There are numerous hill stations and hill resorts, of which Ooty, Udhagamandalam, Kodaikonal, Mahabaleshwar, Khandala, Metheron, Pachmarhi, and Mount Abu are the most important.

(vii) Apart from teak and fuelwood, the forests of Western and Eastern Ghats are rich in medicinal plants.

(viii) The hilly and mountainous areas of the Peninsula are the abodes of many scheduled tribes. South of the Vindhyans is a predominance of Dravidian culture.

THE COASTAL PLAINS

The Peninsular Plateau of India is flanked by narrow coastal plains of varied width from north to south, known as the West-Coastal Plains and the East-Coastal Plains. These coastal plains differ from each other. They were formed by the depositional action of the rivers and the erosional and depositional actions of the sea waves.

According to geologists, the origin of the western and eastern coasts of India may be attributed to the faulting and subsidence of the Arabian Sea and the Bay of Bengal towards the close of the Eocene Period. Consequently, alluvial deposits along these coasts are of very recent origin, ranging from Pliocene to recent times. These coastal plains have the evidence of submergence and emergence. The Indian coastal plains may be subdivided into the following three

divisions: (i) The Gujarat Coastal Plain, (ii) the West Coastal Plain, and (iii) the East Coastal Plain.

(i) The Gujarat Coastal Plain

The Gujarat plain covers almost the entire state of Gujarat, except the districts of Banaskantha and Sabarkantha. It is formed by the alluvial deposits of Sabarmati, Mahi, Luni, and numerous tiny parallel consequent streams. Part of this plain is the product of depositional activity of the winds and recession of the sea. It contains the Gondwana rocks (Umia Series), resting over the marine Jurassic rocks and capped by Lower Cretaceous (Apatian) beds. The Deccan lava lies over the Umia series.

The eastern section of Gujarat Plain is a projected jet of Sindhu-Ganga alluvial tract in Peninsular India. This projection is the outcome of an extensive Pleistocene sedimentation. Present rivers have further advanced this deposition to the Gulf of Khambat. Among the highlands, mention may be made of the Arasur mountains in eastern Gujarat, the Rajpipla Hills (Satpura)-famous for agate quaries, the Parnera Hills in Bulsar district, Sahyadris in the southern side and igneous complex of the Girnar Hills (Gorakhnath Peak, 1117 m) and Mandav Hills in Kathiwad.

The Rann of Katch is an extensive tract of naked tidal mudflats transected by abandoned and live creeks. The Gulf of Katch separates the Rann of Katch from the Kathiawar Peninsula. The salt in the soil makes this low-lying marshy area almost barren and unproductive. The whitish vertebrae of salts appear as white bony structures of the dried creeks. Live creeks form dendritic pattern of drainage and there has been accentuation in this pattern due to earthquakes. South of the Rann lies Katch, formerly an island, which is almost surrounded by the Rann except in the south-west.

(ii) The West Coastal Plain

It lies between the Sahyadris and the Arabian Sea. It is about 1400 km long and 10 to 80 km wide. It has an elevation up to 150 m above sea level, reaching more than 300 m at places. The Western Coastal

Plain is characterised mainly by sandy beaches, coastal sand-dunes, mud-flats, lagoons, alluvial tracts along rivers, estuary, laterite-platforms and residual hills. The Sahaydris (elevation 750–1225 m) run parallel to the plain and present their steep face to the low lands with Thalghaat and Bhorghat (gaps) in the north and the Palghat (Plakkad Gap) in the south of Nilgiri. The northern part of the west coastal plain, known as the Konkan Plain, is about 530 km long and 30 to 50 km wide. Southward is the Karnataka coastal plain which is about 525 km long and 8 to 25 km wide**.** It is the narrowest part of the West coastal plain. The southern part is known as the Malabar coast which is about 550 km long and 20–100 km wide. The maximum extension of the Malabar coast is found in the valleys of the Beypore, the Ponnani (draining through Palghat), the Periyar and Pamba Achankovil rivers. This coast is characterised by sand dunes. Along the coast, there are numerous shallow lagoons and backwaters—*Kayaks* and *Teris***.** These lagoons are linked together to facilitate navigation through small country boats. Here, Vembanad and Asthamudi are the important lagoons of the Malabar coast. It is the homeland to aquatic life like crabs, frogs, mudskippers, birds like terns, kingfisher, cormorants and otters and turtles live alongside the backwaters. The backwaters are important tourist spots, and are of importance for transportation, fishing and even agriculture. The region has deposits of Monazite sands which is reddish brown and rich in phosphate. Monazite is radioactive due to the presence of thorium.

The Eastern Coastal Plain

The eastern coastal plain lies between the Eastern Ghats and the Bay of Bengal, and stretches along the coasts of Odisha, Andhra Pradesh, and Tamil Nadu. These plains are formed by the alluvial fillings of the littoral zone comprising some of the largest deltas of the world. The East-Coastal Plains consist mainly of Recent and Tertiary alluvial deposits. These are gentle, monotonous plains rising gently westward to the foot of the Eastern Ghats. The monotony of the topography is broken by the presence of numerous hills. This coastal

plain has a straight shoreline with well defined beaches of sand and shingles. The most famous is the Marina Beach in Chennai. All along the coast, there are several sandbars generally in front of the river mouths. There are some of the important lagoons of India along the Eastern coast, of which, Chilka in the south-west of the Mahanadi delta is the biggest lake (65 km×8 km) in the country. The Kulleru lake lies between the deltas of Godavari and Krishna, while the Pulicat lake lies further south on the border of Andhra Pradesh and Tamil Nadu.

THE INDIAN ISLANDS

India has a total of 615 islands, of which 572 lie in the Bay of Bengal, and the remaining 43 in the Arabian Sea. Out of the 572 islands of Andaman and Nicobar, only 36 are inhabited. The Bay of Bengal islands include the Andaman and Nicobar Islands which are largely tectonic and volcanic in origin, while the islands of the Arabian Sea are mainly coral formations. Moreover, there are a number of offshore islands along the mouth of the Ganga, eastern and western coasts and in the Gulfs of Khambat, Kachchh and Mannar.

Islands of the Bay of Bengal

The main islands of the Bay of Bengal are the Andaman and Nicobar groups. The Andaman and Nicobar islands are separated by the Ten Degree Channel. The shortest distance of the Andaman Islands from the mainland (Bay of Bengal Head) is about 2000 km and the extreme southern point is the Indira Point—the southern most point of the Great Nicobar Island (**Fig. 2.23).**

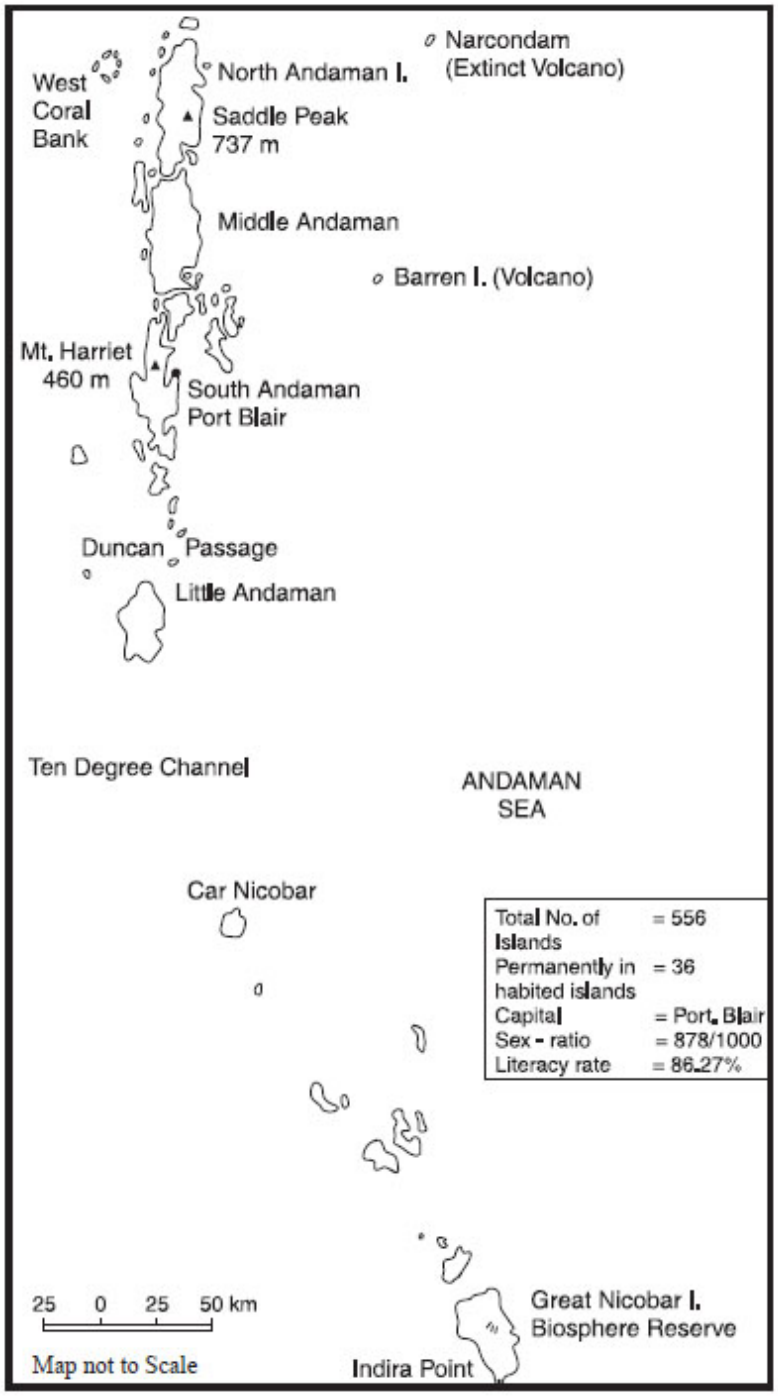


Fig. 2.23 Andaman and Nicobar Islands

The Andaman and Nicobar Islands have a warm tropical climate all year round with two monsoons. Temperatures are around 25^{o} C, but the sea breeze has a cooling effect. The islands receive heavy rain,

with the monsoon season lasting from mid-May to mid-September and from November to mid-December.

The Andaman Nicobar Islands are thickly forested and have a rich marine life among the reefs. The islands are a birdwatcher's paradise with 242 species recorded. The entire region falls in a major earthquake zone. The Barren Island in the Andamans has an active volcano.

In the Bay of Bengal, there are two volcanic islands (Barren and Narcondam) situated within 80 km
east of the Andaman Islands. The Andaman Islands have been formed by the extension of the Tertiary mountain chain of Arakanyoma. The main rocks of these islands are sandstone, limestone and shale. The Nicobar group of islands comprise 18 islands of which only 11 are inhabited. The physiography of the Nicobar islands is mainly of coral origin.

Rice is the main crop in Andaman and Nicobar Islands. Coconut and arecanut are the main cash crops of Nicobar. Tropical fruits like pineapple, a variety of bananas, sweet papaya and mango grow on a smaller scale in the Andaman group of islands.

The Tribal population in the Andaman Islands is fast dwindling. Most of its present inhabitants are migrants from Bangladesh, Myanmar, and India and Tamils from Sri Lanka. Some of the well known surviving tribes of the Andamans and Nicobar are the Onges, Jarawas and Sentinelese.

One of the largest and also the rarest crabs in the world, the *Giant Robber Crab*, can be found in the Wandoor Marine Biosphere Resrve in south Andaman and Great Nicobar Islands. Its powerful claws help it to climb the coconut tree and break the hard shell of its fruit.

The Arabian Sea Islands

There are 43 islands in the Arabian Sea, out of which only 11 are inhabited. The shortest distance from the mainland (Calicut) is about 109 km. Kavaratti, located on the island of this name is the capital of Lakshadweep. Lakshadweep islands are separated from the Maldive Islands by the Eight Degree Channel. Hills and streams are absent

on these islands. The Minicoy is the largest (4.5 sq km)
and has a light house and a weather observatory. Fishing is the main occupation of the people of Lakshadweep. In Lakshadweep coconut is the only major crop, although pulses and vegetables are also grown. The sea around the island is rich in marine life **(Fig. 2.24).**

Offshore Islands

There are numerous islands in the delta region of Ganga and in the Gulf of Mannar. Among the Western coast islands Piram, Bhaisala (Kathiawar), Diu, Vaida, Nora, Pirotan (Pirothan), Karunbhar (Kachchh coast), Khadiabet, Aliabet (Narmada-Tapi mouths), Butchers, Elephanta, Karanja, Cross (near Mumbai), Bhatkal, Pegioncock, St. Mary (Mangalore coast), Anjidiv (Goa coast), Vypin near Kochi, Pamban, Crocodile, Adunda (Gulf of Mannar), Sri Harikota (mouth of Pulicat Lake, Pairkud (mouth of Chilka Lake), Short, Wheeler (Mahanadi-Brahmani mouth), and New Moore, and Ganga-Sagar and Sagar (Ganga Delta). Many of these islands are uninhabited and administered by the adjacent states.

Earthquakes in India

Earthquakes are vibrations of the Earth caused by ruptures and sudden movements of rocks that have been strained beyond their elastic limits. In other words, earthquakes are movements within the earth caused by natural or man-made stresses. Earthquakes are casued by (i) volcanic eruptions, (ii) ruptures and sudden movements of rocks (folding and faulting), (iii) movement of plates (plate tectonics), and (iv) anthropogenic factors.

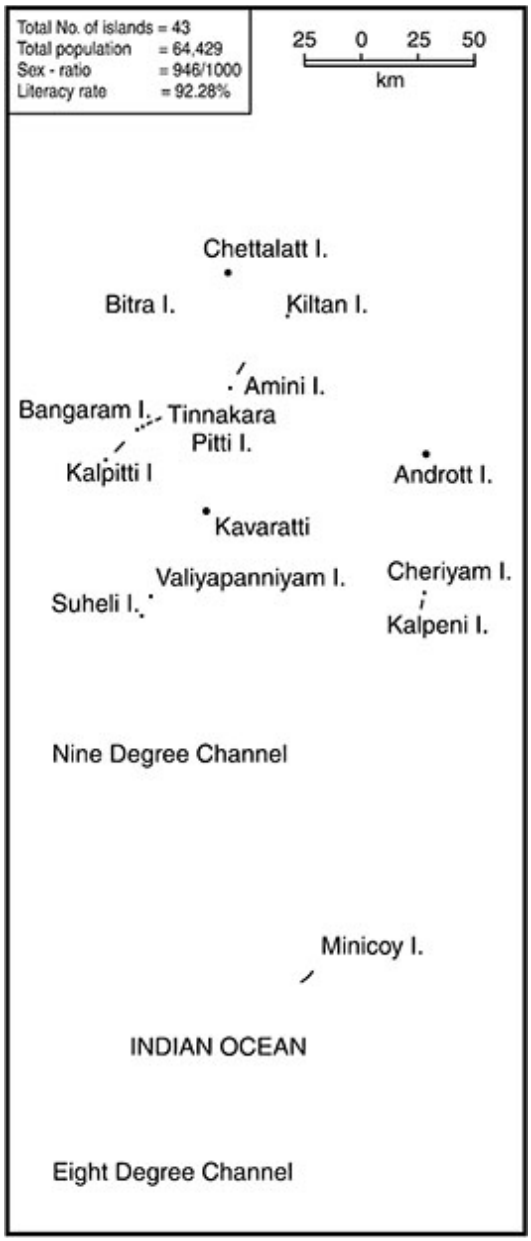


Fig. 2.24 Lakshadweep Islands

The structure of the subcontinent of India is characterised with plate-boundaries in which folding and faultings are a regular feature. Man has constructed numerous dams and reservoirs across rivers. All these factors, individually or collectively lead to earthquakes in the country. Consequently, every year numerous earthquakes are recorded in the Himalayan region, the Great Plains of India, Purvanchal, and the Islands of Andaman and Nicobar. Some of the

important earthquakes of great magnitude have been plotted in **(Fig. 2.25)**. It may be seen from this figure that most of earthquakes of high magnitude occurred in the folded mountains of Himalaya. Their frequency is, however, low in the relatively more stable Peninsular India. On the basis of seismic intensity and frequency, a committee of experts comprising seismologists and geologists have divided India into five seismic zones **(Fig. 2.26).**

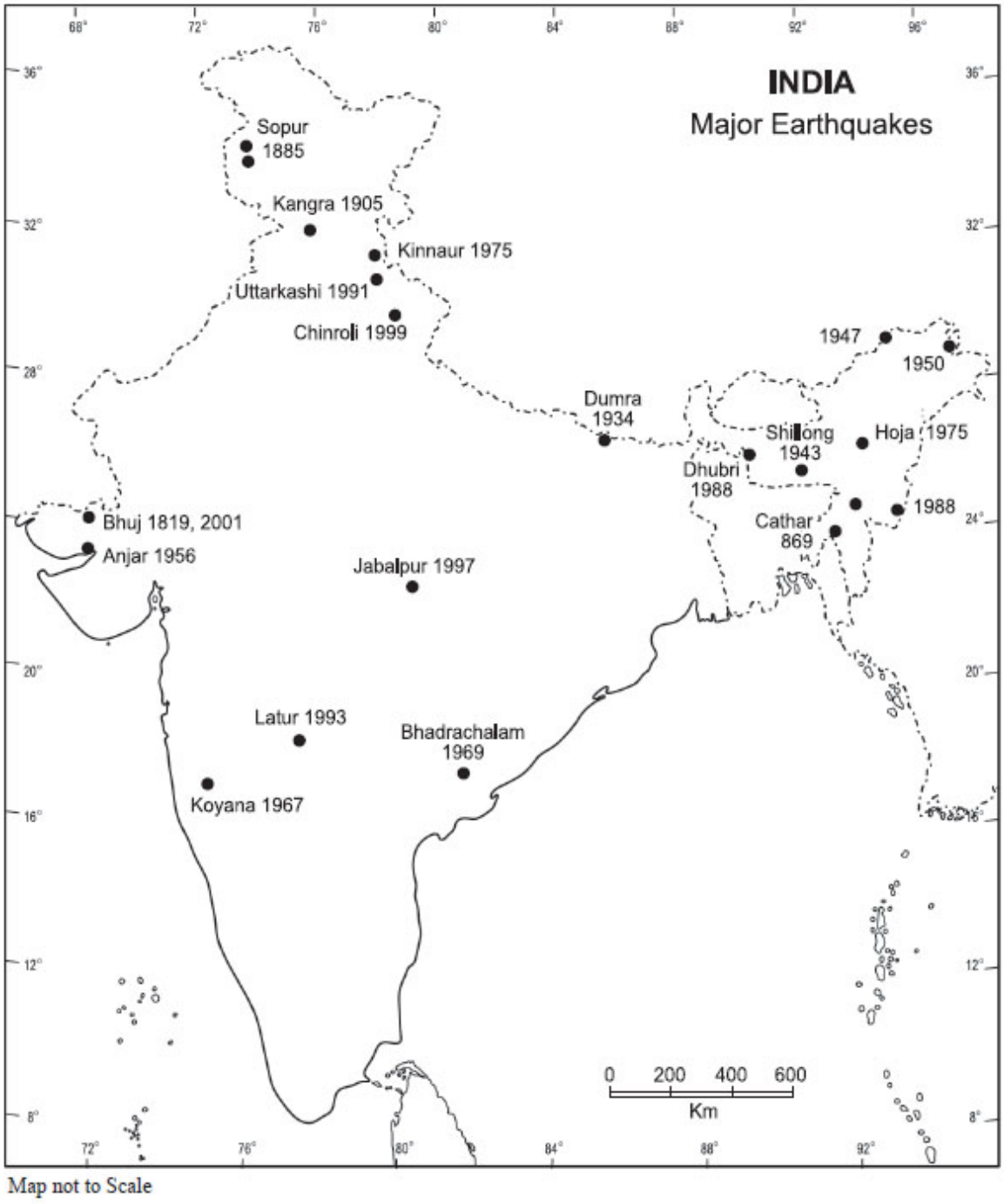


Fig. 2.25 India-Major Earthquakes (1819—2001)

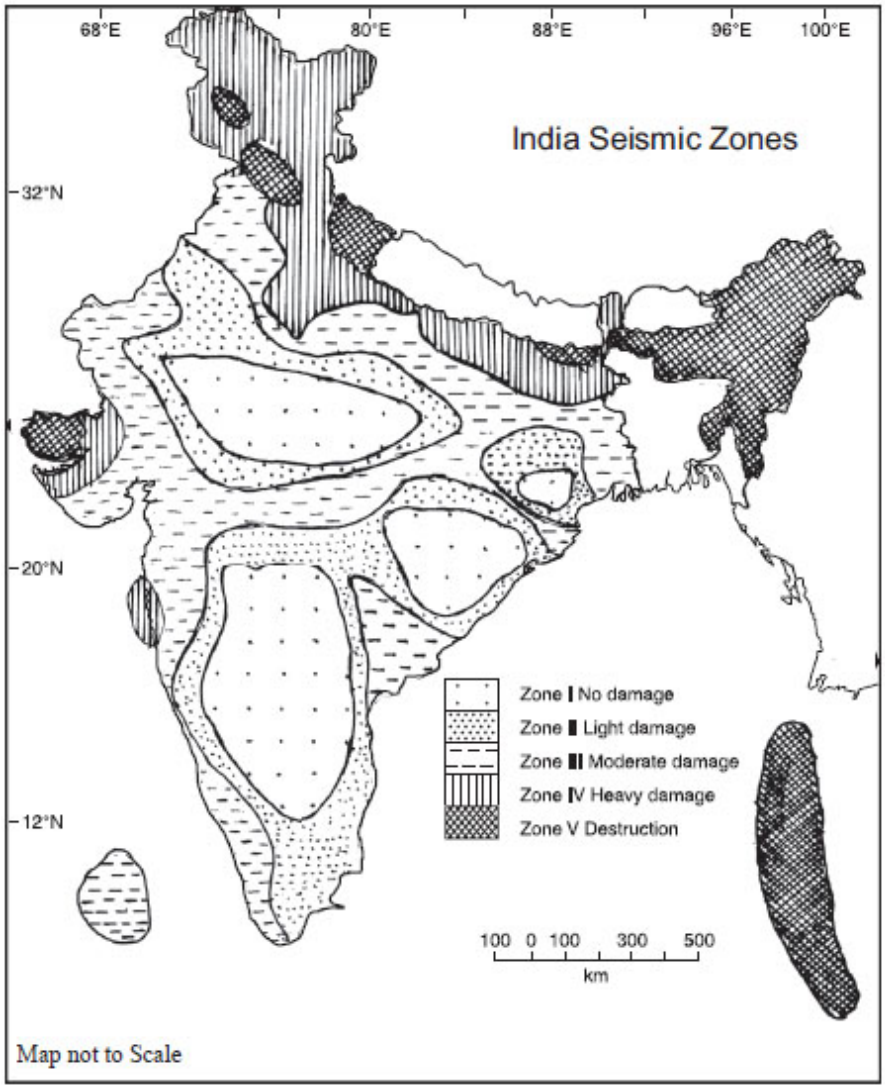


Fig. 2.26 India—Seismic Zones [*Source: Self Creation with the help of NCERT*]

Their intensity is measured on the Richter scale with the help of a seismometer. The scale given below helps in understanding the intensity of the earthquake.

Table 2.5

Richter Scale of Earthquake Magnitude

Magnitude	Category	Effects
Less than 1.0 to 2.9	Micro	Recordable but not felt by humans

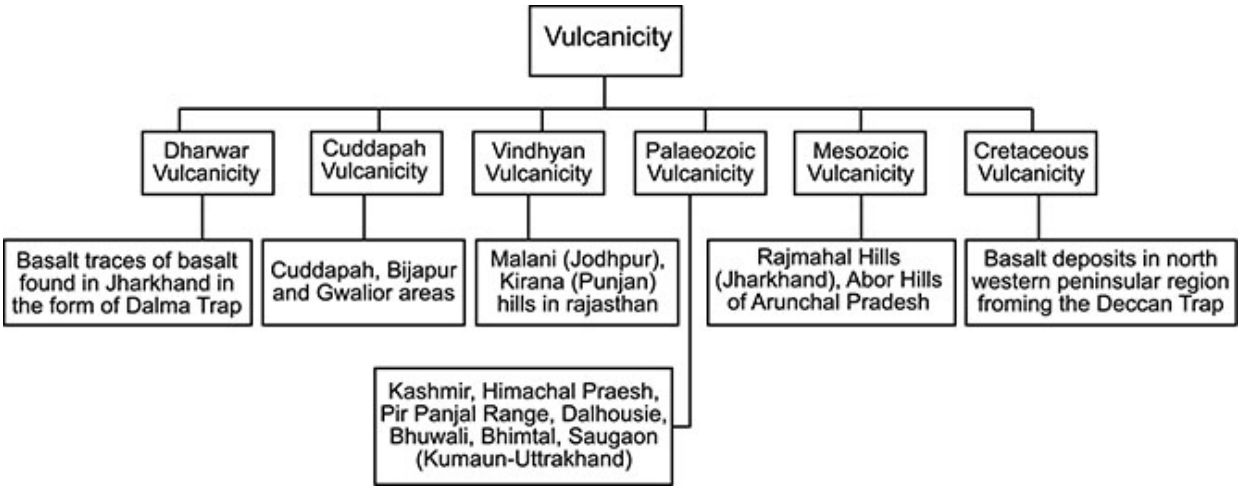
Magnitude	Category	Effects
3.0 to 3.9	Minor	Felt by people without causing any damage
4.0 to 4.9	Light	Felt by everyone with minor damages
5.0 to 5.9	Moderate	Slight damage
6.0 to 6.9	Strong	Moderate damage in populate d areas
7.0 to 7.9	Major	Serious damage and loss of life
8.0 and higher	Great	Severe destruction and loss of life

About 60 % of the total area of the country is under moderate and high seismic zones. A belt of about 3500 km length and about 500 km width in north India is the main earthquake prone zone of India, characterised with devastating earthquakes of high magnitude. Some of the important earthquakes recorded in India during the last 100 years have been plotted in **(Fig. 2.25).**

VULCANICITY

A volcano is an opening in the crust of the earth, connected by a conduit to an underlying magma chamber, from which molten lava, volcanic gases, steam, and pyroclastic materials are ejected. It is usually in the form of a peak which may be cone shaped or dome-shaped depending on the type of volcano and type of material ejected. The main causes of volcanic eruptions are associated with sea-floor spreading, plate-tectonics, and mountain building processes.

At present, the only active volcanoes in India are at the Barren and Narcondom Islands (Andaman group of Islands), but geographical evidence is available for vulcanicity in the past. Prof. H.L. Chibber (1945) has identified six episodes of vulcanicity in the country which are given as follows:



Geologists opine that apart from the Barren Islands, volcanic activity is going on beneath the Himalayan crust due to the subduction and melting of the northern margin of the Indian Plate.

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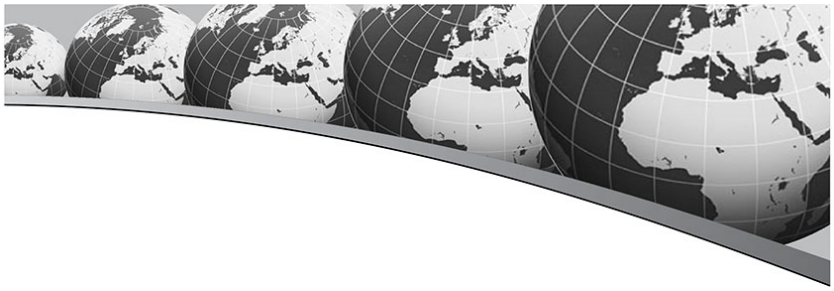
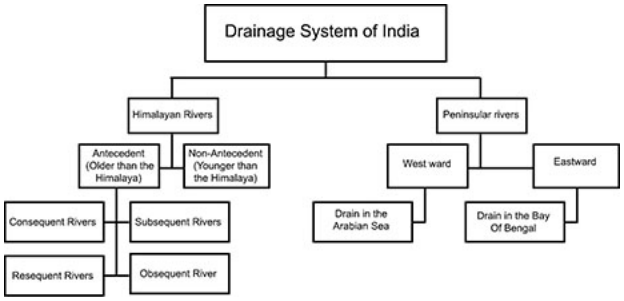


3

The Drainage System of India



THE DRAINAGE SYSTEM

The drainage system is an integrated system of tributaries and a trunk stream which collect and funnel surface water to the sea, lake or some other body of water. The total area that contributes water to a single drainage system is known as a '*drainage basin*'. This is a basic spatial geomorphic unit of a river system, distinguished from a neighbouring basin by ridges and highlands that form divides. Thus, river basins are natural units of land. They are regarded as the fundamental geomorphic as well as hydrological units for a systematic study of the river basins, mainly due to the following three reasons:

(i) They can be placed in an orderly hierarchy,

(ii) They are areal units whose geomorphological and hydroloical characteristics can be measured quantitatively, and

(iii) They can be treated as working systems with energy inputs of climatological variables like temperature and rainfall and output of river discharge as runoff.

The Committee on Runoff of the American Geophysical Union treats the micro-unit within a river basin as the watershed, while the sum of all the micro, meso and macro tributaries of a river is known as a *river basin*.

A watershed is an area of land that drains all the streams to a common outlet such as a stream, river, lake, wetland or coastal bodies. It is the area drained by a river and its tributaries. The size of a watershed is defined on various scales which are referred to as its *Hydrological Unit Code (HUC)*.

A drainage basin is also known as the '*catchment basin*'. It is an area of land where the surface water from rain, melting snow or the ice converges at a lower elevation. A drainage basin is an area drained by the tributary streams that merge with the main stream.

DRAINAGE PATTERNS

A geometric arrangement of streams in a region; determined by slope, differing rock resistance to weathering and erosion, climate, hydrologic variability, and structural controls of the landscape is known as a '*drainage pattern*'. In other words, drainage pattern refers to a design which a river and its tributaries form together, from its source to its mouth. The factors controlling the pattern of drainage in a region include the topography, slope, structural control, nature of rocks, tectonic activities, supply of water, and above all, the geological history of that region. In India, the following types of drainage patterns are found:

1. The Antecedent or Inconsequent Drainage

The rivers that existed before the upheaval of the Himalayas and cut their courses southward by making gorges in the mountains are known as the antecedent rivers. The Indus, Satluj, Ganga, Sarju (Kali), Arun (a tributary of Kosi), Tista, and Brahmaputra are some of the important antecedent rivers, originating from beyond the Greater Himalayas.

1.1 Consequent Rivers

The rivers which follow the general direction of slope are known as the consequent rivers. Most of the rivers of peninsular India are consequent rivers. For example, rivers like Godavari, Krishna, and Kaveri, descending from the Western Ghats and flowing into the Bay of Bengal, are some of the consequent rivers of Peninsular India (Fig. 3.1).

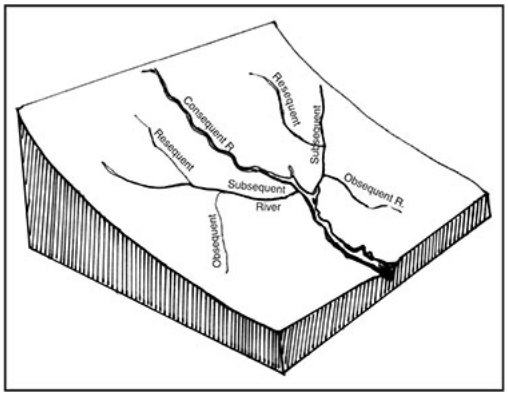


Fig. 3.1 Consequent River and its Tributaries

1.2 Subsequent Rivers

A tributary stream that is eroded along an underlying belt of non-resistant rock after the main drainage pattern (consequent river) has been established is known as a subsequent river. Due to the northward slope of the Peninsula towards the Great Plains, the rivers originating from the Vindhyan and the Satpura ranges flow northward into the Ganga system. The Chambal, Sind, Ken, Betwa, Tons, and Son meet the Yamuna and the Ganga at right angles. They are the subsequent drainage of the Ganga drainage system.

1.3 Resequent Rivers

It flows in the same direction as of the initial consequent stream. However, it develops in response to a new base level. They originate during the initiation of the cycle of erosion in a folded structure. Their origin is recent and thus they are called resequent.

1.4 Obsequent Rivers

It is formed when it adjusts itself to some rock structure. An obsequent river may form perpendicularly to the subsequent stream. It flows opposite to the direction of flow of the original consequent stream. The northward flowing streams of the lesser Himalaya join the subsequent stream Sun Kosi, as obsequent streams. The direction is opposite to that of Ganga and the Yamuna.

2. Superimposed, Epigenetic (Discordant) or Superinduced Drainage

It is formed when a stream with a course originally established on a cover of rock now removed by erosion, so that the stream or drainage system is independent of the newly exposed rocks and structures. In other words, it is a drainage pattern which exhibits discordance with the underlying rock structure because it originally developed on a cover of rocks that has now disappeared due to denudation. Consequently, river directions relate to the former cover rocks and, as the latter were being eroded, the rivers have been able to retain their courses unaffected by the newly exposed structures. The stream pattern is thus superposed on, or placed on, ridges or structural features that were previously buried. The Damodar, the Subarnarekha, the Chambal, the Banas, and the rivers flowing at the Rewa Plateau present some good examples of superimposed drainage.

3. Dendritic Drainage or Tree-like Pattern

A term used for drainage which is branching, ramifying or dichotomising, thereby giving the appearance of a tree. Thus, a dendritic pattern develops in a terrain which has uniform lithology, and where faulting and jointing are insignificant; e.g., massive crystalline rocks or thick plains consisting of clays. Most of the rivers of the Indo-Gangetic Plains are of dendritic type. The term dendritic, coined by I.C. Russel (1898), is used to denote the most common type of drainage pattern and is a distinctive feature of the regions having horizontally bedded sedimentary rocks or massive igneous rocks (Fig. 3.2-A).

4. Trellis Drainage

Trellis drainage pattern is characteristic of folded topography. This pattern exists in nearly parallel mountains, where drainage patterns are influenced by rock structures of variable resistance and folded strata. In a trellis drainage pattern, tributaries are arranged in a pattern similar to that of a garden trellis. In a trellis pattern, the river forms a net like system and the tributaries flow roughly parallel to each other. The old folded mountains of the Singhbhum (Chotanagpur Plateau) have a drainage of trellis pattern (Fig. 3.2-D).

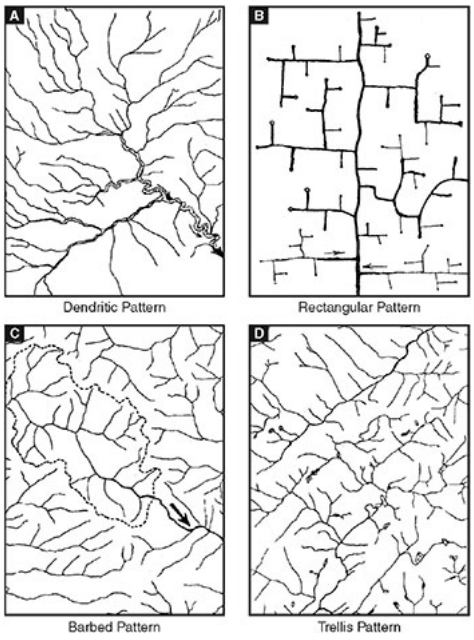


Fig. 3.2 (A, B, C, D) Drainage Patterns

5. Barbed Pattern

A pattern of drainage in which the confluence of a tributary with the main river is characterised by a discordant junction—as if the tributary intends to flow upstream and not downstream. This pattern is the result of capture of the main river which completely reverses its direction of flow, while the tributaries

continue to point in the direction of former flow. The Arun River (Nepal), a tributary of the Kosi is an interesting example of barbed drainage pattern (Fig. 3.2-C).

6. Rectangular Drainage

A rectangular pattern is formed by a faulted and jointed landscape, directing stream courses in patterns of right-angle turns. The drainage pattern marked by right-angled bends and right-angled junctions between tributaries and the main stream is known as rectangular drainage. It differs from the Trellis pattern in so far as it is more irregular and its tributary streams are neither as long, nor parallel as in trellis drainage. A typical example of this drainage pattern is found is the Vindhyan Mountains of India **(Fig. 3.2-B)**.

7. Radial Pattern

Radial patterns of drainage results from streams flowing off a central peak or dome. It is a pattern characterised by outflowing rivers, away from a central point, analogous with the spokes of a wheel. It tends to develop on the flanks of a dome or a volcanic cone. A good example of a radial drainage pattern is provided by the rivers originating from the Amarkantak mountain. Rivers like Narmada, Son and Mahanadi originating from Amarkantak Hills flow in different directions and are good examples of radial pattern. Radial drainage patterns are also found in the Girnar Hills (Kathiwar, Gujarat), and Mikir Hills of Assam (Fig. 3.3-A).

8. Annular Pattern

Annular patterns are produced by structural domes, with concentric patterns of rock strata guiding stream courses. In this drainage pattern, the subsequent streams follow curving or arcuate courses prior to joining the consequent stream. This results from a partial adaptation to an underground circular structure; a dome like igneous intrusion (batholith). The subsequent streams find it easier to erode the concentric, less resistant strata. This is not a very common drainage pattern in India. Some examples of this are, however, found in Pithoragarh (Uttarakhand), Nilgiri Hills in Tamil Nadu, and Kerala (Fig. 3.3-B).

9. Parallel Drainage

The drainage pattern in which the rivers flow almost parallel to each other is known as parallel drainage. The small and swift rivers originating in the Western Ghats and discharging their water into the Arabian Sea provide a good example of parallel drainage pattern in India. The important rivers which are the examples of parallel drainage are Surya, Kalu and Savitri (Maharashtra), Zuvari, Sal and Talpona (Goa), Kalinadi and Sharavati (Karnataka) and Porrani, Periyar and Pemba in Kerala. (Fig. 3.3-C).

S. No.	Type of Drainage	Example of Drainage
1	Superimposed/Epigenetic/Superinduced	The Damodar, the Subarnarekha, the Chambal, the Banas and the rivers flowing at the Rewa Plateau
2	Dendritic or tree-like pattern	Rivers of Indo-Gangetic Plains
3	Trellis	Rivers in mountains of Singhbum (Chhota Nagpur Plateau)
4	Barbed	The Arun River (Nepal)-tributary of River Kosi
5	Rectangular	Found in Vindhyan Mountains
6	Radial	In Girnar Hills (Kathiawar, Gujarat) and Mikir Hills of Assam
7	Annular	Pithoragarh (Uttarakhand), Nilgiri Hills (Tamil Nadu), Kerala
8	Parallel	Surya, Kalu, Savitri in Maharashtra, Zuvari, Sal, Talpona in Goa, Kalinadi and Sharavati in Karnataka, Porrani, Periyar, Pemba in Kerala
9	Deranged	Found in glaciated valleys of Karakoram

10. Deranged Pattern

This is an uncoordinated pattern of drainage characteristic of a region recently vacated by an ice sheet. This is probably due to the irregularities produced by glacially deposited materials, e.g., Kame and Kettle, and by the fact that there has been insufficient time for the drainage to become adjusted to the structures of the solid rock underlying the glacial drift. The picture is one of numerous water courses,

lakes and marshes; some inter-connected and some in local drainage basins of their own. This type of drainage is found in the glaciated valleys of Karakoram (Fig. 3.3-D).

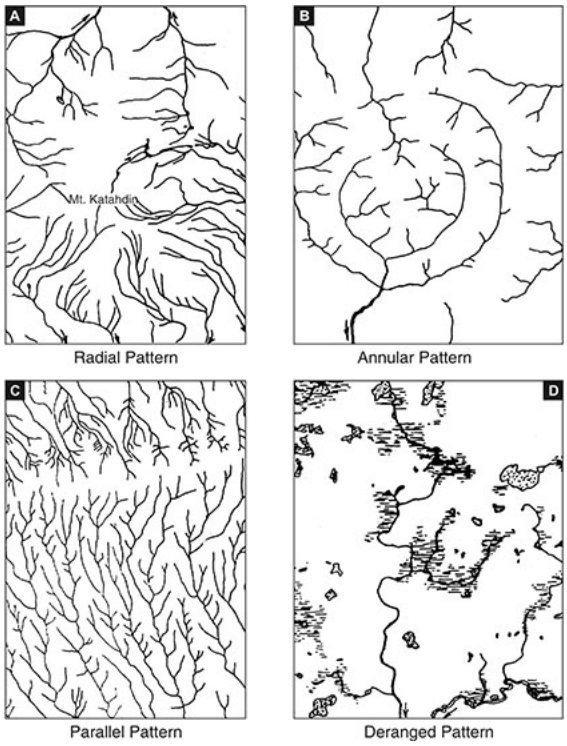


Fig. 3.3 (A, B, C, D) Drainage Patterns

RIVER BASINS OF INDIA

Rivers with their tributaries are the main channels of drainage of the land surface; they are at the same time also the chief agents of land-erosion, and the main lines for transport waste-products of the land to the sea. The area drained by the main river including all its tributaries is known as its drainage basin. On the basis of the area drained, the river basins of India have been classified into three categories:

(i) river basins with catchment area of more than 20,000 sq km known as large river basins;

(ii) river basins having a catchment area between 2000 to 20,000 sq km known as the medium basins, and

(iii) the rivers having a catchment area less than 2000 sq km known as minor river basins.

The major river basins of India in the descending order of area are: the Ganga, Indus, Godavari, Krishna, Brahmaputra, Luni, Mahanadi, Narmada, Kaveri, Tapi, Pennar, Brahmani, Barak, Mahi, Sabarmati, and Subarnarekha. The major river basins form about 81% of the total drainage area of the country (Fig. 3.4).

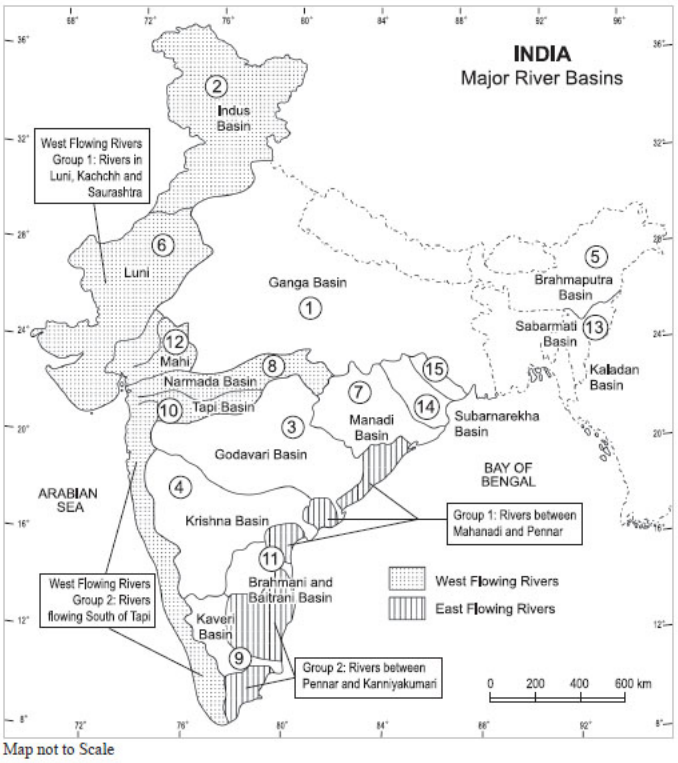


Fig. 3.4 Major River Basins [*Source: Central Institute of Education Technology Government of India*]

Using the SRTM DEM data of NASA, having spatial resolution of 90 m, the country has been divided into 25 major river basins and 103 sub-basins.

The three major river systems (Ganga, Indus, and Brahmaputra) are international rivers. The Indus and some of its important tributaries traverse Tibet (China), India, and Pakistan, while the Ganga and Brahmaputra, and their tributaries cross Tibet, Nepal, Bhutan, and Bangladesh. The main river basins, their basin area, and annual discharge has been shown in **(**Table 3.1).

On the basis of mode of origin, the drainage of India may be divided into (i) Himalayan or the Extra-Peninsular Drainage, and (ii) the Peninsular Drainage.

There is no clearcut line of demarcation between these two drainage systems, as many of the Peninsular rivers like the Chambal, Betwa, Sind, Ken, and Son are much older in age and origin than the Himalayan rivers.

Table 3.1

Major Rivers of India and their Surface Flow

Code on the Map	River Basin	Area (km2)	Percentage Area	Annual Discharge (m^3/km^2)	Percent
1	Ganga	808,334.44	26.2	468,700	25.2
2	Indus	453,931.87	9.8	79,500	4.3
3	Godavari	302,063.93	9.5	118,000	6.4
4	Krishna	254,743.31	7.9	62,800	3.4
5	Brahmaputra	186,421.6	7.8	627,000	33.8
6	Mahanadi	139,659.15	4.3	66,640	3.6
7	Narmada	92,670.51	3.0	54,600	2.9
8	Kaveri	85,624.44	2.7	20,950	1.1

9	Tapi	63,922.91	2.0	17,982	0.9
10	Pennar	54,243.43	1.7	3,238	0.2
11	Brahmani and Baitrani Basin	51,893.68	1.2	18310	1.0
12	Mahi Basin	38,336.8	1.0	11,800	0.6
13	Sabarmati Basin	30,678.59	0.6	7,940	0.4
14	Subarnarekha	25,792.16	0.7	3,800	0.2
15	Medium and Minor Rivers		23.6	–	16.0
	Total (India)		100.00	1,561,170	100.00

*Area means basin area in India.

Source: S.P. Das Gupta, 1989.

The Himalayan Drainage

The Himalayan drainage system comprises all the international rivers of India, i.e., the Indus, the Ganga, and the Brahmaputra. Most of these rivers and their major tributaries are perennial in character, obtaining their water from the glaciers and rains. These rivers are in their youthful stage carving out a number of erosional landforms like waterfalls, cataracts, rapids, gorges, steep slopes, and river terraces. The Himalayan rivers are not only eroding agents, but are also depositing agents in the plains and deltas. The great Himalayan rivers (Indus, Satluj, Ganga, Kali, Karnali, Gandak, Kosi, Tista, and Brahmaputra) are older than the Greater Himalayas. They are antecedent, and drain not only the southern slopes of the Himalayas but to a large extent, the northern Tibetan slopes as well. The Himalayan courses of these rivers are highly tortuous, but in plains they display a strong meandering tendency and shift their courses frequently. The river regimes, although perennial, exhibit wide seasonal fluctuations; causing devastating floods during rainy season but shrinking to the bottom of the valley with a number of shoals during the dry season. These rivers continue to cause intensive erosion and transport heavy loads of sand and silt annually. Several of the Himalayan rivers are older to the Himalayas. Such rivers are known as the antecedent rivers. The Himalayan rivers have great socio-economic and cultural importance in the life of the Indian people. The water of the Himalayan rivers is utilised for irrigation, industries, hydel-power generation, navigation, and domestic purposes. Moreover, the big rivers are navigable in the plain areas of their courses.

Evolution of the Himalayan Rivers

The geologists and geomorphologists are not unanimous about the origin of the Himalayan rivers. The Himalayan rivers have a long geological history. The major rivers of the Himalayas like Indus, Brahmaputra, Satluj, Ganga (Alaknanda and Bhagirathi), Gandak, Kali, Kosi, Tista, Manas, etc., originate on the southern slopes of the Tibetan Highlands. Rivers like Indus, Satluj and Brahmaputra first flowed parallel to the main axis of the moutain in longitudinal troughs, then they take sudden bends towards the south, carving out deep gorges across the mountain ranges to reach the northern plains of India. Such deep gorges created by the Indus, Satluj, Alaknanda, Sarju (Kali), Gandak, Kosi, Tista, and Brahmaputra suggest that they are older than the Himalayan mountains, and are antecedent in character.

E. H. Pascoe (1919) and *G.E. Pilgrim* (1919) attempted to give a tangible explanation to the problem of the Himalayan drainage. According to them, the Tibetan Plateau was drained to the west by the mighty Tsangpo-Indus-Oxus combined with the Tibetan river which merged into Oxus Lake before the upheaval of the Himalayas (Fig. 3.5). Similarly, another big river called the Indo-Brahma River (*Pascoe, 1919*) or the Shiwalik River (*Pilgrim, 1919*) traversed the entire longitudinal extent of the Himalaya from Assam to Punjab from north-east to empty into the Gulf of Sindh, near lower Punjab, during the Miocene Period (Fig. 3.5). The former (Tsangpo- Indus-Oxus) was disrupted by the headward erosion of its left hand tributaries, i.e., the Proto-Sind, the Proto-Satluj, the Proto-Brahmaputra, and later was captured by the present lower Irradwady-Chindwin, the lower Brahmaputra, the lower Satluj, and the lower Indus which are the probable remnants of the Shiwalik River.

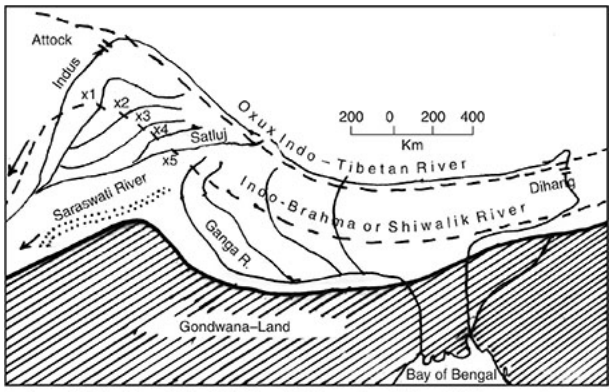


Fig. 3.5 Evolution of Himalayan Drainage

Later, the mighty Shiwalik river was dismembered into three main systems: (i) the Indus and its tributaries in the western part, (ii) the Ganga and its Himalayan tributaries in the central part, and (iii) the Brahmaputra in Assam and its Himalayan tributaries in the eastern part (Fig. 3.5). The dismemberment is attributed to the Pleistocene upheaval in the Western Himalayas including the uplift of the Potwar Plateau (the Delhi Ridge) which acted as a water divide between the Indus and the Ganga systems. Similarly, the down thrusting of the Malda-Gap area between Rajmahal Hills and the Meghalaya Plateau during the mid-Pleistocence period (about 5 lakh years ago) caused the Ganga and the Brahmaputra systems to flow towards the Bay of Bengal. These developments brought a reversal in the direction of flow in the middle section of the severed stream, i.e., the Ganga of today, resulting in its taking a southerly course and eventually annexing the Yamuna and its tributaries. Earlier, the Yamuna had a south-westerly course and was a tributary of the Indus. This entire event was completed by the late Pleistocene period leading to the evolution of the present drainage systems of northern India.

Pascoe's and Pilgrim's concepts about the origin of the Himalayan drainage systems have been criticised on the following grounds:

(i) It is not necessary to postulate a stream of the size of Shiwalik River flowing all along the longitudinal extent of the Himalayas to explain the occurrence of the Shiwalik alluvial deposits and the boulder beds. They might represent a succession of alluvial fans deposited by rivers descending from the Himalayas, which have coalesced over time. Leading geologists, *M.S. Krishnan and N.K.N. Iyangar* (1940), found it difficult to accept the existence of such a mighty river on geological as well as physiographic grounds.

(ii) The evidence furnished by the depositional history of the Ganga Delta does not fit well with the theory. In fact, the alluvium should have been laid down over a much longer period of time than that suggested in the concept.

(iii) The evidence of the Tipam sandstones of Assam, which were deposited in an estuary situated too close to the source of the Indo-Brahma River, also cast doubts over the acceptance of the theory.

Professor Enayat Ahmad (1965–71) has given his own interpretation of evolution of the Himalayan drainage. He opined that the Tethys remained as a basin of sedimentation from the Cambrian to the Eocene Periods, but the major portion of the Himalayan region was occupied by Gondwana Landmass. During the Himalayan upheaval in the Oligocene period, part of the Tethysian geosyncline and probably, part of the Gondwana Land, were uplifted. Most probably, this marked the initiation of the Himalayan drainage. The Tethys Sea was raised into a landmass with a median mass of the high Tibetan Plateau in the centre and the two bordering ranges namely, the Kun-Lun in the north and the Himadari in the south. The drainage started from the southern edge of the median mass and flowed south towards the foredeep. As the formation of east-west ranges created east-west valleys, the rivers partly flowed along these valleys. This is indicated by the upper course of several rivers such as the Indus, the Shyok, the

Satluj, the Ganga, the Brahmaputra, the Arun (Kosi), etc. Since the whole of the Tethys was not fully raised to become land surface, there existed patches of sea along the margins and drainage lines were not fully defined.

The second Himalayan upheaval during the mid-Miocene period increased the altitude of the medium mass and the bordering ranges. The remnant sea was raised to form the landmass. The rise in land resulted in greater and more invigorated drainage. Alongwith these changes, the region to the south of the first Himalayan range was raised as the Lesser Himalayan range. Earlier, streams on the southern margin of the Tibetan Plateau cut down deep valleys to maintain their courses. Along the southern slopes of the Lesser Himalayas, a number of consequent streams also merged which drained into the southern foredeep.

The third Himalayan upheaval during the Pleistocene period resulted into the folding of the Shiwalik foredeep into hill ranges. Also, the height of earlier ranges and the Tibetan Plateau were raised. The rise in the Tibetan Plateau blocked the streams that had gone northward into the Tibetan Sea. These streams were diverted east and west, which probably led to the formation of the Trans-Himalayan master stream. This master stream was broken into two (the Proto-Indus and the Proto-Brahmaputra) by the formation of the Kailash Range. The uplift of the Shiwalik range gave rise to the last set of consequents, originating on the crest of the range, into older streams.

THE MULTIPLE RIVER THEORY

An alternative explanation regarding the evolution of the Himalayan drainage has been offered by the Multiple River Theory. The protagonists of this theory find it difficult to accept the existence of a large river like the Indo-Brahma or Shiwalik on geological and physiographic grounds.

This theory postulates that the Eocene Sea (Tethys Sea) extended from Sindh (Pakistan), Rajasthan, Jammu, and Punjab to Lansdown and Nainital (Uttarakhand). The existence of such a sea is evidenced by the presence of shallow water facies indicative of coast near Lansdown (Garhwal, Uttarakhand). This limit also coincides with the eastern continuation of one of the ridges of the Aravalli Ranges which presumably acted as a barrier. At the same time, another ridge extended from the Rajmahal Hills to the Meghalaya or Shillong Plateau (Rajmahal-Garo Gap) which is now occupied by the Ganga Brahmaputra basin. The sea was broken by the first upheaval of the Himalayas to form an isolated basin in which sediments were deposited. In the next upheaval, a pronounced foredeep all along the southern border of the Himalayas was formed. This foredeep contained numerous lagoons into which flowed streams from the Peninsular area and the newly uplifted Himalayas. These streams brought down sediments which later came to be known as the Shiwalik deposits. The outlet of this foredeep was through the Rajmahal-Garo-Gap in the Bay of Bengal in the east and the Arabian Sea in the west. Later on, the lagoons got dried up and numerous transverse streams flowing from the Himalayan region formed what is now known the Himalayan drainage.

RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

The rivers originating from the Himalayan and Trans-Himalayan regions consist of three river systems, namely: (i) the Indus System, (ii) the Ganga System, and (iii) the Brahmaputra System (Fig. 3.6).

1. The Indus (Sindhu)

The Indus is one of the most important drainage systems of the subcontinent of India. It has a length of 2880 km, of which 709 km lies in India. The catchment area of the Indus is about 1,165,000 sq km, out of which about 45,3931.87 sq km is in India.

The Indus is the westernmost river system in the subcontinent. Jhelum, Chenab, Ravi, Beas, and Satluj are its main tributaries. The Indus originates from the Bokhar Chu (glacier) in northern slopes of Mt. Kailash (6714 m). It drains the largest number of glaciers and mountain slopes of the **Karakoram**.

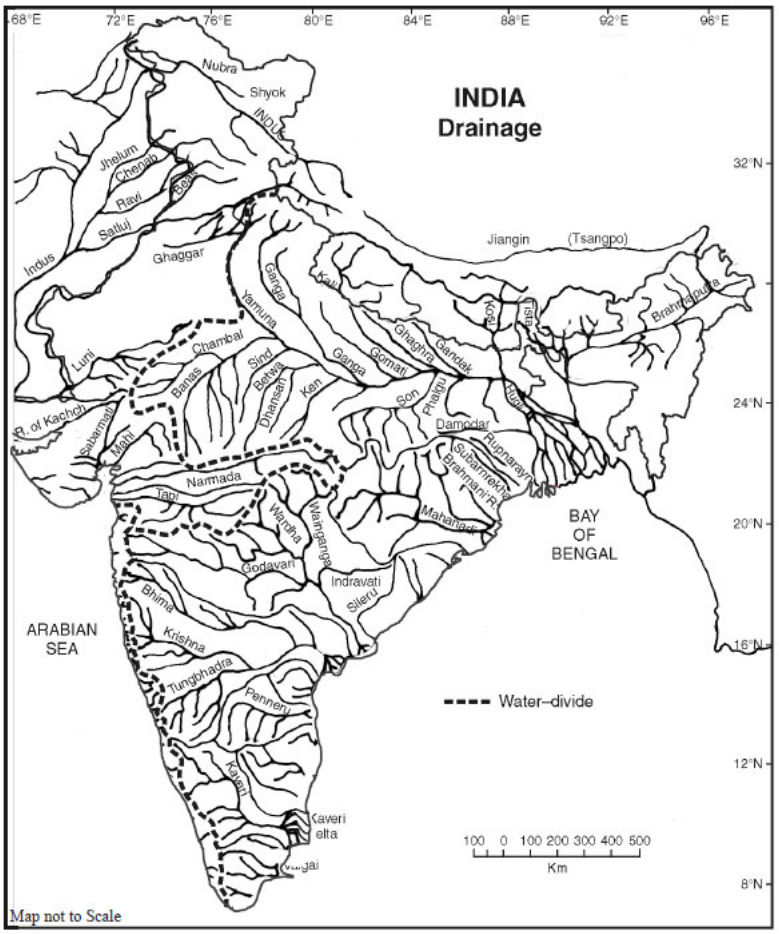


Fig. 3.6 River Systems of India

Ladakh, ***Zaskar***, ***and Himalayan Ranges***. Originating from the Bokhar Glacier lying on the northern slope of Kailash Mountain, it flows in a constricted valley north-westwards through Tibet (China), where it is called as Singi Khamban or Lion's mouth. In Ladakh, it follows a long, nearly straight course between the Ladakh Range and the Zaskar Range. In the first 480 kilometres, it flows along the northern flank of the Zaskar Range over a flat country at heights over 3200 metres where it receives river Zaskar below Leh. Suru and Dras are the other left bank tributaries which join the Indus near Kargil. Moving north-westward, the Indus is joined by the Shyok-Nubra tributaries. These tributaries arise from the Siachen Glacier (Karakoram Range). At Skardu, at a little distance below the Shyok, the Shigar river, which drains the northern slopes of Mt. K2, meets river Indus. In this region, River Indus is more wide than in Ladakh. The transverse glaciers and landslides periodically dam the river. The Gilgit is another important tributary which comes from the west to join Indus. Downwards, the Indus crosses the Central Himalayan Range through a huge synclinal gorge. The Indus makes saveral deep gorges. The deepest of all is at Gilgit, which is 5200 m in height above the level of the water at its bed. The river passes by the Nanga-Parbat and turns south-west to enter Pakistan. With the warming of climate, flow of water from Himalayan glaciers into the Indus is set to increase until atleast 2050 (Hindu 2.6.14, p. 11). Passing through Tibet (China), India and Pakistan, Indus is an international river. India and Pakistan share the water of the Indus river system, according to a treaty (Indus Water Treaty) signed between the two countries on 19th September 1960. India is able to utilise 20% of the total discharge water of river Indus.

The Jhelum (Vitasta)

The Jhelum rises from a spring at Verinag in the south-eastern part of the Valley of Kashmir. It flows north-westward for about 110 km where it enters the Wular Lake. Further downstream from Baramula, it enters a gorge, 2130 m deep, and moves towards Muzaffarabad (Pakistan). It joins the Chenab at Trimmu. Its gradient is gentle in the entire Vale of Kashmir and it is thus navigable between Anantnag and Baramulla. It is the most important river of Kashmir.

The Chenab (Asikni)

It flows in India for about 1180 km draining 26,755 sq km of area in India. The river Chenab is known as Chandra-Bhaga in Himachal Pradesh. The Chandra and Bhaga, the two main upper tributaries of the river, originate on either side of the Bara-Lacha Pass (4843 m) in the Lahul District of Himachal Pradesh. Of these streams, Chandra originates from the glacier east of the Bara-Lacha Pass while the Bhaga originates from the Surya Taal and make a confluence at Tandi. After uniting, the Chenab flows between the Pir-Panjal and the Greater Himalayas. Near Kishtwar, it makes a hair pin bend and flows across the Pir-Panjal at Riasi to enter into Pakistan. The important hydel projects installed across the Chenab are Salal, Baghliar and Dulhasti. The construction of the Baghliar Project in Doda District, financed by the World Bank, was approved in February 2007. The total dam height of the Baghliar Project is 144.5 m. This project costed INR 4500 crore and will produce 450 MW of power. The much delayed project (delayed by 15 years) has brought enough power to the territory of Jammu and Kashmir.

The Ravi (Parushni or Iravati)

It flows for about 725 km and drains 5957 sq km area in India.The Ravi rises near the Rohtang Pass in Kullu, very close to the source of the Beas river. It drains the western slope of Pir-Panjal and the northern slope of Dhauladhar Range. Below the Chamba Town, it turns to the south-west, cuts a gorge in Dhaula Dhar Range and enters the plain of Punjab. In Punjab plains, it runs along the Indo-Pak boundary, along Gurdaspur and Amritsar districts, before entering into Pakistan.

The Beas (Vipasa or Argikiya)

The river Beas has its source at '*Beas Kund*' near the southern face of the Rohtang Pass in Kullu (4000 m), where it runs for a few kilometers and then cuts through the Dhauladhar Range in a deep gorge near Koti and Larji. Then it flows from north to south along Manali and Kullu towns, where the river has a transverse valley popularly known as Kullu Valley. Further down, it flows through Kangra Valley and then turns to the west to enter the Punjab Plain. It finally passes through Kapurthala and Amritsar and joins the Satluj near Harike (within India) after flowing for a total distance of 465 km.

The Satluj (Satadru or Satudri)

The Satluj river rises from the Rakas Lake (Rakas-Tal) which is situated at an altitutude of about 4600 m near the Mansarovar Lake in Tibet (China). This is an antecedent river, called Langechen Khambab in Tibet. It makes gorges through the Zaskar and the Greater Himalayan Ranges. At the pass of Shipki La (4300 m), it enters Himachal Pradesh where it cuts through the Zaskar Range. From here the river flows westward and passes Kalpa, and then crosses Dhauladhar Range near Rampur through a narrow gorge. It passes through the Shiwalik, and at the gorge near Bhakra village, the Bhakra Dam has been constructed across the river. Below the Bhakra Dam, the river comes to the Punjab Plain at Rupar. From Rupar, Satluj River flows westward, and at Harike in the southwest corner of Kapurthala, it meets the Beas. After the confluence, the combined river enters Pakistan. In India, the length of its course is 1050 km, draining an area of about 28,090 sq km.

Ghaggar (the legendary Saraswati)

This is an inland drainage which rises in the talus fan of the Shiwaliks of Sirmaur near Ambala (Haryana). After entering the plains, it disappears, but reappears at Karnal District (Fig. 3.8). Further on, the stream is called Hakra which gets lost near Hanumangarh in Bikaner. The considerably large size of this river bed, 5 to 8 km wide, with the loamy soil in the river bed, led to the belief that in old times the Satluj flowed south through this Ghaggar-Harka river instead of flowing southwestward (in the form of a much bigger river known as river Saraswati in Vedic literature, probably 5000 BC or earlier). This river is traced further on as Eastern Nara which is, at present, an old channel of the Indus in Sind (Pakistan), and flows into the Rann of Kutch which was described as a deep fairly gulf at that time. At present, the

entire area is practically a desert, and the Ghaggar-Hakra are practically ephemeral streams coming into flow only during the season of general rains.

2. The Ganga Basin

With a length of 2510 km and a basin area of 808334.44 sq km, the Ganga basin is the most important and the largest river basin in India. It is an international river as it passes through Bangladesh before merging into the Bay of Bengal (Fig. 3.7).

The Ganga is the most important and sacred river of India. According to Pandit Nehru, “From her source to the sea, from old times to new, Ganga is the story of India’s Civilization.” In fact, this river to beyond water’s material uses, which are critical to life and development, is associated with fertility and spiritual nourishment of the people.

The Ganga originates as Bhagirathi from the Gaumukh (Gangotri Glacier) in Uttar Kashi District of the Uttarakahnad at an elevation of about 7010 m. Alaknanda, originating from the Satopanth Glacier, joins the Bhagirathi at Devprayag. Before making a confluence at Devprayag Alaknanad and its tributaries make five confluences known as Panch Prayag (Fig. 3.7A).

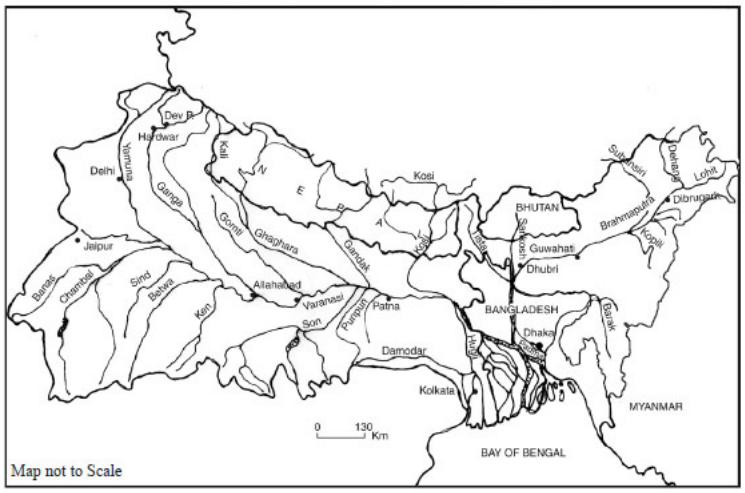


Fig. 3.7 Ganga-Brahmaputra Basins

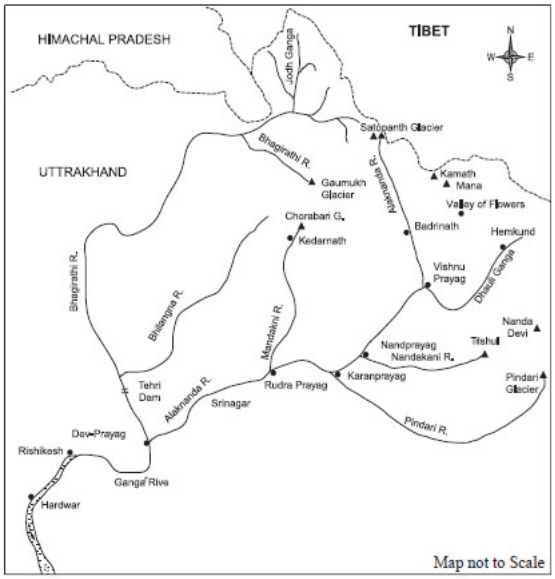


Fig. 3.7-A Panch Prayag (Five confluences) Uttrakhand.

Table 3.2

Five Confluences in the upper reaches of Ganga

Rivers	Confluence
R. Alaknanda (Source: Satopanth Glacier)	Vishnu Prayag
R. Dhauli Ganga (Source: Mana Glacier)	
R. Alaknanda and R. Nandakni (Source: Trishul Glacier)	Nand Prayag
R. Alaknanda and R. Pindari (Source: Pindari Glacier)	Karan Prayag
R. Alaknanda and R. Mandakini or Kali-Ganga (Source: Chorabari Glacier)	Rudra Prayag
R. Alaknanda and R. Bhagirathi (Source: Gaumukh or Gangotri Glacier)	Dev Prayag

Fig 3.7 B: A Confluence in Uttrakhand [*Source: Uttarakhand Tourism Development Board, Department of Tourism, Government of Uttarakhand, India*]

After the confluence of the Bhagirathi and Alaknanda, the river is known as Ganga. After crossing about 289 km from its source, the Ganga reaches Hardwar debouches from the hills and enters plain area. From Haridwar, it flows in south and south-easterly direction for a distance of 770 km to reach Allahabad. Here, it is joined by the Yamuna which is its longest and most important tributary. It crosses another 300 km eastward to reach the plains of Bihar. Near Rajmahal Hills, it turns to the south-east and south of Farakka. Here, it ceases to be known as the Ganga. It bifurcates itself into Hugli in the west and Padma in the east (Bangladesh). After passing through Kolkata and Haora, the Hugli merges into the Bay of Bengal near Haldia. The Bengal delta is the result of the filling-up of geological sag formed between the Rajmahal Hills and the Chhotanagpur Plateau on the west and the Plateau of Meghalaya in the east. This delta, known as the Sundarban Delta stretches in West Bengal and Bangladesh. The Sundarban Delta is the result of depositional action of the Ganga-Brahmaputra and the Meghna Rivers.

This most important and sacred river of India is highly polluted and dying. Pollution, over-extraction of water, emaciated tributaries, and climatic changes are killing the mighty river, on whose fertile plains live one in twelve people of the earth. The Ganga basin makes up almost a third of India's land area and its rich soil is home to millions of people.

However, indiscriminate extraction of water with modern tube wells from the rivers as well as its basin, coupled with the damming of its tributaries for irrigation, have seriously reduced its flow. Climatic change has added to the threat. According to the WWF Programme Director, Sejal Worah: "Glaciers account for 30 to 40% of water in the Ganga and this goes up to 70–80% in the case of Indus. Studies are required to gauge the impact of melting glaciers on the flow."

Apart from humans, many other kinds of lives are in danger due to Ganga's degeneration. The river is home to more than 140 fish species, 90 amphibian species, and the endangered Ganga river dolphin. The Ganga is, of course, sacred to the people of India, besides having spawned many great cities on its banks. The maintenance of this sacred river is imperative from socio-cultural and ecological point of view. The central government has prepared a blue-print for a waterway from Varanasi to Kolkata. This is an ambitious plan to develop the Ganga as a major waterway for transport and tourism. The government decided to make the river stretch between Varanasi and Hugli navigable for boats, while beautifying the major banks and constructing eleven multipurpose terminals, including one at Patna, along the way. The Ministry of Water Resources, Tourism, Environment, power and Shipping will pool in their resources to implement the audacious plan (Source: Hindu–6.6.14, p.10).

According to the Central Pollution Board, the Ganga is one of the most polluted rivers of the world. The water at Hardwar District fails almost all parameters of safety. According to official sources, nearly one lakh devotees bathe at the over 20 Ghats (bathing sites) of Hardwar every day. The biological oxygen demand (BOD) is already 6.4 mg per liter in the river water instead of 3 mg per liter which is normal and safe for bathing.

In India rivers are highly revered, but with rapid industrialization and urbanization, rivers have increasingly become depositories of urban waste and industrial effluents. The Ganga Action Plan (GAP) has seen more than ` 1800 crore spent. The '*Namani Ganga Programme*' is an integrated programme, approved by the Union government in June 2014 with a budget of ` 20,000 crores to accomplish the twin objectives of effective abatement of pollution, conservation and rejuvenation of river Ganga. *(Source: Ministry of Water Resources, River Development and Ganga Rejuvenation)*

The main pillars of the Namani Ganga Programme are:

1. Sewerage treatment infrastructure
2. River front development
3. River surface cleaning
4. Bio-diversity
5. Afforestation
6. Public awareness
7. Industrial effluence monitoring
8. Ganga Gram

A society named as '*National Mission for Clean Ganga*' (NMCG) was established on 12th August 2011. It has the authority of approving projects up to ` 1000 crores. The director general (DG) of NMCG is an additional secretary in the Government of India.

The Yamuna River

This is the longest (1380 km) and the western-most tributary of the Ganga. Its source lies in the Yamunotri Glacier on the western slopes of Banderpunch (6316 m). Downwards, it is joined by Tons river behind the Mussoorie Range (Uttarakhand). From the Mussoorie Range, it debouches into the plains where it flows in a broad curve. Making a boundary between Haryana and Uttar Pradesh, it passes Delhi, Mathura, Agra and flows southward until it joins the Ganga at Allahabad. The important tributaries of the Yamuna are mostly the right bank tributaries, originating from the Aravallis (Rajasthan), Vindhyan Range, and the Malwa Plateau of Madhya Pradesh. The Chambal, Sind, Betwa, Ken and Tons are the main righthand tributaries of the Yamuna River.

It is believed that during the Vedic period, the River Yamuna might have flowed towards south and southwest through Bikaner in Rajasthan, and shared its water with the legendary River Saraswati (Fig. 3.8).

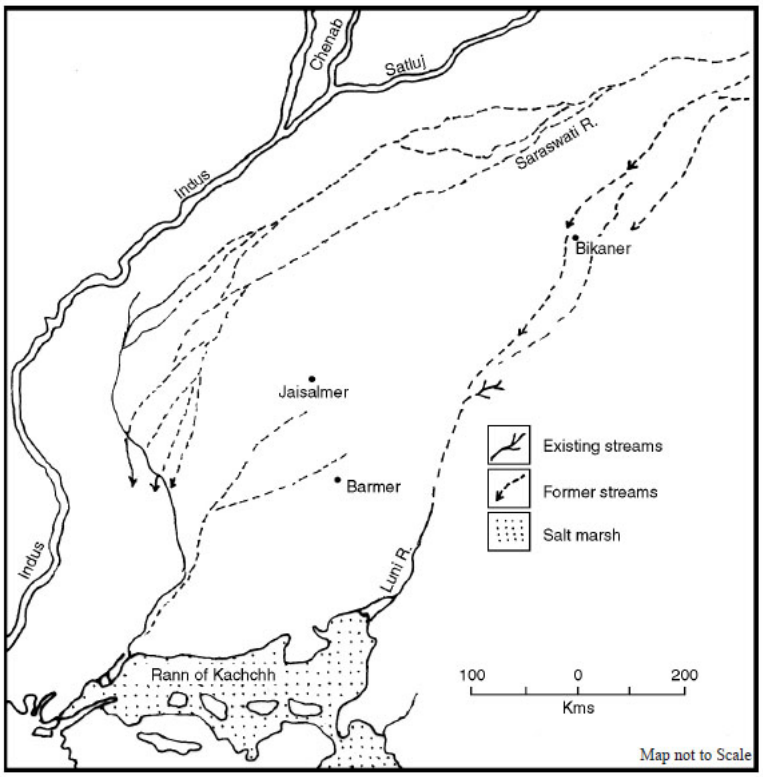


Fig. 3.8 Ghaggar

The Chambal

The Chambal River of length 960 km rises near Mhow Cantt.—south-west of Indore in Malwa Plateau from the Vindhyan Range—and flows towards the north in a gorge upto the city of Kota. Below Kota, it turns to the north-east, and after passing Bundi, Sawai-Madhopur, and Dholpur, it finally joins the Yamuna about 40 km to the west of Etawah. The Banas River, rising from the Aravalli Range is its main left bank tributary. Kali Sind, Parbati and Kunwari originating from the Malwa Plateau are the right bank tributaries of Chambal. The Chambal River is famous for its extensive ravines which it has carved all along in the Lower Chambal Valley. The ravines of the Chambal Basin are attributed to a slight uplift during the recent geological times, and they merge into the Yamuna alluvial plain where the landscape is extensively etched out by other tributaries of the Yamuna to the east and west of Chambal. Multipurpose projects have been constructed across the river. The main dams across the river are Gandhi Sagar, Rana Pratap Sagar (Rawatbhata) and Jawahar Sagar.

Chambal Ravines

A maze of ravines, valleys, and saw-toothed ridges dissect the plateau. These ravines are found in Rajasthan, Madhya Pradesh, and Uttar Pradesh in the lower course of the Chambal River. Infested with dacoits, the ravines are being reclaimed for agriculture, pastures, social forestry and wild life sanctuaries.

The *National Chambal Sanctuary* was established in 1979. It is about 400 km long and one to six km in width of River Chambal on both the sides. It is known for the endangered Gangetic Dolphin. It is also the home to 8 species of turtles. The wild life animals in the sanctuary are Chinkara, Sambar, Blue Bull, wolf and wild boar.

The Chambal is the largest of the cratonic rivers in Ganga-Yamuna drainage basin. The rivers flowing north eastwards, over the Deccan basalts and Proterozoic Vindhyan strata carry large sediment load and when it is deposited in the flood affected areas, it results into floods. This deposition results in the build-up of the flood plain. During heavy rainfall the sediment gets carried to the sea and the floodplain

degrades as erosion along the main river channel and streams cuts gullies and ravines giving rise to the badlands.

The river incision may also have been resulted from the tectonic uplift and tilting of the region.

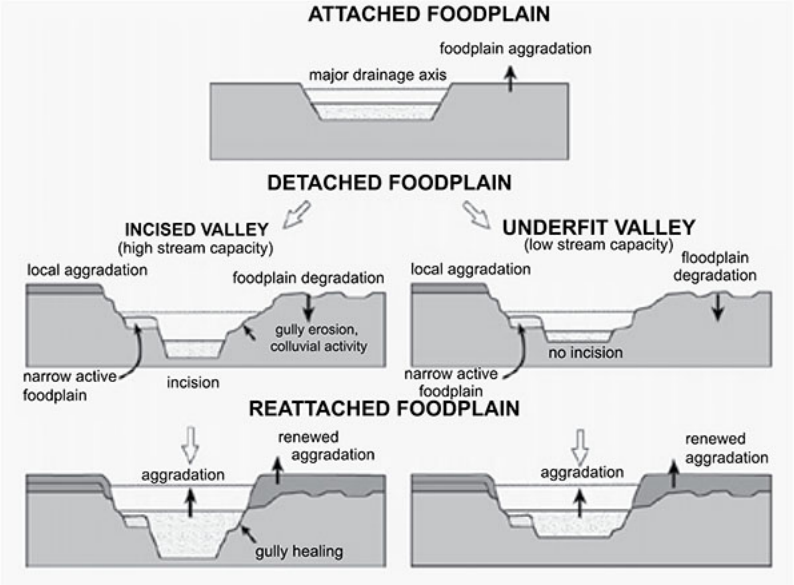


Fig . 3.9 Formation of Badland [*Source: Gibling et al, 2005*]

The Ramganga

This is a comparatively small river which rises in the Kumaun Himalayas. The river is deflected to the south-west by the Shiwalik, which it cuts through, before emerging at the Ganga Plain in Najibabad. It joins the River Ganga in Hardoi district opposite to Kannauj.

The Sharda

This river originates from the Greater Himalaya, at Kalapani (3600 m) near the Lipu-Lekh pass, in the Pithorgarh district of Uttarakhand .The river is known as the Chauka before it joins the right bank of Ghagra near Barabanki. This river originates from the Greater Himalaya, at Kalapani (3600 m) near the Lipu-Lekh pass, in the Pithoragarh district of Uttarakhand. The river is known as the Chauka before it joins the right bank of Ghagra near Barabanki. The river has a potential for the hydroelectric generation. In 1995, the proposal of '*The Pancheshwar Dam*' as a joint venture between India and Nepal was made for irrigation and power generation. The Tanakpur Hydroelectric Power was commissioned by NHPC in April 1993, in the district of Champawat.

The Gomti

It is a tributary of the river Saryu. It has its origin in the higher region near Mainkot from a lake '*Fulhar Jheel*' in Madho Tanda, east of Pilibhit in Uttarakhand. The river flows through the districts of Sitapur, Lucknow, Barabanki, Sultanpur and Jaunpur before meeting river Ganga at Kaithi, Ghazipur district.

The Gomti Valley is also known as '*Katyur Valley*' and makes the major agricultural region of Kumaun. The surface area is flat, sloping towards south and southeast. It is a perennial river. Its drainage area is 30,437 sq km. Sai River is its major tributary.

The river is however getting affected by the pollution assault (effluents from industries and domestic discharge) from Sitapur to Sultanpur, as it meanders through industrial belt of sugar processing, paper and plywood industries.

The Karnali

The Karnali is known as Kauriala in the Nepal Himalayas and as Ghagra in the Ganga Plain. The Karnali is an antecedent river originating from the Gurla Mandhata Peak (7720 m) of the Nepal

Himalayas. Before making a gorge in the Greater Himalayas, the Karnali river traverses a 160 km long tract in the Trans-Himalayan region. This river cuts through the Mahabharata Range in the western part of Nepal through a deep gorge. In the Plains, it is joined by the Sharda river and acquires the name Ghagra meaning rattling or lahnga. Passing through Ayodhya and Faizabad towns, it joins River Ganga at Chapra near Ballia town. Ghagra is a large river with its vagaries of shifting course.

River Gandak

Draining the central parts of Nepal, Gandak River rises in the Nepal Himalayas between Dhaulagiri, Annapurna and Mt. Everest. It enters the Great Plains of India in Champaran District of Bihar, and turning south-east, it joins the Ganga River at Sonpur opposite the city of Patna. This river also changes its course frequently.

The Kosi

The Kosi is also an antecedent river. It is often referred to as the "Sorrow of Bihar". Arun is its main stream which originates from the northern slopes of Mt. Everest in Tibet (China). After piercing the Greater Himalayas in Nepal, it is joined by the Son-Kosi from the west and the Tamur-Kosi from the east. Both these rivers run for a fairly long distance parallel to, and north of, the Mahabharata Range and join River Arun to form the Sapt-Kosi. This river cuts across the Mahabharata Range and the Shiwalik Hills, and emerges into the Bihar Plains near Chatra in Saharsa District. In Bihar Plains, it splits into numerous capricious channels. It is known that about 200 years ago, the Kosi used to flow by the side of Purnea town, but now by its westward migration it is about 160 km to the west of Purnea. In July, 2008, the Kosi River shifted its course towards east by more than 100 km. The devastating flood was declared as a national calamity. The Kosi river joins the Ganga River 30 km west of Manihari. It has however, been tamed since 1962 by the construction of embankments on its two banks.

The Tista

This is the westernmost right bank tributary of the Brahmaputra. Rising from Kanchenjunga, it is a wild mountain torrent in the Darjeeling Hills with a number of tributaries like the Rangpo, Rangit, and Sevak. Situated on its bank, the Jalpaiguri town, was completely swept away in the flood of 1968. At present, river Tista joins the Brahmaputra river in Bangladesh. It shifted its course substantially in *the flash flood of 1787*, prior to which it used to make a confluence with the Ganga.

The Mahananda River

This river rises in the Darjeeling Hills of West Bengal. Near Siliguri, it merges into the duars (Bhabhar) of West Bengal. Making sharp curves, it joins the Ganga river. This is the last north bank tributary of the Ganga.

Ken River

Originating for the Malwa Plateau, Ken passes through Panna District (MP), where it makes a gorge at Gangau. It joins the Yamuna river in Banda District of UP. Sonar and Beawar are its main tributaries.

The Son River

This is a large south bank tributary of the Ganga River of length 780 km and basin area of 54,000 sq km.

It originates from the Amarkantak Plateau not far from the source of Narmada. It leaves the plateau in a series of waterfalls and meets the Kaimur Range which turns its course towards the north-east and allows it to follow a *strike valley*. Below Garwa, it enters the Ganga Plain, where it widens upto about 5 km, and finally flows into the Ganga at Bankipora near Aara to the west of Patna. During the rainy season, the river often flows in spate with an annual peak discharge of about 750,000 cusecs in the rainy season, while in the dry season it has little water.

The Damodar River

The Damodar drains the eastern parts of the Chotanagpur Plateau. This river runs from west to east and, in the narrow bottleneck at Asansol, it emerges on the deltaic plains of Bengal. Barakar is the largest feeder of the Damodar river which meets the main river above Asansol. Below Bardhaman

(Burdwan), the river makes a right-angled bend and meets the Hughli river at Falta to the north of Kolkata.

3. The Brahmaputra River

Originating from the Angsi glaciers lying to the south-east of Kailash Mountain and Mansarovar Lake, at an elevation of 5150 m, the Brahmaputra is known as the Tsangpo in Tibet. It pierces the Greater Himalayas (7755 m) near Namcha Barwa. It passes the Dihang-Gorge in Arunachal Pradesh. At Sadiya, the river comes down to 135 m above sea level.

In India, the river is known as Brahmaputra. The river flows to the west in Assam upto Dhubri (28 m), and further below, it takes a sharp southward bend to enter into Bangladesh.

The catchment area of the Brahmaputra receives heavy rainfall. Consequently, it has numerous tributaries on both of its banks in the 750 km long Assam Valley. Most of the tributaries are large and pour large quantity of water and sediment into the Brahmaputra River. During the rainy season, the river oscillates from one bank to other for a width of 10 km (on an average), and being turbulent with heavy loads of silt, the channel is heavily braided. There is a constant silt movement resulting in the instability of river regime, channel shifting and formation of sandy shoals. Majuli, the largest river island in Asia, lies in this river which is bounded by the Lakhimpur District in the north and the Jorhat District in the south. The Brahmaputra river basin is notorious for flooding and river bank erosion. The floods affect on average, an area of 100,000 hectares annually. The peak annual discharge at Pandu, near Guwahati, is more than 2.5 million cusecs (650,000 cumecs), while the low discharge is 120,000 cusecs (4210 cumecs). Below Pasighat, the river draws a number of tributaries such as Subansiri, Bhareli, Manas, Sankos, Tista, and Raidak on its right bank, while the Dihang, Lohit, and Burhi Dihang are from the east, and Dhansiri, Kalang and Kapili are the left bank tributaries of the Brahmaputra.

Rangit River

Originating from Sikkim, Rangit river has a large number of rapids. This river is well known all over the world for rafting sports. Its banks provide good camping grounds.

The Sankosh River

This is the main river of Bhutan. It meets Brahmaputra River below Dhubri. The Sankosh River forms the boundary between Assam and Arunachal Pradesh. It is known as ‘Puna Tsang Chu’ in Bhutan.

The Manas River

It is an *antecedent river*. Rising from Tibet, it pierces the Greater Himalayas in a gorge with river bed at 3000 m. It collects a number of tributaries in the Lesser Himalayas and debouches into the plains to join the river Brahmaputra.

The Subansiri River

This is a large tributary of the Brahmaputra River. It has a long course in the Himalayas and, after leaving the mountains, it has a long 160 km course in the Upper Assam Plain before joining the Brahmaputra. It separates the Miri Hills from the Abor Hills.

The Dhansiri River

This river rises from the Naga Hills and after flowing for about 300 km through Nagaon, it joins the Brahmaputra River.

The Manipur River

It rises from the northern part of Manipur and flows southwards. On passing through Imphal, this river drains *the Loktak Lake* and joins the Chindwin Valley, a tributary of the River Irrawaddy in Myanmar (Burma).

Kaldan River

This river drains the southern parts of Manipur and Mizoram where it flows southward to the Bay of Bengal.

Barak River

This river rises from Mt. Japov (Nagaland), flows southwards in Manipur and makes a hair pin bend. Several of its affluents, which drain the northern part of Mizoram, combine together to flow past Silchar in the Cachar District. The Barak basin has Mawsynram and Cherrapunji which receive the highest rainfall in the world. Consequently, the Barak river discharges heavy quantity of water. It flows to Bangladesh where it is called as Surma. The Barak river meets the Padma at Chandpur below Dacca, after which the combined river of *Surma (Barak)* and Padma is known as *Meghna*.

Table 3.3

Major Rivers of the Himalayan Region

River	Source	Length (km)	Main Tributaries
The Indus	Glacier Bokhar Chu in the Mt. Kailash	2880	Jhelum, Chenab, Ravi, Beas, Satluj
The Jhelum	Chashma Verinag in the Kashmir Valley	725 km but in India it is 165 km	Sandran River, Bringi river, Aripath, Rambiara Stream, Romuhi stream, Sasara Stream, Nallah Lidder, Nallah Sindh, Nallah Vishow, Dudhganga River, Pohore
The Chenab (Chandrabhaga)	Lahaul Valley of Himachal Pradesh	960	Stream Chandra, Stream Bhaga, The Miyar Nallah, The Bhut Nallah, The Marusudar, The Tawi River, The Niru
The Ravi	Near Rohtang Pass	725	River Bhadal, River Siul, River Baira, River Tant Gari
The Beas	Beas Kund near Rohtang Pass	460	Bain, Bangana, Luni, Uhal, Banner, Chakki, Gaj, Harla, Mamuni, Parvati, Patlikuhlal, Sainj, Suketi, Tirthan
The Satluj	Manasarovar-Rakas Lakes	1450 km in total and 1050 km in India	Baspa, Spiti, Nogli Khad, Soan
The Ganga (Bhagirathi)	Gangotri Glacier in Garhwal, Himalaya	2525	Gomti, Ghaghara, Gandaki, Kosi, Yamuna, Son, Punpun, Damodar
The Yamuna	Yamunotri Glacier near Bandarpoonch peaks in the Mussoorie	1376	Tons, Chambal, Hindon, Betwa, Ken, Giri, Sind, Uttangan, Sengar, Rind
The Brahmaputra	Kailash Ranges	2900	Lohit, Dibang, Subansiri, Jiabharali, Dhansiri, Manas, Torsa, Sankosh, Teesta, Burhidihing, Desang, Dikhow, Dhansiri, Kopili

Source: India WRIS-Wiki Water Resources Systems of India

MAIN RIVERS OF PENINSULAR INDIA

The main rivers of Peninsular India have been described in the following section and shown in **(Fig. 3.10)**.

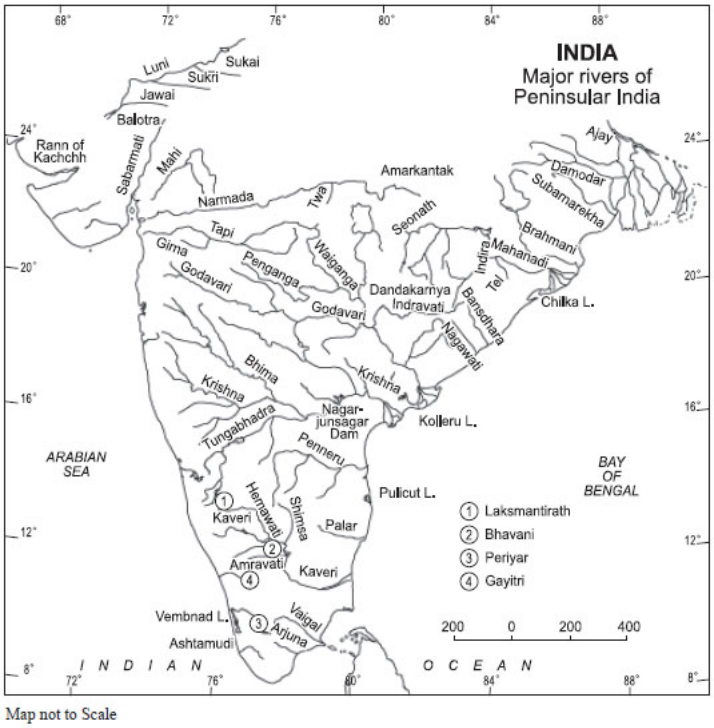


Fig. 3.10 Major Rivers of Peninsular India

Luni River

The Luni River rises near Ajmer in the Aravalli Range and receives a few streams like the Bandi, Sukri and the Jawai only on its left bank. The streams which drain the western flank of the Aravalli Range south of Ajmer are seasonal. The Luni turns brackish below Balotra and loses itself south-westwards into the Rann of Kutch.

Sabarmati River

This river drains the southern slope of the Aravalli Range. It has a narrow long basin with an area of 21,700 sq km. Sabarmati receives its tributaries on its eastern flank and follows a nealy north-south course before merging into the Gulf of Khambat.

Mahi River

The Mahi has its source in the Vindyan Hills of Madhya Pradesh. It flows southwest through the Banswara and Anand districts and finally, making an estuary, merges into the Gulf of Khambat.

Most of the rivers draining the west-facing scarp of the Western Ghats, particularly in the Konkan coastal belt, are short and swift-flowing over a humid tropical landscape at 100 to 280 m above sea level along narrow to broad '*V-shaped valleys*'. These small streams have great erosive capacity. They display a parallel pattern of drainage, characterised by sharp bends, and fall into the Arabian Sea through estuaries. Some of the important streams draining the western slopes of the Western Ghats are Ulhas, Savitri, Vashist, Netravati, Periyar, and Pambiyar. Some of these streams make waterfalls and rapids when they descend their steep gorges.

Sharavati River

Sharavati rises in Shimoga district in Karnataka and creates the Garsopa Falls, the most spectacular waterfalls in India, commonly known as Jog Falls (271 m). The only break in the unbroken watershed of the Western Ghats face is the 13 km wide Palakkad Gap also known as '*Pal-Ghat*'. The river Ponnani flowing westward through this gap appears to be a misfit river.

Narmada River

The Narmada River of length 1300 km and drainage basin area 92670 sq km rises from the plateau of Amarkantak of the Maikal Hills of Chhattisgarh. Moving north-westward, it passes through a complex course near Jabalpur, through some impressive marble gorges, the most spectular being Dhunwadhar Waterfalls (10 m high) near Jabalpur. Moving westward from Jabalpur, it flows through a rift valley between the Vindhyan and the Satpura ranges. It has rich alluvial deposits in its valley. Finally, it widens below Bharuch and makes a 27 km wide estuary to enter the Gulf of Khambat (Arabian Sea).

Tapi River

Having a length of 700 km and a basin area of 66,900 sq km, the river Tapi rises from the Satpura Range and flows westward almost parallel to Satpura. At Khandwa-Burhanpur Gap, the Narmada and Tapi come close to each other. Below Jalgaon, the river, like the Narmada, flows in a rift valley but in a much constricted form between the Satpura Range to the north and the Ajanta Range to the south. Below the city of Surat, it makes an estuary and merges into the Gulf of Khambat.

EASTERLY RIVERS OF THE PENINSULAR REGION

There are a number of rivers originating from the Chotanagpur Plateau and merging into the Bay of Bengal, of which the Brahmani is the most important (Table 3.4).

Subernrekha River

The Subernrekha rises a little to the southwest of Ranchi where it has a number of waterfalls. Its length is 400 km with basin area of 28,000 sq km. In a general easterly direction, it passes through Jamshedpur and flows to the Bay of Bengal near Balasore. Its drainage basin is shared by the states of Jharkhand, Odisha, and West Bengal.

Table 3.4

Major Rivers of Peninsular India

S. No.	River	Source	Length in km	Main Tributaries
1.	Godavari	Trimbak Plateau Near Nasik (Maharashtra)	1465	Sabari Indravati, Penganga, Wardha, Pranhita, Purna, Wainganga, Manjra, Tirna, Pranhita, Moher.
2.	Krishna	Near Mahabaleshwar (Maharashtra)	1400	Musi, Muneru, Koyna, Nira, Muthu-Mula, Sina, Kagna, Bhima, Ghataprabha, Malprabha, Nira, Panchganga, Dudhganga, Tungbhadra
3.	Narmada	Amarkantak	1310	Heran, Orsang, Barna, Kolar, Burhnar, Banjar, Shar, Shakkar, Dudhi, Tawa, Kundi
4.	Mahanadi	Dandakaranya Plateau near Raipur	857	Ib, Mand, Hasdeo, Seonath, Ong, Jonk, Tel
5.	Kaveri	Taal Kaveri	800	Herangi, Hemavati, Yogachi, Lokpavni, Shimsa, Arkavati, Kabani, Bhavani, Noyil, Amravathi, Sanmukta, Nanganji, Laxman-Tiratha, Suvarnavati, Turunananimutta.
6.	Tapi	Multai in Betul Distt. (MP)	730	Purna, Girna, Panjhra, Betul, Patki, Arunavati, Ganjal, Dhatranj, Aner, Bokad, Mon, Guli

The Brahmani River

The Brahmani river of length 420 km is formed by the confluence of the Koel and Sankh rivers. They join together at Rourkela and drain the western parts of the Garhjat Hills. Flowing through Bonai, Talcher, and Balsore districts, it merges into the Bay of Bengal above the Paradwip-port. With the Baitarani river to its north, a delta complex forms below Bhadrak.

The Mahanadi

The Mahanadi is the most important river of Odisha as well as that of Chhattisgarh. Its length is 885 km and basin area 1,39,659 sq km. This river rises in the Chhattisgarh basin, draining the western and eastern parts of Raipur. In the initial stage, it flows towards the north-east, and after receiving a number

of streams such as Seonath and Sandur on both its flanks at heights between 200 m and 700 m, the combined water gets a natural exit towards the east through a gorge which has been impounded to create the Hirakud Dam. A little below the dam at Sambalpur, the river turns eastward and flows through the Eastern Ghats entering the Bay of Bengal through several distributaries in its delta. Cuttack city is located at the apex of the Mahanadi delta.

The Godavari

Godavari is the largest river length 1465 km and basin area of 302063.93 sq km of peninsular India. It rises in the Western Ghats from a spring below Nasik, drains eastern and southeastern Maharashtra, Bastar plateau (Chhattisgarh), and Telengana and Andhra regions of Andhra Pradesh. It receives a large number of tributaries, particularly on its left bank, such as Purna, Maner, Penganga, Pranhita (the combined Wardha and Wainganga), Indravati, Tal, and Sabri. The Manjira is the only important right bank tributary. The Indravati and Sabri are the two streams which rise on the western slopes of the Eastern Ghats, but they flow east and south-eastwards, respectively. Below the confluence with Indravati, it flows in a picturesque gorge through the Eastern Ghats. Below Rajamundry, it has constructed a large symmetrical delta and reaches the Bay of Bengal by its three main ditributaries. The delta of Godavari is characterised by a number of palaeo-channels and mangroves associated with lagoons. The '*Kolleru Lake*', lying to the southeast of Kakinada, is one such '*inland lagoon*'.

The Krishna

The river of Krishna has its origin near Mahabaleshwar in the vertical faces of the Western Ghats. Its length is 1290 km and basin area 2,54,743 sq km. A number of minor streams like Koyna and Ghataparbha join the Krishna river to give a subdendritic pattern. The Bhima in the north and the Tungbhadra in the south are the other important tributaries of the Krishna river. Downwards passing through the quartzite scarps, the Krishna has been dammed to form the Nagarjun-Sagar Reservoir. Further east, beyond the gorge in the Srisailam Hills below Vijaiwada, it has built its fertile bird-foot delta (Mississippi-type).

The Pennar

The Pennar river rises in the Kolar District of the South Mysore Plateau. Its main tributaries are Chitravati and Papaghni. It flows through a gorge of Cuddapah quartzite and enters the Bay of Bengal in the form of an estuary.

The Kaveri

The Kaveri is a sacred river like the Ganga. Hence, it is called the Ganga of South India. Its length is 765 km and has a basin area 85,624 sq km. It rises from the southern part of Mysore Plateau as a rocky mountain stream forming rapids, cataracts and waterfalls. Its drainage basin receives rainfall during the summer monsoon as well as during the retreating and winter monsoon. Only 20 km above Mysore, it has been dammed to form the Krishnasagar Reservoir. It passes through the islands of Srinagapatnam and Sivasamudram. The channel around the Sivasamudram makes a succession of rapids which were harnessed to develop hydel-power in 1902. Below the island, Kaveri River plunges through a succession of beautiful gorges with hair pin bends. This scenic landscape, known as the Hagenakal Fall, may be taken as the end of the plateau course of the river. There is, however, another narrow gorge in the Nilgiri Hills to the east of Dodabetta Peak
(2636 m) which is drained by the Kabari, Amravati and their Bhavani, and tributary, the Moyar, which provides the site for the Mettur Dam. Draining the Coimbatore basin, the Kaveri enters the plains after its confluence with the Bhavani. A few kilometers above Tiruchirappalli, the river fans out forming a quadrant-delta in Thajavur District of Tamil Nadu.

Amravathi River

Amravathi river is a tributary of the Kaveri river in Coimbatore District of Tamil Nadu. Having its origin at the Kerala-Tamil Nadu border, it is 175 km in length. It joins with the Kaveri in Karur District. It irrigates over 60,000 acres of land in Coimbatore. Due to the heavy industrialisation in its basin, the river is highly polluted.

The Tambraparni

This is a river of Tiruneveli District. It rises on the slopes of the Palni Hills of the Western Ghats and passes Madurai to flow into the Gulf of Mannar through the Ramanathapuram Peninsula.

Rivers of the Western Sahyadris

The western slopes of the Western Ghats receive heavy rainfall (200 to 500 cm) during the south-west monsoon. The small and swift flowing rivers traverse the narrow coastal plain. Because of heavy discharge of water and narrow width of the coastal plain they do not make delta. The main rivers merging into the Arabian sea are Surya, Kalu, Savitri (Maharashtra); Mandavi, Terekhol, Chapora, Zuvari, Sal and Talpona (Goa); Kalinadi, Gangawali, Sharavati, Tadri and Natravati in Karnataka; and Beypore, Ponnani, Bharatapuzha, Periyar and Pamba in Kerala. All these rivers have carved out narrow valleys with steep gradient forming cataracts and waterfalls. The famous Jog Falls (721m) lie on Sharavati River. Although with small drainage area (about 2% of India) these rivers contribute about 18% of the water yield of the country and have great potential (62 lakh kw) for hydel power. A number of hydel-power projects have been installed over these rivers.

Comparison of the Peninsular and the Extra-Peninsular Rivers

	Peninsular Rivers		Extra-Peninsular Rivers
1.	The rivers of Peninsular India are much older. Some of them are as old as the Pre-cambrian Period.	1.	The rivers of the Extra-Peninsular India are much younger, excepting a few antecedent rivers like Indus, Satluj, Ganga, Brahmaputra, etc.
2.	They are mostly consequent or rejuvenated rivers.	2.	Many of them like Indus, Satluj, Kali, Kosi, Brahmaputra, Tista are inconsequent rivers.
3.	These rivers, except Godavari, have relatively small basins.	3.	These rivers generally have large basins.
4.	The channels of these rivers are broad.	4.	The channels of these rivers in the upper courses form gorges waterfalls, and rapids.
5.	There is hardly any vertical erosion.	5.	Both vertical and lateral erosions are significant.
6.	These are slow moving.	6.	These are both swift and have sluggish movements.
7.	These have low carrying capacity.	7.	They transport huge quantities of sediments.
8.	These are mainly depositional agents.	8.	These are active erosional and depositional agents.
9.	They make shallow meanders.	9.	In the plains, they make numerous sharp meanders and ox-bow lakes.
10.	These are generally not navigable.	10.	Most of them are navigable in the plains.
11.	These are mostly seasonal.	11.	These are perennial.
12.	Most of them originate from the Western Ghats and plateaux.	12.	Most of them originate from the Himalayan Glaciers.
13.	Most of them are in the senile stage.	13.	Most of them are in the late youth stage.
14.	There is no river capturing.	14.	River capturing is a common phenomenon.
15.	These rivers have been harnessed for the generation of hydel-power, e.g., Hirakund, Koyna, Nagarjunasagar.	15.	There are numerous multipurpose projects on these rivers, e.g., Bhakra, Tehri, Salal, etc.
16.	These rivers make deltas (Godavari, Krishna, Kaveri) and estuaries like Narmada and Tapi.	16.	These rivers make only deltas. The Sundarban Delta is the largest in the world.
17.	Their channels are near the base level.	17.	These rivers are in the ending part of youth stage. In their mountainous course they do river capturing.

River Regimes

The pattern of the seasonal flow of water in a river is called its regime. The main difference in the flow pattern of the Himalayan and the Peninsular rivers is caused by the differences in climate. The Himalayan rivers are perennial and their regimes are dependent on the pattern of water supply both from snow-melt and rainfall. Their regimes are monsoonal as well as glacial. The regime of most of the peninsular rivers, on the other hand, are only monsoonal as they are controlled by rainfall alone. Even regimes of different peninsular rivers are not the same because of the differences in the seasonal distribution of rainfall in various parts of the plateau (Fig. 3.11).

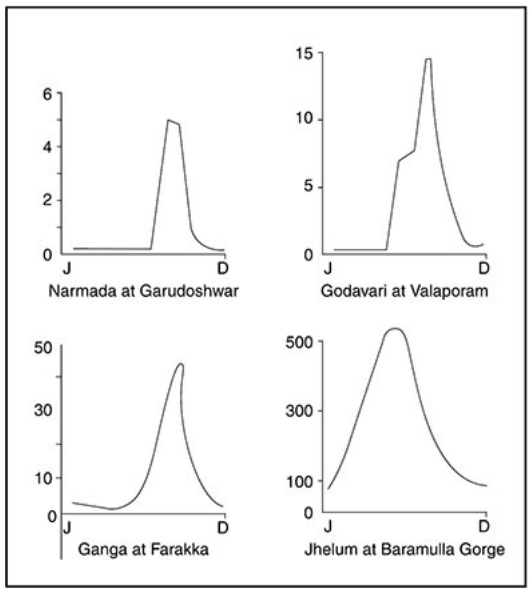


Fig. 3.11 River Regimes

The discharge is the volume of water flowing in a river measured over time. It is measured either in cusecs (cubic feet per second) or cumecs (cu ms per second). The Ganga has its minimum flow during the period of January to June. The maximum flow of water in the Ganga occurs either in July or August. After September, there is a steady fall in the flow of Ganga. The river thus has a typical monsoonal regime. There are striking differences in the river regimes in the eastern and western parts of the Ganga basin. The Ganga maintains a sizable flow in the early parts of summer before the monsoon rains begin. This is largely due to the water supply from the Himalayan snow-melt. The discharge data, however, do not include the volume of water diverted for irrigational purposes at different points before the Farakka Barrage (Malda, West Bengal). The mean maximum discharge of the Ganga at Farakka is about 55,000 cusecs, while the mean minimum is only 1300 cusecs.

The two peninsular rivers display an interesting difference in their regime as compared to the Himalayan rivers. The Narmada has a very low volume of discharge from January to July. It suddenly rises in August when the maximum discharge is attained. The fall in October is as spectacular as the rise in August. The Godavari flows at a low level until May. It has the minimum discharge in May and the maximum in July-August. After August there is a sharp fall in water flow, although the volume of flow in October and November is higher than that in any of the months from January to May. The mean maximum discharge of Godavari at Valaporam is 3200 cusecs. These figures give an idea of the regime of the river. The flow of water in the Narmada, as recorded at Gurudeshwar, shows that the maximum flow is of the order of 2300 cusecs, while minimum flow is only 15 cusecs.

The data on water discharge in different rivers in different parts of the year have important implications to their utilization by states. It is on this count that inter-state disputes arise.

SHIFTING COURSES OF THE RIVERS

An interesting phenomenon of most of the extra-peninsular rivers and that of the peninsular have shifting courses, especially in their lower reaches. In the upper reaches, the Himalayan rivers have the tendencies of river capturing. River capturing is mainly caused by the headward and downward erosion of the river. In the plain areas, generally, the rivers form meanders in their courses. During floods, due to increased quantity of water, the streams try to straigthen their courses. Earth movements do have their role in affecting these processes. Some of the important river captures are as under:

The ancient Saraswati River, which provided an abode for early Aryan settlers, presents a typical example of shifting courses and river capturing. Descending from the Himalayan ranges, its initial

course during the pre-historic period was passing near Churu (about 2000 to 3000 BC), and the Luni river was one of its tributaries. It gradually shifted towards west till it joined the Satluj near Ahmadpur. Later on, the water of its upper course was captured by a tributary of the Ganga River as a result of which its lower course became dry. This gave birth to Yamuna River, an important and the longest tributary of the Ganga System. Even today the dry valley of the Saraswati River is found in Rajasthan area in the form of Ghaggar valley (Fig. 3.8). Similar shifting has also been observed in the rivers of the Punjab during the historical past. The records of the third century BC show that the Indus flowed more than 130 km east of its present course, through the now practically dry beds of the deserted channel, to the Rann of Kutch which was then a gulf of the Arabian Sea. Later on, it gradually shifted towards the west and occupied its present position. During the reign of Akbar the Great, the Chenab and Jhelum rivers joined the Indus near Uch (Pakistan), but their present confluence lies near Mithankot about 100 km downstream of the old place of confluence. Similarly, Multan was formerly located along the Ravi River, but today it is situated about 60 km south of its confluence with the Chenab. About 250 years ago the Beas river changed its old course, traces of which are still found between Montgomery and Multan, and joined the Satluj river near Sultanpur. In the early part of the Christian Era the Satluj had more easterly course and independently discharged its water into the Arabian Sea.

About 250 years ago, the Brahmaputra flowing through Mymensingh was discharging its waters into the Meghna River. In due course of time, it straightened its course and joined the Ganga (Padma) River forming a new stream called Jamuna in Rangpur and Jamalpur districts of Banglasesh. A feeble channel of the Brahmaputra is still flowing along the same old course and retains the old name. This change in the course was associated with 30 m rise in Madhopur forest area between 1720–1830 AD. Even the entry of the Brahmaputra to the plains of Assam is also the outcome of the process of river capturing. According to geologists, during early days, the Tsangpo River of Tibet taking an easterly course used to join the Irrawaddy River (Myanmar) through the Chindwin, which was then a large river, transporting huge quantity of water. Later on, a small river flowing along the southern slopes of the Himalayas through its headward erosion captured the water of the Tsangpo River and, thus, helped in the evolution of the stream of Brahmaputra.

Similarly, Kapili, a tributary of the Brahmaputra, has captured the waters of the Meghna river of Bangladesh. In old days, Meghna originated from the Brail Ranges (between Meghalaya and Manipur) and flowing southward, emptied its waters in the eastern part of the Bay of Bengal. But Kapili, through headward erosion, captured its northern course. The Lumding-Halflong Pass is an evidence of this abandoned valley. In a similar way, the Dhansiri River capturing the water of a tributary, Kapili River has helped in the formation of a new river called Jumuna (Assam).

There can be numerous causes responsible for the shifting of river courses. These include the shifting gentle slope of the Great Plains of India, meandering courses of the rivers, straightening of the river courses during floods, upliftment of the Potwar Plateau (Delhi-Sirhind Plateau), downwarping of the Malda Gap, rise in the Madhopur forest area and uplift of the Barind area.

SARASWATI-THE MYSTERY OF A LOST RIVER

Saraswati is one of the main rivers of the Rigvedic Period. This river has been mentioned in the Rigveda and post-Vedic texts. This sacred river played an important role in the development of Indian culture. The godess Saraswati was originally a personification of this river but later developed an independent identity.

The Saraswati river was revered and considered important for Hindus because it is said that it was on the banks of this river that the Vedic-culture developed and Sanskrit originated. According to the leading historians like Bridget and Raymond Allchin in '*The Rise of Civilization in India and Pakistan*' that the earliest Aryan homeland-Aryavarta/Brahmavarta, was in Punjab and in the valleys of Saraswati and Drishdvati rivers in the time of Rigveda (Fig. 3.11-A and B).

According to the Rigveda, the Saraswati River originated in the hills or mountain and entered the northern plains of India. The all important rivers of that period flowed between the Indus in the west and

the Ganga in the east. The Saraswati was placed between the Yamuna and the Sutlej, which is consistent with the Ghaggar River of the present days (Fig. 3.11-A and B).

The real cause of disappearance of Saraswati is still shrouded in mystery. The major works about the disappearance of the river as given the mythological literature, archaeological, glaciological, and remote sensing and GIS evidences have been presented briefly in the following:

1. Religious and Mythological Evidence: Hindu mythology records several legends and anecdotes that are intertwined with the river's geologically brief existence. Every aspect of the river's life, right from its source to its journey down the Himalayas and over the plains towards the Sindhu (ancient Arabian Sea), have found mention in one religious text or the other, like Rigveda, Yajurveda, Atharveda, Brahmana literature, Manusmriti, Mahabharata and Puranas. These descriptive legends have often proved helpful in cataloguing some of the natural events of the period and linking some of them with the rivers changing course.

   Over a 3000 year-long period since the Vedic times, the drainage pattern of many rivers had changed much from that described in the earlier religious literature. The decline of Saraswati appears to have commenced between 5000–3000 B.C. probably precipitated by a major tectonic event in the Siwalik Hills of Himachal Pradesh and Uttarakhand.

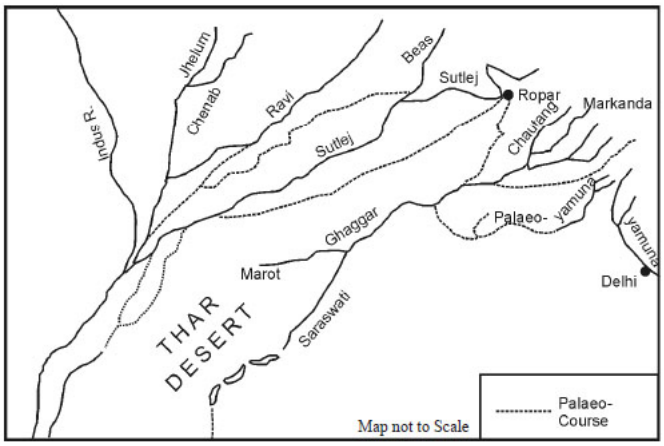


Fig. 3.11-A Progressive Shift of Rivers (Sutlej, Beas, Ravi) and abandoned earlier channels. Note the abrupt termination of Saraswati-Ghaggar at Marot (after Sridhar et. al)

2. Archaeological Evidence: Most of the archaeological sites of the then civilization are located on the Saraswati river basin. There are four Harappan and pre-Harappan sites in Punjab, Rajasthan and Uttar Pradesh. These sites are located at Rupar (present Rupnagar), Nihang Khan, Bara and Sirsa valleys. Harappan culture flourished in the western part of Punjab around 2500 B.C. It is believed that the Harappans entered through the Indus Valley into Kalibagan valley on the left bank of Ghaggar (erstwhile Saraswati) and spread to Punjab along the Saraswati River. Carbon dating of the material at Kalibagan suggests that Harappan culture flourished around 2500 B.C. in India and existed for 1000 years. So the present day geomorphological set-up did not exist in 1500 B.C. and thus Indus, the Sutlej and the Beas followed independent courses to the sea. At present, more than 8000 years since the Vedas came into existence, some of the rivers mentioned therein have become defunct or have shifted from their original courses.
3. Glaciological Evidence: According to the experts of glaciers and geomorphology, the Saraswati, Shatadru (Sutlej), and Yamuna derived their waters from the glaciers which had extensively covered the Himalayas during the Pleistocene Period. The Pleistocene is the first epoch of the Quaternary, preceded by the Pliocene and succeeded by the Holocene (from 1.64 million to 10,000 years ago). The warm period that followed, generated many rivers, big and small, coursing down the Himalayan slopes. The enormity of waters available for agriculture and other economic activities during those times had prompted the religiously bent

ancient inhabitants to describe reverentially seven mighty rivers or ‘Sapt-Sindhu’, as divine rivers arising from glaciers.

The Saraswati was supposed to have originated in the Bandarpanch Massif (Sarawati-Rupin Glacier confluence at Naitwar in the western Garhwal). The river crossing the Shiwalik entered the plains and took a roughly southerly course, passing through the plains of Punjab, Haryana, Rajasthan and Gujarat and finally debouched into the ancient Arabian Sea at the Great Rann of Kutch. In this long journey, Saraswati was believed to have had three tributaries i.e. Shatadru (Sutlej) arising from Mount Kailash, Drishadvati and Palaeo-Yamuna from the Siwalik (Fig. 3.10-B). Together these three rivers flowed along a channel, presently identified as that of the Ghaggar. The rivers, Saraswati and Ghaggar, are supposed to be one and the same, though a few researchers use the name Ghaggar to describe Saraswati’s upper course and Harka to its lower course (Fig. 3.11-B).

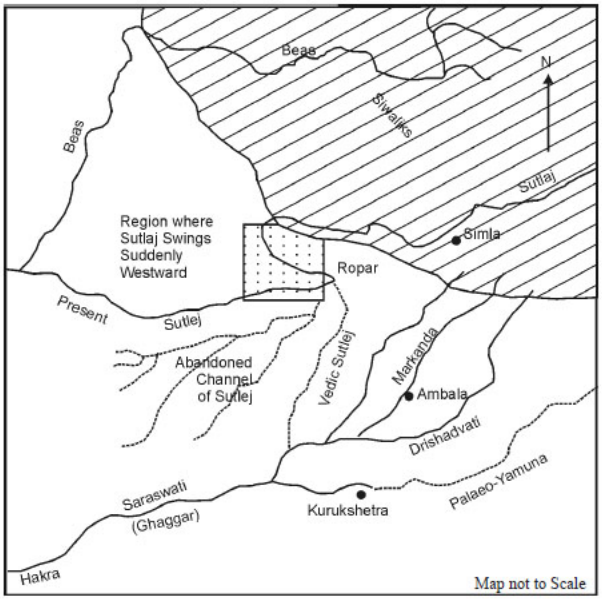


Fig. 3.11-B Abrupt Shift of Sutlej (Shatadru) river westward near Ropar (Rupnagar) cutting off waters to Saraswati River (after Valdia. k.c. et. al.)

Geologic studies indicate the destabilising tectonic events had occurred around the beginning of Pleistocene, (about 1.4 million years ago) in the entire Siwalik region, resulting into massive landslides, avalanches and mass-wasting. These disturbances, which continued intermittently, were all linked to uplift of the Himalayas. Presumably, one of these events must have severed the glacier connection and cut off the supply of glacier melt-waters to the Saraswati River. As a result, the Saraswati became non-perennial and dependent on monsoon rains. Consequently, the Saraswati River dwindled and became a seasonal river.

Leading scholars, like Sridhar and his colleagues have classified the rivers of north western plains of India into four main groups, i.e. (i) Sindhu (Indus) and its tributaries, Vitasta (Jhelum, and Askini (Chenab), (ii) Shatadru (Sutlej) and its two tributaries-Vipasa (Beas) and Parusani (Ravi), (iii) Saraswati and its tributaries (Markanda, Ghaggar, and Patialewali), and (iv) Dishadvati and Lananavati. Baldev Sahai grouped them into (i) Sutlej, (ii) Ghaggar, and (iii) Yamuna systems of the Saraswati.

For about 2000 years, between 6000 and 4000 B.C., the Saraswati flowed as a great river. R.D. Oldham (1886) was the first geologist who argued logically pointing to the great changes in the drainage pattern of the rivers of Punjab and western Rajasthan, covering a once fertile region into a desert.

4. Evidence from Remote Sensing and GIS: A remote sensing study of the Indian desert reveals numerous signatures of palaeochannels in the form of meanndering courses, which is identified by the total variations. The Saraswati River could be traced through these

palaeochannels as a migratory river. Its initial course flowed close to the Aravalli ranges and the successive six stages took west and north-westerly shifts till it coincides with the dry bed of the Ghaggar River.

The course of the Saraswati River in the states of Punjab, Haryana and Rajasthan is clearly highlighted in the LANDSAT imagery by the vegetation cover thriving on the rich residual loamy soil along its earlier course. The digital enhancement studies of IRS-1C data (1995), combined with RADAR imagery from European Remote Sensing Satellite ERS 1/2, identified subsurface features and recognised the palaeochannels beneath the sands of the Thar Desert. A study of NRSA, based on satellite derived data, confirming that the two were independent rivers; also, the three palaeochannels, south of Ambala, seen to swerve westwards to join the ancient bed of Ghaggar, are inferred to be tributaries of Saraswat/Ghaggar. Moreover, the satellite imagery identified two palaeochannels trending NE-SW in Jaisalmer District of Rajasthan, which are presumed to be the lost river Saraswati.

Prof. Yash Pal and B. Sahai have traced the palaeochannel of the river through Punjab, Haryana and Rajasthan. They found that its course in these states is clearly highlighted in the LANDSAT imagery by the lush-green vegetation thriving on the rich residual loamy soil along its earlier course. According to their findings, the river disappears abruptly in a depression in Pakistan, instead of reaching the Arabian Sea. The subsequent studies based on IRS-1C however, identify sub-surface features of the Saraswati River and thus recognise palaeochannels beneath the sand in the Thar desert (Fig. 3.11-A). These channels are seen to extend up to Marot in Pakistan and appear in a line with the present dry bed of Ghaggar. This river continues as Nara River in Sindh (Pakistan) and opens into the Rann of Kachchh .

5. Geophysical Evidence: The Geo-physical surveys carried out by the Geological Survey of India to assess the groundwater potential in Bikaner, Ganganagar and Jaisalmer districts in western Rajasthan desert have brought out several zones of fresh and less saline water in the form of arcuate shaped aquifers, similar to several palaeochannels elsewhere in the state. All these studies prove that the Saraswati was passing through the desert of Thar (Rajasthan and Sindh).

   The wanning period of Vedic Civilization around 3700 B.C. was also the period that disrupted both the Saraswati and Drishadvati rivers. In the opinion of experts, the rivers of this area changed their courses often in the last 5000 years. Moreover, Prof. Yas Pal and his colleagues observed a sudden widening of the Ghaggar River near Patiala. They argued that the widening of the Ghaggar River near Patiala was possible only if some major tributary had joined it. According to them, ancient Satadru (Sutlej) swung suddenly westward near Ropar/Rupnagar (Fig. 3.11-B) to join the Indus with its tributaries (Beas and Ravi), deserting its earlier channel to the sea. This sudden diversion of Sutlej as well as depletion of waters from Drishadvati due to loss of its feeding streams, appear to be major events leading to the drying up of Saraswati. Several geomorphologists attribute this event to tectonism involving the rise of Delhi-Hardwar Ridge and uplift in the Aravallis.

   In the lower part of the Saraswati was the Rann of Kachchh. The topography of the Great Rann of Kachchh is typically deltaic, developing usually at the mouth of rivers, confirming entry of a few rivers in the sea in this region. The satellite imagery, as well as detailed mapping, have revealed a network of distributaries and extensive graded deposits, products of Holocene marine regression.

   It appears that Indus (Sindhu), Satadru (Sutlej), Saraswati, Drishadvati, Palaeo-Yamuna and Lavanavati (possibly an ancestor of present day Luni River) had independent courses and opened into the Rann separately.

   According to Malik et.al. at least three rivers-proto-Satadru (Sutlej), Saraswati and Drishadvati must have drained into the Rann around 2000 B.C. of which only Sindhu (Indus) has survived. The original delta complex with relict channels, including that of Nara, a continuation of Ghaggar, is today better preserved on the western side but covered by wind-borne deposits on the eastern part of the Great Rann.

   On the basis of the above studies, it may be summarised that the disappearance of the Saraswati River was due to the tectonic activities in the western Himalayas and the Northern

Pains; the drying up of glaciers in the Himalayas because of climatic change; rivers piracy; and the shifting in the courses of rivers. All the above studies prove that the Saraswati River in the Vedic Period was moving through Punjab, Haryana, and Rajasthan and was merging into the Arabian Sea.

TYPES OF LAKES

A lake is a natural depression filled with water. In this period of population explosion and scarcity of water, lakes are an important source of water supply to humanity. They also help in the prevention of sudden floods and droughts. The lakes of India have different origins. Some of the important lakes of India have been described briefly in the following section:

1. The Tectonic Lakes

These are formed due to the fractures and faults in the earth's crust. Most of the lakes in the hilly areas of Kashmir and Kumaun belong to this type of lakes. The Tso Moriri and Pangong Tso (Ladakh) are some of the examples of such lakes.

2. The Crater Lakes

Crater lakes are formed when the craters and calderas are filled with water. They are very few in number in India. The '*Lonar Lake of Buldhana'* (Maharashtra) is one of the examples of crater lakes.

3. Glacial Lakes (Tarns)

These lakes are the result of glacial erosion. Tarn is a small mountain lake, especially one that collects in a cirque basin behind rises of rock material. Most of the glacial lakes in India are small in size. The Gangabal lake of Kashmir, located in the Greater Himalayas of Kashmir, is one such examples. Glacial lakes are also found in Ladakh (J&K), Himachal Pradesh, and Kumaun region of Uttarakhand.

4. Fluvial Lakes

Rivers form different types of lakes through their erosional and depositional work. In general, rivers are the destroyers of lakes. In fact, lakes are often obliterated due to filling of sediments and headward erosion by rivers. Fluvially originated lakes are generally temporary and are soon obliterated. Fluviaslly originated lakes include plunge-pool lakes (in front of a waterfall), ox-bow-lakes,
alluvial fan lakes, delta lakes, flood plain lakes and raft-dammed lakes. All these lakes may be observed in the upper, middle, and lower courses of the Ganga and the Brahmaputra rivers. Lakes formed to the meandering of rivers in the plains of gentle gradient are known as alluvial lakes. The ox-bow lakes found in the Middle and Lower Ganga Plains are some such examples.

5. Aeolian Lakes

These are small temporary hollows or depressions lying on the wind blown sand surface. Western Rajasthan has several lakes of this type. The desert lakes generally have a larger proportion of salt content and are often been called salt lakes. The Dhands of western Rajasthan are some of such examples.

6. Dissolution Lakes

These lakes are formed due to a depression of the surface by underground dissolution of soluble rocks like limestone and gypsum. Such lakes are found in and around Cherrapunji, Shillong (Meghalaya), Bhimtal (Kumaun), and Garhwal (Uttarakhand).

7. Lagoons

These are formed by the deposition of sandbars along the sea coast. The Lake Chilka of Odisha; Pulicat (Andhra Pradesh); Vembanad, Asthamudi, and Kayals of Kerala are some of the examples of lagoons.

8. Landslide Lakes

These lakes are produced by landslides and rock-falls causing obstruction in the course of streams. The Gohna Lake of Garhwal was formed due to a huge landslip across a tributary of the Ganga.

MAIN LAKES OF INDIA

Ashtamudi Lake (Astamudi Kayal)
It is a lagoon in the Kollam district of Kerala. Ashtamudi means “eight branches”. In fact, this is a lake with multiple branches. It has been registered as a wetland of international importance under the Ramsar Convention.

Bhimtal
Situated near the town of Bhimtal in Kumaun Divison of Uttarakhand, it is a fascinating lake with an island in the centre. The Bhimtal town, though ancient, was never prominent because of the greater popularity of Nainital. At present, it attracts a large number of the domestic and international tourists.

Bhoj Wetland
Located in the city of Bhopal (capital of Madhya Pradesh), it consists of two lakes, namely the Upper Lake and the Lower Lake. Being in the vicinity of the capital city, it is one of the highly polluted lakes of India.

Chandra Tal
It is a high altitude lake in Lahaul and Spiti district of Himachal Pradesh. It is about 4300 m above sea level. The Kunzam Pass, which connects Lahaul and Spiti, is only about 6 km from this lake.

Chembarambakkam Lake
It is located in the Chengalpattu district of Tamil Nadu, about 40 km south of Chennai. The Adyar River originates from this lake. A part of the water supply of the Chennai metropolis is drawn from this lake.

Chilka Lake (Chilika Lake)
Situated in the state of Odisha, it is a brackish water coastal lake. It is the largest coastal lake in India. The lake was formed due to the silting action of the Mahanadi River which drains into the northern end of the lake. The area of the lake varies from 1175 sq km in the monsoon season to 900 sq km in the dry season.

Dal Lake
Dal is a famous lake in Srinagar. Stretching over an area of 18 sq km, it is divided by causeways into four basins; namely, Gagribal, Lokut Dal, Bod Dal, and Nagin. It is well known for approximately five hundred houseboats. Apart from houseboats, the lake provides a good opportunity to the tourists for canoeing, water-surfing, and kayaking. Being highly polluted, the lake is shrinking at a faster pace. The lake has some interesting flora like lotus flowers and water lilies and water-chestnut.

Dhebar Lake (Jaisamand)
Situated in the state of Rajasthan, about 45 km to the east of Udaipur, it is the largest artificial lake of India. It stretches over an area of about 87 sq km. It was built in the 17th century when Rana Jai Singh of Udaipur built a marble dam across the Gomati River. This lake has three islands; the Jaisamand Resort is located on the biggest island.

Himayat Sagar
Located at a distance of 20 km from the city of Hyderabad, it was named after the youngest son of the seventh Nizam, Himayat Ali Khan. It is an artificial lake constructed across the Musi river in 1927.

Hussain Sagar
Hussain Sagar is in the city of Hyderabad. It was built across a tributary of the Musi River by Hussain Shah Wali in 1562. Its water is supplied to the city of Hyderabad.

Kaliveli Lake
It is a coastal lake in the district of Viluppuram in Tamil Nadu. It lies about 10 km to the north of Pondicherry. The lake is one of the largest wetlands in the peninsular India. It is being encroached by agricultural fields. Its area is shrinking fast.

Khajjiar Lake

Located in the Chamba district of Himachal Pradesh, it is only 24 kms from the important hill station of Dalhousie. Surrounded by the giant deodar trees, it presents a panoramic view for the tourists.

Khecheopalri Lake
It lies in West Sikkim. It is considered a holy lake, both by the Buddhists and the Hindus. The lake is surrounded by thick forests of bamboo. The placid waters of the lake are visited by many pilgrims and tourists.

Kolleru Lake
Situated in Andhra Pradesh, it is the largest fresh water lake of India. It is located between the deltas of the Krishna and Godavari rivers in the Krishna and Godavari districts. The lake serves as a natural flood-balancing reservoir for the two rivers. The lake was an important habitat for an estimated 20 million residents and migratory birds Grey or Spot-billed pelicans. Rich in flora and fauna, it attracts birds from Siberia and eastern Europe between the months of October and March. The lake was notified as a wildlife sanctuary in 1999 under India's Wild Life Protection Act. It was declared a wetland of international importance in 2002 under Ramsar convention. Increasing pollution has reduced the bio-diversity in the lake.

Loktak Lake
It is the largest fresh water lake in north-east India. It is also called the only "Floating Islands Lake" in the world due to the floating Phundis (floating islands) on it. It was designated a wetland of international importance under Ramsar Convention in 1990. It serves as a source of water for hydro-power generation, irrigation, and drinking. The lake is also a source of livelihood for the rural fishermen who live in the surrounding areas and on Phundis (floating islands), which are actually heterogeneous mass of vegetation, soil and organic matter.

Nako Lake
Situated in the district of Kinnaur (Himachal Pradesh), it is a high altitude lake. This lake is surrounded by willow and poplar trees. Near the lake there are four Buddhist temples. It is considered a sacred lake.

Osman Sagar
It is an artificial lake in Hyderabad. It was created by damming the Musi River in 1920 by the last Nizam of Hyderabad (Osman Ali Khan) for providing a drinking water source to Hyderabad. A guest house called "Sagar Mahal" overlooking the lake, now a heritage building, was the summer resort of the erstwhile Nizam.

Pongong Tso
Pong Tso is an endorheic lake that is situated in Ladakh at a distance of five hours from Leh. The road traverses the third highest pass in the world. It is Ladakh's tourist attraction. It is a transboundary lake between India and China for about 134 km. One third of this lake lies in India. Its average width is around 5 km and is the world's largest brackish water lake. The lake is a breeding ground to migratory birds like Bar-headed goose and Brahmini duck. During the winter the roads get blocked and the lake freezes.

Pulicat Lake
It is the second largest brackish water lake on the Coromandal Coast. It lies on the border of Andhra Pradesh and Tamil Nadu. The barrier island of Sriharikota separates the lake from the Bay of Bengal. The lake is 60 km long and varies 0.5 to 18 km in width. It is the habitat of numerous local and migratory birds. Nearly 15,000 flamingos visit the lake every year, along with pelicans, kingfishers, herons, painted storks, spoonbills, and ducks.

Pushkar Lake
Situated in the district of Ajmer, it is an artificial lake. The lake was created in the 12th century when a dam was built across the headwaters of the Luni River. Thousands of pilgrims come to bathe in the waters of the lake during the festival of Kartika Poornima in November.

Renuka Lake
Situated in the Siarmaur district of Himachal Pradesh, this lake has been named after the goddess Renuka. A lion safari and a zoo are major attractions at Renuka. It is a site for the annual fair in the month of November.

Roopkund
Situated in Uttarakhand, it is a lake around which 600 skeletons were found at the edge of the lake. The location is uninhabited and is located at an altitude of about 5030 m. The skeletons were discovered in 1942. Radio-carbon dating suggest that these people died in an epidemic.

Sambhar Lake
Situated about 70 km to the west of Jaipur city, it is the largest salt lake of India. On the eastern end, the lake is divided by a 5 km long dam made of stones. To the east of the dam are salt evaporation ponds where salt is being produced for more than a thousand years. The water depth varies from a few cm during the dry season to about 3 m after the monsoon rains. Sambhar has been designated a Ramsar site (recognised wetland) of international importance. Thousands of Siberian birds reach the lake during the winter season.

Sasthamkotta Lake
It is a large fresh water lake in Kerala state. It is located near Sasthamkotta in Kollam district, about 30 km from Kollam. It is a great attraction for the tourists.

Satta or Sat Tal
It is the calm, quiet group of seven lakes near Bhimtal town of the Kumaun Division of Uttarakhand. These lakes are situated at an altitude of 1370 m above mean sea level. These lakes are a paradise for migratory birds.

Suraj Tal
Located below the summit of the Baralacha Pass, it is a high altitude lake, 4980 m above sea level. This lake is the source of the Bhaga River, one of the main branches of the Chenab River.

Chandra Taal Lake
It has a crescent shaped basin. It is located in Spiti district of Himachal Pradesh at an elevation of 14,100 ft. It was a resting point for the travellers and traders coming from Tibet or Ladakh. The lake is accessible only during June-September months and remains frozen for the rest of the time. It is one of the famous but least trekked lake of Himachal Pradesh. It is flanked by the green carpeted Chandra Bhaga mountain range.

Tawa Reservoir
Located in Hoshingabad on the River Narmada (MP), it was created as a result of the Tawa Dam. It forms the western boundary of Satpura National Park and Bori Wild Life Sanctuary.

Tsongmo Lake
Situated in the state of Sikkim, about 40 km away from Gangtok, it is a glaciated tarn lake. It is oval shaped. Being situated at an altitude of about 3780 m, it remains frozen during the winter season. It is a sacred lake for the Buddhists and the Hindus.

Veeranam Lake
It is located in Cuddalore district of Tamil Nadu, about 235 km from Chennai. It is one of the water reservoirs from where water is supplied to Chennai.

Vembanad Lake (Vembanad Kayal or Vembanad Kol)
Covering an area of about 200 sq km, it is the largest lake in Kerala. The lake lies at sea level, and is separated from the Arabian Sea by a narrow barrier island. Several rivers flow into the lake including the Pamba and Periyar. The lake surrounds the islands of Pallipuram and Perumbalam.

Veeranpuzha Lake

Located in Kochi, it is the northern extension of Vembanad Lake. It attracts a large number of tourists from different parts of the country and abroad.

Vembanattu Lake

About 16 km from Kottayam is a vast network of rivers and canals forming Vembanattu Lake. An enchanting picnic spot and a fast developing back water tourism spot providing boating, fishing and sightseeing experiences that are truly exhilarating. Kumarakom Bird sanctuary is on the banks of Vembanattu Lake.

Wular Lake

Situated in the Valley of Kashmir between Sopore and Bandipore, it is the largest fresh water lake in India. The lake was formed as a result of tectonic activity during the Pleistocene Period. Depending on the season, the size of the lake varies between 30 and 250 km. The River Jhelum feeds the lake, which acts as a natural reservoir. The Tulbul Project is a *"navigation lock-cum-control structure*" at the mouth of the Wular lake. It envisages regulated water release from the natural storage in the lake to maintain a minimum draught of 4.5 feet in the river up to Baramulla during the lean winter months.

Water Resources of India

Water is a primary natural resource, a basic human need and a precious national asset. Water is critical to our daily lives and a principal compound in nature. A human can survive 50 to 60 days without food, but only 2 or 3 days without water. The water we use must be adequate in quantity as well as quality for its many tasks—everything from personal hygiene to vast national water projects. Water indeed occupies that place between land and sky, mediating energy and shaping both, the lithosphere and atmosphere. The significane of water is increasing with the tremendous increase in population. We need water for drinking, personal hygiene, domestic consumption, irrigation, industrial purposes, generation of hydel-power, transport and recreation.

The rivers and canals run throughout the country with a length of about 1.9 lakh kilometers and the other water bodies cover an area of around 7.31 million hectares (M.ha) **(Table 3.5).**

Table 3.5

Inland Water Resources of India

Rivers and Canals (in km)	195,095
Reservoirs	2.93
Tanks and ponds	2.43
Flood plain lakes & derelict water bodies	0.80
Brackish water	1.15
Total	7.31

According to the International norms any country can be categorized as 'Water Stressed' for water availability lesser than 1700 m3 per capita per year and as '*water scarce*' if it is lesser than 1000 m3 per capita per year. The water availability in India in 1991 and 2001 were 2309 m3 and 1902 m3. It is expected that per capita surface water availability is likely to be reduced to 1401 m3 and 1191 m3 by the years 2025 and 2050 respectively.

Of the total world's surface area, India accounts for 2.45%, 4% of the world's water resources. The total available water from precipitation is about 4000 km3. The availability from the surface water and replenishable groundwater is 1869 km3. Merely 60% can be used beneficially.

There are two major sources of water namely:

i) Surface water resources

ii) Ground water resources

Surface water resources include the rivers, ponds, lakes and tanks. The mean flow of water in all the river basins of India is approximately 1869 km3. Owing to the tropological and hydrological constraints only 690 km3 (32%) of available surface water is utilized. The Ganga, the Brahmaputra and the Barak rivers account for one third of the total surface water resources.

* The Ganga and Brahmaputra basins have about 46% of the total underground water.

According to one estimate the total ground water reserve up to a depth of 300 m is about 3700 mham, almost ten times of the annual rainfall. In terms of exploitation of ground water potential, Punjab comes on top (about 94%) followed by Haryana (84%), Tamil Nadu (61%), Rajasthan (51%), Gujarat (42%), Uttar Pradesh (38%), Maharashtra (31%), West Bengal (25%), and Andhra Pradesh (24%). States like Assam, Odisha, Madhya Pradesh, and Bihar have not been able to utilise even 20% of their total ground water potential.

The maximum reserves of the ground water are accounted by the Indo-Gangetic and Brahmaputra plains **(Table 3.6)** and peninsular region also offers potential for the exploitation of the ground water resource in India.

Table 3.6

India's Water Resources

S. No.	Water Resource	Quantity (km3)	Percentage
1	Annual precipitation (inclusive of snowfall)	4000	100
2	Precipitation during monsoon	3000	75
3	Evaporation + Soil water	2131	53.3
4	Average annual potential flow in rivers	1869	46.7
5	Estimated usable water resources	1123	28.1
	Surface water	690	17.3
	Groundwater (replenishable)	433	10.8
	Storage created of utilized water	253.381	22.52
	Storage (under construction) of utilizable water	50.737	4.5
6	Estimated water need in 2050	1450	129
7	Estimated deficit	327	29
8	Interlinking can give us	200	17.8

Source: India-WRIS Wiki Water Resources Information System of India

Table 3.7

Ground Water Aquifer System of India

System	Region	Groundwater Potential
Unconsolidated Formations-Alluvials	Indo-Gangetic, Brahmaputra plains	Enormous reserves down to 600 m depth
		High rainfall ensures the recharge
		Supports development through the tubewells
	Coastal region	Extensive aquifers but there is a risk of saline water intrusion
	Desert region-Rajasthan and Gujarat	Negligible recharge due to scanty rainfall
		Salinity hazards
		Availability at deeper depths
Consolidated/Semi-Consolidated formations-sedimentaries, basalts and crystalline rocks	Peninsular region	Availability depends on porosity developed due to weathering, fracturing etc.
		Scope of geothermal wells at shallow depths (20–40 m) in some areas and (100–200 m) in other areas
Hilly	Hilly states	Low storage capacity owing to faster runoff

Source: India-WRIS Wiki Water Resources Information System of India

It may be noted that the demand of water is growing in all the sectors **(Table 3.8)**.

Table 3.8

Estimated Water demand in BCM (Billion Cubic Metre) in India in Various Sectors

Sector	Water Demand in BCM (Billion Cubic Metre)		
	2010	2025	2050
Irrigation	688	910	1072
Drinking water	56	73	102
Industry	12	23	63
Energy	5	15	130
Other	52	72	80
Total	813	1093	1447

Source: Report of the Standing Sub-Committee on 'Assessment of Availability and Requirement of Water for Diverse Uses-2000', Ministry of Water Resources

The main problems associated with the utilisation of water are: (a) high fluctuations in the regimes of rivers, (b) uneven spatial distribution of precipitation, (c) wastage of water during floods, (d) unscientific utilisation of water, (e) pollution of rivers, and (f) dispute on water distribution among the states.

India's rivers are highly polluted. The rapid growth of population, agricultural modernization, industrialization and urbanization have reduced the per capita availability of water. The pollution of rivers, lakes and ponds have aggrevated the situations.

Water Harvesting in India

Water is a prime national resource, a basic human need and a precious asset. The U.N. report published in 2003 at the time of the Third World Water Forum in Kyoto says that water reserves are drying up fast, and booming population, and global warming will combine to cut the average person's supply by a third by 2020. The report also ranked 122 countries on the quality of their water provision. India was placed at the bottom of the list.

The agricultural output depends on monsoon as nearly 60% of the area sown is dependent on rainfall. In the areas of rain-fed agriculture, watershed harvesting has great economic and ecological significance. The greater parts of Rajasthan, Gujarat, parts of Haryana, Punjab and Uttar Pradesh, western Madhya Pradesh, eastern Maharashtra, northern Andhra Pradesh, western Karnataka, parts of Tamil Nadu, Odisha, Jharkhand, and Chhattisgarh frequently suffer from droughts. These regions face two critical issues:

(i) They have severe droughts in summer with an acute scarcity of water and a declining water-table, resulting in the loss of crop productivity for major and minor crops.

(ii) Heavy and intensive rainfall and surface run-off leads to soil erosion and sedimentation in the lower courses of the rivers, resulting in the formation of ravines and gulleys across the region.

These are the areas which are economically poor, struggling around the vicious cycle of underdevelopment. These areas require appraisal of land and soil resources based on micro watersheds (within an area of 10 to 1000 hectares), and macro watersheds (area over 1000 hectares) to draw plans for in-situ water harvesting and conservation through conjunctive uses, and thus monitor the development mode in a harmonised manner. The entire watershed management involves innovation in group decisions followed by action involving joint decisions. Thus, starting with a watershed (which is a manageable hydrological unit) as a land unit, it takes up water as the most important resource for development. It aims at containing the deterioration of land and its approach to development is not confined just to agricultural land alone, but covers the watershed area, starting from the highest point of the area to the outlet or the natural drainage line. This involves the implementation of ameliorative measures for barren hill slopes, marginal lands, agricultural lands, gullies, and ravines.

Optimal sustainable development can occur only with the maintenance of quality and efficient use of the country's water resources to match the growing demands on this precious natural resource with the active involvement of all in order to achieve accelerated, equitable economic development of the country.

It is rightly said that water resources hold the key to the socio-economic development of any country, and India is not an exception to this. It is a fact that water, a fragile and finite resource, is fast becoming scarce day by day. A recent study by the world bank indicates that per capita availability of water in India, which was to the order of 5000 cu m per year at the time of independence, has drastically come down to 2000 cu m per year at present. The average annual rainfall of 110 cm in the country, though fairly high, is marked by wide variations, both spatial and temporal. About 67% of the water resources are reported to be available in the Indo-Gangetic alluvial basins covering 33% of the geographical area of the country, as against 33% of the potential in the hard rock regions, occupying 67% of the geographical area. Looking at the existing scenario of water availability in the country, there is an urgent need for propagating water harvestring and transforming it into a mass movement. With regard to the water resource, the main challenges that need to be addressed are:

(a) ground water depletion,
(b) water quality deterioration,
(c) low water use efficiency,
(d) expensive new water resources,
(e) resource degradation,
(f) development of new water resources
(g) reformed price incentives,
(h) appropriate technology,
(i) tradable water rights, and
(j) international cooperation.

Rain-water Harvesting

Rainwater is biologically pure, soft in nature, and free from organic matter. Rainwater harvesting is an effective technique of conserving water by guiding the rainwater that falls on rooftops to storage tanks or underground pumps for future usage. Groundwater recharging, on the other hand, is undertaken by guiding water through pipes to wells, bore wells or recharge pits to ensure recharging water in the underground aquifers, for later use, whenever the need arises. Tamil Nadu is the first and only state in India which has made roof-top rain-water harvesting structure compulsory to all the houses across the state. There are legal provisions to punish the defaulters.

Rain water harvesting (RWH) is a simple, low cost technique. The most common technique in the urban areas is rooftop rainwater harvesting where the rainwater is collected on the top of the roofs and collected into reservoirs. The surface run off can be used for various domestic uses or even the groundwater recharge.

The collection of water from a catchment surface is referred to as rainwater harvesting. Water harvesting also includes activities aimed at harvesting surface and groundwater and other techniques aimed at conservation and efficient utilisation of limited water endowment. In general, water harvesting is the activity of direct collection of rainwater. The collected water can be stored for direct use or can be used to recharge the groundwater.

The quantum of harvested rainwater depends on:

1. Frequency, duration and intensity of rainfall;
2. Nature of catchment, and
3. Run-off characteristics.

Merits of Rainwater Harvesting

Following are the main advantages of rainwater harvesting:

1. Improves soil moisture;

2. Increases groundwater level;
3. Improves the quality of water;
4. Allows drought-proofing;
5. Prevents flooding of storm water drains;
6. Saves energy required to lift water;
7. Reduces flooding of roads;
8. Reduces soil erosion, and
9. An ideal solution to water problem in areas having inadequate water resources.

Rainwater Harvesting Practices

There are two main practices of rainwater harvesting:

1. Storage of rainwater on surface for future use. It is a traditional practice and structures used are underground tanks, ponds, check-dams, weirs, etc.
2. Recharge of groundwater. It is a new concept of rainwater harvesting and the structures generally used are:

(a) Pits-recharge: Pits are constructed for recharging the shallow aquifers.

(b) Trenches: These are constructed when the permeable strata is available at shallow depths.

(c) Dug wells: Drainpipes carry the water to a filtration tank from which it flows into the dug well. Rainwater that is collected on rooftop of buildings is also diverted to the dug wells.

(d) Hand pumps: The existing hand pump may be used for recharging the shallow/deep aquifers, if the availability of water is limited.

(e) Recharge well: Recharge wells are generally constructed for recharging the deeper aquifers and water is guided through filter media to avoid choking of recharged well.

(f) Recharge shafts: For recharging the shallow aquifers which are located below clayey surface, recharge shafts are used.

(g) Lateral shafts: Lateral shafts with bore wells for recharging the upper as well as deeper aquifers.

(h) Spre techni perm strata avai then spre techni used. is spre streams/ by m check- cement or perco p

Government Strategy

The total renewable water resources of India are estimated at about 1900 sq km per annum. It is predicted that by 2025 large parts of India will join countries or regions having absolute water scarcity. Groundwater has emerged as the prime source of drinking and irrigation. About 77% of present groundwater withdrawl is being used for irrigation purpose, thus contributing largely in food security of the country.

The following steps have been taken by the government to implement the water harvesting programme:

1. Since sustainability of drinking water-source is of paramount importance for smooth functioning of rural water supply, 25% out of 20% of the allocation under Accelerated Rural Water Supply Programme (ARWSP) has been earmarked exclusively for water harvesting schemes to make implementation of such schemes mandatory.
2. Similarly, 25% out of the allocation under Prime Minister's Gramodaya Yojana has also been earmarked for funding schemes under submission on sustainability.
3. MPs are requested to utilise Local Area Development Fund in their respective constituencies to take up water harvesting schemes.
4. Preparation of pilot projects on water harvesting in selected states have already been

undertaken.

5. Further, preparation of user-friendly atlas type of document on traditional water-harvesting structures in various parts of the country has been initiated for popularising the concept of water harvesting amongst all concerned, including the community.

By adopting watershed as a unit, different location-specific measures are adopted and executed carefully in each of the topo-sequences according to capability. Considering the fact that the rainfed area in India is about 60% of the country's net sown area, and the vast area should not suffer from neglect and poverty, investment in watershed management for water-scarce regions (receiving rainfall below 75 cm) of India is an appropriate development intervention which warrants top priority from the point of social justice and containing the widening spatial imbalance between irrigated wet farming and dryland farming systems.

In brief, watershed development approach being an intensive one, appears to be infinitely expensive in a relative sense over the seed and fertilizer approach, but economic evaluation conducted at the Central Soil and Water Conservation Research Institute at Dehradun shows that this is not so. On the other hand, the realisation that a crop-based approach, or an approach which treats the country as a single unit, would not address the major issue for agricultural development in different location-specific conditions, watershed management (or alternative drainage, flood control and conjunctive uses of water of different sources, or again, a more appropriate management of hill and forest-based agriculture) are alternative regimes, each having a different investment and policy support strategy.

There are a number of successful watershed management experiences like Sukhomajri, near Kalka and Pani-Panchayats (water collectives) at Ralegaon in Maharashtra, where the basic problems of food and fuelwood requirements of poor rural communities have been largely solved by water harvesting.

It is being suggested that in many rural or agricultural situations in our country, we require community participation interfaced with institutional support at the level of, say, "watershed" land and water managements in difficult ecological regimes to develop the slender resource base of the areas. The replication of successes like Sukhomajri would be for the better.

Studies to develop a baseline data for better understanding of the existing and emerging situations need to be undertaken. Recycling of water and water conservation will be a critical component of our daily lives in the new millennium. As far as possible, the technologies should be indigenously developed so as to make them socially acceptable and economically viable.

THE INTER-STATE WATER DISPUTES

Water being the most precious resource is required for domestic, irrigation and industrial purposes. River is more than an amenity, it is a treasure. It offers a necessity of life that must be rationed among those who have power over it. Most of the Indian rivers traverse more than one state. Each of the states of a river basin tries to obtain the maximum quantity of water. This has resulted in many water disputes in the country. In the Indian Constitution (1950), water is a state subject. Consequently, the state governments virtually exercise full control on planning, development, regulation, distribution, and control of water flowing through their territories. Under Article 262 of the Indian Constitution, the Parliament is empowered to provide the adjudication or control of water on any interstate river. Most of the rivers of India travese several states. Under the Water Dispute Act, 1956, amended in 2002, a tribunal of three sitting judges of the Supreme Court has to be constituted by the Central Government for the settlement of an interstate water dispute when a request is received from a state government. According to the Interstate Water Dispute Act, 1968, the Central Government has also been given the responsibility of regulation and development of interstate rivers and river valleys to the extent to which such regulation and development under the control of the Union is declared by the Parliament, by law, to be expedient in public interest. Some of the important interstate water disputes in India are as under:

1. The Kaveri water dispute between Karnataka, Kerala, and Tamil Nadu.
2. The Krishna water dispute between Maharashtra, Karnataka, and Andhra Pradesh.
3. The Tungbhadra water dispute between Andhra Pradesh and Karnataka.

4. The Godavari water dispute between Andhra Pradesh, Madhya Pradesh, Maharashtra, and Odisha.
5. The Narmada water dispute between Gujarat, Madhya Pradesh, Maharashtra, and Rajasthan.
6. The Ravi, Beas and Satluj water dispute between Punjab, Haryana, Rajasthan, Delhi, and Jammu and Kashmir.
7. The Yamuna River water dispute between Uttar Pradesh, Haryana, Himachal Pradesh, Punjab, Rajasthan, Madhya Pradesh, and Delhi.
8. The Periyar (Mullaperiyar Dam), dispute between Kerala and Tamil Nadu.

Many of these water disputes have been settled on the basis of equitable apportionment which is the universally accepted principle. However, there are some interstate water disputes whose final solution, acceptable to all parties, has not yet been worked out.

Kaveri Water Dispute: The Kaveri rises from Brahmagiri (Kodagu District-Karnataka) at an elevation of 1341 m and flows 805 km, draining an area of about 87,900 sq km. Nearly 3% of its catchment area lies in Kerala, 41% in Karnataka and 55% in Tamil Nadu. It is joined by the Hemavati, Lokpavni, Shimsa, Herangi and Arkavati on the left bank, and the Lakshman-Tirath, Kabani, Noyil, Suvarnavati, Bhavanisagar and Amravati on the right bank. since the catchment area of the river records rainfall from the south-west and north-east monsoons, the river carries plenty of water almost throughout the year. About 44,000 sq km area of Kaveri Basin lies in Tamil Nadu and about 32,000 sq km in Karnataka. The Water Dispute Tribunal said that Tamil Nadu has a share of 419 TMC, Karnataka 270 TMC and Kerala 30 TMC. The Kaveri basin reservoirs are Kabini, Hemavati, Harangi and KRS. These reservoirs have a combined storage of 51 TMC ft (Fig. 3.11-A).

The Kaveri dispute between Karnataka and Tamil Nadu is more than a century old. An agreement between princely state of Mysuru (Mysore) and Madras Presidency in 1892 and 1924 over sharing water entitling a major share to Tamil Nadu. To solve this problem, the Kaveri Dispute Tribunal was formed in 1990. The Kaveri Water dispute was resolved by the Supreme Court in February, 2007, but the government and people of Karnataka were not satisfied with the decision of the court.

In September, 2016, Tamil Nadu had sought a direction to Karnataka to release 50.5 tmc feet of Kaveri water to save 40k acres of Samba (Rice) crop. It also sought court help to ensure Karnataka stuck to the water release in keeping with 2007 order of Kaveri Water Dispute Tribunal starting August 20 in irrigation year 2016–17. Moreover, Tamil Nadu says that Karnataka is diverting water meant for farmers, violating the trubunal's orders of 2007. The Karnataka Government says, it is facing a severe stress year and has a deficit of 80 tmc feet in its four reservoirs.

The Supreme Court directed Karnataka to realse 15,000 cusecs of Kaveri water per day to Tamil Nadu for the next 10 days.

The issue of water-sharing should be left in the hands of the technical experts.

Other major inter-state river disputes:

1. Ravi and Beas Rivers Water Dispute between Punjab, Haryana and Rajasthan.

2. Narmada River Water Dispute: Madhya Pradesh, Gujarat, Maharashtra, and Rajasthan.

3. Krishna River Water Dispute: Maharashtra, Andhra Pradesh, and Karnataka.

4. Godavari River Water Dispute: Maharashtra, Telengana, Andhra Pradesh, Karnataka and Madhya Pradesh.

5. Periyar River Water Dispute: Tamil Nadu and Kerala.

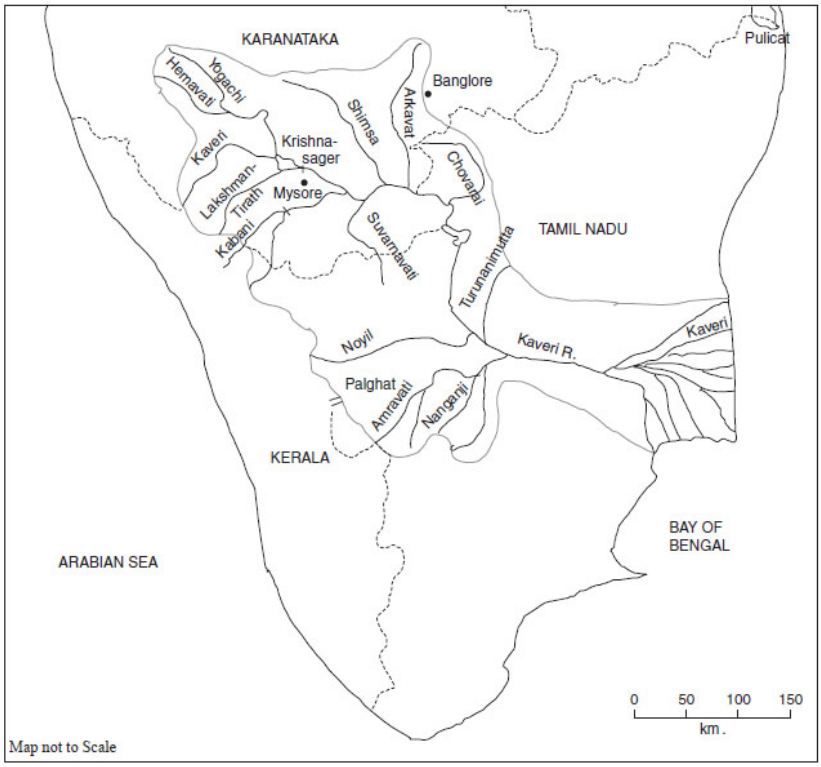


Fig. 3.11-A Kaveri Catchment Area

In developing countries like India, the interstate water dispute must be resolved quickly so that water resources could be utilised and harnessed properly for economic development. One of the measures could be to declare all the major rivers as national property, and national schemes under Central assistance should be launched for the development of total command area of the concerned states. Establishment of separate corporations on the pattern of the Damodar Valley Corporation may be immensely useful in this direction.

INTERNATIONAL AGREEMENTS FOR SURFACE WATER RESOURCES

Several of the Indian rivers flow through Tibet (China), Pakistan, Bangladesh, Bhutan, Nepal, and Myanmar. India has many international agreements with these neighbouring countries regarding the sharing of international river waters.

The Indus Water Treaty

This treaty was signed between India and Pakistan on 19 September, 1960, regarding the sharing of water of the Indus and its tributaries. It was reached through the arbitration of the International Bank for Reconstruction and Developemnt which has set up a permanent Indus Commission to look after the disagreement arising between India and Pakistan. Under this agreement, India has been given exclusive rights to use the waters of the three eastern rivers (Ravi, Beas, and Satluj), leaving out the remaining three (Chenab, Jhelum, and Indus) to Pakistan, which will also take into account the water needs of the Indian territory of Jammu and Kashmir. The agreement between India and Pakistan of the Salal, Baghliar, and Dulhasti projects (Chenab River) are an excellent example of co-operation between the two countries.

Similarly, a thirty-year agreement was also reached between India and Bangladesh on 12 December, 1996, for the sharing of Ganga waters. Under the agreement, India and Bangladesh would share alternately for 10 days each, 35,000 cusecs of water during the lean season (1 March to 10 May) to fulfill their water needs. A similar agreement has been concluded between India and Nepal for sharing

river waters. Moreover, India is helping Nepal in executing many river valley projects. The Mahakali Project is the outcome of one such Indo-Nepal joint ventures. Mutual agreement has also been reached between India and Bhutan for harnessing the waters of the international rivers affecting these countries. The king of Bhutan has agreed to join a sub-regional plan for sharing river waters and power with India and Bangladesh. Bhutan is willing to divert 12,000 cusecs of water from the Sankosh River to Tista and from Tista to Farakka Barrage to be shared by India and Bangladesh. The plan also envisages India purchasing 4000 MW of hydel power from Bhutan to strengthen its National Power Grid and to meet the power needs of the north-eastern region.

The Sankosh Hydel Project

The '*Sankosh Hydel Project*' is a joint venture near Kalikhola area of Bhutan, between India and Bhutan. It is estimated to cost `15,000 crores. The dam of height 250 m shall channelize the water through 141 km long canal across the West Bengal in the River Tista. Its spread area includes the Buxa Tiger Reserve and it is a danger to the wildlife of the region especially the elephants. The deforestation has caused denudation of about 700 ha of forest land.

Mullaperiyar Dam Water Dispute

The Periyar is only marginally an inter-state river, i.e., a very small portion of the catchment falls within Tamil Nadu. Tamil Nadu's right over Periyar water arises not from this fact, but from an agreement between the Madras Presidency and the Princely state of Travancore in 1886. The court in 2006 gave judgment on which the people of Kerala were not satisfied. The main issue was about the height of Mullaperiyar Dam. The issue in question was whether the dam was safe enough to be filled to the level that Tamil Nadu desires, or whether Kerala's fears are justified and the water level should be kept lower. On this Supreme Court gave a vedict in favour of Tamil Nadu. This might be a part of the larger question whether the safety of dam is a matter of judicial determination. The matter should have been decided on the basis of experts opinion. This river water dispute is still a serious issue between the two states.

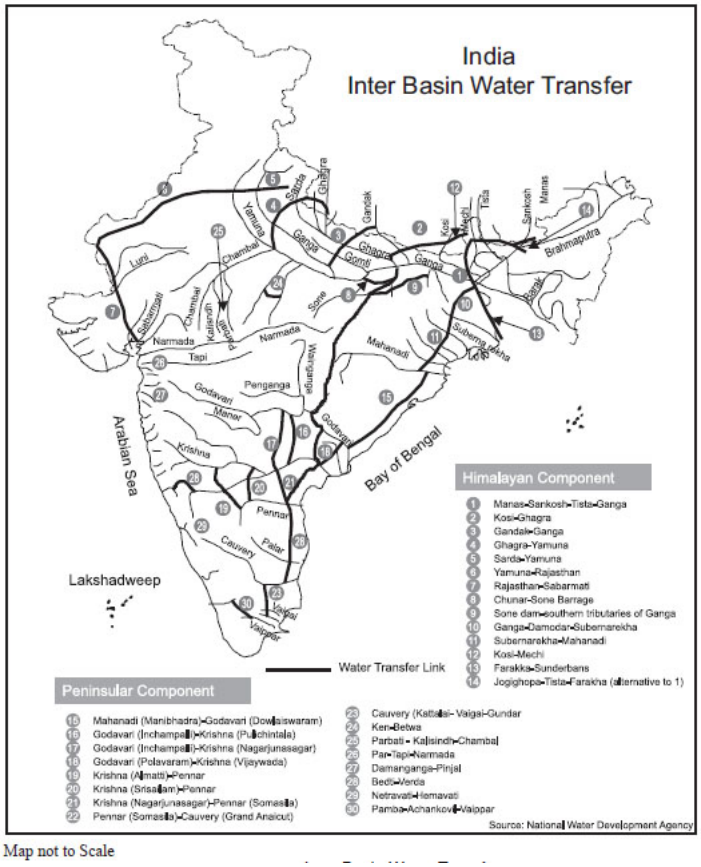


Fig. 3.12 Inter Basin Water Transfer

NATIONAL WATER GRID

The distribution of rainfall in India is highly unequal and seasonal. The rivers having their origin in the Himalayas are perennial, while those of Peninsular India are generally seasonal. During the months of general rains, much of the water is wasted during floods and flows down to the sea, but in the dry months of the year there is scarcity of water. Consequently, there are droughts and famines in one part of the country and floods in the other regions. The problems of droughts and floods can be minimised through the inter-basin linkages or through national water grid, under which, water from one river basin can be transferred to another basin for optimum and judicious utilisation. The government expects to spend a massive `5,60,000 crore on various river interlinking. But there are interstate water disputes and ecological problems which may come in the way of interlinking of rivers (The Hindu August 9, 2016 p. 13). The National River Linking Project (NRLP) aims at linking the Indian rivers through a network of reservoirs and canals. It aims at reducing persistent floods in the eastern India and the water shortages in the southern and the western India. It aims to link 37 rivers and will have 30 links and 3000 storage dams.

The NRLP has two components namely Himalayan component and Peninsular component. The major donor river basins are-The Brahmaputra, Mahanadi and Godavari. Few important components of the National Water Grid are as follows:

1. The Ganga-Kaveri Link Canal passing through the basins of Son, Narmada, Tapi, Godavari, Krishna, Penner, and Kaveri.
2. The Brahmaputra-Ganga Link Canal passing through Bangladesh.
3. The Narmada Canal passing through Gujarat and Rajasthan.
4. The Canal from Chambal to Central Rajasthan.

5. The Link Canals between the rivers of the Western Ghats towards the east.

1. The Ganga-Kaveri Link Canal

At the request of the Government of India, a UNO team prepared a project report on the Ganga-Kaveri Link Canal. The main objectives of the project are to safeguard against the recurring floods in the Ganga Basin and to assure more water to the comparatively less rainfall receiving areas of central India. Under this project, the Ganga is to be linked with Kaveri by a man-made canal; 2636 km or 1650 miles long. This canal was proposed not only to provide drinking water to its command area, but also to provide water for irrigation and sanitation. In addition to these, the canal is supposed to help in the generation of hydel-power, navigation, flood control, tourism promotion, and recreation. The Ganga-Kaveri Link Canal is thus, a multi-purpose project of immense size. If completed, the country will no longer have to depend so much on monsoon, the vagaries of which are well known.

The scheme proposes to draw 1700 cumecs (60,000 cusecs) of water from the Ganga, constructing a barrage near Patna, and lift its water by large pumps to a point near the boundary of the basins of Ganga and the Narmada from where it will be possible to distribute the water by gravity via dug-up canals or through existing rivers to the west or south. The flood waters of the Narmada (flowing into the Arabian Sea) and the Godavari (flowing into the Bay of Bengal) could also be used profitably by a separate water grid. Water for the inter-basin transfer would be derived from the Ganga only during the four months of the rainy season (July to October) when the flow exceeds on an average by 2850 cumecs (100,000 cusecs).

The length of the Ganga-Kaveri Link Canal will be between 2400–3200 km, depending upon the actual alignment finally chosen, with smaller secondary branches connecting areas chronically prone to droughts. It is also proposed to supply the Ganga water to Bihar, Uttar Pradesh, Jharkhand, Chhattisgarh, Madhya Pradesh, and Rajasthan by pumping additional water during the lean water season. Similarly, 1410 cumecs (50,000 cusecs) of water pumped for 150 days or more, depending upon the surplus waters in the Ganga, would be diverted during the high flow period only, and would be transferred outside the basin to meet partially the water demand of the chronically drought prone areas of Rajasthan, Gujarat, Madhya Pradesh, Chhattisgarh, Maharashtra, Andhra Pradesh, Karnataka, and Tamil Nadu.

A site near Patna having an elevation of about 46 m above sea level will be the starting point which would collect the surplus water from the Ganga. From here, the water would be pumped into a series of reservoirs between the watersheds of the Narmada and Son, involving a pumping lift of 335 to 400 m. From this elevation (*Bagri Reservoir on the Narmada 423 m above sea level*), a lined aquaduct will convey the water to the south utilising the natural water resources of the Wainganga, Pranhita, and Godavari, and crossing the Krishna and the Penner to Kaveri, upstream of the Upper Anicut.

Storage would be provided enroute, especially on the ridge regions to conserve the water for the dry season. These storages would be located inside valleys, which do not have sufficient catchment area of their own to provide adequate run-off which would be utilised during the dry season. On the way, water would also be released into the basins of the Narmada and the Tapi rivers which flows westward into the Arabian Sea, and the Godavari, the Krishna, the Penner, the Kaveri which flow eastward into the Bay of Bengal. From the selected points on the Son River itself, water pumped separately for the basin will be diverted to the drought prone areas in the Ganga basin.

The project involves huge expenditure, massive survey operations and strong administrative decisions. Since it will take a decade or more to complete the project which has several administrative and socio-ecological constraints, the government has not yet taken any decision to execute the project.

2. The Brahmaputra-Ganga Link Canal

The Brahmaputra is a mighty river which carries a discharge of 3500 to 5000 cumecs even during the dry season. Much of this water is beyond the requirement limit of the basin and is wasted. On the other hand, there is a scarcity of water in the lower Ganga basin, especially during the dry months. Hence, the diversion of excess water from the Brahmaputra to the Ganga may meet this water deficit, which shall help in the economic development of the region.

The '*Brahmaputra-Ganga Link Canal' Project* involves the construction of a diversion barrage at Dhubri (Lower Assam), and a 320 km long feeder canal linking the Dhubri Barrage to the Farakka Barrage. A portion of this feeder canal will lie in Bangladesh for which an interenational agreement between India and Bangladesh has to be signed. This canal will provide irrigatioin water to Bangladesh also. The canal may augment the flow of water in the Padma River (Ganga in Bangladesh) during the lean months of the year. Besides, the link canal would provide cheap inland navigation facility to both the countries. Due to lack of concurrence from Bangladesh and involvement of huge financial expenditure, the scheme has not yet been started. The *Damanganga and Pinjal rivers* involving the states of Gujarat and Maharashtra is not easy to implement because of water dispute.

3. The Narmada Link Canal to Gujarat and Rajasthan

Under the Sardar Sarovar Project, there is a proposal to build a terminal storage dam across the Narmada River near Navagam, and a diversion canal linking the place to regions of Kutch (Gujarat) and western Rajasthan. This link canal will be of immense help to the drought prone areas of Gujarat and western Rajasthan.

4. The Chambal Link Canal

A canal of about 500 km connecting the Chambal River with the Indira Gandhi Canal has also been proposed. The canal would provide water to the central parts of Rajasthan. It will involve a lift of 200 to 250 m. Moreover, Chambal-Parbati, Betwa-Ken, Ghagra-Yamuna, Gandak-Ganga, Mechi-Kosi-Ganga, Par-Tapi-Narmada, Mahanadi-Godavari are also under consideration
(Fig. 3.11).

5. Links between the Rivers of the Western Ghats to the East

The rivers of the Western Ghats carry enormous quantity of water during the rainy season. Due to steep gradient and the narrow coastal plains much of the water goes to the Arabian Sea as waste. This water may be diverted to the rain-shadow areas of the Western Ghats through the diversion canals where it can be utilised for irrigation. The Periyar Diversion Scheme, constructed several years ago, is such a type of model scheme where the surplus water of the west-flowing Periyar river has been collected in a barrage and diverted through a tunnel across the Sayadri, so as to meet the water needs of the drought prone areas of Tamil Nadu in the east. Other projects of Western Ghats include: Damanganga with Pinjal, Bedti with Varda, Netravati with Hemavati, and Pamba with Vaippar
(Fig. 3.11). Similar schemes may also be executed in case of other rivers of the Western Ghats.

GROUND WATER RESOURCES OF INDIA

India is rich in underground water. Its spatial distribution, however, is most uneven. For example, the average annual rainfall in India is about 110 cm. While Mawsynram and Cherrapunji receive more than 1000 cm rainfall annually, the average annual rainfall in Ganganagar is only about 20 cm.

The underground water resource is a function of geological structure, topography, slope, precipitation, runoff, soils and hydrological conditions of a region. In the opinion of Prof. R.L. Singh (1971), India may be divided into eight ground water provinces (Fig. 3.13). A brief account of these provinces has been given in the following section:

1. The Precambrian Crystalline Province

The Precambrian province stretches over about 50% of the total area of the country, especially over Peninsular India. It sprawls over Tamil Nadu, Karnataka, Andhra Pradesh, Maharashtra, Chhattisgarh, Chotanagpur Plateau, and Aravallis. This province is generally poor in underground water resource.

2. Precambrian Sedimentary Rocks Province

The Archaean, Dharwar, Cuddappah, and Vindhyan formations are included in this province. These regions are also not rich in ground-water resource.

3. The Gondwana Sedimentary Province

Stretching over the Godavari basin, this province is fairly rich in underground water resource.

4. The Deccan Trap Province

Stretching over the states of Mahrashtra, parts of Gujarat, Madhya Pradesh, Chhattisgarh, Andhra Pradesh, and Karnataka, this province is also deficient in underground water resource. In fact, the Deccan Trap is a deep lava formation of the Cretaceous Period in which the seepage of water is possible only through the cracks and fissures. The limited under-ground water is however, of immense help for the irrigation of cereal crops, orchards, sugarcane, vegetables, flowers, and grasses.

5. The Cenozoic Sedimentary Province

This province includes the coastal areas of Andhra Pradesh, Tamil Nadu, Kerala, and Gujarat. The sandstones of the Tertiary Period in these regions are rich in the underground water resource.

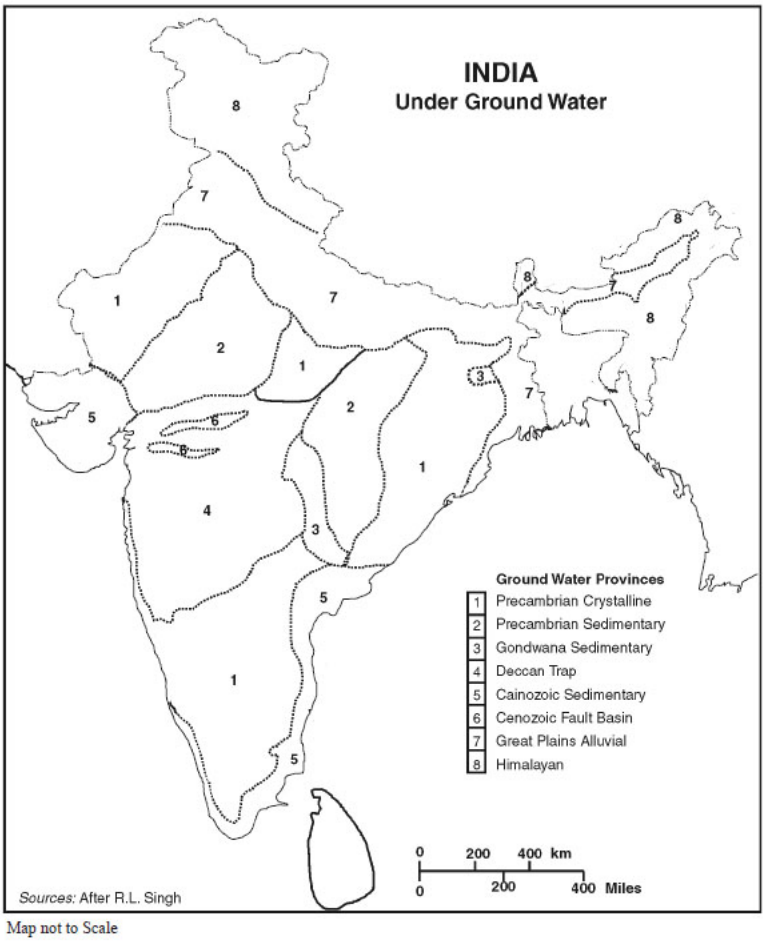


Fig. 3.13 Ground Water Provinces (Modified after R.L. Singh)

6. The Cenozoic Fault Basin

The rift valleys of the Narmada and Tapi rivers are included in this province, which are quite rich in underground water.

7. The Ganga Brahmaputra Alluvial Province

Sprawling over Punjab, Haryana, Uttarakhand, Uttar Pradesh, Bihar, West Bengal, Assam and parts of Rajasthan, this province has the richest underground water resource. About 44% of the Indian underground water resource is found in this province. Thousands of tube wells have been installed in the Northern Plains of India to utilise this resource for irrigation, domestic, and industrial purposes.

8. The Himalayan Province

This is complex structural, physiographic and geographical region of India. In general, it is deficient in underground water except the intermontane valleys, like the Dun, Kashmir, Kangra, Kullu, and Manali valleys. There are numerous springs in this region.

THE NATIONAL WATER POLICY 2012

The government of India designed the National Water Policy for the optimum and judicious utilisation, management, and conservation of water resources in the country. The first policy was adopted in 1987. The Ministry of Water Resources gave the National Water Policy (NWP) on
June 7, 2012. The policy intends to address the issues of scarcity of water, inequalities in its distribution, lack of perspective in planning, management and utilization of water resources. The main features of the policy are given in the following paragraphs:

According to one estimate, out of the total precipitation of about 400 million hectare metres in the country, only 178 million hectare metres is the available surface water. Out of this, about 50% can be put to beneficial use because of topographical and other features. Moreover, there is a ground water potential of about 42 million hectare metres. The distribution of water resource as stated above is highly uneven in terms of both space and time. Precipitation over the greater part of the country is seasonal, mainly concentrated to three or four months of the rainy season. It varies from about 20 cm in Ganganagar (western Rajasthan) to over 1000 cm at Mawsynram (Meghalaya). Hence, there is a water surplus in one part of the country and water deficit in another part. The National Water Policy proposes to evolve a national water grid for the proper management and utilisation of the water of the country. The main objective of the policy is to provide water from the surplus areas to the deficit areas. The national water policy proposes to initiate investigation for a national plan for inter-basin transfer for water from the water surplus areas to the water deficit areas. Its aims are enumerated below:

1. Water is a primary natural resource, a basic human need and a precious national asset. Hence, the planning and development of water resources need to be governed by national perspective.
2. The National Water Policy aims at planning the surface water resources of the country on the basis of hydrological units such as drainage basin as a whole for a sub-basin. This will ensure an optimum utilisation of the water resources of the country.
3. India’s average annual precipitation is about 4000 BCM of which only 1,123 BCM is usable. There is an increasing demand of water for irrigation, industries, and domestic uses. The national water policy aims at taking suitable measures for proper management and conservation of water resources of the country, minimising losses at the storage and diversion points in the distribution system.
4. The policy aims to reduce water pollution and to improve the quality of water of the rivers. There is an emphasis on the recycling of water also.
5. It proposes an adequate maintenance of canals and distribution systems inter-basin transfers by making adequate financial allocations for maintenance of canals and their distributaries.
6. It suggests a detailed survey for the preparation of new projects on priority basis for tribal areas, drought prone areas as well as for the economically weaker sections of society.
7. The policy lays emphasis on the supply of good quality drinking water to rural areas besides meeting the needs of the urban areas.
8. It proposes a proper survey of the underground water resources and how to utilise them judiciously. The conservation and recharging of ground water is also a priority of the national water policy. This shall ensure increase of soil moisture, reduction of sediment field, increase overall land and water productivity.
9. The policy also focuses attention on devising suitable strategies for such problems as waterlogging, salinisation, degradation of fertile arable lands, deteriorating water quality, over exploitation of ground water resources, and uprooting of families due to development projects.
10. The policy also aims at reducing the runoff, soil erosion and silting of river beds.

MAIN WATERFALLS OF INDIA

Agaya Gangai Waterfall
Located in the Kolli Hills (Eastern Ghats, Tamil Nadu). It provides serene solitude, and is a great attraction in the Tamil Nadu tourism. The approach is, however, tortuous.

Ayyanar Waterfall
It is situated in the Virudhunagar district of Tamil Nadu in the Western Ghats. It gets water mainly during the retreating monsoon rain. The water from the falls is mainly used for drinking purposes by the people of Rajapalayam. It is a famous tourist spot for the people of neighbouring districts.

Barakana Falls
Situated in the Shimoga district of Karnataka, Barakana is one of the highest water falls of the country. Currently, Barkamna Falls are the prime source of one of the hydro-electric projects of Karnataka.

Dudhsagar (Goa)
Dudhsagar (The Sea of Milk) is a tiered water fall located in the upper reaches of Mandovi River in the state of Goa. It is a great attraction for the national and international tourists.

Duduma Waterfall (158 m)
Situated about 92 km from Koraput, it lies on the Machhkund River in Odisha. A large hydro-electric project has been constructed at this waterfall. Machhkund is an important place for pilgrimage.

Gokak Falls (53 m)
They are located in the upper reaches of the Ghataprabha (a tributary of the Krishna) in Belgaum District of Karnataka. The waterfall is about 6 km. away from Gokak, a nearby town. It resembles Niagra Falls. It is a great attraction for the domestic and international tourists.

Jog Falls (253 m)
Located on the Sharavathi River in the Shimoga District of Karnataka, they are the highest untiered waterfalls in India. Jog falls is one of the major attractions in Karnataka tourism. It is also called by alternative names of Gerusoppe Falls, Gersoppa Falls, and Jogada Gundi.

Kiliyur Falls
Kiliyur are the waterfalls in the Servary Hill of Eastern Ghats (Tamil Nadu). Having an elevation of about 100 m, it is a great attraction in the tourism of Tamil Nadu.

Kurtalam Falls
Situated in the Tirunelveli district, Tamil Nadu, the Kurtalam waterfalls is a great attraction for the domestic and international tourists.

Lodh Waterfalls (also known as Buddha Ghagh Falls)
The Lodh Waterfalls are located on the Budh River about 40 km from Ranchi (Jharkhand). The falls, named Gautam Budha, also have a Buddha Temple there.

Shivasamudram Waterfall
Shivasamudram Falls formerly known as the Cauvery Falls are the second highest waterfall in India. It is located 80 km from Mysore and 120 km from Bangalore. It is a major attraction in Karnataka tourism.

Siruvani Waterfall
It is situated at the Siruvani River at a distance of about 40 km from Coimbatore in the Western Ghats. It is one of the main water sources of Coimbatore city. The panoramic view of the dam and the falls is enchanting.

Thalaiyar Waterfall (Rattail)
Also known as Rat-tail, it is located near Kodaikanal in Tamil Nadu. With an elevation of 297 m, it is the highest waterfall in Tamil Nadu State. The waterfall is, however, not connected by road and the

approach is tiresome.

Vattaparai Waterfall

Located at the Pazhayar River in the Kanyakumari district of Tamil Nadu, it is a great attraction for tourists. The surrounding area is proposed to be developed into a Wildlife Sanctuary.

Vazhachal Waterfall

Located in the Thrissur district of Kerala, these are one of the best waterfalls in India. They are a great attraction for the domestic and international tourists.

Some of the other important waterfalls in India

Table 3.9

Some Important Waterfalls of India

State	Name of the Waterfalls
Chattisgarh	Teerathgarh
Madhya Pradesh	Dhunwadhar (River Narmada, near Jabalpur)
Himachal Pradesh	Bundla and Palani Falls
Jharkhand	Ghagri, Hundru, Johna Falls
Karnataka	Abbey Falls, Arisina Gundi Falls, Hebbe Falls, Irupu Falls, Kalhatti Falls, Keppa Falls, Koosalli Falls, Kudumari Falls, Kunchikal Falls, Magod Falls, Mekedaatu Falls, Muthyala Falls, Sathodi Falls, Simsa Falls, Chunchi Falls, Unchalli Falls
Kerala	Athirappilly Falls, Meenmutty Falls, Palaruvi Falls, Soochipara Falls, Thusharagiri Falls
Maharashtra	Chchai Falls, Gatha Falls, Keoti Falls, Kune Falls, Marleshwar Falls, Pandavgat Falls
Meghalaya	Beodon Falls, Bishop Falls, Elephant Falls (Shillong Peak), Kynrem Falls, Langshiang Falls, Margaret Falls, Nohkalikai Falls, Spread Eagle Falls, Sweet Falls
Mizoram	Vantawng Falls
Odisha	Joranda Falls, Khandadhar Falls near Buguda
Tamil Nadu	Aintharvi (five falls), Beman Falls (Palani Hills), Bear Shola Falls (Kotagiri), Chitraruvi Falls (Small falls), Courtallam Falls, Elk Falls (Palani Hills National Park), Vaideki Falls, Thakkam Thootam Falls (near Palani), Thenaruvi Falls (Honey Falls), Pazhaya Courtallam Falls (Old Falls), Pykara Falls (Nilgiri), Neptune Falls (in Palani National Park), Puli Aruvi (Tiger Falls), Puthu Aruvi (New Falls-Milk falls), Suruli Falls (Near Madurai)

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4

Climate of India

INTRODUCTION

The climate of India is essentially sub-tropical monsoonic. The word “monsoon” has been derived from the Arabic word ***“mausim”*** which means “season”. Originally, the word “monsoon” was used by Arab navigators several centuries ago, to describe a system of seasonal reversal of winds along the shores of the Indian Ocean, especially over the Arabian Sea, in which the winds blow from the south-west to north-east during the summer season and from the north-east to south-west during the winter season. In other words, monsoons are periodic (seasonal) winds in which there is a complete reversal of the wind direction after every six months.

In the opinion of Chang-Chia-Cheng, Monsoon Meteorology, Oxford University Press, Oxford: “Monsoon is a flow pattern of the general atmospheric circulation over a wide geographical area, in which there are clearly dominant winds in one direction in every part of the region concerned, but in which this prevailing direction of wind is reversed or almost reversed from winter to summer, and from summer to winter.”

Monsoons are especially prominent within the tropics on the eastern sides of the great landmass, but in Asia, it also occurs outside the tropics in China, Korea, and Japan, and may be observed up to 40° north. South-East Asia, especially the subcontinent of India, however, is a typical example of a monsoon region. It is only around the Indian Ocean that monsoonal circulation, in true sense of the term, is observed. There is complete reversal of wind in India, and the monsoons “appear as truely interruptive and reverses normal global atmospheric circulation.” Other areas which experience similar but less pronounced seasonal changes of wind direction include south-eastern USA, the Caribbean Islands, Madagascar, East Africa, the Guinea coast of West Africa, South-east Asia, Philippines, south-eastern China, South Korea, Japan, and North Australia **(Fig. 4.1).**

INDIAN MONSOON

Monsoons are a complex meteorological phenomenon. Experts of meteorology have developed a number of concepts about the origin of monsoons. Some of the important concepts about the origin of monsoons have been given as under.

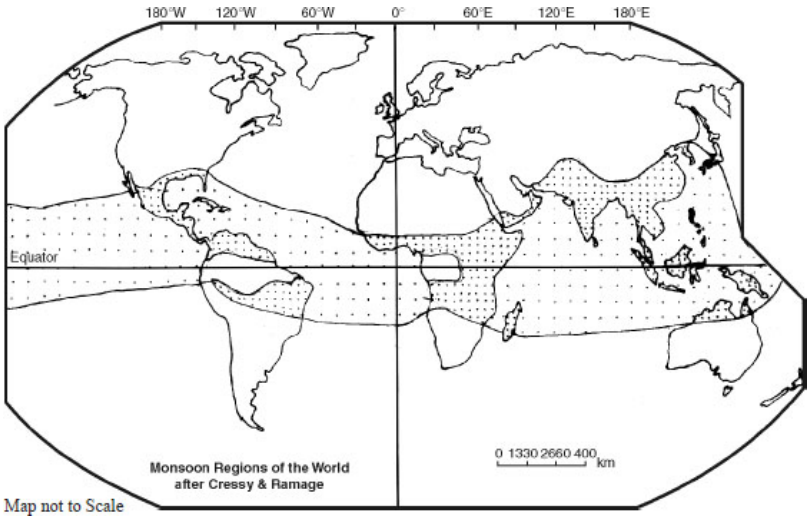


Fig. 4.1 Monsoon Regions of the World

The Thermal Concept of Halley

Halley, a noted astronomer in 1686 hypothesised that the primary cause of the annual cycle of the Indian monsoon circulation was the differential heating effects of the land and the sea. According to this concept *monsoons are the extended land breeze and sea breeze on a large scale, produced by the differential heating of continents and ocean basins* **(Fig. 4.2)**. During the summer season in the Northern Hemisphere, when the sun's rays are vertical over the Tropic of Cancer, the huge landmass of Asia heats quickly and develops a strong low pressure centre near Lake Baikal (Siberia) and Peshawar (Pakistan). This thermal low extends up to 700 mb. Moreover, the pole-ward shift of the Inter-Tropical Convergence Zone (ITCZ) to a position over southern Asia reinforces the thermally induced low pressure centre. In comparison to this, the pressure over the adjacent water of the Indian and the Pacific Oceans is relatively high.

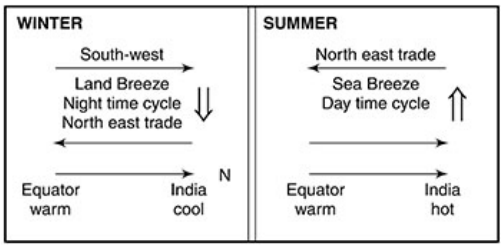


Fig. 4.2 Land Breeze and Sea Breeze: Monsoon Cycle

Under these conditions, a sea-to-land pressure gradient develops. Consequently, the surface air flow is from the high pressure over the oceans towards the low pressure areas over the heated landmass. Under the extreme low pressure condition on land, the wind from the southern part of the Indian Ocean (south of Equator) is attracted towards the subcontinent of India. The air coming from oceans towards land is warm and moist. When land barriers like mountain ranges and plateaus come in the way of the moisture-laden winds, they ascend and result into saturation, condensation, and precipitation **(Fig.4.3).**

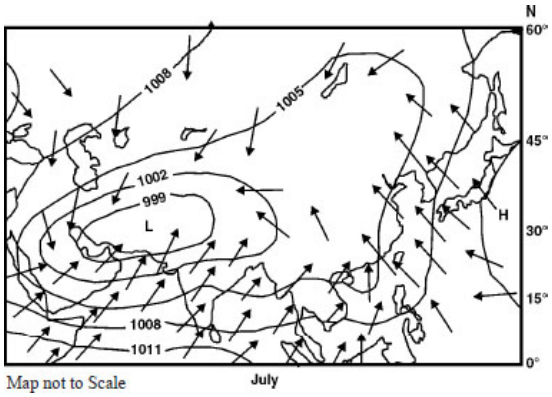


Fig. 4.3 The Summer Monsoon

Contrary to this, in the Northern Hemisphere during winter season, there develops high pressure areas near Baikal Lake (Siberia), and Peshawar (Pakistan). As compared to these high pressures, the Indian Ocean and the Pacific Ocean (south of Japan) remain relatively warm, having low pressure areas. Consequently, there is an outflow of air from the high pressure of the land to the low pressure areas of the oceans. The air blowing from high pressure areas of land towards the sea is cold and dry. This cold and dry air is incapable of giving precipitation unless it comes into contact with some water body (ocean/sea) **(Fig. 4.4).**

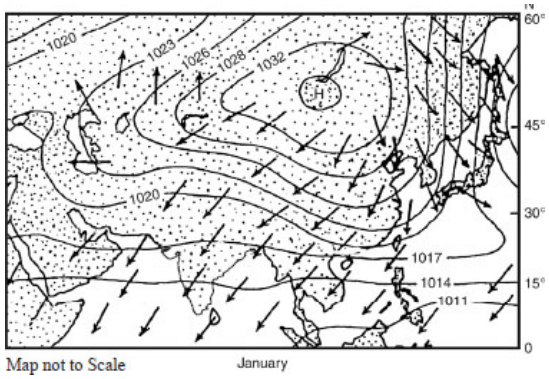


Fig. 4.4 The Winter Monsoon

The thermal concept about the origin of monsoon has, however, not been accepted universally as it fails to explain the intricacies of monsoon. Besides differential heating, the origin and development of monsoon are also influenced by the shape of the continents, orography, and the conditions of air circulation in the upper troposphere. The Halley's concept has been criticised on more than one count as follows:

Criticism:

1. The low pressure areas that develop over the continents during the summer season in the Northern Hemisphere are not stationary. These low pressure areas change their position (location) suddenly. This sudden change in the low pressure areas are not exclusively related to low thermal conditions. The low pressure areas stabilises in June in the north-eastern parts of the subcontinent. In fact, they represent the cyclonic lows associated with the dynamic factors, and therefore, these low pressure areas cannot be termed as only thermally induced.

2. Had the monsoon been thermally induced, there would be anti-monsoon circulation in the upper air of the troposphere, which is lacking.
3. Although high temperature and the consequent low pressure takes the north-west in its grip from the middle of April, no rain starts in northern India till the middle of June.
4. The modern researches in meteorology have shown that the monsoon rainfall is not wholly orographic. They are an amalgamation of convectional, orographic and cyclonic rainfall.
5. Instead of two broad seasons (winter and summer) the monsoon climate has more seasons (four in India), due to the highly variable characteristics of temperature and precipitation.
6. Halley did not take into consideration the Coriolis effect of rotation of the earth on its axis. On a rotating earth, the wind has a tendency to move towards its right in the Northern Hemisphere and towards its left in the Southern Hemisphere.
7. The role of latent heat passing into the atmosphere through water vapour was also not considered by Halley; water vapour also plays an important role in the origin and development of monsoons.

The Dynamic Concept by Flohn

The dynamic concept about the origin of monsoons was put forward by Flohn in 1951. According to this concept, monsoon is the result of seasonal migration of planetary winds and pressure belts**.** The Inter-Tropical Convergence Zone (ITCZ) is formed due to the convergence of north-east and south-east trade winds near the equator. The northern and the southern boundaries of the ITCZ are called NITC (Northern Inter-Tropical Convergence) and SITC (Southern Inter-Tropical Convergence), respectively. There is a belt of doldrums within the Inter-Tropical Convergence, characterised by equatorial westerlies. At the time of the summer solstice (21st June), when the sun's rays are vertical over the Tropic of Cancer, the NITC is extended up to 30° N latitude, covering south and south-east Asia. Thus, equatorial westerlies are established over these areas. The equatorial westerlies become south-west or summer monsoons. On a rotating earth, the trade winds of the Southern Hemisphere after crossing the equator turn towards their right (Coriolis effect) **(Fig. 4.5)**. The NITC is associated with numerous atmospheric storms (cyclones) which yield heavy rainfall during wet monsoon months (July to September). Similarly, the north-east or winter monsoon does not originate only due to low pressure in the Southern Hemisphere during winter solstice (when the Sun's rays are vertical over the Tropic of Capricorn). In fact, the north-east monsoons are north-east trade winds which are re-established over south-east Asia due to southward shifting of pressure and wind belts. It is obvious that due to southward movement of the Sun at the time of winter solstice, the NITC is withdrawn from over south and south-east Asia, and north-east trade winds occupy their normal position. These north-east trades, thus, become winter monsoons. The north-east monsoons having their origin on land are generally dry and devoid of rains **(Fig. 4.6)**.

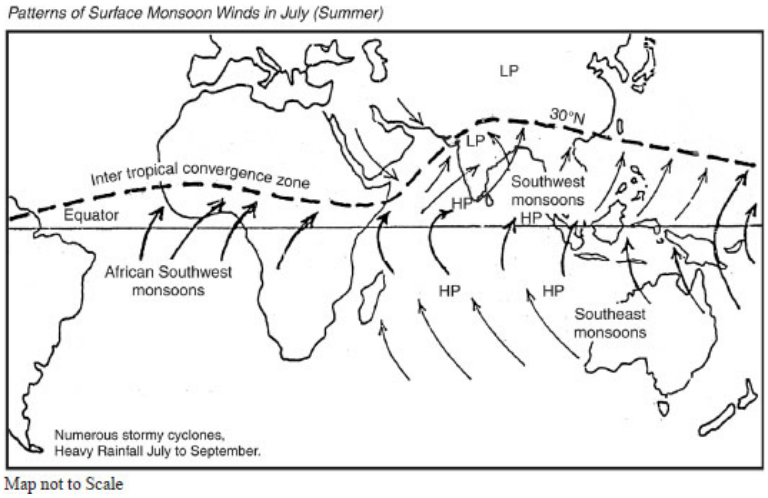


Fig. 4.5 South–West Monsoon (Patterns of Surface Monsoon Winds in July (Summer))

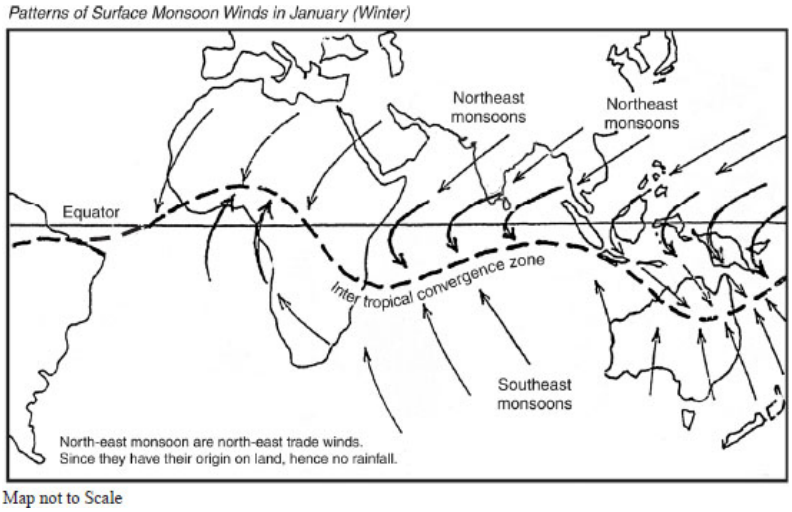


Fig. 4.6 North–East Monsoon (Patterns of Surface Monsoon Winds in January (Winter))

In brief, according to Flohn, the existence of monsoons in Asia, especially in the subcontinent of India, is not due to temperature contrasts between land and sea, but mainly due to the annual migration of thermally produced planetary winds and pressure belts. Despite the relative shifting of Inter-Tropical Convergence (thermal equator) and pressure belts, Flohn seems to have ignored the upper atmospheric circulation (jet streams) and the southern oscillation, which make the Asiatic monsoon a fairly complex system. He also could not explain the causes of early arrival of Indian monsoons in the states of north-east India. The dynamic concept, therefore, was also not taken as the sole explanation of the origin of monsoons.

Recent Concepts about the Origin of Indian Monsoons

During the last five decades, the upper atmospheric circulation has been studied significantly, as a result of which meteorologists have raised certain doubts about the validity of the classical concept of the origin of Indian monsoon. It is now believed that the differential heating of land and sea cannot produce the monsoon circulation. More recent theories have laid greater emphasis on the circulation in atmosphere over the subcontinent and the adjoining areas. Apart

from the upper atmospheric circulation, recent concepts rely heavily on the role of the Tibetan Plateau, jet streams, and the El-Nino (Southern Oscillation).

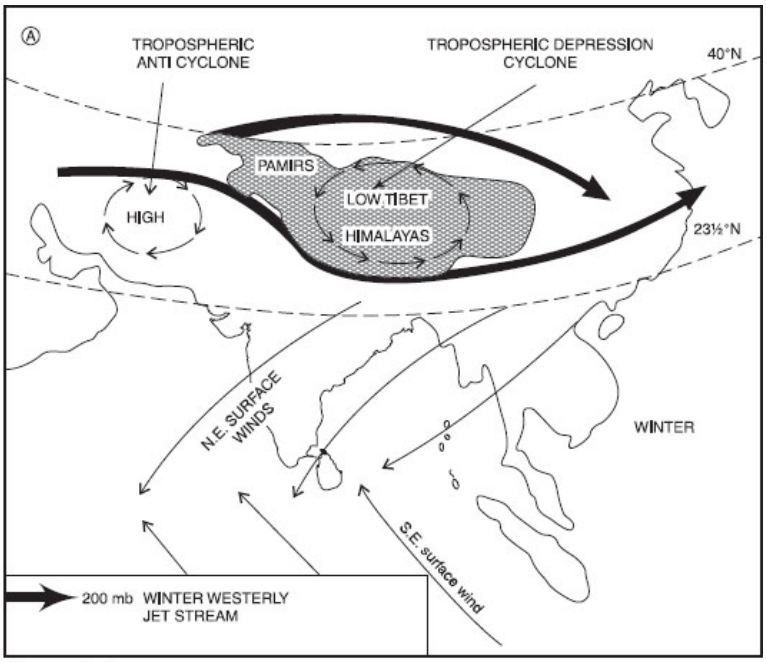
The data gathered by meteorologists after the Second World War have revealed that the origin and mechanism of monsoons are related to the following phenomena:

(i) The role of the Himalayas and Tibetan Plateau as a physical barrier and a source of high-level heat.
(ii) The circulation of upper air jet streams in the troposphere.
(iii) The existence of upper air circum-polar whirl over north and south poles in the troposphere.
(iv) The differential heating and cooling of the huge landmass of Asia and the Indian and the Pacific Oceans.
(v) The occurrence of El-Nino in the South Pacific and Indian Oceans.

INDIAN MONSOONS AND THE TIBET PLATEAU

In 1973, the Monsoon Expedition (Monex) was organised as a joint venture of the Soviet Union and India. Under this expedition, four Russian and two Indian ships, equipped with modern scientific instruments were pressed into service in the Indian Ocean and the Arabian Sea to investigate the phenomenon of Indian monsoons. The period of investigation extended from the month of May to July, 1973. On the basis of the data obtained, the Soviet meteorologists arrived at the conclusion that the Tibet Plateau plays a crucial role in initiating the monsoon circulation over the Indian subcontinent. In 1958, Dr. P. Koteswaram, the Director General of the Indian Meteorological Observatories, while participating in the international symposium on “The Monsoons of the World” had expressed views that the summer-time heating of the Plateau of Tibet was the most important factor in the causation and maintenance of monsoonal circulation. The Indian as well as the Soviet scientists were unanimous in their views on this point.

The Plateau of Tibet is 600 km wide in the west and 1000 km in the east. Its length from west to east is about 2000 km. The average height of the plateau is about 4000 m. Thus, it is an enormous block of high ground acting as a formidable barrier. Due to its enormous height it receives 2°C to 3°C more insolation than the neighbouring areas. It is also one of the most important geographical controls on the general atmospheric circulation in the region. The Plateau of Tibet affects the atmosphere in two ways, acting separately or in combination: (i) as a physical barrier, and (ii) as a high-level heat source. The meridional cross-section of the Indian summer monsoon and its relationship with the Tibetan-Himalayan massif has been shown in **(Fig. 4.7 and Fig. 4.8).**



Map not to Scale

Fig. 4.7 Cross Section of the Indian Summer Monsoon

According to Maung Tun Yin, the Tibetan Plateau acts as a physical barrier. In the beginning of June, the subtropical jet stream disappears completely over northern India **(Fig. 4.8).** At this time, the jet stream shifts to the north of the Himalayas and Tibet, and takes up a position at about 40° N. Yin considers that there is a correspondence between the shifting of the jet and the slowing down of the westerlies over the whole of Eurasia. In fact, the Plateau of Tibet becomes very cold in winter, and proves to be the most important factor in causing the advance of the jet far to the south in the middle of October. Thus, he opines that the abrupt onset of summer monsoon at the beginning of June is prompted by the hydro-dynamic effect of the Himalayas and not by the thermally induced low pressure centre over northwest India. In the middle of October, the plateau proves to be the most important factor in causing the advance of the jet south of the Himalaya or bifurcate it into two parts **(Fig. 4.7).**

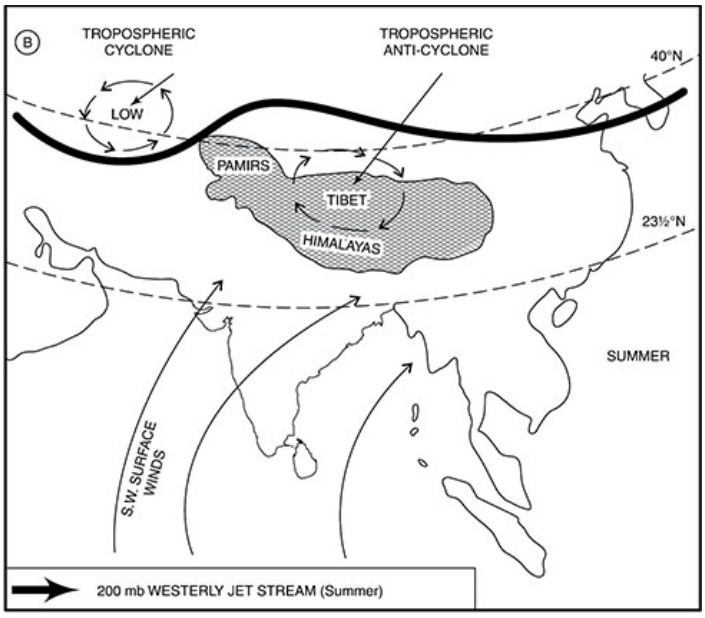


Fig. 4.8 Monsoon and the Tibetan-Himalayan Highlands

The summer-time heating of the Tibetan Plateau makes it a high-level heat source. This “Heat Engine” produces a thermal anticyclone over this region. A warm core anticyclone (high pressure) is formed over this plateau during the summer monsoon period. The formation of this anticyclone takes place in the middle part of the troposphere at 500 mb level. It is the result of a process called anti-cyclogenesis. The anticyclone at 500 mb at Tibet weakens the western sub-tropical jet-stream south of Himalayas, but produces tropical easterly jet on the southern side of the anticyclone. This tropical easterly jet stream first develops in longitudes east of India and then extends westwards across India and the Arabian Sea to eastern Africa. Blowing along Kolkata-Bangalore axis the air under the easterly jet descends over the Indian ocean near Masscarness and Zanzibar Islands of Tanzania. and intensifies its high pressure cell, so as to finally move as south-west monsoon.

The data collected under Monex support that higher the inensity of the tropical easterly jet greater would be potency of the high pressure cell over the Indian Ocean and stronger would be the impact of south-west monsoon. The path of the easterly jet stream from the Tibetan Plateau upto Masscarness Island and Tanzania in the South Indian Ocean has been shown in **(Fig. 4.9)**.

JET STREAM AND INDIAN MONSOON

Jet stream is the most prominent movement in upper level westerly wind flows; irregular, concentrated, meandering bands of geo-strophic wind, travelling at speeds of 300 to 400 kmph. The jet streams are high altitude (9000–12000 m) westerly winds between middle latitudes (summer 35°N–45°N; winter 20°N–35°N) in the Northern Hemisphere. Recent researches have shown that these winds exert considerable impact on surface weather conditions.

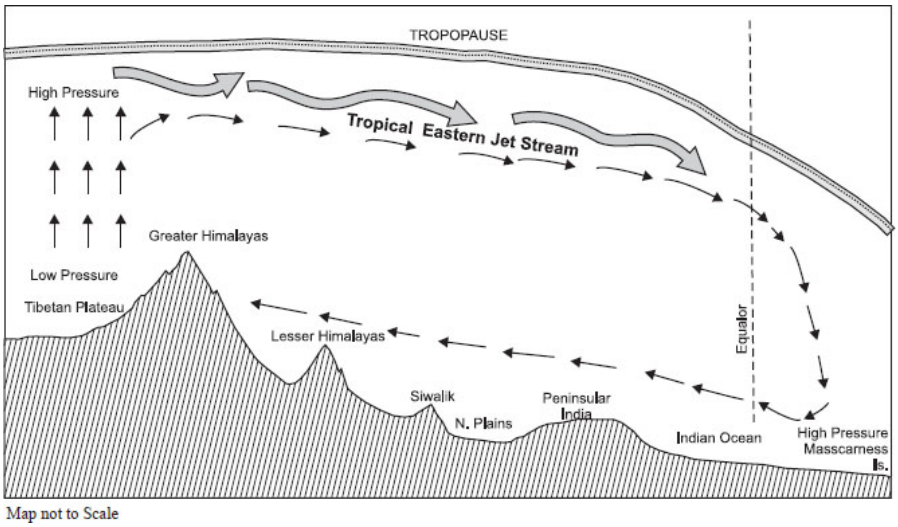


Fig. 4.9 Tropical Easterly Jet Stream (Meridional Cross-section of Indian Monsoon)

The influence of jet streams on the origin and development of Indian monsoons may be appreciated from the following description of weather phenomena during the summer and the winter seasons. The upper air westerly jet streams are extended upto 20°N–35°N (Nagpur, Raipur latitudes) due to equator-ward shift of upper air north polar whirl during northern winter (October to February). In the winter season, the upper air westerly jet streams are bifurcated into two branches due to physical obstruction of the Himalaya and Tibetan Plateau. One branch is located to the south of the Himalayas, while the second branch is positioned to the north of the Tibetan Plateau **(Fig. 4.7)**. The upper air high pressure and anticyclonic (with clockwise air circulation) conditions are developed in the troposphere over Afghanistan and Pakistan. Consequently, the winds tend to descend over the north-western parts of India, resulting into the development of atmospheric stability and dry conditions. Besides, the upper air westerly jet streams also cause periodic changes in general weather conditions because they lie over the temperate low pressure (cyclonic wave) which moves from west to east under the influence of upper air westerly jet streams across the Mediterranean Sea and reach Afghanistan, Pakistan and north-west India. These storms are not frontal cyclones, but waves which move at the height of 2000 m from the mean sea level, while at the surface they are north-east trade winds.

The tropical easterly jet stream as stated above, extends far to the north of Tibet and the air flow is roughly along the Kolkata-Bangalore axis. These upper air easterlies descend into the permanent high pressure area formed over the southern Indian Ocean near the Masscarness Island (Tanzania). This naturally intensifies the “High” already present there **(Fig. 4.9)**. It is from this high pressure cell that the onshore winds start blowing towards the thermally induced low pressure area, developed in the northern part of the Indian subcontinent. After crossing the equator such winds become south-westerly and are known as the south-westerly summer monsoons. These surface winds have vast potentiality for south-westerly summer monsoon and precipitation. It is, therefore, clear that the strength of the easterly jet stream is directly related to the intensification of permanent “High” formed over the southern Indian Ocean. Since this high pressure makes the pressure gradient steeper, so it is the main causative factor for determining the vigour of the summer monsoon.

During the summer season in the Northern Hemisphere, low pressure areas develop at the ground surface near Peshawar (Pakistan) and north-west India due to intense heating of ground surface during April, May, and June. But as long as the position of the upper air jet stream is

maintained above the surface low pressure (to the south of Himalayas), the dynamic cyclonic conditions persist over Iran, Afghanistan, Pakistan, and north-west India. The winds descending from the upper air high pressure obstructs the ascent of winds from the surface low pressure areas, with the result that the weather remains warm and dry. This is why the months of April and May are generally dry and rainless in spite of high temperatures (low pressure on land) and high evaporation. It may be pointed out that monsoon arrives in Myanmar and north east Indian states in May or early June. Upper-air low pressure (anticyclone) is formed to the east of the eastern limit of the Himalayas due to upper air easterly jet streams, with the result that the winds coming from the south of Myanmar are forced to ascend and yield copious rainfall. The Myanmar monsoon also affects Bangladesh and the hilly states of north-east India which receive pre-monsoon showers in the months of April, May, and the first week of June.

EL-NINO AND THE INDIAN MONSOON

The Indian monsoon is also influenced by El-Nino, Southern Oscillation and the Somalian Ocean Current. El-Nino, meaning "*Child Christ*", is a warm ocean current appearing along the Peru coast, generally in December. It replaces the cold Peru ocean current which flows along the Peru coast in normal years. Under normal conditions, the Peru is a cold water current, while over the western Pacific (Borneo, Indonesia and Eastern Australia) the ocean current is warm and deep **(Fig. 4.10(a)).** The appearance of El-Nino "*reverses the condition"* there, and develops warm conditions over the eastern Pacific (Peru and Chile coasts) and cool conditions in the western Pacific (Australia, New Guinea, and Indonesia). Whenever this warm ocean current (El-Nino) is produced near the Peru coast, the amount of precipitation along the coastal areas of (Peru and Chile) South America is usually high, while the eastern coast of Australia, New Guinea, and Indonesia record drought conditions **(Figs. 4.10(a) and 4.10(b))**.

As stated above, normally, the cool Peru (Humboldt) Current flows northward off the South American coast, and then at about equator it turns westward across the Pacific as the South Equatorial Current. The Peru Current is characterized by upwelling of cold, deep water, bringing with it nutrients that act as food for plankton. Fish feed on these high concentrations of plankton. The anchoveta, a small fish used commercially to produce fish meal for animal feed, thrives in great numbers and is harvested by the Peruvian fishermen. With the onset of El-Nino, upwelling ceases, the cool water is replaced by warm water from the west, and the plankton and their anchoveta predators disappear. Vast number of birds that feed on the anchoveta die of starvation.

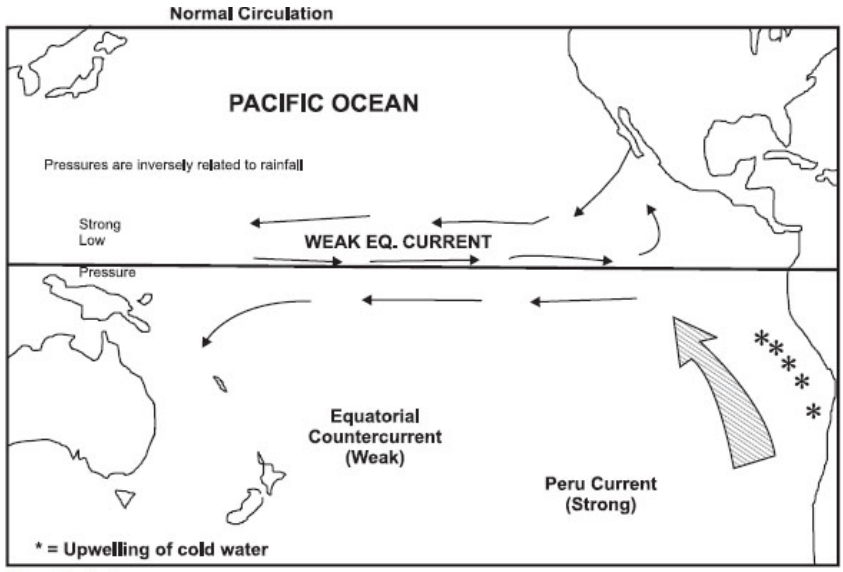


Fig. 4.10(a)

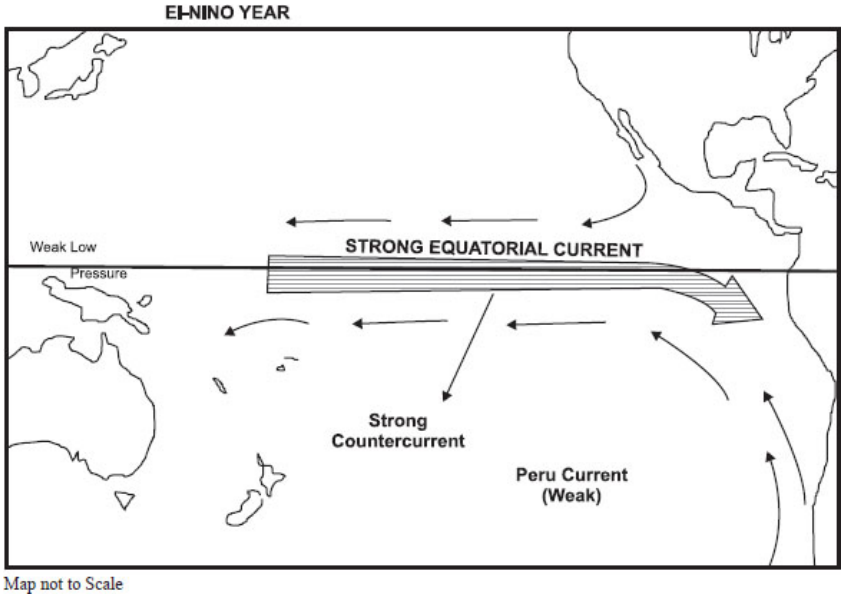


Fig. 4.10(b) El–Nino and Southern Oscillation

The Southern Oscillation

The shifting of atmospheric circulation over the southern Pacific region is known as Southern Oscillation. It is a precursor to the occurrence of an El-Nino event. In an El-Nino year, a major change occurs across the entire stretch of the equatorial zone as far west as south-eastern Asia. Normally, low pressure prevails over northern Australia, New Guinea, and Indonesia, where the largest and warmest body of ocean water can be found. This low pressure is associated with a pool of warm water and a deep thermocline. Abundant rainfall normally occurs in this area during December, which is the high sun period in the Southern Hemisphere **(Figs. 4.10(a) and 4.10(b))**.

During El Nino event, the low pressure system is replaced by weak high pressure and drought ensues. In contrast, pressure becomes lower than normal in the equatorial zone of the eastern

Pacific (Peru and Chile Coast), strengthening the equatorial trough. Rainfall is abundant in this new low pressure region along the coast of Peru, Chile, and Ecuador.

Surface winds and currents also change with this change in pressure. During normal conditions, the strong, prevailing trade winds blow westward, causing very warm ocean water to move to the western Pacific and to “pile up” near the western equatorial low. This westward motion causes the normal upwelling along the South America coast, as bottom water is carried up to replace the water dragged to the west. During an El Nino event, the easterly trade wind die with the change in atmospheric pressure. A weak westerly wind flow sometime occurs, completely reversing the normal wind direction. Without the pressure of the trade winds to hold them back, warm water surge eastward. Sea-surface temperatures and actual sea levels rise off the tropical western coast of the Americas.

The major change in sea-surface temperature that accompanies an El Nino can also shift rainfall patterns dramatically in large regions across the globe. During a typical El Nino event, southeastern and interior north western United States, southeastern South America, and the southern tip of India receive more rainfall. Intense rainfall can also occur along the southern California coast in winter and in the Andean highlands of Peru, Bolivia, Colombia, Chile, and Ecuador. Drought regions include the western Pacific, northern South America, southeast Africa, and northern India.

In brief, a positive value of Southern Oscillation (SOI) indicates:

(i) a cold Peru Current;
(ii) strong trade winds;
(iii) accumulation of warm water in western Pacific, which is balanced by the Equatorial Counter Current and high rate of evaporation;
(iv) a rise in depth of thermo-cline as we proceed from the western half of the Pacific; and
(v) an ascending branch of Walker Circulation over Australia and Indonesia with its descending branch over the western side of South America (coasts of Peru and Chile). Such a normal condition leads to normal south-west monsoon.

La Nina: The strengthening of the normal circulation over the southern Pacific Ocean is known as La Nina (the girl child). This event presents a situation roughly opposite to El-Nino. During La Nina period, sea surface temperature in the central and western Pacific Ocean fall to lower than average levels. This happens because the South Pacific subtropical high becomes very strongly developed during the high sun season. The result is abnormally strong south-east trade winds. The force of these winds drags a more than normal amount of warm water westward, bringing cooler water to the surface off western continental coasts. La Nina conditions were recognized during 1988 and may have been related to drought experienced in some parts of North America and Peru coast in summer 1988.

Fig. 4.11 shows the surface temperature observed during El Nino and La Nina years. During the normal and La Nina conditions, the cool water of the Peru current, moving northward along the coast of Chile and Peru, is carried westward into the Pacific in a long plume. Much cold, upwelling water is brought to the surface along the Chile and Peruvian coast. During an El Nino year, the eastward motion of warm water holds the Peru Current in check. Some light upwelling, but the amount is greatly reduced compared to normal or La Nina years. It adversely affects the planktons growth and marine food-web in the east Pacific Ocean.

El Nino and the Indian Ocean: According the meteorologists and oceanographers, the Indian ocean has been warming at a rate faster than thought before ($1.2^{o}C$ during the past century). It is also the largest consistent contributor to the global ocean warming trends.

Recent studies show that a warm Indian Ocean can in turn modulate the Pacific conditions including the El Nino events. So basically, such large warming over the Indian Ocean has implications on the global climate.

The western Indian Ocean, traditionally thought to have cooler sea surface temperatures than the central and eastern Indian Ocean, is surprisingly showing an even stronger summer warming trend over the whole of the 20 century than the central and eastern Indian Ocean.

The warming is significantly so large that it may alter the monsoon circulation, monsoon rainfall, marine food web, and fisheries (western Indian Ocean is one of the most productive oceans) and global climate including the El Nino.

A recent study published in the *Journal of Climate* found that the western Indian Ocean becomes warmer mainly due to El Nino events, which are getting stronger and more frequent during recent decades, possibly due to changing climate.

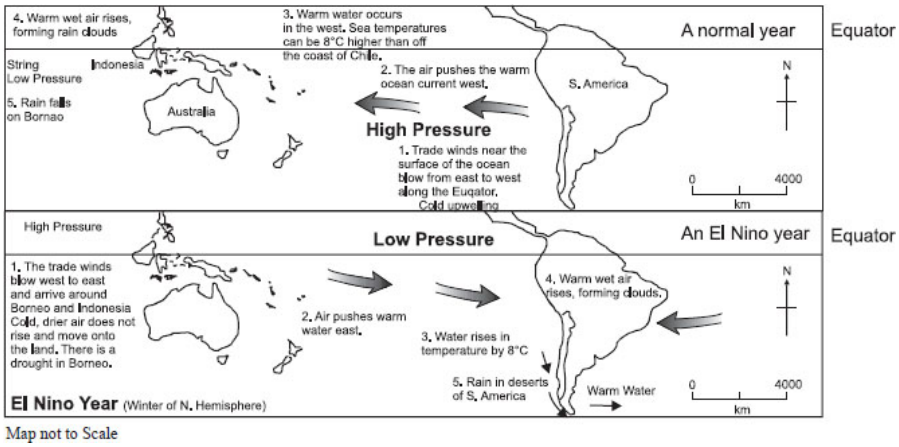
These El Nino events weaken the summer westerly (blowing from west to east) winds over the Indian Ocean. Winds have the effect of cooling the sea surface. Strong winds cause evaporation and loss of latent heat from the ocean leading to cooling. When the winds are weak, the opposite happens, the ocean warms (because of less evaporation and less loss of latent heat from the ocean leading to warming of the sea surface).

During El Nino, the wind direction reverses and the whole circulatory system shifts eastward leaving only weak surface south-west monsoon winds in the western Indian Ocean. This results in further ocean warming in western Indian Ocean.

Unlike the Pacific or the Atlantic Oceans, the Indian Ocean is landlocked on the north. This means that the ocean circulation which carries the tropical heat towards the poles is restricted in the Indian Ocean, withholding the piled up heat. This also helps in making the Indian Ocean relatively warmer and affecting the pattern of trade winds and weakening the south-west monsoon.

Moreover, the warm water of Humboldt (Peru) Current at the event of El Nino is carried to the south Indian Ocean by the West Wind Drift of the south Atlantic Ocean, resulting into relatively warm conditions or weak high pressure over the south Indian Ocean **(Fig. 4.11)**. The warm water of the South Indian Ocean is not conducive for the successful summer monsoon in the sub-continent of India.

According to a study by the Indian Institute of Tropical Meteorology (IITM), the frequency and duration of heat waves in India is expected to increase. '*El-Nino Modoki*' could be responsible for this increase. Depletion of soil moisture (causes reduced evapo-transpiration) and the radiation of heat from the earth to atmosphere also contributes to the increase in heat waves. The regions of southern India and coastal regions would be largely affected.



Map not to Scale

Fig. 4.11 El Nino–Southern Oscillation

In the El-Modoki events, the Central Pacific ocean warms up more with cooling in the eastern and western parts and contributes to the prolonged heat effect. It is expected that the heat wave duration would rise by 0.5 events and by 4–7 days per decade. (*Source: Report in The Times of India, May 18, 2019*)

Possible Causes of El-Nino

What causes the El-Nino/Southern Oscillation (also known as ENSO)? At first, meteorologists tried to explain the changes in winds and pressures by the changes in ocean surface currents, while oceanographers tried to explain the changes in surface currents by changes in winds and pressures. It was not until the 1950s that scientists realized that the two phenomena were linked. One view is that the cycle is simply a natural oscillation caused by the way in which the atmosphere and oceans are coupled by energy exchange. Another possibility has been proposed by geologists, who have tried that volcanic activity along the East Pacific Rise (Ridge), an underwater zone of sea-floor spreading off the west coast of South America. Correlates with El Nino events. They hypothesize that the emergence of upwelling lavas from the sea floor releases vast quantities of heat into the ocean, and that this heat in some way triggers an El-Nino event. In any event, scientists now have good computer models that accept sea surface temperature along with air temperature and pressure data and can predict El Nino events reasonably well some months before they occur.

As stated above, the Southern Oscillation is closely linked with the Walker Circulation. With a high positive Southern Oscillation, there would be a zone of low atmospheric pressure over Australia and Indonesian Archipelago. It will be accompanied by large convective clouds, heavy rainfall and raining air motion. This air eventually runs eastward, and after traversing the Pacific as a high level westerly wind at 200 mb, it descends over South America.

In terms of global winds, the Walker circulation suggests a strong belt of convergence between the trade winds of both the hemispheres at a location slightly to the north of the equator. These trade winds pile up a huge quantity of warm water in the western Pacific, produce equatorial counter-current from the Indonesian coast and facilitate upwelling of cold water from below, near Peru coast, giving rise to cold Peru current.

The appearance of El-Nino leads to a *warm phase* of the Pacific or negative SOI. Now, the ascending branch of the Walker Cell shifts to the central regions of the Pacific Ocean and the descending air branch to the south-eastern parts of the ocean **(Fig. 4.12)**. As upwelling off the South American coast decreases, the sea surface temperature rises. This leads to weaker trade winds, less accumulation of warm water on the western half of the Pacific Ocean, weakening of the Equatorial under current, heavy rain and floods along the South American coast and poor monsoon or monsoon failure over the subcontinent.

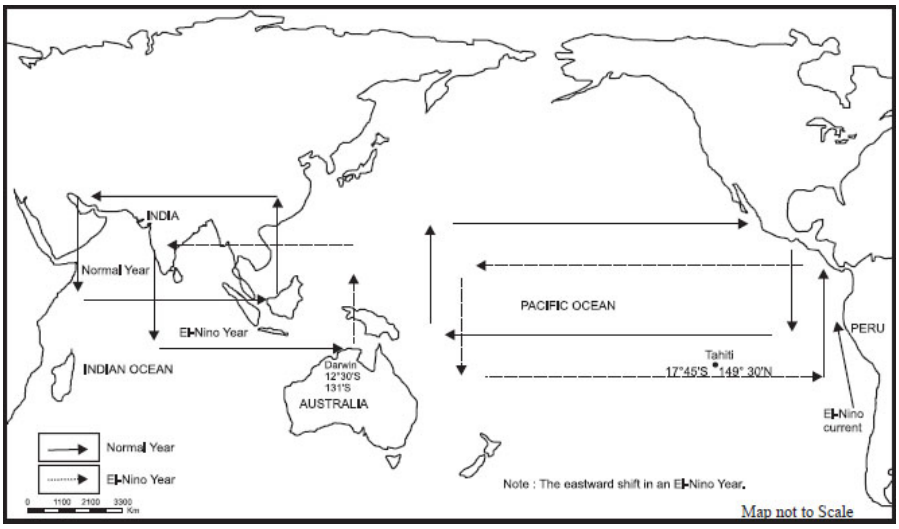


Fig. 4.12 Walker cells [Southern Oscillation (SOI) and Indian Monsoon] *(Reference: Indian Monsoon; Origin Mechanism; IJRAR; Vol 4, Issue 2)*

BURST OF MONSOON

The suddenness and abruptness of the Indian monsoon is known as the *burst of monsoon*. The onset over the Indian subcontinent is abrupt and dramatic. It is always accompanied by turbulent weather. The so called burst of monsoon is associated with certain basic changes in the general upper-air circulation over southern Asia. In April and May, the insolation heating of the subcontinent tends to establish the south-westerly monsoon flow from the adjacent ocean, but northward surge of the same is retarded by the westerly zonal flow associated with the subtropical jet stream over northern India. However, in late May or early June, when the thermal conditions are satisfied, the jet stream disappears completely from the south of the Himalayas and shifts to a position to the north of the Himalayas and Tibet. At the same time, the upper trough low (low pressure) also moves westward from 85° E to 75°E. It may be pointed out that the jet stream does not retreat slowly. The process of this shift is rather quick. Now, with the disappearance of the jet, a definite monsoon circulation from the sea on to the land is established. The summer monsoon generally begins in late May in most parts of south-east Asia. But over India, it is delayed until the middle or late June. The change from one regime to another is abrupt. It is well to remember that the onset of monsoon occurs in several stages depending on the periodic advance and withdrawal of the equatorial convergence zone.

In the opinion of Koteswaram, the burst of monsoon is closely related to the development of a warm-core upper anticyclone (high pressure) over the extensive and lofty Tibetan Highland. This upper level anticyclone produces an easterly jet over India which is positioned at about 15°N. It is definitely a part of readjustment in the general planetary circulation pattern. Gradually, the easterly jet covers the entire region extending from India to east Africa. The above mentioned upper air condition paves the way for the advance of south-westerly monsoon current over the subcontinent. The monsoon current, therefore prevails throughout India. The depth of the monsoon in India is about 6.5 km, while over the Gangetic Plain it is only about 5 km. The current is overlain by a layer of easterly winds (easterly jet stream).

The abrupt arrival of monsoon is of great climatic and social significance to the people of the subcontinent of India. The onset of monsoon puts an end to the scorching weather and the local hot winds *(loo)* in the northern plains of India. The relative humidity increases in the atmosphere

tremendously. The arrival of monsoon is also the beginning of agricultural operations for the kharif crops in the rain-fed areas. The high temperature and high relative humidity are, however, oppressive and injurious to health. It is in the season of general rains (July to September) that people suffer from many diseases and epidemics.

BREAKS IN THE MONSOON

The migration of the monsoon rainfall zone is one of the major sub-seasonal variations of the summer (or south-westerly) monsoon. Thus, the monsoon is not a continual deluge of a number of months, duration, but has inter-seasonal variability; being made of a series of discrete events, both pluvial and dry. Viewed locally, these are the active and break monsoons respectively which exist on a time scale ranging from a few days to few weeks. Thus, while the monsoon appears to have a well-defined annual cycle, closer examination shows that the monsoon has substantial variability which becomes evident as the intensity of monsoon rains wax and wane through the wet season. Periods during which there is a rapid succession of weather disturbances or storms lasting a few days are referred to as *active periods* of the monsoon. Periods during which there is no rainfall for few days are the *break periods* of the monsoon. During an active phase, the Tropical Easterly Jet Stream (TEJ) remains very strong in the upper troposphere indicating strong convection and latent heating. But, when the maximum cloudiness remains locked up in the foothills of the Himalayas and the monsoon rainfall zone moves in this direction, subsidence occurs to produce a weak easterly flow in the upper troposphere. This creates the condition of break in monsoons.

In break monsoon condition, there is a general rise of pressure (as well as temperature) over the country and the isobars show marked refraction along the west coast. Cloudiness decreases and the south-easterlies at the surface levels over northern India are replaced by hot westerly air which blows over the plains, since the broad-scale surface pressure (the monsoon trough) shifts to the Himalayas and the rainfall practically ceases over the country outside the Himalayan regions and the southern slopes of the Himalayas, leading to high floods in the plains of these Himalayan rivers. Thus, though there is no rain over the plains, all the major northern and eastern Indian rivers rise and floods ensue.

Under weak monsoon conditions and in the years when the eastern end of the axis of the monsoon trough is oriented southward in Odisha, Jharkhand, Chhattisgarh, and Madhya Pradesh, a low valley trough develops over the Assam Plain aligned along river Brahmaputra between the eastern Himalayas and the Shillong Plateau. The vertical extent of this low valley trough is 2 to 3 km
with the south west monsoon lying to the south of the trough, remaining independent of the main monsoon trough. But, when the latter moves northwards and extends to the Himalayas, it joins the trough over the Assam Plain to cause heavy rainfall there.

The break in monsoon conditions generally occurs in the peak months of July and August, and lasts for at least 3 to 5 days over 500 to 1000 km length in these months. The breaks occur on two time scales, usually from 10 to 20 days which on some occasions gets locked to a 40 to 50 days' break configuration. Depending upon their timing and duration, the break monsoon conditions are the harbingers of regional droughts.

Onset of the Monsoon

The onset of monsoon rains is viewed in monsoon literature as the burst of the monsoon, commonly in the form of a great thunderstorm. Prediction of the onset of monsoon helps farmers to select the most suitable crops to sow and to determine when to prepare the land. Predicting the timing of the monsoon has been an onerous task for the Indian Meteorological Department

(IMD), almost from its beginning. Since 1878, forecasts of monsoon rainfall has been issued by this department regularly.

Progress of the Monsoons

The normal date of the onset of the monsoon is towards the end of May in Sri Lanka and over Andaman and Nicobar Islands in the Bay of Bengal **(Fig. 4.13)**. Thereafter, it bursts on the Malabar Coast in Kerala a week later by June 1, and the isolines on the onset chart display a pronounced curve over the Bay of Bengal. The subsequent progress of the monsoon in the country is divided by the tapering peninsula into two bifurcating currents, i.e., (i) the Bay of Bengal Current, and (ii) the Arabian Sea Current. The Arabian Sea branch gradually advances northwards reaching Mumbai by 10 June and Central India by 15 June.

In the meantime, the progress of the Bay of Bengal branch is no less spectacular. It moves northwards into the central Bay of Bengal and advances to nearly 20° N latitude in the Bay by the third week of May.

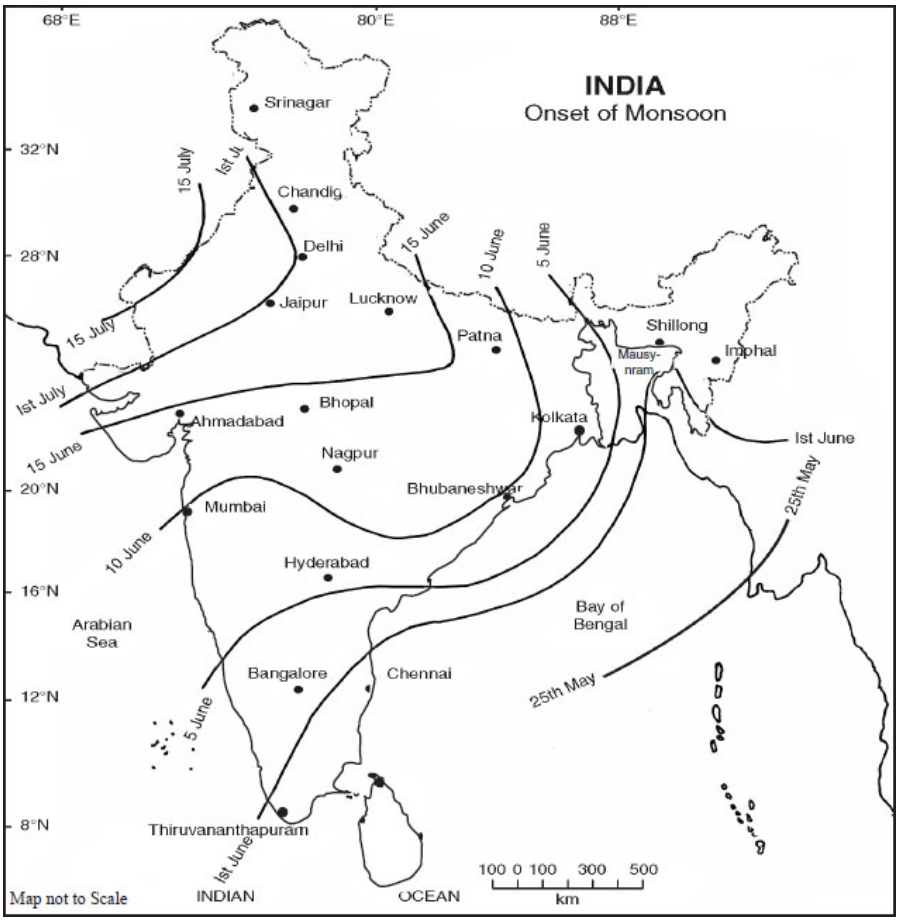


Fig. 4.13 Normal Dates of Onset of South-west Monsson

It rapidly strikes the Arakan Coast and Chittagong Hills with full force, the direction being southwesterly, but the greatest force is felt in the angle between the east-west Meghalaya Plateau of the Khasi Hills and the north-eastern Lushai Hills; Mizoram (Blue Mountains). Here Mawsynram to the west of Cherrapunji at an elevation of about 1313 m receives nearly 1200 cm of rainfall annually. The Brahmaputra Valley is in the rainshadow of the Khasi Hills and receives

less than 250 cm. The rainfall increases once again in the Himalayas, but the Bay of Bengal monsoon branch is deflected westwards due to the combined effect of the monsoon trough and the channelling by the Himalaya. As a consequence, its further progress is towards the Gangetic Plain of India, giving more rain to Himalaya and less in the plains. The arrival of the monsoon over Kolkata is 7 June, where, as mentioned earlier, the Arabian Sea branch of monsoon normally strikes Mumbai on 10 June. Both the branches of the monsoon current merge in the Gangetic Plain and gradually extend over Bihar, Jharkhand, Chhattisgarh, Madhya Pradesh, Uttar Pradesh, Uttarakhand, Haryana, Himachal Pradesh, Punjab, Jammu and Kashmir, and finally to Rajasthan, so that by the first week of July, i.e., within a mean interval of 29 days since the onset in Kerala, the south-west Monsoon is established all over India, excluding the Thar Desert. In the Thar Desert, the monsoon reaches by the middle of July, but only as a descending, drying wind, since by this time it has almost shed most of its moisture. It can be noted from **(Fig. 4.13)** that for parts of the country west of 80°E and south of 25°N, the advance of monsoon is from south to north, while in the rest of the country it is from south-east to north-west.

SEASONS IN INDIA

The subcontinent of India has great latitudinal dimensions: There are different seasons from Jammu and Kashmir to Kanyakumari (Cape-Camorin). The Meteorological Department of India, however, divides the seasons of India into the following four seasons:

A. Seasons of North-East Monsoon

1. Winter season, mid-December to mid-March, and
2. Hot weather season, mid-March to May.

B. Seasons of South-West Monsoon

3. Rainy season, June to September, and
4. Season of retreating monsoon, October to mid-December.

1. The Cold Weather Season

The cold weather season in the greater parts of India begins in the later part of November in the north, and by the beginning of December in the rest of the country. The cold weather season is characterised by out-flowing winds, dry and stable air, and clear skies. There develops a high pressure (anti-cyclone) area over north India, and a north-westerly flow prevails down the Indus and Ganges Valleys. During this season the southern branch of the subtropical jet stream is positioned over northern India. The middle latitude westerlies reach down to the surface north of about 25°N. South of this latitude the general movement of air is from the north-east. This north-easterly wind is called the winter monsoon. In Peninsular India, the general direction of wind is from east to west. Because of its trajectory over the Bay of Bengal, the easterlies are full of moisture and yield some precipitation along south-east coastal regions.

During winter season there is a general decrease in temperature from south to north. The isotherms run almost parallel to the latitudes. The 18° C isotherm for the month of January runs in a east-west direction through the middle of the country, connecting the Tapi estuary in the west and the Mahanadi delta in the east **(Fig. 4.14)**.

In the month of January, the north-western parts of the Great Plains of India (Punjab, Haryana, western Uttar Pradesh, and Rajasthan) experience less than 15°C mean monthly temperature. The night temperature in the plains of Punjab, Haryana, and Rajasthan (Amritsar, Hissar, and Jodhpur) occasionally reads below the freezing point, producing ground frost condition. Often, there is a decrease of more than 6°C in the mean temperature, resulting into **cold wave** in the northern plains of India. In south India, the isotherms tend to bend to the south and run almost parallel to the coast. The western coast is warmer than that of the eastern coast by about 2°C.

The diurnal range of temperature is about 15°C in the Great Plains and only about 5°C in the coastal areas of the Peninsula. January is the coldest month in India, especially in northern India. Peninsular India, however, does not have a well defined cold weather season. The mean maximum temperature for the month of January at Thiruvananthapuram and Chennai reads 31°C and 30°C, respectively **(Fig. 4.14)**.

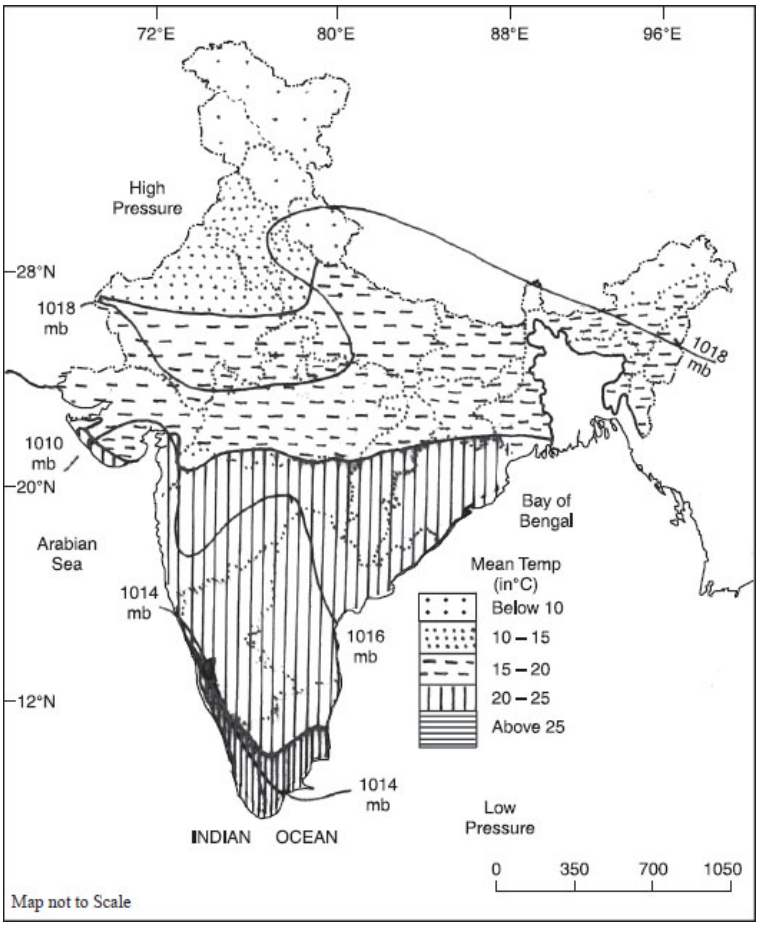


Fig. 4.14 Mean January Temperature

A characteristic feature of the cold weather season is the inflow of western disturbances originating from the Mediterranean Sea. The frequency of these disturbances is 4 to 6 per month between December, January and February. In north-western region of the subcontinent, winter precipitation is caused by the depressions that are associated with the westerly disturbances moving out from the Mediterranean Sea **(Fig. 4.14A)**. The cold weather precipitation, though small, is highly beneficial to *Rabi* crops. Besides, snowfall from the western disturbances feed the glaciers of the Western Himalayas.

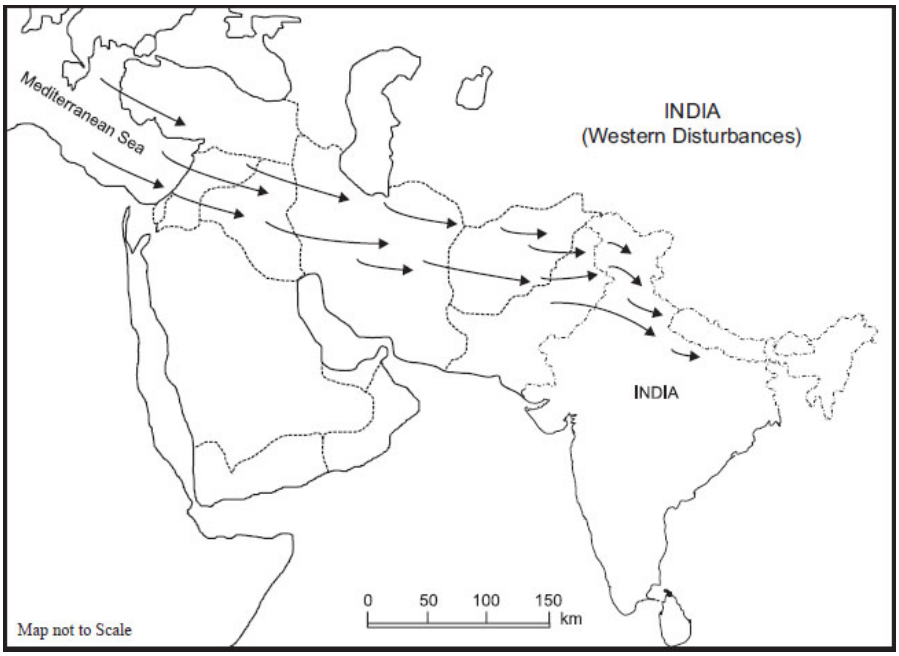


Fig. 4.14A Western Disturbances

The north-eastern parts of India also get some rainfall during the winter season. Arunachal Pradesh, Nagaland, and Assam may get about 50 cm of rainfall during these months **(Figs. 4.15)**.

2. The Hot Weather Season

The north Indian region experiences a well defined hot weather season from mid-March to mid-June. With the northward march of the sun towards the Tropic of Cancer after the vernal Equinox, the temperature begins to rise. Thermal heating over north-western India gradually establishes a thermal “low’ at the surface but, while the jet stream remains south of the Himalayas, it maintains its dynamic anticyclone aloft over Afghanistan and the borderland of Pakistan. This “lid” of subsiding warming dry air prevents the surface thermal “low” from having sufficient effect as a lifting agent to carry air aloft and so to bring about precipitation (Johnson, 1969, p.l7).

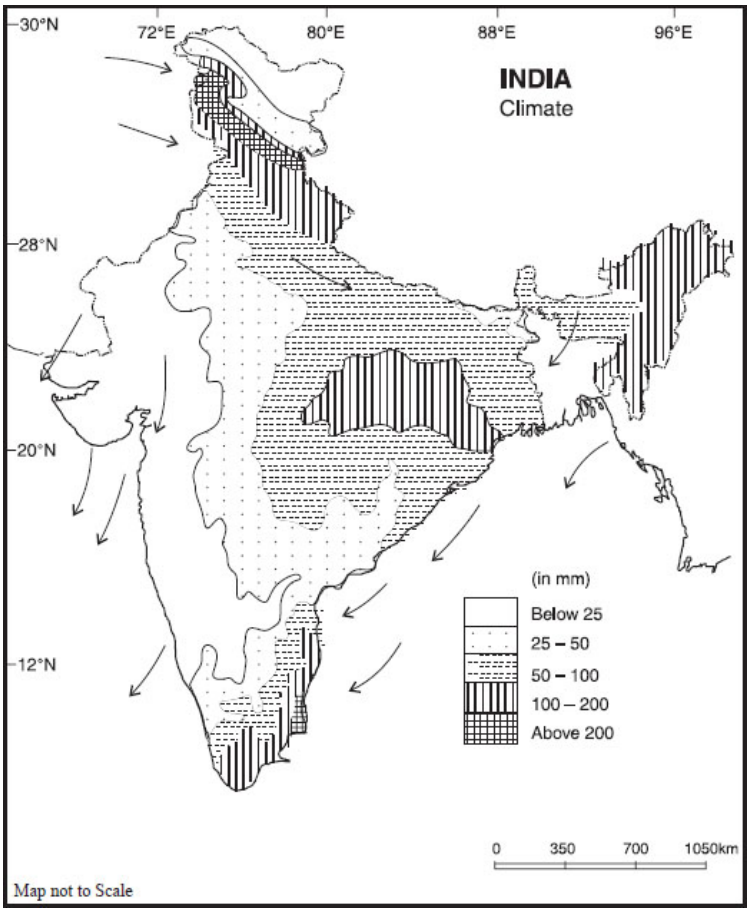


Fig. 4.15 Rainfall and Winds in January

At the advent of March, the temperature starts rising abruptly. By April, the Peninsular regions south of the Vindhyan Range heat up with mean maximum temperature of 40°C. In May, the mean maximum temperature reaches 42°C in Rajasthan, Delhi, west Uttar Pradesh, south Punjab, Jammu and Haryana.

Temperature exceeding 54°C was recorded at Sri Ganganagar in 1967. At some places, particularly in north-western India, day temperature may be as high as 45°C or 47°C. The mean minimum daily temperature in May also remains quite high being about 26°C at Delhi and Jaipur. The temperature in the eastern states of India and in the hilly regions in the month of May is generally cool and invigorating the minimum average in Shillong being 15°C and 17°C in Kohima **(Fig. 4.16).**

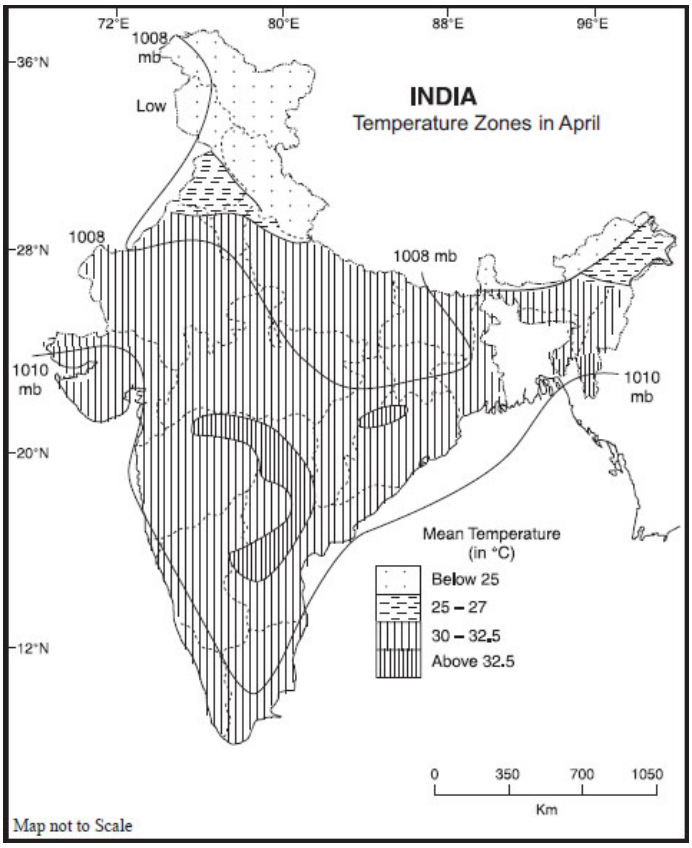


Fig. 4.16 Mean Temperature in April

In the month of April the 30°C isotherm of average temperature encloses a vast area of the country between 10°N and 26°N latitudes (except the west coast and the hilly states of north-east India). The diurnal range of temperature ranges between 5°C and 6°C in coastal areas, but reaches 15°–20°C in the interior parts of the country and in the north-west Satluj Ganga Plains.

Being a transitional season between the north-east and the south-west monsoon (rainy season), it is characterised by unstable air pressure and wind circulation. With the northward movement of the sun, the low pressure area also moves from south-east to north-west. It finally settles over north-western India in the end of May or early part of June. The pressure generally increases towards south in the neighbouring sea. The general direction of winds is from north-west and west in north-western India; from south-west in the Arabian Sea and the adjoining coasts. The tornado-like dust storms of Punjab, Haryana, and western Uttar Pradesh, the hot winds (*Loo*) in western India, the Norwesters (*Kalbaisakhis*) of West Bengal are the characteristics of summer season.

During summer season, the sun is scorching and the relative humidity is generally below 30%, occasionally reaching below 10%. The total rainfall of the season is below 2 cm in Rajasthan, Gujarat, Madhya Pradesh; between 5 and 15 cm in the sub-montane region of Uttarakhand, Himachal Pradesh, Punjab, Uttar Pradesh, Bihar and Odisha; and between 15 and 25 cm in the Malabar coast and over 50 cm in Assam, Meghalaya, Mizoram, and Nagaland. The rains caused by thunderstorms in Karnataka are called as “Cherry Blossoms” (where these are beneficial for coffee plantation) and elsewhere in south India as “mango showers”.

3. The Season of General Rains

By the end of June, a low pressure area develops over Punjab and Rajasthan. The southern branch of the jet stream weakens and is finally withdrawn from the southern slopes of the Himalayas by mid-June, leading to the formation of a dynamic depression over the surface thermal low. The Inter-Tropical-Convergence Zone (ITCZ) moves further northwards occupying a position of 25°N by mid-June and allowing equatorial westerlies to gush in the subcontinent. The cyclonic vortices developed in the ITC cause rains in the country. The tropical easterly jet streams originating due to thermal heating of Tibet intensifies Indian Ocean high pressure cell from which south-east trade winds are pushed by the Antarctic Circumpolar Whirl to develop as south-west monsoon.

In the northern plains of India, the temperature reaches its maximum in June to break the monsoon. At places, the day temperature touches 46°C and more. The mean maximum temperature in the month of June at Jodhpur reaches 41°C , Delhi 40°C, Allahabad 39°C, Kolkata 33°C, 29°C at Kochi and 23°C at Shimla and Srinagar. The diurnal range of temperature is high being about 30°C at Leh, and 15°C at Delhi. The diurnal range of temperature is, however, less than that of the month of May **(Fig. 4.16)**.

Generally in the afternoon of a scorching day, rains begin suddenly. This is known as a “monsoon burst”. The Arabian Sea current advances northwards by 1st June on the Kerala coast and reaches Mumbai by about 10th June. By mid-June, it spreads over Saurashtra, Kutch and Madhya Pradesh.

The Bay of Bengal current first strikes Andaman and Nicobar islands by about 25th of May and reaches Meghalaya, Mizoram and Tripura by about 1st June. It rapidly spreads over most of Assam, and reaches Kolkata around 11th June. In Delhi, western Uttar Pradesh, Haryana, and Punjab, the monsoon reaches around 1st July.

During the season of general rains, most of the country experiences good cloud cover (from 1/8 to overcast sky). During this season, the relative humidity is generally over 65%. Assam and Kerala record the highest percentage of relative humidity; over 80% over the greater part of the rainy season.

The greater parts of the subcontinent of India receive over 85% of their total rainfall during the season of general rains. The Arabian Sea current causes rainfall all along the Malabar coast, Konkan coast, Western Ghats, Maharashtra, Gujarat, and parts of Madhya Pradesh. Mumbai records about 190 cm of rainfall during this season. As the Arabian current crosses the Sahyadris, the places situated on the leeward side of the Western Ghats receive substantially less rainfall. For example, Pune (about 160 km to the east of Mumbai) gets only 125 cm of rainfall. Nagpur records 125 cm and Thanjavur only 85 cm of rainfall. The sub-branch of the Arabian Sea current which moves northwards through Kutch, Gujarat and Rajasthan gives little rainfall to these regions due to the parallel alignment of the Aravallis. The Arabian sea current goes straight up to the western Himalayas, where it gives appreciable rainfall in Uttarakhand and Himachal Pradesh.

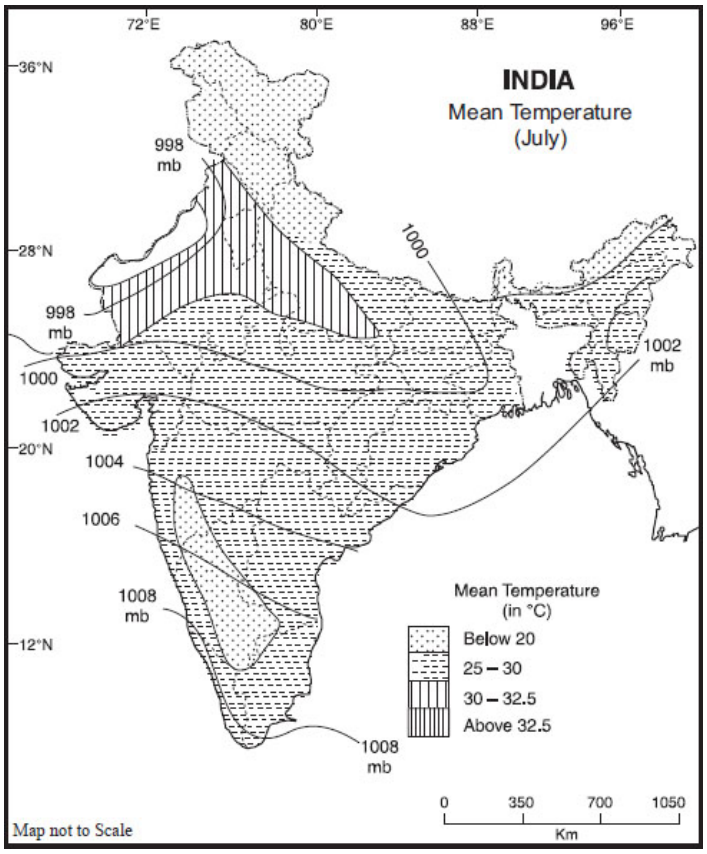


Fig. 4.17 Mean Temperature in July

The Bay of Bengal current first dashes against the Myanmar coast, and obstructed by the eastern hills, is deflected westward towards the Ganga Plain. This current causes the heaviest rainfall at Mawsynram (about 1200 cm) and Cherrapunji (1100 cm) annually. A major part of this rainfall occurs in the morning hours. Here also, the leeward sides of the Meghalaya Plateau receive lower rainfall; 150 cm and 160 cm in Shillong and Guwahati, between June and September respectively. Moving westward, the Bay of Bengal current gives Kolkata 120 cm, Patna 105 cm, Allahabad 90 cm, Delhi 65 cm, Bikaner 25 cm and Srinagar (J&K) only 20 cm of rainfall between June and September **(Fig. 4.18)**.

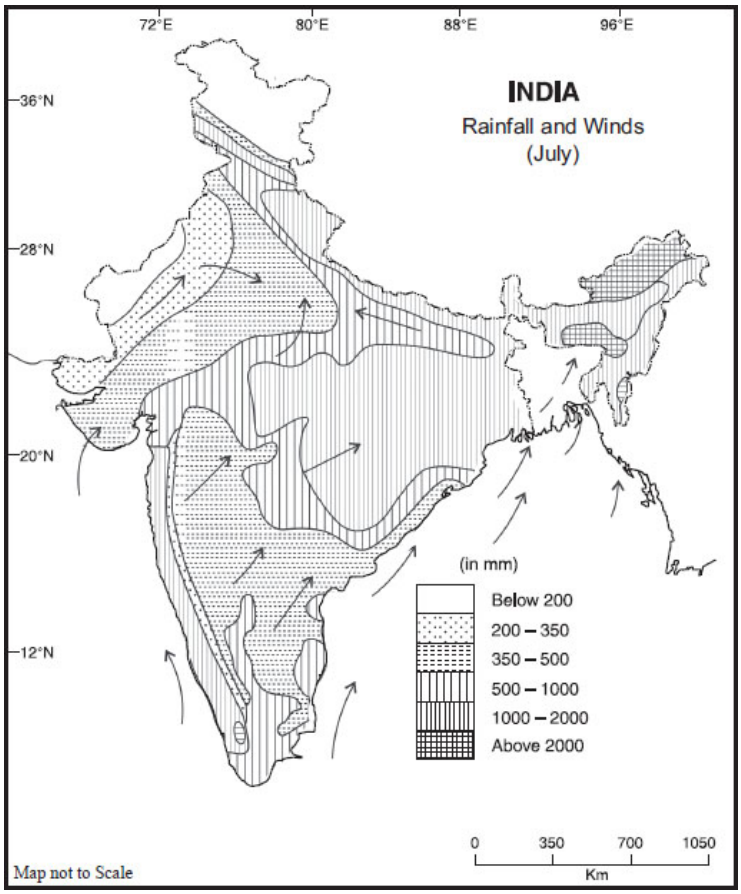


Fig. 4.18 Rainfall and Winds in July

The weather and amount of rainfall received are also affected by a number of cyclonic depressions which enter the country through the Bay of Bengal and the Arabian Sea **(Fig. 4.19).** About 20 to 25 such depressions are developed during the monsoon period, of which some are stronger, causing immense damage to life and property of the people residing in the coastal areas of West-Bengal, Odisha, Andhra Pradesh, Tamil Nadu, and Gujarat. The cyclone of 29 October, 1999 with a velocity of over 300km/h killed over one lakh people. More than two million houses were washed away and about 12 million people were rendered homeless. These cyclones are generally accompanied with heavy rain, the amount of which decreases going away from the coastal areas.

The distribution of rainfall during the rainy season has been plotted in **(Fig. 4.18).** It may be seen from this figure that the western coast, Sahyadris, Meghalaya, Arunachal Pradesh, Mizoram, Nagaland, Sikkim, and Darjeeling hills get more than 200 cm of rainfall.

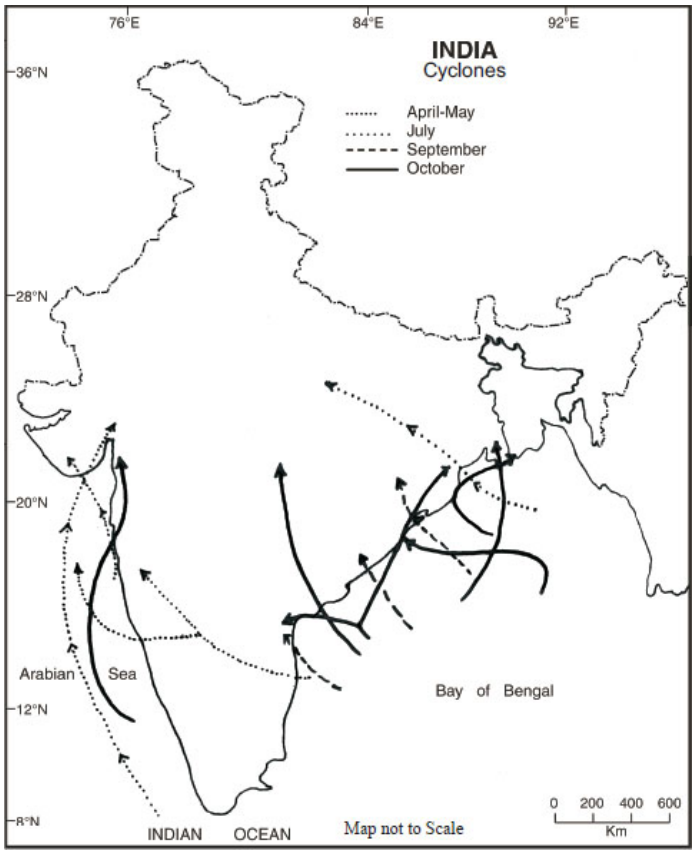


Fig. 4.19 Cyclones

The remaining parts of north-eastern India, West Bengal, Odisha, Jharkhand, Bihar, Chhattisgarh, the Tarai region and hills of Uttarakhand and U.P. receive rainfall between 100 to 200 cm. Similarly, the southern and western Uttar Pradesh, northern and western Madhya Pradesh, eastern Maharashtra and Gujarat, and northern Andhra Pradesh experience rainfall between 50 and 100 cm. Rajasthan, western Gujarat, southern Andhra Pradesh, Karnataka plateau, Tamil Nadu, plains of Haryana, Punjab, Jammu and Kashmir, ladakh receive less than 60 cm of rainfall. The lowest rainfall is recorded in the Thar desert along the border of Pakistan, and the Ladakh region of Jammu and Kashmir state **(Fig. 4.22)**.

4. Season of Retreating Monsoon

The south-west monsoon begins to retreat from northern India by the third week of September with the southward migration of the sun. By the end of September, the south-west monsoon retreats from the Punjab plains and adjacent regions. However, unlike the sudden burst of monsoon, the retreat is steady and gradual **(Fig. 4.19).** By mid-October, the southerly branch of the jet stream returns to its winter position south of the Himalaya.

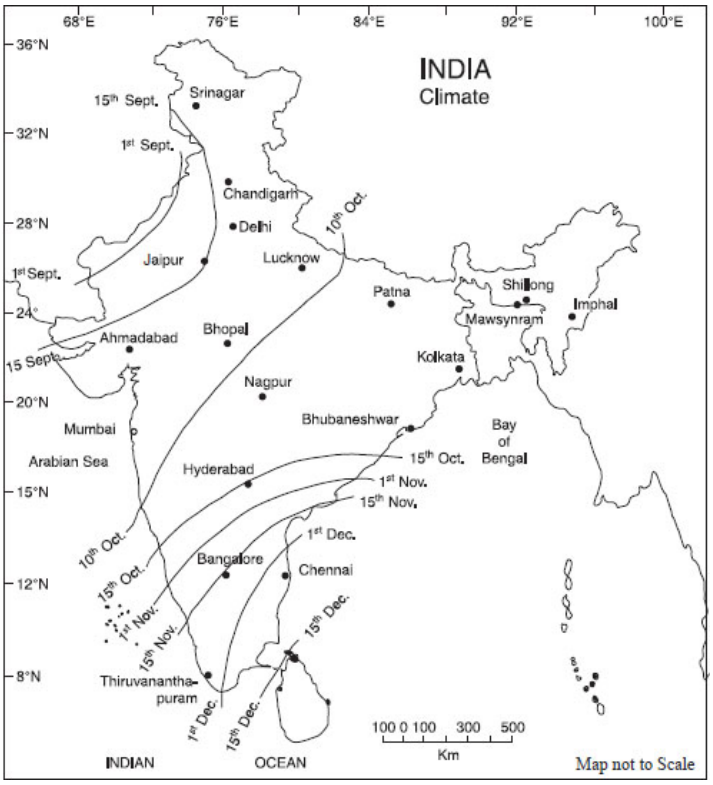


Fig. 4.20 Normal dates of Withdrawal (Retreat) of South-West Monsoon

The temperature during the season of retreating monsoon is uniformally high being about 25°C in the beginning of October in Northern India. The day temperature is generally high, but nights become pleasant with the mean going down to 20°C or even lower. The temperature begins to decline in November and by December cold weather sets in with about 15°C average temperature in the north and north-west parts of Indian plains, about 20°C in the central parts of India, and about 25°C along the coasts**.**

With the advent of October, the low pressure area over the north-western parts of India is dissipated and its place is taken by the low pressure cell located over the northern parts of Bengal. By the beginning of December it moves further southwards and by month end, it merges with the equatorial low. The winds are westerly in the north-western parts of the country and in the Ganga Plain, north-easterly in the Peninsular region and north-westerly in the east coast.

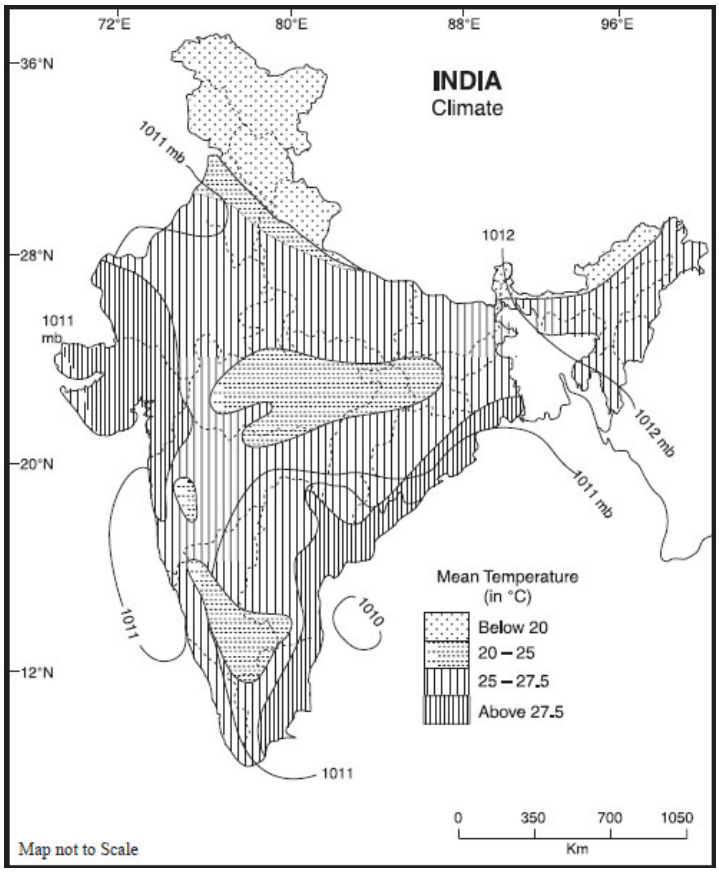


Fig. 4.21 Mean October Temperature

TROPICAL CYCLONES

Tropical cyclone is an intense low pressure traveling storm that develops over tropical or sub-tropical oceans, accompanied by high winds and heavy rainfall. They are potentially the most destructive cyclones experienced by the humans which cause damage to life and property.

Tropical cyclone is a circular storm that rotates counterclockwise in the Northern Hemisphere. The tropical cyclone gets energy from the latent heat of condensation. The energy in an average tropical cyclone of the Bay of Bengal may be equivalent to more than 10,000 atomic bombs of the size of Nagasaki atomic bomb. The tropical cyclone range in size from a few kilometers to several hundred kilometers in diameter. In the middle is an eye that can be about 20 to 65 km across.

The wind system associated with tropical cyclone in one of contrasts. In the eye of the storm, the winds are light and variable and the velocities not usually exceeding 25 km. The wind velocity increases rapidly away from the eye, reaching their highest velocities just outside the eye and at a height of about 0.8 km. When the cyclone moves ashore, or makes *landfall*, it pushes storm surge of seawater inland. The strongest winds of a tropical cyclone are usually recorded in its right front quadrant (relative to the storm's direction path).

Origin of Tropical Cyclones

There is no consensus yet as to why the tropical cyclones originate. The following geographical conditions are required for the formation of a tropical cyclone:

1. Large bodies of warm water and air. When both the air and water temperatures are

higher than normal, generally around 27^{o}C.

2. They develop only on oceans.
3. They drive their energy from the latent heat of condensation.
4. They develop when there is large and continuous supply of warm and moist air.
5. Tropical cyclones occur when the equatorial low pressure trough is farthest from the equator. This condition occurs after summer solstice.
6. At a height of about 9000 meters above the surface disturbance, there must be an anticyclonic circulation.
7. There must be an unstable vertical temperature structure.
8. There must be a mechanism present to start rotation or higher coriolis force.
9. They originate generally between 8^{o} N to 25^{o}N in the Northern Hemisphere.

The above conditions are ideally available in the Bay of Bengal as compared to the Arabian Sea. The surface area of the Bay of Bengal is smaller (2,172,000 sq km) as compared to that of the Arabian Sea (3,862,000 sq km). Moreover, the eastern parts of the Indian Ocean in which the Bay of Bengal is located is relatively more warm as compared to the Arabian Sea. The Somalian water current also flows in the Arabian Sea which mitigates the surface temperature of the Arabian Sea. All these conditions favour the origin of tropical cyclones in the Bay of Bengal.

The prevailing temperature conditions in the Arabian Sea and the Bay of Bengal have been shown in **(Fig. 4.22)**. It may be seen from **(Fig. 4.22)** that the mean temperature in the months of September and October in the Bay of Bengal is about 27^{o}C which is highly conducive to the origin of tropical cyclones.

The strong south-west monsoon winds in the Arabian Sea also help in cooling the surface temperature of the Arabian Sea which is not favourable for the origin of a tropical cyclone.

Tropical cyclones are disastrous natural hazards which cause heavy loss to human lives and property. They destroy the transport systems, water and power supply, disrupt communication system, destroy crops, domestic and wild animals, natural vegetation, mangroves, etc. In 1999 more than 100,000 (one lakh) people lost their lives in Odisha and the coastal areas were submerged under saline water, making the productive agricultural land unproductive. In 2013 a very severe cyclonic storm *Philani* and in 2014, another severe cyclonic storm *Hudhud* killed several hundred people, cattle and wild animals, and damaged the property and means of transportation and communication along the Andhra, Odisha and West Bengal coasts. In the first weak of August, 2015 the Komen Cyclone resulted into deluge and land slides in Manipur and West Bengal, killing around 80 people. The heavy rainfall in Rajasthan and Gujarat in the last week of July, 2006 caused heavy damage to crops, life and property.

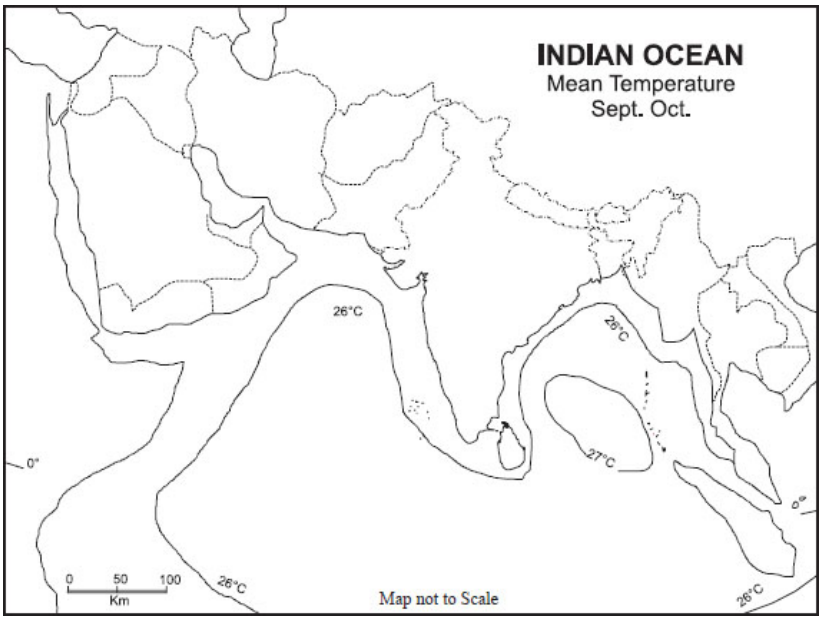


Fig. 4.22 September-October Mean Temperature in Bay of Bengal & Arabian Sea

THUNDERSTORMS

Thunderstorm is an intense, local convectional storm associated with a tall, dense cumulonimbus cloud which yields heavy precipitation, along with lightning and thunder, and sometimes hail. There are very strong updrafts of air in a thunderstorm.

According to meteorologists, a thunderstorm begins when thunder is heard or overhead lightning or hail is observed. Severe thunderstorms may result into three quarter-inch hail and/or wind gusts of 50 knots. The storm is considered ended 15 minutes after the last thunderclap is heard. Thunderstorm in dry climates, often occur without measurable precipitation.

Thunderstorms develop from an initial uplift of moist, unstable air and the release of sufficient latent heat to cause continued uplift. Thus, the keys to storm formation are source of moist air and a mechanism to produce the required uplift. Given moisture and convective heat source, the greatest number of thunderstorms occur in moist, tropical realms, especially in Africa. The second requirement, which is a mechanism to initiate cumulonimbus cloud development.

When a number of individual thunderstorms grow in size and organize into a large convective system, it is termed a meso-scale convective complex. They are often large enough to cover a large area and can persist for more than 12 hours.

In India thunderstorm may occur in any season but their frequency is more during the winter season and the season of general rains. At the event of thunderstorm the standing crops, orchards, floriculture and vegetable cultivation are seriously damaged.

Lightning and Thunder

An estimated 8 million lightning strikes each day on Earth. Lightning refers to flashes of light caused by enormous electrical discharges–tens to hundreds of millions of volts–which briefly superheat the air temperatures of 15,000^{O}C to 30,000^{O}C.

The violent expansion of this abruptly heated air sends shock waves through the atmosphere – the sonic bangs known as thunder. The greater the distance a lightning stroke travels, the longer the thunder echoes. Lightning at great distance from the observer may not be accompanied by thunder and is called *heat lightning*.

Lightning poses a hazard to aircraft and to people, animals, trees and structures. Individuals should acknowledge that certain precautions are mandatory when the threat of lightning discharge is near. When such lightning is imminent, the Meteorological Departments issue severe storm warnings and caution people to remain indoors.

Hail

Hail is a type of precipitation formed when a raindrop is repeatedly circulated above and below the freezing level in a cumulonimbus cloud, with each cycle adding more ice to the hailstone until it becomes too heavy to stay aloft.

Pea-sized and marble sized hail are common, although hail of the size of golf balls and baseballs have also been reported at least once or twice a year in India. Hail precipitation may occur in the hilly and mountainous areas and plains and they may occur in the tropical and temperate climates. They, however, occur in narrow belts but cause heavy damage to property and standing crops.

Tornado

Tornado is an intense, destructive cyclonic rotation, developed in response to extremely low pressure, associated with mesocyclone formation. Tornadoes often occur with thunderstorms. For the formation of a tornado, several prerequisites are essential:

1. There must be a mass of very warm, moist air present at the surface.
2. There must be an unstable vertical temperature structure.
3. There must be a mechanism present to start rotation.

A mesocyclone can range up to 10 km in diameter and rotates over thousands of meters vertically within the parent cloud. As a mesocyclone extends vertically and contracts horizontally, wind speeds accelerate in an upward vortex (much as ice skaters accelerate while spinning by pulling their arms in closer to their bodies). A well developed mesocyclone most certainly will produce heavy rain, large hail, blustery winds, and lightning.

As more moisture-laden air is drawn up into the circulation of a mesocyclone, more energy is liberated, and the rotation of air becomes more rapid. The narrower the mesocyclone, the faster the spin of converging parcels of air being sucked into the rotation.

A tornado can range from a few meters to a few hundred meters in diameter and can last anywhere from a few moments to tens of minutes. Pressures inside a tornado usually are about 10% less than those of the surrounding air. The inrushing convergence created by such a horizontal pressure gradient causes severe wind speeds that can exceed 485 kmph.. The damage produced by such wind is catastrophic. When tornado circulation occurs over water, a waterspout forms, and surface water is drawn up into the funnel some 3-5 m.

RAINFALL DISTRIBUTION

The distribution of rainfall in India is highly uneven. Its distribution is largely controlled by the nearness of the sea and the orographic features. The influence of the Western Ghats, the Plateau of Meghalaya, the north-eastern Hills, and the Himalaya Mountain is quite significant. The average annual distribution of rainfall in India has been shown in **(Fig. 4.22)**. It may be observed from **(Fig. 4.22)** that the regional variations in the distribution of average annual rainfall over India are quite pronounced. In the southern parts of the Meghalaya Plateau (Mawsynram and Cherrapunji), the average annual rainfall is more than 1200 cm. However, it drops to 200 cm or less in the Brahmaputra Valley and the adjoining hills.

In Peninsular India, the highest rainfall occurs along the Konkan and Malabar coasts. The isohyte of 150 cm rainfall runs southwards from the Gujarat coast, roughly parallel to the crest of

the Western Ghats upto Kanyakumari. In nortern India, it includes the hills of Himachal Pradesh, Uttarakhand, Chhattisgarh, eastern Maharashtra, and northern Andhra Pradesh. The regions lying to the west and south-west of this line have generally deficient rainfall where agriculture is largely vulnerable to drought. Along the Coromandal Coast of India the average annual rainfall is over 100 cm. In Punjab, Haryana and northern Rajasthan the mean annual rainfall is 60 cm or less. The lowest rainfall is recorded in western Rajasthan, the north-western parts of Gujarat, and Ladakh where it is less than 20 cm **(Fig. 4.22).**

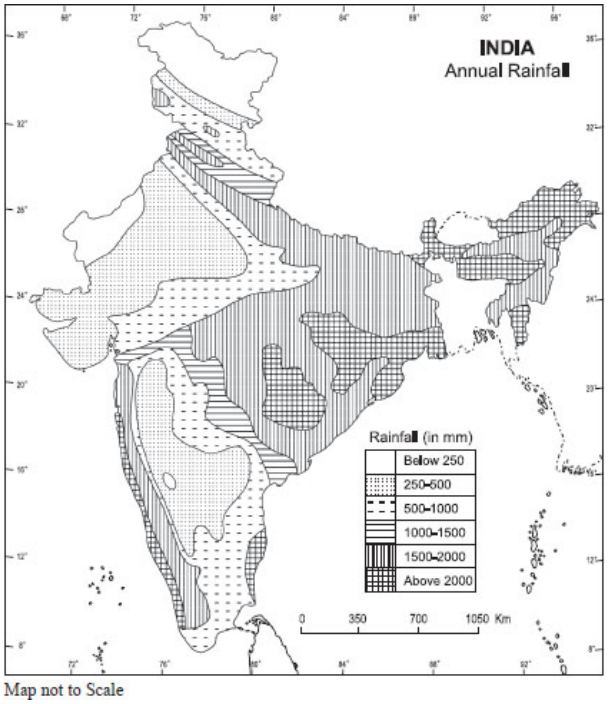


Fig. 4.22 Annual Rainfall (1950–2015) [*Source: Government of India (Ministry of Statistics & Programme Implementation)*]

VARIABILITY OF RAINFALL

The rainfall in India is highly variable. The actual rainfall of a place in a year deviates from its average rainfall by 10 to over 60%. The mean annual variability of rainfall in India has been plotted in **(Fig. 4.23)**.

It may be noted from **(Fig. 4.23)** that the highest variability is found in the areas where the average annual rainfall is the lowest. For example, the desert areas of Barmer, Ganganagar, Jaisalmer, Jodhpur, etc. have less than 20 cm of average annual rainfall. In these areas the variability of rainfall is around 60%. Contrary to this, in the areas where the average annual rainfall is over 200 cm (Mawsynram and Cherrapunji, Meghalaya Plateau), the annual variability of rainfall is less than 10%. The Western slopes of Western Ghats, the Lesser Himalayas, the Shiwaliks and the Tarai belt also record between 100–200 cm of average annual rainfall. The variability of rainfall in these regions is around 10 to 20%. Thus, there is an inverse relationship between the average annual rainfall and variability in rainfall.

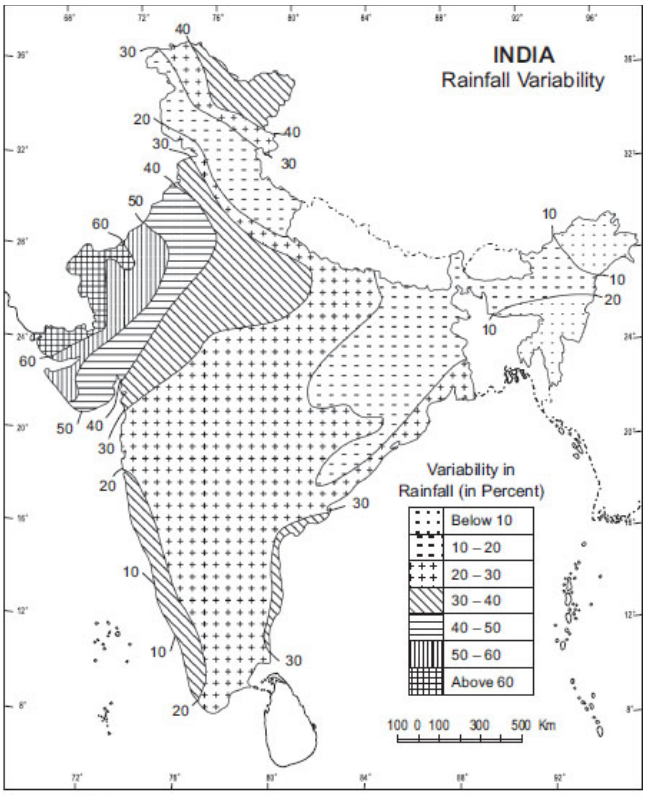


Fig. 4.23 Variability of Rainfall (1950–2015) [*Source: Government of India (Ministry of Statistics of Programme Implementation)*]

The variability of rainfall has a significant role in the agricultural operations and other economic activities of the country. The areas showing high variability of rainfall have chronic deficiency of water. Such regions are highly prone to droughts, floods and famines, while the areas with high average annual rainfall are less affected by droughts; though flood is a regular feature in flood prone areas.

CLIMATIC REGIONS OF INDIA

India is often referred to as a country with tropical monsoon type of climate. The large size of India, its latitudinal extent, the presence of the Himalayas in the north, and the Indian Ocean, Arabian Sea and Bay of Bengal in the south have resulted in great variations in the distribution of temperature and precipitation in the subcontinent of India.

A number of attempts have been made by climatologists, geographers and experts of agriculture to divide India into climatic regions. While some of these classifications have been suggested for world climates, others are exclusively applied to Indian conditions. Some of the important climatic divisions of India were made by the following experts:

1. H.E. Blandford, 1889
2. W. Koppen, 1918, 1931, 1936
3. C.W. Thornthwaite 1931,1933, 1948
4. L.D. Stamp and W.G. Kendrew, 1953
5. S.P. Chatterji, 1953
6. G.T. Trewartha, 1954

7. V.P. Subramanyam, 1956
8. B.L.C. Johnson, 1969
9. K.L Rao, et.al., 1971
10. R.L. Singh, 1971

A systematic study of the climatic divisions of India was attempted for the first time by H.E. Blandford—the first Director General of the Indian Meteorological Department—in 1889, who discovered that all types of climates found in the world are present within the subcontinent of India. This classification based on temperature and rainfall of a few selected stations of India was almost an overgeneralisation. A brief description of some of the important classifications of Indian climate has been given in the following:

KOPPEN'S CLASSIFICATION OF INDIAN CLIMATE

A Koppen's classification is empirical in nature based on climatic data. Koppen, for the delineation of climatic regions took into consideration (i) the mean monthly temperature, (ii) the mean monthly rainfall, and (iii) the mean annual rainfall. Koppen divided the country into three broad climatic zones:

1. Humid (A)
2. Arid (B)
3. Semi-Arid (C and D)

These three broad climatic divisions were sub-divided into sub-types on the basis of seasonal variations in the distribution pattern of precipitation and temperature for which the symbols S, W, m, f, w, s, c, and h have been used. Based on Koppen" s climatic scheme, India can be divided into the following nine climatic regions **(Fig. 4.24 and Table 4.2)**.

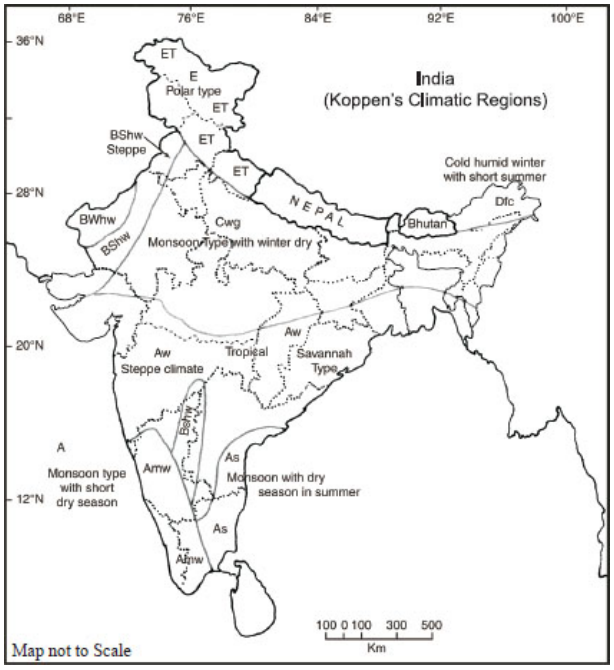


Fig. 4.24 Koppen's Climatic Regions

Table 4.2

Koppen's Classification of Indian Climate

Code	Climate Type	Features	Temperature	Rainfall	Regions of India
Aw	Tropical savanna	Tropical savanna grassland, monsoon deciduous vegetation	Hottest month–May (max. temp.: 40oC and min. temp.: 27oC) Coldest month temperature is greater than 18oC High ranges of annual and diurnal temperatures	Occurs during season of southwest monsoon–July to September Dry winters	Peninsular India, Jharkhand, Chattisgarh, Odisha, Andhra Pradesh, Telangana, Maharashtra, Karnataka, Purulia district of West Bengal
Amw	Tropical monsoon	Short winter season, luxurious evergreen rain forests	Annual temperature ~24oC to 27oC	Heavy rainfall over 200 cm during the season of southwest monsoon	Konkan, Malabar Coast, adjoining areas of Western Ghats, Plateaus of Tamil Nadu, southern areas of Tripura and Mizoram
As	Tropical moist	Dry summers	Mean monthly temperature >18oC	Average rainfall varies between 75 and 100 cm 75% of total rainfall occurs between September and December	Narrow zone along the Coromandel Coast
BShw	Semi-arid steppe	Hot and dry summers from March to May	Maximum temperature as high as 40oC	Seasonal rainfall (in summers)	Stretches over rain shadow zone of Karnataka, Tamil Nadu, eastern Rajasthan, Gujarat, south-western Haryana
Cwg	Mesothermal climate; Gangetic Plain Type	Hot desert type of climate; dry winters	Average temperature <18oC and temperature of coldest month >15oC Max. temperature is recorded in May or first half of June	Below 40 cm	Thar desert
Dfc	Cold humid winter	Short and humid summer and cold humid winter	Winter temperatures are about 10oC and summer temperature 30 to 35oC	Over 200 cm	Sikkim, Arunachal Pradesh, Assam, Tripura, Mizoram
E	Polar	Areas remain under ice during greater part of the year	Temperature of warmest month <10oC Temperature falls by 0.6oC for every 100 m rise in altitude in the Himalayas	Lesser rain on leeward side of the mountains and heavy rainfall on the well exposed slopes	Higher mountainous areas of Jammu and Kashmir, Himachal Pradesh, Uttarakhand

Code	Climate Type	Features	Temperature	Rainfall	Regions of India
ET	Tundra	Polar ice caps, soil is frozen to hundreds of metre depth, vegetation constitutes of mosses, lichens, dwarf trees, scattered woody shrubs	Average temperature of warmest month is between 0oC and 10oC		Higher altitudes of Ladakh, Kashmir, Himachal Pradesh, Uttarakhand

CLIMATIC DIVISIONS BY STAMP AND KENDREW

Professor L.D. Stamp and W.G. Kendrew divided India into several climatic divisions. This classification is arbitrary and subjective. Stamp used 18°C isotherm of mean temperature for January to divide the country into two broad climatic regions, namely:

A. the subtropical or continental zone lying to the north of this isotherm, and
B. the tropical zone lying to the south of this isotherm **(Fig. 4.25)**. It may be seen from **(Fig. 4.25)**
that the isotherm of 18°C runs roughly parallel to the Tropic of Cancer.

The two major climatic divisions, on the basis of rainfall, have been further divided into eleven regions:

A. The Subtropical or Continental climate has been divided into the following five divisions:
    1. The Himalayan Region
    2. The North-Western Region
    3. The arid lowland (dry plains)
    4. The region of moderate rainfall
    5. The transitional zone
B. Tropical India has been divided into the following six regions:
    1. Region of very heavy rainfall
    2. Region of heavy rainfall
    3. Region of moderate rainfall
    4. The Konkan Coast
    5. The Malabar Coast
    6. The Tamil Nadu region

A. Subtropical Climate

The subtropical climate has been subdivided into:

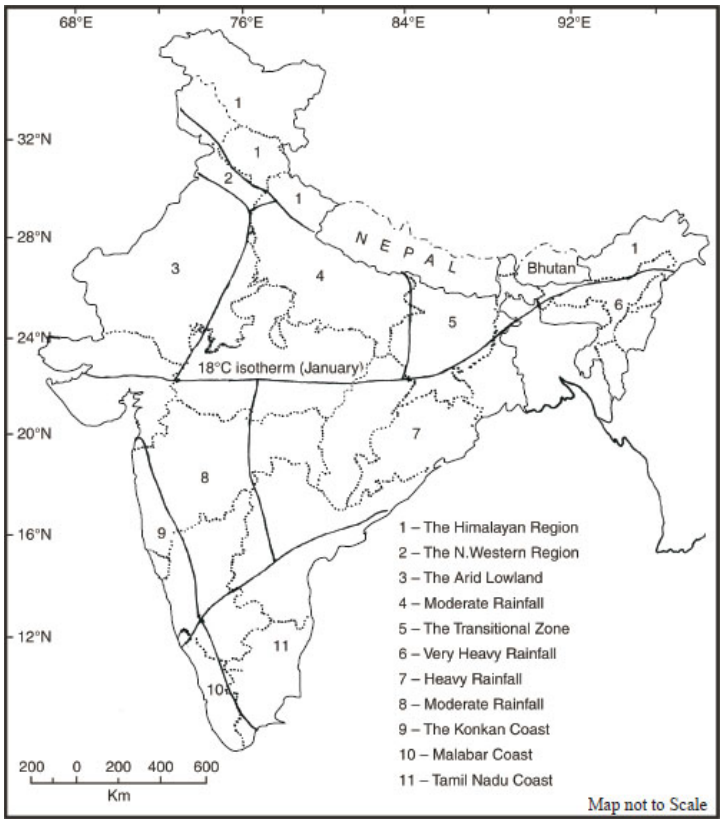


Fig. 4.25 Climatic Divisions (Stamp)

1. The Himalayan Region

Stretching over the mountainous parts of Jammu and Kashmir, Ladakh, Himachal Pradesh, Uttarakhand, Sikkim and Arunachal Pradesh, it has mild summers and cold winters. The regional distribution of temperature is largely controlled by the topography and aspect of slope. Upto a height of 2500 metres, the average temperature of winter season ranges from 4°C to 8°C, and the summer temperature varies from 12°C to 18°C. Rainfall is recorded from the South-West Monsoon in summer and from the western disturbances in the winter season. The amount of rainfall decreases from east to west. For example, in eastern Uttarakhand the average annual rainfall is over 200 cm, while in western parts of Jammu and Kashmir it falls below 150 cm. Snowfall and sleet are the common features in the Greater and Lesser Himalayas during the winter months (December to March).

2. The North-Western Region

This type of climate is found to the north-west of Satluj river where the average temperature of the winter season reads around 15°C. The mean minimum temperature in the month of January may occasionally fall below the freezing point at Amritsar. The mean maximum temperature in the month of May and June may cross 45°C. Hot winds (*Loo*) and dust storms during the scorching summers are the other features of this climatic region.

3. The Dry Plains of North-West India

This climate stretches over Rajasthan, Kutch, and south-western parts of Haryana. In this region the winter temperature ranges between 15°C and 25°C, but occasionally, the maximum temperature crosses 47°C in the months of May and June. The average annual rainfall is below 25 cm and occurs mainly during the months of July and August; the season of general rainfall.

4. Areas of Moderate Rainfall

This climatic division stretches over Punjab, Haryana, the Union Territory of Delhi, western Uttar Pradesh, eastern Rajasthan, and western Madhya Pradesh. The average rainfall in this climatic division varies between 40 to 100 cm. The mean temperature of January varies between 15°C to 18°C. Over 85 % of the total rainfall is recorded during the season of general rains. Delhi, Ludhiana and Meerut are typical examples of this climate.

5. The Transitional Zone

This climate is found in eastern Uttar Pradesh, western and north-western Bihar, and north-western Jharkhand. The average annual rainfall in this zone varies between 100 to 150 cm. Over 90% of the total annual rainfall is recorded during the season of general rains from the Bay of Bengal stream of the South-West Monsoon. The mean January temperature reads about l5°C while the mean maximum in the month of July reads over 40°C.

B. Tropical India

1. Region of Very Heavy Rainfall

This climatic divison stretches over Assam, Meghalaya, Nagaland, Manipur, Tripura and Mizoram. The average annual rainfall in these areas is over 200 cm. The heaviest rainfall in the world is recorded in this region at the stations of Mawsynram and Cherrapunji. Over 90% of the average annual rainfall is recorded during the season of the South-West Monsoon. There are significant variations in the mean monthly temperature of January and July owing to undulating and mountainous topography.

2. Region of Heavy Rainfall

This region covers West-Bengal, Odisha, Jharkhand and eastern parts of Andhra Pradesh. The average annual rainfall in this region varies between 100–200 cm. There is a general decrease in the amount of rainfall from east to west. The mean January temperature is over 18°C, while about 30°C is recorded during the months of June and July.

3. Region of Moderate Rainfall

This region lies to the east of the Western Ghats and includs Gujarat, south-western Madhya Pradesh, Maharashtra, Karnataka and greater parts of Telangana and Andhra Pradesh. Being in the rain-shadow area of the Western Ghats, this region receives relatively less rainfall of about 75 cm. The average temperature in the months of January and July varies between 18°C and 32°C, respectively.

4. The Konkan Coast

It stretches from the mouth of Tapi river to Goa. The average annual rainfall is more than 200 cm, of which over 90% is recorded from the Arabian stream of the South-West Monsoon. The mean January temperature remains around 24°C while the mean July temperature reads about 27°C. The average annual range of temperature varies between 3°C to 6°C depending on the distance from the coast and the equator. In general, the annual range of temperature increases from south to north.

5. The Malabar Coast

This climatic division lies between Goa and Cape Camorin (Kanniyakumari). The Malabar coast records over 250 cm of rainfall. The average annual temperature reads around 27°C with 3°C being the annual range of temperature. Kochi is a typical example of this region.

6. The Tamil Nadu Region

This region includes the greater parts of the state of Tamil Nadu and the Coromandal Coast. The average annual rainfall varies between 100–150 cm. Most of the rainfall is recorded during the

season of retreating monsoon (October to December). The average temperature for the month of January is 24°C, while the July temperature reads around 30°C.

TREWARTHA'S CLASSIFICATION OF INDIAN CLIMATE

G.T. Trewartha modified the climatic classification of Koppen in 1954. His classification of climate is empirical, based on the temperature and precipitation data. He also used English alphabets as the symbols to show the different types of climate. His scheme, applied to India, divides the country into four major climatic regions which are further sub-divided into seven meso-climatic divisions **(Fig. 4.26)**.

1. The Tropical Rainy Climate (Am)

The mean annual temperature in this climate is about 27°C and the mean annual rainfall exceeds 250 cm. The Western Ghats and the state of Tripura exhibit this type of climate.

2. The Tropical Savannah Climate (Aw)

In this climatic type, the mean annual temperature remains around 27°C. The mean annual rainfall is less than 100 cm. It has a marked dry season. The greater parts of the Peninsular India, excluding the coastal plains and the western slopes of the Western Ghats, have been included in this climate type.

3. The Tropical Steppe Climate (BSw)

The mean annual temperature in this climatic zone is about 27°C. It covers peninsular India east of the Western Ghats. In fact, it is the rain-shadow area of the Western Ghats including parts of Maharashtra, Telangana, Andhra Pradesh, Karnataka, and Tamil Nadu.

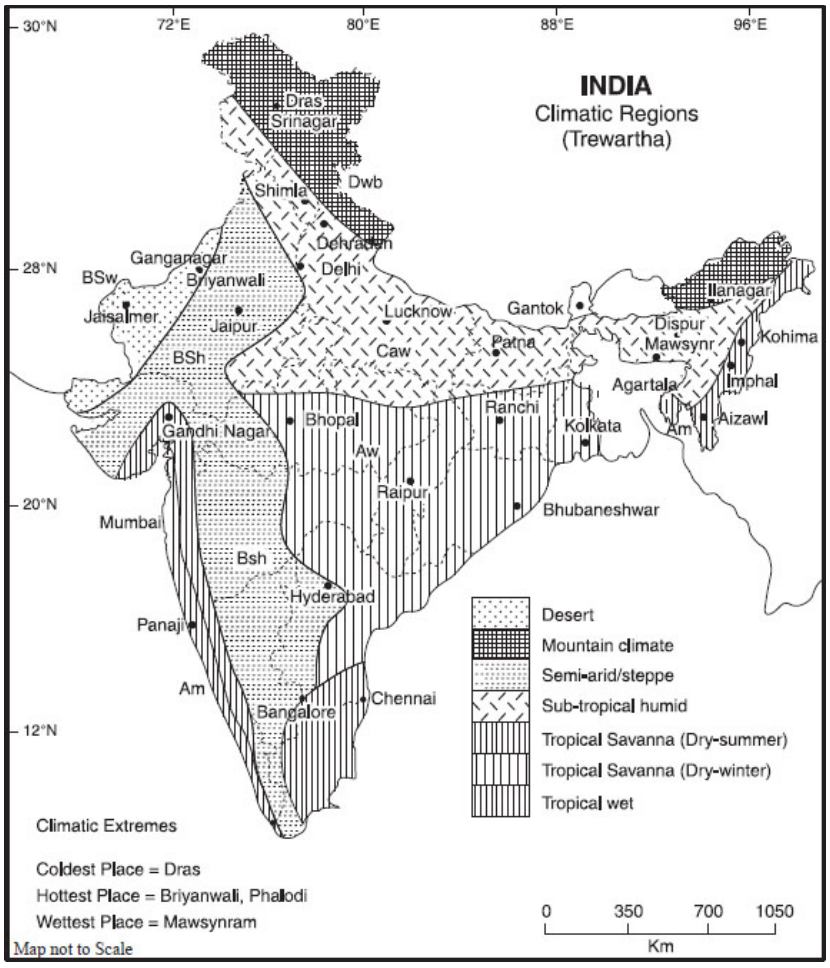


Fig. 4.26 Trewartha's Climatic Regions

4. The Sub-tropical Steppe Climate (BSh)

This is a semi-arid climate stretching over parts of Gujarat, eastern Rajasthan, Maharashtra, Andhra Pradesh, and southern Haryana. The mean annual temperature in this climatic region is over 27°C, though the mean monthly January temperature reads only about 15°C. The annual range of temperature is significantly high. The mean annual rainfall varies between 60 to 75 cm.

5. The Tropical Arid Climate (BWh)

It lies to the west of the Aravallis, stretching over the Thar Desert. The mean maximum temperature during the months of May and June occasionally crosses 48°C. The mean annual rainfall is less than 25 cm. The lowest rainfall in the country is recorded in this climate in the district of Ganganagar. Consequenty, the natural vegetation is in the form of thorny bushes.

6. The Humid Sub-tropical Climate (Caw)

This climate occupies the greater parts of the Great Plains of India, stretching from Punjab to Assam. The mean January temperature for the coldest month of January is less than 18°C, while the mean maximum in the summer season may cross 45°C. The average annual rainfall varies from 250 cm in the east to only about 65 cm in the west.

7. The Humid Mountain Climate (Dwb)

This climate is found in the hilly parts of the states of Jammu and Kashmir, Himachal Pradesh, Uttarakhand, Arunachal Pradesh, and the other hilly parts of north-east India. In this climate, the

average temperature for the summer season reads around 17°C, while the average January temperature is generally around 8°C. The average temperature of all the months is however, closely influenced by the topographical features and slope. In general, the rainfall decreases from east to west. The Western Himalayas record some amount of rainfall from the western disturbances during the winter season.

CLIMATIC DIVISIONS OF INDIA BY R.L. SINGH (1971)

Professor R.L. Singh modified the climatic divisions of Stamp and Kendrew in 1971. Prof. Singh has divided India into 10 major climatic divisions. His classification is quantitative as well as qualitative, largely based on the amount of rainfall and temperature. In fact, the modifications have been done on the basis of variations in temperature **(Fig. 4.26(b)).** A brief account of Singh's classification has been presented in the following section:

I. The Humid North-East

This climatic division includes the whole of north-east India (except Tripura), Sikkim and north-western West Bengal. The average annual rainfall in this climatic division is more than 200 cm. The mean July temperature varies between 25°C and 33°C, while mean January temperature reads between 10°C and 25°C.

2. Humid Sahyadri and Western Coast

This climatic zone includes the Konkan and Malabar coasts from the mouth of Tapi river to Kanyakumari. The average annual rainfall in this zone is about 200 cm. The mean July temperature ranges between 25°C and 30°C, while that of January ranges between 18°C and 28°C.

3. The Humid South-East

This climate spreads over the greater part of West Bengal, Jharkhand, Chhattisgarh, Odisha and parts of Andhra Pardesh. It records an average annual rainfall between 100–200 cm. The mean July temperature varies between 25°C to 35°C, while the mean January temperature reads between 12°C–27°C.

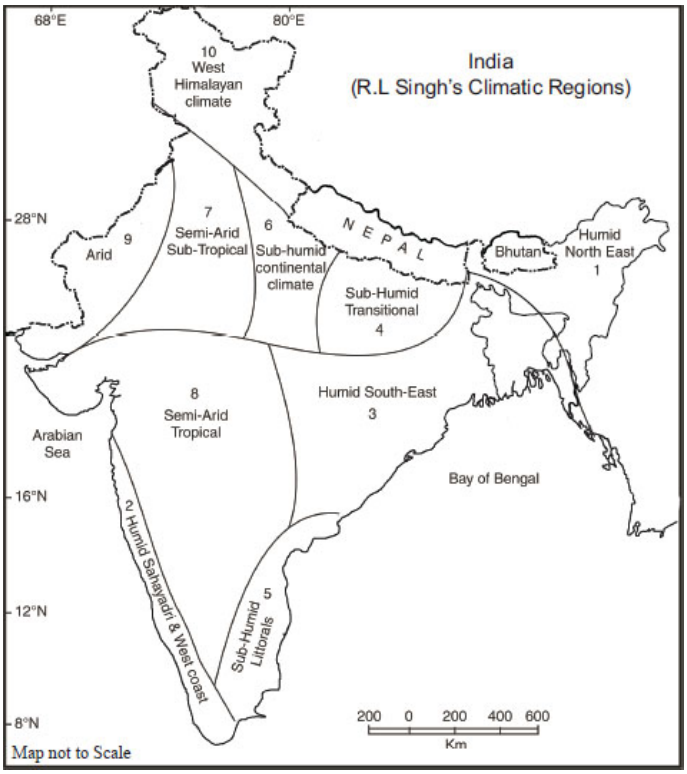


Fig. 4.26(b) R.L. Singh's Climatic Divisions

4. The Sub-Humid Transition

This climate includes those parts of the Gangetic Plain in which the average annual rainfall is between 100 and 200 cm, Average temperature for July varies between 25°C, and 40°C and January temperature between 10°C and 25°C.

5. The Sub-Humid Littoral

This type of climate is found along the Coromandal coast. The mean annual rainfall varies between 75–150 cm. The May temperature reads between 28°C and 38° C, while the January temperature ranges between 20°C–30°C.

6. The Sub-Humid Continental

Covering mostly the Upper Ganga Plain, it receives average annual rainfall between 75–150 cm. The average temperature for July ranges between 25°C and 40° C, and the average January temperature recorded varies between 17°C and 25°C.

7. Semi-Arid Subtropical

This type of climate spreads over eastern Rajasthan, Haryana, and Punjab. The average annual rainfall in this climate varies between 25 to 75 cm. The average temperature for January ranges between 25°C to 28°C, while the average temperature for January ranges between 15°C and 25°C.

8. The Semi-Arid Tropical

It covers the largest area of the central and western Peninsula covering Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Telangana, and western Andhra Pradesh. The average annual rainfall in this region varies between 50 and 100 cm, the mean monthly temperature in July varies between 26°C and 40° C, while the mean average temperature for January reads between 15°C and 28°C.

9. Arid-Climate

Covering the greater part of Kutch, western Rajasthan, and south-western Haryana, it receives rainfall less than 25 cm. In fact, the Thar desert of India has been included in this type of climate. June is the hottest month here in which the average temperature varies between 30°C and 40°C. In January, the mean minimum monthly temperature fluctuates between 10°C and 15°C. It is the most drought affected area of the country in which every third year is a drought year.

10. The Western Himalayas

It includes the mountainous areas of Jammu and Kashmir, Ladakh, Himachal Pradesh, and Uttarakhand. The average annual rainfall in this climatic division is up to 150 cm. The rainfall however, decreases from east to west and south to north. Rainfall from the western disturbances during the winter season is also a unique characteristic of this climatic zone. This small amount of rainfall is highly beneficial for the standing ***Rabi*** (winter) crops. The mean July temperature ranges from 5°C to 30° C, while the average temperature of January varies from 0°C to 5° C.

DROUGHTS

Drought is a continuous and lengthy period during which no significant rainfall is recorded. In India, the Meteorological Department of India defined drought as a period of at least 22 consecutive days on none of which is there more than 0.25 mm (0.01 inch) of rainfall. This definition however, does not apply to the whole of India. In areas like Mawsynram and Cherrapunji (recording over 1200 cm of average annual rainfall) even one week recording less than 0.25 mm of rainfall may be considered as a drought period. In general, the areas recording less than 60 cm of rainfall annually and in which the variability of rainfall is more than 20% are the drought prone areas in India. Areas where the variability of rainfall varies between 20 to 60% (like Rajasthan, west of Aravallis, and east of the Western Ghats) are the chronic drought prone areas. In India, droughts are more frequent in the areas where the average annual rainfall is less than 60 cm and the variability of rainfall is over 20%, provided canal and tube-well irrigation is not available. The variability of rainfall over greater parts of the country is more than 25%, and thus over 50% of the country is vulnerable to droughts.

According to the Ministry of Agriculture and the Ministry of Environment, a drought prone area is defined as one in which the probability of a drought year is greater than 20%. A chronic drought prone area is one in which the probability of a drought year is greater than 40%. A drought year occurs when less than 75% of the average annual rainfall is recorded.

On an average, one in every five years is a drought year in India, while in western Rajasthan every third year is a drought year. Although over 60% of the area has high variability of rainfall, there are three areas which are highly vulnerable to droughts **(Fig. 4.27)**. The drought prone areas of India are:

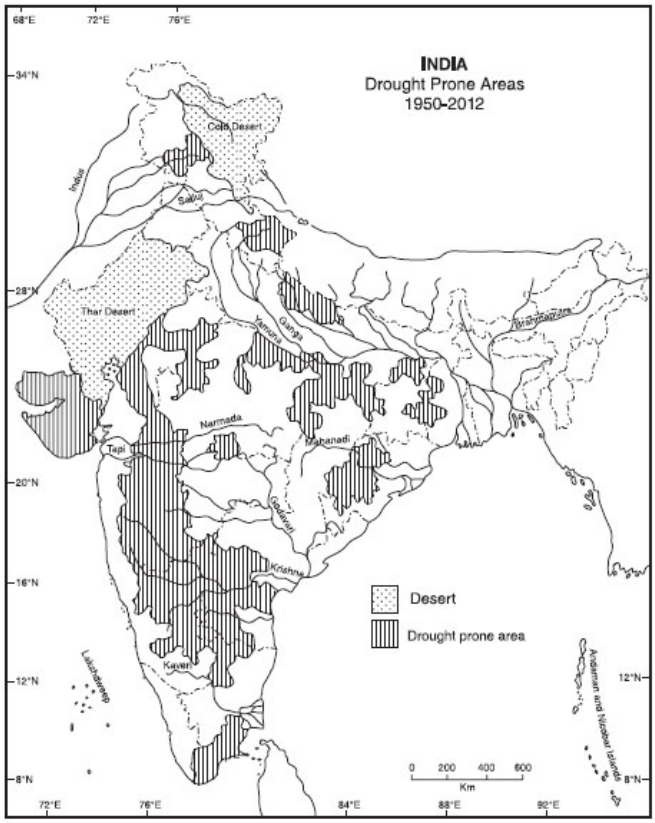


Fig. 4.27 Drought Prone Areas (1950–2012)

1. **The Arid and Semi-arid Areas of Rajasthan:** There is a contiguous region, covering greater parts of northern Gujarat, Rajasthan, south-west Haryana, southern parts of Punjab, western Madhya Pradesh, and parts of the Agra Division of Uttar Pradesh. This drought prone area, having a variability of rainfall of over 40%, stretches over 65 lakh sq km. Parts of these contiguous drought prone areas, irrigated by canals and tube-wells are, however, less affected by droughts.
2. **The Rain-shadow Areas of Western Ghats:** This is the area lying to the east of the Western Ghats (Sahyadri Mts), stretching from Jalgaon (Mahrashtra) to Hyderabad and Mahbubnagar (Telangana), Bidar, Bijapur, Gulbarga, and Raichur of Karnataka, and Coimbatore, Madurai, Salem and Remanathapuram of Tamil Nadu. The variability of rainfall in this region varies between 30 to 40%. It is in this areas that several thousand farmers have committed suicide during the last decade.
3. **Other Drought Prone Areas:** There are isolated tracts, covering an area of about one lakh sq km in different parts of the country, which are drought prone areas. These areas include Kalahandi region of Odisha, Purulia District of West Bengal, Palamu plateau of Jharkhand, Mirzapur Plateau, and Bundelkhand (U.P.), Baghelkhand (Madhya Pradesh), Coimbatore, Madurai and Tirunelveli District of Tamil Nadu. These scattered pockets cover over one lakh sq km.

Though droughts are recurring physical phenomena and their severity was more during the medieval period, some of the serious droughts of the 20th century have been briefly described in the following section:

The Maharashtra drought of 1965–66, the Bihar drought of 1966–67, the Kalahandi drought of 1996–97 and the continuous deficient rainfall in Andhra Pradesh, Karnataka and western Tamil Nadu during the last decade forced a number of the farmers to commit suicide. The monsoon failure in 2009 over greater parts of the country resulted into drought. Consequently the Kharif cropped area decreased by about 40%. The drought of 2016 is historic in which thousands of farmers committed suicide. Upto 33 crore people in 2.5 lakh villages of 266 districts in 11 states were affected in drought in 2015-2016. (*The Hindu, Business Line; November 8, 2018*)

The adverse effects of droughts can be minimised by an extension in the irrigation projects, adoption of sprinkle irrigation, water harvesting, suitable rotation of crops, deviation from the more water requiring crops, and dry-farming techniques. If all these steps are taken together, the miseries of the farmers and inhabitants of the drought prone areas may be reduced substantially.

Droughts are disasters which affect the people and economy of the country adversely. Their frequency, according to experts of climatology, is likely to increase in future due to greenhouse gases, climatic change and ecological imbalances created by interference of man.

The years of 1965–67, 1972–73, 1979–80, 1985–88, 1999, 2000, 2001, 2002, 2009 and 2016 have been the drought years in India after Independence. According to the experts the drought of 2016 was one of the worst droughts in the last one hundred years. In these years, the states of Haryana, Punjab, Uttar Pradesh, Uttarakhand, Bihar, Madhya Pradesh, Gujarat, Tamil Nadu, Karnataka, Jharkhand, Odisha, parts of Kerala, and the whole state of Rajasthan were the worst affected.

Drought Management

The government of India has paid enough attention to droughts and drought affected areas right from the first Five Year Plan. Before 1987, the government tried to provide relief to the people of the drought affected areas by providing employment opportunities to the affected population and distribution of food-grains through Public Distribution System (PDS). The contingent drought relief expenditure imposed a serious strain on public finances as huge amounts had to be divested from development for undertaking relief projects. In some states like Rajasthan, the drought relief outlays exceeded the developmental outlays.

In order to overcome this problem, the drought management strategy was adopted in the early 1970s. The drought management approach differed from drought relief approach with regard to objectives, reliance of early warning indicators and timing of public intervention. The objective of relief approach was to protect entitlement of the affected population by ensuring physical and economic access to food through relief projects and public distribution of food-grains. As against this, the drought management approach aimed at ensuring entitlement to produce food so as to obviate the need for taking up ad hoc relief projects. While drought relief approach relied on socio-economic indicators like crop production data, price rise, migration of the drought affected people and increased rate of crimes for drought declaration and intervention; drought management approach relied on hydro-agriculture indicators, like rainfall, water level of reservoirs and progress of cropping pattern to detect early signs of developing drought situation. While drought relief approach enabled the government to intervene only in the months of November and December when the rainy season is over and the kharif crops have been harvested, the drought management approach enabled the government to intervene in the monsoon season itself.

According to the Committee of Disaster Management (2002), drought management is generally done by focusing on employment generation, water conservation, and power supply, standing crop saving and public distribution of supplies of essential commodities.

Drought Prone Area Programme (DPAP)

The drought prone area programme which covers 615 blocks, spread over ninety districts in the western, central and southern parts of the country, is an integrated area development programme in agricultural sector. It is the earliest area development programme launched by the Central government in 1973-1974 to tackle the problems of the drought hit areas of the country.

Main Objectives:

(i) to make judicious and scientific utilisation of agricultural land, water and livestock resources,

(ii) to enhance and stabilise the income of the people of drought prone areas, particularly of the weaker section of society, and

(iii) restoration of ecological balance.

Some of the important elements of the programme include:

(a) Integrated watershed management and management of water resources.

(b) Soil and moisture conservation measures.

(c) Afforestation with special emphasis on social and agro-forestry.

(d) Development of pasture lands and forest range management in conjunction with development of sheep husbandry.

(e) Livestock development and dairy development.

(f) Restructuring of cropping patterns and changes in agronomic practices.

(g) Adoption of scientific rotation of crops with an emphasis on the leguminous (pulses) crops.

(h) Development of subsidiary occupations.

(i) Water harvesting.

(j) Development of minor irrigation projects.

(k) Construction of underground canals and lined canals.

(l) Desalination of sea-water for irrigation and domestic use.

(m) Diversification of agriculture.

(n) Development of cottage and household industries.

(o) Development of alternate sources of energy (solar, wind and biogas) for domestic and industrial purposes.

(p) More research is required to increase agricultural productivity in the dry farming regions.

Table 4.3

Current Status of DPAP

Year	Regions
1977-78	Hot areas of Rajasthan, Gujarat, Haryana, cold deserts of Jammu and Kashmir and Himachal Pradesh
1989	Integrated Watershed Development Programme was launched under the aegis of National Wasteland Development Board for the development of wastelands on water shed basis
1995	Guidelines for the watershed development were framed and brought into practice on April1, 1995.

Year	Regions
2001 and 2003	Guidelines for the watershed development were renamed as 'Haryali Guidelines'
2008	Common guidelines for watershed development programme have been made effective since April1, 2008
2009	Three watershed programmes (DPAP, DDP, IWDP) have been consolidated under a comprehensive programme called 'Integrated Watershed Management Programme' covered under the Ministry of Rural Development.

These steps, if taken together, can go a long way in minimising the miseries of people living in the drought prone areas of the county.

Some of the important achievements of the Drought Prone Area Programme include the Indira Gandhi Canal Project, Sardar Sarovar Project (Narmada), and the Central Arid Research Institute, Jodhpur to promote drought resistant plants, trees and crops.

FLOODS

Floods occur when peak discharge exceeds channel capacity, and this may be brought about naturally by intense precipitation, snow and ice-melt, storm surges in coastal regions, and the rifting of barriers, such as ice-dams, or by the failure of man-made structures, by deforestation, urbanisations, (which reduce infiltration and interception), and by engineering works such as land drainage or the straightening of embankments of rivers.

Flood has also been defined as a state of high water level along a river channel or coast that leads to inundation of land which is normally submerged. Flood is an important component of hydrological cycle of a drainage basin. In fact, droughts and floods are the two extremes of the hydrological cycle. While droughts occur due to the failure of rainfall; floods generally occur in the event of excessive rainfall. Thus, flood is a natural hazard which occurs in response to heavy rains and it becomes a disaster when it inflicts heavy loss to life and property.

Causes of Floods

The main factors responsible for the occurrence of floods are: (i) Meteorological, (ii) Geomorphic, and (iii) Anthropogenic.

Meteorological factors include heavy rainfall, snowfall due to tropical cyclones, and cloud burst. Most of the floods in India are the result of heavy precipitation, especillay during the season of south-west Monsoon.

Geomorphic factors like the large catchment area of a river, gentle slope and the low gradient of river course, and poor drainage also lead to floods in the Northern Plains and Coastal Plains of India. The Jodhpur Barmer, and Bikaner flood in July 2006 mainly occurred due to cloud burst and poor drainage in the region.

The Anthropogenic causes including deforestation, encroachment for agriculture into pastures and forest areas, shifting cultivation, unscientific rotation of crops, bursting of dams and embankments, urbanisation, and construction of houses in the river beds are also responsible for

a number of floods in the country. The July 2005 flood in Mumbai was mainly due to the construction of houses and structures in the bed of the Methi River.

Flood Prone Areas of India

Out of the country's total geographical area (329 millions hectares) about 45 million hectares (13.6 %) is prone to floods (2012). The flood prone areas of India have been plotted in **(Fig. 4.28).** An examination of **(Fig. 4.28)** shows that the flood prone areas are well scattered in different parts of the country, ranging from the heavy rainfall areas to the scanty rainfall areas. Flood is, however, a recurring feature during the rainy season in Assam, West Bengal, Bihar, Uttar Pradesh, Haryana and Punjab. Floods also occur along the eastern coastal plains, the valley of Kashmir, the lower reaches of the rivers discharging their water in the Bay of Bengal and the Gulf of Khambat (Arabian Sea). The flood affected areas of India are as under:

1. The Ganga River Basin
2. The Brahmaputra River Basin
3. The Punjab Haryana Flood Plain
4. Flood Prone Areas of the Coastal Plains.

1. The Ganga River Basin

Stretching over an area of about 861,400 sq km, the Ganga and its tributaries drain the states of Uttarakhand, Himachal Pradesh, Haryana, Uttar Pradesh, Bihar, West Bengal, Jharkhand, Chhattisgarh, Madhya Pradesh, and Rajasthan. Its large left hand tributaries are Sharda, Kali, Gomti, Ghagra, Gandak, Kosi, and Mahananda; while the right hand tributaries include, the Yamuna, Chambal, Sind, Betwa, Ken, Son, and Damodar. Thus, from the mighty Himalayas and the northern Peninsular India, the Ganga and its tributaries carry enormous quantity of water. These rivers, due to heavy rains in the rainy season and melting of snow in the spring season, cause heavy floods in the middle and lower reaches of their catchments. Kosi (means *cuise-Kosna*) has often been called the "Sorrow of Bihar". Its flood in July 2008 caused heavy loss to life and property.

The cloud burst on 16 June, 2013 in Uttarakhand resulted into disaster in which the sacred places such as Kedarnath, Rambara, Rudra-Prayag, Srinagar, Joshimath, etc., were seriously damaged. Except the Kedarnath temple almost the entire settlement was burried under sediments, boulders and coarse-sand.

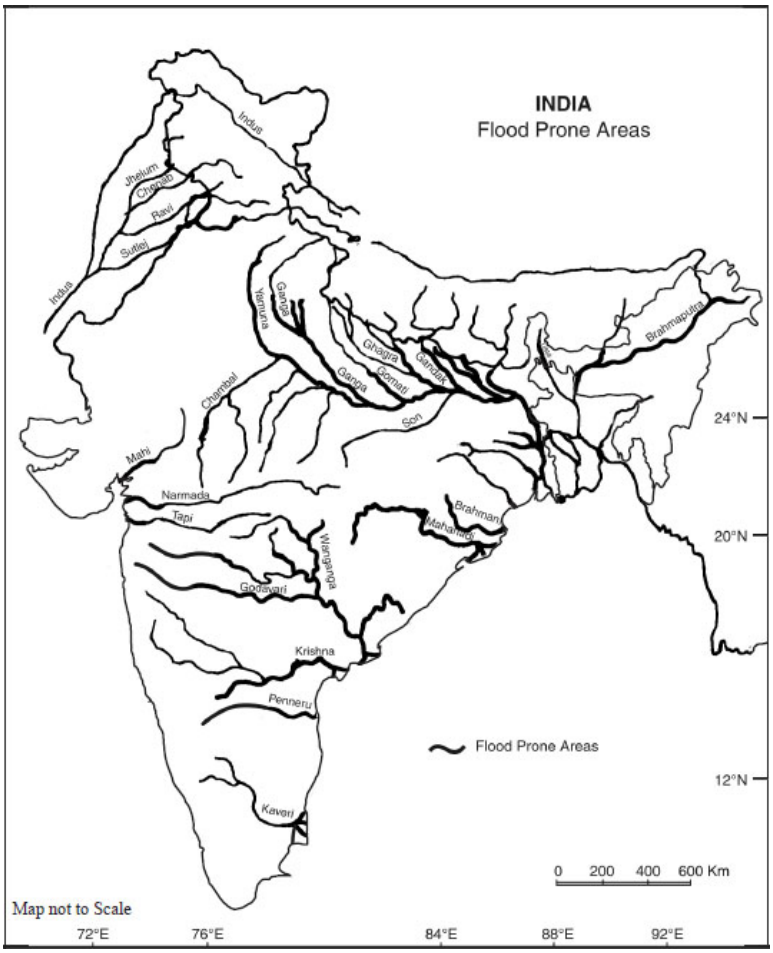


Fig. 4.28 *Flood Prone Areas (1950–2012) (Reference: World Meteorogical Organization)*

Table 4.4

States in India Liable to Floods

State	Area Liable to Floods (Million ha)
Uttar Pradesh	7.34
Bihar	4.26
Punjab	3.70
Rajasthan	3.26
Assam	3.15
West Bengal	2.65
Haryana	2.35
Orissa	1.40
Andhra Pradesh	1.39
Gujarat	1.39

The state of Uttar Pradesh alone constitutes about 22% of the flood prone area of the country. The river channels of these rivers are increasingly getting shallower, especially in their middle and lower reaches. In the event of heavy rainfall, these rivers cross their banks and inundate the neighbouring khadar areas.

In Bihar, floods are largely confined to the northern part of the state where occurrence of flood is almost an annual feature during the season of general rains. The rivers like Burhi Gandak, Bagmati, Kosi, and the lower reaches of the Mahananda spill over their banks and inundate the low lying khadar areas. The Kosi flood 2009, created havoc in the districts Saharsa, Medhapur, and Khagaria of Bihar. Many of the cities and towns were submerged under water causing enromous loss to life and property.

In West Bengal, the southern and central parts are frequently flooded during the season of south-west monsoon. The Mahananda, Bhagirithi, Ajoy, and Damodar (Sorrow of Bengal) are often in floods in rainy season. Even after the formation of the Damodar Valley Corporation (DVC), there occur occasional floods in the rivers.

2. The Brahmaputra River Basin

The maximum discharge of water among the Indian rivers is in the Brahmaputra River. The Assam Valley is considered to be one of the worst flood affected areas of India. In fact, in Assam, floods are almost an annual feature. The *Jaldakha*, *Teesta*, and *Torsa* rivers often overflow their banks to flood the regions of northern West Bengal, Manipur and Sikkim. The main cause of floods in the Brahmaputra basin are: (i) heavy and torrential rainfall—during the rainy season over 200 cm of rainfall is recorded over greater parts of its middle and lower reaches, (ii) silting of the river course due to heavy soil erosion, (iii) landslides, (iv) heavy pressure of population, and (v) shifting cultivation on the surrounding hilly areas. All these factors collectively result in flooding of the vast areas in and around the Brahmaputra Valley.

All the districts of Assam are inundated almost every year. The worst affected areas in Assam are “Majuli”—India’s largest river island, Dhubri, Kokrajhar, Barpeta, Guwahati, Mangaldoi, Sibsagar, Jorhat, Dibrugarh, and Tezpur. According to one estimate, about 80 lakh hectares, i.e., over 45% of the total area of Assam, is flood prone.

3. The Punjab Haryana Flood Plain

This area falls under the scanty rainfall recording areas of India. Despite low rainfall, Punjab and Haryana are adversely affected by floods because of the inadequate surface drainage. In Punjab and Haryana, the rainfall water is waterlogged. Consequently, the ill-drained areas are inundated. The rivers, like Satluj, Beas, Ghaggar, and Markanda record floods almost annually. The main reasons of floods in Punjab and Haryana are: (i) deforestation in the Lower Himalayas and the Siwalik, (ii) increase in soil erosion in the upper reaches of the rivers leading to silting of river beds, (iii) construction of structures in the ***Bet*** (Khadar) areas of the rivers, (iv) unscientific land use, cropping patterns and rotation of crops, and (v) obstruction of natural drainage by the construction of roads, railways and human establishments.

Floods are frequent in the Ghaggar plain of Punjab, Haryana and also Rajasthan. The Ghaggar river is seasonal in character. Its course disappears in the sand-dunes of Rajasthan after crossing Punjab and Haryana.

In the Valley of Kashmir, floods are caused by the Jhelum river. Tawi river is the main cause of floods in Jammu. In 2004, the release of huge quantity of water in Satluj river from the Rakas Lake threatened many areas of Himachal Pradesh.

4. Flood Prone Areas of the Coastal Plains

The coastal areas of Odisha, Andhra Pradesh, Tamil Nadu, Kerala, Maharashtra and Gujarat are included in this region. The magnitude of floods becomes serious in the lower reaches of the Mahanadi, Godavari, Krishna, and Kaveri. The beds of these rivers are shallow which inundate after a heavy downpour in their catchments. Indiscriminate felling of trees in the catchment areas of the main rivers and their tributaries has accelerated soil erosion and silting of river beds. The estuaries of Narmada and Tapi lead to floods at the occurrence of spring tides. The coastal areas

also submerge due to the surge of sea water at the time of tropical cyclones in the Bay of Bengal and the Arabia Sea. The occurrence of Tsunamis, as on 26 December, 2004, lead to serious damage to life and property of the coastal areas. The small swift moving rivers of the western coastal plains caused considerable damage when in spate.

Table 4.5

Damages due to Heavy Floods/Rains from 1953-2016

	1953	1963	1973	1983	1993	2003	2013	2016
Area affected (in m. ha)	2.290	3.490	11.790	9.020	11.439	6.120	7.546	7.065
Population affected (in million)	24.280	10.930	64.080	61.030	30.409	43.201	25.927	26.555
Crop damage area (in m.ha)	0.930	2.050	3.730	3.290	3.206	4.268	7.484	6.658
Houses damaged (in numbers)	264,924	420,554	869,797	2,393,722	1,926,049	775,379	699,525	278,240
Cattle lost	47034	4572	261,016	153,095	211,193	15,161	60,196	22,367
Human Life Lost	37	432	1349	2378	2864	2166	2180	1420
Damage to Public Utilities	2.9	2.74	88.489	873.429	1445.534	3262.154	38937.843	1507.926
Total Damages (Crops, Houses, Public Utilities in Rs. crore)	52.400	36.611	569.016	2491.606	3282.485	11,325.866	47,348.751	5675.325

Source: Central Water Commission, Flood Forecast Monitoring Directorate, Government of India

It is to be noted that the maximum damage to houses (3,95,9191), damage to the crops (Rs 17,043.948 crore), and gross total damage of Rs. 57,291.098 occurred during the 2015 floods. Though, the maximum population (70,450) was affected in 1978 .

Flood Control and Management

The main steps taken by the government of India are briefly given below:

1. Flood Forecasting

The Central Water Commission started flood forecasting since November 1958, when the first forecasting station was established at the Old Railway Bridge of Delhi. Since then, it has extended the flood-forecasting stations to cover almost all the major flood prone rivers of the country. At present, there are 175 flood forecasting stations on different major and minor rivers of the country. The centre issues daily flood forecasts and warnings throughtout the season of general rains.

The Central Flood Forecasting Organisation monitors floods all over the country and issues warnings. The main flood forecasting centres are located at Surat (Tapi), Bharuch (Narmada), Varanasi, Buxar, Patna, Hathidah, and Aimabad (Ganga), Dibrugarh and Guwahati (Assam),

Jalpaiguri (Tista), Delhi (Yamuna), Lucknow (Gomti), Bhubaneshwar (Subarnrekha, Burtha, Balang, Brahmani and Baitarni), Sahibi (Rajasthan), and Gandhi-Sagar (Chambal). In addition to these, there are 157 flood forecasting stations from which about 6000 forecasts are issued every year.

2. Reduction of Runoff

Reduction in runoff is one of the most effective methods of flood control. Runoff can be reduced by inducing and increasing the infiltration of the surface water into the ground. Afforestation in the catchment areas of rivers is a very effective step to check the runoff and to increase the percolation of water.

3. Construction of Dams and Reservoirs

The floods can be reduced by construction of dams and reservoirs. Dams have the capacity of holding huge quantity of water during the flood period and help in reducing the menace of floods. Water stored in reservoirs of dams can be released under controlled conditions, depending upon the carrying capacity of the river channel. A number of dams across the Damodar, Kosi, Satluj, Rihand, Mahanadi, Krishna, Kaveri, Chambal, Narmada, Tapi, and Sabarmati have been constructed to check floods.

4. Control of Flood Levels

A number of steps may be taken to control the flood levels. Among these steps, stream channelisation, channel improvement, and flood diversion are important.

5. Construction of Embankments

Construction of embankments is a very effective method against floods. Construction of embankments has been taken in a big way after the establishment of the Central Water Commission and the Flood Control Commission. Over 12,000 km long embankments have been constructed after the establishment of the Flood Control Commission. Most of these embankments have been constructed along the Brahmaputra, Ganga, Yamuna, Satluj, Beas, Ravi, Mahanadi, Kosi, Godavari, Krishna, and the Kaveri.

6. Flood Plain Zoning

Flood plain zoning is another very effective method of flood management. It is based on information regarding land use in flood plains. Some areas are more prone to floods than others. Construction of buildings, factories, etc. in the zone adjacent to the river channels should be prohibited. The areas occasionally flooded after a few years should be under green-belts in which social forestry is a priority.

7. Other Measures

The following additional measures are suggested by the Flood Control Commission: (i) restriction on indiscriminate felling of trees in hilly regions, (ii) protecting one kilometre tract along the major rivers for massive afforestation, where agriculture and house construction prohibited, (iii) Regular dredging of river beds, (iv) formation of National Water Grid through which flood waters could be diverted to dry areas through diversion channels or could be stored in reservoirs, (v) to develop suitable drainage channels in water-logged areas, and (vi) engineering effective methods to protect the coastal areas from tides and sea-surges.

Flood Control Programme and Strategies

The National Flood Control Programme was launched in India after the devastating flood of 1954. The programme consists of three phases which have been briefly described in the following section:

1. **The Immediate Phase:** This phase extends over two years and includes the collection of basic hydrological data and execution of immediate flood protection measures like

construction of embankments, improvement of river channels and raising the vulnerable villages above the flood level.

2. **The Short-Term Phase:** This phase lasts for the next five years. In this phase there is emphasis on improvement of surface drainage, establishment of effective flood warning system, shifting or raising of villages above flood level, construction of diversion channels, construction of embankments and raised platforms to be used during the period of floods.
3. **The Long-Term Phase:** The long-term phase includes construction of dams and storage reservoirs, digging large diversion channels, taking suitable steps for land-use improvement, and soil conservation in the catchment area of the main river and its tributaries.

In order to overcome the problem of floods in the country, the Government of India has set up a National Flood Commission ***(Rashtriya Barh Ayog)***. This Commission has taken a holistic view of the flood problem. Many multi-purpose projects and large dams have been constructed to overcome this problem. Recently, the Brahmaputra River Board has been constituted to control floods in the Brahmaputra Valley.

Ministry of Environment, Forests and Climate Change (MoEFCC) is the nodal agency that takes care of implementation of India's environment and forest policies, Programmes for the conservation of the natural resources like lakes, rivers, biodiversity, forests and wildlife.

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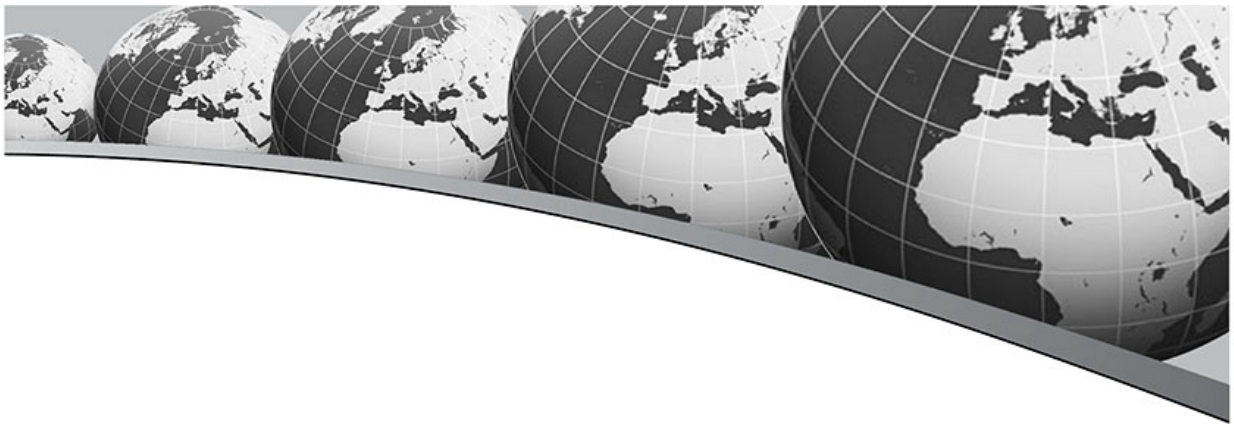

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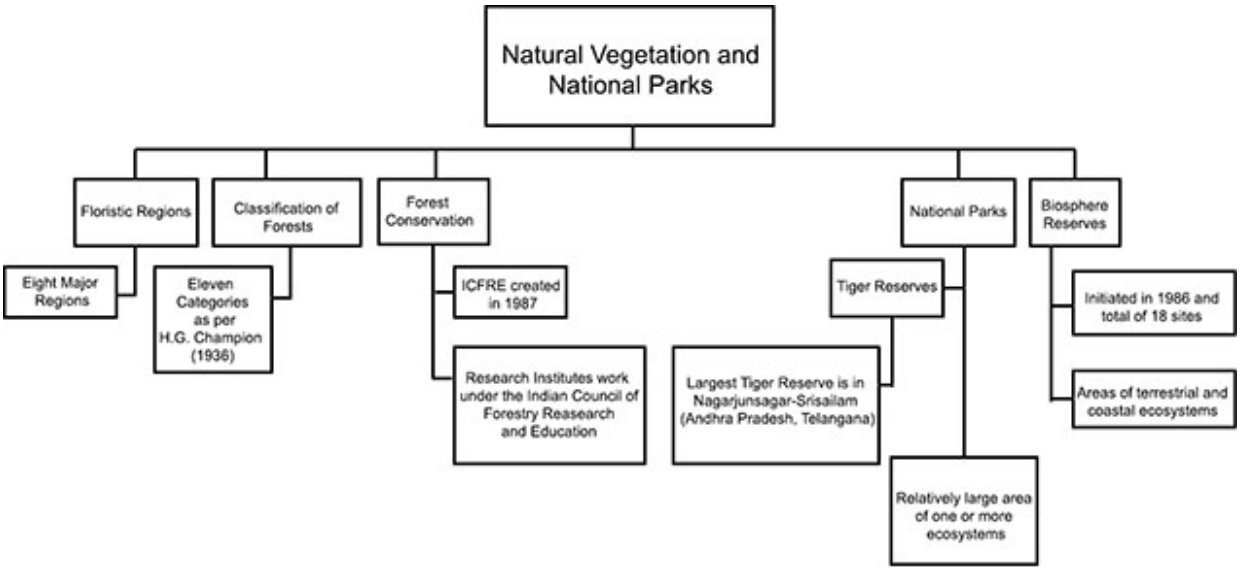
Natural Vegetation and National Parks



INTRODUCTION

Natural vegetation of India reflects a state of perfect harmony with the relief and climatic conditions of the subcontinent. In fact, this correspondence is so perfect that if one superimposes the two maps showing the annual rainfall and the altitude above the mean sea level, one can easily infer the types of vegetation that will be found in each major region of the country.

The present vegetal cover has a long history. According to Palaeo-botanists, most of our Himalayan and peninsular areas are covered with indigenous or endemic flora, while the Indo-Gangetic Plain and the Thar desert contain plant species that have come generally from outside. Here, the plant species are exotic and have migrated from the Trans-Himalayan areas (Tibet and China). This natural vegetation is classified as boreal. Plants which have come from the adjacent tropical regions are known as palaeo-tropical. Those plants which came from north Africa have influenced the vegetation of the arid and semi-arid regions, such as the Thar, as well as a good deal of the Great Plains of India. Those immigrating from Indo-Malaysia have influenced the vegetal cover of the hilly regions of north-eastern India. This process of the immigration of uninvited plant species is not only continuous, but has actually become more marked with the increase in communication with other lands, both by sea and air. Some of the exotic varieties are troublesome weeds. They thrive under conditions of tropical sun with abundant moisture, multiply rapidly and spread out as there are no ‘natural’ enemies to curb them in the new habitat. In course of time, their eradication becomes difficult; they invade the land and reduce the area for other uses, prevent the growth of plants which are economically important and become a hazard to public health by indirectly helping the spread of several diseases. We can cite two striking examples: lantana (*Lantana camara varauyleata Mold*) and water hyacinth (*Eichhornia crassipes Solms*).

Both were brought into India as decorative garden plants; the former having now spread out in forests and pasture lands, and the latter choking up our rivers, lakes and ponds so much so as to earn its nickname ‘terror of Bengal’ because of its phenomenal growth in

that region. It is spreading to almost all water sources, ponds and canals in the rest of the country.

It may be ascertained from the above description that much of our natural vegetal cover is not that natural, except perhaps in the inaccessible parts of the Himalayas and the interior of the Thar desert. A considerable part of it has been replaced or destroyed as a result of human occupancy of the land. Much of the plant cover is in a degraded condition, that is, low in quality and content. What we usually designate as 'natural' vegetation now refers to a plant community that has been left undisturbed over a long time, so as to allow individual species to adjust themselves to geo-climatic conditions, as far as possible.

FLORISTIC REGIONS OF INDIA

Depending on the geo-climatic conditions, the flora of India differs from region to region and altitude to altitude. In 1937, C.C. Calder identified eight floristic regions of India **(Fig. 5.1).** These floristic regions are:

1. The Eastern Himalayan Region
2. The North-Western Himalayan Region
3. The Assam Region
4. The Gangetic Plain
5. The Indus Plain
6. The Deccan Region
7. The Malabar Region
8. The Andaman and Nicobar Islands

1. The Eastern Himalayan Region

Stretching over the hilly regions of Sikkim, Northern West Bengal and Arunachal Pradesh, this is an undulating and mountainous region, recording over 200 cm of average annual rainfall. This region has over 4000 species of plants which vary from tropical to temperate and Alpine. The main trees of this floristic region are sal, oak, chestnut,

magnolia, pyrus, bamboo, silver fir, pine, birch, rhododendrons, and alpine grasses.

2. The North-Western Himalayan Region

The Western Himalayan Floristic region stretches over Jammu and Kashmir, Himachal Pradesh and Uttarakhand. This region records relatively less rainfall and temperatures. The effect of altitude is quite visible on the vegetation of Western Himalayas. Here again, one finds the sub-tropical
(upto 1525 m), temperate (1525 m to 3650 m) and Alpine vegetation from 3650 m to 4575 m. In the sub-montane region the main vegetation is sal, semul, and savanna type. Among the temperate vegetation are chir (pine), oak, deodar, alder, birch, and conifers. At higher altitudes, trees are replaced by alpine pastures and trees like juniper, silver fir, birch, and larch are seen.

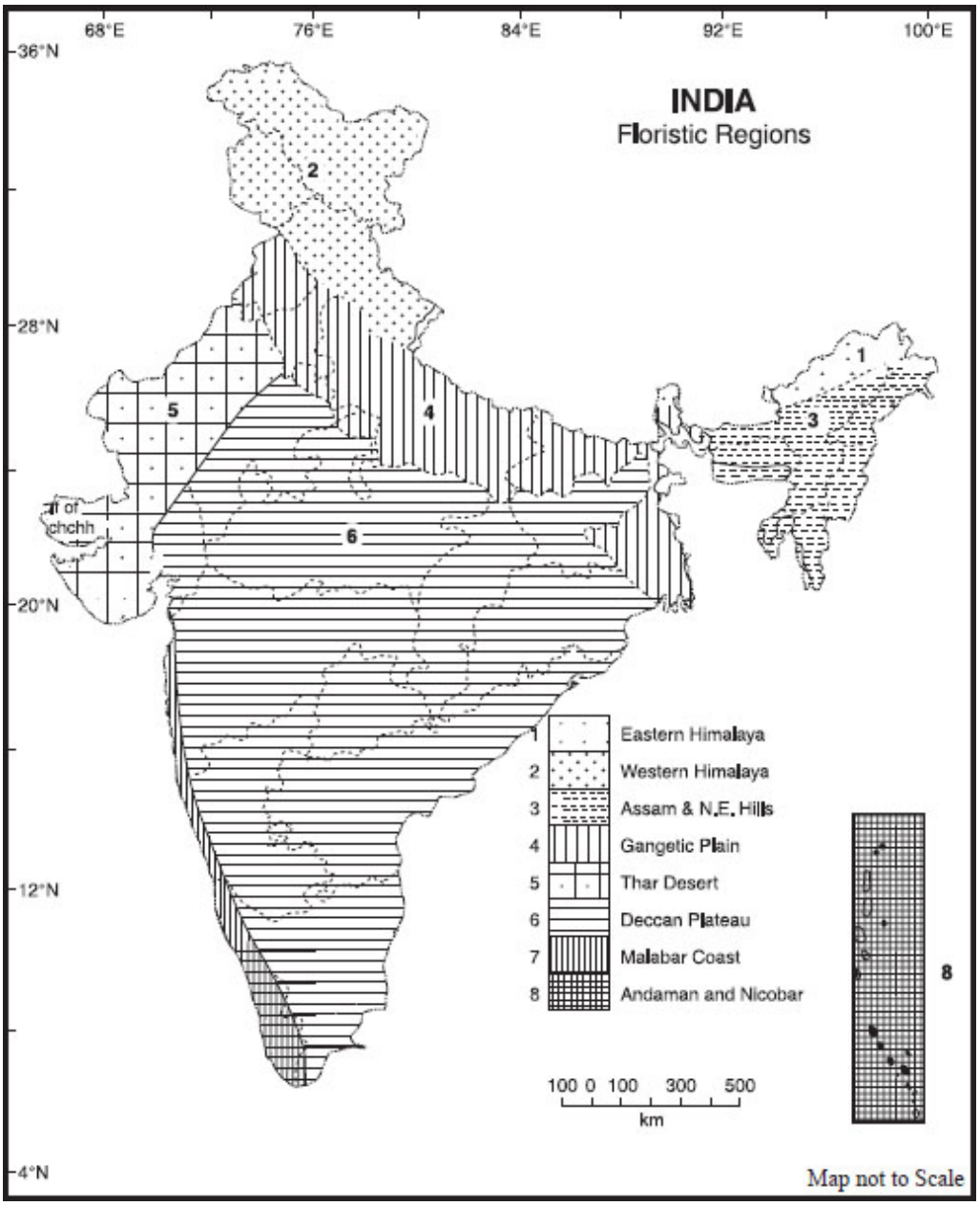


Fig. 5.1 Floristic Regions

3. The Assam Region

The Assam region includes the whole of North-East including Assam, Meghalaya, Nagaland, Manipur, Mizoram and Tripura. This region is rich in various types of bamboos and palms with Nilgiri type of grasslands at higher altitudes.

4. The Gangetic Plain

The flora of The Gangetic plain has been substantially transformed by human activities and cultivation of crops. The vegetation type, however, ranges between the semi-arid shrubs of the Aravalli region to evergreen mangroves of the Sundarban Delta. Sal and Arjun of the Tarai region of Bihar and West Bengal are the representative species of the primordial vegetation. The vegetation of Uttar Pradesh is mainly dry deciduous type which changes to moist deciduous type in Bihar and West Bengal. Sheesham, neem, mahuwa, jamun, acacia, ber, bel, peepal, bargad, etc., are the examples of this type of vegetation. In addition to these, there are numerous types of grasses found in The Gangetic Plain.

5. The Indus Plain

This floristic region spreads over Punjab, Haryana, Rajasthan, west of the Aravallis, Kuchchh, and north-western parts of Gujarat. In this region, the average annual rainfall is less than 75 cm. Consequently, its vegetation is of the type which can bear the arid and severe drought conditions. Acacias, cacti, wild-palms, khejra, and palas, etc., are the main trees of this region. During the rainy season, numerous grasses develop which wither during the dry season.

6. The Deccan Region

This region covers the greater parts of Peninsular India. This region has teak, tendu, sal, palm, and thorny shrubs.

7. The Malabar Region

This region stretches all along the western coast from the Gulf of Khambat (Cambay) to Cape Camorin (Kanniyakumari). Here, the vegetation type ranges from moist tropical evergreen to broad leafed mixed and monsoon deciduous type. The Nilgiri Hills show temperate forests at higher altitudes. The region also contains several species of plants of the Malay origin.

8. Andaman and Nicobar Islands

The Andaman and Nicobar Islands are covered by the equatorial evergreen forests of heavy wood.

SPATIAL DISTRIBUTION OF FORESTS IN INDIA

The total geographical area of India is 3,287,263 km2, of which about 675,500 km2—equal to 22.50%—is under forests. According to the National Forest Policy (1952) about 33% of the geographical area should be under forest. However, the existing forest area is much below the desired level. The areas under forest cover in India have been shown in **(Fig. 5.2)**. It may be observed from **(Fig. 5.2)** that the Himalayan mountains, Bhabhar and Tarai, Western Ghats, Eastern Ghats, Bundelkhand, Baghelkhand, Chotanagpur Plateau, North-eastern Hills, Nilgiris, and the hills of Peninsular India are the main areas of Indian forests. Unfortunately, about 5 to 6% of the total forest area of the country is under the category of degraded forests.

The forest area in India is much below the world average of 34.5% and that of Brazil (57%), Sweden (58%), USA (42%), Germany (41%), and Canada (36%). Similarly, the per head forest area in India is only 0.07 hectare as against the world average of 1.10 hectares, Canada 23 hectares, Brazil 8.6 hectares, Australia 5 hectares, Sweden 4 hectares, and the USA 3.5 hectares per head of population.

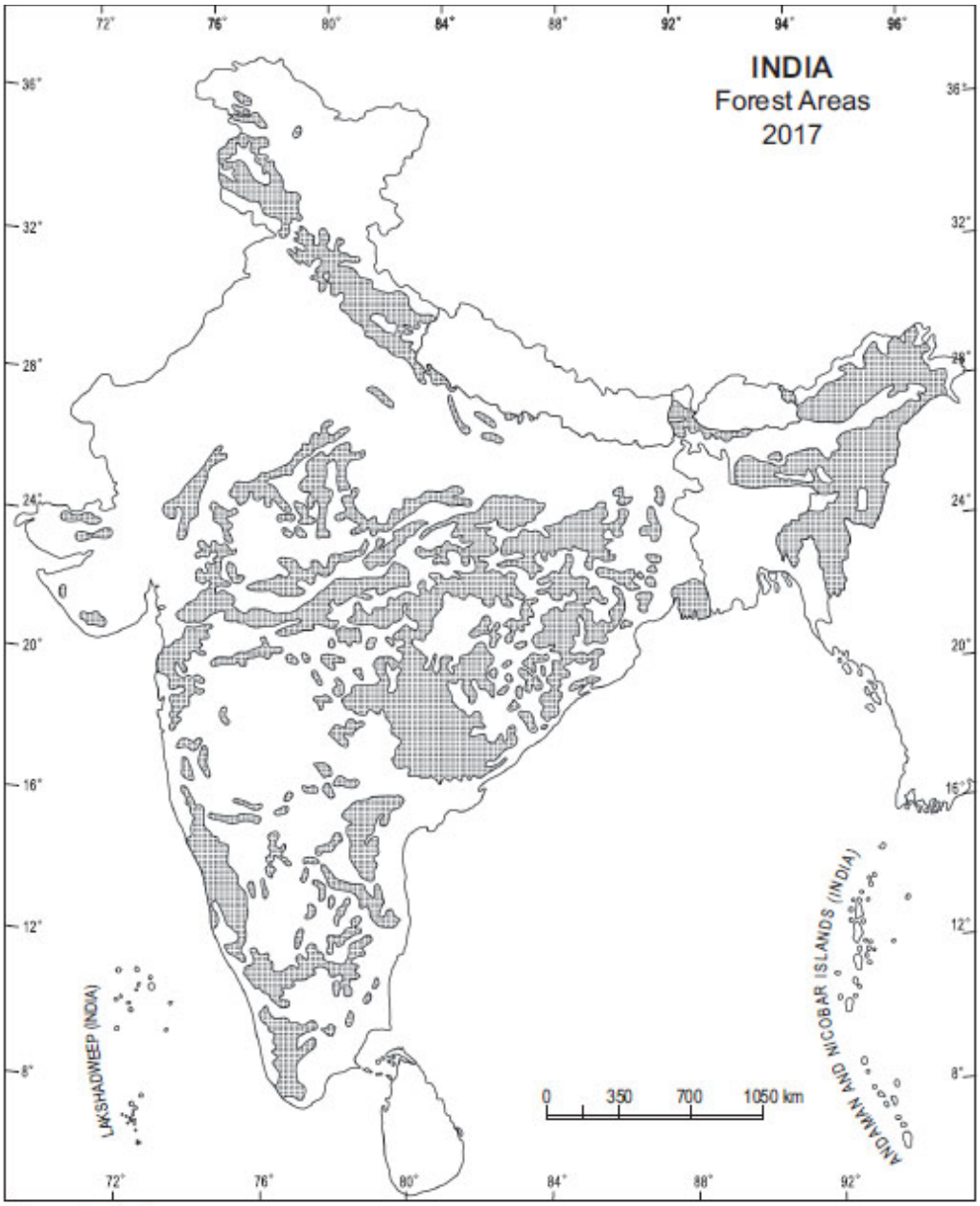


Fig. 5.2 Forest Areas (2017)

Table 5.1

Area Under Different Forest Type Group of India

S. No.	Type Group	Area in Sq Km.	Percent of Grands Totals
1.	Group 1 Tropical Wet Evergreen Forests	20054	2.61

S. No.	Type Group	Area in Sq Km.	Percent of Grands Totals
2.	Group 2 Tropical Semi Evergreen Forests	71171	9.27
3.	Group 3 Tropical Moist Deciduous Forests	135492	17.65
4.	Group 4 Littoral and Swamp Forests	5596	0.73
5.	Group 5 Tropical Dry Deciduous Forests	313617	40.80
6.	Group 6 Tropical Thorm Forests	20877	2.72
7.	Group 7 Tropical Dry Evergreen Forests	937	0.12
8.	Group 8 Sub Tropical Broadleaved Hill Forests	32706	4.26
9.	Group 9 Sub Tropical Pine Forests	18102	2.36
10.	Group 10 Sub Tropical Dry Evergreen Forests	180	0.02
11.	Group 11 Montane Wet Temperate Forests	20435	2.66
12.	Group 12 Himalyan Moist Temperate Forests	25743	3.35
13.	Group 13 Himalyan Dry Temperate Forests	5627	0.73
14.	Group 14 Sub Alpine Forests	14995	1.96
15.	Group 15 Moist Alpine Forests	959	0.13
16.	Group 16 Dry Alpine Forests	2922	0.38
17.	Group 17 Plantation/TOF	64839	8.45
	Total Forest covers scrub	754252	98.26
18.	Group 18 Grassland in different Forest types Group (without Forest covers)	13229	1.74
	Grand Total	**76581**	**100**

It may be observed from **(Table 5.2)** that the largest forest area lies in Peninsular India accounting for 57% of the total forest area of India. The Himalayan region has the second largest forest area having about 12.5% of the forest area of the country. The Eastern and Western Ghats have 10% each, while the Gangetic Plains have only 5% of the forest area of the country.

Table 5.2

India-Statewise Distribution of Forest-2019

State	Geographic Area	Forest Cover	Percent of Forest Cover w.r.t Geographical Area
Andhra Pradesh	162,968	29,137	17.88
Arunachal Pradesh	83,743	66,688	79.63
Assam	78,438	28,327	36.11
Bihar	94,163	7,306	7.76
Chhattisgarh	135,192	55,611	41.13
Delhi	1,483	195.44	13.18
Goa	3,702	2,237	60.43
Gujarat	196,244	14,857	7.57
Haryana	44,212	1,602	3.62
Himachal Pradesh	55,673	15,430	27.72
Jharkhand	79,716	23,611	29.62
Karnataka	191,791	38,757	20.11
Kerala	38,852	21,144	54.42
Madhya Pradesh	308,245	77,482	25.14
Maharashtra	307,713	50,778	16.50
Manipur	22,327	16,847	75.46
Meghalaya	22,429	17,119	76.33
Mizoram	21,081	18,006	85.41
Nagaland	16,579	12,486	75.31
Odisha	155,707	51,691	33.15
Punjab	50,362	1,849	3.67
Rajasthan	342,239	16,630	4.86
Sikkim	7,096	3,342	47.10
Tamil Nadu	130,060	26,364	20.27
Telangana	112,077	20,582	18.36
Tripura	10,486	7,726	73.68
Uttarakhand	53,483	24,303	45.44
Uttar Pradesh	240,928	14,806	6.15
West Bengal	88,752	16,902	19.04
UNION TERRITORIES			
A & N. Islands	8,249	6,743	81.74
Chandigarh	114	22.03	19.32

State	Geographic Area	Forest Cover	Percent of Forest Cover w.r.t Geographical Area
Dadra & Nagar Haveli	491	207	42.16
Daman & Diu	111	20.49	18.46
Lakshdweep	30	27.10	90.33
Pudducherry	490	52.41	10.70
Jammu & Kashmir (Inclusive of Ladhak)	222,236	23,610	10.62
Grand Total	**3,287,469**	**712,249**	**21.67**

Source: ISFR Report 2019, Government of India
Data separately on Jammu Kashmir and Ladakh to be update in next edition.

According to the ISFR 2019 report, the total forest cover area is 712,249 km2 which is 21.67% of the total geographical area of India. The forest and tree combined is 80.73 million hectare or 24.56% of the total geographical area.

Madhya Pradesh has the largest forest cover with 77,82 km2 followed by Arunachal Pradesh 66,688 km2 and Chattisgarh with 55,611 km2. In terms of percentage of forest cover area with the total geographical area, Mizoram has 85.41% and Arunachal Pradesh 79.63% and Meghalaya 76.33%.

The lowest percentage of forest area is in Haryana (3.62%) , followed by Punjab (3.67%) and Rajasthan (4.86%). In general, the hilly and mountainous states have more percentage of area under forest while the forest cover in the states with plains is almost insignificant.

The top five states that have shown an increase in maximum forest cover are Karnataka (1207 km2), Andhra Pradesh (990 km2), Kerala (823 km2), Telangana (433 km2), and Odhisha (346 km2).

15 states and Union Territories have above 33% of the total forest cover. India has about 712,249 km2 of the forest which is 21.67% of the geographic area of the country.

CLASSIFICATION OF FORESTS

The forests of India have been classified in a number of ways.

A. On the basis of **administration**, the forests have been classified into the following three categories:

1. **Reserved Forests:** These forests are under the direct supervision of the government and no public entry is allowed for collection of timber or grazing of cattle. About 53% of the total forest area falls in this category.
2. **Protected Forests:** These forests are looked after by the government, but the local people are allowed to collect fuel-wood/timber and graze their cattle without causing serious damage to the forests. These forests occupy about 29% of the total forest area of the country.
3. **Unclassified Forests:** The unclassified forests are those in which there is no restriction on the cutting of trees and grazing of cattle. About 18% of the total forest area of the country falls under this category.

B. In the **Constitution of India**, forests have been classified under the following categories:

1. **State Forests:** These are under the full control of the government (state/central) and include almost all the important forest areas of the country. They constitute about 94% of the total forest area of the country.
2. **Commercial Forests:** These forests are owned and administered by the local bodies (municipal corporations, municipal boards, town areas, district boards, and village-panchayats). They constitute about 5% of the total forest area of the country.
3. **Private Forests:** These are under private ownership and cover slightly more than one% of the total forest area of the country.

C. On the basis of **merchantability**, Indian forests may be grouped under two categories:

1. **Merchantable:** which are accessible. About 82% of the total forest area belongs to this category.

2. **Non-Merchantable:** These are not accessible being situated in high mountainous areas with inaccessible topographical features. About 18% of the total forest area (especially conifers) of the country fall in this category.

D. Based on Composition: Based on composition and types of leaves, Indian forests fall into two broad groups:

1. **Conifer Forests**: These are temperate forests occupying about 6.50% of the total forest area of the country.
2. **Broad-leaf Forests**: These are tropical and subtropical monsoon forests. About 94% of the country's forests belong to this category. They are found in the plains, plateau, and mountainous areas of the country.

E. Based on Exploitability: On the basis of exploitability, the Indian forests may be classified into:

1. **Exploitable Forests:** These forests contribute 58% of the total forest area of the country.
2. **Potentially Exploitable**: These forests are reserved to be utilised in future. These forests cover about 22% of the total forest area of the country.
3. **Other Forests:** These forests cover about 20% of the total forest area of the country. There is no restriction on their exploitation.

F. Based on Canopy Density: The forest cover in terms of canopy density has been classified into the under-given categories

G. On the basis of Average Annual Rainfall: On the basis of average annual rainfall, Indian forests have been classified by L.D. Stamp into the following four categories **(Table 5.3):**

Table 5.3

Forest Cover in terms of Canopy Density Classes

Class	Description

Class	Description
Very dense forests	All land with tree canopy density of 70% and above
Moderately dense forest than 70%	All land with tree canopy density of 40% and more but less
Open Forest than 40%	All land with tree canopy density of 10% and more but less
Scrub	Degraded forest lands with canopy density less than 10%
Non Forest	Land not included in any of the above classes (includes water)

Source: ISFR Report 2019, Government of India

The natural vegetation of India has been shown in **(Table 5.4)**.

Table 5.4

India-Classification of Forests on the basis of Rainfall Distribution

S. No.	Vegetation Type	Average Annual Rainfall (cm)	Zone
1.	Evergreen forests	Above 200	Humid
2.	Monsoon forests	100–200	Semi-Humid
3.	Dry forests	50–100	Dry
4.	Desert forests	Below 50	Very Dry (deserts)

The total forest cover of India has been classified by MoEFCC in the following classes given in **(Table 5.5)**

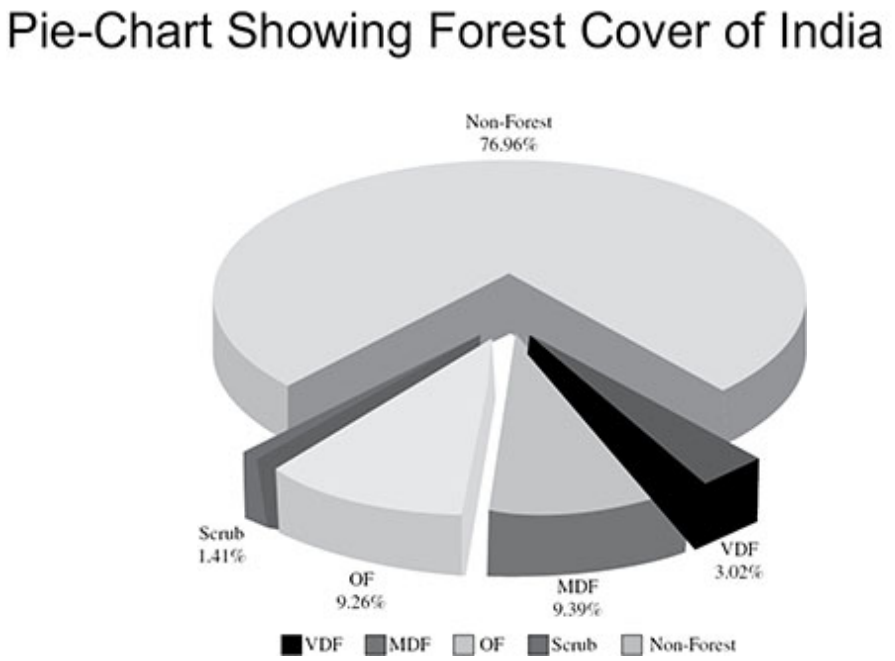
Table 5.5

Forest Cover of India-2019

Class	Area (km2)	Percentage of Geographical Area
Very dense forest	99,278	3.02
Moderately dense Forest	308,472	9.38

Class	Area (km2)	Percentage of Geographical Area
Open Forest	304,499	9.26
Total Forest Cover *	712,249	21.67
Scrub	46,297	1.41
Non Forest	2,528,923	77.06
Total Geographical Area	**3,287,469**	**100.00**

Source: ISFR Report 2019; Released by MoEFCC



Source: ISFR Report 2019 released by MoEFCC

The Indian forests were also classified by H.G. Champion (1936) into eleven categories. The main categories according to this classification are as under:

1. **Tropical Evergreen:** These forests are mainly found in the areas recording over 150 cm of average annual rainfall where the temperature varies between 20^{o}C to 27^{o}C. North-East India, parts of Western Ghats, the Andaman and Nicobar, upper Assam, lower slopes of Eastern Himalayas, Odisha, along the foot-hills of Himalayas, Bhabhar and Tarai regions

(Fig. 5.3). In the areas where the average annual rainfall is more than 250 cm, the forests are dense; composed of tall trees (45–60 m) epiphytes, parasites, lianas and rattans so as to look like a green carpet when viewed from above. Trees have multi-storeyed structures with good canopies. These trees do not shed their leaves annually and are hence evergreen. The floor lacks grasses because of deep shade. There are, however, canes, palms, bamboos, ferns, and climbers which make passage difficult. The important species of these forests are white cedar, toon, dhup, palaquinum, mesua, collophyllum, hopea, and canes, gurjan, chaplas, agor, muli, and bamboo. Due to poor accessibility these forests have not been properly exploited.

In areas where rainfall varies between 200 to 250 cm and the mean monthly temperature varies between 20^{o}C to 27^{o}C , the evergreen forests degenerate into semi-evergreen forests. These forests are found along the Western Ghats, upper Assam, slopes of the Himalayas, and Odisha. The important varieties include aini, semul, gutel, mundane, hopea, kadam, irul, rosewood, laurel, haldu, kanju, holloch, champa, and mesua.

2. **The Tropical Moist Deciduous**: These are typical monsoon forests with teak (*Tectona grandis*) and sal (*Shorea robusta*) as the dominant species. They form the natural vegetation all over the country where the average annual rainfall ranges between 100–200 cm. The tropical moist deciduous forests are found in Sahyadris, the north-eastern parts of the peninsula and along the foothills of the Himalayas **(Fig. 5.3).** These forests on the whole have gregarious species. The typical landscape consists of tall teak trees with sal, bamboos, and shrubs growing fairly close together to form thickets. Both teak and sal are economically important and so are the Sandalwood (*Santalum album)*, Shisham (*Dalbergia*

sissoo), Hurra (*Terminalia chebula*), and Khair (*Acacia catechu*).

3. **The Tropical Thorny Forests:** The tropical thorny forest is a degraded version of the moist deciduous forest. Varies between 75 and 100 cm and the average annual temperature between 16^{o}C and 22.5^{o}C. These forests are found in peninsular India, Rajasthan, Haryana, Punjab, western Uttar Pradesh, Kutch, Madhya Pradesh and the foothills of the Himalayas **(Fig. 5.3).** The important trees of these forests are acacia, wild-palms, euphorbias, jhad, tamarix, khair, kokko, dhaman, erunjha, cacti, kanju, and palas.

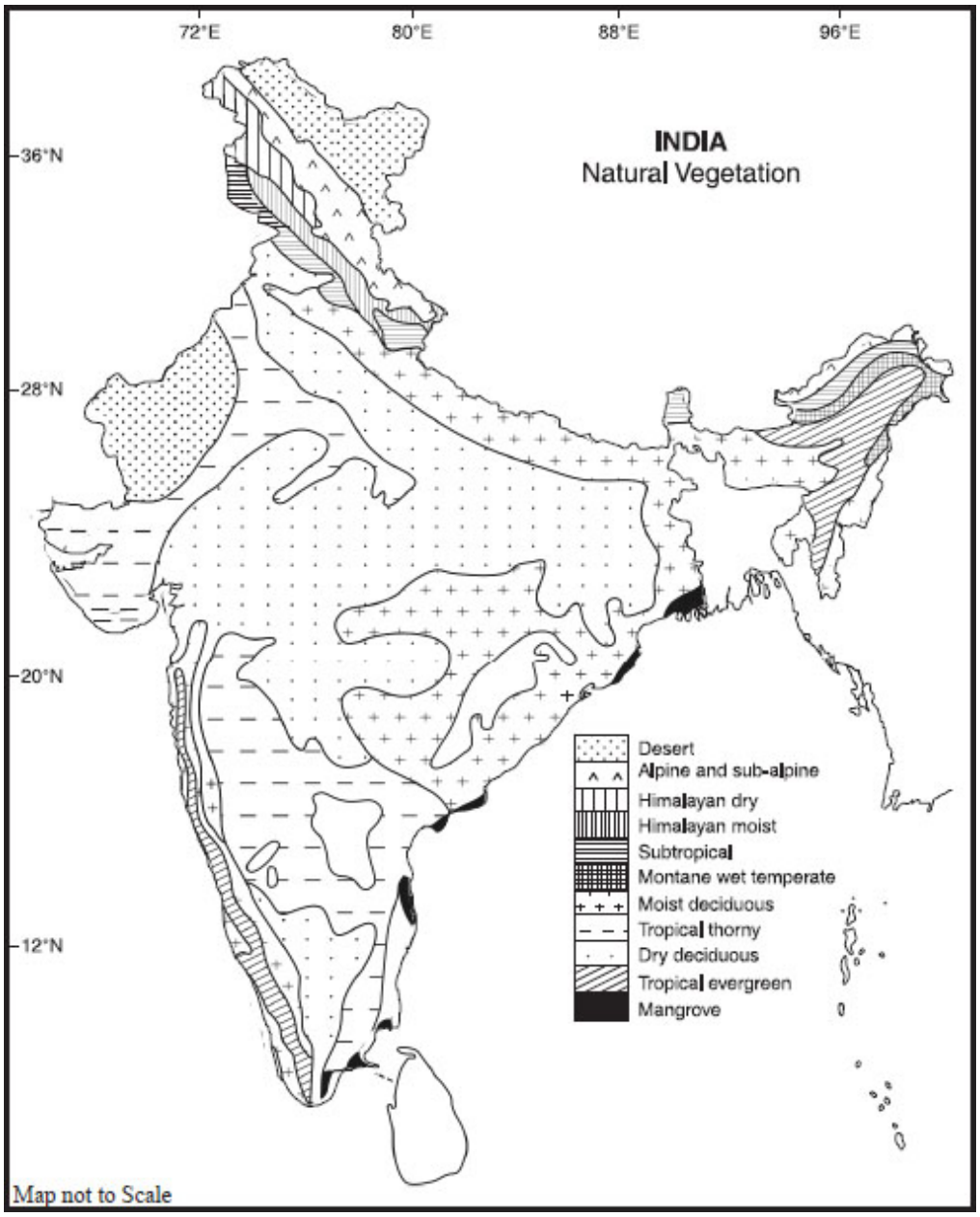


Fig. 5.3 Natural Vegetation

4. **The Subtropical Montane Forests**: These forests are found in areas where the average annual rainfall varies between 100 to 200 cm and the temperature varies between 15^{o} and 22^{o}C. These forests are found in the north-western Himalayas (except in Ladakh and Kashmir), Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh and on the

slopes of north-eastern hill states **(Fig. 5.3).** Chir (pine) is the main tree but broad leaved trees are also found in these areas. Oak, jamun, and rhododendron are the other varieties in these forests.

5. **The Dry Deciduous Forests:** These forests are found in areas where the average annual rainfall ranges between 100–150 cm. These forests are characterised by closed and rather uneven canopies. Enough light reaches the ground to permit the growth of grasses and climbers. Acacia, jamun, modesta, and pistacia are the main trees. Grasses and shrubs appear during the season of general rains.
6. **The Himalayan Moist Forests**: These forests are found in Jammu and Kashmir, Himachal Pradesh, Uttarakhand, and northern hilly parts of North Bengal **(Fig. 5.3).** The wet temperate type is found in a belt where the altitude varies between 1000 and 2000 m. They occur largely as bands of crested dark green landscape of coniferous varieties. The important varieties are oak, chestnut, chir, sal, shrubs, and nutritious grasses.
7. **The Himalayan Dry Temperate Forests:** These forests are found in Jammu and Kashmir, Lahul, Chamba, Kinnaur (Himachal Pradesh), and Sikkim **(Fig. 5.3)**. These are predominantly coniferous forests with shrubs. The important varieties of trees are deodar, oak, chilgoza, ash, maple, olive, mulberry, willow, celtis, and parrotia.
8. **Montane Wet Temperate Forests:** These forests are found in the entire Himalayas from Jammu and Kashmir to Arunachal Pradesh between the altitudes of 1500 m to 3500 m where the temperature varies between 12^{o}C to 15^{o}C, and the mean annual rainfall is between 100 to 250 cm. Oak, fir, spruce *Picea*, deodar, (*Cedrus deodara*), magnolia *(Magnoliaglandifora*) celtis, chestnut, cedar (*Chamaecyparis*) and maple, spruce, deodar, silver-fir (*Abies alba)*, kail, and yew are found here. These forests also contain scrubs,

creepers, and ferns. The woods of these forests are durable. At higher attitudes, above 3500 m, are the alpine pastures known as **Margs** in Kashmir and **Bugyals** in Uttarkhand.

9. **Alpine and Sub-alpine Forests:** The Alpine forests are found all along the Himalayas at altitudes ranging between 2500 to 3500 m. These areas are characterised with short dwarf conifers and lush green nutritious grasses during the summer season. The trees found in the zone are kail, spruce, yew, firs, birch, honeysuckle, artemesia, potentilla, and small scrubs.
10. **Desert Vegetation:** The desert vegetation is confined to the west of Aravallis in the states of Rajasthan and northern Gujarat **(Fig. 5.3).** The average annual rainfall in this zone is less than 50 cm, the diurnal and annual range of temperature are high. Acacia, cacti, jhar and khejra, kanju, and wild palms are the main trees of the desert.
11. **Tidal (Mangrove):** These forests are found along the coastal areas of the Bay of Bengal in the states of West Bengal, Odisha, Andhra Pradesh, and Tamil Nadu, and along the coastal areas of Kutch, Kathiwar, and Gulf of Khambat. Their main concentration is found in areas where tides are frequent. The mangrove which attains a height up to 30 m is the most important tree. It is utilised for fuel. The famous delta of Sundarban is covered by the

Sundri (*Heritiera minor*) trees which supply hard durable timber for construction and boat making. Here, higher grounds *support screw-pines* (*Pennanus spp*.). Palms occupy creeks, and epiphytes are predominant all over the region **(Fig. 5.3).**

The percentage share of the different categories of forests have been given in the **(Table 5.6)**.

Table 5.6

Percentage Share of Different Types of Forests (2010)

S. No.	Type of Forest	Percent (Total Forest Area)

1.	Tropical Moist Deciduous	37.0
2.	Tropical Dry Deciduous	28.8
3.	Tropical Evergreen	12.1
4.	Subtropical	9.5
5.	Montane Wet Temperate	7.0
6.	Alpine and Sub-alpine	2.9
7.	Tropical Thorny	2.6
8.	Mangrove	0.06

Source: Oxford School Atlas, 2011.

It may be seen from **(Table 5.6)** that tropical moist deciduous forests cover 37% of the total forest area of the country followed by the tropical dry deciduous at 28.8%. The tropical evergreen forests occupy 12.1% and the subtropical 9.5%. The rest of the forest area is under montane wet temperate (7.0%) and alpine and sub-alpine (2.9%). The tropical thorny forest occupy about 2.6% and about 0.06% is under mangrove vegetation.

IMPORTANT SPECIES OF TREES AND THEIR UTILITY

On the basis of type of species and utility, the trees of India may be classified under the following categories:

1. Woods from Evergreen Forests.
2. Woods from Monsoon Forests.
3. Woods from Subtropical (Himalayan Forests).

1. Woods from the Evergreen Forests

The main trees from the evergreen forests are:

(i) **Rosewood:** It grows well along the slopes of the Western Ghats (Tamil Nadu, Karnataka, and Kerala) and in some parts of Andhra Pradesh, Odisha, Jharkhand, and Chhattisgarh. The wood from these forests is hard and fine-grained, dark purple in colour, widely used in the manufacture of furniture, floor boards, and ornamental plyboards.

(ii) **Gurjan:** It occurs in the evergreen forests of Assam, West Bengal, and Andaman and Nicobar Islands. The wood is dull

reddish to brown in colour. It is extensively used for internal construction work of houses. It is also used for packing cases, tea boxes, flooring, and wagons.

(iii) **Telsur or Irupu:** It is mainly found in West Bengal, Kerala, Karnataka, Maharashtra, and Andaman and Nicobar Islands. Its wood is very hard, strong and durable which are largely used for the manufacturing of boats, bridges, piles, masts, carts, and railway sleepers.

(iv) **Toon:** It is obtained from the foothills of the Himalayas. Although its wood is not very hard, it is durable. It is used for making tea boxes, toys and furniture.

(v) **Ebony (Diospyros Ebenum):** It is found in the dry evergreen forests of Karnataka, Kerala, Tamil Nadu, Malabar Coast, Goa, and Maharashtra. Its wood is lightly yellowish-grey and often streaked with black. The heartwood (inner core) is jet black, rarely with brown golden streaks. It has a metallic lusture when smoothed. It is one of the most valuable woods as it is resistant to attack by insects. It is used for ornamental carving and decoration. It is also used for veneers, musical instruments, sports goods, piano keys, and caskets. Many species of black ebony tree or blackwood is on the verge of extinction and is an endangered tree. Black ebony comes form trees as old as 150 years.

(vi) **Chaplas:** The *chapla* forests mainly occur in north-east India and the Western Ghats. Its timber is strong and durable and hence, is in great demand for ship-building, furniture-making, and packing boxes.

(vii) **Nahar:** It is found in Assam and the Malabar coast. The wood is fairly strong and durable. Its wood is used for railway sleepers, piles, and boats.

(viii) **Poon:** It is found in the Western Ghats, Kerala, Nilgiri, and Tamil Nadu. Its wood is very hard, can be easily seasoned and is mainly used as structural timber for house making.

2. Woods of the Monsoon Forests

The main trees of the monsoon forests are as under:

(i) **Sal (Shorea Robusta):** It occurs in the sub-Himalayan region from Kangra (Himachal Pradesh) to Darrang (Assam), Meghalaya and the northern parts of Tamil Nadu. Its wood is very heavy, hard and durable. It is much in demand for piles, doors, beams, planking and railway sleepers. Sal forests occupy 11.6 lakh hectares, accounting for about 16% of the total forest area of the country.

(ii) **Teak (Tectona Grandis):** Teak is the most popular tree of the monsoon climate found mainly in Chhattisgarh, Jharkhand, Madhya Pradesh, Odisha, the foothills of Himalayas, Tamil Nadu, Karnataka, Kerala, the western and eastern Ghats, and Banswara (Rajasthan). Its wood is moderately hard, duarable, easy to work and takes a good polish. It is an expensive timber used for doors, cupboards, and furniture. Teak forests cover about 9 million hectares of the total forest area of the country.

(iii) **Shisham (Dalbergia Sissoo):** It occurs throughout the Himalayas from Jammu to Assam up to an altitude of 1500 m. It grows extensively in Punjab, Haryana, Uttarakhand, Uttar Pradesh, Bihar, and West Bengal. On account of its great strength, elasticity and durability, its wood is mainly used in furniture making, musical instruments, and agricultural equipments.

(iv) **Haldu:** It is found all over the monsoon area. Its wood is hard, durable, and light in colour. It is used for toy making and wood carving.

(v) **Palas:** It occurs mainly in Chotanagpur Plateau, Chhattisgarh, and south-eastern parts of Rajasthan. Its leaves are used for rearing shellac worms.

(vi) **Arjun:** It is also an important tree of monsoon forests which is used for the making of agricultural equipments and bullock carts.

(vii) **Mahua (Madhuca Indica):** It is largely found in Madhya

Pradesh, Chhattisgarh Bundelkhand (U.P), Jharkhand, Bihar, Gujarat, Maharashtra, Uttarakhand, and south-eastern parts of Rajasthan. Its fruits are used for the extraction of oil and flowers for wine making.

(viii) **Semul:** It is widely found in Assam, Bihar, and Tamil Nadu. Its timber is soft and white and is used for toy making, packing cases, match boxes, and plywood. Its fruits yield soft fibre for pillows and *lihafs*.

(ix) **Mulberry:** It grows widely in monsoon areas. Its wood is soft and durable, used mainly for the manufacture of sports goods (hockey, cricket bats, tennis rackets, badminton and squash rackets, and cricket stumps).

(x) **Jamun (Syzygium cumini):** It is a large tree of monsoon region. Its timber is moderately strong and used for the construction of houses and furniture. Its fruits are highly beneficial in controlling diabetes and high blood pressure.

3. Woods from the High Altitudinal Forests of the Himalayas

The main woods of the Himalayan forests are deodar, chir (pine), blue-pine, silver fir, spruce, walnut, white willow, Indian birch, and cypress.

(i) **Chir (Pinus Longifolia):** Chir occurs in the Himalayas between 900 m and 1800 m, from Jammu to Arunachal Pradesh. The wood is light and reddish brown in colour and is moderately hard. It is used for furniture, for making tea-boxes, match industry, and railway sleepers. It yields resins, gums, and turpentine oils.

(ii) **Deodar (Cedrus Deodara):** It grows in the north-western Himalayas in the states of Jammu and Kashmir, Himachal Pradesh, and Uttarakhand, between the heights of 1500 m and 2500 m. Its wood is of light brown to yellow colour. Its wood is very sturdy and duarable. It is also an easy timber to saw and work to smooth finish. The timber is used for construction-work, and for railway sleepers. It is also suitable

for beams, floor-boards, ports, doors, window frames, light furniture, and shingles.

(iii) **Blue-Pine (Pinus Excelsa):** It grows along the entire length of the Himalayas from Chumbi Valley to Sikkim between the elevation of 1800 m and 3600 m. The wood is pink in colour, moderately hard and of good quality. It is used for making doors, windows, furniture, and railway sleepers. It also yields resins and turpentine.

(iv) **Silver-fir (Abies):** It is found in the north-western and north-eastern Himalayas between 2200 m and 3000 m. The food is soft but not very durable. It is mostly used for planking, packing boxes, containers, wood-pulp, paper, and match sticks.

(v) **Spruce (Picea Mithiana):** It is found in the western Himalayas between 2100 m and 3600 m. Its soft white wood is used for construction of houses, railway sleepers, cabinets, packing, and pulp making.

(vi) **Walnut (Juglans Regia):** It is found in Kashmir, Himachal Pradesh, Uttarakhand, and Khasi hills. It is a relatively light wood on which work can be done easily and the finish is fine and attractive. Once dried it does not shrink, swell or split. The wood is used for musical instruments, gun-butts, and cabinet works.

(vii) **White Willow (Salix Alba):** It is a small tree found in north-western Himalayas including the Kashmir Valley. Its twigs are used for making baskets. The wood is used for making cricket bats and other sports goods.

(viii) **Indian Birch:** It is obtained from the higher slopes of the Himalayas. The wood is grayish in colour, even textured and straight grained. It is largely used for the making of furniture, plywood, and radio cabinets.

(ix) **Cypress:** It mostly occurs in Uttarakhand, Himachal Pradesh and Jammu & Kashmir. Its wood is durable and used for making furniture.

FOREST PRODUCTS AND THEIR UTILITY

In addition to fuel-wood, timber, and charcoal, the forests provide a number of other products. These include bamboos, canes, tendu leaves, grasses, oils, nuts, chilghoza, lac, resins, gums, medicinal herbs, tanning material, dyes, honey, bees wax, ivory, vegetables, fruits, roots, and tubers.

1. Bamboo

Bamboo is found in most of the monsoon regions where the average annual rainfall is more than 150 cm. Some of the bambooes may attain a height of 30 m. It is used for a variety of purposes—basket making, roofing, and thatching, construction, paper, and pulp making. In India more than 100 varieties of bamboos are found. According to the Planning Commission bamboo occupies 100,210 km2 of the forest area of the country with an annual production of four million tonnes. Different types of decorative items such as flower pots, trays, vases, caskets, and even ornaments are made of bamboo in states like Mizoram, Nagaland, Meghalaya, Manipur, and Tripura. Bamboo also finds a place in cultural activities of the Mizo people, i.e., Cherraw (bamboo) dance.

2. Canes

Canes grow in moist and wet forests in Assam, Kerala, Karnataka, West Bengal, Gujarat, Uttarakhand, Himachal Pradesh and Jammu and Kashmir. It is used mainly for making strings, ropes, mats, bags, and baskets.

3. Tendu

The leaves of tendu are obtained from the forests of Chhattisgarh, Jharkhand, Odisha, Madhya Pradesh, south-east Rajasthan, Telangana, and Andhra Pradesh. Tendu leaves are used for *bidi*-making.

4. Grasses

A large variety of grasses are found in the Indian forests. Some of the important grasses are *Sabai* (sub-Himalayan Tarai tracts), elephant-grass (Assam), spear-grass, *ulla*, and *panni* grasses, etc. *Khus-Khus*

grass (Bharatpur, and Sawai-Madhopur, Rajasthan) is used for making cooling screens during summer season. *Rosha, lemon*, and ginger-grasses yield medicinal and perfumed oils.

5. Oils

The raw materials for a number of perfumes and oils are also obtained from the herbs, plants, and trees. Some of them are *camphor, clove, ylang, cinnamon-oil, cypress-oil, eucalyptus-oil, jasmine-oil, khus-oil, lavender-oil, lemon-grass-oil, mint-oil, sandalwood-oil, patchouli-oil, turpentine oil, nutmeg-oil, and champaca-oil*.

6. Medicinal Herbs and Plants

A number of medicinal plants, herbs and trees are found in Indian forests. The leaves, stems, flowers, fruits, barks, roots, and seeds of different plants and scrubs are used as raw materials for the manufacture of a number of medicines. Some of the important herbs used for medicines are *aconite*, *Keera-jari (insect herb)*, *celery*, *belladone*, *colocynth, sarasaparilla*, *jalap, leadwort*, *chitraka*, *serpentine, and liquorice*. The root of serpentine is an antidote for snake and insect bite. Barks of mountain ebony, Indian oak, quinine, Spanish-cherry, bay-berry, lodh-tree, Indian-red-wood, ashoka, arjuna, and Indian barberry have medicinal value. The stem of ephedrine, white sandal-wood, catechu, and long needle-pine are also of great medicinal importance. Leaves of *vasaka*, *Indian aloe*, *poison-bulb*, *fever-nut*, *life-plant*, *swallow wart*, *Indian-penny-word*, *Tasmanian blue-gum*, *physic-nut*, *holy-basil*, *betel*, *pepper*, *and typlophora* yield different drugs. Similarly, flowers of saffron, iron-wood, violet and fruits of bel, fish-berry, purging-cassia, coriander, cumin, fennel, emblic, opium, long-pepper, black and white pepper, belleric, myrobalan, ammi and solanum are used in medicine making. Keera-jari (insect herb) known as cordyceps sinensis is highly valued as a tremendous energy booster which cost about Rs. 2.5 lakh per kg.

7. Shellac

It is secreted by an insect called '*Kerria lacca'* which feed on the saps of host trees like palas, peepal, kusum, sissoo, kul, gular, siras, and banyan. These trees are extensively found in the Gangetic plains, Madhya Pradesh, Chhattisgarh, Jharkhand, Madhya Pradesh, Maharashtra and Assam. India has a monopoly in lac production in the world. It is used for dyeing silk, making bangles, paints, munitions, fire-works, gramophone records, sealing wax, electrical insulation material, shoe-dressing, plastic-moulding, spirit , baking enamels, and anti-moulding compository for ships.

8. Resins

Resin is the exudation of paints belonging to phaneorogamic families. It is a yellowish solid, insoluble in water, but soluble in alcohol. In the conifer pine forests of the Himalayan region, resin is collected on commercial scale. The resin is used for soap, and sizing paper and cloth. It is also used in the manufacture of sealing wax, linoleum, lubricating compounds, paints, and several kinds of inks.

9. Gums

Gum is obtained from acacia, carob, mesquite and kateera-gum. These are used as adhesives in printing and finishing textiles, in the paint and candy industries, and drugs.

10. Tannins and Dyes

Tannins are used for coagulating the protein in hides and skins, so that resistant leather can be produced. Lighter vegetable tannins dominate in the production of leather. *Tannins* are obtained from the bark of mangrove, sundri trees, wattle, avaram, sumac, arjun, Indian almond, jujube, Cuddapah-almond, hog-plum, chestnut, and leaves of smoke trees.

The dyes are coloured compounds. On being fixed to fabrics, they do not wash out with soap and water or fade on exposure to light. About 150 dye yielding plants are available in Indian forests, but only a few have been utilised so far.

11. Katha

Katha is extracted from the inner wood of '*khair tree*' which is largely grown in Uttarakhand, Uttar Pradesh, Himachal Pradesh, Madhya Pradesh, Chhattisgarh, Jharkhand, Gujarat, and Bihar. Its important factories are at Bareilly (U.P.) and Shivpuri (M.P.).

12. Fruits and Vegetables

The fruits and vegetables obtained from the forests are jamun, bel, ber, gular, jack-fruit, amla, tamarind, khirni, karonda, khajur, and chilgoza. Chinch, munga, chkoora, arvi, ratalu, kanhi, akana, kirchi, jarungi, sua, saijan, saidu, mushrooms, and guchchhi are obtained as vegetables.

13. Valuable Things

In addition to the given benefits from forests, the collection of ivory, honey, bees wax, hides, horns, and furs is also done from the forests.

14. Grazing Grounds

Forests are the grazing grounds for domesticated animals and the abode for over 30,000 species of plants, animals, and micro-organisms.

15. Forests and Climate

Climate, temperature, and precipitation are also directly affected by the presence or absence of forests. It is said that larger the area under forests, greater is the amount of precipitation.

PROBLEMS OF INDIAN FORESTRY

The specific problems of Indian forestry are the following:

1. Low Forest cover

(i) The forest cover in India is only 21.54% as against the world average of about 35%. The overall desired forest area as recommended in the National Forest Policy of India 1952, should be 33% (25% in plains and 60% in hilly regions) of the total geographical area of the country. This percentage is missing in almost all the states and UTs except Andaman and Nicobar, Assam, Goa, Kerala, Meghalaya, Sikkim, Uttrakhand, Mizoram, Arunachal Pradesh, Nagaland, and

Manipur. The states of Haryana, Punjab, and Rajasthan have less than 5% of their areas under forest. The total number of protected areas in India is 730, out of which 105 are in Andaman and Nicobar Islands. India has 103 National Parks, 517 wild-life sanctuaries, 18 Biosphere Reserves, 26 Community Reserves and 66 Conservation Reserves.

(ii) Most of the forests of India are not gregarious which creates problems in their exploitation. Teak, sal, bamboo, pines, oak, deodar, fir, spruce and larch are, however, exceptions.

(iii) About 40% of the total forest area is not easily accessible.

(iv) In about 50% of the total forest area, tribals have been given the rights of free grazing and cutting fuel-wood and timber for their personal consumption. This right is often misused. Nearly 275 million people in India especially the tribal community depend on the forest for their sustenance and livelihood.

(v) The falling of trees is still primitive and indigenous in most of the forests. This damages the ecosystem, leads to more soil erosion and delays the regeneration of forests.

(vi) There are inadequate trained personnel in forestry. Much of the energy of the available man-power is used in the protection and conservation of forests instead of their regeneration.

(vii) There is inadequate protection against forest fire, insects, pests and plant diseases.

(viii) The shifting cultivation in the wet mountainous regions of the country is another serious problem of the Indian forests.

(ix) Forest areas suffer a loss during the fires during the dry season with winds blowing. Recently massive fire broke in Almora and Nainital districts. The inflammable pine jungles on Bhowali road have left the environmentalists into worry. Nearly 3000 acres of forest have been destroyed.

2. Open Grazing

Extensive damage to the Indian forests is being done by the grazing of cattle, sheep and goats, especially in the hilly and mountainous areas, by the local people. Nomadic tribes like Bakarwals, Bhutias, Gaddis, Gujjars, and Lepchas, practicing transhumance (seasonal migration) damage the forest ecosystems.

3. Shifting Cultivation

Tribals in the hilly areas, where rainfall is more than 100 cm, generally practice shifting cultivation (*Jhuming*) in the hilly and mountainous areas. The increasing pressure of population has reduced the Jhum Cycle to only five years in many parts of Nagaland, Meghalaya, Mizoram, Manipur, and Tripura. Consequently, the forests do not have sufficient time to regenerate.

The Jhum cultivation is a primitive and unscientific land use that depletes forest , water and soil resources. The felling of trees , and clearing of bushes shall eventually lead droughts or floods and a decrease in soil fertility.

The state forest Departments have implemented afforestation programmes to stop shifting cultivation. It has been noticed that in the places where 'Mahatama Gandhi National Rural Employment Guarantee Act, (MGNREGA) has been implemented , the dependence on Jhum has declined.

4. Growing Demand for Agricultural Land

With the tremendous increase in population during the last 65 years, the demand for cereals and agricultural raw materials has increased significantly. Consequently, the forest area has been brought under cultivation leading to a continuous shrinkage of the forest area.

5. Urbanisation and Industrialisation

Fast urbanisation and industrialisation in the forest and hilly areas is also an important cause of forest degradation. The size of cities like Shimla, Mussoorie, Dehra-Dun, Nainital, Ranikhet, Chamba, Dalhousie, Darjeeling etc. has increased over ten times during the last thirty years. There has been rapid expansion of roads in the Himalayas and other forest areas as a result of which the valuable forests have been exposed to tourists and pleasure-seekers.

Due to economic development , 14,000 km2 of forests were cleared to set up 23,716 industrial projects in India over the last 30 years.

6. Construction of Multi-Purpose Projects

The construction of reservoirs of big dams like Bhakra-Nangal, Rihand, Hirakud, Tehri, Koteshwar, Salal, Dulhasti, Sardar-Sarovar, etc. has resulted in the submergence of large forest tracts.

7. Commercial Activities

Commercial activities like resin extraction, mining, quarrying, oil-extraction, plantation, orchard development have also led to large-scale deforestation. Unfortunately, paper mills, and saw-factories have been located in the forests areas which accelerate the process of deforestation.

THE NATIONAL FOREST POLICY

India is one of the very few countries of the world where forest policy is in operation since 1894. In 1952 and 1988, revisions were made in the forest policy of 1894. The National Forest Policy of 1952 recommended that the country should aim at a coverage of one-third of the total land area under forest (60% in hilly and mountainous areas, and 25% in the plains). It has suggested the extension of tree-lands on river/canal banks, roads, railways, culturable waste and in such areas which are not suitable for cultivation.

The National Forest Policy 1952 classified the forests of the country into four categories:

(i) **Protected forests** essential for physical and climatic needs.

(ii) **National forests** to be utilised for the economic needs of the country.

(iii) **Village forests** to meet the fuel and domestic needs of villages and neighbouring towns.

(iv) **Tree lands**. The policy envisaged the annual organisation of *Van-Mahotsava* and tree plantation week in the month of July/August.

The National Forest Policy 1952 lays emphasis on :

(i) Weaning the tribal people by persuasion to desist from shifting cultivation.
(ii) Implementation of forest laws more effectively.
(iii) To provide adequate facilities for the management of forest resources.
(iv) To control grazing of cattle, sheep and goats in forest areas.
(v) Providing fuel-wood to rural areas.
(vi) To improve the availability of timber wood for industrial purposes.
(vii) To increase the area under social forestry.
(viii) To promote research in forestry.

The National Forest Policy 1988

The main emphasis of the National Forest Policy 1988 is on the protection, conservation, regeneration and development of forests. The main points of the National Forest Policy 1988 are:

(a) Maintenance of environmental stability through the preservation and restoration of ecological balance.
(b) Conservation of forests as a national heritage with vast varieties of flora and fauna.
(c) Control of soil erosion and denudation in catchment areas of rivers, lakes and reservoirs.
(d) Check on the extension of sand-dunes in desert areas of Rajasthan and along sea-coasts.
(e) Substantial increase in forest cover through massive afforestation and social forestry programmes.
(f) To meet the needs of fuel-wood, fodder and minor forest products for the rural and tribal people.
(g) Augment the productivity of the forests to meet national needs.
(h) Encouragement of efficient utilisation of forest produce and optimum substitution of wood.
(i) Steps to create massive movement of people with the involvement of women folk to achieve these objectives and to

minimise pressure on existing forests.

(j) Involvement of people in forest management under joint forest management.

The National Forest Policy 2018

a) The emphasis in this policy has been laid on the timber and forest based industries.

b) The sections of 'Rights and Concession' and 'Tribal People and Forests' have been replaced by 'Production Forestry', 'Increase the productivity of forest plantations' and ' Facilitate Forest industry interface' that focus on increasing the timber yield.

c) It aims to use the degraded land available with the forest corporations for the production of quality timber.

d) It emphasizes on bringing at least one third of India's total geographical area under forest cover through scientific approach

e) Focus is laid also on the international challenge of climate change.

f) Monitoring of wild life to be adopted on long term basis inside and outside the forests for maintain a balance.

g) The commercial important trees '*Poplar and Euclyptus*' have been encouraged in the draft ,though these are deep rooted and water demanding. These cause depletion of the groundwater.

h) The policy has not highlighted in detail the issues of diversion of forest land for mining and other purposes and issues for projects in the region.

SOCIAL FORESTRY

Social forestry refers to the forests (trees) planted by the people of a society. It has been defined as *the forestry of the people, for the people by the people*. The significance of social forestry has been emphasised in the National Forest Policy 1952 and 1988. The main

objective of social forestry is to reduce pressure on traditional forests by plantation of fuel-wood, fodder, timber, and grasses. The two types of social forestry include:

Agro-forestry which includes community forestry and agro-forestry (commercial and non-commercial farm forestry).

Objectives

(i) To meet the need for fuel wood, small timber, bamboo, fodder and other minor forest produce on sustainable basis.
(ii) To release cow dung as manure for increasing agricultural production,
(iii) To provide gainful employment opportunities to the rural population,
(iv) To develop cottage industries,
(v) To provide efficient soil and water conservation,
(vi) To provide efficient soil and water conservation, and
(vii) To improve aesthetic value of an area and to meet the recreational needs of the population.

Community Forestry

Community forestry is a part of social forestry. It involves the raising of trees on community lands with the set objective to provide benefits to the community as a whole. Although the plants and seedlings are provided by the forest departments, the protection of planted trees is primarily the responsibility of the community as a whole.

India has performed superbly in community forestry and stands only next to China in this respect. The states in which community forestry is a big success are Gujarat, Haryana, Himachal Pradesh, Karnataka, Kerala, Odisha, Punjab, Rajasthan, Tamil Nadu, Uttarakhand, and Uttar Pradesh.

Agro-Forestry

Agro-forestry is a sustainable management for land that increases overall production, combines agricultural crops, tree crops, forest

plants and animals simultaneously and applies management practices that are compatible with cultural patterns of local population.

Agro-forestry is a type of social forestry in which individual farmer undertakes tree-farming and grows fodder plants, grasses and legumes on his own land. In agro-forestry, trees (forest) are considered as a crop and they (trees) become a part of crop combinations.

There is an increasing number of farmers who plant trees either along the boundaries or in the whole field/farm as a crop **(Fig. 5.4).** In fact, in the Northern Plains of India, trees are planted by most of the farmers irrespective of the size of their holdings, but the large farmers and absentee landlords put part of their holdings or total agriculture area under tree crops.

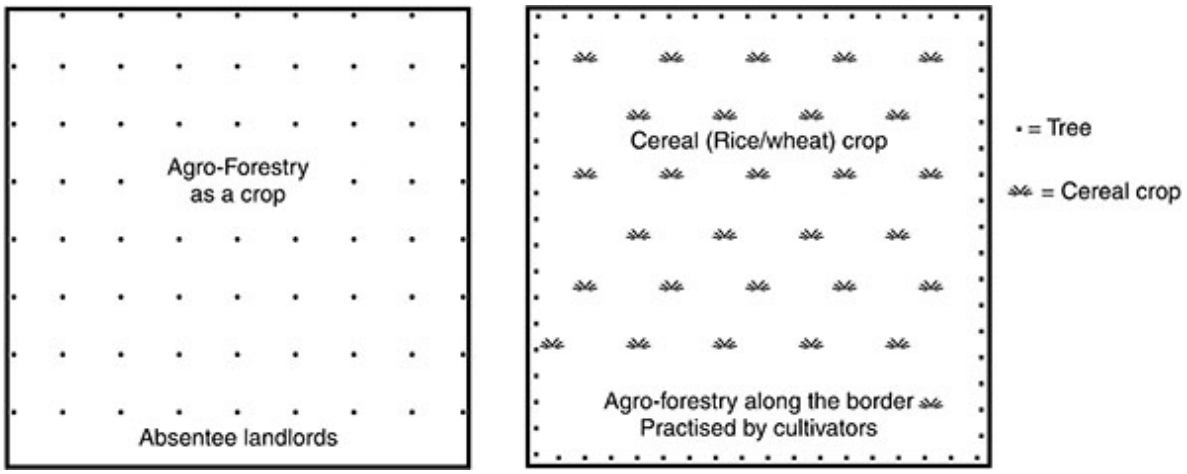


Fig. 5.4 Patterns of Agro-Forestry

Under agro-forestry, a farmer generally uses his degraded or useful land to plant trees for domestic use or commercial use. Such land may be his own or obtained on lease under social forestry programme.

Agro-forestry involves both the big and the small farmers. It fetches additional income to farmers, improves their income and thereby, their standard of living, and provide them employment during lean agricultural seasons. These trees are normally harvested after 6 to 10 years from the date of plantation, depending on the needs and

requirements of the farmer. The main advantages of agro-forestry are:

(i) The absentee landlords go for agro-forestry to retain title of the land and to increase their income.
(ii) To manage the agricultural land even without the availability of family labour.
(iii) To ensure better land use.
(iv) To generate employment.
(v) To conserve soil moisture.
(vi) To meet the needs of fuel-wood, fodder and timber.
(vii) To protect the arable land from winds and water erosion.

Despite numerous material and geo-climatic benefits, agro-forestry has some adverse effects on agricultural land. Some of its shortcomings are given below:

(i) The market-oriented trees are preferred which damage the ecosystem. Instead of poplar and eucalyptus, the farmers should go for the plantation of Neem, Mahua, Karanj, Arjun, and acacia.
(ii) Fuelwood and fodder trees are generally neglected.
(iii) The exotic varieties planted by the farmers in the form of agro-forestry are soil-moisture and water exhaustive. Consequently, the underground water-table is adversely affected.
(iv) The land under agro-forestry becomes unproductive as the roots of trees become so dense that their digging and removal to bring the land under cultivation becomes very difficult unless heavy investment is made in the digging and removing of roots.
(v) In the fields along which trees have been planted, the productivity per unit area decreases, as at least in about two m from the trees the moisture content in the soil is significantly reduced.
(vi) The trees become the habitat of many pests and diseases,

adversely affecting the crops.

(vii) Unscientific method of spacing of trees, reduces the growth and mass of the trees.

In brief, Agro-forestry is a system of agricultural land utilisation which not only provides fuel-wood, fodder, and grasses, but helps in the promotion of forests and their conservation. The Indian Council of Agricultural Research and the Forestry Department should jointly undertake agro-forestry research in order to develop suitable systems of land management which involves integration of silviculture, with horticulture, agriculture and animal husbandry. Agro-forestry, thus, integrates agri-silviculture, silvi-pastoral system, and medicinal plants culture. Under agro-forestry, a farmer generally uses his degraded or useful land to plant trees for his domestic use or for commercial use.

Agro-forestry has become very popular in Andhra Pradesh, Bihar, Gujarat, Haryana, Karnataka, Kerala, Punjab, Rajasthan, Uttarakhand, and Uttar Pradesh. The main species of trees planted by farmers in their fields are eucalyptus, poplar and casuarinas. The wood of these forests is used mainly for fuel-wood, plywood, paper and pulp manufacturing, and match industries.

Agro-forestry has benefited the big farmers more than the marginal and small farmers. Many of the absentee land-lords plant commercial trees in their agricultural land-holdings to save their land from dispossession. Thus, the agricultural labourers are thrown out of employment. The diversion of good agricultural land from cereal and commercial crops may create the problem of scarcity of food stuffs and industrial raw material. The programme, therefore, needs a new strategy and reorientation to achieve its real objectives.

EXISTING POSITION OF FOREST ECOSYSTEMS

The continuing exploitation of our forest resources has damaged the forest ecosystem almost beyond repair. The depletion of forest cover is due to expansion of agriculture, habitat destruction,
over-exploitation, pollution, toxic imbalances in community structure, epedemics, floods, droughts and cyclones. As a consequence of the

denudation of hill slopes, soil erosion and recurrent floods have emerged as major problems. According to one estimate by the National Remote Sensing Agency, India has been losing about 1.3 million hectares of forest cover every year. This is an alarming rate indeed. It is now generally realised that forests are life sustaining agents. They play a vital role in maintaining the ecological balance. The recent concern for environment has resulted in rethinking on these issues. Looking at the present state of our forest resources and their degradation, it is imperative to utilise them in a judicious manner, so that they may be regenerated in a short time to make them sustainable.

FOREST CONSERVATION

The utility of forests, their social relevance and climatic importance have been discussed in the preceding paras. The conservation of forest resources is imperative for our survival. Some of the steps which can go a long way in making forests healthy and sustainable are as under:

(i) *Afforestation:* There should be massive afforestation programmes with main emphasis being on the production of fuel-wood, timber, grasses, and small trees to cover up degraded and denuded lands.

(ii) Plantation of trees along the roads, railway lines, rivers, and canal banks, and along lakes and ponds.

(iii) Development of Green-belts in the urban areas and plantation of trees on community lands.

(iv) Plantation of community forests on *Gram-Sabha* lands.

(v) Villagers should be given loans at easy interest rates to revive degraded forest.

(vi) Encroachment of agriculture in forests should be made punishable.

(vii) The customary rights and concessions like grazing, collection of fuel-wood and fodder from forests by the local people

should not be allowed to exceed the carrying capacity of the forests.

(viii) Rural population should be provided alternate sources of fuel-wood and wood-based products.

(ix) The development projects including mining and industrial activities should be so planned to cause minimum damage to forest ecosystems.

(x) Mining contracts should have an obligatory clause of reforestation when the process of mining is over.

(xi) Industries should adopt anti-pollution devices and must develop and compensate the forest loss by new plantation.

(xii) Tribal and local people should be directly involved in the protection, regeneration, and management of forests.

(xiii) Shifting cultivation should be gradually replaced by terraced farming and orchards development and silviculture.

(xiv) Scientific methods should be adopted to check and contain forest fires. There should be strict control in issuing license for the establishment of industries in forest areas.

(xv) There should be more research on forestry in agricultural universities, for which facilities and funds should be provided by the Central and State governments. Forestry should be made an important part of the course structures in schools, colleges and universities.

(xvi) There should be arrangements to protect the forests from pests and diseases for which trimming and spraying of the trees should be a regular practice.

(xvii) There should be perfect co-ordination between the forest department and other departments of the government for effective and judicious utilisation of forests and their conservation.

(xviii) People should be encouraged to participate in the *Van-Mahotsav* and should be made aware about the *Chipko Movement*.

(xix) There is a need to change our outlook towards forests. A

forest should not be treated as a perennial resource and a source of revenue only. The planning and conservation of forests is not only the duty of the government, but also all the citizens of the country.

(xx) There should be special audio-visual programmes, demonstrations, seminars and workshops to develop awareness among the people about the social relevance of forests.

The Indian Council of Forestry Research and Education (ICFRE) was created in 1987 under the Ministry of Environment and Forests. The following forestry research institutes are working under the Indian Council of Forestry Research and education **(Fig. 5.5)**:

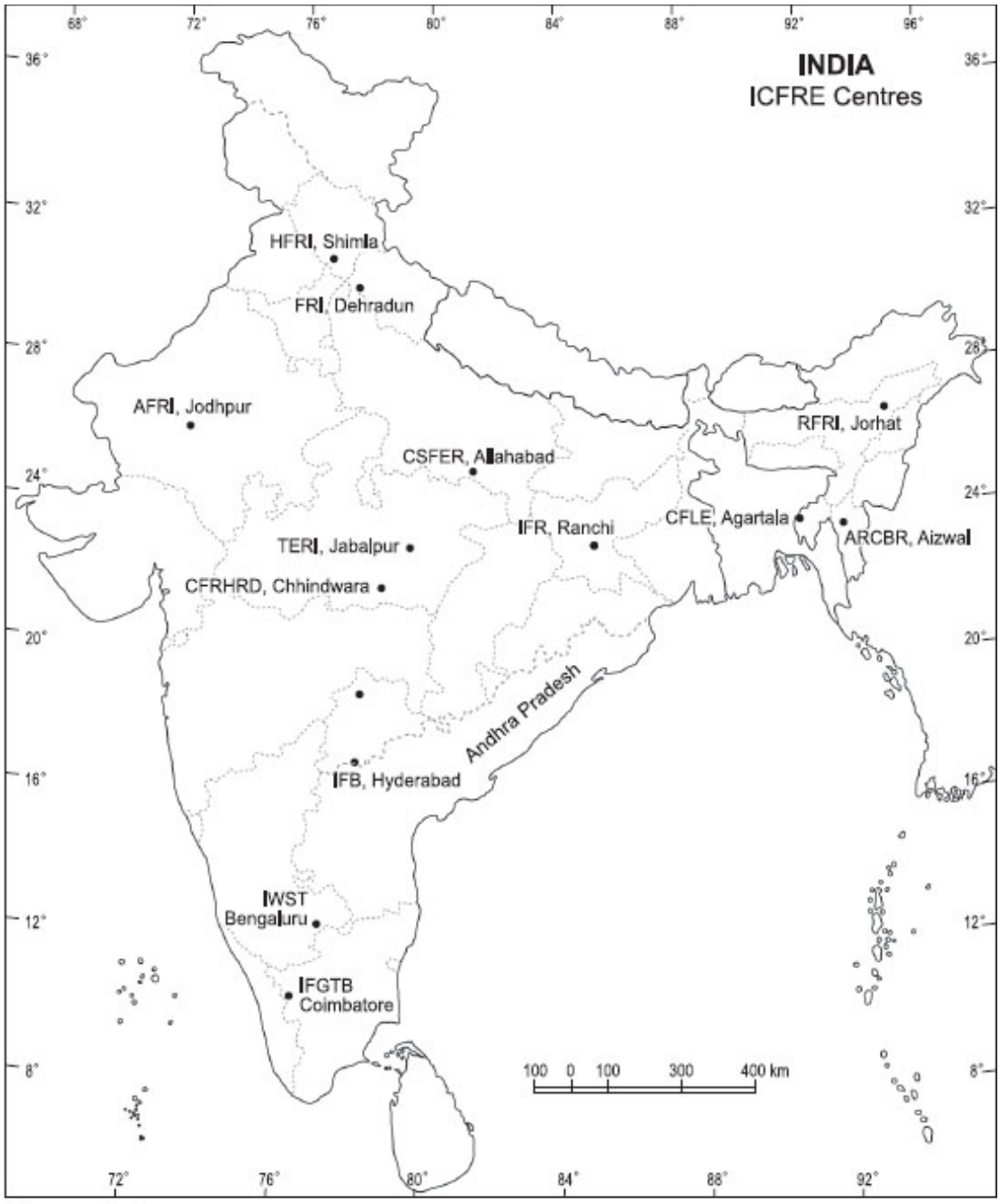
(i) Forest Survey of India, Dehra Dun
(ii) The Central Arid Zone Research Institute, Jodhpur
(iii) The Institute of Rain and Moist Deciduous Forests, Jorhat
(iv) The Institute of Wood Science and Technology, Bangalore
(v) The Tropical Forestry Research Institute, Jabalpur.
(vi) The Institute of Forest Genetics and Tree Breeding, Coimbatore
(vii) The Himalayan Forest Research Centre, Shimla
(viii) The Centre for Forest Productivity, Ranchi
(ix) The Centre for Social Forestry and Environment, Allahabad
(x) Institute of Forest Biodiversity, Hyderabad
(xi) Centre for Social Forestry and Eco rehabilitation, Allahabad
(xii) Centre for Forest based livelihoods and extension, Agartala
(xiii) Directorate of Forest Education , Dehradun

WILDLIFE

India has a great diversity in its geo-climatic environment. This diversity of environment provides natural habitats for wild animals, birds and insects. Out of the world's total of about 15 lakh species of

animals, India has 81,251 or 6.7% species. According to S.H. Prater (1934), India can be divided into six zoo-geographic regions. India has 350 different mammals, 1200 species of birds, 453 species of reptiles, and 45000 plant species. Moreover, India has 50,000 known species of insects, including 13,000 butterflies and moths. They are: (i) The Himalayan region (ii) The Northern Plains, (iii) The Thar Desert, (iv) The Peninsular Plateau, (v) The Malabar Coast, and (vi) The Nilgiri.

Under the pressure of a fast increasing population and heavy demand of food and industrial raw material, the forest cover of India has shrunk substantially. Moreover, the density of trees in the forests is very low as compared to the forests of other countries. Consequently, the natural habitat of wild animals are disappearing. The growing trade of furs of big cats and tiger bones has diminished their number. There is increasing demand of rhino-horns and deer musk which are used in medicines and aromatic substances. Moreover, due to the continuous hunting of wild animals and illegal poaching, the number of wild animals is decreasing day by day. Some of the species are on the verge of extinction. Asiatic lion, clouded leopard, tiger, musk-deer, rhinoceros, great Indian bustard, Nilgiri langur, python and vultures are the species in danger.



Map not to Scale

Fig. 5.5 ICFRE Centres

In order to conserve wildlife, the Government of India passed the Wildlife Protection Act in 1972. Large tracts in various parts of the country covering 1.56 lakh km2—4.75% of the total area of the country—were declared as national parks, sanctuaries and biosphere

reserves. At present (2016), the number of national parks and sanctuaries has gone up to 103 and 535, respectively.

The National Biological Diversity Act 2002 ensures the protection of threatened species and their habitats. The Wild life Crime Control Bureau (WCCB) takes care of the crimes of poaching and illegal trade in wildlife and its products.

To increase the number of animals classified under endangered species, Project Tiger (1973), Crocodile Breeding Project (1975), Rhinoceros Project (1987), Snow-leopard Project, and Project Elephant (1992) have been launched.

National Park: A national park is a relatively large area of one or several ecosystems that are not being materially altered by human exploitation and occupation. Here, plant and animal species, geomorphological sites and habitats for special scientific education and recreation are preserved. The National Parks of India have been plotted in **(Fig. 5.7)** and some of the important national parks have been described in the following section:

Table 5.7

Popular National Parks of India

S.No.	National Parks	Year of Establishment	State
1	Jim Corbett National Park	1936	Uttar Pradesh
2	Kanha National Park	1955	Madhya Pradesh
3	Kaziranga National Park	1974	Assam
4	Gir National park	1975	Gujarat
5	Bandhavgarh National Park	1968	Madhya Pradesh
6	Keoladeo Ghana National Park	1981	Rajasthan
7	Valley of Flowers national Park	1982	Uttrakhand
8	Periyar Wildlife sanctuary	1982	Kerala
	Kanger Valley National Park	1983	Chattisgarh
9	Fossil National Park	1983	Madhya Pradesh

S.No.	National Parks	Year of Establishment	State
10	Mahatama Gandhi Marine National Park	1983	Andaman & Nicobar Islands
11	Namdapha National Park	1983	Arunachal Pradesh
12	Silent Valley National Park	1984	Kerala
13	Saddle Peak National Park	1987	Andaman & Nicobar Islands
14	Sariska Wildlife Sanctuary	1992	Rajasthan
15	Bison National Park	2007	Tripura
16.	Khirganga National Park	2010	Himachal Pradesh
17.	Jaldapara National Park	2014	West Bengal

Wildlife Sanctuary: Similar to a national park, a wildlife sanctuary is dedicated to protect wildlife and particular species. In a sanctuary, human activities are allowed, but in a national park human interference is totally prohibited.

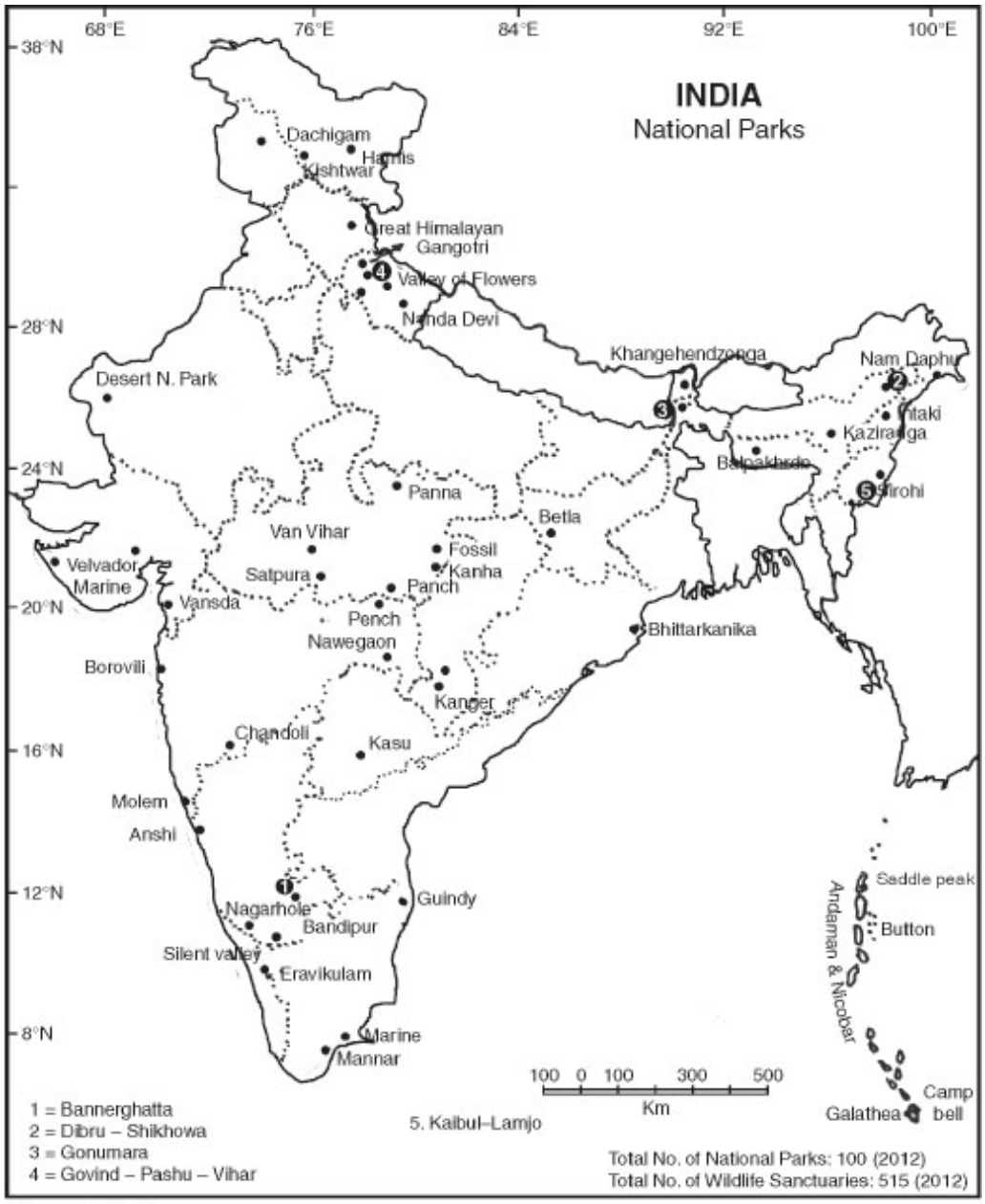
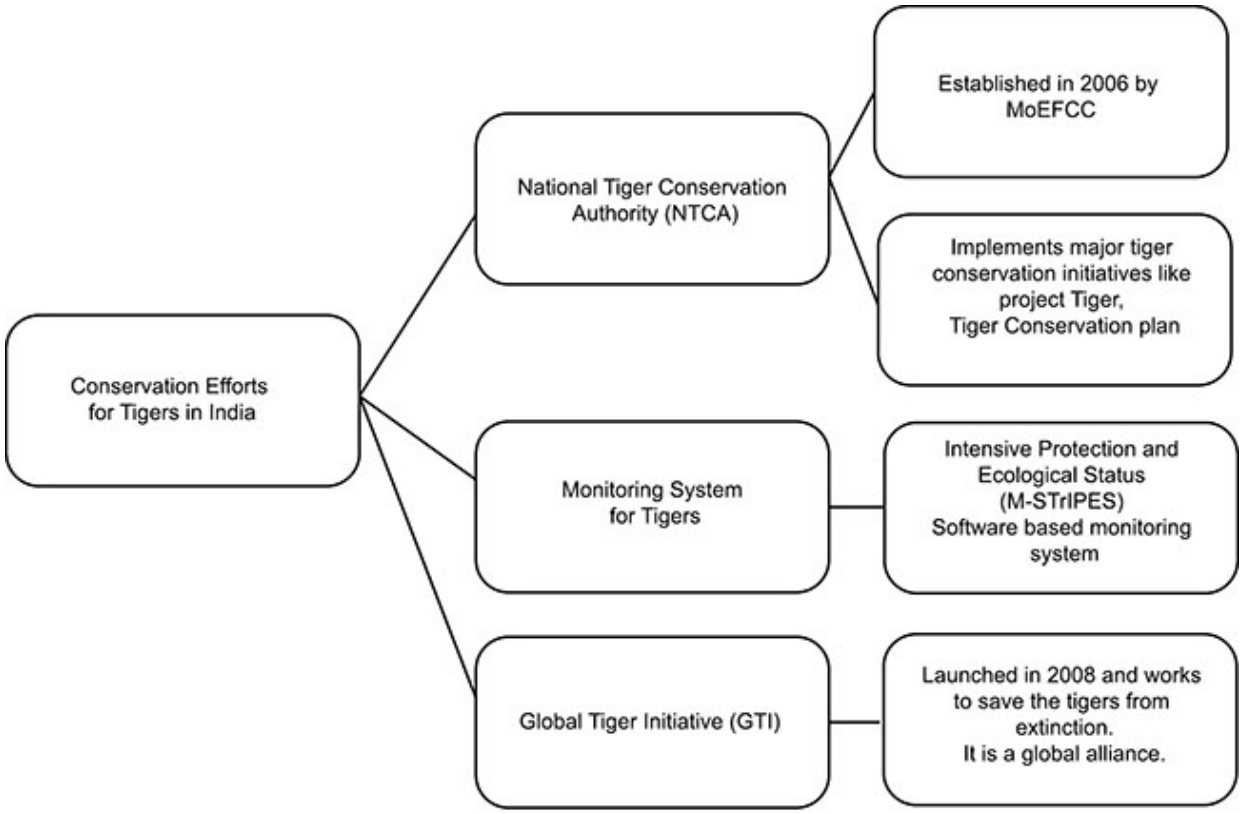


Fig. 5.6 National Parks [*Source: ENVIS Center on Wildlife & Restricted area.*]

Tiger Reserves: Some of the important Tiger Reserves of India have been shown on the map in the subsequent pages. The number of tigers in India has increased 1706 in 2010 and 2226 in 2014 as compared to 1411 in the year 2006. The largest tiger reserve in India is '*Nagarjuna Sagar-Srisailam Tiger Reserve*' (Andhra Pradesh,

Telangana)with a total area of 3296.31 km2, while the smallest is '*Bor Tiger Reserve*' in Mahrashtra with a total area of 138.12 km2. (*Source: ENVIS Centre on Wildlife and protected Areas*). The total number of tiger reserves in July 2014 was 47 and it has got increased to 50 in 2018. The total area under Tiger reserves is 71027.10 Km2 The tigers are maintained for their scientific, economic, cultural, aesthetic and ecological values and to preserve for all time areas of biological importance as a national heritage for the benefit, education, training and enjoyment of the people. Their geographical distribution has been shown in **(Fig. 5.7).**



BIOSPHERE RESERVES

Biosphere reserves are areas of terrestrial and coastal ecosystems which are internationally recognised within the framework of UNESCO's *'Man and Biosphere (MAB) Programme'.* These reserves are required to meet a minimal set of criteria and adhere to a minimal set of conditions before being admitted to the *World Network of*

Biosphere Reserves designated by the UNESCO. These reserves are rich in biological and cultural diversity and encompass unique features of exceptionally pristine nature. The scheme is a pioneering effort at pursuing the increasing difficult yet urgent task of conserving ecological diversity under mounting pressures.

The main objectives for the delineation of biosphere reserves are:

I. To conserve biological and cultural diversity and integrity of plants animals, birds, reptiles, micro-organisms and unique features of pristine nature.
2. To promote research on ecological conservation and other environmental aspects, and
3. To provide facilities for education, research, awareness and training.

The Biosphere Reserves Programme was initiated in India in 1986 and till date, 18 sites have been designated as Biosphere Reserves (BR) in different parts of the country. The names of the biosphere reserves of India have been given in **(Table 5.8)** and their locations have been shown in **(Fig. 5.8).**

Fig. 5.7 Tiger Reserves [*Source: ENVIS Center on Wildlife & Restricted area.*]

The programme was initiated in 1986 and till date, 18 sites have been designated as Biosphere Reserves in different parts of the

country. Out of the 18 biosphere reserves, seven biosphere reserves have been included in the World Network of Biosphere Reserves so far. These 7 include: Sundarban, Gulf of Mannar, Nilgiri, Nanda Devi, Panchmarhi, Simlipal and Nokrek.

These biosphere reserves include:

1. Nilgiri, 2. Nokrek, 3. Nanda Devi, 4. Sundraban, 5. Gulf of Mannar, 6. Manas,
7. Great Nicobar, 8. Simlipal, 9. Dibru-Saikhova, 10. Dehang-Dibang, 11. Panchmarhi,
12. Kanchanjunga, 13. Agasthamalai, 14. Achanakamar, 15. Great Rann of Kutch, 16. Cold Desert (Pin-Valley-H.P.), 17. Seshachalam (Andhra Pradesh), 18. Panna (M.P.).

Table 5.8

Biosphere Reserves of India

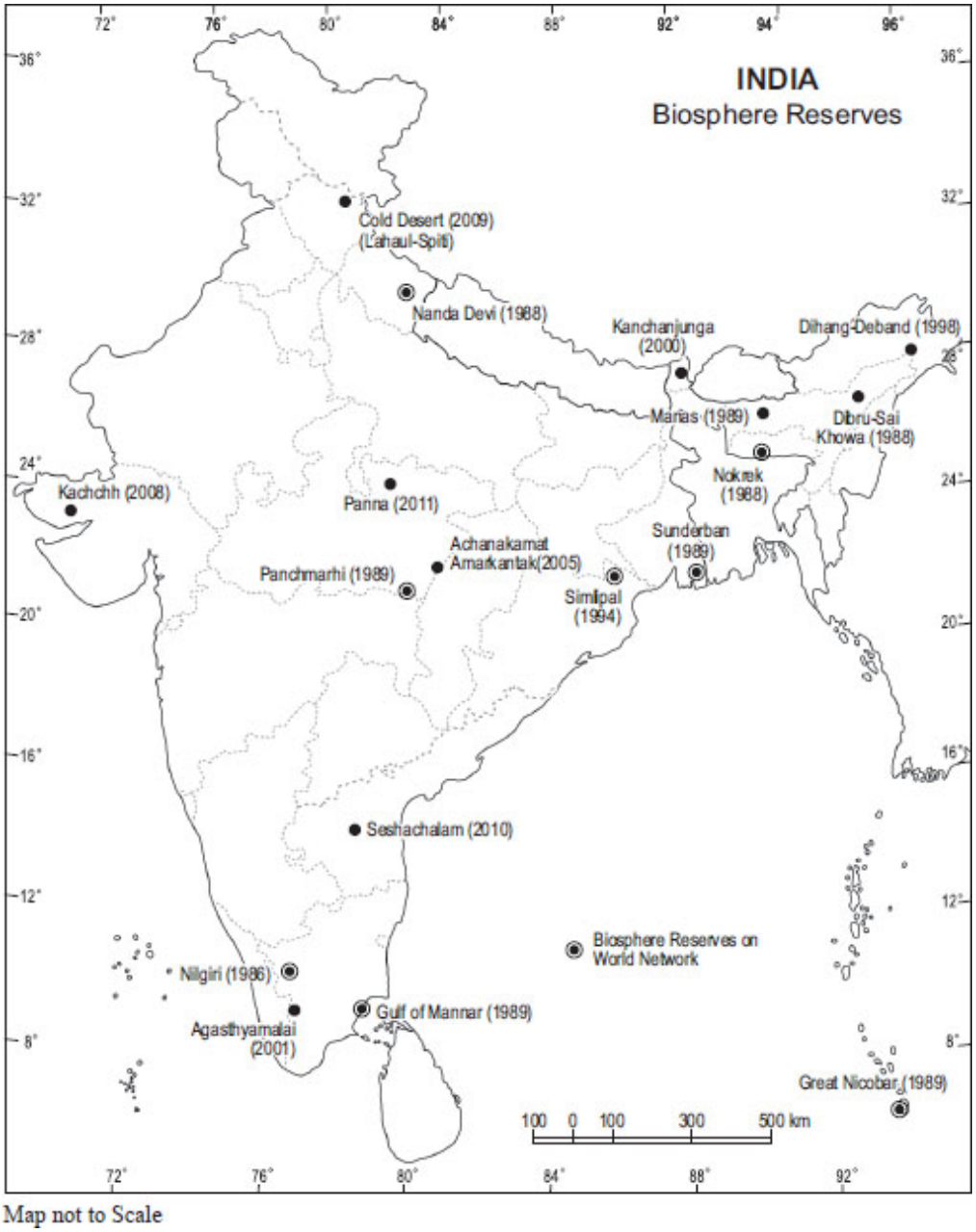
	Biosphere Reserve		Geographical Area in km2	States	Main Fauna
1.	Nilgiri*	1986	5520	Tamil Nadu, Kerala, Kamataka	Nilgiri tahr, lion-tailed macaque
2.	Nanada Devi*	1988	6407	Uttarkhand	Nilgiri tahr, lion-tailed macaque
3.	Nokrek*	1988	0820	Meghalaya	Red panda
4.	Manas	1989	2837	Assam	Red panda, golden langur
5.	Sundarban*	1989	9630	West Bengal	Royal Bengal tiger
6.	Gulf of Mannar*	1989	10,500	Tamil Nadu coast	Dugong or sea cow
7.	Great Nicobar	1989	0885	Andaman and Nicobar Islands	Salt water crocodile
8.	Simlipal*	1994	5569	Odisha	Gaur, Royal Bengal tiger, wild elephant

	Biosphere Reserve		Geographical Area in km2	States	Main Fauna
9.	Dibru-Saikhowa	1997	0765	Arunachal Pradesh	Golden langur
10.	Dihang-Dibang	1998	5111	Arunachal Pradesh	Musk deer, Mishmi takin, red goral, Asiatic black bear
11.	Panchmarhi*	1999	4928	Madhya Pradesh	Giant squirrel, flying squirrel
12.	Kangchendzonga	2000	2931	Sikkim (Kanchenjunga)	Red panda, snow leopard
13.	Agasthymalai	2001	3500	Kerala	Nilgiri tahr, elephants
14.	Achanakmar Amarkantak	2005	3835	Madhya Pradesh	Blackbuck, chinkara, wolves, giant squirrels
15.	Kutch	2008	12,454	Gujarat	Indian wild ass
16.	Cold Desert	2009	7770	Lahaul-Spiti and Ladakh	Snow leopard
17.	Sheshachalam	2010	4756	Andhra Pradesh	Slender loris and pangolin (critically endangered species)
18.	Panna	2011	2999	Madhya Pradesh	Tiger, chital, chinkara, sambhar and sloth bear
	Total		**84,668**		

*Biosphere Reserves on the UNESCO Network of Biosphere Reserves.

Source: ENVIS Center on Wildlife & Restricted area.

Seven out of the 18 biosphere reserves are a part of the World Network of Biosphere Reserves, based on the UNESCO Man and Biosphere Programme List. Their names are: 1. Nilgiri, 2. Sundarban, 3. Gulf of Mannar, 4. Nanda Devi, 5. Nokrek, 6. Panchmarhi, and 7. Simplipal.



Map not to Scale

Fig. 5.8 Biosphere Reserves

MANGROVES

Mangroves are large flowering shrubs or trees that grow in dense thick forests along muddy or silty tropical coasts. They act as green shields buffering the coastline against sea enosion, sea-surge, cyclones and tsunami. They are highly productive ecosystems which are capable of exporting energy and materials to adjacent communities. They support a diverse heterotrophic food-chain and act as nurseries in the life-cycle of some organism. The dominant trees are '*Sundri'* also known as **Kandla**. The sundri flowers are used for dyeing of cloths. According to the latest assessment, mangrove covers 4650 km2 or 0.14% of the geographical area of the country. The government of India has identified 28 mangrove areas. The statewise distribution of mangroves has been given in **(Table 5.9)**.

Table 5.9

Mangroves in India-2019

S.No.	Mangrove	State/UT	Total Area (in km2)
1.	Kendrapara	Odisha	201.49
2.	Uttar Kannada	Karnataka	8.50
3.	Goa	Goa	26
4.	Godavari Delta	Andhra Pradesh	188.06
5.	Gulf of Kutch	Gujarat	794.77
6.	Krishna Delta	Andhra Pradesh	137.08
7.	Guntur	Andhra Pradesh	68
8.	Bhadrak	Odisha	34.87
9.	North Andaman	Andman & Nicobar Islands	424.52
10.	Thiruvarur*	Tamil Nadu	12.85
11.	Thanjavur	Tamil Nadu	12.25
12.	Udupi, N. Kannada	Karnataka	10.04
13.	South 24 Parganas	West Bengal	2082.07

Source: ISFR-Mangrove Cover-2019 *New District according to Census, 2011

Coral Reefs

The four major coral reef areas identified for intensive conservation and management are: (i) Gulf of Mannar, (ii) Gulf of Kutch, (iii) Lakshdweep, and (iv) Andaman and Nicobar Islands.

Wetlands (Ramsar Convention)

The convention on wetlands (Ramsar-Iran, 1971)-called the Ramsar Convention is an intergovernmental treaty that embodies the commitment of its member countries to maintain the ecological character of their wetlands of international importance and to plan for the 'wise-use' or sustainable use. The Ramsar convention works with collaboration of International Union for Conservation of Nature (IUCN) International Water Management Institute (IWMI), Wetlands International, WWF International.

Any wetland, to be declared a wetland of international importance, should support vulnerable, endangered or threatened species and attract more than 20,000 water birds.

In India, the scheme on conservation and management of wetlands was initiated in 1987. Over the years, based on the recommendations of National Wetlands Committee, 115 wetlands have been identified for conservation under the programme. The main wetlands which have been included in the Ramsar List are given in **(Table 5.10)**.

Table 5.10

India Wetland included in Ramsar Convention List

State/UT	S. No.	Name of Ramsar Site	Area (in km2)
Andhra Pradesh	1	Kolleru	901
Assam	2	Deepar Beel	40
Gujarat	3	Nalsarovar Bird Sanctuary	
Himachal Pradesh	4	Pongdam	156.62
	5	Renuka	2
	6	Chandratal	49
Jammu & Kashmir	7	Wullar	189
Ladakh	8	Tso Morari	120
Jammu & Kashmir	9	Hokera	13.75

State/UT	S. No.	Name of Ramsar Site	Area (in km2)
Jammu & Kashmir	10	Mansar & Surinsar	3.5
Kerala	11	Ashtamudi	614
	12	Saathamkotta	3.75
	13	Vembanad Kol	1512.5
Madhya Pradesh	14	Bhoj	8
Manipur	15	Loktak	266
Orissa	16	Chilika	1165
	17	Bhitarkanika	650
Punjab	18	Harike	41
	19	Kanjli	1.83
	20	Ropar	13.65
Rajasthan	21	Sambhar	240
	22	Keoladeo NP	28.75
Tamil Nadu	23	Point Calimere	385
Tripura	24	Rudra sagar	2.4
Uttar Pradesh	25	Upper Ganga	266
West Bengal	26	East Kolkatta wetland	125

Source: Wildlife Conservation using Spatial Techniques, Ministry of Environment, Forest and Climate Change, Government of India(2015)

Strategy for the Conservation of Wildlife

The following steps can go a long way in the conservation of wildlife:

(i) Hunting should be strictly prohibited in the national parks, sanctuaries, tiger reserves and biosphere reserves.

(ii) Poachers and herdsmen should not be allowed to enter the reserved parks and sanctuaries.

(iii) More national parks and wildlife sanctuaries should be established.

(iv) The existing national parks and sanctuaries should be further developed by providing more infrastructural facilities.

(v) Captive breeding of wildlife should be encouraged.

(vi) Adequate medical facilities should be provided in the national parks and sanctuaries for the treatment of wildlife.

(vii) Conducive habitat and environment should be created for living and breeding in the national parks and sanctuaries.

(viii) Seminars, conferences, workshops and exhibitions should be organised in national parks and sanctuaries to improve general awareness among caretakers about wildlife.

(ix) All the degraded forest land should be taken up for afforestation.

(x) The marginal lands that are not suitable for crop production should be brought under social forestry to increase the ecosystems and habitats for wildlife.

Western Ghats: A World Heritage Site

The Coorg or Kodagu is a part of Western Ghats that has been included in the UNESCO World Heritage List in the meeting of the World Heritage Committee held at St. Petersburg in Russia on 1 July, 2012.

The Western Ghats has “outstanding examples representing significant ongoing ecological and biological processes in the evolution and development of terrestrial, freshwater, coastal and marine ecosystems and communities of plants and animals”. It is also the most significant natural habitat for in-situ conservation of biological diversity, including those containing threatened species of outstanding universal value from the point of view of science and conservation. Apart from the World Heritage Site, it is one of the eight hottest **hot spots** of biological diversity in the world. The Ecologically Sensitive Area (ESA) is 56,825 km2 (2014).

The Western Ghats or the Sahyadri is a mountain ranee sec *‘re western* see of the Peninsular India. It separates the Deccan Plateau from the narrow coastal plain along the Arabian Sea **(Fig.5.8).**

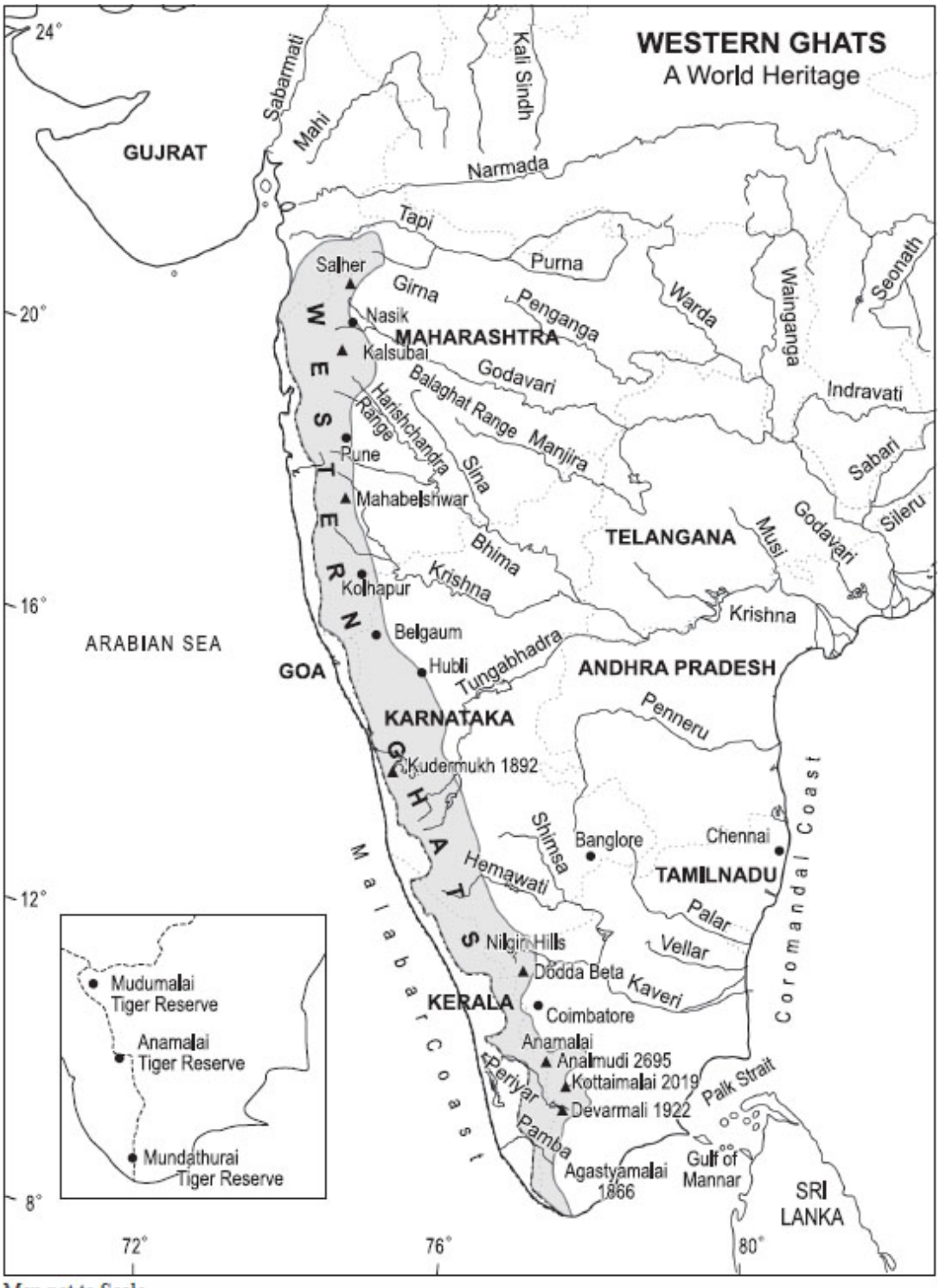
The Western Ghats starts south of the Tapi river in Gujarat and runs about 1600 km through the six states of Gujarat, Maharashtra, Goa, Karnataka, Tamil Nadu and Kerala ending at Kanyakumari, at the southern tip of India.

The main peaks of the Western Ghats are: Anaimudi (2695m), Doddabetta (2636m), Mukurthi (2554m), Kodaikanal (2133m),

Bababudangiri (1895m), Kudremukh (1894m), Agasthymalai (1866m), Pushpagiri (1712m), Kalsubai (1646m), and Salher (1567m). The important hill stations of Ooty (2500m) and Kodaikanal (2285m) are also located in the Western Ghats.

The area has 5000 species of flowering plants, 139 mammal species, 508 bird species and 179 amphibian species. According to one estimate 325 globally threatened species occur in the Western Ghats.

The Western Ghats are covered with the tropical and subtropical forests that provide food and natural habitats for the native tribal people. The area is ecologically sensitive to development. The Government of India has established many protected areas including 2 biosphere reserves, 13 national parks and several wildlife sanctuaries to protect specific endangered species. The Nilgiri Biosphere Reserve (5500 km2) of the evergreen forests of Nagarhole, the Bandipur National Park covered with deciduous forest, the Mudumalai National Park and the Mukurthi National Park in the states of Kerala and Tamil Nadu are the other important protected areas. Moreover, there are Mudumalai (Coimbatore), Anamalai (Nilgiri District) and Munda Thurai (Tirunelveli and Kanniyakumari districts) tiger reserves in the southern parts of the Western Ghats **(Fig. 5.9)**. A judicious use of resources and conservation practices can improve the resilience characteristics of the ecosystems of this important world heritage site.



Map not to Scale

Fig. 5.9 Western Ghats: A World Heritage Sites (UNESCO)

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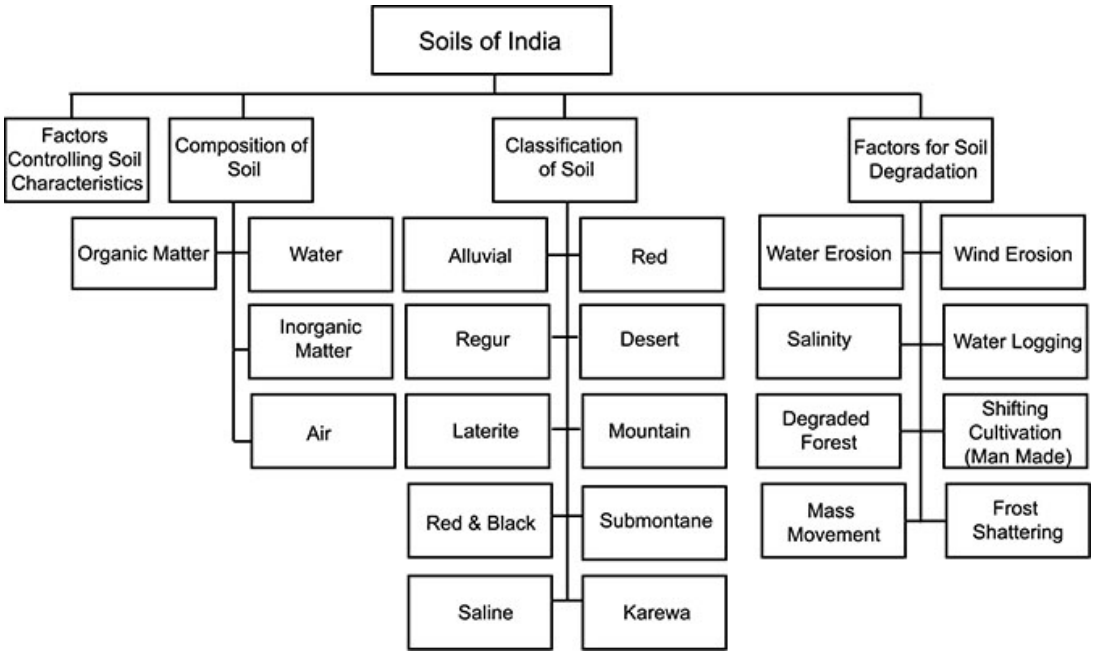
6

Soils



INTRODUCTION

The loose material or the upper layer of the mantle rock (*regolith*—a layer of lose, heterogeneous material covering solid rock) consisting mainly of very small particles and humus which can support the growth of plants is known as

soil. Soil mainly consists of mineral/rock particles, a certain proportion of decayed organic matter, soil water, soil air, and living organisms which exist in a complicated and dynamic relationship with each other. The naturally occurring soil is influenced by (i) parent material, (ii) relief, (iii) climate, (iv) physical, chemical and biological agents (micro-organisms) in it, (v) land use practices, and (vi) time. In general, soil is made up of four elements: (a) inorganic or mineral fraction (derived from the parent material), (b) organic matter (decayed and decomposed plants and animals), (c) air, and (d) water.

Soil is formed under specific natural conditions and each of the elements of the natural environment contributes to this complex process described by soil scientists as the process of pedogenesis or soil formation. The soil of a place is closely influenced by the nature of parent rock, surface features of relief, climatic conditions, natural vegetation, land use practices, organisms, insects, micro-organisms (bacteria), and time. These factors do not act on soil independently or in isolation, but in close association with each other leading to a whole network of inter-relationships of quite a complex nature. The material for soil formation, termed by soil scientists as the parent material, is derived from the rocks exposed on the surface. The relief and characteristics of slope along with the work of the various agents of weathering, determine conditions for the disintegration of the rock materials. Thus, the original soil characteristics, including the chemical constituents, are borrowed from the rock below. Soils may be transported by running water, wind or other agents of erosion, or may remain in the original position. When soil remains in its original position, it is said to be *in situ*, and in that state it is further modified by the climate, particularly moisture supply, plant growth and bacterial activity dependent on these factors.

CONTROLLERS OF SOIL CHARACTERISTICS

Parent Material

The parent material includes both, hard, and resistant rocks such as granite and marble as well as slate, apart from less resistant rocks such as recent volcanic lavas and ashes, the metamorphic (schist, gneiss) and sedimentary rocks (sandstone, clay, silt, and limestone). The term ‘rock’ is strictly applied not only to granite, sandstone and the like, but also to gravel, clay, unconsolidated sand, loess, and alluvium which are less compact and less resistant soils.

Humus

The end product of breakdown of dead organic material is known as humus. It is a structureless, dark-brown or black jelly found beneath or within the soil surface. The humus of ordinary soil is black, and is thus, responsible for making the top soil darker than the subsoil. It helps in the maintenance of soil fertility. The amount of humus in different soils varies considerably.

Soil Texture

A soil is generally characterised by the size of its particles. A clayey soil may thus be described as fine, sandy as coarse, while silt is an intermediate **(Fig. 6.1)**. The standard unit for the measurement of soil particles is millimetre, but a smaller unit is the micron (1 micron = 0.001 mm), which is applicable, for instance, to measure soil colloids. In sandy soil, the size of individual grains varies between 0.05 and 0.2 mm which are visible to the naked eye. The individual grains of clayey soil are 0.002 mm in diameter. Silty soil is finer than sand but coarser than clay. Its particles are found to have a diameter between 0.02 and 0.002 mm.

Soil Structure

Soil structure refers to the arrangement of soil particles. The way in which sand, silt, clay, and humus bond together to form beds is known as soil structure.

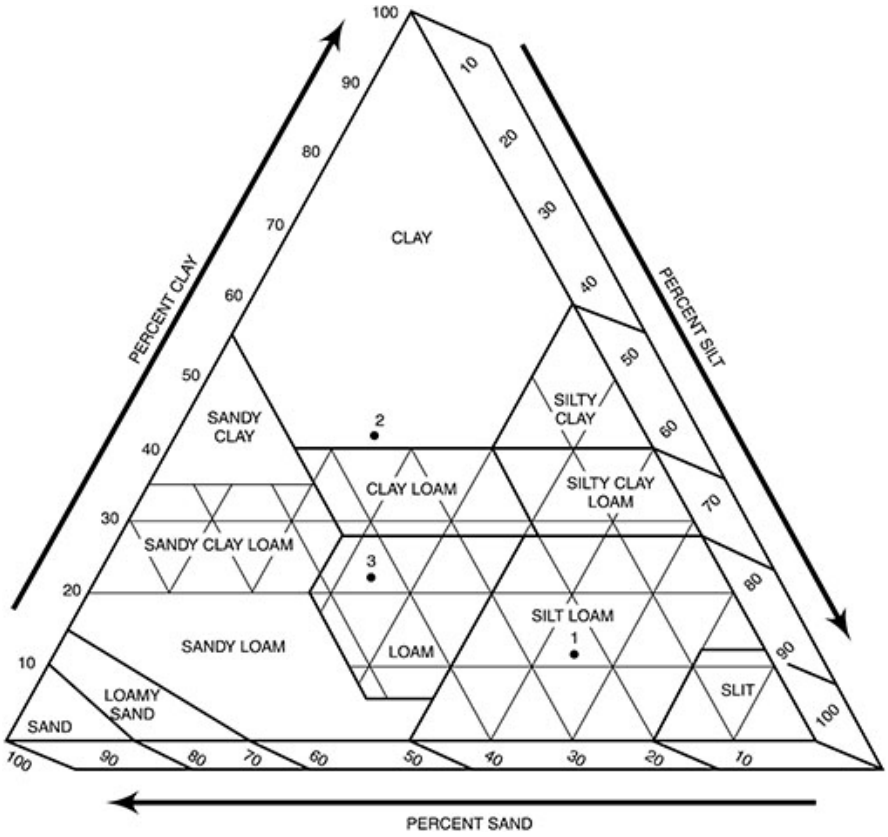


Fig. 6.1 Soil Texture Triangle

Soil Acidity

The acidity and alkalinity of soils is expressed in the pH value, which is a scale that measures the concentration of hydrogen ion held by the soil colloids (particles). In pure water, one part in 10 million is dissociated to form hydrogen ions, i.e., 10^{-7}, and the pH is thus 7; this is a neutral state on the scale of acidity. If a strong alkali such as caustic soda is dissolved in water, the solution is marked as alkaline (pH 14). By contrast, hydrochloric acid has a pH value of 3. A neutral soil has a pH value of about 7.2, and acid soil less than 7.2 (sometimes as low as 3). A strongly alkaline soil has a pH value of about 8 or higher.

Soil Air

The air content of soil is vital, both to itself and to organic life within it. The air in the soil helps in the process of oxidation which converts nitrogen into a form readily available to the plants. On the other hand, too high degree of oxidation (in the tropical regions) may consume so much organic material that the soil becomes increasingly sterile. Moreover, most bacteria, present in the soil in infinite numbers, require oxygen and are said to be *aerobic.* As these

organisms are partly responsible for breaking down plant and animals remains, absence of air limits their activity.

Soil Water

Depending on the texture of the soil, water moves downward by percolation. The amount of water in the soil varies from nil in arid climates, which makes life virtually impossible for organisms, to a state of complete water-logging which excludes air, causes a reduction in bacteriological activity, and limits decomposition.

Soil Horizon

It is a distinct layer within soil which differs chemically and/or physically from the layer below or above. It is a layer of soil which lies more or less parallel to the surface and has fairly distinctive soil properties **(Fig. 6.2)**.

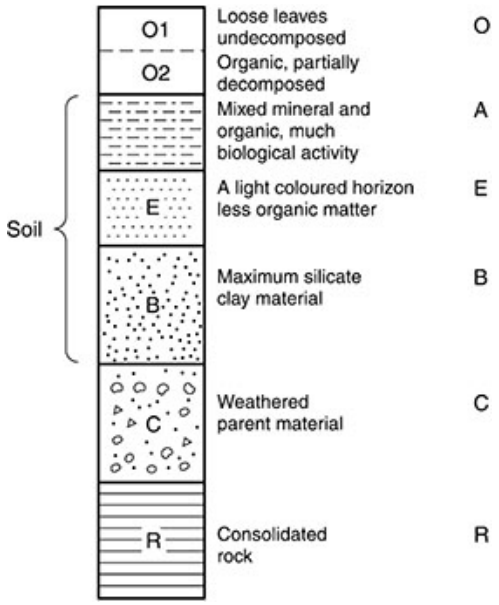


Fig. 6.2 Soil Profile and Soil Horizons [*Source: SLUSI, Government of India*]

Soil Profile

It is a vertical series of five soil horizons from the ground surface to the parent rock. The profile results from the same parent rock having similar horizons and soil profiles, but with varying characteristics according to their location **(Fig. 6.2)**. It extends from the soil surface to the parent rock material .

India is a vast country having great variations in its terrain and climatic conditions. The geo-climatic conditions of India have affected the general distribution of soils, their texture, structure, colour, pH value, and porosity. In general, the soils of India follow the climatic and vegetation belts.

As a natural resource, soil is of immense value to humans. Its nature and fertility determines the agricultural productivity on which depends the carrying capacity of the soil and the level of development of the rural communities.

CLASSIFICATION OF SOILS OF INDIA

A number of attempts have been made to classify the soils of India during the last century. The first scientific classification of Indian soils was made by Voeleker (1893) and Leather (1898). According to them the Indian soils may be classified into four categories, namely (i) alluvial,
(ii) regur (black-earth), (iii) red soil, and (iv) lateritic soil.

Subsequently, on the basis of texture, structure, colour, pH value and porosity, the All India Soil and Land Use Survey Organisation attempted a classification of soils of India in 1956. In 1957, the National Atlas and Thematic Mapping Organisation published a soil map of India in which Indian soils were classified into six major groups and eleven sub-groups. In 1963, the Indian Council of Agricultural Research, under the supervision of S.P. Ray Chaudhry, published a soil map of India in which the soils have been divided into seven groups. More recently, the Indian Council of Agricultural Research, on the basis of texture, structure, colour, pH value, and location porosity and location has identified the following types of soil groups **(Table 6.1 (Fig. 6.3)).**

1. Alluvial soils
   a. Khadar
   b. Bangar
2. Red soils
3. Regur (Black-earth) soils
4. Desert soils
5. Laterite soils
6. Mountain soils
7. Red and Black soils
8. Grey and Brown soils
9. Submontane soils
10. Snowfields

1. Alluvial Soils

The alluvial soil covers about 143.1 million sq km accounting for about 43.4% of the total reporting area **(Fig. 6.2).** These soils are mainly derived from the debris brought down from the Himalayas or from the silt left out by the retreating sea. The colour of the alluvial soils varies from light grey to ash grey and the texture is from sandy to silty-loam. These soils are both well drained

and poorly drained. In general, they have an immature profile in undulating areas, while in the leveled areas they have a well developed and mature profile.

These soils may be divided into the (i) *Khadar* soil: The khadar soils are low-lying, frequently inundated by floods during the rainy season. Thus, the khadar occupies the flood plains of the rivers and is enriched by fresh silt deposits every year. In the drier areas, it also exhibits stretches of saline and alkaline efflorescences locally known as '*reh*', *kallar* or *thur*, (ii) The *Bhangar* soil is above the flood level. It is generally well-drained but contains concretion (kankars) of impure calcium carbonate. The soil texture varies from loamy soil to clayey-loam. It is well drained.

Table 6.1

Types of Soils of India

Soil type		Area (in Million Hectares)	Percentage	Composition	Regions	Crops
Alluvial	Khadar	143.1	43.36	Rich in phosphoric acid, poor in potash	Indus-Ganga–Brahmaputra plains, Narmada–Tapi Plains	Wheat, maize, sugarcane, pulses, oilseeds, berseem (fodder)
	Bhangar					
Red (Omnibus group)		61.0	18.49	Ferric oxide gives it red colour; deficient in lime, phosphate, manganese	Western Tamil Nadu, Karnataka, southern Maharashtra, Andhra Pradesh, Chattisgarh, Jharkhand, Odisha, some areas of Bundelkhand, Mirzapur, Sonbhadra, Banswara, Bhilwara, Udaipur	Wheat, cotton, pulses, tobacco, millets, oilseeds, potato, orchards

Soil type	Area (in Million Hectares)	Percentage	Composition	Regions	Crops
Black (*Regur*); internally known as *Tropical Chernozems*	49.8	15.09	Iron, lime, calcium, potassium, aluminium, magnesium; deficient in nitrogen, phosphorous, organic matter	Gujarat, Maharashtra, western Madhya Pradesh, north-western Andhra Pradesh, Karnataka, Tamil Nadu, Rajasthan, Chattisgarh, Jharkhand up to Rajmahal	Cotton, pulses, millets, linseed, castor, tobacco, sugarcane, citrus fruits and vegetables
Mountain	18.2	5.51	Acidic and low humus content	Mid-latitude zone of Himalaya, regions of Assam, Darjeeling, Uttarakhand, Himachal Pradesh, Jammu and Kashmir	Maize, rice, legumes, fodder, orchards
Desert	14.6	4.42	High percentage of soluble salts, low organic matter, varying percentage of calcium carbonate	Rajasthan, West of Aravallis, northern Gujarat, Saurashtra, Kutch, western parts of Haryana, south-western parts of Punjab	Bajra, pulses, guar, fodder and lesser water requiring crops like Aloe vera

Soil type	Area (in Million Hectares)	Percentage	Composition	Regions	Crops
Laterite	12.2	3.70	Rich in iron and aluminium; poor in nitrogen, potash, potassium, lime, organic matter	Western Ghats, Eastern Ghats, Rajmahal Hills, Satpura, Vindhya, Odisha, Chattisgarh, Jharkhand, West Bengal, North Cachar Hills of Assam, Garo Hills of Meghalaya	Rice, ragi, sugarcane, cashew nuts
Submontane	5.7	1.73	Neutral to alkaline	Jhelum and Indus Valley of Tarai region of submontane from J & K to Assam	Rice
Snowfields	4.0	1.21	Covered with ice and glacier	Greater Himalayan Region, Karakoram, Ladakh, Zanskar	Unsuitable for cultivation of crop
Grey and Brown	3.6	1.09	Iron oxide	Rajasthan and Gujarat	Cotton and oilseeds
Saline and alkaline (known by local names as *reh*, *kallar*, *usar*, *rakar*, *thur*, *karl*, *chopan*)			Sodium chloride and sodium phosphate	Rajasthan, Haryana, Punjab, Uttar Pradesh, Bihar, Maharashtra	Berseem, *dhaincha*, leguminous crops

Soil type	Area (in Million Hectares)	Percentage	Composition	Regions	Crops
Peaty and marshy			Rich in organic matter, highly saline, deficient in phosphate and potash	Kottayam, Alappuzha districts of Kerala, Sundarban Deltas of Mahanadi, Krishna, Kaveri, Rann of Kutch	Rice cultivation
Karewa			Characterised with fossils of mammals and peat	Kashmir Valley, Bhadarwah Valley of Doda district of Jammu	Saffron, almond, walnut, apple, orchards
Red and Black	17.8			Isolated parts of Bundelkhand, East of Aravallis in Rajasthan, Gujarat	Maize, bajra, millets, pulses, oilseeds

2. Red Soils

Red soils occupy the second largest area of about 61 million hectares or 18.5% of the total reporting area. They are found mainly over the Peninsula from Tamil Nadu in the south to Bundelkhand in the north, and Rajmahal in the east to Kathiawad and Kutch in the west **(Fig. 6.3).**

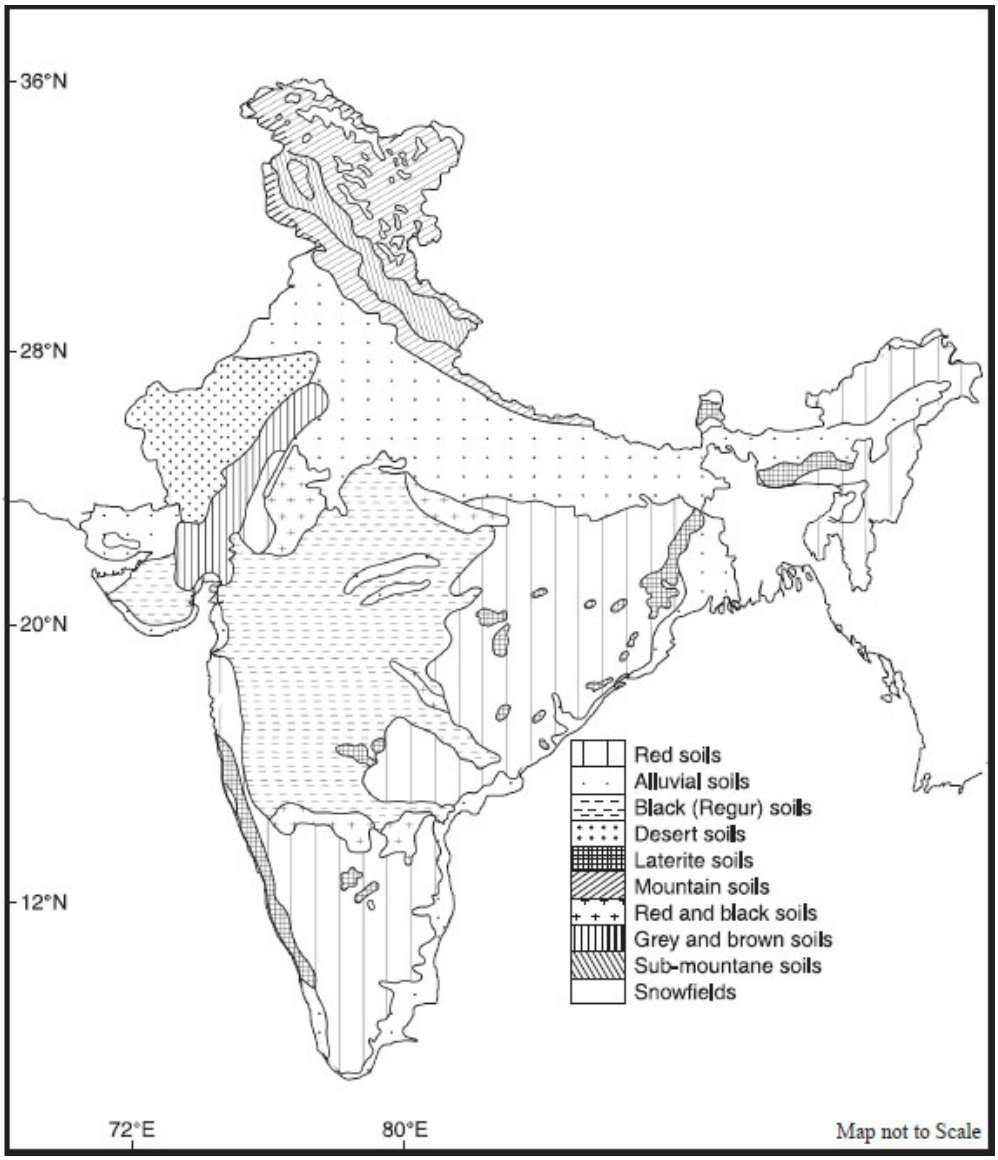


Fig. 6.3 Soils

Developed on Archaean granite, these soils are also known as the '*omnibus group*'. Their colour is mainly red because of the presence of ferric oxides. Generally, the top layer is red, while the horizon below is yellowish in colour. The texture of red soils varies from sand to clay and loam. Their other characteristics include porous and friable structure, absence of lime, *kankar* and carbonates and small quantity of soluble salts. In general, these soils are deficient in lime, phosphate, magnesia, nitrogen, humus, and potash. Intense leaching is a menace to these soils. In the uplands, they are thin, poor, gravelly, sandy, or stony and porous, light-coloured soils, but in the lower plains and valleys, they are rich, deep, dark coloured fertile loams. In places where

irrigation water is available, they are devoted to wheat, cotton, pulses, tobacco, millets, oilseeds (linseed), potato, and orchards.

3. Black or Regur Soils

Black soils, also known as *Regur* (cotton-soil) and internationally as 'tropical chernozems', are the third largest soil group in India. They sprawl over about 50 million hectares accounting for 15% of the total reporting area of the country. They get their parent material from the weathered rocks of Cretaceous lava. They are mature soils. Over the greater parts of the black earth soil, the average annual rainfall varies between 50 and 75 cm.

The colour of these soils varies from deep black to light black. In general, these soils have clayey texture. Moreover, these soils have a high water retaining capacity. They are extremely compact and tenacious when wet, and develop wide cracks when dry. In other words, they swell greatly and become sticky when wet in rainy season. When the soil is wet, it becomes difficult to plough the field as the plough gets stuck in mud. In the dry season, the moisture evaporates, the soil shrinks and develops wide cracks, often 10–15 cm deep.

4. Mountain Soils

Covering an area of about 18.2 million hectares or about 5.5% of the total reporting area of the country, these soils are found in the valleys and hill slopes of the Himalayas between 200 and 300 m. These soils are generally immature and are still to be probed systematically. In structure and texture, they vary from silt-loam to loam. Their colour is dark brown. These soils can be divided into: (i) loamy podzols , and (ii) high altitude soils.

Podzols occupy the mid-latitude zone in the Himalayas.

The High altitude soils, depending on the forest cover, slope, and rainfall are classified as brown earth type and red-loam. The sub-soil surface of these soils remains frozen under snow. Their soil profile is generally less developed.

5. Desert Soils

Sprawling over about 15 million hectares, the desert soils account for over 4.42% of the total reporting area of the country. These soils are developed under the arid and semi-arid conditions and deposited mainly by wind action.

The desert soils are sandy to gravelly in texture organic matter, but have low moisture content and low water retaining capacity. If irrigated, they give high agricultural returns. The availability of water from the Indira-Gandhi Canal has transformed the agricultural landscape of the desert soils of western Rajasthan.

6. Laterite Soils

These soils were studied first by the British geographer F. Buchanan in 1905. Their name has been derived from the Latin word *Later* which means brick. These soils, when wet, are as soft as butter, but become quite hard and cloddy on drying. These are the typical soils of the monsoon climate which is characterised by seasonal rainfall. The alternating wet and dry seasons lead to the leaching away of the alkalis and siliceous matter than iron and aluminum of the rocks leading to the formation of such soils. Laterite soil hardenss rapidly and irreversibly on exposure to the air, a property which lead to its use as building brick in southern India (Buchanam 1807). The red colour of the soils is due to the presence of iron-oxide. These soils developed mainly in the highland areas of the plateau. The soils in the higher areas are generally more acidic than those in the low-lying areas. The soil profile varies from a few centimeters to tans of meters thick and below this saprolita, teached or unleached of iron, varies a few cm to over 100 m.

Laterite soils cover an area of about 12.2 million hectares accounting for about 3.7% of the total reporting area of the country.

Although they have low fertility, they respond well to manuring.

7. Submontane Soils

These soils are found in the Tarai region of the sub-montane stretching from Jammu and Kashmir to Assam in the form of a narrow belt. These soils have been formed by the deposition of the eroded material from the Shiwaliks and the Lesser Himalayas. The soil is fertile and supports luxuriant growth of forests. The clearing of forests for agricultural purposes has made this area highly susceptible to soil erosion.

8. Snowfields

The area under snow and glaciers is about 4 million hectares. The soils in these areas is immature, generally without soil erosion.

9. Grey and Brown Soils

These soils have been formed by the weathering of granite, gneiss, and quartzite. These are loose friable soils. Due to the presence of iron-oxide (haematite and limonite) these soils vary from red to black and brown in colour. These soils are found in Rajasthan and Gujarat **(Fig. 6.3).**

10. Red and Black Soils

They developed over the granite, gneiss, and quartzite of the Archaean and Pre-Cambrian period. These soils are relatively less productive, but perform well under irrigation conditions.

11. Saline and Alkaline Soils

The saline soils are characterised by the presence of sodium chloride and sodium sulphate. In these soils, the saline and alkaline efflorescence appears on the surface as a layer of white salt through capillary action. These soils are known by different names in different parts of the country. They are called as *reh, kallar*, *usar, rakar, thur, karl,* and *chopan.* These soils are found in Rajasthan, Haryana, Punjab, Uttar Pradesh, Bihar, and Maharashtra.

Texturally, these soils vary from sandy to sandy-loam. They are deficient in nitrogen, calcium and have very low water bearing capacity. These soils can be reclaimed by improving drainage, by applying gypsum and/or lime and by cultivating salt resistant crops like barseem, *dhaincha*, and other leguminous crops.

12. Peaty and Marshy Soils

Peaty soils originate in the areas of heavy rainfall where adequate drainage is not available. These are generally submerged during the rainy season and utilised for the cultivation of rice.

13. Karewa Soil

These are the flat topped mounds of lacustrine deposits that border the Kashmir Valley on all sides. They are composed of fine silt, clay, sand, and bouldery-gravel. They are characterised with fossils of mammals and at places by peat. According to geologists, during the Pleistocene Period, the entire Valley of Kashmir was under water. Subsequently, due to endogenetic forces, the Baramullah Gorge was created and the lake was drained through this gorge. The deposits left in the process are known as karewas **(Fig. 6.4).**

According to Middlemiss, the thickness of karewas is about 1400 m. In fact, the karewas have been elevated, dissected and in great measure removed by subaerial denudation as well as by the Jhelum river giving them the present position. The karewas of Palmpur, Pulwama, and Kulgam are well known for their production of superior quality of saffron.

The ICAR has classified the Indian soil on the basis of their nature and characteristics as per the United States Department of Agriculture (USDA) Soil Taxonomy as given in the table below:

Table 6.2

Classification of Indian Soil as per USDA

S. No.	Order	Area (in km^2)	Percentage
1	Inceptisols	130,372.90	39.74
2	Entisols	92,131.71	28.08
3	Alfisols	44,448.68	13.55

S. No.	Order	Area (in km^2)	Percentage
4	Vertisols	27,960.0	8.52
5	Aridisols	14,069.0	4.28
6	Ultisols	8250.0	2.51
7	Mollisols	1320.0	0.40
8	Others	9503.0	2.92
		Total	**100**

Source: Soils of India, National Bureau of Soil Survey and Land Use Planning, Publication Number 94.

India is facing the problem of soil erosion with a loss of soil at the rate of 16 tonnes per hectare (ha) annually. According to the National Bureau of Soil Survey and Land Use Planning about 146.8 Mha is degraded. The annual loss of soil is 5.3 billion tonnes throughout the country.

In the year 2011–13, degradation or desertification of land in India was 29.3% of the total land (1.87 million hectares) as per ISRO's Space Application Centre.

PROBLEMS OF INDIAN SOILS

The forces of nature often destroy the soil cover of an area. The process of soil destruction is the result of natural forces. Soils are, however, also being damaged and destroyed by human activities, such as deforestation, over-grazing of animals and unscientific use of agricultural land. The main problems of the Indian soils are (i) soil erosion, (ii) declining fertility of soil, (iii) salinity and alkalinity, (iv) water-logging, and (v) desertification.

(i) Soil Erosion

Soil erosion refers to the removal of top soil. Soil erosion is a growing menace in many parts of India. When the top soil is removed, it is known as sheet erosion, and when the runoff makes gullies, it is known as gully erosion. In India, soil erosion is a universal problem. In the areas where rainfall is heavy, water is the main agent of soil erosion, while in the arid and semi-arid areas, wind is responsible for soil erosion. According to one estimate about 180 million hectare (about 60% of the total area of the country) is adversely affected by soil erosion.

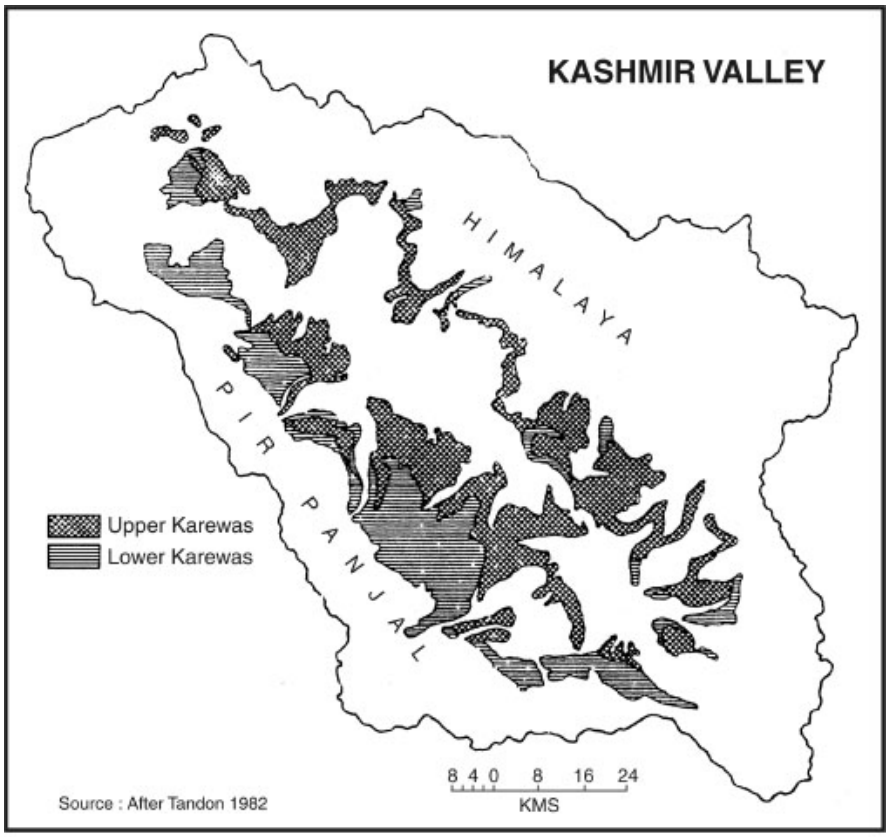


Fig. 6.4 Karewas

The main agents of soil erosion are water, wind, sea-waves, glaciers, and shifting cultivation. The areas adversely affected by the different agents of erosion have been shown in **(Fig. 6.5).** Out of these, water is the most important agent of erosion. Water erosion may be classified under three categories: (a) surface erosion or the uniform removal of soil from the surface, (b) Rill erosion in which the running water makes finger-shaped grooves in the land, and (c) gully formations, in which the rills are enlarged, making the land bad and unsuitable for cultivation. A typical example of gully erosion is provided in the Chambal valley in Madhya Pradesh. Rajasthan, and Uttar Pradesh also provide typical examples of gully erosion. **(Fig. 6.6).** Gully erosion is also significant in the Shiwalik tracts of Punjab, Haryana, Himachal Pradesh, Jammu and Kashmir, Uttarakhand, Uttar Pradesh and along the southern slopes of Himalayas, and the Western and Eastern Ghats.

Wind erosion is significant in the arid and semi-arid areas of Rajasthan, Haryana, Punjab, western Madhya Pradesh and Gujarat. Wind erodes soil along the coastal plains of Peninsular India. Thousands of hectares of fertile

lands of Uttar Pradesh, Gujarat, Haryana, Punjab, and western Madhya Pradesh have been adversely affected by this process.

The tidal waters of the Arabian Sea and the Bay of Bengal cause considerable damage to the soils along the coastal areas. Severe erosion of beaches along the Kerala, Tamil Nadu, Andhra Pradesh, Odisha, West Bengal, and Gujarat coasts are the examples of sea-wave erosion.

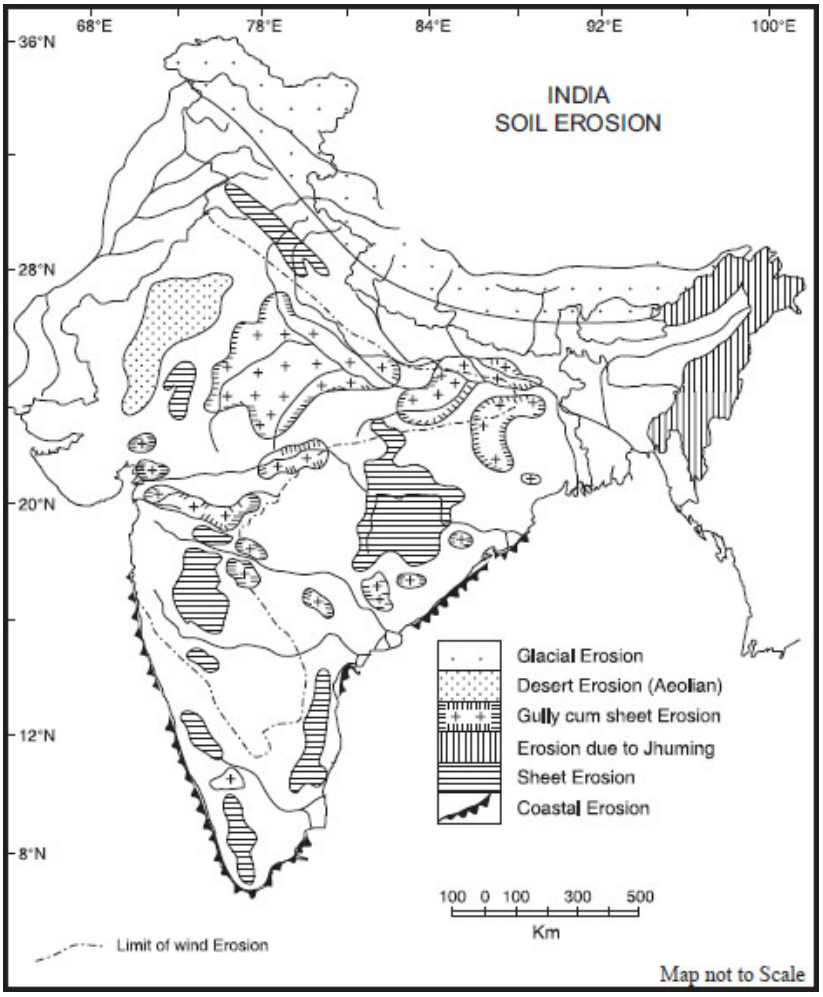


Fig. 6.5 Soil Erosion

The largest area affected by soil erosion is in the state of Rajasthan, followed by Madhya Pradesh, Maharashtra, Uttar Pradesh, Gujarat, Andhra Pradesh and Karnataka. The worst affected areas of soil erosion include: (i) Chambal and Yamuna rivers **(Fig. 6.6),** (ii) the southern slopes of Shiwaliks, Lesser and Greater Himalayas, (iii) the Western and Eastern Ghats, (iv) the Chotanagpur Plateau, and (v) the arid and semi-arid areas of Rajasthan, Gujarat, Haryana, and Punjab.

(ii) Declining Soil Fertility

Being utilised for centuries, and multiple cropping without fallowing the agricultural land, the natural fertility of soil is depleting fast. The farmers of Punjab, Haryana, and western Uttar Pradesh often complain about the decreasing fertility of their soils. It has been reported that the farmers are using more inputs each subsequent year to get the return to the level of the previous year. This testifies to the diminishing fertility of land. In fact, the unscientific rotation of crops (wheat and rice) over several decades has depleted the soil fertility in the Great Plains of India substantially. The cultivation of leguminous crops after a soil exhaustive crop can improve the soil fertility affected regions.

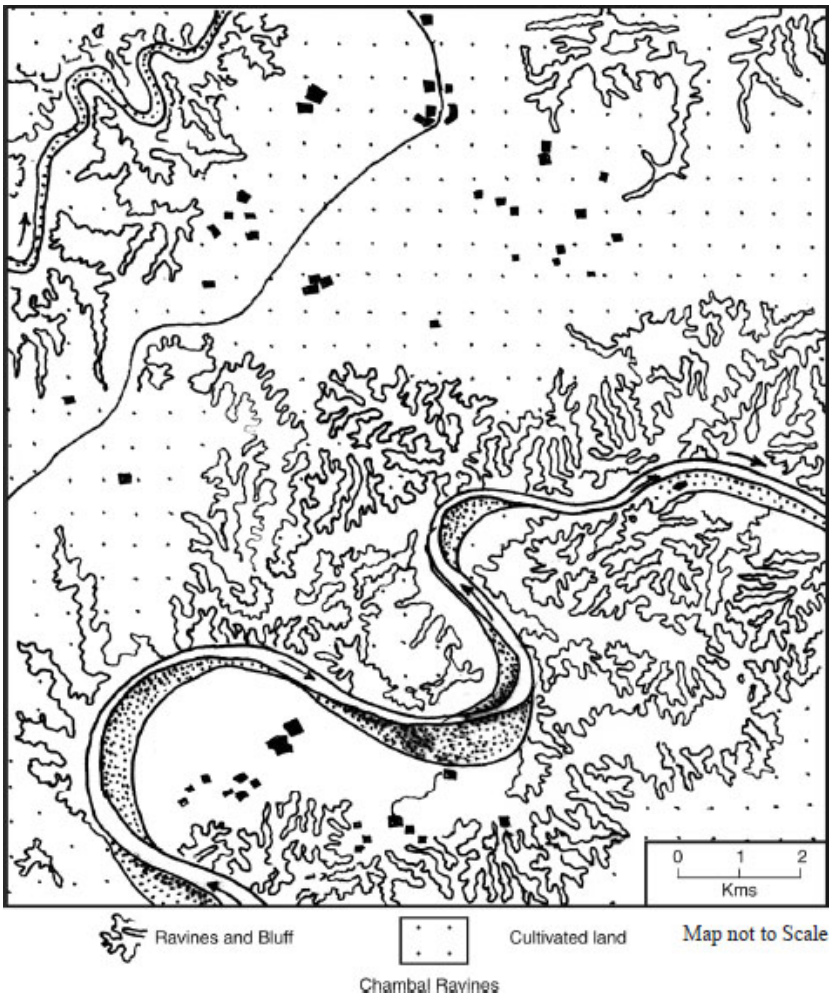


Fig. 6.6 Gully Erosion in Chambal Valley

(iii) Water-Logging

An area is said to be waterlogged when the water table rises to the extent that soil pores in the root zone of a crop become saturated, resulting in the

restriction of normal circulation of the air, decline in the level of oxygen, and an increase in the level of carbon dioxide**.**

Apart from soil erosion, there are many ill-drained and low lying areas in India where the soil has been damaged by water-logging **(Fig. 6.7)**. The main causes of waterlogging are: (i) seepage of water from canals, (ii) faulty on farm water management, (iii) lack of drainage, (iv) interception of natural drainge, (v) indiscriminate cultivation in bed of drainage channel, and (vi) inundation of marine delta cycles, and (vii) inundation in coastal areas during cyclonic storms. Water logging has affected substantial tracts of land along the Indira Gandhi Canal (Rajasthan) and the canals of Punjab, Haryana, and Uttar Pradesh**.** Adequate development of drainage and lining of the canals to reduce water seepage can go a long way in the reclamation of water-logged areas**.**

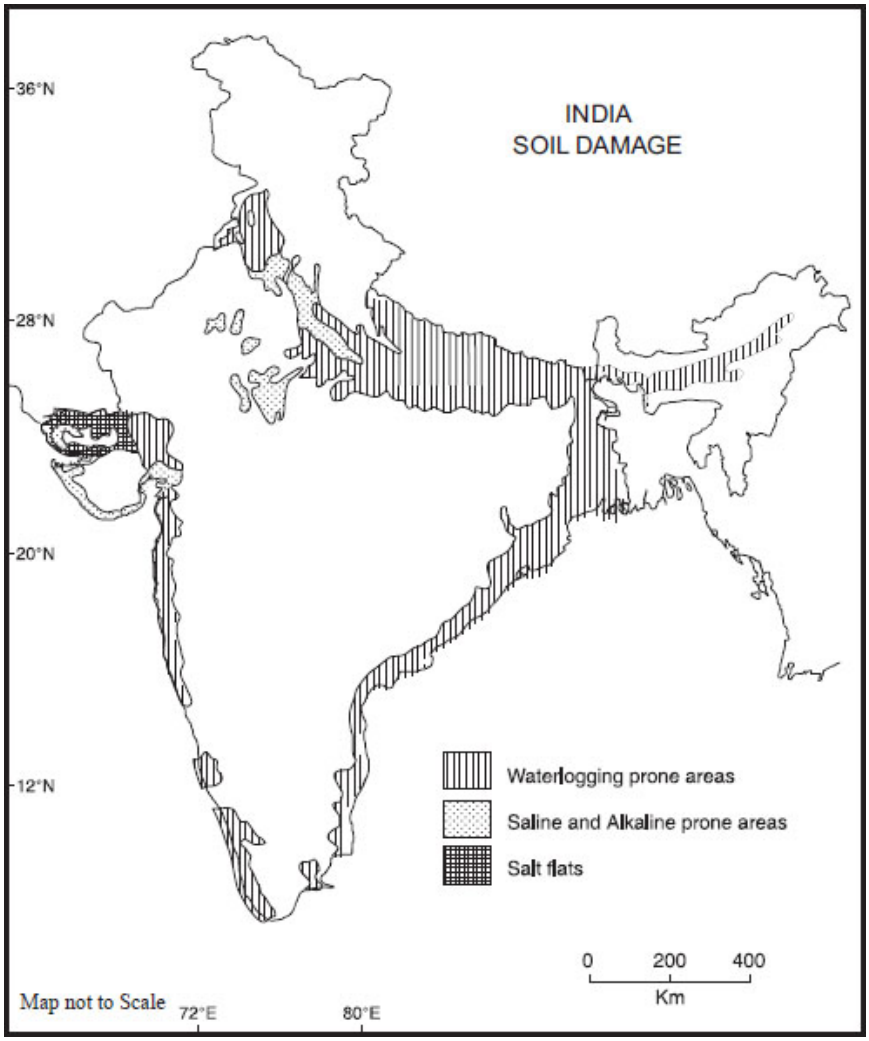


Fig. 6.7 Soil Affected Areas

(iv) Saline and Alkaline Soils

Soil salinity and alkalinity are found in the relatively less rainfall recording areas where the rate of evaporation is generally higher than the rate of precipitation. They also develop in the *Khadar* lands and the canal irrigated areas. Under such conditions, the ground water level rises and saline and alkaline efflorescence consisting of salts of sodium, calcium, and manganese appear on the surface as a layer of white salt through capillary action. According to one estimate, about 80 lakh hectares (2.4% of the country's reporting area) has been adversely affected by saline and alkaline formations.

(v) Salt Flats

The soils seriously damaged by the excess of calcium chlorides are found in the Rann of Kutch **(Fig. 6.7)**. These soils are saline, marshy and infested with tall grasses, bushes and scrubs. They are almost useless from agricultural point of view. A sound strategy needs to be developed to bring the salt flats under agriculture or pastures.

The Indian council of Agricultural Research, New Delhi has in its report degraded and 'Wasteland of India' given the classification degraded and the cause. These categories of land and their causes have been given in **(Tables 6.3 and 6.4).**

Table 6.3

NRSA Wasteland Classes

Categories	Area in Million Hectares
Gullied lands	2.06
Land with/without scrub	19.40
Water logged and marshy lands	1.66
Land affected by salinity and alkalinity	2.05
Shifting cultivation area	3.51
Under utilised/degraded forests	14.07
Degraded pastures and grazing lands	2.60
Degraded land under plantation crops	0.58
Sand (inland/coastal)	5.00
Mining/industrial wasteland	0.12
Barren rock/strong waste sheet rock area	6.46
Steep sloping area	0.77
Snow covered/glacial area	5.58
Total	63.85

Source: NRSA and Mors 2000

Table 6.4

Processes Contributing to Desertification/Land Degradation Between 2003-05 and 2011-13

Process of Degradation	Area Under Degradation (Percent)	
	2011–13	2003–05
Vegetation degradation	8.91	8.60
Water erosion	10.98	10.83
Wind erosion	5.55	5.58
Salinity	1.12	1.22
Water logging	0.20	0.18
Frost shattering	1.02	0.95
Mass movement	0.28	0.26
Barren/Rocky	0.57	0.57
Settlement	0.57	0.45
Manmade	0.12	0.11
TOTAL	**29.32**	**28.75**

Source: ISRO's Desertification and Land Degradation Atlas of India

The causes of soil degradation differ from region to region and from place to place. In some places, it is due to water erosion, while at others it is due to wind erosion, or salt and toxic chemicals.

It may be seen from **(Table 6.3)** that over 2 million hectares of land is gullied and about 1.66 million hectares are waterlogged and marshy lands. Moreover, 3.51 million hectares are adversely affected by shifting cultivation. The degraded notified forests cover over 14 million hectares and over 19 million hectares of land is without scrub. These lands need immediate attention for reclamation.

Water and wind erosion are the most potent causes of land degradation. Water erosion produces gully erosion, while water and wind erosion lead to sheet erosion. Wind erosion becomes active when the vegetation is removed from the land and land is exposed to wind action. Soil degradation by wind and water erosion have been shown in **(Fig. 6.8)**. It may be observed from this figure that wind erosion is mainly in the Thar desert while soil erosion by water is well spread in the Himalayas, Gangetic plains and the Peninsular India **(Fig. 6.8)**. Intensive agricultural practices that rely heavily on water, chemical fertilisers, and insecticides and pesticides have caused water-logging and salinity in many parts of India. The expansion of irrigation without proper drainage has accentuated this. The total area under degradation has increased

by 1.97% and Uttar Pradesh has shown an improvement of 1.27% as can be seen from, **(Table 6.5).**

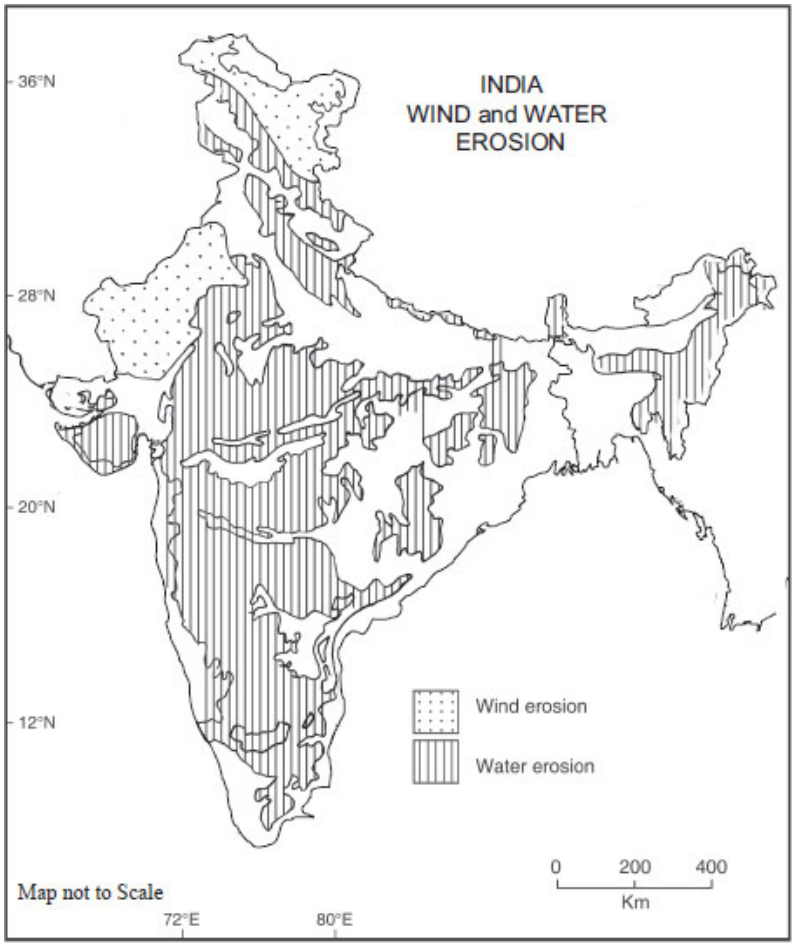


Fig. 6.8 Soil Degradation

Table 6.5

Analysis on Soil Degradation in India

S. No.	Description (wrt TGA of the Country)	Area (Million ha)		Area (Percent)	
		2011–13	2003–05	2011–13	2003–05
1	Total area under desertification/land degradation	96.40	94.53	29.32	28.76
2	Total area under no apparent degradation	226.73	228.68	68.97	69.57
3	State under highest desertification-Rajasthan	21.53	21.63	6.55	6.58
4	Land reclaimed from degraded in 2003–05 to no apparent degradation in 2011–13	1.95 Mha			

S. No.	Description (wrt TGA of the Country)	Area (Million ha)		Area (Percent)	
		2011–13	2003–05	2011–13	2003–05
5	State under highest increase in degradation (2011–13 and 2003–05)–Delhi	0.01		11.03	
6	State showing highest improvement (2011–13 and 2003–05)–Uttar Pradesh	0.30		1.27	
7	Area under desertification in arid region	30.54	30.35	9.29	9.23
8	Area under desertification in semi-arid region	35.40	34.85	10.77	10.60
9	Area under desertification in sub-humid region	16.70	16.28	5.08	4.95
10	Area under desertification in drylands	82.64	81.48	25.14	24.78

CONSEQUENCES OF SOIL EROSION

Soil erosion is responsible for the removal of productive nutrients from the soils and causes ecological imbalances. Some of the adverse consequences from soil erosion are:

(i) Loss of fertile top soil from the top layer leading to gradual loss of soil-fertility and agricultural productivity.
(ii) Loss of important nutrients from soil through leaching and water-logging.
(iii) Lowering of the underground water-table and decrease in soil moisture.
(iv) Drying of vegetation and extension of arid lands.
(v) Increase in the frequency of droughts and floods.
(vi) Silting of rivers and canal beds.
(vii) Recurrence of land slides.
(viii) Adverse effect on economy which retards cultural development.
(ix) Increase in crimes and anti-social activities through the formation of natural hideouts for criminals and dacoits.
(x) Burden on the exchequer to reclaim the bad lands.

There is no uniform strategy to reclaim all the wasteland and degraded soils of different types. Some strategies that might help in the reclamation of wasteland are given below:

(i) All the degraded forest lands should be planted with trees. Marginal lands which are not suitable for agriculture should be brought under

social forestry and agro-forestry.

(ii) Degraded soils and degraded lands can be reclaimed with the help of watershed programmes.

(iii) Rainwater harvesting and conservation should be the focus of development planning. A series of small projects of water harvesting in the watershed area should be undertaken to maximise benefits from watershed projects.

(iv) Soil conservation practices should be adopted which have been briefly described in the following pages.

SOIL CONSERVATION

Looking at the importance of soil resources for a country of over a billion people, a judicious utilisation and conservation of soil is of parmount importance.

The farmers in the drier parts of Gujarat, Haryana, Punjab, Rajasthan, and western Madhya Pradesh have successfully protected their fields from soil erosion by planting rows of trees to reduce the velocity of winds which continually erode soil cover. Soil conservation includes reduction in soil erosion, afforestation, rational utilisation of soils and ways to enhance their sustainability. Some of the important steps which can go a long way in the conservation of soils are as under:

1. Afforestation

Tree plantation helps in the reduction of soil erosion. Trees reduce the intensity of runoff and increase the seepage of water to the underground water-table. Social forestry can be developed along the banks of rivers, canals, lakes, roads, and railway tracks.

2. Restriction on the Felling of Trees

Apart from afforestation, it is equally important to check the indiscriminate felling of trees. People's awareness that resulted in the launch of the *Chipko Movement* can help in achieving this objective.

3. Contour Ploughing and Strip Cultivation

In the hilly and mountainous areas, ploughing should be done according to the contours and not in an up-down direction of slope. The contour ploughing is an effective way of checking soil erosion. Similarly, small strips can be developed on gentle slopes for sowing crops which help in overcoming the menace of soil erosion.

4. Control of Floods

In India, the problem of soil erosion is closely associated with floods. The floods generally occur during the rainy season. Efforts, therefore, need to be made for the storage of flood water or the diversion of additional rain-water. The inter-connecting of rivers as in the Garland Canal Project or the Ganga-Kaveri Link Canal Project can be of immense help in this direction.

5. Reclamation of Ravine and Badlands

Reclamation of gullies and ravines is also necessary to overcome the problem of soil erosion. Several such schemes involving plugging of gully mouths, construction of bunds across the gullies, levelling of gullies, afforestation, restriction on grazing are under implementation in the Chambal ravines of Madhya Pradesh, Uttar Pradesh, and Rajasthan.

6. Restriction on Shifting Cultivation

In the states of north-east India and Western and Eastern Ghats, shifting cultivation (slash and burn) is one of the main causes of soil erosion. The shifting cultivators need to be persuaded to stop the practice of shifting cultivation and should be trained and motivated to adopt terraced farming. A scheme to control shifting cultivation has been launched in the seven states of North-East India. This is a beneficiary oriented programme which aims at the rehabilitation families of the Jhumias (shifting cultivators). There is a need to extend this programme to other states of the country and gradually replace this agricultural system by sedentary farming.

7. Restoration of Long Fallow

There are 96 lakh hectares of fallow land. The old fallow land is mainly found in Andhra Pradesh, Telangana, Karnataka, Madhya Pradesh, Maharashtra, Gujarat, Rajasthan, Tamil Nadu, Bihar, Uttarakhand, and Uttar Pradesh. This old fallow land can be brought under cultivation, grazing and orchards which shall be helpful in the reduction of soil erosion.

8. Reclamation of Saline and Alkaline (usar) Soil

The saline and alkaline affected area in the country is more than 80 lakh hectares. It may be seen that the state of Kerala (24.47 lakh hectares) followed by Chhattisgarh (23.55 lakh hectares) have the soils majorly affected by salinity, alkalinity and acidity **(Table 6.6)**. A complete analysis is presented in **(Table 6.6)** as per the report 'Degraded and Wastelands of India-Status and Spatial Distribution' published by ICAR (2010).

Table 6.6

Affected Soil Areas in the States of India

S. No.	State	Area Affected by Alkali, Saline and Acidic Soils (in Lakh Hectares)
1	Andhra Pradesh	2.55
2	Arunachal Pradesh	17.69
3	Assam	19.95
4	Bihar	1.87
5	Chhattisgarh	23.55
6	Goa	1.03
7	Gujarat	21.04
8	Haryana	2.32
9	Himachal Pradesh	0.76
10*	Jammu & Kashmir (inclusive of Ladakh)	0.78
11	Jharkhand	7.35
12	Karnataka	2.40
13	Kerala	24.47
14	Madhya Pradesh	6.06
15	Maharashtra	8.61
16	Manipur	15.97
17	Meghalaya	10.23
18	Mizoram	11.63
19	Nagaland	15.16
20	Odisha	3.34
21	Punjab	1.52
22	Rajasthan	2.34
23	Sikkim	0.58
24	Tamil Nadu	7.90
25	Tripura	7.09

S. No.	State	Area Affected by Alkali, Saline and Acidic Soils (in Lakh Hectares)
26	Uttarakhand	4.00
27	Uttar Pradesh	13.42
28	West Bengal	8.26
29	Others	1.76
	Total (lakh hectares)	**243.63**
	Total (million hectares)	**24.36**

Source: Degraded and Wastelands of India-Status and Spatial Distribution published by ICAR (2010).
Data seperately on Jammu & Kashmir and Ladakh shall be updated in the next edition.

This land needs reclamation. Application of cow-dung and gypsum are quite helpful in the reclamation of salt-affected soils.

9. Other Measures of Soil Conservation

The government of India has been attempting to check the soil erosion through out the planning period. The following steps can go a long way in reducing the rate of soil erosion.

(i) Construction of small dams across the tributaries of rivers in their upper reaches to control floods and soil erosion.
(ii) Lining of canals to stop seepage of water which leads to waterlogging.
(iii) Solving the problem of waterlogging by improving the surface and vertical drainage.
(iv) Formation of windbreak and shelter belts in arid and semi-arid regions.
(v) Increasing use of organic and compost manure.
(vi) Popularising the application of cowdung and green manure.
(vii) Conversion of human waste and city garbage into manures.
(viii) Scientific rotation of crops.
(ix) Filling up gullies and forming terraces along the slopes.
(x) Levelling of ravines and planting of trees and grasses in the slopes.
(xi) Check on shifting cultivation and conversion of Jhum lands into sedentary agriculture.
(xii) Promotion of afforestation in the degraded soils.
(xiii) Adopting the techniques of sustainable agriculture.
(xiv) To educate public about the adverse effects of soil erosion through seminars, conferences, and workshops in the regions of degraded

soils.

As per the report of Ministry of Agriculture, the Government of India aims to make the India land degradation neutral by 2030. In India 69% (about 105 million hectares) is dry land and 32% of land is undergoing desertification. About 1 mm of top soil is being lost and due to soil erosion the land being lost is 5334 million tonnes annually. The integrated land use planning can stop or reverse the desertification of the Indian land.

The non non-judicious and excessive use of inorganic fertilizer (NPK) is deteriorating the soil fertility and causing nutrient deficiencies. The use of irrigation water from the canal might lead to secondary salinization that also affects the quality of the soil. A number of schemes and projects have been initiated by the Government of India. Some of them are listed below in **(Table 6.7).**

Table 6.7

Schemes and Projects by the Government to Control Land Degradation and Soil Improvement

S.No.	Scheme/Project		Objective	Activities Undertaken
1.	State Plan Schemes	Soil and water conservation in general areas	To reduce soil erosion hazards, land degradation, conservation of water	Terracing and reclamation, erosion control, water conservation & distribution, afforestation, farm ponds, cash/horticulture crops
		Watershed management programme	Aims to treat the micro watershed on an integrated approach	Treatment of arable land, non-arable land, drainage lines, afforestation, cash/horticulture crops

S.No.	Scheme/Project		Objective		Activities Undertaken
2.	Centrally Sponsored Schemes	Integrated Wasteland Development Programme (IWDP) (funded by Department of Land Resources, Ministry of Rural Development) Government of India	Intends to involve the village communities in the implementation of this programme adopted the guidelines of watershed development programme 1995 which were revised in 2001, and in 2003 (*Hariyali*)		112 projects have been sanctioned to treat 439 micro watershed covering an area of 2,21,225 hectares with a cost of Rs. 13,053.25 lakhs
		Integrated Wasteland Management Programme (IWMP)	2009–10	Approval treatment of 30,000 hectares of land in 18 watershed projects of the state was made	i) to dissipate the soil , water erosion and surface water run off ii) Harvesting of surface run off and rainwater iii) To enhance the water holding capacity of the soil iv) To promote the sub surface flow, base flow and groundwater recharge v) To improve soil texture and its productivity vi) To promote generation of employment opportunities
			2010–11	52000 hectares of land	
			2011–12	37,500 hectares of land	
			2012–13	38,870 hectares of land	
3.	Other Schemes of the Government		1. Soil conservation for enhancing the productivity of the degraded land in the catchment area of River Kopili in Jaintia Hills district, Meghalaya 2. Rashtriya Krishi Vikas Yojna (RKVY)		
4.	Special Plan Assistance		Cherrapunjee Ecological Project–Restoration of degraded land under Sohra Plateau		

Source: Soil and Water Conservation Department, Government of India

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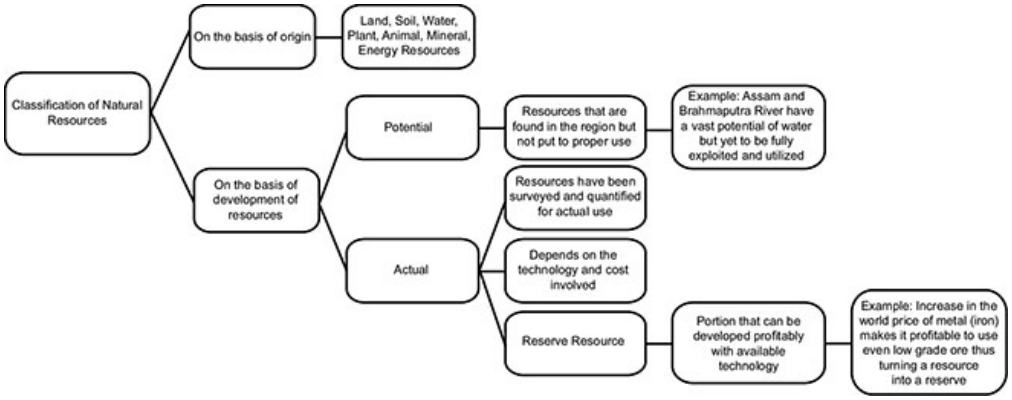
7

Natural Resources of India



NATURAL RESOURCES

Any matter or energy derived from the environment (nature) that is used by living things including humans is called a '*natural resource*'. Resources are the basis for the development of any country. India, one of the largest countries in the world, is blessed with diverse and abundant resources. Only a judicious use of resources will help the development of a country. Over exploitation and unscientific land-use practices leads to environmental problems and resource depletion. Natural resources include air, water, soil, minerals, fossil fuels, plants, and wild life. Many of these natural resources are essential for human survival; e.g., air, water, and plants. While others are used for satisfying other material needs and desires.

Natural resources have been classified in many ways.

Natural resources may also be classified as **renewable** and **non-renewable** resources. The main characteristics of these resources are given below:

Renewable Resources: Resources which get renewed or replenished fast, are called renewable resources. Some of these resources are always available (continuous) and do not get affected by human activities; e.g., solar and wind energy. Many resources, on the other hand, get depleted after use. These may, however, be replenished without endangering future use, provided that the rate of consumption does not exceed the rate of renewal or replenishment. Hence, they maintain a flow. Some resources like crops take short time for renewal. Others like water can be renewed in a comparatively longer time. Some other resources like forests take even longer.

Non-renewable Resources: Non-renewable resources are built over a long geological time span. Minerals and fossil fuels are the examples of non-renewable resources. Since the rate of their formation is extremely slow, they cannot be replenished within a time frame meaningful to human beings. Though these resources are normally found in large quantities, they are distributed most unevenly. Their economic use is viable only when they are found in sufficiently large concentrations and are extractable. Some of these resources like gold, silver and iron are recyclable in nature. It means that the metal content obtained from the ore may be used again and again after necessary processing. Fossil fuels such as coal, mineral oil and natural gas get exhausted. Hence, they are non-recyclable.

MINERAL RESOURCES

A mineral is an aggregate of two or more than two elements. A mineral has a definite chemical composition, atomic structure and is formed by inorganic processes. In economic geography, the term mineral is used for any naturally occurring material that is mined and is of economic value.

Minerals generally occur in the earth's crust in the form of ore. It is extracted, processed, and utilised for the economic benefits of society. The availability and per capita consumption of minerals is taken as an important indicator to assess the economic development of a country.

India is fairly rich in mineral resources but their distribution is highly uneven. The distribution of minerals in India has been described in the following section:

Distribution of Minerals

The mineral wealth of India is largely confined to the igneous and metamorphic rocks of Peninsular India, while the Great Plains of India and the Himalayan region are almost devoid of the metallic minerals. The states which are rich in the metallic and non-metallic minerals are Jharkhand, Chhattisgarh, Odisha, Bihar, West Bengal, Madhya Pradesh, Karnataka, Kerala, Maharashtra, Tamil Nadu, Rajasthan, Gujarat, Uttarakhand, Andhra Pradesh, Tripura, and Assam. The states of Uttar Pradesh, Haryana, Punjab, Himachal Pradesh, Jammu and Kashmir, and Gangetic West Bengal are, however, poor in mineral resources.

Mineral Belts of India

The following mineral belts may be identified in India **(Fig. 7.1)** & **(Table 7.1):**

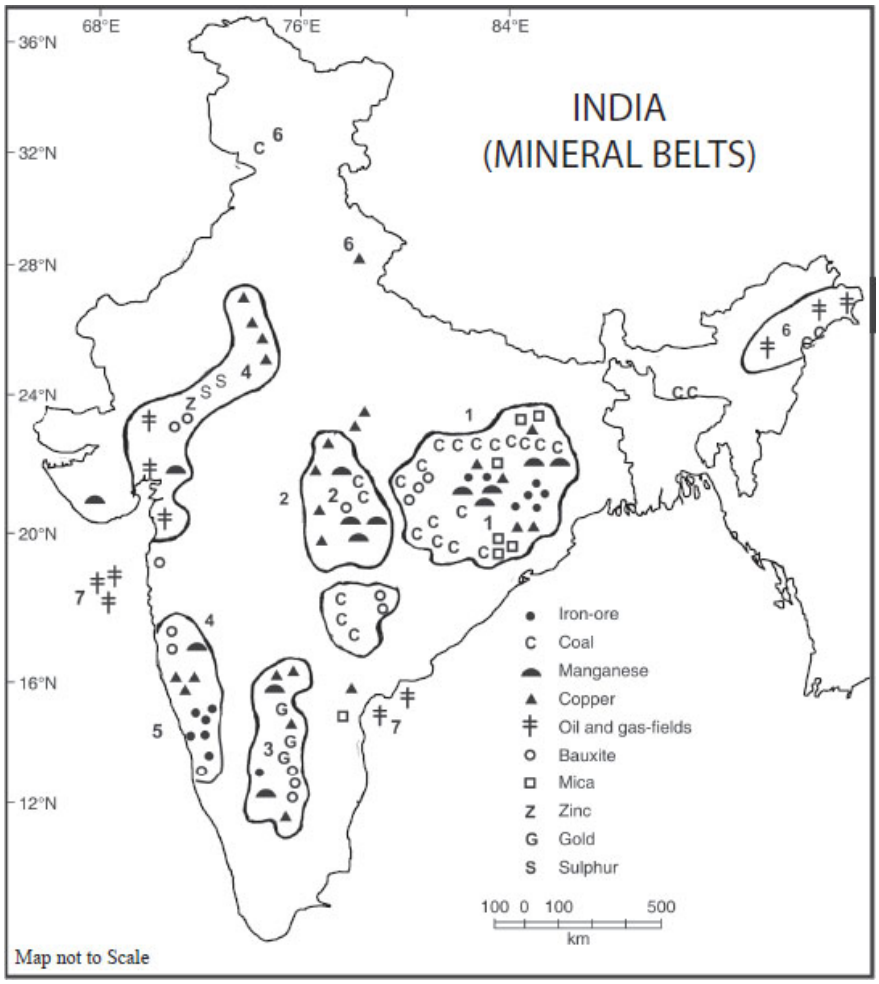


Fig. 7.1 Mineral Belts

S.No.	Name of the Belt	Region	Minerals Found	Features
1	The Chotanagpur Belt	Dhanbad, Hazaribagh, Palamu, Ranchi, Santhal-Pargana, and Singhbhum in Jharkhand; Cuttack, Dhankenal, Kendujhar (Keonjhar), Koraput, Mayurbhanj, Sambhalpur, and Sundargarh in Odisha; and Bankura, Birbhum, Medinipur and Purulia in West Bengal	coal, mica, manganese, chromite, ilmenite, bauxite, iron, uranium phosphate, copper, dolomite, china-clay, and limestone	Region contains almost 100% of kyanite reserves, 93% of iron ore, 84% coal, and 70% of chromite of the country
2	The Midland Belt	Chhattisgarh, Madhya Pradesh, Andhra Pradesh, and Maharashtra	Manganese ore, bauxite, mica, copper, graphite, limestone, lignite, marble, and limestone.	

S.No.	Name of the Belt	Region	Minerals Found	Features
3	The Southern Belt	Andhra Pradesh, Karnataka, and Tamil Nadu.	Gold, iron ore, chromite, manganese, lignite, mica, bauxite, gypsum, asbestos, dolomite, ilmenite, china-clay, and limestone	
4	The Western Belt	Rajasthan, Gujarat, and Maharashtra.	Non-ferrous metals like copper, lead, zinc, uranium, mica, manganese, salt, asbestos, building stonnes, precious stonnes, mineral oil, and natural gas.	
5	The South-Western Belt	Goa, Karnataka, and Kerala	Iron ore, ilmenite, zircon, monazite sands, garnet, china-clay, bauxite, mica, limestone, and soapstone.	
6	The Himalayan Belt			
7	The Indian Ocean	Continental shelf of the Arabian Sea and the Bay of Bengal	Mineral oil and natural gas, seabed also contains high grade nodules (at depth of 4000 m) of manganese, phosphate, barium, aluminium, silicon, iron, titanium, sodium, potassium, chromium, monazite, ilmenite, magnetite, and garnet	

The following (Table 7.2) provides the details on production and value in crores for the various minerals in India. The metallic minerals add to the economy of the country with Rs **53,028.83** crores and the non metallic minerals are resources of value Rs 7702.13 as in the FY 2017-18.

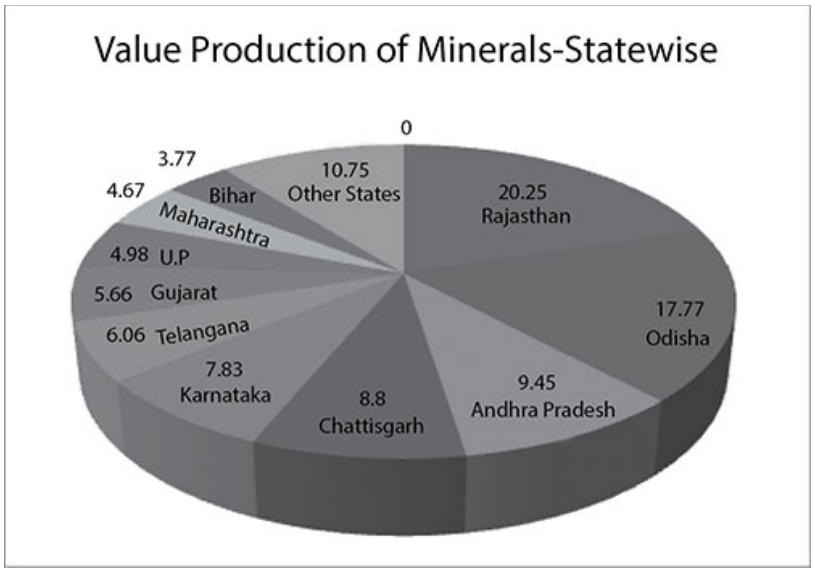
Table 7.2

Production of Selected Minerals (2013–14 to 2017–18) Excluding the Atomic and Fuel Minerals

		2013–14	2014–15	2015–16	2016–17	2017–18	2017–18
Minerals	**Unit**	**Quantity**	**Quantity**	**Quantity**	**Quantity**	**Quantity**	**Value in Crores**
Metallic							
Iron ore	M. tonnes	152.18	31,649.18	158.11	192.08	210.47	32,751.55
Bauxite	Th. tonnes	22,319.15	22,493.67	28,123.79	24,664.63	20,630.33	1292.55
Chromite	Th. tonnes	2878.32	2164.16	2915.58	3727.78	3461.80	4017.53

		2013–14	2014–15	2015–16	2016–17	2017–18	2017–18
Copper	Th. tonnes	139.31	107.60	151.84	134.79	149.97	766.52
Gold	kg	1564.00	1441.00	1323.00	1594.00	1609.00	433.07
Lead Conc.	Th. tonnes	194.43	197.67	261.86	268.05	349.33	1267.05
Manganese	Th. tonnes	2626.29	2369.48	2166.95	2393.18	3100.21	2148.09
Zinc Conc.	Th. tonnes	1490.66	1489.37	1473.81	1484.25	2662.43	8519.11
Total value (in crores)		**42,389.36**	**37,909.11**	**33,621.71**	**40,017.29**		**53,028.83**
Non-metallic							
Diamond	crt	37,517.00	36,107.00	36,044.00	36,516.00	43,936.00	76.58
Garnet	Th. tonnes	483.56	91.39	82.10	85.41	88.82	86.85
Limeshell	Th. tonnes	18.75	16.35	10.35	12.34	29.58	7.12
Limestone	M. tonnes	280.86	293.27	307.00	313.20	325.34	7129.93
Magnesite	Th. tonnes	196.94	285.01	327.66	299.17	172.48	43.64
Phophorite	Th. tonnes	1453.58	1607.22	1571.86	1181.31	914.86	216.83
Sillimanite	Th. tonnes	67.27	66.27	69.94	68.14	87.75	71.73
Wollastonite	Th. tonnes	192.71	186.52	175.35	166.19	136.58	10.67
Other non-minerals							58.78
Total value (in crores)		**7516.05**	**6496.23**	**7572.03**	**7414.53**		**7702.13**

Source: Annual Report, Government of India, Ministry of Mines (2017–18).



Source: Annual report, Government of India, Ministry of Mines (2017–18)

Classification of Mineral Resources

The minerals may be classified under the following three categories:

1. Mineral Fuels (fossil fuels)
2. Metallic Minerals
3. Non-metallic Minerals

1. Mineral Fuels

This group includes energy resources like coal, mineral oil (petroleum), natural gas, and atomic minerals.

Energy Resources (*For details see Chapter 8*)

2. Metallic Minerals

Metallic minerals constitute the second most important group of minerals after fossil fuels. These minerals provide a strong base for the development of metallurgical industry, and thereby help the process of industrialisation and urbanisation. India has a substantial reserve of these minerals **(Fig. 7.3).**

India is a producer of about 95 minerals of which 3 are fuel minerals, 3 atomic minerals, 10 metallic, 23 non-metallic minerals and 55 minerals.

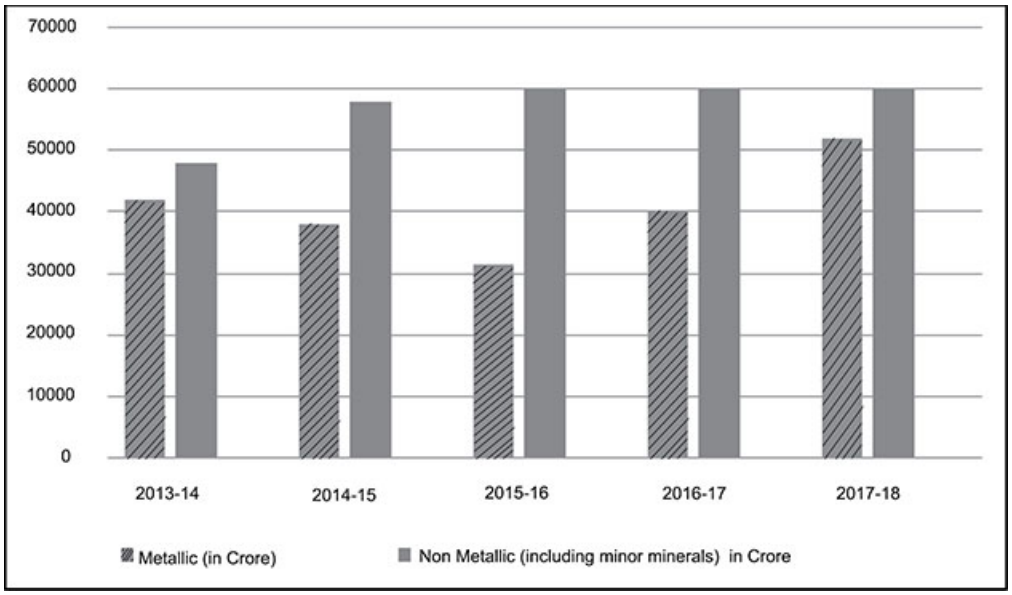


Fig. 7.2: Value of mineral production (by groups) excluding the atomic & mineral fuels [*Source: Annual report, Government of India, Ministry of Mines (2017–18)*]

The total value of mineral production (excluding atomic & fuel minerals) during 2017–18 has been estimated at 113,541 crores, which shows an increase of about 13% over 2016–17.The value of metallic minerals in 2016–17 at Rs 40,017 increased by 19.03% and that of non metallic minerals at Rs 7414.53 crores during 2016–17 decreased by 2.07%.

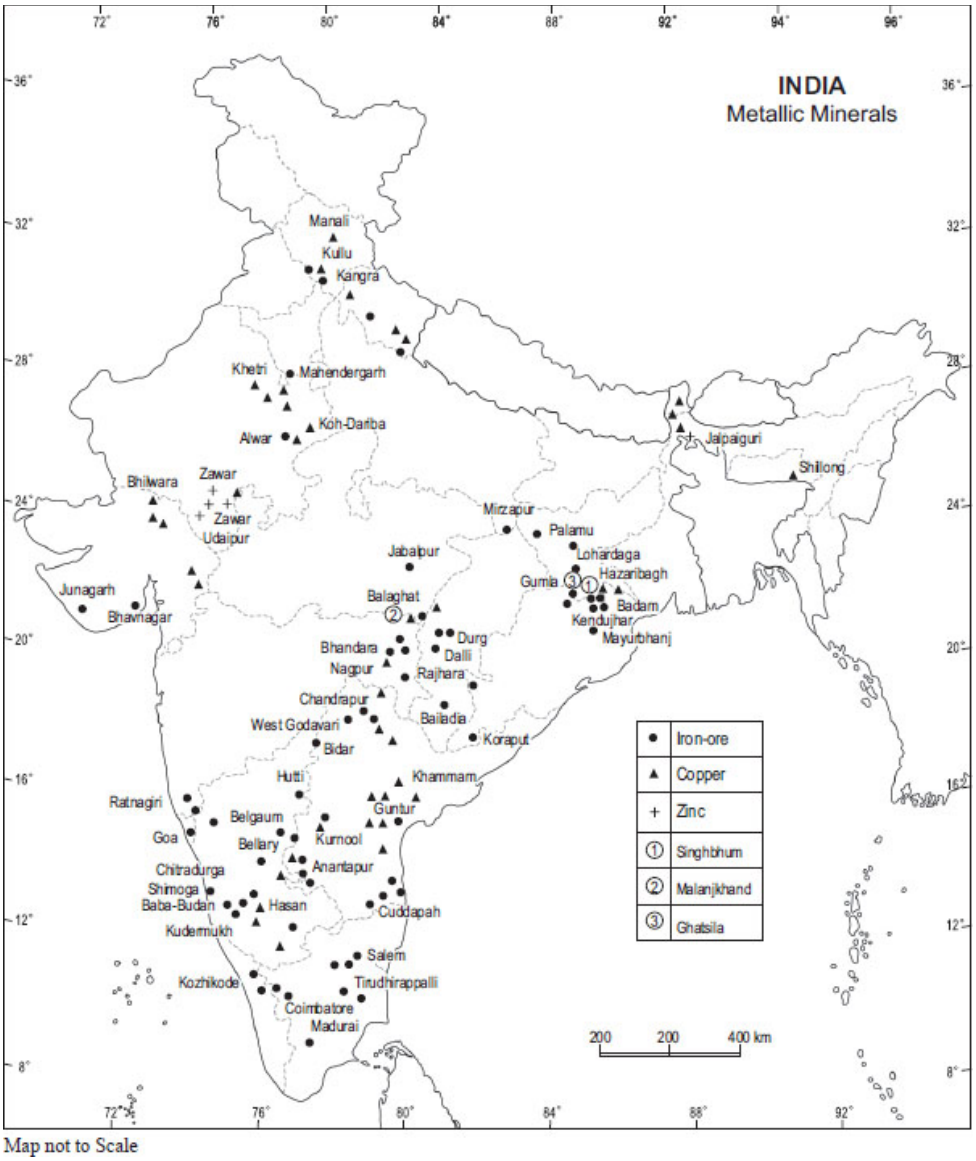


Fig. 7.3 Metallic Minerals (Iron, Copper, and Zinc)

Iron-Ore

Iron ore is the most important mineral on which hinges the economy of a country. Iron ore is not found in pure form in the earth's crust, rather it is often found mixed with lime, magnesium, phosphorus, silica, sulphur, and copper.

Types of Iron Ores There are four main types of iron ores found in India. They are: (i) Haematite, (ii) Magnetite, (iii) Limonite, and (iv) Siderite.

(i) **Haematite ore** (red-ochre): This is called *oxide of iron.* Its metallic content varies between 60 to 70%. It is a massive, hard, compact and lumpy ore with reddish or coral-red in colour. It is mainly found in the Dharwarian rocks. The main deposits of haematite ore are in Jharkhand (Iron-Series), Odisha

(Mayurbhanj), Chhattisgarh (Bailadila Dalli-Rajhara), Madhya Pradesh, Karnataka (Kudermukh, Baba-Budan), Goa, Maharashtra, and Andhra Pradesh.

(ii) **Magnetite**: The magnetite ore is known as *black ore*. The metal content of magnetite varies between 60 to 65%. The ore is either igneous of metamorphic. It is mainly found in the Dharwar and Cuddapah Systems of Karnataka (Dharwar, Shimoga districts), Andhra Pradesh (Bellary district), Tamil Nadu (Salem, Tiruchirappalli districts), and Kerala states.

(iii) **Limonite:** It is yellowish in colour and is known as the *hydrated iron-oxide*. It is inferior and contains 35 to 50% of metal. It is found in the iron-stone shales of the Damuda Series in Raniganj coal fields, Mirzapur district of Uttar Pradesh, Garhwal region of Uttarakhand, and the Kangra Valley of Himachal Pradesh.

(iv) **Siderite:** It is called as *iron carbonate*. Its iron content varies between 10 to 40%. It is an inferior variety of iron ore and not economically extractable at most of the places.

Reserves Iron Ore According to the Geological Survey of India, the total reserves of all types of iron ore in the country are 31323 million tonnes. The production of iron ore form 1951 to 2018 has been given in **(Table 7.3 (Fig. 7.2)).**

Table 7.3

India: Production of Iron Ore (1951–2018)

Year	Production of Iron Ore (Million Tonnes)
1951	4.15
1961	18.70
1971	34.30
1981	41.60
1991	54.60
2001	86.20
2016	156.18
2017	192
2018	210

Source: Year Books & the Statistical Abstracts of India, 2018.

It may be seen from **(Table 7.1)** that total production of iron ore in 1951 was only 4.15 million tonnes which rose to over 86.20 million tonnes in 2001 and reached 156.10 million tonnes in 2016. The growth trend of the iron ore production is significantly increasing.

Output has reached over 210 mt, mostly with an increased production in Odisha which is a 9% higher than 192 mt in 2016–17. Odisha accounts for 50% of the national output with Chattisgarh, Karnataka, Jharkhand, and Goa contributing to the rest of 45%. (Ref: *Business Standard;*

May 3, 2018). The total reserves of iron ore (magnetite) is 10,644,060 thousand tonnes and of haematite are 17,882,097 thousand tonnes in the country. It is expected that India shall produce about 225 million tonnes of iron ore in 2020. India exported about 24.2 million tonnes of iron ore and concentrates in 2018.India is ranked fourth in the world for its iron ore production.

The state-wise production of iron ore of some of the important states has been given in the chart below

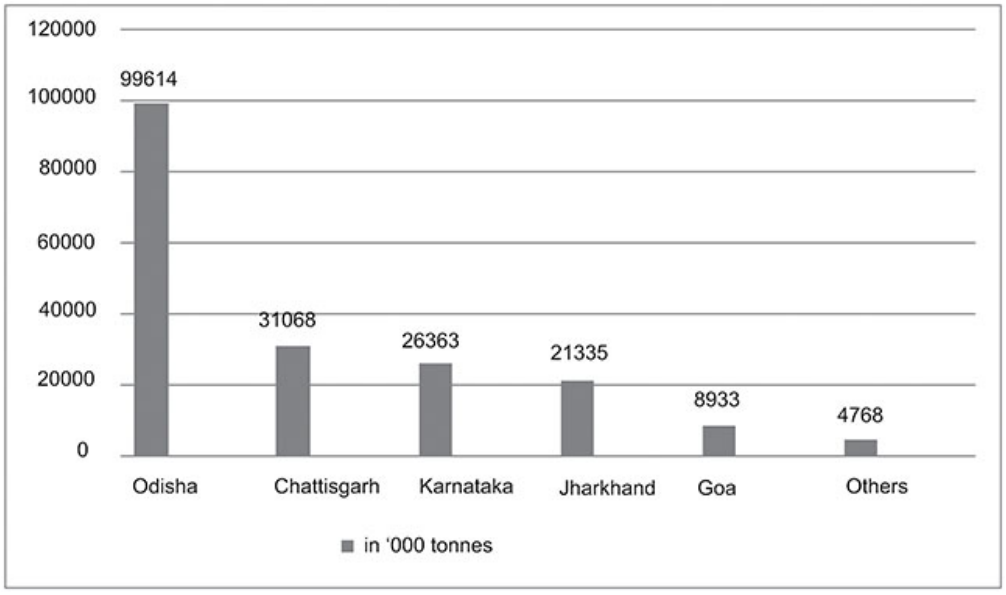


Fig 7.4: Production of Iron Ore in 2017 (in '000 tonnes)

It may be seen from **(Fig 7.4)** that Odisha is the leading producer of iron ore contributing about 50% of the total production. The iron-ore of very high grade are limited to Bailadila (Chhattisgarh) and to a lesser extent in Bellary-Hosepet area of Karnataka and Barajamada sector in Jharkhand and Odisha.

The important iron ore deposits and mining centres have been briefly described in the following section:

1. **Odisha:** The contribution of Odisha in the total production of iron ore in the country is about 50%. The most important deposits are found at Mayurbhanj (Badampahar), Banspani and Toda in Kendujhar (Keonjhar), Tomka Range in Cuttack, Kandadhar Pahar in Sundargarh, Sambalpur, and Hirapur Hills of Koraput district.

The major mines in the year 2017 were Rungta (Odisha), Jajang (Odisha), Balda Block (Odisha), Joda east (Odisha) Bailadilla iron ore mine and Kirandul Complex (Chattisgarh)

(i) ***Badampahar:*** Situated in the Mayurbhanj district of Odisha, Badam Pahar has rich deposits of iron ore. Its height is about 825 m above sea

level. It has 30 million tonnes of iron ore. Iron-ore from Badampahar is supplied to Bokaro, Durgapur, Jamshedpur, and Raurkela.

(ii) ***Bonaigarh Range:*** Situated in the district of Sundergarh, it is one of the most important iron ore bearing ranges. Iron ore is of haematite category which is supplied to Bokaro, Durgapur, Jamshedpur, and Raurkela.

(iii) ***Mayurbhanj:*** Situated in Odisha, it is well known for the iron ore deposits of haematite type. The metal content is more than 65%. Iron ore from Mayurbhanj mines is sent to the iron and steel plants of Bokaro, Durgapur, Jamshedpur, and Raurkela.

2. **Chhattisgarh:** This state has about 16.77% of the total iron ore deposits of the country. Bailadila in the Bastar District and Dalli-Rajhara in the Durg district are the main iron ore producing regions of Chhattisgarh state. The iron ore belongs to the haematite and magnetite categories in which the metal content varies between 60 to 65%. The Bailadila mine is the largest mechanised mine in India. A 270-km long slurry pipeline has been constructed to bring the ore from Bailadila to Vishakhapatnam Plant. The Bailadila iron ore is largely exported to Japan through the port of Vishakhapatnam.

   The Dalli Rajhara range is about 32 km long with iron ore reserves of about 125 million tonnes. The ferrous content is about 65%. The deposits of this range are being worked by the Hindustan Steel Plant of Bhilai. Bilaspur, Jagdalpur, Raigarh, and Surguja are the other iron ore producing districts of Chhattisgarh state.

   (i) ***Dalli Rajhara****:* The Dalli Rajhara Range, well known for the iron ore depsoits, lies in the Durg district of Chhattisgarh. It is 32 km long with an estimated iron ore deposit of 120 million tonnes. The iron ore is supplied to the Hindustan Steel Plant at Bhilai.

   (ii) ***Bailadila:*** Situated in the Bastar district of the Chhattisgarh state, it is known for the rich deposits of iron ore of the haematite category. This iron ore was deposited during the Dharwar Period about 2500 to 1800 million years back. The ore is mostly exported to Japan through the seaport of Vishakhapatnam.

3. **Karnataka**: Karnataka is the third largest producer of iron ore. With 26,363 million tonnes of production, Karnataka ranks third in India and 13.72% of the total iron ore production in the country. The high grade deposits belonging to the haematite and magnetite categories are found in *Kemmangundi* in Bababudan Hills of the Chikmagalur districts. The other important iron ore producing districts of Karnataka are Chitradurga, Dharwar, North Kannad, Shimoga, Bellary, and Tumkur.

   (i) ***Bababudan Hills*****:** Lying in Chikmagalur district of Karnataka they stretch over 22 km in length and 20 km in width. They are rich in haematite

deposits with ferrous content of 60 to 65%. The iron ore is mainly exported to Iran through the port of Mangalore.

(ii) ***Kudermukh Deposits*****:** The Kudermukh iron ore deposits lie in the Chikmagalur district of Karnataka. They contain iron ore of the magnetite category with a metal content of 50 to 65%. The Kudermukh deposits were developed under an export agreement with Iran. The iron ore is exported through the seaport of Mangalore.

(iii) ***Sandur Range*****:** The Sandur Range stretches in the Bellary and Hosepet districts of Karnataka. The iron ore of this range is generally hard, compact and steel-grey. The ferrous content varies between 50–65%. Its ore is supplied to the Vijayanagar Steel Plant.

4. **Jharkhand:** Jharkhand has 25% of the iron ore reserves and accounts for about

   11.1% of the total production of iron ore of the country. Iron ore mining was first of all started at Singhbhum in 1904. The main iron ore deposits lie in Bonai Ragne extending for about 50 km. The famous mines are Naomandi, Daltenganj (Palamu district). Iron ore is also mined at Dhanbad, Hazaribagh, Ranchi, and Santhal Pargana.

5. **Goa**: Goa is the fifth largest producer of iron ore in India. Goa produces about 4.65% of the total iron ore of the country. The iron ore of north Goa is of superior quality. The main deposits and mining centres are at Pirna-Adolpale-Asnora, Sanquelim ponda, Kundem-Surla, and Sirigao-Bicholim-Dalda in north Goa. The nearby Marmagao seaport is a big advantage to these mines for the export of iron ore. The iron ore is exported mainly to Japan and Iran.

Other States: Iron ore in small quantities is also mined in Maharashtra, (Chandrapur, and Ratnagiri); Tamil Nadu (Coimbatore, North Arcot, Ambedkar, Madurai, Salem, Tiruchirapalli, Tirunelveli); Andhra Pradesh (Anantapur, Cuddapah, Guntur, Nellore); Khamman and Kurnool in Telangana; Rajasthan (Alwar, Bundi, Bhilwara, Jaipur, Sikar, Udaipur); Uttar Pradesh (Mirzapur), Uttarakhand (Almora, Garhwal, and Nainital); Himachal Pradesh (Kangra and Mandi); Haryana (Mahendergarh); West Bengal (Birbhum, Burdwan, Darjeeling); Jammu & Kashmir (Udhampur and Jammu); Gujarat (Bhavnagar, Junagarh, Vadodra); and Kerala (Kozhikode).

Export: India is the fifth largest exporter of iron ore in the world. About 55% of India's total iron ore production is exported to Japan, South Korea, West European countries, Iran, United Arab Emirates and other Gulf countries. Most of the export is made through the ports of Vishakhapatnam, Paradwip, Marmagao, and Mangalore.

Manganese India has the largest reserves of manganese in the world after South Africa, Australia, China, Brazil, Gabon, Ukraine and Zimbabwe, and is the fifth largest producer after Brazil, Gabon, South Africa, and Australia **(Fig. 7.3).** It is used mainly for the manufacturing of iron and steel, bleaching powder, insecticides, pesticides, paints, dry-batteries, photography, etc. The total reserves of manganese-ore in the

country are placed at 430 million tonnes (India-2016, p. 585). The statewise production of manganese has been given in **(Table 7.4).**

Table 7.4

India: Distribution of Manganese (2014–15)

S. No.	State	Quantity in Thousand Tonnes	Percentage of All India	Districts/Mining Centres
1.	Maharashtra	6,49670	29	Nagpur, Bhanda, Ratnagiri
2.	Madhya Pradesh	6,48884	37.68	Balaghat, Chhindwara
3.	Odisha	565326	14	Sundargarh, Kalahandi, Koraput, Kendujhar
4.	Andhra Pradesh	3,22470	10.8	Srikakulam, Vishakhapatnam, Cuddappah, Guntur, Viziaragram
5.	Karnataka	136194	8.27	North Kannads, Shimoga, Bellary, Tumkur, Chitradurga
6.	Others	28	0.33	Goa, Panchmahal and Vadodara (Gujarat), Udaipur, Banswara (Rajasthan), Singhbhum, Dhanbad, (Jharkhand)
	All India	2348	100.00	

Source: Indian Minerals Year book 2015, Govt. of India.

1. **Maharashtra:** Maharashtra is the largest producer of Manganese, accounting for about 29% of the total production. In Maharashtra, manganese is found in Bhandara, Nagpur, and Ratnagiri districts. The Ratnagiri ore is, however, of superior quality. The state contributes 39.01% of the total value of manganese ore produce.
2. **Madhya Pradesh:** About 37.68% of the total manganese production comes from the state of Madhya Pradesh. Balaghat and Chhindwara are the main districts in which it is mined. It contributes to 37.86% of the total value of Rs 1362 crore in India.
3. **Odisha:** The state of Odisha is the third most important state in the production of manganese accounting for over 14% of the total production and 13.97% of the total value. The Gondite deposits in Sundargarh and khondolite and kodurite deposits in Kalahandi and Koraput are rich in manganese. Manganese is also mined in Bolangir and Sambhalpur districts of Odisha.
4. **Andhra Pradesh:** About 11% of the total manganese production of India is done in Andhra Pradesh. The contribution of this state to the value of manganese ore production is 3.45%. Srikakulam and Vishakhapatnam are the leading producer districts of manganese in Andhra Pradesh. It is also mined in Cuddapah, Guntur, and Vijayanagram.
5. **Karnataka:** The state of Karnataka produces about 8% of the total production of manganese of the country. Its main deposits are in the districts of North Kannada, Shimoga, Bellary, Chitradurga, and Tumkur.

6. Others: In addition to these, manganese is also mined in small quantities in Goa, Gujarat (Panchmahal and Vadodara); Rajasthan (Banswara and Udaipur); Jharkhand (Dhanbad and Singhbhum). The growth pattern and production of manganese in India has been shown in **(Table 7.5**., **Fig.7.5)**.

Table 7.5

India: Production of Manganese Ore (1950–2016) (in lakh tonnes)

Year	Production (in Lakh Tonnes)
1950–51	13.98
1960–61	14.05
1970–71	18.41
1980–81	15.32
1990–91	13.88
2000–01	15.95
2005–06	19.06
2015–16	21.48

Source: Indian Bureau of Mines, Ministry of Mines Statistical Year Book-2017(MOSPI). Visualize.data.gov.in

It may be seen from **(Table 7.5)** that in 1950–51 the total production of manganese was only about 14 lakh tonnes which rose to over 21.4 lakh tonnes in 2015–16.

Nearly 80% of the manganese production is consumed within the country, while the remaining is exported to USA, UK, Germany, France, Italy, Netherlands, Belgium, Czech Republic, Slovakia, and East European countries including Ukraine.

Fig. 7.5 Metallic Minerals

Copper

Copper is highly ductile, strong, and good conductor of electricity. It is mainly used in electrical machinery, automobile, stainless steel. When alloyed with zinc, it is known as brass and with tin as bronze. Madhya Pradesh has the largest deposits of copper-ore followed by Rajasthan, and Jharkhand. In the production of copper-ore, Madhya Pradesh stands first, followed by Rajasthan and Jharkhand. The total resources of copper are about 1.39 billion tonnes with a metal content of 1228 million tonnes thousand tonnes and the ore content of 1558 million tonnes (Annexure–2.1; Report 2017–18, Ministry of Mines). The distribution of copper in India has been given in **(Table 7.6)**.

Table 7.6

India: Distribution of Copper (2015–16)

State	Production in Thousand Tonnes	Percentage of All India	Districts/Mining Centres
1. Madhya Pradesh	89	55.97	Malanjkhand (Balaghat), Bargaon (Betul)
2. Rajasthan	65	40.88	Jhunjhunu (Khetri-Singhana), Bhilwara, Ajmer, Chittorgarh, Alwar District Dungarpur, Jaipur, Pali, Sikar, Sirohi
3. Jharkhand	05	03.15	Hazaribagh, Santhal Pargana, Singhbhum
All India	159	100.00	

Source: India, 2016.

1. **Madhya Pradesh**: The state of Madhya Pradesh is the largest producer of copper in India. The state has a large deposit of copper in Taregaon in Malanjkhand Belt of Balaghat district. It is also found in Bargaon of the Betul District.
2. **Rajasthan:** Copper is found at Khetri-Singhana belt in Jhunjhunu district. It is also mined in Ajmer, Alwar, Bhilwara, Chittorgarh, Dungarpur, Jaipur, Pali, Sikar, and Sirohi districts. The Koh-Dariba (mountain), about 48 km to the south-west of Alwar city and Delwara-Kirovli area about 30 km from Udaipur are the other important producers of copper ore.
3. **Jharkhand:** This state is the third largest producer of copper in the country. Copper is mined in Hazaribagh, Santhal Pargana (Jharkhand), Gaya, and Palamu districts (Bihar).

Hindustan Copper Limited (HCL), a public sector undertaking is the leading company involved in the mining of this metal. It may be seen from **(Table. 7.7)** that the production of copper is declining during the last 25 years.

Not being self-sufficient in copper, India is importing substantial quantity of copper from Zimbabwe, Australia, USA, Mexico, and Japan.

***Chilpi Series*:** It stretches over parts of Balaghat, and Chhindwara districts of Madhya Pradesh. The series consists of quartzite, copper-pyrite, mica schist, and marble. The copper obtained from this series is used in the Malanjkhand Copper Plant.

Table 7.7

India: Production of Copper (1950–2018)

Year	Production in Thousand Tonnes
1950–51	375
1960–61	423
1970–71	666
1980–81	2110
1990–91	5255
2000–01	164

Year	Production in Thousand Tonnes
2015–16	152
2016–17	135
2017–18	150

Source: India, 2016.

Ghatsila: Located in Jharkhand, it is a copper smelting plant. It is an electrolytic refinery. It manufactures brass sheets. It also obtains gold, silver, and nickel in the processing of copper.

Khetri: It is an integrated copper mining-cum-ore refining plant in the Jhunjhunu district of Rajasthan. It was established in 1967. It also obtains copper ore from the Malanjkhand copper mines of Madhya Pradesh. It also has a sulphuric acid plant, and a fertiliser plant.

Korba: Bharat Aluminium Company Limited (BALCO) has an aluminium plant located at Korba, Bilaspur district of Chhattisgarh. It obtains bauxite deposits from the Amarkantak region and electricity from the Korba Thermal Power Plant. The government has disinvested its share to a private company, Sterlite.

Malanjkhand: It is an open cast copper mine in Balaghat district of Madhya Pradesh. A copper plant has been established at Malanjkhand. The copper ore is also sent to the Khetri Copper Plant of Rajasthan.

Rakha Project: The Rakha copper plant is located in the Rakha District of Singhbhum of Jharkhand. It obtains copper ore from the mines of Rakha.

Tajola: The Tajola Copper Plant is located in the Raigadh town in Maharashtra. The plant has imported copper cathodes. It manufactures copper rods.

Chromite Chromite is an oxide of iron and chromium. It is widely used in metallurgical and chemical industries. The total reserve of chromite is estimated at 210 million tonnes (2016). Odisha, accounting for about 99% of the total production, is the largest producer of chromite. It is mined in Cuttack, Dhenkanal and Keonjhar districts. Karnataka is the second largest producer. In Karnataka, it is mined in Hassan district. Some chromite has been discovered in the Krishna district of Andhra Pradesh and the Tamenglong and Ukhrul districts of Manipur.

Uranium Uranium and thorium are the main atomic minerals. Uranium is mined at Jaduguda, Bhatin, Narwapahar and Turamdih (Singhbhum East), Jharkhand. Only one kg. of uranium can produce as much electricity as is done by 1500 tonnes of coal. The total uranium reserve of India is 120 thousand metric tonnes.

Lead

Lead is widely used because of its heaviness, malleability, softness and bad conductivity of heat. It is used in alloys, cable cover, lead-sheeting, ammunition, glass making, paints making, automobiles, aeroplanes, type-writers, calculating machines, printing and rubber industry. Lead does not occur free in nature. It is obtained from galena which is found in association with limestone, sandstones and calcareous slates. According to an estimate, India has lead reserves of about 715 million tonnes

(India 2016). In 2015–16 the total production lead and zinc ore was 104.53 thousand tonnes **(Table 7.8)**.

Table 7.8

Production of Lead in India (1950–2018)

Year	Production in Thousand Tonnes
1950–51	1.81
1960–61	5.53
1970–71	4.26
1980–81	19.95
1990–91	44.23
2000–01	54.49
2015–16	261.86
2016–17	268.05
2017–18	349.33

Source: Annual Report 2017–18 Annexure 2.1 (Pg 12) Ministry of Mines.

Rajasthan is the leading producer of lead. It is mined in Udaipur (Zawar, Rikhabdeo, and Debari), Dungarpur (Ghugra and Mando), Banswara, and Alwar districts.

In Andhra Pradesh and Telangana, its deposits are in Cuddapah, Guntur, Khammam, Kurnool, North Arcot-Ambedkar and South Arcot (Tamil Nadu). It is also mined in Uttarakhand (Tehri, and Pithoragarh); Jharkhand (Hazaribagh, Palamu, Ranchi, and Singhbhum); in Madhya Pradesh (Gwalior, Hoshingabad and Shivpuri); Himachal Pradesh (Kangra and Kullu); Jammu and Kashmir (Baramulla and Udhampur); and West Bengal (Jalpaiguri and Darjeeling districts).

India is not self-reliant in lead and therefore, has to import about 75% of its requirements from Australia, Canada, and Myanmar.

Zinc Zinc is found in association with lead and silver. It is mainly used for alloying and for manufacturing galvanised sheets. It is also used for dry-batteries, white pigments, electrodes, textiles, die-casting, rubber industry, and for making collapsible tubes containing drugs, and pastes.

Rajasthan is the leading producer of zinc accounting for about 99% of the total production. Small quantities of zinc are obtained from Sikkim, Jammu, Bihar, M.P., Maharashtra, Tamil Nadu, Meghalaya and Kashmir (Udhampur district), and South Arcot Vallalar of Tamil Nadu. India imports about 80% of its requirements from Australia, Canada, Russia, and Zaire. The total production of zinc in 2016–17 was 268.05 thousand tonnes and increased to 349.33 thousand tommes in 2017–18.

Tungsten

Tungsten is obtained from the wolfram ore. It is a self hardening mineral and therefore, used in steel industry, manufacturing of ammunition, armour plates, heavy guns, hard-cutting tools, etc. The total resources of Tungsten ore in the country are about 88 million tonnes.

Tungsten deposits are found at Degana near Rawat Hills in Rajasthan, Bankura district of West Bengal, Sakoli basin in Bhandara and Nagpur districts of Maharashtra, and Kolar mines in Mysore. It is also found in Chittoor, and East Godavari district of Andhra Pradesh, Ahmedabad district of Gujarat, and Singhbhum district of Jharkhand.

Bauxite Aluminium is obtained from bauxite. Bauxite is an oxide of aluminium. It is not a specific mineral but a rock consisting mainly of hydrated aluminium oxides. It is a clay like substance which is pinkish, whitish or reddish in colour depending on the amount of **iron content.** The total reserves of bauxite are about 3896 million tonnes (2016). As per the data from the Indian Minerals Book 2017, the reserves of bauxite in India are 3896 million tones. Odisha accounts for 49% of the bauxite resources followed by Gujarat (24%), Jharkhand (9%),Maharashtra (8%) and Chattisgarh (8%). Major bauxite resources are found in the eastern coast of Odisha and Andhra Pradesh.

In 2016, the production decreased by 12% and it was 24,665 thousand tonnes. The total number of mines were 157 in 2016–17.

It may be observed from **(Table 7.9)** that the production of bauxite has increased from 68 thousand tonnes in 1950–51 to 28,124 thousand tonnes in 2015–16 and increased to 20,630 thousand tonnes in 2017–18. Still, India has to import a substantial proportion of its requirements from abroad.

Table 7.9

India: Production of Bauxite (1950–51 to 2017–18)

Year	Production in Thousand Tonnes
1950–51	68
1960–61	476
1970–71	1517
1980–81	1955
1990–91	4984
2000–01	8689
2015–16	28,124
2016–17	24,664
2017–18	20,630

Source: Statistical outline of India, 2016.

The production of bauxite in the different states of India in 2016, has been given in **(Table 7.10) (Fig. 7.3).**

Table 7.10

Major Bauxite Mining areas in India (1950–2017)

State	Main Districts/Mining Centres
1. Odisha	Kalahandi, Koraput, Sundargarh, Bolangir Sambalpur.
2. Gujarat	Amreli, Bhavnagar, Jamnagar, Junagarh, Kachchh

3. Jharkhand	Dumka, Lohardaga, Aumla, Munger, Palamau, Ranchi.
4. Chhattisgarh	Amarkantak-Plateau, Bilaspur, Durg, Raigarh, Surguja
5. Maharashtra	Kolhapur, Pune, Ratnagiri, Satara, Thane
6. Madhya Pradesh	Balaghat, Katni, Jabalpur, Maikal Range, Mandla, Shahdol.
7. Goa	North Distt. Ponda
8. Tamil Nadu	Madurai, Nilgiri, Salem
9. Others	Andhra Pradesh, J & K, Karala
India	

Source: Statistical Outline of India, 2017–18.

1. **Odisha:** Odisha stands first in the production of bauxite, producing more than 49% of the total bauxite. The Kalahandi-Koraput belt which extends into Andhra Pradesh is the main bauxite deposit region. In addition to this, bauxite is also obtained from the districts of Bolangir, Sambalpur and Sundargarh. The new aluminium plant located at Damanjoli and Doragurha provide a good market for bauxite in this region.
2. **Gujarat**: About 24% of the total bauxite production is from the state of Gujarat. In Gujarat, the main bauxite depsoits are found between the Gulf of Kutch and the Gulf of Khambat (Arabian Sea) through the districts of Bhavnagar, Jamnagar, Junagarh, Amreli and Kachchh. It is mined in Kheda and Sabarkantha.
3. **Jharkhand**: In Jharkhand, bauxite is obtained from Dumka, Gumla, Lohardaga, Munger, Palamu, and Ranchi districts. The Lohardaga mines are, however, known for high grade bauxite deposits.
4. **Chhattisgarh:** In Chhattisgarh, bauxite is obtained from the Maikal Range, Amarkantak Plateau, Bilaspur, Raigarh, and Surguja. Its share in the total production is about 8.0%.
5. **Maharashtra**: About 8% of the total bauxite production comes from Maharashtra. Kolhapur, Pune, Ratnagiri, Satara, and Thane are its main producing centres.
6. **Madhya Pradesh**: In Madhya Pradesh, bauxite is mined in the districts of Balaghat, Jabalpur, Katni, Mandla, and Shahdol. It produces about 2.42% of the total production of baurite.
7. Goa: Goa's contribution in the total production of bauxite is 1.11%.
8. **Tamil Nadu:** The Madurai, Nilgiri and Salem districts are known for the production of bauxite accounting for about 0.1% of the total production.

Bauxite is also mined in the states of Andhra Pradesh (East Godavari, West Godavari, and Vishakhapatnam districts); Kerala (Kannur, Kollam, Thiruvananthapuram); Rajasthan (Kota); Uttar Pradesh (Banda, Lalitpur), Goa, and Jammu and Kashmir (Jammu, Poonch and Udhampur districts).

Nearly 80% of the total bauxite produced is used for the production of aluminium. Italy and UK are the largest importers of Indian bauxite accounting for 60% and 25%

of the total export respectively. The remaining is exported to Germany, Belgium, and Japan. Some of the important aluminium plants are as under:

Balco: The Bharat Aluminium Company Limited (BALCO) was incorporated on 27 November, 1965 as a Central Public Sector Undertaking with an integrated Aluminium Complex at Korba (Chhattisgarh). The Government of India disinvested 51% equity in the company along with the transfer of management control in favour of M/s Sterlite Industries (India) Limited with effect from 2nd March 2001, and consequently, the company has ceased to be a public sector undertaking.

Renukoot (HINDALCO): The Renukut aluminium plant was commissioned in 1958 about 160 km away from Mirzapur (Uttar Pradesh). It obtains bauxite from Lohardaga (Jharkhand), electricity from the Rihand Dam, and the thermal power station near the plant.

Madras Aluminium Company (MALCO): This aluminium plant was commissioned in 1965 near the Mettur Dam (Salem district). It obtains bauxite from the Shevaroy Hills (Tamil Nadu) and electricity from the Mettur Hydel Project.

Koraput Aluminium Plant (NALCO): The Koraput aluminium plant was commissioned in the Koraput district in 1981. This plant was running in loss and therefore, the central government disinvested in this plant in 2006.

Gold

Gold is a precious metal used for making ornaments, and is known as an international currency. India has about 494 million tonnes with a metal content of 491 tonnes of gold ore reserves (India 2016, p. 584). The production of gold in India has been shown in **(Table 7.11).**

Table 7.11

India: Production of Gold (1951–2018)

Year	Production (in kg)
1950–51	7041
2006–07	2361
2007–08	2969
2008–09	2438
2013–14	1564
2014–15	1441
2015–16	1323
2016–17	1594
2017–18	1609

Source: Communitydata.gov.in (Open Government Data Platform, India).

It may be seen from the **(Table 7.11)** that maximum production of 6200 kg was reported in the year 2000 and a record low production of 1323 kg in 2015. There is an increase in the production of gold to 1609 kg as on 2017–18.The Southern India has

gold favoured geology. About 90% of the total gold is produced in the Karnataka state followed by Rajasthan, West Bengal, Jharkhand, Bihar and Andhra Pradesh. There are three important gold fields in the country, namely, (i) Kolar Gold Field, Mysore (Karnataka), (ii) Hutti Gold Field in Raichur (Karnataka), and (iii) Ramgiri Gold Field in Anantapur district (Andhra Pradesh).

Kolar Gold Mines: Mining from the Kolar Gold Fields was started first in 1871. This field still contributes about 60% of the total gold production in the country. The gold mines of Kolar Field are more than 3000 m deep. Most of the gold has already been taken out and it is not economically viable to extract gold from a depth of more than 3 km.

1. **Karnataka:** Karnataka stands first in the reserves and production of gold in India. It is obtained mainly from the Kolar, Dharwar, Hassan and Raichur districts. Some gold deposits have also been found in Belgaum, Bellary, Chikmaglur, Gulbarga, Mandya, and Shimoga districts. '*Hutti Gold Mine*' Company is India's historic mine in which the production started in 1902.
2. **Andhra Pradesh**: Andhra Pradesh is the second largest producer of gold in India. The main deposits of gold in Andhra Pradesh are found in Ramagiri (Anantapur district). In addition to this, gold is also found in Chittoor and Kurnool districts.

Placer or Alluvial Gold: The gold obtained from the sand and sedimentary deposits of the rivers is known as placer gold. '*Placer gold*' is found in the Subarnrekha (Gold Streak) river of Jharkhand. Placer gold is also found near Lowa in Singhbhum district and some other parts of the Chotanagpur Plateau. **(Fig. 7.7**)

Placer gold is also found in the sand of Dras, and Suru rivers of Kargil (J & K), Shimla and Bilaspur in Himachal Pradesh, Punna-Puzha and Chaiyar Puzha rivers of Kerala, Balaghat and Seoni districts of Madhya Pradesh, Bastar, Raigarh, and Raipur of Chhattisgarh and and Purulia district of West Bengal.

Champion Series: It is named after *Champion reef* in the *Kolar Gold Field deepest gold mine in the world*. It is the oldest gneiss in Karnataka. In fact, it is one of the oldest metamorphic sedimentary deposits in India. Known for its gold deposits, it has quartz and muscovite.

Silver Silver is also a precious metal. India, however, is not very rich in silver deposits. It is an important currency metal, and used in the manufacture of chemicals, electroplating, photography, and coloring for glasses.

It is found in association with lead and zinc. Zawar mines of Udaipur (Rajasthan) are the largest producer of silver. In Hindustan Zinc Smelter (Udaipur), it is obtained as a by-product of zinc and lead smelting. The Tundoo Lead Smelter in Dhanbad (Jharkhand) is also an important source. Silver is also produced by Kolar Gold Fields and Hutti Gold mines (Raichur) of Karnataka during refining of gold. The Hindustan Copper Limited at Maubhandar Smelter (Singhbhum, Jharkhand) obtains silver from copper mines. Silver is also produced at the Vishakhapatnam Smelter in Andhra Pradesh from lead concentrates.

Non-Metallic Minerals

India is fairly rich in the non-metallic minerals also. The geographical distribution of some of the important non-metallic minerals has been shown in **(Fig. 7.4)**.

Mica (Abhrak)

Mica is an important non-metallic mineral used mainly in electrical industry as it has great insulating properties, can withstand high voltage and has low power loss factor. It is obtained from muscovite, biotite and phlogopite ores. The total reserves of mica in the country are estimated to be about 6.3 lakh tonnes (India 2016, p. 591) **(Table 7.11)**. The total reserves of mica in India, as per NMI report is 635,302 tonnes (2015). Andhra Pradesh leads with 41% of the total production followed by Rajasthan (28%), Odisha (17%), Maharashtra (13%) and Bihar (2%).
Source : *(Indian Minerals Yearbook-2018; Part-III)*.

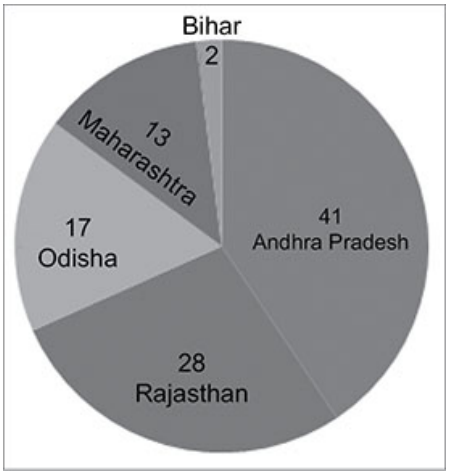


Fig 7.6: Mica Production in India, State-wise (in Percentage)

1. **Andhra Pradesh**: Andhra Pardesh is the largest producer of mica. The micabelt of Nellore is about 100 km long and 25 km wide. Nellore mica is generally light green in colour. Mica is also obtained from Khamma, (Telangana), Krishna, West Godavari, and Vishakhapatnam (Andhra Pradesh).
2. **Rajasthan:** The main mica belt of Rajasthan extends from Jaipur to Udaipur. It is also Wantd at Ajmer, Bhilwara, Dungarpur, Sikar, and Tonk.
3. **Jharkhand and Bihar:** Mica in Jharkhand and Bihar is found in a belt extending for about 150 km in length and 32 km in width from the eastern part of Gaya through Hazaribagh, Giridh, and Munger to Bhagalpur. The main centre of mica production in Jharkhand and Bihar are Kodarma, Dhorhakola, Dhab, Tisri, Chakai.

Fig. 7.7 Metallic Minerals

Table 7.12

India: Major mines/districts of Mica in India

State	Districrts/Mining Centres
1. Andhra Pradesh	Nellore, Krishna. Khamma, Vishakhapatnam, West Godavari.
2. Rajasthan	Ajmer, Bhilwara, Dungarpur, Jaipur, Sikar, Tonk, Udaipur.
3. Jharkhand	Dhanbad, Gaya, Gridh, Hazaribagh, Munger, Ranchi, Singhbhum
4. Bihar	Bhagalpur

Source: Statistical Abstracts of India, 2017.

Mica is also found in Gujarat (Banaskantha, Vadodara and Sabarkantha); Tamil Nadu (Nilgiri, Coimbatore, Salem, and Tiruchirapalli), Chhattisgarh (Bilaspur, Bastar, and Surguja); Madhya Pradesh (Balaghat and Chhindwara); and Uttar Pradesh

(Mirzapur). Some deposits of mica are also found in Haryana, Himachal Pradesh, Odisha, and West Bengal.

It may be seen from **(Table 7.13)** that the production of mica was about ten tonnes in 1950–51 which rose to 3806 tonnes in 2000–01 and 1920 tonnes in 2015–16.

Table 7.13

India: Production of Mica (1950–2016)

Year	Production in Tonnes
1950–51	10.0*
1960–61	28.35
1970–71	1510
1980–81	8534
1990–91	4746
2000–01	3806
2015–16	1920

Source: Statistical Abstracts, 2016, India 2017.

India is the largest producer and exporter of mica in the world. It exports mica to Japan, USA, UK, Norway, Russia, Germany, France, Belgium, Netherlands, Poland, Czech Republic, Slovakia, and Hungary.

Saucer Series: It sprawls over Nagpur, Bhandar (Maharashtra), and Chhindwara district (Madhya Pradesh). Saucer series belongs to the Dharwarian Group. It largely consists of quartzite, mica-schist, marble and magniferous rocks. Its mica is of light green colour.

***Sakoli Series*:** It spreads over Jabalpur and Rewa districts of Madhya Pradesh. It is rich in mica schist, quartz, dolomite, and marble. The famous Jabalpur marble is obtained from the Sakoli Series. It also contains gneisses of the Dharwar period.

Limestone

Limestone is an aggregate of calcium carbonate, carbonate of calcium and magnesium or a mixture of the two. Limestone also contains small quantities of silica, alumina, iron-oxides, phosphorus and sulphur. Limestone deposits are of sedimentary origin and exist in almost all the geological formations from the Pre-Cambrian to Recent except in Gondwana. The total estimate of limesstone reserve in 2010 was 184,935 million tonnes (India, 2016, p. 590) and in 2017–18 it is reported as 2,03,224752 thousand tonnes.

Limestone is used in cement, iron and steel, and chemical industries. The rapid industrialisation and urbanisation has resulted into heavy demand of this mineral, especially for the manufacturing of cement. The production trend of limestone in the country has been given in **(Table 7.13)**.

Table 7.14

India: Production of Limestone (1950–2017)

Year	Production in Lakh Tonnes
1950–51	30.00
1960–61	148.00
1970–71	151.00
1980–81	324.50
1990–91	685.00
2000–01	1275.00
2005–06	1620.00
2010–11	1849.00
2015–16	3038.00
2016–17	3132

Source: Statistical Abstract of India, 2007 and India 2016, p. 590.

***The Semri Series*:** This is the main series of the Lower Vidhyan formations in the Son valley. Its thickness is about 900 m of limestonnes, shales and sandstone with intrusive dolerites and basalt. The upper most stage, known as Rohtas Stage is composed of limestone and shales which provides raw material to the cement industry in the region.

It may be seen from **(Table 7.14)** that in 1950–51 the total production of limestone was only 30 lakh tonnes which rose to 1620 lakh tonnes in 2005–06, and 3038 lakh tonnes in 2015–16.

Limestone in some quantity is produced in almost all the states of India. Its main producing states are Rajasthan, Madhya Pradesh, Andhra Pradesh, Chhattisgarh, Karnataka, Gujarat and Tamil Nadu. The production and percentage of limestone in the important states of the country have been given in **(Table 7.15).**

Dolomite

Dolomite is a type of limestone which contains more than 10% of magnesium. When the percentage of magnesium rises to 45, it is true dolomite. It is used mainly in the metallurgical industry, especially in the iron and steel industry. The total reserves of all grades of dolomites are 7730 million tonnes (India 2016, p. 587). The states of Odisha, Chhattisgarh, Andhra Pradesh, Jharkhand, Rajasthan, Gujarat, Maharashtra and Karnataka are its main producers contributing 90% of the total production **(Fig. 7.8).** The state-wise distribution of dolomite has been given in **(Table 7.16).**

Table 7.15

India: Distribution of Limestone (2016–17)

State	Production in Lakh Tonnes	Percentage of All India Production	Main Districts/Mining Centres
1. Rajasthan	667	21.96	Ajmer, Alwar, Bikaner, Chittaurgarh,

			Dungarpur, Kota, Nagaur, Pali
2. Madhya Pradesh	378	12.44	Betul, Damoh, Jabalpur, Rawa, Satna, Sagar
3. Andhra Pradesh	331	10.90	Cuddaph, Guntur, Kurnool, Krishna
4. Chhattisgarh	275	9.05	Bastar, Bilaspur, Raigarh
5. Karnataka	270	8.88	Belgaum, Bijapur, Chitradurga, Mysore, Uttar-Kannad, Tumkur
6. Gujarat	252	8.29	Amreli, Junagarh, Kheda, Surat
7. Telangana	240	7.90	Nalgonda, Khammam
8. Tamil Nadu	227	7.47	Coimbatore, Madurai, Salem
9. Maharashtra	130	4.28	Chandrapur, Nanded, Ahmednagar, Yavatmal
10. Himachal Pradesh	123	4.05	Bilaspur, Chamba, Kangra
Others	145	4.78	Dehradun, Mussoorie, Darjeeling, Jammu, Ambala, Mirzapur
India	3038	100.00	

Source: Statistical Abstracts of India, 2016–17.

Table 7.16

India: Distribution of Dolomite (2015–16)

State	Production In Thousand Tonnes	Percentage of All India Production	Districts / Mining Centres
1. Odisha	1075	28.71	Gangapur, Koraput, Sambalpur, Sundargarh
2. Chhattisgarh	1025	27.38	Bastar, Bilaspur, Durg, Raigarh
3. Andhra Pradesh	687	18.35	Anantapur, Kurnool and Khammam
4. Jharkhand	295	07.88	Chaibasa, Singhbhum Ajmer
5. Rajasthan	215	05.74	Alwar, Bhilwara, Jaipur, Jai-salmer, Jhunjhunu, Jodhpur, Nagaur, Pali, Sawai Madhopur, Sikar and Udaipur
6. Karnataka	212	05.66	Belgaum, Bijapur, Chitradurga, Mysore, Uttar Kannada, and Tumkur
Others	235	06.28	Arunachal Pradesh, Haryana, Himachal Pradesh, Maharashtra, Tamil Nadu, Uttar Pradesh, Uttarakhand, West Bengal
All India	3744	100.00	

Source: Statistical Abstract of India, 2017 and Mineral Wealth of India, 2017.

1. Odisha is the leading producer of dolomite accounting for about 29% of the total

production in the country, followed by Chhattisgarh with over 27%. The share of Andhra Pradesh and Jharkhand is 18.35% and about 8% respectively in the total production. Rajasthan and Karnataka contribute about 6% each. It is mined in several other states also as given in **(Table 7.16).**

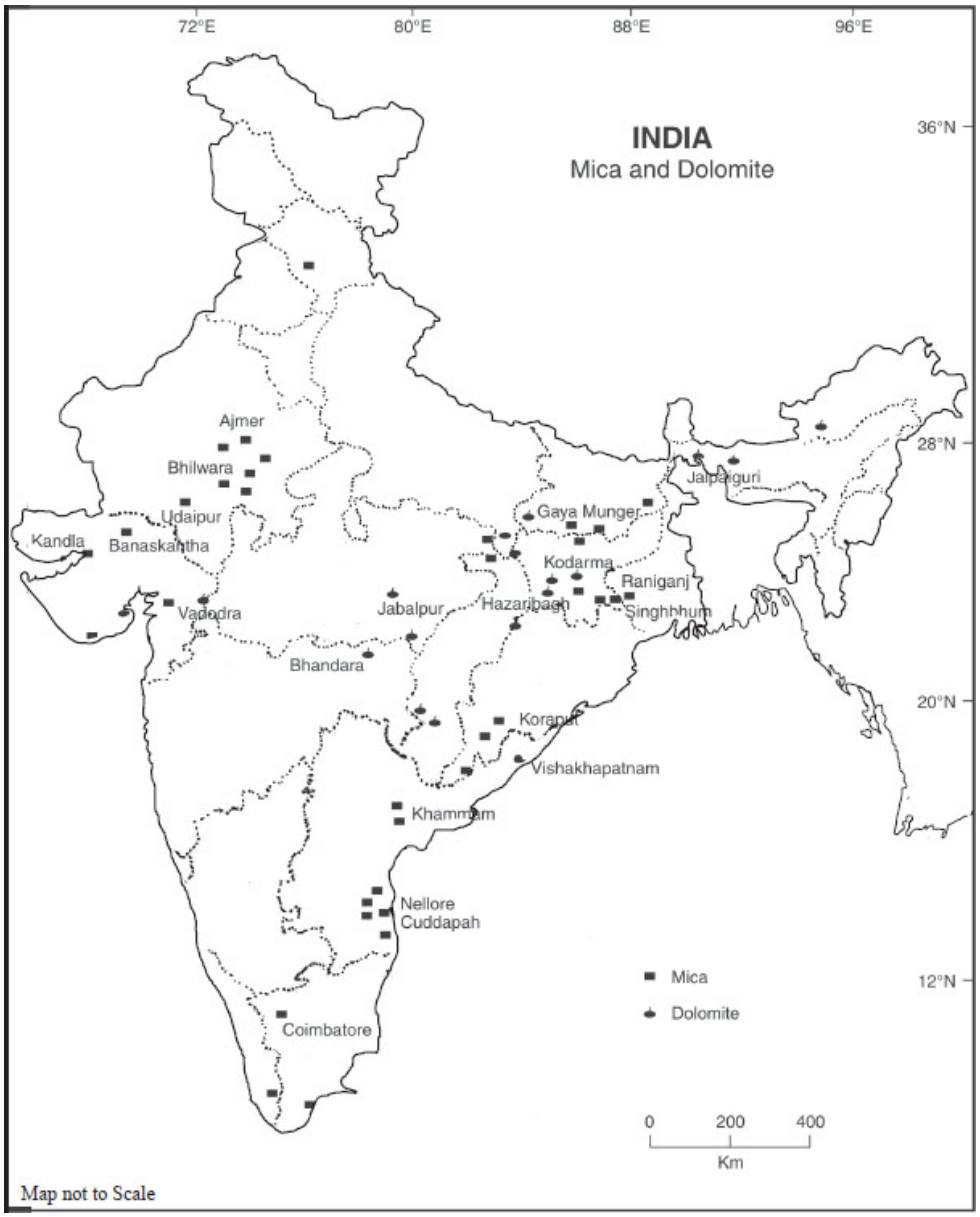


Fig. 7.8 Non-metallic Minerals

Asbestos

Asbestos has great commercial value due to its fibrous structure, and its resistance to fire. It is widely used for making fire-proof clothes, rope, paper, sheeting, belt, fireproof safes, insulators, felts, aprons, gloves, curtains, brake linings in automobiles, and insulating mats. Asbestos cement products like sheets, slates, pipes and tiles are

used for building purposes. Mixed with magnesia, it is used for making ‘magnesia bricks’ used for heat insulation.

Rajasthan is the leading producer accounting for about 95% of the total asbestos production of India. It is mined in Ajmer, Alwar, Dungarpur, Pali and Udaipur districts. Andhra Pradesh is the second largest producer. It is produced in Cuddapah district. It is also mined in Karnataka, Jharkhand, Madhya Pradesh, Chhattisgarh, Tamil Nadu, Uttarakhand, and Nagaland.

Magnesite

Magnesite is used for manufacturing refractory bricks, special type of cement, tiles, fire-proof flooring and for extraction of the metal magnesium, and in steel industry. The total reserves of magnesite in India are about 335 million tonnes (India 2016, p. 591). Its major deposits are found in Uttarakhand (68%), Rajasthan (16%) and Tamil Nadu (13%). Its deposits have also been found in Jammu and Kashmir, Karnataka, Himachal Pradesh, and Kerala.

Tamil Nadu is the largest producer accounting for over 74% of the total magnesite production, followed by Uttarakhand (20%) and Karnataka (6%).

Kyanite

Found in the metamorphic rocks, kyanite is used in metallurgical, ceramic, refractory, glass and electrical industries. The total reserves of Kyanite are 103 million tonnes (India 2016, p. 590). India is the largest producer of kyanite in the world. Kyanite deposits are located in Jharkhand, Maharashtra, and Karnataka. These three states contribute almost the whole production of kyanite of the country.

Gypsum

Gypsum is a hydrated sulphate of calcium which occurs as a white opaque mineral in beds of bands of sedimentary rocks like limestone, sandstone and shale. It is mainly used in making ammonia sulphate, fertilisers and in cement industry. It is also used in making plaster of Paris, ceramic industry, nitrogen-chalk, partition-blocks, sheets, tiles, and plastics.

The total reserves of gypsum are about 1286 million tonnes. Rajasthan is the leading producer of gypsum accounting for about 99% of the total production of the country. It is obtained mainly from the districts of Barmer, Bikaner, Churu, Ganganagar, Jaisalmer, Jodhpur, Nagaur, and Pali. The remaining one per cent is mined in Tamil Nadu, Jammu and Kashmir, Gujarat, Uttarakhand, Andhra Pradesh, Himachal Pradesh, Karnataka, and Madhya Pradesh. The total reserves of gypsum in India were estimated to be about 1286 million tonnes (India 2016, p.588).

Sillimanite

Sillimanite is used is ceramics, metallurgy, glass, refractory, automobiles and cement manufacturing industries. Its main characteristic is that it can withstand high temperatures.

The total reserves of sillimanite are 67 million tonnes (India 2016, p. 590). Odisha, contributing about 57% of the total production, is the largest producer of sillimanite in India. Kerala is the second largest producer accounting for about 33% of the total production. It is also produced in Maharashtra, Rajasthan, Meghalaya, Assam (Karbi-Anglong), Madhya Pradesh, (Sidhi), West Bengal (Darjeeling, Bankura and Purulia), and Tamil Nadu (Kanniyakumari, Tirunelveli, and Tiruchirappalli).

Diamond

Diamond is a precious stone. It is known for its brilliance, luster, transparency and hardness. Diamond is mainly found in the Vindhyan formations of Bundelkhand, (M.P.), Andhra Pradesh (Kurnool, Anantapur), and Karnataka (Raichur). Panna district of Madhya Pradesh is the main diamond producing district in India. In India, the total diamond reserves is about 43.93 million crates (Annexure 2.1; Annual Report 2017–18, Ministry of Mines).

Cutting and polishing of diamond is mainly carried on in Surat, Ahmedabad, Navasari, Palanpur, Bhavnagar, Mumbai, Khambat, Jaipur, Trichur, and Goa.

***Ajabgarh Series*:** Lying in the Rajasthan state, the Ajabgarh Series belongs to the Cuddapah and Lower Vindhyan group. It is rich in biotite-schist, quatzites, and impure limestonnes. It has inferior quality of iron ore, manganese, asbestos, slate, marble, and jasper.

***Bhander Series*:** It belongs to the Vindhyan formation. The main rocks in the Bhander Series are sandstone, shale and limestone. The series provides good quality of building material besides diamond mines. The diamonds from the Bhander series are sent to Surat and Jaipur for polishing and finishing.

Bijwara Series: It occupies parts of Chhatarpur and Panna districts of Madhya Pradesh. It is composed of sandstone, quartzite and limestone. It has basaltic intrusions whose dykes are rich in diamonds. The Panna diamond is famous all over the world for its transparency, brilliance, and hardness.

***Rialo Series*:** This series stretches from Delhi to Alwar. It belongs to the Archaean and Dharwarian groups. The famous marbles of Makrana, Rajnagar, and Bhagwanpura belong to this series. Limestone, marble, quartzite, and building material are the main minerals found in this series.

Atomic Minerals

Uranium and thorium are the main atomic minerals.

Uranium

deposits occur in Singhbhum and Hazaribagh districts of Jharkhand, and Gaya district of Bihar, and in sedimentary rocks of Saharanpur district of Uttar Pradesh. The largest source of uranium comprise the monazite sands, both beach and alluvial. Monazite sand rich in uranium is found in Kerala. Some uranium is found in the copper and zinc mines of Udaipur (Rajasthan). The total reserves of uranium as estimated by the Department of Atomic Energy, Government of India, are about 31,000 tonnes. Uranium Corporation of India Limited, exploits the atomic minerals commercially.

The important uranium mining centres of India are: (i) Jharkhand–Bagjata, Banduhurang, Bhateen, Jaduguda, Mohuldeeh, Narwapur and Turamdeeh, (ii) Meghalaya-Keleng-Pindeng, Maothabah, Shahiyong, and Vakheen, (iii) Andhra Pradesh-Lambapur, Paddagtu and Tummalapalle.

Thorium

Thorium is derived from monazite. It is produced in Kerala, Jharkhand, Bihar, Tamil Nadu, and Rajasthan.

In addition to uranium and thorium, beryllium and lithium are also the atomic minerals found mainly in Jharkhand, Madhya Pradesh, and Rajasthan.

Salt

Salt is used mainly in chemical industry. Common salt (sodium chloride) is used as a food item. Salt is obtained from sea water, brine springs and salt pans in lakes. The main producers of salt are Gujarat, Maharashtra, Tamil Nadu, and Rajasthan. Gujarat coast accounts for about 50% of the total salt production of India. Sambhar lake of Rajasthan contributes about 10% of the total salt production of the country. In addition to this, rock salt is obtained from the Mandi area of Himachal Pradesh. The Mandi salt is hard and massive and has to be blasted.

India exports small quantities of salt to the neighbouring countries like Bangladesh, Bhutan, Indonesia, Japan, Maldive, Nepal, Singapore, South Korea, and Taiwan.

Problems of Mining Industry: India is rich in mineral resources. The mining industry is however, facing a number of problems. Some of the problems have been described briefly in the following:

1. **Ill-Defined Government Policy:** There is no well defined government policy about the prospecting, extraction and processing of mineral resources. A large number of lessees and contractors look upon mines as a quick money-making proposition and use unscientific techniques for the extraction of mineral ores.
2. **Obsolete Technology**: The technology used in mining is generally old and obsolete. Consequently, there is great wastage of mineral wealth.
3. **Inadequate Transport Facilities**: The minerals in the country are not uniformly distributed. For their transportation, the railways are used which are

not very efficient, leading to bottlenecks, scarcity, and higher cost of transportation.

4. **Inadequate Exploration and Prospecting of Minerals**: In the absence of trained geologists, there are many areas in Chhattisgarh, Himachal Pradesh, Jammu and Kashmir, Jharkhand, Karnataka, Madhya Pradesh, Maharshtra, Rajasthan, and Uttarakhand which have not been properly surveyed to explore the minerals.
5. **Inadequacy of Funds**: In the absence of inadequate funds, the infrastructure could not be developed in the areas of isolation and relative isolation where many of the most valuable minerals are found.
6. **Lack of Awareness about Conservation**: There is very little awareness about the exhaustible nature of the minerals and their conservation.
7. **Export of Mineral Ores:** The legacy of the colonial period is continuing and many of the basic minerals are exported to the developed countries of the world. Consequently, the minerals fetch low price in the international market.
8. **Strikes and Naxalites:** The mining industry is adversely affected by the frequent strikes of the miners and the Naxalites. The mineral rich parts of the country are unfortunately infested with Naxalites.
9. Mining is a hazardous industry in India. Hundreds of miners are killed every year. For example between 2008–15, about 450 coal miners lost their life.

Conservation of Minerals: Minerals are the most valuable resources which are imperative for the economic development of the country. They are, however, non-renewable, and can not be replenished in short geological period. A judicious utilisation of mineral resources is necessary to meet the growing demand of our population and to provide the minerals for the future generations. Some of the steps which can go a long way in the conservation of mineral resources have been given in the following:

1. **Judicious Use**: The available minerals need to be utilised judiciously.
2. **Efficient Technology:** For the mining, processing and consumption, efficient technology is required for which continuous research is to be done.
3. **Alternatives to Minerals:** Research should be done to find the alternative to the mineral resources.
4. **Development of Infrastructure:** There should be emphasis on the development of roads and other infrastructure to obtain minerals from the areas of isolation and relative isolation.
5. **Location of Industries near the Mining Sites:** The mineral processing and metallurgical plants should be located near the mining centres to reduce the transportaion cost and problems.
6. **Recycling of Scrap**: The scrap should be recycled and people, especially the students, should be made aware about the exhaustible nature of minerals.
7. **Sustainable Mining**: There should be emphasis on sustainable mining. The

miners should be properly trained in the new technology of mining.

National New Mineral Policy 2019

After the liberalization and globalization, the mining industry has been opened to the private sector. The main Objectives of the New Mineral Policy 2019 are as under:

1. The policy shall ensure more effective regulation that shall lead to sustainable mining sector development in the future.
2. Introduction of Right of First Refusal for RP/PL holders.
3. **Public sector mining**: It shall encourage and promote the private sector to take up exploration by creating dedicated mineral corridors to boost the private sector mining areas. There is a proposal of granting industry status to mining activity to boost the financing for the private sectors and acquisition of mineral assets in the countries. The policy mentions that the long-term import policy for mineral shall help the private sector in a better planning and stability.
4. **Auction of unused reserved areas given to PSUs:** The policy mentions the auction of the unused reserved areas that were given to PSUs, in order to promote the participation by the public sector.
5. **To harmonize taxes, levies and royalty:** Efforts shall be taken to harmonize the taxes, levies and royalty at the world's benchmark to promote the growth of the private sector in the mining industry.
6. **Intergenerational Equity:** This concept shall be introduced to ensure the well-being of the future generations alongwith the present ones.There is also a proposal to constitute an inter-ministerial body to institutionalize the sustainable development in mining.

Thus, the New Mineral Policy, 2019, makes a significant departure from the exclusive control of the Government on the exploration and exploitation of major minerals. The entry of private sector may aggrevate the situation by over-exploitation of minerals. Therefore, the government should take adequate and effective measures to overcome such a problem.

Sources for useful information on mineral resources of India

Ministry of Mines: https://mines.gov.in

Geological Survey of India: www.portal.gsi.gov.in

Indian Bureau of Mines: www.ibm.nic.in

Indian Minerals Yearbook–2018; Part-III)

Communitydata.gov.in (Open Government Data Platform, India)

BIOTIC RESOURCES

The biotic resources include natural vegetation, fisheries, livestock (cattle, buffaloes, goat, and sheep rearing, pig rearing) fisheries, poultry farming. The livestock sector, which contributes about 27% to the GDP from agriculture and allied activities, is of special importance in the arid and semi-arid regions.

Cattle

Animal husbandry and dairy development plays an important role in the rural economy and regional development. Cattle keeping supplements the income of rural households, especially that of marginal farmers and landless workers. It provides a subsidiary occupation in semi-urban areas, and more so for the people living in the hilly and drought prone areas.

Table 7.17

Livestock and Poultry Production in India

Species	Livestock Census (2007) (Number in Millions)	Livestock Census (2012) (Number in Millions)	Growth Rate
Cattle	199.1	190.9	–4.10
Buffalo	105.3	108.7	3.19
Sheep	71.6	65.1	–9.07
Goat	140.5	135.2	–3.82
Pigs	11.1	10.3	–7.54
Poultry	648.8	729.2	12.39

Source: Krishi.jagran.com; May 25, 2019

The total livestock population in the country is 512.05 million as of 2012 and this population has decreased by about 3.33% over the previous census of 2007.

There has been a substantial increase in the population in Gujarat (15.36%), Uttar Pradesh (14.01%), Assam (10.77%), Punjab (9.57%), Bihar (8.56%), Sikkim (7.96), Meghalaya (7.41%) and Chattisgarh (4.34%).Punjab has the highest productivity of the cross bred and indigenous cattle while Tamil Nadu has the highest productivity of exotic cattle.

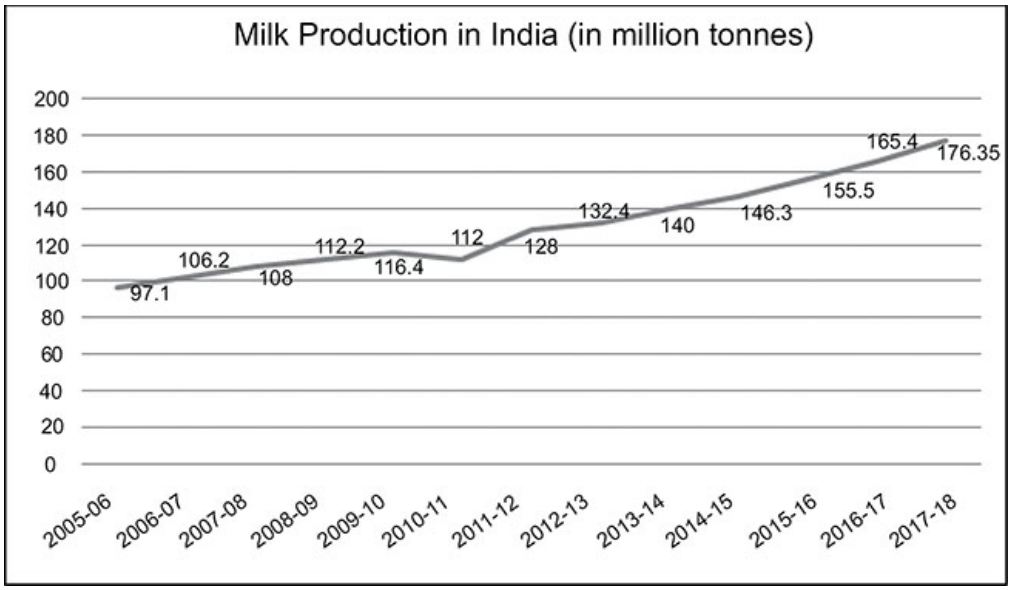


Fig. 7.9 : Milk Production in India (in Million Tonnes) [*Source: Data Collected from Department of Animal Husbandry and Dairying, Ministry of Agriculture and Farmers Welfare*]

Since 1998, India ranks first in the world milk production. Milk production in India during the period 1950–51 to 2017–18 has shown an increase from 17 million tonnes to 176.4 million tonnes. In 2016–17 it was 165.4 million tonnes and has recorded a growth of 6.65%. 36% of the milk production is contributed by indigenous buffaloes followed by 26% by cross bred cattle.

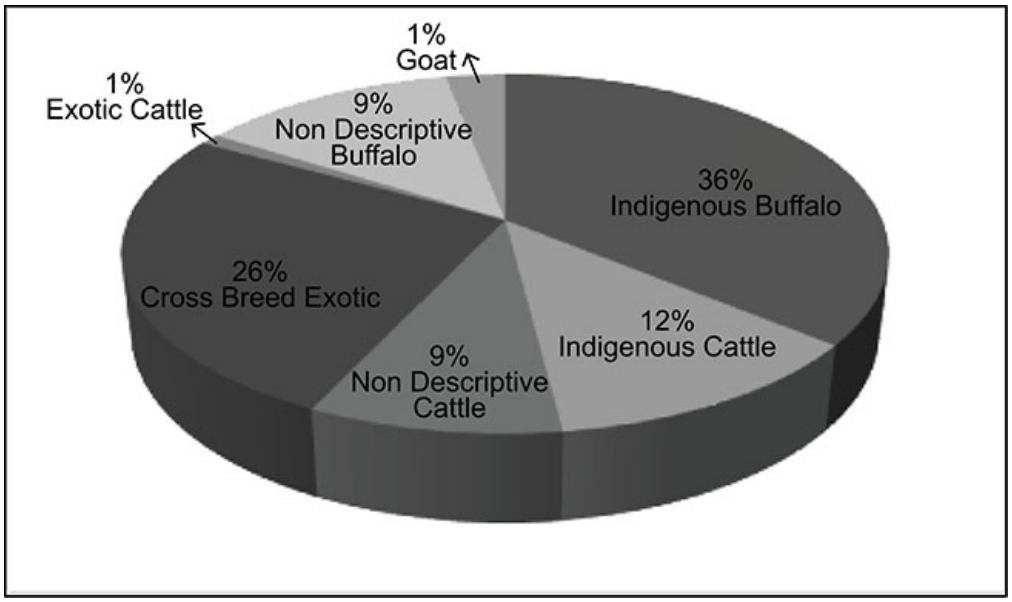


Fig. 7.10: Species wise Milk Contribution (%) 2015–16

Cattle population in India can be classified into: (i) milch breed, (ii) draught breed, and (iii) mixed or general breed.

Milch Breeds: The cows which give relatively higher quantity of milk are known as milch breeds. The famous milch breeds in India are Deoni, Gir, Sahiwal, Sindhi, and Tharparkar. The Deoni breed is a native of the north-western parts of Andhra Pradesh which gives about 2000 kg of milk per lactation. The Gir breed is a native of Saurashtra which yields over 3000 kg of milk per lactation. The Sahiwal breed (formerly known as Montgomery in Pakistan) yields about 3000 to 4500 kg of milk per lactation period. The Sindhi breed is red in colour and produces about 5000 kg of milk per lactation.

Draught Breeds: The bullocks of the draught breeds are excellent draught animals. The main draught breeds in India are: (i) Nagori, and Bachaur, (ii) The Kathiawari, Malvi and Kherigarhi, (iii) the Mysore type characterised by prominent forehead with long pointed horns which are close together, e.g., Hallikar, Amritmahal, Kangyam, and Killari, and (iv) the small black and red coloured breeds of the Himalayan region known as Ponwar and Siri.

Dual Purpose Breed: The cows of these breeds give good quantity of milk and bullocks are good quality draught animals. Some such important breeds are Hariana (popular in Haryana, Uttar Pradesh, and Punjab), Ongole (belongs to Gantur and Nellore districts of Andhra Pradesh), Gaolo (Nagpur and Wardha), Rath (Haryana and Mewat), Dangi (Nasik), Krishna Valley and Nimari.

In order to improve the breed of the Indian cattle, seven central breeding farms have been established. Some of the exotic breeds yielding higher quantity of milk like Jersey, Holstein-Erie, Sain, Swiss brown, German Flekvich and Ayreshire have been introduced in the country which are becoming popular amongst the dairy farmers. The seven Central Breeding Centres are at Suratgarh (Rajasthan), Dhamrod (Gujarat), Alamadhi (Tamil Nadu), Chiplima, Similigude (Odisha), Andeshnagar (Uttar Pradesh), and Hessarghatta (Karnataka).

Buffaloes

Buffaloes are the major suppliers of milk to the Indian population. They constitute about 17% of the total livestock of the country, but contribute about 55% of the total milk production. In fact, the dairy industry of India is largely dependent on buffaloes.

As stated earlier, India has more than 56% of the total buffaloes of the world. The better breeds of Indian buffalo are *Murrah, Bhadwari, Jaffarabadi, Saurti, Mehsana, Nagpuri, Rohtak, and Nili Ravi.* Murrah is an indigenous breed of Rohtak, Hissar and Gurgaon (Haryana). These buffaloes have short horn and massive body. The average lactation yield about 2000 kg. He buffaloes are suitable for draught and hard work. The Bhadawari with light colour is an indigenous breed of Etawah and Agra (UP). It yields about 1600 kg milk per lactation. (read-White Revolution, Chapter 9).

The Indian cattle and buffaloes are weak in health and suffer from many diseases. In order to control livestock diseases “Livestock Health: Disease Control” has been

implemented. The scheme was launched to eradicate the diseases of Rinderpest and Bovine Pleuro-pneumonia.

In the third week of May 2017, the Ministry of Environment, Government of India, introduced rules to regulate cattle trade. This law has been made to reduce cruelty towards livestock. The restrictions on cattle trade will hurt farmers and threaten jobs. Thus this law will have an adverse effect on farmers and industries which use livestock products as raw materials. India's thriving buffalo meat exports, which recorded Rs. 26684 crore, will be crippled. Other industries such as meat, leather-goods, soap, automobile-grease will suffer. Moreover, vigilante groups which have functioned unchecked may choke even the limited livestock trade permitted by the new law. Irrationality in laws promotes overall levels of irrationality in society. The Indian economy today produces few jobs relative to the number of young people entering into the labour market. The people of Meghalaya, Nagaland, Kerala, Tamil Nadu, and Puducherry are describing the move as 'undemocratic, unconstitutional and unethical.' Consequently, protest have been held in Chennai, Kolkata, and many of the states declared that they will not implement the ban and passed resolutions in the assemblies to register their protest against the notification. Moreover, this move has caused consternation in the Christian majority states of Meghalaya and Nagaland. It is therefore, imperative that the government need to review its decision about the cattle trade (Times of India, New Delhi, May 29, 2017).

Goat Rearing

Goat provides milk, meat and hide. It is the main source of meat as about 35% of the meat consumed in India is that of goat. In 1951, India had about 47 million goats which increased to 135.1 million in 2012. Bihar including Jharkhand has the largest number of goats followed by Rajasthan, West Bengal, Uttar Pradesh, Maharashtra and Madhya Pradesh.

Over 90% of the goats are *desi*. The important breeds of goats are Angora or Himachali (known as Gaddi or Chamba breed of Himachal Pradesh and Jammu and Kashmir). This breed provides soft Pashmina, (wool) meat and hide. The Jamunapari is found between Yamuna and Chambal Valley. In Rajasthan, Gujarat and Madhya Pradesh, the main breeds of goats are Marwari, Mehsanwi, Kathiawari, and Zalwadi. Recently, a number of foreign breeds like Alpine, Nubian, Saanen and Angora (Turkey) have been used for cross-breeding with the indigenous breeds so as to improve the quantity of milk and meat production.

Sheep Rearing

India has about 4% of the total population of sheep in the world. They are an important source of mutton, wool and hide in the country. There were about 39 million sheep in the country in 1951 which increased to 65.1 million in 2015–16. The total production of raw wool in 2015–16 was 52 thousand tonnes. Wool Production has shown a negative growth (−9.47%) during 2015–16.

Sheep rearing in India is done mainly in Rajasthan (25%), followed by Andhra Pradesh (16%), Tamil Nadu (12%), Karnataka (11%), and Maharashtra (6%) of the total sheep of the country. The important breeds of sheep are *Lohi, Kutchi, Bikaneri, Marwari, Kathiawari, Jaisalmeri, Sonadi, Malpuri, Magra, Shekhawati, Pugal, Deccani, Nellori, Bellary, Gureji, Karna, Bakkarwal, Gaddi.* The Indian wool is, however, inferior to that of Australia, Turkey, and South Africa in quality and is called the coarse carpet wool. India exports wool to USA and UK.

The sheep breeds in India are generally poor. Efforts are being made to improve sheep breeds by crossing local breeds with the imported quality breeds like Australian Merino, Russian Merino, Spanish Merino, Cheviot, Leicester, and Lincoln (UK).

Poultry Farming

Poultry includes domestic fowls like chickens, ducks, geese, Japanese quil/emu, and turkey. These are kept to obtain meat, eggs, and feathers. Poultry farming requires small capital investment and provides good additional income and job opportunity to the rural population. The poultry farming practices in India have evolved from unscientific practices to commercial production system. The production of the eggs during 2006–07 was 50.70 billion and it increased to 66.45 billion during 2011–12 (Eleventh Year plan).The current total population of poultry in India is 729.21 million and the egg production during 2015–16 was 82.93 billion. The poultry meat production is approximated to 3.26 million tonnes. It went upto 27.95 billion tonnes during 2017–18. The leading states were Tamil Nadu, Andhra Pradesh, Telangana, West Bengal and Haryana.

Government Policies:

The Government of India is making efforts to further improve the production of quality milk, procurement, its processing and marketing of the milk and its products by implementing various Dairy Development Schemes:

1. National Programme for Dairy Development (NPDD)
2. National Dairy Plan (Phase-I)
3. Support to Dairy Cooperatives
4. Dairy Processing and Infrastructure Development (DIDF)

Some of the important schemes for enhancing the cattle population and production are:

1. **Rashtriya Gokul Mission (RGM):** It was launched in 2014, with an outlay of Rs. 2025 crores; 80% of the cattle in India are indigenous and non-descript breeds whose milk is high in fat content. However, the number of indigenous cattle is declining and some breeds like ‘Punganur’ have become threatened. Hence this RGM scheme shall promote the development and conservation of indigenous breeds through selective breeding.

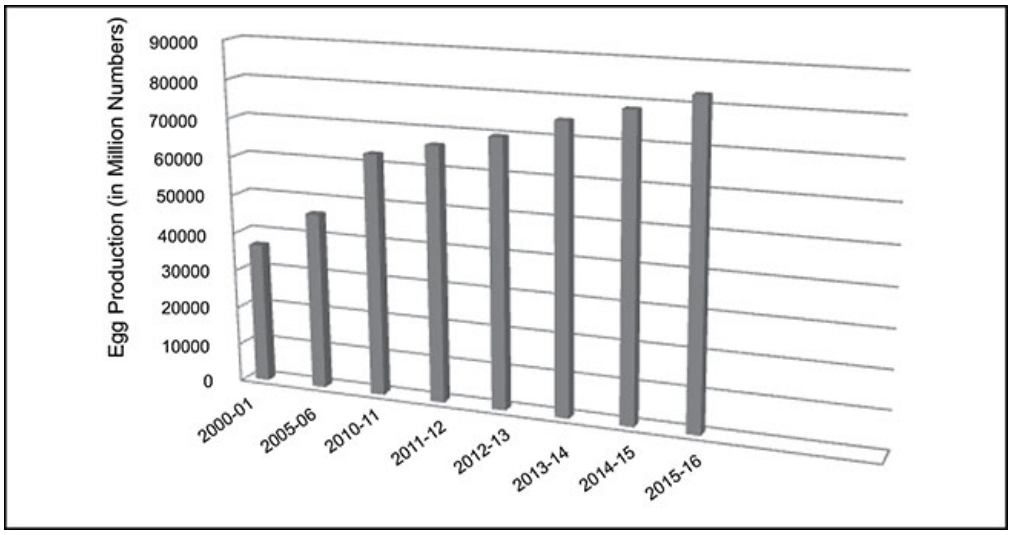


Fig. 7.11: Egg Production from 2000–01 to 2015–16 [*Source: http://dahd.nic.in*]

2. **National Kamdhenu Breeding Centre (NKBC):** It is a centrally operated scheme approved in 2014–15. It has an allocation of Rs. 50 crores. Its objective is to conserve and preserve the bovine breeds, enhance production, to upgrade the genetic merit, to supply the elite germplasm and to protect the threatened breeds from extinction.
3. **RGM Award:** To enhance the healthy competition among the states of India for the veterinary practices. All the states/UT that participates in the implementation of schemes under the Rashtriya Gokul Mission are eligible for this award.
4. **Epashuhaat:** It is a digital platform that offers a market for the bovine sellers and buyers. This minimizes the involvement of the middleman and increases the farmer's income. It also makes availability of the disease free germplasm (genetic merit) to the farmers.
5. **Embryo Transfer Technology:** This technology has revolutionized the breeding strategies. It is a vital tool to improve the genetic constitution of the cattle through techniques like MOET (Multiple Ovulation and Embryo Transfer).
6. **National Awards:**

6.1: Farmers Gopal Ratna Award is given to the farmers maintaining the best herd of the indigenous breed and practicing the best management practices.

6.2: National Kamdhenu Award is given for the best managed indigenous herds by institutions or trusts.

The poultry sector, with a total value of output exceeding Rs.15,000 crore and providing direct and indirect employment to over three million people, produced around 2.5 million tonnes of chicken meat in 2015–16.

Poultry farms have been developed around almost all the important urban centres like Mumbai, Kolkata, Delhi, Chandigarh, Chennai, Hyderabad, Pune, Bangalore, Nagpur, Bhubaneswar, Shimla, and Ajmer.

The Indian fowls belong to two categories: (i) desi, and (ii) exotic or imported. The Desi breed include Chittagong, Punjabi, Brown, Chagas, Lolab, Naked-neck, Titre, Bursa, Tillicherry, etc. The imported breeds include White Leghorn, Rhode Island Red, Black Minorca, Plymouth Rock, New Hampshire, Light Sussex, Brown Leghorn and Australorp, etc.

The Central Poultry Farms are located at Mumbai, Hassarghatta (near Bangalore), Chandigarh and Bhubaneswar. These farms are established to improve poultry breed to produce more eggs.

Export of products such as live poultry, eggs, hatching eggs, frozen eggs, egg-powder, and poultry meat are made to Bangladesh, Sri Lanka, South West Asian countries, Japan, Denmark, Poland, the USA, and Angola.

The bird influenza has created numerous problems for poultry development in India. The first outbreak was in 2006 in a small area in Maharashtra. The second outbreak was also reported from Maharashtra a few months later. In order to overcome this problem, an active surveillance programme is being carried out all over the country focusing on an early detection of avian influenza. The Government of India maintains a strategic reserve of poultry vaccine. India has a fully equipped Bio-Security Level 3 laboratory at Bhopal.

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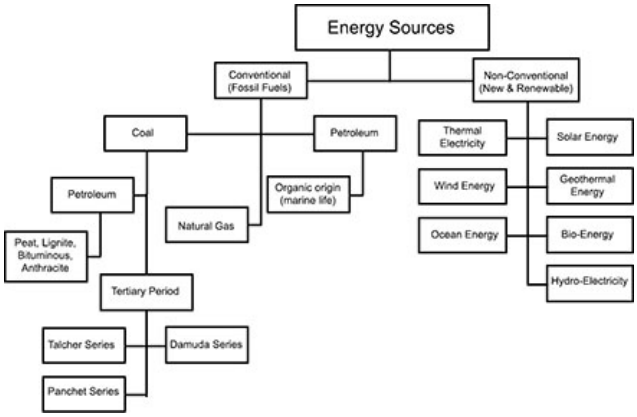
11. megahvt.gov.in/policy.html: Policy and Programmes, Animal husbandry and Veterinary Programmes

8

Energy Resources



Energy is an essential input for economic development and improving the quality of life. Energy may be classified into two categories, namely: (i) conventional (coal, petroleum, natural gas, and electricity), and (ii) non-conventional energy (solar, wind, tidal, geothermal, and biogas energy). Energy can also be classified into non-commercial (fuel-wood, charcoal, dried cow-dung, animal waste and animal power), and commercial energy (coal, mineral oil, natural gas, hydro-power, nuclear power wind energy, solar energy). It is the commercial energy which plays a vital role in the economic development of a country. A brief description of the sources of commercial energy, their distribution and production has been given in the following section.

SOURCES OF CONVENTIONAL ENERGY

Coal

Coal is the main source of energy in the country. It fulfills about 67% of the total commercial energy. It is a fossil fuel also known as '***Black Gold***'. The demand for coal was estimated to be 908.40 million tones while the actual supply was 621.26 million tonnes .

In the year 2017–18, Coal India Limited produced 567.366 MT (84.0%) and SCCL 62.010 MT (9.2%) of coal. In that year main producer of lignite was Neyveli Lignite Corporation and produced 25.153 MT (54.5%).

Odisha registered highest coal production of 143.328 MT (21.2%) followed by Chhattisgarh 142.546 MT (21.1%), Jharkhand 123.296 MT (18.3%) and Madhya Pradesh 112.127 MT (16.6%). In 2017–18, Tamil Nadu was the largest producer of lignite and produced 23.569 MT (51.0%) followed by Gujarat 13.392 MT (29.0%) and Rajasthan 9.294 MT (20.1%) in the year 2017–18. (Source: *Coal Controller; Provisional Coal Statistics; Govt. of India, Ministry of Coal)*

Classification

The coal of India may be classified under two categories: (i) Gondwana coal, and (ii) Tertiary coal.

Gondwana Coal: The Gondwana coal belongs to the Carboniferous Period (570 million years to 245 million years back). It is found in the Damodar , Mahanadi, Godavari, and Narmada valleys. Raniganj, Jharia, Bokaro, Ramgarh, Giridih, Chandrapur, Karanpura, Tatapani, Talcher, Himgiri, Korba, Penchghati, Sarguja, Kampte, Wardha Valley, Singareni (A.P.) and Singrauli (M.P., U.P.), are some of the important coal mines of the Gondwana formations. The Jhingurda coal mine (Madhya Pradesh) is the thickest coal seam 132 m of the Gondwana Period, followed by the Kargali seam near Bokaro coalfield which is about 37 m in thickness. Over 98% of the total coal reserves of India belong to the Gondwana Period (317433 million tonnes). The Gondwana coal is mainly bituminous or anthracite in which the carbon content varies between 60 to 90%. The bituminous coal is converted into coke before being used in the iron and steel industry.

The Tertiary: Tertiary coal is found in the rocks of the Oligocene period of the Tertiary Era. It is about 15 to 60 million years old. The Tertiary coal is also known as the *brown coal*. The Tertiary coal contributes only about

2% of the total coal production of the country. It is an inferior type of coal in which the carbon varies between 30% in Gujarat and Rajasthan to 50% in Assam. Lignite coal is found in Arunachal Pradesh, Assam, Gujarat (Kutch) Kerala, Jammu and Kashmir, Nagaland, Tamil Nadu, Uttar Pradesh, and West Bengal (Darjeeling District). The largest lignite deposits of the country are at Neyveli in the state of Tamil Nadu. The different types of coal and their characteristics have been given in the following:

(i) **Peat**: It contains the highest percentage of moisture, gives more smoke, has less than 40% carbon and, therefore, is the lowest and most inferior quality of coal. It represents the first stage of coal formation.

(ii) **Lignite (Brown-Coal)**: Lignite is superior to peat. Under the increasing pressure and heat, with the passage of time, peat is converted into lignite. It contains 40 to 60% carbon. It is mainly found in Neyveli (Tamil Nadu), Palna (Rajasthan), Lakhimpur (Assam), Jaintia Hills (Meghalaya), Nagaland, Kerala, Jammu and Kashmir, Uttar Pradesh, and the union territory of Pondicherry. Lignite deposits in India have been estimated around 38930 million tonnes, out of which 4250 million tonnes are in Neyveli area of Tamil Nadu (2016). Lignite is also found in Gujarat, and Kerala.

(iii) **Bituminous (Black-Coal):** When coal is buried very deep, the moisture gets expelled. The seam subjected to increased temperatures results into the formation of bituminous coal. It is dense, compact and black in colour. The traces of original vegetation from which it has been formed are found in this coal. Containing 60 to 80% carbon, it is the most popular coal in commercial use. The name is derived after a liquid called bitumen released after heating. Bituminous coal is also used in making coke (coking coal), gas coal, and steam coal. Coking coal results from the heating of coal in the absence of oxygen, which burns off volatile gases and is mainly used in iron and steel industry. Most of the bituminous coal is found in Jharkhand, Odisha, Chhattisgarh, West Bengal, Maharashtra, and Madhya Pradesh.

(iv) **Anthracite (Hard Coal):** It is the highest quality of coal containing 80 to 90% carbon. It has very little volatile matter and insignificant proportion of moisture. It has short blue flame. Of all the coals, it is the most expensive.

Distribution

The distribution of coalfields has been shown in **(Fig. 8.1)**, while the state-wise coal reserves and their percentage share have been given in **(Table 8.1)**.

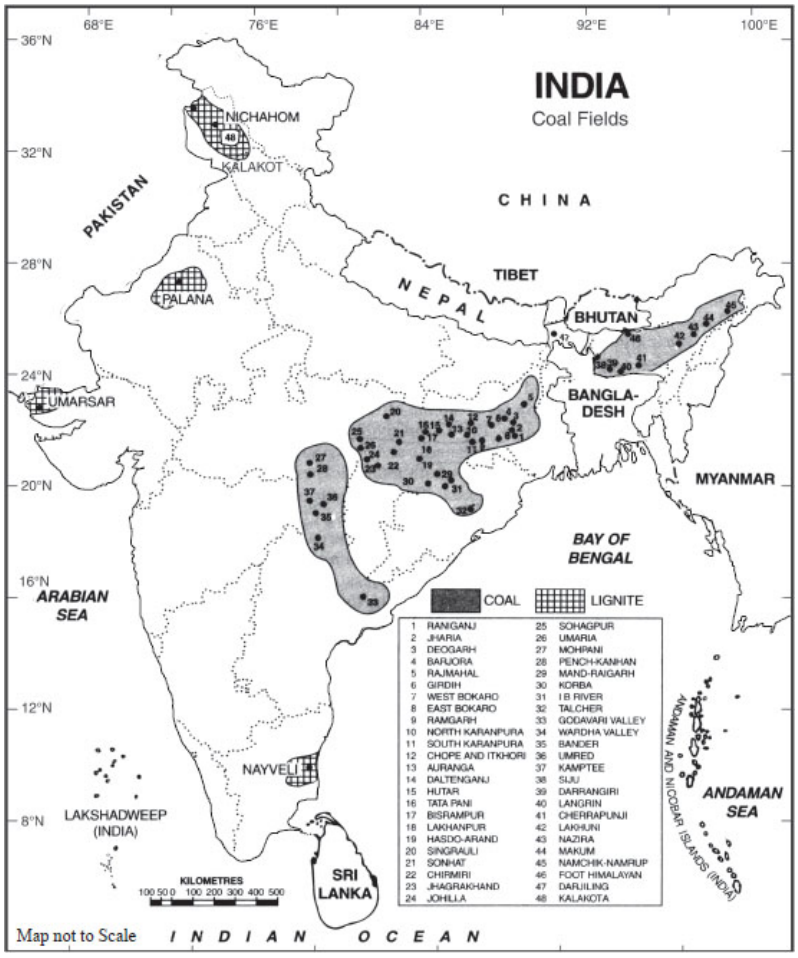


Fig. 8.1 Coal Fields

Table 8.1

Coal Reserves in India 2017–18

State/Coalfield	Total Reserves (in Million Tonnes)	Percentage of All India Reserves
Gondwana Coalfields	**313,561**	**99.49**
Jharkhand	82,440	26.16
Odisha	77,285	24.52
Chattisgarh	56,661	17.98
West Bengal	31,667	10.05
Madhya Pradesh	27,673	8.78
Andhra Pradesh	1581	0.501

State/Coalfield	Total Reserves (in Million Tonnes)	Percentage of All India Reserves
Telangana	21,464	6.81
Uttar Pradesh	1062	0.33
Sikkim	101	0.03
Tertiary Coalfields	**1588**	
Assam	511	32.17
Arunachal Pradesh	90	5.67
Meghalaya	577	36.33
Nagaland	410	25.81

Source: Provisional Coal Statistics 2016–17, Coal Controller's Organization, Kolkata.

The major states having large proportion of the coal reserves of the country are Jharkhand, Odisha, Chhattisgarh, West-Bengal and Madhya Pradesh **(Fig. 8.2)**. A brief account of coal reserves in these states has been given in the following:

(i) **Jharkhand:** The state of Jharkhand, accounting for about 29%, has the first rank in coal reserves 80.716 million tonnes and its production. Most of the coal belongs to the Gondwana period. The districts of Dhanbad, Dumka, Hazaribagh, and Palamu are very rich in coal deposits. The main coal mining centres are Auranga, Bokaro, Daltenganj, Dhanbad, Giridih, Hutar, Jharia, Karanpur, and Ramgarh **(Fig. 8.3)**.

(a) **The Jharia Coalfield:** Out of all the coal mines of Jharkhand, Jharia is the largest and most important coal producing mine, which sprawls over an area of about 460 sq km. It contains the best metallurgical coal (bituminous). Nearly 90% of the coking coal is produced from the Jharia mine. Its coal is mainly supplied to the iron and steel plants of Asansol, Bokaro, Durgapur, and Jamshedpur.

(b) **The Bokaro Coalfield:** This coalfield stretches in the valley of Bokaro river in Hazaribagh district, about 32 km to the west of Jharia. The Kargali seam (37 m) of the Bokaro coalfield is one of the thickest of the Gondwana period in (India). Its coal is mainly supplied to the iron and steel plant of Bokaro.

(c) **The Giridih or Karharbari Coalfield**: The Giridih coalfield stretches in the district of Hazaribagh. Its seams are very close to the surface. It provides one of the finest quality of bituminous coal used for the metallurgical industry. Its coal is supplied to the Bokaro and Jamshedpur steel plants.

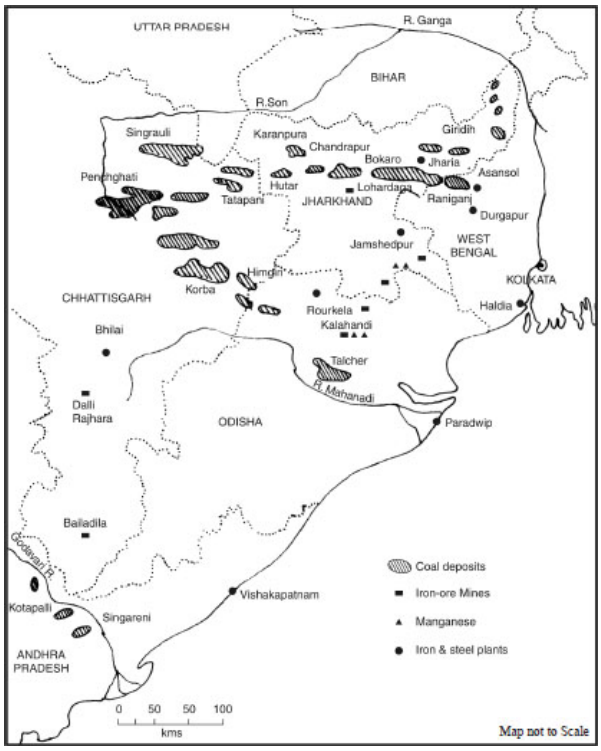


Fig. 8.2 Gondwana Coal Deposits

(d) **The Karanpur Coalfield**: The Karanpur coalfield is divisible into the North and the South Karanpur coalfields. It lies only about 30 km to the west of Bokaro. The thickness of its seam is about 25 m. Much of the coal is, however, non-coking.

(e) **The Ramgarh Coalfield**: Stretching over an area of about 100 sq km, the Ramgarh coalfield is only about 9 km to the west of the Bokaro coalfield. The coal of Ramgarh is of relatively inferior quality containing a high proportion of ash (about 30%) and carbon 35%.

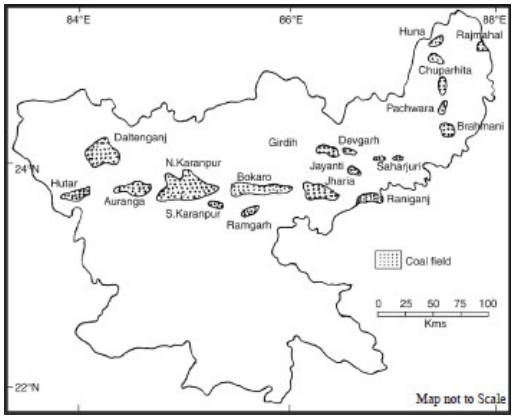


Fig. 8.3 Major Coal-Fields in Jharkhand

(f) **The Hutar Coalfield**: Stretching over about 200 sq km the Hutar coalfield lies in the Palamau district. Its seams are, however, thin and the coal is of inferior quality containing about 50%,

carbon, 30% volatile matter and 20% ash.

(g) **The Daltenganj Coalfield**: Sprawling over 55 sq km, this coalfield lies in the Palamau district. Its coal is either semi-anthracite or non-coking, of inferior quality which can not be used in metellaurgical industries.

(h) **Deogarh Coalfields**: This coalfield lies in the Dumka district and stretches over an area of about 20 sq km. The coal is of inferior quality containing about 40% carbon, 25% volatile matter and 35% ash content. Its coal is mainly used in the brick kilns.

(ii) Odisha: The state of Odisha has more than 24% of the total coal reserves and produces about 15% of the total coal production of the country. It has the coal invertory of 77.285 billion tonnes. In Odisha most of the coal deposits are found in Dhenkanal, Sambalpur, and Sundargarh districts.

(a) **The Talcher Coalfield:** Stretching over Dhenkanal and Sambalpur districts, the Talcher coalfield covers an area of about 500 sq km. It has the second largest coal reserves with 51.163 billion to in the country after Raniganj. The coal is, however, of lower grade containing only about 35% of fixed carbon, 40% volatile matter and about 25% ash content. The coal is mainly utilised in the thermal power and fertiliser plants of Talcher.

(iii) **Chhattisgarh and Madhya Pradesh:** The state of Chhattisgarh has the third largest coal reserves (about 17% of all India) 52,533 million tonnes in the country after Jharkhand and Odisha, but it holds the first rank in its production. Coal deposits are found in Bilaspur, Betul, Chhindwara, Narsinghpur, Raigarh, Rewa and Surguja districts. Singrauli, Sohagpur and Umaria (Rewa district), Bisrampur, Jhilmilli, Ramkola, Tattapani (Surguja district), Korba and Sandurgarh (Bilaspur district), Pench and Kanha valley (Chhindwara district), Mand river (Raigarh district), and Patakhera and Dulhara (Betul district) are the main mining centres of coal in Chhattisgarh.

(a) **The Singrauli Coalfield:** Stretching over the Sidhi and Shahdol districts is the largest
coal-field of Madhya Pradesh. Its Jhingurda coal seam with a thickness of 132 m is the thickest coal-seam of the country. The coal belongs to the Gondwana period, which contains 40 to 50% of fixed carbon. Its coal is mainly utilised in the thermal powerplant of Obra.

(b) **The Korba Coalfield**: The Korba coalfield lies in the Bilaspur district. Two of its coal seams are more than 30 m. The coal is of average quality and is used mainly in the Korba thermal power plant.

(c) **The Pech-Kanha-Tawa Coalfield**: It lies in the Chhindwara district. In quality its coal is of coking and semi-coking category.

(d) **Umaria Coalfield:** This coalfield lies about 60 km south of Katni. The coal contains a higher percentage of ash and moisture. It is used mainly in the generation of thermal power. In addition to these, coal is also mined at Jhilmili, Sonahat, and Tattapani coalfields of Surguja district.

(iv) **West Bengal:** West Bengal has about 11% of the total coal reserves of India **(Table 8.1).** The coal deposits of West Bengal lie in Bankura, Bardhman, Birbhum, Darjeeling, Jalpaiguri, and Puruliya districts. The most important of coal reserves and mining coalfield of West Bengal is Raniganj.

(a) **Raniganj Coalfield**: Stretching over 185 sq km in the Bardhman and Birbhum district to the north-west of Kolkata, it is the most important coalfield of West Bengal. It is known for the good quality of coking coal. It contains 50 to 65% of carbon. It is used in the metallurgical industry, especially in the Durgapur iron and steel plant.

(b) **The Darjeeling Coalfield:** The coal of Darjeeling district belongs to the Tertiary Period. It is exposed in the Mana and Mahanadi valleys. The coal is in powder form with coking quality.

(v) **Madhya Pradesh:** About 8% of the coal reserves of India are found in Madhya Pradesh. The main coal deposits lie at Singrauli, Muhpani, Satpura, Sohagpur, and Pench-Kanhan.

(vi) Andhra Pradesh: About 7% of the coal reserves of India are found in Andhra Pradesh. Its main coal deposits are found in the Godavari valley. The districts of Adilabad, Khammam, Nellorem, and Warangal are known for its production. Coal is mainly used in thermal power plants of Kottagudem, Nellore, Ramagundam, Errazada, Husain-Sagar, and the fertiliser plant at Ramagundam.

The Singareni coalfield lying about 185 km to the east of Hyderabad is the main mining area of coal in Andhra Pradesh. Another important coal producing centre is at Kottagudam. Its coal seam is of about 18 m and the coal is of good quality.

(vi) Maharashtra: The main coal deposits of Maharashtra lie in the Wardha valley, stretching over the Nagpur (Kampte-coalfield), and Yavatmal districts. The coal from these coalfields is utilised by the railways and the thermal power stations of Trombay, Chola (Kalyan), Khaperkheda, Paras, Ballarshah, Nasik, and Koradi.

COAL DEPOSITS OF THE TERTIARY PERIOD

The Tertiary coal deposits came into existence during the Eocene, the Oligocene, and Miocene periods. Coal of this period is found in Arunachal Pradesh, Assam, Meghalaya, Nagaland, Rajasthan, Gujarat, Tamil Nadu, Kerala, West Bengal, and Jammu and Kashmir states. It is also known as brown coal. Containing more moisture, it has less carbon content. The total tertiary reserves of coal are 1588 million tonnes.

Tamil Nadu: The state of Tamil Nadu has the largest deposits of lignite at Neyveli in the South Arcot district. The seams are 10 to 12 m in thickness. Its carbon and moisture contents are 30–40% and 20%, respectively, while the volatile matter varies between 40 to 45%.

Rajasthan: Lignite deposits are found in the districts of Bikaner (Palana and Khari mines). The thickness of Bikaner seams varies from 5 to 15 m. It is of inferior quality and used mostly in the thermal power plants and railways.

Gujarat: In Gujarat coal is found in Bharauch district and Kutch. The coal is of poor quality with about 35% carbon and more moisture.

Jammu and Kashmir: The Tertiary coal in Jammu and Kashmir is found at Raithan of the Shaliganga, Handwara, Baramulla, Riasi and Udhampur districts, and the karewas of Badgam and Srinagar. It is of inferior quality with less than 30% of carbon, over 15% moisture and 30% volatile matter.

West Bengal: In West Bengal lignite deposits of the Tertiary period are found in Burza Hills of Jalpaiguri and Darjeeling districts. Scattered deposits of lignite have also been discovered in Pondicherry.

India is the third largest producer of coal in the world after China and U.S.A, contributing about 5% of the total coal production. An examination of **(Table 8.2)** shows that in 1950–51 the total production of coal and lignite was 323 and 0.4 lakh tonnes which rose to 6319 and 612 lakh tonnes in 2015–16 respectively. The coal mines were nationalised in 1972, after which there had been a tremendous increase in the coal production **(Table 8.2)**.

Table 8.2

India: Production of Coal—1950–51 to 2008–09

Year	Production (in Million Tonnes)		Total (in Million Tonnes)
	Coal	Lignite	
1950–51	32.26	0.04	32.3
1970–71	72.0	3.650	75.65
1990–91	210.00	14.0	224
2000–01	328.0	23.0	351
2010–11	466	35.0	501
2012–13	556.40	46.45	602.85
2013–14	565.77	44.27	611.04
2014–15	609.18	48.3	657.48
2015–16	639.23	43.84	683.07
2016–17	657.868	45.230	703.098
2017–18	675.4	46.26	721.66

Source: Provisional Coal Statistics 2016–17, Coal Controller's Organization, Kolkata.

The Talcher Series: It is the series of the Gondwana Systems, named after the Talcher and Dhankenal districts of Odisha. It rests on the glaciated boulder bed of igneous rocks. It is known for its coal deposits. The coal from here is supplied to the Raurkela and Jamshedpur Steel Plants.

The Damuda Series*:* It is the most important series of the Gondwana System**.** The Damuda Series is well developed in Jharkhand and West Bengal. The Damuda series is known for good quality coal seams. The Raniganj, Jharia coal seams lie in the Damuda series. The Superior quality coal (anthracite and bituminous) is obtained from this series. The bituminous which has

carbon over 60% is used for metallurgy, especiallay the iron and steel plants of Jamshedpur, Raurkela, and Bokaro.

The Panchet Series: It is the youngest series of the Lower Gondwana System. It lies to the south of Raniganj. The series consists of greenish sandstone, shales and iron rich rocks, but is devoid of coal-seams. An outlier of this series is known as '*Mangli beds*' in the Wardha valley of Maharashtra.

Problems of Coal Industry

The main problems of the coal mining industry are as under:

1. **Unequal Distribution of Coal:** The distribution of coal in India is confined mainly to Jharkhand, Chattisgarh, Odisha, Madhya Pradesh, Tamil Nadu, and West Bengal. The transportation cost is consequently heavy which makes this vital source of energy expensive.
2. **Poor Quality of Coal:** Most of the mines of India are producing non-coking coal which cannot be utilised for metallurgical industries.
3. **Less Efficient Transport System**: Most of the coal in India is transported by trains. Adequate number of wagons are not available and railway system is not efficient enough to deliver the coal at distant places in a short time.
4. **Obsolete Method of Mining:** The mining technology is outdated and the per worker production is low. The per tonne cost of production is high.
5. **Shortage of Power Supply**: The shortage of power supply is a big barrier in the mining of coal.
6. **Fires and Water-logging:** There occur heavy losses of coal because of fires and waterlogging in the mines and at the pit-heads.

Conservation of Coal

Coal is an exhaustible resource. It needs to be utilised judiciously. The following steps can go a long way in the conservation of coal in the country.

1. The coking and good quality coal should be reserved only for metallurgical industry.
2. Low grade coal should be washed and impurities removed by modern techniques.
3. Selective mining should be stopped by an act of law. All possible grades of coal should be obtained from all the mines.
4. Environmental safety laws should be effectively implemented.
5. The thermal power plants should be located at the pit-heads to enhance power generation.
6. The pilferages and theft of electricity should be minimised.
7. New reserves should be discovered.
8. The non-conventional sources of energy should be popularised.

Governmental Initiatives

The Ministries of Power, Coal, New & Renewable energy and mines have taken the initiatives to make the mission of 24x7 Affordable Clean 'Power for All' as well as 'Ujwal Bharat' successful .

The principles involved are :

(i) Sulabh (Accessible Power)
(ii) Sasti (Cheap Power)
(iii) Swachi (Clean Power)
(iv) Sunyojit (well planned Infrastructure)
(v) Sunishchit (Assured Power for All)

(vi) Surakshit (Empowering each citizen of India with Transparent Governance and securing their Future)

In order to ensure adequate coal for electricity, shortage to surplus, the Government aims to produce 100 crore tones of domestic coal by 2019–20.The coal imports have been reduced . The principle of “less coal for more power” in 2016–17 resulted in use of 0.63 kg of coal to produce 1 kWh of electricity.

The government also has a plan to bring transparency to the mining sector and optimize the utilization of natural resources. The ‘*National Aero-Geophysical Mapping Project’* will acquire data on 27 lakh line kms of aerogeophysical data. The Space technology ***(****Mining Surveillance System)* keeps a check on illegal mining.

The ***Pradhan Mantri Khanij Kshetra Kalyan Yojana (PMKKKY)*** shall help the people affected by mining. The ***District Mineral Foundation (DMF)*** under PMKKKY has collected about Rs. 7150 crores from mining during 2016–17 to be used for the welfare of the workers involved in mining.

Petroleum

Petroleum is an important source of energy which is much in demand to accelerate the economic development. Apart from an important fuel resource, it provides lubricants and raw materials for a number of chemical industries. Its products include kerosene, diesel, petrol, aviation-fuel, synthetic rubber, synthetic-fibre, thermoplastic resins, benzene-methansol, polystertene, acrylates, detergents, aromatics, gasoline, carbon-black, dyes, colours, food-colours, pigments, explosives, printing ink, film-photography, greases, cosmetics, paints, lubricant oils, paraffin, and wax.

Crude oil is obtained mainly from the sedimentary rocks of marine origin. All Sedimentary rocks, however, donot contain crude-oil. In India, crude oil is found in the sedimentary rocks of the Tertiary period **(Fig. 8.4).** Normally it does not occur at its place of formation. Being lighter than water, crude oil overlain with gas, gets accumulated in the anticlines above the water surface. The geologists propounded two theories about the origin of crude oil.

Origin: The origin of petroleum and natural gas is considered to be organic. According to organic origin, the living organisms (marine life, like fish) and vegetal matter got buried under the accumulated sediments of mud, silt and sand, etc. Due to pressure and heat, this undergoes chemical changes so as to form crude oil and natural gas after millions of years.

Oil in India was discovered near Margherita (Upper Assam); for the first time in 1860 by the Assam Railway and Trading Company. Subsequently, oil was discovered at Digboi in 1889. In the beginning of the 20^{th} century (1917), oil was discovered at Badarpur (Assam). In 1954, production of oil was started in Naharkatiya region. The Oil and Natural Gas Commission (ONGC) was established in 1956. With the efforts of the Oil and Natural Gas Commission (ONGC), oil was discovered in the Gulf of Cambay (Khambat) in 1961 and in Bombay High in 1976.

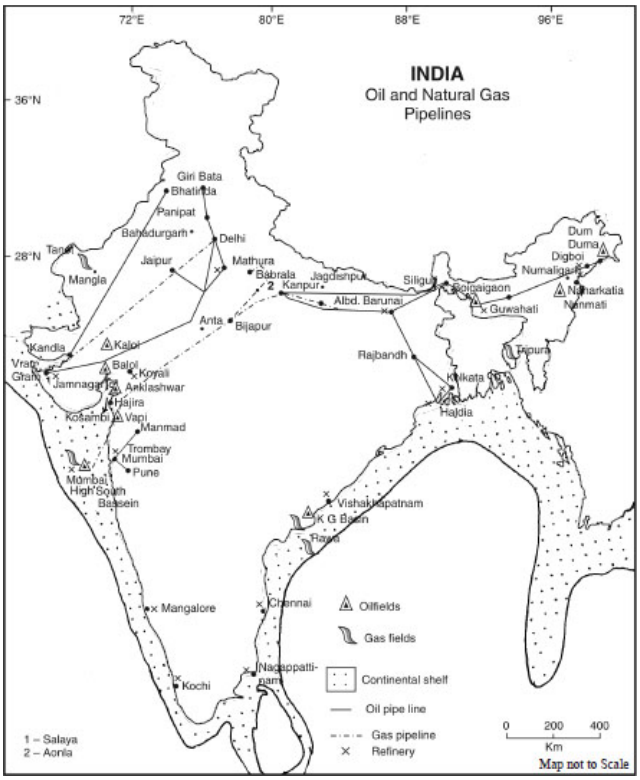


Fig. 8.4 Oil and Natural Gas Deposits

In India, the petroleum and natural gas has been discovered in the following ten basins:

1. The Upper Assam Basin (60,000 sq km)
2. The Western Bengal Basin (60,000 sq km)
3. The Western Himalayan Basin (100,000 sq km)
4. The Rajasthan Saurashtra-Kutch Basin (95,000 sq km)
5. The Northern Gujarat Basin (140,000 sq km)
6. The Ganga Valley Basin (385,000 sq km)
7. The Coastal Tamil Nadu, Andhra & Kerala Basin (75,000 sq km)
8. The Andaman and Nicobar Coastal Basin (2000 sq km)
9. Offshore of the Khambat, Bombay High & Bassein (2000 sq km)

The production trend of oil has been given in **(Table 8.3).**

Table 8.3

India—Production of Crude Oil—1951–2018

Year	Production in Million Metric Tonnes (MMT)
1950–51	0.27
1960–61	0.50
1970–71	7.20
1980–81	15.5
1990–91	33.0
2000–01	32.5
2009–10	34.0

2014–15	37.46
2015–16	36.942
2016–17	36.00
2017–18	35.68

Source: Economic Survey of India, 2015–16, and India 2016, p. 312.

It may be seen from **(Table 8.3)** that the total production of crude oil in 1950–51 was only 0.27 million metric tonnes which rose to 36.942 million metric tonnes in 2015–16. India's oil production has dropped 4.5% to 34.2 million tonnes in 2019. It was 35.68 million tonnes in 2017–18. A marginal increase has been observed in the production of natural gas. The reason for the decline is the decline in the pressure or potential in the GS-4 gas cap reservoir in Gandhar field and closure of two wells in D1D3 fields by Reliance Industries. (Ref: Business Standards, April 25, 2019)

As on March 31, 2018, the reserves of crude oil in India were 594.49 million tones and natural gas reserves of 1339.57 billion cubic meters. 82% of oil is imported to meet the demand. The government aims to reduce it to 67% by 2022, by local exploration, replacement by renewable energy and indigenous ethanol fuel. The produced oil is, however, only about 40% of the domestic need. The production of crude oil from the different regions of the country from 2011 till Dec 2017 has been shown in **(Table 8.4).**

Table 8.4

Statewise Crude Oil Production (in thousand metric tones)

State/Source	2011–12	2012–13	2013–14	2014–15	2015–16	2016–17 (Upto Dec)
Andhra Pradesh	305	295	297	254	295	203
Arunachal Pradesh	118	120	111	68.88	57	41
Assam	5,025	4,863	4,709	4,473	4,185	3,141
Gujarat	5,778	5,332	5,061	4,653	4,461	3,455
Rajasthan	6,553	8,593	9,180	8,848	8,602	6,192
Tamil Nadu	246	238	226	241	261	202
Total Onshore	18,025	19,441	19,584	18,538	17,861	13,234
Total Offshore	20,061	18,421	18,204	18,923	19,089	13,810
Gross Total	**38,086**	**37,862**	**37,788**	**37,461**	**36,950**	**27,044**

Source: Annual Report 2016–17

Before independence, India was not listed in major oil producing countries till Mumbai High started the production . The onshore production was contributing to major oil production till 1980–81.

From 1990–91 to 2009–10 about two thirds of the production came from off shore sources. The offshore crude oil production make the major contributions. The offshore field are located in the Brahmaputra valley,Barmer region of Rajasthan, Gujarat coast in western India and Cauvery on shore basin in Tamil Nadu. Andhra Pradesh has both types of oil reserves. Assam is the oldest oil producing state in India. The strata that bears oil is a 320 km stretch in the upper Assam region.

Table 8.5

Estimated Crude oil and Natural Gas reserves in India, 2017

State	Crude Oil Reserves (In MIllIon Tones)	Contribution to oil (Percent)	Natural Gas Reserves	Contribution to Oil (Percent)
Western Offshore	239.20	39.60	302.35	23.44
Assam	159.96	26.48	158.57	12.29
Gujarat	118.61	19.63	62.28	4.83
Eastern Offshore	40.67	6.73	507.76	39.37
Rajasthan	24.55	4.06	34.86	2.70
Tamil Nadu	9.00	1.49	31.98	2.48
Andhra Pradesh	8.15	1.35	48.31	3.75
Arunachal Pradesh	1.52	0.25	0.93	0.07
Tripura	0.07	0.01	36.10	2.80

Source: Energy Statistics; Ministry of Statistics and Programme Implementation, 2017

The crude oil production in India has been shown in **(Fig. 8.5)**. It may be seen from **(Fig. 8.5)** that the crude oil production over 34.0 million tonnes in 2008–09, while in 2017–18, crude oil production is 35.68 MS.

Crude-Oil Producing Regions

The major oilfields of India are as follows:

1. The Western Coast Offshore Oilfields

(i) **The Bombay High Oilfields:** This is the largest petroleum production oilfield contributing over 65% of the total production of crude oil. This oil field lies about 176 km to the south-west of Bombay. It has about 35 million tonnes of crude oil and about 40,000 million cubic m of natural gas. Production of oil from this field started in 1976. Owing to over exploitation, the production of this oil-field is declining fast.

(ii) **Bassein Oilfield**: This oilfield lies to the south of Bombay High. Here oil occurs at a depth of over 1900 m. It has rich deposits of oil and natural gas.

(iii) **Aliabet Oilfield:** The Aliabet oilfield is located about 45 km to the south of Bhavnagar.

2. The Gujarat Coast

This is the second largest oil producing area of the country. Its main oilfields are in Ankleshwar, Cambay-Luni area and Ahmadabad-Kalol region.

(i) **Ankleshwar:** Situated in the district of Bharauch, it stretches over an area of about 30 sq km.

The oil of this region belongs to the Eocene period. Oil production in this region was started in 1961. Ankleshwar oil is rich in gasoline and kerosene. The crude oil from this region is sent to the Koyali petroleum refinery.

(ii) **Cambay-Luni Region**: This oilfield lies about 60 km to the west of Vadodara. The drilling operations in this region were started in 1958. The estimated reserves of crude-oil are over 30 million tonnes. The oil of this region is very light with a sulphur content of less than 0.1%.

(iii) **The Ahmadabad-Kalol Region:** This crude-oil region lies to the north of Gulf of Khambat (Cambay) around the city of Ahmadabad and extends up to Mehsana. Kalol, situated about 25 km to the north of Ahmadabad is an important oilfield of the region. Oil

production from this region was started in 1961. Crude oil from this region is supplied mainly to the Koyali refinery.

3. The Brahmaputra Valley

As stated above, crude oil was first discovered in the Brahmaputra valley. The oil-bearing rocks spread from the Dehang Basin up to the Surma valley. The main oil producing wells, however, lie in the Dibrugarh and Sibsagar districts of Upper Assam. Some of the important oil producing centres of this region are given below:

(i) **The Digboi Oilfield:** Stretching over an area of about 15 sq km, the Digboi oilfield is one of the oldest oil-fields of the country. The oil of this region belongs to the Eocene and Miocene periods. There are 85 oil wells in this region. Most of the oil is sent to the refinery of Digboi. Since 1959, the Digboi oilfields are worked by the Oil India Limited (OIL).

(ii) **The Naharkatiya Oilfield:** This oil-field lies about 35 km to the south-west of Digboi. Oil production from the Naharkatiya oilfields was started in 1954. Crude oil from this region is supplied to the refineries of Noonamati, New Bongaigaon (Assam), and Barauni (Bihar).

4. The Eastern Coast Oil-fields

Petroleum and natural gas have been discovered in marine delta regions of Mahanadi, Godavari, Krishna, and Kaveri rivers. The Rawa field in the Godavari-Krishna offshore is expected to produce about 3 million tonnes of crude-oil annually. Petroleum has also been discovered in the Kaveri delta.

In addition to these, crude oil has been discovered in the Bilaspur Tehsil of Rampur district of Uttar Pradesh, Jawalamukhi area of Punjab, and in the Barmer district of Rajasthan. There are strong possibilities of petroleum and natural gas deposits to be found on the offshore of Andaman and Nicobar, Gulf of Mannar, Baleshwar coast, Punjab, Haryana and Uttar Pradesh.

The Ministry of Petroleum and Natural Gas is entrusted with the responsibility of exploration and production of oil and natural gas including import of Liquefied Natural Gas (LNG), refining, marketing, distribution import, export and conservation of petroleum products.

Oil and Natural Gas Corporation Limited (ONGC) and Oil India Limited (OIL), the two national companies (NOCs), and private and joint venture companies are engaged in the exploration and production of oil and natural gas in the country. Crude oil production during 2005–06 was 32190 MMT by ONGC, OIL, and Private/Joint Venture companies.

The natural gas production during the year 2005–06 was 32.20 Billion Cubic Metres (BCM) from ONGC, OIL, and Private/Joint Venture Companies.

IMPORT

India is the third largest importer of crude oil after USA and China, mainly from Iran .The import of the crude oil during 2015–16 has been reported to be 202.850 MMT that valued Rs.416,579 crores. India has imported 213.93 million tonnes in 2016–17 worth Rs. 4.7 lakh crores. In 2017–18 the imports were 219.15 million tonnes of the value Rs. 5.65 lakh crores. *(Source: Oil Ministry's Petroleum Planning and analysis cell*). India relies more than 80% on imports to meet its soil demand.

During the year 2016–17, the public sector refineries purchased crude oil on term contract and spot basis. The countries from where term contract purchases were made included Saudi Arabia, Kuwait, UAE, Iran, Oman, Sudan, etc.

Pipelines

Crude oil from oil-wells and finished products from refineries are generally transported through pipelines. Transportation of oil and petroleum through pipelines is cheap, effective, and considered to be safe. Looking at these advantages, a network of pipelines has been developed in India. Some of the important pipelines are as under:

1. **Pipelines of North-East India**
   (i) ***Noonmati-Siliguri***-Pipeline to transport petroleum products from Noonmati to Siliguri.
   (ii) ***Lakwa-Rudrasagar-Barauni Pipeline***: completed in 1968 to transport crude-oil from Lakwa and Rudrasagar to Barauni Oil Refinery (Bihar).
   (iii) ***Barauni-Haldia Pipeline***: This pipeline was laid down in 1966 to carry refined petroleum products to Haldia port and bring back imported crude-oil to Barauni refinery.
   (iv) ***Barauni-Kanpur Pipeline***: This pipeline was completed in 1966 to transport refined petroleum products to Kanpur city.
   (v) ***Noonmati-Bongaigaon Pipeline***: This pipeline was constructed to transport crude-oil to Bongaigaon petro-chemical complex.
   (vi) ***Haldia-Maurigram-Rajbandh Pipeline***: This pipeline was completed in 1998.
2. **Pipelines of Western India:** Bombay-High Mumbai-Ankleshwar-Koyali Pipeline: This pipeline connects the oilfields of Bombay High and Gujarat with the Koyali refinery of Gujarat. The city of Mumbai has been connected with a pipeline of 210 km length double pipeline to Bombay High to transport crude oil and natural gas. The Ankleshwar-Koyali pipeline was completed in 1965 to transport crude oil to Koyali refinery.
3. **The Salaya-Koyali-Mathura Pipeline:** This pipeline, 1075 km in length was laid down from Salaya (Gulf of Kutch) to Koyali and Mathura via Viramgram to supply crude oil to the Mathura refinery. From Mathura, it has been extended to the oil-refinery at Panipat (Haryana) and Jalandhar in Punjab. It has an offshore terminal and the Sayala-Koyali sector of the pipeline was completed in 1978, while the Viramgram-Mathura sector was completed in 1981.
4. **The Mathura-Delhi-Ambala-Jalandhar Pipeline:** This 513 km long pipeline was constructed to transport refinery products of Mathura to the main cities of north and north-west India.
5. **Pipelines of Gujarat:** In Gujarat, there are a number of short distance pipelines to transport crude-oil and natural gas to the refineries and the refined products to the market. These include the Kalol-Sabarmati Crude Pipeline, the Nwagam-Kalol-Koyali Pipeline, the Cambay-Dhuravan Gas Pipeline, the Ankleshwar-Uttran Gas Pipeline, the Ankleshwar-Vadodara Gas Pipeline, and the Koyali-Ahmadabad products Pipeline **(Fig. 8.4).**
6. **Mumbai Pipelines**: From Mumbai, pipelines have been laid up to Pune and Manmad to distribute petroleum products.
7. **The Haldia-Kolkata Pipeline:** Through this pipeline, the Haldia products are sent to Kolkata and neighbouring urban places.
8. **The Hajira-Bijaipur-Jagdishpur (HBJ) Gas Pipeline:** Having a length of 1750 km, this is the longest pipeline of India. It crosses 75 big and small rivers and 29 railway crossings. This pipeline was laid down by the Gas Authority of India. This gas pipeline connects Kawas (Gujarat), Anta (Rajasthan), Bijaipur (M.P.), Auraiya and Jagdishpur (U.P.). It provides gas to the fertiliser plants at Bijaipur, Sawai Madhopur, Jagdishpur,

Shahjahanpur, Aonla, and Babrala. Each one of these fertiliser plants has the capacity to produce about 1400 tonnes of ammonia per day.

9. **The Kandla-Bhatinda Pipeline:** This pipeline transports imported crude-oil from the Kandla seaport to the Bhatinda refinery.

Oil Refineries of India

The oil refineries are the processing factories of crude-oil. The impurities from the crude oil are removed to obtain petroleum, diesel, kerosene, bitumen, and aviation fuel. Petroleum industry contributes over 15% of the GDP (2015–16). The main refineries of India, their year of commission and production capacity are given in **(Fig. 8.5) (Table 8.6)**.

Fig. 8.5 Oil Refineries

Table 8.6

India: Refineries and their Production Capacity

Refinery	State	Year of Commissioning	Capacity (Lakh Tonnes Per Year)
Digboi, IOCL	Assam	1901	5.0
Trombay, HPCL	Maharashtra	1954	55.0
Trombay, BPCL	Maharashtra	1955	60.0
vishakhapattanam, HPCL	Andhra Pradesh	1957	45.0
Noonamati, IOC	Assam	1962	8.5
Barauni, IOC	Bihar	1964	33.0
Koyali, IOC	Gujarat	1965	95.0
Kochi, CRL	Kerala	1966	45.0
Chennai, MRL	Tamil Nadu	1969	56.0
Haldia, IOC	West Bengal	1975	27.5
Bongaigaon, BRPL	Assam	1979	13.5

Refinery	State	Year of Commissioning	Capacity (Lakh Tonnes Per Year)
Mathura, IOC	Uttar Pradesh	1982	75.0
Numaligarh, IOC	Assam	1999	30.0
Jamnagar, RP	Gujarat	1999	270.0
Karnal, IOC	Haryana	1998	60.0
Mangalore, HPCL	Karnataka	1998	30.0
Panagundi, IOC	Tamil Nadu	1999	5.0
Pachpadra, HPCL	Rajasthan	2013–17	9.0
Bina, BORL	Madhya Pradesh	2011	6 MMT
Total			913.5

The total refining capacity of crude oil was 2 lakh tonnes per annum in 1901 which rose to more than 1,000 lakh tonnes per annum in 2015–16. All the refineries except the Digboi and the Jamnagar are in the public sector, or the joint section. Three public sector companies of India namely Indian Oil Corporation (IOC), Hindustan Petroleum Corporation (HPCL) and Bharat Petroleum (BPCL) aim to develop the world's largest green field refinery by 2020 at the cost of Rs. 3 lakh crores. It is planned at the western coast. India has the fourth largest refining capacity in the world.

Natural Gas

The exploration of natural gas is being done by the Oil and Natural Gas Commission. According to one estimate, India has a total natural gas reserve of about 450 billion cubic m. Out of the total reserves, about 75% lies in the Bombay High and the Bassein oilfields, about 12% in Gujarat, 7% in Andhra Pradesh and 6% in Assam. Apart from Bombay High, there are rich deposits of natural gas in Ankleshwar and Gulf of Khambat (Gujarat), Godavari and Krishna basins, the Thanjavur and Shingleput districts of Tamil Nadu. Natural gas is also found in Barmar district of Rajasthan, Kangra district of Himachal Pradesh, and Firozpur district of Punjab. Natural gas production for the year 2014–15 was 36.62 Billion Cubic Metre. The trend of natural gas production in India has been shown in the following **(Fig 8.6)**.

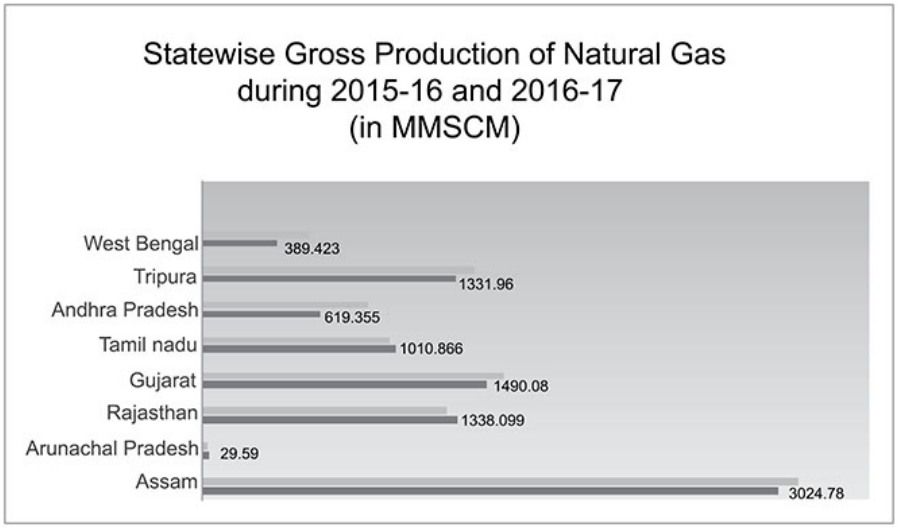


Fig. 8.6 *Source: Visualize.data.gov.in, Government of India*

The largest share of natural gas is consumed in the production of chemical fertilisers accounting for about 40%, about 30% is used in power generation, and about 10% in L.P.G. (cooking gas).

After 1990, the production of natural gas has increased phenomenally to 12,870 million cubic m.

yet the production falls short of demand. Consequently, natural gas in large quantities has to be imported. Natural gas is imported from Iran, Saudi Arabia, UAE., and the other Gulf countries.

Governmental Initiatives in Oil Industry

The government has been active in enhancing the strength of India in the mineral oil. Some of the initiatives and schemes have been given below:

1. **Bio-Diesel Programme**

   - Commissioned on January 16, 2015.
   - Allows direct sale of biodiesel (B100) by the manufacturers, suppliers or dealers of joint venture of OMCs as authorized by MoPNG to the consumers.

2. **Pradhan Mantri Ujjawala Yojna (PMUY)**

   - Launched on May 1, 2015 by the Prime Minister of India.
   - Intends to provide LPG connections to 5 crore women below the poverty line in three years from 2016–17.

3. **Initiative 'Give It Up'**

   - It shall encourage domestic LPG consumers who can afford to pay the non-subsidized price for LPG , to surrender it voluntarily.
   - Till January 1, 2017, 1.05 crore customers gave up the LPG Subsidy.

4. **Execution of Haldia & Bokaro–Dhamra Natural Gas Pipeline Project**

   - It is of 2540 km stretch with an investment of Rs. 12,940 crores.
   - 10 million households are aimed to be connected to the PNG network by the Ministry.

5. **Ethanol Blending Programme**

Oil PSUs are setting up the second generation ethanol plants at 12 locations in 11 states to enhance the farm income and also reduce environmental degradation.

Electricity

Electricity is a clean source of energy. It is generated from water, coal, mineral oil, natural gas, and atomic minerals. Electricity can also be generated through wind energy, solar energy, bio-gas, sea-waves, geothermal and dry batteries. Electricity is relatively cheap, transportable, pollution free and renewable. Because of these advantages it is increasingly becoming popular day by day. The per head consumption of electricity is often considered an important indicator of socio-economic and human development. It is about 350 kWh which is much below the per head consumption in the world 1000 kWh and USA 7000 kWh.

It may be seen from **(Table 8.7)** that the total installed capacity of electricity by different sources was only 2.3 thousand MW in 1950–51 which rose to over 264 thousand MW in 2015–16. The highest installed capacity is that of thermal, 223 thousand MW accounting for over 75% of the total installed capacity.

Table 8.7

India—Installed Capacity of Electricity (in thousand MW)

Year	Hydro	Thermal	Nuclear	RES* (MNRE)	Grand Total
1950–51	0.6	1.1	—	0.6	2.3
1960–61	1.9	2.7	—	1.0	5.6
1970–71	6.4	8.0	0.4	1.6	16.3
1980–81	11.8	17.6	0.9	3.1	33.3
1990–91	18.8	45.8	1.5	8.6	74.7
2000–01	25.0	76.0	2.9	15.2	119.1
2015–16	41.5	201	4.9	18.0	265.4
2018–19	45.4	223	6.7	74.08	350.162

RES* Include small hydro projects, Biomass gasifier, Biomass power

Source: The Economic Survey, 2015–16.

The electricity production in India was started in 1898 when the Darjeeling hydel power project was commissioned. The first thermal power plant was installed in Kolkata in 1899. Subsequently, the Mettur project in Tamil Nadu and the Sivasamudram project in Karnataka were commissioned. In the initial phase the use of electricity was confined only to the big urban centres. The National Thermal Power Corporation (NTPC) was established in 1975, which established a number of thermal power stations based mainly on coal in different parts of the country.

It may be seen from **(Table 8.8)** that the power generation from the conventional sources has increased from 771.551 BU to 1141.988 BU from 2009–10 to 2018–19. In the year 2018–19, the major contribution is from coal of 223 thousand MW followed by RES (MNRE) and hydroelectric generation contributes 45.4 thousand MW of electrical power energy in India. India has the fifth largest power generation capacity in the world.

Table 8.8

Generation and Growth in Conventional Energy Generation in India (2009–10 to 2018–19)

Year	Energy Generation from Conventional Sources (BU)	Generation Growth (in Percentage)
2009–10	771.551	6.6
2010–11	811.143	5.56
2011–12	876.887	8.11
2012–13	912.056	4.01
2013–14	967.150	6.04
2014–15	1048.673	8.43
2015–16	1107.822	5.64
2016–17	1160.141	4.72
2017–18	1206.306	3.95
2018–19	1141.988	−5.33

Source: Ministry of Power, Government of India, www.powermin.nic.in

Hydro-Electricity

As stated in the preceding paragraph, the first hydro-electric plant in India was commissioned in 1898 at Darjeeling. It was followed by the Mettur (Tamil Nadu) and Sivasamudram (Karnataka)

projects which were commissioned in 1899 and 1902, respectively. The fourth hydro-power station was built on the Jhelum river at Mohara in 1909. The real progress in the production of hydro-power took place after independence when a number of multipurpose projects were constructed. Some of these projects are the Bhakra-Nangal Dam, Rihand-Dam, Hirakud-Dam, Nagarjun-sagar, Damodar Valley Corporation, Chambal, Tungbhadra, Tehri, and Koyna projects.

Regional Patterns

The development of hydro-electricity depends on the perennial and seasonal character of rivers, their monthly regime, undulating topography, suitable rock formations for the construction of dam and the demand for energy. Because of these factors, the development of hydro-power is not uniform. Andhra Pradesh ranks first, followed by Karnataka, Tamil Nadu, Punjab, Maharashtra, Kerala, and Madhya Pradesh. The major Hydro-electric Power Plants have been given in **(Table 8.9)**. and **(Fig 8.7)**.

After Independence, especially in the First Five Year Plan, great emphasis was laid on the development of electricity. To achieve this objective, a number of multipurpose projects were installed. Some of the important multipurpose projects have been described in the following:

Table 8.9

State-wise Main Hydro-Electric Power Plants

States	Names of the Hydro-Electric Power Plants
1. Andhra Pradesh	Machkund, Nagarjun-Sagar, Nizam-Sagar, Sileru, Srisalem
2. Bihar	Kosi
3. Gujarat	Akrimota, Sardar-Sarovar, Ukai (Tapi), Hathmati (Sabarmati), Bhadra (Kathiawad)
4. (U.T) Jammu, Kashmir & Ladakh	Dool-Hasti, Lower Jhelum, Salal, Baghliar
5. Jharkhand	Maithon, Panchet, Tilaiya, Balpahadia, Bokaro, Konar, Aiyar, (all three under DVC), Mayurakshi
6. Karnataka	Mahatma-Gandhi (Jog Falls), Sivasamudram (Kaveri), Bhadra, Munirabad, Saravati, Tungbhadra, Krishnaraja-Sagar
7. Kerala	Idukki (Periyar), Kallada, Kuttiaddy, Pallivasal, Parambikulam, Poringal, Panniar, Sabarigiri, Periyar
8. Madhya Pradesh	Indira Sagar and Pratap-sagar on Chambal, Twa (M.P.)
9. Maharashtra	Bhola, Bhatnagar-Beed, Girna, Khopali, Koyna, Purna, Paithon, Vaiterna
10. North-eastern States	Dikhu, Doyan (both in Nagaland), Gomuti (Tripura), Loktak (Manipur), Kopali (Assam), Khandong and Kyrdemkulai (Meghalaya), Sirlui and Barabi (Mizoram), Ranganadi (Arunachal Pradesh), New Umtru (Meghalaya)
11. Odisha	Hirakud (Mahanadi), Balimela, Rengali (Brahmani), Indiravati
12. Punjab and Himachal Pradesh	Bhakra-Nangal on Sutlej, Dehar on Beas, Giri-Bata, Harike Binwa, Chamera, Pong, Siul, Bassi, Nathpa Jhakri, Ranjit Sagar Dam, Chanju (H.P), Sainj (H.P)
13. Rajasthan	Ranapratap Sagar and Jawahar Sagar on Chambal river
14. Tamil Nadu	Bhavanisagar, Mettur, Periyar, Aliayar, Kodayar, Moyar, Suruliyar, Papnasam
15. Uttarakhand	Tehri-dam and Koteshwar-dam on Bhagirathi River
16. Uttar Pradesh	Rihand, Ramganga, Chibro on Tons
17. West Bengal	Panchet, Purulia Pass
18. Sikkim	Teesta Dam, Ranjit, Tashiding

1. **Bhakra Nangal Project**: The Bhakra-Nangal Dam is a joint venture of the Punjab,

Haryana, and Rajasthan governments. Constructed across the Satluj river near Bhakra gorge, it is one of the highest straightway gravity dam in the world. The dam is 518 m long and 226 m high. Its reservoir is known as the Gobind Sagar (named after Sikh Guru Gobind Singh, the tenth Guru of Sikhs). It is a multipurpose project funded by the Central Government, built to generate electricity, provide irrigation, flood control, soil conservation, silt control, recreation, navigation, pisci-culture, preserving wild-life, and cattle rearing.

2. **Damodar Valley Project**: The Damodar River is a tributary of the Hugli River. It used to be called the "Sorrow of Bengal". The Damodar flows through Jharkhand and West Bengal. The Damodar Valley Corporation was established on 18 February, 1948. Under this project, seven dams were constructed namely, Tilaiya, Balpahari, Maithon, Konar, Bokaro, Aiyar, and Panchet Dams.

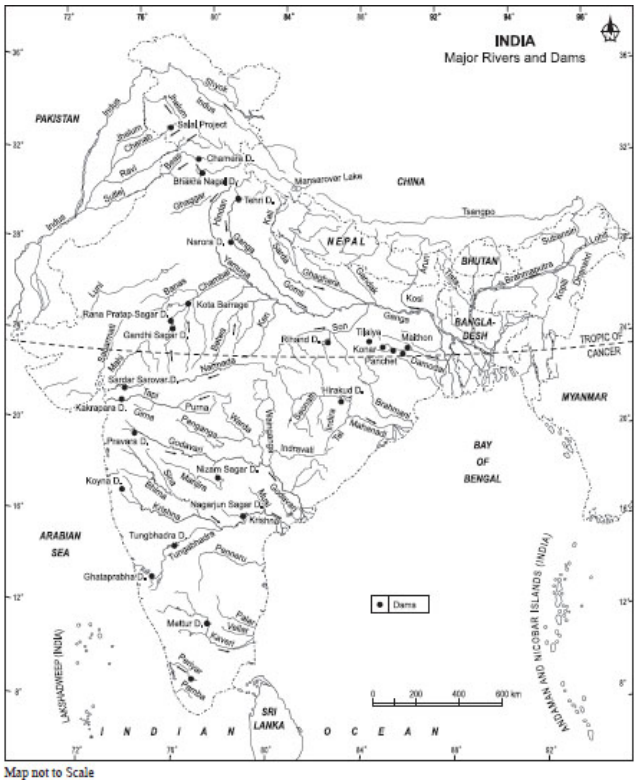


Fig. 8.7 Power Projects

(i) **Tilaiya Dam**: The Tilaiya dam has been constructed across the Barakar river. It is the only concrete dam in the area. Two power stations of 2000 kW each have been set-up here. The dam provides irrigation to forty thousand hectares of land. It has helped in the reduction of floods. This dam was completed in 1953. Its underground power station with installed capacity of 60,000 kW provides cheap power to the mica mines of Kodarma and Hazaribagh.

(ii) **Konar Dam**: The Konar dam has been constructed across the Konar river—a tributary of the Damodar River in the Hazaribagh district. It was completed in 1955. It is an earthen dam with concrete spill-way . Beside irrigation and power, it provides

cooling water to the Bokaro Steel Plant. The hydel station located near the dam generates about 40,000 kW of electricity.

(iii) **Maithon Dam**: Constructed across the Barakar river near the confluence of Barakar with Damodar river, it is a 56 m high dam. The dam completed in 1958, provides irrigation to 50,000 hectares of arable land. The underground power station generates 60,000 kW of electricity.

(iv) **Panchet Hill Dam**: The Panchet dam has been constructed across the Damodar river, about 20 km south of the Maithon Dam. It is 45 m high and 2545 m long. The power station near the dam has an installed capacity of 40,000 kW. It irrigates about 3 lakh hectares of agricultural land.

(v) The Aiyar, Bokaro and Balpahri dams are located on the Damodar and Barakar rivers.

3. **Dool Hasti**: The Dool Hasti Project has been constructed across the Chenab river in the Doda district of the Jammu Division. The main objective of this project was to harness the water of Chenab river and to generate electricity to be supplied to the main cities of the state including the cities of Srinagar and Jammu. It was commissioned in 1986.
4. **Gandhi Sagar**: The Gandhi Sagar project has been constructed across the Chambal river. The installed capacity of the Gandhi Sagar Dam is 115 MW. Five generators have been installed at Gandhi Sagar; four with a capacity of 2300 kW and one with a capacity of 2700 kW. It is providing power and irrigation to the surrounding regions of Rajasthan and Madhya Pradesh.
5. **Hirakud Project**: Constructed across the Mahanadi river, this project was funded by the Central Government. It is a 14 km long dam, considerd to be the longest in the world. The project involves the construction of three dams across Mahanadi, at Hirakud, Tikrapara, and Naraj. The Hirakud Project, according to recent study, has increased floods and droughts in the region. Due to increasing siltation , the storage capacity of the reservoir has been reduced, causing floods in the lower catchment area of the Mahanadi.
6. **Nangal Project**: Nangal dam has been constructed at Nangal, about 13 km downstream of the Bhakra dam. It is about 30 m high, 305 m long, and 121 m wide. Its main function is to generate electricity. It also supplies water to the Bhakra canals.
7. **Jawahar Sagar Dam**: This dam has been constructed to the north of Rana Pratap Dam in the state of Rajasthan. It is about 40 km to the north of the Rana Pratap Sagar. It is a multipurpose project constructed to generate electricity, control floods and provide irrigation water to the catchment area.
8. **Kosi Project**: The Kosi river, often been called ‘the “Sorrow of Bihar”, is an outcome of the joint agreement between the Nepali and Indian governments reached in 1954. Its main objective is to construct a barrage near Hanumannagar in Nepal, to built embankments on both the banks of the river to control floods, to construct canals for irrigation and to generate hydro-power. Kosi in July, 2008 shifted its course about 100 km towards east and caused great damage to life and property.
9. **Koyna Project**: It is a multi-purpose project in the Satara district of Maharashtra state. Its installed capacity is 880 MW. Its hydro-electricity is being supplied to the cities of Satara, Sholapur, Sangli, Kolhapur and Pune.
10. **Machkund Project**: This is a joint venture of the Andhra Pradesh and the Odisha states. The Machkund Dam is 54 m high and 410 m long. It is mainly a hydro-electric project which shall generate 115 MW electricity.

11. **Mahi Project**: This project has been constructed across the Mahi river which originates from the Vindhyan Hills of Madhya Pradesh. The project on completion will generate 40 MW hydro-power and shall irrigate 80,000 hectares of agricultural land.
12. **Mayurakshi Project**: Mayurakshi is a tributary of the Hugli. It rises from the Chotanagpur Plateau and flows through Jharkhand and West Bengal. It is a multipurpose project, generating 4000 kW of electricity and providing irrigation water to 3 lakh hectares. Electricity from this project is supplied to Murshidabad, Birbhum (West Bengal), and Santhal Pargana (Jharkhand).
13. **Mettur Dam**: The Mettur Dam was built in 1937 across a tributary of the Kaveri river in the Nilgiris. The project has a capacity to produce 240 MW of hydro-electricity. The Mettur Dam is not only generating electricity and providing irrigation water, but it has also helped in the flood control in the basin.
14. **Nagarjuna Sagar Project**: The Nagarjunsagar project has been constructed across the Krishna river in Nalgonda district of Telangana state. Its right and left bank canals have been named after Jawaharlal Nehru and Lal Bahadur Shastri respectively. The power plant generates 210 MW of hydro-electricity. The power is supplied to Hyderabad, Khammam, Mahbubnagar, Nalgonda and Vijaiwada.
15. **Pochampad Project**: This project has been constructed across the Godavari river in Adilabad district (Telangana). Its 115 km long canals irrigate about 2.5 lakh hectares in Adilabad, and Karimnagar districts of Telangana.
16. **The Periyar Project**: Originating from the Cardamom Hills, Periyar is an important river of Kerala. A dam has been constructed across its course in hilly gorge. Its installed capacity is 140 MW. It is a multi-purpose project helping in the prevention of floods, soil erosion and generating electricity being supplied to Ernakulam, Kochi and neighbouring cities.
17. **Rampad Sagar Dam**: This dam has been constructed in the lower reaches of the Godavari river, about 30 km to the north of Rajamundry. It is a multi-purpose project designed to check floods, to provide irrigation in the delta region of the river and to generate electricity.
18. **Rana Pratap Sagar Dam**: This dam has been constructed across the Chambal river, about 25 km to the north of the Gandhi Sagar Dam in the Kota district of Rajasthan. It is a multipurpose project designed to generate electricity, to control floods and to provide irrigation water to the surrounding agricultural land. Its installed capacity is 99 MW.
19. **Rihand Project**: Funded by the Central Government, it is the largest multi-purpose project of Uttar Pradesh. It has been constructed across the Rihand river, a tributary of the Son river, near Pipri village in the Sonbhadra district. The reservoir of this dam has been named after Gobind Ballabh Pant. It is connected with the Obra hydro-power station and the Obra thermal power plant located in its vicinity. The power generated from this project is supplied to eastern Uttar Pradesh, western Bihar and northern parts of Madhya Pradesh. Flood control in Son valley, control of soil erosion in Baghelkhand, tourism and pisci-culture are the other benefits from this project.
20. **The Salal Project**: This project has been constructed across the Chenab river in the Riasi district of Jammu Division of Jammu & Kashmir State. The installed capacity of the project is 750 MW. It was inaugurated in 1986. The electricity is supplied to Riasi, Udhampur, Jammu, and other neighbouring urban centres.
21. **Sardar Sarovar Dam**: This has been constructed across the Narmada river near Navagaon. The project when completed, will generate 1450 MW of hydro-electricity and

will irrigate about 18 lakh hectares of agricultural land. It will also promote dairy farming, livestock keeping, animal husbandry and allied occupations.

22. **Shivasamudram Dam**: The Shivasamudram project was built in 1902 across the Kaveri river in Karnataka. The main objective of the project was to supply electricity to the Kolar Gold Mines, the city of Mysore and its neighbouring urban centres. It helped in flood control and became a centre of tourists attraction.
23. **Tawa Dam**: This project has been constructed across the Tawa river; a left bank tributary of the Narmada river. It is a multipurpose project which has been designed to provide irrigation water to more than 50 thousand hectares and has the installed capacity to produce 150 MW hydro-electricity.
24. **Tehri Dam**: The Tehri Dam is being constructed across the Bhagirathi river just below the confluence of Bhagirathi and Bhilaganga in the Tehri district of Uttarakhand. Conceived by the Planning Commission in 1972, the work on the project was started in 1975. It is the biggest, highest and tallest dam in India. The project is being implemented with Soviet (Russian) technical and economic aid. The project will provide irrigation to 2.74 lakh hectares in Uttarakhand and western Uttar Pradesh and will generate 1000 MW of hydro-electricity. Some serious objections were raised about this project as the environment and ecology may be adversely affected by this project which has been constructed in a highly earthquake prone area of the country.
25. **Tungbhadra Project**: Tungbhadra is a right hand tributary of the Krishna river which originates from the Western Ghats (Sahayadri Hills) of the Chikmagalur district of Karnataka. The Tungbhadra Dam has been constructed at Mallapuram near Hosepet in the Bellary district. Three power houses have been constructed in this project to generate 126 MW of electricity. The Tungbhadra canals irrigate more than 4 lakh hectares of arable land.
26. **Ukai Dam**: Ukai is a tributary of the Tapi river. The Ukai project was launched mainly to harness the Tapi water. The installed capacity of the Ukai project is 300 MW. Its electricity is supplied to Surat and other neighbouring urban centres.

Governmental Initiatives: Power Sector

- ***KrishangangaHydroelectric Plant*** (110×3 MW) has been commissioned in Jammu and Kashmir
- A capacity of 60 MW Turial Project in Mizoram also is a step in this direction.
- The potential has been leveraged in the neighbouring countries. Example. **Mangdechhu HEP (720MW)** in Bhutan.
- About 6379 un-electrified villages have been electrified and 26,284,350 households have been given connections of the electricity as per the '***Saubhagya Yojna***' since 2017.
- ***Integrated Power Development Scheme (IPDS)*** shall ensure a 24×7 for all in the urban areas.
- ***Deen Dayal Upadhyaya Gram Jyoti Yojana (DDUGJY)***: It shall provide separate feeders and also strengthen the sub-transmission and districution infrastructure at all levels in the rural areas.
- ***Ujwal DISCOM Assurance Yojana (UDAY)***: To obtain the operational and financial turnout of state owned power distribution companies .
- ***Unnat Jeevan***: It was launched in 2015 and is taken by the Energy Efficiency Services Limited . The project is also known as '*Prakash Path*'. LED bulbs are freely distributed to

decrease the power consumption.

Thermal Electricity

Thermal electricity is produced with the help of coal, petroleum, and natural gas. About 70% of the total electricity produced is thermal in character. The main advantages of thermal electricity are as under:

(i) It can be generated in the areas not suitable for the generation of hydro-electricity.
(ii) Coal, diesel and natural gas can be transported to the areas of isolation and relative isolation.
(iii) It can be generated even when the weather is adverse.
(iv) The gestation period of the thermal power stations is short.

The production of thermal energy is, however, not eco-friendly as the release of carbon dioxide pollutes the atmosphere. Moreover, it consumes the valuable exhaustible resources like coal, petroleum and natural gas.

Some of the important thermal power stations of India have been given in **(Table 8.10).**

Table 8.10

India—Thermal Power Stations

States/U.T.		Thermal Power Stations with Installed Capacity in MW
1.	Andhra Pradesh and Telangana	Bhadrachalam, Kothagudam (680), Manuguru, Nellore (300), Ramagudam (200), Vijayawada (420), Sembcorp Gayatri
2.	Assam	Bongaigaon (250), Chandarpur (100), Namrup (62.25)
3.	Bihar	Barauni (255), Kahalgaon, DVC (780), Durgapur (285)
4.	Chhattisgarh	Korba (750), Ratija TPP (50), Nawapara (300)
5.	Delhi	Badarpur (720), Indraprastha (285), Rajghat (350)
6.	Gujarat	Ahmadabad (180), Banas, Dhuvaram (534), Gandhinagar (240), Kutch, Kandla, Mahuva, Porbandar, Sabarmati (110), Shahpur, Sikka, Ukai (640), Utaran (200), Wankbori,
7.	Haryana	Faridabad (180), Panipat (220), Yamuna Nagar (150)
8.	Jammu & Kashmir	Kalakot
9.	Jharkhand	Bokaro (248), Chandrapur (780), Subarnrekha
10.	Madhya Pradesh	Amarkantak (3000), Satpura (1142), Singrauli, Vindhyachal (1260)
11.	Maharashtra	Ballarshah (500), Bhusaval(485), Chandrapur (210), Chola (95), Dhobal (350), Khapar-Kheda (250), Koradi (660), Paras (95), Parli (240), Trombay (838), Ujjaini, Uran (240), Mauda STPP (660), Mangaon, Nashik (270)
12.	Manipur	Loktak (Imphal)
13.	Punjab	Bhatinda (440), Rupnagar
14.	Rajasthan	Anta, Banswara, Kota (240), Palana, Sawai Madhopur
15.	Odisha	Balimela, Talcher (470)
16.	Tamil Nadu	Ennore (450), Mettur, Neyveli (2300), Tuticorin (630), cuddalore ITPCL (600)
17.	Uttar Pradesh	Bahraich, Dorighat, Gorakhpur, Harduaganj (215), Jawaharpur, Kanpur (325), Mau, Moradabad, Obra (450), Panki (345), Parichha (240), (565), Tundla.
18.	West Bengal	Birbhum, Bundel (540), Durgapur (285), Farakka (1600), Gauripur (350), Kalaghat, Kolkata (250), Murshidabad, Titagarh, Santaldih (480), Teesta IV & V (80)
19.	Karnataka	Kudgi TPP (800) U1 & U2
20.	Uttrakhand	Garna CPP (225), Kashipur (225)
21.	Himachal Pradesh	Kashang (195)

It may be seen from **(Table 8.10)** that the distribution of thermal power stations is highly unequal. Most of them, however, have been located near the source of inferior coal or lignite mining centres. The state of Maharashtra ranks first in the production of thermal power, followed by Uttar Pradesh, Gujarat, and West Bengal. The rank of Madhya Pradesh (including Chhattisgarh) and Tamil Nadu are sixth and seventh respectively. Arunachal Pradesh, Goa, Himachal Pradesh, Kerala, Meghalaya, Mizoram, Nagaland, Sikkim, Tripura and Uttarakhand are the states which are devoid of the production of thermal electricity. On the zonal basis, the western zone is the largest producer of thermal energy, followed by the northern zone, eastern and southern zones. Thermal power stations have been installed at Raichur (Karnataka) and Loktak (Manipur). **(Fig 8.8).**

The contribution of National Thermal Power Corporation is quite significant in the generation of thermal energy. The NTPC thermal power stations in the country have been given in **(Table 8.11).**

Table 8.11

India—Major Thermal Power Stations

Projects/Stations	Installed Capacity (MW)
1. Badarpur	720
2. Farakka (West Bengal)	1600
3. Kahalgaon (Bihar)	840
4. Korba (Chhattisgarh)	2100
5. National Capital Thermal Power Plant Dadri (U.P.)	840
6. Ramagundum (Andhra Pradesh)	2100
7. Rihand /Obra (Uttar Pradesh)	1000
8. Singrauli (Madhya Pradesh)	2000
9. Talcher (Odisha)	720
10. Unchahar (Uttar Pradesh)	720
11. Vindhyachal (Madhya Pradesh)	1260
12. Gas Based Projects:	
(i) Anta (Rajasthan)	430
(ii) Auraiya (Uttar Pradesh)	600
(iii) Kawas (Gujarat)	600
Total	15530

Fig. 8.8 Thermal Power Projects [Source: Annual Report 2017–18; Ministry of Power Govt of India]

Nuclear Energy

Looking at the exhaustible nature of the fossil fuels, nuclear energy development has become very vital for the economic development of the country. In India, it has a vast potential for future energy development. It is produced from uranium and thorium. At present there are 17 nuclear plants across the country. The atomic power stations are given in **(Table 8.12)**.

Table 8.12

India—Atomic Power Stations till 2018

S. No.	Power Station	State	Year of Commissioning	Capacity (MW)
1	Tarapur	Maharashtra	1969	1400
2	Rawatbhata	Rajasthan	1972	1180
3	Kalpakkam	Tamil Nadu	1984	440
4	Narora	Uttar Pradesh	1989	440
5	Kakrapara	Gujarat	1993	440
6	Kaiga	Karnataka	1993	880
7	Kudankulum	Tamil Nadu	2002	1864
8	Jaitapur	Maharashtra	2017	9900
9	Bargi-Chutka	Madhya Pradesh	Proposed but facing strong opposition from locals since 2015	1400

S. No.	Power Station	State	Year of Commissioning	Capacity (MW)
10	Kawada-Chutka	Madhya Pradesh	Proposed but facing strong opposition from locals since 2015	4×700
11	Chayya Mithi Virdi	Gujarat	Might not start	6×1000
12	Kumharia	Gorakhpur, Haryana	Proposed	2×700
13	Mahi Banswara	Rajasthan	Proposed	2×700

Source: Department of Atomic Energy, Government of India, March 2018

At present, nuclear power constitutes only less than 4% of the total energy production. It requires highly sophisticated technology and technical knowhow. Moreover, for the cooling of plant there is heavy need of fresh water.

The Atomic Energy Institution at Trombay was established in 1954. This was renamed as the Bhabha Atomic Research Centre (BARC), in 1967. The first nuclear power station with a capacity of 320 MW was set up at Tarapur near Mumbai in 1969. Subsequently, the Rawatbhata Atomic Plant (300 MW) near Kota was set up in 1969 which was followed by the establishment of Narora (1989), Kaiga (Karnataka), and Kakrapara in Gujarat in1993. Thus, at present, nuclear energy is produced from eleven units located at six centres **(Table 8.12)**. The new sites of nuclear power plants include Bargi or Chutka (M.P.), Haripur (W. Bengal), Jaitapur (Maharashtra), Kawada (Andhra Pradesh), Kudankulam (Tamil Nadu), Kumharia (Haryana), and Mithi-Verdi (Gujarat).

The govt. of India on 17th May, 2017, cleared the building of ten new nuclear power plants to add 7,000 MW to India's power generation capacity. These nuclear power plants will be indigenous with a capacity 700 MW each. The nuclear reactors will make it easier to develop the domestic industry.

Development of nuclear energy is imperative for the economic development of the country. But the disaster's like Fukushima and Chernobyl have proved that it is full of risk. Thus it is only a parochial solution of the Indian energy crisis. Unfortunately, in India, in case of nuclear accident, the maximum fine that can be imposed by the regulator on an offending **nuclear plant is Rs. 500/-**. This amount is too low to serve as a deterrent against such infringements.

NON-CONVENTIONAL ENERGY

The non-conventional energy is also called as renewable energy. The non-conventional sources of energy include solar energy, wind energy, bio-mass energy, fuel-cells, electric vehicles, tidal energy, hydrogen energy, and geo-thermal energy.

Solar Energy

Solar energy is one of the most important sources of non-conventional energy. Solar energy is non-exhaustible, reliable, and pollution free. It may be utilised for water heaters, power generation devices, air-conditioning, space heating, development of pisci-culture, and multifarious uses of water and refrigeration.

The average amount of solar energy received in the earth's atmosphere is about 1353 kW per sq metre. It is 1000 times the total consumption of the global energy. Being situated in the sub-tropical latitudes, India receives higher amount of solar energy. The greater part of the country has more than 300 solar days. The total amount of energy received from the Sun is about 5000 trillion kWh per year.

The Solar Photovoltaic (SPV) technology enables the conversion of solar radiation into electricity without involving any moving part like turbine. Over 650,000 solar PV systems have been installed in the country.

In many parts of the country, the solar energy programmes have been implemented. One such example is the Rural Energy Co-operative at Sagar Island in the Sundarban Delta of West Bengal. Similar programmes have been implemented in the other islands in the Bay of Bengal, the desert of Jodhpur (Rajasthan), Kalyanpur (Aligarh), and Coimbatore **(Fig. 8.8)**.

Wind Energy

Wind is an important source of non-conventional energy. It is cheap, pollution free, eco-friendly and can be developed away from the sources of fossil fuels (conventional sources of energy). Since ancient times wind energy was utilised in sailing ships and wind mills. India ranks 4th in the world after the U.S.A., Spain, Germany, and China.

For the generation of wind energy, a wind speed of more than five km per hour is considered to be suitable. Wind speeds above 10 km per hour are prevalent over parts of the coastal regions of Gujarat, Andhra Pradesh, Karnataka, Madhya Pradesh, Rajasthan, Tamil Nadu, Maharashtra, Kerala, Odisha, West Bengal, Uttarakhand, Jammu & Kashmir, Lakshdweep, and Andaman and Nicobar Islands (**Fig. 8.8**). The state of Rajasthan and Ladakh also record suitable consistency of wind speed. Wind mills can be operated there to harness wind energy.

In India, the total potential of wind energy is estimated to be about 302,000 MW at 100m . Wind energy projects have been implemented in 22 states of the country **(Table 8.13)**. Muppandal in Kanyakumari district of Tamil Nadu is one of the most important sites for wind farm. The maximum potential of wind energy lies in the states of Karnataka, Gujarat , Rajasthan and Tamil Nadu. The potential for wind power generation is estimated to be 102,788 MW. A power capacity of 71 MW has been established at 33 locations in 9 states. It is expected that in 2019, the wind power energy capacity shall improve by 3–3.5 GW. Jaisalmer is India's largest onshore wind farm. Its installed capacity is 1064 MW. The largest wind farm in Odisha is Damanjodi with a 99 MW wind power project.

Table 8.13

Statewise and year wise wind power installed capacity (MW)

State	Upto 31.03.2002	2002–03	2005–06	2008–09	2010–11	2012–13	2014–15	Total potential (MW)	Total number of wind power commisioned at 100 m height
Andhra Pradesh	93.2	0.0	0.5	0.0	55.4	202.2	166.30	912.5	10
Gujarat	181.4	6.2	84.6	313.6	312.8	208.3	126.90	3581.3	12
Karnataka	69.3	55.6	143.8	316.0	254.1	201.7	230.50	2548.7	13
Kerala	2.0	2.0	0.0	16.5	7.4	0.0	0.0	35.1	–

State	Upto 31.03.2002	2002–03	2005–06	2008–09	2010–11	2012–13	2014–15	Total potential (MW)	Total number of wind power commisioned at 100 m height
Madhya Pradesh	23.2	0.0	11.4	25.1	46.5	9.6	143.90	567.3	7
Maharashtra	400.3	2.0	545.1	183.0	239.1	288.6	273.45	4369.8	8
Rajasthan	16.1	44.6	73.3	199.6	436.7	615.4	267.70	3052.7	11
Tamil Nadu	877.0	13 3.6	857.1	431.1	997.4	174.6	124.45	7394.0	8
Others	4.3	0.0	0.0	0.0	0.0	0.0	0.0	4.3	–
Total	**1666.8**	**242.0**	**1716.2**	**1484.9**	**2349.2**	**1700.4**	**1333.2**	**22465.6**	**69**

Source: Ministry of New & Renewable Energy, Government of India; https://mnre.gov.in

Tamil Nadu has the largest installation of wind turbines (offshore) in the country in the Muppandal Perungudi area near Kanniyakumari. This is one of the largest concentrations of wind farm capacity of 1500 MW at a single location anywhere in the world. A Centre of Wind Energy Technology has been set up in Chennai. The Wind Energy Estates are being set up in the joint sector or in private sector. The various softwares involved in the wind power technology are Windographer, WAsP, Wind farmer, Meteodyn, Openwind, Windsim, GIS, etc.

Ocean Energy

The tidal waves and sea-waves are the main sources of ocean energy. India has a very long coastline, more than 6100 km, but the ocean energy production is limited. The suitable areas for the generation of tidal and sea-waves energy are:

(i) The Gulf of Khambat
(ii) The Gulf of Kutch
(iii) The Estuary of Hugli

According to one estimate, the Indian coasts have the potential to generate over 40,000 MW of electricity. A plant has been established near Thiruvanantapuram (Vizhingam) which is generating about 150 MW of electricity through sea-waves.

The main problem in the exploitation of ocean energy is the high cost involved in the construction of civil works.

Geothermal Energy

India has very limited potential of geothermal energy. According to one estimate, the total geothermal energy is about 600 MW. There are 115 hot water springs in the country and 350 sites from which geothermal energy can be produced. The Puga Valley in Jammu and Kashmir, the Manikaran area in Himachal Pradesh, the western slopes of the Western Ghats in Maharashtra and Gujarat, the Narmada-Son Valley, and the Damodar Valley are the main areas which have potential for the generation of thermal energy.

Bio-Energy

Bio-energy is a clean source of energy which improves sanitation, hygiene and the living style of the rural population. The technique is based on the decomposition of organic matter in the absence of air to produce gas. Bio-gas is used for cooking, and lighting fuel in specially designed stove and lamps respectively. According to one estimate, India has a capacity to produce bio-gas to the extent of 25,000 million cubic m. The left over digested slurry serves as manure. This can meet 50% of the rural domestic fuel requirements. Moreover, it can produce 7 million tonnes to nitrogen, 3 million tonnes phosphate, 5 million tonnes of potassium, and over 50 million tonnes of compost manure.

It may be seen from **(Table 8.14)** that Uttar Pradesh has the highest potential in bio-gas, followed by Madhya Pradesh, Andhra Pradesh, and Bihar. The highest production of bio-gas is, however, in the state of Maharashtra 11.90% followed by Gujarat U.P. 9.56%.

Table 8.14

Bio-Gas Development in Major Selected States of India—2015–16

State	Estimated Potential (MW)	Production (MW)	Percentage of Total Production
1. Uttar Pradesh	1620	650	9.56
2. Madhya Pradesh	1460	580	8.53
3. Andhra Pradesh	1185	385	5.66
4. Bihar	1135	570	8.38
5. Rajasthan	1120	575	8.45
6. Maharashtra	1060	810	11.90
7. West Bengal	410	230	3.38
8. Karnataka	1235	515	7.57
9. Tamil Nadu	1260	490	7.21
10. Gujarat	1248	525	7.72
11. Odisha	280	95	1.39
12. Punjab	3510	635	9.34
13. Others	2728	742	10.91
14. Total	15343	6802	100.00

Source: Ministry of Non-Conventional Energy.

The development of bio-gas is adversely affected because of the non-availability of cattle dung, water, labour, space, and low temperatures in certain parts of the country, especially during the winter season.

ENERGY CRISIS

With the rapid growth of population and increase in the per capita income, there is an increasing demand for energy, especially that of conventional sources of energy. The consumption of energy in the country is increasing at the rate of more than 12% per annum. In the absence of energy, there are frequent power failures, load-shedding, closure of factories, etc., resulting in a decrease in industrial and agricultural production.

In comparison to the developed countries, the per head consumption of electricity in the country is very low. For example, the per capita consumption of electricity in India is 350 kWh as against the world average of 1000 kWh and 7000 kWh in the U.S.A. According to one estimate, the country's peak demand projected for 2010 is 175,500 MW against the actual installed capacity of 90,000 MW. This requires an additional installed capacity of 85,500 MW.

Since the coal resources are highly unequally distributed, the transportation cost of coal to the distant thermal power stations is quite expensive.

Mismanagement of power sector, low efficiency of power houses, power theft, labour problem, pilferage, and power wastage are also aggravating the power crisis in the country.

ENERGY CONSERVATION

The conventional sources of energy (fossil-fuel) are fixed and exhaustible and the non-conventional sources of energy are not adequately developed. Energy conservation is, however, imperative for our survival and for the survival of the future generations. Some of the steps which can go a long way in the conservation of energy are as under:

(i) Emphasis on the development of non-conventional sources of energy. This will conserve the fossil fuels (coal, petroleum and natural gas).
(ii) Reduction in the consumption of energy.
(iii) Use of latest technology for cooking stoves and heating lamps.
(iv) Privatisation of electricity.
(v) Reduction in pilferages.
(vi) Severe punishment for power theft.

All these steps, if taken collectively, can ease the power crisis substantially.

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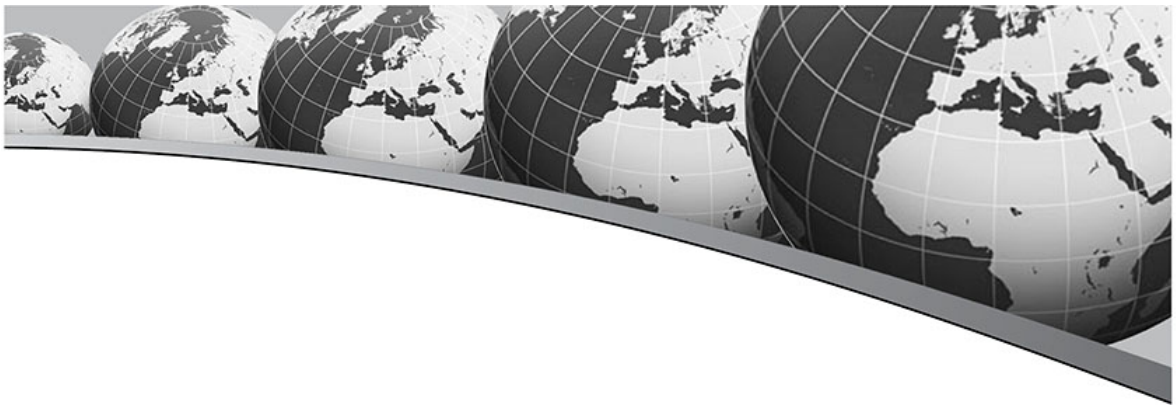
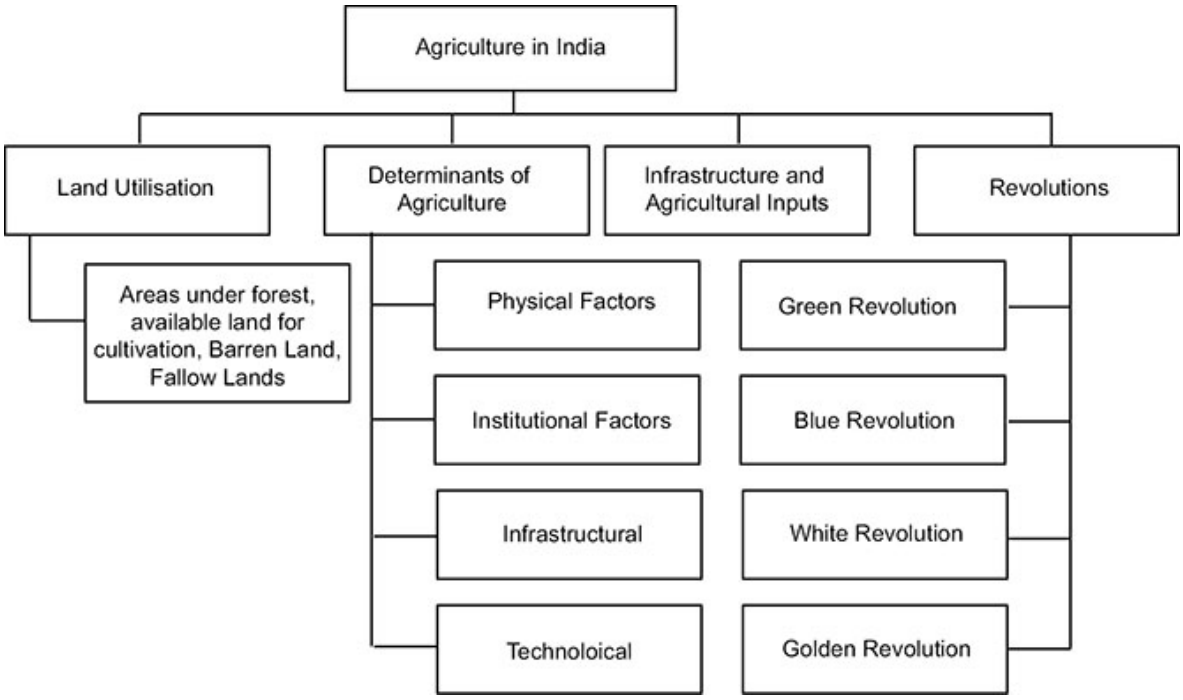


9

Agriculture



Indian economy hinges on agriculture. Over 58% of Indian population is directly or indirectly dependent on agriculture. Agriculture and allied sectors contribute nearly 15.87% of gross domestic Product (GDP) of

India (Statistics Times Feb 8, 2019). Besides, agriculture is an important source of raw material for industrial production, and serves as a huge market for the industrial products. As per the theory of Gunnar Myrdal in 1956“ It is in the agriculture sector that the battle for long term economic development will be won or lost.” **If agriculture goes wrong, nothing else will have a chance to go right in India (M.S. Swaminathan)**. The agricultural output, however, depends on monsoon as nearly 55% of area sown is dependent on rainfall. It not only provides food to its teeming millions; the agro-based industries for their raw material are dependent on agriculture. Moreover, agriculture fetches substantial amount of valuable foreign exchange.

The domestication of plants and animals is known as agriculture. It includes cultivation of crops, animal husbandry, horticulture, pisciculture, sericulture, silviculture, floriculture, etc. Being located in tropical and subtropical latitudes, the greater part of the agricultural land of India can produce two or more than two crops in a year.

LAND UTILISATION

Land utilisation statistics are available for about 93% of the total geographical area (328.75 million hectares) of the country. The net sown area accounts for about 46.15% of the total reporting area of India, against the world average of about 32%. The general land use of the country has been given in (**Table 9.1) (Fig. 9.1)**.

Table 9.1

India-General Land Use (2017–18)

S. No.	Land Categories	Area (in Million Hectares)	Percentage
1.	Total geographical area	328.75	
2.	Total reporting area for land use	308	93.68
3.	Area under forests	80.20	24.39
4.	Area not available for cultivation		
	a) Area out to non-agricultural uses	22.45	6.82
	b) Barren and uncultivable land	19.09	5.81

S. No.	Land Categories	Area (in Million Hectares)	Percentage
5.	Other uncultivated land excluding fallow		
	a) Permanent pastures and grazing lands	11.04	3.36
	b) Land under tree crops and groves	3.57	1.10
	c) Culturable waste	19.09	5.81
6.	Fallow Lands		
	a) Current fallow	13.33	4.05
	b) Other fallow	10.11	3.08
7.	Net sown area	141	46.15
8.	Area sown more than once	48.51	32.60
9.	Total cropped area	189.74	57.71
10.	Area under irrigation	73	22.20

Source: Ministry of Agriculture, Government of India.

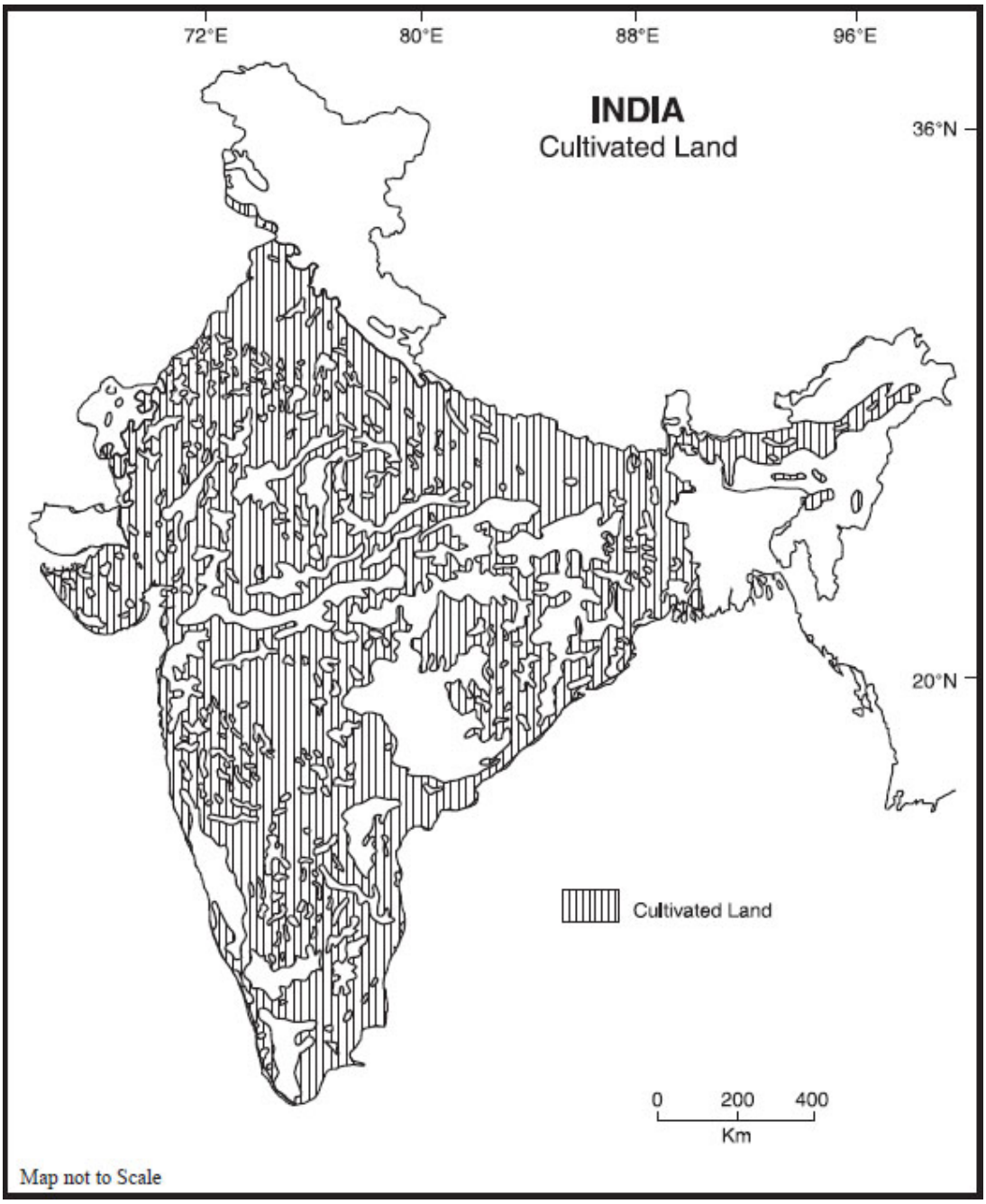


Fig. 9.1 Arable Land (2017–18)

It may be seen from (**Table 9.1)** that over 55.3% of the total reporting area was net area sown in 2015–16, and 32.6% was double cropped area **(Fig. 9.2)**. The forest occupied 22.55% while 6.25 was barren and uncultivable land. The barren and wasteland occupied 6.25% of the total reporting area, while 12.83% was culturable waste (**Table 9.1) (Fig. 9.3)**.

The various categories of degraded lands have been given in (**Table 9.2).**

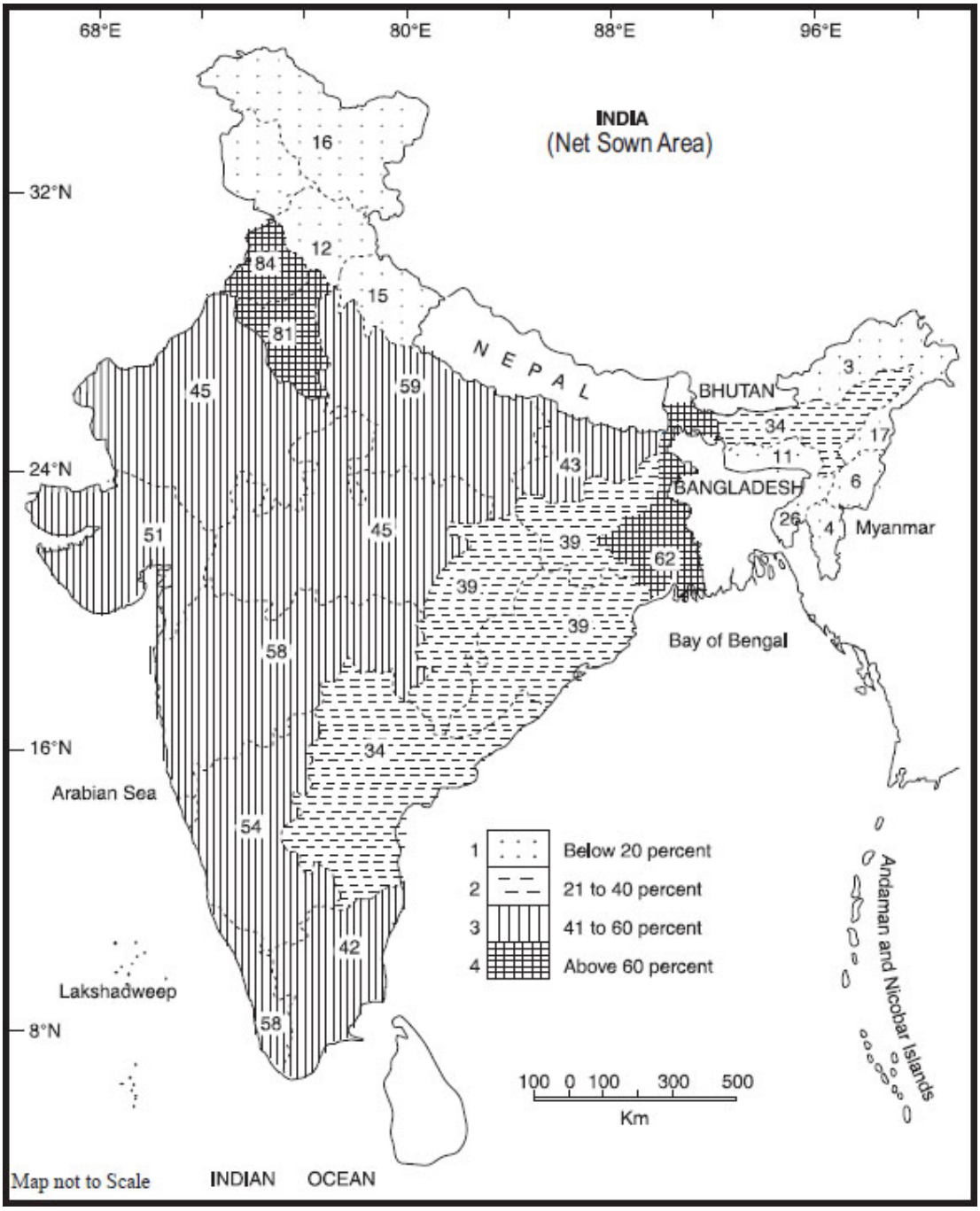


Fig. 9.2 Net Sown Area of India

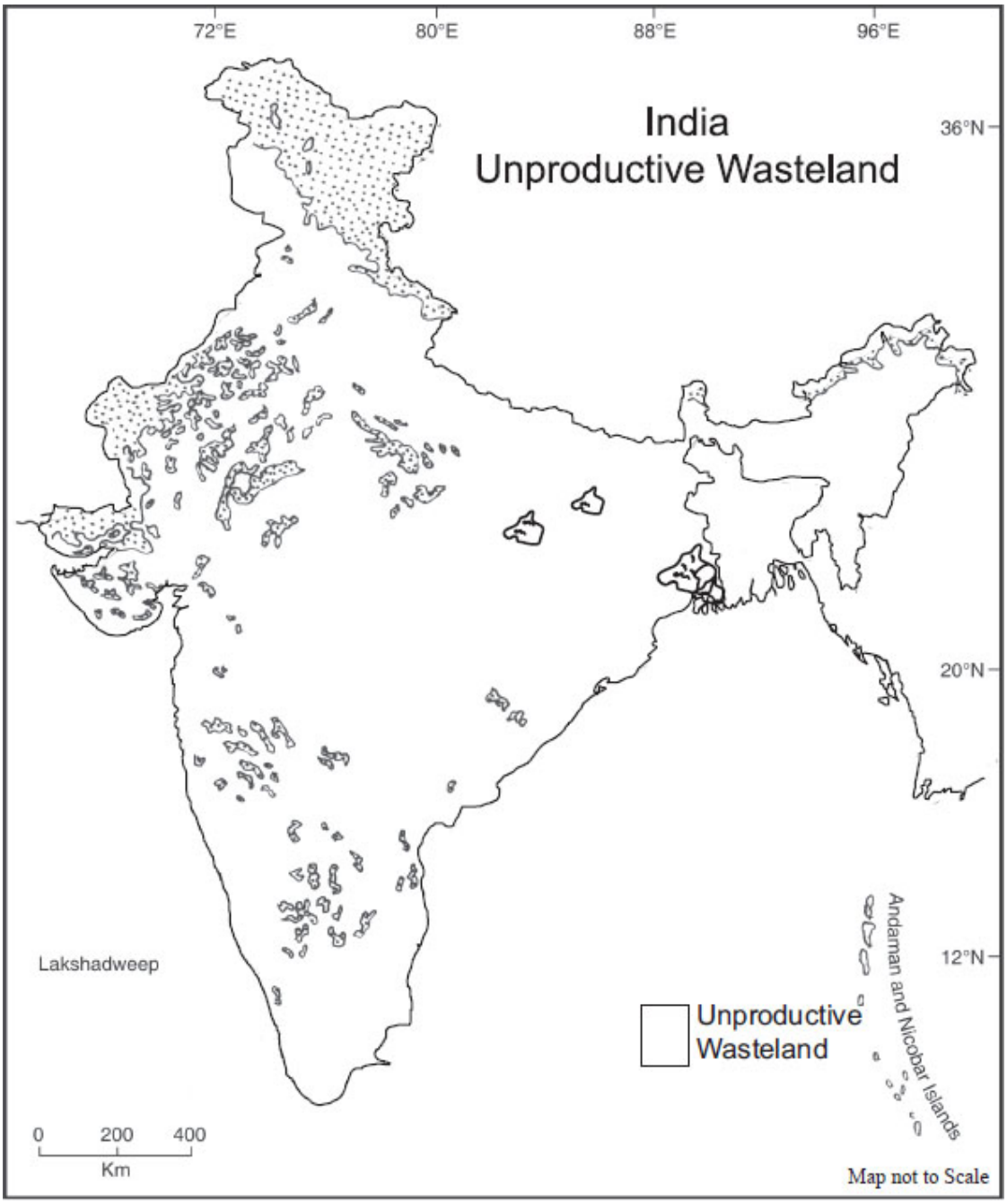


Fig. 9.3 Wasteland (2017–18)

An examination of (**Fig 9.2)** reveals that land with or without scrub and degraded notified forests occupied 19.40 and 14.06% of the waste land, respectively. The reclamation of degraded and waste lands can help in increasing the agricultural land as well as in improving the ecology and environment.

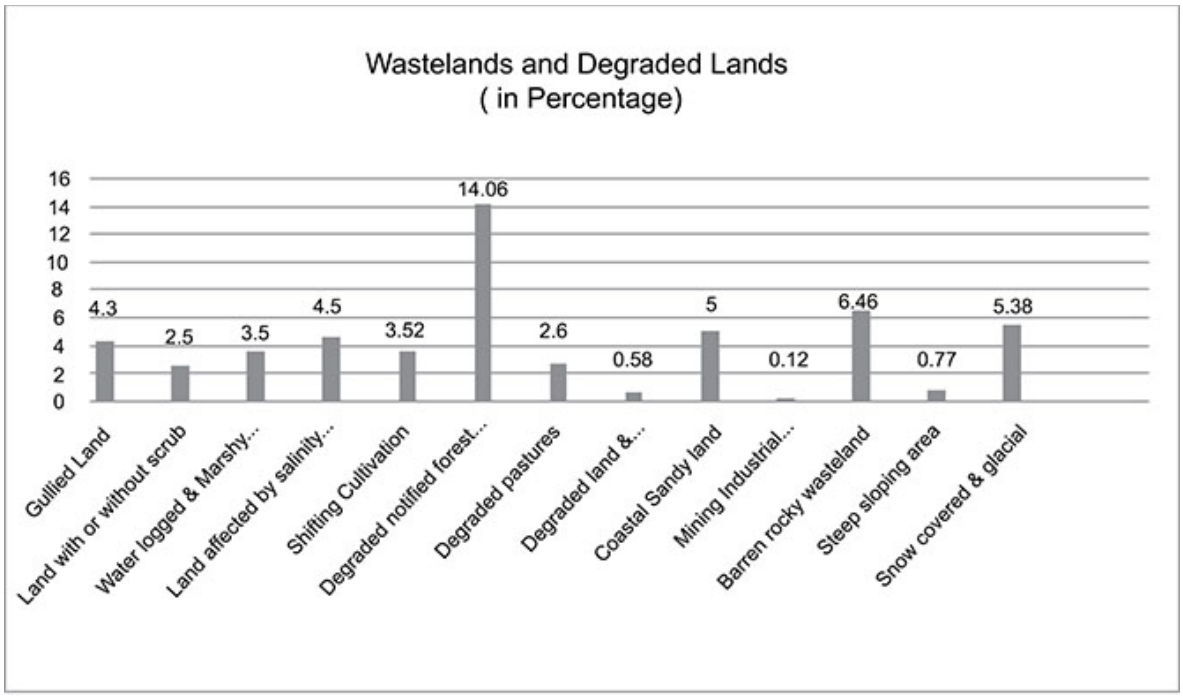


Fig. 9.4 Wastelands and degraded lands *[Source: Planning Commission, Government of India].*

Table 9.2

Causes of Land Degradation

Causes of Degradation	Area (in Million Hectares)	Percentage of Total Area
Water erosion	107.12	61.7
Wind erosion	17.79	10.24
Ravines	3.97	2.28
Salt affected	7.61	4.38
Waterlogging	8.62	4.90
Mines and quarrying	–	–
Degraded land due to shifting cultivation	4.91	2.82
Degraded land due to shifting cultivation	19.49	11.22
Special problems	2.73	1.57
Coastal sandy areas	1.46	0.84
Total	173.64	100

Source: Ministry of Agriculture, Government of India.

The uncovered land (no trees or plant cover) suffers from maximum degradation due to the wind blowing the top cover of the soil as can be seen from (Table 9.2) against the coastal area where there is only 0.84% of damage possibly due to rich green cover and wet sands along the coast.

CHARACTERISTICS AND PROBLEMS OF INDIAN AGRICULTURE

As stated at the outset, Indian economy hinges on agriculture. The socioeconomic status of the people, the national polity and the gamut of life of the people is directly controlled by agriculture. The Indian agriculture, however, has its own characteristics. Some of the important characteristics and problems of Indian agriculture have been described briefly in the following section:

1. Subsistent in Character

Despite eleven five year plans, in greater parts of the country, Indian agriculture is subsistent in character. The cultivators and farmers grow crops mainly for the family consumption. It is only in the controlled irrigated parts of the country like Punjab, Haryana, western Uttar Pradesh, and Kaveri delta where agriculture has become an agri-business or is market oriented.

2. Heavy Pressure of Population

The Indian agriculture is characterised by heavy pressure of population. About 58% of the total population of the country is dependent on agriculture. At present, the per capita agricultural land is only about 0.10 hectare as against 0.30 hectare in 1951. The world average of per head availability of agricultural land is about 4.5 hectares. The fast growth of population industrialization and urbanization are putting enormous pressure on arable land.

3. Predominance of Food Grains

In both the *Kharif* (summer) and the *rabi* (winter) seasons, grain crops occupy the greater proportion of the cropped area. In fact, rice, maize, millets, *bajra, ragi*, oilseeds, and pulses are the dominant crops in the *kharif* season, and wheat, gram, peas, lentil, and barley occupy over three-fourth of the total cropped area in the *rabi* season.

4. Mixed Cropping

In the rain-fed areas of the country, mixed cropping is a common practice. The farmers grow rice, mix millets, maize, and pulses in the *kharif* season and wheat, gram and barley in the *rabi* season. In the areas of *Jhuming* (shifting cultivation), ten to sixteen crops are mixed and sown in the same field. The rationale behind mixing of crops is to get good agricultural return. In case the monsoon is good, the rice crop will give better production and in case of failure of monsoon, the less water requiring crops like maize, millets, bajra, and pulses will give good harvest. Mixed cropping is a characteristic of subsistent agriculture.

5. High Percentage of the Reporting Area under Cultivation

In India, about 55% of the total reporting area is under cultivation of crops and pastures. This is much higher when compared with about 4% in Canada, 12% in China, 15% in Japan, and 16% in USA.

6. Small Size of Holdings and Fragmentation of Fields

Over 70% of the holdings are either small or marginal, i.e., less than one hectare. The small size of holdings is mainly due to the law of inheritance and other sociocultural and economic factors. Moreover, the fields are scattered and fragmented. The small size of holdings and fragmented fields are unsuitable for the modern methods of agriculture.

Table 9.3

Size of Holdings

Category	Area (Hectares)	Percentage of Holdings	No. of Operation Holdings (in Millions)	Percentage of Area	Measure in Hectares
Marginal	Below one	50.6	30	9.0	15
Small	1–2	20.0	14	11.9	19
Semi-medium	2–4	15.2	–	18.5	–
Medium	4–10	11.3	–	20.7	–
Large	Above 10	04.0	–	30.9	–

Source: Statistical Abstracts of India 2015–16.

It may be seen from (**Table 9.3)** that about 51% holdings of the country are marginal farmers having less than one hectare of land. About 20% are in the small category, and 11% in the medium category, while only 4% are the large category farmers, having more than 10 hectares. As per a report published in Economic Times, the Indian farm size has shrunk by 6% in 5 years to 2015–16.

7. Limited Intensive Agriculture

In India, only about one-third of the total cropped area is under double and multiple cropping. Increase in the double cropped area is difficult unless heavy investment is made in development of canal and tube-well irrigation.

8. Primitive Technology

Most of the farmers of the country, especially in the rain-fed areas, use draught animals (bullocks, male buffaloes, and camels) for ploughing and other agricultural operations. The health and efficiency of draught animals is low which often retards the timely operations of sowing, weeding, and harvesting.

9. Indian Agriculture is Labour Intensive

In India, agriculture is a labour based enterprise in which most of the agricultural operations, like ploughing, levelling, sowing, weeding, spraying, sprinkling, harvesting, and threshing are carried on mainly by human hands. The use of machinery is still confined only to the rich farmers of Punjab, Haryana, western Uttar Pradesh, plains of Uttarakhand, Bihar, Madhya Pradesh, Gujarat, and Maharashtra.

10. Rain-fed Agriculture

In the greater parts (over 55%) of the country, agriculture is largely dependent on rainfall, especially the summer monsoon. Unfortunately, the behaviour of summer monsoon is highly erratic. Consequently, the variability of rainfall is high which affects the agricultural return adversely. Only about 55% of the total cropped area is under irrigation in which the farmers are more confident about their agricultural returns even at the failure of monsoon, as it happened in 2009 and 2015.

11. Less Area under Leguminous and Fodder Crops

The nitrogen fixing crops like pulses are getting less area under their cultivation. Consequently, the natural fertility of the soil is depleting and

the soils are losing their resilience characteristics. Moreover, less than 4% of the cropped area is under fodder crops. This, together with lack of good pastures, has detrimental effect over the development of dairy farming and milk-production. India has the largest number of cattle in the world, but it occupies an insignificant place in respect of cattle products in the world.

12. Tradition Bound

By and large, the Indian agriculture is tradition bound. Established several centuries ago, the structures of a self-contained rural economy were founded in caste-derived occupational land tenures, made complex by absentee and parasitic landlords. These institutional factors and tradition bound institutions are a major obstacles in the path of innovations and modernisation of agriculture.

13. Low Productivity

One of the main problems of Indian agriculture is its low productivity. In comparison to the other agricultural countries, the Indian agricultural yields are among the lowest in the world (**Table 9.4) and (Table 9.5)**. The main cause of low yield per hectare is the low fertility of soil and less care to replenish it through green-manure, fertilisers, fallowing, and scientific rotation of crops. The consumption pattern of chemical fertilizers has been shown in (**Fig. 9.5)**. It may be seen from this figure that Punjab with 175 Kg/ha is the leading consumer of chemical fertilisers followed by Haryana 160 kg/ha. Uttar Pradesh, Andhra Pradesh, Tamil Nadu and West Bengal. In general the fertiliser consumption level is very low in the areas of dry farming.

Table 9.4

International Comparisons of Yield of Rice, 2014–15, (in metric tonnes/hectare)

Rice/Paddy (in Metric Tonnes/Hectare)	Yield	Rank
Egypt	9.8	1st
USA	7.3	2nd
Korea	6.73	3rd
Japan	6.42	4th
India	2.9	5th

Thailand	2.63	6th
World	3.96	

Source: Economic Survey 2016–17.

Table 9.5

International Comparison of Yield of Wheat, 2014–15, (in metric tonnes/hectare)

Country	Yield	Rank
UK	7.70	1st
France	7.58	2nd
China	4.25	3rd
India	2.71	4th
Pakistan	2.37	5th
Iran	2.06	6th
Australia	1.64	7th
World	2.87	

Source: Economic Survey 2016–17.

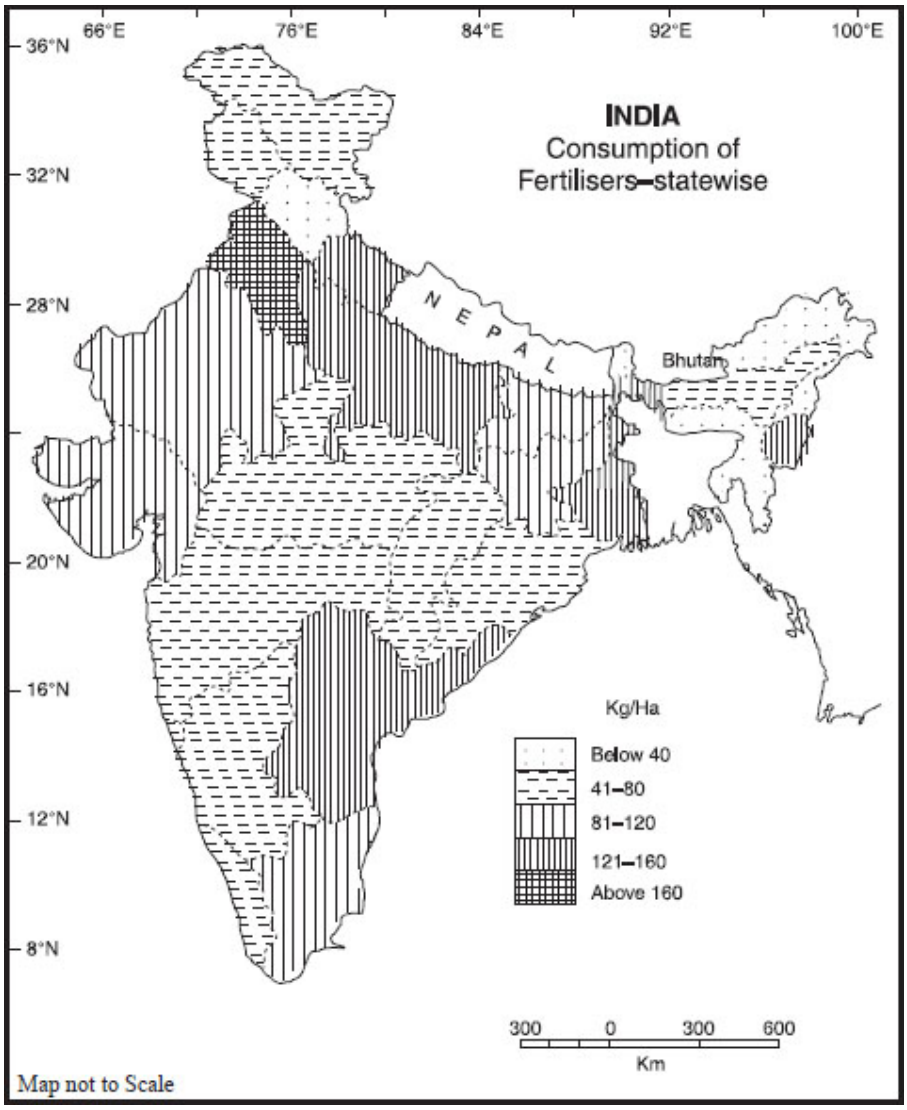


Fig. 9.5 Consumption of Fertilisers (2015–16)

14. Government Policy

After the First Five Year Plan, Indian agriculture got a step-motherly treatment. The farming community has been ignored, while there has been more emphasis on industrialisation and urbanisation. The growth rate of agriculture is only about 2.5%, while the overall growth rate of the country is about 8% (2016). The farmers are not getting remunerative prices, most of them are under debts and in several parts of the country, farmers are committing suicides. This dismal picture is the result of continuous careless agricultural land use planning. Much emphasis has, however, been laid on the rural and agricultural development in the

Eleventh Five Year Plan to remove the rural, urban inequality. Creation of 580 lakh jobs has also been proposed in this plan to overcome the problem of unemployment and to check the rural-urban migration. The real challenge for the government is in trying to boost food output at home, and increase investment in rural and agricultural infrastructure for the same, while at the same time not letting its guard down on fiscal prudence or inflation management. The severe drought of 2009 and 2016 over greater part of the country has and floods increased the miseries of the farmers, which is a set-back in the revival of Indian economy. Because of drought, and less effective price policy, thousands of farmers committed sucide in 2016.

The frequency of natural calamities has increased .The devastating earthquake of Gujarat is an example of national disaster. Special efforts in application of technologies are required. Effective information systems to increase the preparedness and managing the disasters are required.

The globalization of agricultural trade shall provide a method of employment and income generation. The liberalization of agricultural markets will be beneficial to the country in the long run.

The following main issues for sustainable development in the agricultural sector need to be considered while policy formulation.

1. Population pressure and demographic transition
2. Investment in agriculture, structural changes and impact on the poor.
3. Impact of globalization on the poor.
4. Rapid urbanization and declaration in rural poverty

The agricultural policy must lay emphasis in its vision 2020 on the following parameters:

1. Organization of agriculture
2. Sustainability and management of natural resources
3. Institutional changes
4. Investment priorities
5. Incentives
6. Risk management

15. Lack of Definite Agricultural Land Use Policy

In the absence of a definite land use policy, the farmers grow crops according to their convenience. This sometimes leads to excess of production and sometimes scarcity. Many a times the farmers have to burn their sugarcane crop and often get less remunerative price of vegetables (onion, potato, tomato, and other vegetables).

16. Lack of Marketing and Storage Facilities

Lack of marketing and storage facilities and the role of brokers deprive the farmers to fetch remunerative prices for their agricultural products. Except a few states like Punjab, Haryana, Maharashtra, Gujarat, and Andhra Pradesh, marketing and storage facilities are inadequate. In greater part of the country, farmers are still at the mercy of unscrupulous traders and are easily exploited by secret brokerage, false weights and payment of inflated commissions. Moreover, due to lack of proper pricing policy, farmers fail to obtain fair price for their agricultural products.

17. Low Status of Agriculture in the Society

In greater parts of India, agriculture is not considered as a dignified and honourable occupation. This leads to disappointment and lack of enthusiasm among most of the farmers. The younger generation of farmers prefer a petty government job to agriculture. Moreover, rich farmers invest their agricultural profits in non-agricultural sectors which are more remunerative. In fact, there is a mass exodus of people from rural to urban areas in search of lucrative jobs. There is a constant flow of human and material resources from villages to the cities. This has led to fast growth of urban centres which are infested with slums, ghettoes, and shanty colonies.

18. Land Tenancy

In many parts of the country, there are absentee land lords and the tillers do not have the rights on agricultural land. The big landlords who own big farm houses are rich urbanites. The tillers and share croppers who actually cultivate the land of absentee land lords are not much interested in the development, proper management, utilisation of agricultural land, and modernisation of agriculture. This system leads to lack of interest on the part of the tiller and consequently, the per unit yield of most of the crops is low.

19. Poverty and Indebtedness of the Farmers

Although cultivators indebtedness is universal in subsistent farming, its impact is perhaps nowhere as crushing as in India. Unfortunately, over 85% of all the cultivating families are under debt. It is because of heavy indebtedness that several thousand farmers in Andhra Pradesh, Karnataka, Tamil Nadu, Maharashtra, Odisha, Gujarat, Punjab, and Uttar Pradesh have committed suicide during the last ten years. The small and marginal farmers are still dependent on money lenders who charge exorbitant interest on loans (25 to 40% per annum). In the case of non payment, the money lenders grab their mortgage property making them pauper. Some special provisions have been made in the draft of the Eleventh Five Year Plan to overcome the problem of farmers indebtedness. A scheme of debt waiving for small and marginal farmers and debt relief for other farmers was announced by the government in the Union Budgets of 2008–09, and 2016–17.

20. Inadequacy of Extension Service

For the diffusion of agricultural innovations both in the irrigated and rain-fed areas, a team of skilled village level workers is required. There is much to be done in this area. Training of workers and their dedication can help the tradition bound farmers to modernise their agriculture.

21. Inadequate Agricultural Research and Education, Training, and Extension

Though enough progress has been made in the field of agricultural research, there is no co-ordination between the farm and research laboratories in the different agro-climatic regions of the country. Hence, gains of new agricultural researches are not reaching the common cultivators, especially the marginal and small farmers. Very little attention is being paid for educating and training farmers for the adoption of new agricultural innovations and techniques to increase their agricultural production.

22. Soil Erosion and Soil Degradation

Soil erosion is a universal phenomenon. It is, however, significantly high in the areas of heavy rainfall with undulating topography and in the areas of scanty rainfall (deserts and semi-desert areas). The indiscriminate felling of trees, cattle grazing, unscientific land use practices have greatly accelerated the rate of soil erosion in the different parts of the country.

Although soil conservation programmes were initiated in 1953, their impact has not been very encouraging. The people's awareness and their active participation in the soil conservation is essentially required.

23. Other Characteristics and Problems

There are numerous other problems also which are affecting the agricultural production and rural economy and society adversely. For example, unscientific methods of agriculture, inadequate irrigation facilities, less use of chemical fertilisers, insecticides, pesticides, less remunerative prices of agricultural products, poverty, hunger, and malnutrition of farmers and lack of infrastructural facilities like roads, water, irrigation, electricity, credit, banking, and crop insurance.

DETERMINANTS OF AGRICULTURE

The agricultural practices, cropping patterns and their productivity are closely determined by the geo-climatic, socioeconomic, and cultural-political factors. In fact, the agriculture of any region is influenced by the following factors:

1. **Physical factors:** Terrain, topography, climate, and soil.
2. **Institutional factors:** Land tenure, land tenancy, size of holdings, size of fields, and land reforms.
3. **Infrastructural factors:** Irrigation, electricity, roads, credit and marketing, storage facilities, crop insurance and research.
4. **Technological factors:** High Yielding Varieties (new seeds), chemical fertilisers, insecticides, pesticides, and farm machinery.

These factors individually and collectively have their impact on the cropping patterns, level of agricultural development and yield of crops in a region. A brief account of these factors has been given below.

1. Physical Factors

(a) Terrain, Topography, and Altitude

The agricultural patterns are strictly dependent on the geo-ecological conditions; terrain, topography, slope and altitude. While paddy cultivation requires leveled fields, tea plantations perform well in the undulating topography in which water does not remain standing. Orchards of coconut are found at low altitudes, preferably closer to the

sea level, while the apple orchards in the tropical and subtropical conditions perform well above 1500 m above sea level. Moreover, cultivation of crops is rarely done 3500 m above sea-level in the tropical and subtropical latitudes. The highly rarified air, low-pressure, low temperature, and shortage of oxygen at high altitudes are the serious impediments not only in the cultivation of crops, but also in keeping dairy cattle. The soils of high mountainous tracts are generally immature which are also less conducive for agriculture. The topographical features also affect the distribution of rainfall. Normally, the windward side gets more rainfall than the leeward side. The amount of rainfall received in a region determines the selection of crops to be sown.

Apart from altitude and aspects of slope, the nature of the surface also affects the agricultural activities. The gullied land is least conducive for cropping. The Chambal ravines in Madhya Pradesh, Rajasthan, and Uttar Pradesh have put over thousands of hectares of good arable land out of agriculture.

(b) Climate

Of all the physical factors, climate is one of the most significant determinant of agricultural land use and cropping patterns.

(i) **Temperature:** The crops to be grown, their patterns and combinations are closely controlled by the prevailing temperature and precipitation conditions. The agricultural scientists have proved that each crop has a specific zero temperature below which it cannot be grown. There is also an optimal temperature in which the crop is at its greatest vigour. For each stage of crop life, i.e., germination, foliation, blossoming or fructification a specific zero and optimum can be observed in temperature.

The upper limit of temperature for plants growth is 60°C under high temperature conditions, i.e., at over 40°C, crops dry up, if the moisture supply is inadequate. In contrast to this, the chilling and freezing temperatures have a great adverse effect on the germination, growth and ripening of crops. Crops like rice, sugarcane, jute, cotton, chilli and tomatoes are killed or damaged at the occurrence of frost. The minimum temperature for wheat and barley is 5°C, maize 10°C, and rice 20°C.

The impact of temperature on cropping patterns may be seen

from the fact that the northern limit of the regions in which date-palm bear ripe fruit coincides almost exactly with the mean annual temperature of 19°C. The essential factor in the limit of grape orchards seem to be temperature. Grapes ripen only in those countries in which the mean temperature from April to October exceeds 15^{o} C. Crops like winter-wheat and barley perform well when the mean daily temperature ranges between 15^{o}C and 25^{o}C. Contrary to this, tropical crops like cocoa, coffee, spices, squash, rubber and tobacco require over 18^{o} C temperature even in the coldest months, while crops like wheat, gram, peas, lentil, potato, mustard, and rapeseed require a temprature of about 20^{o}C during the growth and development, stage and relatively higher (over 25^{o}C) during the sowing and harvesting periods.

(ii) **Moisture**: All crops need **moisture**. They take water and moisture from the soil. This moisture may be available from the rains or from irrigation systems. Within wide temperature limits, moisture is more important than any other climatic factor in crop production. There are optimal moisture conditions for crop development just as there are optimal temperature conditions. The excessive amount of water in the soil alters various chemical and biological processes, limiting the amount of oxygen and increasing the formation of compounds that are toxic to plant roots. The excess of water in the soil, therefore, leads to stunted growth of plants. The problem of inadequate oxygen in the soil can be solved by drainage practices in an ill-drained tract.

Heavy rainfall may directly damage plants or interfere with flowering and pollination. Cereal crops are often lodged by rain and this makes harvest difficult and promotes spoilage and diseases. Heavy rainfall at the maturity stage of wheat, gram, millets, oilseeds, and mustards causes loss of grains and fodder. Indian farmers all over the country have often suffered on account of failure of rains or fury of floods.

(iii) **Drought: Drought** has devastating consequences on the crops, their yields and production. Soil drought has been described as a

condition in which the amount of water needed for transpiration and direct evaporation exceeds the amount of water available in the soil. Drought damages the crops when plants are inadequately supplied with moisture from the soil.

The drought prone areas of India lie in the states of Rajasthan, Gujarat, Madhya Pradesh, Chhattisgarh, Jharkhand, Maharashtra, Andhra Pradesh, Karnataka, Tamil Nadu, Odisha, Bundelkhand (U.P.), Uttarakhand, Himachal Pradesh, J&K, south-west Punjab, and Haryana. In the areas where the average annual rainfall is less than 75 cm, agriculture is considered a gamble on monsoon. In 2009, the erratic monsoon resulted into drought in more than 200 districts of the country.

The incidence of drought and its intensity can be determined from the annual, seasonal, and diurnal distribution of rainfall. Moreover, different plants have different moisture requirements. In the drought prone areas of India, dry farming is practiced, while in the more rainfall recording regions, intensive agriculture of paddy crop is a common practice.

(iv) **Snow**: The occurrence of **snow** reduces the ground temperature which hinders the germination and growth of crops. Land under snow cannot be prepared for sowing because of permafrost. Melting of snow may cause hazardous floods in the summer season, affecting the crops, livestock, and land property adversely.

(v) **Winds:** Winds have both, direct and indirect effects on crops. Direct winds result in the breaking of plant structure, dislodging of cereals, fodder and cash crops and shattering of seed-heads. Fruit and nut crops may be stripped from the trees in high winds. Small plants are sometimes completely covered by wind-blown dust or sand. The indirect effect of winds are in the form of transport of moisture and heat in the air. In fact, the movement of winds increases evaporation and transpiration.

(c) Soils

In agricultural operations, soil is probably the most important determining physical factor. It determines the cropping patterns, their associations and production. The fertility of soil, its texture, structure and humus

contents have a direct bearing on crops and their productivity. In general, the alluvial soils are considered to be good for wheat, barley, gram, oilseeds, pulses, and sugarcane; while the clayey loam gives good crop of rice. *Regur* soil is known for cotton, and sandy soil for *bajra*, guar, pulses (green-gram, black-gram, red-gram, etc.). The saline and alkaline soils are useless from the agricultural point of view, unless they are reclaimed by chemical fertilisers and biological manures and fertilisers.

2. Institutional Factors

The institutional factors include land tenancy, land-tenure, and ownership. These factors have their bearing on field size, field patterns, farming type, crop land use, crop associations, and productivity of crops. A brief account of the institutional factors has been given in the following:

(i) **Land Tenure and Land Tenancy:** The ownership of agricultural land is determined by the law of land tenure and land tenancy. In the primitive societies like those of the shifting cultivators (*Jhumias*), land belongs to the community. Subsequently, in India, the land ownership rights were vested in the king and the government. The British Government introduced the land settlement in 1793. The Permanent Settlement was introduced in the Province of Bengal and then gradually extended to other states. Later on the Mahalwari System was introduced in Agra and Awadh and the Ryotwari System was introduced in Bombay and Madras. The new land system of British created a class of absentsee landlords, making way for exploitation of the peasants. Thus, the new land systems introduced by the Britons led to the concentration of economic power in the hands of few. This resulted total depression in agriculture and industry. After Independence, the Zamindari System was abolished and the right of the tillers over cultivated land were restored.

After the abolition of Zamindari system, a number of land reform legislations have been passed by the central and state governments, but still there is enough scope to restore the rights of the actual tillers and landless labourers. Still, there are numerous absentee landlords, many of them possessing land more than the ceiling act permits. Only Punjab, Haryana, and

Uttar Pradesh are the states in which consolidation of holdings has been completed.

(ii) **Land Holding:** In India the size of holding is too small. Due to the rapid growth of population during the last few decades and the existing law of inheritance, the agricultural land is divided equally among the male children of the deceased farmer. At present, the per capita available land is only about 0.10 hectare which is much below the world average of about 4.50 hectares. Over 70% of the land holdings are less than one hectare. Such small holdings are not economically viable. In fact, small holdings cannot produce enough to meet the costs of irrigation, improved seeds, chemical fertilisers, insecticides, pesticides and farm machinery. The average size of holdings and their percentage share in India have been given in the (**Table 9.6) (Fig. 9.6).**

Table 9.6

The Average Size of Land Holdings in the Selected States of India, 2015–16

State	Size of Holdings (Hectares)	No. of Size Holdings (in Thousands)
Nagaland	5.00	197
Rajasthan	4.00	7655
Punjab	3.45	1093
Gujarat	2.85	5320
Madhya Pradesh	2.75	10003
Haryana	2.45	1628
Maharashtra	2.40	14707
Karnataka	2.15	8677
Andhra Pradesh	1.50	8524
Odisha	1.30	4866
Himachal Pradesh	1.15	996
Assam	1.15	2742
Tamil Nadu	1.00	7938
West Bengal	0.80	7243
Jammu Kashmir & Ladakh	0.75	1417

Uttar Pradesh	0.72	23822
Bihar	0.70	16413
Kerala	0.18	7583
India	1.08	145727

Source: The Hindu, Oct 3, 2018; Agricultural centus (2015–16)

An examination of (**Table 9.6)** shows that the average size holding in India is about 1.08 hectares which is too small for mechanisation and application of modern technology. The largest size of operation holdings is in Nagaland being 5.00 hectares, Rajasthan being 4.00 hectares, followed by Punjab at 3.45 hectares, and Gujarat at 2.85 hectares. The lowest size of land holding is in Kerala, being only 0.18 hectares and Uttar Pradesh 0.72 hectares. In rest of the states, it varies between 0.8 to 2.50 hectares. The number of small and marginal land holdings (operational landholdings) has shown a marginal increase. The percentage of land holders who are women has increased from 12.79% in 2010–11 to 13.87% in
2015–16, indicating an increased participation of women in the field of agriculture.

The marginal, small and medium land holdings constitute the major share of 86.21% of the holdings whereas the large land holdings account for only 9% of the total operational area. It is highest in Nagaland (5 hectares) and lowest in Kerala (0.18 hectares).

However the average size of agricultural land has declined to 1.08 hectares in 2015–16. On the whole, in most of the states, the size of holdings is not economically viable.

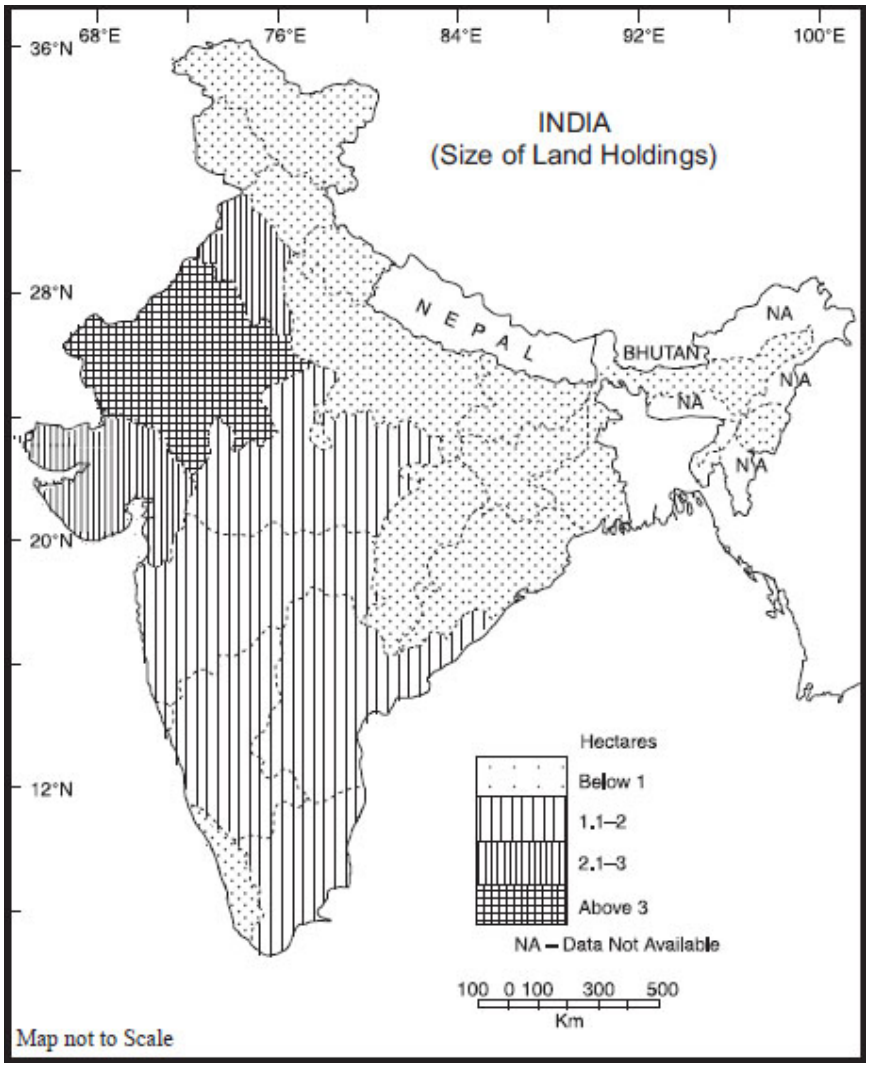


Fig. 9.6 Size of Holdings (2014–15)

LAND REFORMS

The basic objective of land reform is to do social justice with the tillers, land owners, landless labourers, and rural community with the set objective to provide security to the cultivators, to fix a rational rent, the conferment of title to the tiller and to increase the agricultural productivity. The entire concept of land reforms aims at the abolition of intermediaries and bringing the actual cultivator in direct contact with the state. The scheme of land reforms includes:

(i) abolition of intermediaries,

(ii) land tenancy reforms, i.e., regulation of rent, security of tenure

for tenants, and confirmation of ownership on them,

(iii) ceiling on land holdings and distribution of surplus land to landless labourers and small farmers,

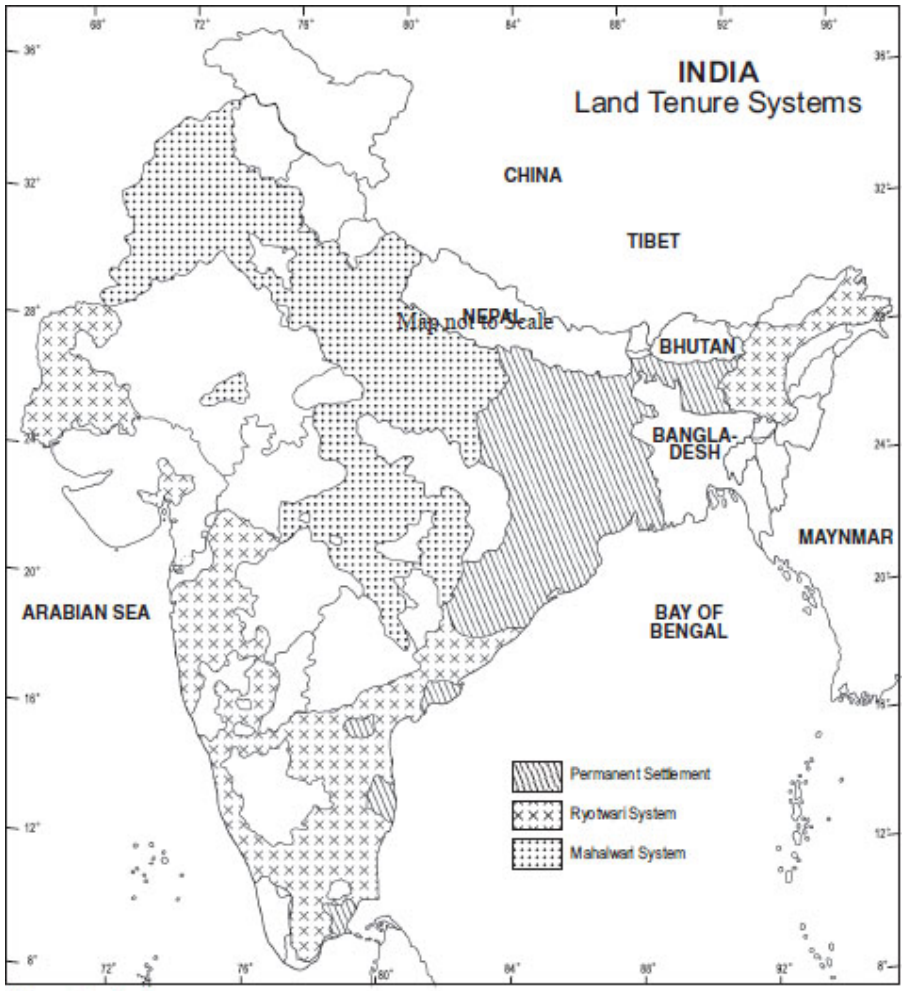
(iv) agrarian reorganisation including consolidation of holdings and prevention of subdivision and fragmentation,

(v) organisation of co-operative farms, and

(vi) improvement in the system of land record keeping.

1. Abolition of Intermediaries

During the British period, three categories of land tenure systems, namely, *Zamindari, Mahalwari,* and *Ryotwari* came into existence. (**Fig. 9.7**)



Map not to Scale

Fig. 9.7 Net Sown Area under Irrigation (2005–06)

(i) **Zamindari System:** Under the Zamindari system, which was introduced by Lord Cornwallis in 1793 in Bengal, land was held by one person or at the most by a few joint owners who were responsible for the payment of land revenue (*Malguzari*). Under the Zamindari system of tenure, these revenue collectors were raised to the status of land owners. The Zamindari settlements were of two types: (i) permanent with fixed land revenue in perpetuity, and (ii) temporary in which land revenue used to be assessed for a period ranging between 20 to 40 years and was subject to revision. Thus, between the state and the actual tiller, there grew an intermediary who was interested in land only to the extent of extraction of exorbitant rent. So, Zamindari symbolised oppression and tyranny and agriculture was degraded to subsistence farming with low productivity.

(ii) **Mahalwari System:** This system of tenure is the product of Muslim tradition. Under the Mahalwari tenure, the village agricultural lands were jointly held by the village communities, the members of which were jointly and severally responsible for the payment of land revenue. The system was first introduced in Agra and Oudh and later in undivided Punjab. Here, the village *Lumbardar* collected revenue for which he received 5% as commission.

(iii) **Ryotwari System:** The Ryotwani tenure system was first introduced in 1772 in Madras and later on this system was also extended to Bombay, Berar, and Madhya Pradesh. This tenure system was an outcome of Hindu tradition. Under Royatwari tenure, every individual registered; and holder is recognised as the proprietor of land is responsible for the payment of land revenue to the government. The Ryot was at liberty to enjoy permanent rights over land and to sub-let it so long as he paid the land revenue to the government.

In order to stop the tyranny and exploitation by Zamindars and to restore tiller's right over the land, measures were taken for the abolition of intermediaries after independence. The Zamindari Abolition Acts were passed in different states of the country between 1948 and 1955. As a consequence, over 260,000 Zamindars were abolished.

Due to several loopholes in the Zamindari Abolition Act, the land could not be transferred to the actual tillers and the landless agricultural labourers. In certain parts of the country, the intermediaries are yet to be abolished, i.e., **water rights in Bhagalpur (Bihar), Jotdari in Meghalaya, Communided in Goa, Trust Estate (Zamindari of religious institutions) in Odisha, and Devasthan Enam in Maharashtra**.

2. Tenancy Reforms

The term land reform means reforms of institutional factors. These institutional factors include land tenure system, land holdings, size of fields, land distribution land ceiling, rend and intermediaries, etc.

During the British period, the *Zamindari, Mahalwari*, and *Ryotwari* systems, tenancy cultivation was quite common in India. The small and marginal farmers as well as the landless labourers were interested in doing cultivation in the fields of Zamindars. There were three types of tenants, namely: (i) occupancy or permanent tenants with permanent and heritable rights, (ii) tenants at will or temporary tenants, and (iii) sub-tenants or *Shikmi-kisan.* The condition of temporary and sub-tenants was adhoc and they were subject to ruthless exploitation. Frequent enhancement of rent, eviction on petty grounds and *begari* (free service) were some of the prevalent ways of exploitation. In India about 20% of the agricultural land is devoted to share-cropping (*batai*), where 50% of the produce is the normal rent. On several occasions, the peasant have to forego even two-thirds of the produce as rent. Under the land tenancy reforms, more than 11.5 million cultivators have been given tenancy rights. The tenancy reforms cover the following points:

(i) regulation of rent,
(ii) security of tenure, and
(iii) conferment of ownership on tenants. States have made the following provisions to achieve the third objective:
    (a) All tenants have been given full security of tenure, without giving the owners the right of personal cultivation.
    (b) Owners have been given the right to resume a limited area (not more than a family holding in any case) subject, however, to conditions that a minimum area is left with the tenant.

(c) A limit has been placed on the extent of land with a land-owner may resume, but the tenant is not entitled to retain minimum area of cultivation in all cases.

3. Rent Control

At the time of independence the rate of rent on agricultural land was 50% or more of the agricultural produce. It was too high for ordinary farmers. Hence, legislation were enacted to bring down the rate of rent to 20 to 30% in Andhra Pradesh, 33% in Punjab and Haryana, and 33 to 40% in Tamil Nadu. Since the Fourth Five Year Plan, the system of payment of rent in agricultural produce was fully abolished and the same was replaced by the system of cash payment.

4. Ceiling of Landholdings

Under the land reforms programme, it was envisaged that beyond a certain specified limit, all lands belonging to the landlords would be taken over by the state and allotted to small proprietors to make their holdings economic or to landless labourers to meet their demand for land. Ceiling on landholdings is, therefore, an effective measure for redistribution of land and achieving the goal of social justice. The ceiling legislations were initiated in late 50s and 60s, with Jammu & Kashmir being the first state to pass the act on ceiling. It was passed in West Bengal, Himachal Pradesh and Maharashtra in 1961.However after criticism guidelines were issued in 1972 to decide on ceiling limits. The land ceiling limits, both in the irrigated and dry lands, vary from state to state have been given in (**Table 9.7):**

The ceiling on land holdings was intended to:

(i) Meet the land needs of the landless

(ii) To reduce the inequalities in land ownership for the development of co-operative rural economy.

(iii) To increase self employment in owned land as distinguished from subletting and tenant cultivation

It may be observed from (**Table 9.7)** that in J&K and Kerala, only 3.6 and 4.86 hectares, respectively, are the ceiling limits of irrigated agricultural land, while in Uttar Pradesh and Haryana it is about 7.30 hectares for the irrigated about 18 hectares for the unirrigated drylands. The unirrigated agricultural land ceiling limit is over 24 hectares in

Rajasthan, over 20 hectares in Tamil Nadu, about 22 hectares in Jammu and Kashmir state, and over 21 to about 71 hectares in Haryana and Punjab, respectively. The actual ceiling limits for lands having two crops and single crop respectively irrigated in Karnataka and Uttar Pradesh are slightly higher owing to land classification whereas in dry land in Himachal Pradesh and Rajasthan are higher due to hilly terrain and being desert

Table 9.7

India—Ceiling Limits on Land Holdings in Hectares-1972

State/U.T.	Irrigated with Two Crops	Irrigated with One Crop	Dryland
As Suggested in National			
Guidelines of 1972	5.05–7.28	10.93	21.85
Actual Ceiling			
Andhra Pradesh	4.05–7.28	6.07–10.93	14.16–21.85
Assam	6.74	6.74	6.74
Bihar	6.07–7.28	10.12	12.14–18.21
Gujarat	4.05–7.28	6.07–10.93	8.09-21.85
Haryana	7.25	10.90	21.80
Himachal Pradesh	4.05	6.07	12.14–28.33
Jammu Kashmir & Ladakh	3.6–5.06	…	21.85
Karnataka	7.28	10.93	21.85
Kerala	4.86–6.07	4.86–6.07	4.86

			– 6.07
Madhya Pradesh	7.28	10.93	21.85
Maharashtra	7.28	10.93–14.57	21.85
Manipur	5.00	5.00	6.00
Odisha	4.05	6.07	12.14–18.21
Punjab	7.28	10.93	21.85–70.82
Rajasthan	4.86	12.14	24.28
Tamil Nadu	5.06	…	20.23
Sikkim	5.06	…	20.23
Tripura	4.00	4.00	12.00
Uttar Pradesh	7.30	10.95	18.25
West Bengal	5.00	…	7.00

Source: Agricultural Statistics at a Glance—Ministry of Agriculture.

5. Consolidation of Holdings

Consolidation of holdings was one of the important steps towards land reforms. It was envisaged with the set objective to increase the agricultural efficiency and production of all category of farmers. Consolidation of holdings means to bring together in compact block, all the fields of land of a farmer which are well scattered in different parts of the village. Under the scheme, all land in the village is first pooled into one compact block and it is divided into smaller blocks called ***chaks***, and allotted to individual farmer. This is a useful scheme which helped in overcoming the problem of fragmentation of holdings. But unfortunately, the scheme has not been implemented in all the states of the country. There are many hurdles in the implementation of consolidation of

holdings in some of the states. Some points which are coming in the way of implementation of consolidation of holdings are as under:

(i) Farmers are emotionally attached to their ancestral land, and therefore, they are not willing to take advantage of the scheme of consolidation of holding.

(ii) Those farmers who own good quality of land do not like the scheme for fear of getting the inferior and poor quality of land after the consolidation.

(iii) Consolidation of holdings is a cumbersome process. The government officials who implement the scheme are generally slow and often corrupt.

(iv) In general, the scheme did not receive the desired support and co-operation from the farmers.

(v) The scheme has paved way for litigation and court cases, many of which are pending in different courts for a long time. This vitiates the serene atmosphere of the rural areas.

(vi) Under the existing law of inheritance, the fields continue to be smaller and fragmented.

(vii) In every consolidation, about 5 to 10% of the village land is taken out for providing house sites to the weaker sections of society, approach roads (*chak-roads*) and village utility services. Hence, if the process is repeated three or four times, a sizable portion of the agricultural land would go out of agriculture.

(viii) The cost of consolidation is realised from the farmers which has adverse effect on their resources and economy.

(ix) It has been observed that the small farmers are generally allotted inferior quality of land, and due to lack of money power, they are neither able to please the officials nor get justice in the court.

Looking at these drawbacks, efforts should be made to remove these barriers and pitfalls in the scheme of consolidation of holdings to modernise the system of keeping revenue records. In the Seventh Five Year Plan emphasis was laid on (i) scientific survey of the un-surveyed land, (ii) registering the name of tenant and share cropper in land records, (iii) strengthening the revenue system at the lowest level, and (iv) providing training facility to revenue officials to improve their efficiency. During the Eighth Five Year Plan, it was decided to use

computer and new techniques for keeping and maintaining and updating revenue records, proper recording of land rights, tenancy and causes of rural unrest and restriction on the sale and purchase of agricultural land for non-agricultural purposes. Land reforms also include computerisation of records of rights with maps, and land passbook.

Thus, a number of legislations have been enacted in the country for land reforms after independence, but due to socioeconomic and cultural complexities, loopholes in the land reform laws, laxity in implementation, and political and legal interference, these land reforms have not been able to achieve the desired success.

6. Computerised Land Records

The centrally sponsored scheme on computerisation of land records was started in 1988–89. At present, the scheme has been implemented in 582 districts out of the 640 districts of the country, leaving those districts where there are no proper land records. The successful implementation of land reforms needs a strong and efficient administrative machinery in villages where officials are easily influenced by powerful vested interests.

The land reform (LR) division has implemented two centrally sponsored schemes:

(i) Computerisation of Land Records (CLR).

(ii) Strengthening of Revenue Administration and Updating of Land Records (SRA & ULR).

In 2008, the two schemes have been merged into one Scheme named as ‘Digital India Land Records Modernization Programme’ (DILRMP).It shall update the land records and also integrate the textual and spatial records. It shall inter connect the revenue and registration as well.

INFRASTRUCTURE AND AGRICULTURAL INPUTS

Provision of quality and efficient infrastructure is essential to realise the full potential of agriculture. In other words, infrastructural development is imperative for the agricultural development of a country/region. Infrastructure includes the facilities of irrigation, availability of electricity, roads, marketing credit facilities and crop insurance.

Irrigation

The process of supplying water to crops by artificial means such as canals, tube wells, tanks, etc., is known as irrigation. In greater part of India, agriculture is rain-fed. In the incidence of failure of monsoon, the crop fails. The behaviour of Indian monsoon is highly erratic. Excess rainfall may cause floods, but scanty rainfall may reduce the crop yield substantially, and in acute cases the crop may be a complete failure. This problem may be solved by increasing the irrigated area in the country. Irrigation helps in bringing the new area under cultivation on the one hand, and increases the double and multiple cropping on the other. Moreover, the per hectare yield of the irrigated area is much higher to that of the unirrigated areas. About 84% of the water resources of India is used for irrigation. The net sown area in the country in 2015–16 was 140,130 thousand hectares whereas the net sown area was 68,383 thousand hectares giving a percentage of 48.79.

Sources of Irrigation

Depending on the availability of surface and underground water, slope of the land, nature of the soil, and the types of crops grown in a region, a number of sources of irrigation are utilised. The main sources of irrigation used in different parts of the country are: (i) canals, (ii) wells and tube-wells, (iii) tanks, and (iv) other sources (springs, *kuhls, Himachal Pradesh*, *swing-basket; dhenkli*, *dongs Assam*, and *bokka Telangana*, etc.). The areas irrigated by the different sources have been plotted in (**Fig. 9.8)** while (**Table 9.8)** gives the temporal changes in the irrigated area under different sources.

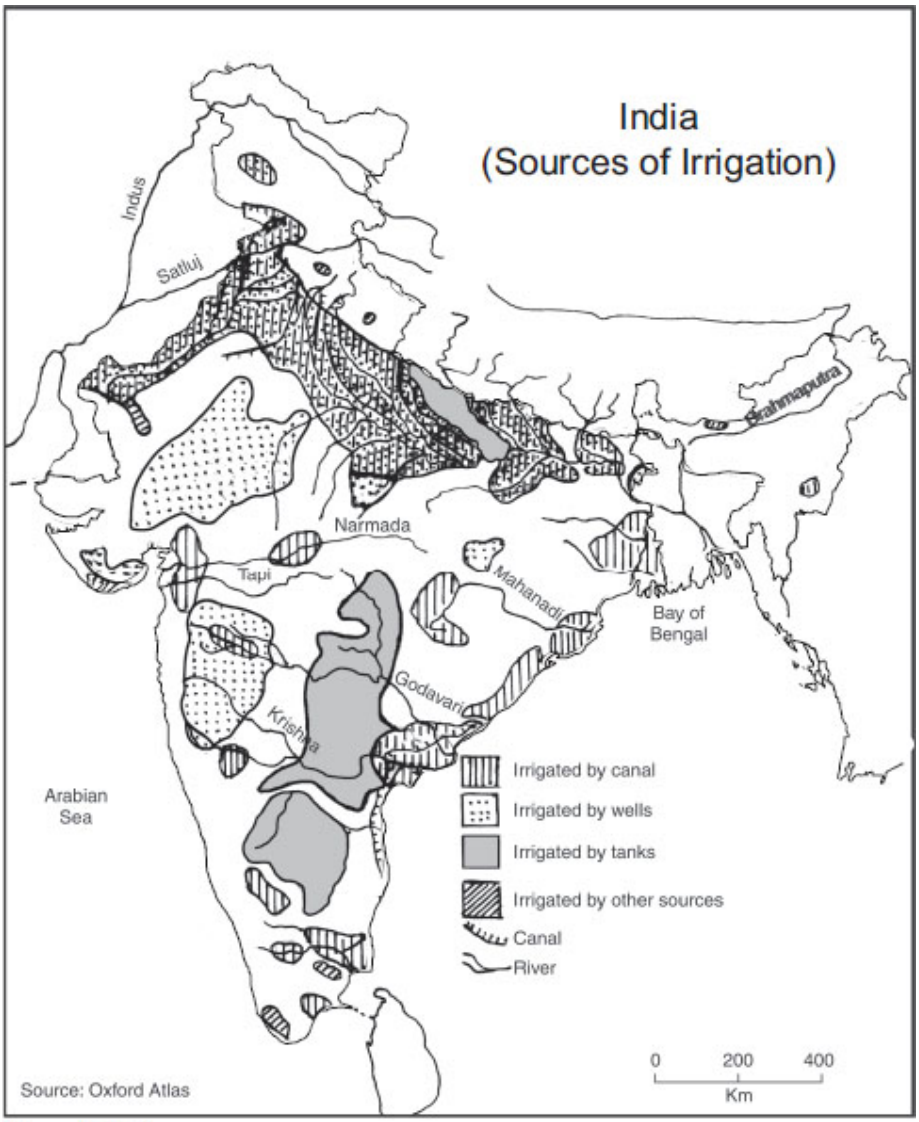


Fig. 9.8 India: Sources of Irrigation

Table 9.8

Area and Sources of Irrigation (area in thousand hectares)

Year	Canals	Wells, including tube-wells	Tanks	Other sources	Total
1950–51	(44.0%)	(31.7%)	(19.1%)	(5.20%)	(32.0 %)
1990–91	(29.17%)	(60.91%)	(4.53%)	(5.37%)	(55.7%)
2000–	(28.98%)	(61.07%)	(4.63%)	(5.32%)	(100.00%)

01					
2014–15	25.00%	63.15%	4.50%	6.35%	100.00%

Source: Statistical Abstracts of India, 2014–15.

It may be seen from (**Table 9.8)** that only 18,855 thousand hectares of the cropped area was under irrigation in 1950–51 which increased to 68383 thousand hectares in 2015–16, an increase of almost three times. While the canal area has been doubled over the given period, the area under wells and tube-well irrigation has increased by more than ten times. Despite these achievements, a lot of area of the agricultural land of the country is un-irrigated and needs a controlled and assured supply of moisture to the crops through irrigation.

The statewise net sown area under irrigation has been marked in (**Fig. 9.9)**. An examination of (**Fig. 9.7)** shows that Punjab cover 95% and Haryana cover 85% have the highest percentage of irrigated area followed by Uttar Pradesh 87% and Tamil Nadu—between 50 to 75%. In the remaining states the irrigated area is below 50% **(Fig. 9.9)**.

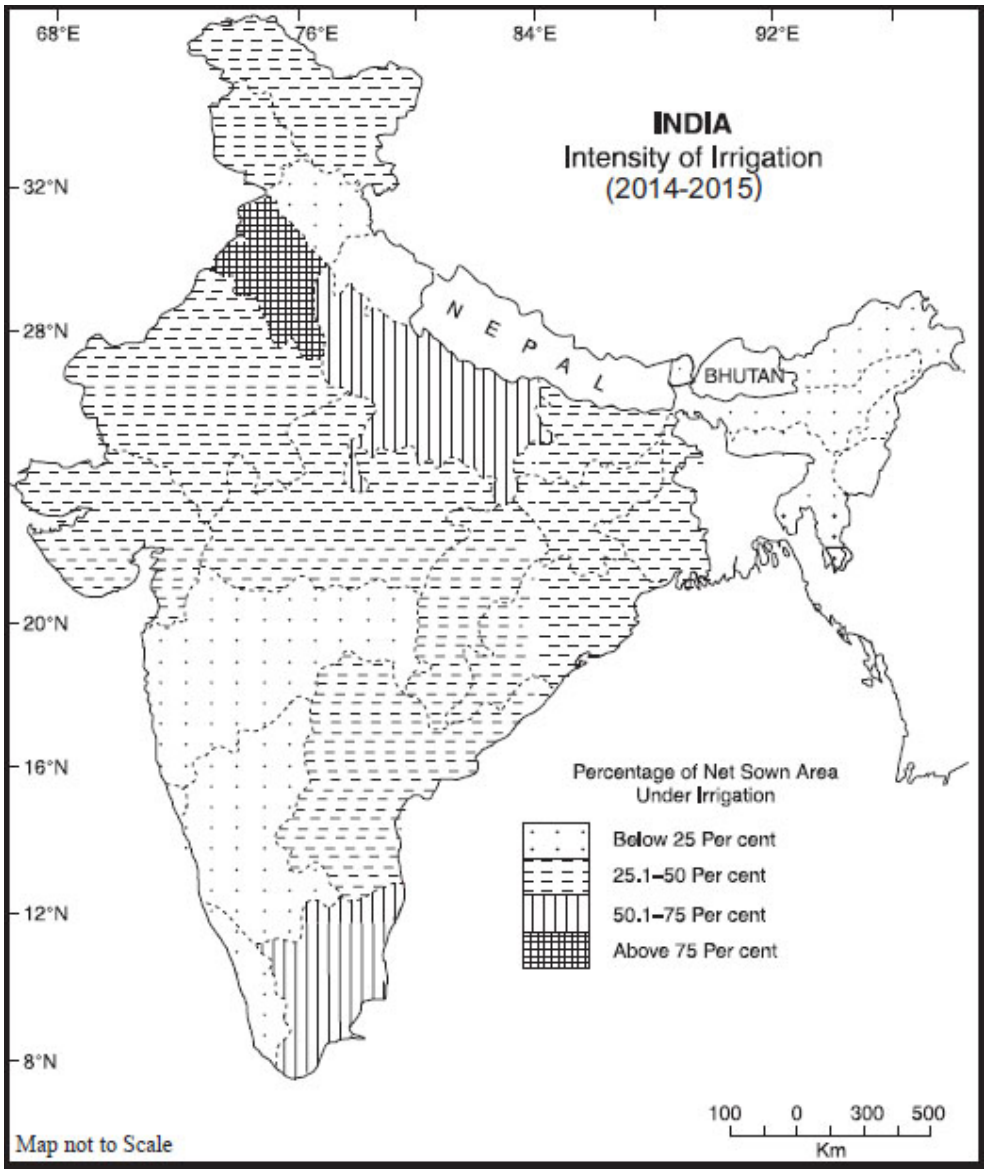


Fig. 9.9 Net Sown Area under Irrigation (2014–15)

Canal Irrigation: Canals used to be the main source of irrigation in 1950–51, irrigating almost 50% of the total irrigated area, but in the Third and Fourth Five Year Plans, there was a tremendous increase in the tube-well irrigated area. Consequently, the percentage of canal irrigated area declined to less than 29% in 2000–01, wells and tube-wells emerged as the major source of irrigation, covering over 63.15% of the irrigated area.

Canals are an effective source of irrigation in the areas of low and leveled relief, productive plain areas where perennial source of surface

drainage is available. These conditions are ideally found in the Northern Plains of India, Kashmir and Manipur Valleys and the Eastern Coastal Plains of India. The main concentration of canals in India is found in Uttar Pradesh, Punjab, Haryana, and western Rajasthan. After the construction of multi-purpose projects, a number of small canals have been dug in the Damodar, Mahanadi, Godavari, Krishna, Kaveri, Narmada, Tapi rivers, and their tributaries. Some of the important irrigation projects have been shown in (**Fig. 9.9)**.

It may be seen from **(Table 9.9)** that in India, about a little over 29% of the total irrigated area was under canal irrigation. Uttar Pradesh with 3090 thousand hectares under canals has the first rank in canal irrigation, followed by Andhra Pradesh with 1650 thousand hectares, and Haryana with 1475 thousand hectares. The states of Rajasthan and Bihar rank fourth and fifth in canal irrigated area. The rank of Punjab was however, 12th in canal irrigation, followed by Gujarat (**Table 9.9).**

Table 9.9

Net Area under Canal Irrigation—(2014–15)

S. No.	State	Net Area Under Canal Irrigation ('000 Hectares)	Percentage of Canal Irrigated Area to Net Irrigated Area of the State/Country	Percentage of Canal Irrigated Area in the State to Total Irrigated Area of India
1	Uttar Pradesh	3090	25.42	21.33
2	Andhra Pradesh	1650	36.42	10.83
3	Haryana	1475	49.89	9.68
4	Rajasthan	1355	27.59	8.90
5	Bihar	1135	31.34	7.45
6	Maharashtra	1050	35.38	6.89
7	Karnataka	975	36.55	6.40
8	Odisha	875	24.38	5.74
9	Tamil Nadu	830	28.81	5.44
10	Madhya	825	19.49	5.41

	Pradesh			
11	Chhattisgarh	800	68.90	4.25
12	Punjab	675	18.76	4.43
13	Gujarat	495	16.52	3.25
	India	**15,230**	**29.24**	**100.00**

Source: Statistical Abstracts of India–2016.

The existing irrigated area and the sources of irrigation are not adequate to meet the growing demand. There are a number of problems Indian agriculture is facing. There is a gap between the developed and utilised irrigation potential. Moreover, in the case of big irrigation projects like canals, there are interstate disputes. There is wastage of water, water-logging in canal command areas and other ecological problems. Hence, there is a need for a judicious and scientific development of irrigation, full utilisation of its potential and rational pricing of irrigation water.

Wells and Tube-Wells: This type of irrigation has been practised in India since the time immemorial. The widely used methods for the utilization of underground water are persian-wheel *Rahat*, *Charas*, or *mot* and tube-wells. Out of these, tube-well irrigation is more popular in the Satluj-Ganga plains. Tube-well irrigation accounts for more than 62% of the total irrigated area of the country.

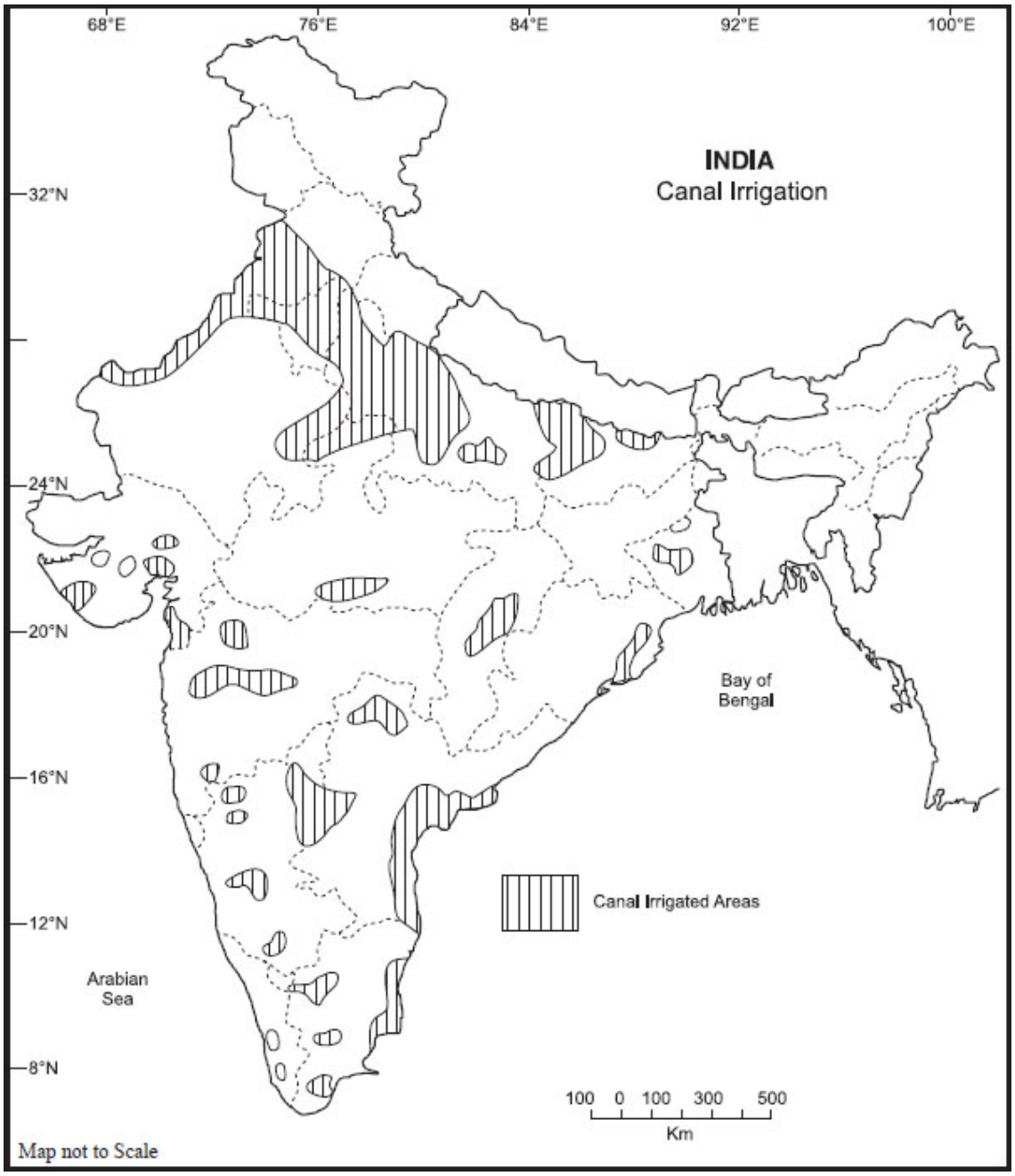


Fig. 9.10 Canal Irrigation (2014–15)

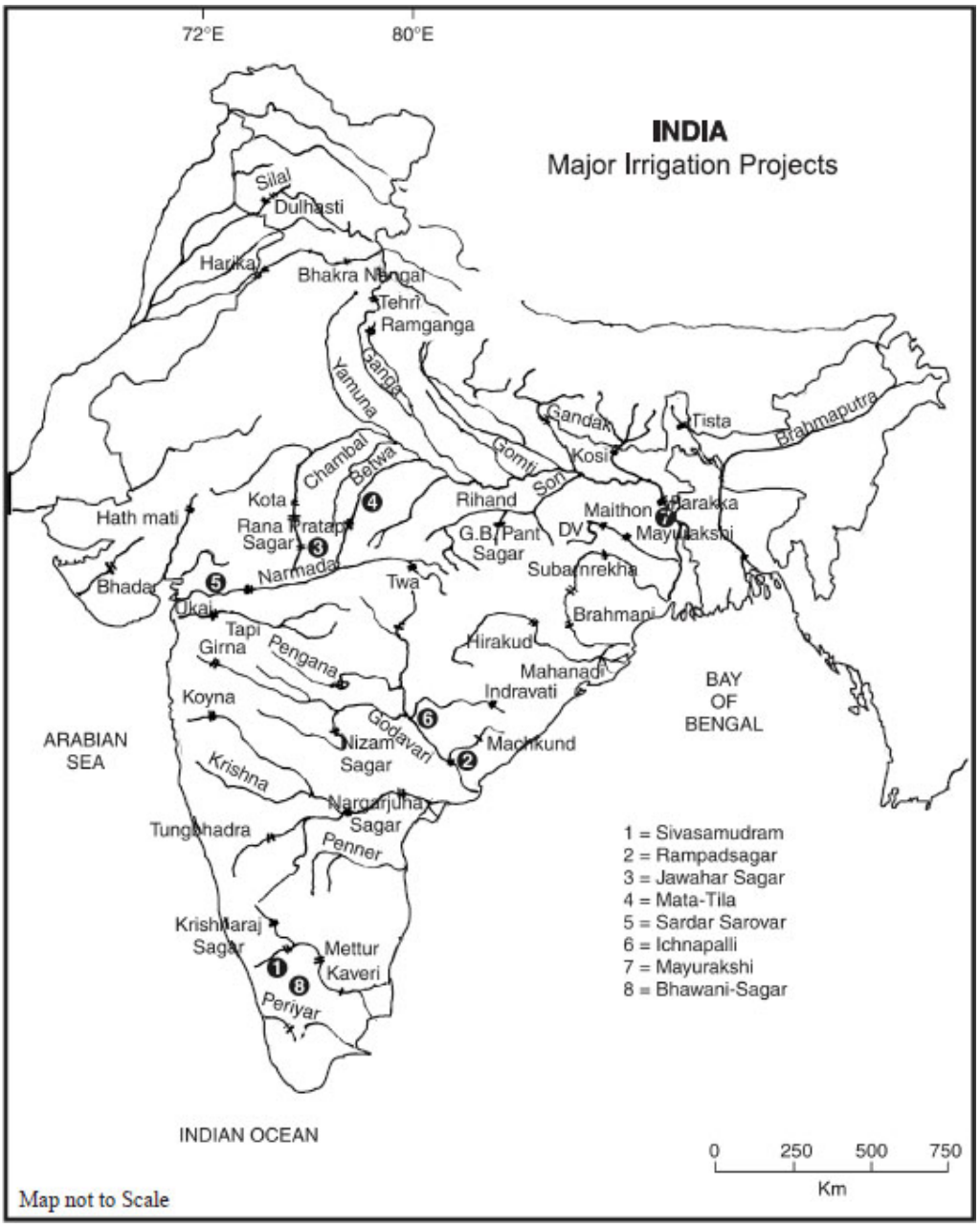


Fig. 9.11 Major Irrigation Projects

Tube-well irrigation is the easiest source of irrigation. It can be installed in a short duration of time. It is however, expensive and diminishes the underground watertable, especially in the years of drought like 2009. The most important problem of tube-well irrigation is the high energy costs for pumping groundwater because of farmers dependence on diesel and electricity. The prices of these energy resources have increased rapidly. The largest area under tube-well irrigation is in Uttar Pradesh followed by

Rajasthan, Punjab, Madhya Pradesh, Gujarat, Bihar, Andhra Pradesh, and Haryana.

Tank Irrigation: In some parts of the country, especially in the peninsular India tank is an important source of irrigation. About 4.5% of the total irrigated area is under tank irrigation. Andhra Pradesh (32%) has the largest area under tanks followed by Tamil Nadu (28%), West Bengal (14%), Karnataka (9.3%), U.P. (7%), M.P. (6.5%), Odisha and Maharashtra. Many of the tanks however, dry up during the summer season when more irrigation is required. It is an expensive and time-consuming technique which can command only small area.

Power: Availability of electric power is also an important determinant of cropping patterns and agricultural productivity. In general, higher the supply of electricity for irrigation and agricultural operation, higher is the yield of crops.

Roads: A well knit and co-ordinated network of metalled roads play a significant role in the agricultural development of a region. In India, the road density is the highest in Punjab, Haryana, and Kerala these states have higher yields of cereals to that of the national average.

Credit: The cultivator needs credit for a variety of purposes and for different periods. Firstly, he needs short-term credit (up to 15 months) for the purchase of requirements of current production**.** (seeds, fertilisers, insecticides, pesticides, electricity, diesel, cattle-feed, repair of farm implements, wages to hired labourers, etc.) and to meet the consumption needs of his family till the harvest of the crop. Secondly, he requires medium-term finance (up to 5 years) for such purposes as buying cattle and implements, undertaking digging of wells or other minor irrigation works, and effecting substantial improvement to land. Finally, he requires long term finance (for over 5 years) for such purposes as purchase of land, substantial investment in tube-wells, farm machinery and orchards, and the repay of old loans.

In the absence of this facility, the cultivators mostly used to borrow from the private money lenders, traders, commission agents and relatives. The All India Credit Survey 1951–52 showed that of the total rural credit secured by the cultivators, money lenders accounted for almost 70%, relatives 14.2%, traders and commission agents 5.5%, co-operatives 3.1%, government agencies 3.3%, and others 4.2%. This led to a lot of

exploitation of the cultivators by money lenders charging exorbitant interest, illegally and fraudulently grabbing the property of the borrower and forcing him to act as bonded labour. At present, farmers are getting agricultural credit facilities through co-operatives, commercial banks and regional rural banks.

The National Bank for Agricultural and Rural Development (NABARD) founded on 2 July, 1982 took over the functions of the erstwhile Agriculture Credit Development, Rural Planning, and Credit Cell of the Reserve bank of India and Agriculture Refinance and Development Corporations. The '***Kisan Credit Card***' scheme was announced in the budget of 1998–99 and it was prepared by the National Bank for Agriculture and Rural Development (NABARD). A crop loan of upto Rs. 3 lakh is given to assist the farmers and the card is valid for 5 years. A sum of Rs.20,339 crores has been approved by the Government of India in 2017–18. It has been emphasised that the largest proportion of the agricultural credit should go to marginal and small farmers.

Research and Extension: Development of new seeds and extension service also enhance significantly aggricultural productivity. According to agronomists the HYVs become outdated after every three years. The old seeds need to be replaced by new varieties, for which research and extension services are required.

Crop Insurance: Crop insurance was introduced by the Government of India in 1985. It is operated through the General Insurance Corporation of the State Government. The '***Pradhan Mantri Fasal Bima Yojana***' was launched on 13th January 2016. As reported, for Kharif 2017, 34.77 crores of farmers were insured and 13.79 crores were benefitted (*india.gov.in*, *portal of India*).

The main objectives and features of the scheme are given below:

1. To provide insurance coverage and financial support to the farmers in the event of failure of any of the notified crop due to natural calamities, pests and diseases.
2. To stabilise the income of farmers.
3. To encourage farmers to adopt innovative and modern agricultural practices.
4. To ensure flow of credit to the agriculture sector.

Salient Features:

- There will be a uniform premium of only 2% to be paid by farmers for all Kharif crops and 1.5% for all Rabi crops. In case of annual commercial and horticultural crops, the premium to be paid will be only 5%.
- The premium rates to be paid by farmers are very low and balance premium will be paid by the Government to provide full insured amount to the farmers against crop loss in any natural calamities.
- There is no upper limit on Government subsidy. Even if balance premium is 90%, it will be borne by the Government.
- Earlier, there was a provision of capping the premium rate which is low claims being paid to farmers. Now this is removed and farmers will get claim against full sum insured without any reduction.
- The use of technology will be encouraged to a great extent. Smart phones, remote sensing drone and GPS technologies will be used to capture and upload data of crop cutting to reduce the delays in the claim payment.
- Allocation of the scheme presented in budget 2016–17 is Rs. 5,550 crores.

Technological Factors: Moderisation of agriculture requires appropriate machiney for ensuring timely field operations, effective application of agricultural inputs and reducing drudgery in agriculture. Traditionally, farmers in India have been using manual and animal operated farm equipment but due to increased cropping intensity, this power is no longer adequate to ensure timeliness. The research institutions and industries together have helped the farmers in developing suitable farm machinery to mechanised field operations. The farm mechanisation increases cropping intensity, timely operations increases crop productivity and profitability. Technological factors include HYVs, chemical fertilisers, insecticides and pesticides tractors and other agricultural machinery. Like combine harvester, sugarcane harvester, potato combine, paddy transplanter, laser guided land leveller etc. The use and availability of these inputs also enhance the agricultural productivity of a region.

Examples of recent innovations are as under:

Farming as a Service: It is a concept introduced in India by EM3 Agri Services. It provides machinery rentals and farmer's services.

Market Linkage Models for the Farmers: Indian agriculture is supply driven more as compared to the market driven. The Government policy in supporting the price for crops would help the farmers to decide the crop they should sow depending on the prices of the previous harvest. Hybrid models involving Big data and aggregation will be helpful. '***Sabziwala***, ***Dehat***, ***Mera Kisan***' are start ups which are significant in horticulture.

Apps for agriculture: Various apps like ***Kisan Suvidha***, ***Pusa Krishi***, ***MKisan Application***, ***Farm-o-pedia***, ***Shetkari Masik***, ***Crop insurance mobile app*** have been launched to facilitate teh farmers by connecting even on their mobile sets.

Climate Smart Agriculture (CSA): Climate-Smart Agriculture (CSA) is an approach that helps to guide actions needed to transform and reorient agricultural systems to effectively support development and ensure food security in a changing climate. CSA aims to tackle three main objectives: sustainably increasing agricultural productivity and incomes; adapting and building resilience to climate change; and reducing and/or removing greenhouse gas emissions wherever possible.

Green Revolution in India

India has faced the food problem since a long period. To meet the growing demand of food the Government of India entered into the first PL-480 agreement with the U.S.A. in 1956 to import 3.1 million tonnes of wheat and 0.19 million tonnes of rice, for the next three years. Thereafter the government resorted to continuous import of foodgrains to meet the deficiency of food supply. It was because of grim food situation that the leading American economists. Paddock Brothers, declared in 1962 that Malthus will prevail in India by 1975.

Green Revolution is a term coined to describe the emergence and diffusion of new seeds of cereals. Norman-e-Burlaug is the Father of Green Revolution in the world, while Dr. M.S. Swaminathan is known as the Father of Green Revolution in India.The new cereals were the product of research work and concentrated plant breeding with the objective of creating High Yielding Varieties (HYVs) of use to the developing countries. New varieties of wheat were first bred in Mexico in the 1950s and that of rice, like IR-8 (miracle rice) at the International Rice

Research Institute, Manila, (Philippines in the 1960s). The increase in the yield from the new seeds has been spectacular as during the last forty years, agricultural production, particularly of wheat and rice, has experienced a great spurt and this has been designated as the Green Revolution.

The Green Revolution has been used to mean two different things. Some experts of agriculture use it for referring to a broad transformation of agricultural sector in the developing countries to reduce food shortages. Others use it when referring to the specific plant improvements, notably the development of HYVs. Whatsoever the meaning of Green Revolution may be taken as, the adoption and diffusion of new seeds of wheat and rice has been considered as a significant achievement as it offered great optimism. In fact, these varieties of seeds have revolutionised the agricultural landscape of the developing countries and the problem of food shortage has been reduced.

In India, hybridisation of selected crops, i.e., maize, *bajra* (bulrush millets), and millets began in 1960. The Mexican dwarf varieties of wheat were tried out on a selected scale in 1963–64. Exotic varieties of rice such as Taichung Native I were introduced in India in 1964. The diffusion of HYVs, however, became fully operational in the country in the *Kharif* season of 1965–66. The diffusion of the new seeds was mainly in the Satluj-Ganga Plains and the Kaveri Delta. Subsequently, a number of varieties of wheat and rice were developed by the Indian scientists and adopted by the Indian farmers.

Merits of the High Yielding Varieties

The High Yielding Varieties have certain advantages over the traditional varieties of cereals which are given as under:

1. Shorter Life Cycle

The High Yielding Varieties have shorter life cycle, thereby enabling the farmers to go for multiple cropping. For example, the new seeds of rice and wheat complete their life cycles in about 100 and 120 days, respectively. Contrary to this, the traditional varieties of rice and wheat take about 130 to 150 days, respectively, to harvest. The new seeds thus enable the farmers to economise on land.

2. Economise on Irrigation Water

The High Yielding Varieties need a lot of water for better returns. The yield of these varieties per unit area is significantly high. If it is considered in terms of water required per quintal of wheat and rice, the new seeds require less water as compared to traditional varieties. Thus, the new seeds economise on water also as the crop remains in the field for a shorter period of time.

3. Generate more Employment

The High Yielding Varieties under optimal conditions require more labour per unit area and thus, help in generating more employment. Prior to the introduction of the new seeds, the farmers over greater parts of the country, especially in the Northern Plains of India, were largely dependent for their agricultural operations on the arrival of monsoon. In fact, they used to remain unemployed during the summer months (April to mid-June), i.e., after the harvest of the *rabi* crops (wheat, gram, etc.). But now, the farmers and the dependent labourers get work in various agricultural operations throughout the year.

4. The High Yielding Varieties are Scale Neutral

One of the main advantages of the High Yielding Varieties is that they benefit all category of farmers in the same proportion. In other words, the new seeds are not biased towards the big and the small farmers.

5. Easy to Adopt

The adoption of High Yielding Varieties does not require any special skill for adoption. The farmers of different socioeconomic and cultural backgrounds can adopt the new seeds without any difficulty. Only a minor adjustment in the dates of sowing of wheat is required as the new seeds require relatively cool temperature at the time of sowing. Being short duration, the wheat crop is to be sown late in the month of November and December instead of October.

When the new seeds were diffused in the mid-sixties, it was expected that the problems of food shortage, unemployment, poverty, hunger, malnutrition, undernourishment, and regional inequalities will be largely solved. But these objectives could not be fully achieved. The geographical conditions required for the successful cultivation of High Yielding Varieties have been given in brief in the following section:

Geographical Constraints in the Adoption of New Seeds

The new seeds are undoubtedly land substituting, water economising, more labour using, and employment generating innovations. Nevertheless, they are very delicate and therefore, require a great deal of care for obtaining a successful harvest. For example, the new seeds are less resistant to droughts and floods and need an efficient management of water, chemical fertilisers, insecticides and pesticides. Any lapse on the part of the farmer in the application of these inputs may reduce the production substantially. In order to obtain a satisfactory agricultural return, the farmer should be in a position to arrange the costly inputs on time for which sufficient surplus capital should be available. The conditions required for the good harvest of new seeds have been described below:

1. Irrigation

Irrigation is the most important input required for the successful cultivation of new seeds. The new seeds need copious irrigation. Adoption of High Yielding Varieties and intensification of agriculture in a country like India without the availability of irrigation is not possible. The new seeds need controlled irrigation, i.e.m they need irrigation at the specific periods of growth, development and flowering in the prescribed quantity. Over irrigation and under-irrigation, both are injurious to the crop. Thus, the timings of irrigation and the quantity of water supplied are decisive for the satisfactory performance of the crop. In the case of wheat for example, appropriate timing and spacing of irrigation raise the yield as much as 50% even if other inputs (fertilisers, etc.) are not applied. The first irrigation of wheat around the third week of sowing alone raises the yield as much as 30%. Moreover, the associated inputs like chemical fertilisers, insecticides and pesticides also perform satisfactorily only if timely irrigation is provided to the crop. Without irrigation, a surfeit of other inputs would be to no avail. In the absence of controlled irrigation, the farmer may not get even a reasonable return from their fields.

2. Availability of Chemical Fertilisers

The natural fertility of the soil decreases with the passage of time. In a region like the Great Plains of India, in which agriculture is being carried out for the last five thousand years, the soils are generally depleted and are increasingly losing their resilient characteristics. For the recuperation

of fertility, the soils need to be rested in the form of fallowing or they have to be enriched by applying manures (cowdung, compost, and green) and chemical fertilisers.

The High Yielding Varieties give rise to short stemmed, stiff-straw plants that respond well to heavy doses of fertilisers. These dwarf varieties are known as the hungry varieties which need more energy in the form of chemical fertilisers. Contrary to this, the traditional varieties, if given heavy doses of fertilisers, get lodged at the occurrence of rains. The lodging of the crop reduces the yield per unit area.

In the areas of controlled irrigation, the recommended dose of chemical fertiliser for the new seeds of wheat and rice in terms of NPK is 90–45–45 kg per hectare. Some of the well off farmers of Punjab, Haryana, and western Uttar Pradesh are applying the chemical fertilisers to the crop in the prescribed quantity. The all India average fertilizer consumption is 140 kg per hectare per annum in 2014–15. India is the third largest producer of fertilizer after China and USA and the second largest consumer after China in the world. Besides the chemical fertilizers, the sales projection for 2017–2018, of the organic fertilizers is given below:

V. Neem – 130 MT, V. Bio fertilizer – 300 MT, V. Organic manure – 6000 MT, V. City Compost – 8000 MT

which is on increase as compared to 2016–17 data of 80.55 MT, 110.74 MT, 712 MT and 7974.30 MT respectively.

The trend indicates an inclination of the produce shifting towards organic due to the ill health effects caused by the chemical fertilizers

The Government of India has introduced Direct Benefit Transfer (DBT) system for fertilizer subsidy on Pilot Basis with effect from October 2016. Under the proposed fertilizer DBT system, 100% subsidy on various fertilizer grades shall be released to the fertilizer companies on the basis of actual sales made by the retailers to the beneficiaries.

3. Plant Protection Chemicals

The new seeds are very delicate and highly susceptible to pests and diseases. The irrigated fields enriched with heavy energy input (fertilisers-NPK) create a micro climate (hot and humid) in the field which helps in the fast growth of plants. The same environment is conducive for the fast growth and multiplication of insects and pests. These insects and pests attack the crop, hamper their growth and reduce the yield

substantially. The danger of pests and insects may be reduced by using plant protection chemicals. The problem may be tackled either by developing the disease resistant seeds or by spraying insecticides and pesticides at the appropriate time prescribed or advised for different crops.

The problems of crop disease and pests may also be tackled by timely application of insecticides and pesticides. Thus, the farmer must have adequate knowledge of plant disease and their controlling chemicals. At the outbreak of a disease in the crop, the entire area should be sprayed. If the timely spray of the insecticides and pesticides is not done, the crop of the entire village/region may vanish. Since the plant protection chemicals are quite expensive, and often adultrated, they are generally out of reach of the small and marginal farmers. And if the crops by small and marginal farmers are not sprayed, the insects may creep in the neighbouring fields and the disease may adversely affect the larger area. At present plant protection is available to only 35% of the total cropped area.

4. Capital Constraint

Availability of capital is also a vital constraint in the adoption and successful cultivation of the High Yielding Varieties. The farmer must have sufficient capital for the purchase of seeds, installation of tube-wells, drilling of pumping sets, chemical fertilisers, plant protection chemicals, tractors, harvesters, threshers, sprayers, and other accessories of agriculture. In case the farmer does not possess the operational capital, he should have an easy access to credit. In India, most of the farmers have no surplus over consumption, and therefore, no saving or operational capital at their disposal. The agrarian institutions like banks and co-operative credit societies have great responsibilities. They should advance loans to the farmers at a reasonable rate of interest. Unfortunately, the credit agencies in India, generally, serve the big farmers who are economically well off and politically well connected. The poor and the small farmers are thus deprived of the required inputs, so essential for the successful cultivation of High Yielding Varieties of crops like wheat and rice. Thus, there is a need of strengthening credit disbursting agencies.

5. Mechanisation

Modern farming tools and technology like tractors, leveller, seeder, planter, threshers, harvesters, winnower and sprayers are also imperative for the successful cultivation of the High Yielding Varieties. These varieties require adequate arrangements of controlled irrigation. Raising of two or three crops from the same field is possible only if the modern technology is available to the farmer. The indigenous plough and bullock/buffalo carts are less efficient to complete the agricultural operations on time. Machinery like tractors, threshers, sprayers, tillers, chaff cutters, leveller, pumping sets, etc., are required for the timely operations of sowing, weeding, spraying, and harvesting. The mechanisation of agriculture also helps in the judicious utilisation of complementary inputs like chemical fertilisers, insecticides and pesticides. For example, a farmer with a tractor and blade-terracer manages to grade his field to much better level in the course of time as compared to a farmer not having the similar equipments at his disposal. Among many useful aids which increase the efficiency of the farmers are seed-cum-fertiliser drills, well designed plant protection equipments, dunlop-cart, trolley, threshers, sprayers and tractors. Availability of electric power which is the nucleus of all technological development, is imperative for multiple cropping and intensification of agriculture. As a matter of fact, electric power has a vital role in the development and diffusion of High Yielding Varieties. It supplies the mechanical power to tube-wells, pumping sets, threshers, crushers, grinders and chaff-cutters. Availability of cheap power for agriculture helps in the adoption of new technology.

6. Marketing and Storage Facilities

The infrastructural facilities, like roads, marketing and storage facilities are also very crucial for the successful cultivation of High Yielding Varieties. It is the transportation cost which determines the cropping pattern and crop intensity in given region. Once a village is linked with a town/market by a metalled (*pucca*) road, its economy undergoes a remarkable transformation. The farmers can market their produce with ease and are also able to purchase fertilisers, plant protection chemicals, and other agricultural equipments from the neighbouring town. Cultivation of perishable commodities like vegetables, fruits, flowers and dairy

products can also be done efficiently and profitably if adequate transport and marketing facilities are available.

7. Extension Service

For the successful adoption and spread of High Yielding Varieties, there should be an efficient extension service which may guide and help the farmers about the various agricultural operations and precautions. The fuller use of inputs may be made only if proper guidance to the farmers is available. The cultivation of delicate and highly sensitive varieties needs the services of qualified and dedicated extension agents. In the tradition bound society of India, the efficiency of extension machinery determines largely the efficiency of the farmers. There should, therefore, be a perfect understanding and co-ordination between farmers, extension agents, farm supervisors, researchers, and agricultural scientists. Any slackness in the work of agricultural machinery may make agriculture a less profitable pursuit. It is a well-established fact that investment in research brings ten times profit to the investor.

8. Human Factor

In the adoption of High Yielding Varieties, the role of human factor is also very important. As a matter of fact, in many cases, man behind the machine becomes more important than the machine itself. Within an agricultural community, individuals vary in their receptivity to innovations and new agricultural techniques. The personal qualities of the farmers, education, progressiveness, attitude towards life, aspirations, life style and family values determine his capacity to adopt the new agricultural technology. There are innovative and progressive farmers within a village who perform better than their orthodox and less progressive brothers. In other words, in all societies, there are rational and irrational farmers. The progressive farmers have largely improved their production and thereby their standard of living, while the conservative and less hard working farmers could not adopt the new agricultural technology successfully and they are in the grip of poverty and undernourishment.

The life style of the farmers and their aspirations for better standard of living also determine their efficiency. It is mainly because of the human factor that the agricultural income of the farmers, having almost the same size of holdings in a village, vary from each other. Human development of

cultivators by focusing on education health and skill programmes is required at a priority basis.

Performance of the High Yielding Varieties

Green Revolution performed well in Punjab, Haryana, and western Uttar Pradesh where the High Yielding Varieties were adopted first. The introduction of the High Yielding Varieties have increased the production of cereal crops substantially. For example, the total food production in 1950–51 was 97.3 million tonnes which rose to about 263.2 million tonnes in 2012–13. The production of rice was 112.9 million tonnes and wheat 99.7 million tonnes in (2017–18) and estimated to be 113 MT & 100 MT for rice & wheat respectively in 2018–19. Thus, there was a substantial increase in the production of wheat and rice after the diffusion of the High Yielding Varieties in Indian agriculture (**Table 9.11**). Although, the production of rice has increased in all the states of the country, it has recorded a phenomenal increase in the states of Punjab and Haryana, mainly because of the adequate and controlled irrigation which are imperative for its cultivation.

Rice is the staple food for about 60% of the total population of India. It is grown under diverse climatic and soil conditions. It, however, does not perform satisfactorily if the temperature falls below 20°C. Availability of moisture either from rains or from irrigation is, however, the main determinant of its cultivation. Rice cultivation is carried on in almost all parts of India except the un-irrigated tracts of Rajasthan, Malwa, Maharashtra, and Gujarat. After the adoption of High Yielding Varieties, its cultivation has assumed great significance in the cropping structure of Punjab, Haryana, and western Uttar Pradesh **(Fig. 9.12) and (Fig. 9.13).**

A comparison of (**Fig. 9.12) and (Fig. 9.13)** shows that rice cultivation has been diffused substantially in the regions of Punjab, Haryana, and western Uttar Pradesh. Interestingly enough, the highest yields of rice are being obtained by the farmers of Punjab who are doing its cultivation with the help of canal and tube-well irrigation. In fact, controlled irrigation in these regions is helpful in achieving high yields per unit area of rice. The percentage share of the area of rice, wheat and other main crops has been shown in (**Table 9.10)**.

Table 9.10

Cropping Pattern in 2016–17

Crops	Area in Million Hectares	Production (MT)
Rice	43.2	110.15
Wheat	30.6	98.38
Jowar	5.1	4.57
Bajra	7.5	9.8
Maize	9.9	26.26
Gram	9.6	9.33
Cereals & Pulses	12.8	275.68
Cotton*	10.8	32.12
Sugarcame	4.4	305.32

**** Production in billion boles of 170 kg each***

Source: Directorate of Economics & Statistics, Department of Agriculture, Cooperation & Farmers Welfare.

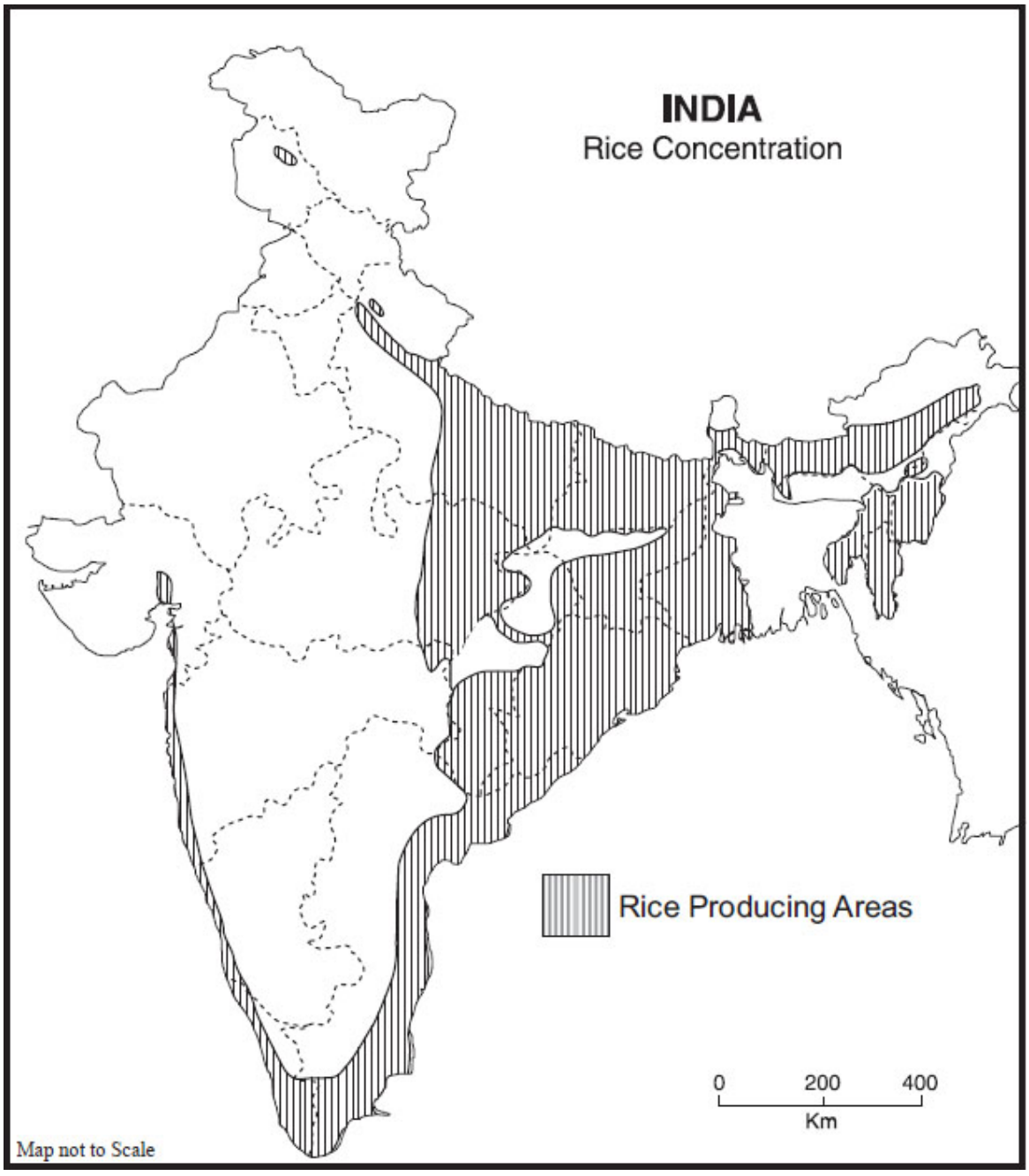


Fig. 9.12 Rice Concentration

It may be seen from (**Table 9.10)** that over 26% of the total cropped area was under rice and 16% under wheat in 2005–06. *Jowar* (millet), *bajra* (bulrush-millet), and *maize* occupied about 6%, 5%, and about 4% of the total cropped area respectively, while the share of pulses was over 12%.

A comparative picture of cereals production in 1950–51 to 2010–11 has been given in (**Table 9.11)**:

Table 9.11

Production of Cereals, 1950–51 to 2016–17 (in million tonnes)

Crops	1950–51	2016–17	2017–18
Rice	30.8	110.2	12.91
Wheat	9.7	98.51	99.7
Pulses	12.5	23.0	24.5
Coarse-grains	15.5	43.77	46.99
Total food grains	97.3	275.7	284.83

Source: Ministry of Agriculture and Economic Survey, 2010–11, and India 2012, and Statewise Estimates of Principle crops–2015–16.

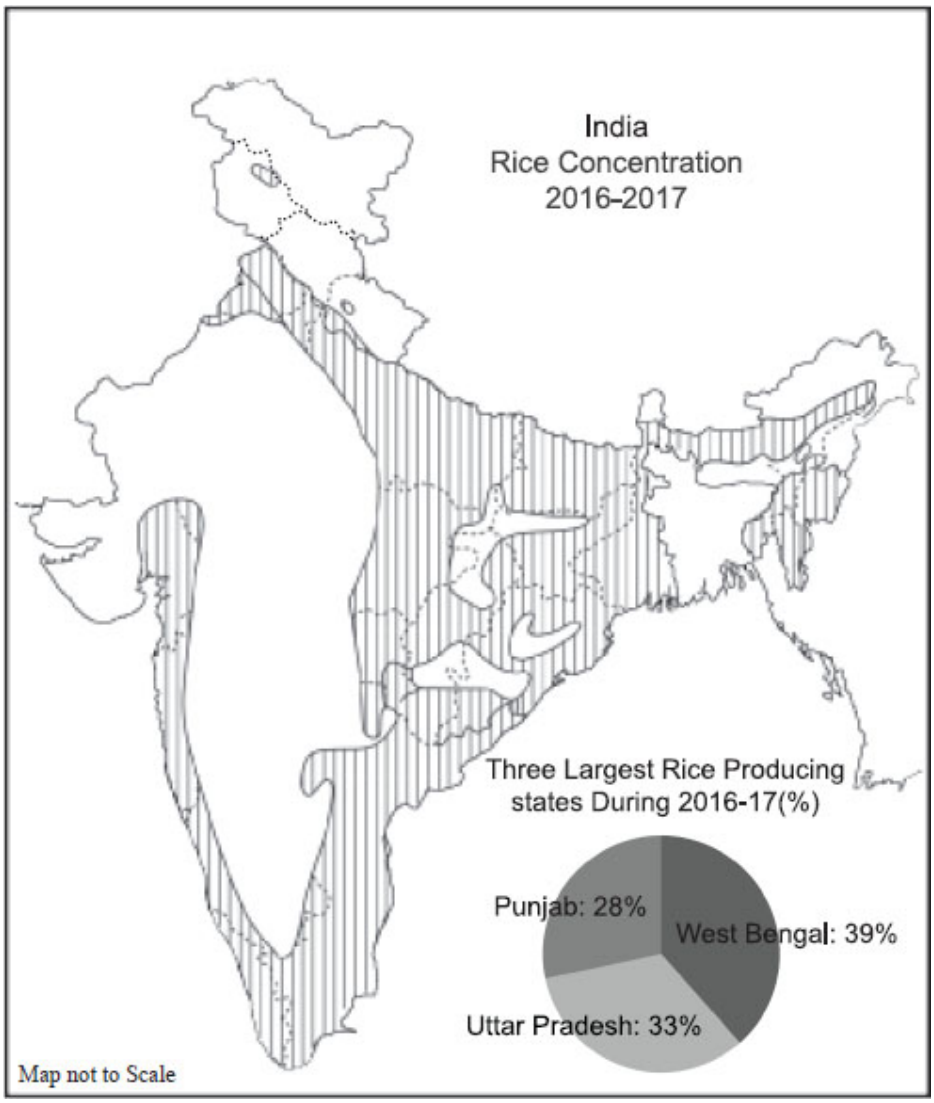


Fig. 9.13 Rice Concentration (2014–15)

Recently, the Indian Rice Research Institute has discovered new varieties of rice that are capable of producing 6–7 tonnes of rice per hectare. This is about three times the average production of rice per hectare.

Wheat is the second most important staple food in India after rice. It contributes about 34% of the total food grain production in the country. The regional distribution of wheat during the pre-Green Revolution and post-Green Revolution periods have been shown in (**Fig. 9.14) and (Fig. 9.15),** while (**Table 9.11)** shows the growth of its production.

It may be observed from (**Table 9.11)** that wheat is the only crop in which the production has increased to the maximum. The production of wheat has gone up 88.93 million tonnes in 2014–15 as against only 9.7 million tonnes in 1950–51.

The regional pattern of wheat distribution shows an overall expansion of wheat area from the Ganganagar District of Rajasthan in the west to Dimapur (Nagaland) in the east, and from the Suru and Nubra Valleys (tributaries of the Indus in Ladakh) in the north to Maharashtra, Andhra Pradesh, and even Karnataka in the south **(Fig. 9.14).**

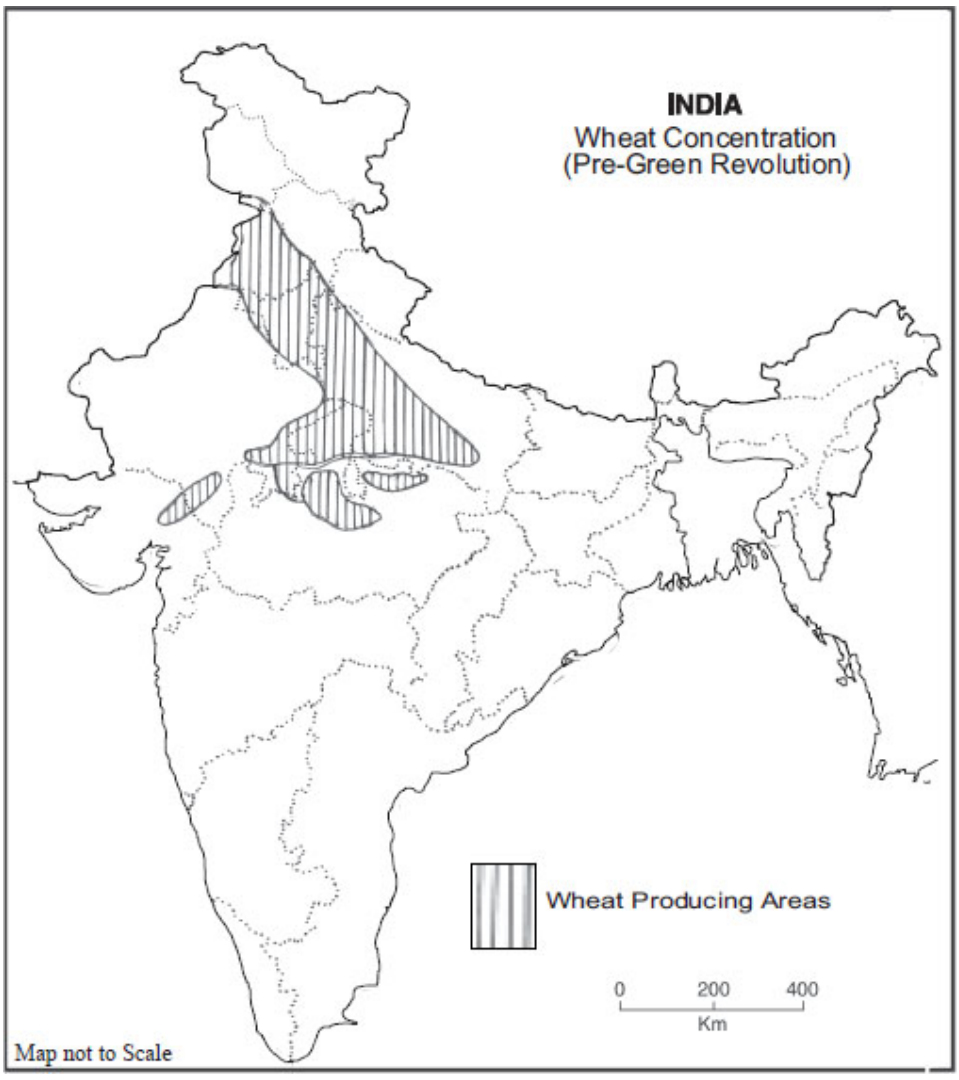


Fig. 9.14 Wheat Concentration (1960–61)

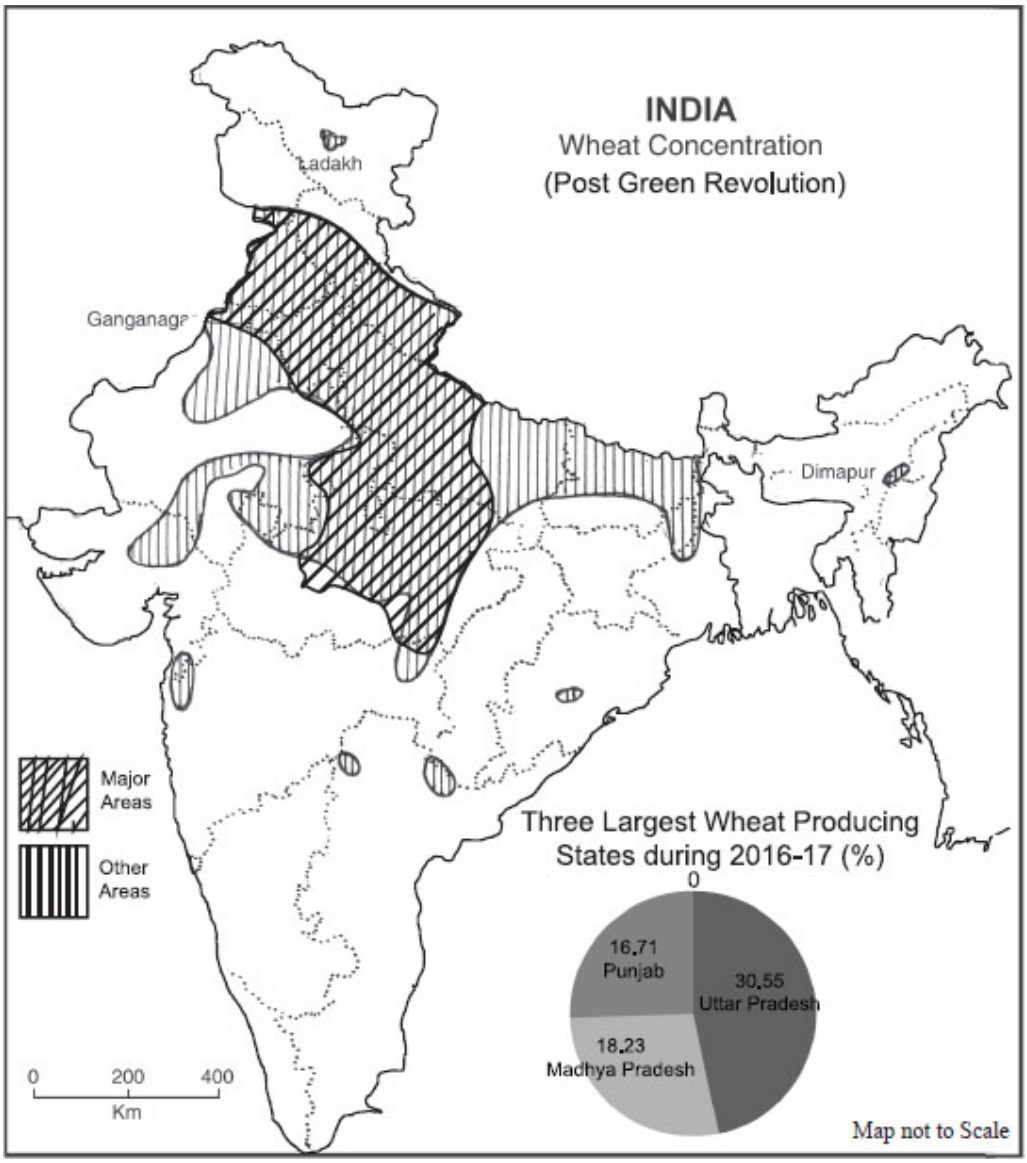


Fig. 9.15 Wheat Concentration (2014–15)

Looking at the spread of wheat and its excellent performance in the Satluj-Ganga Plains, it may be said that Green Revolution is most successful in the case of wheat. It is because of its high production that wheat has become a commercial crop in Punjab, Haryana, and western Uttar Pradesh.

It may be observed from the (**Fig. 9.16)** that Punjab, Haryana, and western Uttar Pradesh have emerged as the areas of major concentration of rice, while wheat has been diffused in all the directions from its traditional heartland of the north-west India.

Among the other crops, the area yield and production of maize and *bajra* have gone up. The area and production of pulses have, however, could not have any significant impact of Green Revolution.

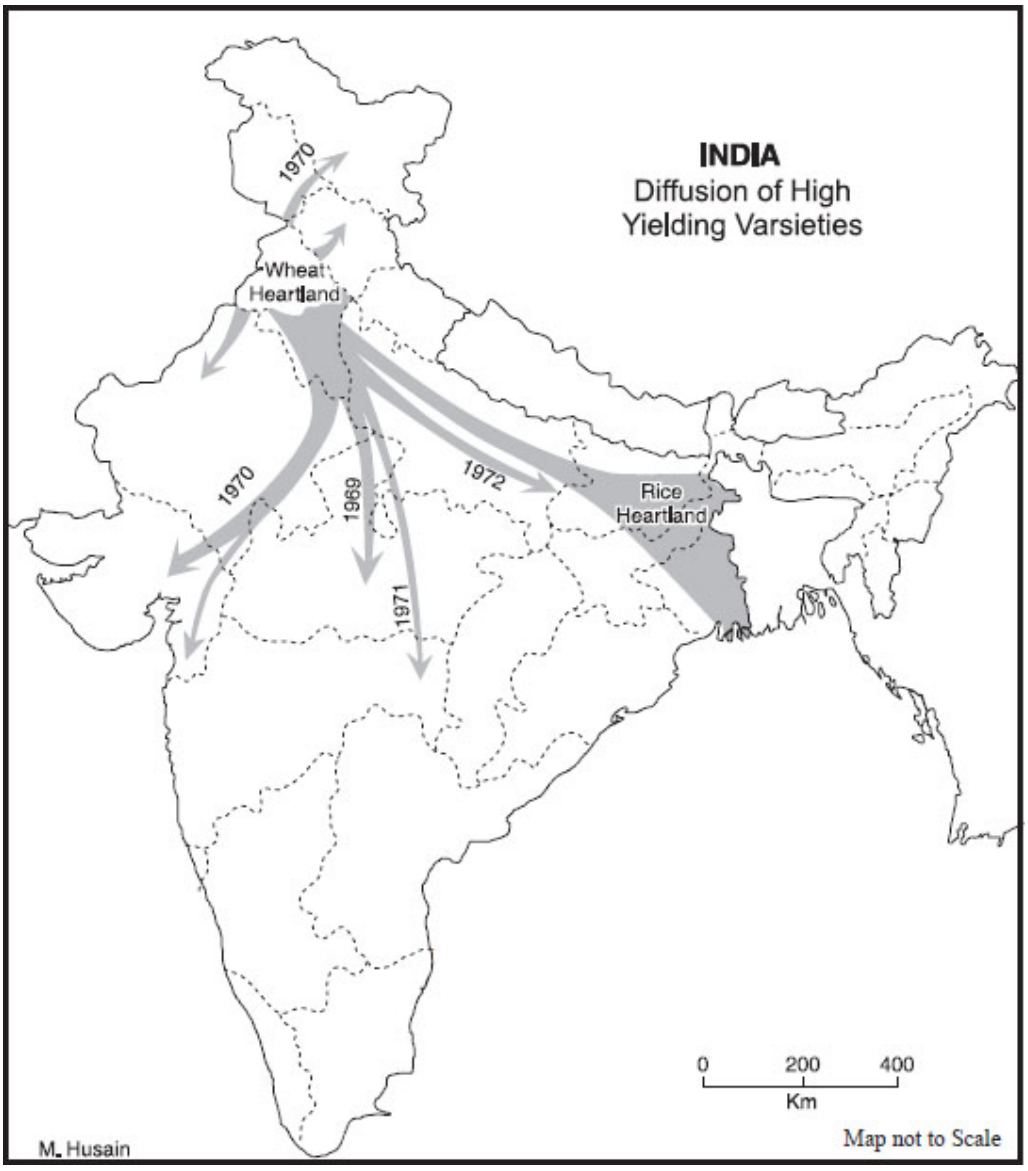


Fig. 9.16 Diffusion of High Yielding Varieties of Rice and Wheat

Green Revolution and Rotation of Crops

In India, rotation of crops is practiced by most of the farmers. The main objective of rotation of crops is to obtain higher agricultural returns on the one hand and to maintain the soil fertility on the other. Thus, rotation of crops helps in making agriculture sustainable. The importance of rotation

of crops is more in the areas where the farmers grow two or more than two crops in a year in the same field.

Prior to the Green Revolution most of the Indian farmers were subsistent in character, growing crops mainly for the family consumption. In the areas, where Green Revolution is a success, agriculture has become agri-business and market oriented in which the farmers are concentrating on a few number of crops. Most of the farmers are devoting their lands in the Green Revolution areas to rice in the kharif season and wheat in the rabi season. Both these staple crops are soil exhaustive. The area under leguminous crops has shrunk and the farmers have given up their empirical practice of fallowing to recuperate soil fertility. The rotation of crops of some of the villages from western Uttar Pradesh and Haryana, where Green Revolution is a success have been given in (**Table 9.12)**.

Table 9.12

Traditional Rotation of Crops (1960–65) in Banhera (Tanda) Village, District Hardwar

Year	Kharif Season (Mid-June to Mid-October)	Rabi Season (Mid-October to Mid-April)	Zaid Season (April-June)	No. of Days Land Left Fallow
1960	Millet/fodder/rice	Gram	Fallow	90
1961	Fallow	Wheat	Fallow	210
1962	Millet/fodder/rice	Gram	Fallow	90
1963	Fallow	Wheat	Fallow	210
1964	Millet/fodder/rice	Gram	Fallow	90
1965	Fallow	Wheat	Fallow	210

Source: Field work by the author, 1960–65, and 2015–16.

It may be observed from (**Table 9.12)** that fallowing was an important practice each year in western Uttar Pradesh before the adoption of the High Yielding Varieties. Moreover, wheat and gram (a leguminous crop)

used to be sown in the same field in alternate years. This type of rotation of crops was helpful in maintaining the fertility of the soil.

Under the pressure of growing population on arable land, the adoption of new seeds have resulted in a new pattern of rotation of crops, which has been given in (**Table 9.13).**

In the new rotation of crops, the farmers are largely concentrating on the cultivation of High Yielding Varieties of rice and wheat, and depending on the circumstances, they grow a cash crop of sugarcane. All these crops are soil exhaustive. In opposition to this the area of pulses, mustard, rapeseed, barley, groundnut, maize, millets, bajra and sunflower has decreased significantly. Moreover, the following practice has been abandoned.

Table 9.13

Rotation of Crops in 2014–15 in Village Banhera (Tanda) Village, District Hardwar

Year	Kharif	Rabi	Zaid	No. of Fallow Days
2000	Rice	Wheat	Fodder	20
2001	Rice	Wheat	Fodder	5
2002	Rice	Wheat	Fodder	20
2003	Rice	Wheat	Fodder	5
2004	Sugarcane	Sugarcane	Sugarcane	20
2005	Sugarcane	Sugarcane	Sugarcane	5
2006	Rice	Wheat	Fodder	20

Source: Field work by the author, 2015–16.

Green Revolution and Regional Inequalities

The High Yielding Varieties adopted under the new agricultural strategy are quite delicate and highly sensitive which require the timely application of costly inputs (water, fertilisers, insecticides, and pesticides). Moreover, they perform better in the areas where infrastructural facilities (electricity, irrigation, and roads) are adequately available. Contrary to this, the areas of extreme climate (precipitation and temperature) where controlled irrigation is not available the new seeds are not performing satisfactorily. This has created regional imbalances in the agricultural development in the irrigated and unirrigated parts of the country.

Intra-Regional Inequalities

The adoption of High Yielding Varieties have created intra-regional inequalities also. All the farmers, even in the states of Punjab, Haryana, western Uttar Pradesh, and Kaveri Delta have not been benefited equally. It is the large, progressive and educated farmers who gained much from the High Yielding Varieties. The early adopters reaped much dividends from the new seeds. By the time the majority came to adopt the new seeds, income gains realised by the early adopters generally disappeared. The average, small and marginal farmers could not gain much, while the late adopters gained almost nothing.

Thus, the High Yielding Varieties have had a discriminatory impact in which the large and early adopters were benefited and the small and marginal farmers who adopted these seeds late could not achieve much.

Negative Impact of High Yielding Varieties

The introduction of HYVs and intensification of agriculture resulted into the following adverse consequences:

1. Decline in soil fertility and caused deficiency of Zn, Mn in Punjab and Haryana due to introduction of paddy-wheat rotation.
2. More incidences of insects-pests and diseases.
3. More use of fertilizers has caused environmental pollution and salinity problems.
4. The indiscriminate use of pesticides produced adverse effects on micro-organisms, soil fertility, and environment.
5. The rice-wheat crop combination reduced the area under pulses which are leguminous and soil enriching crops.
6. The area under oilseeds and fodder crops also declined.

Impact of Green Revolution on Farmers and Landless Labourers

When the package programme for the development of agriculture to increase agricultural production was introduced in the mid-sixties, it was expected that the new seeds will be neutral to the scale. This assumption has, however, been proved wrong as the new seeds are no longer scale neutral.

In India, there are four categories of farmers, i.e., (i) large farmers, (ii) medium farmers, (iii) small farmers, and (iv) marginal farmers. Each one of them has not been equally benefited from the new seeds. It would be

worthwhile to examine the impact of Green Revolution on the various categories of farmers and the dependent labourers.

Large Farmers

The definition of large farmer differs from state to state in India. For example, a farmer having 10 acres in Kerala is a large farmer, while in Rajasthan, Punjab, and Haryana he falls under the category of medium or small farmer. All the studies conducted in areas where Green Revolution is a success show that the large farmers have been the main gainers of the package programme. In the initial phase of the Green Revolution, the large farmers were able to adopt the High Yielding Varieties easily. The new varieties increased their savings, both to buy machinery that can displace labour and to purchase more land of the marginal and small farmers. This trend increased the income base of those who were already relatively well off and better placed in society. The large farmers, in fact, are in a position to make the best use of tractors, threshers, and sprayers. They installed tube-wells and pumping sets for an effective utilisation of underground water. For the purchase of agricultural machinery and installation of tube-wells and other costly inputs, credit was necessary. Since the large farmers have more risk taking capacity, they could modernise their agriculture easily. Small and marginal farmers being constrained by financial resources could not adopt the High Yielding Varieties in the early phase of their diffusion. Consequently, they lagged behind in the adoption of new seeds.

In general, the complexity of farming increases with multiple cropping as more inputs and timely operations are required for good harvests. Intensification of agriculture and multiple cropping means more risk on the part of the cultivator. The agrarian institutions, credit agencies, and extension service, generally serve the large and powerful farmers as a result of which the small and marginal farmers are deprived of adequate inputs so essential for the successful cultivation of High Yielding Varieties. The big farmers who have close connections with the money economy and influential political persons are able to complete their agricultural operations more easily than the small farmers who rely on family labour for agricultural operations. This process accentuated the income inequalities in the rural society and led to polarisation of rural masses.

Small Farmers

The small farmers generally have less than two acres of land in most of the states of India. These farmers are not well placed technologically and financially. Moreover, they do not have an easy access to the credit agencies. For the irrigation of their crops, they have to depend on the tube-wells of large farmers. It has been observed that at the time of peak irrigation demand, the tube-well owners (large farmers) either do not give water to the small farmers or they charge exorbitantly for the water, which is often beyond the reach of small farmers. In the absence of timely irrigation, the crops of the small farmers suffer adversely. Over a period of time, small and marginal farmers, by taking the advantage of rising land prices, sell out their land and attempt a new start in life.

Moreover, the agrarian institutions which are supposed to assist the small and marginal farmers, are not very helpful. The credit agencies as stated above, are serving largely the big farmers who are economically well off and politically powerful. The big farmers could easily pre-empt for their own use the bulk, if not, the entire supply of costly inputs like electricity, water, fertilisers, insecticides, and pesticides. Thus, the poor farmers have been deprived of enough inputs so essential for the successful cultivation of the High Yielding Varieties of crops.

Tenant Farmers

The diffusion of High Yielding Varieties also affected the tenant farmers adversely. In general, the tenant farmers have a low tendency to adopt the new innovations in their cropping patterns as they are not very sure for how long the land will be available to them for cultivation. The difficulties of tenant farmers have multiplied by the astronomical rise in the value of land in recent decades. The tenants want to lease more land while land owners are reorganising the gains to be achieved by direct management of their fields. Under these circumstances, the landlords are reluctant to get into a position where their tenants might be given title of the land. Numerous evasive tactics have been adopted by the landlords. Some of them have directly evicted their tenants from establishing security of tenure by shifting them frequently. In the absence of more effective land reforms, the prospect is for large number of tenant farmers to join the rank of landless labourers. Compelled by financial constraints,

they migrate to big cities in search of employment and to start a new career.

Share Cropping

It is a type of agricultural practice in which a landowner allows the tenant to use the land in return of the share of the crops produced on that land. It is dominant in the lesser developing countries. It is an institutional arrangement in which risks are shared and incentives given in situation where the labour supply is costly.

Landless Labourers

One of the assumptions of the High Yielding Varieties that they will generate more employment could also not be achieved. Undoubtedly, the wages of the unorganised agricultural workers have risen by about 20 times. In the areas where Green Revolution is a success, the labourers are finding employment throughout the year, while in many areas the rural employment has declined. The main cause of decline in labour employment is the natural growth of labour and mainly because of the mechanisation of agriculture by the big farmers which displaces labour.

The impact of Green Revolution has been shown in (**Table 9.14).** It may be observed from this Table that the large farmers who have better risk taking capacity adopted High Yielding Varieties quickly. They installed tube-wells and pumping sets in their fields and purchased tractors, threshers and harvesters, etc. from the loans they got from the funding agencies and co-operative societies. Consequently, their production and productivity went up substantially. Better income helped them in improving their food and nutrition. Improvements also occurred in their housing. Realising the importance of education, many of them sent their children to the English medium schools of the neighbouring towns and cities. The economic prosperity also made them increasingly conscious about health and sanitation. It was at this stage that some of the big farmers started desiring small families. These steps led to a decline in the fertility rate of big farmers which ultimately reduced their dependency ratio.

Moreover, the economic prosperity and interaction with the well-off urban people inspired them to construct elegant and spacious *pucca* houses. They started consuming more comfort and luxury goods, which

in a sense, brought consumerism in the rural society of the regions of successful green revolution.

The traditional farmer became economic farmer who stated thinking all the time in terms of optimising his profit. Being too busy and conscious of the value of his time, he started ignoring the interest of the neighbours and small farmers. On the other hand, he started purchasing the agricultural land of the small and marginal farmers. This broke the well establish reciprocal aid system and *bhai-chara* (brotherhood) in the village community. With better agricultural income, the standard of living of the farmers went up, their longevity increased and life became more enjoyable.

Contrary to this, the small and marginal farmers, having less risk-taking capacity, could not adopt the High Yielding Varieties rapidly as they did not like to mortgage their lands for obtaining loans from the funding agencies and the money lenders. Their production and productivity increased only marginally. Consequently, there was no improvement in their food, nutrition, education, sanitation, and health. Being poor, they could not afford to bear the expenses of school education and thought it better to engage their children in agriculture. Realising the importance of additional hands, these farmers have no desire of small families and did not adopt family planning. Being under debt, poor nutrition, and mental stress they could not maintain their health and many of them, unfortunately, committed suicide.

Under the changed socioeconomic conditions, the income gap between the big and small farmers increased. This broke the traditional rural society and divided them into the rich and the poor. The social tension increased leading to the polarisation of the society. The social implications of Green Revolution has been given in a tabular form in (**Table 9.14):**

Table 9.14

Impact of Green Revolution on Big and Small Farmers

Big Farmers	Small and Marginal Farmers
Fast adoption of High Yielding Varieties	Slow adoption of High Yielding Varieties
Rapid increase in agricultural productivity and	Little increase in agricultural productivity and

Big Farmers	Small and Marginal Farmers
production	production
Improvement in:	*Little or no improvement in:*
Food and nutrition	Food and nutrition
Housing	Housing
Education	Education
Sanitation	Sanitation
Health	Health
Some of the big farmers went for family planning	No or little desire for smaller family
Decrease in dependency ratio	Little or no family planning Increase in dependency ratio
Increase in the consumption of comfortable and luxury goods	Little or no improvement in consumption
Multiplication of immovable assets	Decrease in immovable assets
Better standard of living	Decline in standard of living
Increase in longevity	Little or no increase in longevity
Big farmers became rich	Small farmers became poor
Polarisation of the rural society	Small farmers united
Increase in social tension	Increase in social tension

Source: Fieldwork by the author, 1962–63 and 2015–16.

Environmental and Ecological Implications of Green Revolution

Some of the environmental and ecological problems that emerged out of the cultivation of the High Yielding Varieties are depletion of forests, reduction in pastures, salination, water-logging, depletion of underground water-table, soil erosion, change in the soil chemistry, reduction in bio-diversity, decline in soil fertility, silting of rivers, increase in weeds, emergence of numerous new plant diseases, and health hazards. An overview of these environmental and ecological problems has been given here.

1. Salination

The High Yielding varieties of rice and wheat require several waterings, especially in an area like Punjab and Haryana in which the average annual rainfall is about 65 cm. The continuous supply of moisture through irrigation during the summer and winter seasons have changed the soil

chemistry. In the arid and semi-arid areas, owing to capillary action, the soils are becoming either acidic or alkaline. The saline and alkaline affected tracts, locally known as *kallar* or *thur* in Punjab and *kallar* or *reh* in Uttar Pradesh, have expanded and increased in area. According to one estimate, about 50% of the total arable land of Punjab and Haryana has been harmed by soluble salts. The saline and alkaline affected areas have been shown in (**Fig. 9.17) and (Fig. 9.18).**

It may be observed from (**Fig. 9.17) and (Fig. 9.18)** that some of the best agricultural tracts have been adversely affected and rendered useless from the agricultural point of view**.** If the existing pattern of crops and their rotation is not changed more agricultural land may become unproductive.

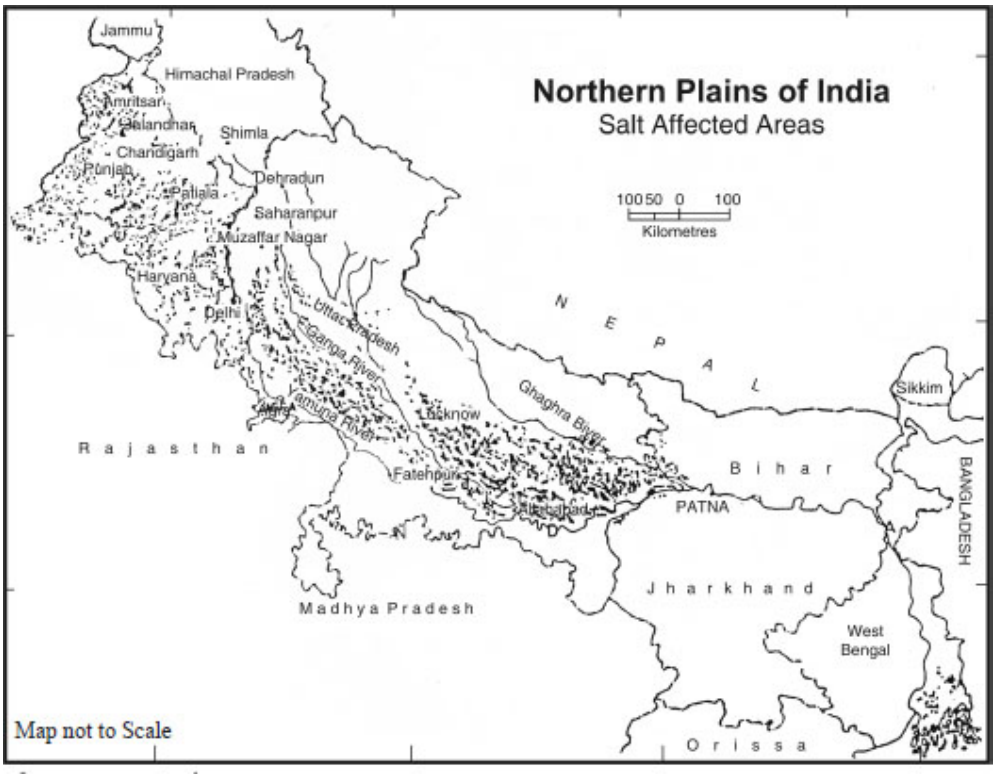


Fig. 9.17 ***Satluj–Ganga Plain: Salt Affected Land [Source: Waste Land Map of India, 1980–82 Prepared by N.R.S. a Govt. of India]***

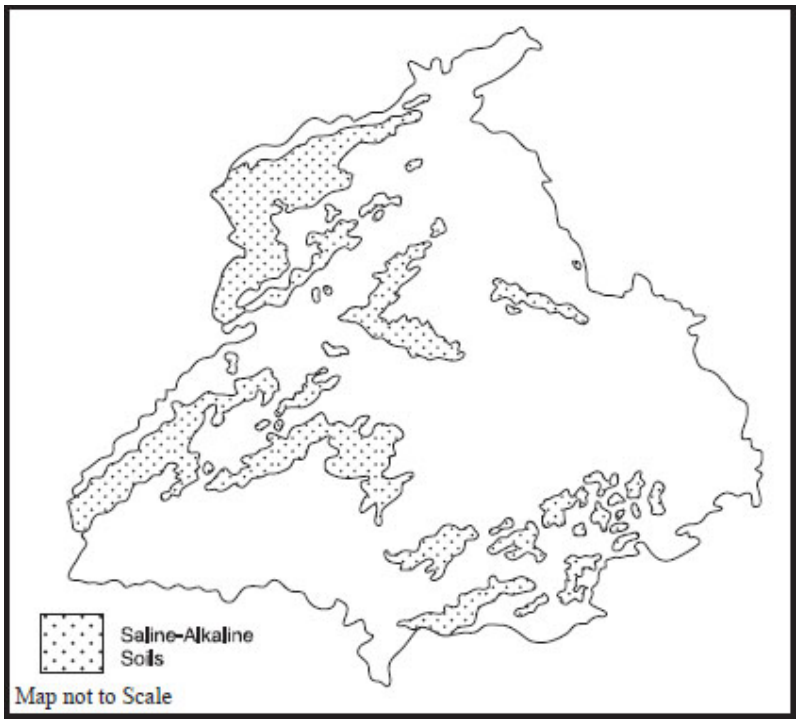


Fig. 9.18 ***Punjab—Soil Salinity (2015–16) [Source: Punjab Agri Univ. Ludhiana]***

The problem of salinity and alkalinity can be solved by use of manure (cowdung, compost, and green manure) and by a judicious selection of leguminous crops in the rotation. Cultivation of salt tolerant crops like barley, sugar-beet, salt grass, asparagus, spinach, and tomato may also help to a great extent and may improve the fertility of such lands.

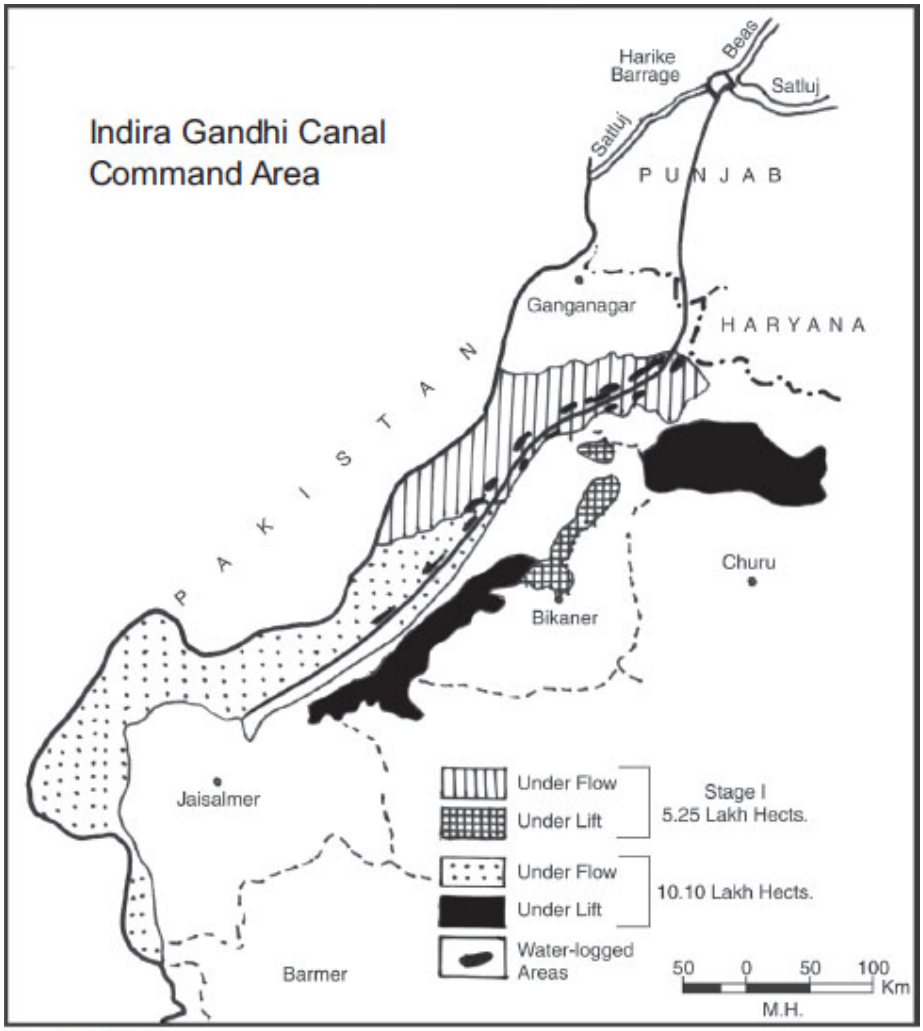
2. Waterlogging

Water logging is the other major problem associated with over-irrigation. In all the canal irrigated areas of Punjab, Haryana, and western Uttar Pradesh, waterlogging is a serious problem. The Indira Gandhi Canal command area is a recent example in which waterlogging is progressively becoming a serious menace to the arable land. Several thousand acres of productive agricultural land and pastures in the districts of Ganganagar, Bikaner, and Jaisalmer (Rajasthan) have been submerged under water **(Fig. 9.19).** The progressive and ambitious cultivators of the irrigated areas of these districts have changed their cropping patterns and have introduced rice and wheat in place of bajra, pulses, cotton, barley, millets, groundnut, sunflower and fodder.

Repeated irrigation of these crops in the summer and winter seasons have resulted into waterlogged condition, especially along the canals.

3. Soil Erosion

Soil erosion is a universal phenomena. It may be observed to some extent in all parts of the country, its intensity, however, is more in the arid, semi-arid, and mountainous areas. The presence of forests reduces the danger of soil erosion significantly. In recent years, the agricultural area has been expanded by indiscriminate felling of trees. The increase in the rate of soil erosion is not only damaging the agricultural lands, it is also affecting adversely the areas where the eroded soil is deposited.



Map not to Scale

Fig. 9.19 Indira Gandhi Canal Command Area

In order to minimise the danger of soil erosion, afforestation is imperative. Moreover, the farmers should apply more manures and develop wind breakers in the desert areas. Development of terraces in the hills, leveling of gullies, and contour ploughing in the hilly areas can also go a long way in reducing soil erosion.

4. Pollution

The High Yielding Varieties perform better if heavy doses of chemical fertiliser, insecticides, and pesticides are applied. Application of heavy doses, of these inputs destroy the micro-organisms which are so necessary to maintain the fertility of the soil. The use of manures in place of chemical fertilisers can go a long way in overcoming the problem of soil pollution.

5. Lowering of the Underground Water-Table

The High Yielding Varieties of rice and wheat are water-relishing crops. Rice, being sown in the low rainfall recording area of Punjab and Haryana, demands several irrigations and same is the case with wheat crop. The continuous lifting of water through tube-wells and pumping sets has lowered the water table in the eastern districts of Haryana **(Fig. 9.20)**. Many farmers have to lower their tube-wells in the years of inadequate monsoon rainfall. If the cropping pattern is not changed, and irrigation of rice and wheat continues at the present level, the underground water table may not be sufficiently recharged and may get substantially depleted. In opposition to this, the underground water table in western Haryana is rising as there is a gypsum layer in that part of the state which does not permit the percolation of water through this layer. The watertable in the Jhajjar District of Haryana has risen significantly. The crops of millets, bajra, arhar are damaged. In fact, people in this district pray for drought so that the waterlogged areas may be sown. Consequently, there are water-logged conditions in several tracts in the western parts of Haryana. This rise in the water table is resulting into capillary action, leading to the occurrence of saline and alkaline formations.

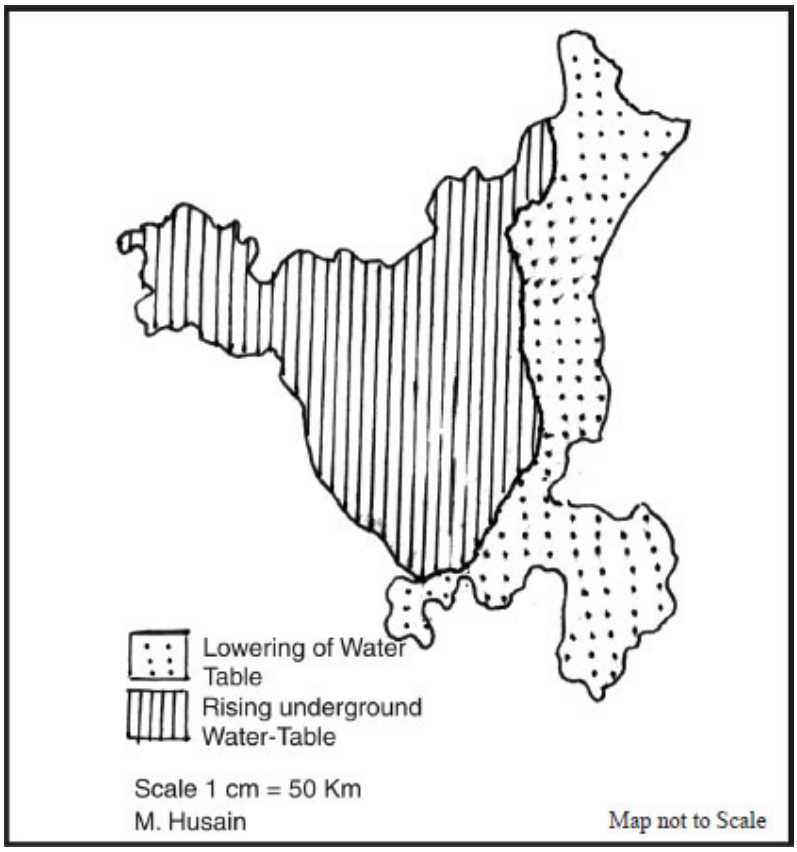


Fig. 9.20 Haryana Green Revolution and Underground Water–Table (2015–16) *[Source: Created by the author M. Husain]*

6. Deforestation

There had been heavy felling of trees to bring the forest area under cultivation. In Punjab and Haryana, less than 5% of their area is under forest. This is affecting the environment and ecology adversely.

7. Noise Pollution

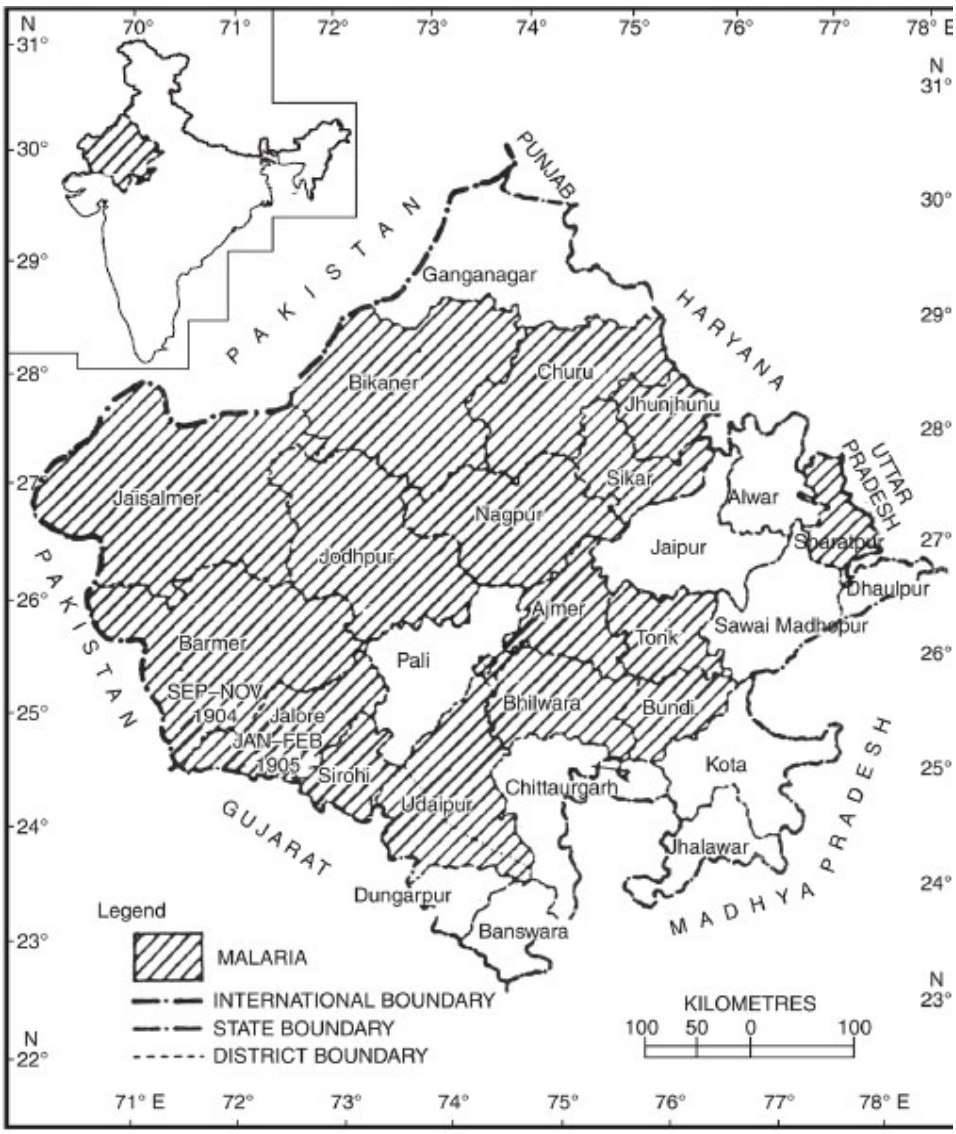
The change in the agricultural technology, the use of tractors, terracers, harvesters, threshers, and crushers have increased noise pollution which have disturbed the rural tranquility.

8. Health Hazards

Application of heavy doses of insecticides, pesticides, and chemical fertilisers are health hazards. The application of these poisons on vegetables, fruits, and grasses are health hazards. The Indian Council of Medical Research (AIMS) established that traces of lead, zinc and copper are found in the milk and vegetables on which the fertilisers,

insecticides, and pesticides are sprayed. The incidence of cancer have increased in the state of Punjab significantly. Paddy and wheat straw burning in Punjab has contributed to air pollution and loss of soil fertility which has a negative impact on the health of people because of respiratory, skin and eye diseases.

The recurrence of malaria in irrigated tracts of arid and semi-arid regions of Rajasthan and Punjab is the result of heavy irrigation and water-logged tracts along the canal which have become the breeding grounds for mosquitoes **(Fig. 9.21).**



Map not to Scale

Fig. 9.21 Rajasthan—Malaria Affected Areas

Green Revolution—Achievements

The diffusion of High Yielding Varieties has transformed the rural landscape. The main achievements of Green Revolution may be summarised as under:

1. The production and productivity of wheat, rice, maize, and *bajra* has increased substantially.
2. India has become almost self sufficient in the matter of staple foods.
3. The double cropped area has increased, thereby intensification of the Indian agriculture has increased.
4. In the areas where Green Revolution is a success, the farmers have moved from subsistent to market oriented economy, especially in Punjab, Haryana, western Uttar Pradesh, and the plain districts of Uttarakhand (Hardwar and Udhamsinghnagar).
5. The adoption of High Yielding Varieties under the Green Revolution has generated more rural and urban employment.
6. Green Revolution has increased the income of farmers and landless labourers, especially that of the big farmers and the semi-skilled rural workers. Thus, Green Revolution has increased rural prosperity.
7. Green Revolution has created jobs in the areas of biological (seed fertilisers) innovations, and repair of agricultural equipments and machinery.

Green Revolution—Problems and Prospects

1. Depletion of soil owing to the continuous cultivation of soil exhaustive crops like rice and wheat.
2. Depletion of underground water table due to overirrigation of more moisture requiring crops like rice and wheat.
3. Green Revolution has increased the income disparity amongst the farmers.
4. Green Revolution led to polarisation of the rural society. It has created three types of conflicts in the rural community, namely, between large and small farmers, between owner and tenant farmers, between the employers and employees on agricultural farms.

5. Green Revolution has displaced the agricultural labourers, leading to rural unemployment. The mechanical innovations like tractors have displaced the agricultural labour.
6. Agricultural production in the Green Revolution areas is either stationary or has shown declining trend.
7. Some valuable agricultural lands have submerged under water (water-logging) or are adversely affected by salinity and alkalinity.
8. Green Revolution is crop specific. It could not perform well in the case pulses and oil-seeds.
9. The traditional institution of *Jajmani system* has broken. Consequently, the barbers, carpenters, iron-smith, and watermen have migrated to the urban areas.
10. The soil texture, structure, soil chemistry, and soil fertility have changed.
11. About 60% of agricultural land in the country remains unaffected by Green Revolution.
12. Green Revolution technologies are scale neutral but not resource neutral.
13. Punjab feeds the nation but farmers in the state, especially in the Malwa region fall prey to cancer. The take '***Cancer Train***' to Bikaner for cheap treatment.

SECOND GREEN REVOLUTION

The first Green Revolution had its own share of prosperity and problems. In the Food Security Atlas by the M.S. Swaminathan Research Foundation, Chennai, it was pointed out in 2004, "The Punjab which is India's granary today will become food insecure in 15–20 years from now, if the present unsustainable land and water use practices continue. The International Food Policy Research Institute, Washington, opined, with the rising demand for food in the coming decades, India will depend on the nainfed areas to help increase supply. In fact, the overall production of the cereal and non-cereal crops has reached almost the plateau stage. The growth rate of agricultural sector is only about two percent. Looking at the growing demand of agricultural produce, there is an urgent need for undertaking agriculture to a higher trajectory of four

percent annual growth rate. In order to achieve these objectives, various governments have undertaken important steps towards agricultural reforms. These reforms aim at efficient use of resources and conservation of soil, water and ecology on a sustainable basis, and in holistic framework. The main objectives of the second Green Revolution are: (i) To raise agricultural productivity to promote food security (ii) More emphasis on bio-technology (iii) To promote substainable agriculture (iv) To become self sufficient in staple food, pulses, oil seeds, and industrial raw material (v) To increase the per capita income of the farmers and to raise their standard of living. The holistic framework, thus, must incorporate financing of rural infrastructure such as irrigation, roads and power.

The Eleventh Five Year Plan has aptly highlighted such a holistic framework and suggested the following strategy to raise agricultural output:

1. Doubling the rate of growth of irrigated area.
2. Improving water management, rainwater harvesting, and watershed development.
3. Reclaiming degraded land and focusing on soil quality.
4. Bridging the knowledge gap through effective extension.
5. Diversifying into high value outputs, e.g. fruits, vegetables, flowers, herbs and spices, medicinal plants, bamboo, bio-diesel, but with adequate measures to ensure food security.
6. Providing easy access to credit at affordable rate of interest.
7. Improving the incentive structure and functioning of markets, and
8. Refocusing on land reforms issues.
9. Laying emphasis on the cultivation of pulses. With the limited availability of pulses overseas, development of hybrid varieties becomes a pre-requisite for increasing domestic production.
10. Focusing on the development of area specific seeds and their application.
11. Attention has to be focused on areas such as rainfed, drought-prone crops, and drought resistant crops, and those amenable to biotechnological application.

The National Commission on Farmers has already laid the foundation for such a framework. Moreover, the National Agricultural Innovation Project initiated in July, 2006, for enhancing livelihood security in partnership mode with farmers' groups, Panchayati-Raj institution and private sector would go a long way in strengthening basic and strategic research in frontier agricultural sciences.

WHITE REVOLUTION

The package programme adopted to increase the production of milk is known as White Revolution in India. The White Revolution in India occurred in 1970, when the ***National Dairy Development Board (NDDB)*** was established to organise the dairy development through the co-operative societies. Prof. Verghese Kuerin was the father of White Revolution in India. The dairy development programme through co-operative societies was first established in the state of Gujarat. The co-operative societies were most successful in the Anand District of Gujarat. The co-operative societies are owned and managed by the milk producers. These co-operatives apart from financial help, also provide consultancy. The increase in milk production has also been termed as ***Operation Flood***. ***Varghese Kurien (1921–2012)*** who is considered as the '***Father of White Revolution in India***' was one of world's great agricultural leaders of the 20th century.

Objectives

1. The main objectives of the co-operative society is the procurement, transportation, storage of milk at the chilling plants.
2. To provide cattle feed.
3. The production of wide varieties of milk products and their marketing management.
4. The societies also provide superior breeds of cattle (cows and buffaloes), health service, veterinary treatment, and artificial insemination facilities.
5. To provide extension service.

The technology of White Revolution is based on an extensive system of co-operative societies. Milk, after being collected at a village collection centre, is promptly transported to the dairy plant at the milk chilling

centre. Timing of collection is rigidly maintained by the village society, truck operators, and the quick transport to the dairy plants. Milk tankers, each, normally carry 14,000 litres of milk. The chilling centres are managed by producers' co-operative unions to facilitate the collection of milk from producers who live at some distance from the chilling centres and thus, the middlemen are eliminated.

Phases of the White Revolution

The White Revolution may be examined under the following three phases:

Phase I (1970–81)

During this period, the dairy development programme was set up in ten states to provide milk to the cosmopolitan cities, i.e., Mumbai, Kolkata, Delhi, and Chennai. The important step in this phase was the setting up of 4 Mother Dairies in Mumbai, Kolkata, Delhi, and Chennai.

Phase II (1981–85)

During this phase, the dairy development programme was extended in the states of Karnataka, Madhya Pradesh, and Rajasthan. In this phase, within 25 contiguous milk-shed areas (in 155 districts) a cluster of milk producers' union was established. The Research Institute at Hyderabad developed a vaccine called 'Raksha' to control cattle diseases. The programme also involved the improvement in milk marketing in 144 more cities of the country. The Dairy Co-operative societies were set up in 35,000 villages and the membership exceeded 36 lakhs.

Phase III (1985–2000)

A number of co-operative societies were set up in most of the major states of the country and the number of co-operatives went up by 1,35,439 with a membership of 14 million. The following (**Table 9.15)** shows the spurt in milk production in India.

Table 9.15

The Phenomenal Increase in Milk Production in India—1950–51 to 2017–18

Year	Milk Production in Million Tonnes	Per Capita Availability Grams/Day

Year	Milk Production in Million Tonnes	Per Capita Availability Grams/Day
1950–51	17.0	124
1960–61	20.0	126
1970–71	22.0	128
1980–81	31.6	128
1990–91	53.9	176
2000–01	80.6	220
2005–06	97.1	241
2009–10	112.54	263
2015–16	155.5	337
2016–17	165.4	355
2017–18	176.3	375

Source: *Govt. of India, India 2016, p. 97, and Department of Animal Husbandry, Dairying and Fisheries, Ministry of Agriculture & Farmers Welfare, Govt. of India, 2018.*

Achievements

White Revolution is as important to dairy development as Green Revolution has been to grain production. Its outcome is based on the improvement in cattle breeding and adoption of new technology.

Today, India has earned the first position in milk production in the world. In 2017–18, livestock sector produced 176.3 million tonnes of milk as compared to 54 million tonnes in 1990-91. (**Table 9.15).** Dairying has become an important role in providing employment and income generating opportunities. The contribution of milk alone in 2005–06 was over 1.25 trillion, which is much higher than that of paddy (0.75 trillion). The statewise distribution of milk production has been shown in (**Fig. 9.22)**. It may be observed from this figures that Gujarat, Maharashtra, U.P., Punjab, Haryana, Madhya Pradesh, Rajasthan, West Bengal,

Andhra Pradesh, Karnataka, and Tamil Nadu are the main milk-producing states of the country.

Some of the important achievements of the White Revolution are as under:

1. The White Revolution made a sound impact on rural masses and encouraged them to take up dairying as a subsidiary occupation.
2. India has become the leading producer of milk in the world. The milk production that was about 17 million tonnes in 1950–51 rose to over 145.7 million tonnes in 2014–15. The production of milk has gone up by more than six times when compared with that of the Pre-Independence situation.
3. The per capita availability of milk per day at present is about 337 gm as against 126 grams before the White Revolution.

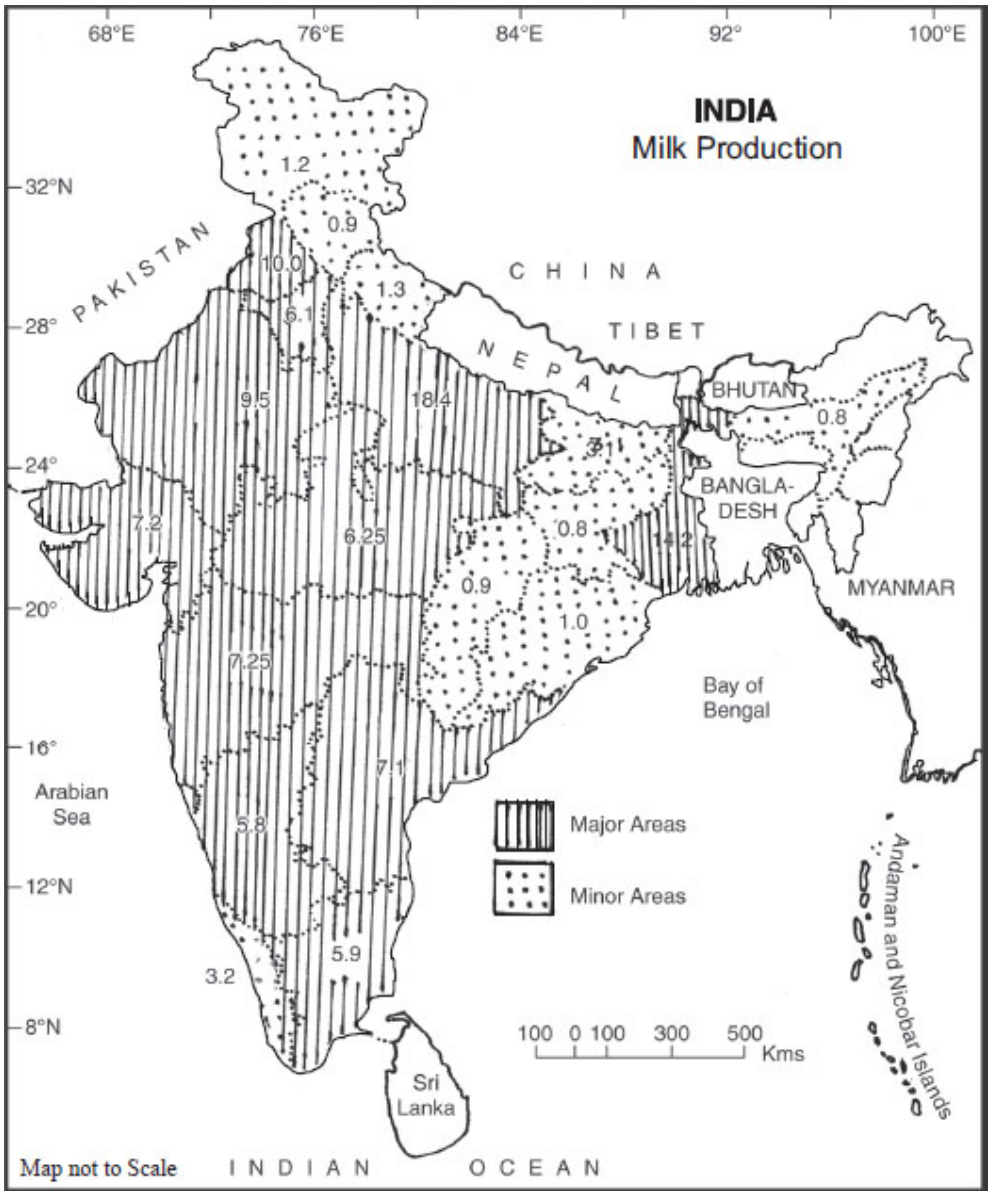


Fig. 9.22 Milk Production Percentage (2016–17)

4. The import of milk and milk production has been reduced substantially.
5. The small and marginal farmers and the landless labourers have been especially benefitted from the White Revolution. About 14 million farmers have been brought under ambit of 1,35,439 village level dairy co-operative societies. At present (2014-15), there are 145.7 million tonnes across India producing 20 million litres of milk every day.

6. To ensure the success of Operation Flood Programme, research centres have been set up at Anand, Mehsana, and Palanpur (Banaskantha). Moreover, three regional centres are functioning at Siliguri, Jalandhar, and Erode. Presently, there are metro dairies in 10 metropolitan cities of the country, beside 40 plants with capacity to handle more than one lakh litres of milk.
7. Livestock Insurance Scheme was approved in February 2006 and in 2006–07 on a pilot basis in 100 selected districts across the country. The scheme aims at protecting the farmers against losses due to untimely death of animals.
8. To improve the quality of livestock, extensive cross breeding has been launched.
9. For ensuring the maintenance of disease-free status, major health schemes have been initiated.
10. The government implemented livestock insurance on pilot basis in 2005–06.

The world's cattle inventory in 2018 is at 1.002 billion heads. India possesses the largest livestock population 30.44% in the world followed by Brazil 23.16%. It accounts for about 56% of the cattle population of the world's buffalo population, and 14% of the cattle population. It ranks first in respect of buffalo and second in respect of cattle population, second in goat population, and third in respect of sheep in the world.

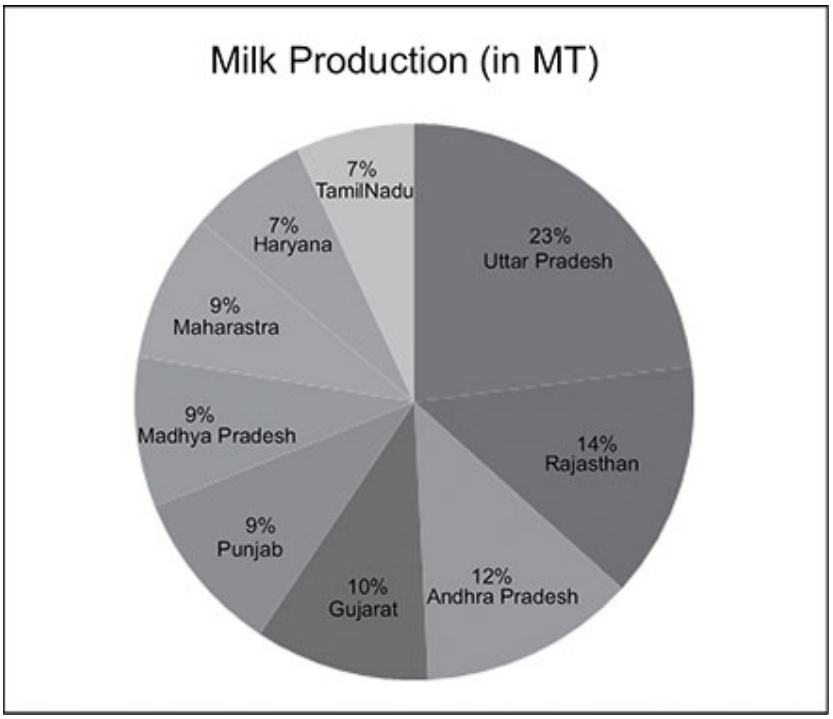


Fig. 9.23 Ten leading states of India in milk production

The chart in (Fig. 9.21) depicts the top 10 milk producing states of India. **'*Kamdhenu Scheme*'** was launched to develop quality *Milch Animals* as well as to develop entrepreneurship in animal husbandry. Loan was sanctioned upto 240 and 162 units of Kamdhenu scheme. The government invested an additional Rs. 596 crores in the scheme.

Problems and Prospects

Some of the important problems of the White Revolution are as under:

1. Collection of milk from the remote areas is expensive, time consuming, and not viable economically.
2. In most of the villages the cattle are kept under unhygienic conditions.
3. There are inadequate marketing facilities. The marketing infrastructure needs much improvement.
4. The breeds of cattle is generally inferior.
5. The extension service programme is not effective.

In India, dairy development has a great future. It should take the advantage of liberalisation in the global trade and should try to capture international market. Many corporate sector firms like Indana (plants at

Nagpur, Hyderabad, and Bangalore), The Sheel International and Milk and Food, and the Amrut Industries are taking advantage of the existing situation of liberalisation and globalisation. The government has constituted Technology Mission for dairy development and Amul Model Co-operatives are being promoted to cover about 60% of the total area of the country.

BLUE REVOLUTION

Blue Revolution means the adoption of a package programme to increase the production of fish and marine products. The Blue Revolution in India was started in 1970 during the Fifth Five Year Plan when the Central Government sponsored the ***Fish Farmers Development Agency (FFDA)***. Subsequently, the Brakish Water Fish Farms Development Agency were set up to develop aquaculture. The Blue Revolution has brought improvement in aquaculture by adopting new techniques of fish breeding, fish rearing, fish marketing, and fish export. India is the second largest producer of fish both in total, and from aquaculture. Under the Blue Revolution programme, there had been a tremendous increase in the production of shrimp. Andhra Pradesh and Tamil Nadu have developed shrimp in a big way. The '*Nellore District*' of Andhra Pradesh is known as the **Shrimp Capital of India**.

There are more than 1800 species of fish found in the sea and inland waters of India, of which a very few are commercially important. The important sea fish include catfish, herring, mackerels, perches, mullets, Indian salmon, shell fish, eels, anchovies, and dorab. Similarly, the main fresh water fish include catfish, loaches, perches, eels, herrings, feather backs, mullets, carps, prawns, murrels, and anchovies. Sea bass, Pompano, Milk fish, Cobia are 'brackish water' fish, whereas, Catla, Rohu, Mrigal, Silver Carp, Grass Carp are 'fresh water' fish. The production of fish in the country has been shown in (**Table 9.16)**, while (**Fig. 9.24)** shows the statewise percentage– production of fisheries.

It may be observed from (**Table 9.16)** that the fish production in the country has increased from 0.75 million tonnes in 1950–51 to 68.69 million tonnes in 2006–07. Fishing, aquaculture and a host of allied activities, a source of livelihood to over 14 million people as well as a major foreign exchange earner, in 2005–06 contributed about one

percent of the total GDP and 5.3% of the GDP from agriculture sector. India is the second largest fish producing country in the world with a production of about 12.60 MT and it aims to increase to 15.00 MT by 2020.

Table 9.16

India—Fish Production (production in million tonnes)

Year	Marine Fish	Inland Fish	Total
1950–51	0.535	0.240	00.75
1970–71	10.86	6.75	17.56
1990–91	23.00	15.36	38.36
2000–01	28.11	23.23	56.75
2010–11	29.89	48.62	78.51
2015–16	33.70	58.85	92.55
2016–17	36.41	77.68	114.09

Source: India 2012, pp 118–119, and Agricultural statistics at a Glance, 2016.

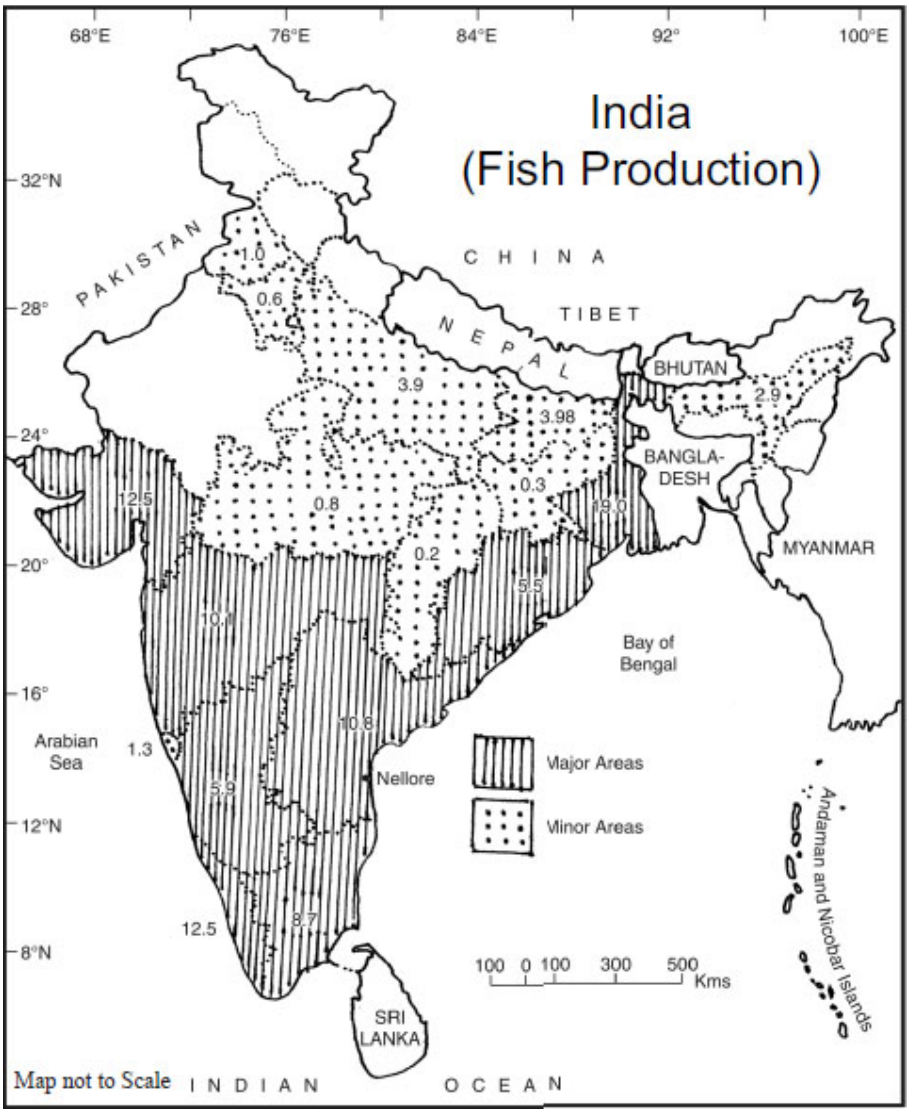


Fig. 9.24 Percentage of Fish Production

The geographic base of Indian marine fisheries has 8118 km coastline, 2.02 million sq of Exclusive Economic Zone including 0.5 million sq km of continental shelf, and 3937 fishing villages. There are 189 traditional fish landing centres, 59 minor fishing harbours, which serve as bases for about 2,80,000 fishing craft consisting of 1,81,000 non-motorised traditional craft and 54,000 mechanised boats. Out of 180 deep sea fishing vessels, only 60 are in operation at present.

About 50% of the country's total fish production comes from the inland fisheries including the freshwater fisheries like ponds, tanks, canals, rivers, reservoirs, and fresh water lakes.

Marine fisheries contribute about 50% of the total fish production of the country. Kerala is the leading producer followed by Maharashtra, Karnataka, Gujarat, and Goa. The fishing season extends from September to March. The higher fish production in the Arabian Sea is due to the broader continental shelf. The important fish varieties include sardines, mackerel and prawn.

The East Coast contributes about 28% of the total production of marine fish in the country. The fishing activity along the East Coast is mainly carried on from Rameswaram in the south to Ganjam in the north, with fishing season from September to April along the Coromandal Coast.

The ***National Fisheries Development Board was established in 2006*** to realise the untapped potential of fishery sector with the application of modern tools of research and development including biotechnology.

Strategies for the Fisheries Development

1. The Indian Council of Agricultural Research has established eight fisheries research institutes. These institutes are developing strategies for the exploitation of various aquatic resources.
2. Refrigeration and cold storage facilities have been provided in Chennai, Cuddalore, Kochi, Kollam, Kozhikode, Mumbai, Pune, Ratnagiri, and Thiruvananthapuram.
3. Training centres for fishermen have been established at Satpati (Maharashtra), Veraval (Saurashtra), and Kojan and Tutukandi (Tamil Nadu).
4. Fishing farm docks have been established at Cuddalore, Royapuram (Tamil Nadu), Kandla, Veraval (Gujarat), Vijinjam (Kerala), and Port Blair.
5. Various programmes have been launched by the government for the development of inland fisheries. Over five hundred fish farms have been established by the central government in collaboration with the state governments.
6. The Indian Council of Agricultural Research (ICAR) has established 422 district level Fish Farms in different parts of the country with due emphasis on development.

7. Under the Jawahar Rozgar Yojna, village panchayats have been authorised to carry out fisheries development programmes in respective villages.
8. Under the programme of Development of Model Fishermen Villages, basic civic amenities such as housing, drinking water and construction of community halls for fishermen villages are provided.
9. Brackish Fish Farmers Development Agencies (BFDA) functioning in the coastal areas of the country are providing a package of technical, financial and extension support to shrimp farmers.
10. Insurance facilities have been extended to fishermen for the insurance and security of their life.
11. The government is collecting data on the micro-climates of various water bodies to promote fisheries in the country.
12. Development of Fishing Harbours: Six majors fishing harbours (Cochin, Chennai, Vishakhapatnam, Roychowk, Paradwip and Mumbai), 62 minor fishing harbours and 194 fish landing centres have been constructed in various coastal states (India, 2012, p. 120).

Problems and Prospects

Despite tremendous success in the development of fisheries in the country during the last four decades, pisciculture is facing a number of problems.

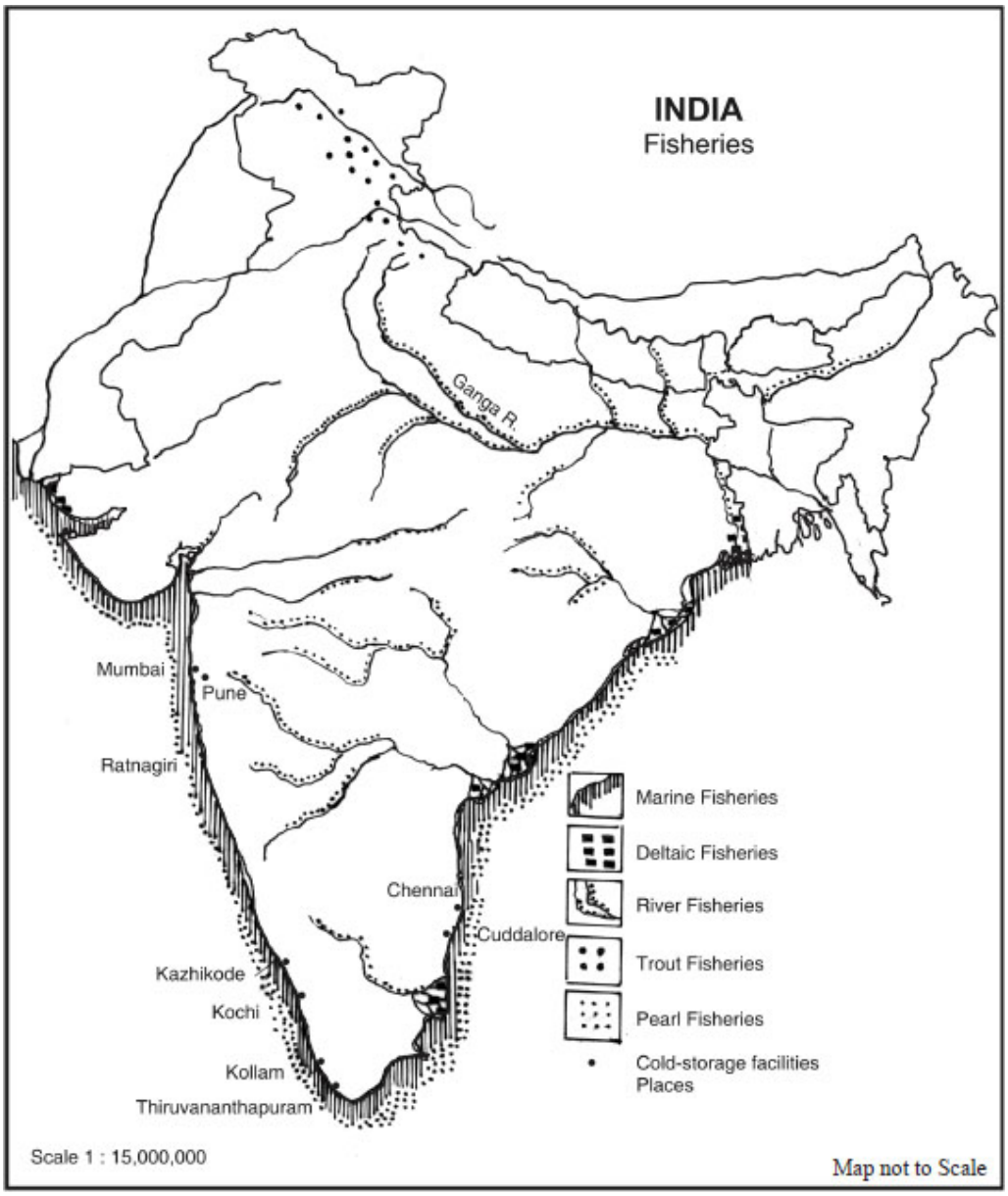


Fig. 9.25 Inland and Marine Fisheries

1. Most of the fishermen are poor. They are not able to purchase good equipment to improve the harvest of fish.
2. The water bodies (rivers, lakes, ponds, and coastal areas of the seas) are increasingly polluted.
3. The area of paddy fields in which fisheries used to be kept is also decreasing under the impact of fast growth of population, industrialisation, and urbanisation.
4. Adequate information about the environment of water-bodies

(ponds, lakes, rivers, and sea is not available).

5. Unpredictable nature of monsoon as a result of which the inland fisheries suffer adversely.
6. Problem of marketing, storage, and transportation.
7. Inadequacy of research and extension service facilities.
8. There is need of Pink Revolution (Prawns) in the coastal regions of the country.

AQUACULTURE

Aquaculture is the cultivation of aquatic organisms. Unlike fishing, aquaculture, also known as aquafarming, implies the cultivation of aquatic populations under controlled conditions. Mariculture refers to aquaculture practiced in marine environments. Particular kinds of aquaculture include agriculture (the production of kelp, seaweed, and other algae), fish farming, shrimp farming, shellfish farming, and growing of cultured pearls.

Growth and Development of Aquaculture

Aquaculture has been used in China since circa 2500 BC. The practice of aquaculture gained prevalence in Europe during the Middle Ages since fish were scarce and thus expensive. Americans were rarely involved in aquaculture until the late 20^{th} century but California residents harvested wild kelp and made legal efforts to manage the supply starting circa 1900, later even producing it as a wartime resource. In contrast to agriculture, the rise of aquaculture is a contemporary phenomenon.

Types of Aquaculture

1. Algaculture

Algaculture is a form of aquaculture involving the farming of species of algae.The majority of algae that are intentionally cultivated fall into the category of microalgae, also referred to as phytoplankton, microphytes, or planktonic algae.

Macroalgae, commonly known as seaweed, also have many commercial and industrial uses, but due to their size and the specific requirements of the environment in which they need to grow, they do not

lend themselves as readily to cultivation on a large scale as microalgae and are most often harvested wild from the ocean.

2. Fish Farming

Fish farming is the principal form of aquaculture, while other methods may fall under mariculture. It involves raising fish commercially in tanks or enclosures, usually for food. Fish species raised by fish farms include salmon, catfish, tilapia, cod, carp, trout, and others.

Increasing demands on wild fisheries by commercial fishing operations have caused widespread overfishing. Fish farming offers an alternative solution to the increasing market demand for fish and fish protein.

3. Freshwater Prawn Farming

A freshwater prawn farm is an aquaculture business designed to raise and produce freshwater prawn or shrimp for human consumption. Freshwater prawn farming shares many characteristics with, and many of the same problems as, marine shrimp farming. Unique problems are introduced by the development life cycle of the main species (the giant river prawn, *Macrobrachium rosenbergii*).

4. Integrated Multi-Trophic Aquaculture

Integrated Multi-Trophic Aquaculture (IMTA) is a practice in which the by- products (wastes) from one species are recycled to become inputs (*fertilisers, food*) for another. Fed aquaculture (e.g., fish, shrimp) is combined with inorganic extractive (e.g., *seaweed*) and organic extractive (e.g., *shellfish*) aquaculture to create balanced systems for environmental sustainability (biomitigation), economic stability (product diversification and risk reduction), and social acceptability (better management practices).

5. Mariculture

Mariculture is a specialised branch of aquaculture involving the cultivation of marine organisms for food and other products in the open ocean, an enclosed section of the ocean, or in tanks, ponds or raceways which are filled with seawater. An example of the latter is the farming of marine fish, prawns, or oysters in saltwater ponds. Non-food products produced by mariculture include fish meal, nutrient agar, jewelleries (e.g. *cultured pearls*), and cosmetics.

6. Shrimp Farming

A shrimp farm is an aquaculture for the cultivation of marine shrimp for human consumption. Commercial shrimp farming began in the 1970s, and production grew steeply, particularly to match the market demands of the US, Japan, and Western Europe.The total global production of farmed shrimp reached more than 2.5 million tonnes in 2014–15, representing a value of nearly USD 10,500 million. About 75% of farmed shrimp is produced in Asia, in particular in China and Thailand.The other 25% is produced mainly in Latin America, where Brazil is the largest producer.The largest exporting nation is Thailand. Shrimp farming on modern lines is being done in Andhra Pradesh (Nellore District), a state of India (see *Blue Revolution*).

APICULTURE (BEEKEEPING) OR GOLDEN REVOLUTION

Apiculture is the science and culture of honeybees and their management. Beekeeping (or apiculture, from Latin *apis*, a bee) is the practice of intentional maintenance of honey bee colonies, commonly in hives, by humans. A beekeeper (or apiarist) may keep bees in order to collect honey and beeswax, or for the purpose of pollinating crops, or to produce bees for sale to other beeskeeper. A location where bees are kept is called an apiary.

History of Beekeeping

Globally, there are more than 20,000 species of wild bees, many of which are solitary or which rear their young in burrows and small colonies, like mason bees or bumblebees. Beekeeping, or apiculture, is concerned with the practical management of the social species of honey bees which live in large colonies of upto 100,000 individuals.

Wild Honey Harvesting

Robbing honey from wild bee colonies is one of the most ancient human activities and is still practiced by aboriginal societies in parts of Africa, Asia, Australia, and South America.

Domestication of Wild Bees

At some point humans began to domesticate wild bees in artificial hives made from hollow logs, wooden boxes, pottery vessels, and woven straw baskets or *skeps*.

Traditional Method of Honey Collection

Early forms of honey collecting entailed the destruction of the entire colony when the honey was harvested.The wild hive was crudely broken using smoke to suppress the bees, the honeycombs were torn out and smashed up, along with the eggs, larvae, and honey they contained.The liquid honey from the destroyed brood nest was crudely strained through a sieve or basket.This was destructive and unhygienic but for honey-gathering societies this did not matter since the honey was generally consumed immediately and there were always more wild colonies to exploit.

Traditional Beekeeping

Fixed Frame Hives

There are considerable regional variations in the type of hive in which bees are kept. A hive is a set of rectangular wooden boxes filled with moveable wood or plastic frames, each of which holds a sheet of wax or plastic foundation. The bees build cells upon the sheets of foundation to create complete honeycombs. Foundation comes in different sizes: '*worker foundation*' which enables the bees to create small, hexagonal worker cells, and '*drone foundation*' which allows the bees to build much larger cells—the drone cells for the production of male bees.

The bottom box, or brood chamber, contains the queen and most of the bees ; the upper boxes, or supers, contain just honey.Only the young nurse bees can produce wax flakes which they secrete from between their abdominal plates; they build honeycomb using the artificial wax foundation as a starting point, after which they may raise brood or deposit honey and pollen in the cells of the comb.These frames can be freely manipulated and honey supers with frames full of honey can be taken and extracted for their honey crop.

Modern Beekeeping

Movable Frame Hives

In the USA, the Langstroth hive is commonly used. The Langstroth was the first successful top-opened hive with movable frames, and designs of hive have been based on it. Langstroth hive was, however, a descendant of Jan Dziezon's Polish hive designs. In the United Kingdom, the most common type of hive is the British National Hive, but it is not unusual to

see some other sorts of hive (Smith, Commercial, and WBC, rarely Langstroth). Straw skeps, bee gums, and unframed box hives are now unlawful in most of the US states, as the comb and brood cannot be inspected diseases. However, straw skeps are still used for collecting swarms by hobbyists in the UK, before moving them into standard hives.

Top Bar Hives

A few hobby beekeepers are adopting various top bar hives of the type commonly found in Africa. These have no frames and the honey-filled comb is not returned to the hive after extraction, as it is in the Langstroth hive. Because of this, the production of honey in a top bar hive is equal to only about 20% of the production of a Langstroth hive, but the initial costs and equipment requirements are far lower. Top-bar hives also offer some advantages in interacting with the bees and the amount of weight that must be lifted is greatly reduced. Top Bar Hives are being widely used in developing countries in Africa and Asia as a result of '***Bees For Development***' programme.

Types of Beekeepers

Beekeepers generally categorise themselves as:

- Commercial beekeeper—Beekeeping is the primary source of income.
- Sideliner—Beekeeping is a secondary source of income.
- Hobbyist—Beekeeping is not a significant source of income.

The Colony of Bees

A colony of bees consists of three classes of bee: a queen, which is normally the only breeding female in the colony; a large number of female worker bees, typically 30,000–50,000 in number; and a large number of male drones—ranging from thousands in a strong hive in spring to very few during death or cold season

The production and consumption of honey in some of the selected countries of the world has been given in (**Table 9.17).**

Table 9.17

World Honey Production and Consumption in 2016

Country	Production (1000 Metric	Consumption (1000 Metric

	Tonnes)	Tonnes)
Russian Federation	65.0	70
United States of America	95.2	175
Argentina	98.5	65
China	310.8	290
Turkey	85.45	75
India	89	50

Source: Food and Agriculture Organisation of the United Nations, 2016.

It may be seen from the (**Table 9.17)** that China is the largest producer and consumer of honey in the world with a production of about 300 thousand metric tonnes annually. Argentina stands second with an annual production of 93.42 thousand metric tonnes, followed by Turkey, USA, and India. India produced over 52 thousand metric tonnes in 2005.

As per the 2016–17 data, 94.5 thousand MT of honey was expected to be produced. In order to increase the honey production in India, in 1953, 'All India Khadi & Village Industries Board' started organizing the honey industry and in 2015–16, KVIC targeted to train about 2 lakh people in beekeeping and support them under the 'Entrepreneurship Development Programme'. The major states that produce honey are West Bengal, Uttar Pradesh, Punjab and Bihar contributing about 61% of the total production as reported in 2015–16.

India exports honey to U.S.A., Saudi Arabia, UAE, Morocco, Bangladesh, Canada, etc. In 2015–16, India exported 38.2 thousand MT of honey worth Rs. 706 crores.

SERICULTURE

Sericulture or silk farming is the rearing of silkworms for the production of raw silk. Although there are several commercial species of silkworms, *Bombyx mori* is the most widely used and intensively studies. According to Chinese records, the discovery of silk production from *mori* occurred about 2700 BC making the start of history of silk. Today, China and Japan are the two main producers, together manufacturing more than 50% of the world production each year. India has a unique distinction of

producing all the five known varieties of silk viz. mulberry, Oaktsar, tropical tsar, eri, and muga.

Production

Silkworm larvae are fed on mulberry leaves and after the fourth molt, they climb a twig placed near them and spin their silken cocoons. The silk is a continuous-filament fibre consisting of fibroin protein, secreted from two salivary glands in the head of each larva, and a gum called *sericin*, which cements the two filament together. The sericin is removed by placing the cocoons in hot water, which frees silk filaments and readies them for reeling. The immersion of cocoons in hot water also kills the silkworm larvae.

In India, silk worms thrive on the leaves of mulberry, *mahua, sal*, *ber,* and *kusum* trees. India ranks third among the silk producing countries of the world. In India, about 4.5 lakh hectares of area is under mulberry cultivation. Silk production is mainly confined to areas between 15° and 34° N latitudes. About 55 lakh people are engaged in this industry.

The state of Karnataka is the largest producer of raw silk (65%) follwed by Andhra Pradesh (17%) West Bengal (8%), Tamil Nadu (5%), Assam (2.5%) and Jammu & Kashmir (1.2%), Limited quantity of mulberry silk is also produced in Arunachal Pradesh, Chhattisgarh, Himachal Pradesh, Jharkhand, Kerala, Madhya Pradesh, Maharashtra, Manipur, Odisha, Punjab, Tripura, Uttarakhand, and Uttar Pradesh.

The growth pattern of mulberry cultivation has been given in (**Table 9.18).**

Table 9.18

Trend in Raw Silk Production

Year	Silk (Lakh kg.)
1950–51	52.50
1960–61	78.32
1970–71	231.90
1980–81	459.30
1990–91	114.85
2000–01	165.00
2014–15	178.00

Year	Silk (Lakh kg.)
2015–16	243.0
2016–17	281.9
2017–18	326.8

Source: Statistical Abstracts India, years 2017–18

It may be seen from the (Table 9.18) that in 1950–51 the raw silk production was 52.50 lakh kg which has risen to 326.8 lakh kg in 2017–18. Silk production has a great future in India.

POULTRY FARMING (SILVER REVOLUTION)

Poultry farming is the practice of raising poultry, such as chickens, turkeys, ducks, geese, as a subcategory of animal husbandry, for the purpose of farming meat or eggs for food. It requires small capital and provides additional income and job opportunities to a large number of rural population in the shortest possible time. The vast majority of poultry are farmed using factory farming techniques; according to the Worldwatch Institute, 75% of the world's poultry meat, and 70% of eggs are produced in this way.

The contrasting method of poultry farming in free range and friction between the two main methods, has led to long term issues of ethical consumerism. Opponents of the factory farming argue that it harms the environment and creates health risks, as well as abuses animals. In contrast, proponents of factory farming highlight its increased productivity, stating that the animals are looked after in state-of-the art confinement facilities and are happy; that it is needed to feed the growing global human population; and that it protects the environment.

Poultry Farming in India

Poultry farming in India is quite old. At present, more than three million people are directly or indirectly employed in poultry farming. It produced around 2.4 million tonnes of chicken meat in 2010–11. Between the 1970 and 2011, the annual per capita availability of eggs has quadrupled from 10 to 45, while the corresponding increase in chicken meat has been faster from, 145 grams to 1.6 kgs. India produced more than 82.92 billion in 2015–16 (India 2016, p. 97).

Poultry sector, besides employment generation and subsidiary income, increase provides nutritional security especially to the rural poor. Further, landless labourers derive more than 50% of their income from livestock, especially poultry.

Exports of products such as live poultry, eggs, hatching eggs, frozen eggs, egg powder, and poultry meat to countries including Bangladesh, Sri Lanka, South West Asia, Japan, Denmark, Poland, USA, and Angola augurs well for industry. The value of output from poultry sector is nearly Rs. 20,000 crore.

In India, there are over 260 million hens in the country which laid down about 30 billion eggs during 2010–11. The largest number of poultry population is in Andhra Pradesh followed by Bihar, West Bengal, Tamil Nadu, Maharashtra, Assam, Karnataka, Kerala, Odisha, Madhya Pradesh, Uttar Pradesh, Punjab, and Haryana. Most of the important poultry farms are being developed around almost all the important urban centres like Mumbai, Kolkata, Delhi, Chennai, Hyderabad, Bangalore, Pune, Nagpur, Shimla, Bhubaneshwar, Ajmer, Chandigarh, and Bhopal.

The production of eggs in Tamil Nadu was 161.25 billion and accounted for 19.44% of the total egg production in the country. Tamil Nadu, Andhra Pradesh, Telangana, West Bengal, Maharashtra, Haryana, Karnataka, Punjab, Kerala and Uttar Pradesh are the top states in the egg production. (https://data.gov.in/catalog/production-wool-and-eggs)

Uninterrupted supplies of feed as well as ***avian influenza*** are critical for the continued robust growth of the poultry sector. The first outbreak of avian influenza occurred in India in the state of Maharashtra in the Nandurbar district on 18 February. 2006. The Government of India initiated immediate steps to control and contain the outbreak.

The Central Poultry Development Organisation has been playing a pivotal role in the implementation of the policies of the Government with respect to poultry as a tool for alleviating nutritional hunger and palliating the impecuniosities of the resource-poor farmers, especially the women. The mandate of the Central Poultry Development Organisation has been specifically revised, by restructuring all poultry units of this Department to focus on improved indigenous birds, which lay on an average 180–200 eggs per annum and have a vastly improved FCR ratio in terms of feed consumption and weight gain. The Central Poultry Development

Organisations have been entrusted with the responsibility of producing excellent germplasm in the form of day-old chicks and hatching eggs of these varieties like Nierbheek, Hitkari, Vanaraja, Shyama, Cari, Chabro, etc. Besides, these organisations are also playing a crucial role in analysing feed samples.

These organisations, besides conducting the activities stated above, also work for scaling-up of diversification of other avian species like Ducks/Turkeys/Guinea fowl/Japanese Quail, and upgrading of Training Unit into International Tropical Avian Management Institutes in which private-public partnership is envisaged. Presently these organisations are also supporting and hand-holding the centrally-sponsored schemes related to assistance to state poultry farms.

A new centrally-sponsored scheme called Assistance to State Poultry, is being implemented during the Tenth Plan where one time assistance is provided to suitably strengthen the farms in terms of hatching, brooding, and rearing of birds with provision for feed mill and their quality monitoring and in-house disease diagnostic facilities.

A subsidy of 25% on loans is provided for poultry farming under the '*Poultry Venture Capital Fund Scheme (PVCFS)*'. The nodal agency for the implementation of this scheme is NABARD through nationalised commercial bank.

The Department of Agriculture and Cooperation, Ministry of Agriculture under the Mission for Integrated Development of Horticulture (MIDH) has launched a project called '***Chaman***' with an objective to develop and firm up scientific methodology for estimation of area and production under horticulture crops.

HORTICULTURE

Horticulture is a branch of agriculture relating to the cultivation of fruits, vegetables and ornamental plants. Horticulture is a capital and labour intensive agriculture. India is bestowed with varied agro-climates, which is highly favourable for growing large number of horticultural crops such as fruits, vegetables, spices, root tuber, ornamental, aromatic plants, medicinal species and plantation crops like coconut, arecanut, cashew and cocoa. Presently, horticulture crops occupy about 10% of the gross cropped area of the country, producing about 305.4 million tonnes in

2017–18. India is the second largest producer of fruits and vegetables with a sown area of 24.92 million hectares. The total production of fruits has been estimated at about 63 million tonnes from 5.7 million hectares. India's share in world fruit and vegetable production is 10% and about 13.25% respectively. The Haryana goverment lanched 'Bhavantar Bharpai' scheme for potato, onion, tomato and cauliflower. The goverment fixes best price for the vegetables.

Fruits

Indian climate favours the development of a large range of varieties of fruits. Indians' share in the total fruit production of the world is 10%. Mango, banana, citrus, pineapple, papaya, guava, sapota (*cheekoo*), jackfruit, litchi, and grapes, among the tropical and subtropical fruits; apple, pear, peach, plum, apricot, almond, walnut, among the temperate fruits; and aonla, ber, pomegranate, fig, phalsa, among the arid fruits are important. India leads the world in the production of mango, banana, sapota (*cheekoo*) and nimboo (*acid lime*), and in productivity of grapes per unit land area.

India is the largest producer of mango, banana, sapota, and acid-lime. About 40% of world's mango and 23% of world's banana are produced in the country. In grapes, India has recorded the highest productivity per unit area in the world. The overall production of the horticulture crops registered an increase of 8% during 2004–05 as compared to 2003–04, while the percentage increase in fruit production has been about 1.5% during the same period.

Vegetables

More than 40 kinds of vegetables are grown in India. Important vegetable crops grown in the country are potato, tomato, onion, chillies, carrot, raddish, turnip, beans, ladiesfinger, guard, letuce, brinjal, cabbage, cauliflower, spinach, okra, and peas.

India is next only to China in area and production of vegetables, and occupies the first position in the production of cauliflower, second in onion, and third in cabbage in the world. The contribution of vegetables is the highest (59–61%) in the horticulture crop production, as can be seen from (**Fig. 9.26)**. The area under vegetable during 2016–17 was about 10.3 million hectares with a production of 175 million tonnes. Uttar

Pradesh was the leading state (26.4 MT) followed by West Bengal (25.5 MT).

Flowers

Though flower cultivation has been practised in India since times immemorial, floriculture has blossomed into a viable business only in recent years. The increased growing of contemporary cut-flowers like rose, gladiolus, tuberose, carnation, etc., has led their use for bouquets and arrangements for gifts, as well as decoration of both home and workplace. A growing market, as a result of improvement in the general level of well-being in the country and increased affluence, particularly among the upper and the middle classes, has led to transformation of the activity of flower growing into a well developed industry. Availability of diverse agro-climatic conditions in the large country, facilitates production of all major flowers throughout the year in some part or the other, and improved transportation facilties, have increased the availability of flowers all over the country.

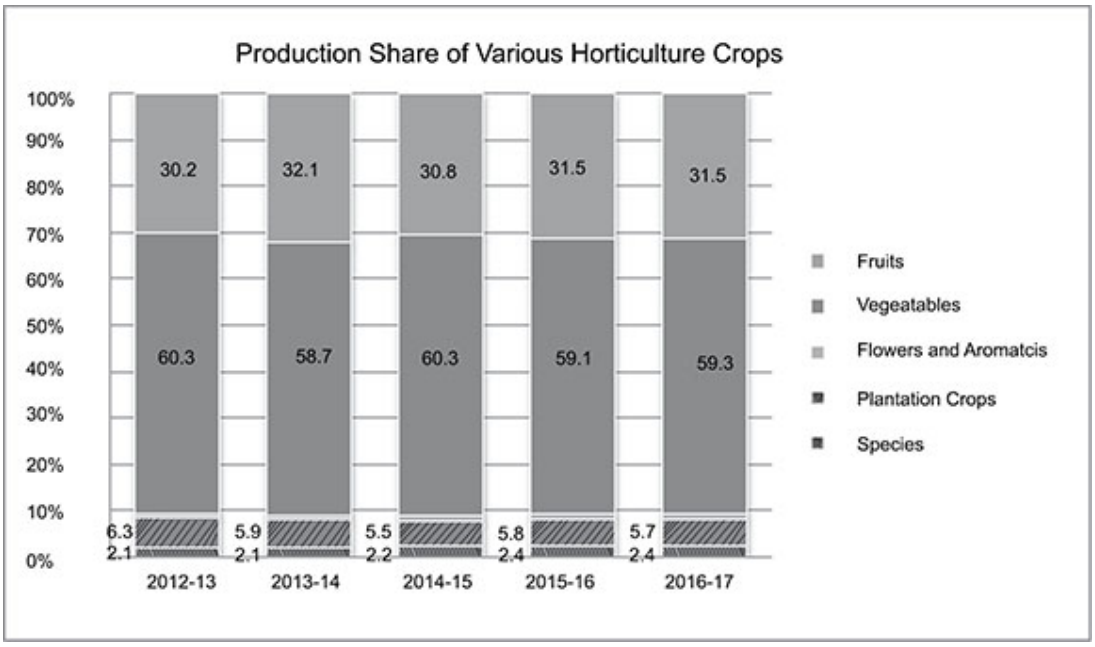


Fig. 9.26 Production Share of Various Horticulture Crops in India
[*Source: Horticulture Statistics at a Glance 2017, Government of India*]

India has made noticeable advancement in the production of flowers. Floriculture is estimated to cover an area of 1.14 lakh hectares with a

production of 670,000 million tonnes of loose flowers and 13,010 million tonnes of cut flowers.

A major programme, namely '***National Horticulture Mission (NHM)***' was launched in the country during the Tenth Five Year Plan with effect from 2005–06. The main objectives of the Mission are to enhance horticulture production through area based regionally differentiated strategies to improve nutritional security and income support to farm households, and to promote and disseminate technologies.

Poly House Revolution

Poly House Revolution is a high tech-agriculture, using precision methods to boost the production of vegetables. The high tech poly house is made out of a polycarbonate double layer transparent sheet, and installed with digital water showers, digital heaters and digital clocks, along with a special green coloured net to protect the sapling or crops from excess heat or cold.

In poly houses, the saplings and seedlings are prepared of onion, lettuce, garlic, cauliflower, cabbage, potato, tomato, cucumber in the months of cold season (January, February, and March). The poly houses can grow vegetables during the off-season.

Poly houses are very common in the developed countries like Germany, France, Italy, Japan, New Zealand, U.K., and U.S.A. In India, in the valley of Kashmir, there are 9000 poly-green houses. Poly houses in fact, has brought a revolution in the Valley of Kashmir. These houses are becoming increasingly popular in the states of Himachal Pradesh and Uttarakhand. Hopefully, the poly-house revolution will bring more posperity to the growers and will provide adequate vegetables in the off-season.

DRY FARMING

The spread of the dry farming is in the regions where the average annual rainfall is less than 75 cm and irrigation facilities are not available. About 60% of the net-cultivated area is under dryland and rainfed cultivation in India, which contributes 40% of the total agricultural production. In these areas the rainfall is scanty and uncertain, where hot and dry conditions prevail. It is not only that the average annual rainfall is low, the variability

of rainfall in these areas varies between 25 to 60%. Agriculture in the dry farming regions belongs to fragile, high risking and low productive agricultural ecosystem. The areas in which more than 75 cm of average annual rainfall is recorded are known as the '*areas of rain-fed agricultur***e**' **(Fig. 9.27).**

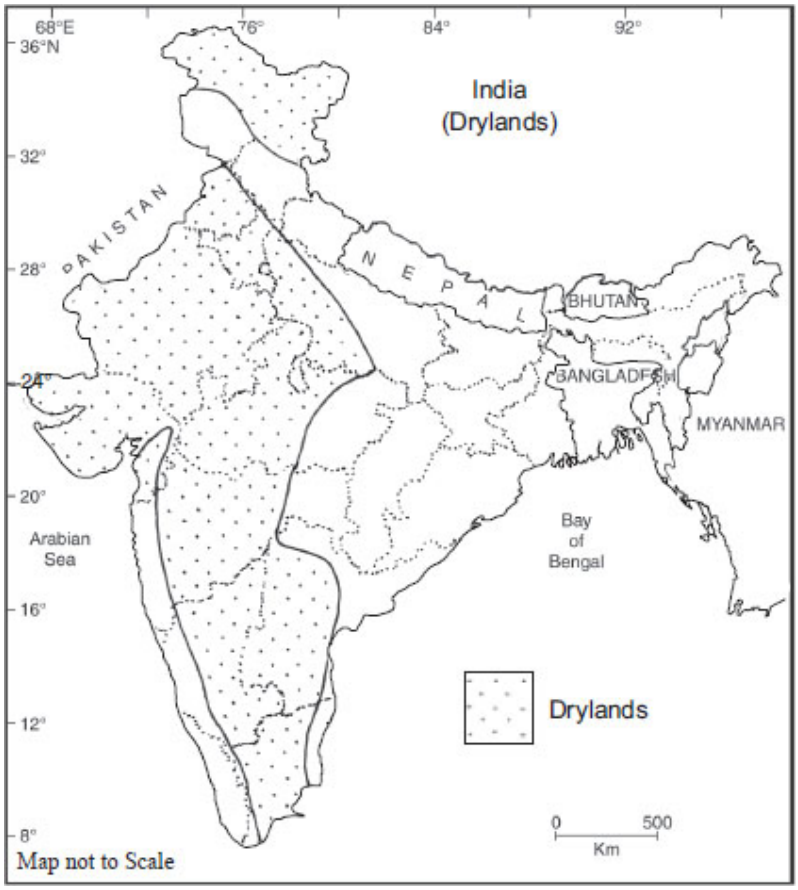


Fig. 9.27 Dry Farming Areas

In India dry-lands cover about 108 million hectares at rainfed area (75%). The dry farming areas cover the greater parts of Rajasthan and Gujarat. Moreover, there are small tracts of dry land farming in Punjab, Haryana, Maharashtra, Andhra Pradesh, Karnataka, Himachal Pradesh, Jammu and Kashmir, Jharkhand, Chhattisgarh, Odisha, Uttarakhand, Uttar Pradesh, West Bengal, and Tamil Nadu. These areas having scanty rainfall and high variability of rainfall are adversely affected by erratic precipitation, frequent droughts, high temperature, and high wind velocity

resulting in soil erosion. More than 75% of the peasants involved in day farming are small and marginal.

Salient Features of Dry Farming

Dry farming in India is characterised by the following:

1. Moisture conservation is basic to dry farming. In order to achieve this objective, the field is ploughed repeatedly, especially during the rainy season.
2. Sowing of crops in alternate years or fallowing of land after each harvesting of crop. The fallowing of agricultural land helps in the recuperation of soil fertility.
3. Pulverisation of the soil before sowing.
4. Regular hoeing and weeding of the crop to control weed growth and to conserve moisture. Hoeing is generally done before sun-rise so that the night dew may be mixed into the soil to provide moisture to the crops.
5. Covering of the land with straw to prevent evaporation of the soil moisture and to control soil erosion.
6. Livestock keeping and dairying are also important allied agricultural activities in the dry farming regions.
7. Where ever possible, thrifty use of water.
8. Development and efficient use of solar energy.

Crops

The main crops grown in the dry farming areas are coarse, grains (maize, millets, bajra), wheat, barley, pulses, groundnut, oilseeds and fodder. Though 75% of the total population of dry-farming regions are directly or indirectly dependent on agriculture, their per capita income, and standard of living are significantly low.

The cropping patterns and their combinations in the dry-farming regions of India have been given in (**Fig. 9.28).**

Main Problems of Dry Farming

The main problems of dry farming agriculture are as under:

1. Scarcity of precipitation, erratic occurrence of rains leading to famines, droughts, and floods.
2. The soils, being sandy, lack in humus and organic nutrients.

3. The dry farming areas are highly vulnerable to soil erosion.
4. These are low yields per unit area.
5. In the absence of moisture and irrigation, the use of High Yielding Varieties and new technology is not possible.

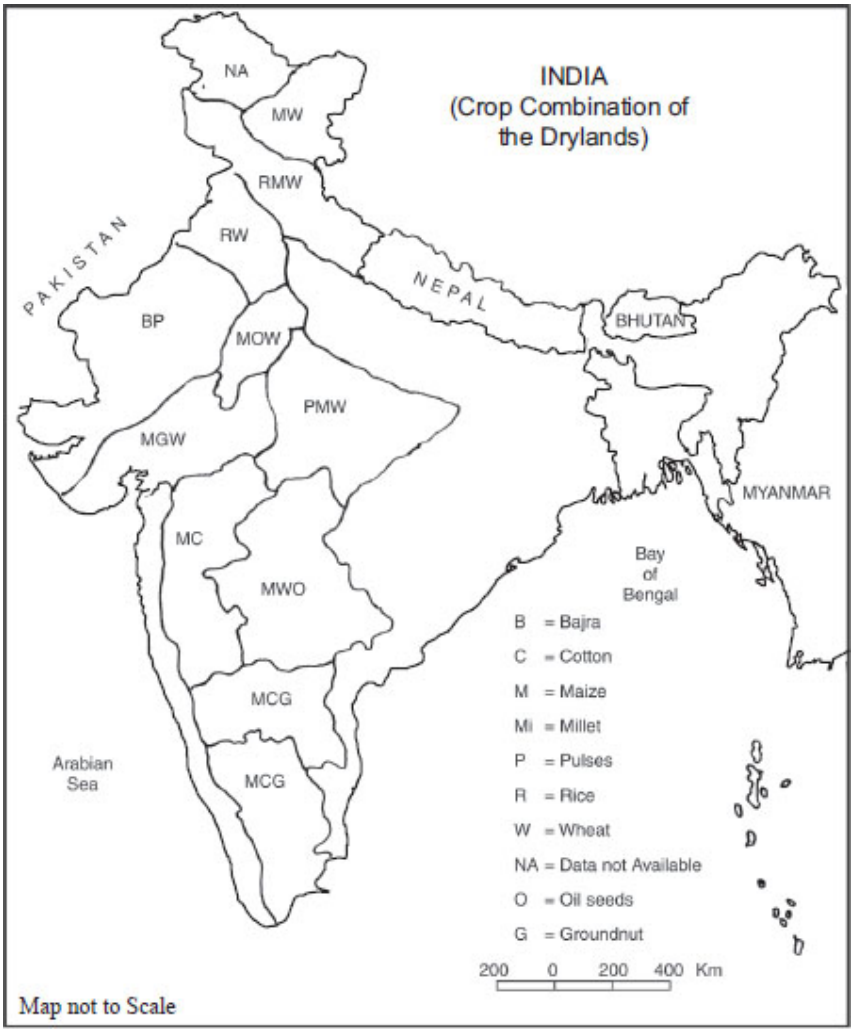


Fig. 9.28 Dry Farming Areas Crop Combination

6. Most of the farmers in the dry farming regions being poor, are not able to apply the new costly inputs.
7. These areas are not having the basic irrigation and other infrastructural facilities, like roads, marketing and storage.

Strategy for Development

As stated earlier, agriculture is a highly vulnerable occupation in the scanty rainfall recording areas in which dry farming is practiced.

1. In dry farming areas, water harvesting should be done. The government and other non-government agencies should provide the necessary guidance to the people.
2. Seeds of food crops which are drought resistant should be provided to the farmers at a subsidised rate.
3. Efforts should be made to check soil erosion by adopting soil conservation practices.
4. The farmers should space their crops at a wide gap and there should be regular weeding and hoeing.
5. Seeds of the quick and short duration maturing crops should be developed.
6. Cultivation of crops requiring more moisture should be done in the low lying areas, especially in the lower parts of the catchment.
7. Cotton should be grown only in the areas where rainfall is more dependable or sprinkle irrigation is available. Drip irrigation should be practiced to save water.
8. Soil fertility should be enhanced by applying cowdung and compost manures.
9. Repeated tilling of the field is required during the rainy season.
10. Research should be promoted in the dry land farming.
11. Agronomic practices (mixed cropping and crop rotation need to be adopted)

In addition to these, there are many other practices like contour-ploughing, contour-bunding, and field-bunding that help water conservation measures. Practice like mulching prevents evaporation from the soil. Deep placement of manures and fertilisers would help the roots to penetrate deep layers. This, along-with weed control, will help in increasing the yield. The latest advance technology of dry farming is to lay stress on soil moisture and its conservation. It should be noted that in the dry areas, soils suffer from nutrient deficiency, particularly nitrogen. Band placement of fertilisers in sub-soil layers is a good method of helping the roots to go deep for exploiting the conserved moisture. The

World Congress on Conservation Agriculture was organised in February 2009 at New Delhi in order to address the issues of improving efficiency, equity and environment in which the problems and prospects of dry farming were also discussed.

AGRIBUSINESS

Agribusiness means agriculture for commercial purpose. Johnston and colleagues have defined agri-business as farming organisation applying modern management techniques and accounting methods with the aim of maximising final profit. The concept is applied differently in the USA and Western Europe. In the USA, agribusiness has grown as a result of increased involvement by food processing companies in the actual production of their own raw material inputs. Such companies have purchased farms and run them as subsidiary elements within their overall production system, while in Western Europe such integration of farm production with processing is less common. In Western Europe and also in some developing countries like India, agribusiness constitutes large farming companies independent of food processors. But there are certain characteristics which are common to all agribusiness. More specifically, that part of modern national economy devoted to production, processing, and distribution of food and fibre products and by-products falls under the category of agribusiness.

Characteristics of Agribusiness

Agribusiness essentially has the following characteristics:

1. Hierarchical system of management—financial administrators and accountants with farm managers to carry out day-to-day business
2. Large farms
3. Farming operations organised in sizeable production units
4. An extension of the plantation system

In highly industrialised countries, many activities essential to agriculture are carried out at places away from the farm. These include the development and production of equipment, fertilisers, and seeds. In some countries, the processing, storage, preservation, and delivery of agricultural products have also been separated from basic farming. In

consequence, farming itself has become increasingly specialised and business-like. Some business firms even raise crops, as in the case of winery, that operates its own wine-yards, or large commercial producers who maintain their own farms. Many of these farms are extensively mechanised with computer technology to increase production.

In recent years, conglomerate companies that are involved in non-agricultural business have entered into agri-business by buying and operating large farms. Some food processing farms that operate farms have begun to market fresh produce, under their land resources.

India has recently entered into agribusiness, which consists of processed and unprocessed agricultural products. In the processed category are processed fruits and vegetable juices, meat, and meat and fish products, sugar and molasses, coffee and tobacco, spices, and wheat, tea, and rice. Entry into agribusiness requires polishing and parboiling. Commodities like fruits, vegetables, and eggs are sorted out according to size, shape, and colour. Canning of fruits and vegetables constitute an important part of agribusiness. However, there is need in India to develop facilities for research to evolve new techniques and standardised recipes and methods of manufacture of fruits and vegetable products.

It is also necessary to develop sophisticated allied industries such as manufacture of containers and closures (covers) equipments and machinery used by fruits and vegetable processing. Further, grant of subsidies and incentives like the removal of duty on tin plates, sugar, and railway freight to processors is necessary to encourage exports.

India can have good agribusiness in mango products and other processing industries based on fruits like apple, pineapple, oranges, lemon, lime, *aonla* (*Phyllanthus emblica sp.*), and guava, and vegetables like peas, tomato, and potatoes (canned). Other fruits and vegetables which are exported to some extent in the processed form are apricot, *bel* (*Aegle marmelos*), peach, plum, strawberry, ladies finger, brinjal, cabbage, carrot, *karela* (bitter gourd), *tinda* (*citrullus volgaris*), mustard green, *parwa*l, spinach, turnip, chillies, and ginger.

NATIONAL COMMISSION ON FARMERS

The National Commission on Farmers (NCF) Constituted on 18th November, 2004 under the chairmanship of Dr. M.S. Swaminathan gave the following recommendations to improve the conditions of farmers. The recommendations include:

1. Assest reforms covering land, water, livestock, and bio-resources
2. Farmer-friendly support services covering extension, training and knowledge
3. Credit and insurance
4. Assured and remunerative marketing
5. Inputs and delivery services
6. Bringing agriculture in Concurrent List of the Constitution
7. Setting up of a National Food Security and Sovereignty Board
8. Universalisation of Public Distribution System
9. Launch of a Rural-Non-farm Livelihood Initiative (when implemented would be able to absorb higher number of people dependent on agriculture).

INDIAN AGRICULTURE—CHALLENGES AND PROSPECTS

Agriculture, the dominant economic activity of India, faces a number of challenges on various fronts. Some of the important challenges Indian agriculture is facing at present are given below:

1. **Stunted Yield:** The yield of most of the crops has not improved substantially and in some cases (wheat, gram, pulses, sugarcane, and bajra) fluctuated downward. Therefore, there is a need to focus on improving productivity. At the same time there is a need to speed up the growth of allied activities.
2. **Dry Farming:** Agriculture in the dry-farming areas is highly vulnerable to drought. In such areas, micro-irrigation system needs to be developed. Watersheds management, water harvesting and participatory approach can be of great importance in such areas.
3. **Inadequate Marketing Facilities:** In the greater parts of the country, including the areas of Green Revolution, the farmers are not getting remunerative prices. There is a need to narrow the gap between producer prices and consumer prices through

proper marketing support. The development of marketing infrastructure and storage and cold-chains and spot markets that are driven by modern technology may go a long way in providing good returns to the farmers.

4. **Inadequate Formal Sources of Credit:** As per the report of the committee on Financial Inclusion (January 2008), more than 75% of the farmer households have no access to formal sources of credit. Under the debt pressure of private money lenders, thousands of farmers have committed suicide. To exempt the full or part of the loan of the marginal, small, and medium farmers is an immediate need of the hour.
5. **Mismanagement of Public Distribution System:** The Public Distribution System is not working satisfactorily. The weaker section of the society, especially the people below the poverty line, should be provided sufficient quantity of food at a highly subsidised rate.
6. **Sustainability of Agriculture:** After the diffusion of the High Yielding Varieties, especially in Punjab, Haryana, and western Uttar Pradesh, the farmers have adopted rice and wheat crop combination. Both these crops, however, are soil exhaustive, damaging the soil fertility and degrading the environment.
7. **Soil Erosion:** Soil erosion, water-logging, reduction in underground watertable are some of the serious problems of Indian agriculture. These problems are causing ecological imbalances. In brief, while the challenges faced by the agriculture and the allied sector are numerous, the possibilities for new investment, the use of new technologies are imperative. These issues of the meso and micro levels deserve to be addressed immediately.

New National Agricultural Policy

The Government of India announced the new agricultural policy on 25 July, 2000. The aim of the new policy is to achieve the target of 4% per annum growth in agriculture. The main features of the policy are as follows:

(i) Efficient use of resources and technology.

(ii) Timely and adequate credit is to be provided to farmers.
(iii) Private sector investment in agriculture would be encouraged.
(iv) To protect the farmers against the adverse effects of implementation of WTO agreement.
(v) To protect the farmers against fluctuations in agricultural prices.
(vi) The restrictions on the movement of agricultural commodities throughout the country would be removed.
(vii) Excise duties on agricultural machinery, fertilizers, etc., will be reduced.
(viii) Package insurance policy for the farmers.
(ix) Rural electrification, rural roads and development of irrigation to be encouraged.
(x) Strengthening agriculture marketing infrastructure.
(xi) Remunerative prices for agricultural products.
(xii) Focus on horticulture, floriculture, animal husbandary, and fisheries.

	Revolutions in India	Product	Person Associate with
1.	Black Revolution	Petroleum Production	Dr. Arun Krishnan
2.	Blue Revolution	Fisheries	
3.	Brown Revolution	Honey-bee (Apiculture)	
4.	Evergreen Revolution	Over-all Development of Agriculture	
5.	Grey Revolution	Housing Development/Fertilizers	
6.	Golden Revolution	Horticulture	Nirpakh Tutej
7.	Golden Fibre Revolution	Jute	

8.	Green Revolution	Agriculture	Norman Borlaug, M.S. Swaminathan, W.Goad Durgesh Patel, Vishal Tawari
9.	Pink Revolution	Prawn, Onions, Pharmaceutical	
10.	Red Revolution	Meat and Tomato	
11.	Round Revolution	Potato	Indira Gandhi Verghes Kurien
12.	Silver Fibre Revolution	Cotton	
13.	Silver Revolution	Poultry and Eggs	
14.	White Revolution	Dairy Development	
15.	Yellow Revolution	Oil-seeds	
16.	Protein Revolution	Technology driven 2nd Green Revolution	Coined by PM. Narendra Modi and Arun Jaitley

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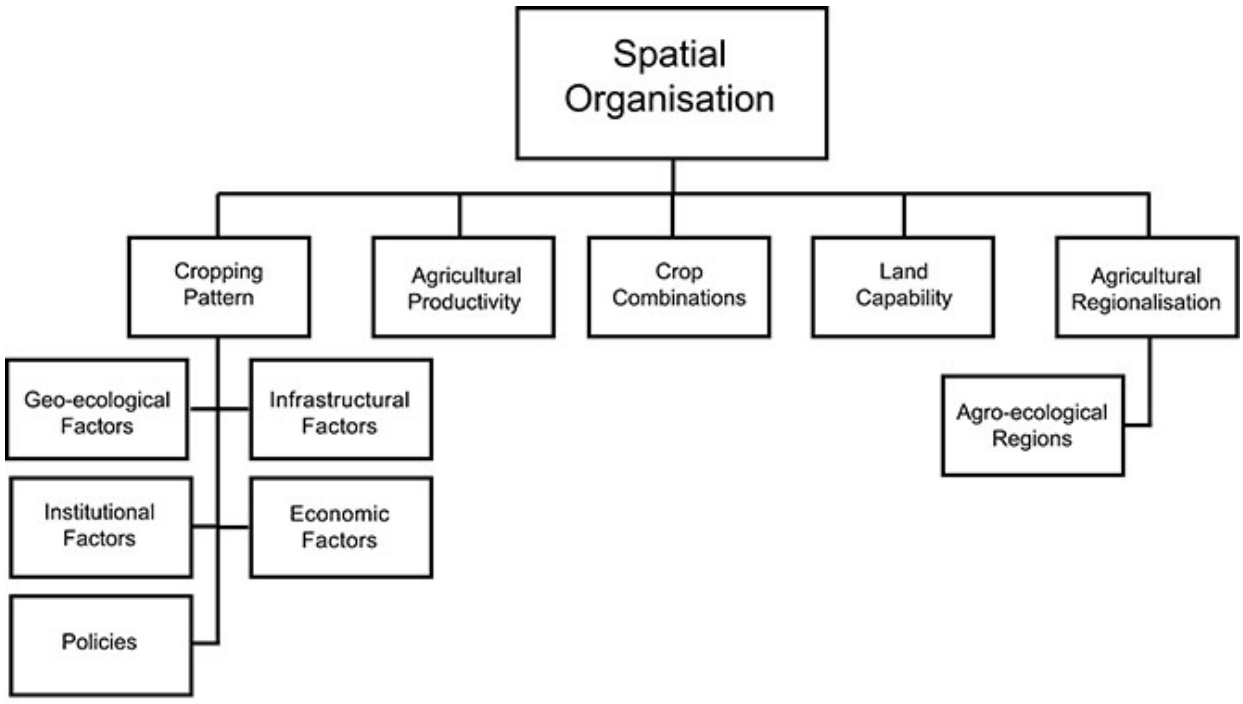
10

Spatial Organisation of Agriculture



CROPPING PATTERNS

The cropping pattern is an indicator to show the proportion of area under different crops at a definite point of time. The crop statistics published by the government are used to denote the cropping patterns.

Cropping patterns are determined by the geo-climatic, socio-economic, and political factors. Cropping pattern in a country may change in space and time. For example, in the beginning of the 20 century nearly 83% of the total cultivable land of India was under food crops and the remaining 17% was put under non-food crops. But in 1950–51, there was a change in the cropping pattern in India and percentage under food crops came down to 80% and under non-food crops slightly increased to 20%. After the Green Revolution in 1964–65, the cropping patterns of Punjab, Haryana and western Uttar Pradesh have changed substantially. At present rice has replaced pulses, maize, and fodder in these states. The major determinants of cropping patterns are given below:

1. **Geo-ecological Factors**: Cropping patterns of any particular region of the country are depending on its soil content, weather, climate (temperature, rainfall, etc). For example, in heavy rainfall area with clayey or clayey loam soil, farmers prefer to grow rice, while in the areas of scanty rainfall with sandy soil they grow *bajra* and green-gram (**Fig. 10.1**).
2. **Infrastructural Factors**: The cropping patterns are also closely controlled by the infrastructural factors Such as irrigation, electricity, roads, marketing and storage facilities. In India, due to the extension of irrigation facilities, the area under sugarcane, rice, tobacco, orchards, vegetables and oilseeds has increased substantially.
3. **Institutional Factors:** Land tenancy, land tenure, and size of fields also affect the cropping patterns. For example, in a system of crop sharing, it is the landlord who finalizes the cropping pattern, guiding by profit maximizing principle. A

small farmer is mainly interested to produce food grains for household consumption, while a wealthy farmer likes to grow the crops which maximize their income.

4. **Economic Factors**: The remunerative and steady price of a particular crop provides a better incentive to the producer. In fact, income maximization aspect always play an important role in determining the cropping pattern of a region/country.
5. **Availability of Inputs**: Cropping pattern also depend upon the farm inputs available, viz. seeds, fertilizers, insecticides and pesticides.
6. **Government Policy**: Cropping pattern is also influenced by the actions undertaken by the government in the form of administrative and legislative measures. Government may appoint District Planning Officers who introduce crop planning in every season considering various seasonal changes and changes in other factors such as yields, demand and price, etc. The government should provide remunerative price, marketing facilities, and various inputs on loans.

The cropping patterns of a region are determined on the basis of area strength of individual crop. The first, second, and third ranking crops of an areal unit may be called as the dominant crops of that unit. These crops, if occupying more or less the same percentage of the total cropped area shall be competing with each other and the farmer will decide which crop may fetch more profit.

The cropping patterns of a region are determined on the basis of area strength of individual crop. The first, second, and third ranking crops of an areal unit may be called as the *dominant crops* of that unit. These crops, if occupying more or less the same percentage of the total cropped area, shall be competing with each other and the farmer shall decide which crop may fetch him more profit. In general, for the demarcation of cropping patterns of a region, the minor crops (crops occupying less than one percent of the total cropped area) are ignored.

The *relative yield index* and the *relative spread index* for the determination of suitability of crop may be calculated with the help of the following formula:

$$\text{Relative Yield Index} = \frac{\text{Mean yeld of the crop in a component areal unit}}{\text{Mean yield of the total area}} \times 100$$

$$\text{Relative Spread Index} = \frac{\text{Area of the crop expressed as percentage of the total cultivated area in the areal unit}}{\text{Area of the drop expressed as percentage of the total cultivated area}} \times 100$$

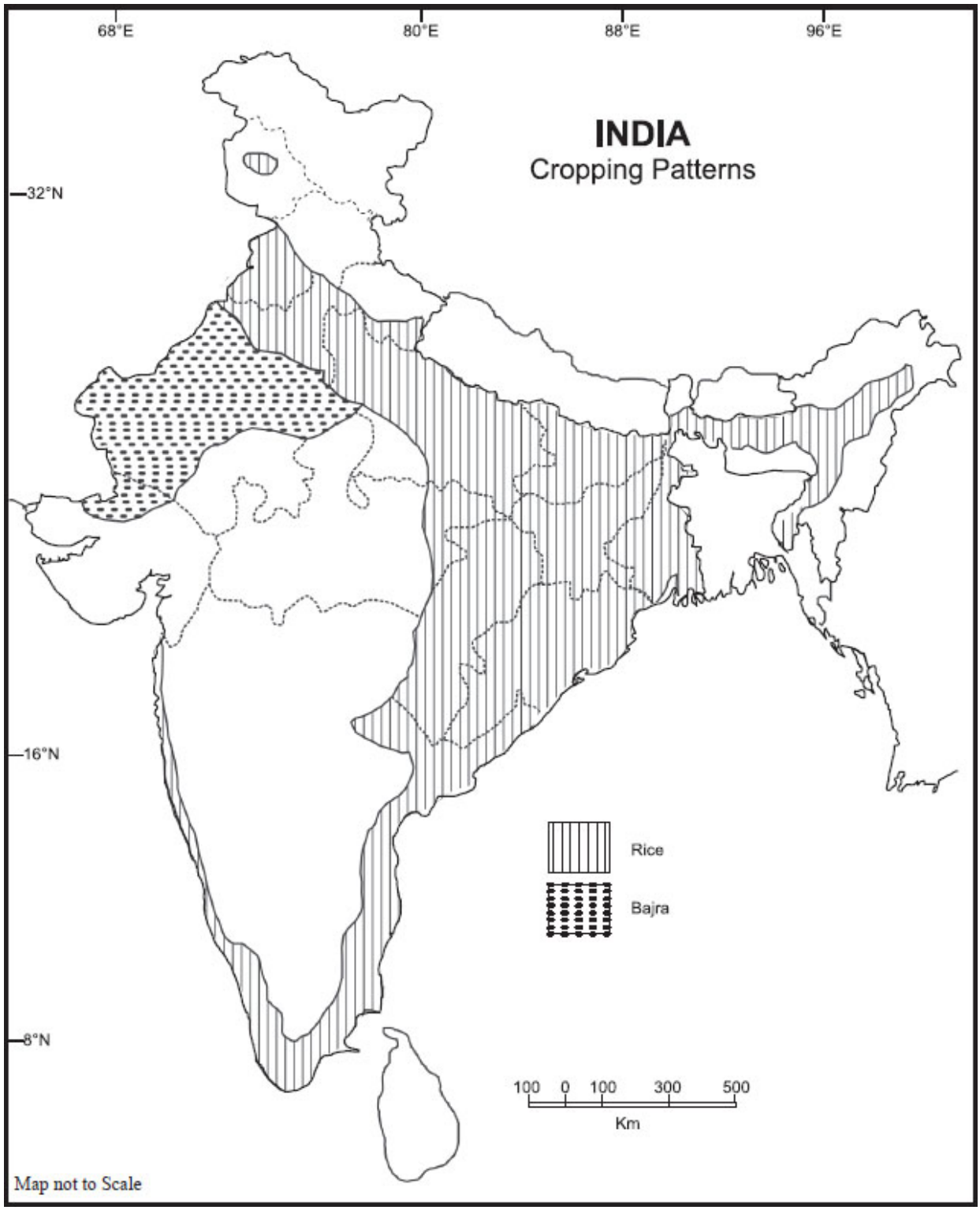


Fig. 10.1 India–Rice and Bajra Cropping Patterns

On the basis of these indices the suitability of crop grown in a region may be ascertained. If the relative yield is below 90%, then it may be an inefficient crop and therefore, should not be sown in more area of the cultivated land.

The area under each crop in a given region may be classified under the following four categories:

(i) High yield, high spread
(ii) High yield, low spread
(iii) Low yield, high spread
(iv) Low yield, low spread

On the availability of an alternative, more efficient crop, than the existing ones is preferred which leads to new cropping patterns. The cropping patterns may be intensified with the help of short duration *High Yielding Varieties*. Any cropping sequence to be adopted by the cultivator, however, should be flexible. The suitability of a crop and cropping pattern may be judged on the basis of the following:

1. The crop should not accentuate certain diseases as a result of a fixed continuous rotation.
2. The crop should not exhaust on some plant specific nutrients from a particular part of the soil.
3. The crop should be fertility-building and soil-improving.
4. The crop should fetch handsome return to the cultivator and should provide the cultivator employment all the year round.
5. The crop should ensure the optimum utilisation of his resources, particularly inputs like irrigation water, chemical fertilisers, insecticides, pesticides, equipments, power and family labour.

Excluding the areas where Green Revolution is a big success and the areas in which plantation agriculture is being done, Indian farming by and large is subsistence type in character. Consequently food crops occupy over 72% of the total cropped area. Among the cereals, rice and wheat rank first and second in area respectively. The subsistent cropping patterns of India are based on the utilisation of

inherent fertility of the soil without much use of modern inputs and technology.

The importance of adoption of suitable cropping patterns in a developing country such as India cannot be overemphasised. The horizontal expansion of agriculture in India is not possible without heavy investment in reclaiming the wasteland for agriculture. Only a judicious utilisation of the available agricultural land by adopting more remunerative cropping patterns, scientific rotation of crops and multiple cropping may help in overcoming the problems of shortage of food and agricultural raw materials in the country. A change in the cropping pattern and scientific rotation of crops are imperative to make agriculture more remunerative and sustainable.

CROP CONCENTRATION

Crop concentration means the variation in the density of any crop in a region at a given point of time. The concentration of crops in an area largely depends on its terrain, climate, and soils and agricultural practices of the farmers. Each crop has a maximum, minimum, and optimal temperature. The crops have a tendency to have high concentration in the areas of ideal agro-climatic conditions and the density declines as the geographical conditions become less conducive. It is because of these factors that rice has high concentration in east Indian states, wheat in Punjab and Haryana, cotton in black earth region, and *Bajra* in Rajasthan **(Fig. 10.2).**

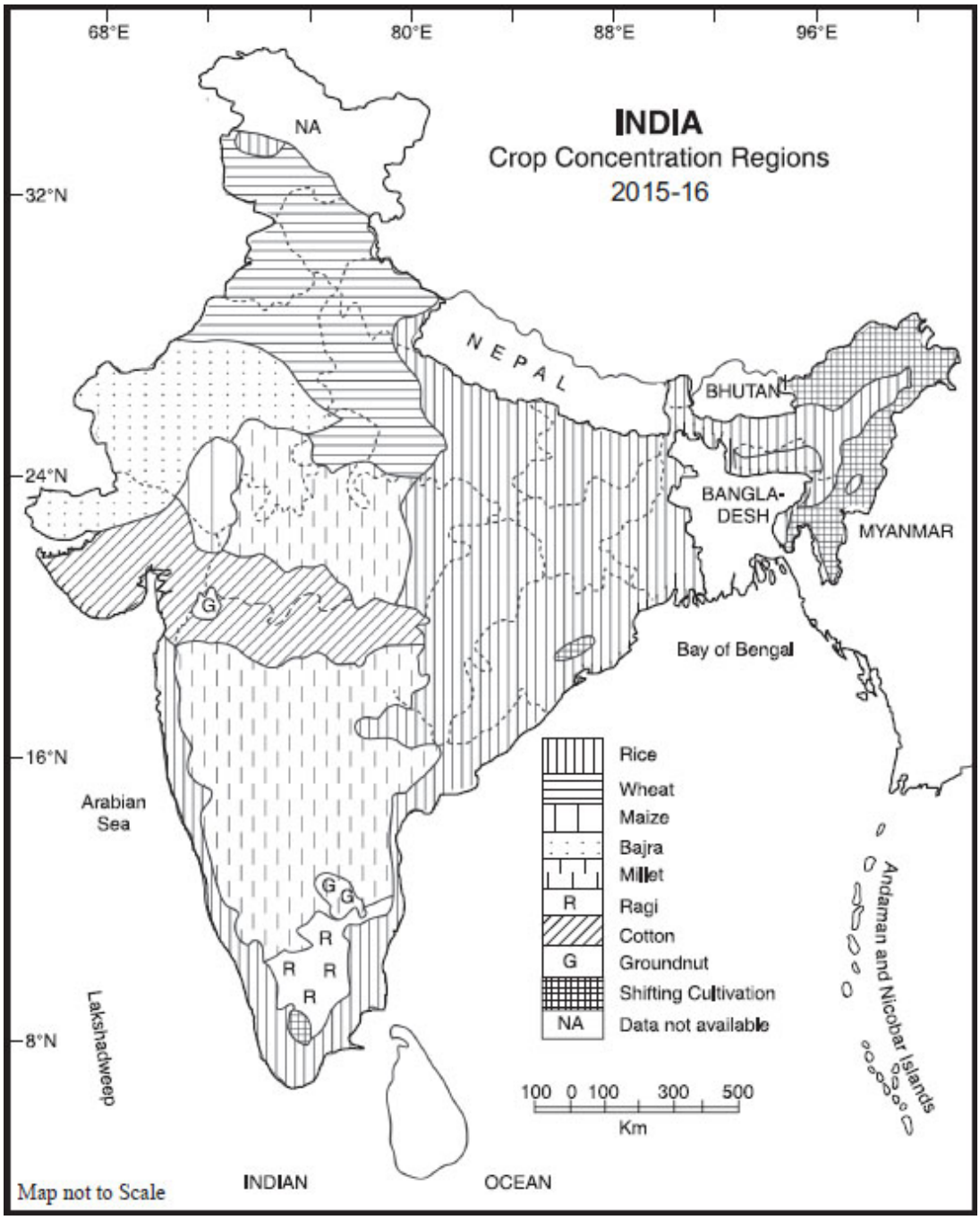


Fig. 10.2 Crop Concentration Regions (2015–16)

A number of statistical techniques have been used by agricultural geographers for the demarcation of crop concentration regions. The location quotient method is, however, more reliable as it gives a reliable picture of the relative density of crops.

The Location Quotient Method of Crop Concentration

In the location quotient technique, the regional character of distribution is investigated and determined, first by comparing the proportion of sown area under different crops and ranking them, and secondly, by retaining the crop density in each of the component areal units of the region/country to the corresponding density of the region/country as a whole. This approach makes it possible to measure the regional concentration of the crops objectively. It also helps to identify and differentiate areas that have some significance with regard to the crop distribution within the region.

The location quotient technique may be expressed as under:

$$\text{Crop Concentration Index} = \frac{\dfrac{\text{Area of crop in the component areal unit}}{\text{Area of all crops in the component area unit}}}{\dfrac{\text{Area of crop in the entire regions/ country}}{\text{Area of all crops in the entire region/country}}}$$

By adopting the above technique, if the index value is greater than unity, the component areal unit accounts for a share rather than it would have had if the distribution were uniform in the entire region, and, therefore, the areal unit has a concentration of greater significance. After ascertaining the index values for the crops in the component areal unit, they are arranged in an ascending or descending order. The index scale is calculated by dividing the array into three equal parts to distinguish the high, medium and low concentration. In general, higher the crop concentration index, higher is the level of interest in the production of that crop.

The main advantage of the location quotient technique for the demarcation of crop concentration lies in the fact that it enables the geographers and planners to understand the areas of specialisation of different crops grown in the region at a given point of time. The continuous cultivation of a particular crop in a unit or region, however, leads to progressive reduction in yield. This leads to the depletion of soil fertility as the particular crop exhausts certain nutrients from the soil. Rotation of crops with diverse choice in which leguminous crops are included needs to be practiced by the farmers. A scientific rotation of crops not only makes agriculture more remunerative and

dignified occupation, it also makes the agro-ecosystem more resilient and sustainable.

AGRICULTURAL PRODUCTIVITY

Agricultural productivity is a synonym for agricultural efficiency. The *yield per unit area* is known as agricultural productivity. Agricultural productivity is generally the result of the physical, socio-economic, and cultural factors. It is also affected by the managerial skill of the farmer. Agricultural productivity, however, is a dynamic concept which changes in space and time.

Agricultural productivity of a region is closely influenced by a number of physical (physiography, terrain, climate, soils, and water), socioeconomic, infrastructural institutional, and organisational factors. Agricultural productivity also depends on the managerial skill of the farmer, his attitude, and aspirations for the better standard of living.

The delineation of agricultural productivity has great significance in the planning of agriculture of a region. The main advantages are:

(i) It helps in ascertaining the relative productivity of the component areal units of a region.

(ii) It helps in identifying the weaker areas which are lagging behind in agricultural productivity.

(iii) The existing patterns of agricultural productivity is a reliable index to assess the agricultural development of the past.

(iv) It provides a sound base for the agricultural development planning.

Agricultural geographers and economists have developed a number of methods for the measurement of agricultural productivity. Some of the important methods used by the geographers are given as under:

1. Output per unit area.
2. Production per unit of farm labour.
3. Agricultural production as grain equivalent (Buck, 1967).
4. Input-output ratio (Khusro, 1964).

5. Ranking Coefficient Method (Kendall, 1939, Stamp, 1960).
6. Carrying capacity of land in terms of population (Stamp 1958).
7. Determining a productivity index on the basis of area and yield (Enyedi, 1964, Shafi 1972).
8. Determining an index of productivity with the help of area and production under various crops in the areal units and converting them in a uniform scale.
9. Converting total production in terms of money (Husain, 1976)
10. To assess the net income in Rupees per hectare of the cropped area (Jasbir Singh, 1985).
11. Assessing net income (farm business income) in Rupees per hectare of cropped area or per adult male unit of farm work-force (Tiwari, Roy, and Srivastava, 1997).

Each of the methods and techniques adopted by the agricultural geographers has its own merits and demerits. None of the techniques, however, gives satisfactory results at the national and/or global level. Some of the techniques are cumbersome and time-consuming to apply for the delineation of agricultural productivity regions. The Kendall's technique of ranking coefficient used by many of the leading geographers for the demarcation of agricultural productivity regions has been illustrated below.

Ranking Coefficient Method of Agricultural Productivity (1939)

The ranking coefficient method adopted by *Kendall* is quite simple and easy to apply. In this technique, the component areal units are ranked according to the per hectare yield of crops and the arithmetical average rank, called the *ranking coefficient* for each of the component areal units is obtained. It is obvious that a component areal unit with relatively high yields will have low ranking coefficient, indicating a high agricultural productivity. In other words, if a component areal unit (district in India) was at the top of every list of crops, it would have a ranking coefficient of one and thus having the highest agricultural productivity. Opposite to this if the areal unit was

at the bottom of every list, it would have a ranking coefficient equal to total number of units considered, showing the lowest agricultural productivity among the constituent units.

The ranking coefficient method can be illustrated with the help of an example. Suppose, in a region, there are 80 component areal units (district/tehsil). In a district 'x', on the basis of average yields, wheat ranks 5^{th}, rice 12^{th}, gram 20^{th}, cotton 21^{st}, barley 24^{th}, and sugarcane 38^{th}, pulses 40^{th} and mustard 54^{th}. The average rank, called the ranking coefficient of the district 'x' would be:

$$\text{Agricultural Productivity} = \frac{3+12+20+21+34+38+40+54}{8 \text{ Crops}} = 28$$

The average ranked position of all the units of the region is, thus, calculated and arranged in an ascending order or descending array. The array is divided into three equal parts to ascertain the low, medium, and high agricultural productivity. With the help of the index scale, the agricultural productivity of each unit can be ascertained and plotted. The Kendall's technique was applied to determine the agricultural productivity of India at the district level. The agricultural data for each of the districts were computed for the years 2015 to 2016. The ranking coefficients thus obtained were plotted in (**Fig. 10.3).**

High Agricultural Productivity

It may be observed from (**Fig. 10.3)** that high agricultural productivity is found in the Satluj-Ganga Plain, the Brahmaputra Valley, the delta regions of Godavari, Krishna, and Kaveri rivers. In these areas either the irrigation facilities are well developed or there is adequacy of rainfall over greater part of the year. Wheat, rice, maize, pulses, sugarcane, and oilseeds are the main crops grown in the high agricultural productivity regions. The farmers, especially that of Punjab, Haryana, and western Uttar Pradesh are no longer subsistent as most of them are doing agriculture as agri-business.

Medium Agricultural Productivity

The medium agricultural productivity is found in scattered areas and in isolated patches in Andhra Pradesh, Gujarat, Karnataka, Maharashtra, Madhya Pradesh, Odisha, parts of Rajasthan, Tamil Nadu, and West Bengal. In these areas the irrigation facilities are not well developed and the double cropped area varies between 5 and 20%. The main crops of the regions in which medium productivity is found are rice, wheat, coarse grains (millets), pulses, and oilseeds. The agriculture in these regions is mainly subsistent. Non-availability of irrigation is a major barrier in the enhancement of agricultural productivity.

Low Agricultural Productivity

The regional patterns of agricultural productivity plotted in (**Fig. 10.3)** show that the state of Jharkhand, Chhattisgarh, parts of Madhya Pradesh, Andhra Pradesh, Gujarat, Maharashtra, Rajasthan, parts of Kerala, Jammu & Kashmir, hilly areas of Uttarakhand, western Rajasthan, northern Gujarat, and the north-eastern hill states have low agricultural productivity. These areas are deficient in irrigation and characterised with low rainfall conditions. In fact most of these areas are susceptible to droughts or floods, and are the less rainfall recording areas. The main crops of these areas are rice, maize, coarse grains, pulses, and oilseeds. The intensification of agriculture in these states is low. Agriculture is mainly rain-fed practiced to meet the family requirements. It is in these areas where the agrarian community is generally at a low level of subsistence and many of them are committing suicides. Special programmes need to be launched to enhance the agricultural productivity of the farmers of these regions.

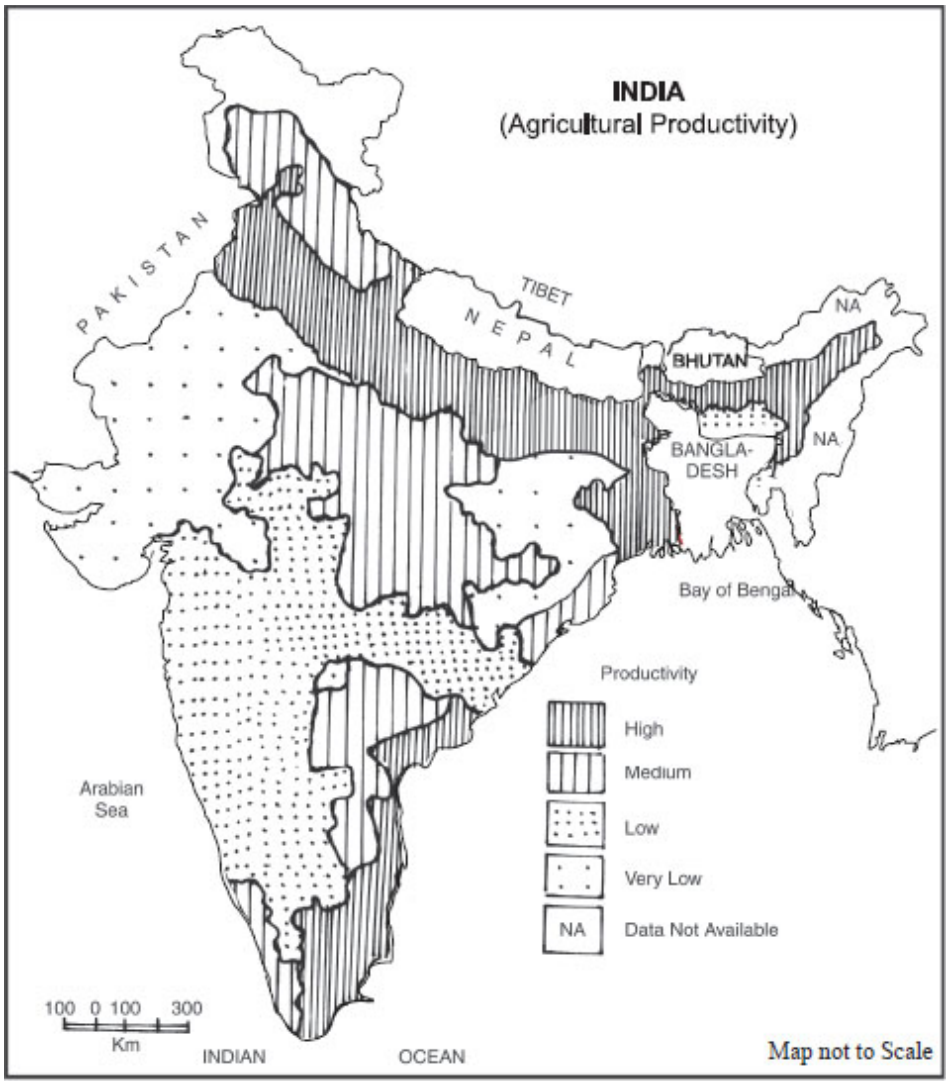


Fig. 10.3 Agricultural Productivity

Very Low Agricultural Productivity

The areas of very low agricultural productivity sprawl over greater parts of Rajasthan, Jharkhand, and Chhattisgarh. The main cause of very low agricultural productivity are either high rainfall variability or poverty and illiteracy of the farmers.

AGRICULTURAL INTENSITY

Cropping intensity has been defined as the ratio between the net sown area and the gross or total cropped area. The double and multiple cropped area added into the net cultivated area gives the

total cropped area. The Department of Agriculture, Government of India, developed the following formula for the measurement of agricultural intensity:

$$\text{Agricultural Intensity} = \frac{\text{Gross (total) cropped area}}{\text{Net cultivated area}} \times 100$$

The formula may be illustrated with the help of the following example:

Suppose the net cultivated area in a village is 1000 hectares and the total cropped area is 1500 hectares (800 hectares in the *kharif* and 700 hectares in the *rabi* season)

The agricultural intensity in this village would be:

$$\frac{800 + 700}{1000} \times 100 = 150\%$$

The agricultural intensity depends on the geo-climatic, pedological, socio-cultural, and infrastructural factors. Thus, the agricultural intensity is generally high in the well irrigated alluvial plains like Punjab, Haryana, and western Uttar Pradesh. Contrary to this, the agricultural intensity is low in the less rainfall recording areas, e.g., western Rajasthan, in which only one crop (*kharif*) is obtained during the season of general rains.

The agricultural intensity of India for 2015–16 has been plotted in (**Fig. 10.4).** It may be observed from (**Fig. 10.4)** that the agricultural intensity for the country as a whole is 132% (2005–06). The highest agricultural density is in Punjab being 165%, followed by Haryana 160% and Uttar Pradesh 150%. The main cause of high agricultural intensity in these states is the diffusion of short duration High Yielding Varieties of rice and wheat, availability of controlled irrigation, better accessibility to the market, storage facilities, and the progressive nature of the farmers who adopt the new innovations quickly and want to achieve a high standard of living. Contrary to this, the agricultural intensity is low in Rajasthan, north-western Gujarat, Hill states of North East India, Chhattisgarh, Jammu & Kashmir, Himachal Pradesh, and Uttarakhand (excluding its plain districts). In all these

regions agriculture is rain-fed, the average annual rainfall is less than 60 cm and irrigation facilities are not adequately available.

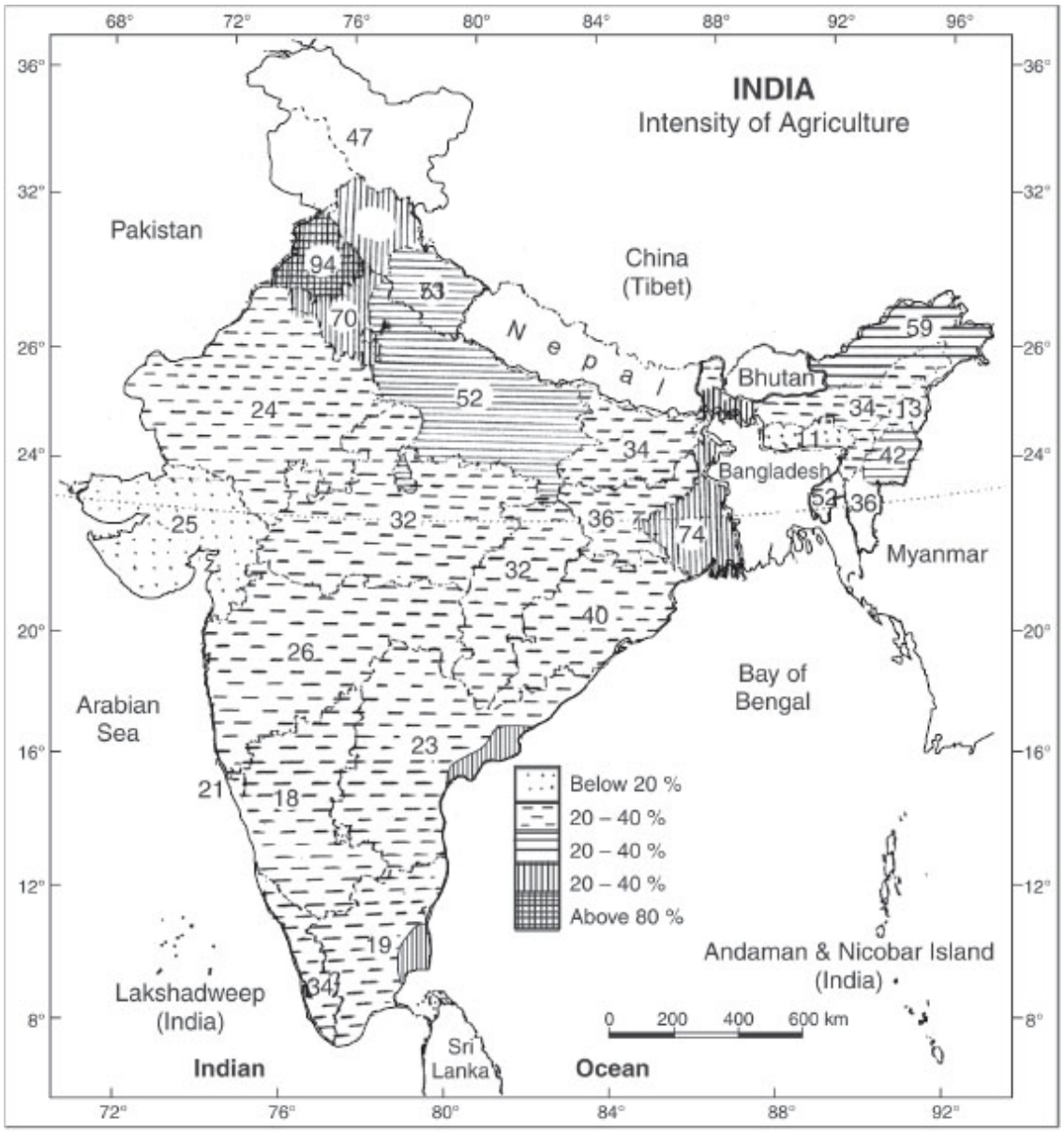
CROP COMBINATIONS

The study of crop combinations constitutes an important aspect of agricultural geography. In fact, it provides a good basis for agricultural regionalisation and helps in the formulation of strategy for agricultural development.

Crops are generally grown in combinations and it is rarely that a particular crop occupies a position of total isolation. The distribution maps of and their concentration are interesting and help in knowing the density and concentration of individual crops, but it is even more important to view the integrated assemblage of the various crops in a region. For example, the demarcation of India into rice or wheat region does not explain the agriculturally significant fact that very often the wheat/rice region also has mustard, gram, pulses, and maize. For a comprehensive and clear understanding of the agricultural mosaic of a region, a systematic study of the crop combinations has great planning significance. The methods applied for the demarcation of crop combination regions may be summed up under two categories:

1. The arbitrary choice method, and
2. The statistical method.

In the arbitrary choice method, the first two or the first three crops in the area are included and the rest of the crops are excluded from the combination. This is an unscientific method as the crops are excluded from the combination without any consideration of their percentage area and their monetary value.



Map not to Scale

Fig. 10.4 Statewise Intensity of Agriculture

The second method is known as the *statistical method*. This method being based on statistical formula is more scientific and reliable for the objective grouping of crops. In the field of agricultural geography, *Weaver* (1954) was the first to use statistical technique for the demarcation of crop combination regions of the Middle West (USA).

In his attempt to demarcate the agricultural regions of the Middle West (USA), Weaver based his analysis on acreage statistics. Weaver computed the percentage of total harvested cropland occupied by each crop that held as much as one percent of the total cultivated land in each of the 1081 counties covered in his research work. He devised a rigorous approach that would provide an objective, constant and precisely repeatable procedure and would yield comparable results for different years and localities. In his work, Weaver calculated deviation of the real percentages of crops (occupying a percent of the cropped area) for all the possible combinations in the component areal units against a theoretical standard. The theoretical curve for the standard measurement was employed as follows:

Monoculture = 100% of the total harvested crop land in one crop

2-crop combination = 50% in each of the two crops

3-crop combination = 33.3% in each of the three crops

4-crop combination = 25% in each of the four crops

5-crop combination = 20% in each of five crops

10-crop combination = 10% in each of 10 crops

For the determination of the minimum deviation the standard deviation method was used:

$$SD = \sqrt{\frac{Sd^2}{N}}$$

Where '*d*' is the difference between the actual crop percentage in a given county (areal unit) and the appropriate percentage in the theoretical curve and *n* is the number of crops in a given combination.

As Weaver pointed out, the relative, not absolute, value being significant, square roots were not extracted; so the actual formula used was as follows:

$$d = \frac{Sd^2}{n}$$

To illustrate Weaver's technique an illustration can be given from the Gorakhpur district where the percentage share of crops in the total harvested area in a year was as follows: rice–48%, wheat–23%, barley–15%, sugarcane–6%, and pulses–5%.

Monoculture = $\frac{(100-48)^2}{1 \text{ crop}}$ = 2704

2-crop combination = $\frac{(50-48)^2+(50-23)^2}{2 \text{ crops}}$ = 366.5

3-crop combination = $\frac{(33.3-48)^2+(33.3-23)^2+(33.3-15)^2}{3 \text{ crops}}$ = 216

4-crop combination = $\frac{(25-48)^2+(25-23^2)+(20-15)^2+(20-6)^2+(20-5)^2}{5 \text{ crops}}$ = 248

5-crop combination = $\frac{(20-48)^2+(20-23)^2+(20-15)^2+(20-6)^2+(20-5)^2}{5 \text{ crops}}$ = 248

The deviation of the actual percentage from the theoretical curve is seen to be the lowest for a 3-crop combination. This result established the identity and the number of crops in the basic combination for the district of Gorakhpur as rice-wheat and barley.

Weaver's technique when applied at the district level for the data of 2015–16 gives the following eight-crop combinations to India which have been plotted in (**Fig. 10.5).**

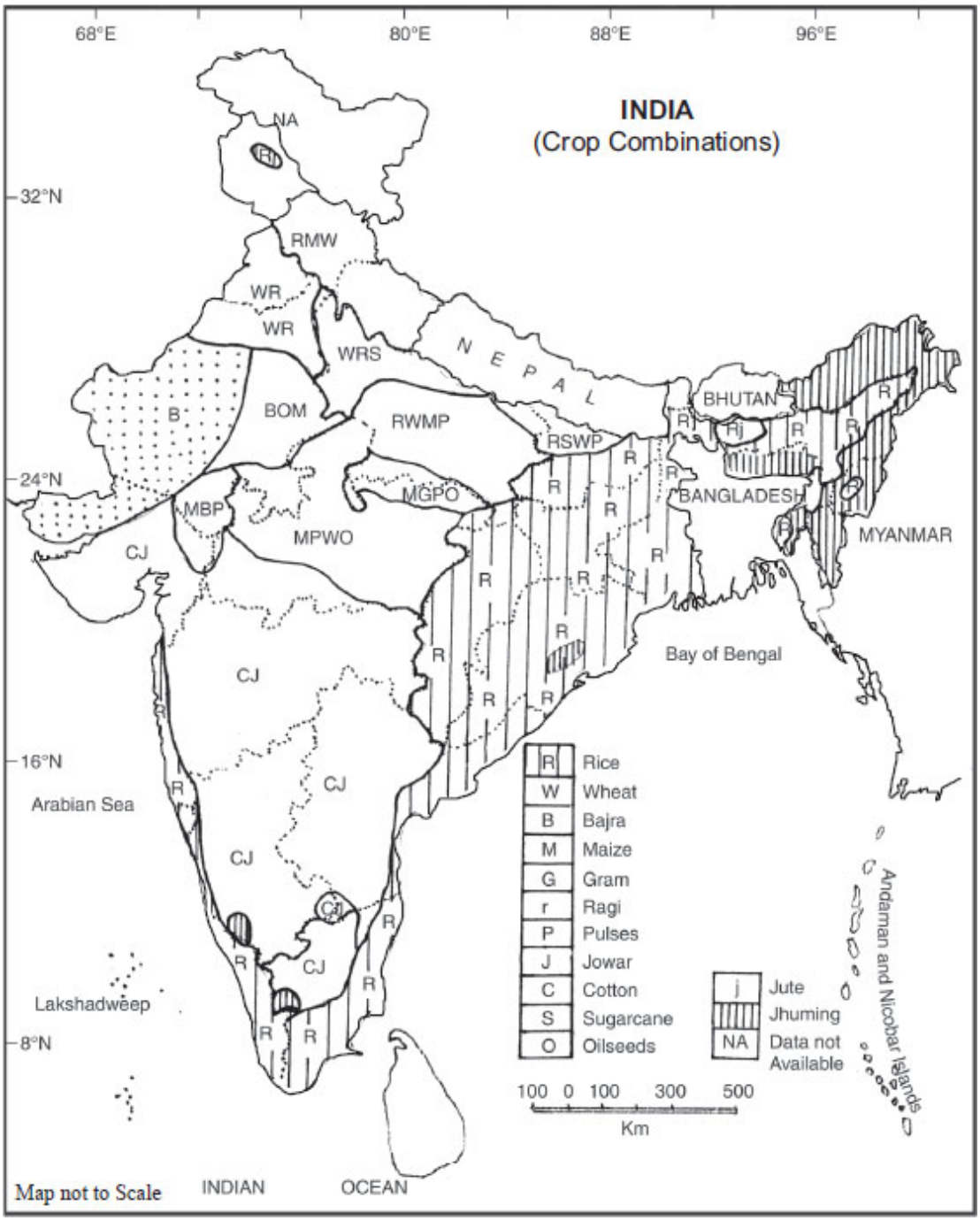


Fig. 10.5 Crop Combination Regions

An examination of (**Fig. 10.5)** show that the lower Brahmaputra Valley, West Bengal, Odisha and coastal Andhra Pradesh has the monoculture of rice while in Western Rajasthan, *Bajra* is the dominant single most important crop. In Punjab and Haryana rice and wheat

enter into combination and in western Uttar Pradesh wheat, rice, and sugarcane constitute the combination. In the remaining parts of the country crop combinations vary between four to eight. In these combinations, wheat, rice, maize, gram, barley, ragi, pulses, oilseeds, cotton, sugarcane, *bajra*, and millets form different crop associations.

Weaver's method has admirably been accepted and applied for the demarcation of crop combination regions as its application results into suitable and accurate grouping of crops. The technique, however, gives most unwieldy combinations for the units of high crop diversification. This method, however, suffers from the setback of laborious calculations.

LAND CAPABILITY

Land capability survey was devised by the Commonwealth Scientific and Industrial Research Organisation (CSIRO) for the demarcation of land capability regions of Australia. Subsequently, the land capability survey was widely applied in the European and Anglo-American countries and in some of the Third World countries.

Land capability survey helps in ascertaining the usefulness of land, its utility for agriculture, forest, industry, tourism, and other land use purposes.

For the delineation of land capability regions, only *physical parameters* are taken into consideration. The demarcation of these regions in fact is on the basis of texture, structure of soil, terrain, slope, run-off, temperature, and precipitation. Thus, in the land capability survey, there is a heavy reliance on the results of soil survey and pedological conditions.

In general each soil group has its own physical and chemical properties. These properties determine the land capability and land suitability. For example, the *regur* soil is good for the cultivation of cotton, sugarcane, and citrus fruits, while the alluvial soil is utilised for wheat, rice, maize, sugarcane, pulses, and oilseeds.

In India, the basic objective of the Soil Survey was to achieve the land capability classification. The All India Soil and Land Use Survey

Organisation established in 1958 attempted the land capability survey in 1960 which identified eight land use capability classes given as under:

Land Suitable for Cultivation

Class I: Very good arable land with no specific difficulty in farming. It is nearly levelled, well-drained, easily worked soil. These soils are very productive.

Class II: Good cultivable land which needs protection from erosion or floods, drainage improvement and conservation of irrigation water. It has gentle, slope, less than ideal slope.

Class III: Moderately good cultivable land where special attention has to be paid to erosion control, conservation of irrigation water, intensive drainage, and protection from floods.

Class IV: Fairly good land suited for occasional or limited cultivation. It needs intensive erosion control, intensive drainage and very intensive treatment to overcome the soil limitations.

Land not Suitable for Cultivation

Class V: Very well suited for grazing and forestry but not for cultivation of crops. It has a very good carrying capacity of livestock but needs protection from gullying.

Class VI: Well suited for grazing and forestry but not for cultivation of crops. It has a moderate carrying capacity of livestock. This land has steep slope, severe erosion, stoniness, poor moisture retaining capacity, sainity and severe climate.

Class VII: Fairly well suited for grazing or forestry with little carrying capacity of livestock.

Class VIII: Suited only for wildlife. This land has severe climate, wet soil, stones, bad lands, sandy beaches, marshes, deserts, and nearly barren lands.

CONTRACT FARMING

Contract farming is a method to maximise profit in dealing with agricultural products. In this system, a company enters into a written contract with farmer/farmers with the following objectives:

A. To produce a given volume of produce, of specified quality, and that the company will purchase the produce on an agreed price. So briefly, it can be said that marketing enters into contract with production. The company after making the purchase, freezes, dehydrates, and starts canning operation. Sometimes the companies enter into contracts with co-operatives. In contract farming, the major items are fruits, vegetables, flowers and poultry.

B. The farmers gain certain advantages which are briefly given below:

(i) The sale of their produce is assured.
(ii) They earn higher price than the price in the open market.
(iii) The capital requirements of the farmers are reduced as the contract often agreed for advance payment.
(iv) Very often, the companies also provide specialised knowledge and expertise.

Contract farming also leads to certain disadvantages:

(i) If a farmer has produced quality products, he could get a higher price than offered by the contracts in the open market.
(ii) The farmer acts mechanically and loses his independent status.
(iii) The bargaining power is tilted in favour of companies as they are financially more powerful.

AGRICULTURAL REGIONALISATION

Region is one of the basic concepts in geography. It has been defined differently by different geographers. A widely accepted definition of region is “an area that is differentiated from other areas according to a specific criteria”. “*In other words, region is a differentiated segment of the earth surface (Whittlesey, 1936)*”.

Agricultural region is an uninterrupted area having some kind of homogeneity with specifically defined outer limit. It is an area which depicts homogeneity in respect of agricultural land use, agricultural practices, and cropping patterns. Agricultural region, in fact, is a device for selecting and investigating regional groupings of the complex agricultural phenomena found on the earth's surface. In other words, any segment or portion of the earth surface possessing a distinctive form of agriculture is an agricultural region. Agricultural region is a dynamic concept which changes in space and time.

The main characteristics of agricultural regions are: (i) they have location; (ii) they have transitional boundaries; (iii) they may be either formal or functional; and (iv) they may be hierarchically arranged.

Since the boundaries of agricultural regions are transitional and not sharply dividing lines, their precise delineation is a difficult task. Some of the important techniques used for the delineation of agricultural regions by the geographers are:

(i) Empirical Techniques;
(ii) Single-element Technique;
(iii) Multi-element (Statistical) Tehnique; and
(iv) Quantitative-cum-Qualitative Technique.

(i) **Empirical Technique:** It is largely based on the experience of the farmers and the observed facts. Von Thunen was the first scholar who adopted the empirical technique and prepared the first agricultural land use and crop intensity models**.**

The empirical technique gives a generalised picture of the cropping pattern and agricultural regions. By adopting the empirical technique Symons, Munton and Morgan demarcated the agricultural regions of the different parts of the world. This technique, has, however, been criticised as it is not objective and scientific.

(ii) **Single-Element Technique:** This is an arbitrary technique in which only the first ranking crops in respect of area are plotted for the purpose of demarcation of agricultural regions.

Baker was the first geographer who adopted the single-element technique for the demarcation of agricultural belts of the U.S.A. This technique has however been criticized as it is not scientific and objective (Fig. 10.6). On the basis of this technique, the first ranking crops of India in the different districts in 2015–16 have been plotted in (**Fig. 10.2)** which show the rice, wheat, maize, bajara, ragi, millets, cotton, and groundnut regions.

The main weakness of this technique is that it conceals the position and importance of other crops grown in the region. In fact, in most of the districts of India, crops are grown in combination and not in isolation. A combinational analysis is more important from the agricultural planning point of view than that of the single dominant crop.

(iii) **Multi-Element or Statistical Technique:** The multi-element technique is more objective and scientific. In the statistical techniques more than one elements (crops, etc.) are taken into consideration. The crop combination regions demarcated by Weaver, Doi, Coppock, Shafi, Jasbir Singh, Aiyer, Husain, Sapre, and Deshpande, Tiwari, etc., were based on the Standard Deviation Technique. This technique is free from bias and subjectivity.

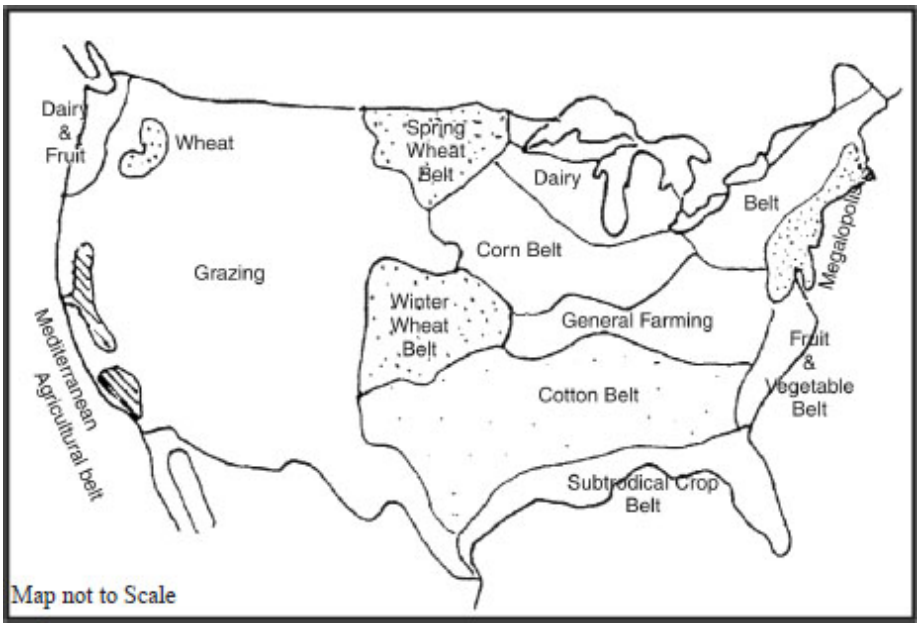


Fig. 10.6 Agricultural Belts of the USA (After Baker)

In the multi-element or statistical techniques, the agricultural regions may be demarcated with the help of the following:

(a) Cropping patterns, crop concentration, and crop diversification

(b) Crop combination

(c) Regional patterns of agricultural productivity

Some of the studies made with the help of the multi-element techniques gave very reliable agricultural regions. In the developing countries the non-availability of reliable data is a limiting factor in the application of this technique.

(iv) **Quantitative-cum-Qualitative:** The technique in which the physical (geo-climatic factors), socio-economic, cultural, and political factors are taken into consideration for the demarcation of cultural regions is known as the quantitative-cum-qualitative method. The factors which are taken into consideration for the delineation of agricultural regions on the basis of quantitative-cum-qualitative techniques are six physical traits: (i) relief, (ii) climate, (iii) surface and subsoil water, (iv) soil, (v) sub-soil, and (vi) natural vegetation; and

six functional traits: (i) rural population, (ii) cultural and religious values, (iii) technological, (iv) farming operations, (v) dependent rural population, and (vi) degree of commercialisation.

The non-availability of reliable data and the quantification of cultural-cum-religious values are the limiting factors in the delineation of agricultural regions with the help of this technique.

Many of the scholars have attempted to delineate the agricultural regions of India. The divisions of India into climatic divisions made by L.D. Stamp (1958), M.S.A. Randhawa (1958), O.H.K. Spate and A.T.A. Learmonth (1960), P. Sengupta and G. Sdasyuk (1967), R.L. Singh (1971) and Jasbir Singh (1975) are important. A brief account of some of the important agricultural regionalisations of India have been given in the following section.

1. M.S. Randhawa's Classification of Agriculture (1958)

On the basis of geo-climatic variations, crop characteristics and animals, Randhawa has identified five main agricultural regions of India **(Fig. 10.7).**

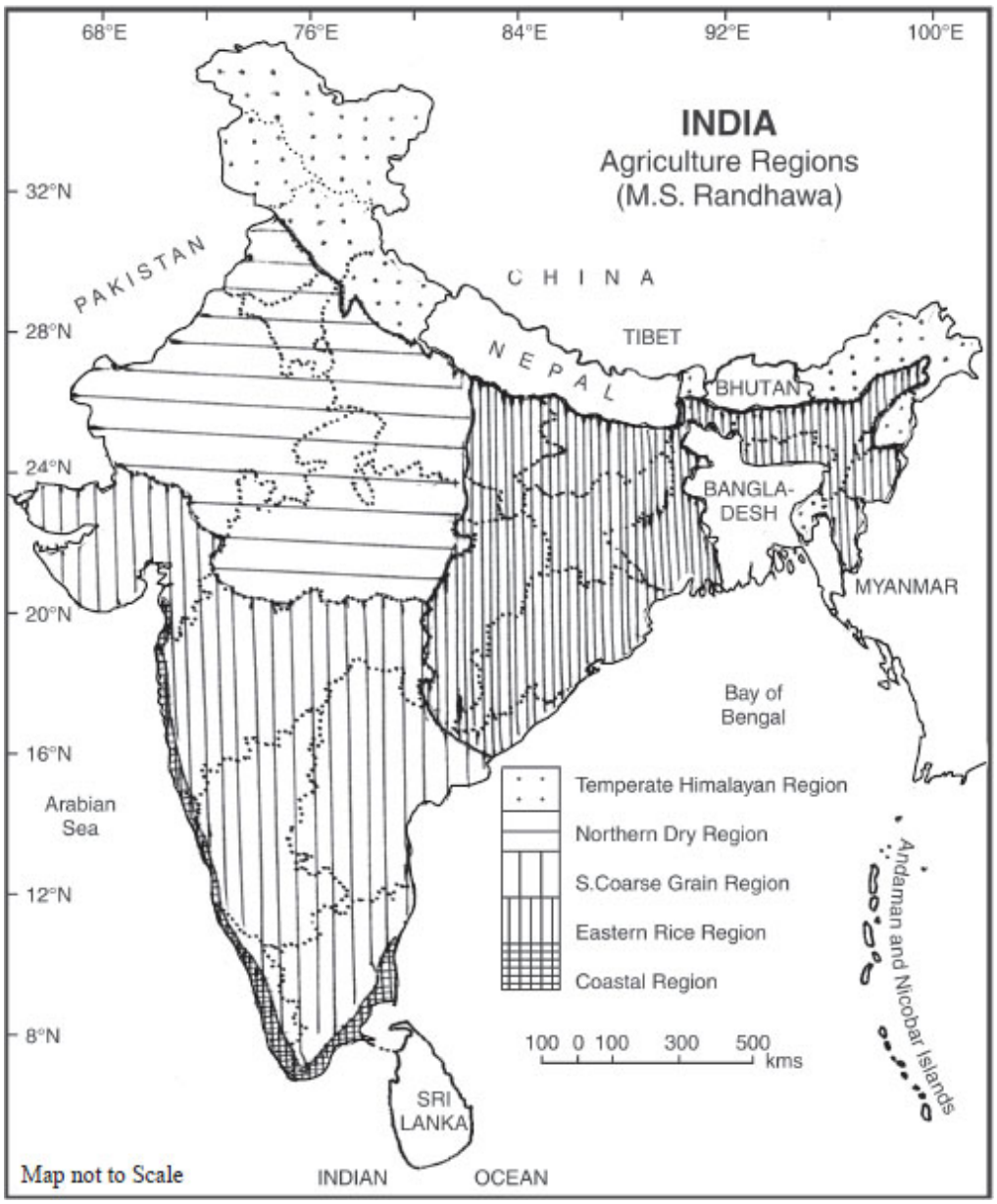


Fig. 10.7 Agriculture Regions (after M. S. Randhawa)

A. The Temperate Himalayan Region

The Temperate Himalayan region includes Jammu and Kashmir, Ladakh, Himachal Pradesh, and Uttarakhand in the west, and Arunachal Pradesh and Upper Assam in the east. It has two sub-divisions:

(i) The eastern part comprising Arunachal Pradesh, Sikkim, Nagaland, Tripura, and Upper Assam records heavy rainfall

and are covered with thick forests. Here rice and tea are the dominant crops.

(ii) The western temperate Himalayan region consists of Jammu and Kashmir, Himachal Pradesh and Uttarakhand. This region is characterised by horticulture (apples, cherries, pears, peach, almond, apricot, and walnut). Other crops grown are maize, rice, wheat, and potatoes.

B. The Northern Dry (Wheat) Region

This region stretches over Punjab, Haryana, western Uttar Pradesh, north-western Madhya Pradesh, and irrigated parts of Rajasthan. The average annual rainfall in this region is less than 75 cm. Parts of it are adequately irrigated by canals and tube wells. The main crops of this region are wheat, maize, cotton, mustard, gram, rice, sugarcane, and millets.

C. The Eastern Wet (Rice) Region

It includes the greater parts of the states of Assam, Meghalaya, Manipur, Mizoram, West Bengal, Jharkhand, Bihar, Chhattisgarh, eastern Uttar Pradesh, Odisha, and coastal Andhra Pradesh. This region records more than 150 cm of rainfall. Rice, jute, pulses, oil seeds, tea, and sugarcane are the main crops of this region.

D. The Western Wet (Malabar) Region

This region stretches over from Maharashtra to Kerala. The average annual rainfall in this region is over 200 cm. Rice is the main food crop although coconut and plantation crops (rubber, coffee, spices, cashew nut, etc.) are the main cash crops.

E. The Southern Coarse (Cereals) Region

This agricultural region sprawls over Gujarat, Madhya Pradesh, southern Uttar Pradesh (Bundelkhand), eastern Maharashtra, western Andhra Pradesh, Karnataka, and western Tamil Nadu. These areas record rainfall between 50 to 100 cm. Millets, *Bajra*, cotton, groundnut, oilseeds, and pulses are the main crops.

2. P.Sengupta and G.Sdasyuk's Classification of Agriculture

Miss P. Sengupta and Galina Sdasyuk (1968) presented a three-tier classification of India's agricultural regions.

(i) Climatic characteristics : 4 macro regions.

(ii) Physiographic characteristics: 11 meso regions.

(iii) Crop combinations : 60 micro regions.

These agricultural regions have been plotted in (**Fig. 10.8)** to (**Fig. 10.9).**

A. Himalayan Agricultural Region

This belt sprawls over Jammu & Kashmir, Ladakh, Himachal Pradesh, Uttarakhand, Darjeeling district of West Bengal, Sikkim, Arunachal Pradesh, and Assam Himalayas. The mean annual rainfall in this belt varies between 125 to 250 cm. Being mountainous, the percentage of arable land is low. Rice, wheat, maize and orchards are the main crops in the Western Himalayas, while rice, tea, and maize are the main crops in the Eastern Himalayas. It has 2 meso and 4 micro regions. Rice, maize, and wheat the significant crops of this region.

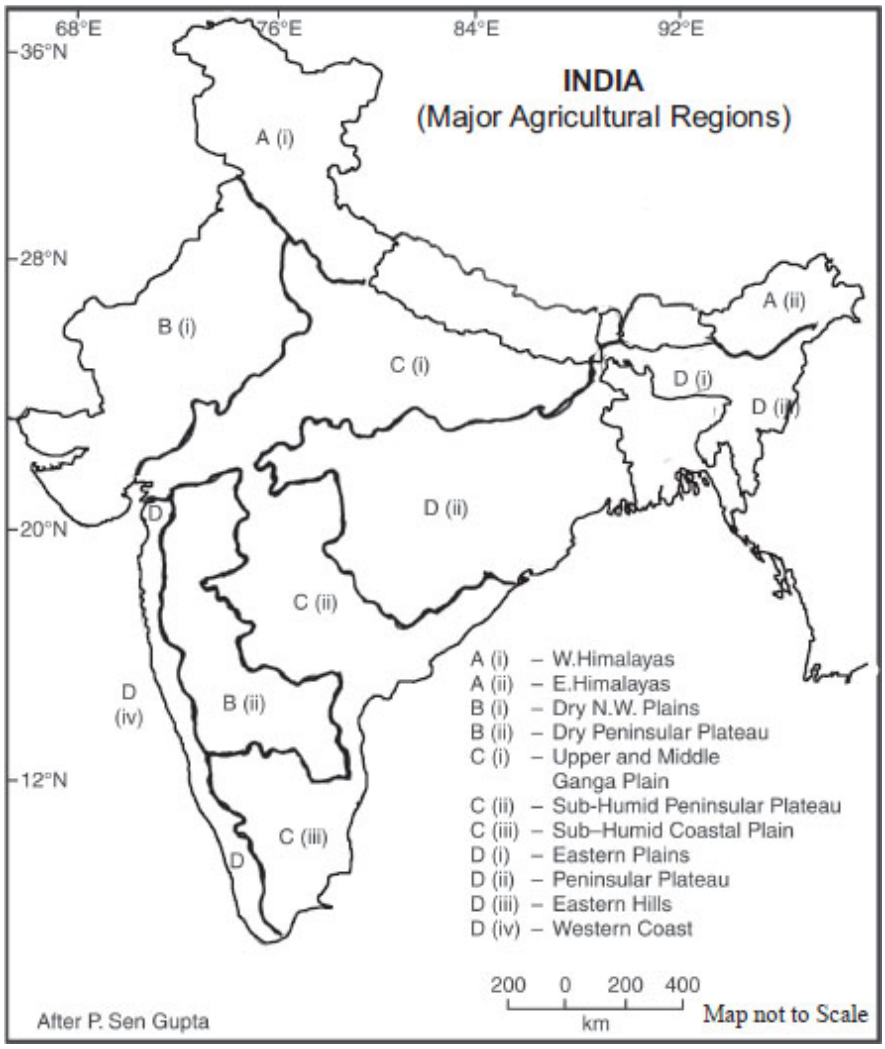


Fig. 10.8 Major Agricultural Regions

B. The Dry Agricultural Region

The dry agricultural belt stretches over Rajasthan, western Uttar Pradesh, Haryana, and the leeward side of the Western Ghats (Sahaydris). This region records less than 75 cm of average annual rainfall.

In Punjab, Haryana, and western Uttar Pradesh, irrigation by canals and tube wells is adequately available. The main crops of this region are wheat, rice, sugarcane, maize, pulses, oilseeds, millets, and *bajra*. It has 2 meso and 11 micro agricultural regions.

C. The Sub-Humid Region

The sub-humid belt stretches over the Upper and Middle Ganga Plains and a narrow central belt running from Bundelkhand Andhra Pradesh to Tamil Nadu Plateau and the Eastern Coastal plain. The average annual rainfall in this region varies between 75 cm to 100 cm. The main crops are wheat, rice, sugarcane, gram, maize, millets, cotton, groundnut, oilseeds, and tobacco. It has three mesodivisions and 24 micro-agricultural regions.

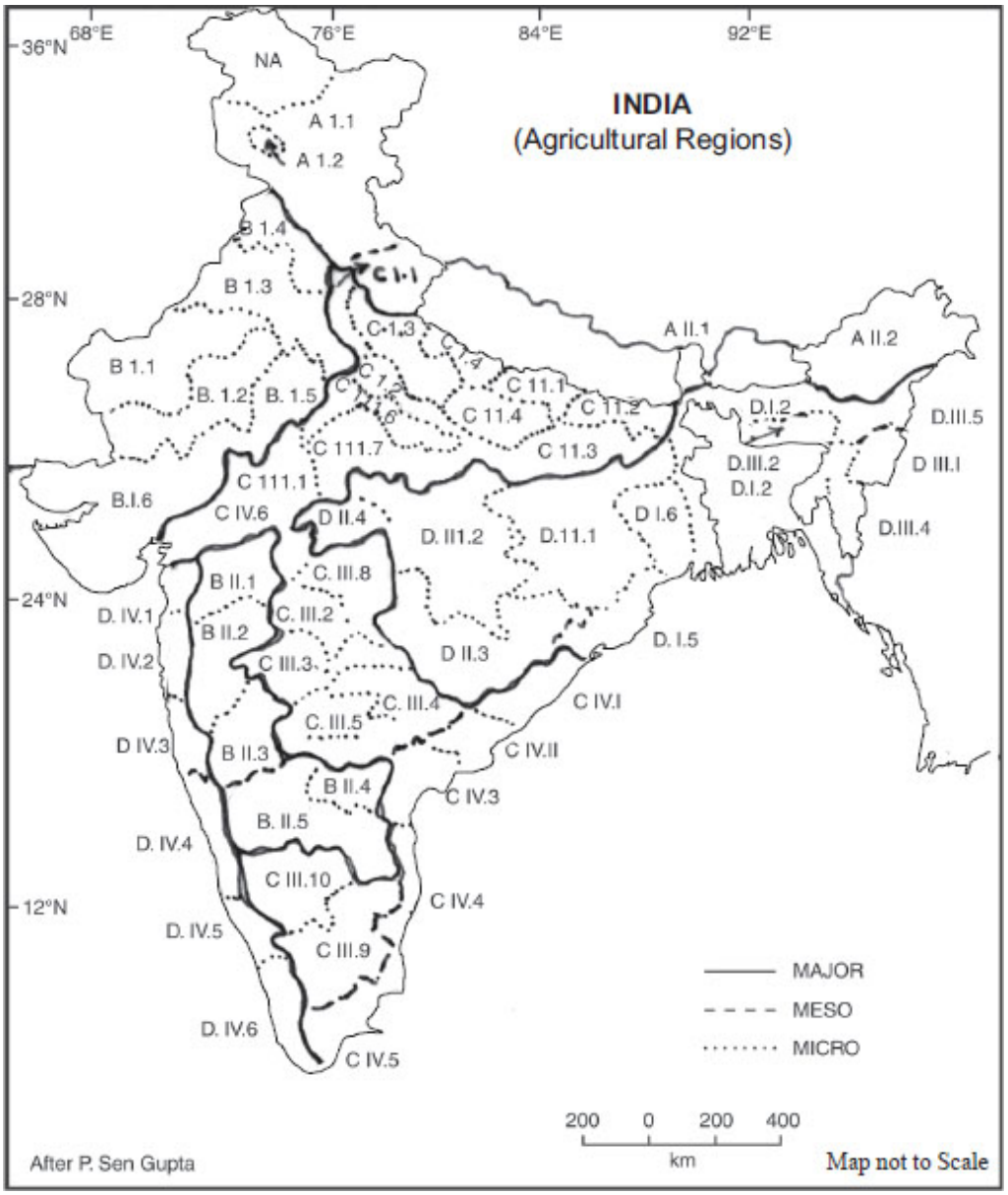


Fig. 10.9 Agricultural Regions

D. The Wet Agricultural Region

The wet belt sprawls over West Bengal, Jharkhand, Meghalaya, Assam, Manipur, Nagaland, Mizoram, and Tripura. It receives 100 to 250 cm average annual rainfall. The main crops are rice, jute, tea, rubber, spices, and oilseeds. It consists of 4 meso and 21 micro-level agricultural regions. Rice, jute, oil seeds, maize, coconut, sugarcane are the dominant crops in this region.

AGRO-CLIMATIC REGIONS OF INDIA

India has great variations in the geo-climatic, socioeconomic, and agricultural practices. The variations in the geo-ecological and socioeconomic conditions have closely influenced the agricultural activities. For the planning of agriculture, the Planning Commission and the National Remote Sensing Agency (NRSA) have divided the country into 15 agro-climatic regions **(Fig. 10.10)**. The main objectives of agro-climatic regions are

(i) to optimise agricultural production;
(ii) to increase farm income;
(iii) to generate more rural employment;
(iv) to make a judicious use of the available irrigation water; and
(v) to reduce the regional inequalities in the development of agriculture.

These agro-climatic regions are as under:

Fig. 10.10 Agro-Climatic Regions

1. The Western Himalayan Region
2. The Eastern Himalayan Region
3. The Satluj-Yamuna Plain
4. The Upper Ganga Plain
5. The Middle Ganga Plain
6. The Lower Ganga Plain

7. The Eastern Plateau and Hills
8. The Aravalli Malwa Upland
9. The Plateau Maharashtra
10. The Deccan Interior
11. The Eastern Coastal Plain
12. The Western Coastal Plain
13. The Gujarat Region
14. The Western Rajasthan
15. The Islands (Andaman and Nicobar and Maldives)

1. The Western Himalayan Region

The Western Himalayan region stretches over the Jammu Kashmir and Ladakh, Himachal Pradesh, and Uttarakhand. There is great variation in its topography. The average temperature of July varies between 5°C and 30°C, while the January temperature reads between 15°C to –5°C. The mean annual rainfall varies between 75 cm to 150 cm except Ladakh where it is below 20 cm. The valley of Kashmir, Kullu, and Dun have thick alluvial soils while the hills and mountains have thin cover of brown soils. This region has perennial rivers, some of which are utilised for irrigation and hydel-power generation.

Rice, maize, wheat, barley, oilseeds, pulses, potato, and vegetables are the main crops of this region. Temperate fruits like apples, peach, pears, almond, apricot, walnut, etc., are the source of cash for the cultivators. Though the geographical conditions of this region are well suited for orchards and plantation crops, there is a need of improvement in the old less productive varieties of fruits.

A more judicious land use planning is required for the region. Land should be rationally allotted for cultivation of crops, horticulture, pastures, and forestry. Better quality of seeds and planting materials should be made available for growing fruits, vegetables, flowers, saffron, ginger, and potato.

2. The Eastern Himalayan Region

This region sprawls over Arunachal Pradesh, Hills of Assam, Nagaland, Meghalaya, Manipur, Mizoram, Tripura, Sikkim, and Darjeeling District of West Bengal. It has rugged topography. The temperature of July varies between 25°C to 30°C and of January between 10°C and 20°C. The soil is red-brown and less productive. About 33% of the total cultivated area is under *Jhuming* (shifting cultivation).

Rice, maize, potato, tea, and orchards (pineapple, lime, litchi, oranges, etc.) are the main crops. Shifting cultivation should be controlled by developing terrace farming. Soil erosion is another important problem which needs immediate attention of the planners.The region needs marked improvement in infrastructural facilities to accelerate the pace of agricultural development. Moreover, a long term quality seed production suitable for the region needs to be implemented.

3. The Satluj Yamuna Plain

It stretches over Punjab, Haryana, Chandigarh, Delhi, and the Ganganagar district of Rajasthan. The climate has the semi-arid characteristics with July's mean monthly temperature between 25°C and 35°C and that of January between 10°C and 20°C. The average annual rainfall varies between 50 cm and 100 cm. The soil is alluvial which is highly productive. Apart from the canals, there are lakhs of tube wells and pumping sets installed by the cultivators and the governments. The intensity of agriculture (165%) is the highest in the country.

Wheat, rice, sugarcane, cotton, maize, gram, millets, pulses, oilseeds, and orchards are the main crops. The farmers of this region are most innovative and progressive. The high yielding varieties of rice and wheat were adopted first by the farmers of this region which led to Green Revolution. At present, contract farming has been started by the corporate sector in some of the districts of this region.

Development of saline and alkaline patches, water-logging, and soil erosion are the main problems of the region.

Agriculturally, this is the most developed part of the country. Agriculture may be made more productive and sustainable if the following steps are taken:

(i) Less area be devoted to rice as it is depleting the underground water table and the fertility of soil. Nearly 5% of the rice and wheat area may be diverted to pulses, oilseeds, maize, fodder, vegetables, flowers, and orchards.
(ii) Pests and disease resistant seeds of rice and wheat need to be developed.
(iii) Horticulture should be promoted.
(iv) Emphasis should be made on the cultivation of vegetables.
(v) High quality fodder crops should be developed.
(vi) Steps should be taken for the diversification of agriculture.

4. The Upper Ganga Plain

The Upper Ganga Plain stretches over the western Uttar Pradesh and the Hardwar and Udham-Singh Nagar districts of Uttarakhand upto Allahabad of U.P. The July temperature varies between 25°C and 35°C and that of January between 15°C and 25°C. The soil varies from sandy loam to clayey loam. Adequate facilities of irrigation by canals and tube well are available. Agriculture is well developed and Green Revolution is a big success in this region also.

Wheat, rice, sugarcane, maize, millets, fodder, pulses, oilseeds, and vegetables are the main crops. Mango and guava orchards are also quite significant in the agricultural landscape.

In order to make agriculture more productive and sustainable there should be emphasis on mixed cropping, horticulture, floriculture, judicious use of irrigation water, lining of canals, reclamation of saline/alkaline soils.

5. The Middle Ganga Plain

It sprawls over the greater parts of eastern Uttar Pradesh and Bihar. The July temperature varies between 25° C and 35°C and that of January between 150°C and 25°C. The average annual rainfall varies

between 100 cm and 150 cm. The soil is alluvial and there is great potential of underground water.

Rice, wheat, barley, maize, millets, pulses, mustard, sugarcane and oilseeds, are the main crops. Agriculturally, these plains are less developed than the Upper Ganga Plains.

Agriculture can be made more productive and sustainable to improve the input delivery system and diversification of crops by giving more area to vegetables, fruits, and floriculture. Dairying, silviculture, agro-forestry, and pisciculture can also go a long way in improving the income of the farmers.

6. The Lower Ganga Plains

This region stretches over the Brahmaputra Valley, West Bengal (excluding the Darjeeling District) and eastern Bihar. The July temperature varies between 25°C and 35°C and that of January between 18°C and 25°C. The mean annual rainfall varies between 100 cm and 200 cm. The soil is alluvial and highly productive.

Rice is the main crop. From the same field the farmers obtain two or three successive rice crops (*Aman, Aus, and Boro*) in a year. Jute, maize, pulses, and potato are the other important crops of the region.

Development of horticulture (banana, mango, and citrus fruits), pisciculture, poultry, and livestock can enhance the income of the cultivators substantially.

7. The Eastern Plateau and Hills

The Chotanagpur Plateau, stretching over Jharkhand, Odisha, Chhattisgarh, and Dandakarnaya are included in this agro-climatic region. The mean monthly temperature of July ranges between 25°C and 35°C and that of January between 18°C and 25°C. The mean annual rainfall ranges between 100 cm and 150 cm. Soils are red and yellow.

The agriculture in this region is mainly rain-fed The main crops are rice, maize, millets, *rag*i, oilseeds, gram, and tobacco.

The cultivation of high value crops like oilseeds, pulses, vegetables can increase the income of the farmers. Water-harvesting and

watershed development as well as soil conservation practices can make the agriculture more remunerative.

8. The Aravalli-Malwa Upland

This region includes Bundelkhand, Baghelkhand, Mahabharat Plateau, Malwa Plateau, and Vindhyachal Hills. This region is characterised by semi-arid climatic conditions. The mean July temperature reads between 25°C and 35°C, while the mean January temperature ranges between 15°C and 25°C. The soil is mixed yellow, red, and black. It has scarcity of water, both surface and underground.

Millets, wheat, gram, pulses, oilseeds, cotton, and sunflower are the main crops. Agriculture returns can be improved by farming, sprinkle irrigation, water-harvesting, dairy development, and poultry farming.

9. The Plateau of Maharashtra

It stretches largely on Deccan Plateau. This is a region of *regur* soil (black earth soil). The July temperature ranges between 25°C and 35°C, while the mean monthly temperature in January ranges between 15°C and 25° C. The average annual rainfall varies between 50 cm and 75 cm.

Wheat, gram, millets, cotton, pulses, groundnut, and oilseeds are the main crops in the rain-fed areas, while in the irrigated areas, sugarcane, rice, and wheat, are cultivated.

In place of coarse grains, value crops like pulses and oilseeds need to be cultivated. About 5% of the cotton area may be transferred to oilseeds. Dairy development, poultry farming, and social forestry are the other aspects of agricultural landscape which deserve immediate attention.

10. The Deccan Interior

The Deccan interior sprawls over the greater parts of Karnataka, Andhra Pradesh, and Tamil Nadu uplands from Adilabad district in the north to Madurai district in the south. The mean monthly temperature

of July varies between 25°C and 35°C, and the mean January temperature reads between 20°C and 25°C. This region records rainfall between 50 cm and 100 cm.

This is essentially a dry-farming area where millets, pulses, oilseeds, groundnut, coffee, tea, cardamom, and spices are the main crops. The area under millets and *bajra* can be devoted to pulses. Development of horticulture, poultry, and dairy-farming can help the farmers in the augmentation of their income.

11. The Eastern Coastal Plain

This region sprawls over the Coromandal and northern Circar coasts of Andhra Pradesh and Odisha. The mean July temperature ranges between 25°C and 35°C and the mean January temperature varies between 20°C and 30°C. The mean annual rainfall varies between 75 cm and 150 cm. The main crops of this region are rice, jute, tobacco, sugarcane, maize, millets, pulses, groundnut, and oilseeds. Area from coarse grains need to be shifted to high value crops like pulses and tobacco. Development of dairying, poultry, fisheries, and social forestry deserve more attention and investment.

12. The Western Coastal Plains

This region stretches over the western fringes of the states of Maharashtra, Karnataka, and Kerala. In the north, it is known as the *Konkan coastal plains* and in the southern parts as the *Malabar coastal plains*. The mean July temperature varies between 25°C and 30° C and mean January temperatures between 18° C and 30°C. The mean annual rainfall is more than 200 cm.

Rice, oilseeds, sugarcane, millets, pulses, cotton, and coconut are the main crops. Moreover spices are raised along the hill slopes of the Western Ghats.

Raising of high value crops (pulses, spices, and coconut) should get more agricultural area. Development of infrastructural facilities and promotion to prawn culture in brackish water should get preference.

13. The Gujarat Plains and Hills

This region consists of the hills and plains of Kathiawar, and the fertile valleys of Mahi, and Sabarmati rivers. It is an arid and semi-arid region where the July temperature reads about 30^{o}C and that of January about 25^{o}C. The mean annual rainfall varies between 50 cm and 100 cm.

Groundnut, cotton, rice, maize, millets, oilseeds, wheat, pulses, and tobacco are the main crops. It is an important oilseeds producing area. The major thrust of development in the region should be on canal and ground water management, rain-water harvesting, agro-forestry in the region, and brackish/backwater aquaculture in coastal areas.

14. The Western Region

This region stretches over Rajasthan west of the Aravalli mountains, and northern Gujarat. The mean July and January temperatures vary between 40^{o}C and 15^{o}C, respectively. The mean annual rainfall is less than 25 cm.

Bajra, pulses, and fodder are the main crops. Livestock contributes greatly to the desert ecology. Rain water harvesting horticulture of date-palm, watermelon, and plums deserve more attention.

15. The Islands Region

This region includes Andaman-Nicobar, and Lakshadweep which have marine equatorial climates. At Port Blair, the mean July and January temperatures read 30^{o}C and 25^{o}C, respectively. Rice, maize, millets, *bajra*, pulses, arecanut, turmeric, and cassava are the main crops. The main thrust in the development of agriculture should be on crop improvement, water management, and fisheries. Brackish water prawn culture should be promoted in the coastal areas.

AGRO-ECOLOGICAL REGIONS OF INDIA

The concept of “agro-ecological region” is a modification and improvement on agroclimatic regions. The concept of agro-ecological

region is perhaps hazy to our scientific communities. People generally confuse these two terms (agroclimatic and agro-ecological) and use them in more or less the same way. But there is a distinct difference between these two terms. According to FAO (1983), agro-climatic region is a land unit in terms of major bioclimate and length of growing period and which is climatically suitable for certain range of crop cultivation. But agro-ecological region is the land unit carved out of agro-climatic region when superimposed on land form and soil condition that acts as modifier of the length of growing period. Therefore, within an agro-climatic region there may be a few agro-ecological regions depending on soil condition. This approach has been used in delineating agro-ecological regions of India.

Methodology

The methodology used in the delineation of agro-ecological region is given below:

(i) Mean monthly temperature and mean monthly precipitation;
(ii) Type of soil; and
(iii) Length of growing period.

In order to prepare an agro-ecological region map, the soil-scap (soil-map) was superimposed on bio-climatic map. On the resultant map, the growing period was incorporated, using GIS technology.

In the demarcation of agro-ecological regions of India, the agro-climatic regions of India have been sub-divided on the basis of soil type. In the meso-regions thus obtained, the length of growing period (LGP) has been superimposed. This method has resulted into 20 agro-ecological regions and 60 agro-ecological sub-regions. The 20 agro-ecological regions of India have been plotted in (**Fig. 10.11).**

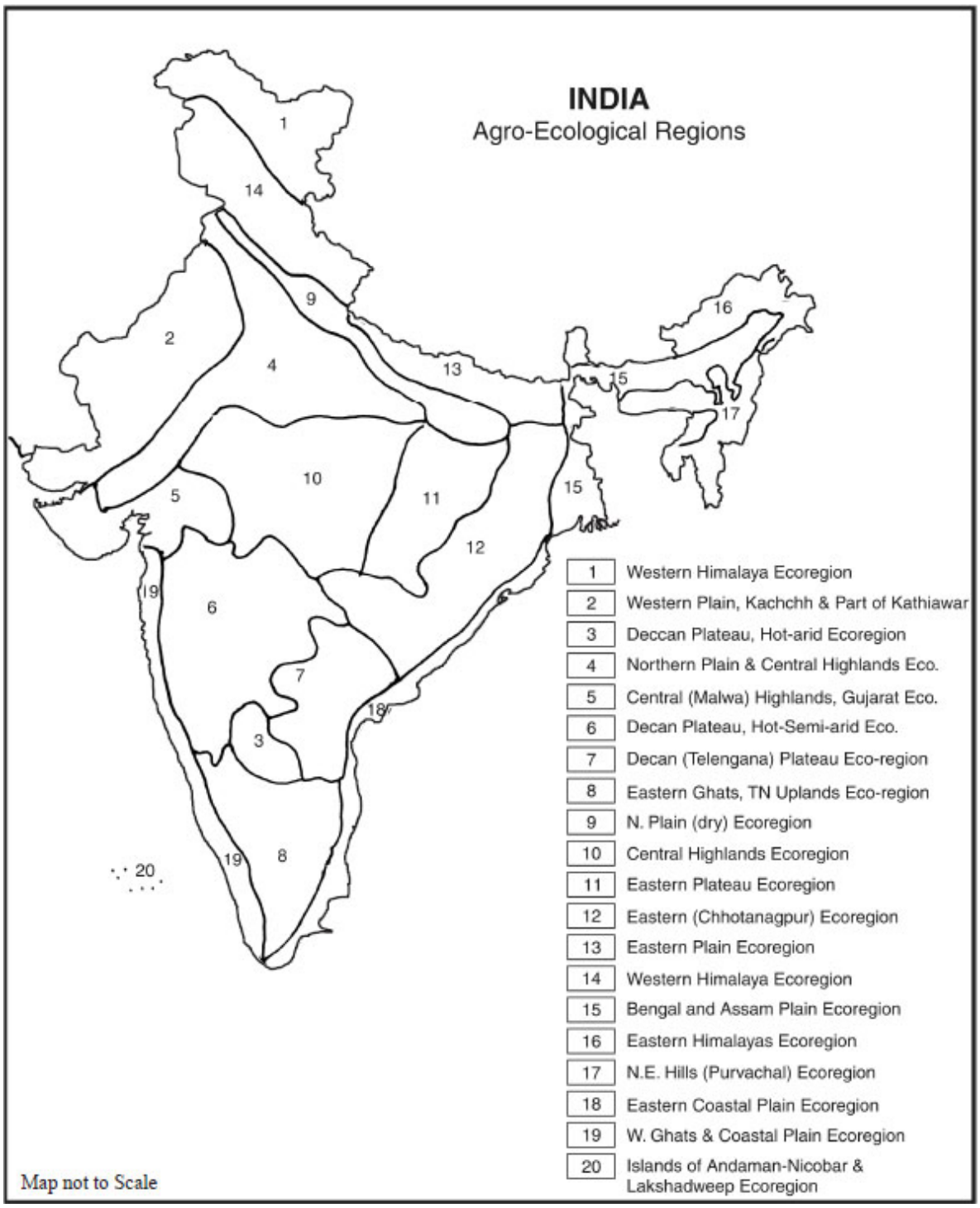


Fig. 10.11 Agro-Ecological Regions

The agro-ecological regions of India are:

1. Western Himalayas, cold arid eco-region with shallow soil

and length of growing period less than 90 days.

2. Western Plain, Kutch, and part of Kathiwara Peninsula, hot and arid ecoregion with desert and saline soils and growing period less than 90 days.
3. Deccan Plateau, hot arid region with red and black soils and growing period of less than 90 days.
4. Northern Plain and Central Highlands including Aravallis, hot semi-arid ecoregion, with alluvium-derived soils and growing period between varying between 90–150 days.
5. Central Malwa Highlands, Gujarat Plains, and Kathiawar Peninsula, hot semi-arid ecoregion with medium and deep black soils and growing period varying between 90 to 150 days.
6. Deccan Plateau, hot semi-arid ecoregion, with shallow and medium black soils and growing period varying between 90 to 150 days.
7. Deccan (Telengana)Plateau and Eastern Ghats, hot semi-arid ecoregion with red and black soil and growing period from 90 to 150 days.
8. Eastern Ghats, Tamil Nadu Plateau and Deccan (Karnataka), hot semi-arid ecoregion with red-loamy soils, and growing period of 90–150 days.
9. Northern Plain, hot sub-humid (dry) ecoregion, with alluvium soils, and growing period of 150–180 days.
10. Central Highlands (Malwas, Budelkhand, and Eastern Satpura), hot subhumid ecoregion, with black and red soils, and growing period 150–189 days.
11. Eastern Plateau (Chattisgarh), hot sub-humid ecoregion, with red and yellow soils, and growing period of 150–180 days.
12. Eastern (Chotanagpur) Plateau and Eastern Ghats, hot subhumid ecoregion, with red and lateritic soils, and growing period of 150–180 days.
13. Eastern Plain, hot sub-humid (moist) ecoregion, with alluvium

soils, and growing period of 180–210 days.

14. Western Himalayas, warm sub-humid to humid with inclusion of perhumid ecoregion, with brown forest and podzolic soils, and growing period of 180–210 days.
15. Bengal and Assam Plain, hot sub-humid (moist) to humid (inclusion of perhumid) ecoregion with alluvial soils, and growing period 180 to 210 days.
16. Eastern Himalayas, warm perhumid ecoregion with brown and red soils, and growing period over 210 days.
17. North-Eastern Hills (Purvanchal), warm perhumid ecoregion with red and lateritic soils, and growing period over 210 days.
18. Eastern Coastal Plain, hot sub-humid to semi-arid ecoregion with coastal alluvium soils, and growing period varying from 90 to 210 days.
19. Western Ghats and Coastal Plain, hot humid per humid ecoregion, with red lateritic and alluvium soils, and growing period varying from 90 to 210 days.
20. The Islands of Andaman and Nicobar and Lakshadweep, hot humid to perhumid island ecoregion, with red loamy and sandy soils, and growing period over 210 days.

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11

Industries

EVOLUTION OF INDUSTRIES

Industrial development is considered as one of the important indicators of socio-economic and human development. Before the rise of modern industries, India was known all over the world for its cottage and household industries. Indian muslin, silk goods, and artistic pottery were in great demand the world over. The arrival of English resulted in the decay of traditional handicrafts. The East India Company developed a policy of export of raw material from India to Britain and the import of finished products to India from Europe. It was because of this policy that no industry could be developed in India before 1854. In the later part of the 19 century the growth and development of industries in India was stunted and slow.

The industrial development in India started after 1854 when some cotton and jute mills were established by the British in Mumbai and Calcutta (Kolkata), respectively. The beginning of modern industrial

phase may be traced back to the '*establishment of the first iron and steel charcoal plant at Port-Navo (Tamil Nadu)'*. This mill was closed down in 1886. The first cotton textile factory was established at Mumbai in 1854, followed with the opening of a jute mill at Rishira near Calcutta (Kolkata) in 1855. The cotton textile industry expanded during 1870s when there was civil war in America. The number of cotton mills grew to 275 before the First World War. The progress in jute industry was, however, not satisfactory.

The beginning of iron and steel industry was even more delayed. The first iron and steel mill was established by Bengal Iron Works Ltd. at Kulti in 1874. It resulted in a failure and was shut down in 1881. The real development of iron and steel industry started in 1908 when a steel plant was established at Jamshedpur.

The *first successful paper mill started at Ballygunj Circular Road in Kolkata in 1870*. Two more paper mills in Kolkata and one in Lucknow (1879) started in 1905. It was in 1870, when the woollen textile mills were established in Bangalore (1886) (Karnataka), Dhariwal (1881) (Punjab), and Kanpur (1876) (Uttar Pradesh).

After the First World War, the Indian industries got a good boost as India became the main supplier of cotton and woollen textiles and liquors. The government gave protection to some of the industries. The period during the Second World War was a time of crisis as India also got involved into war. After the Second World War, the production fell down due to decreasing demand for industrial products, lack of capital, political unrest, transport bottlenecks, and labour strikes. The partition of the country in 1947 gave a severe blow to jute and cotton textiles as the raw material producing areas of jute and good quality cotton went to Pakistan.

As stated above, the post Second World War period was characterised by industrial turmoil. Production in most of the industries declined. The condition of cotton textiles, cement, paper, iron and steel industries, and consumer goods was worst due to the non-availability of raw material.

After Independence, the Government of India realised the importance of an appropriate industrial policy, which led to the Industrial Policy Resolution, 1948. According to this policy, the concept of mixed economy was introduced in which the state and the private enterprise were allowed to co-exist and co-prosper in the fields demarcated for them. This resolution divided the industries between public and private sectors.

INDUSTRIAL DEVELOPMENT DURING THE FIVE-YEAR PLANS

The real growth and development of the industrial sector in India started during the period of Five-Year Plans.

First Five-Year Plan (1951–56)

The main thrust of the First Five-Year Plan was on agricultural development. Therefore, the emphasis was on increasing capacity of the then existing industries rather than the establishment of new industries. Cotton, woollen and jute textiles, cement, paper, newsprint, power-looms, medicine, paints, sugar, *vanaspat*i (vegetable oil), chemical and engineering goods, and transport equipments showed some progress.

Second Five-Year Plan (1956–61)

Great emphasis was laid on the establishment of heavy industries during the Second Five-Year Plan. The second industrial policy was announced in 1956. The main thrust of industrial development was on iron and steel, heavy engineering, lignite projects, and fertiliser industries. Moreover, there was an emphasis on the expansion of existing steel plants, like Jamshedpur, Kulti-Burnpur, and Bhadravati. Three new iron and steel plants were located at Bhilai, Durgapur, and Raurkela. The Chittranjan Locomotive Workshop, the Hindustan Ship-building Yard (Vishakhapatnam), the Sindri Fertiliser Factory, and Hindustan Machine Tools Ltd. (HMT) at Bangalore were expanded. Many of the targets, however, could not be achieved because of the war with China in 1962 and the failure of monsoon over greater parts of the country.

Third Five-Year Plan (1961–66)

There was an emphasis on the expansion of basic industries like iron and steel, fossil-fuel, power, and machine building. The Ranchi Machine Tool and three more HMT units were established. Machine building, locomotive and railway coach making, ship-building, air-craft manufacturing, chemicals, drugs, and fertiliser industries also made steady progress.

Fourth Five-Year Plan (1969–74)

The period between 1966 and 1969 was the period of annual plans. The industrial period could not make much progress during the annual plans period. During the fourth Five-Year Plan there was much emphasis on the agro-based industries such as sugar, cotton, jute, *vanaspati*, metal-based, and chemical industries. It was during this plan when much progress was made in alloys, aluminium, automobile-tyres, electronic goods, machine tools, tractors, and special steel. Efforts were also made to accelerate the process of industrial dispersal.

Fifth Five-Year Plan (1974–79)

The main stress in this plan was on rapid growth of steel plants, export-oriented articles, and goods of mass consumption. The steel plants at Salem, Vijaynagar, and Vishakhapatnam were proposed to create additional capacity. The Steel Authority of India (SAIL) was constituted. Moreover, drug manufacturing, oil refining, chemical fertilisers, and heavy engineering industries made steady progress.

Sixth Five-Year Plan (1980–85)

The main emphasis in the Sixth Five-Year Plan was on producing goods to exploit the domestic and international markets. To achieve this objective industries like aluminium, automobiles, electric equipments, thermostatics were given the priority. Production targets were achieved in industries like commercial vehicles, drugs, T.V. receivers, automobiles, cement, coal, jute industry, non-ferrous metals, textiles, railway wagons, sugar industry, etc.

Seventh Five-Year Plan (1985–90)

The main thrust of the Seventh Five-Year Plan was on “high tech” and electronic industries. Industrial dispersal, self employment, exploitation of local resources, and proper training were the preference areas of the plan.

Eighth Five-Year Plan (1992–97)

The period between 1990 and 1995 was the period of annual plan. There was a major change in the industrial policy of the Government of India which was initiated in 1991. The policy of liberalisation was adopted for the investment of foreign multi-nationals. Emphasis was given on the removal of regional imbalances and encouraging the growth of employment in small and tiny sectors.

Ninth Five-Year Plan (1997–2002)

The main emphasis during this plan was on cement, coal, crude oil, consumer goods, electricity, infrastructures, refinery, and quality steel products.

Tenth Five-Year Plan (2002–07)

During the Tenth Five-Year Plan, the main emphasis was on:

(i) the modernisation, technology upgrading, reducing transaction costs, and increased export;

(ii) to enhance export and to increase global competitiveness; and

(iii) to achieve balanced regional development.

To achieve these objectives and reduce the regional inequalities Special Economic Zones (SEZs) policy has been adopted. Moreover, industrial packages have been given to Himachal Pradesh, Jammu and Kashmir, and Uttarakhand. In addition to these, production of basic foods, capital goods, intermediate goods, and consumer goods has been encouraged.

Eleventh Five Year Plan (2007–12)

The Eleventh plan document entitled “Towards Faster and More Inclusive Growth” gave priority to: agriculture, irrigation, and water

resources, education, health, infrastructure, and employment, along with programmes for SCs/STs, other backward classes, minorities, women and children. Government has realised that in recent years although economic growth has accelerated but it has failed to be 'inclusive'. In other words benefits of growth have not reached all sections of population particularly small farmers, landless labourers and persons working in unorganised sector have remained beyond the benefits of development. Eleventh plan stresses that benefits of development should reach all sections of population. This plan emphasizes on social justice. Eleventh plan recognizes that we need a growth process that brings faster reduction in poverty, generate employment and ensures essential services such as health, and education to all sections of the society.

Twelfth Five Year Plan (2012–17) and Niti Ayog

This plan was approved on October 4, 2012.The objective of this plan was "*Faster, More inclusive and Sustainable Growth*". The target of the growth rate was 8%. The twelfth five year plan was the last plan to be implemented. It ended on March 31, 2017.

The Niti Ayog (National Institute for Transforming India) was founded on January 1, 2015. It replaced the Planning Commission. The three-year agenda is further divided into seven parts, with a number of specific action points for each part to boost economic growth. The key feature of this document is to reduce fiscal deficit to 3% by 2018–19 and revenue deficit to 0.9% of the GDP by 2019–20.

INDUSTRIAL POLICY

The first industrial policy was announced by the government of India on 6 April, 1948. In this policy both the public and private sectors were involved towards industrial development. Accordingly, the industries were classified into four broad categories:

1. **Exclusive State Monopoly**: This includes the manufacturing of arms and ammunition; production and control of atomic energy; and the ownership and management of railways. In

June 2016 the Government has allowed 100% Foreign Direct Invesment (FDI) in many industries including the defence goods production.

2. **State Monopoly for New Units**: This category included coal, iron and steel, aircraft manufacturing, ship-building, manufacturing of telephone, telegraph and wireless apparatus (excluding radio receiving sets), and mineral oil. New undertakings in this category could, henceforth, be undertaken only by the state. In June 2016, the Govt. of India has allowed 100% Direct Foreign Investment (FDI) in Airlines and Air-craft industry.

3. **State Regulation**: This category included industries of such basic importance like machine tools, chemicals, fertilisers, non-ferrous metals, rubber manufactures, cement, paper, newsprint, automobiles, electric engineering, etc., which the central government would feel necessary to plan and regulate.
4. **Unregulated Private Enterprise**: The industries in this category were left open to the private sector, individual as well as co-operative.

Industrial Policy, 1991

A major shift in the industrial policy was made by the Congress Government, led by Shri. P.V. Narasimha Rao on 24 July, 1991. The main aim of this policy was to unshackle the country's industrial economy from the cobwebs of unnecessary bureaucratic control, induced liberalisation with a view to integrate the Indian economy with the world economy, to remove restrictions on direct foreign investment and also to free the domestic entrepreneur from the restriction of MRTP (Monopolies and Restrictrive Practices Act). Moreover, the policy aimed at shedding the load of public enterprises which have shown a very low rate of return or incurred losses over the years. The salient features of the 1991 industrial policy were:

1. Except some specified industries (security and strategic concerns, atomic energy, coal, social reasons, environmental issues, railways, atomic minerals, hazardous projects, and

articles of elitist consumption), industrial licensing would be abolished.

2. Foreign investment would be encouraged in high priority areas upto a limit of 51% equity.
3. Government would encourage foreign trade companies to assist Indian exporters in export activities.
4. With a view to inject desired level of technological dynamism in Indian industries, the government would provide automatic approval for technology agreements related to high priority industries.
5. Relaxation of MRTP which has almost been rendered non-functional.
6. Dilution of Foreign Exchange Regulation Act (FERA) making Rupee fully convertible on trade account.
7. Disinvestment of Public Sector Units' shares.
8. Closing of such public sector units which are incurring heavy losses.
9. Abolition of (C.C.I.) and wealth tax on shares.
10. General reduction in custom duties.
11. Provide strength to those public sector enterprises which fall in reserved areas of operation or in high priority areas.
12. Constitution of special boards to negotiate with foreign firms for large investments in the development of industries and import of technology.

A new Industrial Policy has replaced the 27 year old existing policy, with the aim of promoting new technology and reduced regulations. The policy aims at creating jobs over the next 10 years, promote the foreign technology transfer and attract $ 100 billion FDI annually.

The constraints to the industrial growth are:

(i) Inadequate infrastructure
(ii) Restrictive Labour Laws
(iii) Complicated business environment

(iv) Slow technology adoption
(v) Low Productivity
(vi) Challenges for trade
(vii) Inadequate expenditure on R esearch and development

COTTON TEXTILE INDUSTRY

Textile industry includes cotton, jute, wool, silk, and synthetic fibre textiles. India is one of the leading producers of textile goods. It is one of the largest and most important sector in the economy in terms of output, foreign exchange earning, and employment in India. Its contributioin forms 14% (2015–16) of the industrial production, 10% of the excise collection, 18% of employment in the industrial sector, 20% of the country's total export earning and 4% of the GDP. India is the second largest producer and exporter of cotton in the world at $6.3 billion, marginally close to China.India has emerged as the largest producer of cotton in the world with the production of 345 lakh bales in 2016–17 and second largest exporter after China.

India enjoyed monopoly in the production of textile goods from 1500 BC to 1500 AD. Indian cotton and silk textiles were in great demand all over the world. It was the arrival of the British in India and the Industrial Revolution in Britain in 1779 which led to the downfall of the Indian manufacturing. The British after the consolidation of their rule in India encouraged the export of raw material from India to Britain and import of manufactured goods from Britain to India. The first textile *mill was establish ed in 1854 in Mumbai by C.N. Dewar*. The fast growth of cotton textile occurred in 1870 when there was much demand of Indian goods in the wake of American Civil War. Before the First World War the number of Indian textile mills rose to 271. The demand for cloth during the Second World War led to further progress of the textile industry.

The industry suffered a setback in 1947 as good quality cotton growing areas went to Pakistan. Consequently, India had to import cotton from the African countries.

Cotton accounts for 25% of the total global production. India occupies the first position in the world in cotton acreage with about 105 laks hectares under cotton cultivation.

Cotton being a pure raw material provides a chance to establish textile mill either in the areas of raw material or in the market. In India, most of the textile mills are in the cotton growing areas or in the neighbouring cities and towns. The location of cotton textile industry is mainly affected by: (i) raw material, (ii) proximity to market, (iii) moist weather, (iv) capital, (v) skilled, efficient and cheap labour, (vi) transport, (vii) sea-port, (viii) export facility and the domestic and international markets. The trend of growth in cotton textile industry has been given in (**Table 11.1)**.

Table 11.1

Growth of Cotton Textile Industry in India

Year	Production	Production (in Lakh Bales of 170 kg Each)
2008–09	290	229
2009–10	305	259
2010–11	339	259.61
2011–12	367	375.28
2012–13	370	283.16
2013–14	398	299.55
2014–15	386	309.44

Year	Production	Production (in Lakh Bales of 170 kg Each)
2015–16	332	315.28
2016–17	345	306.36
2017–18	377	334.00

Source: Economic Survey 2017–18.

(i) **Cotton mills and factories:** In the earlier days of its development, mills were the main producers of cotton goods. For example, in 1950–51 over 80% of the total cotton goods production was from the mills and factories, which dropped to only about 10% in 2015–16. The production and employment is increasing steadily in the powerloom sector.

(ii) **Powerlooms:** The decentralised power-loom sector played a vital role in meeting the clothing needs of the country. The powerloom industry produces a wide variety of cloth with intricate designs and accounts for almost 83% of the total cloth production.

(iii) **Handlooms:** The handloom sector provides over 12% of the total production.

Most of the cotton mills and factories are located in the cotton producing regions of the country. The main cotton producing areas of the country have been shown in (**Fig. 11.1).**

Distribution and Production of Cotton Goods Statewise

The cotton textile production in India between 1980 to 2015 has been shown in (**Fig. 11.2)**. It may be seen from **Table. 11.1** that the total production of cotton textile in 2010–11 was 339 Lakh bales of and it has risen to 377 lakh bales of 170 kg each in 2017–18.

The statewise distribution and production of cotton textile have been shown in (**Table 11.2)**. It may be seen from (**Table 11.2)** that Maharashtra is the leading producer of cotton textile, followed by

Gujarat and Telangana. During 2017, the total area under cotton cultivation was 122.35 lakh hectare with a production of 345 lakh bales of 170 kg each.

India's textile and clothing industry is one of the main stays of the national economy. Cotton accounts for about 25% of the total global fibre production. Cotton sustains the livelihood of about 5.8 million cotton farmers. India has the largest area under cotton in the world with around 105 lakh hectares making 35% of the world's total.

Table 11.2

State wise Production of Cotton Cloth in India (2017)

State	Area (Lakh Hectares)	Production (in Lakh Bales of 170 kg Each)	Yield (kg/Hectare)
Punjab	3.85	12.00	529.87
Haryana	6.56	25.00	647.87
Rajasthan	5.03	22.00	743.54
Gujarat	26.18	104.00	675.32
Maharashtra	41.98	85.00	344.21
Madhya Pradesh	5.99	20.00	567.61
Telangana	18.24	57.00	531.25
Andhra Pradesh	5.44	22.00	687.50
Karnataka	5.65	19.00	571.68
Tamil nadu	1.48	6.00	689.19
Odisha	1.45	3.00	351.72
Others	0.50	2.00	680.00
All India	**122.35**	**345.00**	**523.83**

Source: Cotton Advisory Board (CAB)

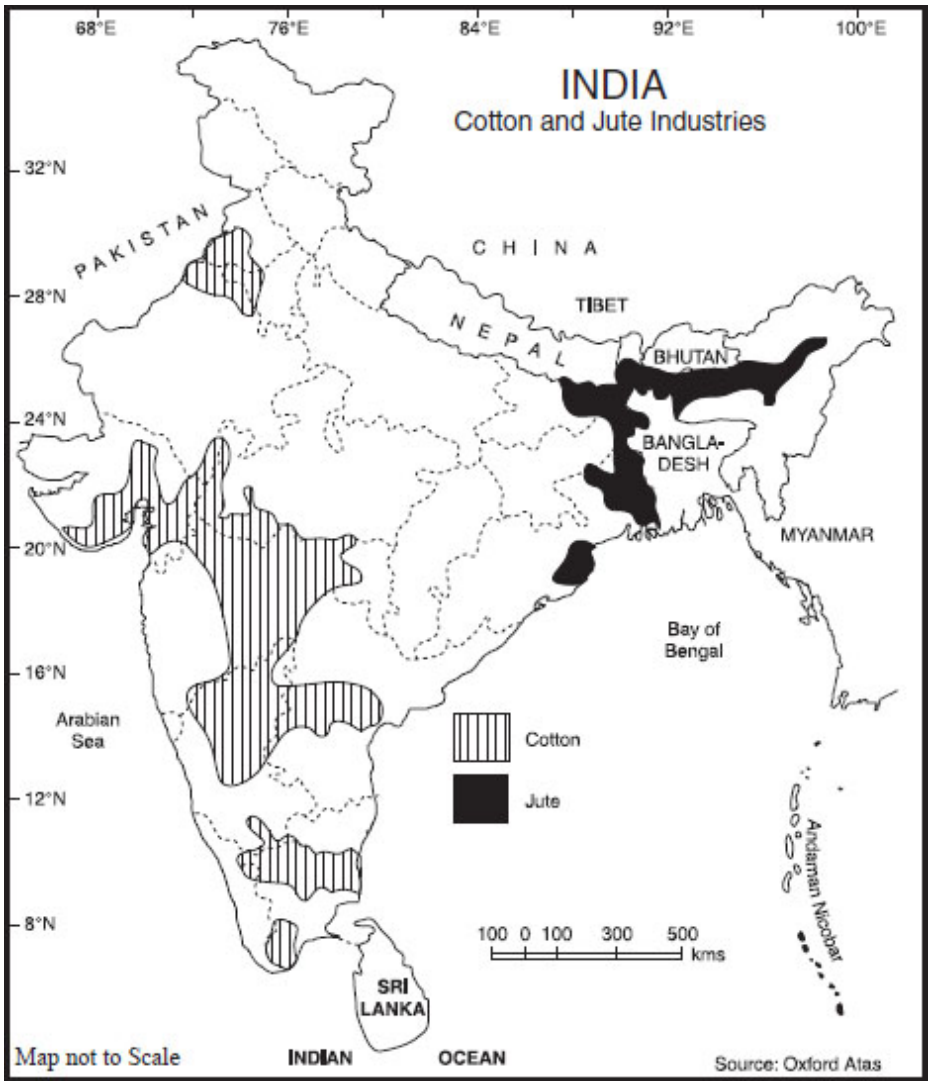


Fig. 11.1 Concentration of Cotton & Jute

Fig. 11.2 Cotton Textile

Maharashtra

The state of Maharashtra is the largest producer of cotton goods. The locational factors in the high concentration of cotton mills in the state of Maharashtra are:

(i) **Availability of raw material:** The state of Maharashtra is one of the leading producers of cotton.

(ii) **Climate:** The city of Mumbai where most of cotton mills are located has a mild climate with enough moisture in the air; so the thread does not break frequently.

(iii) Mumbai is close to Egypt, Sudan, and east African countries from where the long staple cotton is imported for the production of good quality of cloth.

(iv) **Labour:** Cheap skilled labour is available in the state.

(v) **Electricity:** Cheap electricity is available and there is not much power breakdown in the state.

(vi) **Market:** There is a large market of cotton products, both in India and abroad.

(vii) **Seaport:** The seaport of Mumbai is well connected by rails and highways.

(viii) **Investment:** There is no dearth of money investment in this industry.

(ix) **Entrepreneurs:** Mumbai being the commercial capital of the country enjoys the presence of entrepreneurs who are always willing to invest in this industry.

(x) **Early Start:** The state of Maharashtra and the city of Mumbai got the advantage of an early start in cotton textile industry.

With 122 cotton textile mills in Maharashtra, this industry contributes about 39% of the total cloth production of the country. The city of Mumbai with 63 cotton mills is the largest producer of cotton in the country. The bulk production is in the form of light textured cloth, long cloth, shirting, domestics, sheets, *dhotis*, and coloured clothes. Due to the high concentration of cotton mills Mumbai is called the Cottonopolis of India.

In Maharashtra, Sholapur is the second largest producer of cotton goods. It lies in the cotton growing area of south Maharashtra. Pune, Nagpur, Jalgaon, Akola, Sangli, Miraj, Kolhapur, Kalyan, and Thane are the other important cotton goods producing centres **(Fig. 11.2)**.

Gujarat

Gujarat has 118 mills producing about 35% of the total cotton goods of the country. The city of Ahmedabad is the second largest cotton

producing centre after Mumbai in the country. It specialises in the production of fine qualities of *dhotis* and *saris* and a large variety of bleached, coloured, or printed fabrics. In addition to this Bhavnagar, Bharuch, Kalol, Kadi, Kelot, Khambat, Nanded, Porbander, Rajkot, Surat, Vadodra, and Viramgam are the other important cotton producing centres of Gujarat.

Tamil Nadu

Tamil Nadu has the largest number of cotton mills in the country. It is the third largest producer of cotton textile in the country. The largest number of mills are, however, in the city of Coimbatore which has over 200 small and big factories. Tamil Nadu's mills are however, smaller in size. This state produces about 45% of the total yarn of the country. Chennai, Madurai, Perambur, Salem, Tiruchirappalli, Tirunelveli, and Tuticorin are the other important cotton textile centres in the state.

Uttar Pradesh

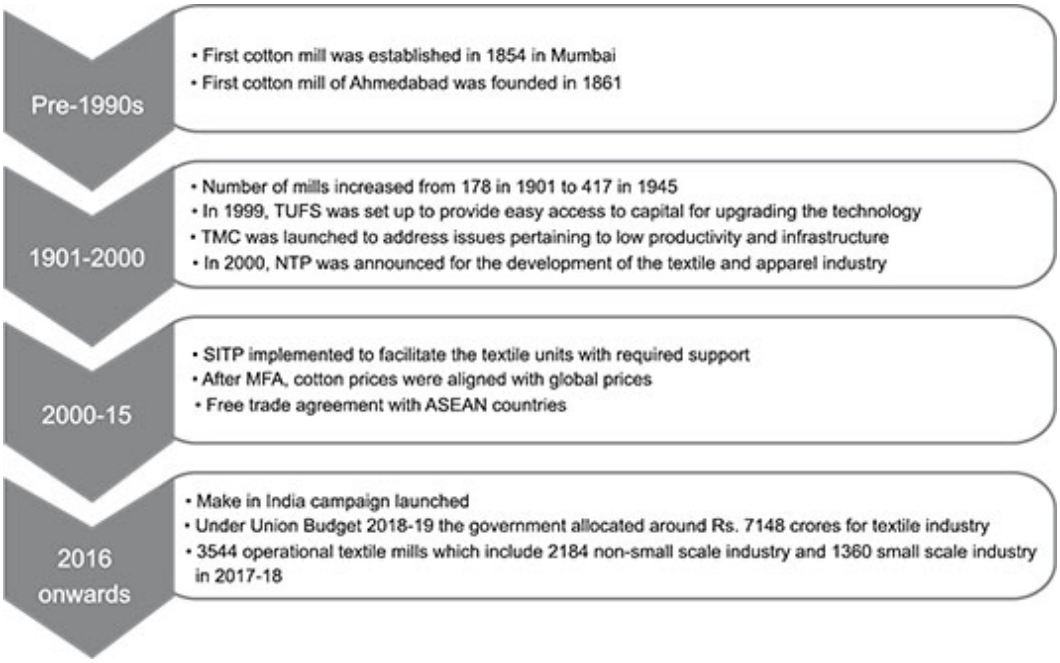
Kanpur is the most important cotton textile centre of Uttar Pradesh. In addition to this, cotton textile industry is located in the cities of Agra, Aligarh, Bareilly, Etawah, Lucknow, Mirzapur, Modinagar, Modipuram, Moradabad, Saharanpur, and Varanasi.

West Bengal

Kolkata is the most important cotton textile producing centre of West Bengal. Cotton goods are also produced in Haora, Hugli, Murshidabad, Panihar, Sirampur, and Shiampur.

In addition to these, Ajmer, Beawar, Bhilwara, Ganganagar, Kishangarh, Jaipur, Kota, Pali, and Udaipur in *Rajasthan*; Amritsar, Ludhiana, and Phagwara in *Punjab*; Bhopal, Dewas, Gwalior, Indore, Jabalpur, Ratlam, and Ujjain in *Madhya Pradesh*; East Godavari, Guntur, Hyderabad, Secunderabad, and Udaygiri in *Andhra Pradesh*; Gaya, Bhagalpur and Patna in *Bihar*; Bhiwani, Hissar and Rohtak in *Haryana*; Bangalore, Belgaum, Chennapatnam, Chitradurga, Gulbarga, Mangalore, and Mysore in *Karnataka*; Alleppey, Alwaye, Kollam, Thiruvananthapuram, and Trichur in *Kerala* are the important cotton textile centres in the country.

In India, cotton goods are produced in the following three types of sectors: (i) Mills, (ii) Power-looms, and (iii) Handlooms. The evolution of the Indian textile sector has been summarised in the flow diagram given below.



Source: *Indian Brand Equity Foundation; www.ibef.org*

The production of yarn has grown to 5676 million kg in FY2018 from 4712 million kg in FY2011. There has been a growth of 2.69%. The fabric production in India has risen to 66,514 million square metres in FY2018.

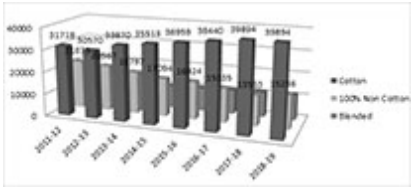


Fig. 11.3 Fabric Production (Million Square meter)

Problems of the Cotton Textile Industry

The cotton textile industry is suffering from many serious problems. Some of the problems of the cotton textile industry have been briefly presented here:

1. Shortage of Raw Material

There is a shortage of raw material especially of good quality cotton to meet the growing demand of the Indian textile industry. The fluctuating prices and uncertainties in the availability of raw material lead to low production and sickness of the mill. Consequently, long staple cotton is imported from Egypt, Sudan Kenya, Peru, Tanzania, Uganda, and the USA. There is a need of Silver Fibre Revolution in the country.

2. Obsolete Machinery

Most of the Indian textile mills are working with obsolete machinery. According to one estimate 70% of the spindles are more than 30 years of age. The outdated machinery cannot compete with the machinery of countries like China, Japan, South Korea, Taiwan, and the USA which have the latest sophisticated machinery in their textile mills.

3. Erratic Power Supply

Power supply to most of the factories is inadequate and erratic which adversely affects the production of goods.

4. Strikes and Lockouts

The cotton textile industry suffers seriously because of frequent strikes by the workers. Occasionally the owners lockout the mills which decreases the production of goods.

5. Competition in Foreign Market

The Indian cotton goods are facing a stiff competition in foreign markets, especially from China, Egypt, Japan, South Korea, and Taiwan.

6. Heavy Excise Duties

The cotton textile industry suffered because of heavy excise duties. The high rate of duty on imported cotton has increased the cost of production of clothes which has created problems in selling the cloth in the international market.

7. Competition with Synthetic Fibres

The poor people of the country prefer to use synthetic fibre clothes which are more durable and attractive.

8. Sick Mills

In India more than 130 mills are sick. The machinery of these mills is obsolete which can only be replaced with heavy investment. The slow pace of modernisation, outdated machinery, and old technology are some of the long-term problems which need to be addressed.

JUTE TEXTILE

India is the largest producer and second largest exporter of jute goods in the world, contributing about 35% of the total output of the world. It is a labour intensive industry which directly and indirectly provides job to more than 4 lakh people. The industry is, however, facing a tough competition from synthetics and its export market is shrinking.

The first jute factory in India was established at '***Rishira***', about 20 km north of Calcutta in 1854. The industry made tremendous progress in the later part of the 19^{th} century. Subsequently, the industry was boosted by the two world wars. The industry suffered a serious setback in 1947 due to the partition of the subcontinent. After partition about 80% of the jute growing areas went to East Pakistan (Bangladesh), while nearly 90% jute mills remained in India. In 1959, the international demand of jute products decreased substantially as a result of which 112 jute factories were closed down. At present there are only 79 jute producing mills in India. Most of these mills are along the Hugli river, especially to the north of Kolkata. Out of total 79 jute mills, 62 are located in West Bengal, 3 each in Bihar and UP, seven in Andhra Pradesh and one each in Assam, Odisha, Tripura and Chhattisgarh.

The jute mills are integrated units consisting of both spinning and weaving units. The main products of jute industry are gunny bags, canvas, pack-sheets, cotton-jute, paper-lined Hessians, Hessian cloth, carpets, carpet-backings, rugs, cordage, and twines. The

maximum installed capacity in Jute mills is to be more than 2.64 million tonnes per annum.

Jute industry is mainly a raw material based industry. Most of the jute mills are in the jute producing areas of the country. West Bengal alone accounts for 85% of the total jute production of the country **(Fig. 11.1) and (Fig. 11.4).** The high concentration of Jute mills in West Bengal **(Fig. 11.5)** is because of the following factors:

1. **Availability of raw material:** Jute cultivation needs highly productive, well-drained soils and hot and humid climate. These conditions are ideally available in many tracts of West Bengal and Lower Assam.
2. Cheap and skilled labour is available in West Bengal, one of the most densely populated state of the country.
3. Cheap water transport through the Hugli river.
4. Availability of coal from the Raniganj coal-mines.
5. Export facility through the port of Kolkata and Haldia.

In addition to West Bengal, jute mills are also located in Eluru, Guntur, Ongole and Vishakhapatnam (Andhra Pradesh); Darbhanga, Gaya, Katihar and Samastipur (Bihar); Kanpur (Uttar Pradesh); and Raigarh (Chhattisgarh). The jute mills outside the Hugli belt are generally smaller in size. The total production from these 25 mills is only about 3% of the total production of jute goods.

Jute industry in India is essentially export-oriented. India stands second after Bangladesh in the export of jute and jute products. Jute goods are exported to the USA (30%), Russia (25%), UAE (10%), and UK, and Germany (2% each). Among the other importers of Indian jute goods, Argentina, Australia, Russia, Cuba, Indonesia, Japan, Myanmar, Malaysia, Singapore, Sudan, and Thailand are important.

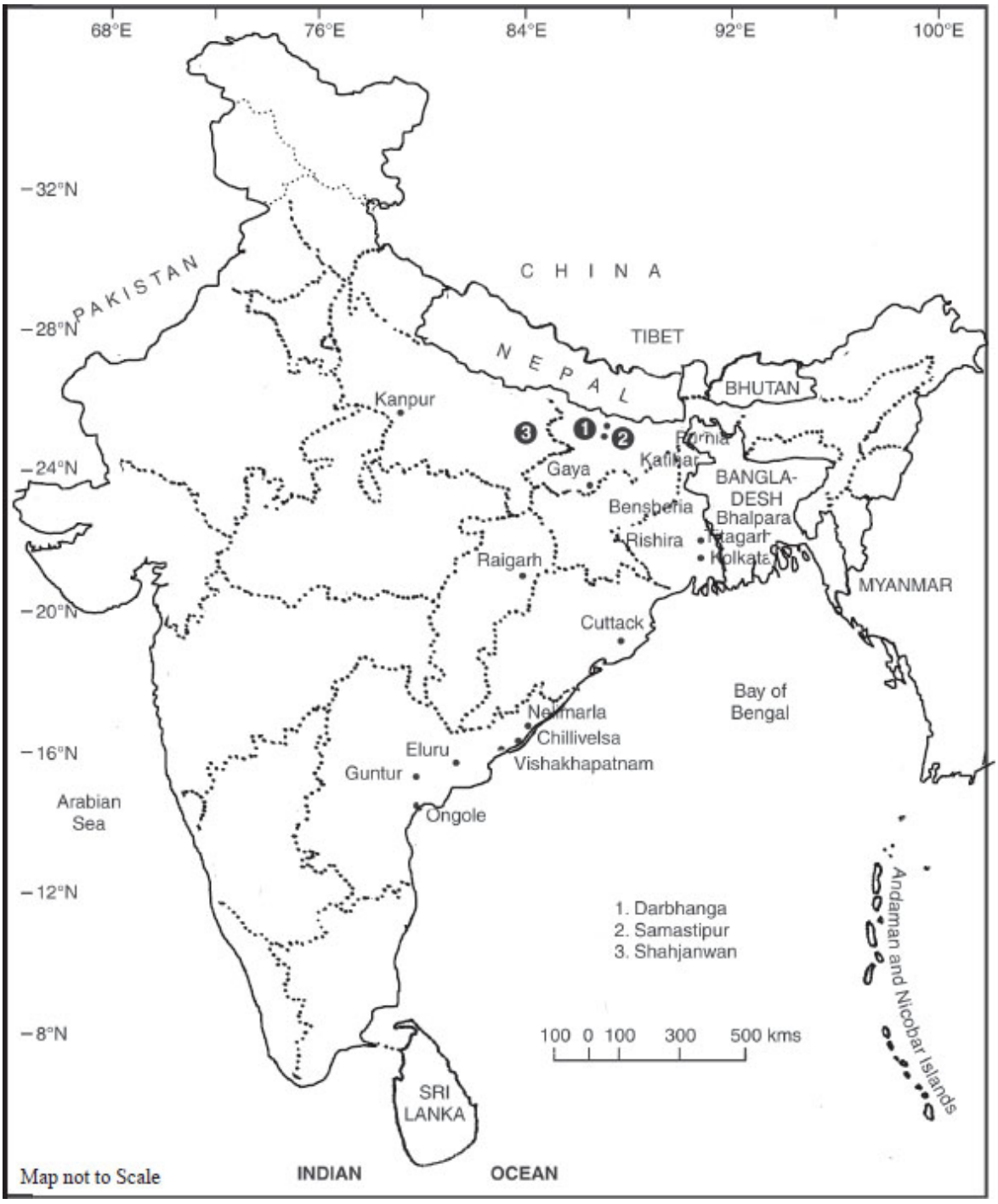


Fig. 11.4 Jute Industrial Centres

India is the largest producer of jute in the world with an average production of about 80 lakh bales of raw jute annually. The government supports the jute growers through MSP operation by the

jute Corporation of India and also through direct purchase of jute sacking at around Rs 6000 crores annually to pack the foodgrains.

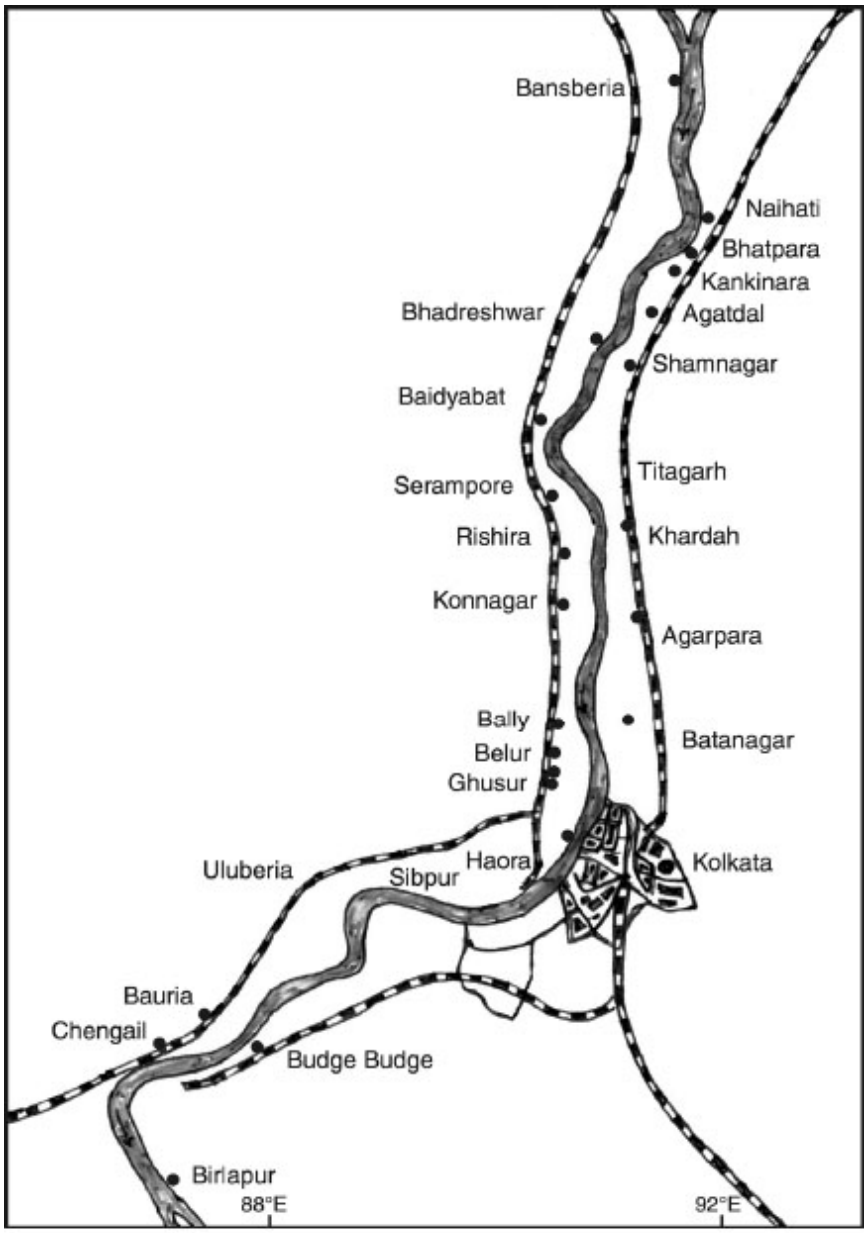


Fig. 11.5 Hugli Valley Jute Mills

As an initiative of e-governance, a software platform "*Jute-SMART*" (Jute Sacking Supply Management & Requisition Tool) was implemented for procurement of jute sacking from 2016. Through this

software, different state government agencies are procuring jute bags of around 27.0 to 30.0 lakh bales (68.0% of the total production of jute goods) annually valued at about Rs. 7,000 crores.

Problems of the Jute Textile Industry

The jute industry of India is confronted with many problems. Some of the important problems of jute industry are given here:

1. Shortage of Raw Material

Despite expansion of jute growing area and intensification of its cultivation, India is not self-sufficient in the supply of raw material. To meet the growing need of the industry, raw material is imported from Bangladesh, Brazil, and Philippines. There is a need of Golden Fibre Revolution in the country.

2. Obsolete Machinery

Most of the machinery in jute mills is more than 25 years old. These machineries are outdated and lead to low production.

3. International Competition

India is facing a tough competition from Bangladesh, Brazil, Japan, Philippines, and South Korea.

4. High Prices

The high price of raw material is making the jute products expensive. Consequently, a number of jute mills are becoming sick units.

5. Decrease in the Demand of Jute Products

The overall demand of jute products is steadily decreasing in the international market.

6. Strikes and Lock-outs

The trade unions frequently go on strike which hamper the routine production of the jute goods.

7. Competition from Substitutes

Jute industry is facing a tough competition from synthetic bags.

WOOLLEN TEXTILES

Woollen textile is one of the oldest industries of India. During the ancient and medieval periods, woollen clothes were manufactured at the cottage industry level. The modern woollen textile industry started with the establishment of *Lal Imli* at Kanpur in 1876. It was followed by the setting up of *Dhariwal* at Punjab in 1881. Subsequently woollen mills were established in Mumbai (1882) and Bangalore (1886). The industry made tremendous progress after Independence. At present there are 625 big and small woollen textile factories in the country.

The main concentration of woollen textile industry is found in Punjab, Maharashtra, and Uttar Pradesh. Other states which are producing woollen goods are Gujarat, Jammu and Kashmir, Karnataka, and West Bengal **(Fig. 11.6)**.

In Punjab, there are more than 260 small and big mills which are located mainly in Amritsar, Dhariwal, Kharar, and Ludhiana. In Maharashtra, Mumbai is the leading producer having 45 woollen factories. In Uttar Pradesh, woollen factories are located at Kanpur, Agra, Mirzapur, Modinagar, Shahjahanpur, and Varanasi. In Gujarat, Ahmedabad, Jamnagar, Kalol, and Vadodra are the main producers of woollen goods. In Haryana, Bahadurgarh, Faridabad, Gurgaon, and Panipat are the centres of woollen textile production. In Rajasthan, Alwar, Beawar, Bikaner, Bhilwara Jaipur, Jodhpur, Sikar and Nagaur are the important woollen textile centres. Bangalore, Kolkata, Haora, Srinagar, Kullu (Himachal Pradesh), Salem and Chennai are the other woollen textile centres. This industry provides employment to 27 lakh workers in the wide spectrum of activities (India 2012, p. 726).

Fig. 11.6 Woolen Textile

India is not self sufficient in quality wool production. Good quality wool is, however, imported from Australia. The important items of export are blankets, caps, carpets, cardigans, druggets, gabbas, hosiery, jerseys, mufflers, *Namdas*, pullovers, shawls, and socks. These goods are exported to the USA, UK, Australia, Belgium, Canada, Denmark, France, Germany, the Netherlands, Russia, South West Asian, and African countries.

Problems of the Woollen Textile Industry

The woollen textile industry is facing a number of problems regarding:

(i) Shortage of quantity and quality of wool;
(ii) Obsolete machinery;
(iii) Competition with more advanced countries;
(iv) Competition with synthetic fibres;
(v) Shortage of power;
(vi) Low quality of goods;
(vii) Lack of market; and
(viii) Strikes by the workers.

SILK TEXTILE

India's position in silk production is number two in the world after China, contributing 18% of the total silk production of the world. India has the monopoly in the production of '***muga***' silk, produced in Assam and Bihar. Silk textile was essentially a household industry in the early stage of its development. The Mughals were very much fond of silk clothes. The cotton goods used to be exported to the countries of south-west Asia and Europe. ***The first silk mill was, however, located at Haora by the East India Company in 1832***. The industry made tremendous progress after Independence. In 2013–14, the silk production in the country increased to 26, 480 million tonnes (India–2016, p. 555).

Distribution

The states of Andhra Pradesh, Assam, Bihar, Jammu and Kashmir, Karnataka, Tamil Nadu, Uttar Pradesh, and West Bengal are the leading producers of silk textile goods. The silk manufacturing centres have been shown in (**Fig. 11.7)**.

The state of Karnataka is the largest producer of silk textile. Bangalore, Belgaum, Gokak, Kolar, Mandya, Mysore, and Tumkur are the main producing centres. Channapatna and Mysore are the main centres of silk textile in the state. Silk yarn from these centres is exported to Arani, Dharmavaram, Kancheepuram, Kumbakonam, Surat, and Varanasi.

In West Bengal, the main centres of silk textile are Basawa, Bishenpur, Chak-Islampur, Kolkata, and Madhu-Ghat (Maldah District). In Andhra Pradesh, silk textile is concentrated in Anantapur, Chittor, Karimnagar, Vishakhapatnam, and Warangal. In Bihar, Bhagalpur; in Jharkhand, Hazaribagh and Ranchi; in Assam, Barpeta, Goalpara, Kamrup, Nalbari, and Naogaon are known for silk textile. In Tamil Nadu, Coimbatore, Dharmapuri, Kancheepuram, Nilgiri, Salem, and Tirunelveli; while in Madhya Pradesh, Balghat, Bastar, Bilaspur, Raigarh, and Surguja are the silk textile centres. Silk textile is also concentrated in Anantnag, Baramulla, Doda, Jammu, Riasi, Srinagar, and Udhampur in Jammu and Kashmir.

India is one of the important exporters of silk textile. Silk and silk products are exported to the USA, UK, Kuwait, Russia, Oman, Saudi Arabia, Singapore, and UAE.

Governmental Initiatives for Textile Development:

The government has been implementing various policy initiatives and programmes for development of textiles and handicrafts, particularly for technology, infrastructure creation, skill development, including:

- Amended Technology Upgradation Funds Scheme (ATUFS)
- Power Tax India Scheme
- Scheme for Integrated Textile Parks
- SAMARTH-Scheme for Capacity Building in Textile Sector
- Silk Samagra-Integrated Silk Development Scheme
- North Eastern Region Textile Promotion Scheme (NERTPS)
- National Handicraft Development Programme (NHDP)
- Comprehensive Handicrafts Cluster Development Scheme (CHCDS)
- Outreach programmes for handloom weavers: Hastkala Sahyog Shivirs
- E-Dhaga App
- Bunkar Mitra helpline
- Weavers Mudra Scheme
- Hastkala Samvardhan Sahayata Yojna

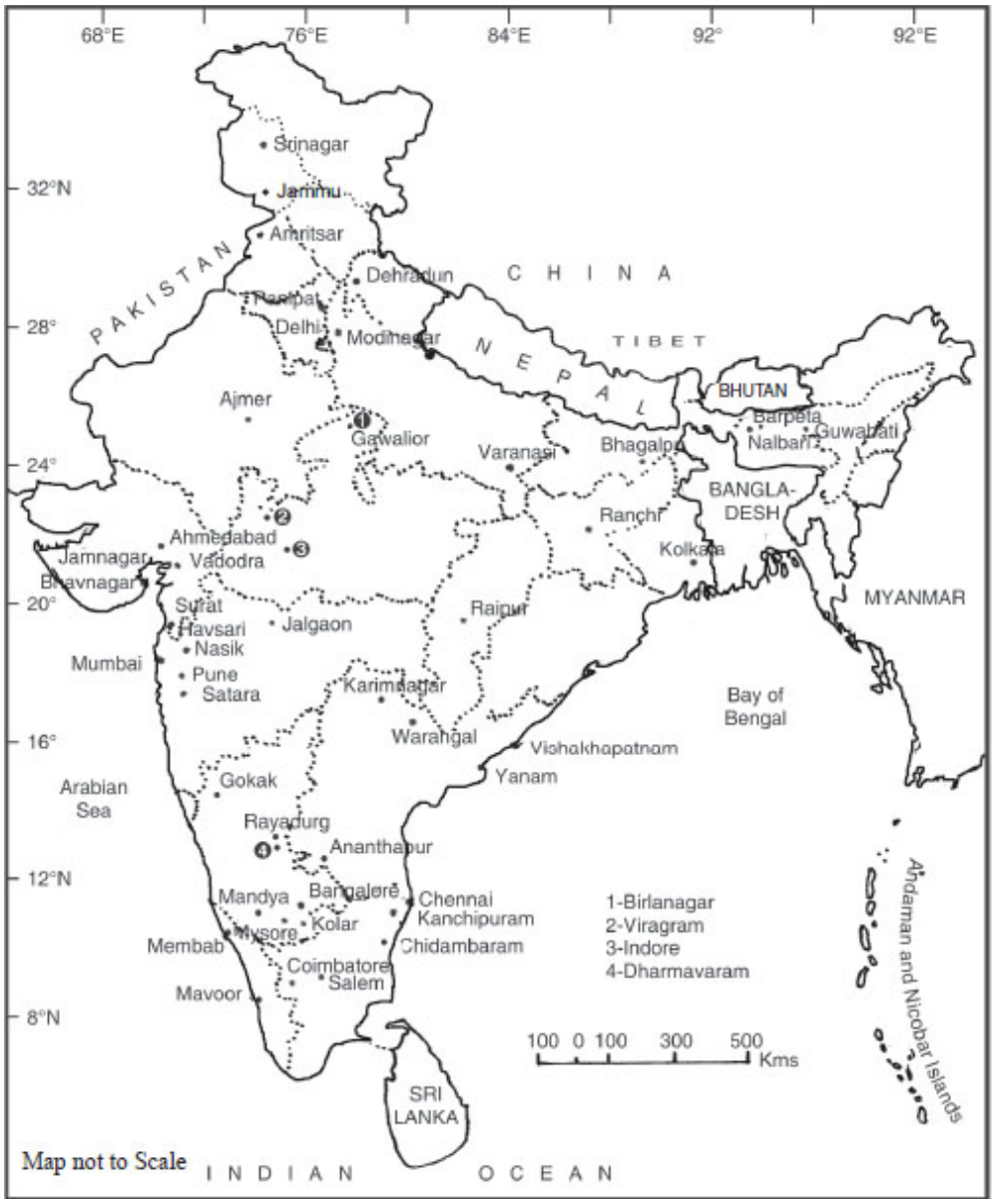


Fig. 11.7 Silk Textile

IRON AND STEEL INDUSTRY

The history of iron and steel industry in India is nearly 4000 years old. The famous iron pillar near Qutab Minar dates back to 350 AD. The first attempt to produce iron and steel on modern lines was made in 1830 at Porto Nova, near Chennai (Tamil Nadu). But it was not successful as the smelting used to be done with the help of charcoal. '***Pig-iron***' was produced in 1874 for the first time by the Bengal Iron

Works. The real progress in the iron and steel industry was made in 1907 when ***J.N. Tata*** established the smelting factory at Sakchi (the former name of Jamshedpur). Subsequently, the ***Indian Iron and Steel Company Ltd. (IISCO)*** was set up at ***Hirapur in 1918***, and the ***Visveswaraya Iron*** and ***steel Limited*** (VISL) at ***Bhadravati*** in 1923.

It was in the Second Five-Year Plan 1956–61, when tremendous progress was made by the iron and steel industry. It was during this plan, when integrated iron and steel plants were established at Bhilai, Durgapur, and Raurkela. The Bokaro Iron and Steel Plant was established with the help of Russian government in 1964. The (SAIL), was set up in January 1973. In the Fifth Five-Year Plan, it was resolved to set up four new iron and steel plants at Paradwip, Salem, Vijaynagar, and Vishakhapatnam. After China, Japan and U.S.A, India was the third largest producer of steel in the world with the production of 106.5 MT in 2018. India's crude steel and finished steel production increased to 103.13 MT and 104.98 MT in 2017–18.

Locational Factors

The raw material used in the iron and steel industry are iron ore, manganese, flux, limestone, fuel (coking coal), and fire-clay. It consumes heavy quantity of coal and iron-ore. Both these minerals are weight-losing. Consequently, it is located either at the site of coal mines or near the iron-ore deposits. All these raw materials including water, bauxite, dolomite, etc., and coal are found in the Chotanagpur Plateau. The states of Bihar, Chhattisgarh, Jharkhand, Karnataka, Madhya Pradesh, Odisha, Tamil Nadu, and West Bengal are rich in the raw material required for the smelting of iron-ore.

Iron and Steel Plants in India

There are 14 important iron and steel plants in India **(Fig. 11.8)**. In private sector, Tata Iron and Steel is the oldest plant. In recent decade a number of steel plants have been located at Kalinganagar (Odisha). During and after the Second Five Year Plan, plants were established under public sector. Plants were established under public sector. The steel plants under public sector are supervised by the

Steel Authority of India Limited (SAIL). A brief description of the main steel plants of India has been given in the following Section:

1. The Tata Iron and Steel Company (Jamshedpur)

TISCO Plant is located at the confluence of Subernrekha and Kharkai rivers about 240 km north-west
of Kolkata in the Singhbhum district of Jharkhand. It was established in 1907; the production of pig-iron started here in 1908 and that of steel in 1911.

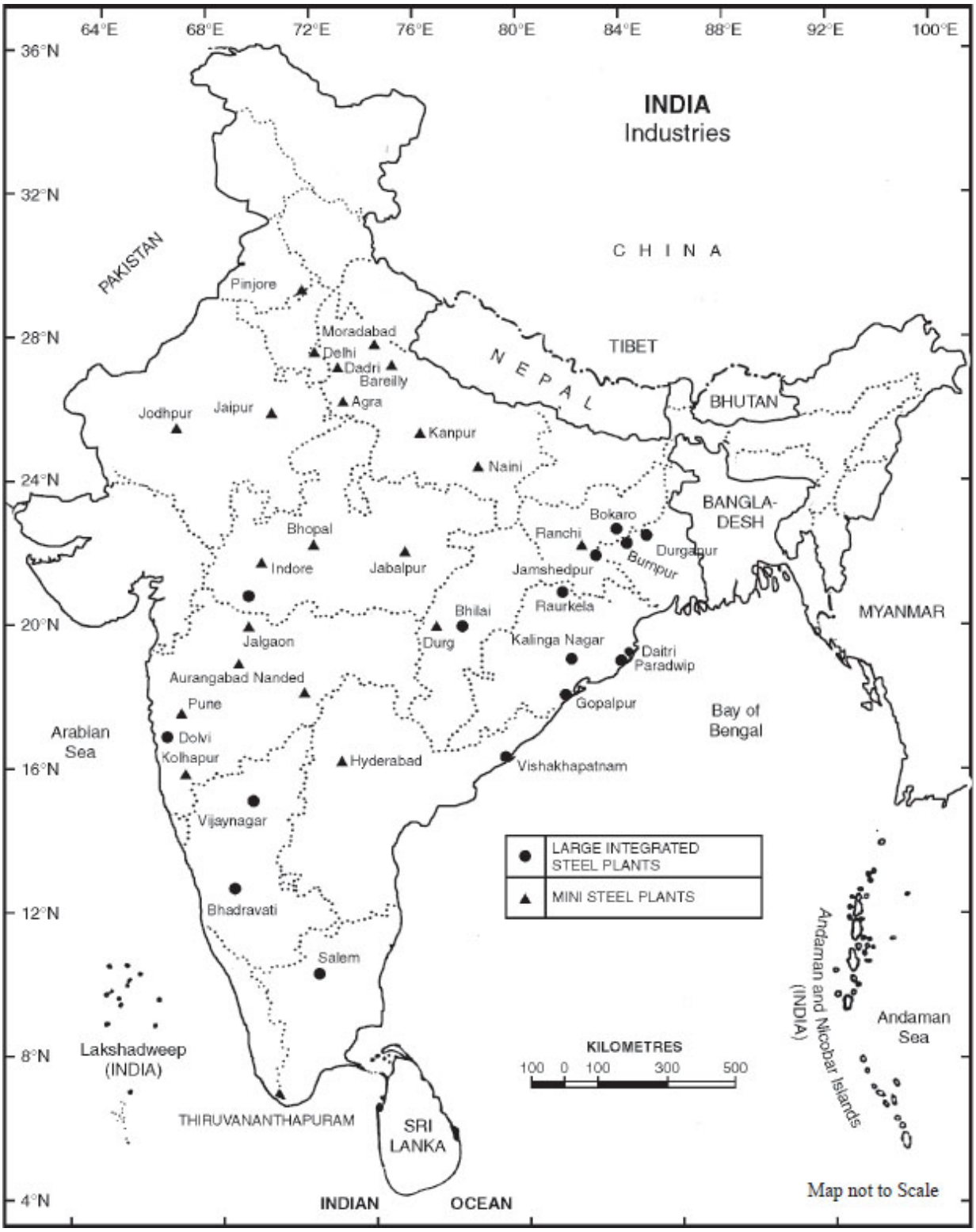
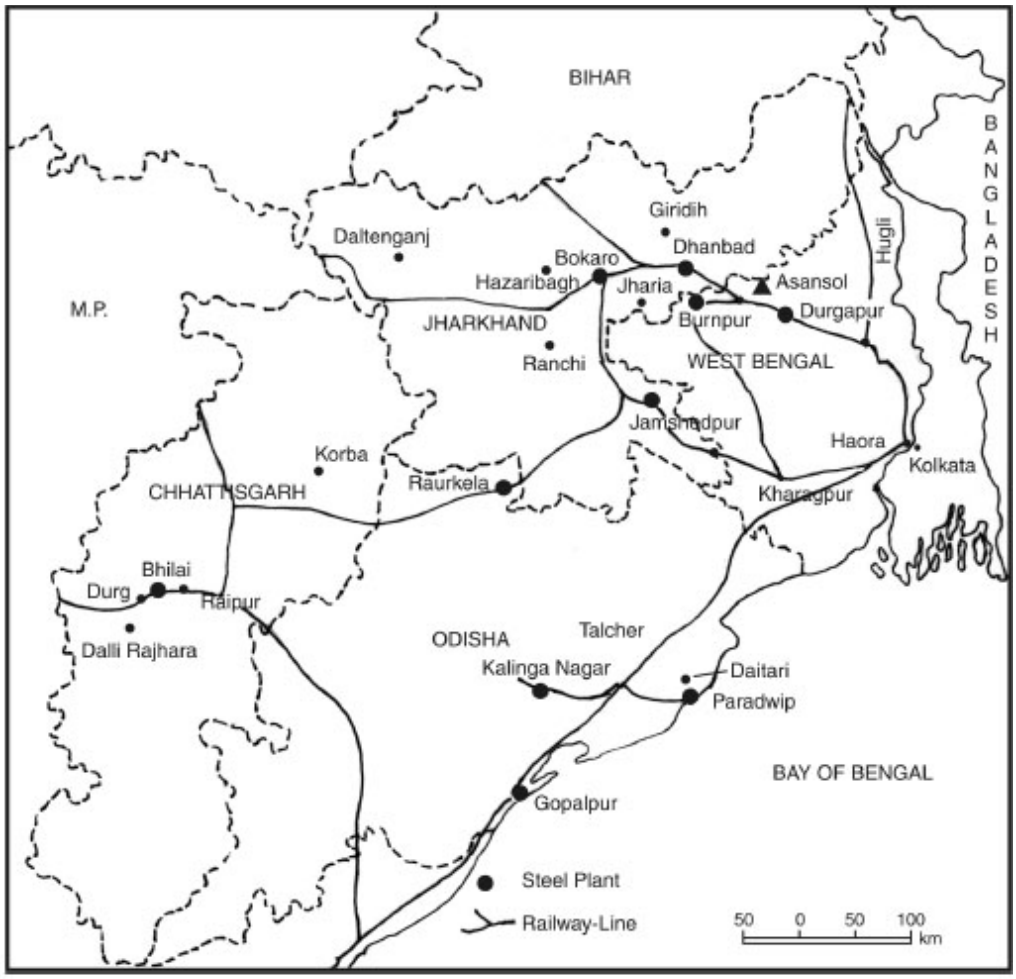


Fig. 11.8 Iron and Steel Industry

The Jamshedpur Steel Plant has an ideal location at which the transportation cost is the least **(Fig. 11.9)**. It obtains the supply of iron-ore from Badam-Pahar (Mayurbhanj-Odisha), Noamundi (Singhbhum District of Jharkhand); coking coal from Jharia and Bokaro, manganese from Keonjhar (Odisha); limestone and fire-clay from Sundargarh district (Odisha), and fresh water from the Subernrekha and Kharkai rivers. Being situated in the tribal belt, cheap labour is also available and the finished products can be exported through the port of Kolkata.

TISCO produces pig iron, high grade steel, acid steel for making railway wheels, axles, bars, bolts, corrugated-sheets, nails, nuts, rods, sheets, steel-casting, tinplates, tyres, and wires. Special alloy steel produced by the plant is used for making bullet-proof armour plates and for armour-piercing bullets.



Map not to Scale

Fig. 11.9 Chotanagpur Plateau Iron and Steel Plants

2. Indian Iron and Steel Company (IISCO), Burnpur

The Indian Iron and Steel Company, founded in 1918, and the Steel Corporation of Bengal, founded in 1927, were merged under the former name in 1952. It has three separate plants at Burnpur (about 5 km south-west of Asansol), Hirpur (about 6 km south of Asansol) and Kulti (about 16 km west of Asansol). The management of these steel plants was taken over by the government in 1972.

These steel plants obtain iron ore from their mines at Goa. The coal is obtained from Ramnagar mines of Jharia, which is only about 3 km from Kulti. Manganese is obtained from Bihar, Jharkhand, Madhya Pradesh, and Odisha; fire-clay from Singhbhum. Moreover, cheap

hydro-electricity is available from the Damodar Valley Corporation (DVC) and water from the Damodar river. Cheap labour is available from within the tribal belt of the region and Bihar, while the seaport of Kolkata helps in the export of the finished product.

3. The Bhadravati Iron and Steel Plant (VISL)

The Visveswaraya Iron and Steel Limited at Bhadravati, formerly known as **Mysore Iron and Steel Limited** was **established in 1823**. The plant was taken over by the **Central Government in 1962**. It obtains **iron ore** from **Kudermukh, Baba Budan Hills** (Chikmaglur district of Karnataka). Manganese, limestone, dolomite, and fire-clay are also available within a distance of 50 km. Electricity is available from the **Jog and Shravati Power Projects**.

This plant is one of the major producers of alloy and special steel in the country. It also produces casting iron, ferro-silicon, mild-steel and, spun-pipes.

4. Bhilai Iron and Steel Plant (HSL)

The Bhilai Iron and Steel Plant was established with the technical co-operation of Russain Government **(erstwhile Soviet Union)** in the **Durg district of Chhattisgarh in 1959**. It obtains its iron ore from the **Dhalli-Rajhara mines** about 95 km south of the plant, coal from **Korba (Chhattisgarh)**, **Bokaro and Jharia (Jharkhand)**, **manganese from Balaghat** and **Bhandara**, **limestone from Nandani** mines only 18 km from Bhilai, water from Tandula Canal and Reservoir, electricity from the Korba Thermal Power Station. As it is located in the tribal belt of Chhattisgarh, cheap labour is locally available. The finished products are exported through the port of Vishakhapatnam.

Bhilai steel plant specialises in the production of pig-iron, crude-steel, and plates for ship-building industry. The plant also produces byproducts like ammonium sulphate, benzol, coal-tar, and sulphate-acid.

5. Raurkela Iron and Steel Plant (Hindustan Steel Limited)

The Raurkela Steel Plant is located in the Sundargarh district of Odisha along the Kolkata-Nagpur railway line. It was built with the

technical co-operation from the **German firm**, **Krupps** and **Demang in 1959**. The plant obtains its iron-ore from Mayurbhanj; coal from Bokaro, Jharia, Talcher, and Korba; manganese from Sundargarh; water from the Sankha and Koel rivers (tributaries of Brahmani); and hydro-power from the Hirakud Dam. Cheap labour is available from the Jharkhand and densely populated Bihar state.

It specialises in the production of flat products. The main products of this steel plant are cold-rolled-sheets, hot-rolled-sheets, galvanised sheets and electrical steel plates. The plant also releases large quantity of nitrogen as by-product of fertilisers and various chemicals like anthracite-oil, benzole, crude-anthracite, crude-phenol, naphtha, naphthalene, and wash-oil.

6. Durgapur Iron and Steel Plant (HSL)

The **Durgapur Iron and Steel Plant** was established with the help of **British companies in 1956**; production started in 1962. The city of Durgapur is located along the Damodar River. It obtains iron-ore from Singhbhum (Jharkhand), and Kendujhar (Odisha), coal from Raniganj, manganese from Balaghat (M.P.), and water from the Damodar river. It produces ingot steel.

7. Bokaro, Bharat Steel Limited (BSL)

The Bokaro Steel Plant was located with the help of the Soviet collaboration in 1964. It obtains its iron ore from Keonjhar district; coal from Bokaro, Jharia, and Kargali coal mines; lime from Daltonganj in Palamu district; dolomite from Bilaspur district; and water from the Tenu Dam across the Damodar river.

Bakaro is essentially a flat product plant, and the hot and cold rolling mills are its main production units. Its sludge and slag are being used in making fertilisers at Sindri.

8. Salem Steel Plant

This steel plant was **commissioned in 1982**. It obtains iron-ore from the neighbouring areas, manganese, dolomite and limestone are also available within a distance of 60 km. Cheap power and labour and enormous market are the added advantage. It produces iron and steel of special grade.

9. Vijayanagar Steel Plant

This steel plant is located near Hosepet in the Bellary district of Karnataka. It obtains iron-ore from Hosepet; coal from **Kanhan valley (Chhattisgarh)** and Singareni (Andhra Pradesh); limestone and dolomite are also available within a distance of 150 km; water from the Tungbhadra Reservoir (about 35 km); and cheap hydel-power from the Tungbhadra Project. In this plant steel is produced with the Corex process which makes use of non-coking coal.

10. Vishakhapatnam Steel Plant

The foundation of this steel plant was laid in 1971 and the actual production was commissioned in 1992. This steel plant was established by the Rashtriya Ispat Nigam Limited (National Steel Corporation Ltd.). This is the oldest steel plant of India which has a coastal location. It obtains iron-ore from Bailadila (Chhattisgarh); coal from Bokaro, Raniganj, and Jharia; limestone and dolomite from Bastar (Chhattisgarh), Madhya Pradesh, and Odisha. It specialises in the production of steel and the quality of its steel can be compared to global standards.

11. Daitari Steel Plant

A steel plant has been located at Daitari near Paradwip in Odisha. Initially it was to be established in collaboration with the British and South Korean companies. Its responsibility has, however, been given to the Tata Group. It has a capacity of 2.6 million tonnes.

12. Dolvi Steel Plant

A new steel plant is being set up by the Ispat Industries Limited at Dolvi in the Ratnagiri district of Maharashtra. This steel plant is equipped with the latest technology in steel manufacturing. It requires less space, less energy, high labour productivity, and less cost of production. The plant is capable of producing strips as thin as 1.00 mm. Its annual capacity is 3 million tonnes.

13. Kalinganagar Steel Plant

In the Jaipur District of Odisha, a new industrial hub has been developed in recent years. The main steel plants are that of Tata

group, Jindal, Maithan Group, Dinbandhu, J.K. Spat, POSCO, etc. in Kalinga industrial area. This steel plant has been undertaken by the Tata Steel Company. The first phase of the project was completed in 2005.

14. Pasco Steel, Paradwip

The Pohang Steel Company of South Korea has agreed with the Odisha Government to establish an iron and steel plant at Paradwip. The project is likely to be commissioned in 2016. It is one of the biggest foreign direct investment (FDI) in India. When complete, it will require over 600 million tonnes of iron-ore for the production of iron and steel.

15. Mini-Steel Plants

In addition to these, there are more than 230 mini-steel plants with a capacity of 10,000 tonnes to 5 lakh tonnes. The main factors responsible for the growth and establishment of mini-steel plants are:

(i) Heavy demand for iron and steel;
(ii) Lower cost of production;
(iii) Controlled price of the steel;
(iv) Easy availability of scrap at lower prices from home and abroad;
(v) Lower capital investment; and
(vi) Shorter gestation period.

While (**Table 11.3)** gives the production of pig-iron and steel. It may be observed that the production of finished was only about one million tonnes in 1950–51 which rose to 100.74 million tonnes in 2016–17.

Table 11.3

Trends in the Production of Iron and Steel in India (in million tonnes)

Year	Pig Iron	Finished Steel
1950–51	1.69	1.04
1960–61	4.31	2.39

1970–71	6.99	4.64
1980–81	9.55	6.82
1990–91	12.15	13.53
2000–01	03.39	29.27
2005–06	04.25	32.50
2013–14	07.950	87.67
2014–15	9.69	92.16
2015–16	9.22	90.98
2016–17	9.39	100.74

Source: Joint Plant Committes www.Steal.gov.in.

National Steel Policy–2017

The policy aims to enhance domestic steel consumption and transform Indian steel industry into a technologically advanced and globally competitive steel industry.

Its key Features are enlisted below:

1. The policy aims to provide support to private manufacturers and MSME steel producers.
2. It promotes the development of steel manufacturing capabilities.
3. It facilitates the foreign investment and enhance the domestic steel demand.
4. The policy promotes the cost efficient production.
5. It also aims to ensure at availability of iron ore, coking coal, natural gas at competitive prices.
6. It targets a production of 255 MT and robust finished steel per capita consumption of 158 kg by 2030-31.

PROBLEMS OF IRON AND STEEL INDUSTRY

1. Heavy Investment

The establishment of iron and steel industry requires huge amount of capital. Moreover, the plant has a long gestation period. Construction of iron and steel plants is a difficult proposition, and therefore, most of the steel plants are in the public sector.

2. Obsolete Technology

The iron and steel plants established during the Second Five-Year Plan are not working to the full capacity as the machinery is outdated. Moreover, the high cost of coking coal has affected the industry adversely.

3. Inefficient Public Sector

Most of the iron and steel plants in public sector are incurring heavy losses mainly due to poor management and under-utilisation of capacity.

4. Controlled Prices

The government has fixed price for iron and steel which leaves very little margin of profit for the manufacturers.

5. Sickness of Mini-Steel Plants

Due to the inadequate supply of power and sharp increase in the raw materials, many of the small iron and steel plants are either experiencing sickness or are being closed down.

6. Inadequate Supply of Coking Coal

India's coking coal deposits are confined largely to the Raniganj and Jharia coal mines. These coal mines have become fairly deep and the cost of product of coal has gone up. Consequently, the input cost of energy is going up, affecting the output and margin of profit adversely.

7. Competition in the International Market

In the international markets, it is becoming increasingly difficult to compete with the steel of Australia, Brazil, France, Germany, Japan, Canada, Sweden, UK, and the USA.

Aluminium Industry

Aluminium is the second most important metallurgical industry of the country. It plays a very vital role in the overall industrial development of the country. Its elasticity and good conductivity of electricity and heat, and capacity to be moulded into any desired shape has made it a universally accepted metal. It is widely used in the generation and

distribution of electricity, manufacturing of aeroplanes, railway coaches, defence and nuclear accessories, utensils, packing, and for making coins. It is a cheaper substitute of steel, copper, zinc, lead, etc., in a large number of industries. In India, Aluminium is used in power (48%) automobiles (15%), contruction (13%), packaging (8%), industrial (7%). Consumption of aluminium has near by (4.3%).

Locational Factors: Availability of *bauxite* (raw material) and hydro-electricity are the basic requirements for the establishment of aluminium industry. For the production of one tonne of aluminium about six tonnes of bauxite is required. Hydel power is used in this metallurgy which is relatively cheap and clean.

About 30 to 40% of the production cost of aluminium is accounted for electricity alone. Bihar, Chhattisgarh, Jharkhand, Madhya Pradesh, Maharashtra, Odisha, Rajasthan, and Tamil Nadu are the major producers of bauxite in India.

Development: The aluminium industry was started in India during the Second World War at Alupuram (Alwaye) by the Aluminium Company in 1938. It was later on converted into a public sector company in 1944. Another company, namely Aluminium Corporation of India started production of alumina in 1942 at Jaykaynagar (West Bengal). At the time of Independence, there were only two plants in the country with a total installed capacity of 4000 tonnes of ingots. During the Second Five-Year Plan two new aluminium plants were established at Hirakud (INDAL) and Renukoot (HINDALCO). Another plant was established in the Third Five-Year Plan at Mettur (MALCO) in 1967. The INDAL established a new plant at Belgaum in 1970. Later on the Bharat Aluminium Plant was established at Korba. Consequently the annual production increased to more than 3 lakh tonnes in 1979. At present, there are seven major aluminium producing plants in the country. The location of aluminium plants have been shown in (**Fig. 11.10).** A brief account of the important aluminium plants has been given here:

Important Aluminum Plants

1. The Indian Aluminium Company Ltd. (INDAL), Hirakud

This company started production in 1938 as a private company and was converted into a public company in 1944. It is an integrated plant having three units at five different places for the production of alumina and aluminium sheets. The plants for the extraction of alumina from bauxite are located at Muri (Jharkhand), near the bauxite mines. Its three smelting units are located at Alupuram (Alwaye in Kerala), Hirakud (Odisha), and Belgaum (Karnataka). The rolling mill at Belur (West Bengal) manufactures aluminium sheets, rod, aluminium paste, electrical conductors, and domestic utensils. The plant gets bauxite from the Bagru Hills near Lohardaga, coal from Damodar valley, and hydro-electricity from Hirakud. The plant has an installed capacity of one lakh tonnes of aluminium ingots.

2. The Aluminium Corporation of India, Jaykaynagar (near Asansol)

The production from this plant was started in 1942. The plant gets bauxite from Ranchi (Jharkhand) and Unchera (MP). It has its own coal-mine, a thermal power plant and an alumina plant, a reduction plant, a sheet rolling plant and a utensils producing plant. It has a capacity of producing 90,000 tonnes of aluminium ingots annually.

3. The Hindustan Aluminium Corporation Ltd. (HINDALCO), Renukoot

This plant was set up at Renukoot, about 160 km south of Mirzapur, in 1958. It obtains bauxite from Lohardaga (Jharkhand and Amarkantak region of Madhya Pradesh), and power from the Rihand Dam. Its installed capacity is 1.26 lakh tonnes of ingots per annum, manufacturing mainly aluminium sheets and wires.

Fig. 11.10 Aluminium Industry

4. The Madras Aluminium Company Ltd. (MALCO), Mettur

This company set up its plant at Mettur near Salem in 1965. It obtains bauxite from the Shevaroy Hills and electricity from the Mettur Hydel Project. Its installed capacity is 25,000 tonnes of aluminium ingots.

5. The Bharat Aluminium Company Ltd. (BALCO), Korba

This is a public sector company which set up its **plant at Korba (Bilaspur District, Chhattisgarh) in 1965**. It obtains bauxite from the Amarkantak (Shahdol District of Madhya Pradesh) and electricity from the Korba Thermal Power Plant. The plant has an installed capacity of 2 lakh tonnes of ingots per year. The government has recently disinvested its share to a private company namely, Sterlite Industries, India (March 2001).

6. The National Aluminium Company Ltd. (NALCO), Koraput

Located at Koraput, it is the largest aluminium plant of the country. The company was incorporated in 1981. It obtains bauxite from the bauxite mines at Panchpatmali (district Koraput). It has an installed capacity of 1.6 million tonnes of ingots per year. There is an alumina refinery at Damanjodi (district Koraput) and alumina smelter at Angul. It obtains hydro-electricity from the Angul Power Plant and the port facilities from the Vishakhapatnam for export of alumina and import of caustic soda. It has a capacity of production of alumina of 8 lakh tonnes per annum. The Central Government has disinvested about 45% of NALCO's shares.

The Government approved the second phase expansion of NALCO's Integrated Aluminium Complex in 2004 at the outlay of INR 4,091.51 crore to be completed by 2008. With this expansion, capacity of Bauxite Mines Refinery, Smelter, and Captive Power Plant will increase from 4.8 million tonnes per year to 6.3 million tonnes per year.

The aluminium industry has made a commendable progress during the planned period. It may be seen from (**Table 11.4)** that the total production of aluminium was only 4000 tonnes in 1950–51 which rose to 585 thousand tonnes in 2005–06 and in 2017–18 it has been 2800 KT. Despite appreciable progress in the production of aluminum, India is not self-sufficient in matter of aluminium.

Table 11.4

India: Trends of Aluminium Production (000 tonnes)

Year	Production
1950–51	4.0
1960–61	18.5
1970–71	169.0
1980–81	199.00
1990–91	450.00
2000–01	560.00
2005–06	585.00
2016–17	2425.00
2017–18	2800.00

Trade

India is almost self-sufficient in the matter of aluminium. Except some high quality aluminium which it imports from the foreign countries. The demand for good quality of aluminium is on the increase and consequently, India is importing aluminium and its products from the developed countries.

Problems

The major problem of the aluminium industry is international competition. India has the best grade of bauxite but unfortunately, the quality of products are not at par to that of the countries like Australia, Canada, France, Germany, Japan, UK, and the USA.

Non-availability of power at a cheaper rate, strikes, and labour unrests are the other problems this industry is facing.

AUTOMOBILE INDUSTRY

Automobile indusry in india is one of the largest in the world and one of the fastest growing globally. India manufactures about 12 million vehicles (including 2 wheeled, 3 wheeled, and 4 wheeled) and exports more than 1.5 million every year. By 2050, India is expected to top the world in cars number with 600 million on the nation’s roads.

The automobile industry of India made a steady progress after independence. It was in 1947 when the Premier Automobiles Ltd. was established at Kurla (Mumbai). In 1948, the Hindustan Motors

Ltd. Uttarapara was established near Kolkata. Within a short period of about fifty years, automobile industry in India has made a tremendous progress. At present, there are 17 manufacturers of passenger cars, 9 manufacturers of commercial vehicles, 16 producers of two and three wheelers and 14 producers of tractors, besides 5 manufacturers of engines. India's annual production was 29.08 million vehicles in FY 18.and it was 25.33 million in FY 17. A sharp growth of 14.8% has been reported. The industry is valued at about 7.1% of the Gross Domestic Product (GDP). A growth rate of 18–20% has occured in CVs, 8–10% in PVs and 17–19% in two and three wheelers.

The automobile industry was delicensed in July 1991 with the announcement of the New Industrial Policy. The passenger car was, however, delicensed in 1993. At present 100% FDI is permissible in automobile industry. The auto industry directly or indirectly employs 12.5 million people and contributes 5% of national GDP (India 2012, p. 704).

The automobile industry is mainly located near the iron and steel producing centres as steel is the basic raw material used in the industry. The automobile industry is also located near the markets and the seaports. Some of the automobile industries have been located in the under-developed areas of isolation and relative isolation **(Fig. 11.11)**.

The main automobile producing centres are: Chennai, Delhi, Gurugram (Gurgaon), Jabalpur, Jamshedpur, Kolkata, NOIDA (UP) and Mumbai. Motorcycles are produced in Faridabad and Mysore. Scooters are manufactured at Akurdi (near Pune), Lucknow, Odhav (near Ahmedabad), Panki (near Kanpur), and Satara in Maharashtra.

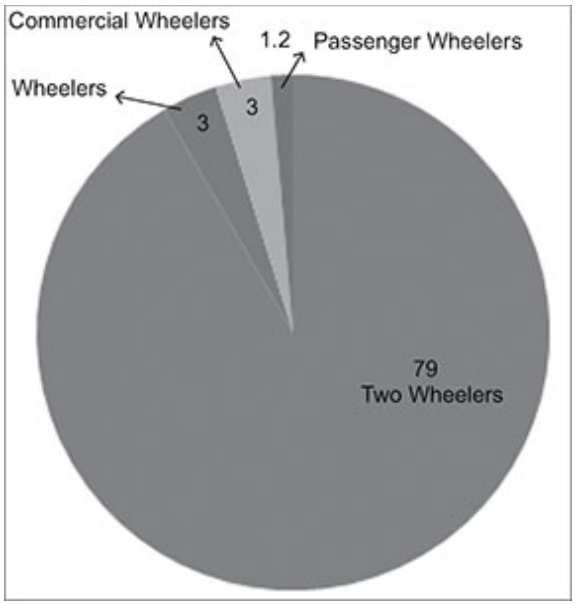


Fig. 11.11 Percantage above in total Production FY 2017 [*Source: IBEF; www.ibef.org*]

The commercial vehicles are produced by Tata Engineering and Locomotive Co. (TELCO). It produces over 70% of the total commercial vehicles. Its four plants are located at Hyderabad, Pithampur (Madhya Pradesh), Rupnagar (Punjab), Surajpur (NOIDA —Uttar Pradesh). The Premier Automobiles and Mahindra and Mahindra are located at Mumbai; Ashok Leyland Ltd. and Standard Motors at Chennai; Hindustan Motors Ltd. at Kolkata, and Bajaj Tempo Ltd. at Pune. Moreover, Shaktiman Trucks are manufactured under the Ministry of Defence and Nissan Jeeps at Jabalpur in collaboration with the Nissan Company of Japan.

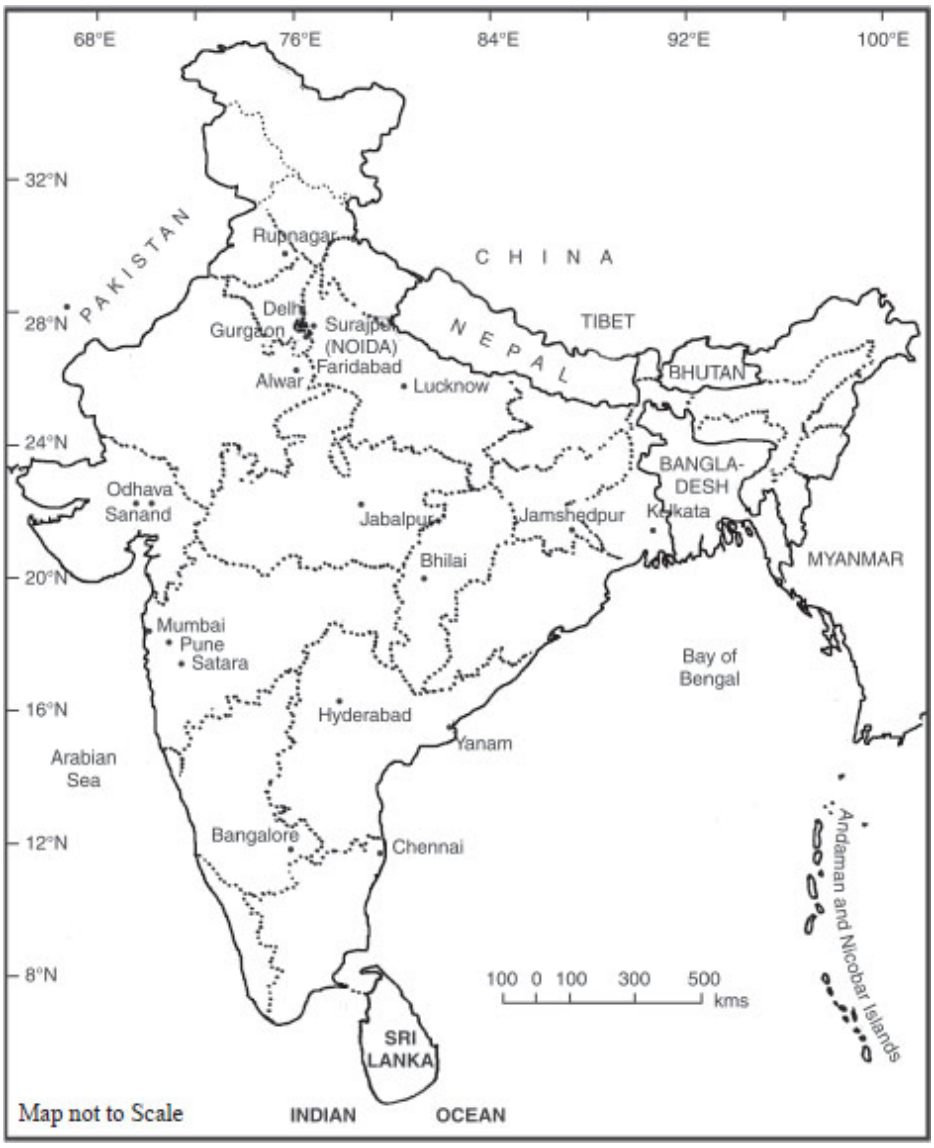


Fig. 11.12 Automobile Industry

Passenger Cars

The demand of passenger cars has increased tremendously during the last two decades

(Fig. 11.12). There are a number of companies producing passenger cars. Some of the important car manufacturers are: (i) Maruti Udyog Limited, Gurgaon (Haryana); (ii) Hindustan Motors

(Chennai and Kolkata); (iii) Premier Automobiles, (Mumbai); (iv) Standard Motor Products, Chennai; (v) Hyundai Motors India: Surajpur (NOIDA-Uttar Pradesh); (vi) Daewoo Motors India Ltd. near

Pune; (vii) Honda City (Uttar Pradesh); (viii) The collaboration between Mahindra and Ford has introduced the Ford Escort; (ix) Mercedes Benz of Germany in collaboration with Telco is producing expensive E220 and 250D; and (x) the Premier Automobiles in collaboration with Fiat-Ind-Auto Limited has introduced Fiat Uno car.

Jeeps

The entire production of jeeps is contributed by Mahindra and Mahindra, Mumbai. It has a capacity to produce 15,000 jeeps annually.

Two and Three Wheelers

There is an increasing demand of two and three wheelers all over the country. In fact, the two-wheeler industry has a great future. The two-wheeler industry comprises moped, scooters, and motorcycles. Mopeds are produced mainly by TVS-Suzuki, scooters by Bajaj Auto and LML at Mumbai, Pune, New Delhi, and Kanpur. Public sector units are located at Alwar, Bangalore, Hyderabad, Lucknow, and Satara. The motorcycle producing units are located at Chennai, Delhi, and Mysore.

Defence Vehicles

Vehicles for defence are produced by the '*Vehicles Factory*', *Jabalpur*. It was located in 1969 in collaboration with Nissan Company of Japan. Heavy Shaktiman trucks are also made with license from the MAN Company of Germany.

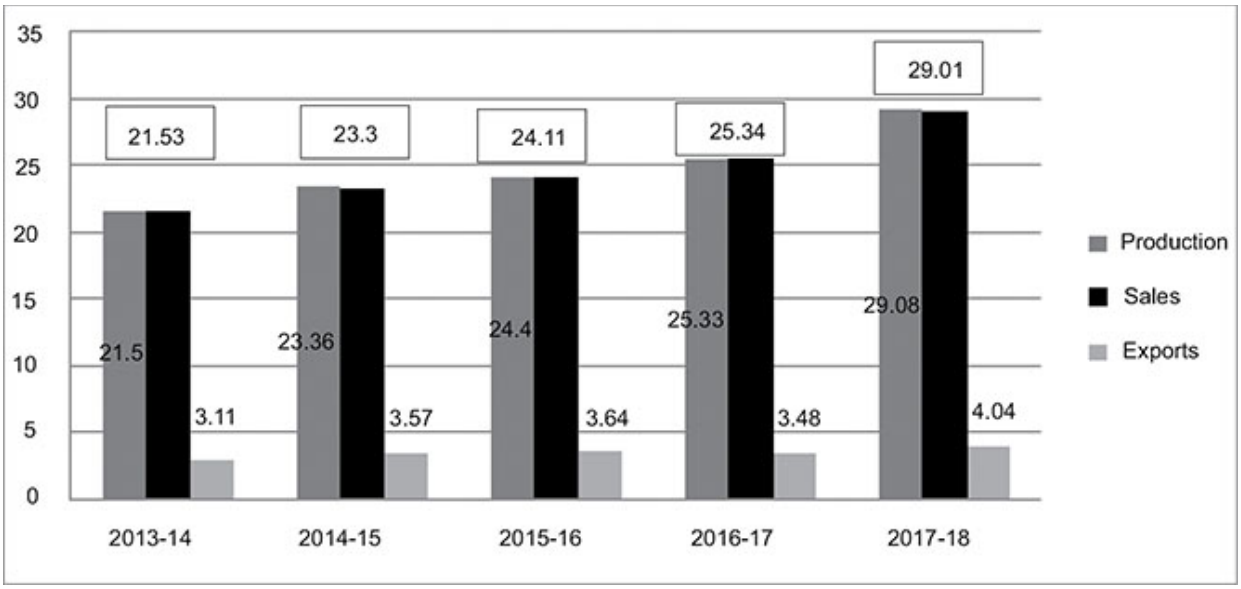


Fig. 11.13 Production, Sales and Exports of Automobiles (million units) *[Source: CMIE; Care Ratings Industry research, Automobile Update-Outlook FY 19]*

The financial year 2018 was an eventful year for the automobile industry . The following were the key features:

(i) Demonetization was announced in 2016.

(ii) Supreme court's BS-III vehicle ban became effective from April 1, 2017.

(iii) Goods and Service Tax was implemented from July1, 2018.

(iv) Continued ban on diesel cars by National Green Tribunal (NGT), September 2018.

(v) Cess increase in September 2017.

Automobile industry was delicensed in July 1991 with the announcement of New Industrial Policy. At present 100% FDI is permissible in this sector.

Automobile industry of India is now finding increasing recognition worldwide. In 2017–18; the number of vehicles exported was 4.04 million units as compared to 3.48 million units in 2016–17.

India exports some passenger cars to the countries of South West Asia, East Africa, South East Asian countries, East European countries, USA, and the UK.

Chemical Industry

Chemical is one of the basic industries. It is the fourth largest group of industries in the country after textiles, iron and steel, and engineering industries. It plays a vital role in the nation's economy. It supplies raw materials to several industries such as iron and steel, textiles, paper, synthetic fibers, rubber, plastics, paints, soaps, detergents, fertilisers, drugs-pharmaceutical, pesticides and dyestuff. chemicals are of two types: (i) heavy chemicals, and (ii) fine chemicals. Soda-ash, caustic soda and sulphuric acid are heavy chemicals. Fine chemicals generally refer to dyestuffs, pharmaceutical, and photographic chemicals and a large variety of products used in analytical and research work.

The beginning of the Indian chemical industry may be traced back to 1901 when the pharmaceutical plant was established near Kolkata. Since then there has been steady progress in the development of the industry. The industry recoded a tremendous progress after independence.

The main chemical industries include (i)Sulphuric Acid, (ii)Nitric Acid, (iii) Soda-Ash,
(iv) Chlorine, (v) Calcium carbide, (vi) Petrochemicals, (vii) Fertiliser Industry, (viii) Drugs and Pharmaceutical (ix) Paper and Pulp, (x)Pesticides, (xi) Dyestuffs, (xii)Paints and Varnishes,
(xiii) Leather industry, (xiv) Glass industry, etc. Some of the important chemical industries have been described in the following:

FERTILISER INDUSTRY

Fertiliser industry has made a tremendous progress after independence. The first fertiliser plant was established in Sindri India in 1906. The five decades of planning and development of fertiliser industry have brought India to the frontline of fertiliser producing countries. India today is the third largest producer of nitrogenous fertilisers in the world after China and the USA. The domestic production of urea in the year 2017–18 was about 24.02 MT which is

85% of the urea requirement of the country, while the per hectare consumption was about 130 kg per hectare.

The public sector has been playing a dominant role in the fertiliser industry. The first state-owned fertiliser unit was set up in 1951 at Sindri in Bihar (Jharkhand) which was followed by another plant at Nangal in Punjab. At present, there are, 57 fertiliser units manufacturing a wide range of nitrogenous and complex fertilisers, 29 units producing urea, and 9 units producing ammonium sulphate as a by-product. Besides there are 68 medium and small scale units in operation producing single superphosphate.

For a fertiliser industry, the basic raw materials are neptha, rock-phosphate, sulphur gypsum, and smelter gases. In case gas is not available, the plant may be operated with the help of coke and coke-oven-gas. In India, we have coal based fertiliser plants at Bhilai and Korba (Chhattisgarh), Durgapur (West Bengal), Ramgundam (Andhra Pradesh), Jamshedpur and Sindri (Jharkhand), and Rourkela and Talcher (Odisha). Neyveli is based on lignite while Hazira (Gujarat) and Thal (Maharashtra) use natural gas from the Bombay High. Most of the fertiliser plants are located close to the petroleum refineries. Plants near sea are based on imported material. Some of the fertiliser plants have, however, been located near the consumer centres. The maximum number of fertiliser plants are in the state of Gujarat and Tamil Nadu **(Fig. 11.14)**.

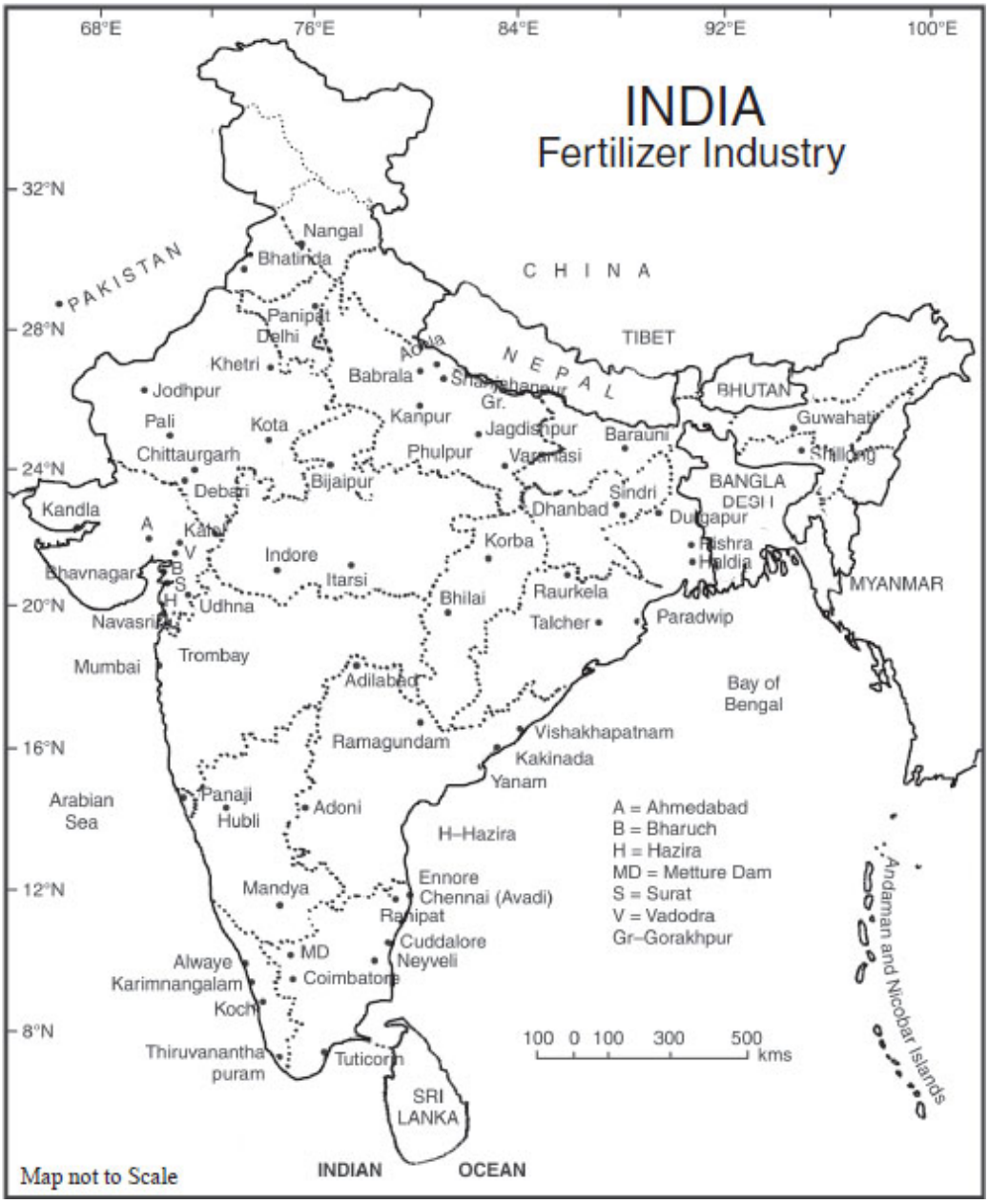


Fig. 11.14 Fertiliser Industry

Fertiliser Units in India

At present, 57 large fertiliser plants, 64 medium and small sized chemical fertiliser units in India. India ranks third at the global scenario Fertiliser industry contributes to 25% of GDP. The Fertiliser Corporation of India (FCI), the National Fertilser Limited, the Hindustan Fertiliser Corporation, the Rashtriya Chemicals and

Fertiliser Limited (RCF), the Fertilisers and Chemicals Travancore Limited (FACT), the Madras Fertiliser Limited, the Paradwip Phosphate Limited (PPL), and the Projects and Development Limited India are some of the important public sector undertakings.

Table 11.5

Main Fertiliser Producing Industries of India

Undertakings	Year of establishment	Location	Remark
The Fertiliser Corporation of India Limited (FCI)	1961	Sindri and Nangal (Punjab)	It has four units: Sindri (Jharkhand), Gorakhpur(UP), Talcher (Orissa), Ramagundam (Andhra Pradesh)
The National Fertilisers Limited	1974	Bhatinda, Nangal, Panipat, Bijaipur	Specialization is production of calcium ammonium nitrate
The Hindustan Fertiliser Corporation Limited (HFC)	1978	5 operation units at Barauni (Bihar), Durgapur (West Bengal) and 3 units at Namrup (Assam)	
The Rashtriya Chemicals and Fertilisers Limited (RCF)	1978	4 units at Trombay, 1 unit at Thal (Maharashtra)	
The Fertiliser and Chemicals Travancore		2 units at Kochi, 1 unit at Udhyogamandalam	
Madras Fertilsers Limited	1996	Chennai, Tamil Nadu	Set up in collaboration with National Iranian Oil Company

Undertakings	Year of establishment	Location	Remark
The Paradeep Phosphate Limited (PPL)	1981	Bhubaneswar, Odisha	First plant was commissioned in 1986 and second phase was completed in 1992 Produces phosphoric and sulphuric acid

It may be seen from the (**Fig. 11.15)** that the Indian fertiliser companies produced 32.4 MT of fertiliser during 2012–13 . The total availability was short of demand and was met through imports. The urea production increased to 22.6 MT during 2012–13 from 22.0 MT in FY 2011–12.

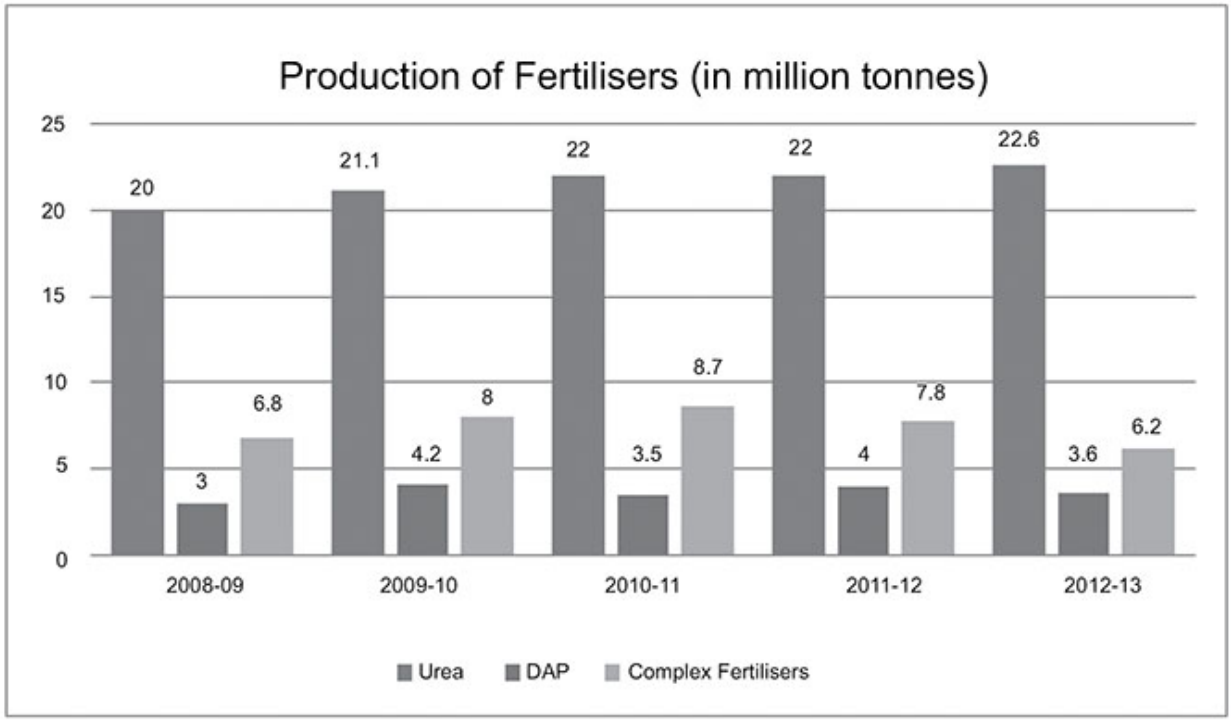


Fig. 11.15: Trends in the growth of Fertiliser Production in MT [*Source: krishijagran.com*]

Co-operative Sector

There are a number of fertiliser plants in the co-operative sector out of which, the Indian Farmers Fertiliser Co-operative Limited (IFFCO) is the most important. IFFCO has six production units one each at Aonla—two units (Bareilly), Kalol and Kandla (Gujarat), and Phulpur —two units (Allahabad). The KRIBHCO has a gas based urea-ammonia plant at Hazira in Gujarat.

India is not self sufficient in the matter of chemical fertilisers. About 50% of the country's requirement is met by import.

The consumption, indigenous production, and imports of fertilisers in terms of fertiliser nutrients (NPK) during the period 2000–01 and 2005–06 are given below in (**Table 11.6).**

Table 11.6

Consumption, Production, and Imports of Fertilisers (lakh tonnes)

Year	Consumption	Production	Imports
2000–01	167.02	147.04	20.91
2007–08	228.00	190.00	77.56

Source: India, 2010, p. 667.

Aonla: Situated in the district of Bareilly, the Aonla Chemical Fertiliser Plant was established by the Indian Farmers Fertilisers Co-operative Limited. The plant was commissioned in 1974. It meets the requirements of the farmers of the Rohilkhand Division of Uttar Pradesh.

Hazira: Situated in the state of Gujarat, Hazira is a modern chemical fertiliser plant. The smelting in this plant is done with the help of natural gas obtained from the Bombay High.

Kalol: Situated in the state of Gujarat, Kalol is one of the important petro-chemical complex of India. In addition to refinery it produces nitrogenous and phosphoric fertilisers. The natural gas for the production of fertilisers is obtained from the Gulf of Khambat.

Sindri: The first chemical fertiliser plant of India in public sector was established at Sindri in 1951 in the state of Jharkhand (erstwhile

Bihar). The chemical fertilisers are produced with the help of coking coal. It is the main supplier of chemical fertilisers to the states of Bihar, Jharkhand, and West Bengal.

PAPER INDUSTRY

Paper industry is one of the core industries. It is a unique product which helps in the preservation of information and propagation of noble ideas and thoughts. In fact, the paper industry has a vital role to play in socioeconomic development of the country. It has great social and cultural significance. The per capita consumption of paper is considered as a bench mark of a country's modernization.

Paper making in India as a cottage industry was started during the Medieval Period, but the first *paper mill was set up in 1832* at *Serampore (West Bengal)*. It, however, resulted in a failure. Subsequently paper mills were set up at *Ballyganj (Kolkata in 1870)*, *Lucknow (1879)*, *Titagarh* (1881), *Pune (1887)*, *and Raniganj (1891)*.

At present there are more than 700 paper mills producing 9 million tonnes of paper and paper-board and around 1.25 million tonnes of newsprint against an estimated operational capacity of 15.0 million tonnes of paper and paper-board and 1.44 million tonnes of newsprint (2014–15). The projected demand for paper, paper-board and newsprint is expected to touch 10.0 million tonnes by the year 2018. The per capita consumption of paper, which is the benchmark of modernisation of any country, stands at 5.5 kg for India, which is far below the global average of 50 kg.

The paper industry was delicensed and decontrolled in 1997. FDI upto 100% was permitted on automatic route. The Indian paper industry employs three lakh persons directly and ten lakh persons indirectly.

The paper and newsprint industry is highly fragmented with the installed capacity ranging from 10 tonnes to 800 tonnes per day. Indian paper industry can broadly be classified into three segments:

(i) Large integrated mills using bamboo and wood.

(ii) Medium mills using agri-residue and recycled fibre.

(iii) Small mills using waste paper/recycled fibres.

All the three sectors are contributing equally in the production of paper, paper-board, and newsprint in the country. The turnover of the industry is about INR 16,000 crore and it contributes INR 2500 crore to the national exchequer.

Paper industry is essentially a raw material based industry. The raw material for the paper industry is soft wood, bamboo, grasses, bagasse, rags, and waste paper. In India bamboo alone constitutes about 70% of the raw material for paper industry. Besides cellulosic raw material obtained from wood, paper industry needs chemicals like caustic soda, chlorine, soda-ash, sodium-sulphate, sulphuric acid, lime, and water. Bamboo is obtained from the states of Andhra Pradesh, Himachal Pradesh, Karnataka, Madhya Pradesh, Maharashtra, North East India, Odisha, Tamil Nadu, Uttar Pradesh, Uttarakhand, and West Bengal.

Distribution

The distribution of location of paper industry in different states of India has been shown in (**Fig. 11.16)**.

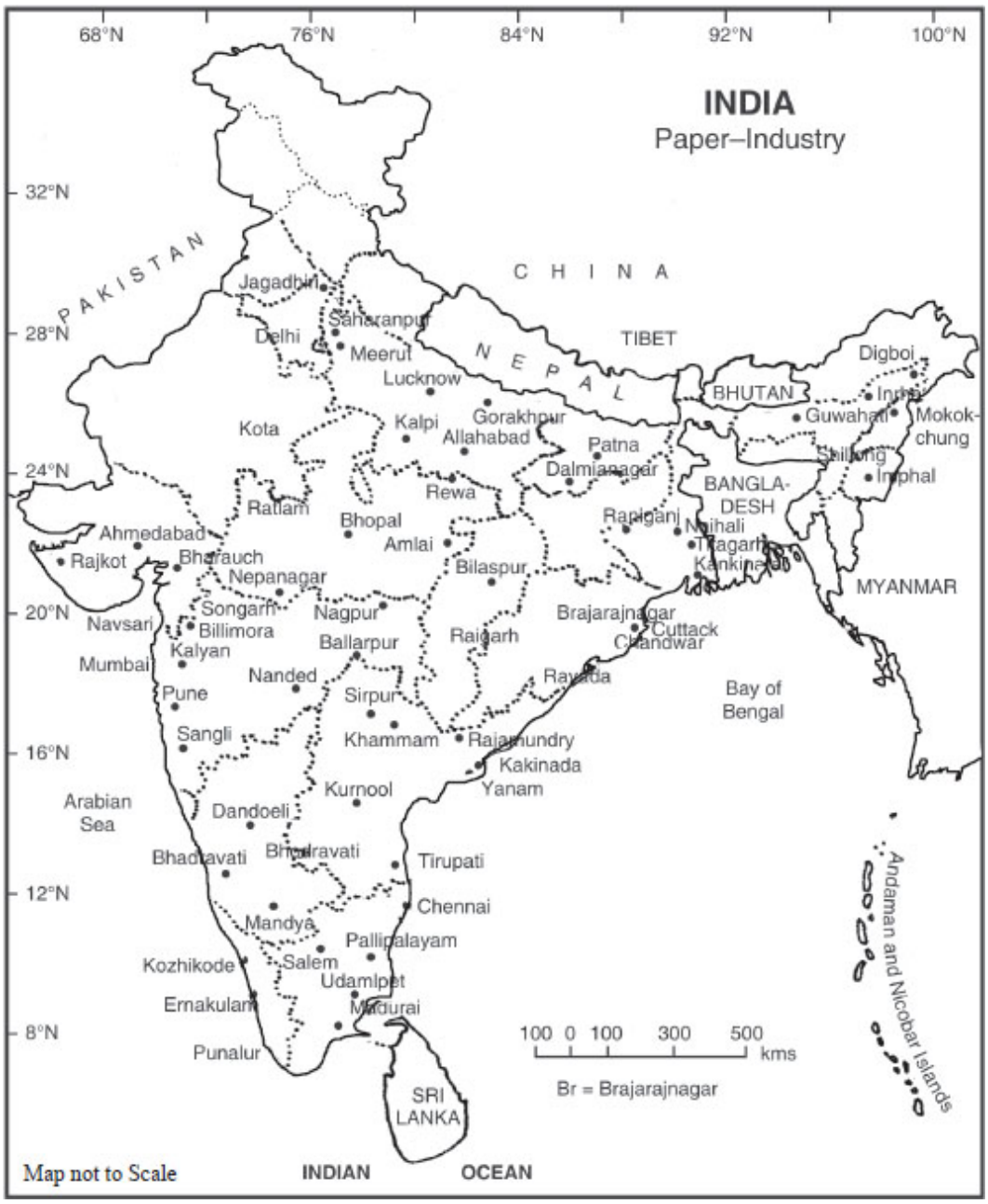


Fig. 11.16 Paper Industry

1. Andhra Pradesh and Telangana

These are the leading producers of paper contributing about 18% of the total production. There are 18 paper mills in the state. The paper mills of Bhadrachalam, Bodhan, Khammam, Kurnool, Rajahmundry, Sirpur, and Tirupati are important.

2. Maharashtra

The state of Maharashtra with 35 paper mills contributes about 14% of the total production. The paper industry is located in Chinchwad (Pune), Goregaon, Jalgaon, Kalyan, Kamptee, Khandala, Khopoli, Paravarnagar, Sangli, and Vikhroli. Sangli and Ballarpur are famous for newsprint. The industry uses a variety of raw materials like bamboo, bagasse, rags, rice bran, and imported pulp.

3. West Bengal

Contributing about 11% of the total production, West Bengal is the third largest producer of paper in the country. The main paper mills are located at Alambazaar, Bansberia, Baranagar, Chandrahati, Dum Dum, Ganganagar, Howrah, Kankinara, Kolkata, Naihati, Raniganj, Sheoraphuli,Titagarh, and Tribeni. Availability of raw material from the north-east, availability of coal and water, and cheap labour have helped in the location of paper industry in West Bengal.

4. Odisha

With over 8% of the total paper production of the country, Odisha is the fourth largest producer of paper in the country. The state has six paper mills in the state. The paper mills of Brajranagar, Chowdwar, and Rayagada are most important. Abundance of raw material and availability of coal are the main advantages which helped in the location of this industry in these areas.

5. Karnataka

The state of Karnataka contributes about 7% of the total production of paper in the country. Its major paper mills are at Bangalore, Bhadravati, Belagula, Dandeli, Mandya, Nanjanguda, and Ramanagaram.

6. Gujarat

Gujarat contributes about 7% of the total production of paper in the country. Its paper mills are located at Barejadi, Bavla, Gondal, Kalol, Khadki, Songarh, Utran, Utvada, Valsad, and Vapi.

7. Madhya Pradesh

There are 12 paper mills in Madhya Pradesh. These are located at Amlai (Shahdol), Bhopal, Dhemka, Indore, Nepanagar, Ratlam, and

Vidisha. Bamboo, salai-wood, eucalyptus, and sabai grass are the main raw materials used in the paper manufacturing.

8. Tamil Nadu

There are 16 paper mills in Tamil Nadu. The main paper mills are located at Chennai, Charanmahadevi, Pallipalayam,Tambaram, Salem, and Udamalpet. The main raw material is bamboo.

9. Uttar Pradesh

There are 48 paper mills in Uttar Pradesh contributing to about 5% of the total paper production of the country. The main paper mills are located at Budaun, Jagadhri, Lal-kuan, Lucknow, Mainpuri, Meerut, Modinagar, Naini (Allahabad), and Saharanpur. The raw materials used by these factories are bamboo, sabai grass, conifer wood, eucalyptus, wheat barn-rags, and scrap paper.

10. Bihar

In Bihar, there are six paper mills, located at Barauni, Dalmianagar, Dumka, Patna, Rameshwarnagar, and Samastipur.

11. Kerala

The paper mills in Kerala are located at Ernakulam, Kozhikode, Punalur, and Rayanpuram.

In addition to these, there are important paper mills at Cachar, Guwahati, Jorhat and Nagaon (Assam), and Mokokchung in Nagaland.

Despite steady progress in the paper production during the planned period, India is not self sufficient in the paper requirement. India is importing huge quantity of paper from Belgium, Canada, France, Germany, Japan, New Zealand, Norway, Romania, Russia, Sweden, UK, and the USA.

India also exports some quantity of printing paper to the South-East Asian and South-West Asian countries.

Problems of Paper Industry

Some of the important problems of paper industry are:

(i) Shortage of raw material;

(ii) Inadequate supply of chemicals;

(iii) Heavy investment;
(iv) Strikes and lockouts;
(v) Tough competition with the foreign paper producers;
(vi) Obsolete machinery;
(vii) High cost of basic inputs; and
(viii) Quality and environmental concern.

PHARMACEUTICAL INDUSTRY

The drugs and pharmaceutical industry has made a phenomenal progress in India during the last four decades. The country now ranks third in terms of volume of production (10% of global share) and 14th largest by value. Indian exports are destined to various countries around the globe including USA, Europe, Japan and Australia (2011-12). There are five Central Public Sector Undertakings and five joint Sector Undertakings in the pharmaceutical Industry Sector under the administrative control of the Department of Chemicals and Petrochemicals. Besides, there are two wholly-owned subsidiaries. A brief profile of these organisations is given in the following paragraphs.

1. Indian Drugs and Pharmaceuticals Limited (IDPL)

It was incorporated in 1961. The company has presently three manufacturing plants one each at Rishikesh in Uttarakhand, Hyderabad (Andhra Pradesh) and Gurgaon (Haryana). IDPL has two wholly owned subsidiaries, namely IDPL (Tamil Nadu) Ltd., Chennai in Tamil Nadu and Bihar Drugs and Organic Chemicals Limited at Muzaffarpur (Bihar). In addition, IDPL has two joint sector undertakings, promoted in collaboration with the respective State Governments. These are Rajasthan Drugs and Pharmaceutical Limited (RDPL), Jaipur and Odisha Drugs and Chemicals Ltd (ODCL), Bhubaneshwar **(Fig. 11.17)**.

2. Hindustan Antibiotics Limited (HAL), Pimpri (Pune)

It was incorporated in 1954. This was the first public sector company in drugs and pharmaceuticals. It has its plant located at Pimpri. There

are three joint sector units promoted by HAL in collaboration with the respective State Governments. These are Karnataka Antibiotics and Pharmaceuticals Ltd. Bangalore (Karnataka), the Maharashtra Antibiotics and Pharmaceuticals Ltd. (MAPL) at Nagpur (Maharashtra), and Manipur State Drugs and Pharmaceuticals Ltd (MSDPL) at Imphal (Manipur). The company produces a wide range of pharmaceutical formulations including agro-vet products.

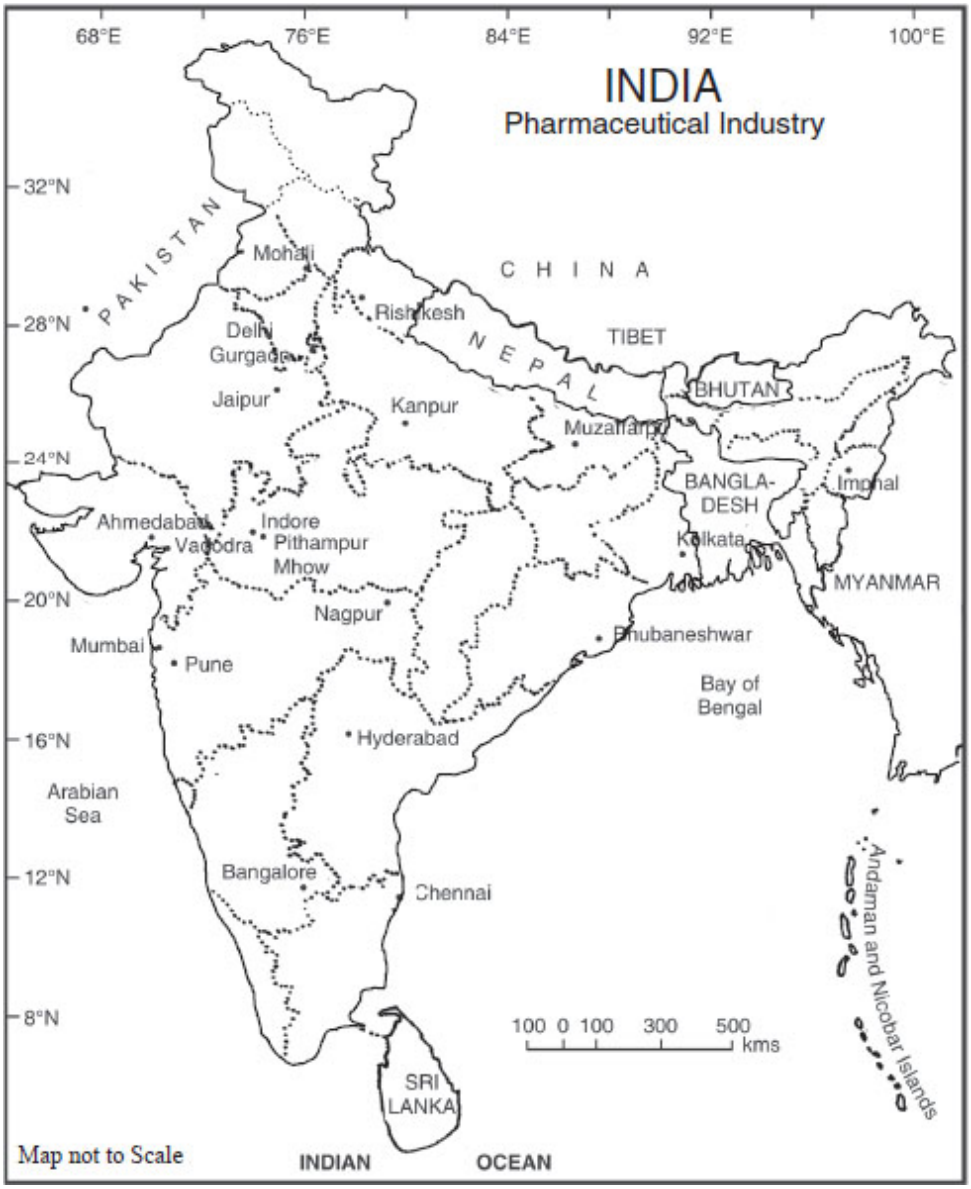


Fig. 11.17 Pharmaceutical Industry

3. The Bengal Chemicals and Pharmaceuticals Limited (BCPL)

It was incorporated in 1981. The company has four manufacturing units located at Maniktala (Kolkata), Panihati (24 Parganas West Bengal), one in Mumbai and one at Kanpur.

Most of the drugs and pharmaceutical units in India are located in Delhi, Gujarat, Maharashtra, Madhya Pradesh, Rajasthan, Tamil Nadu, Uttarakhand, Uttar Pradesh, and West Bengal. The important centres are Ahmedabad, Delhi, Hyderabad, Indore, Kanpur, Kolkata, Jaipur, Mumbai, Muzaffarpur (Bihar), Pune, Rishikesh (Uttarakhand), and Vadodra. The Surgical Instruments Plant at Chennai produces different types of surgical instruments, while the Bengal Chemicals and Pharmaceutical Limited (BCPL) with four manufacturing units is the largest producer of anti-snake venom in India.

Although it is producing a large variety of drugs and is meeting 75% of its requirements by indigenous products. India is importing expensive life saving drugs from Australia, France, Germany, Italy, Japan, Russia, Singapore, Spain, UK, and the USA.

Drug Policy

The salient features of the modified drug policy (1986) as announced on 15 September, 1994 are:

1. Abolition of Industrial Licensing for all bulk drugs, their intermediates and formulations except five bulk drugs of the public sector.
2. Foreign investment up to 51% will be automatically permitted in the case of bulk drugs, their intermediares and their formulations.
3. Automatic approval for foreign technology agreements shall be given in case of all bulk drugs, their intermediares and formulations, except those produced by the use of recombinant DNA technology.
4. Measures to give further impetus to R & D in the drug sector:
   (a) Newly developed drugs would be put outside price control for a period of 10 years;

(b) Department of Chemicals and Pharmaceuticals to set up inter-ministerial group to decide, within a set time frame, on measures to give further impetus to R & D in the drug sector; and

(c) Ministry of Health and Family Welfare to take suitable steps for the quick clearance of new drug application, especially those developed through indigenous R & D.

5. The system of price control would be operated through a single list of price-controlled drugs, selected on the basis of transparent and objective criteria as laid down in the modification in drug policy 1986.
6. In case of basic manufacture, the rate of return would be higher by 4% over the existing rates which are 14% on net worth or 22% on capital employed.
7. To achieve uniformity in prices of widely used formulations there should be ceiling on prices for commonly marked standard pack-sizes of price-controlled formulations.
8. The National Pharmaceutical Pricing Authority to entrust with the task of price
   fixation/revision and other related matters.
9. Government to keep a close watch on the prices of medicines which are taken out of price control.
10. A National Drug Authority to be set up to look after the quality control aspects, national use of drugs, and related matters.
11. To provide better focus to all matters relating to developing and promotion of indigenous and other systems of medicines, a separate department would be created by the Ministry of Health and Family Welfare.

Pharmaceutical Policy, 2002

The first pharmaceutical Policy-'***Drug Policy***' was drafted in 1978 .

The Government of India announced the New Pharmaceutical Policy 2002. The salient features of the policy are:

(i) Industrial licensing for all bulk drugs to be cleared by Drug

Controller General of India.

(ii) Foreign direct investment upto 100% will be permitted subject to the stipulations laid down from time to time.

(iii) Automatic approval for foreign technology agreement.

(iv) Ceiling price may be fixed for any formulation.

(v) Government will keep a close watch on the prices of medicines.

(vi) Price fixation by the Pharmaceutical Pricing Authority, an independent body of experts.

The draft of Pharmaceutical Policy 2017 has laid emphasis to quality concerns that affect the domestic concerns.

The ***Jan Aushadhi Campaign*** allows the selling of medicines in their generic names for providing medicines at cheaper prices . However it requires elaborated efforts. The Pradhan Mantri Bharatiya Janaushadhi Kendra shops also need expansion. Salient Features Key features of this policy are as follows: Price Control to Monitoring The new policy has an aim of reorienting the Drug Price Cont

COTTAGE INDUSTRIES

The production of finished goods by a worker, sometimes together with his or her family, at home is known as cottage industry. The products may be sold directly to the public by the worker, or to an entrepreneur who pays according to number of goods produced. Cottage industry has a long history in India. It exists in almost all the states and regions of the country. Thus, the cottage industry involves the family labour and with the minimum investment the family attempts to increase its income. Some of the important cottage industries in India are *Bidi*-making, sports goods, basket-making, lac industry, manufacture of crackers, paper machie, perfumery, needlework on shawls, and wearings.

The main raw materials used in the *bidi* are *tendu* leaves (*Diospyros melanoxylon*) and *kachnal* (*Bauhinia racemosa*), found in the forests of Chhattisgarh, Jharkhand, Madhya Pradesh, Odisha, and Western and Eastern Ghats. Inferior tobacco which is an

important ingredient of *bidi*-making is obtained from Andhra Pradesh, Bihar, Chhattisgarh, Madhya Pradesh and Uttar Pradesh. The annual production of *bidi* is more than 1000 crores. India exports *bidi* to the Asian, African, and European countries.

India produces the entire equipments for athletics goods: basket-ball, cricket, tennis, badminton, baseball, billiard, carrom, chess, dart, football, golf, gymnastics, squash, volleyball, and water-polo. Many of these products are produced in the cottage industry.

Bidi-making

This is a popular cottage industry. It is mainly produced in Bastar, Belgaum, Bhandara, Bhind, Gondia, Hyderabad, Jabalpur, Jagdalpur, Kamptee, Kheda, Mangalore, Nagpur, Nasik, Pune, and Vadodra.

Sports goods

Sports goods making is a traditional cottage industry in the country. The raw materials used in sports goods are fine quality of wood, leather, cloth, rubber, and metal. These are locally available. Some of the finer quality of raw materials are, however, imported. Sports goods cottage industry, mainly concentrated in the rural and urban areas of Jalandhar, Ludhiana, Ahmedabad, Ambala, Chennai, Delhi, Jammu, Kolkata, Ludhiana, Meerut, Modinagar, Moradabad, Mumbai, Mysore, Nainital, Pune, Saharanpur, Shimla, and Srinagar.

Basket-making

This is generally confined to the hilly and mountainous areas of the country. The important raw materials used in the basket-making are bamboo, cane, and willow. The main producers of baskets are Arunachal Pradesh, Himachal Pradesh, Jammu and Kashmir, Ladakh, Maharashtra, Manipur, Meghalaya, Mizoram, Nagaland, Odisha, Sikkim, Tripura, and Uttarakhand.

Lac Industry

Lac is a natural resin secreted by an insect (*Cerria lacca*). This insect thrives on *babool, bargad, pipal, kusum,* and *palash*. The important products produced with the help of lac as a raw material are

adhesives, electrical insulators, gramophones records, nail-polish, and printing ink.

This cottage industry is concentrated in the districts of Assam, Chhattisgarh, Gujarat, Jharkhand, Karnataka, Madhya Pradesh, Maharashtra, Meghalaya, Mizoram, Nagaland, Odisha, Sikkim, Tripura, Uttar Pradesh, Uttarakhand and West Bengal.

India is the third most important lac producing country in the world after Thailand and Malaysia. About 85% of the total production of lac is exported to France, Germany, Italy, Japan, Russia, UAE, UK, and the USA.

Others

Needlework on cloths, especially on shawls is a speciality of the cottage industry in Kashmir, while Lucknow is famous for needlework on *kurtas* and appeals. In Tamil Nadu, cracker-making is an important cottage industry which has a great future.

INDUSTRIAL REGIONS OF INDIA

Delineation of industrial regions empirically has been attempted by a number of geographers. The parameters used by them, however, differ from each other. It was Trewartha and Burner (1944) who divided India into industrial regions. Subsequently, P.P. Karan and W.M. Jenkrins (1959) demarcated the industrial regions of India. Industrial regions of India were also delineated by Spencer and Thomas (1968), R.L. Singh (1971), B.N. Sinha (1972), M.R. Chaudhry (1976), and the Centre for Monitoring Indian Economy (1971, 1982) CMIE. These scholars adopted one or more than one of the following indicators for the demarcation of industrial regions of India:

1. Number of registered factories in a region.
2. Number of industrial workers.
3. Population engaged in the secondary activities.
4. Percentage of industrial workers to the total workers.
5. Gross industrial output.

6. Production in terms of money.

In the demarcation of industrial regions, most of the experts divided India into six major and six minor industrial regions. A brief description of the industrial regions of India as demarcated by Prof. R. L. Singh has been given here **(Fig. 11.18)**.

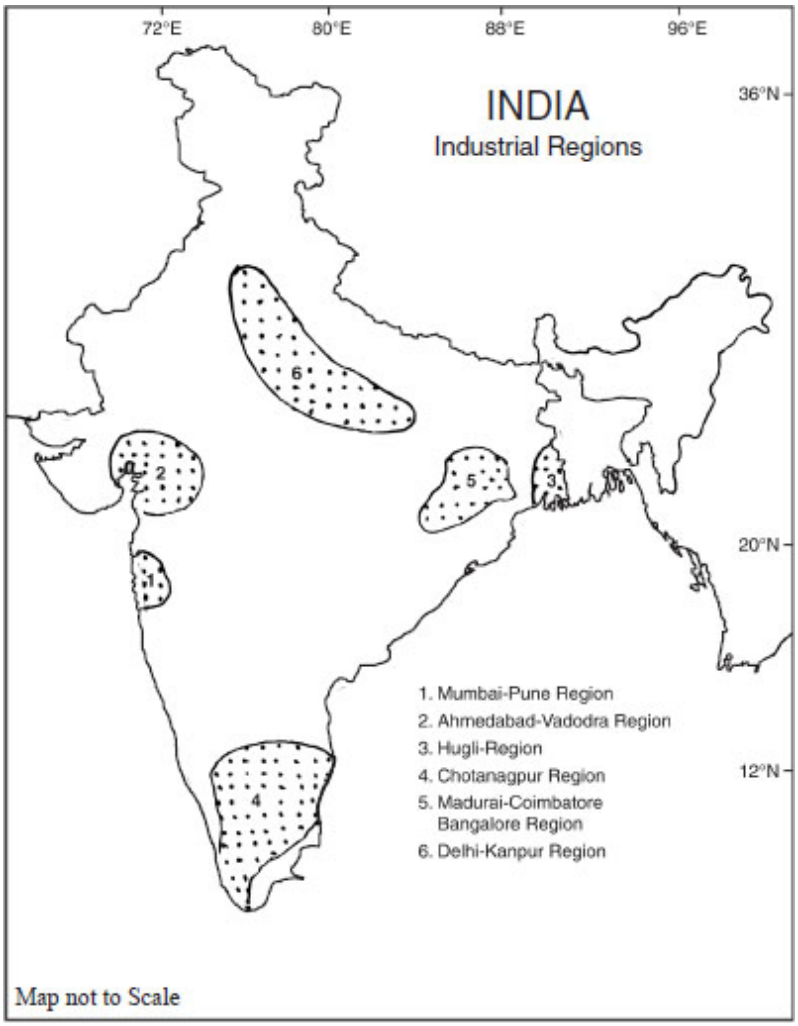


Fig. 11.18 Industrial Regions

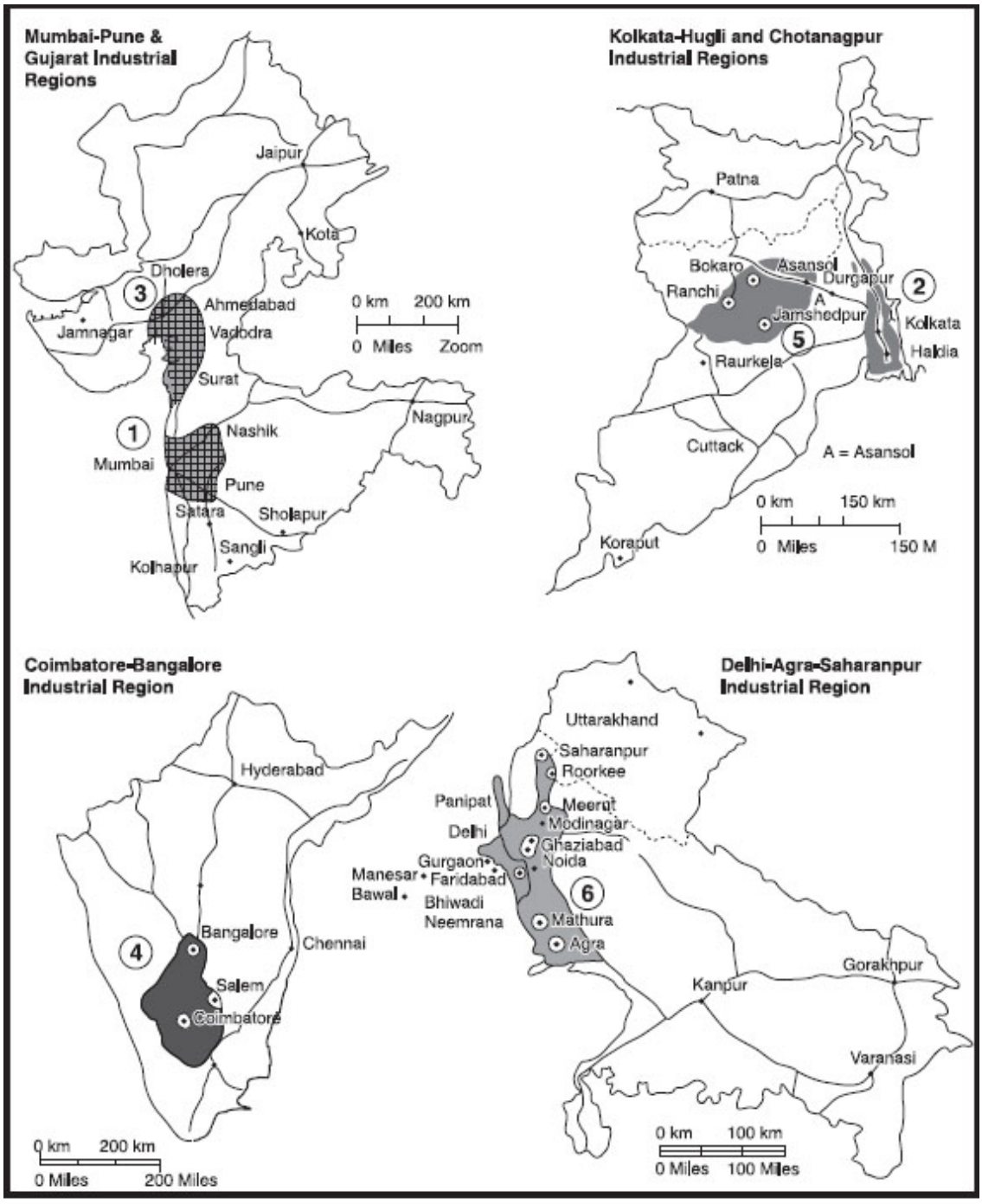
The Major Industrial Regions

1. The Mumbai-Pune Industrial Region

This is the most important industrial region of the country **(Fig. 11.18)** and **(Fig. 11.19)**. The region developed after the arrival of British in India who developed the Mumbai Seaport. After the opening of Suez

Canal in 1869 the sea-route between India and Europe was reduced substantially. The development of this industrial region is closely connected with the history of development of cotton textile industry in India. The humid climate, natural port facilities, availability of hydro-power, skilled labour, and a vast hinterland producing cotton became the main factors in the development of this industrial region.

There are more than 8000 registered factories only in the Greater Mumbai, out of which 350 belong to cotton textile. The other industries of the region are engineering goods, chemical industries, food-processing industries, leather goods, pharmaceutical, and film industries. In Mumbai, the bulk of the production is light textured fine and super fine cotton cloths. Over 15 lakh people are engaged in the industrial sector of this region.



Map not to Scale

Fig. 11.19 Major Industrial Regions of India

Pune is the second most important industrial centre of the region. It has more than 1200 registered factories. Its industries are producing

metallurgical, chemical, engineering, and automobile goods. Pune has two factories producing scooters and mopeds.

In addition to Mumbai and Pune, the industrial centres of this region are: Ambarnath, Andheri, Bhandup, Ghatkopar, Hadapsar, Jogeshwari, Kalyan, Kirkee, Kolhapur, Kurla, Nashik, Sholapur, Thane, Trombay, Ulhasnagar, and Vikroli.

This industrial region has almost reached the saturation level. Some of the important problems of this industrial region facing are:

(i) Inadequate supply of power;
(ii) Obsolete and outdated machinery;
(iii) High cost of land and high rent of commercial space;
(iv) Labour unrest;
(v) Increasing regionalism;
(vi) High rate of crime; and
(vii) Increasing environmental pollution.

2. The Kolkata-Hugli Industrial Region

The Kolkata-Hugli industrial region is located along the banks of the Hugli River. The availability of agro-raw material (jute, indigo, and tea), nearness of coal mines (Raniganj and Jharia), abundance of water, cheap labour, and facilities of export are the main factors which helped in the fast growth of this industrial region. Moreover, Kolkata was the capital of British India from 1773 to 1911. Being the capital, Kolkata attracted many of the industrialists to locate their industries in this region.

There are over 10,000 registered industrial factories in this region in which over 20 lakh people are engaged. This belt specialises in the production of jute, silk, cotton textiles, engineering, electrical goods, automobiles, chemicals, pharmaceutical, transport equipments, leather-footwear, iron and steel and food processing, light machine, locomotives, iron and steel, and spare goods for different types of machines.

The main industrial cities and towns of this region are Bansberia, Naihati, Bhatpara, Kankinara, Jagatdal, Shamnagar, Bhadreshwar, Krishnanagar, Baidyabati, Serampore, Titagarh, Rishira, Konnagar,

Agarpara, Batanagar, Ghusuri, Chunchra, Kolkata, Haora, Budge Budge, Bauria, Chengail, and Birlapur.

The main problems of this industrial region are:

1. Paucity of space and traffic jams;
2. Shortage of drinking water, insanitation and lack of infrastructural amenities;
3. Silting of the Hugli River resulting in the silting of Kolkata port;
4. Obsolete machinery;
5. Naxalites movement and political unrest;
6. Strikes and lockouts; and
7. Shortage of power supply.

In order to overcome these problems the government of West Bengal is pursuing the policy of liberalisation and inviting domestic and foreign entrepreneurs to invest in the region. Some progress has been made in this direction in recent years.

3. The Ahmedabad-Vadodra Industrial Region

This is the third largest industrial region of the country **(Fig. 11.19)**. The main cause for the development of this industrial region is the availability of cotton in the hinterland, availability of cheap land, cheap skilled and unskilled labour, port facilities, and nearness of petroleum (Kioli), thermal (Dhuvaran), Hydel (Ukai Project), and nuclear power station (Kakrapara).

There are over 11 thousand registered factories in this region engaging over 15 lakh workers.

It is the second largest cotton textile industrial centre in the country. It also specialises in chemical industries, engineering goods, and pharmaceutical products. Vadodra is an important centre of woollen textile and petrochemical goods. Surat is well known for silk textile and diamond cutting. The other important industrial centres of this region are Anand, Ankleshwar Bhavnagar, Bharuch, Godhra, Himmatnagar, Jamnagar, Kalol, Kheda, Nadiad, Navsri, Rajkot, and Surendernagar.

Scarcity of water and shortage of good quality of cotton are some of the important problems of the region. For the last few years,

communal tension has adversely affected the investment in industries in this region.

4. The Madurai-Coimbatore-Bangalore Industrial Region

Stretching over the state of Tamil Nadu and the southern parts of Karnataka is an important industrial centre which made great progress after independence. This region is mainly the cotton producing area of the country. The good climate, disciplined skilled and unskilled labour, regular supply of power (from the Mettur, Papanasam, Pykara, Savitri and Sivasamudram), and the nearness of Chennai, Kochi, Mangalore, and Tuticorin seaports have contributed in the fast development of this industrial region **(Fig. 11.19)**.

This industrial region is well known for cotton and silk textiles, leather goods, oil mills, coffee processing industries, chemicals, paper, rubber goods, cement, sugar, and food processing industries. Bangalore has emerged as the industrial centre of the region having the location of Hindustan Machine Tools, Aeronautics, Indian Telephone Industry, Bharat Electronics, Visvesvaraya Iron and Steel Works, electronics, automobiles, and information technology.

5. The Chotanagpur Industrial Region

This industrial region stretches over Jharkhand, Chhattisgarh, Odisha, southern Bihar, and western parts of West Bengal. Having a large concentration of iron and steel industry, it is often called as the "*Ruhr* of India". This region is rich in the fossil fuel, and metallic and non-metallic minerals. Power is available from the Damodar Valley Corporation. There is enormous supply of cheap labour from the states of Bihar, Jharkhand, Odisha, Uttar Pradesh, and West Bengal **(Fig. 11.19)**.

The main iron and steel producing centres of the region are Asansol, Bokaro, Burnpur, Durgapur, Kulti, Jamshedpur, and Raurkela. The other important industrial centres of the region are Sindri for fertiliser, Chittranjan for locomotives, Khalari for cement, Ranchi for HMT, and Ramgarh and Bhurkunda for glass industry. The main problems of the region are shortage of power supply, and

political unrest like those caused by Naxalites. The labour unrest has deterred many of the investors in this region.

6. The Agra-Delhi-Kalka-Saharanpur Industrial Region

This industrial region emerged mainly after independence. Being in and around the national capital and in the vicinity of the Indira Gandhi International Airport it has a more peaceful industrial environment, largely free from the labour unrest and Naxalite movement. It is located in the most productive part of the country providing enough raw material for the agro-based industries. The nearness of market and availability of hydro-power from Bhakra Nangal and thermal power from Badarpur, Faridabad, Dadri, Harduaganj, and Indraprastha (Delhi) have helped largely in the development of this industrial region.

The main industries of this region are engineering, electronics, chemical, glass, textile, sugar, and food-processing and agricultural machinery **(Fig. 11.19)**. The main industrial centres are Agra (textile, tourisms), Ambala (scientific instruments), Chandigarh (electronic and strategic goods), Delhi (textile, chemicals, drugs, pharmaceutical, light machine, electronic goods, food processing), Faridabad (engineering), Ghaziabad (synthetic fibre, chemicals, electronics, pharmaceuticals, agricultural equipments, iron & steel, cycle tyre and tubes), Gurgaon (automobiles), Kalka (HMT), Mathura (petrochemicals), Meerut (sugar and textile), Modinagar (textile, engineering goods, and paper), Modipuram (textiles), Mohannagar (brewery, alcohol), Moradnagar (ordinance), NOIDA (automobile, electronics, etc.), Panipat (textile, chemical and food processing), and Saharanpur (paper, wood-work, sugar, textile and food-processing).

High price of land, traffic jam, and high rate of crimes are the main problems of this region.

Minor Industrial Regions

Apart from the industrial regions described above there are several minor emerging industrial regions in the country. Some of them are as under:

(i) Kanpur-Lucknow Industrial Region: Cotton, woollen and jute

textiles, leather goods, fertilisers, chemical, drugs, pharmaceuticals, electric goods, and light machinery.

(ii) Assam Valley Industrial Region: This region has the industries of petro-chemical, jute and silk textiles, tea-processing industry, paper, plywood, match, and food processing industries. Important industrial centres are: Bongaigaon, Dibrugarh, Digboi, Guwahati, Noonmati, and Tinsukia **(Fig. 11.20)**.

(iii) Darjeeling-Siliguri Industrial Region: Tea-processing industry and tourism.

(iv) North Bihar and Eastern Uttar Pradesh Industrial Region: Sugar, cement, glass, jute, fertilisers, locomotive, paper, and food processing are the main industries of this region. The main industrial centres are Allahabad, Dalmianagar (Bihar), Gorakhpur, Patna, Sultanpur, and Varanasi.

(v) Indore-Ujjain Industrial Region: Main industries are cotton textile, chemicals, drugs, electronic and engineering goods, and food processing.

(vi) Amritsar-Jalandhar-Ludhiana Industrial Region: Sports goods, cotton and woollen, textiles, hosiery, food-processing, and tourism are the main industries of this region.

(vii) Nagpur-Wardha Industrial Region: Textiles, engineering, chemicals, and food processing are the main industries of this region.

(viii) Godavari-Krishna Delta: Main industries are iron and steel, ship-building, fertiliser, rice-milling, cotton textile, sugar, fish processing, engineering, and chemicals. Main industrial centres are Guntur, Machlipatnam, Rajamundry, and Vishakhapatnam.

(ix) Dharwar-Belgaum Industrial Region: Cotton textile, chemicals, spices packing, and food processing are the main industries.

(x) Kerala Coast Industrial Region: Main industries of this region are coconut-oil extraction, rice-milling, fish packing, paper, coir-matting, ship-building (Kochi), petroleum refining (Kochi), and chemical and electronic goods.

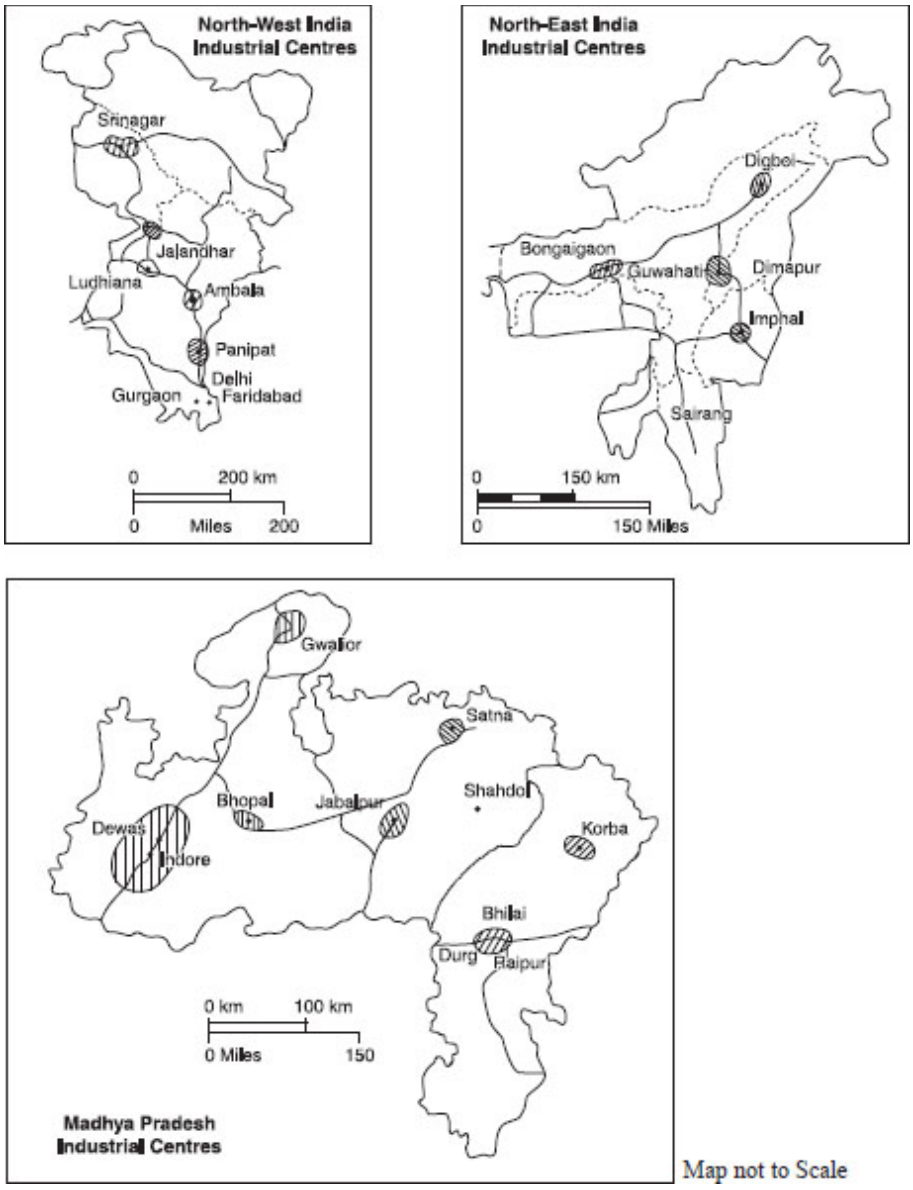


Fig. 11.20 India—Emerging Industrial Regions

MULTINATIONAL CORPORATIONS

A firm which owns or controls production facilities in more than one country through direct foreign investment is known as a multinational or multinational corporation (MNC). These are also called the Transnational Corporations. During the colonial period, the East India Company, the Dutch East Indonesian Company, the Royal African

Company, and the Hudson Bay Company were such multinational companies.

Although multinationals grew most rapidly in the 1960s, the foundations were laid in the inter-war period, notable examples being Ford, Vauxhall, and Phillips. In the mid-1980s, multinationals accounted for 14% of the UK employment and 30% of the UK exports. The corresponding figures for France were 24% and 32%, respectively. Multinationals are made possible by improved international communications which provide rapid containerised transshipment and foreign travel, easy communications of information, and international mobility of capital. When one market is saturated, the multinational can rapidly develop others, since foreign investment cuts transport costs, and makes possible a rapid response to local markets. It also eases tariff barriers. Moreover, multinationals can compare costs at different locations, and can switch activities to different areas as appropriate.

Multinational corporations are probably the major force affecting world wide shifts in economic activity, since the largest MNCs have a turnover greater than the GNP of many less developed countries. Although a developing country may benefit from the construction of a plant for a multinational corporation in terms of jobs and markets, it has been argued that the price is loss of local control.

Impact of Multinational Corporations

The multinationals have affected the Indian economy, society, and ecology favourably and adversely. Some of the important impacts of multinationals have been presented here:

1. **Replacement of Technology:** The obsolete and outdated technology performing below the capacity has been replaced by the latest and more efficient technology by the multinationals.
2. **Damage to Cottage and Small Scale Industries:** The traditional cottage industries as well as small scale industries are not able to compete with the large-scale production of the

multinationals. These industries may disappear completely, unless special steps are taken to promote their interest.

3. **Heavy Remittance Abroad:** According to the Reserve Bank of India, the average rate of profit to the multinationals varies between 20 to 25%. This is a huge profit remitted outside of the country.
4. **Low Foreign Investment:** Most of the foreign subsidiaries have raised financial resources from within India, and the transfer of capital from the parent company has been marginal.
5. **Change in the Initial Activities of the Multinational Corporation:** A number of foreign companies in India are acquiring the character of multi-product and multi-industry enterprises. For example, the Imperial (now India) Tobacco Company (ITC) has diversified its activities to hotel industry, constructing a chain of hotels all over India.
6. **Transfer of Technology—A Myth:** The assumption that the entry of multinational corporation would ensure transfer of latest sophisticated technology to developing countries has not been found valid in practice.
7. Females Employment: In some industries, like readymade garments, electronics, jewellery and toy-making, females got more employement. The family institution has been adversely affected. Crime against females has increased. The younger generation is adopting western values, leading to cultural pollution.

LIBERALISATION

Liberalisation in India is a reform adopted to accelerate the industrial growth and socioeconomic development of the country. The main objective of liberalisation is to permit Indian and foreign entrepreneurs to enter the power, road, and communication as well as in the petroleum sector so that the Central and State governments can lay greater emphasis on the social and economic development

programmes. The salient features of the policy of liberelisation are as under:

1. Under liberalisation the new Industrial Policy, July 1991, was adopted to remove the bureaucratic control, which was a barrier in the industrial development of the country. In fact, liberalisation means deregularisation of the industrial sector by cutting down the minimum administrative interference in the operation, instead, letting the market forces operate through the profit motive of the producers and free competition among them to regulate and guide the future development of the sector.
2. Under the policy of liberalisation, the requirement of industrial licensing has been abolished for all industries except for 16 products, namely, (i) aerospace equipments,
   (ii) alcohol, (iii) asbestos, (iv) chamois leather, (v) coal and lignite, (vi) consumer electric goods, (vii) defence equipments, (viii) electric equipments, (ix) drugs and pharmaceuticals, (x) fur-skin, (xi) hazardous chemicals, (xii) industrial explosives, (xiii) paper and pulp, (xiv) petroleum and natural gas, (xv) plywood, and (xvi) sugar.
3. The liberalisation has opened the economy to direct foreign investment with 51% equity, and started with the process of reducing government subsidies.
4. Abolition of restrictions on most import and export items.
5. Substantial reduction in import tariffs and almost complete removal of restrictions on foreign investments.
6. Disinvestment of shares of public undertakings to reduce government share holding to 51%. This will improve the efficiency of the public sector as the decision-making process will be efficient and prompt.

Impact of Liberalisation

The leading economists of the country differ in their opinion about the socioeconomic and ecological consequences of the policy of

liberalisation. Some of the consequences of liberalisation have been briefly described here:

1. **Increase in the Direct Foreign Investment:** The policy of liberalisation has resulted in a tremendous increase in the direct foreign investment in the industrial and infrastructural sector (roads and electricity).
2. **Enhancement in the Growth of GDP:** There is a significant growth in the Gross Domestic Product (GDP). Prior to the liberalisation, the growth rate of GDP was around 4% which rose to around 8.2% in 2018 (Source: Economic times Aug 31, 2018).
3. **Reduction in Industrial Recession**: The industrial sector of India was passing through a period of recession prior to the policy of liberalisation. The foreign and private investment has checked the recession trend. This happened because of the massive investment in modernisation, expansion, and setting up of many new projects. Industries like automobiles, auto-components, coal-mining, consumer electronics, chemicals, food-processing, metal, petrochemicals, software, sport-goods, and textiles have undergone a growth rate of about 25%. In addition to these, other industries, like crude-oil, construction, fertilisers, and power generation have shown an increase of about 15%.
4. **Employment:** The heavy investments in industries and infrastructure by the Indian and foreign investors have generated great employment opportunities for the professionals, and skilled and unskilled workers.
5. **Development of Infrastructure**: Prior to the liberalisation, the infrastructure (roads and electricity) were in a bad shape affecting the industrial growth and economic development of the country adversely. Heavy investment in infrastructure has improved the efficiency of the industrial sector significantly.
6. **Rise in Export:** There is a phenomenal increase in export after liberalisation. Simultaneously India is importing raw

materials, machinery, and finished products. Despite heavy imports, there has been a tangible improvement in the balance of payment.

7. **Increase in Regional Disparities**: The policy of liberalisation and New Industrial Policy (1991) could not reduce the regional inequalities in economic development. In fact, investments by the Indians and foreign investors have been made in the states of Andhra Pradesh, Gujarat, Haryana, Karnataka, Maharashtra, Rajasthan, Tamil Nadu, and West Bengal. The states like Bihar, Himachal Pradesh, Jammu and Kashmir, Kerala, Meghalaya, Mizoram, Nagaland, Odisha, Tripura, Uttar Pradesh, and Uttarakhand are lagging behind. This has accentuated the regional imbalance. The maximum investment so far has been done in Maharashtra, Gujarat, Karnataka, Andhra Pradesh, West Bengal, and Tamil Nadu. This uneven industrial development has resulted into many socioeconomic and political problems. The Naxal Movement, ULFA, and political turmoil in Jammu and Kashmir may be partly explained as being caused due to the less industrial and economic development of the regions.
8. **Damage to Cottage and Small Scale Industries**: Liberalisation in a country like India has adversely affected the traditional cottage and small scale industries which are unable to compete with the large-scale industries established by the multinationals. The cottage and small scale industries need protection in the form of subsidies, technology, technical access, funds, and network to export their products.
9. **Sophisticated Technology:** The latest technology, being sophisticated, replaces labour and thus results in unemployment. This may be counter productive and detrimental to our industrial structure.
10. **Comparatively Little Direct Investment:** The foreign investors are more inclined to portfolio investment rather than

direct investment. The former may be withdrawn at will at the slightest of hurdles giving a jolt to the economy of the country.

11. **Investment in Selected Industries:** Most of the foreign investment comes to white-goods and not to wage-good sector. Hence, it may be fruitful in improving the high priority sector and bringing in the latest technology. This will be counter productive.
12. **Economic and Political Freedoms are at Stake:** The over-enthusiasm of liberalisation to attract more investors and foreign exchange might lead to gradual handling over of the whole economy to the multinationals. This will affect adversely our economic and political freedom.
13. **Inflation:** Since the new industrial policy and liberalisations, the rate of inflation is continuously increasing. A section of the society is becoming more rich and adopting the lifestyle of consumerism. As opposed to this, the absolute number below the poverty line is also increasing. The gulf between the rich and the poor may be the cause of numerous social problems resulting in social tension.

INDUSTRIAL PROBLEMS OF INDIA

India has made tremendous progress during the plan period and the industrial sector and service sectors have made significant progress. Despite industrial progress, India is not self sufficient in the matter of industrial products. Moreover, the quality of the products are not at par with the products of the developed countries. The main problems of the Indian industries have been briefly described here:

1. **Inadequacy of Industrial Structure:** Despite 65 years of planning, India is not self sufficient in the production of transport equipments, electrical and non-electrical machinery, iron and steel, paper, fertilisers, drugs and pharmaceuticals, and plastic materials.
2. **Low Demand of Industrial Products**: The purchasing power and poor standard of living of a greater section of the India

population (about 85%) has resulted in low demand of industrial products.

3. **Regional Concentration of Industries:** Most of the industries of India are located in and around Mumbai, Kolkata, Ahmedabad, Delhi, Banglore, and Chennai. This is leading to unequal industrial development. Most of the industries are in Maharashtra, Gujarat, Karnataka, Andhra Pradesh, West Bengal, and around Delhi. This has not only created regional disparities but encouraged political unrest, violence, and terrorism.
4. **Industrial Sickness:** In the private sector after the policy of liberalisation, the small and medium as well as some of the large industries are becoming sick. The reasons for industrial sickness may be poor management, obsolete technology, and international competition. Many a times, the government takes over the sick industrial units which further worsen the problems.
5. **Loss in Public Sector:** Most of the public undertakings are running at loss. The reasons may be poor efficiency of the management and workers, strained labour and management relations, and obsolete technology. Every year the government has to incur huge expenditure to cover up this loss and to meet the obligations of paying salaries and wages to the employees. To avoid this burden on exchequer, the government is promoting privatisation and disinvestment of shares of public sector undertakings.
6. **Lack of Infrastructure:** Inadequate infrastructure is a serious barrier in the speedy industrialisation of the country. There is energy crisis in most of the industrial regions of the country. The power cut and load-shedding reduce the efficiency and production of the industries.
7. **Irrational Location of Industries**: Some of the industries in the private sector have been located without considering the transportation cost. Many a times the political expediency

have been the basis of selection of site of industries. This not only makes the product expensive, but also promotes sub-nationalism/regionalism. The location of Barauni Refinery far away from the source of crude oil (Digboi-Assam) is one of such examples.

8. **Shortage of Industrial Raw Material**: Many of the raw materials used in industries are agro-based. Indian agriculture over greater part of the country depends on monsoon. In case of failure of monsoon, the raw materials for the agro-based industries suffer as the raw material is not available. Moreover, the purchasing power of the Indian rural community largely depends on the success or failure of monsoon.
9. **Shortage of Capital:** For the industrial and infrastructural development, India largely depends on loans from the World Bank and the Asian Development Bank. A lot of foreign exchange is lost in the payment of these loans. The situation becomes acute when fresh loans are taken to pay the instalments of the old loans.
10. **Low Quality of the Products:** Our industries are not able to compete in the international market because of the relatively poor quality of goods. In fact, the products of the developed countries are far superior and reasonable in price. China has become an additional competitor with the Indian goods both in the developed and the developing countries. The Indian industries survive on the basis of domestic demands. Since the purchasing power of most of the lower income rural population is low, they cannot go for the purchase of superior industrial goods.
11. **Lack of Institutional Organisation**: There is no well defined planning and policy to attract foreign capital and promote industrialisation.
12. **Change in the Industrial Policy:** The frequent changes in the planning approach— emphasis sometimes on agricultural

sector and rural-employment; sometimes on urban industrial development; and sometimes on tertiary sector—have resulted in stunted growth and development of industries. The political expediency and shortage of capital, energy, and raw materials remain the main barriers in the industrial development of the country.

SPECIAL ECONOMIC ZONE (SEZ)

Special Economic Zone is a geographical region that has economic laws that are more liberal than country's typical economic laws.The category SEZ covers a broad range of more specific zone types, including Free Trade Zones (FTZ), Export Processing Zone (EPZ), Free Zones (FZ), Industrial Estates (IE), Free Ports, Urban Enterprise Zones, and others.

Defining Feature

Broadly SEZ has certain defining features such as special tax incentives; duty free movement of goods and services; world class infrastructure; construction intensive nature; export orientation; differentiated economic management like relaxation in certain basic restrictions applicable to the rest of the economy; and free inflow of foreign capital.

Objectives of Special Economic Zone

The main objectives of SEZs are:

(i) To provide a conducive structure to increase foreign and domestic investment in industries.
(ii) To promote foreign trade.
(iii) To generate more employment.
(iv) To develop the relatively less developed areas.
(v) To accelerate the process of industrialisation and urbanisation.
(vi) To reduce the regional inequalities in socioeconomic development.

Historical Perspective

According to Robert C. Hayword, Director, World Economic Processing Zones Association, the concept of free economic zones dates back to 300 BC. The author notes that such enclaves were found in the Phoenician city of Tyre in the Greek island of Delos. The city became wealthy as a result of such policies and was viewed as a challenger to the centralism of the Roman Empire.

One of the earliest and most famous Special Economic Zones were founded by the government of China under Deng Xiaoping in the early 1960. Following the Chinese example, Special Economic Zones have been established in several countries including Brazil, India, Iran, Jordan, Kazakhstan, Pakistan, Philippines, Poland, Russia, and Ukraine.

Conditions for Ideal SEZ

If the SEZ policy is to provide a real economic impetus especially in the form of creation of world class infrastructure, the minimum size should be 1000 hectares or a minimum investment at INR 10,000 crore.

Special Economic Zones of India

In order to enhance foreign investment and promote exports from the country and to provide liberal facilities to the foreign and domestic investors, the Special Economic Zones were created in India. The Government of India had in April 2000 announced the introduction of Special Economic Zones policy in the country, deemed to be foreign territory for the purpose of trade operations, duties, and tariffs. As of 2007, more than 500 SEZ have been proposed, 220 of which have been created. India passed Special Economic Zone Act in 2005.

The policy provides for setting up of SEZs in the public, private, joint sector or by state governments. It was also envisaged that some of the existing Export Processing Zones would be converted into Special Economic Zones. Accordingly, the government has converted Export Processing Zones located at: (i) Kandla and Surat in Gujarat, (ii) Cochin (Kerala), Santa Cruz, (Mumbai—Maharashtra), Falta

(West Bengal), Chennai, Ilandaikulam, Nanguneri and Tirunelveli (Tamil Nadu), Vishakhapatnam (Andhra Pradesh), NOIDA (Uttar Pradesh), Rajiv Gandhi Info Tech Park, Hinjewadi, Pune-Maharashtra, and Indore (Madhya Pradesh).

At present, India has 81 units in operation in 8 functional SEZs, each with an average of 200 acres. Eight Export Processing Zones (EPZs) have been converted into SEZs. These are fully functional. All these SEZs are in various parts of the country in the private, joint sectors or by the state government.

The process of planning through SEZs is, however, under question, as the state in which the SEZs have been approved are facing intense protests, from the farming community, accusing the government of forcibly snatching fertile land from them, at heavily discounted prices as against the prevailing prices in the commercial real estate industry. Also some reputed companies like Bajaj and others have commented against this policy and have suggested using barren and wasteland for setting up SEZs.

Attempts to set up a Special Economic Zone in Nandigram (West Bengal) have led to protests by the villagers in the area (2008). A Parliamentary Committee to study and give recommendations on SEZs has said that no further SEZs be notified unless the existing law is amended to incorporate the changes related to the land acquisitions.

The state governments have advised that in land acquisition for SEZs, first priority should be acquisition of *waste and barren land* and if necessary single crop agricultural land. If perforce, a portion of double cropped agricultural land has to be acquired to meet the minimum requirements, especially for multi-product SEZ, the same should not exceed 10% of the total land required for the SEZ.

Prospects

If the right price is paid for acquiring land through open bidding, land of any size may be available. Otherwise, the current phenomenon of mushrooming SEZs in and around major cities will only add to the pressure on the already weak urban infrastructure and will result in

migration of people to cities when the objective should be to bring about a reversal of this trend.

PUBLIC SECTOR UNDERTAKING

A public sector undertaking is a corporation in the public sector in India, where management control of the company rests with the Government, it can be Central Government or the state governments. Below is given a partial list of Public Sector Undertakings of the Government of India. There are about 237 Public Sector Undertakings all over India.

The important 16 products which are under the public undertaking after the liberalisation of 1991 are: (i) coal and lignite, (ii) petroleum, (iii) alcohol, (iv) atomic energy, (v) asbestos,
(vi) plywood, (vii) chamois-leather and fur-skin, (viii) paper, (ix) railway engines and railway transport, (x) air-craft manufacturing, (xi) aereospace, (xii) defence equipments, (xiii) industrial explosives, (xiv) hazardous chemicals, (xv) drugs and pharmaceuticals, and (xvi) telephone, telegraphs and wireless apparatus (excluding radio receiving sets).

Growth of Public Sector

In absolute terms, the growth of the public sector in India to date is substantial. This is remarkable, considering the growth in the sector started only in the *fifties* with the emergence of plan-oriented development in the country. The rate of growth has, therefore, been fast, steady, and continual, except that in recent years the growth has been unchecked and many of these are no more proficient profit-centres.

The public sector in India has expanded substantially. These have emerged as important sphere of activity and growth leaders, not only in the production and manufacturing industries but also in the field of trade and commerce, energy, and transport.

The noteworthy features in the growth of public sector are:

(i) The public sector has grown not only under the central

government but also under the ownership of the different state governments.

(ii) The public sector owned by the central government of India extends over 75% of the total scenario, especially in steel, oil chemicals, minerals and metals, railways, and portion of shipping and civil aviation, and considerable proportion of banking, insurance, and financial services.

(iii) Generation and distribution of energy is, by and large, with the state governments, except the Centre has lately entered the field marginally. The State Electricity Boards are almost universally under public section.

Performance

The growth of public sector has been quite appreciable in India. The public sector today controls 'the commanding heights' of industrial development, particularly in terms of the total investments in important sectors like steel and coal, cement and fertiliser, minerals and metals, petroleum and engineering goods industries. It has been estimated that:

(i) The steel groups, with INR 8000 crore cover the investment control nearly 15% of the total public sector investment, the highest for any individual sector.

(ii) The Coal India Ltd and its subsidiaries, controlling 11% of the total investment come next.

(iii) The others in this order are the residual coal sector and the engineering industries.

(iv) The first ten enterprises of the public sector under the Central Government control nearly 56% of the total investment.

Appraisal

Influences of public enterprises have been quite important and expansive:

(i) **Employment ratio and potential:** In public sector the capacity for labour absorption has been and continues to be stupendous. For example, in 2000 the railways were manned

by over 17 lakh people, numbering to nearly 70% employed in the organised sector.

(ii) **GDP:** The contribution of public sector to GDP has been considerable—raising from 7.5% in 1950–51 to 30% in 2000–01.

(iii) **Capital Formation:** The contribution to Gross Domestic Capital formation was only 3.5% in 1950–51 which rose to about 18% in the ninth plan.

(iv) **Savings:** After about 60 years of planning, the public sector contributes only about 24% to the nation's saving. The trend indicates a poor growth.

(v) **Capital Stock:** Capital stock, which represents installed capacity, has risen substantially over sixty years.

Problems of Public Sector

The main criticism of public sector is the lack of productivity and recurrent accumulation and annual losses. By and large, the service, manufacturing, and infrastructural units in India are not viable, mainly due to non-professionalism, external interference, and spendthrift recklessness.

The sustenance, prosperity, and development of the private sector and the rest of the economy today is closely dependent on how well the public sector functions. Castigating the public sector, however, is futile. The public sector, thus, has come to stay as an essential component of the economy. What is required is an objective analysis of what ails the public sector in India and the formulation and enforcement of remedial strategy.

Remedial Strategy

The future of a company or a corporation depends upon its profitability. If it has to depend on state subsidisation, its future is bleak.

Firstly, the unit should yield a surplus and profit.

Secondly, the unit should build such a surplus from one year to another. A viable public sector unit should generate a wholesale net

annual or periodical surplus to be ploughed partly back towards expansion, modernisation, and development.

The public sector has come to such a sorry pass due to the spread of the unfortunate *civil service culture*. In such a culture, routine, red tape, and financial indiscipline rule at the cost of professionalism and profitability. In the opinion of experts:

(i) There should be freedom from red tapism, and financial and administrative recklessness.

(ii) Such freedom and decision-making levels can be useful and pragmatic only where it is matched by appropriate accountability in terms of productivity and profitability and does not proliferate a civil service culture.

(iii) Bureaucratic inroads into professionalism should be curbed and the reins of public sector should be entrusted to the hands of technologists, economists, and specialists. These experts can improve the viability and profitability of the unit.

(iv) The principle of *autonomy with accountability* is to be followed strictly.

TOURISM

Making a holiday involving an overnight stay away from the normal place of residence is known as tourism. This is in contrast to recreation which involves leisure activities lasting less than twenty-four hours. This holiday may be based on cultural, historic, and social attractions of an urban centre, or on the appeal of different environment.

Tourism is based on the tertiary sector, providing services to the tourists. It is one of the most important emerging industry of India. Tourism in India is an important foreign exchange earner. Tourism not only fetches foreign exchange, it has great potential to provide employment to the trained, skilled, and unskilled workers. Tourism is recognised as a powerful engine for economic growth and employment generating in the country. The contribution of tourism to the country's GDP and total jobs was 5.92% and 9.24% respectively

during 2007–08. More importantly, tourism frames massive and continuous flows of people to more and more places of the country which ultimately leads to national integration and promotes international brotherhood.

India, a place of infinite variety, is fascinating with its ancient and complex culture, dazzling contrasts, and breathtaking physical beauty. Tourism in India is growing fast because of its cultural exuberance, and physical diversity. In fact, India is a vast country of great natural beauty and cultural diversity. India has a rich cultural heritage as superbly manifest in many of the architectural wonders like tombs, forts, palaces, temples, mosques, etc. With the great potential available and the development initiatives taken by the government, inbound tourism has shown a substantial growth in the last two decades. In terms of visitors, at present, India is the 10^{th} among Asia-Pacific countries. India's share in international tourist arrival was 5.6 million in 2010, constituting only 0.50% of the world tourists. The World Travel and Tourism Council has identified India as one of the foremost growth centres in the world in the coming decades. Domestic tourism is estimated to be much higher than that of international tourism which has been rising rapidly.

TOURISM IN INDIA

In India tourism has emerged as an instrument of employment generation, poverty alleviation and sustainable human development. During 2008–09, direct imployment in the tourism sector was estimated to be over 50 million. Tourism promotes national integration and international understanding and gives support to local handicrafts and cultural activities.

Tourism in India may be examined under the following headings:

1. Nature Tourism and Hill Stations

India has great geographical diversity, which resulted in varieties of nature tourism. Its snow-covered peaks, waterfalls, valleys, gorges, high altitude lakes, and cliffs, provide enough attraction for the

nature-loving tourists. India has numerous waterfalls in the Himalayas, Chotanagpur Plateau, and Western Ghats, including Jog Falls (the highest in India). The backwaters of Kerala, hill stations, national parks, natural vegetation in wilderness, and biosphere reserves are also fascinating and attract tourists in large numbers.

2. Hill Stations

The mountainous and hilly areas of India are dotted with places of tourist interest. The most famous hill stations have been shown in (**Fig. 11.21)**. Several of these hill stations have served as summer capitals of Indian provinces, princely states, or, in the case of Shimla, of British India itself. Since Indian Independence, the role of these hill stations as summer capitals has largely ended, but many hill stations remain popular summer resorts. Most famous hill stations of India are: Almora, Abu, Amarnath, Amarkantak, Chamba, Dalhousie, Darjeeling, Dharamshala, Gangtok, Gulmarg, Kasauli, Kodaikanal, Kullu, Mahabaleshwar, Lavasa, Manali, Matheran (Maharashtra), Mussoorie, Nainital, Ooty (Tamil Nadu), Pahalgam, Panchmarhi, Ranchi, Ranikhet, Shillong, Shimla, Sonmarg, Udhagamandalam, Yeusmarg, etc.

3. Historic Monuments and Archaeological Sites

India has a large cultural heritage. Its cultural, historical, and archaeological centres are of great interest for the domestic and international tourists. Places like Ajanta, Ellora, and some of the important historical monuments like Taj Mahal (one of the seven wonders of the world), Golden Temple, Qutab Minar, Red Fort, Jama Masjid, Humayun's Tomb, Char-Minar, Birla Mandir, the Victoria Terminus (Mumbai), Rashtrapati Bhavan, Gateway of India, India Gate, Lotus Temple (New Delhi), etc., attract a number of tourists.

4. Cultural and Religious Tourism

The main places of cultural tourism have been shown in (**Fig. 11.22)**. Places of cultural tourism include Allahabad, Ajmer, Amarnath, Ayodhya, Badrinath, Bhadrachalam, Bhubaneshwar, Bodh Gaya, Dwarka, Gangotri, Guwahati, Hardwar, Kanchipuram, Kedarnath, Kochi, Madurai, Mahabalipuram, Mathura, Nashik, Pushkar, Rajgir,

Rameswaram, St. Paul's Cathedral (Kolkata), Somnath, Tirupati, Ujjain, Vaishali, Varanasi, Vaishnodevi, Victoria Memorial (Kolkata), Yamunotri, etc.

5. Sea Beaches

The tourist beaches of India are: (i) beaches of Puri in Odisha, (ii) beaches of Digha in West Bengal, (iii) beaches of Goa, (iv) Kollam and Kovalam beach in Kerala, (v) Marine beach in Chennai,
(vi) beach temples of Mahabalipuram, (vii) beaches of Andaman and Nicobar Islands, (viii) beaches in Mumbai, (ix) Gopalpur (Odisha) (x) beaches of Lakshadweep, and (xi) beaches of Diu.

6. Adventure Tourism

India has enormous potential for adventure tourism. For example: (i) river rafting and kayaking in Himalayas, (ii) mountain climbing in Himalayas, (iii) rock climbing, (iv) skiing in Gulmarg and Auli, (v) boat racing in Kerala, (vi) paragliding in Maharashtra, etc.

Fig. 11.21 Hill Stations and Beaches

Tourism in India has great relevance to regional economic development. Since Independence, Indian tourism, especially the number of foreign tourists, has grown considerably as given in (**Table 11.7)**.

Fig. 11.22 Historical, Archaeological, and Religious Tourist Centres

Table 11.7

Foreign Tourists in India

Year	Number of Foreign tourists
1951	16,830

Year	Number of Foreign tourists
1961	140,000
1971	301,000
1981	1,280,000
1991	1,670,000
2001	2,540,000
2010	5,580,000
2017	over 10 million

Source: India 2017

It may be observed from (**Table 11.7)** that in 1951 the total number of foreign tourists was only 16,830 which grew to 2,450,000 in 2006, an increase of more than about 150 times. Most of the foreign tourists who visited India were from West European countries (30%), South Asia (26%), North America (20%), South East Asia (6%), East Asia (5%), West Asia (4%), Africa (4%), Australia (3%), and East-European countries (2%).

The world Travel and Tourism, council calculated the contribution of tourism as 9.2% of India's GDP in 2018. It supports 42.673 million jobs i.e 8.1% of total employability. There has been an increase of 16.5% from 2011.

Problems of Indian Tourism Industry

The tourism industry in India is confronted with many problems Some of the problems of the tourist industry are given below:

(i) Lack of adequate infrastructure (transport, banking, and hotels);
(ii) Complex visa formalities;
(iii) Multiplicity of taxes;
(iv) Problem of law and order in some of the regions of the country like Jammu and Kashmir, and the states of North East India;
(v) Safety and security of the tourists;
(vi) Inadequacy of qualified tourist guides; and

(vii) Absence of participation of the people.

Despite all these shortcomings and problems, India has great potential for tourism development. The World Tourism and Travel Council (WTTC) estimated that India's travel and tourism potential can provided a substantial resource to economy (INR 500,000 crores to GDP) by 2010. The World Tourism and Travel Council has suggested the following four-fold plan of action to achieve the potentials of tourism:

(i) Make travel and tourism a strategic, economic, and employment priority.

(ii) Move towards open and competitive markets including civil aviation liberalisation.

(iii) Pursue sustainable development.

(iv) Eliminate barriers to growth of tourism, i.e., provide incentives to the private sector and invest in human resources.

ECO-TOURISM

Eco-tourism, also known as ecological tourism, is a form of tourism that appeals to the ecological and socially-conscious individuals. *Generally*, *eco-tourism focuses on volunteering, personal growth, and learning new ways to live on the planet, typically involving travel to destinations where flora, fauna, and cultural heritage are primary attractions.*

Much of what goes on in the name of eco-tourism is business as usual, albeit with a few peripheral changes like notices in your room requesting you to re-use the towel. Some visits are called eco-tourism simply because they take tourists to ecologically interesting areas, such as national parks. Most principles of genuinely sensitive tourism, developed internationally over the last years, are ignored.This includes carrying out assessments of the ecological impact of tourism and whether it actually benefits the local people or not.

According to some experts, eco-tourism is responsible tourism. Responsible eco-tourism includes programmes that minimise the negative aspects of conventional tourism on the environment, and

enhance the cultural integrity of people. Therefore, in addition to evaluating environmental and cultural factors, an integral part of eco-tourism is in the promotion of recycling, energy efficiency, water conservation, and creation of economic opportunities for the local communities.

Ideally, eco-tourism should satisfy several criteria, such as:

(i) Conservation of biological diversity and cultural diversity through ecosystem protection.

(ii) Promotion of sustainable use of biodiversity, by providing jobs to local populations.

(iii) Sharing of socioeconomic benefits with local communities and indigenous people by having their informed consent and participation in the management of eco-tourism enterprises.

(iv) Tourism to unspoiled natural resources, with minimal impact on the environment being a primary concern.

(v) Minimisation of tourism's own environmental impact.

(vi) Affordability and lack of waste in the form of luxury.

(vii) Local culture, flora and fauna being the main attractions.

For many countries, eco-tourism is not simply a marginal activity to finance protection of the environment but a major industry of the national economy. For example, countries like Costa Rica, Ecuador, Kenya, Madagascar, and Nepal represent eco-tourism as a significant industry adding good revenue to their gross domestic product by playing a major role in their economic activity.

Growth and Development of Eco-tourism

Eco-tourism, responsible tourism, and sustainable development have become prevalent concepts since the late 1980s, and eco-tourism has experienced arguably the fastest growth of all sub sectors in the tourism industry. The popularity represents a change in tourist perceptions, increased environmental awareness, and a desire to explore natural environments. Such changes have become a statement affirming one's social identity, education, sophistication,

and disposable income as they are about preserving the Amazon rainforest or the Caribbean coral reef for posterity.

The Niti Aayog (Govt. of India) has identified nine islands in Andaman and Nicobar and Lakshadweep to be developed as eco-tourist destinations for Singapore like tourism boost. The names of the selected islands are 1. Smith Island, 2. Rose Island, 3. Aves Island, 4. Long Island, and 5. Little Island in Andaman and Nicobar Islands, and 6. Bangaram Island, 7. Thinrakara Island, 8. Suheli, and 9. Cheriyam Island and Minicoy Islands in Lakshadweep Islands **(Fig. 11.23)**.

In addition to infrastucture, there will be the development of water-parks to promote tourism. The biggest island will be little Island of Nicobar (Times of India 25.9.16).

With its great potential for environmental protection, the United Nations celebrated the International Year of Eco-tourism in 2002.

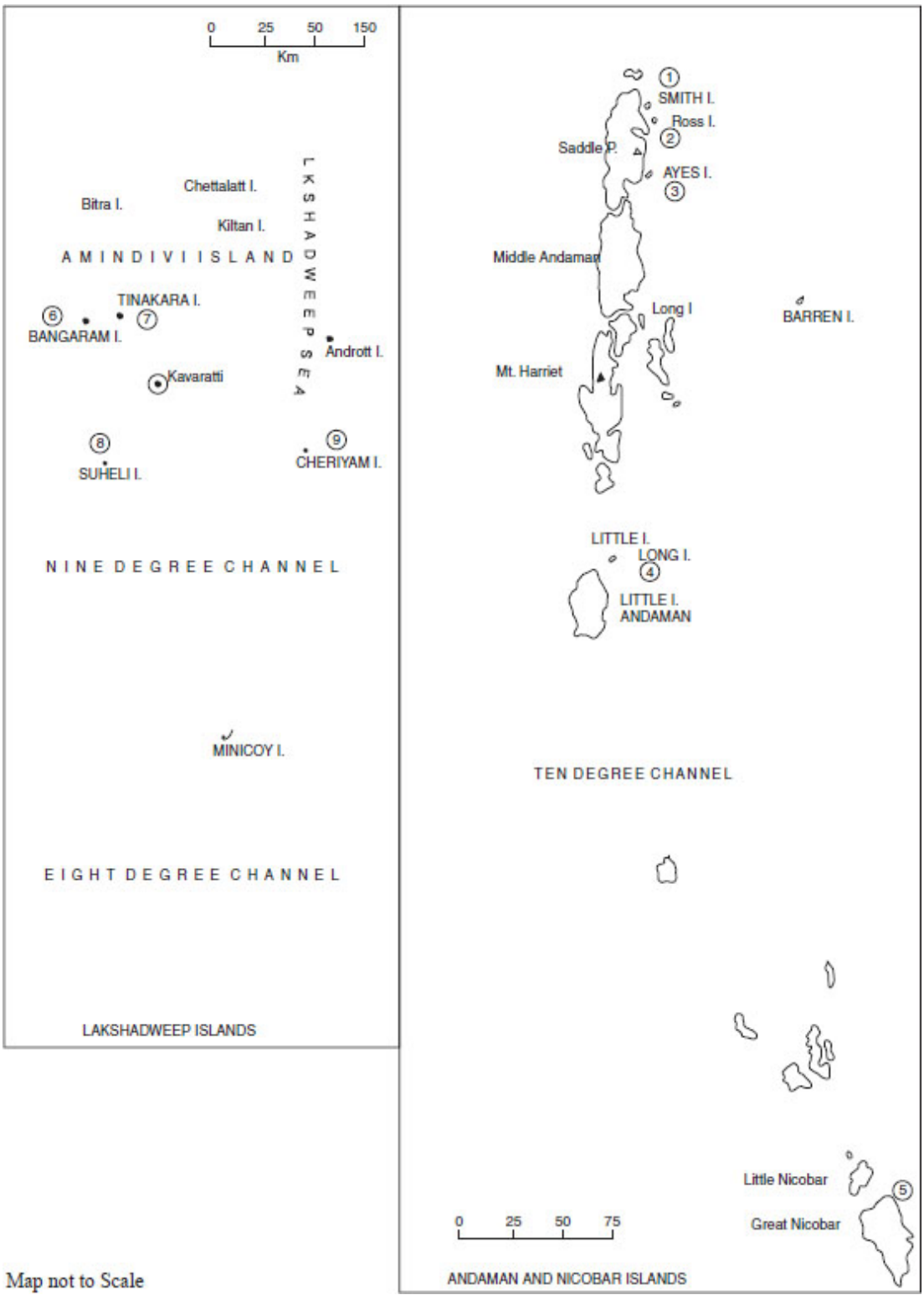


Fig. 11.23 Nine Indian Islands in Andaman & Nicobar and Lakshadweep selected for Singapore like Tourism

Definitional Problems of Eco-tourism

To approach an understanding of the problem, a clear definition must delineate what eco-tourism is, and what it is not. Ideally, eco-tourism satisfies several general criteria, including the conservation of biological diversity and cultural diversity through ecosystem protection, promotion of sustainable use of biodiversity, share of socioeconomic benefits with local communities through informed consent and participation, increase in environmental and cultural knowledge, affordability and reduced waste, and minimisation of its own environmental impact. In such ways, it contributes to the long-term benefits to both the environment and local communities.

However, in the continuum of tourism activities that stretch from conventional tourism to eco-tourism proper, there has been a lot of contention to the limit at which biodiversity preservation, local socio-economic benefits and environmental impact can be considered “eco-tourism”. For this reason, environmentalists, special interest groups and governments define eco-tourism differently. *Environmental organisations have generally insisted that eco-tourism is nature-based, sustainably-managed, conservation-supporting, and environmentally-educated*. The tourist industry and governments, however, focus more on the product aspect, treating eco-tourism as equivalent to any sort of tourism based in nature. As a further complication, many terms are used under the rubric of eco-tourism, and others have been used in literature and marketing. Although they are not necessary synonymous with eco-tourism.

The problems associated with defining eco-tourism have led to confusion among tourists and academics alike. Definitional problems are also subject of considerable public controversy, a trend towards the commercialisation of tourism schemes disguised as sustainable, nature-based, and environmentally-friendly eco-tourism. *According to McLaren, these schemes are environmentally destructive, economically exploitative, and culturally insensitive at its worst*. They are also morally disconcerting because they mislead tourists and manipulate their concerns for the environment. The development and

success of such large scale, energy intensive and ecologically unsustainable schemes are a testament to the tremendous profits associated with being labelled as eco-tourism.

COMMUNITY ECO-TOURISM INITIATIVE

Jammu & Kashmir and Ladakh

In Ladakh, several villages have initiated home stay programmes for trekkers and other tourists, with funds going back into conservation and village development, in the Sham area, the Hemis National Park, Zanskar region, and Tso Moriri Lake.

Nagaland

Khonoma village, close to Kohima, is the site of Green Village Project, set up by the Maharana Kumbha Common Interest Group (CIG), with nine villagers from families below the poverty line. The youth were trained in visitor management. Looking quite simple from outside, the tents inside are almost luxurious, beautifully done up with local designs. The cost is at about INR 2500 per night. The camp offers camel and horse riding, hiking, folk dances, and other activities, but its biggest attraction is the serenity of its environs.

Kerala

In the Periyar Tiger Reserve, forest officials used the Ecodevelopment Project to establish a unique eco-tourism programme by local *adivasis*, employing amongst others several hardcore poachers.

Sikkim

Two community eco-tourism initiatives have been started with the help of the Ecotourism and Conservation Society of Sikkim (ECOSS). At Dzongu, close to the Khangchendzonga Biosphere Reserve, the Lepcha-community-run Dzongu Eco-tourism Committee manages trekking and other tourism. The second is Pastanga village, close to Gangtok, where tourism based on traditional Sikkimese village life is run by the local community NGO Khedi Ecotourism and Eco-development promotion.

West Bengal

In Bali village in Sundarbans, local people have started the Bali Nature and Wildlife Conservation Society with support of Help Tourism, association for Conservation and Tourism, and Worldwide Fund for Nature India. The society runs a “Community Tourism Demonstration Model Camp” that employs 22 local people and benefits over fifty families and seven other villages.

THE NATIVE AND PSYCHO-SOCIAL IMPACT OF ECO-TOURISM

Some of the negative impacts of eco-tourism are:

1. Commercialisation

Eco-tourism is a trend towards commercialisation of tourism throughout the nation. This trend has become one of the fastest growing sectors of tourism industry growing annually 10–15% worldwide. Many of the eco-tourism projects are, however, not meeting these standards. Even if some of the guidelines are being executed, the local communities are still facing other negative impacts. For example, South Africa is reaping significant economic benefits from eco-tourism, but negative effects including physical displacement of persons, gross violation of fundamental rights, and environmental hazards far out weight the medium-term economic benefits.

2. Direct Environmental Impact

Eco-tourism operations typically fail to live up to conservation ideals. It is often overlooked that eco-tourism is a highly concentrated activity, and that environmental conservation is only means to further economic growth.

3. Problem of Garbage and Pollution

Apart from environmental degradation with tourist infrastructure, population pressure from
eco-tourism also leaves behind garbage and pollution associated with the Western lifestyle. Although eco-tourists claim to be educationally sophisticated and environmentally concerned, they rarely understand

the ecological consequences of their visits and how their day-to-day activities have physical impacts on the environment. The eco-tourists rarely recognise the great consumption of non-renewable energy required to arrive at their destination, which is typically more remote than conventional tourism destinations. For instance, an exotic journey to a place 10,000 km away consumes 100 litres of fuel per person.

4. Impact on Fauna and Flora

The eco-tourists disturb fauna and flora. Eco-tourists believe that because they are only taking pictures and leaving footprints, they keep eco-tourism sites pristine, but even harmless sounding activities such as a nature hike can be ecologically destructive. Where the eco-tourism activity involves wildlife viewing, it can scare away animals, disrupt their feeding and nesting sites.

5. Encroachment on Virgin Land

While the term tourism may sound relatively benign, one of its most serious impact is its consumption of virgin territories. These invasions often include deforestation, disruption of ecological life system, and various forms of pollution, all of which contribute to environmental degradation.

6. Heavy Traffic in Ecologically Fragile Areas

To generate revenue you have to have a high number of traffic of tourists, which inevitably means a higher pressure on the environment.

7. Adverse Effect on Local People

Most forms of eco-tourism are owned by rich investors and corporations that provide few benefits to local communities. An overwhelming majority of profits are put into the pockets of investors instead of reinvestment into the local economy or environmental protection. The limited number of local people who are employed in the economy enter at its lowest level, and are unable to live in tourist areas because of meager wages.

8. Trade of Wild Animals

The presence of affluent eco-tourists encourage the development of destructive markets in wildlife souvenirs, such as the sale of coral trinkets and animal products, contributing to illegal harvesting and poaching from the environment.

9. Threat to Indigenous Cultures

Eco-tourism often claims that it preserves and enhances local culture. However, evidence shows that with the establishment of protected areas local people have illegally lost their homes, and most often with no compensation. Pushing people onto marginal lands with harsh climates, poor soils, and lack of water; lands which are infested with livestock and disease does little to enhance livelihoods even when a portion of eco-tourism profits are directed back into the community.The establishment of parks can create harsh survival realities and deprive people of their traditional use of land and natural resources. The local people struggle for cultural survival and freedom of cultural expression while being *observed* by tourists. Eco-tourism has been allowed to develop with virtually no control. Too many lodges have been built, too much firewood is being used, and no limits are being placed on tourism vehicles.

10. Mismanagement of Eco-tourism Sites

While governments are typically entrusted with the administration and enforcement of environmental protection, they often lack the commitment or capability to manage eco-tourism sites effectively. Government regulatory agencies, as political bodies, are susceptible to making decisions that spend budget on politically beneficial but environmentally unproductive projects. Finally, influential groups can pressure and sway the interests of the government to their favour.

Improving Sustainability

Looking at the disadvantages of eco-tourism, a strategy is to be designed to make eco-tourism sustainable. Some of the steps which may help in making eco-tourism sustainable are as under:

1. Environmental Protection Strategy

An environmental protection strategy must address the issues of eco-tourists removed from the cause and effect of their actions on the environment. More initiatives should be carried out to improve their awareness, sensitise them to environmental issues, and care about the places they visit.

2. Qualified Eco-tourist Guides

Tour guides are an obvious and direct medium to communicate awareness. With the confidence of eco-tourists and intimate knowledge of the environment, they can actively discuss conservation issues. A tour guide training programme can go a long way in improving the quality of guides and ultimately making the eco-tourism sustainable.

3. Small Scale Eco-tourism

There is a need to develop eco-tourism on a small scale. Local people have a vested interest in the well being of their community and are, therefore, more accountable to environmental protection than multinational corporations. The lack of control, westernisation, adverse impact to the environment, and loss of culture and traditions outweigh the benefits of establishing large-scale eco-tourism.

4. Participation of Local People in Eco-tourism

Involvement of local people is imperative for making the eco-tourism sustainable. The increased contribution of communities to locally managed eco-tourism create viable economic opportunities, including high level management positions, and reduce environmental issues associated with poverty and unemployment.

5. Establishment of Training Centres for Guides

Qualified guides and helpers can help in overcoming many of the setbacks of eco-tourism by providing the necessary information about the fragility of ecosystems to the eco-tourists.

If the above steps are taken together, the problems of eco-tourism can be reduced substantially.

INDUSTRIAL HOUSES IN INDIA

India today has vibrating industries, their number running in thousands. Some industrial houses in India have played a pivotal role in the industrialisation of the country which was an agrarian economy primarily till mid 20th century. Many of the industrial houses have existed since the pre-independence days, yet there are some which have come up in the post-independent India and have played praiseworthy roles in the direction of industrial development in the country. As new opportunities surfaced, many of these industrial houses went for successful diversification too. We may have a precise look at ***some of the important industrial houses*** which are considered not only harbingers of industrialisation in India but also ***motivational*** examples of ***entrepreneurship development*** in the country:

1. The Tatas

Being one amongst the oldest industrial houses in India this industrial house has iron and steel as well as automobile (heavy) as its traditional areas of interest. But in the coming times it has diversified in a great many areas/sectors such as pharmaceuticals, cement, publishing, power, finance, hotel, insurance, software, refrigeration, air-conditioning, telecommunication, soft drinks, consumer non-durables, etc. After its recent acquisition of the Chorus iron & steel company, it has catapulted on international horizon overnight. Its Tata Consultancy Services, a software development and export company, is the largest software services companies in the country. Its recent foray into the four-wheel segment has been able to make a historic news around the world with the launching of the USD 2500 car—the ***Nano.***

2. The Birlas

The Birlas, though they have gone for divisions, are among the oldest industrial houses in India like the Tatas. Their traditional areas of activities have been textile, paper, paraffin, cement, aluminium, and automobiles. In the last few decades, they have diversified in other areas too such as machine tools, pharmaceuticals, telecommunications, consumer durables and non-durables, etc.

3. The Singhanias

Among the oldest industrial houses of India, this has business interests in the areas of textile, pharmaceuticals, tyres and tubes, and consumer durables and non-durables.

4. The Goenkas

One among the oldest industrial houses of the country, this has business interests in the areas such as power generation and distribution, textile, pharmaceuticals, machine tools, entertainment, etc.

5. The Modis

The Modis have been among the oldest of the industrial houses in the country with their industrial interests in the areas such as textiles, garments, carpets, tyres & tubes, hospitality, pharmaceuticals, etc.

6. The Bajajs

Among the oldest business families of India, they are known around the world for their commendable product the Bajaj Scooter, which is competent enough to fight out the international competition in the segment. Of late, they diversified into the areas of bikes also. Their areas of activity expands to electric and home appliances, entertainment, etc. Recently, they announced their intentions of entering the automobile sector with one of the cheapest cars in the world (after the Tata's Nano).

7. The Srirams

Among the oldest industrial houses of India, the interest of this house runs across a broad spectrum of industries—textiles, garments, fertilisers, electric and home appliances, generator sets, hospitality, pharmaceuticals, etc.

8. The Oberois

Among the oldest of the industrial houses in India, they are considered as the doyen of hospitality industry in the country. They run one of the finest hotels in the world across a number of countries.

9. The Escorts

One among the oldest industrial houses, this organisation has been popular in the country for its contributions in three areas specially—manufacturing the most popular and lone motor-bike brand “Rajdoot”, a heart-oriented hospital, and the “Eicher” brand of tractor.

10. The Krisloskars

A highly reputed and amongst the oldest industrial houses in India, this establishment has enough contributions in the areas of heavy and light machine tools industries, with speciality in the areas of locomotion. It is credited to produce the first and the reputed ‘diesel pump sets’ in the country. It has played an important role in farm mechanisation by producing the ‘first’ and the most trusted ‘tractors’ in India. The company produces farm implements also. It has entered the fast-growing automobile industry with Toyota, the second largest car-maker in the world.

11. The Firodias

Being one of the oldest industrial houses of the country, the company has made important contributions in the areas of automobile, two-wheelers, generator sets, etc.

12. The UB Group

This group is amongst the oldest and the leading industrial houses of India with its traditional interest in the alcoholic drinks sector. Of late, it has also started diversifying in the areas such as airlines, infrastructure, hospitality, and real estate segments.

13. The Sarabhais

This is one of the important industrial houses in the country. The company has led the drug and pharmaceutical industries in India with a reputation well-established around the world.

14. The Jagatjit Group

This is amongst the oldest industrial houses of India and its focus remains the liquor industries.

15. The Godrejs

Amongst the top, oldest, and reputed industrial houses of the country, this organisation has contributed in the areas of detergents,

refrigeration, almirahs, furniture, air-conditioning, and lock industries providing highest of the standards.

16. The Reliance Group

Though the group is now divided between two brothers, the industrial house is considered a rags-to-riches story, and represents the rise of a new entrepreneurial class in the country. It has paraffin and synthetic yarn as its traditional interests; today it has diversified in more than a dozen traditional and new industrial areas such as power, petroleum, textile, garments, hospitality, telecommunication, pharmaceuticals, software, banking, insurance, etc.

17. The Wipro Group

The doyen of sunrise industries in the country, the group functions in the areas of software development and export including the production of computer peripherals. This is one of the leading companies in the areas of business process outsourcing (BPO) and is also seriously involved in social work. This company is amongst the top names in the software around the world.

18. The Infosys Group

This is another big name in the areas of electronics and software development, which is similar to Wipro with a stronghold in the areas of business process outsourcing. It has its serious aims in the areas of education and is committed to social work.

19. The Biocon Group

This is a big name in the areas of new industries and is considered to be the torchbearer in the biotechnology segment.

20. The Bhartis

This is a typical by-product of the economic reform process in the country. The industrial house catapulted into popularity with its ‘Airtel’ brand in the area of mobile telephony. Now the group is diversifying in the areas of agriculture, hospitality software, insurance, internet, trading (with *Walmart*), etc.

21. The Ranbaxy Group

This is among the leading drug and pharmaceutical industries of India, having a reputation of international standards. The industrial house has been an inspiration for a great number of new drug-makers in the country.

22. The Hamdard Group

This industrial house is credited to preserve and propagate the Unani tradition of medicines in India. The largest Unani certified company with an ISO 9001:2000 certificate, it is popular around the world for its beverage *(sherbat), Rooh Afza* and child tonic, *Sinkara*.

23. The Arya Vaidyashala

This is among the oldest Ayurvedic prepratory medicine-makers of India. It has been the harbinger of the Ayurvedic tradition and its preserver, too. It runs a world-renowned ayurvedic-and- naturopathy-based hospital also.

24. The Dabur Group

This is another harbinger of the Ayurvedic tradition in the country. The company, of late, has diversified in the areas of consumer non-durables such as fruit-based drinks, toilet soaps, shampoo, hair oil, etc.

25. The Baidyanath Group

This is another renowned, old, and well-established Ayurvedic preparatory medicine-maker of India. This industrial house has a great name in the field of entrepreneurship while preserving the tradition of the Ayurveda in the country.

26. The Jaypee Group

This is one of the major real estate and construction companies of India. Jaypee has the reputation of successfully completing major road and bridge projects in the stipulated time frame.

27. The Ansals

The Ansals are one of the oldest real estate developers of India. The activities run across all the major urban settlements in the country.

28. The Arvind Mills

This is one of the oldest textile industries of India. The group has earned a reputation of producing one of the finest and the cheapest cotton clothes and garments in the world. A leading garment and textile exporting company, it has the credit of supplying the Jeans cloth to almost all major brands of the World. The company is the largest Jeans cloth maker.

29. The Bombay Dyeing Group

This is a highly reputed brand in suiting-shirting and cotton cloth. The company is known around the world for manufacturing the finest as well as the cheapest bed-sheets and towels with a market grip over the Euro-American countries.

30. The ACC Group

This is the largest and the most trusted cement manufacturing company of India. The company (Associated Cement Company) is basically a conglomerate of many cement manufacturers in the country selling their product under one brand name. They have the credit of promoting the very construction works of the whole country.

31. The Pioneer

One of the most reputed industrial houses in the area of seed production, they get the credit for promoting the hybrid seed-based farming in India, thus playing a great role in the Green Revolution. The company also produces various pesticides and herbicides.

32. The Pantnagar Group

This is amongst the reputed names in seed production. The company propelled the Green Revolution and new farming in the country. It also produces different kinds of pesticides and herbicides required by the agriculture sector.

INDUSTRIAL COMPLEXES

The industrial complexes are made up of a set of specific industries, which have important production, marketing, and other interrelationships among them, strengthened by their togetherness and innovations. These complexes are geographically localised.

Developed as an important tool for development planning, the industrial complexes gained popularity in the wake of vigorous process of industrialisation and planning, which started in the developing world under the Soviet influence, taking immediate clues from the planned industrial development of France. The idea is associated with the concept of 'the growth pole' (Francois Perroux, 1955) and 'the growth centre' (J.R. Boudeville, 1966).

Once the governments of the planned economies started developing the industrial complexes as part of their conscious policy, there developed the concept of 'industrial complex analysis'. Developed by W.Isard, industrial complex analysis is a method of analysing the *linkages* between industries in an industrial complex with the objective of identifying the type of industrial complex which would be most suitable for a given development plan or region—it is based on inter-industry relations of *input-output analysis* and *comparative cost analysis.*

In the industrial complexes, the localised industries do significant purchases from each other using one's output as another's input, saving the cost of transportation and assuring quality as well due to immediate feedbacks. As a result, not only these industries grow faster but the whole industrial region grows too, via the *spillover effect* and the *multiplier effect,* ultimately benefitting the whole economy through the *trickle down effect.*

Industrial Complexes in India

We see thousands of such industrial complexes in India, which have been developed in all the ***major*** and ***minor*** industrial regions. There are vibrating iron and steel based industrial complexes in the vicinity of iron and steel plants in the country. Similarly, petro-complexes, pharma complexes, hosiery complexes, garment complexes, electronics complexes, etc., with their strengths and strongholds in the specific industries are dispersed around the country, and developed around the important *vibrant* and *dynamic* major industries functioning as *growth poles* and *growth centres*.

In the era of economic reforms, the ongoing process of setting up such complexes are being continued by the central as well as the state governments. The Export Processing Zones (EPZs), Export Oriented Units (EOUs) of past, and the Special Economic Zones (SEZs) of present are among the best examples of the industrial complexes in India. The Technology Parks, Hardware Parks, Software Parks, Biotechnology Parks, etc. are all the examples of the industrial complexes in India. Till the process of planning remains relevant, the industrial complexes are going to be relevant in India. To the extent private sector industries are concerned, the idea is not different.

Conceived as the catalysts of growth and development, the industrial complexes in India faced the following problems hampering their expected functioning in general:

1. Difficulties in selecting a suitable industry around which the industrial complexes could be grown.
2. Frequent sicknesses of industries, which leads to breaking down the chain of *inter-linkages* hampering the proper growth of the industrial complexes.
3. Infrastructural facilities in general and power in particular have done a great damage to the prospects of the complexes.
4. Underdeveloped and lopsided market of the country has also hampered proper growth and development of industrial complexes.
5. Proper support of external sector has also been one major problem.
6. Lack of skill and entrepreneurial acumen together with the problem in the availability of adequate and timely finance have hampered the prospects of industrial complexes in India.
7. Also, the labour laws have not been conducive to the growth of industrial complexes.
8. Last but not the least, the industrial complexes also suffered

because the selection and location of the growth poles and growth centres in India has not been smooth.

Now as the economy is going for a process of reforms, many of these problems are being attended by the government with differing levels of success. As the process of economic reforms gets success, we may hope for the success of the industrial complexes in India, too.

Make in India

The Prime Minister of India launched the 'Make in India' global initiative on 25th September, 2014 to invite both domestic and foreign investors to invest in India. The initiative was simultaneously launched in the capital of all states and several Indian Embassies/High Commissions where time-zones permitted. Few other Indian Embassies also organized 'Make in India' interaction after launch.

The 'Make in India' initiative is based on four pillars, which have been indentified to give boost to entrepreneurship in India, not only in manufacturing but also other sectors. The four pillars are:

1. New Processes: 'Make in India' recognizes 'ease of doing business as the single most important factor to promote entrepreneurship.
2. New Infrastructure: Government intend to develop industrial corridors and smart cities, create world class infrastructure with state-of-the art technology and high speed communication.
3. New Sectors: 'Make in India' has identified 25 sectors in manufacturing, infrastructure and service activities and detailed information is being shared through interactive web-portal and professionally developed brochures.
4. New Mindset: In order to partner with industry in economic development of the country, Government shall act as a facilitator and not a regulator.

The programme 'Make in India' highlights Government's commitment to chart out a new path wherein business entities are

extended red carpet welcome in a spirit of active co-operation.

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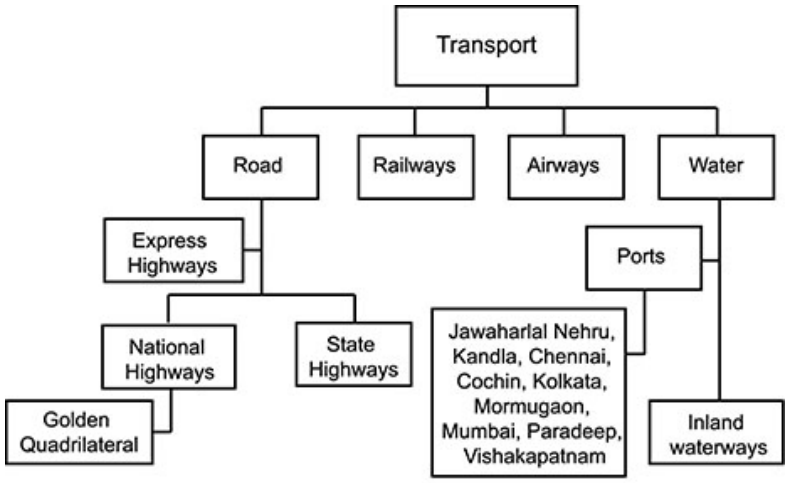
12

Transport, Communications and Trade



TRANSPORT

Provisions of quality and efficient infrastructure services are essential to realise the full potential of growth impulses surging through the economy. In fact, a well-knit and coordinated system of transport plays an important role in the sustained economic growth of a country. The present transport system of India comprises several modes of transport including (i) road, (ii) rail, (iii) inland waterways, (iv) coastal shipping (coastal and international), and (v) airways. Transport has

recorded a substantial growth over the years both in the spread of its network and in the output of its system. The Ministry of Transport is responsible for the formation and implementation of policies and programmes for the development of various modes of transport except the railways and the civil aviation.

Roads

India is second largest road networks in the world with an aggregate distance of 4.3 million km (India 2016, p. 789). Roads in India connect village to village and village to urban centres. Moreover, roads offer door-to-door service and their construction can be undertaken even in areas of difficult terrain and steep slopes. The movement of goods is safer through road transport. They help the farmers to move their perishable commodities (flowers, fruits, milk, and vegetables) to the urban markets. The role of roads in the economic development and regional planning cannot be underestimated. India's National Highways constitute about 2% of road network (70,000 km).

The country's road network consists of: (i) Expressways (1000 km), (ii) National Highways (131,899 km), (iii) State Highways, (iv) Major District Roads, (v) Other District Roads, and (vi) Village Panchayat Roads. The road network comprises 122,434 kms of National Highways and Expressway, 147,800 km of State Highways, 4,455,000 km of Major District Roads, and Rural Roads. NH–44 is the longest national highway in India with a length of 3745 km. It connect Srinagar to Kanyakumari.

Out of these, the National Highways and State Highways together account for about 244,015 km
length. Though, the National Highways—the construction and maintenance of which is the responsibility of the Central Government—has about 66,590 km of length and comprises only about 2% of the total length of roads, they, however, carry over 40% of the total traffic across the country.

About 65% of freight and 80% of traffic is carried by the roads. The pressure on the road network is increasing day by day. The number of vehicles have been growing at a rapid pace of 10.50% per annum over the last five years. The rapid expansion and strengthening of the road network, therefore, is imperative both to provide for present and future traffic, and for improved accessibility to the hinterland. In addition, road transport needs to be regulated for better energy efficiency, lesser pollution, and enhanced road-safety.

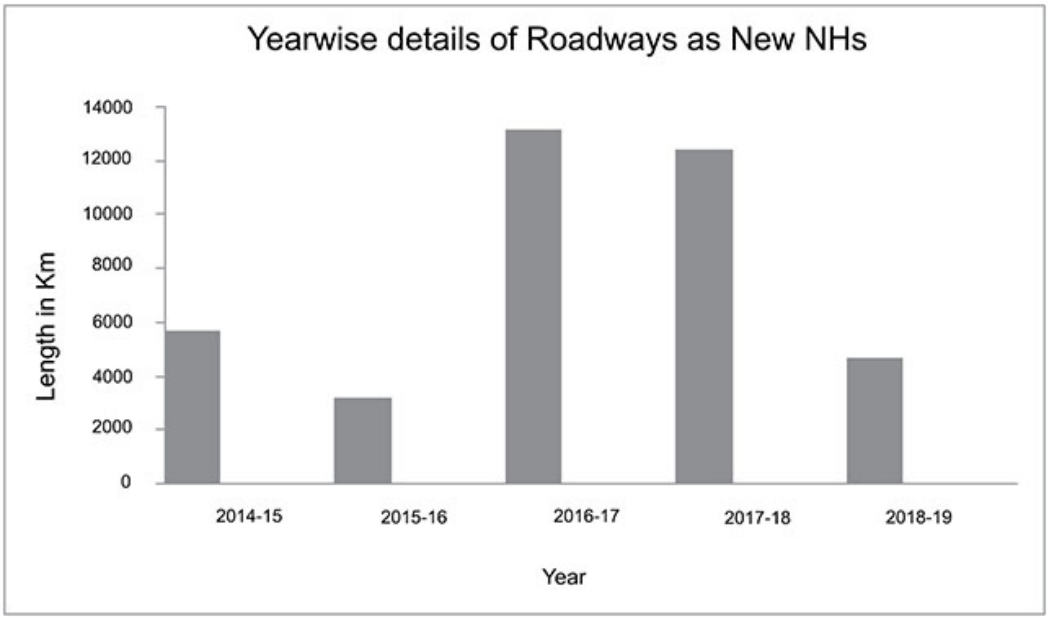
Border Roads Organisation (BRO): The Border Roads Organisation was established in 1960 for the development of roads of strategic importance in the northern and north-eastern borders of the country. The Border Roads Organisation has not only linked the border areas of the North and North-East with the rest of the country, but has also developed the road infrastructure in Bihar, Maharashtra, Karnataka, Rajathan, Andhra Pradesh, Andaaman and Nicobar, and

Chhattisgarh. It has also constructed roads and air fields in Tajikistan, Afghanistan, Bhutan and Myanmar. So far it has constructed more than 35,000 km of roads and surfaced 40,500 km of roads. The Zojila-Kargil and Manali-Leh roads were completed in 1998. The Pathankot-Jammu-Srinagar National Highway is also maintained by the Border Roads Organisation.

Highways and Roads

1. **National Highways:** The Central Government is responsible for the development and maintenance of the National Highways System. The National Highways of India have been shown in **(Fig. 12.1).** The total length of the National Highways in 2017 was 115,435 km. The Ministry is carrying out development and maintenance work of the National Highways through three agencies: (i) the National Highways Authority of India (NHAI), (ii) the State Public Works Department (PWD), and (iii) the Border Road Organisation (BRO). During FY 2018–19, Rs 71,000 crore have been sanctioned for the development of national highways.

The total length of National Highways (NHs) has been enhanced from about 91,287 km as on 31 March 2014 to about 131,326 km at present in view of declaration of about 39,040 km length of state roads as new NHs.



Source: Press Information Bureau, Govt. of India, Ministry of Road Transport and Highways 2018

In order to give boost to the economic development of the country, the Government has embarked upon massive National Highways Development Projects (NHDP) being implemented by the National Highways Authority of India (NHAI). Some of the important National Highways Projects under progress are given below:

Golden Quadrilateral (GQ): The National Highways Development Project (NHDP) has taken up a massive programme of road building in the country. Launched in January 1999, the Golden Quadrilateral Project is perhaps one of the *largest* projects of road building in the country with a length of 5846 km **(Fig. 12.2).** The project is being implemented by the National Highways Authority of India (NHAI). The National Highways Development Project has the following components:

(i) ***Phase I—Golden Quadrilateral:*** This phase comprises connecting Delhi-Mumbai, Chennai, and Kolkata-Delhi by six-lane super highways. It has a total length of 5846 km. The four sides of the quadrilateral have varying length.

Golden Quadrilateral Corridor and its Length

Delhi-Mumbai	1419 km
Mumbai-Chennai	1290 km
Kolkata-Chennai	1684 km
Delhi-Kolkata	1453 km

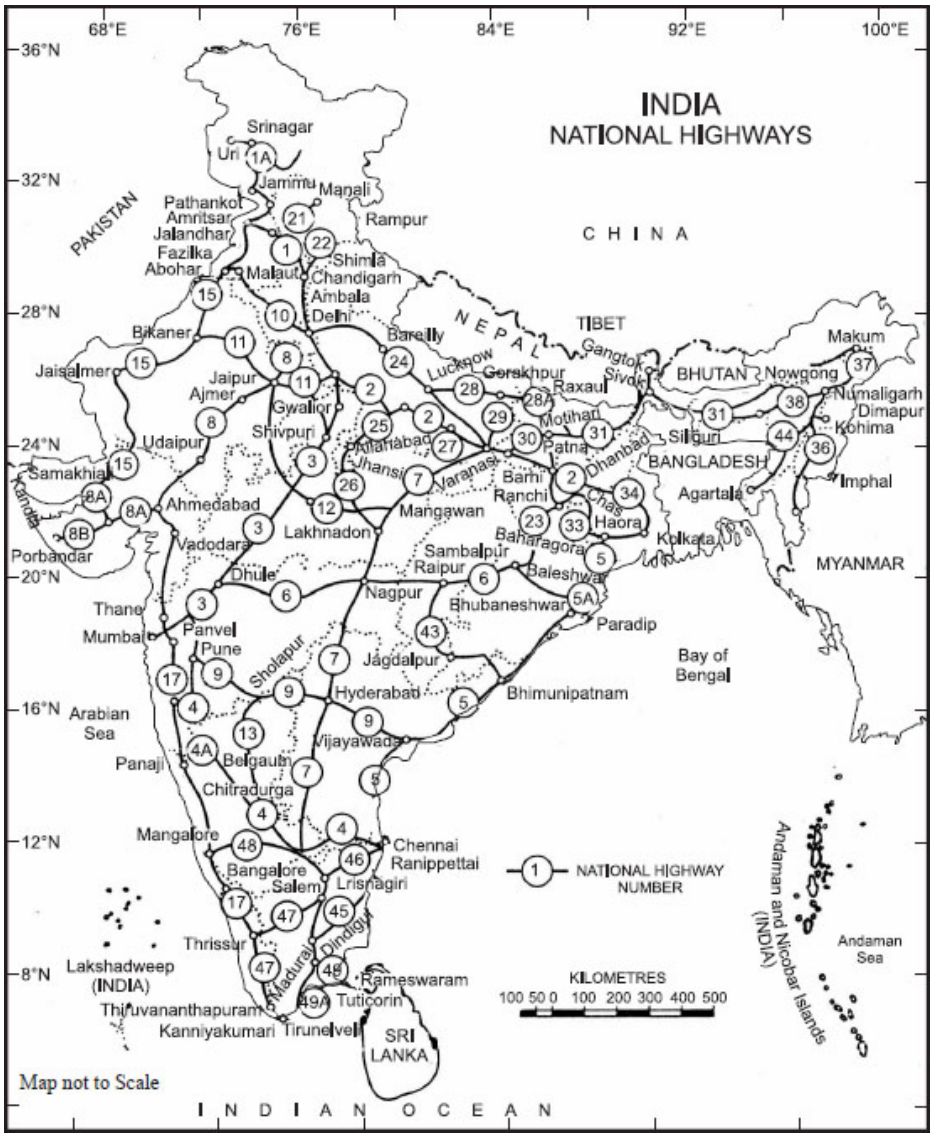


Fig. 12.1 Network of National Highways

(ii) ***Phase II—***

(i) The North South Corridor: This corridor aims to connect the National Highways from Srinagar (J & K) to Kanyakumari including Kochi-Salem; and

(ii) The East West Corridor: This corridor aims to connect the National Highways from Silchar in Assam to Porbandar in Gujarat **(Fig. 12.2).**

(iii) ***Phase III—***Phase three comprises widening of the existing National Highways to 4/6 lane standard. Thus to connect state capitals, seaports, and the important tourist locations with the Golden Quadrilateral.

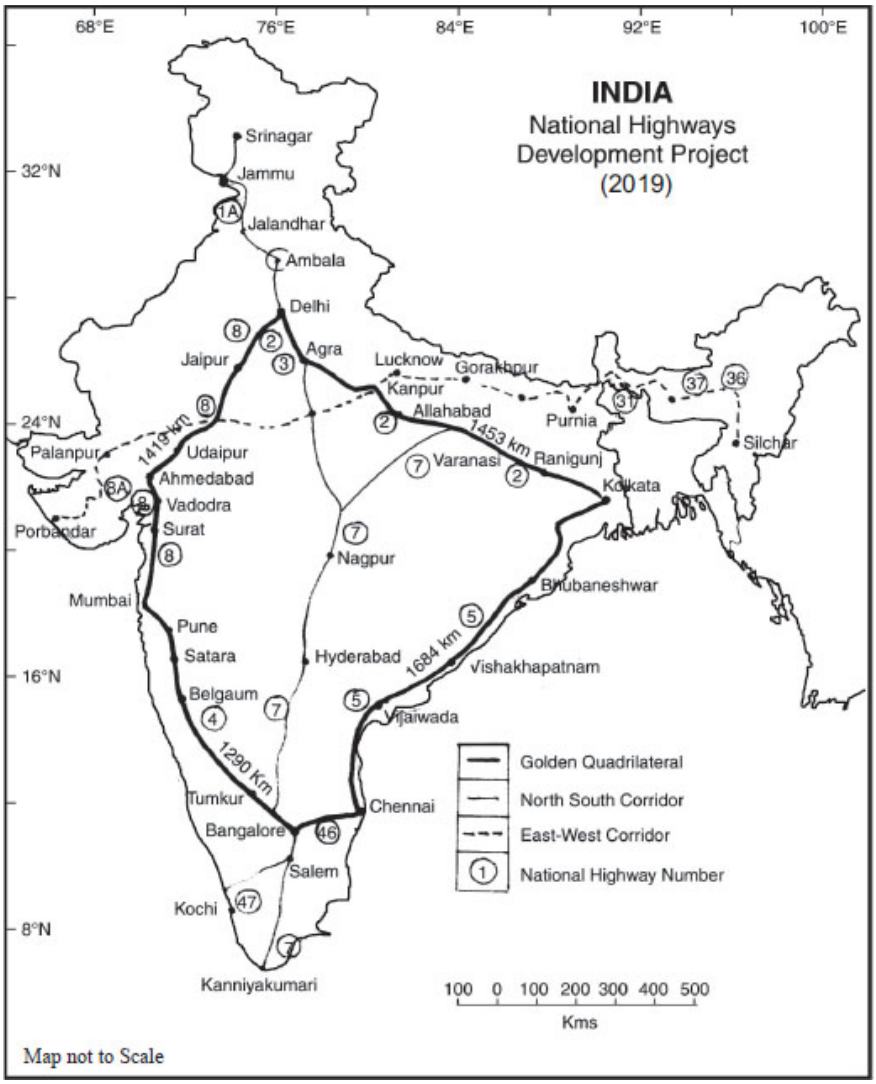


Fig. 12.2 National Highways Development Project (NASP) *(Source: www.nhai.gov.in)*

Traditionally, the road projects used to be financed by the government. But in the last decade after the liberalisation or globalisation, a significant contribution is being made by the private sector. To encourage the private sector, several steps, given below, have been taken by the government.

1. Declaration of the road sector as an industry.
2. Provision of capital subsidy up to 40% of the project cost to make project commercially viable.
3. Full (100%) tax exemption in any consecutive 10 years out of the first 20 years of the project.
4. Provision of encumbrance free site work, i.e., the Government shall meet all expenses relating to land and other pre-construction

activities.

5. Foreign direct investment upto 100% in road sector.
6. Easier external commercial borrowing norms.
7. Higher concession period (upto 30 years).
8. Right to collect and retain toll.

2. **State Highways** The length of state Highway was 154,522 in 2017. The State Highways and Rural Roads are developed and maintained by the various agencies of the state and union territories. The funds, however, are also provided by the Central Government for the development of roads under the following schemes.
   (a) **Improvement of State Roads from the CRF**: The funds from the CRF are provided for improvement of state roads other than the rural roads.
   (b) **Economic importance and Inter State Connectivity Scheme**. To promote inter-state facilities and also to assist the state governments in their economic development through construction of roads bridges of inter-state and economic importance, the Central Government provides 100% grant for inter-state connectivity projects and 50% for projects of economic importance.
   (c) **Rural Roads:** Roads are also being developed in rural areas under the *Pradhan Mantri Gram Sadak Yojna* (PMGSY). The objective of this scheme was to link all the villages with a population of more than 500 with all weather roads by the end 2019.
3. **District Roads:** These roads mostly connect the towns and large villages with one another and with the district headquarters. The construction and maintenance of roads is the responsibility of the Zila Parishad and the PWD.
4. **Village Roads:** Village roads are constructed and maintained by the *village panchayats*. These roads are generally, narrow and zig-zag. They are generally not suitable for heavy mechanised traffic. The length of rural roads is more than 2,650,000 km. In 2018–19 about 24,638 km of road has been constructed in rual areas.
5. **International Highways**: Under the agreement with the Economic and Social Commission on Asia and Pacific (ESCAP) some of the country's highways linking neighbouring countries have been declared international highways. These highways are of two types:
   (a) **The Main Arterial Routes**, linking the capitals of the neighbouring countries like (i) Lahore-Amritsar-Delhi-Agra-Kolkata-Golaghat-Imphal-Mandalay (Myanmar),

(ii) Agra-Gwalior-Hyderabad-Bangalore-Dhanushkodi, and (iii) Barhi-Kathmandu;

(b) **The routes joining the main cities, seaports, and industrial centres with the arterial road-network**, like (i) Agra-Mumbai Road, (ii) Delhi-Multan Road, (iii) Bangalore-Chennai Road, and (iv) Golaghat–Ledo Road.

The World Bank provides funds for the maintenance of these roads.

6. **Express Highways** These are multi-lane well-paved highways used for movement of goods and traffic. Some of the important express highways are (i) Kolkata-Dum-Dum Airport Highway, (ii) Durgapur-Kolkata Express Highway, and (iii) Mumbai-Pune Express Highway.

Some of the important National Highways have been shown in **(Fig. 12.1) and (Fig. 12.2)** while **(Table 12.1)** gives their routes and length in kiloms.

Table 12.1

India—Some of the Important National Highways and their Lengths

NH No.	Length (in km)	Route
44 (Old NH7)	3745	Srinagar-Kanyakumari
27	3507	Porbandar-Silchar
48 (Old NH8)	2807	Delhi-Chennai
52	2317	Hisar, Jaipur, Kota, Indore, Dhule, Aurangabad, Bijapur, Hubli
30	2010	Sitarganj (Uttarakhand)-Ibrahimpatnam in Andhra Pradesh
6	1873	Jorabat (Meghalaya)-Shillong (Mizoram)
53	1781	Hazira (Gujarat)-Paradip Port (Odisha)
16 (Old NH5)	1659	Part of Golden Quadrilateral West Bengal-Andhra Pradesh
66 (Old NH17)	1593	Panvel-Kanyakumari
19 (Old NH2)	1435	Delhi-Kolkata (has historical part of Grand Trunk Road)
34	1426	Gangotri Dam (Uttarakhand)-Lakhnadon (Jabalpur)
2	1214	Dibrugarh, Assam, Nagaland, Manipur, Mizoram (second longest in North Eastern Region)
13 (Old NH229)	1150	Tawang (Arunachal Pradesh) to Pasighat (Assam)
47	1080	Bamanbore (Gujarat)-Nagpur (Maharashtra)
31	968	Uttar Pradesh-West Bengal

Roads and Rural Development Plan

This includes construction of roads under Minimum Needs Programme (MNP), Rural Landless Employment Guarantee Programme (RLEGP), *Jawahar Rozgar Yojana* (JRY), and Command Area Development (CAD) to connect all villages having a population of 500 or more with all-weather roads and those having less than 500 with the link roads.

Density of Roads

The state-wise density of roads has been shown in **(Fig. 12.3)**. It may be observed from this figure that the density of roads is the highest in the states of Goa, Kerala, and Punjab, followed by Maharashtra, Tamil Nadu, Haryana, and Uttar Pradesh. The density of roads is however, very low in Bihar, Chhattisgarh, Jammu & Kashmir, Jharkhand, Madhya Pradesh, and the states of North-East India.

Transformation of Roads & Goverment Initiatives

With the new policy of globalisation of our national economy and emphasis on promotion of exports of agricultural and industrial products, a dense and efficient road network is a must. The Government has opened up road construction and its maintenance to private sector including joint venture with foreign collaboration. One of the policies is to invite private parties to build, operate, and transfer (BOT) roads decided upon by the Government. The parties would bear cost of construction, operate roads for stipulated period, collect road taxes from users and return the roads to the Government at the end of the contracted period. Various international agencies have given funds for this specific purpose.

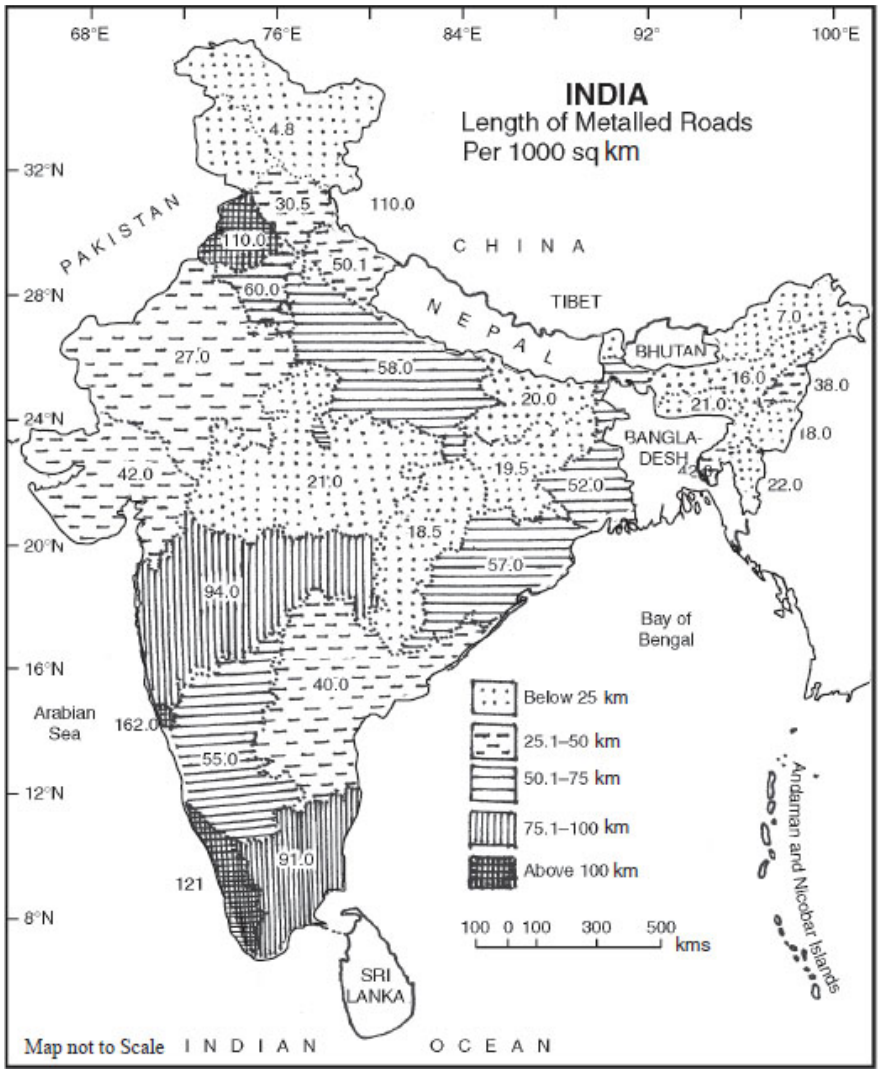


Fig. 12.3 Density of Roads

1. India and Nepal have signed agreements regarding roads and reconstruction projects. An MoU was also signed for Project Management Consultancy Services for upgradation/ improvement of road infrastructure in *Terai region* of Nepal.
2. A new service called '***integration of DigiLocker with Driving Licenses (DL) &3 Vehicle Registration Certificates (RC)***' at New Delhi has been launched.
3. 12 major national highways projects worth Rs. 10,596.19 crores at Ropar, Samrala and Jalandhar have been sanctioned by the Government of India.
4. **Bharatmala Pariyojna**

It is a scheme to develop the road connectivity to border areas, development of coastal roads including road connectivity for non-major ports, improvement in the efficiency of national corridors, development of economic corridors, inter corridors and feeder routes along with integration with Sagarmala.

The Bharatmala Pariyojana envisages development of about 26,000 km length of economic corridors, which along with Golden Quadrilateral (GQ) and North-South and East-West (NS-EW) Corridors are expected to carry majority of the freight traffic on roads.

5. **National Green Highways Project**

The policy (2016) aims to develop eco-friendly national highways with the participation of community, farmers, NGOs and private sectors. The initial plantation drive on 1500 km of national highways at a cost of about Rs. 300 crores was launched as part of the policy.

Main Problems of Road-Transport

Although India has one of the longest road networks in the world, it is facing a number of problems. Some of the important problems of road transport are given here:

1. **Unmetalled Roads and their Improper Maintenance** About 40% of the roads are not metalled. Moreover, the roads are not properly maintained which accentuate the problems, especially during the rainy season.
2. **Mixed Traffic**: Mixed traffic is a serious problem of Indian road transport. In fact, over greater parts of the country including the megacities, the same road is used by cars, trucks, two-wheelers, tractors, harvesters, animal driven carts, cyclists, e-rickshaws, rickshaws, and pedestrians. This increases travel time, congestion, pollution, tension, and road accidents. India has the highest number of accident fatalities (137,572 in 2013) in the world. The magnitude of the problem—380 deaths a day–is equivalent to a crash of a jumbo jet. The maximum number of people died in Uttar Pradesh (11.31%) followed by TamilNadu (11.30%), Andhra Pradesh (10.3%) and Maharashtra (10%). The least number of road accidents were recorded in Chhattistgarh (Times of India, New Delhi, 7.9.2014, p. 12).
3. **Multiple Check Posts:** There are multiple check posts, toll tax, and octroi duties collection points on the roads which bring down the speed of the traffic, waste time, and causes irritation. Moreover, the rate of road tax varies from state to state and in the different regions of the same state.
4. **Inadequate Side Amenities:** Along the roads, repair shops, first-aid centres, telephones, toilets, restaurant, rest-places, and cheap hotels are not adequately developed.
5. **Shortage of Funds:** There is shortage of funds for the construction and

maintenance of roads. Unfortunately, there is insignificant participation of the private sector as a result of which capital for development, extension, and maintenance of roads is not adequately available.

6. Unstable Road Policy There is not a stable policy for the construction, extension, and maintenance of roads. The road policy in the states, generally changes with the change of government. This leads to poor maintenance of roads.

Rail Transport

India has the third largest rail network in the world (66,000 route kms), after the USA (224,792 km),
China (98,000 km). Railways connect different parts of the country. The total length of Indian Railways presently is around 67,368 kms. Indian railway carries two crore people on 18,000 trains daily.

Indian Railways have grown into a vast network of 7030 stations spread over a route length of 63,974 km with a fleet of 8593 locomotives, 51,030 passaenger service vehicle, 6505 other coaching vehicles and 219,931 wagons as on 31 March, 2010 (India-2012, p. 1076). Railways had a target of laying 4100 km in 2018–19.

About 30% of the route kilometer, 41% of running track kilometer and 41% of total track kilom is electrified. About 85% of the total railway route is in broad gauge, 11.3% in meter gauge and only 3.70% in narrow gauge.

Passengers originating had risen from 1284 million in 1950-51 to 7246 Million in 2009–10. Despite constraing of resources, the railways have been able to cope with increasing demand of passenger traffic. Railways are the premier mode of passenger transport both for long distance and suburban traffic.

The revenue growth for Indian railways maked an increase at a CAGR of 9.66% during
FY07–FY18. They provide the principal mode of transportation for freight and passengers. Thus, they unite people from the farthest corner of the country and make possible the conduct of business, sight seeing, pilgrimage, and education. It has also played an important role during the periods of droughts, floods, wars, epidemics, and natural calamities. The process of industrialisation and economic development has also been accelerated with the help of railways.

The railway system started in India in 1853 when the first railway line between Mumbai (Bombay) and Thane (a distance of 34 km) was inaugurated. This was followed by the opening of another railway line between Kolkata (Calcutta) and Raniganj in 1854 , and Chennai (Madras) and Arkonam in 1856. The railway development was, however, quite fast after 1900 and a tremendous progress in the development of railway network was made after Independence (1947). Originally the railways were operated by private companies owned by the British;

but in 1950, the entire railway management was taken over by the Central Government.

Main Features of Indian Railways

Railway Gauges: There are four different types of gauges of the Indian railways. They are: (i) Broad Gauge (width 1.676 m), (ii) Metre Gauge (width 1 m), (iii) Narrow Gauge (width 0.762 m), and
(iv) Lift Gauge (width 0.610 m). These gauges were designed during the colonial period keeping in mind the volume of traffic and goods movement. The major port cities were connected by Broad Gauge to facilitate the export of raw materials from India to Britain and the Commonwealth countries.

After Independence, the Government of India decided to convert all the gauges into Broad Gauge to facilitate the traffic and transport of goods. Prior to Independence, the whole railways system was under steam traction. Since it had poor efficiency and caused great environmental pollution the diesel engines were introduced in 1950s. At present, Diesel Locomotive Works (DLW), Varanasi (UP), Chitranjan Locomotive Works (West Bengal), and Mihijam and Tat Engineering and Locomotive Works, Jamshedpur (Jharkhand), are producing diesel engines for railways. The diesel engines were also the cause of environmental pollution; therefore, in order to overcome the pollution problem and to cope with the growing pressure of traffic and goods, a decision was taken to switch over to electric traction all the important routes. In the electrification process, priority was given to high density sections of railway lines. Of the two main segments of the Indian Railways, ((i) freight and (ii) passenger), the freight segment accounts for about 66% of the revenues. Within the freight segment, bulk traffic accounts for nearly 95%, of which more than 45% is coal.

Distributional Pattern: The distributional pattern of railway network is closely influenced by the terrain, topography, density of population, minerals, fertility of land, and arid and semi-arid climatic conditions The railway network of the country has been shown in **(Fig. 12.4)** while
(Table 12.2) shows the density of railway-line per 1000 sq km of area in the major states
(Table 12.2) (Fig. 12.5). It may be observed from the **(Fig. 12.5)** that the density of roads is significantly high in the Northern Plains of India while in the hilly states their density is very low.

Map not to Scale

Fig. 12.4 Railway Network

Table 12.2

Length of Railways in Major States

S. No.	State	Railway Length per 1000 sq km Area
1.	Punjab	42.78
2.	West Bengal	41.85
3.	Bihar	30.82
4.	Uttar Pradesh	30.20
5.	Tamil Nadu	29.96
6.	Gujarat	28.73
7.	Assam	27.58
8.	Goa	20.72
9.	Andhra Pradesh	17.39
10.	Maharashtra	17.00

Source: Statistical Abstracts 2015–16.

It may be seen from **(Fig. 12.5)** that the highest density of railway per 1000 sq km area is in Punjab (42.78 km) followed by West Bengal (41.85 km), and Bihar (30.82). Uttar Pradesh (30.2 km) and Tamil Nadu (29.96 km) occupy fourth and fifth positions respectively, while Maharashtra has the 10th rank.

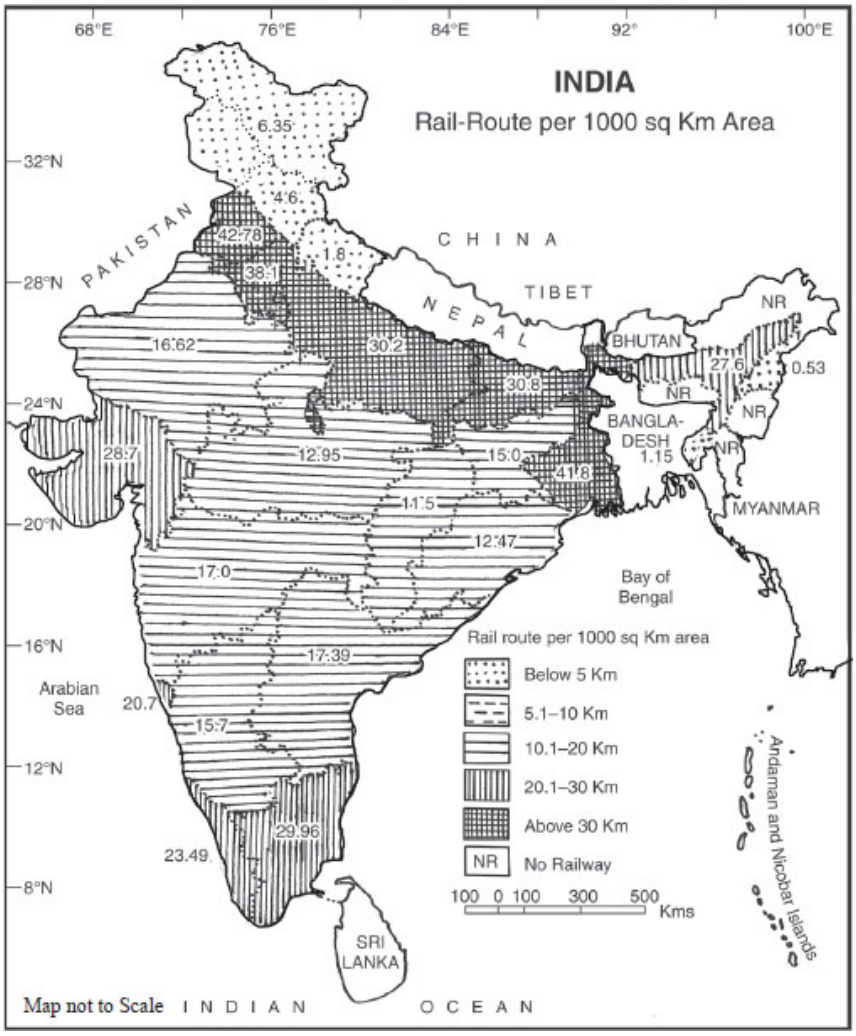


Fig. 12.5 Density of Railways

Railway Management: The entire railway system is managed by the Railway Board. The railway network has been divided into 16 railway zones which are subdivided into divisions **(Table 12.3).** These divisions are the basic operating units for the Indian railways.

Table 12.3

India—Railway Zones

	Zones	Date of Formation	Headquarters
1.	Southern	14.4.1951	Chennai
2.	Central	05.11.1951	Mumbai (CST)
3.	Western	05.11.1951	Mumbai (Churchgate)
4.	Northern	14.04.1952	New Delhi
5.	North-Eastern	14.04.1952	Gorakhpur
6.	South Eastern	01.08.1955	Kolkata
7.	Eastern	01.08.1955	Kolkata
8.	North-Eastern Frontier	15.01.1958	Maligaon (Guwahati)
9.	South-Central	02.10.1966	Secunderabad
10.	East Central	01.10.2002	Hajipur
11.	North-Western	01.10.2002	Jaipur
12.	East Coast	01.04.2003	Bhubaneshwar
13.	North Central	01.04.2003	Allahabad
14.	South-East Central	01.04.2003	Bilaspur
15.	South Western	01.04.2003	Hubli
16.	West Central	01.04.2003	Jabalpur
17.	Metro Railway, Kolkata	31.12.2010	Kolkata

Impact of Railways on Economy and Society

The impact of development and expansion on the economy and society is enormous. The construction and expansion of the railways have improved the economy of the country appreciably. Railways have played a significant role in the development of cotton textile industry in Maharashtra and Gujarat; jute industry in Hugli Basin; coal industry in Chhattisgarh, Jharkhand, Maharashtra, Odisha, and West Bengal; tea-plantation and petro-chemical industries in Assam and West Bengal. In the post-Independence period, railways have helped substantially in setting up of industries in the under-developed and backward areas (Bhilai, Bokaro, Durgapur, Raurkela, etc.), of isolation and relative isolation. Railways have played an important role in the transportation of grains, and chemical fertilizers in the regions of Punjab, Haryana and Western Uttar Pradesh where Green Revolution is a success.

Merits of Railway Transport

Some of the important merits of railway transport are as under:

1. **Easy Mode of Travel:** Railways provide the cheapest and the most

convenient mode of passenger transport, both for short and long distances. Railways are particularly suited for long distance journeys.

2. **Accelerates Industrialisation**: Railways have played a vital role in the process of economic development, industrialisation, and urbanisation of the country.

3. **Agricultural Development**: Railways have played a significant role in the development and intensification of agriculture. In fact, the farmers can sell their perishable and non-perishable commodities to long distant market at a remunerative price. The fertiliser used in the High Yielding Varieties is also generally transported to the farmers in different parts of the country by the railways.

4. **Connects Remote Cultures:** Railways connect people from the areas of isolation and relative isolation with the urban centres and thus accelerate social interaction and national integration and help in the diffusion of new ideas and innovations.
5. **Helps in the Maintenance of Law and Order:** At the time of communal tension and socio-political turmoil, railways play a significant role
6. **Fast Mode of Transportation:** Through super-fast container services in major cities, India has ensured quick movement of people and commodities.

7. Cheapest Mode of Transportation for Bulky Materials: Railways are particularly suited to long haulage of bulky materials like coal, petroleum, and mineral ores.

8. **Connects Hinterland with Seaports:** Railways connect hinterland with the seaports and thus help in the socio-economic development of the coastal areas as well as the hinterland.
9. **Promotes National Integration:** The areas of isolation and relative isolation have been connected by the railways with the mega and metropolitan cities. This has helped in the diffusion of innovations and new technology in the remote areas of the country.
10. **Helps in Administration** Railways are playing an important role in the administration of the country and safeguarding its sovereignty and integrity. Railways facilitate easy movement of police, troops, and defence equipments to different parts of the country.
11. **National Integration** Railways unite the people from different cultural backgrounds and thus lead to cultural and national integration.

Main Problems of Indian Railways

Some of the main problems of railways are given below:

1. **Heavy Pressure:** With the increase in population, industrialisation, and

urbanisation, there is heavy pressure of passengers and goods on the railways.

2. **Competition with Road Transport**: Railways are facing tough competition from road transport. Many a times the road transport is preferred as roads can deliver the goods at the doors.
3. **Safety:** According to Railway Board, 15% of the total railway accidents in the world, occur in India. Thus, no where in the world do so many people die in train accidents as in India. Looking at the frequency of accidents, railway journey is no longer very secure. On some of the routes, frequent loots and robbery incidents have made the railway journey unsafe and full of tension.
4. **Economic Burden** There are about 17 lakh railway employees, out of which about 45% of the total man-power is unskilled. Many of the workers are not efficient and dedicated. This puts an economic burden on the railway budget.
5. **Faulty Planning:** Under the political pressure and interference, several projects of railways have been introduced which are not economically viable.
6. **High Rates of Electricity**: The Electricity Boards and NTPC often make heavy increase in their tariffs. This adds to the financial burden on the electric traction. Railways are the largest consumers of diesel. The increase in the price of diesel affects the finances of the railways adversely.
7. **Slow Decision-Making Process:** The decision-making process is in the hands of the Central Government. Being a lengthy process, often the decisions are delayed and the railways suffer adversely.
8. **Obsolete Machinery:** A number of equipments used by the railways are obsolete and need replacement for which adequate capital is not available.
9. **Obsolete Tracks and Equipments:** Many of the railway tracks are outdated and obsolete. Track renewal, and improvement in signalling, communication system, and safety measures are not adequate.
10. **Miscellaneous Problems**: Late running of trains, filthy trains, choked toilets, lack of passenger facilities, cleanliness at railway stations, lack of security arrangements on the railways result into frequent thefts, robberies, and dacoities.

Looking at the above problems, it will be decades before India gets close to Japan's superfast and clean trains.

WATER TRANSPORT

Prior to the development of railway network, waterways were the main mode of transportation in India. Waterways are the cheapest means of transportation. Moreover, they are the most suitable for carrying heavy and bulky goods. Despite all these merits, water transport in India at present provides only about one per cent of the total transport of the country.

Ports

There are 13 major and 200 minor ports in India. Ports not only play a crucial role in facilitating international trade, they also act as fulcrums of economic activity and development in their surrounding areas. India has about 7517 km of main coastline spread over 13 states/union territories, and studded with 13 major ports and 200 non-major ports (minor and intermediate). Of the non-major ports, around 60 are handling traffic. The major ports are under the jurisdiction of the Central Government, while the minor and intermediate ports are managed and maintained by the respective state governments. The number of cargo vessels handled at major port is about 16,500 per annum.

The total traffic carried by both the major and minor ports during 2005–06 was estimated at around 570 million tonnes. The 12 major ports carry about 75% of the total traffic with Vishakhapatnam as the top traffic handler in each of the last six years (2000–01 to 2006–07).

Major Seaports of India: The major seaports of India are: (i) Chennai, (ii) Cochin, (iii) Ennore,
(iv) Jawaharlal Nehru Port (Nhava Sheva), (v) Kandla, (vi) Kolkata/ Haldia, (vii) Mormugao,
(viii) Mumbai, (ix) New Mangalore, (x) Paradeep, (xi) New Tuticorin, and (xii) Vishakhapatnam **(Fig. 12.6).** The salient geographical features of the major seaports of India have been described briefly here:

1. **Chennai*:*** Situated along the north-eastern coast of Tamil Nadu, it is an artificial seaport. It spreads over an area of about 81 hectares with an average depth of about 16 m. The harbour has an entry from the north. It can accommodate 22 vessels. The main exports from the Chennai port are food-grains, hides and skins, iron-ore, mica, oil-cake, sugar, turmeric, timber, and tobacco. The port imports chemicals, coal, cotton, cotton-goods, edible-oils, fertilisers, iron and steel, machinery, metals, and petroleum.

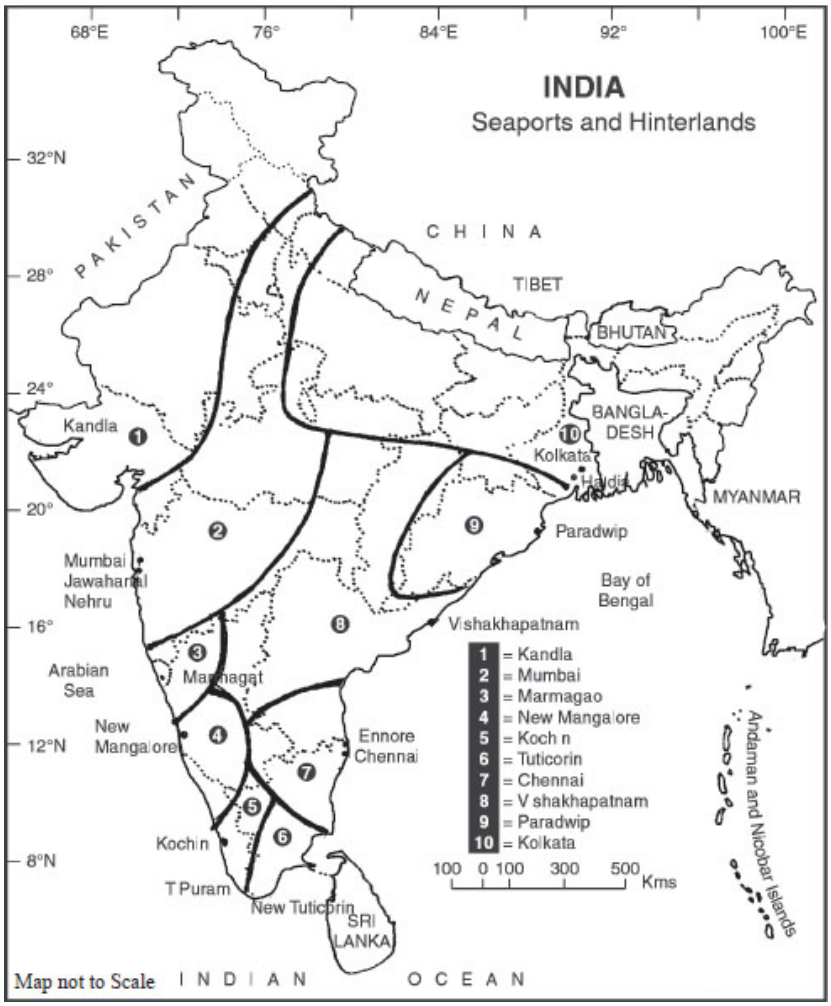


Fig. 12.6 Major Seaports and their Areas of Influence

2. **Ennore:** Situated about 20 km to the north of Chennai, Ennore is a natural harbour. It was developed to ease the pressure of the Chennai-seaport. Ennore exports hides, machinery, mica, rice, and sugar. The principal imports are cement, cotton, edible-oils, fertilisers, machinery, and petroleum products.
3. **Jawahar Lal Nehru (Nhova Shiva) Port:** Situated about 14 km to the south of Mumbai, this seaport was developed to ease the pressure of the Mumbai seaport. It was inaugurated in 1989. It is a world-class port equipped with all the modern facilities. The seaport is connected by a four-lane highway with the hinterland. India's largest container port is Jawaharlal Nehru Port. Located at the Elephanta Island, it handles vessels having draught upto 12.50 m.
4. **Kandla:** Located at the head of the Gulf of Kutch, Kandla seaport was developed after Independence to take up the place of Karachi which went

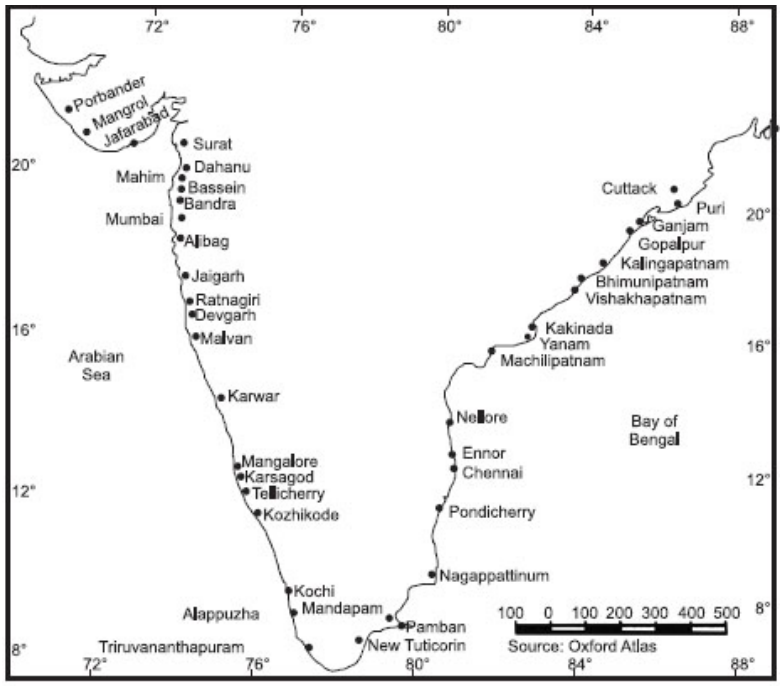
to Pakistan. It is a tidal harbour in the Kandla Creek with an average depth of 10 m. The port has a vast hinterland in the states of Gujarat, Rajasthan, Haryana, Punjab, Himachal Pradesh, and Jammu and Kashmir. Its main imports are petroleum, fertilisers, phosphates, and sulphur, while the export includes bones, cotton, food-grains, naptha, salt, and sugar.

5. **Kochi**: Situated along the coast of Kerala, Kochi is a natural seaport and the largest shipyard in the country. The indigenous Aircraft Carrier for the Indian Navy is also presently under construction in the shipyard. It remains open for cargo traffic throughout the year. Being situated close to the Suez-Colombo route, it has great commercial and strategic importance. The main items of export are cashew-kernel, coconut, coir goods, copra, oil lubricants, rubber, fish, and sea food, while the main imports include chemicals, coal, cotton, edible-oils, fertilisers, iron and steel, machinery and metals. Its other exports are food-grains, hides and skins, iron-ore, oil-cakes, mica, sugar, timber, and tobacco.
6. **Kolkata**: Kolkata is the only riverine major seaport of India, having been in existence for about 138 years. Kolkata port handled 41.38 million tonnes traffic in 2013–14. The Kolkata port is situated along the Hugli River. The port has a large hinterland which covers the seven states of Bihar, Chhattisgarh, Jharkhand, north-east India, Odisha, and eastern Uttar Pradesh. The export and import of Bhutan and Nepal is also done through Kolkata port. The main exports of the port include bones, bone-meal, bunker-oil, electric goods, iron and steel, jute products, lac, leather goods, machinery, mica, scrap, tea, and timber. The main imports include edible-oils, fertilisers, machinery, railway equipments, and petroleum. South of Kolkata river port, Haldia seaport has been developed along the coast of Bay of Bengal to accomodate modern larger ships.
7. **Marmagao*:*** Situated at the entrance of the Zuvari-estuary in Goa, it is the leading iron-ore exporting seaport of the country. It also exports betel-leaves, cashew-nut, manganese, and salt. The main imports include manganese, cement, crude-oil, food-grains, fertilisers, and machinery.
8. **Mumbai:** Mumbai is the largest and most important seaport along the western coast of India. It is a fully integrated multipurpose port handling container, dry bulk, liquid bulk, and break bulk cargo. It was developed by the British in 1672 as a substitute to the Surat seaport. It has a natural harbour about 12 m in depth. It has a vast hinterland, stretching over Maharashtra, Gujarat, Madhya Pradesh, Delhi, Haryana, Punjab, Uttarakhand, and Uttar Pradesh. The main exports of Mumbai seaport include cotton-goods, cotton-yarn, electrical goods, machinery, and vehicles while fertilisers, chemicals, electronic goods, machinery, paper,

pulp, petroleum, petroleum-products, and raw-cotton are the main items of import.

9. **New Mangalore**: Situated along the coast of Karnataka, it was developed about 9 km to the north of old port of Mangalore. It was inaugurated in 1975. The port is linked through Broad Gauge railway line and the National Highway NH-17 with Mumbai. Its main exports are granite cashew-nuts, coffee, forest products, manganese-ore, and timber, while the imports include crude oil, coal, LPG, iron-ore, cement, fertilisers, machiney, and petroleum and petroleum products.
10. **New Tuticorin*:*** This seaport has been developed about 8 km to the south to the old Tuticorin port. The port has an artificial deep sea-harbour. It has a rich hinterland comprising the districts of Kanniyakumari, and Ramanathapuram. It is well connected with railways and National Highway (NH-7A). Its main exports include cardamom, spices, cotton, coconut, cotton goods, hides and skins, while the main imports consist of coal, fertilisers, hardware, and machinery.
11. **Paradwip**: Situated along the coast of Odisha, it is a deep water seaport. In fact, Paradwip has the deepest harbour in the country. It was inaugurated in April 1966. It exports cotton goods, iron-ore, iron and steel, manganese, and scrap, while the imports include petroleum products, edible oils, machinery, and electric goods.
12. **Vishakhapatnam*:*** Developed in 1933, it has the best natural harbour in the country. It has a huge hinterland in the states of Andhra Pradesh, Telangana, Chhattisgarh, Madhya Pradesh, and Odisha. Vishakhapatnam port ranked first in India for the last six years in respect of cargo traffic. Its main imports are petroleum, fertilisers, chemicals, machinery, and metals, while iron-ore, manganese ore, leather goods, timber, and food-grains are the main items of exports.

Some of the intermediate and minor seaports include Bhimunipatnam, Cuddalore, Kakinada, Kalingapatnam, Machlipatnam. Nagappattinam, and Tuticorin along the eastern coast, and Alappuzha, Calicut, Daman, Kollam, Karwar, Kozhikode, Murad, Porbandar, Tellicherry, Veraval along the west coast of India **(Fig. 12.7)**.



Map not to Scale

Fig. 12.7 Major Fishing Ports

Coastal Shipping: Coastal shipping is an energy efficient, environmental-friendly, and economical mode of transportation in Indian transport network. It is a crucial component for the development of domestic industry and trade. Coastal shipping involves movement of goods and passengers from one port to another port within the country**.** India's coastal Shipping Tonnage in December 2005 was 470 vessels with 804,612 GRT.

Coastal shipping has many advantages as given below:

1. It reduces the pressure on rail and road transport systems.
2. It is relatively pollution free.
3. It is less capital intensive.
4. It provides large employment.
5. It is the cheapest mode of transport.
6. It promotes coastal base industries such as fisheries and corals collection.
7. It promotes tourism.

Problems: some of the important problems of the Indian ports are given below:

1. Indian ports are the most congested.
2. The ports are not adequately connected with the hinterland.
3. There is heavy pressure on container traffic. The largest container port in the world is Singapore which handles 23.19 million TEUs (twenty foot

equivalent units). In comparison to this India's largest container port handled roughly 2.67 million TEUs in 2005–06.

Inland Water Transport

India has about 14,500 km of navigable waterways which comprise rivers, canals, lakes, backwaters, and creeks. About 45 million tonnes of cargo is being moved annually by inland water transport. It is a fuel-efficient and environment-friendly mode of transportation. Its operations are currently restricted to a few stretches in the Ganga-Hugli rivers, the Brahmaputra, the Barak river (Assam), the rivers in Goa, the backwaters of Kerala, inland waterways in Mumbai, and the deltaic regions of the Godavari-Krishna rivers. Besides the organised operations by mechanised vessels, country boats of various capacities also operate in various rivers, lakes, and canals. Data of cargo and passenger-movement in unorganised sector (i.e., by country boats, etc.) has not been compiled, but it is a fact that substantial quantum of cargo and passengers are transported in the unorganised sector as well.

The Central Water Transport Corporation (CIWTC): Constituted in 1967, it has its headquarters at Kolkata. It is mainly engaged in the transportation of goods by inland waterways in the Ganga, Brahmaputra, Hugli, and Sundarban regions. It is operating regular cargo service between Kolkata and Karimganj (Assam), Kolkata and Bangladesh, and Haldia and Patna.

Inland Waterways Authority of India (IWAI): The Inland Waterways Authority of India came into existence on 27 October, 1986 for the development and regulation of inland waterways for shipping and navigation. The Authority primarily undertakes projects for development and maintenance of IWT infrastructure on national waterways through grant received from the Ministry of Shipping, Road Transport and Highways. The head-office of the Authority is located at NOIDA (UP near Delhi). The authority also has its regional offices at Guwahati, Kochi, Kolkata, and Patna and sub-offices at Allahabad, Bhagalpur, Farakka, Kollam, and Varanasi. The National Inland Navigation Institute is located at Patna**.**

National Waterways: Following are National Waterways in India:

1. The Ganga between Allahabad-Haldia (1620 km).
2. The Sadiya-Dhubri stretch of the Brahmapputra (891 km).
3. The Kollam-Kottapuram stretch of West Coast Canal along Champakara and Udhyogmandal Canals (205 km) in Kerala.

India's Ist 'freight village' has been planned to be located near varanasi. Spread over 100 acres in Varanasi, it is an specialized industrial estate. It includes an inter-modal terminal, which shall facilitate integration between different modes of transport. It will also help relocation of retailors, ware-house operators and logistic service providers, supply the regional markets.

Factors Affecting Inland Waterways: The following factors have a close bearing on the development and maintenance of inland waterways:

1. **Fluctuation Regime of Rivers:** Many of the rivers are seasonal. Even in the perennial rivers, like Brahmaputra, Ganga, Gandak, Kosi, and Yamuna, the discharge of water decreases substantially during the summer and winter months, while during the rainy season they are in floods. Both the conditions are not conducive for navigation.
2. **Natural Obstacles:** Presence of waterfalls, cataracts, and rapids in the course of rivers hinder the development of inland navigation.
3. **Silting of River Beds:** Silting of river bed reduces the depth of water and creates problems of navigation.
4. **Water for Irrigation:** Diversion of water for irrigation reduces the quantity of water in the river channel.
5. **Economically Unviable:** The demand for waterways is not adequate as the people want fast movement of their commodities. This makes river transport economically unviable.

Sagarmala Programme

In order to harness the coastline, 14,500 km of potentially navigable waterways and strategic location on key international maritime trade routes, the Government of India has embarked this programme to promote port-led development in the country. The objectives of the programme include port modernisation, new port development, port connectivity, coastal community development, etc.

Shipping

Managing about 95% of the country's trade volume is carried by shipping (2019). India has a long history of shipping. The Scindhia Steam Navigation Co. was set up in 1919 on modern lines. At the time of Independence, there were only 59 ships with less than 2 lakh tonnes of Gross Registered Tonnage (GRT). Since Independence, India has made a significant progress in shipping and the shipping fleet had 872 ships [282 overseas and 590 coastal vessels (India 2010)]. Coal, crude oil and natural gas are mainly transported by ship.

At present, shipping plays a significant role in the transport sector of the country's economy. Nearly 95% of the total trade volume is moved by sea making shipping the backbone of trade and economic growth. Today India has one of the largest merchant shipping fleet in the developing world. The major international sea routes of India have been shown in **(Fig. 12.8).**

Ship recycling

Ship Recycling India has 25–30% share in the global ship recycling industry. Ship recycling is carried out mainly at *Alang-Sosiya* in Gujarat. Alang-Sosiya is the largest ship recycling yard in the world that started in February 1983. It has high

recycling potential of more than 450 ships (approximately 4.5 million LDT) per annum.

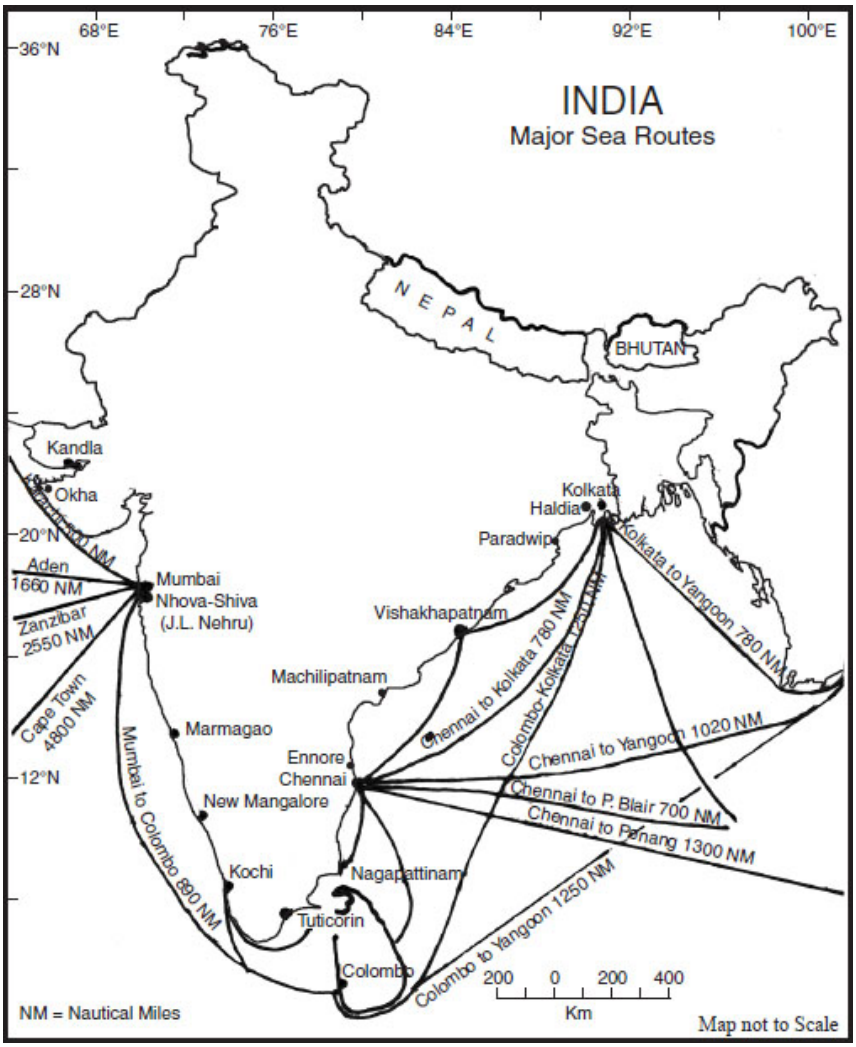


Fig. 12.8 Major Sea routes

Adam's Bridge (Ram Sethu)

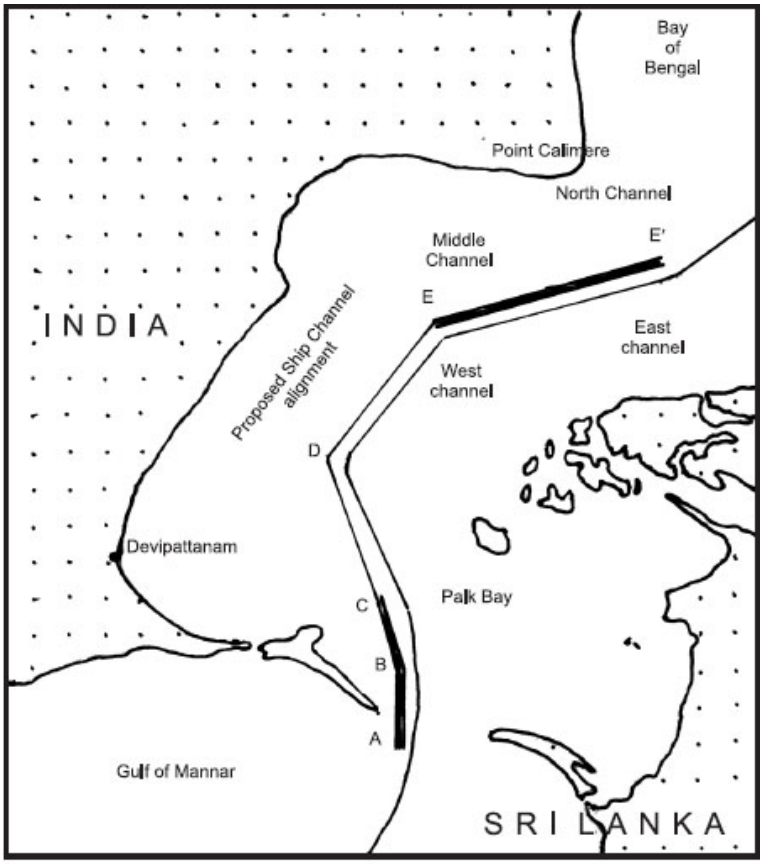
The Adam's Bridge or Ram Sethu is a discontinuous chain of sandbars dotting 30 km stretch in the east-west direction between the southern tip of Rameswaram Island in India and Talaimannar in north-western Sri Lanka. It creates a geological divide between the Palk-Bay and the Gulf of Mannar which form part of the southern Kaveri Basin in the Bay of Bengal **(Fig. 12.8).**

The '***Sethu-Samudram Ship Channel Project (SSCP)***' of the Government of India envisages the dredging of the shallow ocean region in the Bay of Bengal to create an artificial 167 km long 300 m wide and 12 m deep channel like passage

for (10,000–12,000 gross tonnage) ships across the island formations called Adam's Bridge or *Ram Sethu*.

The proposal for channel linking the Palk-Strait and the Gulf of Mannar goes back to the British in 1860 and since then several proposals have been made and six distinct alignments for the passage to go across Ram Sethu have been put forward. But only in 1998, Shri Atal Bihari Vajpayee, the then Prime Minister of the previous National Democratic Alliance (NDA) Government, finally launched the project. It was only inaugurated during the United Progressive Alliances (UPA) Regime in 2005 by the then Prime Minister, Dr. Manmohan Singh.

The Channel project, when developed and completed cut down steaming time and fuel costs, and galvanised traffic for 15 small, neglected seaports in Tamil Nadu, Andhra Pradesh, Odisha, and West Bengal. The fortunes of Tuticorin Port and town improved dramatically from the transshipment of containers that originated from or are destined for ports on the coast of India. Moreover, the Indian Navy needs a channel in India's own territorial waters.



AB (17.3 Km) North	Adam's Bridge.
BC (13.38 Km) North	Adam's Bridge
CD (37.44 Km)	Palk Bay
DE (40.48 Km)	Palk Bay
EE' (54.25 Km)	Palk Strait

Total 167.22 Km

Map not to Scale

Fig. 12.9 India—Ram Sethu Project

Geological Studies: Two independent scientific agencies, namely, the Geological Survey of India (GSI) and the Space Application Centre (SAC) of the India Space Research Organisation conducted detailed studies on the geological formations associated with the Ram Sethu and all have established conclusively that Ram Sethu is not a man-made structure. The agencies concluded that Adam's Bridge (Ram Sethu) is not man made but comprises 103 small coral reefs lying in linear pattern, sand-cays (communities of loose coral sands and beach rocks), and intermittent deep channels.

Dr. Ram Mohan a leading geomorphologist and oceanographer, in his research paper entitled "Geological Evolution of Adam's Bridge" opined that Adam's Bridge consists of sand-bars and coral formations. The possibility of formation of shoals in the shallow continental shelf as barrier bars (sandbars) appear to be the most plausible explanation for the evolution of Rameswaram and Adam's Bridge.

Thus all the scientific studies about the Ram Sethu suggest a consistent picture of natural process that led to the formation of Adam's Bridge. It is not a man-made structure. It is merely a sand and coral formation. The reduction in the distances between Tuticorin and Chennai, Ennore, Vishakhapatnam, Paradwip, and Haldia/Kolkata, etc., has been given in **(Fig. 12.10).**

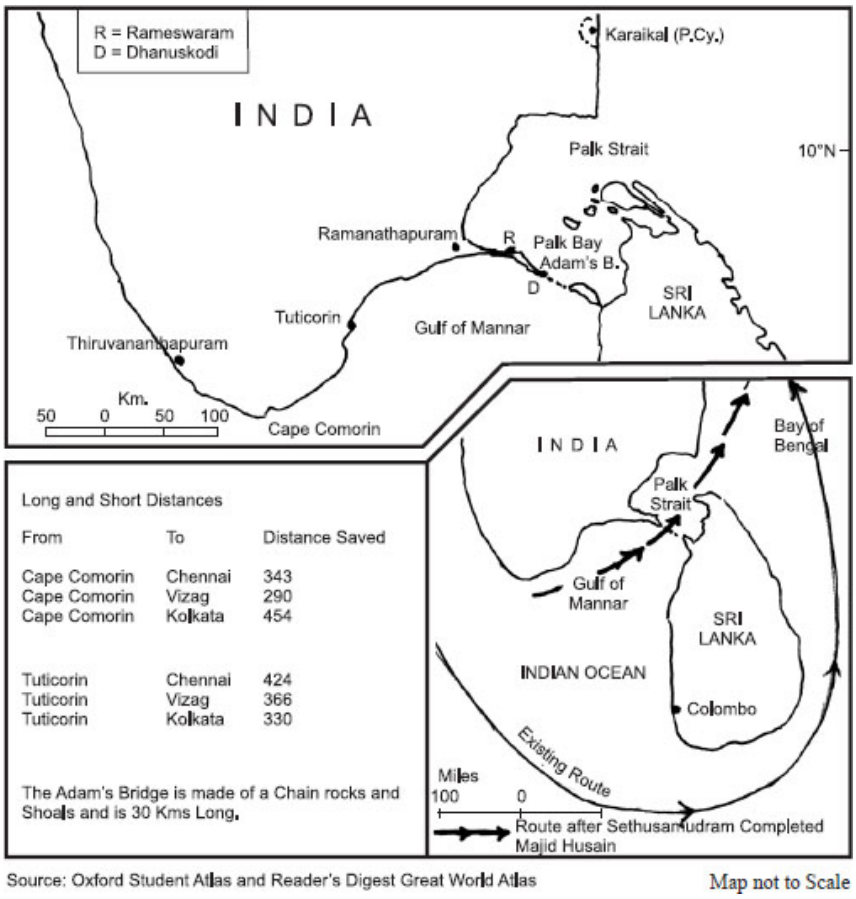


Fig. 12.10 Sea Route through Palk Strait

Proposed Navigation Canal between Sanchore (Rajasthan) and Lakhpat (Gujarat)

The government of Rajasthan is planning to construct a navigation canal between Sanchore (Sirohi District-Rajasthan) and the Lakhpat town located at the head of Kori-Creek (Gujarat). If the proposed navigation canal is constructed, the Thar Desert of Rajasthan will become accessible by water transport to the Arabian Sea **(Fig. 12.10-A)**.

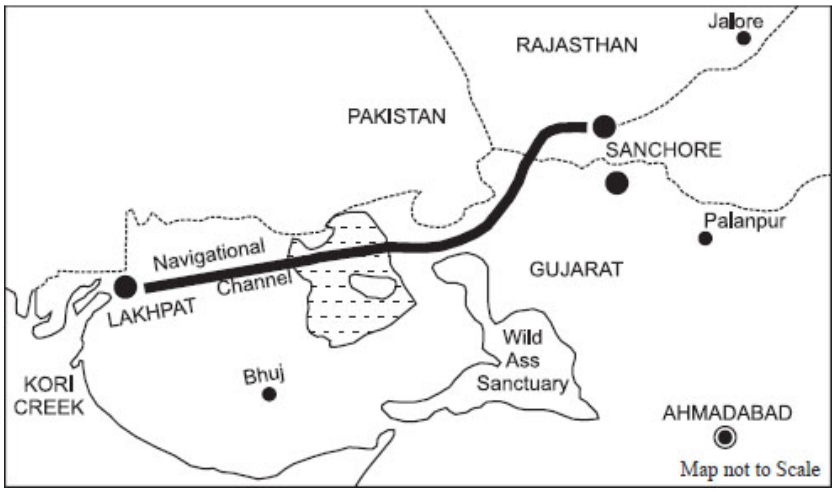


Fig. 12.10-A Proposed Navigation Canal between Sanchore and Lakhpat (Kori-Creek)

Planning of the navigation canal though in the initial phase but the officials claim that the project is economically viable and geographically sustainable. The length of the canal will be about 300 km between Lakhpat and the Sanchore town in Jalore District of Rajasthan. The navigation channel would be about 300 meter wide and about 25 meter deep. It would start from the tip of Lakhpat at Kori Creek in Arabian Sea, run through the Rann of Kachchh and joins Sanchore, about 135 km from the district headquarters of Jalore (Rajasthan).

In the construction of the navigation canal, the government has to examine are geostrategic aspect also as the project is very close to international border. Since navigational channel would run parallel to the border with Pakistan within the Indian territory, it would provide an efficient patrolling medium for the paramilitary forces. Moreover, it will open up more opportunities in the Thar Desert which is bereft of investment so far. The environmentalists however, are of the opinion that the canal may create ecological imbalances in the desert area of the country. The increasing pressure of population along the trading canal in the desert and wetland areas may be the cause of new ecological problems in the long run.

AIR TRANSPORT

India has bilateral Air Service Agreement with 103 countries. Recently, new Air Services Agreement have been signed with Mexico and Chile. Air transport is the fastest mode of transport. It has reduced geographical distances, making the world a village.

The main advantages and disadvantages of the air transport are:

1. With the help of air-transport one can easily reach the remote and difficult terrains, like mountains, deserts, thick-forests, deserts, and marshy lands.
2. It is the fastest mode of transportation.
3. Air-transport plays a vital role at the time of emergency like a war situation and in the event of natural calamities like earthquakes, cyclones, famines, floods, droughts, epidemics, etc.
4. Air-transport is, however, adversely affected at the occurrence of fog, mist, and stormy weather.

Civil Aviation

The Airport Authority of India (AAI), constituted on 1 April 1995, operates upon 127 airports including civil enclave and defence airfields for Commercial Airlines operations. The Ministry of Civil Aviation is responsible for the formulation of national policies and programmes for the development and regulation of civil aviation and for the devising and implementing of schemes for an orderly growth and expansion of civil air transport. Its functions also extend to overseeing the provision of airport facilities, air traffic services, carriage of passengers and goods by air, safeguarding civil aviation operations, regulation of air transport services, licensing of aerodromes, air carriers, pilots, and aircraft maintenance engineers.

Directorate General of Civil Aviation

The Directorate General of Civil Aviation (DGCA) is an attached office of the Ministry of Civil Aviation and is headed by Director General (Civil Aviation). The headquarters is located in New Delhi and there are regional and sub-regional offices all over the country. DGCA is primarily responsible for regulation of air transport services to/from/within India and for enforcement of civil air regulations, air safety and airworthiness standards.

Airports Authority of India

Airports Authority of India (AAI) came into being in April 1995. AAI is a Miniratna Category-I PSE involved in building, upgrading, maintaining and managing airports infrastructure across the country. It owns and maintains 125 airports (96 operational and 29 non-operational) comprising 21 international airports (3 civil enclaves), 8 custom airports (4 civil enclaves), 77 domestic airports and 19 domestic civil enclaves.

Air Sewa

It is an initiative of the Ministry of Civil Aviation launched in 2016 to offer people a convenient and hassle-free air travel experience. It works through an interactive web portal (airsewa.gov.in)
or a mobile app for both android and iOS platforms that includes a mechanism for grievance redressal, back-office operations for grievance, handling flight

status/schedule information, airport Information and FAQs. (Ref: India 2019 Year book)

Airports: There are 450 airports in the country in various stages of development. Out of these Ahmadabad, Amritsar, Bangalore, Chennai, Delhi, Guwahati, Hyderabad, Jaipur, Kochi, Kolkata, Mumbai, Panaji, Srinagar and Thiruvananthapuram are the international airports.

The improvement in the infrastructural facilities at the airports need heavy capital investment which the government cannot afford of its own. Therefore, private domestic and foreign investors including Non-Resident Indians have been encouraged to participate in the process of improvement of the Indian Airlines.

Cargo: In order to help the Indian exports to make their exports more competitive, the Government had introduced in April 1992 an *Open Sky Policy* for cargo. Under this policy, foreign airlines or association of exporters can bring any number of freighters to the country for upliftment of cargo. The Government has also permitted market forces to determine cargo traffic with IATA rates as the floor rates.

Air Services: India has bilateral Air Services Agreement with 103 countries. A revised Air Service Agreement between India and the USA was signed in 2005 replacing earlier Agreement, signed in 1956. The revised agreement grants unlimited access to the designated airlines to any point of call in the territory of the other country as against four airports under the earlier agreement.

Air India: The Air India was constituted in 1947, after which the Air India International launched its first service to London via Cairo and Geneva on 8 June, 1948 with Constellation aircraft. In 1952, the Planning Commission recommended the nationalisation of Air Transport Industry which was effected in March 1953 with the creation of nationalised Corporations—Air India International Limited, which retained its identity and international flag carrier status; and the Indian Airlines, to operate domestic services.

Fleet: Air India owns a fleet of 16 aircrafts consisting of six B747-400, two B747-300, and eight A-310-300. In addition, Air India has inducted six B747-400, eleven A310-300, and four B777-222 on dry lease basis, taking the total number of aircrafts in Air India to 37. Additionally, B737-800 have been leased for Air India Express.

Air India operates 173 flights per week serving 59 stations (45 international and 14 domestic). Air India also has code-share agreements with 12 airlines to offer its passengers more destinations and convenient connections.

In addition to Air India, there are at present six private scheduled operators, namely, Jet Airways (India), Jetlite Airlines, Go Airlines India, Spicejet Ltd, Paramount Airways Ltd., and Inter Globe Aviation Ltd. (IndiGo) operating on the domestic sector providing a wide choice of flights and connectivity to

various parts of India. Some of these flights operate between India and the Gulf countries.

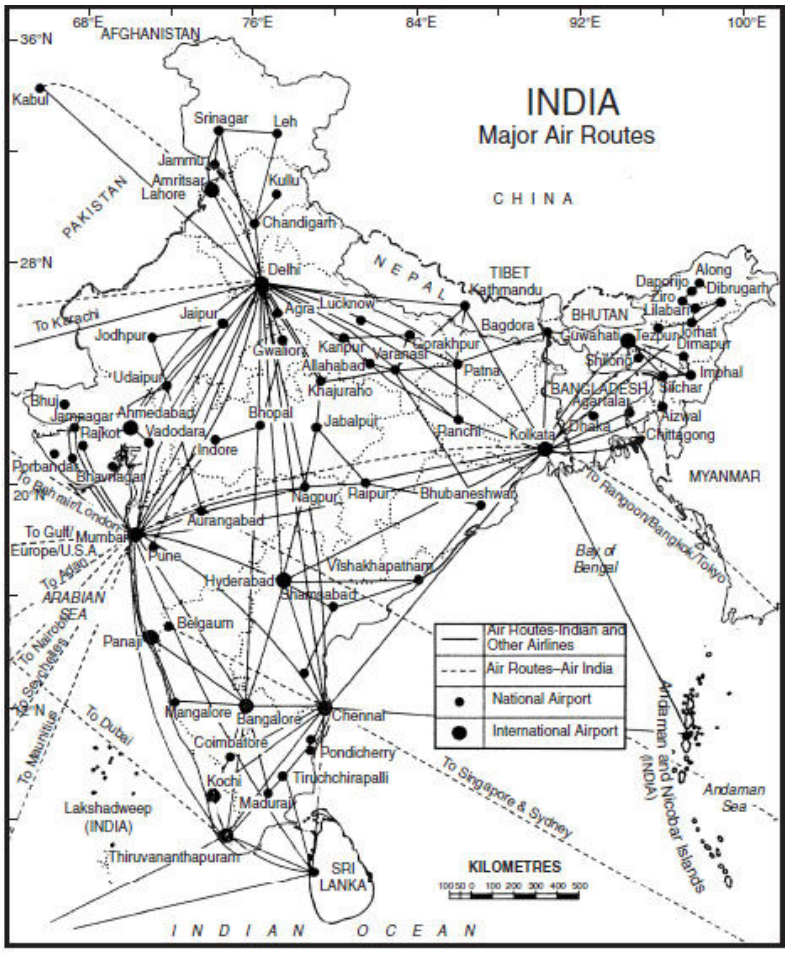
Indian Airlines: Indian Airlines was set up under the Air Corporation Act, 1953 with an initial capital of Rs. 3.25 crore with its headquarters at Delhi. The India Airlines is the major domestic air carrier of the country. The Indian Airlines operates to 55 domestic stations alongwith its wholly-owned subsidiary Airlines, Allied Services Ltd. (Alliance Air). Besides Indian Airlines also operates to 18 international stations.

The Indian Airlines presently has a fleet of 73 aircrafts comprising 03 Airbus, A-300s; 48 Airbus, A-320s, 5 Airbus, A-319s; 11 Boeing 737s; 18 Dornier DO-228; and 04 ATR-42-320. All B-737 and ATR aircrafts are operated by Alliance Air.

In addition to Air India, there are: Jet Airways, Kingfisher Airlines, Deccan Aviation Co; Jet-lite, Paramount Airways, MDLR Airways, Jagson Airlines and Go-Air airlines.

Pawan Hans Helicopters Limited: The Pawan Hans Helicopters Ltd (PHHL) is one of India's leading helicopter companies and is known for its reliable helicopter operations. The company was incorporated in 1985 with the objective of providing helicopter services to the petroleum sector, linking inaccessible areas of the country and operating charters for promotion of tourism.

Private Companies: In addition to these, five more companies provide services in India: out of these Jet Airways and Air Sahara operate on domestic as well as international air-routes, while Air Deccan, Kingfisher, and Spice-Jet operate on domestic routes only.



Map not to Scale

Fig. 12.11 Major Air-Routes

Problems

The air transport of India is facing the following main problems:

1. **Running in Loss:** Both the Air India and the Indian Airlines are incurring constant losses. Some better airlines like Singapore and Lufthansa have shown willingness to join with Air India.
2. **Strikes:** Strikes have become regular feature in the air-transport industry. For example, the September/October 2009 strike of pilots created great inconvenience to the passengers and heavy losses to the Indian Airlines.
3. **Decline in Quality of Service:** There has been a constant decline in the quality of air service offered by the Air India and the Indian Airlines.

4. **High Aircraft-Man Ratio:** The airlines manpower base is one of the highest in the world. Its average is about 700 employees per aircraft as against an average of 225 employees internationally.

5. **Stiff Competition**: The airlines are facing a stiff competition from foreign

airlines which are providing better and efficient service.

6. **Old Aircrafts**: The aircrafts of the airlines are more than 20 years old. The airlines need huge amount for the replacement of the old aircrafts.
7. **Expensive Fares:** The airlines in India are still serving only the elite class. Their fares are beyond the reach of middle and lower-middle class.
8. **Mismanagement and Political Interference:** The inefficient function of the management and political interference are bringing a bad name to the airlines. People prefer to travel by some alternate airlines.

New Aviation Policy 2016

The Government of India launched the new aviation policy which was approved by the cabinet on 15th June, 2016. The salient points of the new aviation policy are:

1. At present the number of about 80 million. The objective is to increase the number of flyers to 300 million by 2022.
2. Fifty new airports handling scheduled commercial flights in addition to the current 77 by 2019.
3. Total fare on one hour-long flights under regional connectivity scheme to unserved airports to cost about 2500 per ticket.
4. A small levy on passengers flying from top 18 airports.
5. Regional connectivity via 50-80 no-frills airport based on lower levies concessions.
6. Requirement of five years flying for starting international operations removed. But airlines should be able to deploy 20 aircraft or 20% of total capacity (in terms of average number of seats on all departures put together), whichever is higher for domestic operations. The 5/20 rule has been replaced with 0/20.
7. There are nearly 8000 pilots holding Commercial Pilot License (CPL) but who have not found any regular employment. The aviation ministry will develop a scheme with budgetary support for type-rating of pilots.
8. Open sky policy for SAARC region and countries beyond 5000 km from Delhi.
9. The new civil aviation policy included a string of other changes–new norms on airport classification, bilateral flying rights, ground handling and deployment for private security personal in non-core security functions.

The decision to scrap the condition of five years of domestic operation before an airline could fly international routes is welcome. It will add to competition in international routes and enhanced competition can only mean better service and lower fares for consumers. The scheme for regional connectivity through the proposed no-frills airports should expand the travel options available to those living in smaller towns, again a welcome development.

PIPELINES TRANSPORTATION

Pipelines provide the most convenient mode of transport of crude-oil, petroleum, petroleum products, natural gas, and solid minerals. The merits and demerits of pipeline transport have been given in the follwing:

Advantages of Pipelines: The main advantages of pipelines are given below:

1. Pipelines can be laid through rough and mountainous terrain as well as under water.
2. The operating and maintenance cost of pipelines are lower than that of other modes of transportation.
3. Pipelines involve low energy consumption.
4. The industrial regions are well integrated by pipeline construction.
5. Pipelines are free from bottle-necks due to poor road coniditions.
6. They are not affected by strikes.
7. They do not pollute the environment.

Disavantages of Pipelines:

1. Construction of pipelines is a very expensive and demand heavy investment.
2. The capacity of pipeline can not be increased.
3. The repair of pipeline is highly technical and difficult.
4. Security threats from terrorist.

The important pipelines of India are as under:

1. Naharkatia-Nunamati-Barauni Pipeline: This was the first pipeline constructed in India to bring to crude oil from Naharkatia oilfield to Nunamati (Upper Assam). Later, it was extended to transport crude oil refinery at Barauni in Bihar. It is about 1167 km long. Subsequently, it was extended up to Kanpur (U.P.). The pipeline between Naharkatia and Nunamati became operative in 1962 and that between Nunamati and Barauni in 1964. It has a number of pumping stations and subsidiary pipelines (Fig. 12.12).

   The subsidiary pipelines are:

(i) Nunamati-Siliguri Pipeline transport oil from Nunamati (Guwahati) in Assam to Sillguri	(ii) Lukwa-Rudrasagar-Barauni Pipeline has been constructed to carry crude oil from Lakwa and	(iii) Barauni-Haldia pipeline was completed in 1966. It transports refined petroleum	(iv) Barauni-Kanpur pipelines carries refined petroleum products from Barauni	(v) Nunamati-Bongaigaon Pipeline is used to transport raw materials for Bongaigaon	(vi) Haldia-Rajbhandh-Maurigrame Pipelines has been constructed to meet the requirements of southern

in West Bengal.	Rudrasagar to oil refinery at Barauni.	ptoducts to haldia Port and bring back crude oil to Barauni Refnery.	to Kanpur.	petro-chemical Complax.	parts of west Bengal.

2. Mumbai High-Mumbai-Ankleshwar Kayoli Pipeline: This pipeline connects oilfield of Mumbai High and Gujarat with oil refinery at koyali. A double line about 210 km long double pipeline connects Mumbai With Mumbai High. It provides facilities for transporting crude oil and natural gas. This pipeline was completed in 1965. It transports crude oil from Ankleshwar oilfield to Koyali Refinery (Fig. 12.12).
3. Salaya-koyali-Mathura Pipeline: An important pipeline was laid from Salaya in Gujarat to Mathura in Uttar Pradesh via Viragrame. This is 1256 km long pipeline which supplies crude oil to refineries at Koyali and Mathura. From Mathura, it has been extended to the oil refinery at Panipat (Haryana) and further to Jalandhar in Punjab (Fig. 12.12).

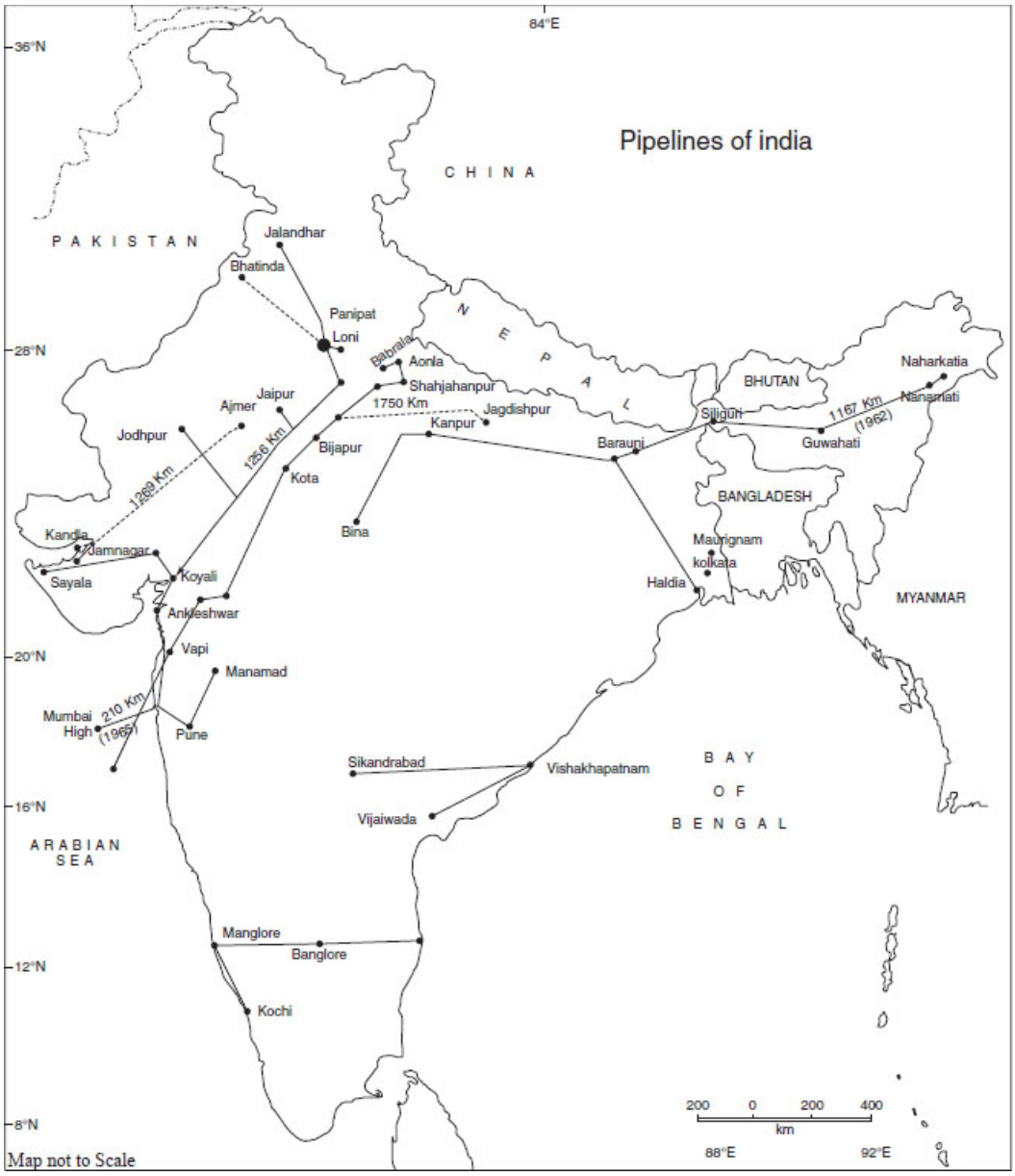


Fig. 12.12 Main Pipelines of India

4. Hajira-Bijapur-Jagdishpur (HBJ) Gas Pipeline: This pipeline has been constructed by Gas Authority of India Limited (GAIL) to Transport gas. It is, 1750 km long and connects Hazira in Maharashtra to Bijapur in Madhya Pradesh and Jagdishpur in Uttar Pradesh. It carries 18 million cubic metres of gas everyday to three power houses at Kawas (Gujarat), Anta (Rajasthan) and Auraiya (U.P.) and Six fertlizer plants at Bijapur, Sawai Madhopur, Jagdishpur, Aonla, Babrala and Shahjahanpur (Fig. 12.12).

   Each of the fertilizer plants has a capacity of producing 1350 tonnes of ammonia per day. This is the largest underground pipeline of the world and has brought a substantial transformation in the economy of rural Gujrat,

Madhya Pradesh, Rajasthan, Haryana and Uttar Pradesh. It has been extended up to Delhi so that enough gas is made available to meet the growing demand of the capital city (Fig. 12.12).

5. Jamnagar-Loni LPG Pipeline: This 1269 km long gas-pipeline was constructed by the Gas Authority of India Limited (GAIL). It connects Jamnagar in Gujrat to Loni near Delhi in U.P and passes through the states of Gujrat, Rajasthan, Haryana and Uttar Pradesh. It is transporting 3.5 Lakh LPG gas cylinders across 1269 km every day.

   This is the first time that cross-country pipeline has been used to transport LPG. LPG is received at various point along the route for bottling is Ajmer and Jaipur (Rajasthan), Piyala (Haryana), Madanpur Khadar (Delhi) and Loni (U.P.).

6. Kandla-Bhatinda Pipeline: This 1331 km long pipeline connects Kandla with Bhatinda (Fig. 12.12).

   Solid Mineral Pipelines:

1. Bailadila (Chattisgarh) to Vishakhapatnam (Andhra Pradesh) for the transportation of iron-ore.	2. Kudarmukh to Manglore Port (transportation of iron-ore).	3. Debari to Udaipur Pipeline constructed to transport zinc-ore and products (Rajasthan).

Urban Infrastructure

Urban infrastructure consists of drinking water, sanitation, sewage systems, electricity and gas distribution, urban transport, primary health services, and environmental regulation. The process of urbanisation has gathered considerable momentum in recent times and this has put urban infrastructure and services under severe strain. For example, several cities (Shimla, Udaipur, Jodhpur, Udaipur, Delhi, Bikaner, etc.) in India have only two-to-four hours of water-supply during the entire day. Smaller cities, because of inadequate financial resources, have found it particularly difficult to cope up with the increasing demands of services.

METRO RAIL SYSTEM

Metro is a rapid mode of transportation in the mega and million cities. It helps in reducing the pressure on the available road transport. It is almost pollution free and eco-friendly transport. The Ist metro system was started in Kolkata in 1984. Metro transport is available in Chennai also. Out of these, the Delhi Metro is

India's first modern metro and the third rapid transit system in India. The salient features of the Delhi Metro Rail system have been concisely given in the following:

Delhi Metro Rail System

The Delhi Metro Rail System, technically known as Delhi Mass Rapid Transit System (DMRTS) is serving Delhi and its satellite cities of Faridabad, Gurugram (Gurgaon), Noida, Ghaziabad in the National Capital region of India. Delhi Metro is the 12th largest metro system of the world, both in length and the number of railway stations. The network consists of five colour-coded regular lines and the faster Airport Express Line, with a total length of 214 kilometers, serving 164 stations 9 including 6 on Airport Express Line. The system has mix of underground, at-grade, and elevated stations using both broad-gauge and standard-gauge. The metro has an average daily ridership of 2.76 million passengers (June, 2017).

Planning for the metro was started in 1984, when the Delhi Development Authority and the Urban Arts the Urban Arts Commission came up with a proposal for developing a multi-modal transport system for the city. The Delhi Metro Rail Corporation (DMRC) was incorporated in May 1995. Construction was started in 1998, and the first section on the Red Line was opened in 2002. The development network was divided in to phases. Phase I containing 3 lines was completed by 2006, and Phase II in 2011. Phase III is scheduled to be completed by 2017.

The Rapid Metro Rail Gurugram (Gurgaon) was opened in 2013, while linked to Delhi Metro by the Yellow Line is a separate metro system (with a different owner/operator than the Delhi Metro), although tokens form the Delhi Metro can be used in this network.

The main routes of the Delhi Metro System have been given in the **(**Table 12.5-A).

Table 12.5-A

A Salient Features of the Major Metro Lines

Line	First Operation	Last Extension	Station	Length (km)	Terminals
Red Line	2002-12-24	2008-06-04	21	25.09	Dilshad-Garden-Ritahala
Yellow Line	2004-12-20	2015-11-10	37	49	Samaypur Badli-HUDA City Centre
Blue Line	2005-12-31	2010-10-30	44	49.93	Noida City Centre-Dwarka Sector 21
Green Line	2011-08-27	–	2	3.32	Ashok park Main-Kirti-Nagar
Violet Line	2010-10-03	2017-05-28	32	40.34	Kashmiri Gate-Escorts Mujesar

Line	First Operation	Last Extension	Station	Length (km)	Terminals
Airport Express	2011-02-23	–	6	22.70	New Delhi-Dwarka Sector 21
Magenta Line	2017-12-25		16	24.82	Janakpuri West-Kalkaji Mandir
Pink Line	2018-03-14		39	58.59	Mukundpur Depot-Shiv Nagar

Source: "Delhi Metro About Us Introduction" (htt:/www.delhimetrorail.)

Transport Planning

It may be ascertained from the description and analysis of the transport network of the country that there is urgent need of coordination in the railways, roads, air, and water transport. There are still several areas in the country which are not easily accessible. In fact, many villages of the country are not linked with metalled roads. Moreover, there is a stiff competition between the railways and the roads transport which exposes a faulty transport planning. Moreover, much attention has not been paid towards the development and maintenance of water transport. Air travel is very expensive for the ordinary citizens. The following attempts, if taken together may improve the transport system of the country appreciably.

1. **Geographical Bases of Transport Planning:** Planning for transport should be based on geographical conditions of terrain, topography, slope, and climate. For example, roads and railways are the good means of transportation in the plains, while roads and airlines/helicopters can be a good alternative in the mountainous areas.
2. **Comprehensive Transport Plan:** A comprehensive transport plan should be chalked out and prepared to promote integration amongst different modes of transportation.
3. **Single Broad Gauge in Railways:** The entire railway transport should be brought under Broad Gauge to avoid transport delay and bottlenecks.
4. **Widening of Roads:** There should be more emphasis on the development of National and State Highways to improve the connectivity and efficiency of the road transport network.
5. **Integration of the Transport System:** There is a need to coordinate the train, road and air transport services.
6. **Water Transport:** Water transport development and its maintenance deserve more attention as its the cheapest mode of transportation.

7. Infrastructure: The social amenities at the railway stations, bus stands, and airports need a substantial improvement.

8. **New Technology:** The new technology of transport is to be taken at a preferential bases.

9. **Involvement of Private Sector:** For the generation of funds there should be more involvement of the private sector in the transport industry.

Traffic accidents

IIn 2018, 4.61 lakh road accidents were reported with almost 1.49 lakh deaths. Uttar Pradesh had 22,256 accidents recorded, Haryana 5118 deaths in 2018 and Tamil Nadu with the lowest count. *(The Times of India, March 12, 2019)*.

According to information from the Indian Railways, nearly 50,000 people have lost their lives between 2015 and 2017 on railway tracks after being hit by trains

The *Seemanchal Express derailment* occurred on February 3, 2019 in the Vaishali district of Bihar in which six people were killed and more than 30 injured

Superior and safer *Linke Hofmann Busch (LHB) Coaches* having anti-climbing features of Indian Railways are necessary steps to reduce the fatalities in case of accidents.

Road Safety

The Government has approved a ***National Road Safety Policy*** so as to minimise road accidents. The various measures included in it are promoting awareness, establishing road safety information data base, identification and rectification of road accident black spots , improving

The important schemes on road safety operated by the Ministry *include publicity measures and awareness campaign on road safety, Scheme for Setting up Institutes for Driving Training, Refresher Training of Drivers in Unorganized Sector and Human Resource Development, National Highways Accident Relief Service Scheme (NHARSS)*, setting up of inspection and certification centres and road safety and pollution testing equipment and programme implementation.

All the passenger buses and taxis are mandated to be fitted with GPS devices and panic buttons to enable real time tracking and interventions in times of crisis to enhance the safety of the passengers.

COMMUNICATIONS

Postal System

The establishment of modern postal system in India can be traced back to the second half of the 18^{th} century. This postal system, established by Lord Clive in the year 1766, was further developed by Warren Hastings by establishing the Calcutta General Post Office (GPO) under the Post Master General in the year 1774. In the other Presidencies of Bombay and Madras, the General Post Office came into existence in 1786 and 1793, respectively. The Act of 1873 first regulated the Post Office on a uniform basis to unite the post office organisation throughout the three presidencies into one All India Service. The Post Office Act 1854 reformed the entire fabric of the Postal System. Post Office of India was

placed on the present administrative footing about one hundred and fifty years ago on 1 October, 1854. The statute presently governing the postal services in the country is the Indian Post Office Act, 1898.

Besides providing postal communication facilities, the post office network has also provided facilities for remittance of funds, banking and insurance services from the latter half of the 19th century.

Postal Network

At the time of Independence there were 23,344 post offices throughout the country. In April 2005, the country had 155,516 post offices, of which 139,120 were in the rural areas and 16,396 in the urban areas. In 2017, the number has increased to 154,965 of which 139,067 are in the rural areas. As a result of this seven-fold growth, today India has the largest postal network in the world.

The postal network consists of four categories of post offices, viz., (i) Head Post Office,
(ii) Sub-Post Offices, (iii) Extra Departmental Sub Post Offices, and (iv) Extra Departmental Branch Post Offices. All the categories of Post Offices retain similar postal services, while delivery function is restricted to specified offices. In terms of management control, accounts are consolidated progressively from Branch Post Offices to Sub Post Offices and finally in Head Post Offices.

The Department had about 2.47 lakh departmental employees and about 2.93 lakh Gramin Dal Sevaks as on March 31, 2005.

Mail System

First class mail, viz., post cards, inland letters, and envelops are given airlift wherever found advantageous, without any surcharge, between stations connected by air. Second class mail, viz., book packets, registered newspapers, and periodicals are carried by surface transport, i.e., trains and road transport.

International Mails

India is a member of the Universal Postal Union (UPU) since 1876 and of the Asian Pacific Postal Union (APPU) since 1964. These organisations aim at extending, facilitating, and improving postal relations among other countries. India exchanges mail with more than 217 countries by air and surface.

Money can be remitted from selected foreign countries to India by way of money orders and postal orders. India has money order service with 27 countries. India has two-way money order service with Bhutan and Nepal wherein money orders can be sent to and received from these countries. With the remaining 25 countries, only inward service is available where money orders booked in these countries can be paid in India. British Postal Orders and Irish Postal Orders are encashable in India at selected post offices.

International EMS, which started in 1986 with five countries, has now been extended to 97 countries. With a view to facilitate export and import to and from

foreign destinations, principal foreign offices of exchanges have been set up at Chennai, Delhi, Kolkata, and Mumbai. In addition, six sub-foreign post offices have been established at Ahmadabad, Bangalore, Cochin, Jaipur, NOIDA, and Srinagar. Export Extension Windows have also been made operative at Guwahati, Kanpur, Ludhiana, Moradabad, Surat, and Varanasi to cater to the needs of the exporters, tourists in these areas.

Telecommunication

India's telecom sector has been one of the biggest success stories of market-oriented reforms, and India is now amongst the fastest growing telecom markets in the world. The total number of telephones has increased to 190 million in 31 December, 2006.

The announcement of the New Telecom Policy, 1999 was a watershed event in telecommunications
in India. Other policy milestones include the opening of the long-distance market in 2002. As a result, telecom tariffs which were among the highest in the world less than four years ago have now dipped among the lowest. The growth of wireless services has been phenomenal, with wireless subscribers growing at a compound annual growth rate of above 90% per annum since 2003.

The total number of telephone has increased from 54.63 million on March 31, 2006 to 189.92 million on December 31, 2006. About five million subscribers are added every month. With this growth, the number of telephones is expected to reach 250 million by the end of 2007. Today the wireless subscribers are not only much more than the fixed subscribers in the country, but also increasing at a much faster pace. The share of wireless phones has increased from 24% in March 2003 to 79% in December, 2006.

Internet

As on December 31, 2006, there are 400 licenses for provision of Internet Services out of which 128 have signed Licenses for Provisions of Internet Services (including Internet Telephony). Based on reports received from Internet Service Providers till March 2006, there are approximately 12.00 million Internet subscribers in India (India 2010).

Manufacturing of Telecom Equipments

The Indian telephone industry manufactures a complete range of telecom equipments using state of the art technologies designed specifically to match the diverse terrain and climatic conditions. Production of telecom equipments has increased from Rs.16,090 crore in 2004–05 to Rs. 17,833 crore in 2005–06. There is heavy demand of the mobile phones from domestic and international markets. It is expected that within the next decade, India is expected to become a manufacturing hub for telecom equipments.

Telegraph and Telephone Service

Telegraph services were introduced at Calcutta in 1851 and a telephone service also began at Calcutta in 1851–82 soon after their invention in the United States of America. Today there are over 45,000 telegraph offices. There are over 25,000 telephone exchanges in the country by 2005–06. Ours is the largest telecom network in Asia.

Radio, Television, and Cinema

Radio, television and cinema (films) are the electronic media of mass communication, unlike postal, telegraphic, telephone, and telex services which are essentially of personal nature. Unlike telephone, radio is a means of wireless communication. It is a very powerful medium to transmit and receive useful information, news and variety of entertainment programmes including sports.

The All India Radio came to be known as '*Akashvani*' from 1957. At the time of Independence, there were six radio (Akashwani) stations. At preset All India Radio is accessible to almost 98.5% of the total population of the country.

Prasar Bharati (*Broadcasting Corporation of India*) is the public service broadcaster in the country. *Akashvani* (All India Radio) and *Doordarshan* are its two constituents. It came into existence on November 23, 1997, with a mandate to organize and conduct public broadcasting services to inform, educate and entertain the public and to ensure a balanced development of broadcasting on radio and television.

The ***News Services Division (NSD)*** of All India Radio has been expanding its presence across various domains with the aim of reaching diverse cross section of people. NSD has various units viz., the Reporting Unit, Talks and Current Affairs Unit, Reference Unit and the Monitoring Unit. The live webcasting of "*Mann Ki Baat*" is done on air.nic.in.

DD National Channel, a public service broadcaster is the largest terrestrial network in the world. It covers about 92.0 per cent population and 81.0 per cent land area of the country.

DD News is the only terrestrial-cum-satellite, multilingual news channel of the country. DDNews channel was launched in November 2003 by converting DD-Metro into a 24-hours news channel. DD News terrestrial reach is 49 per cent area of the country.

Doordarshan opened its windows to the world (DD -India) by launching its international channel on March 14, 1995.

DD Urdu came into existence in 2006 which became 24x7 channel with effect from 2007.Sports Channel of Doordarshan was launched in 1999.

In India, ***Electronic Media Monitoring Centre (EMMC)*** is entrusted with the task of monitoring the content being aired by TV channels. EMMC is among one

of the premier organisations of the world that monitor the broadcasting sector and the content being aired by broadcasters.

Doordarshan is the national television service of India. It was started in 1959. It is one of the largest and essential networks in the world with over 900 transmitters. Its programmes are watched by over 500 million viewers in their homes. Its commercial advertisements brought in a huge revenue of over Rs.10,000 million in 2005–06. Now under the open skies policies a large number of private parties registered in India. It manages radio and television channels and have been assigned to private parties registered in India. It manages radio and television services under the supervision of parliamentary Committee.

Cinema is the most popular means of entertainment in India. It is the largest producer of feature films in the world. Hindi films by far is the most dominant section. Almost all the regional languages bring out their own films almost continuously. Personal computers and Internet services have brought about a new revolution in the age of explosion of information technology.

Print Media

The Office of the ***Registrar of Newspapers for India (RNI)*** is an attached office of the Ministry of Information and Broadcasting. (***Source:*** *India 2019*)

The ***Press Information Bureau (PIB)*** is the nodal agency of the Government of India to disseminate information to the print, electronic and social media on government policies, programmes, initiatives and achievements. It functions as an interface between the government and the media and also provides feedback to the government on the reaction of people as reflected in the media. (***Source:*** *India 2019*)

PM's Publicity and Reference Unit is a dedicated unit for the publicity and media support for the Prime Minister's Office and cabinet. (***Source:*** *India 2019*)

The total number of newspapers and periodicals being published was over 42,000 on 31 December, 2006. Hindi publication has the largest share of over 40% of the total. Books are an equally important means of communication for preserving and propagating knowledge, information and entertainment to posterity.

Social Media

PIB has social media presence on Twitter, Facebook, YouTube and Instagram. The information about various events/activities/programmes of ministries and departments is put out on social media for immediate and wider dissemination.

Twitter: On twitter, PIB's handle, @pibindia and @pibhindi have nearly 1.35 million+followers and 42,000+followers respectively.

Facebook: PIB's Facebook page, www.facebook.com/pibindia has reached 1,20,000+subscribers.

Instagram: PIB's Instagram gallery, www.instagram.com/pibindia has reached 52,000+subscribers.(***Source***: *India 2019*)

INTERNATIONAL TRADE

A flow of commodities from producers to consumers is known as trade. It is an important tertiary sector of economy which is carried out at the local, regional, national, and international levels.

India has a long tradition of trading with countries located far and near. Today we are living in a fast shrinking world mainly because of tremendous advances in both transport and communications. We are living in an economically interdependent world, where the world itself has turned into a global village with a self-contained economy. Because of fast means of transport and communications, India conducts international trade with 203 countries with an endless list of around 8000 commodities. The share of trade in the GDP is about 15%, which engages 7.3% of work force in the country.

There was a time when commodities were imported only for domestic consumption. But now more and more countries including India, have been importing certain raw or semi-processed materials, not for their own domestic consumption, but to process them further and export them after value addition. *Indirectly, what the country exports are items of human skills.* For example, Japan exports cotton and woollen textiles although it imports all its requirements of raw materials from other countries. The same is true of mineral based industries. India imports raw cashewnuts only to re-export them after they are further reprocessed. India imports crude diamonds and other precious stones only to process them further and re-export them as highly finished fine products at a considerable upward margin. India also imports gold and silver and exporting them in the form of attractive and expensive ornaments. International or foreign trade has played a crucial role in India's economic growth.

During the colonial days India used to export almost entirely agricultural raw materials. India's traditional exports consisted of jute, cotton, tea, spices, hides, skins, oil-seeds, especially ground-nut. At present, instead of exporting raw materials, India is exporting jute-packing materials (gunny-bags, linen, and carpets). In place of cotton, India exports quality yarn, cotton fabrics, including hosiery, readymade garments and skilled and semi-skilled workers. India also exports silk woollen, and synthetic textiles, processed marine products, manufactured leather goods, sports goods (cricket bats, hockey sticks, etc.) engineering goods like fans, sewing machines, bicycles, three-wheelers, scooters, cars, commercial vehicles, chemicals and allied products, carpets, rice, processed food, medicines, electric goods, books and films.

Table 12.6 shows India's total export, import, total value of foreign trade and balance of trade from the year 1990–91, 2000–01, and 2005–06.

Table 12.6

India's Foreign Trade (in Rupees crore)

Year	Exports	Imports	Total trade	Trade deficit
1950–51	606	608	1214	– 2
1960–61	642	1122	1764	– 480
1970–71	1535	1634	3169	– 99
1980–81	6710	12,549	19,259	– 99
1990–91	32,558	43,193	75,751	– 10,635
2000–01	203,571	230,873	434,444	– 27,302
2008–09	696,498	870,399	1,566,897	–173,901
2014–15	1,896,348	2,737,087	4,633,435	– 840,738

Source: India 2016, p.158.

India's total external trade (exports plus imports including re-exports) in the year 1950–51 was

Rs. 1214 crore. Since then, this has witnessed continuous increase with occasional downturns. During 2008–09, the value of India's external trade reached Rs. 1,566,897 crore **(Table 12.6).** It may also be noted from this Table that the balance of payment of India is adverse throughout the planning period.

It may be seen from **(Table 12.7)** that the major items of export from India are engineering goods (20.7%), followed by gems and jewellery (15.1%), textiles (14.5%), chemicals and related products (11.6%), and agricultural products (10.2%).

Table 12.7

India—Commodity Composition of Exports

Commodity Group	Percentage Share 2016–17
I- Primary Products	**15.4**
(i) Agriculture and allied	10.2
(ii) Ores and minerals	5.2
II- Manufactured Goods	**65.7**
(iii) Engineering goods	20.7
(iv) Gems and jewellery	15.1
(v) Textile including RMG	14.5
(vi) Chemicals and related products	11.6
(vii) Leather and manufactures	02.6
(viii) Handicrafts including hand-made-carpets	01.2
III- Petroleum, crude and products (including coal)	
	11.5
IV- Others	**07.4**

Commodity Group	Percentage Share 2016–17
Total	100.00

Source: India, 2017.

An examination of **(Table 12.8)** vividly shows that about 34% of the total import consists of crude oil, petroleum, and allied products. Capital goods constituting 12.1% of the total import ranks second, while electric goods and gold and silver rank third and fourth in the list of import, respectively. Other important items of import are chemicals, pearls and precious stones followed by metal scraps and coke.

Table 12.8

India—Imports of Principal Commodities (2016–17)

Commodity	Percentage Share
Petroleum, crude-oil and petroleum products	33.7
Capital goods	12.1
Electronic goods	09.4
Gold and silver	08.6
Chemicals	05.5
Pearls, precious and semi-precious stones	04.1
Metal scrap	03.9
Coke coal and briquettes	02.4
Edible oils	01.3
Professional instruments and optical goods	01.3
Other commodities	17.7
Total	100.00

Source: India, 2017

About three-fifths (60.4%) of Indian exports in 2018 were delivered to the above trade partners given in **(Table 12.9)**.

Table 12.9

India's Major Trading Partners 2018

	Country	2018–2019 (April-March) in Billions	Percentage Share in the India's Total Exports
1.	USA	52.4	15.88
2.	United Arab Emirates	30.1	9.13

	Country	2018–2019 (April-March) in Billions	Percentage Share in the India's Total Exports
3.	China	16.7	5.07
4.	Hong Kong	13.0	3.94
5.	Singapore	11.6	3.51
6.	U. K	9.3	2.80
7.	Bangaladesh	9.2	2.79
8.	Germany	8.9	2.70
9.	Netherlands	8.8	2.67
10.	Nepal	7.7	2.35

Source: Directorate General of foreign Trade, 2018

SALIENT FEATURES OF FOREIGN TRADE

Following are the salient features of the Indian foreign trade:

1. Unfavourable Balance of Trade

India is importing enormous quantity of crude-oil, petroleum, petroleum products, precious stones, gold, silver, copper, machinery, cashewnuts, fertilisers, and mainly exporting agro-based products, engineering goods, commercial automobiles, etc. The balance of trade and balance of payment are still unfavourable as import exceeds the export. India's balance of payment in the post-reform period (1991) is growing from strength to strength.

2. More Export of Manufactured Goods

India is exporting nearly 8000 commodities, and the number of exports of finished commodities is increasing appreciably.

3. Worldwide Trade

India is the 17th largest economy in the world. India exports its products to 203 countries and imports from 195 countries.

4. Change in Import

Earlier India used to import almost all types of machinery, precision instruments, surgical equipments, automobiles, and electrical goods. Now India is exporting all sorts of machinery including vehicles, electronic, and chemical goods, and

importing raw materials like diamonds, precious stones, gold, cashew-nuts, cotton, jute, mineral ores, and raw and semi-processed raw materials.

5. Maritime Trade

About 96% of our foreign trade is carried out through sea routes. Trade through land routes is limited either because of the physical barriers of Himalayas or the less friendly relations with the neighbours.

6. Trade through Selected Ports

India has only 13 major international seaports which handle about 92% of foreign trade. The remaining seaports handle insignificant amount of the foreign trade.

7. Insignificant Position in the International Trade

India has almost 17% of the total population of the world but unfortunately, its share in the international trade is less than one per cent.

8. State Trading

Over 95% of the overseas trade is done in public sector by the state agencies. There is insignificant trade done by the private undertakings.

9. Increasing Import of Raw Material

India is importing cashew-nut, cotton, gems, jute, mineral ores, pearls, precious and semi-precious stones, and raw silk.

10. Increasing Import of Capital Goods

Goods like manufactures of metals, electrical and non-electrical machinery, transport equipments, chemicals, and new technolsogy are being increasingly imported.

Export Processing Zones (EPZ)

In order to promote International trade, the government of India has created seven Export Processing Zones. The zones and their headquarters are as under:

1. Chennai (Tamil Nadu), 2. Falta (West Bengal), 3. Kandla (Gujarat), 4.Kochi (Kerala), 5. NOIDA (Uttar Pradesh), 6. Santa Cruz (Maharashtra), and 7. Vishakhapatnam (Andhra Pradesh).

Each zone provides the basic infrastructural facilities in addition to whole range of fiscal incentives. Customs clearance facilities are offered within the zones. However, EPZs were not able to emerge as effective instruments for export promotion on account of multiplicity of controls and clearances, absence of world class infrastructure, and an unstable fiscal regime.

General Agreement on Tariff and Trade (GATT)

The General Agreement on Tariff and Trade (GATT) was established in 1948 in Geneva to pursue the objective of free trade in order to encourage growth and development amongst all member countries of the world. The main objective of the GATT was to ensure competition in commodity trade through the removal or

reduction of trade barriers. The first seven rounds of negotiations conducted under GATT were aimed at stimulating international trade through reduction in tariff barriers and also by reduction in non-tariff restricts on imports imposed by the member countries.

BALANCE OF TRADE AND BALANCE OF PAYMENT

The exports and imports of a country should be roughly equal in value, since the foreign exchange earned by exports is necessary to finance imports, but such a balance is rarely achieved. This is partly because trade passes through the hands of many different companies working independently, and thus an exact balance can never be reached, but is also due to fluctuations in markets leading to changes in import and export values over a period of years. The difference in value between imports and exports is referred to as the balance of trade. If exports exceed imports a country is said to have a *favourable balance of trade*, while if imports exceed exports it has an *adverse balance of trade*. These terms are relics of an earlier trading era, when if a country exported goods in large quantities they were paid for in gold. Thus, an excess of exports brought an addition to gold reserves. A country which has consistent adverse balance of trade, as Britain has done for more than a century, can enjoy great prosperity, while a favourable balance of trade is often characteristic of underdeveloped countries today, which often have great financial difficulties notwithstanding.

The balance of trade only takes account of visible trade or the value of actual goods transferred from one country to another. But there are many other ways in which foreign exchange can be earned or spent. These are collectively called *invisible trade* which accounts for a quarter of all transactions with foreign countries can be worked out. This is called the *balance of payments*.

Transactions which bring money into the country are called invisible exports and can be of several kinds.

1. Payment for financial services including insurance, banking, brokage, and other services carried out on behalf of foreigners.
2. **Payment of transport services** such as shipping or air transport of passengers or freight. Britain and certain other European countries have large invisible earnings in these two fields because of their importance in trade and financial dealings.
3. **Expenditure by foreign tourists** is often an extremely important source of foreign exchange.
4. **Interest and dividends on foreign investments**. India is earning a

substantial amount in the form of interest and profit on foreign investment, annually.

5. **Remittance from emigrants.** Many emigrants send money to their families and thus countries like India, which have supplied large number of emigrants, may receive considerable foreign exchange in this way.
6. **Loans and aids** from foreign countries or international organisations. Many underdeveloped countries receive aid or loans to finance development, while other countries may obtain loans to cover balance of payments deficits.

Transactions which take money out of the country are called *invisible imports*. Payment for services, payment of interest on investments, remittances of immigrants, repayment and interest of foreign loans, or deficit on tourism, may all be greater than invisible exports in the same fields.

When these invisibles, both exports and imports, are brought into account, many countries which would have an unfavourable balance of trade are found to have a more favourable balance of payments. It is much more important that the total payments and receipts rather than the visible trade, should be well balanced as long-term loans may have to be obtained, which may lead to future difficulties when they have to be repaid; or gold reserves may be reduced; or overseas assets may be sold so that longer-term invisible exports are reduced; or currency values may be changed by devaluation or revaluation to alter the relative value of exports and imports and thus balance total transactions more nearly.

All these measures may have detrimental effects on the economy in the long term but have to be resorted to by many countries. Other ways of balancing transactions are to increase visible or invisible exports, or to apply trading restrictions in order to cut down on imports. Countries may also reduce the mobility of money, restrict tourism, or confiscate foreign property to reduce the outflow of money.

Since the economic reforms of 1991, the Balance of Payment (BoP) of India is continuously showing an upward trend.The growing strength of India's Balance of Payment, observed in the post reform period since the crisis of 1991 continued in 2005–06. This growing strength was in spite of widening of the current account deficit. With a burgeoning trade deficit, primarily on account of rising oil prices, the reversal from current account surplus witnessed between 2001–02 and 2003–04 to a current account deficit in 2004–05 appears to be continuing into 2006–07.

TRADE POLICY

India's exports and imports have been guided, if at all, under different relevant acts and rules and regulations. It was for the first time in 1988, that an Export

Import Policy-that too for three years–was promulgated. It had three-fold objectives:

1. To stimulate industrial growth by providing easy access to essential imported capital goods, raw materials, and components to industry.
2. To sustain the movement towards modernisation and technological upgradation.
3. To make the industry progressively competitive internationally.

The enforcement of the policy, however, has been sectoral and tentative, and it has been more or less like a declaration of intentions and objectives rather than a policy strategy of action. It is also not clear why the formulation of a national trade policy and its declaration had to wait until 2004.

Foreign Trade Policy

To make India a major player in world trade, a comprehensive view is necessary.While increase in exports is of vital importance, we have also to facilitate those imports which are required to stimulate our economy. Thus, independent of the annual EXIM policy, it is necessary to take an overall view of India's foreign trade. This is the context of the new Foreign Trade Policy.

The objectives of the new policies are:

1. To double our percentage share of global merchandise trade within the next five years.
2. To act as an effective instrument of economic growth by giving a thrust to employment generation.

These objectives are proposed to be achieved by adopting the following strategies:

(i) Unshackling of controls and enabling the innate entrepreneurship of our businessmen, industrialists, and traders.

(ii) Simplifying procedures and bringing down transaction costs.

(iii) Neutralisation incidence of all levies and duties on inputs used in export products based on the fundamental principle that duties and levies should not be exported.

(iv) Facilitating development of India as a global hub for manufacturing trading and servicing.

(v) Identifying and nurturing special focus areas which would generate employment opportunities particularly in semi urban and rural areas and developing a series of 'initiatives' for each of these.

(vi) Facilitating technological and infrastructural upgradation of all the sectors of the Indian economy, especially through import of capital goods and equipments, thereby increasing value addition and productivity while attaining internationally accepted standards of quality.

(vii) Avoiding inverted duty structures and ensuring that our domestic sectors are not disadvantaged in the Free Trade Agreements/Preferential Trade Agreements that we enter into in order to enhance our exports.

(viii) Upgrading our infrastructural network, both physical and virtual, related to the entire foreign trade chain, to international standard.

(ix) Revitalising the Board of Trade by redefining its role, giving it due recognition and inducting experts on Trade Policy.

(x) Activating our Embassies as key players in our export strategy and linking our Commercial Wings abroad through an electronic platform for real time trade intelligence.

INDIA—SPACE PROGRAMME

History: India's experience in rocketery began in ancient times when fireworks were first used in the country, a technology invented in neighbouring China, and which had an extensive two-way exchange of ideas and goods with India, connected by the Silk Road. Military use of rockets by Indians during the Mysore War against the British inspired William Congreve to invent the Congreve rocket, predecessor of modern artillery rockets, in 1804. After India gained Indepenedence from British occupation in 1947, Indian scientists and politicians recognised the potential of rocket technology in both defence applications, and for research development. Recognising the fact that a country as demographically large as India would require its own independent space capabilities, and recognising the early potential of satellites in the fields of remote sensing and communication, these visionaries set about establishing a space research organisation.

Phase I: 1960–70

Dr. Vikram Sarabhai was the founding father of the Indian space programme, and is considered not only a scientific visionary by many but also a national hero. After the launch of Sputnik in 1957, he recognised the potential that satellites provided. India's first Prime Minister, Pt. Jawaharlal Nehru, who saw scientific development as an essential part of India's future, placed space research under the jurisdiction of the Department of Atomic Evergy in 1961. The DAE Director Homi Bhabha, who is regarded as the father of India's atomic programme, then established the Indian National Committee for Space Research (INCOSPAR) with Dr. Sarabhai as Chairman in 1962.

From its establishment in 1962, the Indian space programme began establishing itself with the launch of sounding rockets, which was complemented by India's geographical proximity to the equator. These were established from the newly-established Thumba Equatorial Rocket Launching Station (TERLS), built near

Thiruvananthapuram in south Kerala. Subsequently, India developed indigenous technology of sounding rockets called *Rohini Family* of sounding rockets.

Recognising the need for indigenous technology, and possibility of future instability in the supply of parts and technology, the Indian space programme endeavoured to indigenise every material supply route, mechanism, and technology. As the Indian Rohini Programme continued to launch sounding rockets of greater size and complexity, the space programme expanded and was eventually given its own government department, separate from the department of Atomic Energy. In 1969, the India Space Research Organisation (ISRO) was created and finally the Department of Space was established in 1972.

Phase II: 1970–80

Sarabhai had taken part in an early study with NASA regarding the feasibility of using satellites for applications as wide as direct television broadcasting. India started designing and creating an independent launch vehicle. Meanwhile, India also began development of satellite technology, anticipating the remote sensing and communication needs of the future. India's first foray into space began with the launch of its satellite Aryabhata in 1975 by a Soviet booster. By 1979, the SLV was ready to be launched from a newly-established second launch site, the Sriharikota Rocket Launching Station (SRLS). The first launch in 1979 was a failure, attributed to control failure in the second stage. By 1980, this problem had been worked out. *The first indigenous satellite launched by India was called Rohini.*

Phase III: 1980–90

Following the success of the SLV, ISRO was keen to begin construction of a satellite launch vehicle that would be able to put truly useful satellite into polar orbit. The Augmented Satellite Launch Vehicle (ASLV) was tested in 1987, but this launch was a failure. After minor corrections, another launch was attempted in 1988, and this launch again failed.

Phase IV: 1990–2000

It was not until 1992 that the first successful launch of the ASLV took place. The first successful launch took place in 1994, and since then, the PSLV has become the workhorse launch vehicle, placing both remote sensing and communications satellites into orbit, creating the largest cluster in the world, and providing unique data to Indian industry and agriculture.

Phase V: 2000–2010

In 2001, the first development flight of the GSLV took place. India is developing a project to send unmanned probe to the moon in 2008, as the first attempt at exploration of solar system.This project is called *Chndrayaan*.

ISRO has entered the lucrative market of launching payloads of other nations upon its rockets from Indian soil. ISRO is planning a mission to Mars early in the next decade.

Major Events

1962: Indian National Committee for Space Research (INCOSPAR), formed by the Department of Atomic Energy, and work on establishing Thumba Equatorial Rocket Launching Station (TERLS) near Trivendrum began.

1963: First sounding rocket launched from TERLS on November 21, 1963.

1965: Space Science & Technology Centre (SSTC) established in Thumba.

1967: Satellite Telecommunication Earth Station set up at Ahmedabad.

1972: Space Commission and Department of Space set up.

1975: First Indian Satellite, Aryabhatt launched (April 19, 1975).

1976: Satellite Industrial Television Experiment (SITE) conducted.

1979: Bhaskara-1, an experimental satellite, launched. First experimental launch of SLV-3 with Rohini satellite on board failed.

1980: Second experimental launch of SLV-3; Rohini satellite successfully placed in orbit.

1981: Bhaskara II launched on November 20.

1982: INSAT-1A launched (April), deactivated in September.

1983: Second launch of SLV-3; RD-D2 placed in orbit. INSAT-1B launched.

1984: Indo-Soviet manned space mission (April). Rakesh Sharma became the first Indian to reach space.

1987: ASLV with SROSS-1 satellite on board launched.

1988: First Indian Remote Sensing Satellite, IRS-1A launched.

1990: INSAT-1D launched successfully.

1991: Launch of second operational Remote Sensing Satellite, IRS-1B (August).

1992: Third developmental launch of ASLV witth SROCC-C on board (May). Satellite placed in orbit. First indigenously built satellite, INSAT-2A, launched successfully.

1993: INSAT-2B launched in July successfully. First developmental launch of PSLV with IRS-1E on board fails.

1994: Fourth developmental launch of ASLV successful (May). Second developmental launch of Polar Satellite Launch Vehicle (PSLV) with IRS-P2 successful (October).

1995: INSAT-2C launched in June.

1996: Third developmental launch PSLV with IRS-P3 successful in (March).

1997: INSAT-2D launched in June and became inoperational in October. Arabsat-1C, since renamed INSAT-2DT, acquired in November. First operational launch of

PSLV with IRS-1D successful (September).

1998: INSAT system capacity augmented with readiness of INSAT-2DT aquired from Arabsat (January).

1999: INSAT-2D the last satellite in the multi-purpose (INSAT-2D series), launched by Ariane from Kourou French Guyana (April 3, 1999). IRS-P4 (OCEANSAT), launched by Polar Satellite Launch Vehicle (PSLV-C2).

2000: INSAT-3B was launched on March 22, 2000.

2001: Geosynchronous Satellite Launch Vehicle-D1 (GSLV-D1), the first developmental launch of GSLV V with GSAT-1 on board, partially successful.

2002: INSAT-3 CGSLV-D2 launched successfully by Ariannespace (January); launch of KALPANA-1 (September).

2003: GSLV-D2, the second developmental launch of GSLV with GSAT-2, successful (May).

2004: First operational flight of GSLV (F02) unsuccessfully launches EDUSAT (September).

2005: Launch of CARTOSAT and HAMSAT by PSLV-C6 from the second launch pad (Universal Launch Pad) (May 1). ISAT-4A launched successfully on July 10, 2006. GSL V-F02 carried INSAT-4C.

2006: Second operational flight of GSLV (F02) unsuccessful July 2006.

2007: Successful launch of CARTOSAT-2, SRE-1, LAPAN-TUBSAT and PEHUENSAT-1 on PSLV V C7 on January 10, 2007.

2008: • PSLV-C11 successfully launches CHANDRAYAAN-1 from Sriharikota (October 22, 2008).

• PSLV-C9 successfully launches CARTOSAT-2A, IMS-1 and 8 foreign nano satellites from Sriharikota (April 28,2008).

• PSLV-C10 successfully launches TECSAR satellite under a commercial contract with Antrix Corporation (January 21, 2008).

2009: • PSLV-C14 successfully launches Seven Satellites-OCEANSAT-2, FourCUBESAT Satellites and Two RUBIN-9 from Sriharikota (Sept. 23, 2009).

• PSLV-C12 successfully launches RISAT-2 and ANUSAT from Sriharikota (April 20, 2009).

2010: • GSLV-F06 launched from Shriharikota (Dec 25, 2010). GSAT-5P could not be placed into orbit as the GSLV-F06 mission was not successful.

• Successful launch of advanced communication satellite HYLAS (Highly Adaptable Satellite), built by ISRO on a commercial basis in partnership with EADS-Astrium of Europe, by Ariane-5 V198 from Kourou French Guiana (November 27, 2010).

• PSLV-C15 successfully launches Five Satellites-CARTOSAT-2B, ALSAT-2A, two nanosatellites-NLS-6.1 & 6.2 and a pico-satellite- STUDSAT from

Sriharikota (July 12, 2010).

• GSLV-D3 launched from Sriharikota (Apr 15, 2010). GSAT-4 satellite could not be placed in orbit as flight testing of the Indigenous Cryogenic Stage in GSLV-D3 Mission was not successful.

2011: • PSLV-C18 successfully launches Megha-Tropiques, Jugnu, SRMSat andVesselSat-1 from Sriharikota (October 12, 2011).

• PSLV-C17 successfully launches GSAT-12 from Sriharikota (July 15, 2011).

• Successful launch of GSAT-8 by Ariane-5 VA-202 from Kourou French Guiana,
(May 21, 2011).

• PSLV-C16 successfully launches Three Satellites - RESOURCESAT-2,YOUTHSAT, X-SAT from Sriharikota (April 20, 2011).

2012: • Successful launch of GSAT-10 by Ariane-5 VA-209 from Kourou French Guiana (September 29, 2012).

• ISRO's Polar Satellite Launch Vehicle, PSLV-C21 successfully launchesSPOT 6 and PROITERES from Sriharikota (September 09, 2012).

• PSLV-C19 successfully launches RISAT-1 from Sriharikota (April 26, 2012).

2013: • PSLV-C25 successfully launches Mars Orbiter Mission Spacecraftfrom Sriharikota (Nov 05, 2013).

• Successful launch of GSAT-7 by Ariane-5 VA-215 from Kourou French Guiana
(August 30, 2013).

• Successful launch of INSAT-3D by Ariane-5 VA-214 from Kourou French Guiana
(July 26, 2013).

• PSLV-C22 successfully launches IRNSS-1A from Sriharikota (Jul 01, 2013).

• PSLV-C20 successfully launches SARAL and six commercial payloadsfrom Sriharikota (Feb 25, 2013).

2015: • Ariane-5 VA-227 launched GSAT 15 (Communication Satellite)

• GSLV-D6 launched GSAT 6 (Communication Satellite)

2016: • PSLV C35 launched Cartosat 2 (Meteorological Satellite)

• GSLV F05 launched INSAT-3DR (Meteorological Satellite)

2018: • PSLV C41 launched IRNSS-11 (Navigation Satellite)

• PSLV C40 launched Cartosat 2 (Earth Observation)

• PSLV-C33 launched IRNSS-1G (Navigation satellite)

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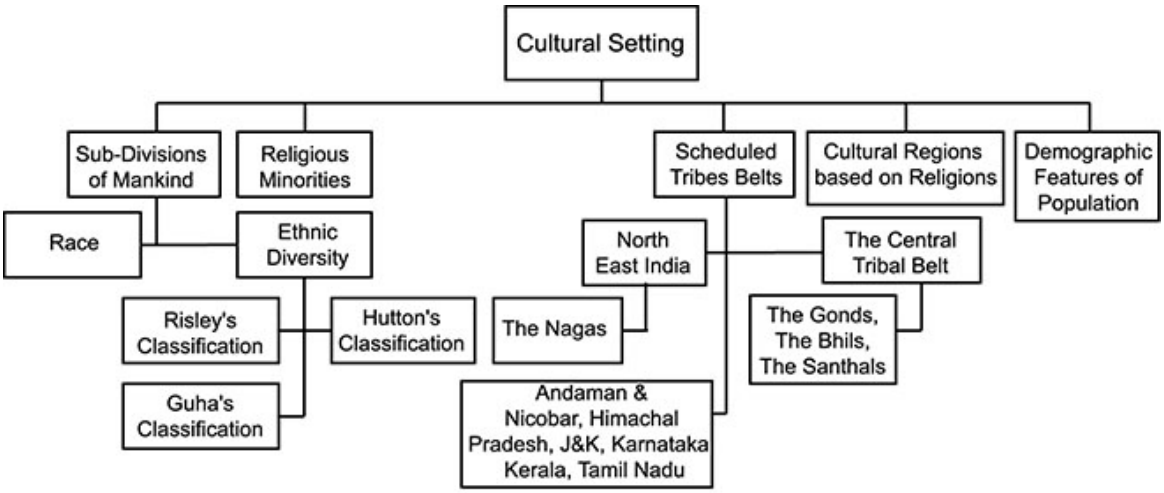
13

Cultural Setting



ORIGIN OF MANKIND

There is no direct evidence to show when the first man appeared on the earth and at which place he was born. In fact, the origin of man is shrouded in mystery. But the protagonists of the Theory of Evolution believe that man is the last product of slow evolution of life on the earth. The anthropologists opine that the ancestors of the first man were born in

the region to the east of the Victoria Lake in east Africa. The oldest fossils of the ancestors of *Homo Sapiens* (Modern Man) are found in that part of Africa. From this region, modern man migrated and entered into Asia, Europe and the Subcontinent of India around one million years back **(Fig. 13.1).**

Subdivisions of the Human Species

Race

It is difficult to give a precise definition of race. Man is one of most variable tool-making animals, and each man is a distinct individual, differing in greater or lesser degree from each of his fellows. But just as all men have sufficient general resemblance to be classified as human, to a lesser degree, certain groups of man have enough characteristics in common to be classed, somewhat arbitrarily, as races.

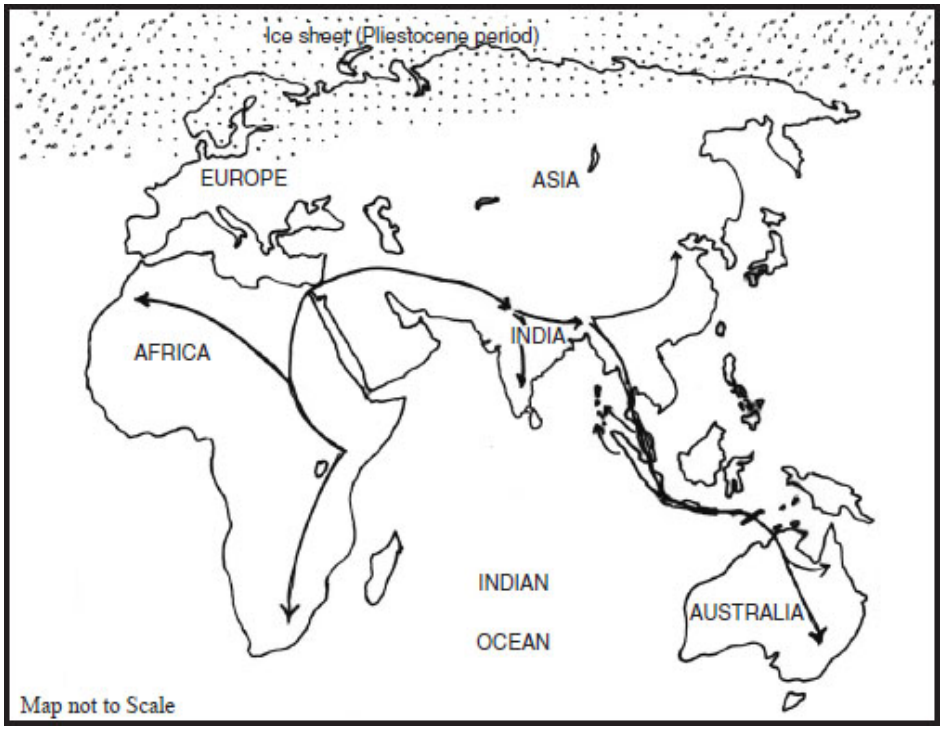


Fig. 13.1 Dispersal of Homo Sapiens

In the opinion of Kroeber, '*A race is a valid biological concept*; *a breed of genetic strain or sub-species*. *It is not a valid socio-cultural concept*'. In the opinion of J.B. Birdsell, 'A race is an interbreeding group whose gene pool is different from other populations.' Haddon considers 'Race as a

biological breed based on a combination of physical traits which are transmitted.'

Ethnic Group

An ethnic group is a distinct category of the population in a larger society whose culture is usually different from that of the society. The members of such a group are, or feel themselves, or are sought to be, bound together by common ties of race ancestry, or nationality or culture. In other words, a group of people having common racial, ancestry, religious, linguistic, or national characteristics is known as an ethnic group.

The cultural elements, i.e., language, religion, faith, belief, nationality, customs, and traditions that have been passed on from one generation to another are of paramount importance in the division of mankind. In fact, division of mankind can be most accurate and precise when based on ways of living and thinking.

Historical Perspective of Indian Society

Indian society is one of the oldest in the world. India's social structure and cultrural patterns are characterised by unity as well as diversity. Historically, India has been hospitable to numerous groups of immigrants from different parts of Asia, Africa, and Europe, but the culture of each group has undergone enough change over the centuries to become an integral part of the Indian mosaic. The institution of caste may be mentioned as a typical example of the paradox that is Indian society. Each caste stands for a way of life that is to some extent distinctive, but at the same time the castes of a region form part of a single social framework. It is important to note that caste is found not only among the Hindus but also among Muslims, Christians, Jews, Sikhs, Buddhists, and Jains. Caste is ubiquitous, and this has resulted in an ideology tolerant of diversity.

Factors making for diversity are apparent even to the casual observer. The population of India is racially and ethnically diverse, containing elements from six main racial types: (i) *Negrito*, (ii) *the Proto-Australoid*, (iii) *the Mongoloid*, (iv) *the Mediterranean*, (v) *the Western Brachycephals*, and (vi) *the Nordic*. All the great religions of the world are represented in this country. The tribal groups enjoy varying degrees of contact with one or the other of the great religions. The major literary languages alone number fourteen. Diversity is seen in the patterns of rural as well as urban settlements, community life, forms of land tenure, and agricultural

operations. In kinship, marriage rites and customs, inheritance, and general mode of living, there are striking differences between groups.

Diversity is, however, only one side of the picture. There are underlying factors as well. India is a political entity, every part of which is under the same constitution. The process of unification developed as several great rulers—Asoka, Samudragupta, Akbar—brought large parts of the country under their power; but it was only during the British rule that India became, for the first time, a single political entity.

The concept of unity of India is inherent in Hinduism. There are sacred centres of Hindu pilgrimage in every corner of the country. India is a sacred land not only of the Hindus but also of Sikhs, Jains, and Buddhists. The Muslims and Christians too, have several sacred centres of pilgrimage in India. The institution of caste cuts across diverse religious groups and gives them all common social idiom.

The declaration of India as a Secular state provides one more evidence of the tolerance of diversity which has been characteristic of Indian history from its beginning. The process of economic development ushered in the Five-Year Plans and the spread of egalitarian ideals have brought about revolutionary changes in the Indian pattern of social life. A single government and a common body of civil and criminal law, a developing economy, and a secular approach to public life and problems are now providing substance and reality to India's claim to be a nation.

THE CASTE SYSTEM

The first literary traces of the caste system are found in the Rig Veda, where three groups are mentioned: *Brahma* (priests), *Ksatra* (kings and rulers), and *Vis* (common people). The *Purusasukta hymn*, however, mentions of four classes originating from four parts of the body of the Creator (God). These four classes—*Brahmana*, *Rajanya*, *Vaisya*, and *Sudra*—are referred to in later literature as *Chaturvarna.*

The post-Vedic period saw the growth and consolidation of the power of Brahmins. Brahmin writers continually discussed and defined the duties and rights of each caste amidst its place in the hierarchy. Justifications and rationalisations of the hierarchy were also produced during this period. In the *Bhagavad Gita*, for instance, the caste system is sought to be justified on the basis of the ideas of *Guna*, *Karma*, and *Dharma.*

The 6th century BC. saw the rise of Buddhism, which is believed to have questioned the basis of the caste system itself. Some scholars, however, have said that Buddhism on its social and political side was chiefly a *Ksatriya* movement against Brahmanical supremacy.

The Bhakti Movement with its long history contained elements which ran counter to caste ideology. The Bhakti saints came from all castes, including Harijans. The movement was more or less continuous in Indian history, and it spread right across the subcontinent. The movement attracted converts from all castes including Harijans. The followers of Kabir (Kabirpanthi) also became a caste. Caste even survived conversion to Christianity and Islam. During the last century, there came into existence the Arya Samaj in the Punjab and Brahmo Samaj in Bengal. These movements, as well as the Ramakrishna Mission, represented a shift towards the liberalisation of castes.

Main Features of Castes

The main features of caste prevailing through the past centuries may be described under nine heads: hierarchy, endogamy, and hypergamy; occupational association; distinction in custom, dress, and speech; pollution; ritual, other privileges, and disabilities; and caste organisation and caste mobility.

Caste and Village Community

The caste living in a village or a group of neighbouring villages, are bound together by economic ties. Generally, peasant castes are numerically predominant in villages and they need carpenter, blacksmith, barber, water-man, watch-man, washer-man and leather worker castes to perform agricultural work. Servicing castes such as priest (Brahmin as well as non-Brahmin), barber, washerman, and water carrier cater to the needs of everyone except Harijans. Artisan castes produce goods which are wanted by everyone. Most Indian villages do not have more than a few of the essential castes and depend on neighbouring villages for certain services, skills, and goods.

In rural India, with its largely subsistent economy, the relationship between the different caste groups in a village takes a particular form. The essential artisan and servicing castes are paid annually in grain at harvest. In some parts of India, the artisan and servicing castes are also provided

with free food, clothing, fodder, and residential site. On such occasions as birth, marriage, and death, these castes perform extra duties for which they are paid a customary sum of money and some gifts in kind. This type of relationship is found all over India and is called by different names: *Jajmani* in the North, *Mirasi* in Madras, *Bara Balute* in Maharashtra, and *Adade* in Mysore. The relationship between the *Jajman* and his *Kamin* is unequal, since the latter is regarded as inferior. The right to serve is hereditary, transferable, saleable, mortgageable, and partible.

The Jajmani system bound together the different castes living in a village or a group of neighbouring villages. The caste-wise division of labour and the consequent linking up of different castes in enduring and pervasive relationships provided a pattern of alliances which cut across the ties of caste. The modern ‘caste problem’ is to some extent the result of the weakening, in the last sixty years or more, of these vertical and local ties and consequent strengthening of horizontal ties over whole area.

The relationship between landowner and tenant, master and servant, creditor and debtor, may all be subsumed under a single category—patron and client. This relationship is widespread and crucial to the understanding of rural India. Voting at elections, local and general, is influenced by the patron-client tie.

Ritual occasions like life-cycle ceremonies, festivals and fairs, require the co-operation of several castes. Certain rituals which are common for all the castes occur at birth, girls puberty, marriage, and death.Several castes are also required to cooperate in the performance of calender, festivals, and festivals of village deities.

The functioning of the village as a political and social entity brought together members from different castes. Every village had a headman usually belonging to the dominant caste. The accountant was always Brahmin in South India. Every village had a watchman and messengers. In the irrigated areas, there was always a man to look after and regulate the flow of water in the canals feeding the fields. The headman and accountant collected the land taxes with the aid of Harijan village servants.

The village council performed a variety of tasks, including the maintenance of law and order, settling of disputes, celebrations of festivals and construction of roads, bridges, and tanks.

The village community consisted of hierarchical groups, each with its own rights, duties and privileges. The castes at the top had power and

privileges which were denied to the lower castes. The lower castes were tenants, servants, landless labourers, doctors, and clients of the higher castes. There was competition among the former to be clients of the rich and powerful patrons while the latter wanted to have as many clients as possible. The patron had duties towards clients and *vice versa.* The caste system together with the inequalities of landownership produced a deeply stratified society, but that did not prevent the village from functioning as a community. Conflict and cooperation went together. In recent years the lower castes have shown an increasing desire to free themselves from the control of the locally dominant caste. This has been assisted by political forces operating from higher levels. The Una-Movement (Gujarat) of the scheduled caste is quite significant from the social point of view which may affect the state and national politics.

RACIAL AND ETHNIC DIVERSITY IN INDIA

The physical characteristics of the Indian population vary from region to region. The population of India consists of several racial strains varying from the Negrito to the Nordic (Caucasoids). Some of the anthropologists like H. de Terra, A.C. Haddon, and P. Mitra regard India as the cradle land of humanity on the evidence of fossils. But most of the anthropologists agree that, in India, there had been the arrival of people in successive waves from the regions of Mediterranean, Central Asia, Mongolia, China, Tibet, and the South East Asia; Malaysia and Thailand **(Fig. 13.2)**. Consequently, there are hardly any pure races; but rather a racial admixture in India. It is only in some remote areas where there are tribes that have maintained their aboriginal features that purity of race is visible. In the opinion of Prof. Aijazuddin Ahmad, "*Those who managed to drift into the isolated and remote parts of the subcontinent could preserve their original ethnic traits which remained by and large unaffected by the fresh waves of incursions witnessed along the main corridor of movement connecting the Kabul Valley with the North Indian Plain*". The Indian population has been classified by a number of scholars. Some of the important classifications are as follows:

Risley's Classification of Indian Races

Sir Herbert Risley, the Census Superintendent of India, 1901, made a pioneering effort and classified the Indian races into seven groups as

follows:

1. Indo-Aryans

The Indo-Aryan tribes migrated into India from Central Asia. They are presently found in Jammu and Kashmir, Punjab, Haryana, Himachal Pradesh, Uttar Pradesh, Uttarakhand, and Rajasthan. Their main physical traits are tall stature, fair complexion, long arms, broad shoulders, and heavy to slim bodies. Their representative communities are the Rajputs, Khatris, and the Jats of northern India.

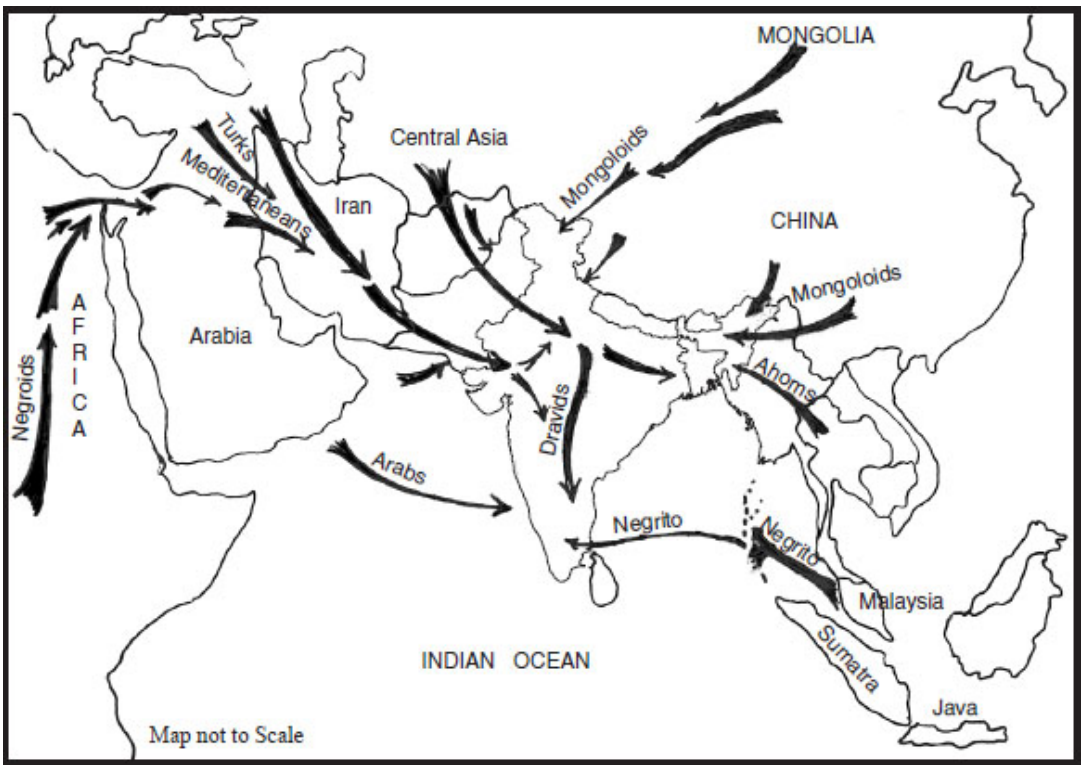


Fig. 13.2 Arrival of Various Racial and Ethnic Groups in India

2. Dravidians

They are found particularly in South India (Andhra Pradesh, Karnataka, Kerala, and Tamil Nadu), Chhattisgarh, Jharkhand, southern Madhya Pradesh, Odisha, and southern Rajasthan. Their representatives are the Santhals of Chotanagpur Plateau, the Todas of Nilgiri, the Juangs of Odisha, the Gonds of Bastar, and the Bhils of Rajasthan. They were probably the ancient people of India, but have been influenced by the admixture of Aryans, the Scythians (Ukrainians) and Mongoloid people.

They are characterised by short stature, dark complexion, hairy bodies, curly hair, long heads, and broad nose.

3. The Mongoloids

They are found in the Himalayan and sub-Himalayan regions stretching from Ladakh (Jammu and Kashmir) to Sikkim, Arunachal Pradesh and the Hill states of North East India. They are of short stature, yellow in complexion, have medium broad nose, flat face, and oblique eyes with an epicanthic fold. Their representatives are the Kinnets of Lahul-Spiti and Kullu, the Lepchas, and Bhutias of Sikkim, and the Tharus of the Tarai region of Uttar Pradesh and Uttarakhand.

4. Aryo-Dravidian

It is an amalgamation of the Aryans and the Dravidians. This group is found in parts of Rajasthan, Uttar Pradesh, and Bihar. Their elements are present in both the lower and the upper castes of Brahmins and the scheduled castes. Their complexion varies from light brown to black and they have long head, short stature, and medium to broad nose.

5. The Mongolo-Dravidians

They are an intermixture of the Mongoloids and the Dravidians. They are mostly found in West Bengal and Odisha (Brahmins and Kayastha). Their characteristics include broad head, dark complexion, flat nose, short to medium stature, and plentiful hair on face.

6. The Scytho-Dravidians

They are an admixture of the Scythians and the Dravidians. Some of the anthropologists hold that the Sakas belong to the Scythian race which came to India from Central Asia. They settled in the Sind Province of Pakistan, Gujarat, Maharashtra, Karnataka, Rajasthan, and Uttar Pradesh. The Scythian features are pronounced in higher castes, while the Dravidian features are associated with the lower strata of society. The Marathas are considered to be the representatives of the Scythian-Dravidian race. Their main features are low stature, long head, moderate nose, light brown to fair complexion, and scanty body hair.

7. The Turko-Iranian

They are found in Baluchistan and Afghanistan. Their representatives are the Pathans who are found in some parts of north-west India as a minority

group. They are well built, tall statured, with fair complexion, long head, and moderate to narrow nose.

Risley's classification of Indian races is largely based on languages rather than on physical traits. It does not recognise the Negrito element in Indian population which has been duly recognised by J.H. Hutton, A.C. Haddon and B.S. Guha.

Huttons's Classification

J.H. Hutton has also based his racial classification of India on the Census findings. However, he does not recognise India as the cradle land of humanity and holds that all the races of India are constituted by foreign immigrants. On the basis of chronological order, he has recognised the following seven races in India:

1. The Negritoes

In the opinion of Hutton, the Negritoes were the oldest inhabitants of India, but have left virtually no trace on the main land of the subcontinent. Some of the representatives of the Negritoes are found in Andaman and Nicobar Islands and among some of the Nagas.

2. The Proto-Australoids

They came in large numbers and spread in different parts of the country. Their representatives are found in the aboriginal peoples of India.

3. The Early Mediterraneans

The Early Mediterraneans reached South India where their representatives are still found in Karnataka and Tamil Nadu. They are also found in the tribal belt stretching from Gujarat to West Bengal. They have medium stature, brown to black complexion, and slight built.

4. The Advanced Mediterranean

They are the early Dravidian people who were associated with the Indus Valley Civilisation. They were the Pre-Aryan people who mingled with the Aryans. They were taller and fairer than the Early Mediterraneans. They spread over Punjab, Haryana, Uttar Pradesh, and Bihar.

5. The Alpines or the Pre-Vedic Aryans

The Pre-Vedic Aryans are found in both Gujarat and West Bengal.

6. The Nordics or the Vedic Aryans

They came to India during the second millennium BC. Their elements are found in Jammu and Kashmir, Punjab, Haryana, Rajasthan, and western

Uttar Pradesh among the higher castes. Their main physical traits are fair complexion, tall stature, and sharp nose.

7. The Mongoloids

The Mongoloids are mainly found in the Himalayan and Sub-Himalayan belts. They have modified Mongoloid features like light yellow colour, oblique eyes, short stature, and flat face.

Guha's Classification

Dr. B.S. Guha, the Ex-Director of the Anthropological Survey of India, has based his racial classification of India on physical measurements carried out by him during the Census operation of 1931. His classification of Indian races is considered as the most authentic. He has recognised the following six racial groups in India **(Fig. 13.3)**:

1. The Negritoes

Most probably, the Negritoes were the earliest arrivals in India. These are characterised by short stature (150 cm), frizzy-hair, bulbous forehead, flat nose, slightly protruding jaws, small chin, black skin colour, weak hands, and long arms. Their representatives are the Andamanese, Nicobaris, and the Irulas, Kadars, Kanikkars, Muthaiwans, Paniyans, Puliyans, Uralis living in the hills of Tamil Nadu, Kerala, and Karnataka. The Angami-Nagas have also been considered as having some of the traits of the Negritoes. Their arrival in the Andaman and Nicobar Islands (Jarawa, Ongs, Sentinelese, Shompen, etc.) is believed to be from the Peninsula of Malaysia. In appearance, culture, and traditions, they are very close to the Semangs and Sakais tribes of Malaysian Peninsula.

2. Proto-Australoids

After the Negritoes, the Proto-Australoids entered the Subcontinent of India, most probably from Australia. Their representatives are found among the Bhils, Chenchus, Hos, Kurumbas, Mundas, Santhals, and Yeruvas. Their common physical traits are dark brown to black-brown complexion, broad nose, wavy to curly hair, short stature, and thick everted lips.

3. Mongoloids

The Mongoloids reached India from China, Mongolia, Tibet, Malyasia, Thailand, and Myanmar in about the first millennium BC. They occupied the Ladakh Division of Jammu and Kashmir, Himachal Pradesh,

Uttarakhand, Sikkim, the Himalayan and sub-Himalayan belts, and the North Eastern states of India.

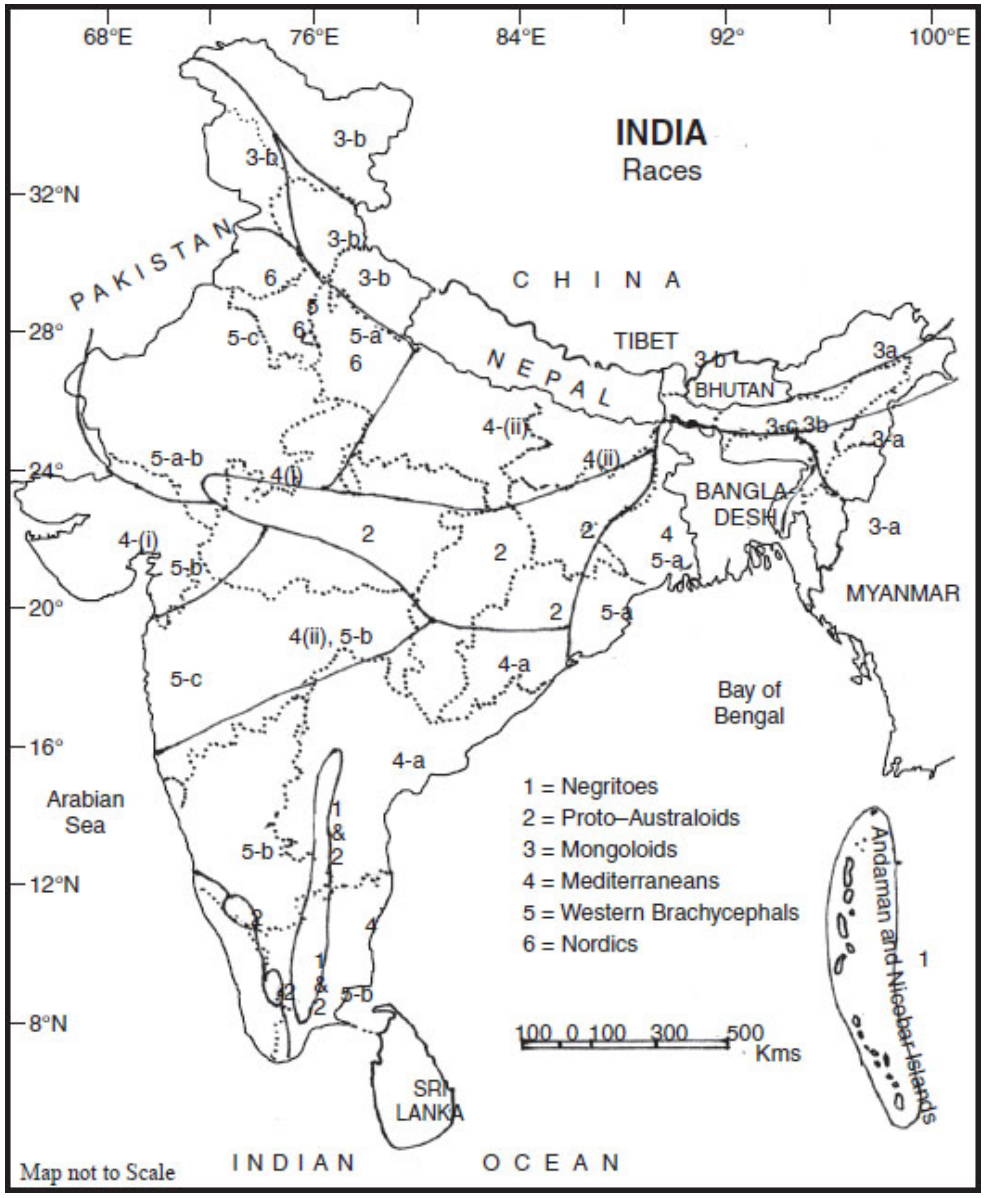


Fig. 13.3 Major Races

Dr. B.S. Guha has recognised two sub-groups of Mongoloids: (a) Palaeo-Mongoloid and

(b) Tibeto-Mongoloid. The Palaeo-Mongoloid are the most primitive type which have long heads, medium stature, light brown colour, medium nose, oblique eyes, and scanty hair on body. Their representatives are found in the sub-Himalayan region, particularly Arunachal Pradesh, Assam, and

Indo-Myanmar (Burma) border districts. Their representatives are Daflas, Garos, Kacharis, Khasis, Kuki-Nagas, Lalung, Machi, Miris, and Tipperas.

The Tibeto-Mongoloids are found in Bhutan, Himachal Pradesh, Ladakh, Nepal, Sikkim, and Uttarakhand. Their typical features are long stature, light yellow colour, hairy body, oblique eyes, long nose, and flat face. The Bhutias, the Gorkhas, the Ladakhis, the Kinnauris are the representatives of the Tibeto-Mongolid racial group.

4. The Mediterraneans

They came to India in successive waves from the Mediterranean region. Dr. Guha has identified three sub-groups of the Mediterraneans, namely, (i) Palaeo-Mediterranean, (ii) Mediterranean, and (iii) Oriental Type.

(i) **Palaeo-Mediterraneans** The oldest group is the Palaeo-Mediterranean. They are characterized by medium stature, long and narrow head, long face, short and medium nose, and brown complexion. Perhaps they entered the subcontinent during the Neolithic Period and were pushed into the area south of the Vindhyan Mountains in areas of isolation and relative isolation.

(ii) **The Mediterraneans** The Mediterraneans have medium stature, olive brown complexion, long head and long-broad open eyes. They are found in Haryana, Punjab, Uttar Pradesh, Maharashtra, and Kerala.

(iii) **The Oriental Mediterraneans** The Oriental Mediterraneans were one of the last to enter India. Most of their physical traits are common, but they are distinguished by long and convex nose and fair complexion. They are limited to Rajasthan, Punjab, Haryana, western Uttar Pradesh, and southern parts of Uttarakhand (district Hardwar and Udhamsingh Nagar).

5. The Western Brachycephals

Dr. Guha has classified them into three sub-groups:

(a) **Alpinoid (the Pre-Vedic Aryans)** They entered the Indus Valley and spread in Gujarat, Maharashtra, Andhra Pradesh, Karnataka, Kerala, and Tamil Nadu. Later on, they entered the Gangetic Valley and reached up to West Bengal and Odisha. They are characterised by medium stature, round face, prominent nose, straight long hair, and fair complexion.

(b) **Dinarics** The Dinarics followed the Alpinoids and reached

Gujarat, Maharashtra, Andhra Pradesh, Karnataka, Kerala and Tamil Nadu. They are characterised by long stature, brown complexion, long face, and sharp nose. They have mixed with Alpinoids. Their main representatives are the people of Kathiawar and Kurg (Karnataka).

(c) **The Armenoids** The Armenoids came to India from Armenia. They are characterised by medium stature, broad head, long nose, and hairy body. Their main representatives are the Parsis of Mumbai and Gujarat.

6. The Nordics

The Nordics or the Vedic Aryans were the last to reach the subcontinent of India from the north-west about the second millennium BC. They overpowered the Dravidians and established their stronghold in the Northern Indian Plain (Aryavarat or Madhydesa). Later on, they reached South India and founded their great kingdoms. Their main physical traits are tall stature, long head, long face, blue or black eyes, and blond hair. Their representatives are found in Punjab, Haryana, Rajasthan, western Uttar Pradesh, and Southern Uttarakhand among the upper castes of Rajputs and Brahmins. They have locally intermixed with other races and ethnic groups in West Bengal and Maharashtra.

Although Guha's classification of Indian races and ethnic groups is considered as the most reliable one, in the contemporary world there has been too much intermingling of blood. The areas of isolation and relative isolation have been connected by roads with the national and state highways which have resulted in more intimate interaction of the people of different races, religions and castes. Under the changed cultural milieu of the 21st century, there are hardly any isolated groups which are the true and typical representatives of their races and ethic groups. The division of people in the racial, ethnic and caste groups, leads to sub-nationalism which may go against the national interest and may retard the process of national integration.

RELIGIOUS MINORITIES

India is a multi-religious country. In India, Hindus are in majority, constituting about 80% of the total population of the country while the

minorities account for 18.42% of the total population. Out of these, Parsi, with a population of about 167,000 is the smallest minority (2011.)

Five communities, namely, Muslims, Christians, Sikhs, Budhists and Parsis, were notified as minority communities under Section 2(C) of the National Commission for Minorities Act, 1992.

The distribution of each of the minorities in the country is highly unequal. For example, the Muslims constitute 5 to 25% of the population in most of the states except Arunachal Pradesh (1.9%),
Chhattisgarh (2.0%), Himachal Pradesh (2.0%), Mizoram (1.1%) and Nagaland (1.8%). The overall percentage of population of Muslims is 14.0% (2011). The literacy level and per capita income of the Muslims is much below the national average. Most of them are below the poverty line.

Christians constitute about 2.3% of the total population of the country. Their percentage, however, is significantly high in Nagaland (90%), Mizoram (87%), Meghalaya (70%) and Goa (27%). Their density, however, is insignificant in the states of north and west India.

Sikh minority is largely confined to Punjab (60%), Haryana (5.5%), and Uttarakhand (2.5%). Their presence is small in the greater parts of the country. The Sikhs are progressive, hardworking and forward-looking people, who perform superbly in all spheres of life. The proportion of Budhists in the total population of the country is only 0.8%—Sikkim (28%) and Arunachal Pradesh with a density of 13% have the largest concentration followed by Mizoram (8%) and Maharashtra (6%). The Parsis are confined to the cities of Mumbai, Navsari, Udvada, Surat, and Ahmedabad. Essentially Parsis are industrialists and businessmen.

For the development of minorities, Rs. 1000 crore have been earmarked. For the development of education of minorities, three scholarships have been launched, with a total provision of Rs. 305 crore in 2008–09. The issues of education, skill development, employment, sanitation, housing, drinking waters, etc., in the minority concentration districts have been addressed. The corpus of Maulana Azad Education Foundation (MAEF) has been enhanced significantly. Several self-employment schemes have been financed. Numerous other economic ventures among backward sections of the minority community have been designed and implemented.

SCHEDULED TRIBES

The tribal people of India, who come under the category of scheduled tribes in term of the provisions of the Constitution of India, number 10.41 million, constituting about 8.6% of the population of the country (Census 2011). According to the Census 1961, there were 365 scheduled tribes. Their number has however, gone up as some new ethnic groups have also been included in the category of scheduled tribes. At present, there are more than 425 scheduled tribes in the country. Many more ethnic groups and communities have applied to the government for being included into the list of scheduled tribes.

The scheduled tribes of India belong to the differential racial, ethnic, linguistic and religious groups. They are mostly occupying the areas of isolation and relative isolation in the hilly and forested areas.

The scheduled tribes comprise the indigenous people. They have often been termed as the people of the *Fourth World*. The scheduled tribes are the people who cherish their own culture. They are the victims of past and present colonialism. They have maintained a close relationship with land. In their societies, there exists a co-operative attitude of give and take, a respect for the earth and life it supports. Their philosophy of life is: '*From everybody according to his capacity, to everyone according to his needs*.

The state-wise percentage of the scheduled tribes population has been shown in **(Fig. 13.4)**
(Table 13.1). According to Niti Ayog, SC and ST account for more than 25 % of India's population.

It may be seen from **(Fig. 13.4)** that the Union Territory of Lakshadweep has the highest percentage of tribal population at 98.8, followed by Mizoram (94.4%) and Nagaland 86.5%. The states of Meghalaya, Arunachal Pradesh, and the Union Territory of Dadar and Nagar Haveli also have more than 52% of their total population as tribal. Contrary to this, the states of Goa, Haryana, Punjab and Uttar Pradesh have less than 0.1% of their population as tribal population **(Table 13.1).**

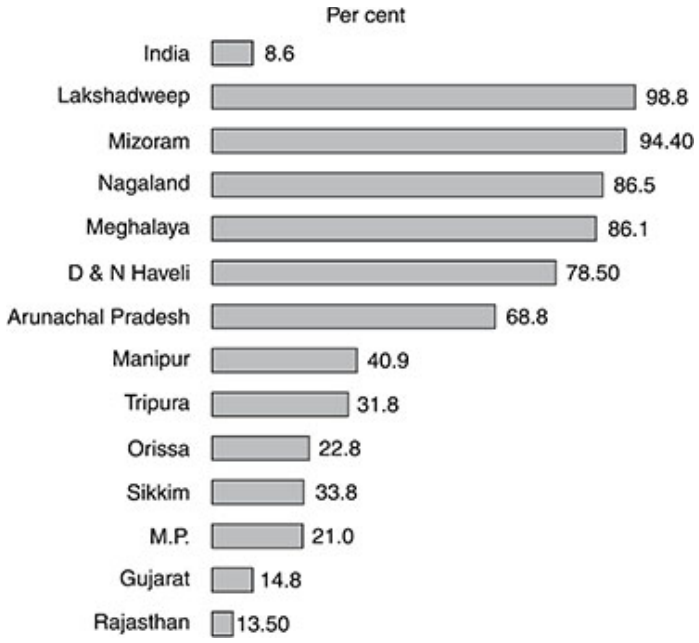


Fig. 13.4 India—Tribal Population 2011 (Statewise)

Spatial Distribution of Scheduled Tribes

The spatial distribution of the scheduled tribes population at the district level also indicates the tendency of clustering and concentration. The geographical distribution of tribal population is highly uneven. They are distributed in 548 districts of the country in varying proportions. About 68% of the total tribal population is confined in 105 districts. There are 16 districts (12 districts in north eastern states of India), Dangs in Gujarat, Jhabua in Madhya Pradesh and the Union Territory of Lakshadweep in which the proportion of tribal population is more than 80%. In the districts of Goa, Haryana, Punjab, and Uttar Pradesh, tribal population is almost absent **(Fig. 13.4A and 13.5).**

Table 13.1

Scheduled Tribes Population, 2011

India: State/Union Territory		Percentage of scheduled tribes Population in the State
1.	Lakshadweep	98.8
2.	Mizoram	94.4

India: State/Union Territory		Percentage of scheduled tribes Population in the State
3.	Nagaland	86.5
4.	Meghalaya	86.1
5.	Arunachal Pradesh	68.8
6.	Dadar & Nagar Haveli	52.0
7.	Manipur	40.9
8.	Tripura	30.6
9.	Chhattisgarh	31.8
10.	Jharkhand	26.2
11.	Odisha	22.8
12.	Sikkim	33.8
13.	Madhya Pradesh	21.1
14.	Gujarat	14.76
15.	Rajasthan	13.5
16.	Assam	12.4
17.*	Jammu & Kashmir	11.9
18.	Maharashtra	9.4

India: State/Union Territory		Percentage of scheduled tribes Population in the State
19.	Daman and Diu	6.3
20.	Andaman and Nicobar	7.5
21.	Andhra Pradesh	7.0
22.	Karnataka	7.0
23.	West Bengal	5.8
24.	Himachal Pradesh	5.7
25.	Uttarakhand	2.9
26.	Kerala	1.5
27.	Tamil Nadu	1.1
28.	Bihar	1.3
29.	Uttar Pradesh	0.6
30.	Goa	0.0
31.	Haryana	0.0
32.	Punjab	0.0

India: State/Union Territory		Percentage of scheduled tribes Population in the State
33.	Chandigarh	0.0
34.	Delhi	0.0
35.	Puducherry	0.0
	INDIA	8.6

Source: Census of India, 2011.
**** Inclusive of Ladakh Tribal population data***

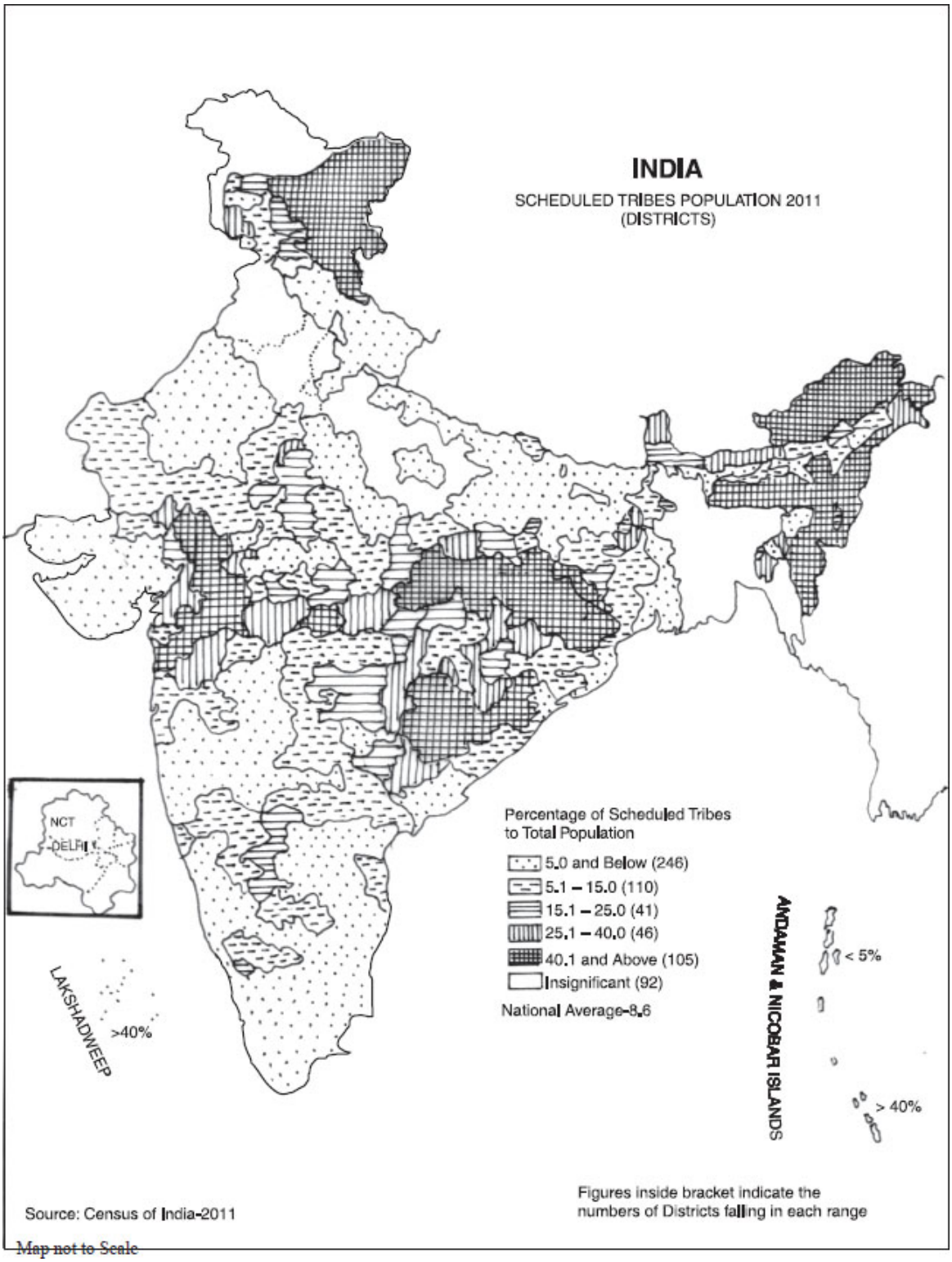


Fig. 13.4A India–Scheduled Tribes Population–2011

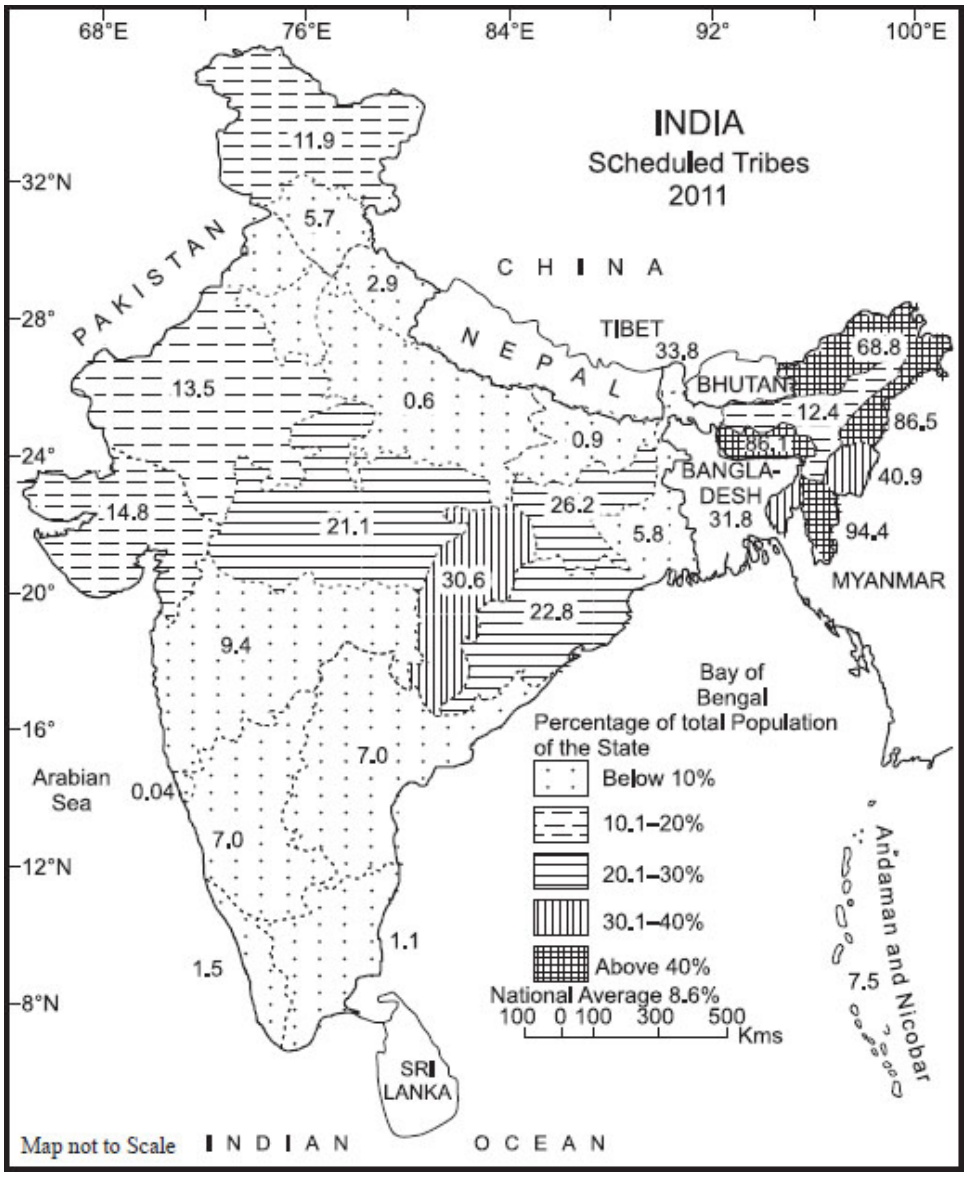


Fig. 13.5 Density of scheduled tribes, 2011

In terms of total scheduled tribes population, Madhya Pradesh ranks first, followed by Maharashtra, Odisha, Jharkhand, Gujarat, and Rajasthan. There is a negative correlation between the distribution of the scheduled tribes population and the scheduled castes, the Muslim and the Christian population.

(Table 13.1). provide an insight to the SC and ST population in the country.

In terms of percentage and absolute population, the population of scheduled tribes is highly unevenly distributed. Their main areas of

concentration of tribals are, however, in three regions as given in **(Table 13.2).** The main concentration of the tribal population has been shown in **(Fig. 13.4A) and 13.5.**

Table 13.2

Main Areas of Concentration of Tribal Population

1.	The North Eastern Region (NER)	Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, and Tripura.
2.	The Central Tribal Belt	Andhra Pradesh, Bihar, Chhattisgarh, Dadar & Nagar Haveli, Daman and Diu, Gujarat, Jharkhand, Madhya Pradesh, Maharashtra, Odisha, Rajasthan, and West Bengal.
3.	Other States and Union Territories	Andaman and Nicobar Islands, Himachal Pradesh, Jammu and Kashmir, Karnataka, Kerala, and Tamil Nadu.

Source: Census of India, 2001.

The tribal communities live in about 15% of the country's area in varying ecological and geo-climatic conditions; mountains, hills, forests, plains and inaccessible areas. They have their presence in almost all the states and union territories of the country except Chandigarh, Goa, Delhi, Haryana, and Punjab **(Fig. 13.6)**.

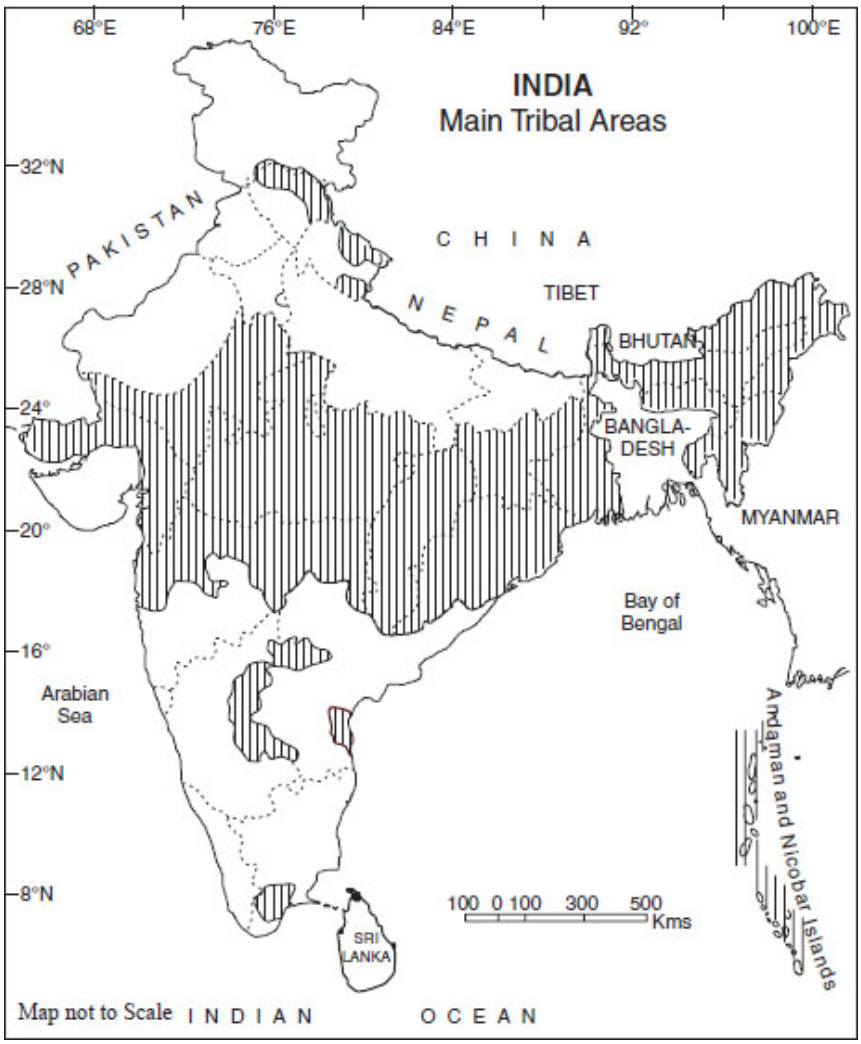


Fig. 13.6 Scheduled tribes—Areas of Main Concentration (2011)

An interesting feature of the tribal population is that it is growing at a faster pace. The temporal change in the growth of population has been shown in **(Table 13.3)**.

It may be observed from **(Table 13.3)** that the total population of the scheduled tribes in 1951 was only 225 lakhs accounting for 6.23% of the total population of India. Their growth rate is, however, higher than the national average. Consequently, in 2011 the tribal population became 1014 lakhs or 8.6% of the total population of the country. The lower standard of living of over 95% of the tribal population may be the main cause of their high birth and high growth rates. The higher growth rate of the tribal population is also because of the inclusion of new tribes in the list of scheduled tribes.

Table 13.3

India: Growth of Scheduled Tribe Population, 1951–2011

Year	Total Population of Scheduled Tribes in Lakhs	Percentage of Scheduled Tribes to Total Population
1951	22.5	6.23
1961	302	6.87
1971	380	6.94
1981	538	6.94
1991	678	8.08
2001	843	8.20
2011	1041	8.60

Source: Census of India, 2011.

According to the Census of 2011, about 84% of the main workers from these communities were engaged in primary activities. A majority of scheduled tribes continue to live below the poverty line. They have poor literacy rates, suffer from malnutrition, undernourishment, water-borne diseases, and are vulnerable to displacement.

The Central and the State Governments have been implementing schemes/programmes, Centrally Sponsored Schemes, etc., however, a lot

more is required to be done to achieve the desired developmental goals for scheduled tribes.

The Main scheduled tribes

The number of scheduled tribes in India is quite large. Some of the main tribes of India are as under:

1. North East India

Bodo, Kachari Miri, Xaxa (Assam), Angami, Ao, Chang, Konyaks, Kuki, Lotha, Mikir, Sema (Nagaland), Apatani, Dafla, Mishmi (Arunachal Pradesh), Garo, Khasi, Jaintia (Meghalaya), Lushai (Mizoram), Bhutia, Lepcha, Serpa (Sikkim), Chakmas and Orang (Tripura)**.**

2. Bihar, Jharkhand, and West Bengal

Asur**,** Birja, Birhor, Bhuiya, Gond, Ho, Khaira, Maler, Malpaharia, Munda, Oraon, Pakiha Pantaram, and Santhal.

3. Andhra Pradesh, Chhattisgarh, Madhya Pradesh, and Maharashtra

Akha, Bhil, Bhilara, Chenchu, Dabla, Dandami, Gadaba, Godwa, Gond, Kol, Kadar, Ketkari, Kotku, Koya, Korva, Maler, Mariya, Mauriya, Lambadi, Paradh, Paliyan, Pariya, Rajgond, Todia, Turi, Yaruba, and Yendi.

4. Odisha and Tamil Nadu

Bagda, Birhor, Khond, Gond, Juang, Kanikkar, Kol, Lambadi, Munda, Santhal, Savara, etc.

5. Gujarat

Bhil, Bhila, Dafar, Dhanka, Garsaya, Dubala, Koli, Patelia, Raibari, Todia, Varali, etc.

6. Rajasthan

Bherat, Bhil, Bhila, Dhanka, Garasiya, Meena, Mira, Rawat, Saha, Sahariya, etc.

7. Kerala

Arandan, Chenchu, Gadaba, Irula, Kadar, Kochu, Malapandaram, Malkurvan, Mallayan, Muthuwan, Paniyan, Paniyan, Sholiga, Sumali, Toda, Yurali, etc.

8. Uttar Pradesh and Uttarakhand

Badkol, Bhotiya, Bhoksa, Bhuiya, Chero, Jaunsaries/Khasa (Chakrata Tehsil, Dehra Dun), Khoha, Manhi, Raji, Tharu, Viyar, etc.

9. Himachal Pradesh and Jammu and Kashmir (inclusive of Ladakh)

Bholia, Bakkarwal, Gujjar, Gaddi, Kinnar, Laholi, Laddakhi, Pangwal, etc.

10. Andaman and Nicobar Islands

Andamani, Aong, Jarawa, Nicobarese, Onges Sentinelese, Shompen, etc. Jarawas also known as Angs is one of the most endangered tribes of the world. According to the Anthropological Survey of India (Andaman & Nicobar Centre), the Jarawas population was 266 in 2001 which rose to 429 in 2016, an increase of 62%.

Economy

The scheduled tribes are essentially dependent on hunting, food-gathering, fishing, shifting cultivation, lumbering, sedentary agriculture, and animal husbandry.

1. Hunting, Fishing, and Food Gathering

The main tribes which are dependent on these economic activities are Chenchu and Yanadi in Andhra Pradesh; Hill-Maria in Chhattisgarh; Bhil, Garasia in Gujarat and Maharashtra; Bhil, Garasia, and Sahariya in Rajasthan; Birgias, Birhor, Kharia, Korwa, Pariha in Jharkhand; Ao, Angami, Changs, Konyak, Kuki, Lotha, Mikir, Sema in Nagaland; Juang in Odisha; Kadar, Koya, Paliyan, Reddi in Tamil Nadu and Raji in Uttar Pradesh.

2. Shifting Cultivation

Bagola, Khond, Kurumba, Saora in Andhra Pradesh; Bhil and Garasia in Gujarat and Madhya Pradesh; Asur, Korwa in Jharkhand; Kathodia in Rajasthan; Baiga, Dhora and Gond in Chhattisgarh and Madhya Pradesh; Chakmas (Tripura), Garo, Khasi, Jaintia, Lushai, Nagas in North Eastern Hill States; Gond, Khond, Kurumba, and Muduwan in Tamil Nadu; Bhutias, Korias, Kharwar and Saharia in Uttar Pradesh and Uttarakhand practice shifting cultivation.

3. Sedentary Cultivation and Animal Husbandry

Badaga, Kota, Koya, Irula in Andhra Pradesh; Baija, Bhatra, Gond, and Parja in Chhattisgarh and Madhya Pradesh; Bhil, Barali, Dubla, Koli, Raiwari in Gujarat and Maharashtra; Bhila, Ho, Korwa, Munda, Oraon, Tamaria and Santhali in Jharkhand; Bhil, Garasiya and Meena in Rajasthan; Badaga, Irula, Malydi and Praga in Tamil Nadu; Bhoksa, Bhotias, Khasa, Kol, and Tharus in Uttar Pradesh and Uttarakhand;

Bhumji, Polia and Santhali in Bihar, Jharkhand, West Bengal are engaged in sedentary cultivation and animal husbandry.

Polygamous Tribes—Baiga, Gond (Chhattisgarh, M.P.), Lushai (Mizoram), Mon (Nagaland), Toda (Kerala, Tamil-Nadu).

Polyandrous Tribas—Kota, Khasa, Jaunsari (U.P., Uttarakhand), Ladakhi, Bota (Ladakh), Nayar and Todas (Kerala, Tamil-Nadu).

Tribal Development

The human development of the scheduled tribes of India is quite low. Most of them are living below the poverty line. The central and the state governments have taken a number of measures over the years to improve the condition of scheduled tribes and for their development. In 1952, five principles known as Nehruvian Panchasheel, guided the administration of tribal affairs namely.

1. Tribals should be allowed to develop according to their own genius.
2. Tribals rights in land and forest should be respected.
3. Tribal teams should be trained to undertake administration and development without too many outsiders being inducted.
4. Tribal development should be undertaken without disturbing tribal, social and cultural institutions.
5. The index of tribal development should be the quality of their life and not the money spent.

The Government of India formed a Ministry of Tribal Affairs in October 1999 in order to promote the tribal development.

The Tribal Sub-Plan (TSP)

The innovative strategy of the '*Tribal Sub-Plan (TSP)*' for scheduled tribes was launched during 1974. This special strategy was expected to ensure that all the general development sectors earmark funds for scheduled tribes in proportion to their population so that adequate benefits from all the concerned sectors flow to this disadvantaged group. The Tribal Sub-Plan strategy has been extended to the following states and Union Territories:

1. Andhra Pradesh, 2. Assam, 3. Bihar, 4. Chhattisgarh, 5. Gujarat. 6. Himachal Pradesh,
7. Jammu and Kashmir, 8. Jharkhand, 9. Karnataka, 10. Kerala, 11. Madhya Pradesh, 12. Maharashtra, 13. Manipur, 14. Odisha, 15.

Rajasthan, 16. Sikkim, 17. Tamil Nadu, 18. Tripura, 19. Uttar Pradesh, 20. Uttarakhand, 21. West Bengal, 22. Andaman and Nicobar, and 23. Daman and Diu.

The ultimate objective of extending the Tribal Sub-Plan is to boost the demand based income generation programmes, and thus raise the economic and social status of tribals in the sectors of agriculture, horticulture, land reforms, watershed development, soil moisture conservation, animal husbandry, ecology and environment, development of forests/forest villages, and development of entrepreneurship.

The main focus of the Tribal Sub-Plan is to make an integrated development effort to develop their agriculture, animal husbandry, irrigation, forestry, education, health, and employment. There is an emphasis on preparing a broad policy framework for tribal development and defining a suitable administrative strategy for its implementation.

In addition to these, the Central Ministers/Departments have to take the following steps for the socioeconomic development of the scheduled tribes:

(a) to formulate appropriate need-based programmes for tribal areas;
(b) suitably adapt the ongoing programmes to the specific requirements of scheduled tribes;
(c) provide sufficient quantity of funds for tribal areas under Central Ministries programmes; and
(d) earmarking of senior officer dedicated exclusively for the welfare of scheduled tribes.

The Tribal Research Institutes

The Tribal Research Institutes are engaged in the work of providing planning inputs to the state governments, collection of data and conducting research and evaluation studies on problems relating to the tribes living in respective states. The Tribal Research Institutes also conduct training, coaching, and guidance seminars and workshops for the cause of tribal development. At present, there are 16 Tribal Research Institutes in the country situated in different states.

The respective state governments have constructed Girls Hostels, Boys Hostels, Ashram Schools, Vocational Training in Tribal Areas, Post-Matric Scholarship for scheduled tribes Students, National Overseas Scholarships for scheduled tribes Students (Non-Plan). In addition to

these, the National scheduled tribes Finance and Development Corporation, and Tribal Co-operative Marketing Development of India are providing useful service and relief to the scheduled tribes.

The National Tribal Policy

The Ministry of Tribal Affairs has formulated the draft of a National Tribal Policy covering various issues such as alienation of tribal land, tribal forest interface; displacement; rehabilitation and resettlement; enhancement of human development index; creating critical infrastructure; violent manifestations; conservation and development of particularly vulnerable tribal groups; adoption of Tribal Sub-Plan strategy; empowerment; gender equality; enlisting support of non-government organisations, tribal culture and traditional knowledge; administration of tribal areas; the Regularity and Protective Regime; scheduling and descheduling of tribes, etc.

Some Important Tribes of India

A brief description of some of the important tribes of India has been given below:

The Gonds

The Gonds with a population of about 8 million constitute the largest tribal group of India. The spatial distribution of Gonds has been shown in **(Fig. 13.7)** They are spread over Chhattisgarh, Jharkhand, Madhya Pradesh, Andhra Pradesh, Telangana, Karnataka, Gujarat, Maharashtra, Bihar, and West Bengal. Their main concentration is in forest and hilly areas between the Vidhyans and Satpura. Their physical traits are dark complexion, flat nose, thick lips, straight hair, and short stature. Their language is Gondi which belongs to the Austric family. They live in small villages of 20–30 families and build their houses along the east west street on both sides. Most of them are agricultural labourers or marginal sedentary farmers. Some of them like Dhimar and Kewats are dependent on fishing, while the Rawats are cattle keepers.

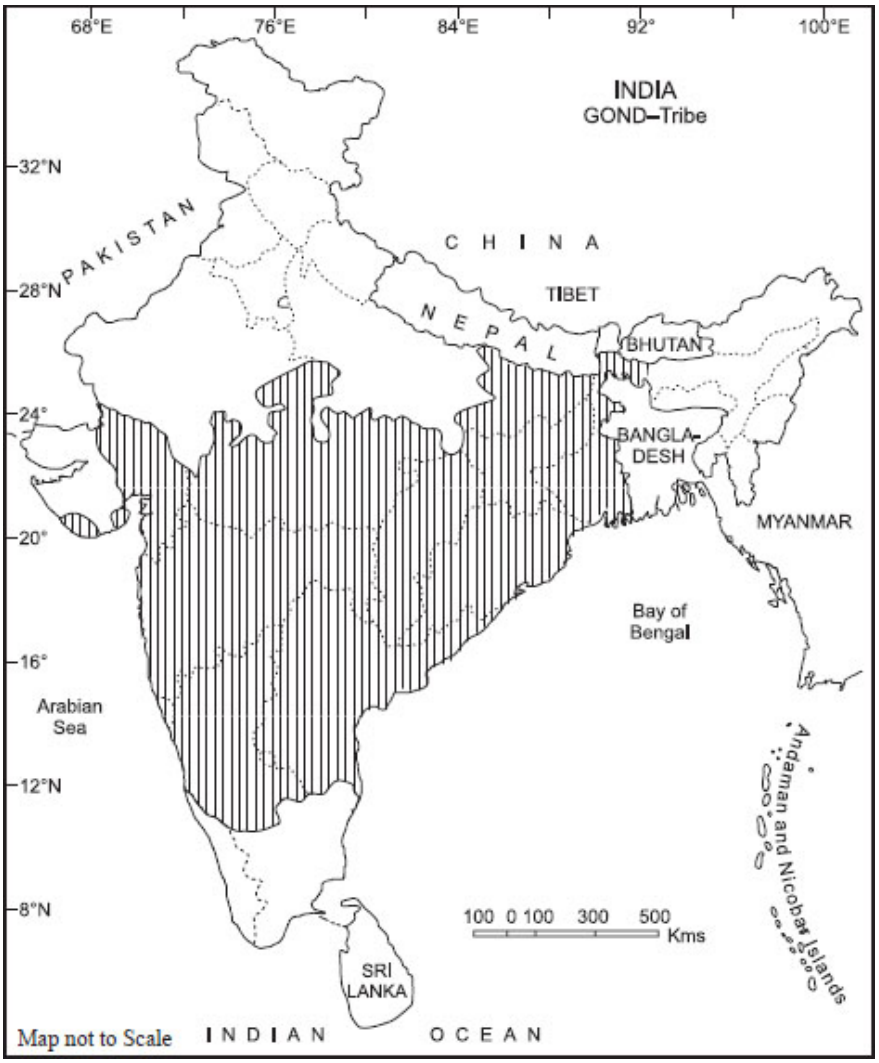


Fig. 13.7 Distribution of Gonds

The Gonds maintain ghotals (youth-house). They practice and participate in music and dance. They also provide security to the village in the night. It is like a club, where both unmarried boys and girls sleep in separate apartments at night and carry on love affairs which culminate in matrimonial relationship. Such affairs are kept secret till their marriage materialises. But it is common knowledge that hardly any girl remains virgin before she is locked in nuptial bond. In fact, the pre-marital sex relations are condoned by the community, but after marriage, sexual fidelity is strictly observed by both husband and wife. They practice endogamous marriage. Polygamy is also in use. Divorce is quite common. They believe in the animistic religion and worship their family deity.

The Gonds are hard working, honest and treat theft as a serious crime. They are, however, superstitious and believe in sorcery and black magic.

The Bhils

The total population of Bhils is over six million (2011). They are distributed over a vast territory extending from Rajasthan, Madhya Pradesh, Chhattisgarh, Gujarat, Maharashtra, Andhra Pradesh, and Karnataka **(Fig. 13.8).**

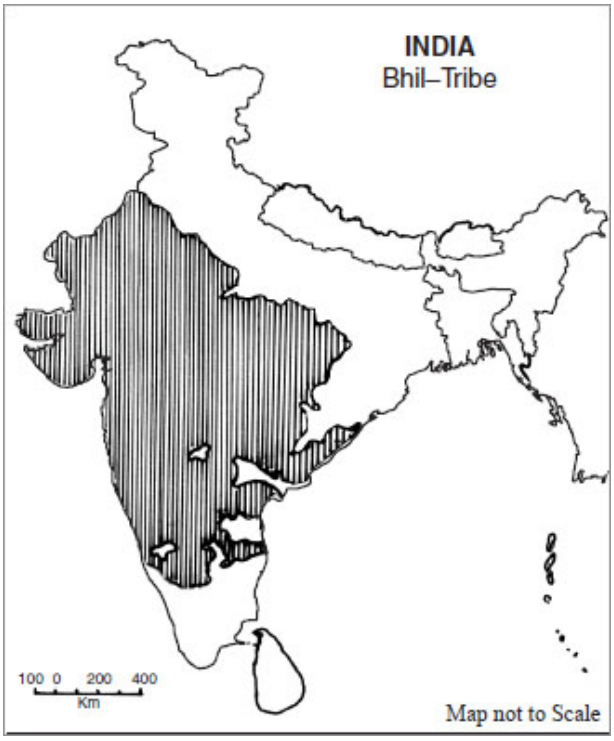


Fig. 13.8 Distribution of Bhils

The habitat of the Bhils is characterised by varied terrain features of plains, hills, plateaus, and forested areas. They are generally living in the areas of isolation and relative isolation.

The origin of the word 'Bhil' is associated with the Dravidian word 'Billu' (meaning bow). In fact, in ancient past, the Bhils were known for their skills in archery. Eklavya was also a Bhil.

According to some anthropologists, the Bhils are a Pre-Dravidian tribe who enjoyed sway over a large territory in Rajasthan before they were subjugated by the Rajputs. The Bhil language is an offshoot of the Rajasthani language.

In the past, the Bhils were mainly engaged in hunting, food-gathering, and pastoral activities. Their economy was based on the local resources. Subsequently, they settled down and started sedentary agriculture. Their other economic pursuits include animal husbandry, poultry farming, grazing, collection of forest products, lumbering, fishing, and services.

They are economically backward. The incidence of poverty is very high. Liquor drinking and gambling have further deteriorated their economic condition. They are often cheated by moneylenders and *Sahukars* of the other communities. Both the males and females participate in agricultural activities.

The Bhil society is divided into 41 *septs* in different states. As there have been many cultural contacts with the Hindus and the Muslims, they have accepted some of their customs and traditions. In many parts, they are virtually Hinduised tribes as they observe the Hindu *Sanskars* of birth, *mundan*, marriage, and death ceremonies.

Among the Bhils marriage is not only universal, but also compulsory and the unmarried men and women are looked down upon by the society. There is ample freedom for marriage; boys and girls can choose their partners and later on get social approbation by the village community. However, the Thakur Bhils of Nimar do not permit such liberty and prefer prior engagement in matrimonial relations.

Divorce can be sought by either party with the consent of the village Panchayat. Child marriage is conspicuous by its absence.

Bhils are animists by nature as reflected in their folk music and art. They believe in rebirth and migration of soul. They stress on good conduct for their salvation, they also believe in evil spirits, ghosts and black magic. The services of Badva (magician) are sought for riddance from disease. They also believe in superstitions. The ladies do a lot of tatooning. They decorate the walls of houses with folk pictorial designs/sketches. They take great interest in dance and music and arrange *Nautanki* for community celebrations.

The society is basically patriarchal. The families have great reverence for gods and goddesses. They also believe in Hindu gods like Hanuman, Ram, and Shiva besides, ‘Baba Deo’ and ‘Ghata Deo’.

They generally live in small hamlets and construct their houses with bamboos. The Bhils make pretentious houses and take care in their maintenance for nuclear families.

The Santhals

Having a population of over 6 millions, the santhals constitute one of the largest tribal groups in India. The anthropological evidence confirms their Dravidian origin which is corroborated by their Austric language (Santhali language and Col Chika script). The Santhal scholars claim that it was the santhals who had developed the Indus Valley Civilisation. The geographical distribution of the Santhals has been shown in **(Fig. 13.9).**

It may be seen from **(Fig. 13.9)** that their present habitat is mainly in the Rajmahal Hills and Chotanagpur Plateau in the state of Bihar, Jharkhand, Odisha, and West Bengal. Tripura also has the Santhal Tribes and they can also be found working in tea gardens of Assam and West Bengal. They are a dominant majority in Santhal Pargana, Giridih, Bankura, Bardhaman, and Birbhum while in more than 16 districts they have a majority. Their demographic growth is about 1.75% per annum.

The Santhals are basically agriculturists (68%). They own landed property and hardly any Santhali is landless. Their subsidiary occupations include hunting, fishing, and collecting forest products. With increasing literacy and cultural contacts, they are also engaged in service, trade and contracting business. Their demand is for a greater Jharkhand which extends over the Santhal areas in West Bengal and Odisha.

The Santhals have a great cultural heritage and have a well defined social order. Widow remarriage is permissible. The Santhal society is patriarchal. The Santhal woman is deemed a strong labour force and contributes to the family income. She participates in the agricultural operations and decorates her house before the Bandana festival.

The Santhals live in permanent settlements. The village settlement generally has a rectangular plan and houses are arranged along the main street of the village. They have their own Panchayats and a well developed penal system.

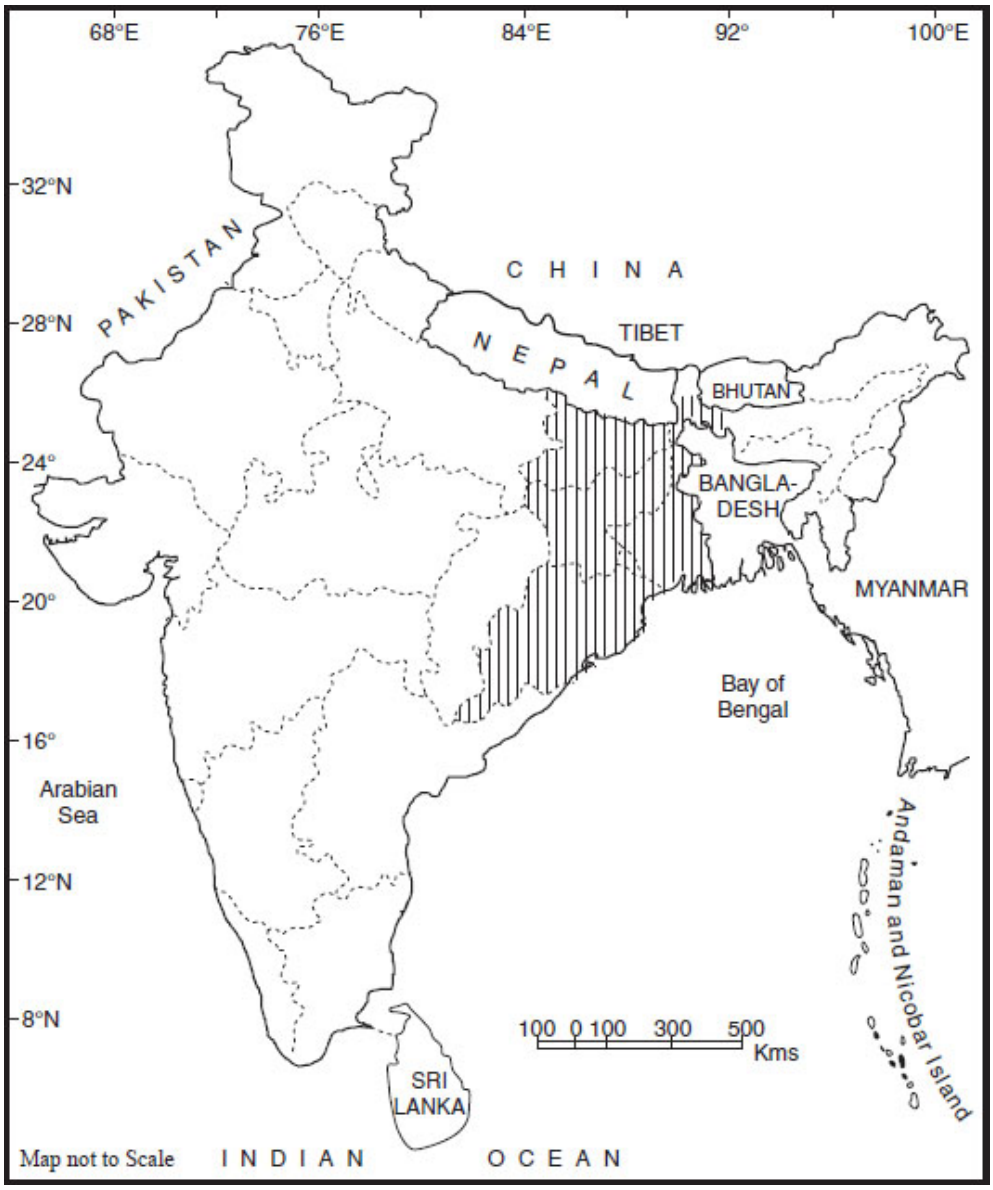


Fig. 13.9 Distribution of Santhals (2011)

Karma is the only festival of the women in the month of Bhadon. It takes place in Manjhithan with music and dance with unmarried girls participating. Their other important festivals are Bandhan or Sohrai (in winter after harvest).

They are animists and believe in many gods and goddesses. Chando (Moon) is the highest power. Their village deity is Jahir-Era.

The Nagas

The Nagas (population about 2 million) are one of the largest and politically awakened tribes of north east India. Their cultural core is Nagaland where they account for about 86.5% of the total population. The spatial distribution of some of the important tribes of north east India including that of Nagas has been shown in **(Fig. 13.10).** They are in good numerical strength in Manipur, followed by Assam, and Meghalaya. They are also found in the border districts of Myanmar. Their terrain is characterised by the serrated ridges of the Brail Range and the Arakan Yoma. Nagaland has a cold winter and moderately hot and humid summer. They have adopted English as the official language of Nagaland. It has helped their cultural and political unification.

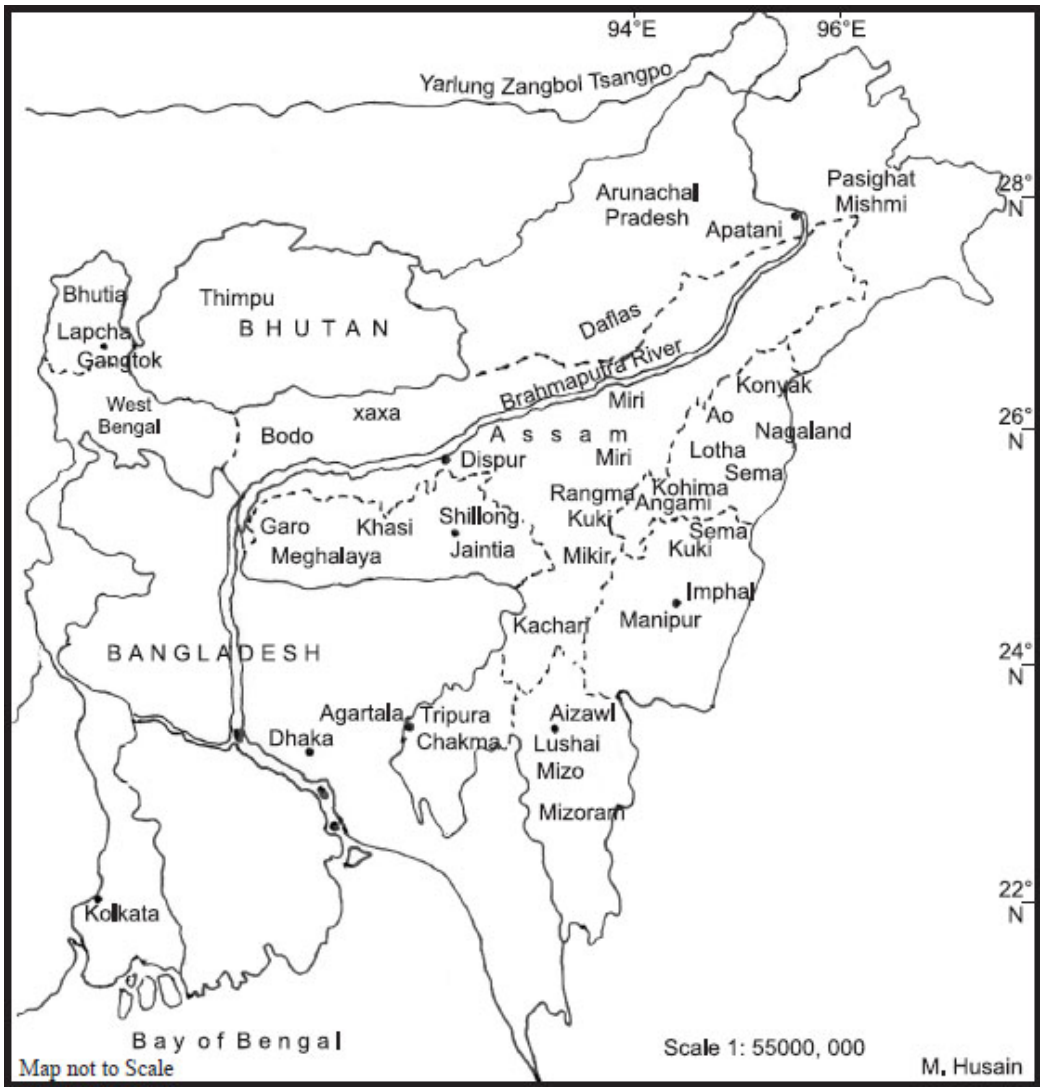


Fig. 13.10 North-East India—Major Tribes

Their villages are located at the top of ridges for reasons of security. They use hill slopes for terraced cultivation. Where the development of terraces is difficult, they practice shifting cultivation.

The village has a *Morung* (communal dormitory) for the bachelors, where both boys and girls stay at night. It functions as a youth club where the youths are initiated into tribal traditions and art of community living. They also receive sex education and often their pre-marital sex relations consummate in marriage with the consent of the village community.

The Nagas are by and large agriculturists (85%). They practice Jhuming (shifting cultivation). The Angami tribe around Kohima has developed beautiful terraces for paddy cultivation. They grow rice, pulses, coarse cereals, vegetables and cotton.The Nagas are known for their artistic handicrafts, particularly weaving and wood carving.

The Nagas have a well knit social organisation which is an excellent example of unity in diversity. They are proud of their cultural heritage. The joint family system is much in vogue. The village headman plays a significant role in their family and communal/ceremonial affairs.

The Nagas do not practice child marriage. Most of the Naga tribes allow their girls great liberty prior to their marriage. They lead a happy life. Both males and females dress smartly. On the occasion of communal feasts and celebrations, they perform dance to music. Traditionally, they are animists, but now a large majority of them is Christian. Educationally, they are more advanced.

The Tharus

The Tharus (population about 1.25 lakh) is the most important scheduled tribe of Uttar Pradesh and Uttarakhand. Their largest concentration is in the Khatima and Sitarganj tehsils of Udhamsingh Nagar (Uttarakhand) accounting for about 55% of their total population. They are also found in Pilibhit, Kheri, Gorakhpur, and Bahraich.

The Tharus are essentially cultivators. During the slack season, they take to fishing as their main occupation. Some of them practice hunting and food gathering. They are the Mongoloid people or predominantly so, who have assimilated non-Mongoloid features as well. According to some experts, they came in this region from the Thar Desert of Rajasthan.

The Tharu village is a well-knit community whose members have developed a unique bond of fellowship and co-operative life. Every village

has a village Panchayat headed by the Pradhan who plays a dominant role in all affairs of the community.

Although Tharus are patriarchal, the status of females is very high. They claim themselves to be Rajputs. The Tharus have accepted most of the Hindu religious practices. They observe Holi, Diwali, and Dashehra. They believe in Sorcery and witchcraft. They are basically honest, hospitable, straightforward and peace loving, but external contacts have affected their morals negatively. They are highly superstitious and have blind faith in their priest.

The Todas

The Todas are a small tribe in the Nilgiris of Tamil Nadu (Ootacamund) and Kurnool district of adjoining Karnataka. They are a classical example of pastoral nomadism. The 'Toda' is derived from the word '*Tandra*' which is a sacred tree for them.

The Todas are essentially pastoralists. Their economy is largely based on buffalo breeding and dairy products. They practice fraternal polyandry. The eldest brother marries a girl who is shared by all younger brothers. Their society is patriarchal; descent and inheritance are determined accordingly. Marriage is a simple ceremony. Divorce is rare. Women are highly respected. The religion is animist. They worship Teikirzi who is their supreme god. Now, Christianity has also reached the Todas.

SCHEDULED CASTES (16.60% OF THE TOTAL POPULATION)

The term Scheduled Caste appeared for the first time in the Government of India Act, 1935 and the Government of India scheduled castes Order, 1936. Dr. B.R. Ambedkar preferred to call them the *Depressed Classes* and Mahatma Gandhi called them the *Harijans*. The Constitution of India has retained the term scheduled castes.

The origin of the scheduled castes lies in the ancient past when the various Hindu scriptures placed them at the lowest ladder of Chaturvarna (four-fold hierarchical social order). Perhaps it was based on division of labour initially, but became hereditary and exclusive practice of the dogmatic doctrine of purity and pollution.

The anthropological evidence suggest their varied origin from an admixture of aboriginal/tribal population with the successive waves of invaders.

According to the Census 2011, the population of scheduled castes in India was 16.60% of the total population, while that of the scheduled tribes population was 8.6%. They constitute the socially and economically deprived caste group of the Indian society. They consist mainly of agricultural labourers, marginal cultivators, artisans, landless labourers, and industrial workers. They include the Malls of Andhra Pradesh, the Meghs of Gujarat, the Mahars of Maharashtra, the Adi-Davidas of Tamil Nadu, the Chamars of Uttar Pradesh and Uttarakhand, and Balais of West Bengal.

Geographical Distribution

The geographical distribution and density of the scheduled castes has been shown in **(Fig. 13.11)**. The state of Uttar Pradesh has the largest number of scheduled castes population, accounting for about 21% of the total scheduled caste population of the country, followed by West Bengal (11.1%), Bihar (7.83%), Andhra Pradesh (7.4%), Tamil Nadu (7.12%), Maharashtra (5.93%), Rajasthan (5.8%) and Madhya Pradesh (5.49%). In opposition to this, the state of Nagaland, and the Union Territories of Andaman and Nicobar Islands and Lakshadweep have no Scheduled Caste population **(Fig. 13.11).**

The highest percentage of scheduled caste population is found in Punjab (28.85%), followed by Himachal Pradesh (24.72%), West Bengal (23.02%) and Uttar Pradesh (21.15%). The states of Arunachal Pradesh, Goa, Mizoram, Meghalaya, Nagaland, Andaman, Nicobar, and Lakshadweep do not have Scheduled Caste population **(Fig. 13.11).**

The percentage of the scheduled castes is low in the areas of schedued tribes, Muslims, and Christians dominate regions/districts. The north-eastern states of India, except Tripura have a very low percentage of the scheduled caste population. Similarly, the percentage of scheduled caste population is low where the percentage of Muslim and Christian population is high. Another important feature about the spatial distribution of the scheduled tribes is that they are mainly concentrated in the rural areas. Despite reservation and special programmes for their upliftment, over 90% of them maintain a very poor standard of living.

CULTURAL REGIONS

The concept of cultural region has been derived from cultural ecology referring to the geographical region over which a degree of homogeneity in cultural traits may be identified. *A cultural region is a contiguous geographical area characterised by cultural homogeneity.*

The concept owes its origin to Ratzel's notion of a *Kulturprovinz*. He was the first to introduce the concept of cultural region in his book Anthropogeography. Subsequently, Sauer developed the Berkeley School of Cultural Geography. He demarcated the cultural regions of the world on the basis of cultural practices. The main focus of Sauer was on relation between apparently static folk culture and their physical environment (resources).

Table 13.4

Scheduled Castes: Geographical Distribution-2011

Rank	State/Union Territory	Percentage of Total Population of the State/Union Territory
1.	Punjab	31.9
2.	West Bengal	25.5
3.	Himachal Pradesh	23.2
4.	Uttar Pradesh	20.7
5.	Haryana	20.2
6.	Tamil Nadu	20.0
7.	Chandigarh	18.9
8.	Uttarakhand	18.8
9.	Rajasthan	17.8
10.	Tripura	17.8
11.	Karnataka	17.1
12.	Odisha	17.1
13.	Delhi	16.8
14.	Telangana	16.5
15.	Andhra Pradesh	16.3

Rank	State/Union Territory	Percentage of Total Population of the State/Union Territory
16.	Bihar	15.9
17.	Puducherry	15.7
18.	Madhya Pradesh	15.6
19.	Chhattisgarh	12.9
20.	Jharkhand	12.1
21.	Maharashtra	11.8
22.	Kerala	9.81
23.*	Jammu & Kashmir	7.4
24.	Assam	7.2
25.	Gujarat	7.1
26.	Sikkim	4.6
27.	Manipur	3.4
28.	Goa	1.7
29.	Meghalaya	0.6
30.	Arunachal Pradesh	0.6
31.	Nagaland	–
32.	Mizoram	–
33.	Andaman & Nicobar Islands	–
34.	Dadae-Nagar Havelli	–
35.	Daman & Diu	–

Rank	State/Union Territory	Percentage of Total Population of the State/Union Territory
36.	Lakshdweep	–
	INDIA	16.6

Source: Census of India, 2011.
**** Inclusive of Ladakh as per census 2011.***

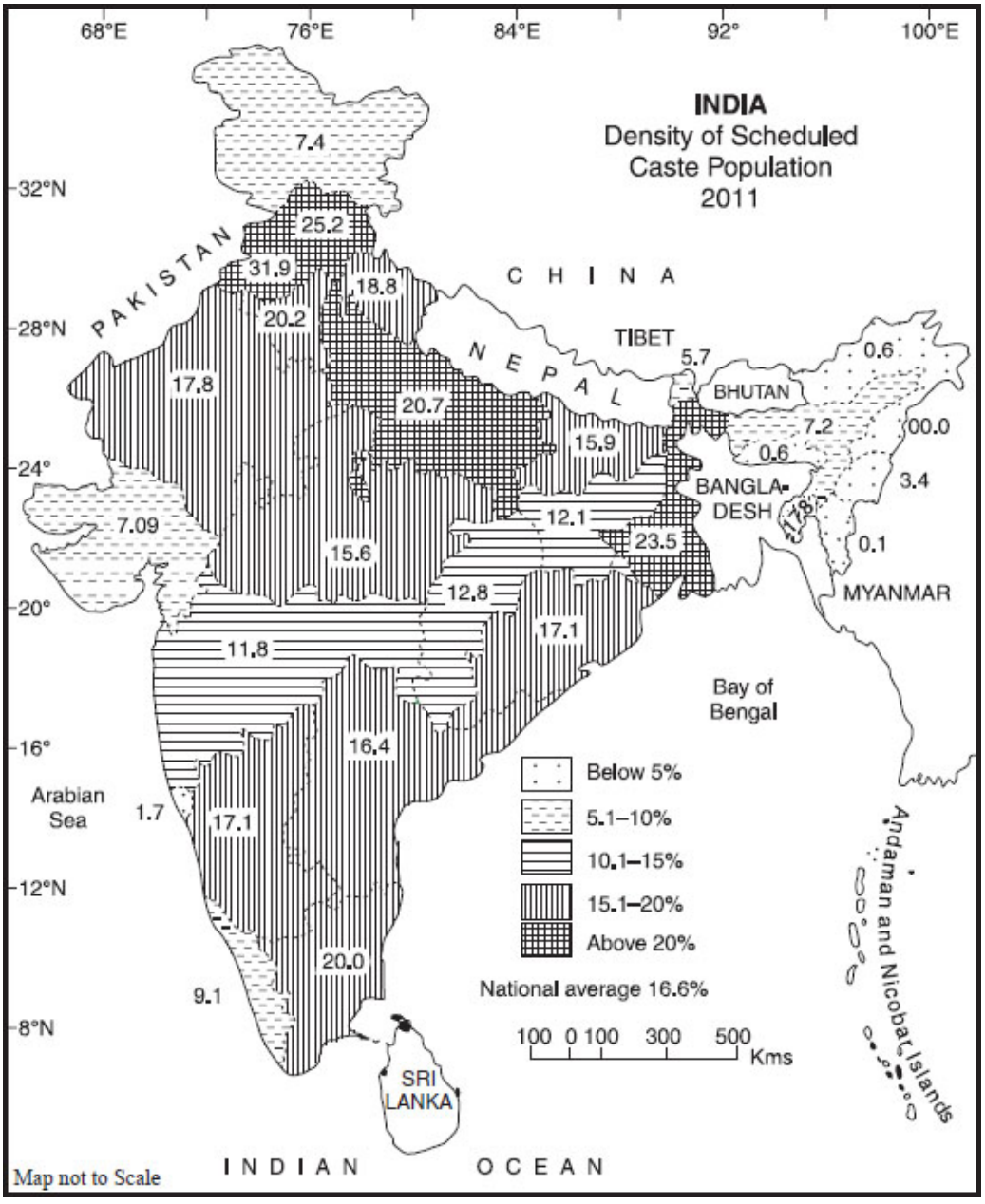


Fig. 13.11 Density of Scheduled Caste Population, 2011

Zelinsky has rightly observed that a ‘Cultural region comprises of a huge mass of learned behaviour, attitudes and ideas that control the greater part of their participants’ thoughts and actions. The possible combinations are so numerous that no two individuals are culturally identical, indeed each constitutes a kind of cultural micro-organism. At the other extreme, certain cultural practices are almost universal … relying on such criteria, the entire human species might be regarded as sharing a cultural macro-region. Between these two extremes, are the cultural meso-regions, having a common heritage. The sense of cultural identity is most pronounced in a micro-region and becomes diffused in a macro-region.

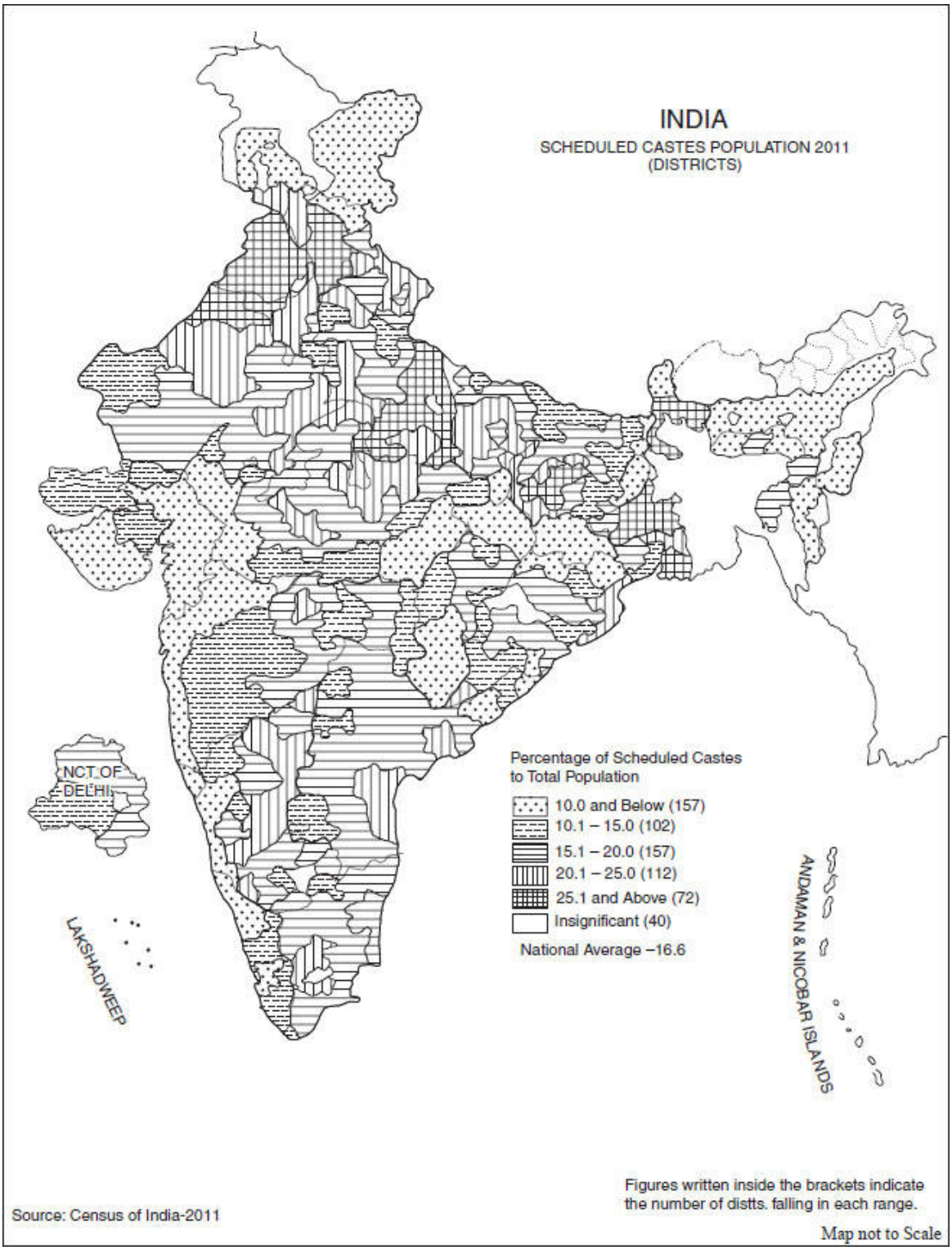


Fig. 13.11-A India-Scheduled Castes Population–2011

Cultural Regions of India

Language, religion, customs, and traditions are some of the important elements of culture. Cultural regions may be delineated on the basis of these cultural traits.

Language as a Determinant of Cultural Region

India is a multi-ethnic, multi-lingual, and multi-religious country. According to anthropologists and historians, the Indian population comprises the people who came here from the Mediterranean region, central Asia, south-west Asia, south-east Asia, Mongolia, Tibet, and China. Each of the racial and ethnic groups has its own language. After coming to India, the cultural mixing led to the mixing of their languages also. These languages have their core and peripheral areas. This broad linguistic regional identity formed the basis for the demarcation of Indian states in 1956. According to the Census of 1961, there were 187 languages spoken by different sections of the Indian society. Of these, 94 were spoken by less than 10,000 people. According to Vadodra-based Bhasa Research and Publication centre, the country had 1100 languages in 1961, but nearly 220 of them disappeared in the past 50 years. The lost languages were spoken mostly by nomads. At present, there are more than 19,500 languages or dialects spoken as mother tongue (Census 2011). The fifteen main languages, as mentioned in the Eighth Schedule of the Indian Constitution, are spoken by over 92% of the total population of the country. The geographical distribution of the main languages of India has been shown in **(Fig. 13.12)**.

The Indian languages belong to the following four linguistic groups:

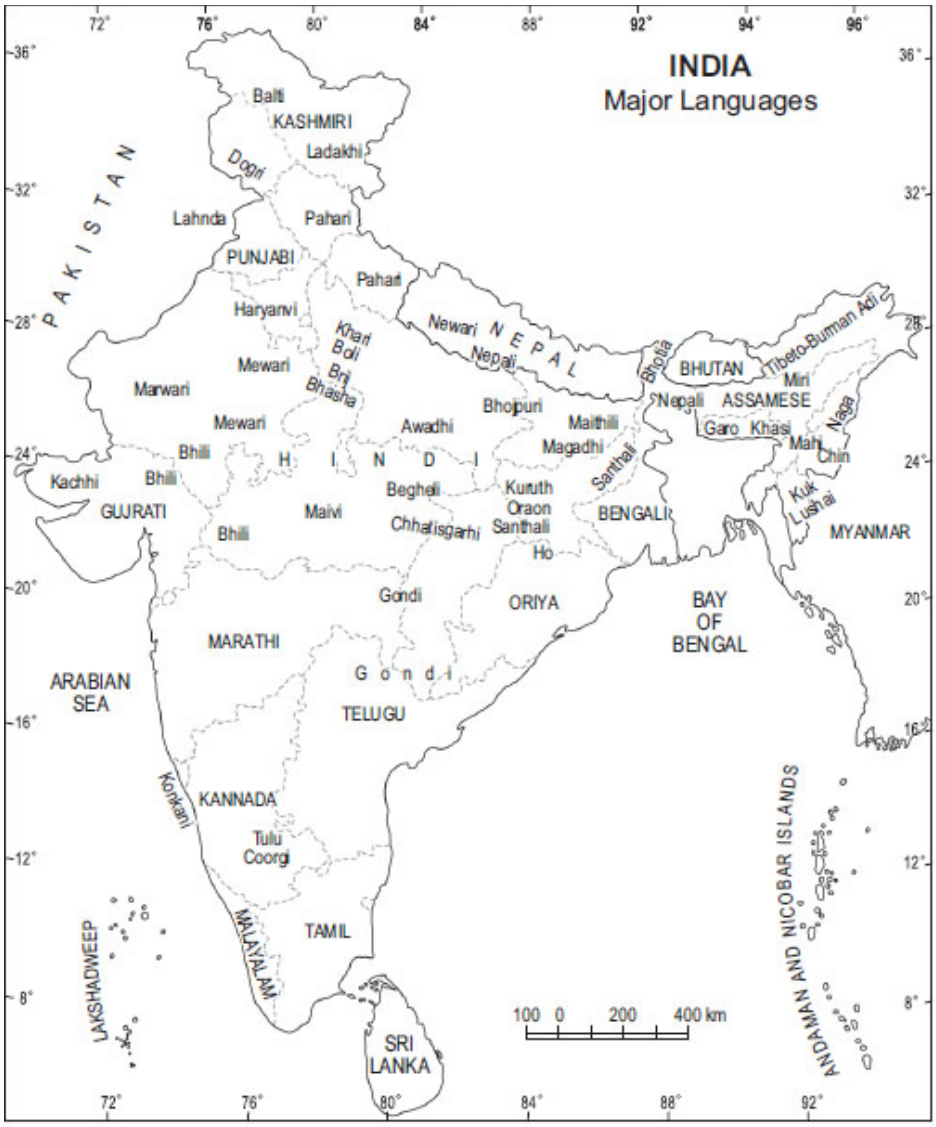
1. The Indo-European family (Arya)
2. The Dravidian family (Dravida)
3. Austric family (Nishada)
4. Sino-Tibetan family (Kirata)

1. **The Indo-Aryan language:** This is the most important group of Indian languages spoken by most of the people of northern India. Its core area is known as the *Khari Boli* region, comprising Haryana and western Uttar Pradesh. Going away from the core, it has different shades and dialects. Prof. A. Ahmad has given a diagrammatic representation of the diffusion of ***Khari-Boli*** (Hindi) in different directions from the core area. **(Fig. 13.13).** Offshoots include Dardi, Kohistani,Kashmiri, Lahnda, Sindhi, Kachhchi,

Gujarati, Marathi, Odiya, Bengali, Assamese, Bihari, Avadhi, Bagheli, Chhattisgarhi, Hindi, Punjabi, Rajasthani, Nepali, and Pahari.

Hindi (the National language) is the principal language of the Indo-European Family spoken by over 40% of the total population of the country. It is mainly spoken in Bihar, Delhi, Haryana, Himachal Pradesh, Madhya Pradesh, Rajasthan, Uttar Pradesh, and Uttarakhand. Urdu is closely akin to Hindi and is popular in Bihar, Delhi, Hyderabad, J&K, M.P., Uttar Pradesh, Uttarakhand and in most of the places of urban India.

2. **The Dravidian Family:** The Dravidian family of the Indian languages is mainly spoken in Andhra Pradesh (Telgu), Karnataka (Kannada), Kerala (Malayalam), and Tamil Nadu (Tamil). These four languages are spoken by more than 22% of the total population of India.



Map not to Scale

Fig. 13.12 Major Languages of India

3. **The Austric Family** The Austric languages are spoken by the tribal groups of Chhattisgarh, Jharkhand, Madhya Pradesh, Meghalaya, Odisha and West Bengal. These languages belong to two main branches: (i) Munda (Santhali), and (ii) Mon-Khmer (Khasi, and Nicobari). Of the two, Mon Khmer (Khasi) is confined to the Khasi and Jaintia Hills of Meghalaya, and the Nicobari to Nicobar Islands, while Munda language is spoken by the tribals of Chhattisgarh, Jharkhand, Madhya Pradesh, Odisha, and West Bengal.

4. **The Sino-Tibetan Family**: The Sino-Tibetan language is spoken mainly in the Himalayan belt. It has three major sub-divisions:

(i) Tibeto-Himalayan: It comprises Chamba, Lahauli, Kannauri, and Lepcha in Himachal Pradesh. The Balti, Bhutia, Ladakhi, and Tibetan languages are spoken in the northern parts of the Jammu and Kashmir state. The Bhutia and Kinnauri are the dominant languages in Himachal Pradesh.

(ii) The North Assami and Arunachal Pradesh: In north Assam and Arunachal Pradesh the main languages are Abor, Aka, Assami, Dafla, Miri, and Mishmi.

(iii) The Assami–Myanmari (Burmese): These languages are spoken by the Assamese, Bodo, Kochin, Kukichin, Miri, Naga, and Xaxa tribes.

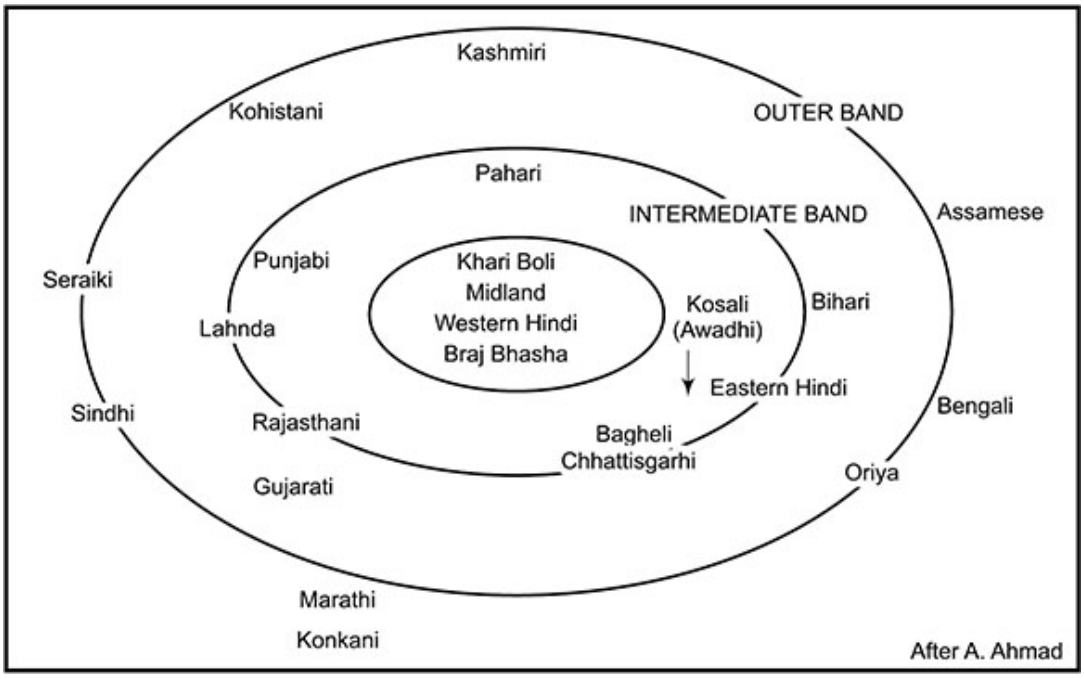


Fig. 13.13 Evolution of the Indo-Aryan Languages: Geographic Patterning (Second Stage)

Table 13.5

Linguistic Regions of India

	Linguistic Region	State/Union Territory

1.	Assamese	Assam and adjacent regions
2.	Bengali	West Bengal and parts of Tripura
3.	Gujarati	Gujarat and adjacent regions
4.	Hindi	Bihar, Chhattisgarh, Delhi, Haryana, Himachal Pradesh, Jharkhand, Madhya Pradesh, Rajasthan, Uttar Pradesh, and Uttarakhand
5.	Kannada	Karnataka and adjacent regions
6.	Kashmiri	Kashmir Division
7.	Malyalam	Kerala and Lakshadweep
8.	Marathi	Maharashtra and Goa
9.	Odiya	Odisha and adjacent regions
10.	Punjabi	Punjab and adjacent parts of Haryana
11.	Tamil	Tamil Nadu and Puducherry
12.	Telgu	Andhra Pradesh and adjacent regions

As the states of India were delineated on the basis of languages in 1956, the scheduled languages are spoken by the majority of the population in respective states. In Kerala for example, over 96% of the population speaks Malayalam, and in Andhra Pradesh over 85% of the people speak Telgu. Thus, every scheduled language has its specific region and the core of these exist in the state. The boundary of a linguistic region is, however, not a demarcated line, but a transitional zone over which one language gradually loses its dominance and gives way to another. There is an intermingling of languages in several areas. Further, in many states, the major language of one of the adjacent states is the second most important language spoken by the second largest group of people in the state. For example, in Kerala, Tamil is the second most

important language, while in Tamil Nadu, it is Telgu, in Karnataka and Andhra Pradesh the second language is the Urdu.

Hindi is the official language of the country spoken by about 40% of the total population. The Hindi belt includes the states of Uttar Pradesh, Bihar, Madhya Pradesh, Jharkhand, Chhattigarh, Rajasthan, Haryana, Himachal Pradesh, Uttarakhand, and Delhi in which over 90% of the total population speaks Hindi.

Urdu is basically a variant of Hindi written in Arabic/Persian script instead of Devnagri script of Hindi. It was born in India but is virtually 'homeless' without a strong regional base. The state of Jammu and Kashmir has adopted Urdu as the official language of the state. It is the mother tongue of about 8% of the total population of the country. It is spoken mainly in Uttar Pradesh, Bihar, Haryana, Andhra Pradesh, Karnataka, Delhi, Jummu & Kashmir and Uttarakhand.

Bengali, the second most spoken language of India has its cultural core in West Bengal, but its periphery extends in Assam, Bihar, Jharkhand, Odisha, and Tripura.

Telugu stands next to Bengali with its linguistic core in Andhra Pradesh and extension in Karnataka and Tamil Nadu. It has been called as the *Italian of the East.*

Marathi stands fourth in numerical strength. Its linguistic core lies in Maharashtra (93%) with speakers also in Karnataka, Madhya Pradesh, Gujarat, and Goa. Konkani, spoken in the Konkan coastal areas and Goa, is an offshoot of Marathi.

Tamil occupies the fifth rank in numerical strength, but best represents the old Dravidian script. It has rich literature commencing with the beginning of the Christian Era. Its linguistic core lies in Tamil Nadu (92%), but it extends its influence in Karnataka, Andhra Pradesh, and Puducherry.

Gujarati emerged from Gujarat and has carved out its influence in Maharashtra and Rajasthan. Kannada stands next to Gujarati. Its linguistic core lies in Karnataka (91%) and it has its extension in Tamil Nadu, Maharashtra, and Andhra Pradesh.

Among the Dravidian languages, Malayalam has the smallest number of speakers. Its linguistic core lies in Kerala (92%) and extends in Tamil Nadu, Karnataka, and Maharashtra.

Odiya has a distinctive character as it is the old Apabhramsa and has enriched itself with Sanskrit.

Assamese has its distinctive pronunciation and grammar, but is often included in the Bengal Assam group.

Religion as a Determinant of Cultural Region

Religion has been defined differently by different scholars. Friedrich Schleimacher defined religion as *feeling of absolute dependence.* According to William James, *religion is the enthusiastic temper of expousal*. Otto defines the essence of religious awareness as awe, a unique blend of fear and fascination before the divine. The main characteristics of religious life are: (i) traditionalism,
(ii) myth and symbol, (iii) concept of salvation, (iv) sacred places and objects, (v) sacred actions (rituals), (vi) sacred writings, (vii) the sacred community (monastic order), and (viii) the sacred experience.

Religion, like language, is a symbol of group identity and a cultural rallying point. All societies have value systems, common beliefs, understandings, and expectations which unite their people. Religion plays a crucial role in the socioeconomic life of the people and even their utilisation of natural resources is closely controlled by the religion of the people. Geographers are concerned with the interaction between religion and landscape (resources). Thus, religion provides a good basis for the demarcation of cultural regions.

India is a multi-religion country. It is the birth place of Hinduism, Buddhism, Jainism, and Sikhism. Subsequently, the successive waves of people of other religious faiths came to India. They maintained their religious identity. For example, the Syrian Christians appeared on the west coast of India in the first century AD. They are still found in Kerala. The Muslims came to India from South-West Asia and Central Asia and maintained their religious identity.

Growth and Composition of Various Religions

The growth rate of population of different religions in India is different during the last decade. In general, the population growth rate of all the religions has come down in the past decade (2001–11). Hindus population growth rate slowed down to 16.76% from the previous decade figure of 19.92% ,
while Muslims witnessed sharp fall in growth rate of 24.60% (2001–11)

from the previous figure of 29.52% (1991–2001). Such sharp fall in population growth of Muslims did not happened in the last 6 decades. Christian population growth was at 15.5% while Sikh population growth rate stood 8.4% . The most educated and wealthy community of Jains registered the least growth in 2001–11 with figure of just 5.4% .

The growth rate of Hindus, Muslims and Christians is expected to fall more in upcoming 2021 Census while other religions like Sikhism, Jainism and Buddhism are expected to remain stable for the next two decades considering already slowed down rate of these religions.

(Source: **All India Religion Census Data 2011.)**

Hindus: According to the Census 2001, about 80% of the total population of India is Hindu by faith. They are predominantly distributed throughout the country, but in a few areas, like the Kashmir Valley, Punjab, Mizoram, Meghalaya, Nagaland, Manipur and parts of Kerala, they are in minority **(Fig. 13.14).**

Hinduism is one of the oldest religions of the world. It is a polytheistic (multiplicity of gods and goddesses) religion. The proportion of Hindu population is the highest in Himachal Pradesh (95%) and lowest in Mizoram (3.6%). It is higher than the national average in Andhra Pradesh, Bihar, Gujarat, Himachal Pradesh, Haryana, Karnataka, Madhya Pradesh, Odisha, Rajasthan, Tamil Nadu, and Tripura, while it is much lower in the states of Jammu and Kashmir, Mizoram, Meghalaya, Nagaland, and Punjab **(Fig. 13.14).**

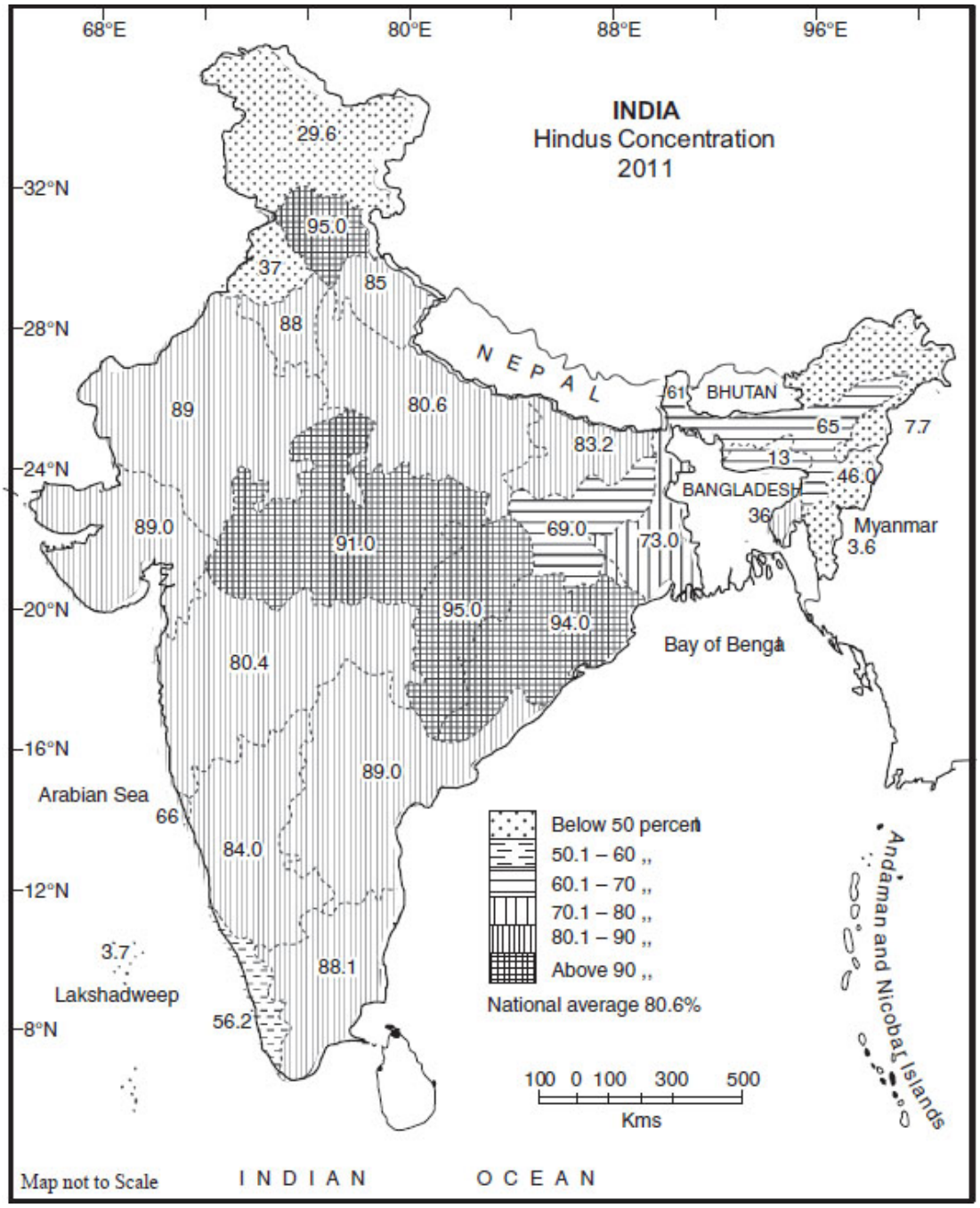


Fig. 13.14 Hindu Population (2011)

Muslims: Islam is a strictly monotheistic (one God) religion. The percentage of Muslims population in India is 14.2% (2011). The Muslims are well spread out in the country, but their high concentration is found in Uttar Pradesh, West Bengal, Bihar, Andhra Pradesh, Karnataka, Maharashtra, Kerala, Jammu and Kashmir, and southern districts of

Uttarakhand. The proportion of Muslim population is, however, the highest in the state of Jammu and Kashmir (over 68%) and insignificant in the state of Mizoram (1.1%). Their proportion is higher than the national average in Assam, Bihar, Jharkhand, Jammu and Kashmir, Kerala, Uttar Pradesh, and West Bengal **(Fig. 13.15).**

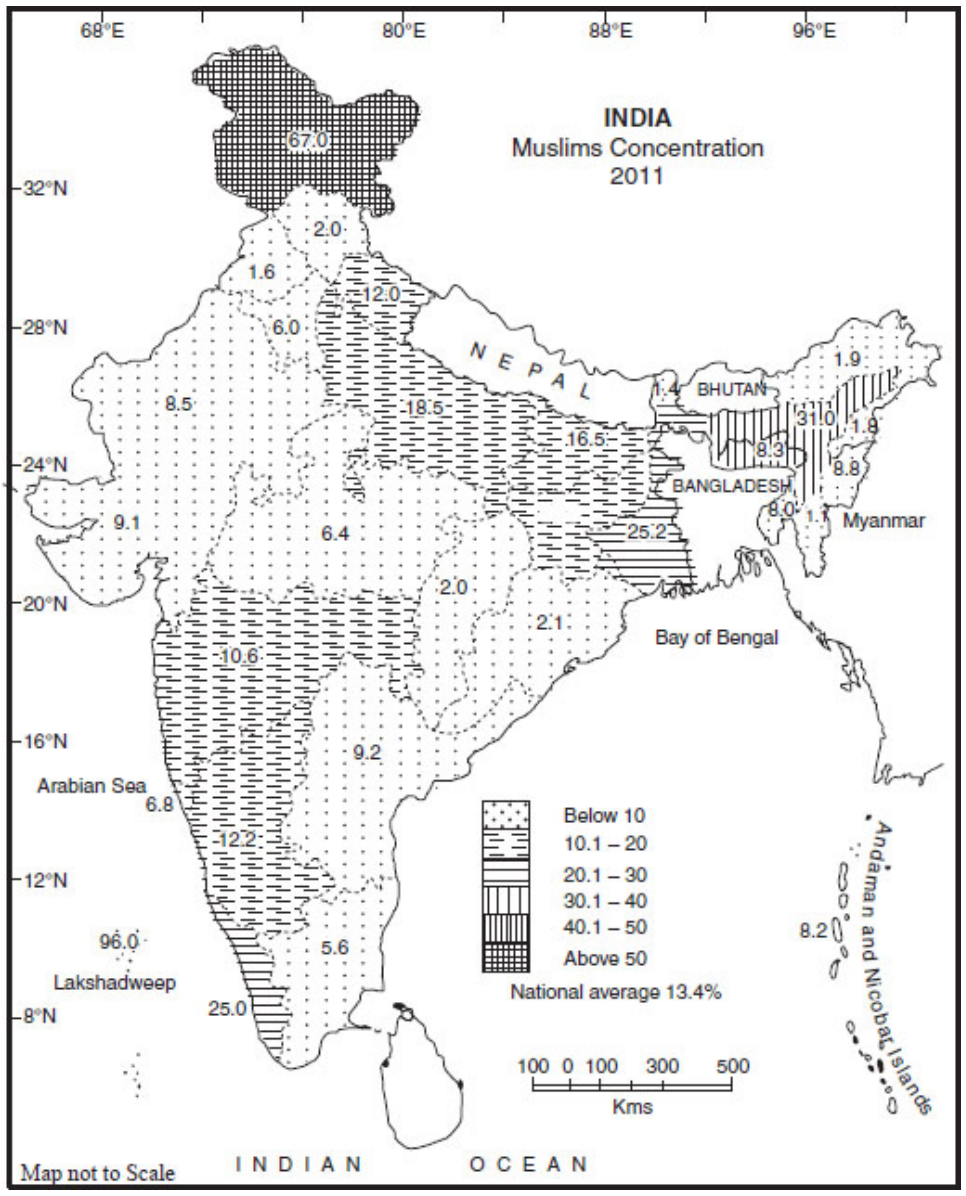


Fig. 13.15 Muslim Population

Christianity: Christianity is a universal religion which has the largest number of followers in the world. As stated above, Christianity came to India in the first century AD when the Syrian Church was established in

Kerala. Christians constitute about 2.3% of the total population of India (2011). The largest number of Christian population is in the state of Kerala —about 29% of the total population. Christians number more than one million in the states of Andhra Pradesh, Meghalaya, Nagaland and Tamil Nadu. Their proportion is significantly large in the states of Mizoram, and Goa. Nagaland (90%) and Mizoram (87%) have the largest percentage in population. **(Fig. 13.16).**

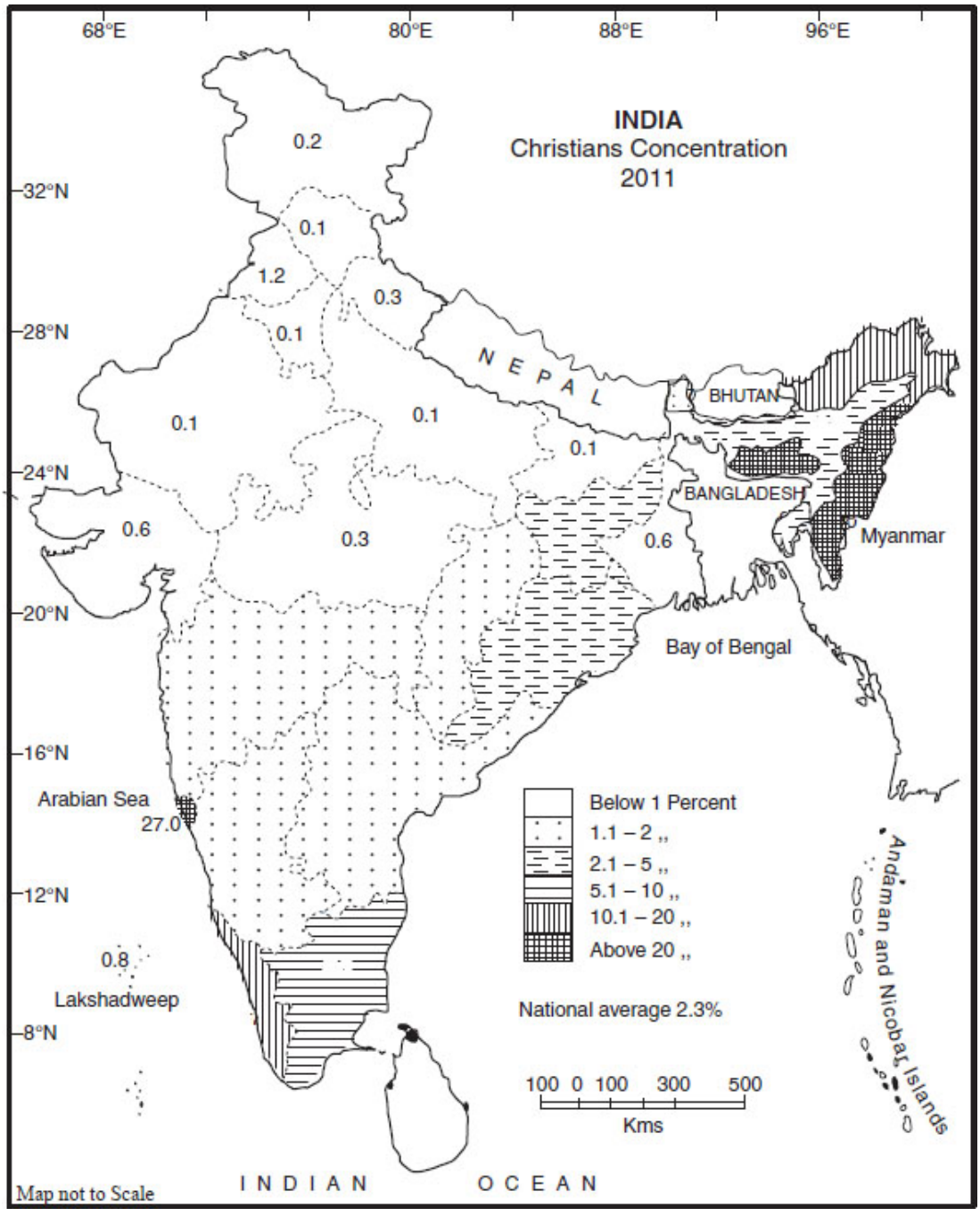


Fig. 13.16 Christian Population (2011)

Sikhs: The religion of Sikhism was founded by Guru Nanak Sahib in the 15^{th} century. The Sikhs constitute about 2% of the total population of the country (Census, 2001). Sikhism attempted to create social harmony by removing the Hindu caste system and permitting widow remarriage. But for a long time it remained confined to Punjab and has accepted Gurmukhi as its language. Nearly 79% of the total population of Sikhs is concentrated in the state of Punjab **(Fig. 13.17).** In addition to Punjab, Sikhs are found in Haryana, Chandigarh, Delhi, Rajasthan and the Terai region of Uttar Pradesh and Uttarakhand (Udhamsingh Nagar). At present, canbe found in any part of the country and have acquired international presence in the U.K., Canada, the U.S.A., Australia, New Zealand, Kenya, Pakistan, Afghanistan, Singapore, and Hong Kong.

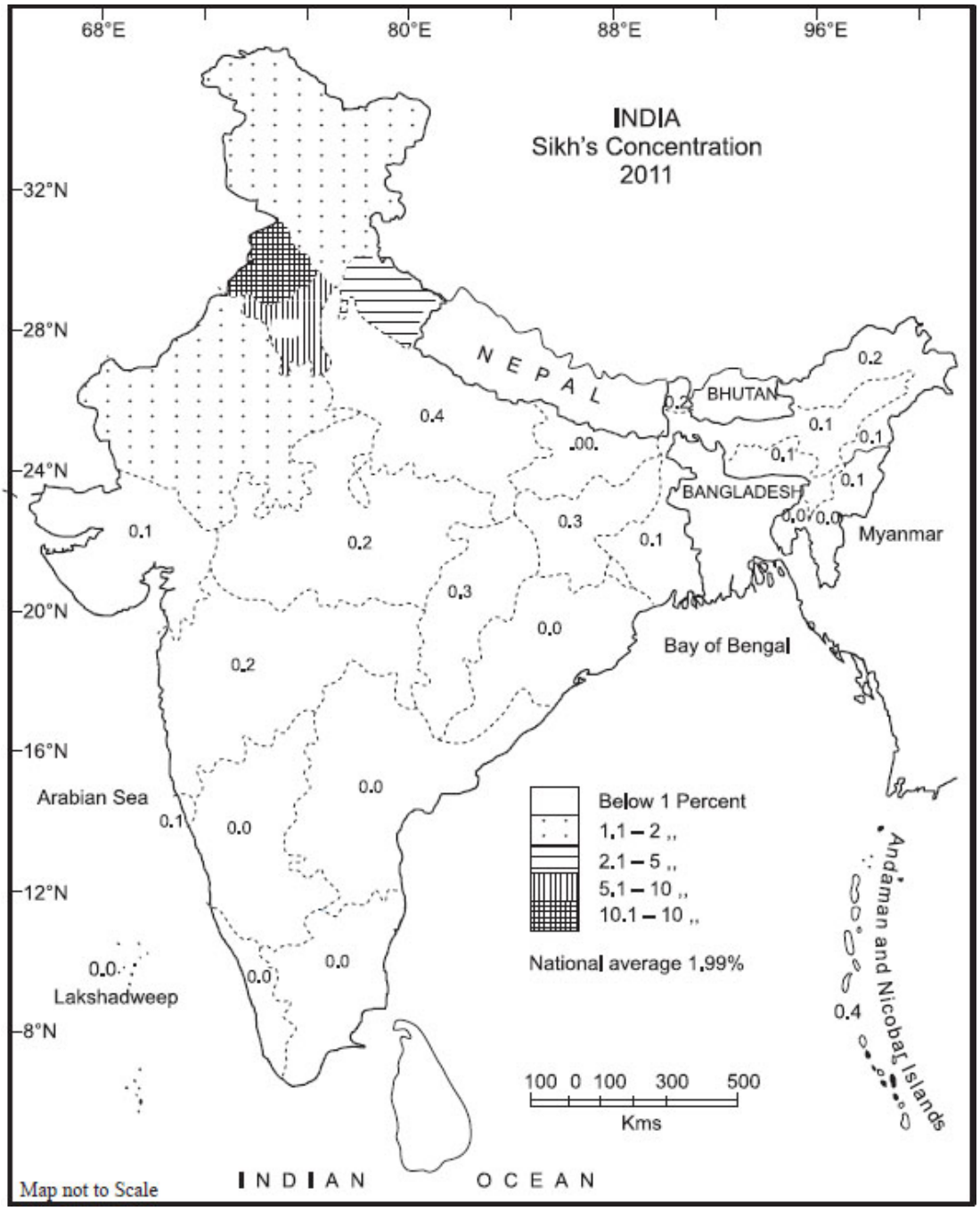


Fig. 13.17 Density of Sikh Population (2011)

Buddhism: Buddhism was founded by Gautam Buddha (563–483 BC) in North India. The Buddhists constitute less than one per cent of the total population of the country. Nearly 80% of the Buddhists live in Maharashtra. The traditional pockets of Buddhism are Ladakh, areas of

Jammu and Kashmir, Himachal Pradesh, Sikkim, Arunachal Pradesh, and Tripura **(Fig. 13.18).**

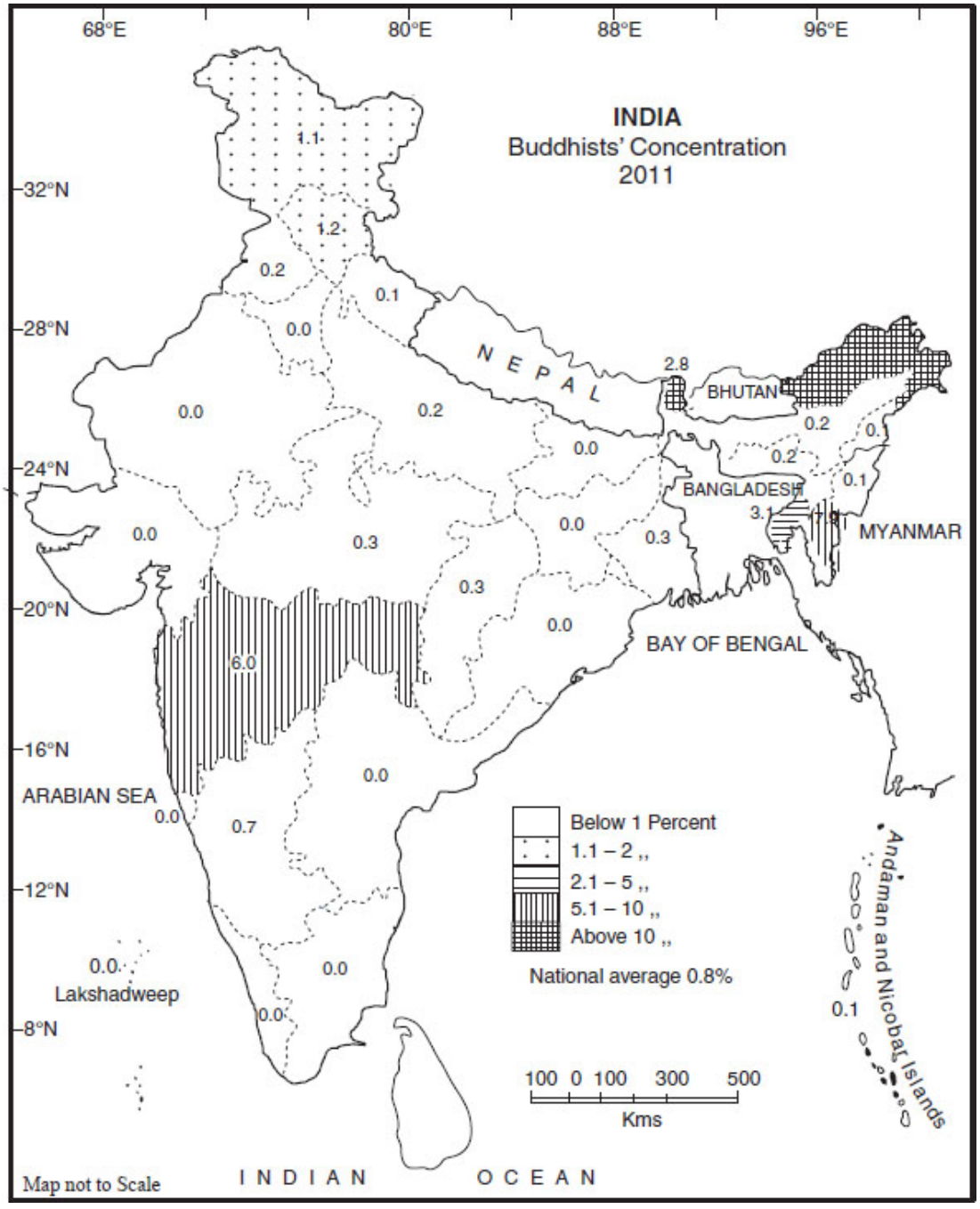


Fig. 13.18 Density of the Buddhist Population (2011)

Jainism: India is the homeland of Jainism which is a minority religion (0.4%) and has no perceptible following in other countries. Its followers

are found in Maharashtra, Rajasthan, Gujarat, Madhya Pradesh, Uttar Pradesh, Uttarakhand, and Karnataka, particularly in urban areas. The Jains have an important influence in business and politics.

Zoroastrianism: The Parsis (population about 1.67 lakh) are the followers of Zoroastrianism (2011). It was a dominant religion in the days of the Old Persian Empire. The essence of its ethics is well summed up in three words: *Humata* (good thoughts), *Hukhta* (good words), and *Huvarshta* (good deeds). Their religious book is *Dinkart*. They first reached Diu in 766 AD and shifted their colony to Mt. Bharhat (1490). From there they spread to Navsari and ultimately to Udvada. They have been influenced by the Hindu customs, but they do not advocate celibacy, and permit remarriage. About 80% of the Parsi population is concentrated in Greater Mumbai and the rest in Navsari, Udvada, Surat, and Ahmadabad. The population of Parsis is decreasing at an alarming rate. Their birth rate is significantly low as compared to the death rate.

Religious Regions of India: The cultural regions of India based on religion have been plotted in **(Fig. 13.19).**

It may be seen from **(Fig. 13.19)** that the greater part of the country has a dominance of Hindu religion and culture with sprinkled populations of Muslims, Christians, Sikhs, Buddhists, and tribals. The hill states of north east India are, however, characterised by the mixed population of Christians, Tribals, Hindus, and Muslims. The Muslim dominated regions are the Kashmir Division and Kargil district of Jammu and Kashmir state. Muslims are quite significant in northern Kerala and in Agra, Meerut, Lucknow, Rohilkhand, and Saharanpur divisions of Uttar Pradesh. Ahmadabad, Bhopal, Hydarabad, Kolkata and Mumbai, Surat. The state of Punjab and the Union Territory of Chandigarh are the Sikh dominated parts of the country **(Fig. 13.19).**

Customs: Customs are a very important component of cultural geography. A custom is a frequent repetition of the same act to the extent that it becomes characteristic of the group of people performing the act. Habit is a similar word which is adopted by an individual, while custom is the act which has been adopted by most of the people of an ethnic group or society. There is a positive correlation between the customs and utilisation of environment (resources). In fact, the tradition bound society has many oral folk traditions. In the delineation of cultural regions,

customs (folk dance, folk lore, folk medicine, etc.) are also important indicators.

Cultural regions of India based on Language, Religion and Customs: The cultural regions of India based on language, religion, and traditions have been plotted in **(Fig. 13.20).**

It may be seen from **(Fig. 13.20)** that on the basis of language, religion, customs, and traditions, India may be divided into the following ten cultural regions:

1. The Ladakhi-Buddhist Cultural Region
2. The Kashmiri-Muslim Cultural Region
3. The Sikh-Gurumukhi Cultural Region
4. The Kinnauri-Dev-Bhumi Cultural Region (Himachal Pradesh and Uttarakhand)
5. The Hindu-Hindi Cultural Region
6. The Mixed Cultural Region of north east India
7. The Bengali Cultural Region
8. The Tribo-Hindu Cultural Region of Chotanagpur
9. The Marathi Hindu Cultural Region
10. The Dravido-Cultural Region (comprising of Telugu, Kannada, Tamil, and Malyalam)

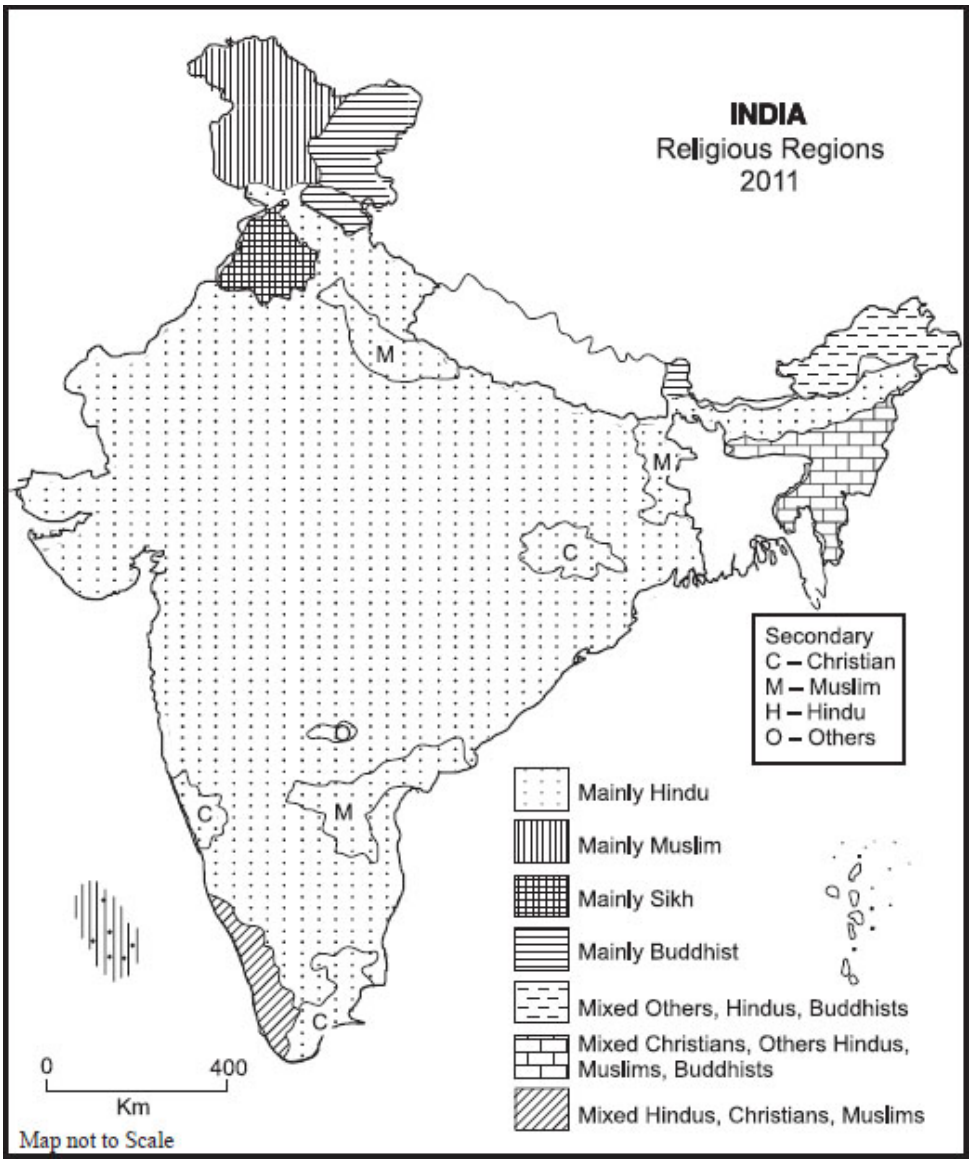


Fig. 13.19 Religious Regions (2011)

1. **The Ladakh Cultural Region:** It has the dominance of Buddhists and Ladakhi language. There are Gompas and monasteries in this region. Leh and Dhramshala are the important sacred and cultural centres of this region.
2. **The Kashmiri-Muslim Cultural Region:** Stretching over the valley of Kashmir and northern parts of Jammu (Doda district, etc.) and southern parts of Ladakh (Kargil) divisions, it is a predominantly Muslim dominated region in which Kashmiri is the

main language. Hindus and Sikhs though in minority, speak Kashmiri and follow the Kashmiri cultural traditions.

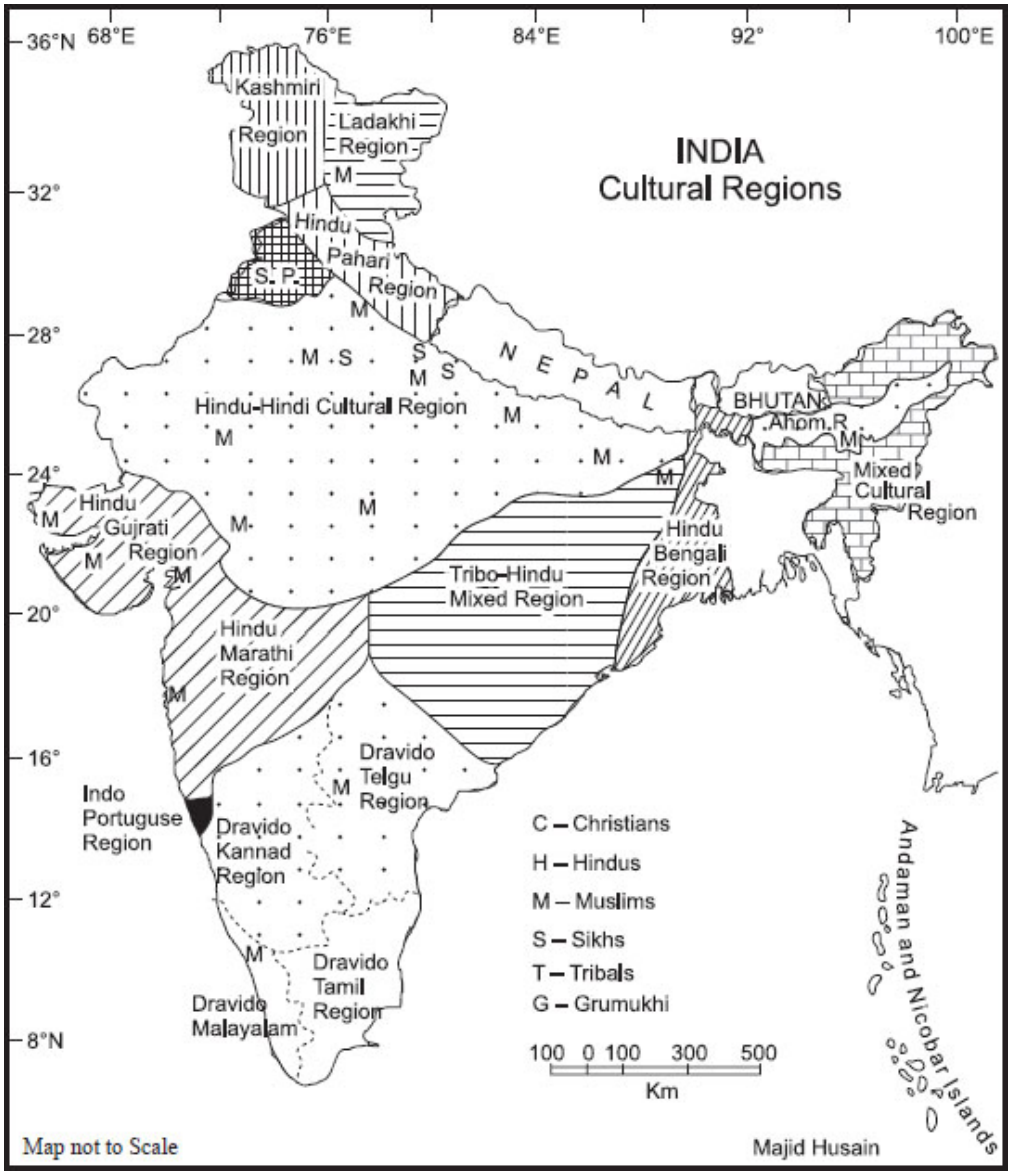


Fig. 13.20 Cultural Regions based on Religion, Language, and Traditions

3. **The Sikh-Gurmukhi Cultural Region:** Stretching over the state of Punjab and the Union Territory of Chandigarh, this region has the majority of Sikhs who speak Punjabi language. The Hindus are in minority. This region is characterised by Gurudwaras in almost all the villages and towns. The Golden Temple situated in

the city of Amritsar is a sacred place and an important pilgrimage centre for religious people.

4. **The Kinnauri-Dev Bhumi Cultural Region**: This region sprawls over the mountainous parts of Himachal Pradesh and Uttarakhand. It is called the Dev-Bhumi in which there are many religious shrines (Kedarnath, Badrinath, Haridwar, Hemkund, Paonta Saheb, etc.). In the region of Himachal Pradesh, Kinnauri is the dominant language, while in Uttarakhand, Hindi is the language of the masses.
5. **The Hindu-Hindi Cultural Region:** This region covers the states of Bihar, Haryana, Madhya Pradesh, Chhattisgarh, Jharkhand, Rajasthan, southern parts of Uttarakhand and Uttar Pradesh. It is the Hindi Heartland with dominance of Hindu religion. In western Uttar Pradesh and in urban centres, Muslims consititute a significant minority. Sikhs and Christians are also sprinkled, mainly in the urban areas like Delhi, Kanpur, Lucknow, Varanasi, Meerut, Agra, and Allahabad etc.
6. **The Mixed Cultural Region of North East India**: Stretching across the states of Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, and Tripura, it is a region of mixed culture in which there are areas of dominance of Hindus, Christians, Muslims and Tribal religion. There is a great diversity in the languages, religion, customs, folk-dances, music, and folk medicine.
7. **The Bengali-Cultural Region:** Spreading over West Bengal and the adjacent regions of Odisha, Jharkhand and Bihar, this region has the dominance of Bengali speaking people. The main religion of the people is Hinduism, while Muslims constitute a significant minority in isolated pockets.
8. **The Tribo-Hindu Cultural Region:** This cultural region spreads over Chotanagpur plateau. Most of the people belong to Hindu religion, while Christians are also significant in number. Most of the people speak Hindi language.
9. **The Marathi-Hindu Cultural Region**: Stretching over Maharashtra, parts of Gujarat, Goa, and the adjacent regions of Andhra Pradesh and Karnataka, this region has a dominance of

Marathi language and Hindu population. Concentration of Muslims, Parsis, and Buddhists is in isolated pockets.

10. **The Dravido-Cultural Region**: This region sprawls over Andhra Pradesh, Karnataka, Kerala, and Tamil Nadu. The people belong to the Paleo-Mediterranean race and speak Dravidian language. The major languages are Tamil, Malyalam, Telgu, and Kannad **(Fig. 13.20).**

DEMOGRAPHIC CHARACTERISTICS OF INDIAN POPULATION

The spatial variations in the distribution, density, composition, sex-ratio, literacy rate, migration, and growth of population have a direct bearing on the utilisation of resources and the levels of development of a region/country. It is, therefore, an important concern of geographers. The distribution, density, composition, growth, and migration of population is directly or indirectly controlled by the physical, socio-cultural, economic, and political factors. India has only about 2.4% of the total area of the world but supports over 17.64% of the total population of the world. The population density in India is 460/km2, (1192 people/m2) 33.6% of the population is urban (460, 249, 853 person as in 2019).

India—Demographic Transition and Population Growth

The Theory of Demographic Transition was propounded by W.S. Thompson in 1929 which was modified by Frank Notestein in 1945. Demographic transition is the transition from a stable population with high mortality and fertility to stable population with low mortality and fertility. During the transition population growth and changes in the age structure of the population are inevitable.

In India, the demographic transition has been relatively slow but steady. As a result the country was able to avoid adverse effects of rapid changes in the number and age structure of the population on social and economic development.

Growth of population is the rate in the number of people living in a particular area between two given points of time. The net change between the two points of time is expressed in percentage and is described as the growth rate of population.

The patterns of growth changed between 1901 and 2011 have been plotted in **(Fig. 13.21)** and the relevant data has been given in **(Table**

13.6).

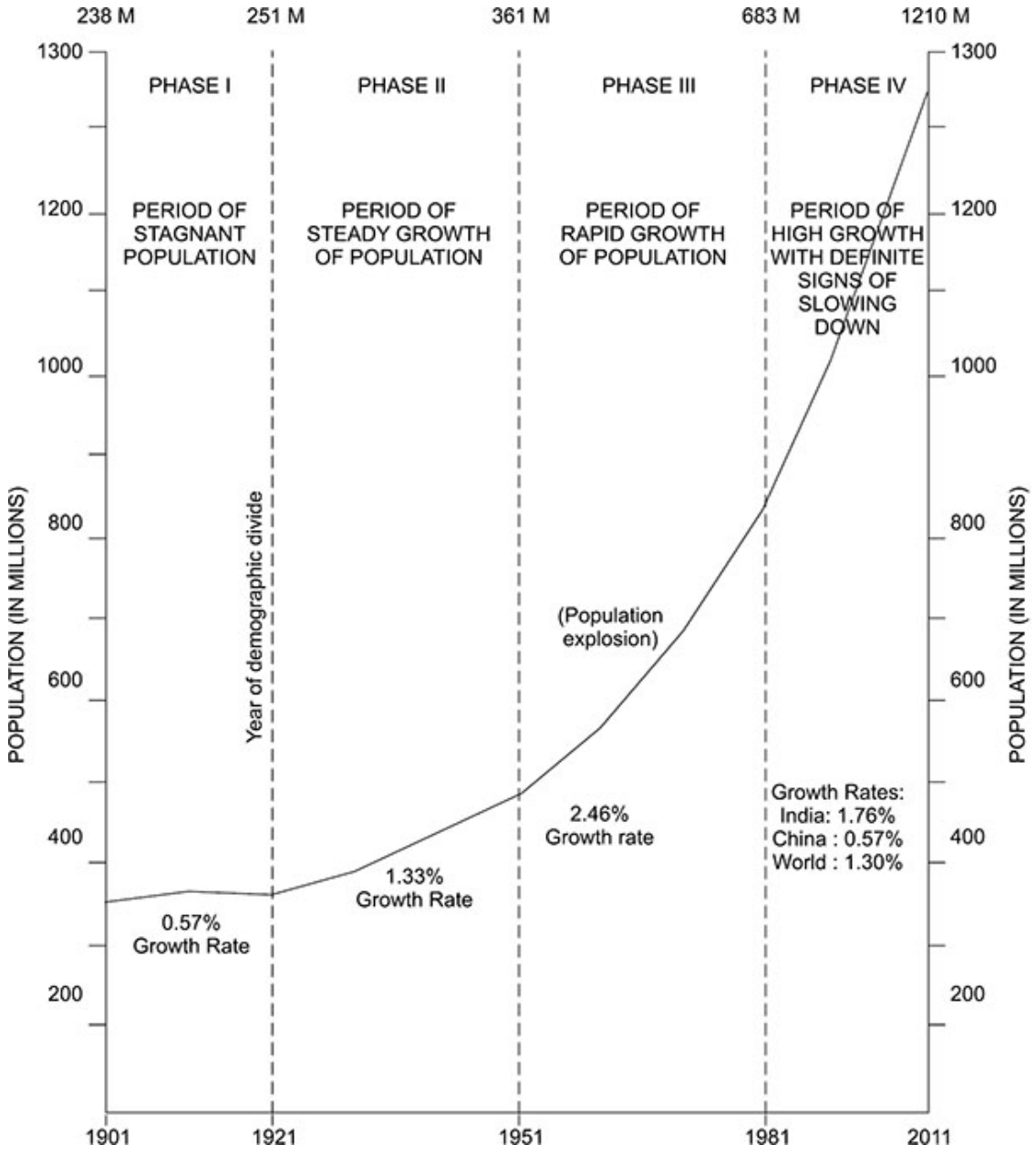


Fig. 13.21 India: Demographic Transition (1901–2011)

Table 13.6

India—Growth of Population 1901–2011

Census Year 1	Population 2	Absolute 3	Decadal Growth Rate 4
1901	238,396, 327	–	–

1911	252,093,390	13,697,063	5.75
1921	251,321,213	–772,177	0.31–Year of Demographic Divide
1931	278,977,238	27,656,025	11.00
1941	318,660,580	39,683,324	14.22
1951	361,088,090	42,427,510	13.31
1961	439,234,771	78,146,681	21.64
1971	548,159,652	108,924,881	24.80
1981	683,329,097	135,169,455	24.66
1991	846,421,039	163,091,942	23.87
2001	1,028,737,436	182,316,397	21.54
2011	1,210,195,000	181,457,564	17.7
2030*	1,510,000,000	299,805,000	24.77

UNO Projection, June, 2017

Source: Census of India, 2011.

As stated at the outset, India has the second largest population in the world after China. The population of India, according to the census of 2011 was 1210.19 million. It is growing at a faster pace. India's 2019, population is estimated at 1.37 billion (1,368,097,497) On an average, Indian population is increasing at a growth rate of over 1.08% per annum as against 1.3% for the world as a whole and 0.57% for China (2001–11). According to Uno projections, India's population, is projected to rise to 1.51 billion by 2030 and further to 1.68 billion by 2050, before declining to 1.52 billion by the end of the 21st century. India will be the most populous by 2024 (Times of India, 23rd June, 2017).

The population of India as recorded at each decennial census from 1901 has grown steadily except for a decrease during the decade of 1911–21. The year 1921 is known as the '*year of demographic divide*' in the

demographic history of India when mortality started to decline leading to acceleration in the rate of population growth. The decadal growth of population of India has been given in **(Table 13.6).**

It may be observed from **(Table 13.6)** that the total population of India in 1901 was about 238 million which rose to 361 million in 1951 and became 1210.19 million in 2011.

It may also be seen from **(Table 13.6)** that there have been significant turning points in the Census Years of 1921, 1951, and 1981. Thus, the temporal changes in the population growth of India during the 20^{th} century may be classified under the following four phases **(Fig. 13.21** and **Fig. 13.22)**.

1. Period of Stagnant Population—1901–21
2. Period of Steady Growth—1921–51
3. Period of Rapid Growth—1951–81
4. Period of High Growth with Declining Trend—1981–2011

1. Period of Stagnant Population

The first two decades (from 1901 to 1921) was the period of stagnant or stable population. During this period, the birth rate and the death rate were high. Consequently, the population showed a stagnant phase of population growth. It may be termed as the first phase of demographic transition in India. The birth rates were very high due to universal and early marriages, wide spread prevalance of illiteracy, traditional social beliefs and customs. Absence of knowledge about family planning techniques, attitude towards children for supplementing family income. There was high mortality during this period because of epidemics like, cholera, influenza, malaria, plague, small-pox, absence of medical and sanitation facilities etc. Food shortage caused severe droughts in the years of 1911, 1913, 1915,1918 and 1920. Moreover, thousands of Indian soldiers lost their lives during the First World War (1914–18). Interestingly enough, the population showed a declining trend between 1911 and 1921. The Census year of 1921 is termed as the year of *demographic divide* in the demographic history of India.

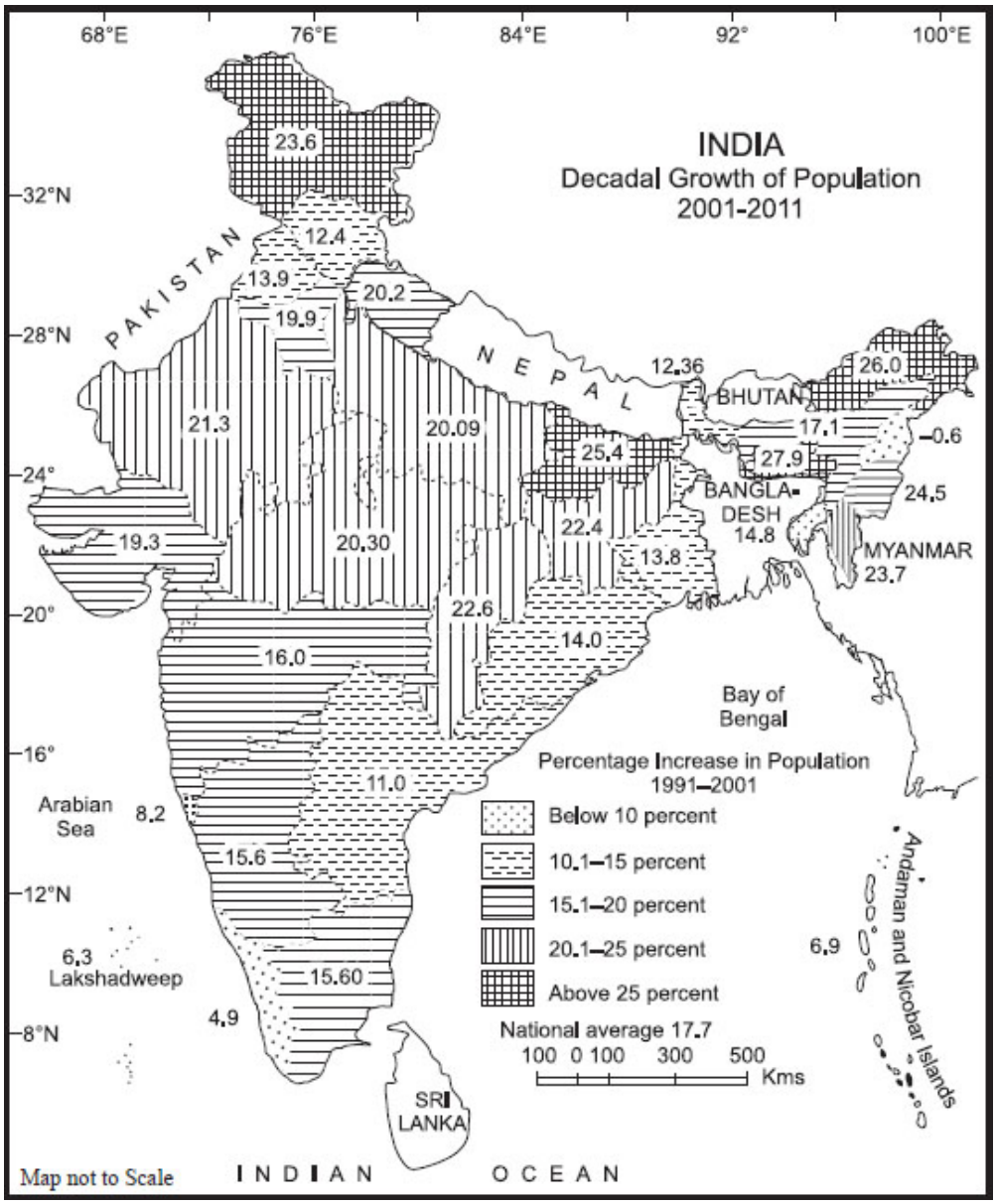


Fig. 13.22 Percentage Decadal Growth of Population—2001–11

2. Period of Steady Growth of Population (1921–51)

The period between 1921 and 1951 is known as the period of steady growth of population. The birth rate during this period remained high but the death rate started to show the declining trend. Moreover, some of the epidemic diseases like cholera, malaria, plague, small pox were significantly controlled. The problems of scarcity of food were also controlled with the help of the expansion and improvements in the roads and railways networks.

3. The Period of Rapid Growth of Population (1951–81)

After Independence, there was a steep fall in the death rate but the birth rate remained significantly high. During this stage some people became conscious about the size of family. The people from rural areas migrated to urban areas in search of jobs. With growing urbanization of the country, the adoption of small family norm became popular. Consequently, there remained high birth rate, low death rate, and high growth of population. This period is often referred to as the *period of population explosion*. The unprecedented growth of population during this period was mainly due to the accelerated developmental activities and improvement in the health facilities. The standard of living and medical facilities reduced the death rate. The birth rate, however, remained high. The high birth rate and low death rate resulted into the rapid growth of population.

4. Period of High Growth of Population with Declining Trend of Population Growth (1981–2011)

The last phase of the 20th century (1981–2011) and the First decade of the 21st century showed a declining trend in the growth rate of population. It is interesting to note that for the first time since 1921, India's urban population went higher than its rural population. This could become possible because of the education, awareness about the size of family, especially among the middle class people of the society. The health care facilities were extended even to the remote rural areas. During this period, the decaded growth rate declined rapidly from 19.7 per thousand in 2001 to 17.7 per thousand in 2011 **(Fig. 13.23).**

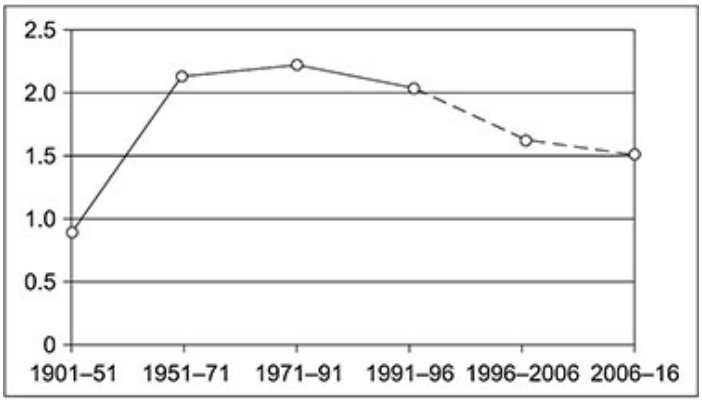


Fig. 13.23 India: Declining Population Growth Rates (Percentage)

It is encouraging to note that 2001–11 is the first decade (with the exception of 1911–21) when the absolute increase in population over the ten-year period has been less than the previous decade.

The percentage decadal growth during 2001–11 has recorded the sharpest decline since Independence.

According to the Population Division of the United Nations Deptt. of Economic and Social Affairs, India will continue to grow to about 1.6 billion by 2060 before beginning to decline. India, with 1.46 billion people, will have overtaken China, with 1.39 billion, as the world's most populous country in 2028.

The UN report, ''World Population Prospects. The 2012 Revision'', has projected that the global population will touch 9.6 billion by 2050 with most of the growth in developing countries. India's population will be a round 1.6 billion in 2060 and then decline slowly to 1.5 billion in 2100.

The United Nations "World Population Prospects", released on 24.02.2005, New York, estimates that there will be 1395 million people in India by 2025, and 1595 in 2005. China will be populated 1441 million by 2025 and 1392 in 2050. Interestingly enough, half of India's population is below 26 years of age.

The changing patterns of birth rate and death rate and the natural increase in population has been shown in **(Table 13.7).**

Table 13.7

India—Birth Rate, Death Rate, and Natural Growth Rate of Population (1911–2011)

Census Year	Crude Birth Rate	Crude Death Rate	Natural Growth Persons Per Thousand of Population
1911	49	43	6
1921	48	47	1
1931	46	36	10
1941	45	31	14
1951	40	27	13
1961	42	23	19
1971	37	15	22
1981	34	12	22

1991	31	11	20
2001	26	9	17.7
2011	23	5.4	17.7

Source: Census of India, 2011.

It may be seen from **(Table 13.7)** that the natural growth rate of population was the highest between 1961–71 and 1971–81, being 22 persons per thousand of population. In 2001, the natural growth of population declined to 17 persons per thousand which is an encouraging factor for the demographers and population planners. In fact, 2001–11 is the first decade (with the exception of 1911–21) which has actually added lesser population compared to the previous decade. The absolute growth of population is still significantly high being about 20 million per year which is mainly because of the large base (over one billion) of population of India.

At present, India is in a state of rapid fertility transition with the pace of decline having accelerated in recent years. India's population is in the late expanding stage in which birth rate is falling, death rate is also falling and population continues to grow. This is because of: (i) family planning, (ii) lower infant mortality, (iii) changing status of women and (iv) increase in standard of living. The pace of decline varies from state to state and experiences of Andhra Pradesh, Nagaland, Goa, Tamil Nadu, W. Bengal, Assam, Tripura, Odisha, and Kerala indicate that the Crude Birth Rate (CBR) can go much below the replacement level of fertility. The rural urban differential in fertility tend to narrow down as fertility declines. In Nagaland, Kerala, Tamil Nadu, Andhra Pradesh, and Goa, the rural urban birth rates are almost the same. There is greater homogeneity in the Crude Birth Rate in urban areas compared to rural areas. Surprisingly, among the larger high fertility states of north India, Bihar has exhibited a comparatively rapid decline in the Crude Birth Rate during the last ten years, probably as a result of relatively greater acceptance of family planning methods. Interestingly enough, Nagaland recorded the lowest growth rate of –0.6% during the decade of 2001–11.

Distribution of Population

The distribution of population in any country/region is closely influenced by the physical (terrain, topography, climate, natural vegetation, minerals, and soils), socioeconomic (religion, customs, education, awareness, per

capita income, and standard of living), and political factors (government policy towards population). The parts of the country having good climate, fertile soils, adequate availability of water, minerals, good connectivity by roads and railways have a high concentration of population.

The spatial distribution of population in India is not uniform. The Indo-Gangetic-Brahmaputra Plains and the Eastern and Western Coastal Plains are the densely populated areas where the density of population generally exceeds 550 persons per sq km **(Fig. 13.24)**.

As opposed to this, the areas with inclement climate, mountainous topography, deserts, barren lands and thickly forested and marshy lands have sparse distribution of population.

The uneven distribution of population in India may be appreciated from the fact that Uttar Pradesh alone with an area of 7.25% has a population of 228.95 million or 17.15% of the total population of the country (2018). Uttar Pradesh is the most populous state with almost 200 million people, which is more than the population of Brazil. The combined population of Uttar Pradesh and Maharashtra (the second most populous state), at 312 million is substantially greater than the population of the USA. About 60% of India's population lives in seven states, namely, Uttar Pradesh, Maharashtra, Bihar, W. Bengal, Andhra Pradesh, Madhya Pradesh, and Tamil Nadu (2011). Dibang valley of Arunachal Pradesh is the least populated. Longleng district of Nagaland registered a negative population growth rate of (-) 58.39. Among the Union Territories, NCT of Delhi is the most populous (11,034,555 people). Lakshadweep has a population of 71,218 and is the smallest union territory. The distribution of population and the percentage share of the states and the union territories have been given in **(Table 13.8):**

Unique Identification Authority of India, 2018

The top metropolitan cities with their population are:

Mumbai (18,394,912), Delhi (16,349,831), Kolkata (14,035,959), Chennai (8,653,521), Bangalore (8,520,435).

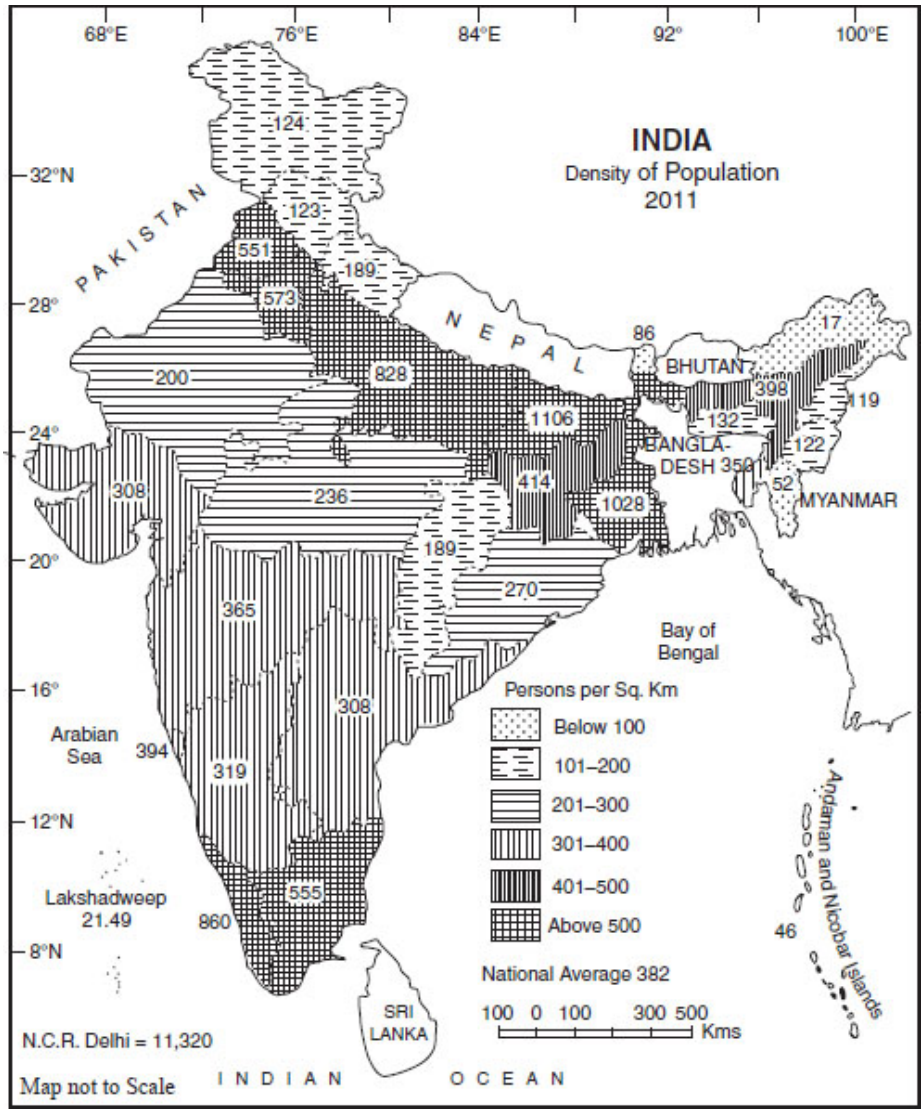


Fig. 13.24 Density of Population (2011)

Table 13.8

State-wise Distribution of Population (2011)

Rank	State/Union Territories	Population 2011	Population 2018	Percentage to the Total Population of India	Density of Populations sq km
1.	Uttar Pradesh	199,58 1,477	228,95 9,599	17.15	829
2.	Bihar	104,09 9,452	119,46 1,013	14.76	1106

Rank	State/Union Territories	Population 2011	Population 2018	Percentage to the Total Population of India	Density of Populations sq km
3.	Maharashtra	96,878,627	120,837,347	7.53	365
4.	West Bengal	91,347,736	97,694,960	7.03	1028
5.	Andhra Pradesh	84,580,777	52,883,163	6.67	308
6.	Madhya Pradesh	72,626,809	72,626,809	13.38	236
7.	Tamil Nadu	72,147,030	76,481,545	6.01	555
8.	Rajasthan	68,548,437	78,230,816	14.12	200
9.	Karnataka	61,095,297	66,165,886	8.30	319
10.	Gujarat	60,439,692	63,907,200	5.74	308
11.	Odisha	41,974,218	45,429,399	8.23	270
12.	Kerala	33,406,061	35,330,888	5.76	860
13.	Jharkhand	32,988,138	37,329,128	13.16	418
14.	Assam	31,205,576	34,586,234	10.83	398
15.	Punjab	27,437,338	29,611,935	6.74	551
16.	Chhattisgarh	25,545,198	28,566,990	11.83	189
17.	Haryana	25,351,462	27,388,008	8.03	573
18.	Delhi (NCT)	16,787,941	18,345,784	9.28	11320
19*.	Jammu and Kashmir	12,541,302	13,635,010	8.72	124
20.	Uttarakhand	10,086,292	11,090,425	9.96	189
21.	Himachal Pradesh	6,864,602	7,316,708	6.59	123

Rank	State/Union Territories	Population 2011	Population 2018	Percentage to the Total Population of India	Density of Populations sq km
22.	Tripura	3,673,917	4,057,847	10.45	350
23.	Meghalaya	2,966,889	3,276,323	10.43	132
24.	Manipur	2,855,794	3,008,546	5.35	128
25.	Nagaland	1,978,502	2,189,297	10.65	119
26.	Goa	1,458,548	1,542,750	5.77	394
27.	Arunachal Pradesh	1,383,727	1,528,296	10.45	17
28.	Pudducherry	1,247,953	1,375,592	10.23	2598
29.	Chandigarh	1,055,450	1,126,705	6.75	9258
30.	Mizoram	1,097,206	1,205,974	9.91	52
31.	Sikkim	610,577	671,720	10.01	86
32.	Andaman and Nicobar Islands	380,581	419,978	10.35	46
33.	Dadar and Nagar-Haveli	343,709	378,979	10.26	700
34.	Daman and Diu	243,247	220,084	−9.52	2191
35.	Lakshadweep	64,473	71,218	10.46	2149
	All India	**1,210,854,977**	**1,335,140,907**	**10.26**	**382**

Source: Census of India, 2011, Primary Census Abstract, Total Population: Table A-5
**** Inclusive of Ladakh as per cencus 2011.***

Density of Population

The density of population shows the land-man ratio. It is calculated on the basis of total population of a country divided by total area of the country. The temporal change in the density of population of India between 1901

and 2001 has been shown in **(Fig. 13.21)**, while a comparative chart of density of population has been plotted in **(Fig. 13.24) (Table 13.9)**.

Table 13.9

India—Density of Population

Year	Density of Population (Persons/km2)	Country's Share of World Population (in Percent)
1955	138	14.76
1960	151	14.82
1970	186	19.8
1980	234	15.63
1990	293	16.32
2000	354	17.14
2010	414	17.69
2019*	460	17.74

Source: Census 2011,Government of India. *https://www.worldometers.info/

It may be seen from **(Table 13.9)** that the average density of population of India has a population density of 460 people per sq km which ranks 31st in the world. The highest density among the major states was recorded in Bihar, being 1106 persons per sq km followed by West Bengal 1028 and Kerala 860 persons per sq km. The lowest density was found in the state of Arunachal Pradesh (17 persons per sq km). Mizoram (52) and Sikkim (86) also had less than 100 persons per sq km (2011).

The states of Gujarat, Andhra Pradesh, Telangana, Karnataka, and Maharashtra have lower density than the states of the plains of North India (Bihar, Haryana, Punjab, West Bengal and Uttar Pradesh).

The areas of low density are Arunachal Pradesh, Mizoram, Sikkim, J&K, Meghalaya, Manipur, Himachal Pradesh, Nagaland, Chhattisgarh, Uttarakhand, Rajasthan, Madhya Pradesh, Assam and the Union Territory

of Andaman and Nicobar Islands. The low density areas are characterised by hilly and mountainous topography, hot and cold deserts, marshy lands, and areas of hot and humid climate like Andaman and Nicobar Islands. In general, the density of population has increased in all the States and Union Territories between 2001 and 2019.

Agricultural or Physiological Density of Population

The agricultural density reveals the pressure of population on agricultural land. It is calculated by dividing the total population by the total agricultural land. The agricultural density of India was 670 persons per sq km as against the arithmetic density of 324 persons per sq km (2001) which rose to 791 persons per sq km in 2011. In other words, the agricultural density of India is more than twice to that of the average density (382) of the country. The highest agricultural density was found in Kerala, followed by Manipur and Goa, while the lowest agricultural density is found in Rajasthan being only 275 persons per sq km of agricultural land in 2001, and 295 pp sq km in 2011.

Some Facts about Population Size and Density

In the case of India, there is no direct relationship between the total population and the density of population. For example, Uttar Pradesh, the most populous state of India, ranks fourth in density of population among the major states of the country. The density of population of Uttar Pradesh is, however, higher to that of China, India, the USA, Indonesia, and Brazil.

Similarly Bihar is the second most populous state with a population of 104 million followed by Maharashtra (96.87 million) and West Bengal (91.3 million)–(Table 13.9).

The state of Kerala which claimed the highest density of population till 1981 lost ground to Bihar and West Bengal in 2011. In 2011 Bihar recorded the highest density (1106 pp sq km) and pushed West Bengal to the second rank with a density of 1029 persons per sq km.

Dibang valley of Arunachal Pradesh has the lowest density (1 Person/sq km) followed by Mizoram (52) and Sikkim (86).

Andaman and Nicobar Islands is the least densely populated Union Territory, while Lakshadweep has the lowest population (64,429 persons) but a very high (2013) density of population. North East Delhi has the highest density (37346 person/sq km).

Sex Ratio

Sex ratio is defined as the number of females per thousand of males. Of all the demographic attributes of population, sex ratio is the most fundamental as it determines the reproductive potential, marital status, work-force, migration pattern, growth of population, and the socio-economic relationship. In fact, it is an important demographic indicator to measure the extent of prevailing equality between males and females in a society at a given point of time. Sex ratio in India is defined as "*number of females per 1000 males in the population*".

$$\text{Sex-Ratio} = \frac{\text{Number of females}}{\text{Number of males}} \times 1000$$

The sex ratio in the country had always remained unfavourable to females. The relative picture of sex ratio in some of the selected countries has been shown in **(Fig. 13.25)**.

The sex ratio in India is 107.48 i.e 107.48 males/100 females in 2019. India has 930 females/1000
males. India is at 191th rank out of 201 countries in terms of female to male ratio. Among Asian countries, India is at 43th position out of 51.

The sex ratio of the country as per Census 2011 has been plotted in **(Fig. 13.26) and**
(Fig. 13.27), while the temporal change in the sex ratio during the last century has been given in **(Table 13.10)**.

It may be observed from **(Table 13.10)** that the sex ratio was 972 at the beginning of the 20th century (1901) and thereafter showed a continuous decline until 1941. The post Independence period, however, recorded a sharp declining trend in the sex ratio, the lowest being in 1991 when it was only 927 females per 1000 of males. According to 2001 census, sex-ratio in India was 933 females to 1000 males which increased to 943 in 2011. In India, Kerala is the only state where the number of females is more than that of the males (1084/1000). Haryana with 877 is at the bottom. Three major states (J & K, Bihar and Gujarat) have shown decline in sex ratio as compared to census 2001. Mahi distrist of Puducherry has highest sex ratio of 1176 females/1000 males

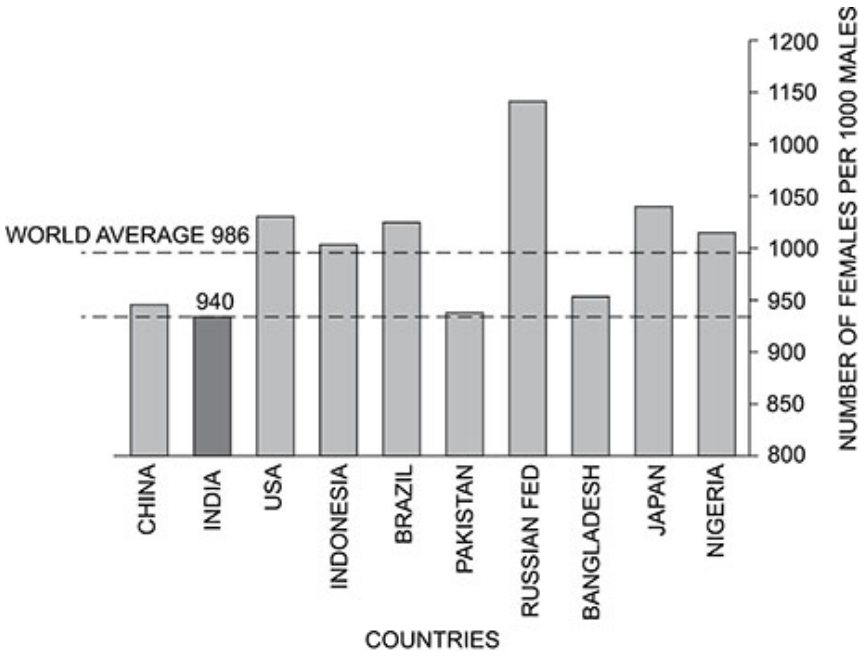


Fig. 13.25 Sex Ratio in India and Selected Countries 2011

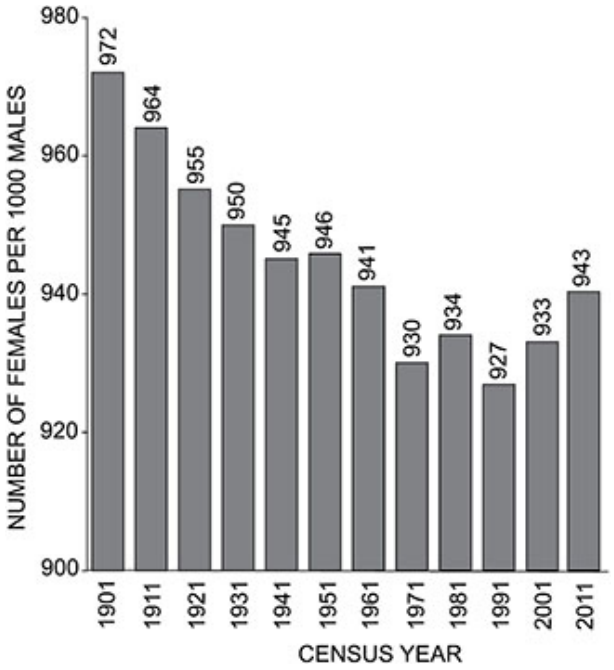


Fig. 13.26 India Sex Ratio (1901–2011)

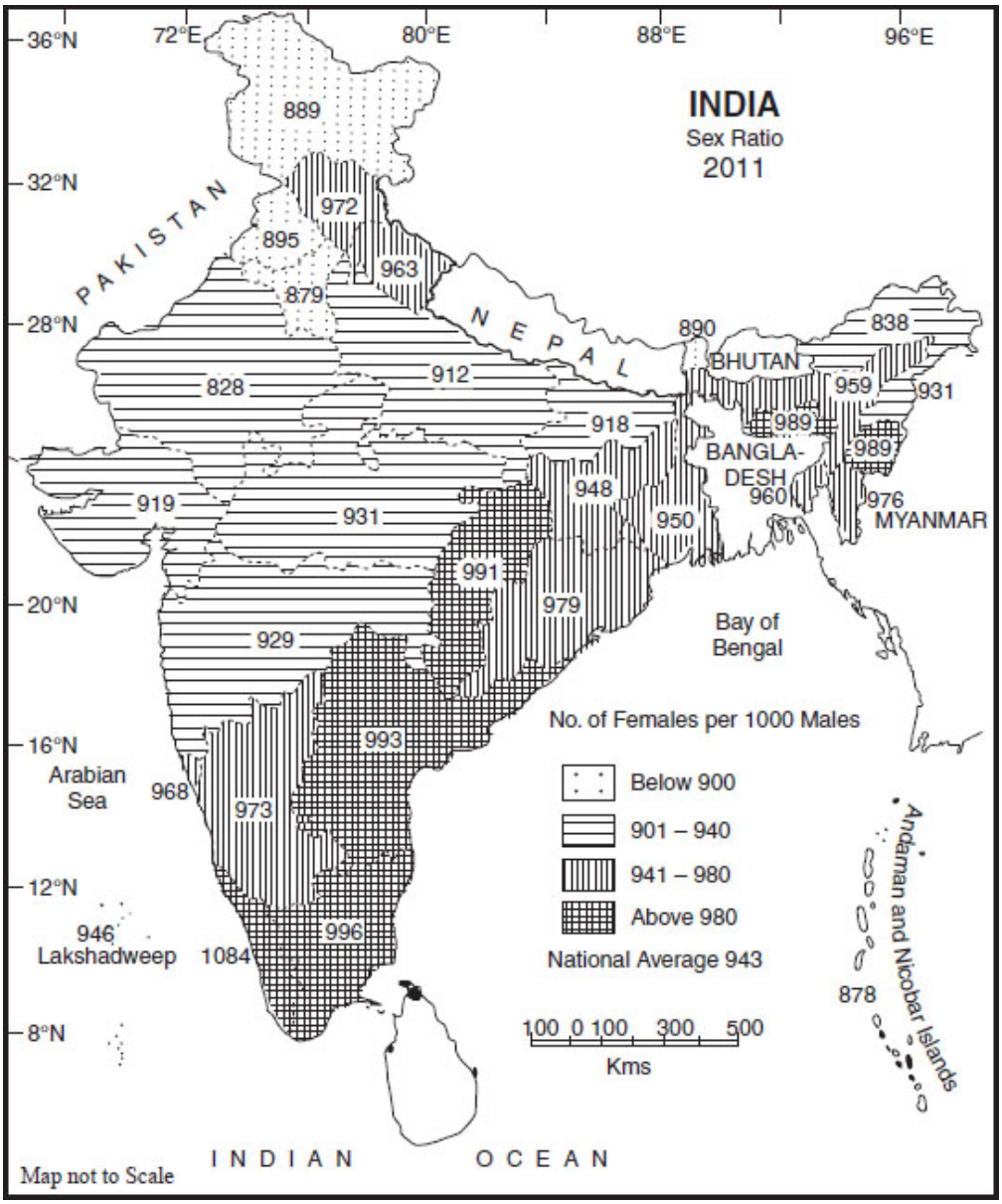


Fig. 13.27 Sex-Ratio (2011)

Table 13.10

India: Sex Ratio (1901–2011)

Census Year	Sex Ratio (Females per 1000 of Males)
1901	972
1911	964
1921	955
1931	950

Census Year	Sex Ratio (Females per 1000 of Males)
1941	945
1951	946
1961	941
1971	930
1981	934
1991	927
2001	933
2011	943

Source: Census of India, 2011.

The child sex ratio at India level (918) is lowest since independence. Increasing trend in the child sex ratio (0–6) seen in Punjab (846), Haryana (834), Himachal Pradesh (909) Gujarat, Tamil Nadu, Mizoram and Andaman and Nicobar Islands. The daughters are alarmingly disappearing from the villages of Gujarat state. The census of 2011 revealed that 20% of the 18,618 villages in Gujarat have a child sex-ratio of less than 800/1000. Thus the problem of low sex ratio has gone from being urban malaise to a rural one as well. Some of the elers are however, trying 'Beti-Bachao' through the awareness movement (Times of India 3.2.14, p. 12). When infant and child mortality are driven by biology, fewer girls die than boys, but the third National Family Health Survey (NFHS-3) found that the postneonatal mortality rate Indian girls is 21/1000, compared with 15/1000 for boys. The country cannot be secure when half of its citizens live in deep insecurity.In all the remaining 27 states/UTs the child sex ratio in 2011 shows decline over census 2001. The socio-cultural factors and the pre-birth sex determination seem to be responsible for the low sex ratio in the country. Interestingly enough, the higher sex ratio is found in the scheduled tribes, Christian and Muslim dominated districts of the country. The state-wise sex ratio has been given in **(Table 13.11).**

Table 13.11

India: State-wise Sex Ratio (2001–11)

States/Union Territories	Sex Ratio-2001	2011
States		
Andhra Pradesh	978	993
Arunachal Pradesh	901	838

States/Union Territories	Sex Ratio-2001	2011
Assam	932	958
Bihar	932	918
Chhattisgarh	990	991
Goa	960	973
Gujarat	921	919
Haryana	861 (Lowest)	879
Himachal Pradesh	970	972
Jammu and Kashmir*	900	889
Jharkhand	941	948
Karnataka	964	973
Kerala	1058 (Highest)	1084
Madhya Pradesh	920	931
Maharashtra	922	929
Manipur	978	985
Meghalaya	975	989
Mizoram	938	976
Nagaland	909	931
Odisha	972	979
Punjab	874	895
Rajasthan	922	828
Sikkim	875	890
Tamil Nadu	986	996
Tripura	950	960
Uttarakhand	964	963
Uttar Pradesh	898	912
West Bengal	934	950
Union Territories		
Andaman and Nicobar Islands	846	878
Chandigarh	773	818
Dadar & Nagar Haveli	811	774
Daman and Diu	709	618 Lowest
Delhi	821	868
Lakshadweep	947	946
Puducherry	1001	1037
India	**933**	**943**

Source: Census of India, 2011, Primary Census Abstract, Total Population: Table A5.
**** Inclusive of Ladakh as per cencus 2011.***

It may be seen from the **(Table 13.11)** that there are great variations in the sex ratio at the state level. Kerala with a sex ratio of 1084 females per thousand of males is the only major state which has excess of females. The high sex ratio is an indicator of less discrimination against the females and low rate of female infanticides. As opposed to this, the sex ratio in Haryana and Punjab is only 879 and 895, respectively, which may be because of the socio-cultural factors. In the opinion of some of the demographers, the pre-birth determination of sex is mainly responsible for the significantly low sex ratio in these two relatively more developed states (Haryana and Pubjab) of the country.

The child sex ratio has continuously declined from 976 in 1961 to 918 in 2011. It is a cause for concern to our leaders, planners and the government. In fact, it is indicative of female foeticide.

Among the Union Territories, Puducherry has the highest sex ratio (1037) while Daman and Diu has the lowest (618) and Chandigarh 818 females per one thousand of males (2011).

The percentage of male and female child population have been shown in **(Figs. 13.28** and **13.29)**. It may be seen from these Figures that the population in the younger age group is significantly low in Haryana, Punjab and Kerala and much higher in the states of Meghalaya, Rajasthan, Bihar and Uttar Pradesh. At the district level Lahul & Spiti (H.P.) and Tawang (Arunachal Pradesh) have 1013 and 1005, respectively, which is the highest child sex ratio, while Jhajhar (774) and Mahendargarh (778) both from Haryana have the lowest child sex ratio (2011).

The average picture of sex ratio is encouraging as it has risen to 943 in 2011 from 933 in 2001. But the sex ratio in the younger age group is declining at an alarming rate. Unfortunately, in some of the districts in the age group 0–6 years of age is around 750 girls per 1000 of boys. The ten bottom and top districts in respect of sex ratio in the younger age group (below 6 years) in 2001 have been plotted in **(Fig. 13.30) and (Fig. 13.31) (Table 13.12).**

It may also be observed from the **(Table 13.10)** that the sex ratio (female to male) has been declining almost steadily in India from 1901 to 1991, except for a small rise in the 1981 Census. This decline has taken

place in all the states, excepting Kerala, where the number of females exceeded that of males all along. It is encouraging that the sex ratio has improved from 927 (1991) to 933 in 2001, and 943 in 2011.

The child sex ratio has continuously declined from 976 in 1961 to 918 in 2011. It is a cause of concern to our leaders, planners and the government. In fact it is indicative of female foeticide.

An examination of **(Table 13.12)** vividly shows that the sex ratio in the children age group (0-6 years) is alarmingly low in some of the districts of Punjab and Haryana. For example, the child sex-ratio in Fatehgarh Sahib, Kurukshetra, Ambala and Sonipat (Haryana) is between 754/1000 to 783/1000. Almost, the sex-ratio is similar in the districts of Patiala, Gurdaspur, Kapurthala, Bhathinda, and Amritsar of the Punjab state. Interestingly enough, both the states of Punjab and Haryana are agriculturally advanced and industrially well developed. The significantly low sex ratio in these states may be attributed to the socio-cultural traditions and the pre-birth sex determination facility. The highly distorted sex ratio in the younger age groups is likely to affect the work force in such areas adversely and the socio-economic implications may be serious.

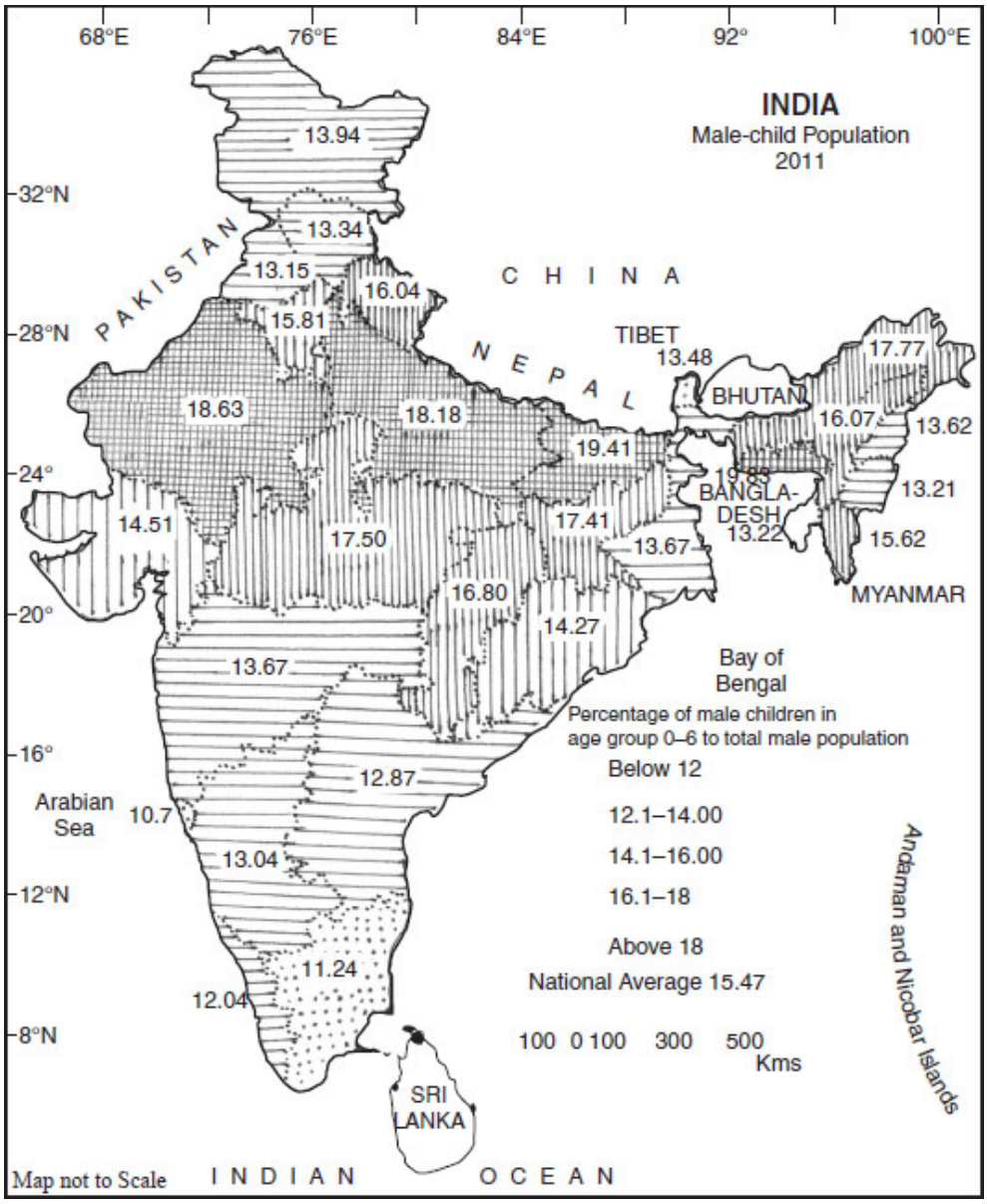


Fig. 13.28 Percentage of Male Child Population in Age Group 0–6 (2011)

Table 13.12

India: Bottom Ten districts in Child Sex Ratio in the Age Group of 0–6 Years (2001)

	District	State	Sex Ratio
1.	Fatehgarh Sahib	Punjab	754
2.	Patiala	Punjab	770
3.	Kurukshetra	Haryana	770

	District	State	Sex Ratio
4.	Gurdaspur	Punjab	775
5.	Kapurthala	Punjab	775
6.	Bhatinda	Punjab	779
7.	Mansa	Punjab	779
8.	Amritsar	Punjab	783
9.	Sonipat	Haryana	783
10.	Ambala	Haryana	784

Source: Census of India, 2011.

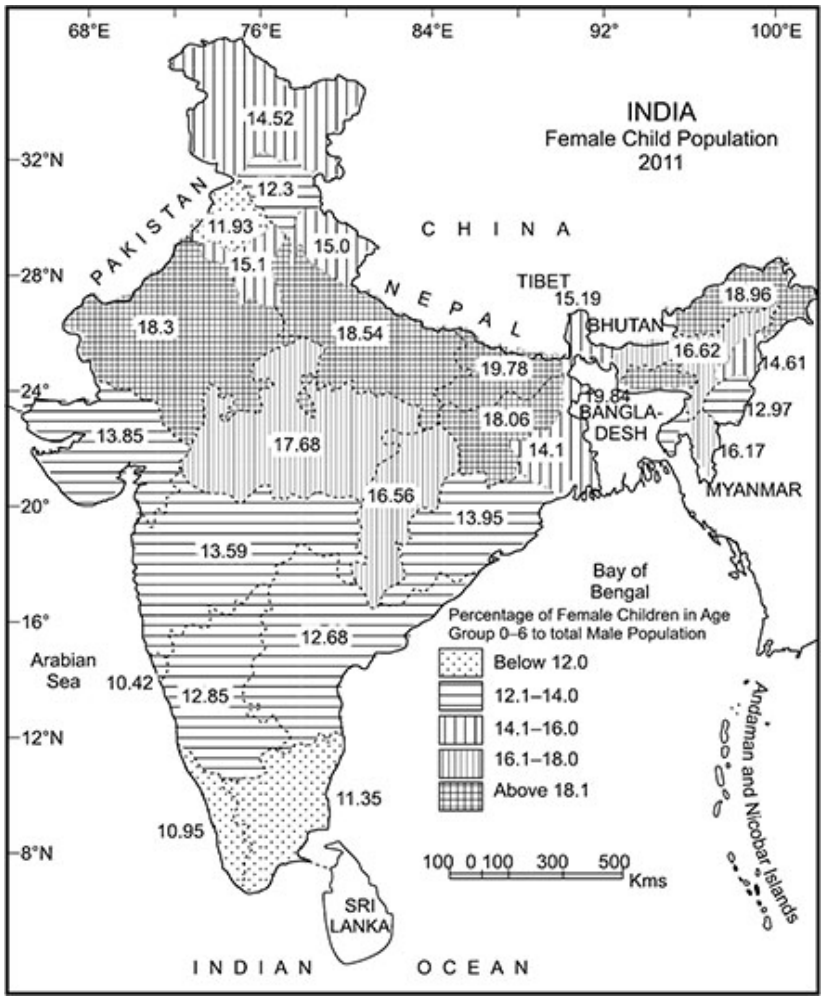


Fig. 13.29 Percentage of Female Child Population in Age Group 0–6 (2011)

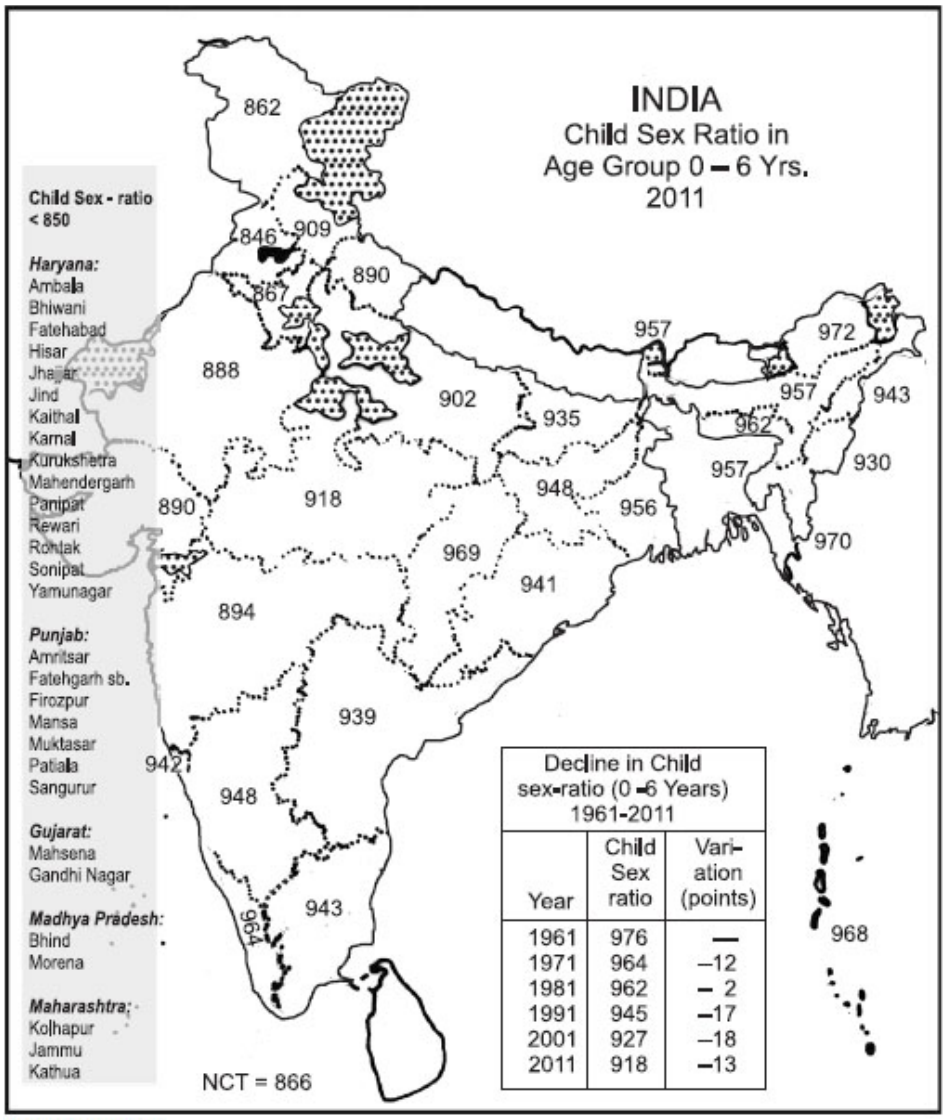


Fig. 13.30 Child Sex Ratio in India-2011

As opposed to this in 2011, the ten districts of the country having the highest sex ratio in the age group between 0–6 have been plotted in **(Fig. 13.31) (Table 13.13)**.

Table 13.13

Top Ten districts in Child Sex Ratio in the Age Group of 0–6 (2011)

	District	State	Sex Ratio
1.	South Sikkim	Sikkim	1036
2.	Upper Siang	Arunachal Pradesh	1018

	District	State	Sex Ratio
3.	Pulwama	Jammu and Kashmir	1017
4.	Bastar	Chhattisgarh	1014
5.	Dantewada	Chhattisgarh	1014
6.	East Kameng	Arunachal Pradesh	1011
7.	Kupwara	Jammu and Kashmir	1010
8.	Senapati	Manipur	1007
9.	Mokokchung	Nagaland	1004
10.	Badgam	Jammu and Kashmir	1003

Source: Census of India, 2011.

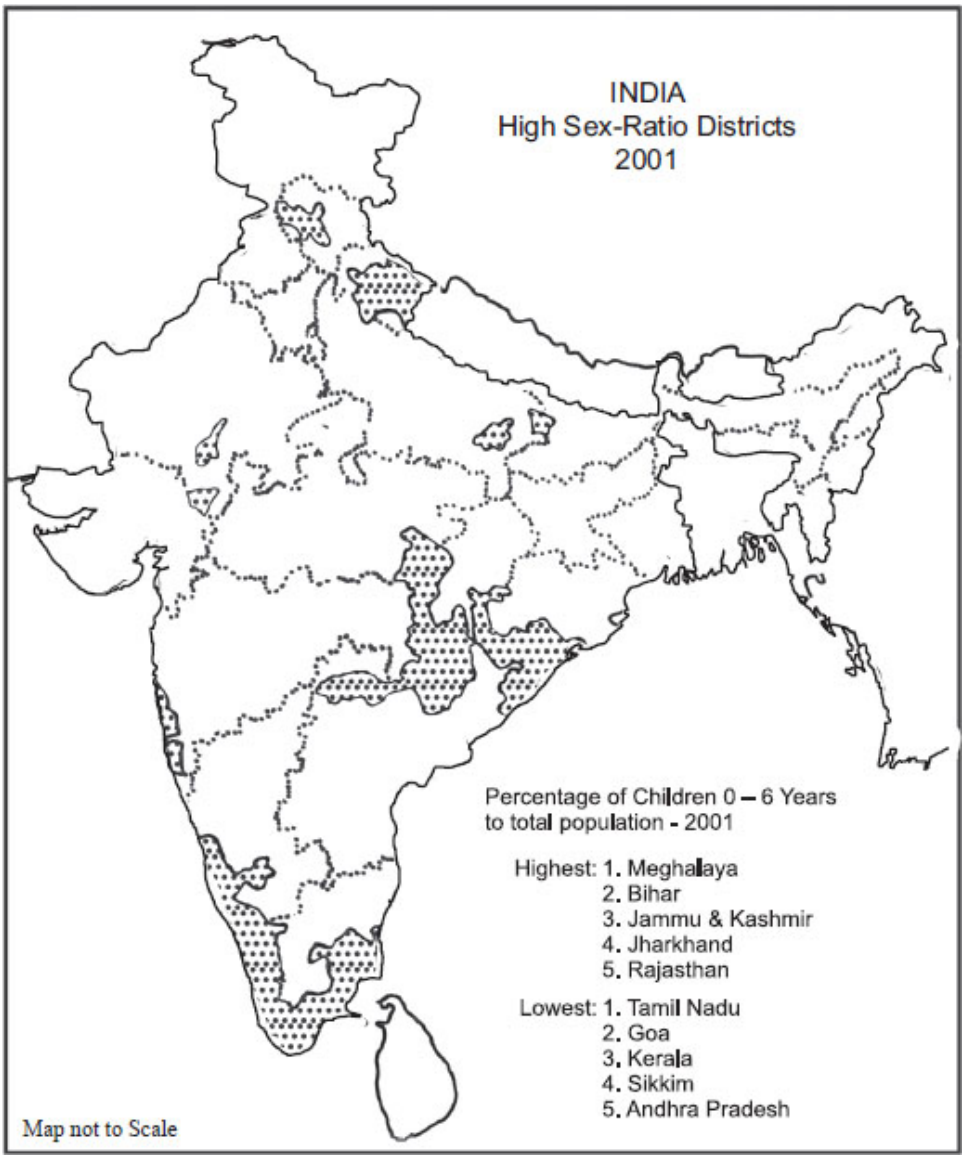


Fig. 13.31 Districts with High Sex-Ratio of 1000 and above (2011)

It may be seen from the **(Table 13.13)** that in 2011, South Sikkim had the highest sex ratio in the age group of 0–6 years (1036), followed by Upper Siang (1018) of Arunachal Pradesh and Pulwama (1017) of the Kashmir Division (J & K). The scheduled tribes dominated districts of Bastar (1014) and Dantewala (1014) of Chhattisgarh have a high sex ratio of more than 1000 females per thousand of males. According to the Census of 2011 Lahul & Spiti (H.P.) and Tawang (Arunachal Pradesh) have a high sex ratio in the child age group (0–6), being 1013 and 1005, respectively.

LITERACY RATE

Literacy level and educational attainments are vital indicators of development in a society. In India the literacy rate in 2001 was 64.84% which increased to 74.04% in 2011.There has been increase of 6.88% in literacy of the males and an increase of 11.79% among the females from 2001 to 2011. (*Source: Census of India*). The overall growth has been slow at
9.2% during the decade 2001–11. The concept of literacy, which varies from country to country, generally refers to the minimum level of literacy skills. In fact, it is an important indicator of socio-cultural development and political consciousness. It is the vehicle of socioeconomic transformation as it facilitates the acquisition of specific skills, and occupational competence, and accelerates the process of social change. Thus education is a long term investment-not an expense.

According to the Census of 2011, literacy has been defined as *any person above the age of 7* years, *who can read and write with understanding in any language is treated as literate.* It is not necessary for a person to have received any formal education or passed any minimum educational standard for being treated as literate. The population below six years is not counted as they are considered illiterate irrespective of their ability to read or write. As per provisional population totals of Census 2011, literates constitute 74% of the total population aged 7 and above and illiterates from 27%. Literacy reflects the socio-economic and cultural set-up of a nation, ethnic group or community. The main advantage of

literacy is that it provides relatively more opportunity of employment. The literate and educated persons have better personality and can interact with the superiors and inferiors in a better and more confident way. The rate of literacy is largely determined by a large number of socio-cultural, political, and physical factors.

Spatial Patterns of Literacy

The changing patterns of literacy rate from 1901 to 2011 have been given in **(Table 13.14)**. It may be observed from this Table that the over all literacy rate in 1901 was only 5.35% while the male and female literacy rates were 9.83 and 0.60%, respectively. It is heartening to note that the literacy rate has continuously increased from 18.33% (males: 27.16% and females:8.86%) in 1951 to 73.0% (males: 82.1% and females: 65.46%) in 2011. Thus the literacy rate increased to 73% of the total population aged seven and above and illiterates form 27%. The literacy rate for males and females works out to 82.14% and 65.46% respectively. The most spectacular increase is in the female literacy and the male-female differential has declined from 24.84% in 1991 to 21.61% in 2001. The literacy rate for males and females work out to 82.14% and 65.46%, respectively. The difference in literacy rate between 2001–11 is reduced to 16.6% **(Fig. 13.32)**.

Table 13.14

India—Literacy Rate: 1901–2011

Census Year	Persons	Males	Females
1901	5.35	9.83	0.60
1911	5.92	10.56	1.05
1921	7.16	12.21	1.81
1931	9.50	15.59	2.93
1941 (undivided India)	16.10	24.60	7.30
1951	18.33	27.16	8.86
1961	28.3	40.40	15.35
1971	34.45	45.96	21.79
1981	43.57	56.38	29.76
1991	52.21	64.13	39.29
2001	64.32	75.26	53.67

Census Year	Persons	Males	Females
2011	73.00	82.1	65.46

N.B. Literacy rates for the years 1951, 1961, 1971 relate to population aged 5 years and above and for the years 1981, 1991, and 2001 to the population aged 7 years and above.

Source: Census of India, 2011

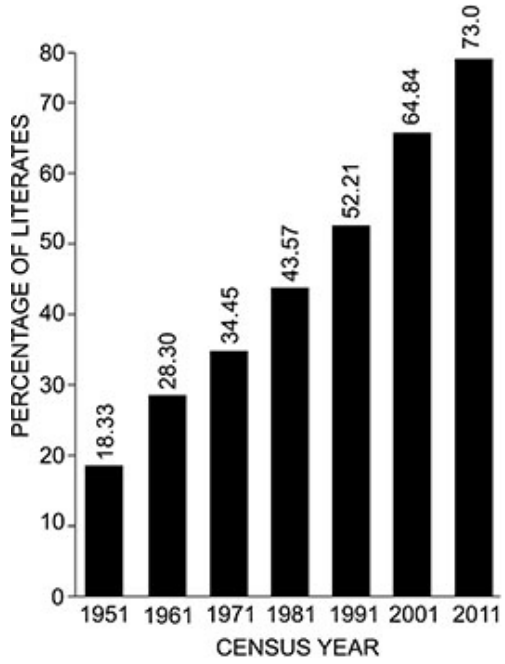


Fig. 13.32 India: Literacy Rate (1951–2011)

During the last 60 years the highest differential (26.62% was in 1981). Incidently, the lowest differential was in 1901 (9.23%) and the lowest literacy rate was also in 1901 (total 5.35% and females:0.6%) when the country had appalling high mass illiteracy (94.65%).

In 2011, the state of Kerala had the highest literacy rate (total 94.0%; males: 96.02% and females: 91.98%), followed by Lakshadweep 91.8%, Mizoram 91.58%, Tripura 87.2%, Goa 88.7%, and Himachal Pradesh 82.8%. The lowest literacy rate is in Bihar (61.8%), followed by Arunachal Pradesh 65.4% **(Fig. 13.33)**. The distribution of male literacy rate has been shown in

(Fig. 13.34). It may be observed from **(Fig. 13.34)** that Kerala with a male literacy rate of over 96.02% ranks first, followed by Mizoram (93.7%) and Goa (92.81%). The states of Bihar (73.39%) and Uttar Pradesh (79.24%)

have significantly low male literacy rates when compared to the national male literacy average of 82.1%.

In female literacy, the leading states are Kerala (91.98%, Mizoram 89.40%, Tripura 83.15%,
Goa 81.84%, Himachal Pradesh 76.60% and Maharashtra 75.48%) **(Fig. 13.35).**

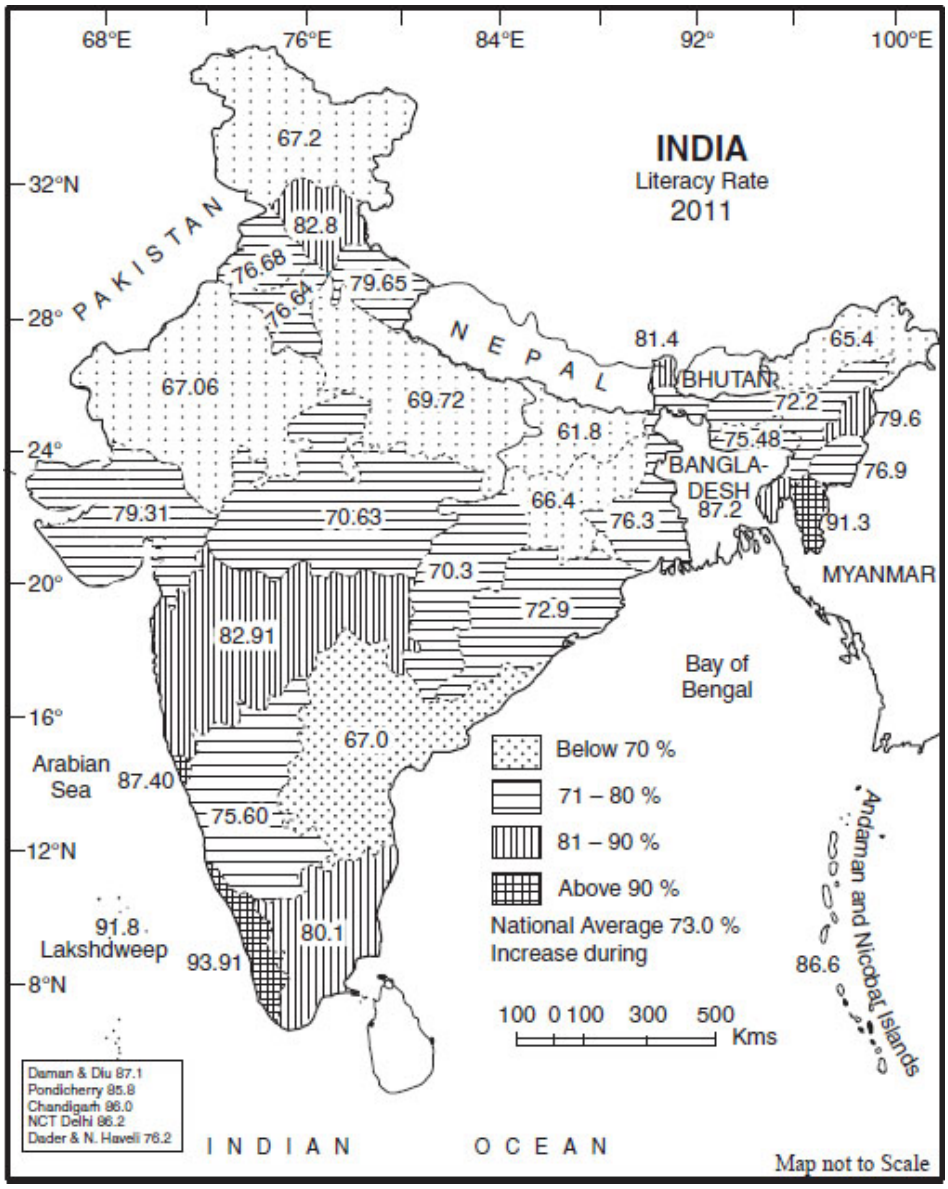


Fig. 13.33 India—Effective Literacy Rate—2011

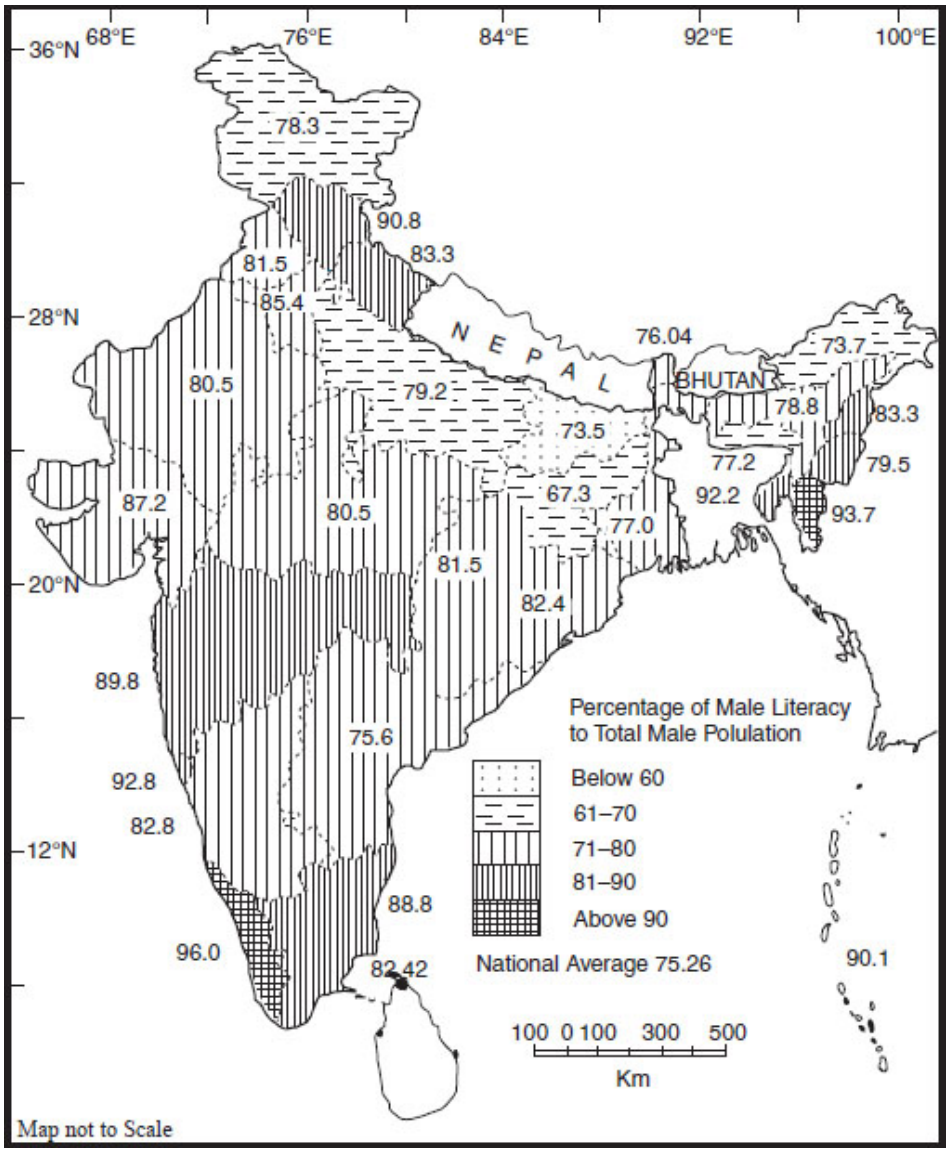


Fig. 13.34 Male Literacy Rate—2011

Among the Union Territories, the highest literacy is in Lakshadweep 91.8% followed by Daman and Diu 87.1%, Puducherry 85.8%, Chandigarh 86.0% and NCT of Delhi 86.2%

(Fig. 13.33).

Bihar has the lowest literacy rate (total: 61.8%; males 73.5%; females 53.33%), Jammu and Kashmir (67.2%), Arunachal Pradesh (65.4%) and Uttar Pradesh (67.7%). Incidently, Uttarakhand has a higher literacy rate of 78.8% than that of Uttar Pradesh (67.7%), Madhya Pradesh (69.3%), Rajasthan (66.1%) and Haryana (75.6%).

The states and Union Territories of Kerala, Lakshadweep, Mizoram, Tripura, Goa, Daman & Diu, Puducherry, Chandigarh, NCT Delhi, and Andaman & Nicobar Islands have achieved literacy rate of above 85%, the target set by the Planning Commission which was achieved by 2011–12.

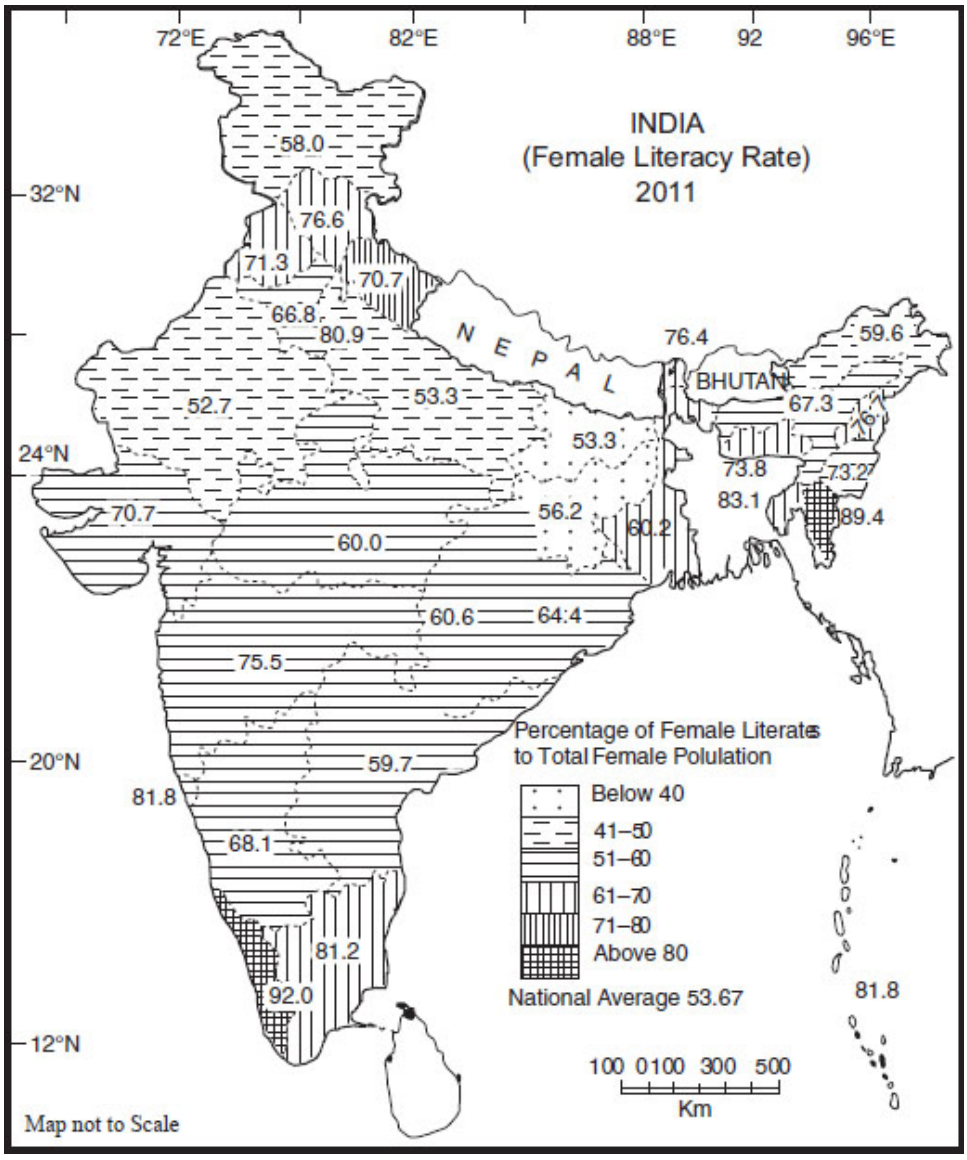


Fig. 13.35 India—Female Literacy Rate—2011

Despite a commondable achievement in the field of literacy during the last century, especially after Independence, India's illiterate population is the largest in the world. The last Global Monitoring Report (GMR) released by the UNESCO on 28 January, 2014, India has the largest population of

illiterate adults (above 7 years of age) with 287 million people. This is 37% of the global total. While India's literacy rate rose significantly between 2011 and 2011, the polulation growth cancelled the gains so there was no change in the number of illeterate adults.

Table 13.15

Ranking of States and Union Territories by Literacy Rate 2011

Rank	Person	
	India/State/Union Territory	Literacy Rate
1	Kerala	94.0
2	Lakshadweep#	91.8
3	Mizoram	91.3
4	Tripura	87.2
5	Goa	88.7
6	Daman & Diu#	87.1
7	Puducherry#	85.8
8	Chandigarh#	86.0
9	NCT of Delhi#	86.2
10	Andaman & Nicobar Islands#	86.6
11	Himachal Pradesh	82.8
12	Maharashtra	82.3
13	Sikkim	81.4
14	Tamil Nadu	80.1
15	Nagaland	79.6
16	Manipur	76.9
17	Uttarakhand	78.8
18	Gujarat	78.0
19	Dadra & Nagar Haveli#	76.2
20	West Bengal	76.3
21	Punjab	75.8
22	Haryana	75.6
23	Karnataka	75.4
24	Meghalaya	74.4
25	Odisha	72.9

Rank	Person	
	India/State/Union Territory	Literacy Rate
26	Assam	72.2
27	Chhattisgarh	70.3
28	Madhya Pradesh	69.3
29	Uttar Pradesh	67.7
*30	Jammu & Kashmir	67.2
31	Andhra Pradesh	67.0
32	Jharkhand	66.4
33	Rajasthan	66.1
34	Arunachal Pradesh	65.4
35	Bihar	61.8

Source: Census of India, 2011.

**** Inclusive of Ladakh as per census 2011 #U.T.***

AGE COMPOSITION

Age composition is also an important characteristic of population. It helps in ascertaining the longevity, work-force, and the dependent population (younger age group and the senior citizens).

Table 13.16 vividly shows that between 37 and 42% was the younger age group in the last century, while the work force varied between 52 and 57%. The highest dependency ratio of the younger age group was in 1971 when it rose to 42%. There is a steady increase in the senior citizens age (Geriatrics) group which was 5.1% in 1901 and rose to about 7% in 2001, and became 8.0% of the total population in 2011.

Table 13.16

India—Age Composition (1901–2011)

Years	Age Group (Younger Age Group) (0–14 Years)	Age Group (Workforce) (15–60 Years)	Age Group (Senior Citizens) (Above 60 Years)
1901	38.1	56.8	5.1
1911	37.8	56.9	5.2

Years	Age Group (Younger Age Group) (0–14 Years)	Age Group (Workforce) (15–60 Years)	Age Group (Senior Citizens) (Above 60 Years)
1921	38.60	56.0	5.4
1931	38.5	56.4	5.1
1941	39.1	55.2	5.7
1951	37.5	56.9	5.6
1961	37.5	56.9	5.6
1971	42.0	52.0	6.0
1981	39.7	54.1	6.2
1991	37.8	55.5	6.7
2001	37.3	55.8	6.9
2011	29.5	62.5	8.0

Source: Census of India, 2011.

Size of Families

According to the census authorities of India, the average size of families in India in 4.8 people per household. Uttar Pradesh has the most crowded households with an average of six people. Tamil Nadu's average family size is now the lowest in the country with 3.9 members, followed by Kerala. After U.P. is J & K that has 5.8 people in a household. In most of the states and Union Territories, the number of people living in a rural household is more compared to the urban counterparts. In the opinion of experts, smaller households are better for girl child, while larger households leads to crowding and a fight for resources. With smaller households, the standard of living increases and more amenities are available for the individual (The Times of India, June 14, 2014, p. 17).

Occupational Strucure

In 2009–10 about 50% of the population was dependent on primary sector, about 20% on secondary sector and 30% on tertiary sector. In

2011, the work force was 62.5% (between 15 and 59 years). The dependent population was 37.5% (29.5% in 0–14 years age group, and 8.0% in the 60 + age group).

Rural-Urban Population

Ratio of urban population to the total population of a country is an index of the level of industrialization of that country. As industries gather momentum in a country, ratio of urban population goes on rising. India is an agricultural country, so ratio of urban population is less than the rural population. Ratio of urban-rural population in different census years is shown in the following table.

It may be seen from the **(Table 13.16A)** that in 1901 the urban population of India was 10.8% which rose to 31.16% in 2011. In 2011 about 69% of the total population was in the village (rural) and 31% in the urban areas.

Table 13.16A

India-Rural-Urban Population

Year	Ratio of Rural Population (Percent)	Ratio of Urban Population (Percent)
1901	89.2	10.8
1951	82.7	17.3
1961	82.0	18.0
1971	80.1	19.9
1981	76.7	23.3
1991	74.3	25.7
2001	72.2	27.8
2011	68.84	31.16

Source: Census of India—Provisional Tables—2011

MIGRATION

Migration, together with fertility and mortality, is a fundamental element in determining the population growth and structure in a country or a region. The permanent or semi-permanent change of residence of an individual or group of people is known as migration. Migration implies spatial mobility of population embracing all sorts of territorial movements—intra-regional, inter-state, and international. It causes redistribution of population and is of great importance in the understanding of changing man-land (resources) relationships. In fact, migration is a universal characteristic of all human groups irrespective of their socio-economic development.

Migration has both temporal and spatial dimensions. In terms of time framework, it may be permanent, semi-permanent, temporary, or seasonal. In spatial context, it may be international or intra-national (internal migration within a country: region to region; urban to urban; rural to urban; and urban to rural).

In the intra-national migration, the migration numbers and rates are affected by the size, shape, and internal population distribution, stimulated by *push and pull* factors. In international migration, the extent of national government intervention and control have an upper hand. Another important difference lies in 'the greater change in socio-cultural environment involved in moving from one country to another, with all its implications, spatial as well as social, for migrant adaptation and assimilation'. Sometimes it is a big exodus due to compelling reasons as happened at the time of independence when nearly 12 million Hindus, Muslims, and Sikhs 'swept up in tumultuous shuffle of citizens between India and Pakistan after the partition of the sub-continent in 1947.'

International Migration

International migration is quite old, and limited in volume and time. It is caused because of a number of special conditions at the place of origin and place of destination. International migration may be short migration of small groups of people or large groups travelling long distances. Contrary to this, the trans-oceanic migration has been observed largely in modern times, affecting the chief centres of population in Europe, America, South-East Asia, South-West Asia, China, Japan, and India.

Migration during the prehistoric times was mainly due to physical factors like harsh environment, inclement climate, floods, droughts, earthquakes, volcanoes, cyclones, transgression of sea, etc.

On the basis of *volume* (*number*), *distance*, and *means of transportation*, *international* migration may be classified into: (i) *intercontinental*, and (ii) *transoceanic*.

International migration is as old as the appearance of man on the surface of the earth. In fact, people have always been on the move. It equally applies to the primitive hunting communities, the agricultural communities of the Neolithic Age, or nomadism or transhumance, or modern man. As population grew, pressure on resources increased and cultural differences widened, people began to move out for better opportunities and greener pastures. But the most important large-scale international migration occurred from 17^{th} to 19^{th} centuries when the Europeans moved to the New World (North and South America), Africa and Asia for colonisation. They also carried indentured labour from their Asian and Negro-slaves (11–12 million) from Africa to the USA.

It has been rightly observed that "with the abolition of slavery during the 19^{th} century, the British and Dutch colonists substituted a semi-slave trade in indentured labour from the huge population and poverty reservoirs of Asia, especially China and India. There were appreciable flows to Guyana, Trinidad, Malaysia, Sri Lanka, Fiji, Natal, East Africa, and Mauritius." Today there are more than 40 lakhs Indians abroad (Myanmar —10 lakhs, Sri Lanka—7.5 lakhs; Malaysia—6 lakhs, Caribbean countries —4.5 lakhs; South Africa—3 lakhs, Mauritius—2.6 lakhs; East Africa—1.30 lakhs; Fiji—1.26 lakhs; etc.). Since Independence a large number of Indians have migrated to Australia, Canada, France, Germany, New Zealand, Saudi Arabia, United Arab Emirates, Scandinavian countries, UK and USA. Similarly, the Chinese are settled in Australia, Canada, Indonesia, Korea, Malaysia, Myanmar, New Zealand, Philippines, Singapore, and Thailand.

The European migration during the 17^{th}, 18^{th}, and 19^{th} centuries occurred mainly because of over-population and industrialisation. The USA has been the cherished destination of the European immigrants during the 19^{th} century. Among the largest contingent was the one from Britain, followed by Germany, Ireland, Scandinavia, Italy, Spain, and Russia. Latin America was the destination of people from the Iberian Peninsula (Spain and Portugal). In fact, over 95% of the total out-migrants were from north-west Europe.

The Europeans emigrated in two different directions:

(i) They entered into the sparsely populated tropical and sub-tropical coastslands which were easily accessible and which possessed potential for the production of exotic crops because of their hot and humid climates. Consequently, commercial production of rubber, sugarcane, cotton, tobacco, coffee, tea, coco, a banana, spices, indigo, and rice, and so on flourished in the eastern coastlands of North and South America. In order to cultivate these virgin lands, labour had to be imported from the African and Asian countries. It gave rise to slave trade from Africa. After the slavery was abolished, the densely populated countries of Asia were exploited by the British and Dutch colonists and they supplied labour to newly developed plantation agricultural areas of East African (Ethiopia, Kenya, Tanzania, etc.), Fiji, Malaysia, Mauritius, Sri Lanka, Java, and Sumatra.

(ii) Secondly, large number of European emigrants moved into the temperate zones of the Americas, South Africa, Australia and New Zealand.

Although people emigrated from most of the countries of Western Europe, the out-migration from Ireland was almost one of the unparallel in the human history. The great exodus was due to extraordinary increase in the population of Ireland. The country was over-populated, the agricultural density was very high and the people were largely dependent on potato cultivation. The failure of potato crops for the successive three years (1845 to 1848) and the higher prices of their meager food meant famine for the whole country. Between 1850 and 1900, more than 4 million people left Ireland. The check on the movement which began in 1885 was due to the depopulation of the country to a dizzy fall in birth rate. The statistics of migration during the 18^{th} and 19^{th} centuries have been given in **(Table 13.17) and (Table 13.18).**

Table 13.17

International Migration (1945–52)

Continent	Number
Emigration from Europe	4,452,000
Immigration into Europe	1,150,000

Other International Migration	710,000
Total	**6,312,000**

Source: Spencer, J.E., et.al. Cultural Geography.

Table 13.18

Migration from Europe (1945–52)

Emigration	Number	Immigration	Number
Great Britain	1,107,000	USA	1,104,000 (27%)
Italy	741,000	Argentina, Brazil & Venezuela	883,000 (21%)
Netherlands	318,000		
Spain	272,000	Canada	726,000 (21%)
Portugal	152,000	Australia	697,000 (17%)
		Israel	526,000 (13%)
		South Africa	125,000 (3%)
		New Zealand	75,000 (2%)

Source: Spencer, J.E. et.al. Cultural Geography.

It may be seen from **(Table 13.17)** that out of the 6.3 million migrants over 4.45 million were the emigrants from Europe and the remaining, about 0.7 million, from the rest of the world. The number of immigrants in Europe during the same period was only 1.15 million .

The migration from the different countries of Europe between 1945 and 1952 has been given in the **(Table 13.18).** It may be noted from this Table that most of the out-migrants from European countries left for the USA (27%), Canada (21%) and South American countries (21%). About 17% of the total emigrants left for Australia and 13% for Israel.

Recent International Migration

International migration which includes both voluntary migration for economic or other reasons as well as the involuntary movement of refugees is on the rise, especially after the Second World War. Data on the recent international migration are uncertain and trends are difficult to

track, but according to the UN at least about 210 million people (excluding refugees) lived or worked outside of their country in 2015. The annual growth rate of immigration has been steepest in the developing countries and about half of all international migration takes place within the developing countries.

So far as the involuntary migration (refugees) is concerned, it was about 40 million in 2005. The refugees problem became quite alarming after the American-Afghan War and the invasion of Iraq by the American forces. The number of refugees is overshadowed by the increase in the number of internally displaced persons—those who have been forced to flee their homes by armed conflicts, persecution, or natural or man-made disaster, but who remained within their national borders. Because of the rising number of civil wars, and local conflicts, the total number of internally displaced persons now adds upto about 45 million worldwide. Africa (Somalia, Sudan), Albania, Macedonia, and South-West Asia (Afghanistan, Iran, Iraq, Syria, Lebanon, Turkey, etc.) are the worst affected regions, with upto 25 million people having been internally displaced.

During the recent decades, the major destinations for the permanent settlers have been the USA, Canada, Australia, and New Zealand. The contributions of the Third World countries to their immigration is nearly 77%.

In recent period, environmental degradation and natural disasters have also forced the people to out-migrate from their places. Population growth, scarcity of agricultural land and cycles of droughts and floods have encouraged the illegal immigration of more than 12 million Bengalis to the neighbouring states of India. The influx has prompted local resentments, and more than 4000 people were killed in a series of violent clashes in the early 1980s in the north-eastern states of India, especially in Assam.

Many governments increasingly view immigration as a problem, despite the fact that immigrant labour often benefits both the home and the host countries. Perception of national identity, cultural differences, and fears of unemployment are all contributing to the actual and potential hostilities between immigrants and nationals. Nonetheless, immigration, either legal or illegal, seems certain to continue.

Indian International Migration

India has the second largest population in the world**.** During the last 65 years the land man ratio has declined substantially because of the population explosion. People during the pre-Independence period were more attached to their birth places and agricultural land. But under the changed socio-economic scenario and great stress on natural resources people have become more mobile and migratory.

The first group of Indian labourers emigrated in 1815. It consisted of convicts and criminals, transported from Kolkata (Calcutta) to Mauritius. In fact, Mauritius became a real training ground for agricultural labourers on the sugarcane-fields. After 1834, many free workers were attracted by the mild and oceanic climate of Mauritius. Today, out of the total population of 12 lakh (1.2 million) persons, there are more than 7 lakh (0.7 million) Indians in Mauritius. A big colony of about 8000 Indians exists in the Reunion Islands situated near Mauritius in the Indian Ocean. Subsequently, people of Indian origin settled in the rubber producing areas of Malaysia and the sugarcane-growing areas of West Indies.

In the continent of Africa, the Indians emigrated during the 18^{th} and 19^{th} centuries. With the help of Arab traders, the Indian merchants established their business in the countries of Africa. The Europeans also encouraged the Indian labourers to work in their fields to establish plantation, to clear forests for agricultural development, and to construct railways. The labourers were followed in large number by the other social elements such as traders, small shop-keepers, and money-lenders. As a result of these emigrations, there developed large Indian colonies in Kenya, Mozambique, Natal, Somalia, South Africa, Tanzania, Uganda, Zambia, and Zimbabwe. Most of these emigrants were from Punjab, Gujarat, Maharashtra, Tamil Nadu, and Kerala.

In the West Indies, the Indians began to reach in 1840. Their number was the largest in British Guyana and Trinidad where there are over two lakh Indians in each of these places. They also settled in Jamaica, Martinique, and Guadeloupe in small numbers. At present, there are nearly three lakh Indians in West Indies who constitute more than half of its total population. These emigrants were mainly from Bihar and Eastern Uttar Pradesh **(Fig. 13.36).**

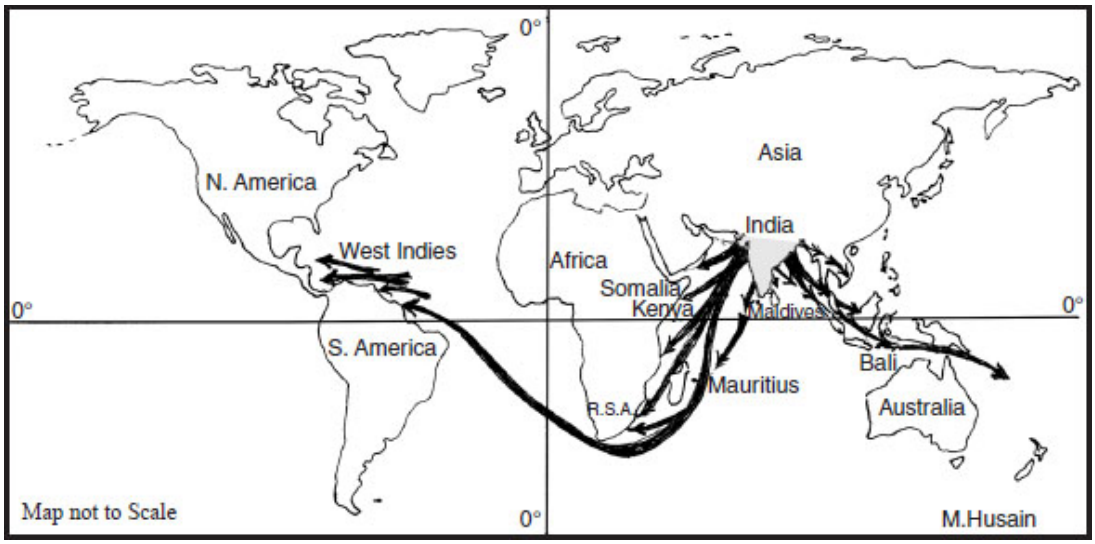


Fig. 13.36 India—Past Migration

By the end of the 19th century, the emigration movement became quite numerous. The tea and rubber plantation estates of Malaysia and Sri Lanka also attracted the Indian workers in large number. In Sri Lanka, there are about four million Indians who constitute about one-third of the total population of Sri Lanka. Most of the immigrants in Sri Lanka and Malaysia are from the state of Tamil Nadu. Moreover, there are numerous Indians who out-migrated from the densely populated parts of Bihar, Odisha, West Bengal, and Uttar Pradesh. A large scale migration occurred at the time of partition of the subcontinent in 1947 **(Fig. 13.37)**.

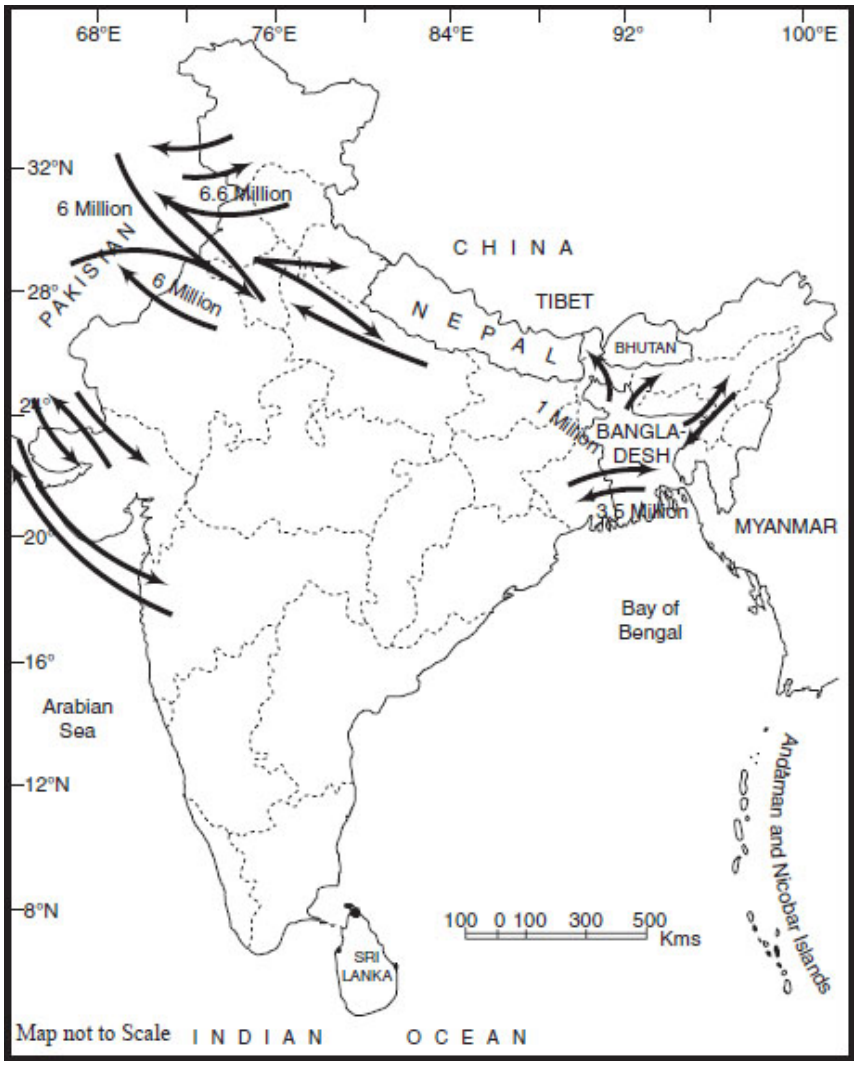


Fig. 13.37 Political Migration in 1947

Internal Migration in India

There is scarcity of reliable data on the internal migration in India. The Census of India provides general information in terms of 'place of birth' as well as 'last residence'. It also provides information for 'streams of migrants' and reasons for internal migration.

Prior to the Second World war, the internal migration in India was free and people voluntarily moved from state to state on a large scale in search of employment. The streams of migrants from Uttar Pradesh moved to east (Calcutta and tea estates of Assam and West Bengal), to west (Delhi and agricultural areas of Punjab) and to south Bombay (Mumbai, Ahmedabad, Surat, Bhopal, Hyderabad, Pune, Kerala, Madras, and

Bangalore). Similarly, streams of migrants also moved from Andhra Pradesh, Bihar, Madhya Pradesh, Odisha, Punjab, Rajasthan, and Uttar Pradesh to other states for better economic opportunities and education. But during 1961–2001, the stream of internal migrants have slowed down mainly due to 'Sons of the Soil' policy of the different states as well as the foreigner's issue (e.g. Assam and Maharashtra). It is estimated that during this period, the percentage of internal migrants has declined from 33% to 30%.

In India, there is a positive relationship between the level of economic development of a district and its intra-state and inter-state migration. In fact, the states with more industrialisation attract more migrants than other states. Immobility is particularly prevalent in the densely populated rural areas of the Ganga Plain, the delta and the coastal plains as well as the sparsely populated hill areas. On the other hand, there has been marked migration from Assam, since about the turn of the century, into West Bengal, where Damodar Coal-fields and Hugli Valley are the areas of great attraction; into Maharashtra state where Greater Bombay is the main destination; into south and south-west Karnataka where plantation and Bangalore draw migrants; and into the territory of Delhi. The productive plains of Punjab and Haryana are also a great attraction for the internal migrants. During the summer season, migrants from the plain areas migrate to the states of Jammu and Kashmir, Himachal Pradesh, Uttarakhand, and Sikkim **(Fig. 13.38).** At present, the rural-to-rural migrations are intra-district as inter-district migrations within the state and inter-state migration have declined. They are mostly for marriage, employment, education, and other socio-religious and cultural reasons. The rural-to-rural migration in search of jobs and education is mainly male selective.

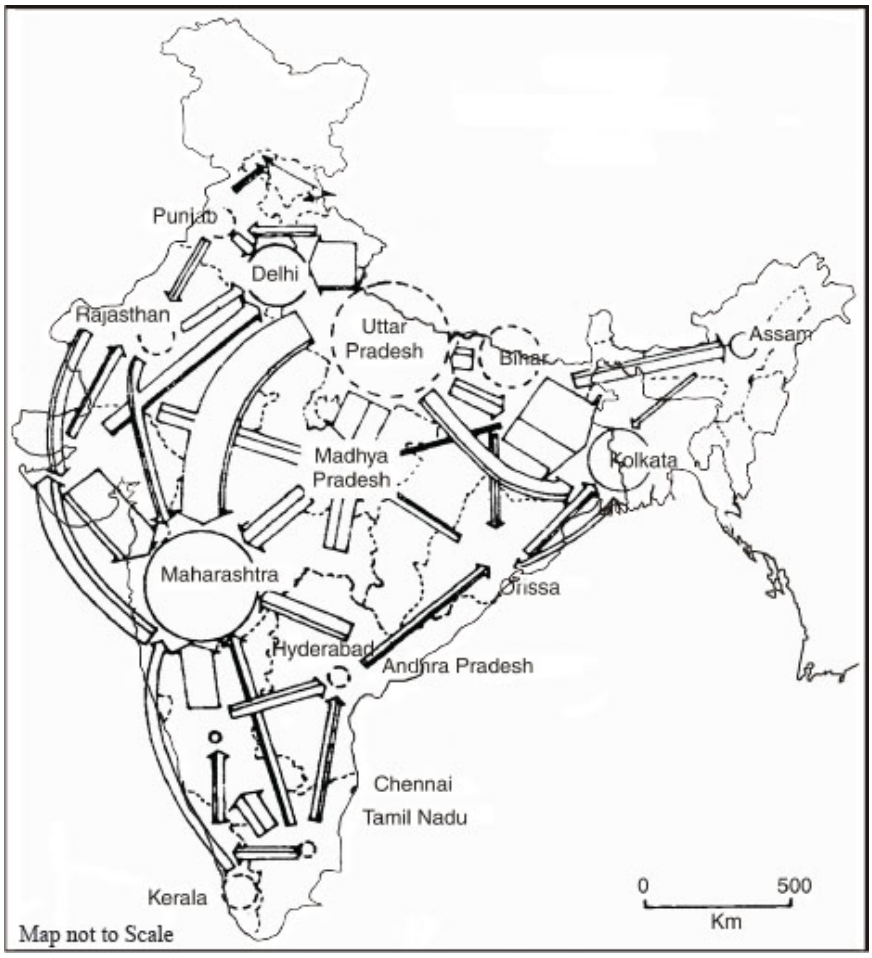


Fig. 13.38 Internal Migration 2011

The rural urban streams are often inter-state and inter-district and have claimed increasingly higher migrants, both permanent and circulatory as well as commuters. They are more prompted by employment opportunities, education and civic amenities. Delhi provides the best example where during the last 40 years, the increasing influx of migrants has posed a serious problem for the city planners. Of the total migrants to Delhi, Uttar Pradesh contributes more than 50%, followed by Rajasthan, Haryana, Punjab, Bihar and Madhya Pradesh. Recent trends indicate slowing down of migration by the neighbouring states and increasing pressure from far off states and the Union Territories. The main reasons are employment, education and marriage. The scheme of the National Capital Region (NCR) has failed to control migration to Delhi. The civic services like electricity, water supply and transport facilities are unable to cope up with the rising tide of migrants. Unfortunately, air, soil, water, and

noise pollution has assumed an alarming proportion, affecting the health and efficiency of the Delhi population adversely.

During the last 40 years, the urban-to-urban (U-U) migration has been rising and migrants have been moving from small towns to large cities and metropolises. But its contribution to the total migration is only 12% as compared contribution of 24% from rural-urban streams. In recent years, the professionally trained, highly educated, and skilled and semi-skilled people have chosen to migrate to Australia, Canada, European countries, Gulf-countires, Japan, New Zealand, U.K., USA, etc. **(Fig. 13.39)**.

The urban-to-tural (U-R) migration is very limited, hardly 5%, and consists of circulatory and semi-permanent migrants basically who return to their homes after retirement.

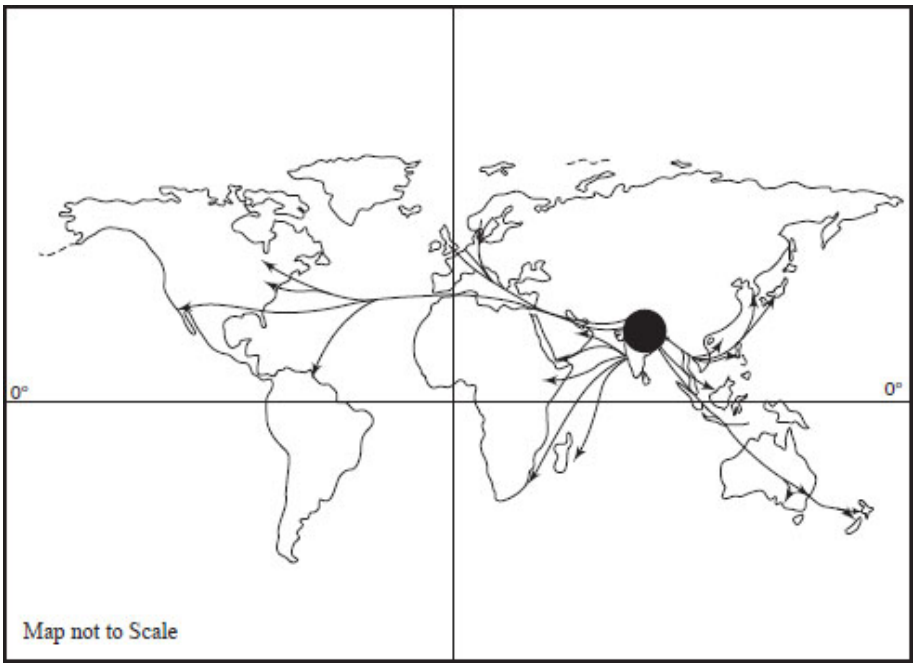


Fig. 13.39 Recent International Migration of Indians

The internal migration in India is still predominantly male-selective where employment is the prime cause of migration. About 41% and 24% of the total male migrants participate in rural-to-urban (R-U) and urban-to-urban (U-U) migration. Women contribute to all migration streams mainly for marriage, followed by employment, and education, but in both intra-state and inter-state rural migration females outnumber males.

At present, the professionally educated and technocrates are migrating either to the developed countries or to the South-West Asia in search of better job opportunities.

Consequences of Migration

Migration affects the population and resources at the place of origin, place of destination, and the migrants also. It is an instrument of cultural diffusion and social integration which exercises three fold impact (i) on the area of in-migration, (ii) on the area of out-migration, and (iii) the migrants themselves.

Some of the important consequence of migration have been given in **(Table 13.19)**:

Table 13.19

Consequences of Migration

	Place of Origin		Place of Destination
1.	The absolute number of population decreases.	1.	The absolute number of population increases.
2.	The density and fertility of the population decreases.	2.	The density and fertility of population increases.
3.	Increase in sex ratio.	3.	Decrease in sex ratio.
4.	Decrease in the proportion of workers.	4.	Increase in the proportion of workforce.
5.	Reduces the population pressure on resources.	5.	Increases the population pressure on resources.
6.	In consequence of out-migration of the work-force, the available resources are likely to be under-utilised.	6.	With the increase in workforce the resources are likely to be better utilised.
7.	The birth rate declines.	7.	The birth rate increases.
8.	As a result of out-migration of the educated youths, the literacy rate declines.	8.	The arrival of the literate youths increases the literacy and education rate.
9.	The families suffer in the absence of the earning youths.	9.	The in-migrants face the problem of adaptation.
10.	The elderly people, sick, females, and children miss their out-migrant members.	10.	The hosts may become aggressive and hostile towards the immigrants.

	Place of Origin		Place of Destination
11.	New values and cultural tradition are adopted.	11.	Enriches the culture.
12.	Leads to national integration and international brotherhood.	12.	Leads to national integration and international brotherhood. This is the greatest benefit of migration.
13.	The new values acquired by the migrants may not be appreciated by the parents and relatives.	13.	The immigrants bring their language, religion and cultural values which lead to integration of cultures. The in-migrants try to preserve their value system which may cause social tension.
14.	The economy flourishes by the transfer of money to the parents and family by the migrants.	14.	With the availability of more and skilled youth labour, the production goes up and the
15.	The crime rate goes up as the youths are away from their families.	15.	Crime against property and famales increases.

Impact of Migration on Migrants

The people who migrate from the rural areas to the urban-industrial areas suffer from the lack of open space, good housing, and fresh air. The highly polluted atmosphere of the place of work (urban centres), the toxic fumes and dust affect the health adversely. They also have to adapt to the new dietary habits and timings of food. There are evidence to prove that the incidence of respiratory diseases, cardiac problems, TB, AIDs and cancer among the in-migrants. Being away from the family, the migrants suffer psychologically. They often miss their family. Thus, the psychological cost is quite high which may lead to nervous breakdown. Sometimes the physical contact of the people belonging to different racial and ethnic groups may become the cause of diseases like AIDS. It has been found that there occurs change in the biological characteristics of the people, because of the amalgamation of blood of the in-migrants and the hosts.

POPULATION PROBLEMS

The growing pressure of population on limited resources, especially on agricultural land, water and forest resources has created many socioeconomic, cultural, political, and environmental problems. The population problems of India vary in space and time. Some of the

important problems of population of India have been briefly presented here.

1. **Rapid Growth of Population**: In India the growth rate of population is significantly high. The birth rate remained high and the death rate has been checked substantially. Consequently, there is rapid growth of population. The rapid growth of population of India may be appreciated by the fact, that the world population grew three times during the last century, while the population of India grew five times. The explosion of population in India during the last fifty years has put enormous pressure on the available natural resources.
2. **Unemployment**: Unemployment is a curse. It symbolizes poverty of an individual, decay of a society and loss of human resource of a country. Unemployment is widespread in the country, and it may become even more grim in the future (Bhagwati Committee Report). At present more than 20% of Indians in the age-group of 15–24 are jobless and seeking work, according to the census of 2011. In absolute terms around 4.7 crore (2.6 crore men and 2.1 crore women) are jobless.

   Over all unemployment rate in the 15–59 working age population was worrisome 14.5% .

   Among Dalits, the unemployment rate in the working age group of 15–59 years was a shocking 18%, much higher than the general population. The educated and even professional technocrats have limited opportunities of jobs. Under such a situation of unemployment, both the educated and uneducated, skilled and unskilled people try to out-migrate to other countries in search of jobs and employment.
3. **Poor Standard of Living and Malnutrition:** The per capita income in India is one of the lowest in the developing countries of Asia. Having a low purchasing power, an average Indian family cannot think of balanced diet. The standard of hygiene and quality of nutrition are low. About 50% of the total population is not getting even 1800 calories per head per day.
4. **Poverty:** According to the National Sample Survey (2009–10), 29.8% of the total population is below the poverty line, down from

37.2% in 2004/2005. The divide between the richest 10% and the poorest 10% has widened. Around 45% of Muslims are poor-higher among religious groups. Sikhs are the least poor is rural areas and Christians in Urban areas. The About 22% of the total population (more than 266 million) is below the poverty line 2011. They cannot purchase two times food for their families. The unfed and ill-nourished population suffer many deficiency diseases. The ignorance of the people, the inadequacies of medical facilities, and lack of financial resources come in the way of improving the housing and health standard of the people **(Fig. 13.40)**.

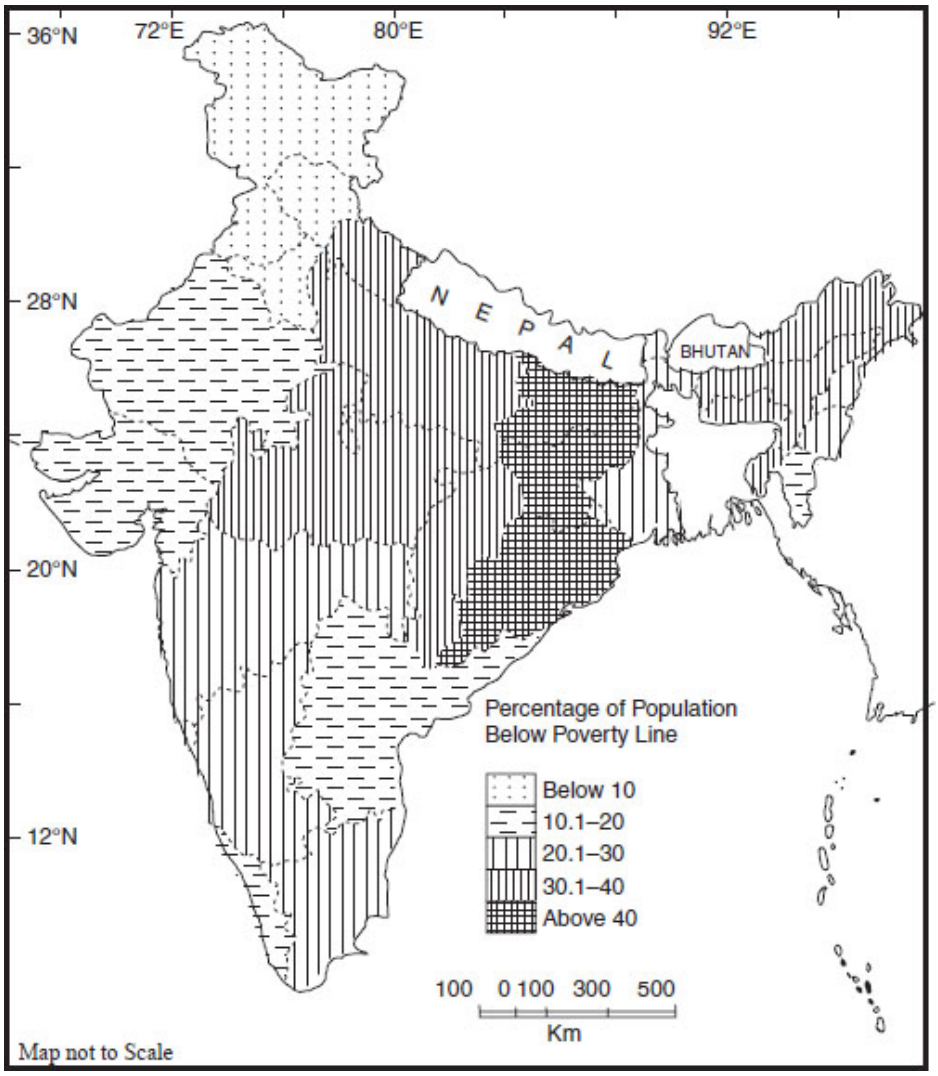


Fig. 13.40 Poverty—2011

5. **Mismanagement of Agricultural Resources**: In India, over greater part of the country, especially in the rain-fed and dry farming areas, agriculture is largely of traditional and subsistence type. In fact, agriculture is being done mostly by obsolete equipments. The farmers are generally poor, who are not able to purchase the costly inputs. Consequently, the production per unit area is low. Moreover, the institutional factors, like land tenancy, law of inheritance, the fragmentation of holdings, and small size of fields are some of the serious barriers in the enhancement of agriculture and its modernisation.
6. **Slow Growth of Industrial Sector**: The industrial sector, though growing fast, is still not strong enough to absorb the surplus labour from the rural areas. There is lack of funds which are required for the setting up of industries. The workforce, though large in number, is generally unskilled. Being poor, the Indian population is not providing a very good domestic market for the industrial goods. The lack of capital and poverty of people create a vicious circle and the growth of industries is affected adversely. The pressure on agricultural land is continuously increasing.
7. **Orthodoxy:** The rural society is tradition-bound and same is the case with many of the urban dwellers. The orthodoxy, ignorance, and staunch religious attitude of the masses come in the way of adoption of innovation and new technology. Being over religious, many of them do not practice family planning, which leads to fast growth of population, putting additional pressure on the available limited natural resources. In order to eradicate orthodoxy, superstition, and blind faith, large scale literacy and mass education movements are necessary.
8. **Frequent Strikes and Bunds:** The unemployment, poverty, hunger, malnutrition, and lack of social amenities have made many of the youths indisciplined. Strikes are frequent in the industrial sector which affect the industrial production and distribution of goods adversely.
9. **Political Upheaval:** Under the existing socio-economic scenario in which some of the people have become very rich and others are unemployed or below the poverty line, the youth in many parts

of the country have joined the ranks of Mao-Wadi-Naxalites, militancy and terrorism. These movements are a manifestation of the frustration of the youths who do not find jobs for their sustenance.

10. **Environmental Degradation:** The stupendous pressure of population on the resources is creating many environmental and ecological problems. These problems vary from region to region. For example, in the hilly states of North-East India, the Jhum cycle has been reduced to about five to ten years; in Punjab, Haryana, and western Uttar Pradesh, the soil and water resources have been polluted; most of the rivers of India are highly polluted; and there is a high degree of air and noise pollution in the metropolitan and big cities of the country. The pollution is the cause of many diseases, putting great pressure on the institutions of health care and hospitals.
11. **Life is not Enjoyable:** Life in India, both in the rural and urban areas is full of tension. The crime rate is going up, and good educational institutions, hospitals, and transport systems are over-crowded. The basic civic amenities like drinking water, electricity, and roads are inadequate. Fresh air, clean water, and unadulterated food are missing. Consequently, life is full of tension and not enjoyable for most of the people of the country.
12. **In India children having children-and dying:** In rural areas of some of the states, marriage occurs at a young age and is often arranged without the consent of girls.

    In brief, the huge size of population and its continuous unparallel growth is putting heavy pressure on the available resource base of the country. The heavy pressure of population leads to economic retardation, and deprivation. The large population also causes poor quality of life, low standard of living, shortage of food, inadequacy of housing, lack of health care and family welfare facilities, illiteracy, unemployment, insanitation, price rise, adulteration, hoarding, black-marketing, *loot,* kidnapping, dacoity, high rate of crimes against females and property, prostitution, indiscipline, strikes, *dharna*, lockouts, environmental pollution, political upheaval, and social tension.

13. **Child Labour:** More than 43.5 lakh kids aged between 5 and 14 are forced to work as labourers, according to the census of 2011. The number marks a drop of 65% over the 2011 census which counted 1.3 crore child workers. It became possible because of tougher laws, judicial intervention, increase in education and NGOs.

The National Population Policy, 2000

The phenomenal growth of Indian population has created numerous problems which find their manifestation in economic development and socio-political unrests in different states and regions of the country. The main objective of economic and social development to improve the quality of life and social well-being of the teeming millions remains a distant dream. In fact, over 26% of the total population, i.e., over 315 million are below the poverty line who are not earning even one dollar a day. It is a serious issue which needs to be addressed immediately. As the family planning programme, launched in 1952 has failed to deliver the goods, the Government of India announced the National Population Policy on 15 *February, 2000* to affirm its commitment for reproductive health care services and adoption of small family norm.

The immediate objective of the National Population Policy, 2000, was to address the unmet needs for contraception, health care infrastructure and to provide integrated service delivery system for the basic reproductive and child health care. The medium objective was to bring the Total Fertility Rate (TFR) to replacement level (2.1) by 2010 through vigorous implementation of inter-sectoral operation strategies. The long-term objective is to achieve a zero growth rate of population by 2045 at a level consistent with the requirements of sustainable economic growth, social development, and environmental protection.

The National Population Policy, 2000 stressed on the following points to reduce the fertility rate and to restrict the growth of population at the level of replacement:

1. Address the unmet needs for basic reproductive and child care services, supplies and infrastructure.
2. Make school education up to the age of 14 free and compulsory, and reduce dropout at primary and secondary school levels to below 20% for both boys and girls.

3. Reduce infant mortality to below 30 per thousand of population.
4. Reduce maternal mortality to below 100 per 1000 of population.
5. Achieve universal immunisation of children against all vaccine-preventable diseases.
6. Promote delayed marriage for girls, not earlier than 18 years and preferably after 20 years of age.
7. Achieve 80% institutional deliveries and 100% deliveries by trained persons.
8. Achieve universal access to information/counseling and services for fertility regulation.
9. Achieve 100% registration of births, deaths, marriages, and pregnancy.
10. Contain the spread of AIDS and promote greater integration between the management of reproductive tract infections and sexually transmitted diseases.
11. Prevent and control preventable diseases.
12. Integrate Indian System of Medicine (ISM) in the provision of reproductive and child health services and reaching out to all households.
13. Promote vigorously the Small Family Norm to achieve replacement level of Total Fertility Rate.
14. Bring out convergence in implementation of social sector programmes so that family welfare becomes a people centred programme.

In pursuance of the above objectives, the following National Socio-Demographic Goals were said to be achieved by 2010.

Strategies

For achieving the above National Socio-Demographic Goals, the Policy Document has outlined the following strategies:

(i) Decentralised planning and programme implementation.
(ii) Convergence of service delivery at village level.
(iii) Empowering women for improved health and nutrition.
(iv) Child care services.
(v) Meeting the urgent needs for family welafare services.
(vi) Operational strategies for urban slum dwellers, tribal communities,

hill area population, and displaced and migrant population; increased participation of men in planned parenthood.

(vii) Diverse healthcare providers.

(viii) Collaboration with the NGOs and private sector.

(ix) Contraceptive technology and research in reproductive and child health.

(x) Information, education and communication (Legislation, Public Support, National Commission on Population and Technology Mission). The Government of India has appointed the National Commission on Population with the Prime Minister as the Chairman.

In brief, there is an urgent need for developing over 1.25 billion people as the greatest asset by Human Resource Development Programme. Population stabilisation is the desired goal for requisite economic and social development. The objective of the National Population Policy, 2000 can be realised by the strategies suggested above by their effective implementation by the Government and whole-hearted support and cooperation of all sections of the society. Sooner this wisdom grows the better.

The Government of India recently started the National Commission on Population (NCP) after a five year gap, with the specific aim of repositioning family planning in India. Repositioning of family planning is directly linked with advancing family planning on national, state, and community agendas, with a renewed emphasis on enhancing the visibility, availability and quality of services provided for increased contraceptive use and healthy timing and spacing of births, and ultimately, improved quality of life. At the national level, policy makers, donors, scientists out work budget line items dedicated to family planning, enact family planning law and policies. At the local level, local leaders should educate and mobilise families to delay, space on limit child bearing.

HEALTH INDICATORS

Health is one of the important indicators of human development. Life free fron illness and ailment and living a a reasonably long span are indicative of a healthy life.

Health indicators include:

(i) life expectancy at birth,
(ii) crude birth rate per 1000,
(iii) total fertility rate per 1000,
(iv) crude death rate per 1000,
(v) infant mortality rate per 1000,
(vi) under five mortality rate per 1000,
(vii) maternal mortality rate%.

Following are the health indicators:

The health of the Indian masses and longevity has increased appreciably as the following indicators reveal:

1. Life Expectancy at Birth Increased from 37 years in 1951 to 67 in 2011
2. Crude Birth Rate Decreased from (per 1000) 33.90 in 1981 to 26.40 to 21.8 in 2011
3. Total Fertility Rate Decreased from (per 1000) 4.50 in 1981 to 2.4 in 2011
4. Crude Death Rate Decreased from (per 1000) 146 in 1951 to 69 in 2011
5. Infant Mortality Decreased from (per 1000 births) 146 in 1951 to 54 in 2011
6. Under Five Mortality Decreased from (per 1000) 326 in 1951 to 55 in 2011
7. Maternal Mortality Rate% 20.20 % in 1946 to 38% in 2011.

It may be observed from the above data that the longevity of the Indian population has increased appreciably as it was only 31.7 years in 1947 which rose to 67 years in 2011.

Indicators of Social Empowerment

'Development is freedom'. Freedom from hunger, poverty, servitude, bondages, ignorance, illiteracy, and other forms of domination is the key to human development. Freedom in the real sense of the term is possible only with the empowerment and participation of the people in the exercise of their capabilities and choices in the society. Access to knowledge about the society and environment are fundamental to freedom. Literacy is the beginning of access to such a world of knowledge and freedom (*for details read literacy pp. 13.58–62).*

HUMAN DEVELOPMENT INDEX IN INDIA

India has 130th rank out of 188 countries in human development in the world (Source: HDR 2015). India has been placed among the countries showing medium human development. India's position is way below China (89th spot) and Sri Lanka 91 rank. The main cause of India's low rank is high degree of inequality in education and healthcare. There are, however, inter-state, and intra-state variations in the human development in the country **(Fig. 13.41)**.

There are several socio-political, economic, and historical reasons for such a scenario of human development. Kerala is able to record the highest value in the HDI largely due to its impressive performance in achieving near hundred per cent literacy (94%) in 2011. In a different scenario the states like Bihar, Chhattisgarh, Jharkhand, Rajasthan, Madhya Pradesh, Odisha, Assam, and Uttar Pradesh have very low literacy. For example, the total literacy rate of Bihar was as low as 61.8% during the same year (2011). States showing higher total literacy rates have less gaps between the male and female literacy rates. For Kerala, it is only 6.34%, while it is 26.75% in Bihar, and 25.95% in Madhya Pradesh.

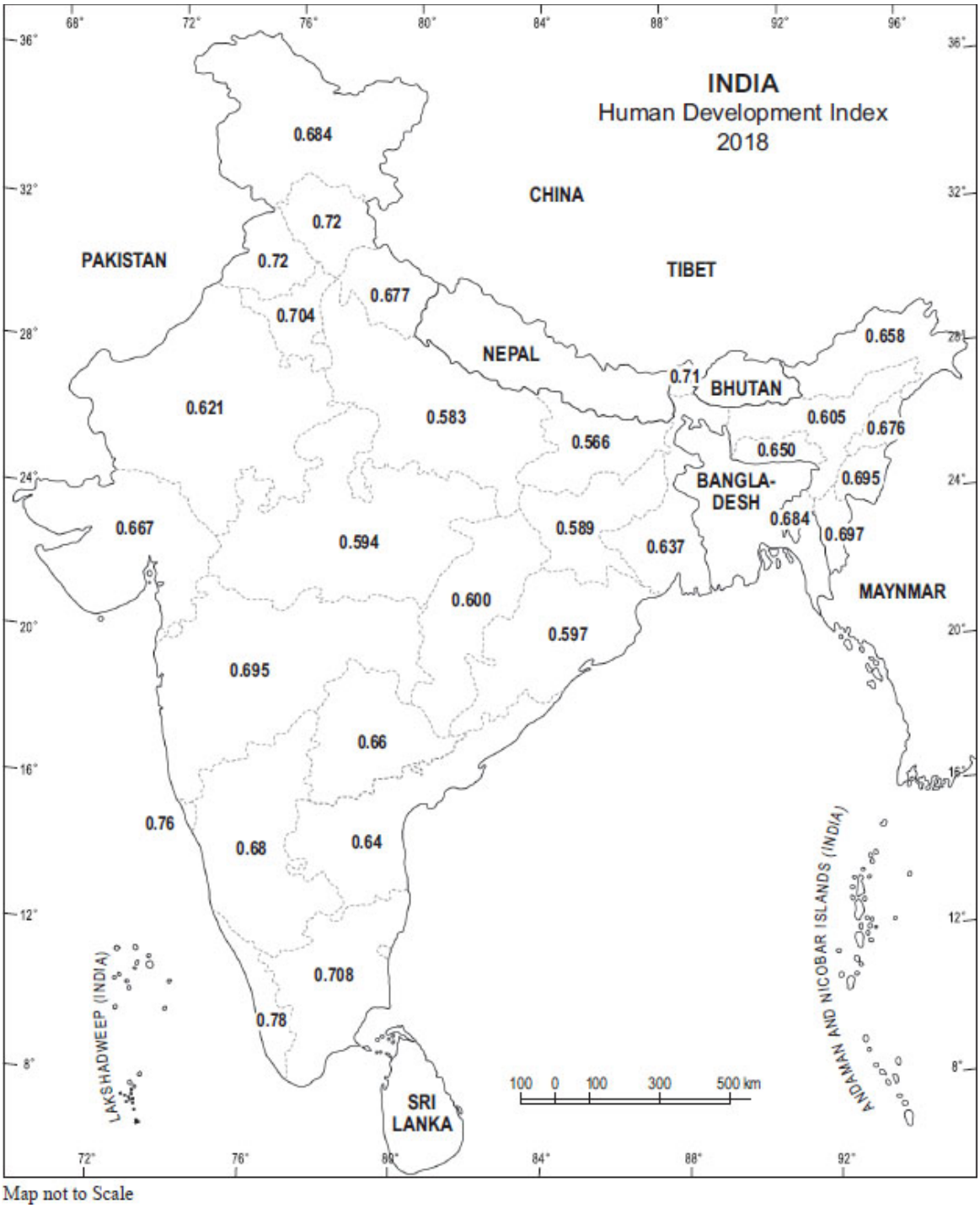


Fig. 13.41 India—Human Development Index 2018

Table 13.20

India: Human Development Index of the selected states of India: 2017

State/Union Territory	Consumption Based HDI 2017
1. Kerala	0.784
2. Delhi	0.744
3. Himachal Pradesh	0.720
4. Goa	0.764
5. Punjab	0.721
6. N.E States	0.662
7. Maharashtra	0.695
8. Tamil Nadu	0.708
9. Haryana	0.704
10. Jammu & Kashmir	0.684
11. Gujarat	0.667
12. Karnataka	0.682
13. West Bengal	0.637
14. Uttarakhand	0.677
15. Andhra Pradesh	0.643

Source: India Human Development Report, 2017.

Apart from the educational attainment, the levels of economic development too play significant impacts on HDI. Economically developed states like Maharashtra, Tamil Nadu, Gujarat, and Punjab have higher value of HDI as compared to states like Assam, Bihar, Madhya Pradesh, Odisha, and Uttar Pradesh.

Regional distortions and social disparities which developed during the colonial period continue to play an important role in the Indian economy, polity and society. The government of India has made concerted efforts to institutionalise the balanced development with its main focus on social distributive justice through planned development. It has made significant achievement in most of the fields but, these are still below the desired level. The causes of unequal regional development are largely corruption and scams.

Government Schemes on SC/ST/OBC

Pre-Matric Scholarship for SC Students: The Scheme was introduced in 2012. A hundred percent assistance from Government of India for expenditure under the scheme is provided.

National Overseas Scholarship for SCs: The scheme provides for fees charged by institutions as per actual, monthly maintenance

allowance, passage visa fee and insurance premium, annual contingency allowance, incidental journey allowance. The total number of awards (maximum) to be given each year is 100 and 30% of the awards have been earmarked for women candidates.

Babu Jagjivan Ram Chhatrawas Yojna: The scheme aims to provide hostel facilities to SC boys and girls studying in middle schools, higher secondary schools, colleges and universities.

Rajiv Gandhi National Fellowship for SC Students: The scheme provides financial assistance to scheduled caste students for pursuing research studies leading to M.Phil, Ph.D.

Venture Capital Fund For Scheduled Castes: Government announced the setting up of a Venture Capital Fund for Scheduled Castes in 2014 in order to promote entrepreneurship among the scheduled castes providing concessional finance to them.

Credit Enhancement Guarantee Scheme : The Government in 2014, announced Rs 200 crores to be allocated towards credit enhancement facility for the young SC entrepreneurs.

Adivasi Mahila Sashaktikaran Yojana (AMSY): It is an exclusive scheme for the economic development of ST women, at a highly concessional rate of interest. Under it National Scheduled Tribes Finance and Development Corporation (NSTFDC) provides term loan for scheme(s)/project(s) costing upto ₹ 50,000/- per individual unit/profit centre.

The Tribal Cooperative Marketing Development Federation of India Limited (TRIFED): It was set up in 1987 as a national level apex body under the Multi State Cooperative Societies Act, 1984 (MSCS Act, 1984). TRIFED functions as a '*market developer*' for tribal products and as 'service provider' to its member federations.

Vanbadhu Kalyan Yojana: It aims at creating an enabling environment for need based and outcome oriented holistic development of tribal people.

Umbrella Scheme For Education of ST Children: This scheme covers the undermentioned schemes: (a) establishing and strengthening of ashram schools; (b) establishing and strengthening of hostels; (c) vocational training in tribal areas; (d) pre-matric and post-matric scholarship.

Pre-Matric Scholarship for OBCs: The aim of this Scheme is to motivate children of OBCs studying at pre-matric stage. The income ceiling for eligibility should not exceed Rs. 250,500/- per annum.

Post-Matric Scholarship for OBCs: The objective of the scheme is to provide financial assistance to the OBC students studying at post-matriculation or post-secondary stage to enable them complete their education. The income ceiling of parents/guardians for eligibility is Rs. 1.00 lakh per annum .

Construction of Hostels for OBC Boys and Girls: This scheme aims at providing hostel facilities to students belonging to socially and educationally backward classes, especially from rural areas, to enable them to pursue secondary and higher education. The cost of construction for boys hostels is shared between the centre and the state in 60:40 ratio and for the girls hostels the ratio is 90:10.

Dr. Ambedkar Scheme of Interest Subsidy on Educational Loans: The objective of the scheme is to award interest subsidy to meritorious students belonging to the Other Backward Classes (OBCs) and Economically Backward Classes (EBCs) in order to provide them better opportunities for higher education abroad and enhance their employability.

National Backward Classes Finance and Development Corporation: NBCFDC was incorporated in 1992 by the Government of India under the aegis of Ministry of Social Justice and Empowerment It assists the poorer sections of these classes in skill development and self employment ventures.

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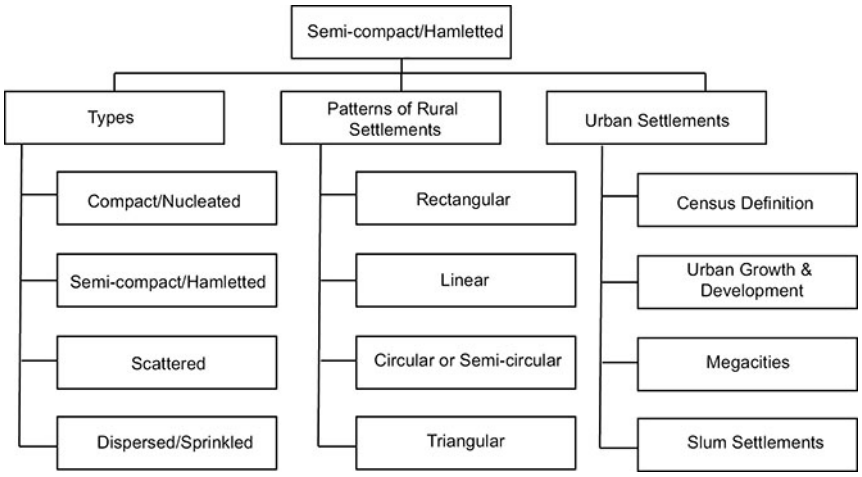
14

Settlements



INTRODUCTION

After food, settlement is the most important need of human being. They construct houses and develop settlements to protect themselves against the vagaries of weather and to enjoy social life. In fact, settlement is man's most important step towards adapting himself to his or her physical and cultural environment. In the opinion of Perpillou, "*Settlement is man's first step towards adapting himself to his environment*". A settlement essentially consists of houses and roads.

CLASSIFICATION OF SETTLEMENTS

On the basis of economic activities settlements can be divided into two general categories: (i) rural settlements, and (ii) urban settlements.

Rural settlements are those where people are mostly engaged in agriculture and allied activities (livestock keeping, dairy farming, poultry, forestry, fisheries, and mining), whereas urban settlements are those whose residents are engaged in non-agricultural activities (industries, services, etc). In fact, rural and urban nomenclatures depict the modes of community life. A village has a community bond, while the city life is heterogeneous, alien, incongruous, and disparate. We may think of a spectrum with the rural community standing at one extreme and the large metropolis at the other. Between the

two are all grades of urbanization. This is why there is increasing emphasis on rural urban continuum. Interestingly enough, the city comes to include a much larger proportion of country bred residents than the country does of those bred in city.

Types of Rural Settlements

On the bases of structure, size, growth of population and functions, the rural settlements may be broadly divided in two categories: (i) The compact or nucleated settlements, and (ii) the dispersed or scattered settlements. The compact settlements have a well developed network of roads, streets, lanes, and *Galis* not only in the settlement, but also in the fields and surrounding areas. In the dispersed or scattered settlement, only one family residence is the core which stands isolated from the other houses. For a systematic and comprehensive study, the Indian rural settlements may be divided into the following categories:

1. Compact settlements
2. Semi-compact or Hamlet-settlements
3. Scattered settlements
4. Dispersed settlements

1. Compact (Agglomerated, Clustered or Nucleated Settlement)

A compact settlement has a well defined form in which the houses are clustered in a small space. They have been termed as *Reefs of population collected at certain points in a process of crystallisation* (Perpillou). The compact settlements are an expression of a common need to centralise agricultural activity at a particular point (**Fig. 14.1).** Nucleated settlements have a separate organisation with its own individual life and a personality which forms part of the landscape. Each village is a well-knit community which strictly adheres to the village agricultural calendar. They are mainly found in the productive plains of the *Old World* (Africa, Asia, and Europe). The compact settlements have following characteristics:

(i) They are the lodging places of large populations, generally more than 500 in Europe and more than 1000 in northern India.

(ii) They develop mainly in the productive plains. Intensive subsistence agriculture is one of the important causes for the development of compact settlements. In fact, they develop in the areas of large tracts of fertile arable lands.

(iii) The site of compact settlements is generally the high banks of rivers, *dry-points (places free from inundation)* in a flood prone area, and *water point* (oases) in the deserts. They also develop along the meander of an ox-bow lake.

(iv) They develop at the place of assured supply of water.

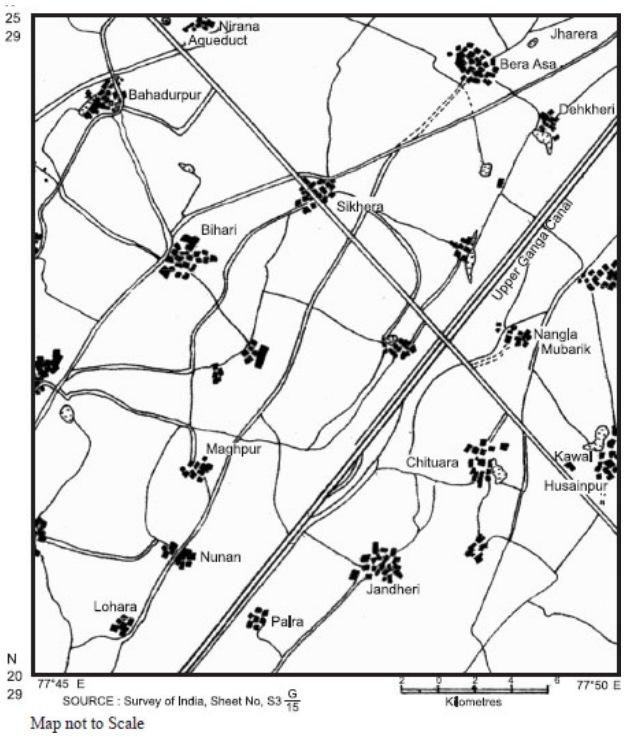


Fig. 14.1 Compact Settlements (District Muzaffarnagar, U.P.)

(v) Availability of water also promotes clustered villages. In Punjab, Haryana, western Uttar Pradesh, eastern Rajasthan, and the Deccan Trap, peasants are grouped together close to water reservoirs, wells and canals. In Karst (limestone) topography where water points are limited, compact settlements develop around the place where there is assured supply of water, but no inundation.

(vi) In some of the tribal areas like Nagaland, compact settlements developed at the top of ridges, spurs, and hills. The development of compact settlements at hill tops are the result of security and defence. In a society in which head-hunting was a practice in the past, defence and community bonds are often the causes of nucleation.

(vii) Besides the close social relationships and warm neighbouring feeling of the village community, agricultural practices have centred round cooperation and strengthened the cohesive character of compact villages. Moreover, the need for defence in the past was a vital factor for building nucleated settlements.

(viii) The depredation of wild animals necessitated group living which found expression in compact settlements.

In India compact settlements are found in the Satluj, Ganga, Brahmaputra plains; the plains of Eastern and Western Ghats; the deltas of Mahanadi, Godavari, Krishna, and Kaveri; the forested areas and *Tarai* in Uttar Pradesh and Uttarakhand; the Chotanagpur Plateau, and the Eastern and Western Ghats.

2. Semi-Compact or Hamletted Settlements

The semi-compact settlement is an intermediary type of settlement, marked by the presence of one easily recognisable site and one or more small hamlets closely linked with the main site by foot-paths, cart-tracks or roads. The attached *Purvas* and *Mazras* (hamlets) have grown either because of increasing pressure of population on the main site which forced the large families to seek living space outside the village, but not too far off due to social links and economic interdependence. Other reasons includes (i) the immigration which congregated into some families of the agricultural

labourers from outside, and (ii) some artisan castes have built their community hamlets in the close vicinity of the main village. The *Purvas* and *Mazras* are located upon some single geographical facility such as an embankment, a river-ford, or a cross-road, and they generally possess a single common thoroughfare. Because of the small size of the *Purvas* and *Mazras*, they ordinarily manifest little or no geographical division into functional units. Such semi-compact settlements are quite common in the *Khadar* lands of the Ganga, Yamuna, Brahmaputra and their tributaries, the *Bet*-lands of Punjab and Haryana, the deltas of the peninsular rivers (**Fig. 14.2).** The semi-compact settlements are also found in the *Tarai* region of Uttarakhand and Uttar Pradesh. A typical semi-compact settlement of the Tharu-tribe in the Tarai region of Uttarakhand has been shown in (**Fig. 14.3).**

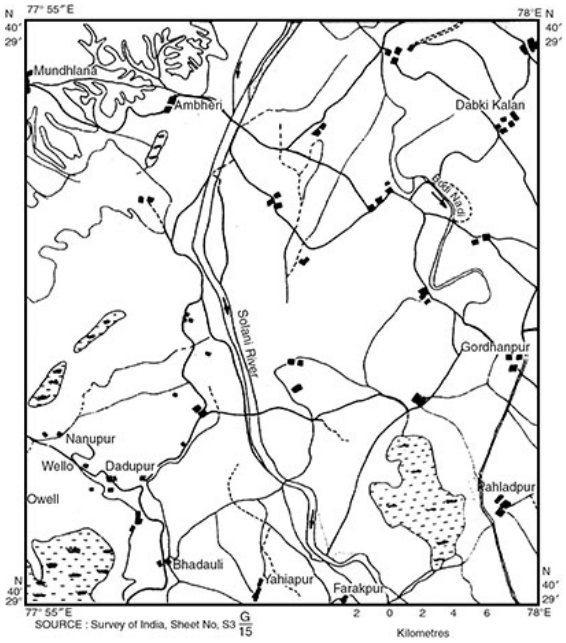


Fig. 14.2 Semi-Compact Settlements in the Khadar of Ganga River (District Haridwar, Uttarakhand)

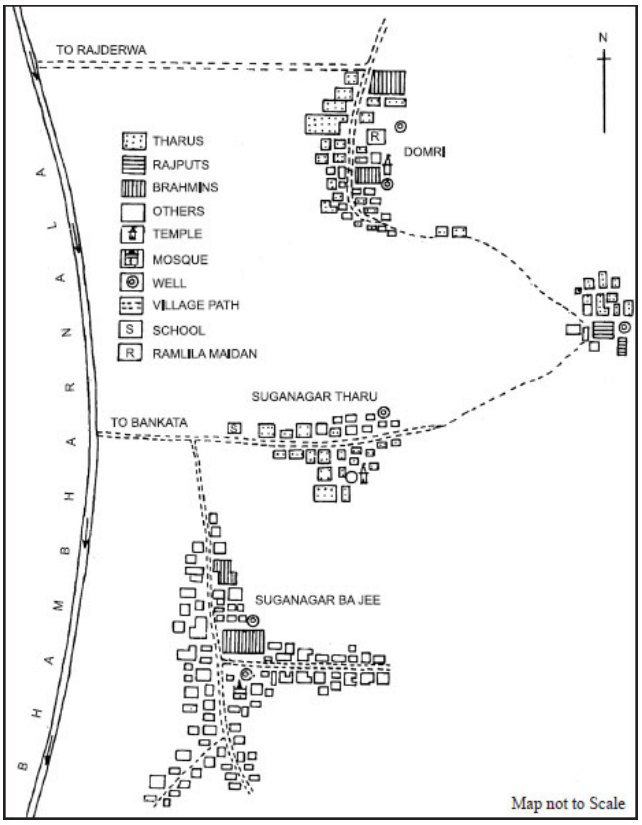


Fig. 14.3 Compact and Semi Compact Tharu Settlements of the Tarai Region of UP

3. Scattered (Semi-Sprinkled Settlements)

Such settlements are characterised by the presence of several smaller *purvas* (hamlets**)** and several individual habitations. Spreading over the entire *Mauza* (village), the main site may not be distinguishable, the number of such hamlets may vary from 5 to 10. In these loosely connected settlements a street pattern is lacking and the straggling residences are separated by cultivated fields. These settlements manifest the development of individualistic tendencies for economic advantages and are surely indicative of peaceful conditions. Economically, they may claim approbation, but socially they are not conducive for the development of community life. The close warmth of communal life of the compact village, the feeling of being an integral part of the co-operative life is missing and the institutions of reciprocal aid-system and mutual cooperation are either absent or very weak. In India they are well marked in the eastern part of the Ghagra plain, Vindhyan upland, south-west Punjab and the command area of the Indira Gandhi Canal in Rajasthan **(Fig. 14.4)**.

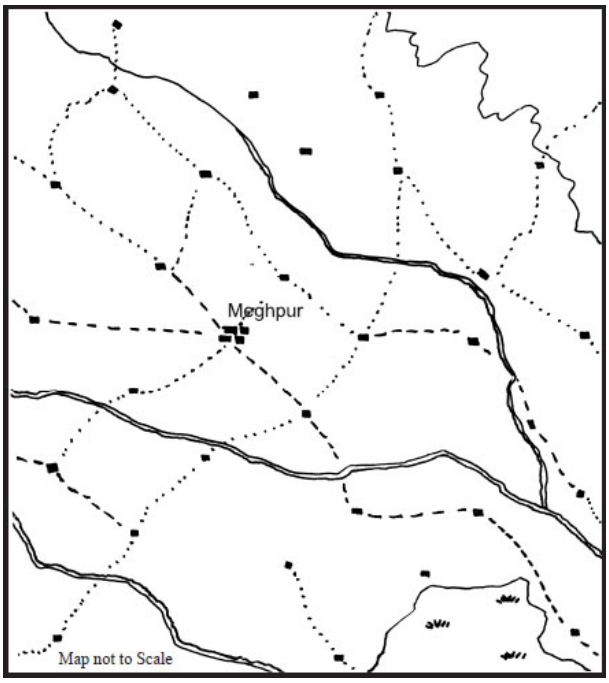


Fig. 14.4 Scattered Settlements in Eastern Ghagra Plain (U.P.)

4. Dispersed or Sprinkled Settlements

The dispersed settlement is characterised by complete diffusion of farmsteads over the entire *Mauza* land. What is significant is the large distance between dwellings compared with the congested homes of the compact settlements. The main causes of dispersed settlements are mountainous and forested areas, poor soil and hydrography, scarcity of arable land, too hot or humid regions or too cold regions. Such dispersed settlements are quite common in the mountainous, semi-desert areas and the areas of less productive soils of Aravallis (**Fig. 14.5)**. The caste prejudice and the existence of low agricultural castes and tribes have been partly responsible for dispersed settlements. The psychological change in social values has also promoted individualistic tendencies and dispersion of rural settlements. They are also found in the newly developed agricultural areas like the Indira Gandhi Canal Command Area and agricultural land of the *Tarai* regions of Uttar Pradesh and Uttarakhand. In the dispersed settlements everything spells isolation and private (Blache). The impression gained from such a landscape is one of melancholy and isolation; even the stranger is lost in the inhospitable labyrinth. Here, the parochial economic considerations have done away with the communal life of compact village (**Fig. 14**.**5**).

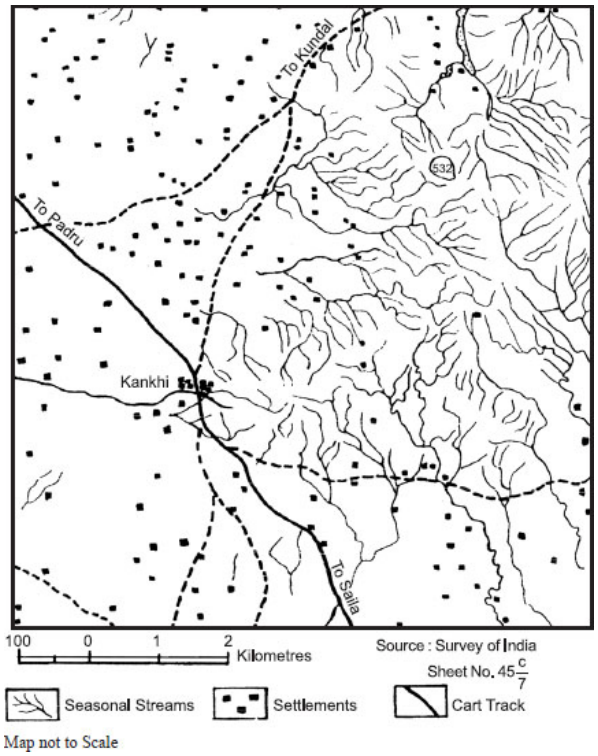


Fig. 14.5 Dispersed Settlements (Barmer and Jalore Districts, Rajasthan)

Patterns (Forms or Designs) of Rural Settlements

Patterns of rural settlements have been defined as the relationship between one house to another. Pattern means the *geometric form* or *design* of the village settlement. The pattern of a settlement may be easily identified by reading and observing a large scale map, like the *topographical sheet* published by the Survey of India, Dehradun. It may be mentioned that the term pattern of settlement deals only with the compact and semi-compact settlements.

The site of the village and the surrounding topographical features have a direct bearing on the form and design of a rural settlement. Moreover, the socioeconomic and cultural factors also give different forms to the rural settlements. The rural settlements have varied patterns in the arrangement of their streets and houses. They are visible in the rural landscape of the country. A definite layout (village plan) is developed only in rectangular and linear settlements where the controlling factor is the physical character of the site which determines its form since the very inception; although the cultural factors of *push* and *pull* have brought tangible modification in their growth process. The rural settlements may be classified under the following broader categories:

1. Rectangular pattern
2. Linear pattern
3. Circular or Semi-circular pattern
4. Star-like pattern
5. Triangular pattern
6. Nebular pattern
7. Terraced pattern
8. Amorphous pattern

1. **Rectangular Pattern:** Rectangular settlements develop generally in the intensively cultivated productive plains of the country. Their geometric shape is either rectangular or square. The lanes in a rectangular settlement are generally straight and meet each other at right angles.

The rectangular rural settlements enjoy universal importance. In India, they are the heritage of our ancient past when villages were mostly planned in a rectangular or square pattern. Our system of land measurement, the *Bigha* has acted as a consistent factor in maintaining their form. When the cultivated plots and orchards, all conform to almost rectangular pattern, the settlement follows their character. The village streets conform to the field patterns and give a rectangular pattern. The nature of the orientation of the dwellings and houses also help in the maintenance of rectangular shape.

The rural settlements of the Northern Plains of India, especially in the Satluj-Ganga-Plains (Punjab, Haryana, and western Uttar Pradesh), and the deltas of the peninsular rivers have numerous villages of rectangular shape (**Fig. 14.6**).

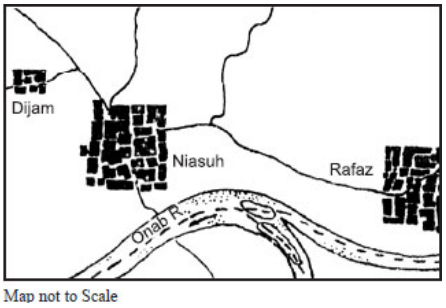


Fig. 14.6 Rectangular Settlements

2. **Linear Pattern:** Linear settlements are those which develop in a line and acquire a linear shape. Such settlements are found in the narrow valleys of the mountains, along the higher levees, ox-bow lakes or abandoned channels. They are also found along the roads, canals and railway-tracks. In the mountainous valleys, at the slopes, and along the river banks the danger of recurring floods are the restrictive forces that restrict the lateral expansion of the settlement. Such settlements are numerous in the Lesser Himalaya, Siwaliks, Hill-states of North-East India, Eastern and Western Ghats (**Fig. 14.7**).

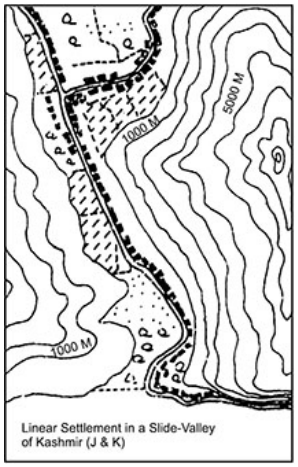


Fig. 14.7 Linear Settlements (Beera-Village)

3. **Circular and Semi-Circular Pattern** Circular and semi-circular settlements generally develop around a pond, a lake, a crater or a sea-coast. Since people prefer to stay closer to the water body, they construct their houses close to the pond or lake. In due course of time, such settlements acquire circular or semi-circular shape. There are many settlements in

Kathiwar, Rann of Kutch where the fisherman and salt producers have developed circular or semi-circular patterns of rural settlements. The crater (mouth of an extinct volcano) is also a place where circular settlements may be seen. People of some of the circular settlements earn their livelihood from the water-bodies either by selling fish, water-nuts, grasses, lotus-roots, or by providing services to the tourists, pleasure seekers, picnic goers and aesthetic beauty lovers **(Fig. 14.8)(a) and (Fig. 14.8(b)).**

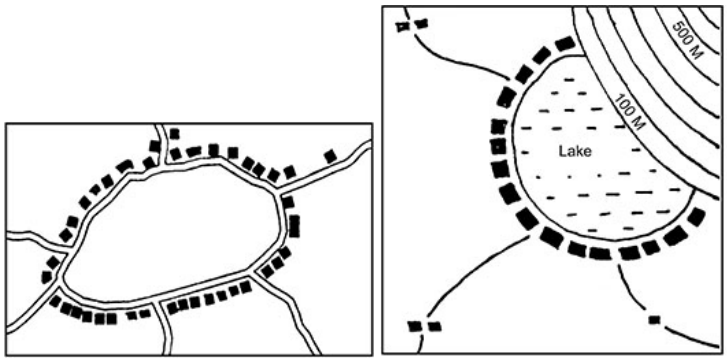


Fig. 14.8(a) Circular Settlements (Uhsba-Village) ***Fig. 14.8(b)*** Semi-circular Settlements (Riazu-Village)

4. **Star-like Pattern:** The star-like pattern of rural settlements develop at the sites and places where several roads converge. In fact, such settlements are associated with the nodal locations of settlements where several roads converge at a site and the rows of the houses spread along the roads in all directions. This pattern is mainly found in the productive agricultural tracts in which the network of roads is dense. Such settlements can be seen in the productive plains of Punjab, Haryana, and the Upper Ganga-Yamuna Doab of western Uttar Pradesh **(Fig. 14.9)**.

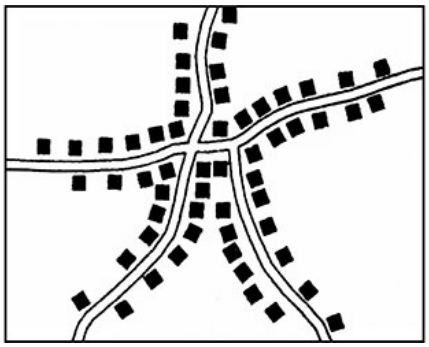


Fig. 14.9 Star-like Pattern (Aheedam-Village)

5. **Triangular Pattern:** The triangular patterns of rural settlements develop at the confluence of rivers or where two roads converge in the form of a triangle. The lateral expansion of dwellings at the confluence is constrained by the rivers. Consequently, the shape of the settlement becomes triangular (**Fig. 14.10)**.

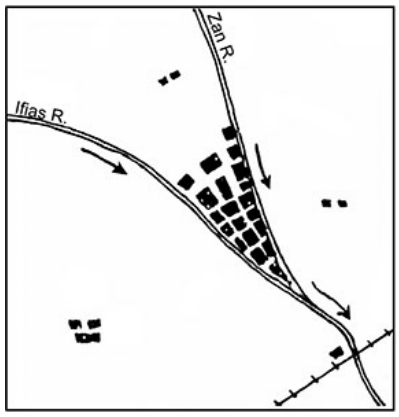


Fig. 14.10 Triangular Pattern (Naarbij-Village)

6. **Nebular Pattern:** When the shape of a settlement resembles a nebula, it is known as a nebular pattern of rural settlement. The arrangement of roads in such a settlement is generally circular which ends at a central location (nucleus) of the village. These settlements may be on a hillock or a mound and their size is generally small. Nebular settlements may be observed in the hilly and undulating tracts of Himachal Pradesh, the Kandi lands of Jammu and Kashmir, and the Lesser Himalaya and Shiwaliks in Uttarakhand (**Fig. 14.11**).
7. **Terraced Patterns:** In the mountainous areas the cultivators develop terraced fields for cultivation and orchards. They construct their houses along the terraces. Such settlements may be seen in Arunachal Pradesh, Himachal Pradesh, Jammu and Kashmir, Mizoram, Sikkim, Western Ghats, and Uttarakhand (**Fig. 14.12)**.

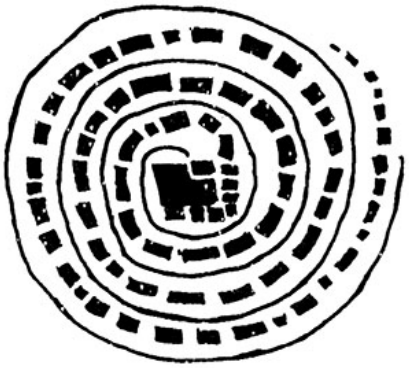


Fig. 14.11 Nebular Pattern (Dahsar-Village)

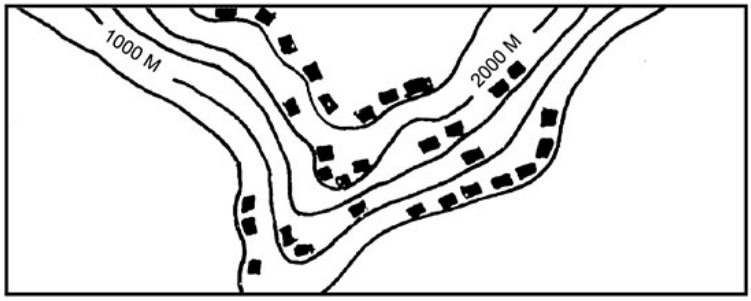


Fig. 14.12 Terraced Pattern (Aheeraf-Village)

8. **Amorphous Patterns:** When the village land is dotted with several hamlets and individual farmsteads, all being linked by footpaths with the main settlement, no definte pattern is discernible. Such a loose pattern may be termed as amorphous settlement. This pattern of settlements is found in West Bengal and the Lower Brahmaputra Valley of Assam **(Fig. 14.13)**.

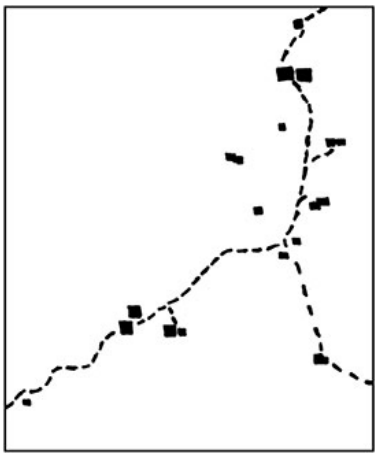


Fig. 14.13 Amorphous Pattern (Abih-Village)

Rural Settlements in the Himalaya

The settlements in the Himalaya are either semi-compact, scattered (semi-sprinkled) or dispersed (sprinkled). In the inter-montane valleys like that of Kashmir, Kullu-Manali, and the Dun Valley, the rural settlements are largely compact and semi-compact. In the higher altitudes, especially in the alpine pastures *(Margs* in Kashmir) the settlements are dispersed. The undulating topography, cold climate and paucity of agricultural land favours the growth of small scattered and dispersed settlements. The Gujjars and Bakarwals of Kashmir, the Bhutias of Himachal and Kumaun and the Lepchas of Sikkim who practice transhumance, construct their permanent houses in the plain areas to pass their winters and temporary houses (*Kothas* and *Bandis*) in the alpine pastures of Kashmir to graze their goats in the summer season. Most of the Gujjars and Bakarwals of Jammu and Kashmir have their permanent settlements in Kathua, Jammu, Rajauri, and Poonch districts where they pass their winters and engage in agriculture, while in the summer season they ascend into the alpine pastures to utilise the nutritious grasses.

In the states of Eastern Himalaya (Arunachal Pradesh, Mizoram, Nagaland, Tripura, etc.) settlements are usually small and widely dispersed due to steep slope, undulating topography, heavy rainfall, dense forests and multiplicity of tribes who speak different dialects. The compact and well organised settlements are however, found in the northern zone of the Indo-Tibetan culture. In Meghalaya, the Khasi and Jaintia villages are located along the hill slopes near the springs and water-bodies which range from isolated homesteads to dispersed and semi-compact settlements. In Nagaland, villages generally occupy flat tops of hills, spurs, and ridges between 1000 and 2000 m above the sea level. Security and defence against invaders is the main consideration for site of settlement in Nagaland. In Manipur, the Kukis (Nagas) practicing shifting cultivation construct their temporary huts in the Jhum fields, but the permanent settlements are there, i.e., the permanent houses in the settlement are not shifted. The Lushai tribes of Mizoram construct their linear settlements in valleys, on the hill tops, and along the roads. Same is the case with the Chakma tribes of Tripura.

Rural Settlements in the Northern Plains of India

The Northern Plains of India are highly productive and intensively cultivated. Irrigation by canals, wells and tube-wells is adequately available to the crops. In the plains of Punjab, Haryana, and western Uttar Pradesh rural settlements are uniformally spaced and have large compact size. In the northern plains the main types of settlements in the *Bhangar* tracts are compact and semi-compact, while in the *Khadar* and *Bet* land of the rivers, which are frequently inundated during the rainy season, the rural settlements are either semi-compact, scattered or dispersed. In western Rajasthan where the landscape is characterised by sand-dune, the rural settlements are generally of semi-compact type. In eastern Rajasthan, the settlements are, however, compact and large.

In the Tarai tracts of Uttar Pradesh and Uttarakhand most of the settlements are semi-compact and their size is medium. The settlements in the *Tarai* region are unevenly distributed due to high percentage of forests, marshy lands and seasonal floods. In the *Ghar*-tract, rural settlements follow the drainage lines and the nature of slope.

In the Middle Ganga Plain, the rural settlements are semi-compact, small sized but closely spaced, while in the Lower Ganga-plains hydrological characteristics have a dominant role in the determination of types and patterns of rural settlements. The scattered settlements are very common in the *Rahr-Plain* (West Bengal), *Duars* and Sundarban Delta.

In the Brahmaputra Valley, villages are generally agglomerated along the river levees and transport arteries. In Brahmaputra Valley, villages are smaller in size in which houses are separated by bamboo fences. *Machan* type of houses on wooden pillars are constructed in low-lying and flood prone areas where boat is the only mode of transport during the rainy season.

Rural Settlements in Peninsular India

Peninsular India has great diversity in its topography and relief. Consequently, mixed type of rural settlements are found in the Peninsular part of the country. Throughout the hilly tract of the Aravallis, huts are widely dispersed within the revenue village land. In the dissected hills of Mewar and Alwar, isolated farmsteads, locally known as *Dhani* are dotted in the long narrow valleys. Jaipur and its vicinity are characterised by compact and semi-compact settlements. The southern parts of Aravallis (Udaipur Divison) have isolated and dispersed types of rural settlements. In Bundelkhand, settlements have been developed near the points of water-supply which are by and large scattered.

Settlements in the Chotanagpur Plateau are generally clustered and semi-clustered type. In the fertile tracts of Baghelkhand the settlements are compact. The lava-Plateau region of Maharashtra has compact settlements near the sources of water, but semi-compact and dispersed between Pune and Solapur.

The *Maidan* of Karnataka and Rayalseema have widely spaced compact settlements, while in the Malnad (hilly-land) the settlements are semi-compact. The Kaveri Delta and coastal regions have compact and semi-compact settlements. The Sahyadris have a predominance of isolated settlements.

Rural Settlements in the Coastal Regions

The patterns of rural settlements differ from each other. In the delta tracts of the Mahanadi, Godavari, Krishna, and Kaveri, the rural settlements are located at the elevated levees which are free from floods. These deltas are highly productive and intensively cultivated. Farming is organised and has thus promoted the tendency of compact settlement. Along the Eastern Coast, just above the high-tide mark, there are several fishing villages of different shapes and patterns.

The Malabar coast is dominated by large sized compact settlements. There are, however, scattered and dispersed settlements in the areas of coconut and cashew plantation. In the Rann of Kutch, there are semi-compact and scattered settlements of small size having one or two rooms. These semi-compact and dispersed settlements are widely spaced.

The significant attributes of Indian villages in the different states of the country have been given in (**Table 14.1)**.

Table 14.1

India: Attributes of Villages–2011

States/UTs	No. of Villages	Rural Population	Population per Village	Rural Area (In sq km)	Area per Village (sq km)	Spacing (km)	Population Density (sq km)	Village Density (100 sq km)
Andaman and Nicobar Islands	555	237,093	427	8,211.08	14.79	3.86	29	7
Andhra Pradesh	27,800	56,361,702	2027	267,190.50	9.61	3.15	211	10
Arunachal Pradesh	5589	1,066,358	191	NA	NA	3.87	NA	7
Assam	26,395	26,807,034	1016	77,178.12	2.92	1.72	347	34
Bihar	44,874	92,341,436	2058	91,838.28	2.05	1.45	1005	48
Chandigarh	5	28,991	5798	4.47	0.89	4.77	6486	4
Chhattisgarh	20,126	19,607,961	974	131,940.34	6.56	2.59	149	15
Dadra and Nagar Haveli	65	183,114	2817	445.30	6.85	2.75	411	13
Daman and Diu	19	60,396	3179	62.48	3.29	2.42	967	17
Goa	334	551,731	1652	2937.21	8.79	3.33	188	9
Gujarat	18,225	34,694,609	1904	188,838.60	10.36	3.28	184	9
Haryana	6841	16,509,359	2413	42,235.92	6.17	2.54	391	15
Himachal Pradesh	20,690	6,176,050	299	55,402.18	2.68	1.64	111	37
Jammu and Kashmir	6553	9,108,060	1390	220,990.10	33.72	5.82	41	3
Jharkhand	32,394	25,055,073	773	77,292.06	2.39	1.57	324	41
Karnataka	29,340	37,469,335	1277	185,776.57	6.33	2.56	202	15
Kerala	1018	17,471,135	17,162	31,253.20	30.70	6.18	559	3
Lakshadweep	21	14,141	673	9.17	0.44	1.20	1542	70
Madhya Pradesh	54,903	52,557,404	957	300,505.59	5.47	2.37	175	18
Maharashtra	43,665	61,556,074	1410	298,619.45	6.84	2.65	206	14
Manipur	2582	2,021,640	783	22,147.50	8.58	2.94	91	12
Meghalaya	6839	2,371,439	347	22,146.11	3.24	1.81	107	30
Mizoram	830	525,435	633	20,494.00	24.69	5.04	26	4
Nagaland	1428	1,407,536	986	16,335.52	11.44	3.41	86	9

States/UTs	No. of Villages	Rural Population	Population per Village	Rural Area (In sq km)	Area per Village (sq km)	Spacing (km)	Population Density (sq km)	Village Density (100 sq km)
NCT of Delhi	112	419,042	3741	326.44	2.91	3.64	1284	8
Odisha	51,311	34,970,562	682	152,355.34	2.97	1.74	230	33
Puducherry	90	395,200	4391	335.44	3.73	2.33	1178	18
Punjab	12,581	17,344,192	1379	47,847.40	3.80	2.00	362	25
Rajasthan	44,672	51,500,352	1153	335,606.04	7.51	2.77	153	13
Sikkim	451	456,999	1013	7057.75	15.65	3.97	65	6
Tamil Nadu	15,979	37,229,590	2330	116,424.47	7.29	2.85	320	12
Tripura	875	2,712,464	3100	10,094.12	11.54	3.46	269	8
Uttar Pradesh	106,774	155,317,278	1455	233,365.71	2.19	1.50	666	44
Uttarakhand	16,793	7,036,954	419	52,581	3.13	1.78	134	31
West Bengal	40,203	62,183,113	1547	83,626.43	2.08	1.49	744	45
India	640,932	833,748,852	1301	3,101,473.97	4.84	2.26	269	19

Source: Census of India, 2011.

It may be observed from (**Table 14.1)** that the average density of villages in India is 19 villages per 100 sq km, the total number of villages is 640,932 and the average size of population per village is 1301.

It may also be seen from (Table 14.1) that the highest density of the villages is in Laksadweep (70), Bihar (48), and Uttar Pradesh (44), each per one hundred sq km. while the lowest density is in Jammu and Kashmir (3 Villages/100 sq km), followed by Mizoram (4 villages/100 sq km). The average population per village is the highest in Kerala (17,162) and Chandigarh (28,991), followed by Pondicherry (4391), NCT of Delhi (3741) and Haryana (2413).

Density of Villages: The density of rural settlements at the state-level has been shown in (Fig. 14.14). It may be seen from (**Fig. 14.14)** that the density of villages is between 31 to 40 per 100 sq km in Assam, Bihar, Jharkhand, Uttar Prades, Odisha and West Bengal, while it is less than or equal to 10 villages per 100 sq km in Andhra Pradesh, Gujarat, Goa, Jammu and Kashmir and the hill states of North East India **(Fig. 14.14)**.

Morphology of Rural Settlements

The morphology of rural settlements deals with the land use and internal structure of the settlement. Each village of India has two components: (i) the built-up area (*Basti*-settlement), and (ii) the open area or the fields.

(i) The built-up area consists of arrangement of houses, house of the village head-man (*Haveli*), cattle-sheds, roads, streets, ponds, cultural places like temples, mosques, guruduwaras or churches, shops, *chowks*, and

(ii) The open area (fields) consists of the fields which are work places where the cultivators grow their crops and the dairy-farmers keep their cattle.

Doxiadis, one of the leading settlement geographers has identified four main parts within the morphology of a settlement. These include: (a) fields or the homogeneous part, (b) roads, streets and lanes, (c) built up area or *Basti,* and (d) especial or cultural area consisting of religious places, *Panchayat Ghar,Chowk, Morung* (youth-club in tribal areas), and other cultural places **(Fig. 14.15)**.

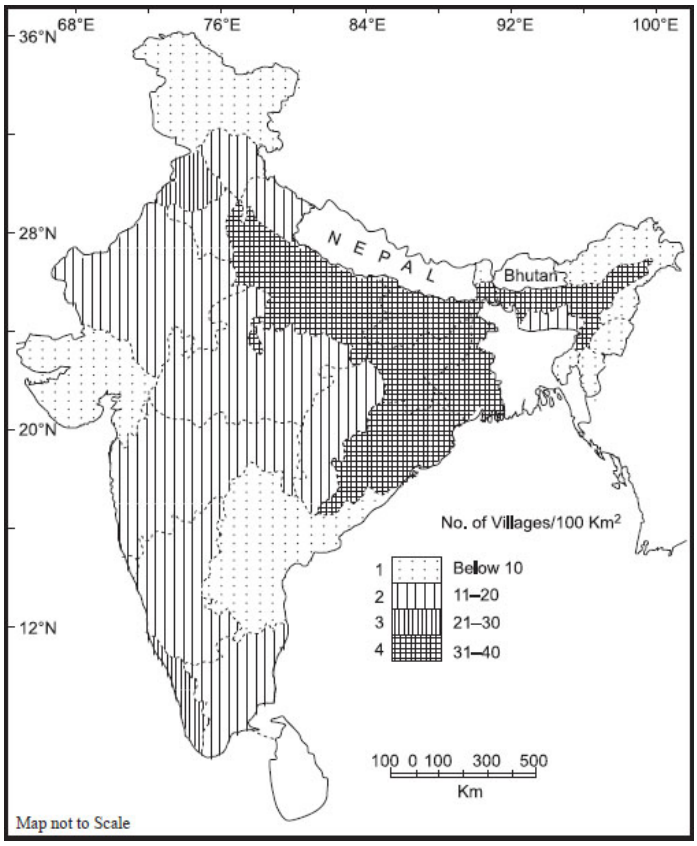


Fig. 14.14 India—Density of Villages (2011)

In almost all the settlements, the built-up area is surrounded by fields and pasture-lands, and the two are connected with each other with *Chak-roads* and *footpaths* (*Pagdandis*). The main site of the village, generally on an elevated area is occupied by the first settlers (upper castes and land lords; *Zamindars*). This area forms the nucleus of the settlement around which expands the village in the subsequent period. In the vicinity of the main settlement, there may develop some hamlets dominated by the lower castes and landless workers. These settlements in the vicinity of the main settlement are known by different names (*Mazra, Purva, Nagla, Tola*, etc.) in the Northern Plains of India. Around each settlement, there are concentric zones of soils. The agricultural land in the vicinity of the settlement is highly productive and is known as *Goira or Gohnda* in western Uttar Pradesh, Punjab and Haryana, and *Goind* or *Gauhan* in eastern Uttar Pradesh and Bihar. The outer fields which are less productive are called *Manjha* in Punjab, *Khalsa, Dahar*, *Raitla* or *Sotla* in Haryana and Uttar Pradesh. In almost all the compact villages of the Great Plains of India, the maximum socioeconomic activities are confined in or around the nucleus. The intensity of socioeconomic activities decreases while going away from the nucleus of a compact settlement.

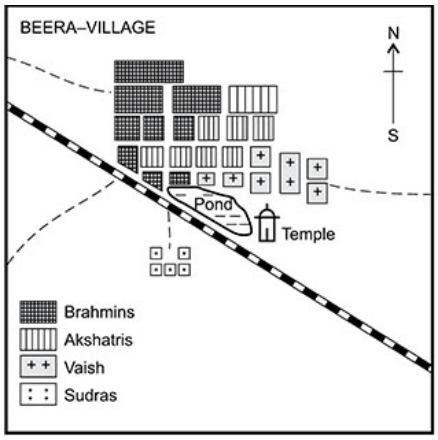


Fig. 14.15 Indian Village: Socio Spatial Structure

Socio-Spatial Structure of Rural Settlements

In the Northern Plains of India, it has been observed by some of the leading settlement geographers that there is a close impact of caste structure on the morphology of rural settlements. For example, the upper-castes like Brahmins, Rajputs, and Kayastha occupy the elevated best sites of the village and their houses are relatively large with courtyards and separate apartments for each married woman. In opposition to this, the low ranking castes, especially untouchables (*Sudras*) have single room huts/houses shared by all members of the family, and at times by cattle, and without much open space or a courtyard. The settlements of this type can be observed in the Ganga-Yamuna *Doab*, in the Rohilkhand Division as well as in Haryana and eastern Uttar Pradesh. Moreover, the lower castes generally occupy the peripheral areas. In fact, the Chamars, Mochis, Pasis, Jatavs, Mehtars, and Dhanuk construct their houses slightly away from the houses of the upper castes. The hamlets and *Purvas* of the lower castes are known as the *Ahiran*, *Chamarkheda, Jatvara, Lodhian, Pasiwada, Mochpura*, etc. These hamlets, together with their inhabitants, are closely linked with the main site under the *Jajmani-System* and act like a single functional unit. These two concepts of social space and functional integration have much relevance in understanding the socio-spatial structure of the Indian villages and have been displayed by Prof. K.N. Singh (1972) through his religio-ritual and secular dominance models **(Fig. 14.16)**.

Religio-Ritual Model of Rural Settlements

The Hindu society is divided into caste system, with each caste performing different socioeconomic functions. The caste system resulted in the labelling of the upper and lower castes in society. Ultimately, it resulted in the concepts of purity-pollution and untouchability. The rural settlements, as a result of the concept of purity and untouchabilty, led to the development of twin-settlements of caste Hindus and out-castes which are separated from each other more by social space than by any appreciable barrier. The Brahmins, being the priestly caste and performers of rituals and ceremonies, find place in almost all villages and so is the case with untouchables, whose labour and services make the foundation of the village economy. The segregation was much pronounced in the past favouring the outgrowth of hamleted settlements. In case of compact settlements, outcastes generally lived on the outer parts of the built-up area in a direction (south) less conducive for wind movement; for even air was believed to get polluted after coming in contact with a *Shudra's* body. How and why these traditions came into being is a matter of debate? **(Fig. 14.16)**.

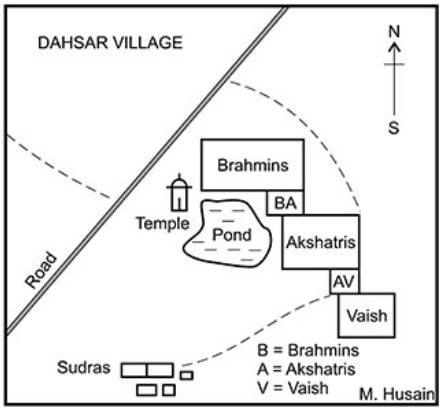


Fig. 14.16 Morphology of Rural Settlements

The Secular Dominance Model

The Secular Model of settlements development as proposed by Prof. K.N. Singh states that different castes constructed their houses closer to each other out of economic needs. The land-holders (mostly the upper castes) for their agricultural work had to depend on landless labourers and workers of low castes, while the lower castes had to depend on the upper castes to earn their livelihood and for their survival. This reduced the distance between these two social groups making the settlement compact, nucleated and unified. There are instances when land owning upper castes invited various service castes to settle within the village territory for carrying out the functions of *Jajmani System.* This functional interdependence is seen even in case of certain Muslim villages.

Either of the two models is notable for explaining the present socio-spatial structure of the villages independently. It is their joint pull which gives distinct pattern to the village morphology, and under the stresses of new socioeconomic orders, the old system is gradually losing its importance. The *Jajmani System* is now a decayed institution and the rigidity of the caste system is fading out. The rise in the socioeconomic conditions of the *Dalits* due to improvement in their education level has made them conscious about their rights due to which the traditional barrier is breaking down.

Environmental Issues in Rural Settlements

Rural settlements are natural growth in their physical environment and cultural setting. They do not have well-defined shapes and a distinct internal structure. The layout, morphology, design, architecture and direction of windows, doors and ventilators are directly controlled by the physical environmental factors and prevailing cultural traditions. In fact, the configuration of site, surface and underground water, nature of soil, water bodies (ponds, rivers), field-patterns, roads, tracks, paths, and security from enemies and wild animals play a vital role in the origin, growth, development, and the sustainability of rural settlements.

In a rural settlement, a group of people, who have common ties with each other, live in the same place. Rural settlement is the smallest social group which caters for the daily social life of the inhabitants.

Rural settlements have traditionally been important to geography. The studies of rural settlements by Paul Vidal de la Blache and Albert Demangeon may be considered as pioneer works. The rural settlements, however, were neglected until the rural geography emerged in the 1970s.

At present, the study of rural settlements, people, places, and their environment with special reference to society and economy is an area of interest in which the geographers of the developed and developing countries are increasingly probing. Now, there is emphasis on rural settlement studies with special reference to land use, agriculture, forestry, conservation of environment, rural employment, energy, housing, recreation, health, education, and tourism.

The rural settlements grow from the available environmental resources, i.e., soil, mud, thatch, stones, wood, bricks, etc. In some economically well off areas, however, houses are now made of burnt-bricks, cement, and concrete.

The impact of environment on the rural house types, their designs, and settlement patterns may be appreciated from the following:

1. In a traditional house in northern India, a courtyard, verandah, and **Kothah** or **Dalan** (room) are made in accordance with the climatic conditions.
2. The walls and roofs are also made according to weather and environmental conditions.
3. The houses have separate space for storing agricultural produce and pens for draught animals and milch cattle.
4. The direction of main door of the houses and the lay-out of the house depends on the environmental conditions. The cultural values, traditions, rituals and superstitions are also considered in the lay-out, placement of doors, and windows of the house.

Physiographic Divisions and Indian Rural Settlements

Predominantly, India is divided into six physiographic divisons, namely:

(i) The Himalayan Region
(ii) The Great Plains of India
(iii) The Indian Desert
(iv) Peninsular India
(v) The Coastal Plains
(vi) The Indian Islands

1. In the hilly and mountainous regions (like Himalaya and north-eastern hills), the roof is slanting so that the water of heavy rains and snowfall may slip downward without damaging the house.
2. The northern plains have three distinctive seasons—summer, **Barsat** (rainy season), and winter. According to the seasons, there is an elaborate courtyard in the houses of northern India, there is a verandah for the rainy season and a cosy room for the winter season.
3. In the desert areas the houses are made up of mud and stones with thick walls. This retains the heat during the daytime and keeps the inside of the houses cool. These houses are such that the heat is reflected in summers and keep the cold outside during the winters.
4. In the peninsular region, the houses usually have two huts, placed generally at right angles and enclosed by a fenced rectangular courtyard. In Kerala, Tamil Nadu, Karnataka, Andhra Pradesh, and Odisha, a typical house consists of two huts facing each other and the courtyard encloses them on two sides.
5. In the coastal areas and the islands, the earthen houses with flat or conical roofs are the common features. The houses and settlements of the cultivators differ from those of the fishermen.

The Salient Features of Indian Villages

Following are the salient features of Indian villages:

1. The domestication of plants and animals in India is more than 7000 years old and same is the history of Indian villages.
2. The Indian villages are considered as the bases of Indian culture.
3. The building materials include mud, unbaked and baked bricks, stone pieces, thatch, wood and grasses.
4. Villages are generally surrounded by grooved of trees and orchards. Coastal villages usually have trees of coconut-palm, arecanut, and banana.
5. Ponds, temples/mosques, well, chaupals, graveyards, and cremation grounds are public places shared by the entire community of the village.

6. Villages are, by and large, unplanned. They do not have planned streets. Heaps of garbage and absence of toilets are still the typical characteristics of Indian villages.
7. The central and elevated parts of the village settlement are generally occupied by the cultivators of upper castes.
8. A courtyard is a typical feature of an Indian village. Provision is made in the courtyard for keeping livestock fodder, agricultural implements, etc.
9. Village settlements are joined with each other with unmetalled or metalled roads.
10. Mutual cooperation and reciprocal aid system are the cornerstone of the Indian rural life.
11. Indian villages are, by and large, tradition bound.

Environmental Degradation and Rural Settlements

It is often believed that rural settlements cause very little environmental degradation. The rapid growth of population and scarcity of infrastructural and social amenities have degraded the rural environment substantially. Apart from water supply and sanitations, some of the important problems of rural India are:

(i) Lack of running water supply and sanitation using sewers.
(ii) Protection and maintenance of open areas.
(iii) Restoration of degraded land.
(iv) Forests and groves management.
(v) Silting of ponds and water bodies.
(vi) Sustainable approach to controlling floods.
(vii) Irrigation related problems—seepage, salinity, and waterlogging.
(viii) Soil erosion and soil degradation.
(ix) Pollution of surface and underground water. Pollution of the underground water due to heavy application of chemical fertilisers, insecticides and pesticides.
(x) Problem of spillovers from urban industrial areas.
(xi) Air pollution from the paddy fields due to release of methane (a greenhouse gas).
(xii) Burning of traditional fuel for cooking and heating leads to indoor and outdoor air pollution.
(xiii) Depletion of common propertyland.
(xiv) Over-exploitation of available resources.

Urban Settlements

Urban settlement is a large compact (nucleated) settlement in which most of the work-force is engaged in the secondary (industries) and tertiary (service) sectors. There is no uniform criterion to define an urban place. It differs from country to country and from one state to another. Thus, to some extent, there is some degree of arbitrariness in the definition of urban and rural places. The urban places are, however, generally defined on the basis population size, population density, occupations of inhabitants, and the type of local government.

Census Definition of Urban Places

The Census of India until 1951 defined an urban settlement as:

1. All places with municipality, corporation, cantonment board, and notified areas.
2. All other places which satisfy the following conditions:
   (i) A minimum population of 5000.
   (ii) At least 75% of the male working population engaged in non-agricultural economic activities (industrial and service sectors).
   (iii) A density of population of at least 400 persons per sq km.

In the Census of 1981 some more changes were incorporated whereby livestock, forestry, fishing, hunting, plantation and orchard settlements were treated as agricultural activities. Places having distinct urban characteristics and civic amenities like industrial area, railway colonies, places of tourist

interest, health resorts, and project sites were regarded as urban places at the discretion of the Director General of Census Operation in consultation with the concerned state government.

It has been said that urban places as defined by the Census Department are vague because some of the marginal cases may be deleted or added at the discretion of the Director General of Census Operation. The state-level officers exercise their discretion in an arbitrary manner which is not objective and scientific.

The above criteria, based on statistics, had certain limitations given below:

(i) There was an element of arbitrariness in the definition of urban place.

(ii) The decisions about the 'big village' and 'small towns' rests largely on the discretion of Census Superintendent.

(iii) The aggregate urban population from decade to decade was based on individual judgement of scores of Census Officials.

(iv) The census definition also laid emphasis on urban characteristics (social amenities) like water, electricity, schools, post-offices, and hospitals etc. But there was no specific list available of these amenities, and everything was left to the discretion of the Census authorities.

(v) The density of population of 400 persons per sq km is unrealistically low in the Indian context, and states like Bihar, Haryana, Kerala, Maharashtra, Uttar Pradesh, and West Bengal had urban densities well over 1000 persons per sq km. A much higher value (around 1000 persons per sq km) would be more appropriate in the Indian situation.

The Census Department of the Government of India has classified the urban places under the following six categories:

Class I : Population of 100,000 and above
Class II : Population of 50,000 to 99,999
Class III : Population 20,000 to 49,999
Class IV : Population 10,000 to 19,999
Class V : Population 5000 to 9999
Class VI : Population less than 5000

Apart from the Census Department of the Government of India, geographers have classified urban places into the following five categories:

1. **Urban Village:** In an urban village most of the working population is engaged in non-agricultural activities. The land use is, however, dominated by the agricultural and allied economic activities. It denotes the transitional stage of urban development wherein both the rural and urban functions are found in a mixed form.
2. **Town:** A general name for an urban place, usually a settlement with a population of less than one lakh. The size of towns differs from country to country and no specific size range is generally accepted to distinguish a town from the city. In India, a town generally has a municipality. It is the smallest unit of urban settlement in which urban functions are quite conspicuous. Some agricultural and allied activities like vegetable cultivation also engages the work force.
3. **City:** A city is a full-fledged urban agglomeration showing the predominance of secondary and tertiary occupations and complex internal structure. In India any town with a population of one lakh or more is termed as a city.
4. **Metropolitan (metro) city:** It is a region of densely populated urban core and its less-populated surrounding territories, sharing industry, infrastructure and housing. They form the economic and political regions. It has jurisdiction and municipalities-neighbourhoods, townships, cities, towns, exurbs, suburbs.
5. **Megalopolis:** A Greek word (combining the terms for great and city) adapted by Jean Gottmann (1964) to describe the urban complex of the north-eastern board of the U.S.A. (see

conurbation).

6. **Conurbation:** A term coined by Patrick Geddes to describe a built up area created by the coalescence of once separate urban settlement through Ribbon Development along the main inter-urban routes. With great urban sprawl, the term has now largely been replaced by concepts such as Daily Urban System, Megalopolis, Metropolitan Area, and Metropolitan Labour Area.
7. **Cosmopolis:** It is a large city inhabited by people from many different countries.

In India, conurbations are found around Delhi, Kolkata, and Mumbai. In all these mega-cities, two or more than two million cities have merged in such a way that an ordinary person cannot differentiate the territorial jurisdiction of either one of them.

In India, due to the paucity of space, transport bottlenecks, power breakdown industrial recession, industrial lockouts, and emergence of slums, the conurbations are facing a number of problems which have their bearing on the urban environment and quality of life of the inhabitants. In fact, life in the conurbations of India is most expensive and less enjoyable.

Urban Growth and Development in India

Development of urban places is quite old in India. India is a country in which urban centres and urbanisation flourished as early as 3000 BC. The urban centres of Mohenjodaro and Harappa may be cited as examples of pre-historic urbanisation in India. During the ancient period the cities of Harappan civilization flourished for about 600 years between 2350 BC and 1750 BC. Important towns of Harappan culture are Kalibangan (Rajasthan), Lothal Rangpur, Rojdi (Gujarat), Banwali (Haryana) and Rupar (Punjab). During the period of Aryans the major cities developed include Ayodhya, Kapilavastu, Kushinagar, Patliputra, Rajgir, Varanasi, Vaishali etc. During the Medival period ths Turks and Mughals developed the cities of Agra, Ahmadabad, Ahmadnagar, Aurangabad, Bijapur, Bikaner, Hyderabad, Jaipur, Jodhpur, Kota, Moradabad, Mysore, Pune, Raipur, Saharanpur, Shahjanpur, etc. The cities of Kolkata, Madras, Mumbai, Dehradun, Mussoorie, Nainital, Darjeeling, Dalhousi, Shillong. Ooty, Panaji, Pondicherry, Kanpur, Asansol, Ranchi, Siliguri, Dhanbad, etc. include Bhilai, Chandigarh, Raurkela, Durgapur, Lavasa, etc.

Throughout the Indian history, a number of towns and cities developed mainly because of the socioeconomic, geo-political and cultural factors. The arrival of the English people and their capture of power through the East India Company led to the development of many cities and towns. The Britons established trading centres to export raw materials from India to Britain and to sell their finished products in the huge market of the subcontinent. They also established many cantonments at strategic points. Under the planned economy of the Government of India after Independence, many new industrial, capital, and planned cities were developed.

The growth of urban places in India may be examined in the following historical phases:

1. **Ancient Cities:** Most of the cities of the ancient period were the capitals of Hindu kings. The centre of the city used to be occupied by the royal palace. Near the palace were the cultural places like temples and the residences of high ranking Brahmins, and the houses of the ministers of the royal cabinet. The *Shudras* (untouchables) were given the southern side during the period of Maurya Empire. Ayodhya, Bodh-Gaya, Hastinapur, Kanchipuram, Kapilvastu, Madurai, Mathura, Patliputra (Patna), Rajgir, Vaisali, Varanasi, Ujjain, and Uraivur are some of the important towns and cities of the ancient period **(Fig. 14.17).**

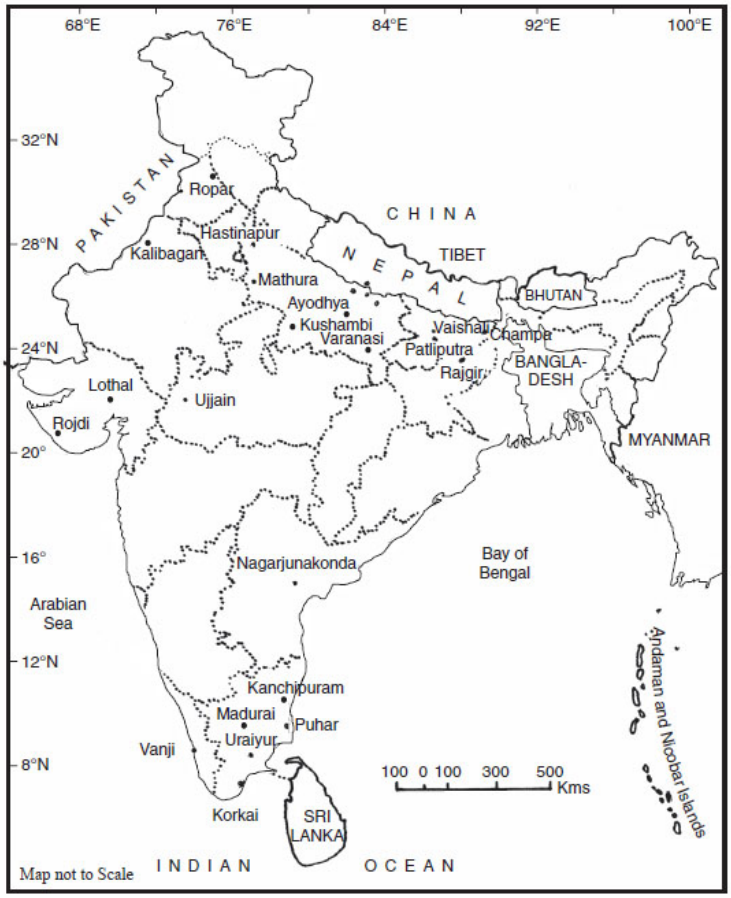


Fig. 14.17 Ancient Cities (2350 B.C. 300 B.C.)

2. **Medieval Cities:** During the medieval period of Indian history, the Muslim imprint on the city structure is significantly conspicuous. The Muslims introduced fortifications, mosques, *bazaar* (markets) and *chowk,* and residential segregation in their cities.

   The walled city of Shahjahanabad (Old Delhi) is a typical example of the medieval town **(Fig. 14.18)**. The city of Shahjahanabad was built on the bank of river Yamuna. Its architecture was a fusion of Islamic and Hindu influence. The cities of the medieval period were surrounded by brick walls without a moat. Every city had a market centre—the main chowk (cross-roads) of the city. The cities of Adilabad, Agra, Ahmadabad, Ahmadnagar, Aligarh, Allahabad, Aurangabad, Bareilly, Bhopal, Bikaner, Bijapur, Burhanpur, Hisar, Indore, Jaipur, Jodhpur, Kota, Lucknow, Ludhiana, Moradabad, Murshidabad, Nizamabad, Rampur, Sagar, Saharanpur, Shahjahanpur, Sikandrabad etc. are some of the important cities developed by the Muslim rulers during the medieval period (**Fig. 14.19)**. All the cities had gates for regurated entries.

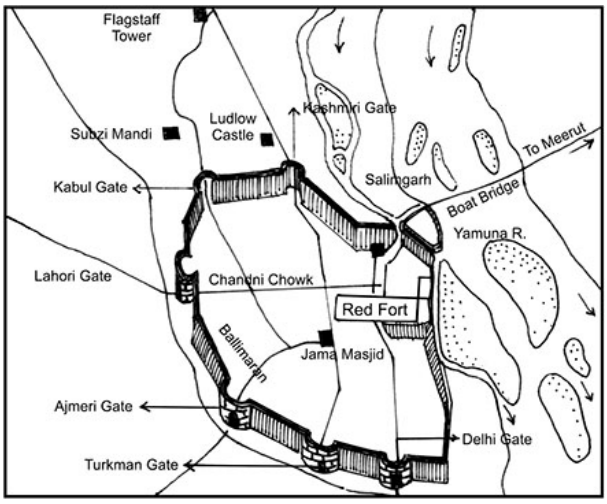


Fig. 14.18 Shahjahanabad—Old Delhi

Kashmere Gate

Built in 1835 by British Major, Robert Smith, it lies towards the northern end of the city near ISBT in Delhi. The emperors used this gate as exit for visits to the Northern India.

Delhi Gate

It is also known as 'Dilli Darwaza' and lies towards southeastern end of the city near Daryaganj. It was built in 1638 by the emperor Shahjahan. It was used by him to visit the Jama Masjid for his prayers. It is deemed as a protected heritage site and maintained by ASI.

Fig. 14.19 Medieval Cities

Ajmeri Gate

This gate lies to the south-western end of the city and is presently located near the New Delhi Railway Station. It was built in 1644 and has witnessed the first battle (Sepoy Mutiny) for independence of 1857.

Turkman Gate

The gate lies on the southern end of the medieval city of Shahjahanabad. It was constructed in 1650. It was named to honour a holy saint Hazrat Shah Turkman Bayabani).

Gates of Red Fort

Delhi Gate and Lahore Gate are maintained gates of the seventh city. The Lahore gate connects the Lahori Bazaar with Chandni Chowk. It has its name since it faced in the direction of Lahore. Delhi Gate was the main entry point of the Mughal army. These gates were secured to the prime minister to hoist the national flag on the Independence Day.

Nigambodh Gate

It lies on the eastern end of the city and is situated near the Yamuna market. The gate was named after the burial ground near the ghats. ‘Nigam’ means ‘Ved’ and ‘Bodh’ means ‘knowledge’.

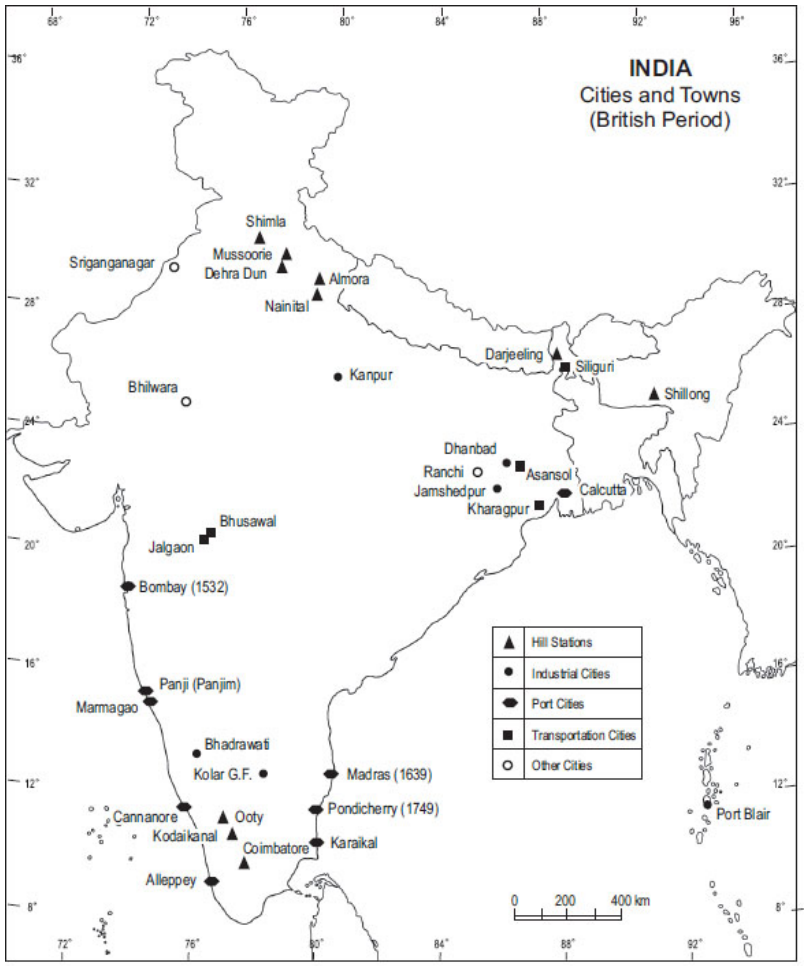
Khooni Darwaza

It is of historical significance. It witnessed the bloodshed during 1857 Sepoy Mutiny. It is located on the Bahadur Shah Zafar road and lies opposite to Feroz Shah Kotla. It was named so because it led to Kabul in Pakistan.

Bahadur Shahi Gate

It was constructed between 1854 and 1855 by the Emperor Bahadur Shah Zafar. It was an entry point from the northern end of the Shalimar Fort. It was made mainly from bricks.

3. **Cities of British Period:** The British modified the traditional cities of India of the ancient and medieval period. The development of railway lines and increasing trade and commercialisation resulted in the development of new patterns of urban settlements. Some of the important cities developed by the Britons were Coimbatore, Dehradun, Dalhousie, Darjeeling, Gulmarg, Jamshedpur, Kanpur, Kharagpur, Kullu, Mumbai, Chennai, Kolkata, Manali, Marmagao, Mussoorrie, Nainital, Panaji, Ranikhet, Shimla, and Shillong. Just to have segregated residential areas for the whites, they developed the civillines and cantonments. The demolition of fortification walls meant removal of barriers, and new extensions of the cities began to grow along the main road and railway lines. Further, commercialisation added new nodes in the city in the form of Central Business Districts (CBDs), like Connaught Place in Delhi. Mall Roads with green-belts on both sides, clubs for recreation, churches and well maintained gardens, parks, and graveyards were added in the urban places by the Britons. Thus, the traditional morphology of the Indian cities was transformed substantially **(Fig. 14.20).**



Map not to Scale

Fig. 14.20 Cities and Towns of British Period

4. Cities Developed after Independence A number of cities and towns were developed by the national and the state governments after Independence. These are well planned cities which have been provided adequate social amenities. The cities of Bhilai, Bhubaneshwar, Bokaro, Chandigarh, Dispur, Faridabad, NOIDA, Gandhinagar, Gurgaon, Lavasa, Panchkula, Pantnagar, Rourkela, etc., are some of the important cities developed during the post-Independence period **(Fig. 14.21)**.

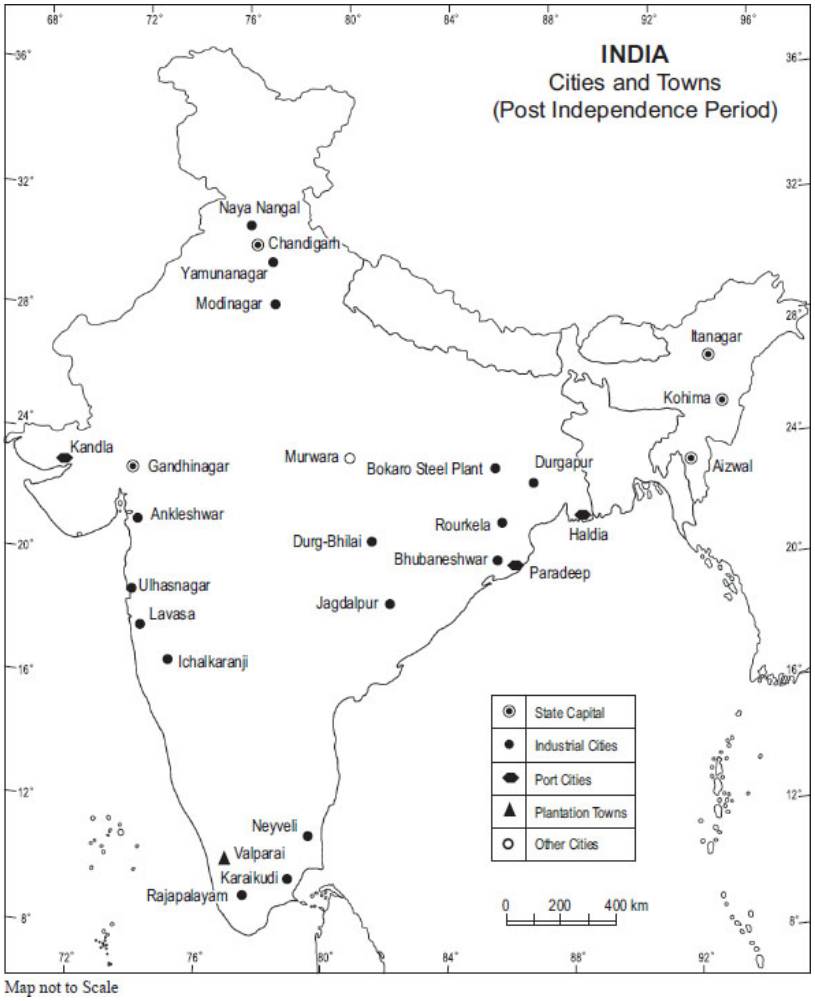


Fig. 14.21 Cities and Towns of Post Independence Period

The growth of urban population and its percentage share in the different years have been given in (**Table 14.2**).

Table 14.2

India: Trend of Urbanisation—1881–2011

Year	Urban Population (in Million)	Decadal Growth (Per cent)	Urvbanisation (Per cent)	No. of Towns
1881	9.3			
1891	9.4			

1901	25.85	–	11.0	1917
1911	25.95	0.35	10.4	1909
1921	28.09	8.27	11.3	2047
1931	33.46	19.12	12.2	2219
1941	44.15	31.97	14.1	2424
1951	62.44	41.42	17.6	3059
1961	78.93	26.41	18.3	2699
1971	109.09	38.22	20.2	3119
1981	156.19	46.02	23.7	4019
1991	217.61	39.32	25.7	4689
2001	286.12	31.48	27.8	5167
2011	377.00	31.00	31.16	7935

Source: Census of India, 2011

It may be seen from (**Table 14.2)** that in 1881 the urban population in India was only 9.3% of the total population which grew to 31.16% in 2011. The growth trend of urban population is however, not steady. Looking at the (**Table 14.2)**, three distinct phases of urban growth may be observed in India:

1. **Period of Slow Urban Growth:** The period from 1881 to 1931 may be termed as the period of slow urban growth. The frequent occurrence of natural calamities like drought, floods, famines, epidemics, and high mortality rate retarded the urban growth of population. During the fifty years period (1881–1931) the urban population grew at an average annual growth rate of less than one per cent.
2. **Period of Medium Growth Rate:** The period between 1931 and 1961 was the period of medium urban growth. The urban population of India grew at a relatively faster pace. For example, in 1931 the total urban population of India was 33.45 million (12.2%) which increased to 78.93 (18.3%) by 1961. During the planned period after Independence a number of new industrial and urban centres were established.
3. **Period of Rapid Urban Growth:** Between 1961 and 2011 the urban population of India grew at a faster pace. It may be seen from (**Table 14.2)** that the total urban population in 1961 was only 18.3% of the total population of the country which grew to 31.16% in 2011. At present the country is passing through a phase of rapid urban growth as the rural population is migrating towards the big urban areas in search of employment and better opportunities in life.

The national average of urban growth during the period of 1961–2011 was over 8%. At the state level, Manipur has recorded the highest increase in urban growth (19%) during 1961–2001, followed by Nagaland (15%) and Kerala (12%). The growth of urban places was stunted in Assam, Bihar, Jammu and Kashmir, Himachal Pradesh, Odisha, and Sikkim, etc.

Regional Patterns of Urbanisation

The regional patterns of urbanisation have been plotted in (**Fig. 14.22)** and the relevant state-wise data has been given in (**Table 14.3)**.

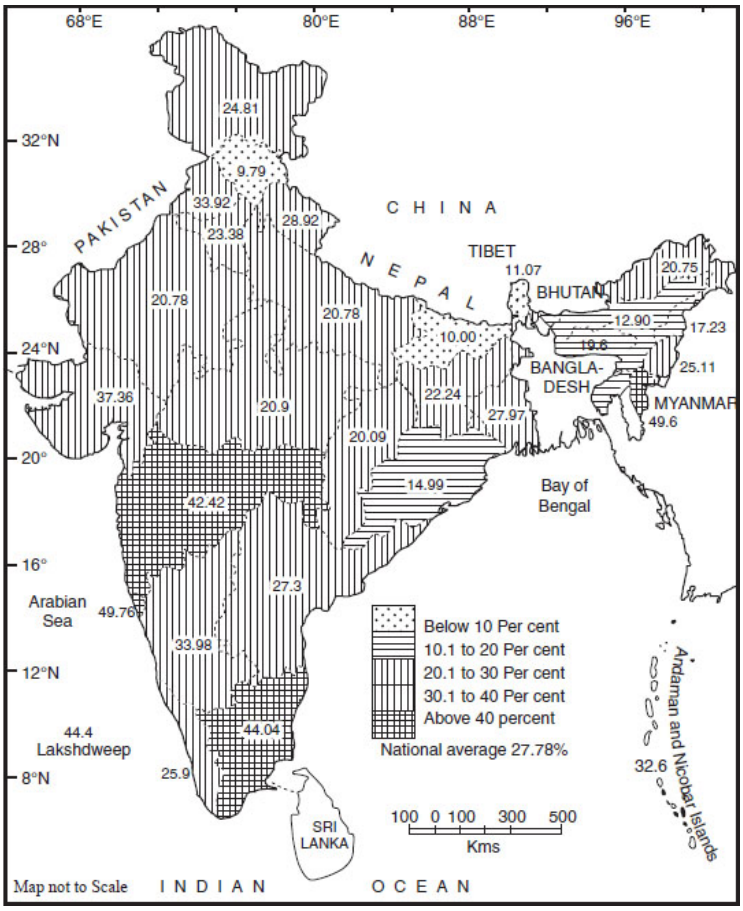


Fig. 14.22 India Urbanization–2011

Table 14.3

India—Urban Population 2011

State/Union Territory	Total Population	Urban Population	Percentage of Urban Population to Total Population
States			
Andhra Pradesh including Telangana	84,665,533	28,353,745	33.48
Arunachal Pradesh	1,382,611	313,446	22.67
Assam	31,169,272	4,388,756	14.08
Bihar	103,804,637	11,729,609	11.30
Chhattisgarh	25,540,196	5,936,538	23.24
Goa	1,457,723	906,309	62.17
Gujarat	60,383,628	25,712,811	42.58
Haryana	25,353,081	8,821,588	34.79
Himachal Pradesh	6,864,602	688,704	10.04

State/Union Territory	Total Population	Urban Population	Percentage of Urban Population to Total Population
Jammu & Kashmir*	12,548,926	3,414,106	27.21
Jharkhand	32,966,238	7,929,292	24.05
Karnataka	61,130,704	23,578,175	38.57
Kerala	33,387,677	15,578,175	47.72
Madhya Pradesh	72,597,565	20,059,666	27.63
Maharashtra	112,372,972	50,827,531	45.23
Manipur	2,721,756	822,132	30.21
Mizoram	1,091,014	561,997	51.51
Meghalaya	2,964,007	595,036	20.08
Nagaland	1,980,602	573,741	28.97
Odisha	41,947,358	6,996,124	16.68
Punjab	27,704,236	10,387,436	37.49
Rajasthan	68,621,012	17.080,776	24.89
Sikkim	607,688	151,726	24.97
Tamil Nadu	72,138,958	34,949,729	48.45
Tripura	3,671,032	960,981	26.18
Uttar Pradesh	199,281,477	44,470,455	22.31
Uttarakhand	10,116,752	3,091,169	30.55
West Bengal	91,347,736	29,134,060	31.89
Union Territories			
Chandigarh	1,054,686	10,25,682	97.25
Delhi	16,753,235	16,333,916	97.50
Daman & Diu	242,911	182,580	75.16
Lakshadweep	64,429	50,308	78.08
Pondicherry	1,244,464	850,123	68.31
Dadar & Nagar Haveli	342,853	159,829	46.62
Andaman & Nicobar Islands	379,944	135,533	35.67
India	1,210,193,422	377,105,760	31.16

Source: Census of India 2011.
**** Inclusive of Ladakh.***

An examination of (**Table 14.3)** vividly shows that the urban population of India is highly unevenly distributed. The state of Goa is the most urbanised state of the country with about 62% of its total population as urban. Moreover, the states of Mizoram (51.51%), Tamil Nadu (48.45%), and Maharashtra (45.23%) are the highly urbanised states of the country, followed by Gujarat (42.58%), Karnataka (38.57%), and Punjab (37.49%). On the contrary, Himachal Pradesh (10.04%), Bihar (11.30%), Assam (14.08%), and Odisha (16.68%) are the least urbanised states of India.

Among the Union Territories, Delhi (97.5%), Chandigarh (97.25%), Pondicherry (68.31%), and Lakshadweep (78.08%) have a high concentration of urban population.

Among the major states, the largest concentration of urban population is in Maharashtra being 13.48% of the total urban population of the country, followed by Uttar Pradesh (11.79%), Tamil Nadu (9.27%), and West Bengal (7.73%).

The urban population of different sizes of urban places and the temporal change in the number of urban centres have been given in (**Table 14.4).**

Table 14.4

India—Temporal Changes in Urban Places:1901–2001

Size Class	Proportion of Urban Population				Number of Urban Centres			
	1901	1961	1991	2001	1901	1961	1991	2001
Class I	22.93	48.37	65.20	62.32	25	107	300	423
Class II	11.84	11.89	10.95	12.02	42	139	345	498
Class III	16.50	18.53	13.19	14.79	136	518	944	1386
Class IV	22.06	13.03	7.77	7.87	395	820	1170	**1660**
Class V	20.38	7.23	2.60	2.73	756	847	740	1057
Class VI	6.29	0.95	0.29	0.27	494	268	198	227
India	100.00	100.00	100.00	100.00	**1848**	2699	3697	**5161**

Source: Census of India, 1971, 1981, 1991, 2001.

It may be seen from (**Table 14.4)** that in 1901 the total number of urban places was only 1848 in the country which rose to 5161 in 2001. It is also interesting to note that the number of Class IV urban places, having a population between 10,000–19,999 was the highest being 1660 in 2001, followed by the Class III (1386) and Class V (1057). The Class VI towns are decreasing fast which were 494 in 1901 and decreased to 227 in 2001 **(Table 14.4).**

Salient Features of Indian Urbanisation

Rapid urbanisation in India is a recent phenomenon. Some of the salient features of Indian urbanisation are as under:

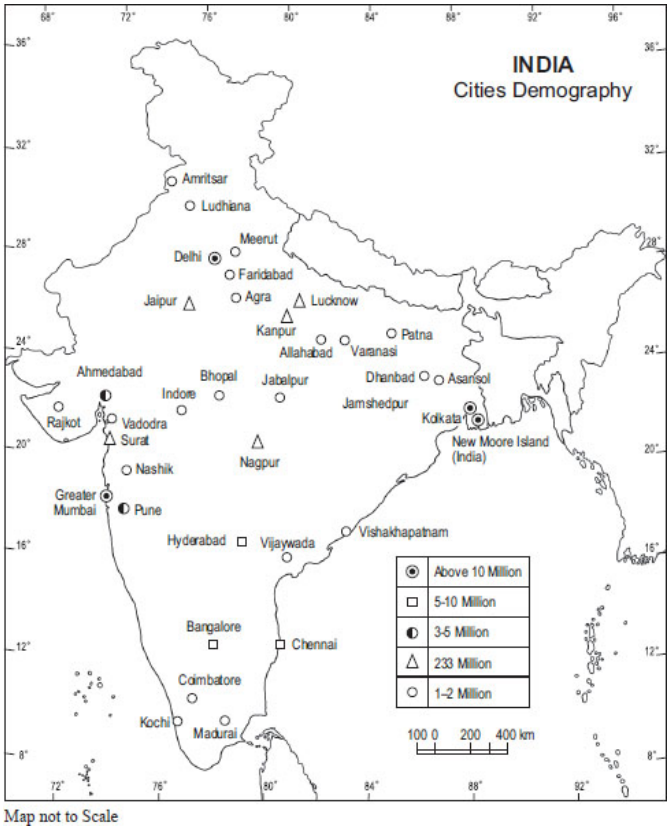
1. Urbanisation in India is as old as the Indus Valley Civilisation.
2. Rapid urbanisation in India was started during the Medieval Period, for which cottage industries and marketing were responsible.
3. Urbanisation during the 20^{th} century showed the fluctuating trends which were affected by droughts, floods, famines, and epidemics in the first part, and partition of the subcontinent in 1947.
4. The Indian urbanisation is of subsistence type whereby rural unskilled and semi-skilled labourers in-migrate in search of jobs and employment.
5. The big cities of India (Mumbai, Kolkata, Delhi, and Chennai) have shown stupendous growth of population.
6. Most of the big urban centres are growing fast because of tertiary sector. The industrial base is limited to a few industrial centres.
7. The southern and western parts of the country are more urbanised than the northern and north-eastern parts of the country.
8. Indian urban centres do not have a primacy at the national level. Primate cities are, however, found at the regional and state levels, e.g., Patna in Bihar, Srinagar in Kashmir Valley, Shillong in Meghalaya and Aizawl in Mizoram.

Mega Cities of India

At present any city with more than ten million population is known as a mega city. In 2019, there cities in India are termed as '*Mega cities*' namely Mumbai, Delhi, Bengaluru, Kolkata namely: Delhi, Mumbai, Kolkata, Bengaluru and Chennai. By the year 2030, the number of mega cities in India will be seven. The new mega cities added in the list will be Hyderabad and Ahmedabad.

Metropolitan Cities of India

According to the Census of India, a metropolitan city is an urban agglomeration/city having a population of one million or more. All the cities and agglomerations which have a population of more than a million are known as the million cities. These big cities are dependent on the surrounding rural areas for the supply of milk, vegetables, fruits, flowers, cereals, and work-force, and there is a lot of inflow of commuters. The metropolitan cities are essentially the products of centralisation of administrative, political, and economic forces in the country at the national and state capitals. There are 53 million cities in India (2011). The million cities (2001) have been plotted in (**Fig. 14.23),** while the ranks and total population of each of the million cities have been given in (**Table 14.5).**



Map not to Scale

Fig. 14.23 Million Cities 2001

Million Cities of India (2011)

As per the census of 2011, there are 53 million cities in India. Delhi with a population of 25 million is the largest agglomeration of India and the second largest is the world after Tokyo (38 million). Delhi's population has more than doubled since 1990, a U.N. report said. Delhi is expected to retain the spot of the world's second most populous city through at least 2030, when its population is expected to rise swiftly to 360 million. Tokyo and New Delhi were followed by Shanghai with 23 million and Mexico, Mumbai and Sao-Paulo, each with around 21 million inhabitants (Times of India, p.5, July 12, 2014).

But, when we consider urban agglomeration as an extended city comprising built up area of central core and any suburbs linked by continuous urban area, we have a change at the top. Delhi NCR, with the inclusion of Gurgaon, Faridabad, Noida, and Ghaziabad became the No. 1 Urban Agglomeration with a population of 21,753,486, ahead of Mumbai Metropolitian Region 20,748,395 comprising Mumbai, Navi Mumbai, Thane, Vasai-Virar, Bhiwandi and Panvel. Kolkata has clocked moderate growth.

Table 14.5

Million Cities of India–2011

Rank	City	2001 Census	2011	2011-Extended UA*
1	Delhi	12,877,470	16,314,838	21,753,486
2	Greater Mumbai	16,434,386	18,414,288	20,748,395
3	Kolkata	13,205,697	14,112,536	14,617,882
4	Chennai	6,560,242	8,696,010	8,917,749
5	Bangalore	5,701,446	8,499,399	8,728,906
6	Hyderabad	5,742,036	4.9 Crore	7,749,334
7	Ahmedabad	4,525,013	6,240,201	6,352,254
8	Pune	3,760,636		5,049,968
9	Surat	2,811,614		4,585,367
10	Jaipur	2,322,575		3,073,350
11	Kanpur	2,715,555		2,920,067
12	Lucknow	2,245,509		2,901,474
13	Nagpur	2,129,500	2,497,777	2,583,911
14	Ghaziabad(NCR)	968,256	2,358,525	(NCR)
15	Indore	1,506,062		2,167,447
16	Coimbatore	1,461,139		2,151,466
17	Kochi	1,355,972		2,117,990
18	Patna	1,697,976		2,046,652
19	Kozhikode	880,247		2,030,519
20	Bhopal	1,458,416		1,883,381
21	Thrissur	330,122		1,854,783
22	Vadodara	1,491,045		1,817,191
23	Agra	1,331,339		1,746,467
24	Vishakhapatnam	1,345,938		1,730,320
25	Malappuram	170,409		1,698,645
26	Thiruvananthapuram	889,635		1,687,406
27	Ludhiana	1,398,467		1,613,878
28	Kannur	498,207		1,642,892
29	Nashik	1,152,326		1,562,769
30	Vijayawada	1,039,518		1,491,202
31	Madurai	1,203,095		1,462,420
32	Varanasi	1,203,961		1,435,113
33	Meerut	1,161,716		1,424,908

Rank	City	2001 Census	2011	2011-Extended UA*
34	Faridabad (NCR)	1,055,938		1,404,653 (NCR)
35	Rajkot	1,003,015		1,390,933
36	Jamshedpur	1,104,713		1,337,131
37	Srinagar	988,210		1,273,312
38	Jabalpur	1,098,000		1,267,564
39	Asansol	1,067,369		1,243,008
40	Vasai-Virar(MMR)	.....		1,221,233 (MMR)
41	Allahabad	1,042,229		1,216,719
42	Dhanbad	1,065,327		1,195,298
43	Aurangabad	892,483		1,189,376
44	Amritsar	1,003,917		1,183,705
45	Jodhpur	860,818		1,137,815
46	Ranchi	863,495		1,126,741
47	Raipur	700,113		1,122,555
48	Kollam	380,091		1,110,005
49	Gwalior	865,548		1,101,981
50	Durg-Bhilainagar	927,864		1,064,077
51	Chandigarh	808,515		1,025,682
52	Tiruchirapalli	866,354		1,021,717
53	Kota	703,150		1,001,365

*urban Agglomeration

Source: Census of India, 2001 and 2011.

It may be seen from (**Table 14.5)** as per census 2011 that Mumbai is the largest urban agglomeration in India with a population of 18,414,288 million (2011), followed by Delhi (16.31 million), Kolkata (14.1 million), and Chennai (86.96 million). The population of Bangalore was more than eight million, while Ahmedabad (6.2 million) and Pune (5.0 million) are also growing very fast. The population of the remaining 27 cities varies between one and 4.5 millions.

Morphology of Indian Cities

Morphology of urban settlements means urban land use. Urban morphology comprises the structure of the city, open areas, industrial estates, business districts (markets), down-town (old city), residential areas, cultural centres, roads, play-grounds, water-bodies, parks, orchards, social forestry areas, and the green belt. The internal structure of all the cities differ from each other. In the context of India, Prof. E. Ahmad has identified some of the components of urban morphology as site characteristics, historical background, sky-line, green, open spaces, water-bodies, residential, and cultural areas. All these when combined constitute the physical morphology of an Indian town.

Morphology of towns in India has distinct phases like ancient, medieval, modern and recent including sprawls with the rapid increase in population after Independence. For a systematic study of the urban morphology of Indian cities, a historical perspective is most appropriate.

1. **Ancient Urban Morphology:** Most of the urban settlements in the Indus Valley had a typical morphology known as ‘grid-iron’ of main streets, running north and south and east and west, dividing the town roughly into blocks of almost equal size. The rectangular plan of the urban morphology was designed to conform with the needs of some sort of ceremony—religious or secular or both in which the terrace or terraces played a dominant role, and to which processional access was required. In the Vedic period, the town plans and their morphology represented ‘rectangular layout’ with its sides oriented to the four quarters and intersected by two avenues terminating in four gateways, the centre being occupied by a temple, palace, tank or garden.The cities like Ayodhya, Kanchipuram, Lumbini, Madurai, Mathura, Rajgir, Varanasi, Vaishali, and Ujjain represent such morphology of India cities.

2. **Morphology of the Medieval Towns:** The medieval towns adopted the grid-pattern of the ancient towns and added the forts, city walls, *Bazars* (markets), mosques, *chowks*, tanks or water-bodies. The cities of Agra, Ahmadnagar, Ahmadabad, Aligarh, shahjahanpur, Moradabad, Rampur, Saharanpur, Aurangabad, Bijapur, Bulandshahr, Chittorgarh, Delhi, etc., are some of the representatives of medieval urban morphology.
3. **Modern Development and Changed Pattern of Urban Morphology:** The rapid industrialisation and urbanisation of the second half of the 20th century has changed substantially the morphology of the India cities. The process of migration from the rural to the urban areas has made the old cities more crowded and rich people from the old cities constructed their houses in the peripheries.

Most of the Indian cities, even after about six decades of planning, display a duality in their morphology—the indigenous pattern as well as the alien pattern.

At present in India, 'a city unit', including areas around its nucleus, form the indigenous oriental type of common patterns visible in its uncontrolled growth and provide a mixed land use. They consist of numerous foci of activity flanked by irregular streets, nearly from all side by narrow tortuous, and at places, blind-curves, and lanes. The indigenous city unit is deeply influenced by phenomenon of caste and community, and living quarters in separate localities or residential areas. The central parts are mostly occupied by higher castes and business communities, while the lower and poorer people are largely confined to the outskirts.

One of the peculiar features of the cities developed by the British was the Central Business District (CBD). The CBD is the heart of an urban area, usually located at the meeting point of the city's transport systems, which contains the highest percentage of shops and offices. Land values in the CBD are high because of high accessibility, therefore, land use is at its most intense in order to offset rent cost. In consequence, in many countries development is upward rather than sideward. Within the CBD, specialist areas such as a jewellery or garment-making quarter, may arise in order to benefit from external economies. Vertical land use zoning is also common, so that retail outlets may be on the ground floor, with commercial users above them and residential users higher up. Connaught Place, one of the important CBDs of Delhi, has been shown in (**Fig. 14.24)**.

The alien pattern has developed into the civil lines, the cantonment, educational institutions, hospitals, and the railway colonies. Here, the areas are monotonously planned in European style with bungalows, barracks, and residential quarters along straight broad roads. The Civil Lines of Allahabad, situated to the east of New Cantonment was established on an extensive grid plan cut through by the main route which was related to the axis of the Old Mughal City of Allahabad**.** Within the area were all the standard representatives of English culture in India; the government offices, the law-courts, the hotels, the university, the educational institutions, the hospitals churches, and the Allahabad Club. It was neatly segregated: to the north was the River Ganga, while the new cantonment was to the west, and the railway line was to the south.

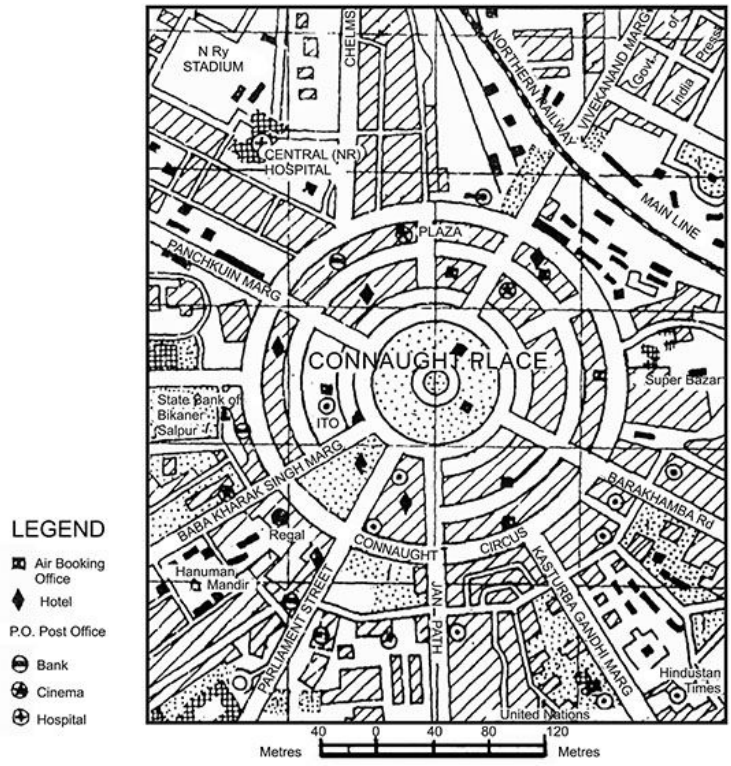


Fig. 14.24 Connaught Place—CBD of New Delhi

The railway, thus, completely shuts off the crowded and irregular Indian city (Old Allahabad) from the Civil Lines **(Fig. 14.25).** Thus, the two different cultures of the colonisers and colonised continued side by side.

It is vividly clear from the above discussion that morphology of a city is the product of both the historical, socioeconomic, and geographical processes. Its site, situation, nature of terrain, spatial limits and water-bodies are the important determinants of the morphology of an Indian city.

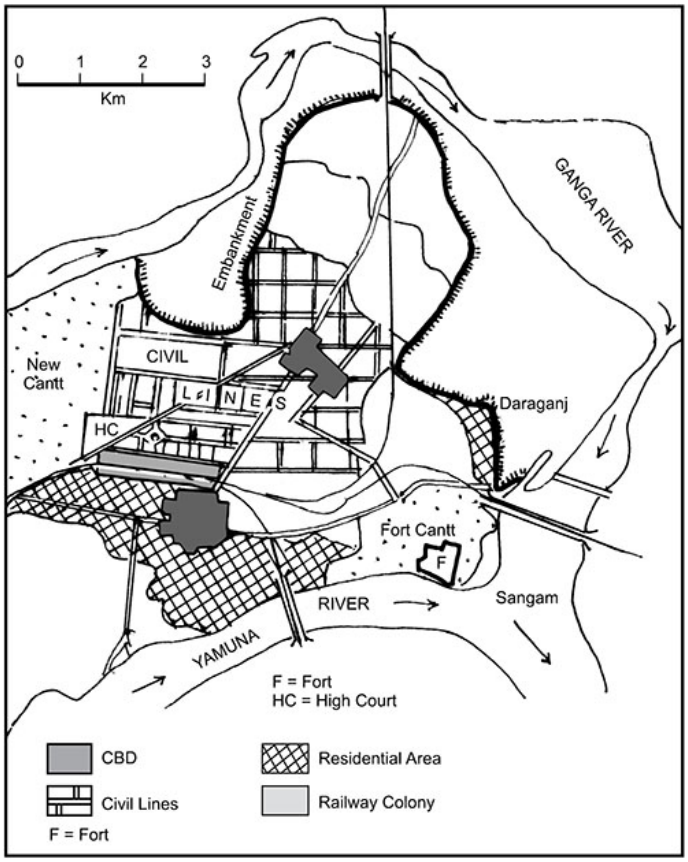


Fig. 14.25 Allahabad—Morphology 1950

Functional Classification of Indian Cities

Functions in geography means economic activities in which the working population is mainly engaged. A categorisation of cities according to the functions is known as the functional classification of towns. The urban functions are mainly non-agricultural, like administration, manufacturing, trade, commerce, defence, provisions of goods and services, communication and recreation. In reality, none of the cities is mono-functional. In other words, any one city may fulfill a number of functions.

The functional classification of cities attempts to categorise towns and cities according to their economic functions, thereby identifying their role within the urban systems.

The urban geographers have applied a number of techniques to classify the urban places on the basis of their functions. Most of the classifications, however, utilised the occupational data. Such data are processed by applying simple and sophisticated statistical techniques to ascertain the main functions of the cities. **Amrit Lal (1959)** was **the first to attempt a functional classification** of Indian cities. He used the location quotient (L.Q.) method to determine the functional classification of the Class I cities of India. According to Lal, all the Class I cities of India, except a few, are multifunctional in nature.

Qazi Ahmad (1965) used 62 variables to classify 102 Indian cities on the basis of their functions. Subsequently, *Ashok Mitra* (1971, 1973) used seven categories of workers as variables grouped into three major functional types, e.g., manufacturing, trade, transport and services.

In addition to Lal and Ahamd, functional classification of Indian cities was also made by Janki, Mahamaya Mukerjee, Anant Padmanabhan, Ashok Mitra, S.M. Rafiullah, O.P. Singh, Raza, and Agarwal. By applying the Maximum Positive Deviation Method of Prof. S.M. Rafiullah, the Indian cities may be classified under the following categories:

S.M. Rafiullah's Technique:

$$d = \sqrt{\frac{SD_N^2 - D_N^2}{N^2}}$$

1. **Administrative Cities:** The main function of the administrative cities/towns is to administer the country, state or any other administrative unit. It includes not only the capital cities of the country, but also all the centres of states, districts and other administrative divisional headquarters of the country. Administrative cities also have the legislative, executive and judiciary of the respective administrative unit. New Delhi, Mumbai, Kolkata, Chennai, Bangalore, Hyderabad, Lucknow, Jaipur, Patna, Gandhinagar, Bhopal, Chandigarh, Shillong, Dispur, Aizawl, Kohima, etc., are essentially administrative cities.
2. **Defence Towns:** The dominant functions in a defence town pertain to security and defence of the country. In fact, such towns are characterised with cantonments, barracks, military training centres, garrisons, air-force bases, air-fields, harbours, strategic locations, and naval headquarters. Adampur, Ambala, Halwara, Jalandhar, Jamnagar, Jodhpur, Khadakwasla, MOHO, Pathankot, Udhampur, Vishakhapatnam, etc., are some of the examples of defence towns.
3. **Cultural Cities:** These cities perform either religious, educational or recreational functions. The cities of Allahabad, Amritsar, Ajmer, Bodh-Gaya, Dharamshala, Gangotri, Hardwar, Kushipur, Nashik, Peerankalyar (Uttarakhand), Pushkar, Varanasi, etc., are the religious centres in which the religious rituals are performed, and the markets are full of religious books and accessories required for the religious rituals.

   The educational cities like Aligarh, Gurukul, Kharagpur, Pantnagar, Shantiniketan, etc. are some of the examples of educational cultural centres.
4. **Collection Centres:** The mining towns, fishing ports, lumbering centres are included in this category. The urban places of Zawar near Udaipur, Digboi in Assam, Ankleshwar in Gujarat, Bailadila in Chhattisgarh; Kathgodam, Haldwani and Kotdwar in Uttarakhand, Machlipatnam, Kakinada, Navsari, Mahe, Kozhikode, Cuddalore, etc., are some of the examples of collection centres.
5. **Production Centres:** The urban places which have manufacturing industries are included in the category of manufacturing cities. The manufacturing cities are generally well connected with the areas of raw material and the markets where the manufactured goods can be sold. Thus, these cities are well connected by roads and railways. Bhilai, Bhadrawati, Bokaro, Coimbatore, Dhanbad, Durgapur, Jamshedpur, Vijainagram, Vishakhapatnam, etc., are some of the important manufacturing centres of India.
6. **Transfer and Distribution Centres:** The main functions performed at the transfer centres are trade, commerce, and services. This category includes several categories of towns. The market towns are characterised by markets containing wide range of goods, godowns, cold storages, and wholesale markets. The most important commercial centres are Mumbai, Kolkata, Chennai, Ahmadabad, Gwalior, Indore, Ludhiana, Muzaffarpur, Phagwara, Surat, etc.
7. **Resorts:** The urban places which cater the recreation needs of people are known as resorts or recreation towns. These towns may be based on health-giving water (hot-springs), seaside-recreation, mountain-climbing, sports facilities, national parks, tiger reserves and places of aesthetic beauty. Bageshwar, Dehradun, Dalhousie, Darjeeling, Dharamshala, Gulmarg, Kullu, Manali, Mt. Abu, Nainital, Pahalgam, Panchmadhi, Ooty, Ranikhet, etc., are some of the examples of resort towns.
8. **Residential Towns:** Some of the towns and cities are developed just to provide residential accommodation to the urban people. In Delhi, Rohini, Indirapuram, Saraswati Vihar,

Zakirnagar, etc., are some of the examples of residential towns. Panchkula near Chandigarh, and Partapur near Meerut are essentially residential towns. Similar residential towns are found at the outskirts of Mumbai, Kolkata, Chennai, Bangalore, Hyderabad, Jaipur, etc.

9. **Seaports:** The basic task of seaports is to export and import goods. Diamond Harbour, Haldia, Kandla, Kochi, New Mangalore, New-Tuticorin, Okhla, Paradeep, etc., may be included in this category.
10. **Cities with Diversified Functions:** As stated, most of the cities and towns of India are multi-functional. The capital cities are also the commercial, manufacturing, cultural and recreational centres. The seaports are engaged in trade and commerce, beside cultural activities. Cities like Mumbai, Delhi, Kolkata, Chennai, Patna, Gauhati, Vishakhapatnam, Jaipur, Kanpur, Allahabad, and Varanasi, are performing highly assorted functions.

Many of the cities have changed their sites in the past. For example, Delhi has shifted its site at Indraprastha, Surajkund, Mehrauli, Sirifort, Kilokri, Firozshah-Kotla, Tughlakabad, Shahjahanabad and New Delhi. Some of the important cities of the past have disappeared as they lost their strategic, administrative, manufacturing or commercial importance.

Conurbation and Metropolitan Regions

The term conurbation was coined by Patrick Geddes to describe the built up area created by the coalescence of once separate urban settlements (million cities), initially through ribbon-development along the main inter-urban routes. With greater urban sprawl, the term has now largely been replaced by concepts such as daily system, megalopolis, metropolitan area and metropolitan labour area. In conurbation there is a substantial number of daily commuters.

Conurbation and metropolitan areas, having their origin in the USA, are used to describe very large urban settlements in which two, or more than two million cities merge with each other in such a way that an ordinary person may not know where one million city ends and the other begins.

The first and the largest conurbation of the world may be observed in the eastern sea-board of USA. In this region, there is a coalescence of the million cities of Boston, New-Haven, New York, Philadelphia, Baltimore, Washington D.C., Richmond, Charlotte, New Port, etc. **(Fig. 14.26**).

In India, a huge conurbation is in the process of development between Agra, Mathura, Hodal, Ballabgarh, Faridabad, Gurgaon, Delhi, Sonipat, Panipat, Karnal, Kurukshetra, Ambala, Chandigarh, Panchkula, and Kalak. From Delhi, one offshoot of this conurbation moves towards Ghaziabad, Muradnagar, Modinagar, Meerut, Modipuram, Muzaffarnagar, Roorkee, Hardwar, and Dehradun **(Fig. 14.27).**

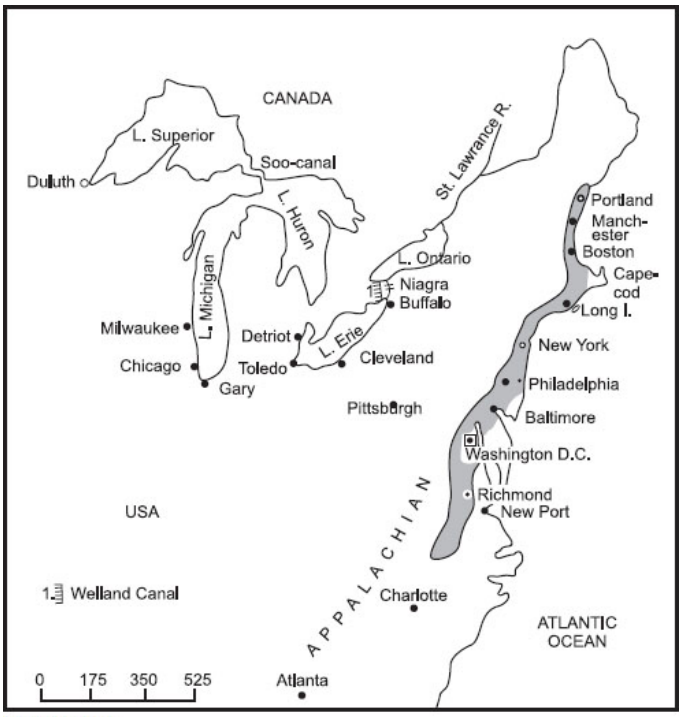


Fig. 14.26 Conurbation—USA

Conurbations and metropolitan areas have their own merits and demerits. Some of the merits and problems of conurbations have been briefly described in the following points:

1. Most of the metropolitan cities and conurbations are growing at an alarming pace. There is inadequacy of civic amenities and infrastructural facilities. Consequently, life in conurbations is not enjoyable.
2. The unplanned growth of population in conurbations is resulting into more urban slums and squatters. The population of slum dwellers in India has already crossed the mark of 200 million.
3. There is decay in the quality of urban environment. In fact, the conurbations in India are characterised by poverty, unemployment, insecurity, and high rate of crime.
4. Owing to the overlapping of urban influence, there is a serious problem of administration. The criminals after commiting the crime, cross the territorial limits of one city to another.
5. The enormous size of conurbation leads to traffic congestion, scarcity of water, electricity and environmental pollution.

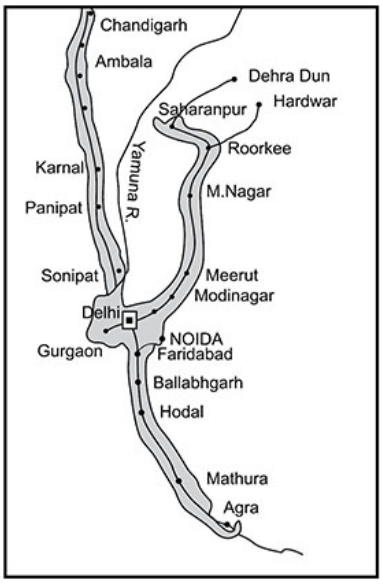


Fig. 14.27 Agra–Delhi–Chandigarh–Kalka–Meerut–Dehra Dun–Hardwar

Despite these shortcomings, the metropolitan areas and conurbations generate more employment to skilled, semi-skilled and unskilled workers. Moreover, the conurbations integrate the rural-urban cultures, and thereby help in solving the problems of unemployment and housing of the workers who commute to their suburbs and rural areas.

Metropolitan Regions of India

A metropolitan region is a geographical area which has functional links with a city. These links may relate to activities such as employment, education, health-care, banking, business, recreation etc. The strength of these links depends mainly on the cost of transport and time involved in travelling. These links also depend on the competition with other cities. A metropolitan area is a very large urban settlement usually with accompanying suburbs. The term is used rather loosely as no precise parameters of size or population density have been established. The metropolis in the USA is the equivalent of a conurbation.

The main metropolises of India are Mumbai, Kolkata, and Chennai. During the colonial period, these cities became India's economic backbone since the early 19^{th} century. To this list one should add Delhi, which became the national capital of British India in 1911. In fact, Delhi's importance in economic and regional development has been spectacular since Independence. At present, Delhi overshadows the colonial cities in the national spatial economy.

The four metropolitan cities, between them, account for over 85% of the richest people of the country. They also attract the greatest volume of traffic in terms of goods and people. Their area of dominance includes several states. The areas of influence of the four metropolitan cities have been demarcated on the basis of: (i) airtravel, (ii) railway passengers and commodity flow, and (iii) long-distance telephone calls. The spheres of influence of these metropolis have been plotted in (**Fig. 14.28).**

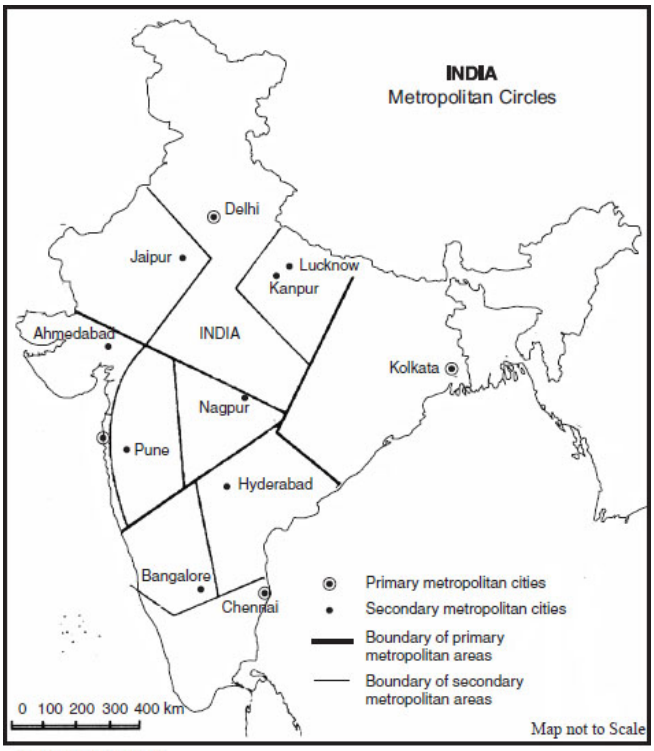


Fig. 14.28 Spheres of Influence of Four Metropolitan Cities

The Delhi metropolitan region covers the northern states/UTs of Jammu and Kashmir, Himachal Pradesh, Punjab, Haryana, Uttarakhand, western Uttar Pradesh, greater parts of Madhya Pradesh, and Rajasthan. The Kolkata metropolitan region includes all the states of North-East India (Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim and Tripura), Bihar, Jharkhand, Odisha, and West Bengal. In addition to these, the eastern parts of Uttar Pradesh and Chhattisgarh also form part of this region.

The Mumbai metropolitan region sprawls over the states of Maharashtra, Gujarat, parts of Rajasthan, and Goa and the Union Territories of Daman and Diu. It also includes the south-western parts of Madhya Pradesh, including the city of Indore, and Ujjain of the Malwa region. The Chennai metropolis region includes the four southern states of Tamil Nadu, Kerala, Karnataka, and Andhra Pradesh as well as the southern parts of Odisha, Chhattisgarh and Madhya Pradesh.

The city region concept provides an ideal framework for the regionalisation of the country into a set of hierarchical territorial divisions. Delineation of city region especially in a country like India is a tedious job, mainly because of the following:

(i) Clustering of Class I cities, having a population of one or more than one lakh.

(ii) Absence of one-lakh cities over vast areas in certain parts of the country.

The clustering of one-lakh cities may be found around some metropolitan areas, such as Delhi, Coimbatore, Patna, Agra, Kanpur, Kolkata, Lucknow, Jhansi, Meerut, Mumbai, etc. In the case of clustering, cities become complementary.

The city region in India, by and large, is much larger than an average district. In the more urbanised states of Tamil Nadu, Maharashtra, and Gujarat, majority of the districts have a big city within their territorial limits, while in the Northern and North-Eastern India, the city region encompasses several districts. The size of the city, therefore, bears an inverse relationship with the level of urbanisation. The hilly and mountainous regions of Himachal Pradesh, Jammu and Kashmir, and Uttarakhand have very few large-sized urban places.

At the meso-level, Srinagar in the Valley of Kashmir, Shillong in Meghalaya, Aizawl in Mizoram, Agartala in Tripura, Imphal in Manipur, and Gangtok in Sikkim are all primate cities in their own way.

CITY REGION AND PLANNING

The city region has often been recommended as a territorial unit for regional development planning. In fact, the city region has a number of advantages as a unit for development planning. Some of the merits of city region as a unit of economic and regional planning have been given in the following section:

(i) The city, being the only focal point, has an easy access to all places within the region. Even at the initial stage, its influence in terms of tertiary services extends to all parts of the region.

(ii) Economic and social planning can take the advantage of the existing links between the city and the countryside.

(iii) Development generally takes place in an uneven manner in which the city may become an initial nucleus of development.

(iv) Cities play an important role in bringing about changes in agriculture, employment, rural education and health.

(v) At present, cities have become the new centres of modern industry. The forces of urban and industrial agglomeration helps in the development of the city and also lead to the industrialisation of the countryside. Thus, the city region provides a convenient and sound spatial framework for such an approach to regional planning.

Urban Sprawl

The extension of the city into the countryside, particularly associated with improvement in mass transport, is known as urban sprawl. It is the physical expansion of the city both in terms of population and area. The main causes of urban sprawl are:

(i) large scale migration from the rural areas and small towns to the big cities,

(ii) migration from smaller towns to big cities, and

(iii) the natural increase in population.

The continuous in-migration of the people from the countryside results into the '*hyper-urban-growth'* of the city which has its own merits and demerits.

In developing countries like India, urban sprawl goes on largely unchecked. It generally takes place along the roads. In some cities, urban sprawl has been checked with the help of '*green-belts'*.

Around all the metropolitan cities and megalopolis, the suburbs develop along the main roads leading into town. This is known as *ribbon settlement*. The roadsides attract people, especially the shop keepers because of easy accessibility. Subsequently, the demand for suburban homes causes the land between ribbon settlements to be built and made accessible by constructing new roads. This type of development is known as *infil* . Simultaneously, small towns and villages within the commuting distance of the major cities are also developed for residential purposes. In this way towns are continuously growing, and in some areas the suburbs of a number of neighbouring cities amalgamate so as to make conurbations. Unfortunately, urban sprawl takes place at the cost of highly productive agricultural land, creating the problem of scarcity of agricultural land, pollution, insecurity, crimes, and environmental degradation.

In India, the process of urban sprawl is a phenomenon of the twentieth century. The first large flow of migration towards the cities was started during the period of *depression* in the 1930s. There were droughts, famines and epidemics during this period and the farmers were not able to sustain themselves in the rural areas. They migrated towards the cities in search of employment. Subsequently, between 1941–51 millions of people moved to urban places in response to war-time industrialisation and partition of the subcontinent. Under the planned economy, there is a steady migration from the countryside towards the city in search of jobs and employment.

Interestingly enough, the impact of influx of in-migration from the countryside to the big cities is felt maximum in the old city (downtown). There is greatest pressure as the immigrants flock to their relatives and friends before they get a job and search for housing.

The urban sprawl is maximum in the big cities like Mumbai, Kolkata, Delhi and Chennai. Such large cities provide more employment and act as magnets which attract large number of people from the rural areas and small towns. The Agra–Mathura–Faridabad–Gurgaon–Delhi–Ambala–Chandigarh and the Delhi–Ghaziabad–Modinagar–Meerut–Muzaffarnagar–Roorkee–Dehradun urban sprawls are the creations of the last fifty years. Similar sprawls may be seen around Mumbai, Hyderabad, Bangalore, Lucknow, Allahabad, Kanpur, Varanasi, etc. The urban sprawls are growths in which qualitative improvement in the form of social amenities is required.

The urban sprawl is depleting the quality of life in the million cities. It is resulting into the creation of new slums. The problem of urban sprawl needs to be addressed at priority basis. Sooner this wisdom grows the better.

SLUMS AND ASSOCIATED PROBLEMS

A slum is an area of poor housing, often characterised by multi-occupance and over-crowding. Schools are poor, items sold in local shops are relatively of inferior quality, and sanitation inadequate. Slum population often exhibits high concentration of drug abusers, alcoholics, criminals and vandals.

Urban slums or squatters are a universal phenomenon. They are the products of urban explosion which has accompanied industrialisation. The growth of slums, especially in the developing countries has caused physical and social stress to the urban environment.

The Indian slums are known by different names in different cities. For example, they are called as *Bastees* in Kolkata, *Chei* in Chennai, *Jhopadpatti* in Mumbai, and *Jhuggi-Jhopri* in Delhi.

The main causes of slum development are:

(i) Rapid growth of population, especially in rural India.
(ii) Heavy pressure on agricultural land, and small size of holdings.
(iii) Unemployment in the rural areas.
(iv) Shortage of cheap residential accommodation in urban areas.
(v) Attraction for better jobs and better quality of life in cities.

Slums in the developing countries like India are characterised by poverty, malnutrition, disease, poor health, poor sanitation, poor standard of living, illiteracy, pollution, crime, and corruption. The unhealthy environment of slums and sub-standard dwellings offer an important habitat for a wide range of insects (fleas, bugs, cockroaches, mosquitoes, flies), rats and rodents. These insects and rodents transmit a number of diseases. The poor quality of drinking water becomes the cause of typhoid, dysentery, diarrhoea, malaria, jaundice, cholera, yellow-fever, dengue, and conjunctivitis. The main characteristics of the Indian slums are:

(i) They generally develop over government land. They come up near industrial areas, railway lines, ports, river banks, parks, large drains, and whole-sale markets.
(ii) The houses in slums are built with mud or bricks with thatched roofs or roofs covered with corrugated sheets, tins, bamboo mats, polythenes, and gunny-bags.
(iii) There is poor sanitation and public latrines. According to a JMP WHO/UNICEF report 2017, in India, 626 million people practice open defecation. On October 2, 2014, Swachh Bharat Mission was launched throughout the country with an aim to achieve the vision of a "*Clean and Open Defecation-Free India*" by October 2, 2019.

The percentage of latrines in households has increased from 88.7% in 2002 to 91.2% in 2012. Many people, especially children, defecate anywhere in the slum area.
(iv) Running water (piped water) is missing.
(v) People wash their clothes and utensils under the hand-pumps.
(vi) There is low sex ratio in the slum areas.

(vii) The slum dwellers are mainly engaged in the tertiary sector. Some of them are self employed petty traders, household servents, vendors and hawkers.

(viii) The slum dwellers belong to the lower income group.

(ix) The slums lack in basic infrastructure such as electricity, running water and sewage and garbage removal.

(x) There is obnoxious smell in the surroundings.

Distribution of Slums in India

In India, slums are created when squatters illegally occupy land, either on the edge of a built-up area or in the interstices of existing development; as alongside railway lines, roads, ponds, drains, and public lands. Such an occupancy may be entirely unplanned and piecemeal, but most slum settlements are the results of planned invasion of land which neither private owners, nor states are likely to resist. In India, the slum areas have grown rapidly, especially during the last four or five decades.

Indian cities have been subject to an ever burgeoning population, and as a result, the problems of finding space and housing for all have intensified. Slums have become an inevitable part of the major Indian metropolises and big cities. The proportion of population living in slums has been increasing over the years.

The state-wise distribution of slum population in 2011 has been shown in (**Fig. 14.29),** while (**Table 14.6)** gives the number of slum towns and their population.

Table 14.6

Number of towns and households and Slum Population-2011

S. No.	India/State/ Union Territory	Total	Statutory	Slum Reporting	Urban Households	Slum Households. Absolute	Percentage
	INDIA	**7,933**	**4041**	**2543**	**78,865,937**	**13,749,424**	**17.4**
1.	Jammu & Kashmir	122	86	40	517,168	96,990	18.8
2.	Himachal Pradesh	59	56	22	166,043	14,240	8.6
3.	Punjab	217	143	71	2,094,067	296,482	14.2
4.	Chandigarh	6	1	1	228,276	22,080	9.7
5.	Uttarakhand	115	74	31	592,223	89,398	15.1
6.	Haryana	154	80	75	1,751,901	325,997	18.6
7.	Delhi (NCR)	113	3	22	3,261,423	383,609	11.8
8.	Rajasthan	297	185	107	3,090,940	383,134	12.4
9.	Uttar Pradesh*	915	648	260	7,449,195	992,728	13.3
10.	Bihar	199	139	71	2,013,671	194,065	9.6
11.	Sikkim	9	8	7	35,761	8,612	24.1
12.	Arunachal Pradesh	27	26	5	65,891	4,005	6.1
13.	Nagaland	26	19	7	115,054	15,268	13.3
14.	Manipur	51	28	NS	171,400	NS	NS

S. No.	India/State/ Union Territory	Total	Statutory	Slum Reporting	Urban Households	Slum Households. Absolute	Percentage
	INDIA	**7,933**	**4041**	**2543**	**78,865,937**	**13,749,424**	**17.4**
15.	Mizoram	23	23	1	116,203	16,240	14
16.	Tripura	42	16	15	235,002	33,830	14.4
17.	Meghalaya	22	10	6	116,102	10,936	9.4
18.	Assam	214	88	31	992,742	48,122	4.8
19.	West Bengal	909	129	122	6,350,113	1,393,319	21.9
20.	Jharkhand	228	40	31	1,495,642	79.200	5.3
21.	Odisha	223	107	76	1,517,073	350,306	23.1
22.	Chhattisgarh	182	168	94	1,238,738	395,297	31.9
23.	Madhya Pradesh	476	364	302	3,845,232	1,845,232	28.3
24.	Gujarat	348	195	96	5,416,315	360,291	6.7
25.	Daman & Diu	8	2	NS	47,631	NS	NS
26.	Dadar & Nagar Haveli	6	1	NS	37,665	NS	NS
27.	Maharashtra	534	256	187	10,813,928	2,449,530	22.7
28.	Andhra Pradesh & Telangana	353	125	124	6,778,225	2,421,268	35.7
29.	Karnataka	347	220	206	5,315,715	728,277	13.7
30.	Goa	70	14	3	198,139	4,846	2.6
31.	Lakshadweep	6	0	NS	8,180	NS	NS
32.	Kerala	520	59	19	3,620,696	54,849	1.5
33.	Tamil Nadu	1097	721	504	8,929,104	1,451,690	16.3
34.	Puducherry	10	6	6	206,143	35,070	17
35.	Andaman & Nicobar islands	5	1	1	34,346	3,053	8.9

Note: NS indicates slum not reported

*Delhi includes 19 Census towns and Uttar Pradesh includes 1 Census town

Source: Census of India, Housing Stock, Amenities and Assets in Slum Series, Series 1, p. ci, Census of India 2011.

The state-wise distribution of slums and slums households have been given in (Table 14.6). It may be seen from this Table that the total number of slum is 2543 in which about 13.7 million people are living under miserable condition. The largest number of slums are in Tamil Nadu (504) followed by Madhya Pradesh (302), Uttar Pradesh (260), Maharashtra (187), West Bengal (122), Rajasthan (107). and Gujarat (96). The number of slums is least reported from Mizoram (1), and Goa (3).

The percentage share of slum population to the total urban population have been plotted in (Fig. 14.29). This Figure vividly shows that slum population is significantly high in Meghalaya, Maharashtra, Andhra Pradesh, Telangana, while the north-east hill states, Kerala, Himachal Pradesh, Uttarakhand and Jammu & Kashmir have very small population in the slum.

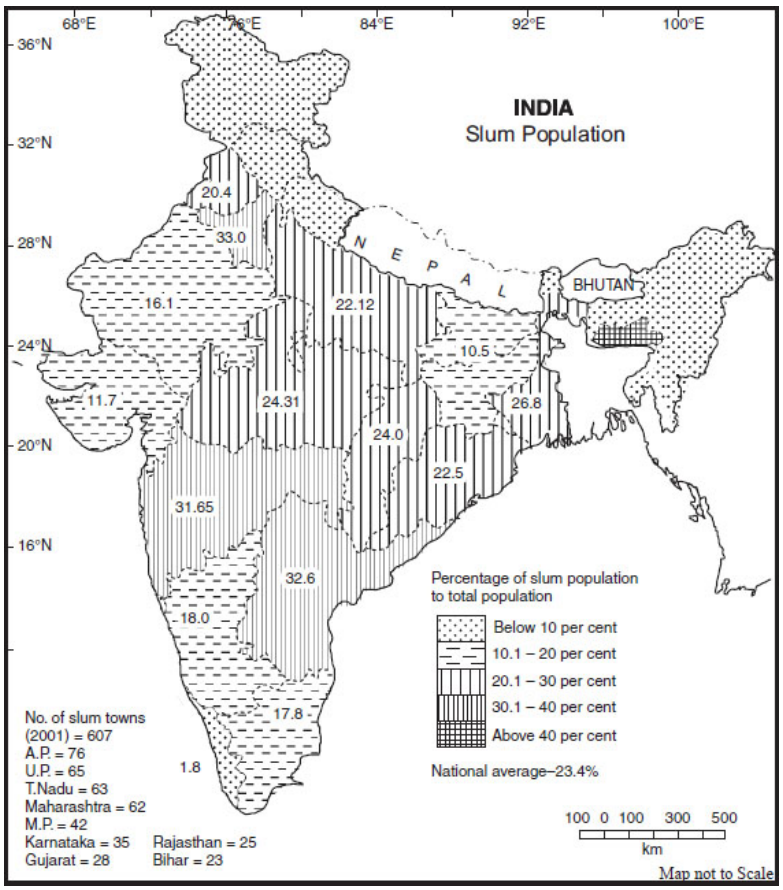


Fig. 14.29 Slum Population—2011

Table 14.7

Slum Population in Million Cities (2001)

	Name of the City	Total Population	Slum Population	Percentage of Slum Population to Total Population
	Total	70,813,906	16,565,459	23.40
1.	Greater Mumbai	11,914,398	5,823,510	48.88
2.	Delhi	9,817,439	1,854,685	18.89
3.	Kolkata	4,580,544	1,490,811	32.55
4.	Bangalore	4,292,223	345,200	8.04
5.	Chennai	4,216,268	747,936	17.74
6.	Ahmedabad	3,515,361	439,843	12.51

Name of the City		Total Population	Slum Population	Percentage of Slum Population to Total Population
7.	Hyderabad	3,449,878	601,336	17.43
8.	Pune	2,540,069	531,337	20.91
9.	Kanpur	2,532,138	368,808	14.57
10.	Surat	2,433,787	406,018	16.68
11.	Jaipur	2,324,319	350,353	15.07
12.	Nagpur	2,051,320	726,664	35.42
13.	Indore	1,597,441	259,577	16.25
14.	Bhopal	1,433,875	126,346	8.81
15.	Ludhiana	1,395,053	314,759	22.56
16.	Patna	1,376,950	3,511	0.25
17.	Vadodara	1,306,035	107,289	8.21
18.	Thane	1,261,517	420,276	33.31
19.	Agra	1,259,979	121,890	9.67
20.	Kalyan-Dombivli	1,193,266	34,854	2.92
21.	Varanasi	1,100,748	138,183	12.55
22.	Nashik	1,076,967	142,234	13.21
23.	Meerut	1,074,229	471,316	43.87
24.	Faridabad	1,054,981	491,131	46.55
25.	Haora	1,008,704	118,235	11.72
26.	Pimprichinchwad	1,006,417	129,357	12.85

Name of the City	Total Population	Slum Population	Percentage of Slum Population to Total Population

Note: In case of Lucknow M.C., no slum population has been reported by the Mahanagar Adhikari (MNA)—the highest executive authority of the Corporation. This claim is being scrutinized by the Census authorities.

The slum population of Patna M.C. is partial and is being subjected to scrutiny.

Source: Census of India, 2001.

It may be seen from the (**Table 14.7)** that Mumbai has the largest percentage of slum population being 48.88% of the total population. Dharavi in Mumbai is the largest slum in Asia, while Delhi and Kolkata rank second and third, respectively. The city of Faridabad has 46.55% of its total population in slums while that of Meerut is 43.87%. The cities of Nagpur, Thane, and Pune also have more than 20% of their population in the slums. The percentage of slum population in the remaining cities varies between 5 and 20%.

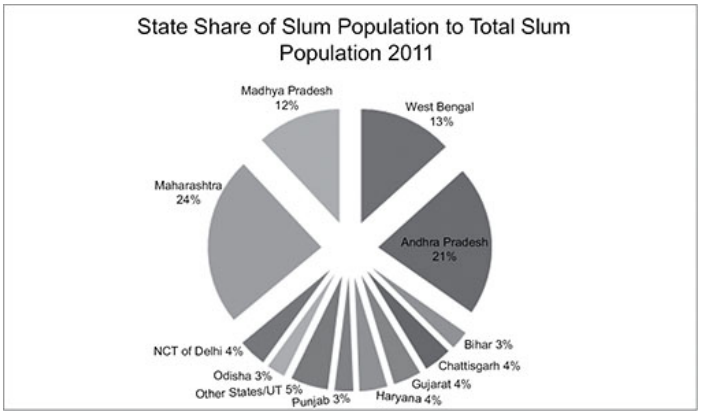


Fig. 14.30: State Share of Slum Population to the total Slum Population (2011) [*Source: Primary Census Abstract for Slums 2011, Office of the Registrar General Commissioner, India*].

The state share of population to the total population can be understood from (Fig. 14.30). It indicates that Maharashtra leads with 24% followed by Andhra Pradesh (21%), West Bengal (13%) and Madhya Pradesh (12%).

The slums are a cancerous growth which deserve immediate attention for the improvement of their environment. The Planning Commission of India has realised that 'adequate attention has not been paid to the formulation of a policy for arresting and decongesting large cities.' In the past, the main stress has been laid on improving conditions in congested metropolitan areas only, and which has not succeeded in arresting slum development. Now the government is aware of its urgency, but financial constraints and legal problems are coming in the way. Such improvements as water supply, community baths and latrines, sewers, and paving of roads and streets and lighting should have been provided on the basis of local conditions. Urban conservation measures could be more rewarding with people's co-operation.

PROBLEMS OF URBANISATION

India has the second largest urban population in the world after China. India's urban population (about 30% of the total population 2011) is almost equal to the total population of USA. The rate of urban growth is on the upswing. People by the hundreds of thousands are arriving in the mega and metropolitan cities, swelling urban India by more than five per cent annually. This stupendous growth

of population is the cause of numerous socio-economic, and environmental problems. Some of the important problems of urban India which have been briefly presented below, are shortage of: (i) Water, (ii) power, (iii) health-care, (iv) transport, (v) roads, (vi) eduation, (vii) housing, and (viii) environment.

1. Problem of Space and Scarcity of Residential Accommodation

Urban growth in India, especially after Independence has been tremendous. The area and size of some of the metropolitan cities, like Mumbai, Kolkata, Delhi, Chennai, Bangalore and Hyderabad, are growing at an unparallel scale. There is urban sprawl in these cities. The urban population has increased more than five times during the last four decades. The growing population demands more space which is not easily available because of physical and geographical constraints. The island nature of Mumbai, and the salt water in eastern Kolkata are some such constraints. The scarcity of space leads to high price of land and high rents for offices and residential accommodations. Since people cannot afford high rents, it is the main cause of unwanted growth of slums and squatters.

According to one estimate, there is an annual shortage of about two million houses in Indian cities. This has forced low income group people to live in slums or occupy footpaths and road pavements. The number of such slums and pavement dwellers is rising by leaps and bounds in the metropolitan cities of India**.**

2. Inadequacy of Social Amenities

In most of the cities of India, there is growth and not urbanisation. In fact, the number of people is increasing in the cities while the infrastructural facilities and civic amenities are quite inadequate. With the greater concentration of people in urban places, the social amenities like housing, electricity, drinking water, transport, sanitation, sewage disposal, educational institutions, hospitals, parks, playgrounds, and recreational facilities are under great stress.

3. Unemployment

Unemployment is the state of being involuntarily out of work. In India, the rate of urban unemployment which is more than 3% annually is increasing progressively. According to one estimate, about 25% of the workers in the urban centres are unemployed. The high rate of unemployment and under-employment often leads to high rate of crime.

4. Problem of Transport

Transport bottlenecks and traffic congestion are the major problems of most of the Indian cities. The larger a town grows, the more important its functions become. The workers and commuters need more transport facilities. Unfortunately, the roads in most of the cities, especially in the old towns (down-towns) are narrow which cannot cope with the growing pressure of passengers, travellers, and commuters. The number of private vehicles is rising steeply. It leads to traffic congestion, delays, irritation, and tension. If the number of vehicles are allowed to increase at the present rate without widening and upgrading the roads, the whole transport system of the major cities may collapse. Ideally, 1500 vehicles should use a single lane during one hour. In Delhi 5000 to 10,000 vehicles occupy the space.

5. The Energy Crisis

The shortage of energy is universal in urban India. The shortage of energy reduces the industrial production of goods and their distribution. In fact, on energy depends the industrial growth, efficiency of the transport and human comfort. The peak power demand in the metropolises, million and class one cities is increasing day by day and power situation is not geared to meet it.

6. Inadequacy of Water Supply

Water is the first and foremost necessity of human life. In fact, water is life, and man cannot survive without it. The average per head consumption of water in Kolkata is 250, in Mumbai 175 and only 80 litres in Delhi as against 1200 litres in Los Angles and 1100 litres in Chicago. The acute scarcity of water in the urban places of India may be appreciated from the fact that in Chennai, Hyderabad,

Jaipur, Jodhpur, Nagpur, Shimla, Solan, Surat, Udaipur, Vadodara, etc., only one to two hours of water supply in a day is permitted. The National Capital (New Delhi) also regulates water supply to only about four hours a day.

7. Environmental Pollution

Environmental pollution is the other serious problem of all the million and mega-cities of India. It is not only air-pollution caused by smoke emitted from vehicles, factories and houses; water and noise pollutions are equally serious. As per the report from International Energy Agency (IEA), about 9 lakh premature deaths in India are expected by 2040 resulting from the rise in air pollution. India has 11 out of the 20 most polluted cities in the entire world. India ranks 141 out of 180 in terms of air pollution. The WHO Report 2016 has shown that Delhi is the 3rd largest polluted city of the world. Moreover, Delhi's air has the highest particulate matter in the world, i.e. 2.5 microns per cubic meter. These can invade lungs and cause serious health complications. The scarcity of dumping grounds makes the rural-urban fringe unhygienic and less conducive for human health. The problem of garbage disposal (hazardous plastics, metal and package) is thus quite serious in most of the Indian cities and urban places. Unfortunately, most of the garbage is dumped into the rivers or along their banks. The cities like Mumbai, Kolkata, and Chennai continue to discharge a major part of their garbage into the sea. In Delhi, through numerous *nallahs* (drains), the sewage and polluted water is drained into the Yamuna River. Occasionally, the highly polluted water of Yamuna flows into the taps of the residents of Delhi causing numeous water borne diseases. The local governments lack the resources and technical expertise to deal with the trash and garbage crisis of an increasingly consumerist behaviour of the urban people.

8. Unemployment and Social Tension

In urban places, especially in the million and Class I cities, people immigrate in search of jobs and economic benefits. Usually, they come alone to cities leaving their families in the villages. In the urban areas, they are exposed to the glitz and insensitivity of urban people who have gross disregard for the feelings and sentiments of their fellow citizens. In such a situation, the immigrants face cut-throat competition for jobs. Many of them may not find jobs and become unemployed.

9. Increase in Crimes

Increasing urban crimes are disturbing the peace and tranquility of modern cities. According to sociologists, unemployment is the main cause of crimes in urban areas. The unemployed youths indulge in crime like abduction, extortion, kidnapping, murder, pick-pocketing, rape, robbery, snatching, and theft. The slums are especially infested with unemployed criminals who, in due course of time, become habitual offenders. Material culture, growing consumerism, selfishness, stiff competition, lavishness, increasing socioeconomic disparities, rising unemployment and loneliness are some of the main reasons of this menace. It is not only the poor, deprived and slum-dwellers who are prone to such crimes, but many youngsters from the well-off families also indulge in criminal activities. In addition to this, gambling, alcoholism, drug-addiction, crimes against females and property, prostitution, and social tension create a new cultural milieu in which traditional norms of life are ignored. The new values and life styles may not be according to our oriental values.

The crimes against women in Mohali have increased from 227 to 273 in 2018. The cybercrime is reported as maximum (one third of the total cybercrimes) in the states of Maharashtra and Uttar Pradesh. More than 24,000 cases were registered under the IT act. Sexual crimes on juveniles are on the increase. The data from National Crime Records Bureau (NCRB) reveals that sexual crime committed by the juveniles (16 years) was 2.4% and by juveniles of all age groups was 3.4% as on 2013. The report from NCRB is indicative of 143% increase in the rape cases by the juveniles from 2002 to 2012.

TOWN PLANNING IN INDIA

Cities and towns are important creations of man on the earth's surface. In fact, cities are human agglomerations showing a lot of economic, social and cultural diversities. Unfortunately, each one of the Indian cities is characterised by numerous problems. These problems can be solved only through proper urban planning, which may make the cities livable, clean, safe and healthy, not only for the present, but also for the future generations. *Tjeerd Deelstra* (1989) has identified the following ingredients in a good city:

1. Good cities are livable places where growing old can be a pleasurable experience, and where children can live and learn.
2. Good cities incorporate nature in simple and complex form. This contributes to balanced urban climates, wild-life and habitat. Nature in cities can contribute to good production, the recycling of organic waste, education and recreation.
3. In good cities, a lot of activities are going on. There is continuous and multiple use of space.
4. Good cities are safe. Pedestrians and cyclists are given priority; women, children, senior citizens and handicapped people can get around easily.
5. Good cities are clean and healthy. They have good air and water and healthy buildings in which it is good to stay.
6. Good cities combine centralised infrastructure with decentralised systems. Management of flow of water, materials, and energy is optimal and attuned to natural life-cycle.
7. Good cities have great experimental activities; movements and flows are visible. This enhances emotional, behavioural and cognitive growth of cities, and facilitates their engagement with urban life and the environment.
8. Good cities have past and future, identifiable in their built form as well as in their social and cultural life.

PRINCIPLES OF URBAN PLANNING

Some of the broad principles of urban planning have been described briefly in the following:

1. Urban Decentralisation

Under this principle decentralisation of industries and residential areas is encouraged through the development of new suburban towns to bring down overcrowding in large metropolitan cities.

2. Garden Suburbs and Garden Cities

The development of gardens and garden belts are imperative to provide the cities a clean and healthy environment.

3. New Towns

This principle suggests the formation of new planned towns outside the municipal limits of the city. Such towns should be equipped with all urban facilities so as to attract industrial and business houses. In India, New Delhi, Bhilai, Chandigarh, Faridabad, Gandhinagar, Gurgaon, Raurkela, etc. were planned on the basis of this principle **(Fig. 14.31)**.

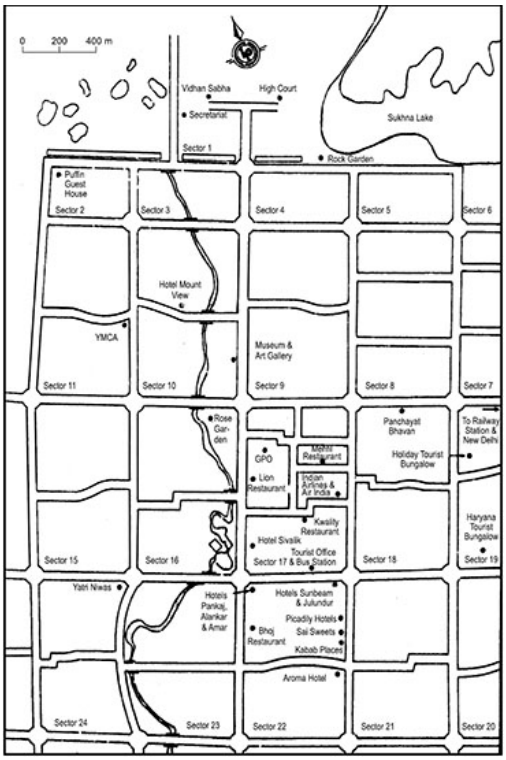


Fig. 14.31 Chandigarh

4. Expanded Towns

These towns attract workers, artisans, businessmen, and industrialists from the neighbouring cities and hence, play a significant role in easing out the burden of population.

5. Urban Renewal

This is an important concept of urban planning. It helps in planning and improvement of land use pattern in the old cities through the development of Master Plan for 10 or 20 years.

6. Slum Clearance

Under this principle, two-pronged strategies are being adopted to improve the conditions of urban slums. These slums are removed and slum dwellers are provided alternative residential accommodation in new planned colonies free of cost or at highly concessional rates. Wherever such slum clearance is not possible, basic civic amenities like drinking water, electricity, garbage disposal, sewerage, roads, etc. are provided to improve the environmental conditions of the slums.

7. Rehabilitation versus Redevelopment

This principle is applied for improving the housing conditions in the old cities. It may involve improvement in old buildings or construction of new ones at their place after demolition.

8. Traffic Segregation

This principle is applied to remove transport bottlenecks and to improve urban traffic. It offers various solutions, like widening of roads, construction of flyovers and subways, bridges, metros, underground railways and parking facilities.

9. City Centre Development

Under this, principle attention is focused on the improvement of Central Business District (CBD).

10. Planning for Future Cities

This principle focuses attention on new towns and cities keeping in mind the future needs of the city region. This may be done through development of satellite towns, linear cities, dispersed cities or

radial cities **(Fig. 14.32)**.

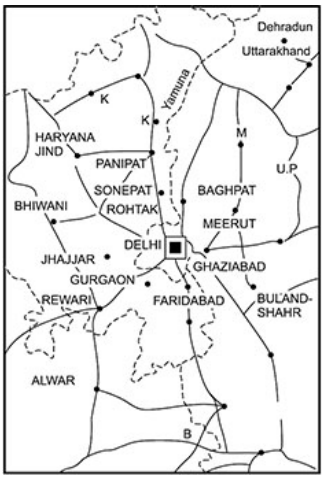


Fig. 14.32 The National Capital Region (NCR)

The National Urbanisation Policy

The main objective of the urbanisation policy of the Government of India is to develop an urban-rural continuum, replacing the existing urban-rural dichotomy. The National Urbanisation Policy gives first priority to urban housing in which both public and private sectors have been invited to contribute. Here, public sector responsibilities have been limited with the improvement of slums, and providing cheap housing to weaker sections of the urban society.

Recently, a major change in national urbanisation policy has been observed towards the slums. The policy puts greater emphasis on the improvement of slums rather than on their demolition or removal. For this purpose, slum beautification programmes have been launched in several cities to provide the basic civic amenities to the slum dwellers.

Under the Urban Land Ceiling Act, 1976, the state governments have been authorised to fix maximum ceiling for urban land. State governments and local bodies have also been granted right to acquire any private property for public use after paying adequate compensation.

The urbanisation policy also puts greater emphasis on the development of infrastructural facilities in small and medium sized towns and cities.

Under the policy, incentives have been given to industrial houses and entrepreneurs to establish new industries in small, medium and intermediate towns. Such facilities include tax concessions, land for factory at a cheaper rate, infrastructural facilities free of cost or at concessional rates.

The urbanisation policy makes it obligatory even for small, medium and intermediate towns to prepare their Master Plan. The management of all the towns and cities has been given to the respective state governments. There is a plan to divide the country into urban regions so as to provide greater opportunity for the solution of local and regional urban problems. Provisions have also been made for research and policy formulation on urbanisation and urban development.

The main objectives of the National Urbanisation Policy may be summarised as under:

1. Balanced urban growth
2. Growth of urban centres
3. Dispersed location of towns
4. Provision of qualitative infrastructure and civic amenities
5. Development of small and medium towns
6. Development of suburbs and rural-urban fringe
7. Imporvement of slums rather than their demolition or removal

8. Land-ceiling and restriction on building plot size
9. Co-ordination in civil amenities
10. Provision of cheap and suitable mass housing
11. Policy to check air, water, and noise pollution
12. Preparation of a master plan for all categories of towns and cities.

At present, despite the National Urbanisation Policy, Indian cities are growing and not developing. How India manages its urbanisation in the coming decades will determine its future. This requires political leadership, vision, capacity building, and institutional reforms.

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15

Regional Development and Planning

PLANNING IN INDIA

Planning is a device of achieving socioeconomic prosperity by optimum utilisation of the available natural and human resources. In other words, planning is a process of socioeconomic development to make space (region) habitable and to keep the ecology and environment in a healthy condition. In India, the main objectives of planning is to eradicate poverty, ignorance, diseases, and inequalities of opportunities and providing a better and higher quality of life. The objectives of planning are also to provide social justice, distribution and judicious utilisation of natural resources and to maintain the resilience characteristics of physical resources.

The process of planning in India was started during the period of the Freedom Movement. The National Planning Committee was set

up in 1938 by the then Indian National Congress under the guidance of Mahatma Gandhi, Subhash Chandra Bose, and Pandit Jawahar Lal Nehru. Planning in India made good progress immediately after Independence. The Indian planning, from its very inception, has been deeply rooted in the concepts of democracy and socialism with the chief objective of providing equal opportunities and good quality of life to all. The three basic directive principles of planning policy of India as given in the Constitution of the country are:

(i) that the citizens, men and women, equally, have the right to adequate means of livelihood;

(ii) that the ownership and control of material resources of community are so distributed as best to sub-serve the common good; and

(iii) that the operation of economic system does not result in the concentration of wealth and means of production to the detriment.

Thus, the planning policy in India was directed mainly towards accelerating the pace of social and economic development; alleviating poverty and unemployment; improving standard of living and quality of life, an egalitarian society with equal opportunities to all individuals of both sexes, of all regions, of all religions and of all castes without any discrimination.

Experience of Regional Planning

Regional planning is a form of planning in which a synthesis is made of physical, social, economic, cultural, and political interests of the region concerned. The scope of regional planning in India is constrained because the states act as independent political entities and have their own planning machinery. In India, during the last six decades, there have been some experiments in regional planning. Some of the examples of regional planning, their success and failure have been discussed below:

1. **Damodar Valley Project:** The Damodar Valley Project was designed on the pattern of Tennessee Valley Corporation (USA) which was implemented in 1948. Originating from the

Chhotanagpur Plateau, the Damodar River has a length of 541 km which drains the states of Jharkhand and West Bengal before merging info the Hugly River. The Damodar river was known as “The Sorrow of West Bengal”. This regional project was designed to control floods, to generate electricity, to provide irrigation in the plain areas, to control soil erosion, to develop tourist places and to use the available mineral and other resources judiciously. It is an important example of river basin management and regional planning. Thus, it is largely a success story of regional planning in India.

2. **Western Ghats:** The Western Ghats stretches over the states of Gujarat, Maharashtra, Goa, Karnataka, Tamil Nadu and Kerala. A regional plan to consider the issues of afforestation, soil erosion and conservation, hydel power development, plantation agriculture, mineral resources development, wildlife protection and biodiversity conservation was prepared. A high level body worked out a set of guidelines for the regional development of the Western Ghats. Being lying in different states the regional plan could not be implemented effectively. The region lost its biodiversity and became a biodiversity hotspot. The region has been included in the UNESCO World Heritage List in July, 2012.
3. **Bundelkhand Region:** This region includes the southern parts of Uttar Pradesh (Banda, Hamirpur, Mahoba, Orai and Chitrakootdham districts), and the northern parts of Madhya Pradesh (Tikamgarh, Chhatarpur, Panna, Satna, Rewa, and Sidhi districts). This region has the problems of scarcity of surface and underground water, low level of agricultural productivity, soil erosion, deforestation, low industrialisation and poor accessibility. Being located in two states, a comprehensive plan could not be implemented successfully for its regional development.
4. **The North-Eastern Region:** This region includes the states

of Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland and Tripura. The problems faced by this region include poor accessibility due to difficult terrain, poor physical infrastructure, floods, shifting cultivation, low agricultural productivity, and soil erosion. In 1971, the North-East Council was set up by an act of Parliament to coordinate and advise on plan efforts of projects of inter-state nature such as those involving transport, communication, power, flood control and training facilities.

5. **The Dandakarnaya Project:** This project includes the tribal areas of Chhattisgarh, Odisha, and Jharkahnd. The objective of this project was to resettle the refugees from Bangladesh (erstwhile East Pakistan) through integrated planning with special reference to tribal interests. The project was implemented by the Dandakaranya Development Authority. This area at present is infested with Naxalites and extremists which shows the unsuccessful story of the project.

In brief, the experiences of regional planning in India has taught that the implementation of regional plan is a difficult proposition. In fact, any inter-state body for planning can only be advisory and not mandatory in nature.

FIVE-YEAR PLANS

Immediately after Independence the government decided to introduce Five-Year Plans on the pattern of the erstwhile Soviet Union. The First Five-Year Plan of the country set out the following five objectives for accomplishment:

(i) increasing the national income and improving the average standard of living of the people;

(ii) increasing the pace of industrialisation of both the basic and heavy industries;

(iii) increasing employment potential of various sectors;

(iv) narrowing down both vertical and horizontal inequalities; and

(v) achieving self reliance so as to minimise dependence on foreign aid.

First Five-Year Plan (1950–51 to 1955–56)

At the time of the First Five-Year Plan, the country was confronted with the:

(i) problems of influx of refugees,
(ii) severe food shortage resulting in import of grains, and
(iii) mounting inflation.

It had also to correct the imbalances in the economy caused by the Second World War and the partition of the country. Accordingly, the First Five-Year Plan emphasised as its immediate objectives :

(a) the rehabilitation of refugees,
(b) rapid agricultural development to achieve self-sufficiency in food, and
(c) to increase the per capita income. The main thrust of the plan was the rapid development of agriculture and irrigation, and establishment of power projects in the form of multi-purpose projects. The general growth rate during the First Five-Year Plan was about 3.7% per annum against the target of 1.2%.

Second Five-Year Plan (1955–56 to 1960–61)

The main focus of the Second Five-Year Plan was on:

(i) rapid industrialisation and establishment of basic and heavy industries; to increase production of iron and steel, and chemical fertilisers,
(ii) an increase of 25 % in the national income,
(iii) to create more employment opportunities, and
(iv) to reduce inequalities in income and wealth.

Third Five-Year Plan (1960–61 to 1965–66)

After the Second Five-Year Plan, the Indian economy entered the *take off stage* as the first two plans had generated the required infrastructure for rapid economic development. The Third Five-Year

Plan gave top priority to agriculture and basic industries. The main objectives of the Third Five-Year Plan were:

(i) to achieve five per cent increase in the national income,
(ii) to achieve self sufficiency in food grains,
(iii) to expand basic industries like iron and steel, chemicals, fuel, and power,
(iv) to utilise fully the manpower resources of the country and to increase substantially the employment opportunities,
(v) establish progressively greater employment opportunities and bring about reduction in income disparities.

The main focus of the Third Five Year Plan was on balanced regional development. Because of the war with China in 1962, and with Pakistan in 1965, the Third Plan was later shifted from development to defence development.

Annual Plans

The war with Pakistan in 1965, severe droughts in successive years, and devaluation of currency delayed the finalisation of the Fourth Five-Year Plan. Consequently, three Annual Plans (1966, 1967, to 1968–69) were described as the *Plan Holiday.* These annul plans were implemented within the framework of the draft outline of the Fourth Five-Year Plan.

Fourth Five-Year Plan (1969–74)

The Fourth Plan was aimed at accelerating the tempo of development and reducing fluctuation in agricultural production as well as the impact of uncertainties of foreign aid. The main objectives of this plan were:

(i) growth with stability, and
(ii) progressive achievement of self reliance. It laid especial emphasis improving the condition of the less privileged and weaker sections of society through the provisions of employment and education. The average annual growth rate during the Fourth Plan was 3.4% against the target of 5.7%.

Fifth Five-Year Plan (1974–78)

The Fifth Five-Year Plan was formulated against the backdrop of severe inflationary pressure. The plan proposed to achieve two main objectives: (i) removal of poverty, and (ii) attainment of self reliance through promotion of higher rate of growth, better distribution of income and very significant step-up in the domestic rate of saving. The plan targeted an annual growth rate of 5.5% in national income while the actual achievement was only 5%.

Sixth Five-Year Plan (1978–83)

There were two Sixth Five-Year Plans. The Janta Party Sixth Plan (1978–83) sought to reconcile the objectives of higher production with greater opportunity for employment. The focus of the plan was enlargement of employment opportunities in agriculture and allied activities, encouragement to household and small industries producing consumer goods for mass consumption, and to raise the incomes of the lowest classes through minimum needs programme.

The new Sixth Five-Year Plan (1980–85) was launched by the Congress (I) with the prime objective of direct attack on the problem of poverty by creating conditions for an expanding economy.

The strategy adopted for the plan was to strengthen infrastructure for both agriculture and industry. Stress was laid on dealing with inter-related problems through a system approach rather than in separate compartments, greater management, efficiency and intensive monitoring, and active involvement of people. The targeted annual growth was 5.2% and the achieved growth rate was 5.4%.

Seventh Five-Year Plan (1985–90)

The Seventh Five-Year Plan emphasised policies and programmes which aimed at a rapid growth in food-grain production, increase in employment opportunities and productivity. The foodgrain production during the Plan grew by 3.23%. To reduce unemployment and the incidence of poverty, special programmes like *Jawahar Rozgar Yojna* were initiated in addition to the already existing programmes. During the Plan period, the Gross Domestic Product (GDP) grew at an average rate of 5.6% exceeding the targeted growth rate by 0.6%.

Annual Plans

The Eighth Five Year Plan could not take off due to the fast changing political situation at the centre. Its place was taken by the Annual Plans with basic thrust on the maximisation of employment and social transformation.

Eighth Five-Year Plan (1992–97)

The Eighth Plan aimed at an average annual growth rate of 5.6% and an average industrial growth rate of 7.5%. The salient features of the plan included: (a) faster economic growth, (b) faster growth of manufacturing, agriculture and allied sectors, (c) significant growth rates in exports and imports, (d) improvement in trade and current account deficit, and (e) significant reduction in Central Government's fiscal deficit.

The main emphasis in the Eighth Plan was on:

(i) generation of adequate employment to achieve near full-employment level by the turn of the century,

(ii) containment of population growth through active people co-operation,

(iii) universalisation of elementary education, and complete eradication of illiteracy among the people in the age group of 15 to 35 years,

(iv) provision of safe drinking water and primary health care facilities,

(v) growth and diversification of agriculture to achieve self-sufficiency in food and generate surplus for exports, and

(vi) strengthening the infrastructure (energy, transport, communication, irrigation) in order to support the growth process on a sustainable basis. However, there was shortfall in the attainment of the Plan objectives due to lack of mobilisation of adequate resources owing to deterioration in the balance of current revenue, erosion in the contribution of state electricity boards and state road transport corporations,

negative opening balance, mounting non-Plan expenditure and shortfalls in the collection of small savings, etc.

Ninth Five-Year Plan (1997–2002)

The main objective of the Ninth Plan was 'Growth with social justice and equality'. It included:

(i) priority to agriculture and rural development,
(ii) accelerating the growth rate of the economy with stable price,
(iii) ensuring food and nutrition security for all,
(iv) providing the basic minimum services of safe drinking water, primary health care facilities, universal primary education, shelter and connectivity to all in a time bound manner;
(v) ensuring environmental sustainability,
(vi) containing growth of population,
(vii) empowerment of women,
(viii) promoting and developing Panchayati Raj institutions, co-operatives and self-help groups,
(ix) strengthening efforts to build self-reliance, and
(x) strengthening sound foreign trade. The Ninth Plan fixed growth rate at seven per cent per annum.

Tenth Five-Year Plan (2002–2007)

The main objective of the Tenth Plan was at doubling the per capita income in ten years and achieving the growth rate of eight per cent of the GDP per annum. The plan aimed at harnessing the benefits of growth to improve the quality of life of the people by setting the following key targets:

(i) reduction in poverty ratio from 26 to 21%,
(ii) bringing down the decadal growth of population to 16.2% in (2001–11),
(iii) growth in gainful employment,
(iv) universal access to primary education,
(v) increase in literacy to 75%,
(vi) reduction in maternal mortality ratio to 2% and infant mortality

rate to 45 per 1000 live birth,

(vii) provision of potable drinking water to all villages,

(viii) increase in forest and tree cover to 25%,

(ix) cleaning of the major polluted rivers,

(x) creation of 50 million job opportunities, and

(xi) sreducing regional inequalities in development. The target was to achieve 10% growth rate in industrial sector and improving people's participation in the process of development.

Eleventh Five-Year Plan (2007–12)

The main focus of the Eleventh Five Year Plan was on towards faster and more inclusive growth. The priority areas of this plan of this plan are: agriculture, irrigation, and water resources; education, health, infrastructure, and employment, along with programmes for upliftment of SCs/STs, other backward classes, minorities, women and children. This Plan stresses that benefits of development should reach all sections of population.

Twelfth Five-Year Plan (2012–17)

The outline of the 12th Five-Year Plan was presented by the Deputy Chairman of Planning Commission-Montek Singh Ahluwalia under the Chairmanship of Prime Minister Manmohan Singh on 21st April 2011. The Planning Commission expected that during the 12th Five-Year Plan the economic growth would reach 9.0 to 9.5%. In order to achieve the targeted growth rate, the Planning Commission felt that the following primary conditions need to be fulfilled:

1. The door for foreign investment should be opened more to attract investment in different important sectors including infrastructure.
2. For the interest of development, the issues of foreign trade regulations need to be revised under the present regime of economic liberalization.
3. Additional importance should be given on development of agriculture for attaining at least 4% growth rate. For that

purpose the area under Rashtriya Krishi Vikas Yojna (RKVY) needs to be expanded.

4. Manufacturing industries have failed to fulfill the target during the Eleventh Plan. Thus, this sector should try to attain 11 to 12% growth rate.
5. More stress be given on health, education and infrastructure.
6. After petrol, the price of diesel needs to be deregulated or controlled and be kept under market determined price mechanism.
7. Burden of subsidy on LPG cylinder and kerosene needs to be reduced. Subsidy should only be given on these two items only to people lying below the poverty line.
8. More importance be given to atomic energy and non-conventional sources of energy.

Niti Ayog

The Niti Ayog (National Institute for transforming India) was founded on January 1, 2015. It replaced the Planning Commission. The three-year agenda is further divided into seven parts, with a number of specific action points for each part to boost economic growth. The key points from this document are as follows:

(i) Reducing fiscal deficit to 3% by 2018–19 and revenue deficit to 0.9% of the GDP by 2019–20.

(ii) To double farmers' income by several means including reform in APMCs; raising productivity through enhanced irrigation; faster seed replacement rates; precision agriculture; and a shift to high value commodities, horticulture, animal husbandry, fisheries, etc.

Achievements of Five-Year Plans

In almost 65 years of planning, the Five-Year Plans have not been able to achieve their objectives in full. The constructive role in the socioeconomic development of the country cannot be under-estimated. During the planning period, the national income has gone up by about eight times, the per capita income has increased by

about 3%, the GDP has increased by about eight times. The main achievements of the Five-Year Plans may be summarised as under:

1. Impressive industrialisation in the capital goods sector through the public sector.
2. Development of economic infrastructure like energy, irrigation, roads, transport, and communication.
3. Diversification of export and import substitution.
4. Rise in life expectancy of the Indian people from 37 years in 1951 to 67 years in 2011.
5. Development of educational system, thereby raising the literacy level from 34% in 1951 to 73% in 2011.
6. Development of science and technology and the nuclear power for peaceful purposes.
7. Achievement of almost self-sufficiency in foodgrains. The per capita cereal consumption increased from 334 grams per day in 1951 to 500 grams in 2014.

Failures of Planning

The Indian planning has miserably failed to achieve some of its important objectives:

1. India failed to evolve a society based on equity and social justice that is free from tyranny of exploitation.
2. India could not eradicate poverty as about 26% of the total population is below the poverty line.
3. The planning process could not eradicate poverty, malnutrition, hunger, unemployment, exploitation, child labour, tyranny, intolerance, and injustice.
4. The planning could not provide an equal status to females. Much efforts need to be made to remove gender bias.
5. Black money has generated a parallel economy.
6. Redistribution of land and quick transition to progressive agriculture have not yet been achieved. The growth rate of agriculture is around 2%, much below the desired level.
7. The pace of progress of land reforms is extremely slow, and

the restoration of title rights over land is a distant dream.

8. The planning has not been able to bridge the gap between economic, social, and regional inequalities.
9. According to WHO Unicef report about 638 million people in India do not have access to toilets. Around 69% of rural Indians defecate in the open. In the cities, the figure is 18% (*Times of India*, 17th March, 2010, New Delhi ed. pp. 1 and 19.
10. The plans have failed to achieve balanced regional development.

REGIONAL DIMENSIONS OF PLANNING IN INDIA

The planning process in India, however, remained highly centralised for all these sixty years with little decentralisation. The main components of the planning machinery in India include:

(a) The Planning Commission of India,
(b) The National Development Council, and
(c) The State Level Planning Boards for each state.

The Planning Commission of India, headed by the Prime Minister and having a full time Deputy Chairman, was constituted in 1950 as an advisory as well as an executive body. The main functions of the Planning Commission of India are:

(i) to formulate five year/annual plan;
(ii) to supervise the work of national planning;
(iii) to consider the national plans formulated by the Planning Commission;
(iv) to recommend measures for achieving the targets set out by the plans;
(v) to review the working and to monitor the plans from time to time;
(vi) to encourage people's participation in the planning process.

Unfortunately, the planning process in India in its first two Five-Year Plans did not incorporate the regional dimension. There were no

serious attempts to identify a regional scheme or a set up of regions, which could form the basis of the planning activity in the country.

The regional or spatial dimension to the planning process in the country was believed to have been incorporated from the Third Five-Year Plan when the philosophy of balanced regional development got appended to the idea of Five-Year Plans.

The Damodar Valley Corporation (DVC) and the National Capital Region (National Capital Region) continue to be the two significant regional planning efforts of the country. Both were outside the domain of Five-Year Plans, as in both the cases inter-state boundaries were crossed. The partial success of the DVC, and the disgustingly poor performance of the National Capital Region, despite being in the capital of the country, reveal the official apathy to the spirit of regional planning in India.

Recently, the 73rd and 74th amendments of the Indian Constitution have given new direction to planning at the grass-root level. These constitutional amendments have given new direction to the planning process in the country. These are aimed at decentralising the entire planning process by introducing the concept of planning at the grassroot level by involving the village Panchayats in the task of identification of problems at the village level.

The 73rd and 74th Constitutional Amendments have provided for the fund collection by the Panchayats at local level for local developmental works, for which the state governments are to provide the matching grant. Despite all these efforts, no one at the administrative level is ready to accept any devolution of power. The success of regional planning depends mainly on:

(i) the quality of planning education,
(ii) the degree of public awareness about the rationale for a planned effort,
(iii) the efficiency of planning institutions, and
(iv) the proper comprehension and appreciation of the basic issues involved in regional planning by society itself. In India,

regional planning still has a long way to go in terms of its content and implementation.

Regional Planning

Planning is a process to achieve certain objectives and goals. In fact, *it is a process of human thought, and action based upon that thought.* Planning is believed to be related fundamentally to human beings, their behaviour, their aspirations, their thoughts, and their abilities.

Regional planning is neither economic planning, nor physical planning alone. It is a synthesis of physical, social, economic, political, and cultural interests of the concerned region. It has to deal with the concerned region not in isolation, but in integration with other regions constituting the regional space as a whole. It may be at the macro, meso or micro level. It varies both in time and space.

In other words, regional planning is a specific type of planning, based on a specific planning structure (regional system), for inducing public action aimed at the well being of society. It implies that regional planning is concerned fundamentally with the society in the context of space. Naturally so as the society does not exist independent of space. It constitutes an integral part of space. Similarly, space too is incomplete without society. The two (space and society) are intimately interwoven with each other.

It implies that regional planning, while confining itself to the planning of regional space in the physical sense, can hardly afford to overlook the economic, sociocultural, and political dimensions of the ever evolving regional organism. Regional planning should provide habitability, meaning ideal living conditions in a regional space.

In brief, regional planning is a geo-technology of re-organising the regional space for its comprehensive development with a view to provide ideal living conditions to all human communities in all regions of human occupance, not in isolation from each other, but in integration with each other.

Principles of Regional Planning

The principles of regional planning are as below:

1. The Principle of Vertical Unity of Phenomena

The principle of vertical unity means that all phenomena (both physical and cultural) that operate in a regional space are fully integrated with each other. Therefore, planning for any single phenomenon independent of other phenomena shall not be in the real spirit of regional planning.

2. The Principle of Horizontal Spatial Unity

This principle implies that each region constitutes a sub-system of the whole regional system. It means that various regions, constituting regional space for planning process, do not exist in isolation from each other, instead they exist in integration with each other as part of the regional system whole.

The human body provides the best illustration of such an integration between sub-systems and the system. The entire human body as a whole functions as a system, which consists of a number of sub-systems like digestive system, renal system, respiratory system, nervous system, etc. Each one of them constitutes a complete system in itself, and yet is a sub-system of the body whole. If something goes wrong with any of these sub-systems, say digestive system, it will affect the functioning of the entire human body as a person may fall sick and may not be able to perform even day to day functions. Thus, planning for any region cannot be done in isolation from the planning of the adjacent region. Similarly, the planning within the national space cannot be done in isolation from the planning in international space. It implies that regional planning amounts to planning in universalism.

3. The Principle of Space-time Continuum

This principle implies that spatial reality is four dimensional consisting of three dimensions of space and one dimension of time, and the two, i.e., space and time; are inseparable. This is considered as an integral part of space. A region is a living dynamic entity that operates simultaneously in the past, present, and future like the human body's DNA. The regional planner must, therefore, recognise the fact that regional space is a continuously growing organic whole. The regional

planner has to be conscious of the fact that while he or she is planning in space, he is also planning in time.

4. The Principle of Comprehensive Development

Comprehensive development means that the regional planning seeks to achieve the comprehensive development of the entire regional space; the regional system in its entirety. It seeks the development of all sectors of economy along with advancement of all segments of society.

5. The Principle of Community Development

The principle of community development is the principle of equal opportunities to all for self development. The entire community is considered as an organic whole. It is only through equal opportunities to each individual (education, health and employment) that the whole society can be developed into community with the sense of belonging to each other. The regional planning, therefore, should be to create socially harmonious communities.

6. The Principle of Equilibrium between Social Desirability and Economic Viability

This principle implies that while planning for comprehensive development of all regions, the regional planner has to maintain a balance between what is socially desirable and what is economically viable. The regional plans should not only be good intentioned, but also be economically viable.

7. The Principle of Ecological Equilibrium

The regional planning should make the ecology and environment sustainable. It means that a regional planner has to develop the regional space at his disposal only within the framework of ecological equilibrium. In other words, all developments in a region should be carried out without disturbing the ecological balance.

To sum up, regional planning can be successful in its operation and practice only when it is guided by the seven basic principles of (i) vertical unity of phenomena, (ii) horizontal spatial unity, (iii) space

time continuum, (iv) comprehensive development, (v) community development, (vi) equilibrium between social desirability and economic viability, and (vii) ecological equilibrium. A strict adherence to these basic principles is the only way for creating habitability.

INTEGRATED AREA DEVELOPMENT

The concept of integrated area development is difficult to define. It has different meanings and interpretations at different points of time. In fact, it is difficult to decide *what* type of integration should be sought and *what* ways should be adopted to accomplish it. In general, four aspects of integration can be differentiated.

1. Sectoral-cum-Temporal Integration

This is to ascertain what activities have to be developed, when they should be developed, and what should be the pace and intensity of their development keeping in view the underlying linkages among various activities. They can be designated as sectoral-cum-temporal integration.

2. Spatial Integration

As there are linkages between various economic activities, similarly, there are linkages between different places which determine the pattern of flow of raw materials, people and goods in different areas.

3. Integration of Development of Individuals and Group of People

It has been found that the fruits of development tend to get concentrated in the hands of a few while a considerable majority of people continue to lead a miserably poor life. It is the task of integrated development to ensure that the benefits of development are conferred more on such poorer sections of society, so that inter-personal disparities of income can be considerably brought down.

4. Integration of the Conflicting Goals of Socio-economic and Environmental Development

In a country like ours in which about 37% of the total population is below the poverty line, and most of the people still have a low standard of living, the basic aim of development should be to develop the human resource by providing food, clothing, shelter, education, health facilities, and employment to all. While achieving these goals, the resources, ecology and environment should be kept in a healthy condition to sustain the present population and to protect the interest of the future generations.

The planning process can be considered satisfactory only when it pays due regard to all the aspects of integration discussed above. Our own planning efforts have failed to achieve the desired results in the socio-economic field, precisely because they did not address the need for integrated planning. Attention was concentrated on: (i) the sectoral-cum-temporal integration, and (ii) to some extent, on the development of individual groups. Unfortunately, for most of the planning period, we have tended to adopt a tentative, ad hoc and piecemeal approach, with the result that national planning has more or less by-passed large backward rural areas. It is in this overall context that one must view the widespread interest generated in recent years in India for adopting an integrated area development approach. Much emphasis on Integrated Rural Development was laid in the draft of the Sixth Five-Year Plan.

INTEGRATED RURAL DEVELOPMENT PROGRAMME (IRDP)

The *Integrated Rural Development Programme (IRDP)* was launched by the Janta government in 1978-79, by bringing together the Community Area Development Programme (CADP), Drought Prone Area Programme (DPAP), Small Farmer Development Agency (SFDA), and Marginal Farmers and Agricultural Labourers Agency (MFALA).

Integrated rural development is one of the important tasks before the Government of India. The National Common Minimum Programme (NCMP) of the Central Government reiterates the cardinal importance of villages to overall development of the country

and commits to work towards the development of rural areas. The main objective of Integrated Rural Development is to eradicate poverty, hunger and unemployment from rural India. The integrated rural development programme was confined in the initial phase to 2000 blocks out of the then 5004 development blocks in the country. Through coverage under this programme, about 3000 blocks and over 5.45 lakh families have been assisted. It is a centrally sponsored scheme with funds shared on 50:50 basis between the Centre and the states. The programme was intended to generate gainful employment for all able-bodied persons in rural areas within a period of about ten years by introducing such schemes as would help development of production potential of each area, and utilisation and upgrading of the human skills available. The aims of the programme are listed below:

1. To provide assistance in self-employment opportunities.
2. To give assistance to a target group of rural poor, belonging to the families below the poverty line, in the form of subsidy. The target group under IRDP includes labourers, artisans, scheduled castes, scheduled tribes, sharecroppers, and marginal and small farmers.
3. To take up measures for livestock and poultry development, fishery, and social forestry in the village.
4. To promote cottage industries in the village to enhance the per capita income of the targeted group and to raise the standard of living of weaker sections of the rural population.

The main elements of the programme during the Sixth Five-Year Plan period were proposed to be as under:

1. A five year development profile will be drawn up for each district, disaggregated into blocks, based on practical (achievable) possibilities of development in agriculture and allied sectors. This will form the ‘framework of action’ for the relevant schemes of development in these sectors.
2. A specific operational programme will be drawn up by the extension agency to provide guidance on a systematic basis

to the small and marginal farmers' families.

3. A special programme of assistance to the poorest of the rural households will be drawn up to raise the level of specific households so identified, above the poverty line.
4. A blueprint for exploiting the available potential in the secondary and tertiary sectors, which also spells out linkages for training and marketing, will be prepared for each block, and families from among the target group identified for assistance based on such blueprint.
5. The IRDP has been conceived essentially as an *anti-poverty* programme. This objective is proposed to be achieved by enabling the poorest families to acquire productive assets, technology and skills as would make their economic activities viable. These families will also need support from social services such as health, education, and housing.

The IRDP is implemented through District Rural Development Agencies (DRDAs) and Block Level Agencies at the grass-root level. The governing body of DRDAs include local MPs, MLAs, Chairman of Zila-Parishad, Heads of District Development Departments, representatives of Scheduled Castes and Scheduled Tribes, and women.

Some of the important integrated rural development programmes include National Rural Development Programme (NRDP), Minimum Needs Programme (MNP), Training of Rural Youth for Self-Employment (TRYSEM, 1979), Development of Women and Children in Rural Areas (DWCRA, 1982), and Indira Awaas Yojna (IAY, 1985), etc.

In 1999, the Government of India launched a restructured poverty alleviation programme for rural areas which replaced the IRDP and its allied schemes by the *Swarnjayanti Gram Swarozgar Yojna (SGSY).*

The programme was implemented through the Panchayat Samitis.

The main objectives of the Swarnjayanti Gram Swarozgar Yojna are as under:

1. A holistic programme covering all the aspects of self

employment such as organisation of the poor into self help groups, credit, training, technology, infrastructure, and marketing.

2. To make every assisted family rise above the poverty line in three years.
3. The income of the assisted families should be more than two thousand per month.

More than six lakh people (Swarozgaris) have been assisted till date, of which 30% were from Scheduled Castes and Scheduled Tribes, 33% women, and one% handicapped.

DEVELOPMENT OF BACKWARD AREAS

Identification of backward areas is a difficult task. In general, any area with relatively low per capita income and low per capita consumption is known as backward area. At the time of formulation of the Fourth Five-Year Plan (1966–71), the Planning Commission appointed a Study Group to suggest the criteria for identifying backward areas. The group suggested the following criteria for the identification of backward areas:

1. Total population and the density of population.
2. Number of workers engaged in agriculture including agricultural labourers as percentage, to total workers.
3. Total area per agricultural worker.
4. Net area sown per agricultural worker.
5. Percentage of total irrigated area, to net irrigated area.
6. Percentage of area sown more than once, to net sown area.
7. Per capita (rural population) gross value of agricultural output.
8. Establishments (manufacturing and repair)using electricity:
   (i) Total
   (ii) Household
   (iii) Non-household
9. Number of workers per lakh of population employed in registered factories.

10. Mileage of surface roads
    (i) per 100 sq km
    (ii) per lakh of population
11. Number of commercial vehicles registered in a district.
12. Percentage of literate population.
13. Percentage of school going children: (i) boys, and (ii) girls.
14. Number of seats per million population for technical training: (i) craftsmen, and (ii) diploma level.
15. Hospital beds per lakh of population.
16. Per capita income.

On the basis of these criteria, 238 districts all over India were identified as backward districts. The various governments gave numerous fiscal and other incentives for the development of agriculture, industries, transport and social amenities in these districts. Some of the fiscal incentives given were as under:

1. Grants for the minor irrigation projects, cottage and small industries, emphasis on roads, and electricity development.
2. Grant of higher development rebates to industries located in backward areas.
3. Grant of exemption from income-tax, including corporate tax for five years after providing the development rebate.
4. Exemption from payment of import duties on plant and machinery, components, etc., imported by units set up in backward areas.
5. Exemption from excise duties for a period of five years.
6. Exemption from sales tax, both on raw materials and finished products to units set up in specified backward areas for a period of five years.
7. Transport subsidy.
8. Industrial estates, like Okhla (Delhi), Naini (Allahabad), Rajkot (Gujarat), Guindy and Virudhunagar (Tamil Nadu), Kanpur and Agra (Uttar Pradesh), Palghat and Trivandrum,

Kottayam, Quilon, and Trichur (Kerala) were established to encourage the growth of small scale industries.

COMMAND AREA DEVELOPMENT

The *Command Area Development Programme* was launched in 1974–75 with the set objectives of improving the utilisation of created irrigation potential and optimising agriculture production and productivity from irrigated agriculture through a multi-disciplinary team under an Area Development Authority. It is about the command areas of the major and medium irrigation projects in the country.

The programme was restructured and termed as “Command Area Development and Water Management (CADWM) Programme” from 1 April 2004.

This programme was initially introduced in 60 major and minor projects including the Indira Gandhi Canal Command Area in 1974. Subsequently, the Command Area Development Programme was spread 110 districts in 13 states, covering about 15 million hectares of irrigated agricultural land. So far it has covered 310 irrigation projects with a Culturable Command Area (CCA) of about 28.45 mha. Financial assistance is provided to the state governments on 50:50 basis for the construction activities.

The main objectives of the Command Area Development and Water Management Programme are as under:

1. *Warbandi* or the rotational system of water distribution: Some of the marginal and small farmers were not getting their share of the irrigation water. In fact, their water was being utilised by the influential big farmers. In order to overcome this problem, *Warbandi* was done to ensure equitable and timely supply of water to all the farmers irrespective of the size of their holdings.
2. Construction of field channels and field-drains to reduce the water pilferages and effective utilisation of water for irrigation. Central assistance up to 50% (limited to the prescribed cost

norms) is provided to the irrigation and field-developmental activities.

3. Land levelling for the uniform spread of irrigation water in the field.
4. Demonstrations and training to farmers to diffuse technical know-how and for the adoption of new agricultural innovations for developing suitable cropping patterns, as well as improved farming practices and maintaining soil fertility.
5. To prepare a plan for the supply of inputs, credit, seeds, fertilisers, insecticides and pesticides.
6. Construction of *chak-roads.*
7. To promote ancillary activities like animal husbandry, forestry, poultry, marketing, and processing facilities.
8. Diversification of agriculture to make it more profitable and sustainable.
9. To give more emphasis to the cultivation of oil-seeds, pulses, and green-manure crops.
10. To introduce participatory management of irrigation.
11. Reclamation of the water-logged areas rendered useless in the irrigated command areas.
12. Adaptive trials and demonstrations, training of farmers, etc.
13. Repair, renovation, and restoration of waterbodies.

Indira Gandhi Canal Command Area Development Programme

The Command Area Development was introduced in the Indira Gandhi Canal Command Area in 1974. The main objectives of the Indira Gandhi Canal Command Area were as under:

1. To Reduce Water Pilferages of Canal Water

The wastage of water results not only in under-utilisation of water, but also leads to water-logging and soil salinity, affecting adversely the yield and production of crops.

2. Development of Social Amenities and Infrastructural Facilities

Since the Indira Gandhi Canal Command Area was newly colonised, there was an urgency to provide civic amenities and infrastructural facilities to supply the necessary agricultural inputs to the farmers. Emphasis, therefore, was laid on the construction of roads, markets and storage facilities to enhance agricultural production.

3. Control of Wind Erosion

There was a serious menace of wind erosion in the Indira Gandhi Canal Command Area which was a potential danger for the silting of the canal and its distributaries. Therefore, efforts needed to be made to reduce the wind erosion in the command area of the canal.

4. Lining of the Canal

The area through which the canal passes being desertic and sandy, there was a strong possibility of water seepage and water-logging. To check and minimise the water seepage, lining of the canal was imperative.

5. Reclamation of Land

Provisions of levelling of land and reclamation of degraded lands.

6. Afforestation and Pastures Development

Afforestation along the banks of the canal, its distributaries as well as along the new rural and urban settlements to stabilise the sand-dunes. The migratory sand-dunes not only damage the arable land and pastures but may also bury the orchards and settlements under sand.

Implementation of the Indira Gandhi Canal Command Area has helped in bringing the land under irrigation rapidly, increase in water use efficiency, agricultural production and productivity. The improvement in irrigation not only led to phenomenal increase in agricultural production, but also introduced change in the patterns and rotation of crops. Due to the availability of canal water, farmers have adopted the cultivation of wheat, barley, groundnut, mustard, cotton, green manuring crops, orchards, and vegetables in place of *Bajra*, millets and pulses.

The Indira Gandhi Canal has brought much prosperity to farmers of the region and the cropping patterns have changed substantially. However, there occurred numerous environmental and ecological problems in the command area. Apart from water logging, the underground water table is rising, rendering the agricultural and pasture lands useless from the agricultural point of view. The water-logged areas have become the breeding grounds for mosquitoes and malaria. These problems need to be addressed on a priority basis.

WATERSHED MANAGEMENT

Watershed is defined as any surface area from which rainfall is collected and drains through a common point. It is synonymous with a drainage basin or catchment area. In other words, watershed is a geohydrological unit, comprising all land and water within the confines of a drainage divide. The size of watershed may vary from a few hectares to several thousands of hectares. Watershed sizes are classified into three: micro, mini, and macro watersheds. In fact, watershed is a biological, physical, economic and social system. It is a land mass bounded vertically by the area influenced by human activities and horizontally by the water that drains into a point in the channel.

Watershed management is the integration of technologies within the natural boundaries of a drainage area for optimum development of land, water, and plant resources to meet the basic needs of the people and animals in a sustained manner. It can tackle the problem of environmental degradation.

Objectives of watershed management are:

- To effectively conserve soil, rainwater, and harness the surplus water to create water sources in addition to groundwater recharge.
- To promote sustainable farming and stabilise crop yield by adopting suitable cropping and crop management systems.
- To cover non-arable area effectively through afforestation, horticulture and pasture land development based on the

capability of land.
- To enhance the income of the individuals by adopting alternative enterprises.
- To restore ecological balance.

Principles of watershed management are:

- Utilising the land according to its capability based on land use classification.
- Conserving as much rainwater as possible at the place where it falls i.e., in situ conservation.
- Draining out excess water with a safe velocity and diverting it to storage structure
  (farm ponds, tanks) for future use.
- Avoiding gully formation and putting up check-dams at suitable places to control soil erosion, to store rain water and recharge groundwater.
- Identifying a suitable cropping pattern for the watershed area.
- Maximising productivity per unit area, per unit time and per unit water.
- Increasing cropping intensity and land equivalent ratio through intercropping and sequence cropping.
- Safe utilisation of marginal and waste lands through alternate land use systems such as agro-forestry.
- Ensuring sustainability of the ecosystem benefiting the man-animal plant-land-water complex in the watershed.
- Stabilizing total income and cutting down risk of during aberrant weather situations.
- Improving infrastructural facilities with regard to storage, transportation and marketing.

Comprehensive planning for water management starts with obtaining suitable maps through remote sensing techniques. Remote sensing provides a synoptic picture of the watershed for characterisation of natural resources, land, water, vegetation and inter-relationship between them. Besides mapping natural resources, satellite imagery can give the estimate of the area covered by major

crops, crop yield, area affected,by pests and diseases and drought conditions, if any. The location of structures like check-dams, farm ponds etc., can also be delineated using satellite imagery pictures Distribution pattern and assessment of status of material resources can be obtained from satellite pictures. Comprehensive planning of various activities is then carried out. Mechanical, agronomical, agro-ecological and forestry measures of soil and rainwater conservation are then planned and implemented.

On July 27, 2016, rain for only two hours paralysed life in Gurugram (Gurgaon), with deep waterlogging. The main cause of this tragedy was an ill-conceived planning. In the planning of Gurgaon Watershed planning was ignored. The Firozpur-Jhirka-Delhi ridge forms the western boundary and Delhi ridge forms the eastern boundary of this watershed. The natural drainage pattern of the city comprises large depressions and streams, tending to converge inland instead of flowing into Yamuna. The Delhi-Jaipur Highway No. 8 crosses several watersheds almost in the middle, and blocks the natural drainge pattern from south to north. Thus the unscientific planning of National Highway No. 8 is largely responsible for Gurgram tragedy. It is thus important to incorporate the watershed approach in the planning of a city.

Water Resource Potential

The average annual water availability of the country is assessed as 1869 billion cubic metres (BCM). Of this, the total utilisable water resource is assessed as 1123 BCM, surface water 690 BCM and ground water 433 BCM. The per capita availability of water at national level has been reduced from 5177 cubic metres in 1951 to estimated level of 1200 cubic metres in 2011 with variation in water availability in different water-basins, catchments, and watersheds.

The All India Soil and Land Use Survey Organisation of the Department of Agriculture, Government of India, has finalised a nationwide system of delineation and codification of watersheds. According to this scheme, various sizes of watersheds have been divided into: (i) basins, (ii) catchments, (iii) sub-catchments, and (iv)

watersheds. In India, there are 35 river basins, 112 catchments, 500 sub-catchments, and 3200 watersheds. For the purpose of planning and development, the 3200 watersheds have been divided on the basis of area into the following three categories:

Sub-watersheds (10,000 to 50,000 hectares)

1. Milli-Watershed (1000 to 10,000 hectares);
2. Micro-watershed (100 to 1000 hectares); and
3. Mini-watershed (1 to 100 hectares).

The watershed management is a comprehensive programme to utilise the available water and other resources in such a way that the productivity of agriculture is enhanced; to protect and enhance water resources, moderate floods and reduce sitting up of tanks and conserve rainwater for crops and thus mitigate droughts; and to utilise the natural local resources for improving cottage and small industries to improve socio-economic conditions of the local people.

The allied activities of industries are promoted; social forestry provides an additional source of income; soil erosion is checked; cottage industries are developed; more employment opportunities are created; the space available may be made more enjoyable, and ecology may be maintained in a healthy and sustainable condition.

The National Water Policy-2002

The National Water Policy 2002 lays emphasis on the integrated water resource development for:

(i) Optimal and sustainable utilisation of available surface and ground water.
(ii) Creation of well-developed information system.
(iii) Water conservation and demand management.
(iv) Quantity and quality aspects as well as environmental consideration.
(v) Involvement of beneficiaries in the project planning, and participatory approach in water resource management.
(vi) Adequate training and research.
(vii) Rehabilitation of project affected persons.

(viii) The policy also encourages private sector participation in the development and management of water resource projects.

AREA DEVELOPMENT PROGRAMMES

In the Fifth Five-Year Plan emphasis was laid on the preparation of sub-plans on the basis of area. The sub-plans prepared for specific areas ensures a certain order of developmental effort from the state plan funds supplemented by special central assistance. The area development plans are prepared to overcome the specific problems of the area with the help of the available physical and human resources. Some of the important area development programmes are: (i) Hill Area Development Programme, (ii) Desert and Drought Prone Area Development Programme, and (iii) Tribal Area Development Programme. Brief description of these programmes has been given as under:

Hill Area Development Programme

Nearly 18% of the total area of the country is hilly and mountainous in which over 70 million people are living. The hilly areas generally have poor accessibility. The soils on the slopes are thin, less productive and subject to severe soil erosion.

For the purpose of area development the hilly areas have been classified under two categories:

(i) **The Hilly Areas which are along the boundaries of the states and the Union Territories:** This category includes the states and union territories of the North-Eastern Region, Sikkim, Jammu and Kashmir, Himachal Pradesh and Uttarakhand. These are termed under *SpecialCategory* whose outlays are met, substantially, out of the Central assistance.

(ii) **Hilly Areas which form a Part of a State**: The developmental plan of such areas is prepared, implemented and financed by the respective state governments. The districts of Rajasthan situated in Aravallis, the hilly districts of

Chhattisgarh, Jharkhand, Madhya Pradesh, Telangana Andhra Pradesh, Karnataka, Tamil Nadu, Kerala, etc., fall in this category.

The hill areas of the country are different from the plain areas. The terrain, topography, slopes, climate, the cultural settings, and occupational structure are unique and diversified. The formulation of region specific development strategies for different hill areas of the country require, as a basic pre-condition, detailed information about the terrain economic, social, cultural, and political features, resource endowment (both physical and human), development potential, and their particular problems.

The Hill Area Development Plans are formulated and implemented with the set objective to provide basic life support system to the people and make a judicious use of the available physical and human resources. The sub-plans are centred around the active participation of people, particularly women, in the fulfillment of their basic needs. The people's involvement can be ensured by encouraging the concept of *social fencing,* which implies a voluntary and self imposed discipline in managing society's resources at a local level.

For the integrated development of the hill states and union territories of the North Eastern region, the Central Government had set up the North Eastern Council in 1971 by an Act of Parliament. The Council takes up such schemes as are of common interest to more than one State or Union Territory and to the region as a whole under its development plan. The North Eastern Council has played an important role in the development of inter-regional programmes of power generation, and transmission, construction of roads, agriculture, animal husbandry, fisheries, forestry, etc. Training infrastructure is being built up for manpower development in the region under the auspices of the Council.

The major thrust of the hill area programme is on the indigenous resources of the hills through the specially designed programmes for the development of the following:

1. Agriculture—shifting, sedentary, plantation (tea, coffee,

spices, rubber, coconut, litche, pineapple, etc.).

2. To develop horticulture, especially banana, apple, grapes, mangoes, and citrus fruits.
3. To promote animal husbandry, poultry, bee-keeping, sericulture, and piggery.
4. Strategy to conserve soil.
5. To promote cottage and village industries.
6. To provide employment to the rural youths.
7. To strengthen co-operatives and farmers' service societies.
8. In the hilly areas having hot and humid climate where shifting cultivation is practiced, special programmes have been designed to stop *Jhuming*.

The hilly regions are particularly suited to industries which require pollution free atmosphere, relatively cold climate, are based on high skill and high value additions like electronics, watch making, optical glasses, medicines, and drugs. Cottage industries like carpet making and handloom are also suitable industrial activities. Along with this, tourism is an important industry which can be developed in the hilly areas.

The Himalayan region is rich in pharmaceutical herbs, orchids, and flowers. The pharmaceutical industry can be developed in the Shiwalik and the Lesser Himalayas.

In order to achieve these objectives, a detailed survey of the hilly areas is required to obtain information about the distribution of minerals, vegetation types, and availability of the surface and underground water. The surface water can be a big asset for the generation of hydro-power.

Tribal Area Development

According to the Census of India 2011, the total population of the Scheduled Tribes was 84.3 million or 8.6% of the total population of India. The Government of India has included 427 communities in the Scheduled Tribes, given in the Eighth Schedule of the Constitution of India. These are the *first people* who are living in areas of isolation

and have been exploited throughout history. Their sense of history is shallow, in the sense that, after few generations, the remembered history tends to shade off into mythology. Their cultural ethos (language, religion, faith, traditions, and customs) are different from the other sections of society.

Numerically, the most important tribes of India are Gonds (8 million) the Bhils and the Santhals, each having a population of more than 35 lakhs. Next to them are the Minas, the Mundas, and the Oraons; each having a population of more than 10 lakhs. Then come the Hos, the Khonds, and the Kols; each having a population of more than five lakhs. Then there are 45 tribes, each having a population of between one and five lakhs. The smallest tribal community is the Jarawa and Sentales of the Andaman and Nicobar Islands.

Economy

The Scheduled Tribes are essentially subsistence cultivators. Over 90% of them (Gonds, Bhils, Santhals, Mundas, Oraons, Nagas, Khasi, Mizo, etc.) are dependent on cultivation of crops. The tribes of Andaman and Nicobar are dependent on food-gathering and hunting. Most of the forest dwellers (Birhor, Mallar, Kharia, Kadar, Chenchu, etc.) gather various roots and tubers, fruits, honey; manufacture ropes from wild creepers, and keep their collection for personal use or exchange it with agricultural produce. They occasionally indulge in fishing and hunting. The Todas, inhabiting the Nilgiris are pastoral tribes subsisting on the products of buffaloes.

Programmes for the Development of Scheduled Tribes

The Scheduled Tribes, as stated above are living in areas of isolation. Their habitat and cultural milieu are quite different from each other, and therefore, their developmental needs and problems are different from each other.

A number of programmes have been initiated by the Central and State governments for the development and welfare of the Scheduled Tribes. They can be classified as under:

Centrally Sponsored Programmes

1. Post-matric scholarships;
2. Girls' Hostels;
3. Pre-examination Training;
4. Tribal Development Blocks;
5. Co-operationx;
6. Research, Training and Special Projects;
7. Improvement of working and living conditions of those engaged in unhygienic occupations;
8. Coaching-cum-Guidance Centres; and
9. Grants for All-India non-official organisations doing welfare work among the Scheduled Tribes and Scheduled Castes.

Programmes in the State Sector

1. Pre-matric scholarships and stipends;
2. Exemption from tuition and examination fee;
3. Provision of educational equipments;
4. Provision of mid-day meal;
5. Setting up of Ashram Schools;
6. Grants for the construction and maintenance of hostel and school buildings;
7. Provision of land and irrigation;
8. Supply of bullocks, agricultural implements, seeds, and fertilisers;
9. Development of cottage industries;
10. Development of communications;
11. Cooperation;
12. Colonisation of shifting cultivation;
13. Supply of poultry, sheep, pigs, goats, etc;
14. Medical facilities;
15. Drinking water supply schemes;
16. Provision of house sites and houses;
17. Legal aid; and
18. Grants-in-aid to non-official organisations working at State

level.

Areas of Development

1. **Education:** The literacy rate of the Scheduled Tribes is only 49.5% as against the national average of 73.0% (2011). Since education plays an important role in bridging the gap between the tribals and non-tribals and in helping the process of change and modernisation, a considerable amount of money is being spent on the provision of educational facilities to the tribals. For this purpose, award of scholarships, hostel facilities, stipends, stationery grants, boarding grants, mid-day meals etc. have been provided by the central and state governments. The coaching and guidance facilities are also provided to tribal students who appear for the Indian Administrative Services (IAS) and other competitions. There are overseas scholarships also for them. Schemes are also being operated to train tribals to work as teachers in their own areas.
2. **Employment:** The Central and State governments have reserved 7.5% of the vacancies for the candidates of Scheduled Tribes. Reservation in direct recruitment to Class III and Class IV posts which normally attract candidates from a locality or region are fixed in proportion to the population of the Scheduled Tribe in the respective States and Union Territories.
3. **Agriculture:** About 90% of tribals are cultivators and about 82% of the main workers are engaged in primary economic activities. Most of them are dependent on shifting and/or rudimentary cultivation. The size of holding and the yield per unit area in tribal areas is generally very low. The Central and State governments are assisting them in the procurement of improved seeds, fertilisers, irrigation, credit facilities, soil conservation and land reclamation, adoption of better implements and new technology.
4. **Co-operation:** To achieve the transformation of tribal

agriculture and to prevent the exploitation of tribals from moneylenders, forest contractors, and other non-tribals, it is necessary to develop ‘co-operation’ and ‘co-operatives’. Because of extreme poverty, most of the tribals have to borrow money from the moneylenders and traders. However, once they incur a debt, they land into the clutches of the moneylenders from which they find it difficult to extricate themselves. There is an urgent need to provide alternate sources of credit facilities. They also need protection from forest contractors also who engage them as bonded labourers at very low wages. Here, the co-operatives can play a vital role.

5. **Communications:** Most of the tribals are living in hilly and mountainous areas or in areas of isolation or relative isolation. It is necessary to provide roads, transport, and other means of communication if they are to be absorbed and assimilated in the national stream. They also need marketing facilities to dispose off their surplus products.
6. **Land Distribution and Land Alienation:** The problem of land alienation is a serious one in tribal areas. Moneylenders, *Sahukars*, and other non-tribal people have continued to grab the land belonging to the tribals on one pretext or the other. To cope with this problem, safeguards along the following lines have been provided: (i) provisions restricting the transfer of tribal land to non-tribals by sale, as found in Gujarat, Odisha and Rajasthan, (ii) provisions restricting transfer of tribal land to any person—tribal or non-tribal, as found in West Bengal, and (iii) provisions restricting the transfer of tribal land by any means. Despite all these legal safeguards, it has not been possible to check land alienation.
7. **Industrialisation:** The state governments carried out programmes for developing cottage industries and subsidiary occupations among tribal people. These include bee-keeping, poultry, sheep rearing, weaving, sericulture, and *palm-gurs.*

In addition to these, some important industrial projects were located in tribal areas. These included Bhilai, Bokaro, Burnpur, Durgapur, Raurkela, Bhadravati, Kalinga, etc. These projects have provided considerable employment opportunities to tribal people as unskilled labour.

8. **Tribal Development Agency Projects:** The Tribal Development Agency Projects were established in some areas on an experimental basis in 1971–72. In the selected districts the work of the tribal development agencies is being implemented under the chairmanship of the district collector. The agency also has a full-time project officer and other district officers as members.
9. **Other Schemes of Development:** A number of schemes were also undertaken to improve the housing, drinking water, and sanitary conditions of tribals and in providing adequate medical facilities to them.

No one can ignore the genuine desire of the government to uplift tribal people and the role played by its efforts in changing the mode of thinking and attitude of many tribals. Thus, the educational policies of the government, provisions of reservation in jobs, policies in the fields of agriculture, co-operation, communications, industrialisation, social services, etc., have brought tribals closer to the main stream of national life. Tribal markets have been linked with urban markets providing greater economic opportunities to the tribals and creating a commercial aptitude in them. Despite all these projects and schemes, the efforts so far have not helped in solving the problems of poverty, malnutrition, hunger and exploitation faced by most tribals.

The benefits, however, have failed to reach the lower strata of the society and got concentrated in the hands of the upper crust of the tribal community. The most important criticism of the Tribal Development Blocks is that the whole programme has been carried out without the participation of tribal people. The tribals had no particularly say in the programmes formulated for their own development, nor was their co-operation obtained in the

implementation of these programmes. The unchanging bureaucracy has failed to fulfill the role of development agency. The role of local politicians and local elite power was also not in the interest of the tribal masses. They perpetuated the situation in their self interest and deprived the tribal poor of their share.

Moreover, industrialisation in tribal areas created two complex problems:

(i) Firstly, establishment of large industrial projects in tribal areas has caused a large scale displacement of tribals, and

(ii) Secondly, tribal population is assuming more and more migratory character and tribal life is getting disintegrated.

It is unfortunate that, in spite of specific safeguards provided in the Constitution for the tribals protection and development, most of them have not only remained neglected, but an exploited lot.

Integrated Development of Tribal Areas

The strategy in the integrated development is (i) sectoral-cum-temporal integration, (ii) spatial integration, (iii) integration between the development of different people and groups of people, and (iv) integration of the conflicting goals of economic, social, and environmental development. Such integrated planning is necessary to ensure popular interest and participation in the planning process and equitable distribution of the fruits of economic growth and development. However, the tribal development projects were generally formulated on a uniform basis for all regions of the country and hence, lacked the necessary appreciation of the local needs and flexibility to adjust according to local situations.

To take account of these factors, the Fifth Plan introduced the concept of sub-plan. The distinctive features of the sub-plan are:

1. It caters to the special needs of the designated areas.
2. It employs a total (as against sectoral) approach to problems of the area.
3. It accepts the uniqueness of the problem of each viable area and community and formulates programmes with reference to it.

4. Its resources will be the sum total of resources of the State Plan for the sub-plan area, the Central sectoral outlays, special central assistance and institutional finance.

The contents of the tribal sub-plan are as under:

(i) Identification and demarcation of areas of tribal concentration.
(ii) Identification of socio-cultural barriers and promoters of change and development.
(iii) Assessment of potentialities, special problems, and felt-needs of the tribal areas.
(iv) Assessment of the resources available for the tribal sub-plan from the normal plan funds, special Central assistance and institutional sources.
(v) Formulation of sectoral programmes.
(vi) Developing a suitable administrative set-up.

The structure of planning proposed in a tribal sub-plan is a three tier structure involving micro, meso, and macro levels of planning. The micro region will be coterminous with a development block, the meso region will be contiguous in the development block involving 3 to 5 lakh people having comparable levels of socio-economic development, and the macro region will be formed of bigger tribal belts for planning of tribal development. The functions at the micro, meso, and macro levels have been visualised as follows:

Micro Level: At the micro level, the functions included are:

(a) education up to higher secondary level,
(b) elementary health services,
(c) agricultural extension,
(d) supply of agricultural inputs,
(e) minor irrigation schemes,
(f) elementary veterinary services,
(g) multi-purpose co-operatives,
(h) local panchayat,
(i) household industry, and
(j) village approach roads.

Meso Level: At the meso level, the functions included are:

(a) higher general education,
(b) technical and vocational training,
(c) manpower planning and employment services,
(d) advanced health services with referral facilities,
(e) agricultural research extension,
(f) seed multiplication farms,
(g) soil conservation and land management,
(h) apex integrated credit marketing structure with adequate storage and buffer stock facilities,
(i) development of road and communication infrastructure connecting market with
state/district highways,
(j) distribution of network of power, rural electrification, etc.
(k) local resource based industries with adequate market linkages,
(l) forest management,
(m) horticulture development,
(n) complementary development programme in hinterland and bigger industries,
(o) medium irrigation projects, and
(p) research statistics and evaluation.

Macro Level: At the macro level, the functions include:

(a) co-ordination of activities in tribal development projects,
(b) agricultural research on regional basis,
(c) direction of the various sectoral programmes in the project,
(d) major irrigation projects,
(e) river valley development,
(f) industrial and mineral development of the region to ensure complementarity of the project level development programme,
(g) marketing support projects, and

(h) evaluation.

The idea behind the integrated area planning is to ensure a collective utilisation of the resources of a particular area keeping in view the problems and needs of development of that area. The spectrum of the problems of development that still await solution in tribal areas include the following:

1. land alienation,
2. indebtedness and exploitation,
3. bonded labour,
4. low educational standard,
5. low agricultural production and productivity,
6. inadequate supply of essential consumer goods,
7. land reclamation,
8. inadequate irrigation facilities,
9. shortage of drinking water,
10. inadequate communication facilities,
11. housing problem,
12. inadequate marketing facilities,
13. inadequate employment opportunities, and
14. under-developed state of health services.

In order to overcome these inadequacies there is a need to bring about institutional changes in the tribal society.

Strategies for Development

The following steps can go a long way in achieving the developmental goals at the micro, meso, and macro levels:

1. **Problem of Agricultural Development**: Developmental programmes aimed at increasing agricultural production and productivity should include: (i) development of irrigation facilities, (ii) schemes of soil conservation, (iii) provisions of modern agricultural inputs at reasonable prices, (iv) making the size of farm holding economic, (v) improving agricultural marketing facilities, and (vi) educating tribal farmers in new agricultural techniques.

2. **Problem of Land Alienation**: Since tribals have lost a considerable extent of land to moneylenders, forest contractors, etc., there is a need to evolve steps such as: (i) making legal protection available to tribals, and (ii) making tribals conscious of their rights. It is also necessary to prepare authentic land records. In most of the tribal areas no land records are available. This appreciably reduces the effectiveness of legal protection guaranteed to the tribals.
3. **Development of Transport and Communication**: In order to increase the personal social interaction, the development of roads and communication facilities is imperative. Such infra-structural facilities are also required to generate economic progress in tribal regions.
4. **Forest Policy**: Since forests have an important place in tribal economy, it is necessary to carry out a review of the forest policy to provide maximum benefits to tribal people.

5. **Development of Allied Agricultural Activities**: Allied agricultural activities such as dairying, poultry, piggery, animal husbandry, honey-bee keeping, sericulture, etc., should be given encouragement.

6. **Development of Public Distribution System**: To supply essential consumer goods to tribals at reasonable prices, the public distribution system should be adequately developed.
7. **Setting up of Cottage and Small Scale Industries**: In selected growth centres, cottage and small scale industries should be developed. The uprooted tribals can be given training in cottage industries and jobs in small scale industries.
8. **Provisions of Social Services:** Social amenities, such as medical and health facilities, drinking water, nutrition, rural housing, etc., should be provided in the rural areas.

9. Co-operation: Since the tribals have a strong community feeling and long tradition of democratic institutions, the co-operative movement and panchayati-raj institutions should be strengthened.

10. **Development of Educational Facilities:** Development of

more educational facilities in the tribal areas is necessary to sustain the programmes of economic development.

11. **Exploitation:** Adoption of protective measures are necessary to prevent exploitation of tribals. Numerous atrocities are committed on tribals by the landlords, forest contractors, government employees (including police), panchayat members, etc. It is necessary to take preventive steps to stop such crimes committed against the tribals.

12. **Special Programmes:** Special programmes should be launched for the benefit of tribals belonging to small, isolated, and primitive tribes that have remained more or less untouched by the plans.

13. **Change in Administrative Set-up:** The administrative set-up should be made more responsive to the needs of development of tribal people.

14. **Periodic Evaluation:** A periodic evaluation of the programmes adopted in tribal areas should be carried out to ensure 'timely corrective action'.

Desertification and Desert Development Programme

The term desertification was coined by the French forester Aubreville in 1949 to describe land degradation. Since then there have been over 100 published definitions of this controversial term which gained prominence after the 1977 UN Conference on Desertification. The term has been confused with drought in some circles, and its social impacts have sometimes been associated with famine that may in fact have non-environmental causes.

In the Convention to Combat Desertification (1995), desertification is defined as *and degradation in arid, semi-arid, and dry-sub-humid areas resulting from various factors including climatic variations and human activities.'* The problem is therefore confined to the susceptible *dry-lands*, with land degradation regarded as soil erosion, internal soil changes, depletion of ground water reserves and irreversible changes to vegetation communities. In many respects, desertification is no different than land degradation occurring world-wide, except that it specifically refers to its occurrence within dry areas.

In environmental terms, the soil degradation component of desertification comprises water and wind erosion, and physical and chemical changes within the soil. These factors reduce the potential productivity of the affected land.

Desertification is a serious global problem, affecting 16% of the total population, about 34% of the land area and 70% of all dry-lands of the world.

Though the '*Convention to Combat Desertification (CCD)*' definition includes possible multiple causes for desertification, it is undoubtedly the case that the principal agent of degradation is human actions. Though desertification has often been viewed as a particular problem of the 20 Century, the ability of humans to detrimentally alter dry-lands is not new. Salinization in Mesopotamia around 2500 BC has been attributed to agriculture, while in central Mexico, prehistoric societies caused severe soil erosion. In the 20 Century it is the growth and changing distribution of human population in dry-lands, and the spread of technological advances, that have increased the propensity for dry-land degradation. Several issues make the dry-lands especially susceptible to degradation by humans.

Desertification involves four processes that have an extensive impact on the biological productivity of desert land. They are : (i) degradation of vegetation cover, (ii) soil erosion, (iii) salinization, and (iv) water-logging including soil compaction. Thus, desertification is a serious threat to the socioeconomic fabric all over the world. Environmentally, it contributes to climate change, water air and soil pollution, desertification and soil loss. Some of the glaring aftermath effects of environmental degradation are:

1. loss of global biological diversity,
2. loss of biomass and bio-productivity,
3. exhaustion of humus reserve,
4. disrupting normal global bio-geochemical turnover and reduction on the global carbon-dioxide sink,
5. desertification contributes to climatic changes by increasing potential and decreasing actual evapo-transpiration rate,

6. changing ground-surface-energy budget and adjoining air temperature, and
7. adding dust and carbon dioxide to atmosphere.

Drought and Desertification: Land in many parts of the world has been degrading into desert like conditions because of drought. Consequently, drought is threatening more people than ever before. Worth noting is the fact that drought is a temporary problem but desertification is not that temporary, creating a more serious problem than drought. Man can do little to prevent drought, but he can halt desertification. Drought is temporary, cyclical and can be expected in all climates, whereas desertification is insidious, unexpected and permanent. In fact, drought is considered as an engine of desertification.

Drought and desertification are closely linked. Water shortages and high rainfall variabilities make desertification more acute. This, in turn, promotes drought and intensifies its effect. Ecological degradation processes are set in through excessive destruction of natural vegetation. It is only during prolonged periods of drought that the above mentioned processes lead desertification.

The ecological consequences of long-term rainfall deficits are:

(i) elimination of grass cover,
(ii) elimination of some bush vegetation,
(iii) drop in ground water-table near the surface,
(iv) an increase in shifting sands and reactivation of some old sand dunes,
(v) more wind erosion of fine soil components, and
(vi) increased evaporation, with drying out of soils cracking on clay soils.

Desertification Control: Regardless of the challenges involved, desertification control and restoration of degraded land are imperative for the survival mankind and sustainability of ecology and environment.

Before taking up desertification control on any dry-land or area, it is necessary to assess and monitor the type and the severity of the land

degradation. The causes of land degradation are also to be ascertained. Based on degradation, the selection and application of appropriate actions, will have to be taken. Whatever steps or actions are taken to combat desertification, they should aim at achieving sustainable development of the area, which can be possible with the management and conservation of the resource-base, the land, so as to meet the present and future needs of the people habitating that area or region. The sustainable development of arid, semi-arid and dry-sub-humid zones usually confronts three challenges: (i) using preventive measures, to check or to prevent desertification of land slightly or not yet degraded; (ii) to regenerate, through corrective measures, the productivity of the moderately degraded land, and (iii) to restore the productivity of seriously degraded land, using rehabilitation and repair measures.

Desertification Control in Arid Lands of India: Desertification is a serious problem in the arid dry-lands in India. It is more severe in the north-western plains of the country, especially in the Thar Desert of Rajasthan, Gujarat, Haryana and Punjab **(Fig. 15.1)**. Wind erosion is a major problem in part of the desert. Higher average wind speeds, dominantly sand terrain, sparse vegetation cover and high human activities on the sand-dunes and sandy-plains lead to accentuation of sand-blowing. It also leads to erosion of top soil, containing nutrients and precious little organic matter as also moisture, thereby reducing soil quality required for sustainable agricultural production. Standing crop-plants are damaged and good agricultural lands and infrastructure are buried as a result of wind erosion. The major activities of wind-erosion control are sand-dune stabilization and shelter belt plantation. The following steps can go a long way in controlling the process of desertification:

1. **Sand-Dune Stabilization**: The sand-dunes are generally categorized into the old and new sand-dunes. Old sand-dunes, usually higher than 10 m, are naturally stabilized with potential to support copious natural vegetation. These dunes have extremely low rates of movement unless their ecology is

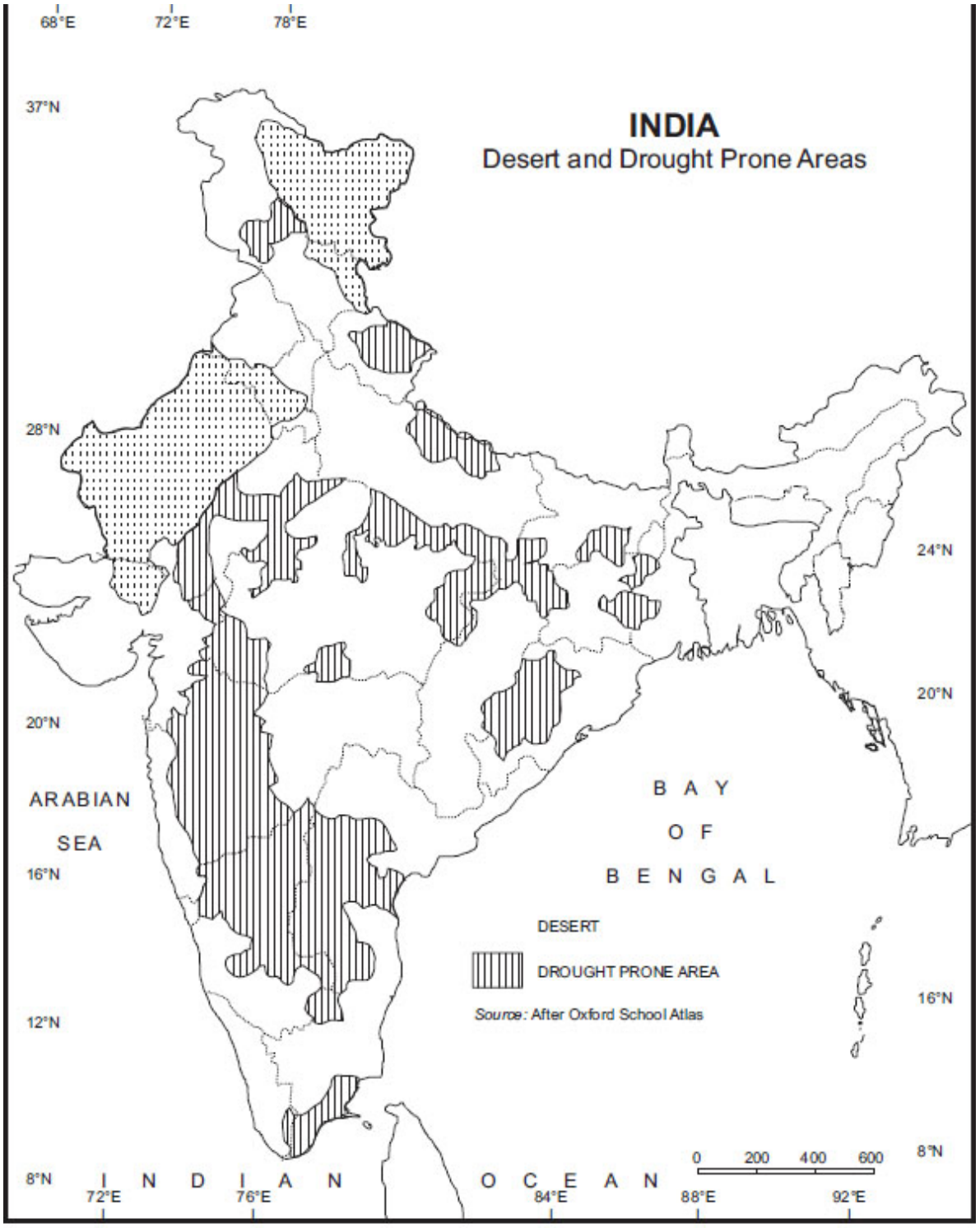
disturbed by human action. The new dunes are mostly smaller than 10 m. They are bare of vegetation and have a high rate of movement.

Most programmes of sand-dune stabilization have been directed towards old dunes, to restore production potential. The activities mainly consist of: (i) protection of the area from human and livestock encroachment; (ii) creation of micro-wind breaks on the dune slopes, (iii) direct seeding or transportation of indigenous or exotic species; (iv) plantation of grass-strips, and (v) management of vegetated sites.

2. **Shelter-belt Plantation**: Erection of shelter belts along the boundaries of crop fields helps reduce injuries to tender seedlings from sand blasting and hot desiccating winds. Wind velocity at the lee side of the shelter belt is reduced and also the losses by about 76%.
3. **Soil and Water Conservation Measures**: The wetter parts of the arid zone and the semi-arid areas are more prone to water erosion. The counter crusting of soil and its subsequent erosion, a number of practices have been suggested, like contour bunding (low rainfall area) and contour tillage and contour sowing in the agricultural fields. Moreover, mixing of crop residue and organic matter with light-textured soils helps to increase moisture and crop yield.
4. **Management of Pastures**: Permanent pastures are most degraded, as they do not have any basal plant-cover. The natural vegetation has been destroyed and the pastures have been converted into barren ground. These degraded pastures need to be rehabilitated.
5. **Management of Waterlogged and Salt Affected Tracts**: Mismanagement of water and over-irrigation have rendered many productive lands to waterlogged and salinity-alkalinity affected areas. The command area of the Indira Gandhi Canal (Rajasthan), and the command areas of small and medium irrigation tanks in the desert are some of the glaring

examples. Lining of canals, judicious use of canal-water, soil management and scientific rotation of crops are necessary to arrest degradation of land.

6. **Rehabilitation of Mine-Spoils**: There is need to plant suitable tree bushes, shrubs and grass species to be planted in the areas of desertification. Development of micro-catchments with 5% slope to ensure run off to single row of plants, coupled with other soil conservation practices like ridge and furrow system.



Map not to Scale

Fig. 15.1 Desert and Drought Prone Areas of India

7. **Hilly Area:** In the hilly and rocky areas like the Aravalli

planting acacia (Babul trees) helps in the combating of desert. In the saline and alkaline affected areas the application of gypsum and farmyard manure along-with salt-tolerant plants can help appreciably in combating deserts.

8. **Contour Farming**: Contour farming and contour bunding, graded bunding, bench terracing on steep slopes, and run-off harvesting, storage and recycling are also important steps in combating deserts, especially in the sub-humid areas. Mixed grass and fuel-wood tree species may be planted on denuded lands.
9. **Ravine Lands Reclamation**: Gully plugging is done to conserve soil moisture, control erosion and provide better growing conditions for natural as well as for planted grasses and forest species.
10. **Rehabilitating Degraded Rural Commons**: In the tropical deserts, Rural Common Property Resources, community pasture, forest, wasteland, watershed drainage etc. form an important asset in the desert of India. Rehabilitation of the community degraded lands and pastures should be done with new technology at the priority basis. The silvi-pastoral system, which not only provides fodder and fuel-wood but also improves the soil fertility.

If the above steps and methods are adopted in collaboration with each other, the problem of desertification can be solved to a great extent.

The Drought Prone Area Programme (DPAP) was launched in 1973–74 to tackle the special problems faced by areas constantly affected by severe drought conditions, while the Desert Development Programme (DDP) was launched in 1977–78 to mitigate the adverse effects of desertification, the Integrated Watershed Development Programme (IWDP) has been implemented since 1989–90 for the development of watersheds/degraded lands. The basis of implementation of all the three programmes has been shifted from sectoral to watershed from April 1995.

As stated above, the Desert Development Programme (DDP) was launched in the year 1977–78. It extends over 227 development blocks of 36 districts with a total area of about 4.6 lakh sq km in seven states (Andhra Pradesh, Gujarat, Haryana, Himachal Pradesh, Jammu and Kashmir, Karnataka, and Rajasthan).

Under the programme a number of projects have been targeted and sanctioned for implementation in the identified desert areas. All these projects are envisaged to be completed within the five year plan period. At present there are over four thousand watershed projects under different stages of implementation in the seven states mentioned above. Presently, the programme is 100% sponsored (75:25 basis in hot arid areas between the Centre and the states). Eighty per cent of the funds are transferred to the bank account of the Watershed Committee/Gram Panchayats which implement the programme.

Desert Development Programme aims at checking further desertification of the desert areas and raising productivity of the local resources to generate employment, to raise the income and standard of living of the inhabitants of the desert.

Main Problems

The main problems of the deserts are:

1. In the desert areas, rainfall is scanty and highly erratic which makes agriculture highly vulnerable to droughts and famines.
2. The soils are sandy which are susceptible to wind erosion.
3. Only the coarse grains like *bajra*, millets, and pulses can be grown. Moreover, the yield per unit area is low.
4. Most of the farmers in the desert areas are poor and they cannot adopt costly inputs.
5. At the occurrence of drought the people have to migrate in search of jobs.
6. The infrastructural facilities like market, storage, and transport are poor.

Main Objectives

The main objectives of the plan are:

1. To promote dry farming to increase agricultural productivity.
2. To make judicious use of the available water resources.
3. To generate more employment to enhance the per capita income and to raise the standard of living of people.

Strategies for Development

Under the Desert Development Programme the following strategies have been adopted:

1. The development of deserts forms part of the 20-point Programme initiated by the government.
2. Under the dry-farming programme, crops like bajra, sunflower, mustard, coarse grains, pulses and the less moisture requiring crops have been promoted.
3. Emphasis has been laid on social forestry, agro-forestry, horticulture, silviculture, bee-keeping, and silvi-pastures.
4. Restoration of soil fertility and control of soil erosion by creating wind-barriers.
5. Emphasis on eco-friendly insecticides, pesticides, and weedicides.
6. To assist the farmers, especially the marginal and small farmers engaged in dairy, poultry farming, sheep-rearing and horticulture.
7. Reclamation of water-logged areas along the canals, especially the Indira Gandhi Canal.
8. Involvement of the Non-Government Organisations (NGOs) in the desert development programmes.
9. Research in water harvesting techniques and judicious use of the available water.

NATIONAL WATERSHED DEVELOPMENT PROJECT FOR RAINFED AREAS (NWDPRA)

The National Watershed Development Project for rainfed areas was launched in the Eight Plan period and has been subsurned under

Micromanagement of Agriculture since November, 2000. The broad objectives are:

(i) Conservation, upgradation and sustainable utilisation of natural resources;
(ii) restoration of ecological balance in the degraded and fragile rainfed ecosystems by greening these areas through appropriate mix of trees, shrubs and grasses;
(iii) reduction in regional disparity between irrigated and rainfed areas; and
(iv) creation of sustained employment opportunities for the rural poor.

The programme is presently being implemented in all the states. During 2011–12 an area of 2.50 lakh hectares is targeted for treatment with an estimated cost of Rs.300.00 crore.

RAINFED AREA DEVELOPMENT PROGRAMME (RADP)

Ministry of Agriculture has initiated farming system and farmer centric interventions under a new scheme 'Rainfed Area Development Programme' to look into the specific needs of the rainfed areas for improving the quality of life of farmers, especially small and marginal farmers by offering a complete package of activities to maximise the farm returns for their food and livelihood security.

Under this progrmame an end to end approach covering production, on farm management, storage, marketing, value addition are being contemplated to assure appropriate returns to the farmers from the rainfed areas.

During 2011–2012, an amount of INR 250 crore has been allocated for RADP for adoption of holistic agriculture development through integrated farming system in 10 identified states across the country on pilot basis.

Integrated Watershed Management Programme (IWMP)

It was implemented by Department of Land Resources of Ministry of Rural Development. Its main objective is to restore ecological balance

by harnessing, conserving and developing degraded natural resources such as soil, vegetative cover and water. It intends to cover 55 million hectares of rain fed land by 2027. It is the world's second largest programme after China.

The major outcomes of IWMP are:

(i) Prevention of soil run off
(ii) Regeneration of natural vegetation
(iii) Rain water harvesting
(iv) Recharging of the ground water table.

MULTI-LEVEL PLANNING

A planning process can be either single-level or multi-level. In the single-level planning, the formulation of plans and decision making are done at the national level; the process is centralised and the lower territorial levels come into the picture only at the implementation stage. On the other hand, in the multi-level planning process, the national territory is divided into small territorial units, their number depending upon the size of the country, the administrative, the geographical and cultural settings.

The concept of multi-level regional planning may be defined as *planning for a variety of regions which together form a system and subordinate systems.* In multi-level planning, the various levels of planning provide bases for higher-level planning. Similarly, the higher-level regional plans provide the basic frame-work for the lower-level plans. In such plans, there is direct participation of the people in the planning process. In multi-level planning, every region/unit constitutes a system and hence, the planning process becomes more effective.

The various levels of multi-level planning in India are: (1) Centre, (2) States, (3) Districts, (4) Blocks, and (5) Villages. Before the Amendment of the Constitution in 1992, the task of plan formulation was basically carried out by the Centre and the State governments. The other levels (district, block, and village) of the planning process came into the picture only at the implementation stage of planning. However, after 1992, the lower levels of planning have been given

more powers in the formulation and implementation of their plans and financial provisions.

1. The First-Level Planning

In the context of multi-level planning in India, the first (and foremost level) is constituted by the Central Government. The Head of the Government is the President who acts under the advice of the Council of Ministers headed by the Prime Minister. The advice of the Council of Ministers is binding on the President. The business of the government is transacted through a three tier set up. The Prime Minister, Cabinet Minister, Minister of State, Deputy Ministers, and Parliamentary Secretaries constitute the first or top layer. The second layer is constituted by the secretariat organisation of the ministry with the secretary as head. The executive organisation of the department, comprising the ministry, forms the third layer. The actual task of plan formulation is done by the Planning Commissions at the Centre which has the Prime Minister as Chairman, and some members of the Cabinet and three or four full-time professional experts as members.

2. The Second-Level or State Level Planning

The second level is constituted by the States. The Executive head of a state is the Governor, who is appointed by the President of India on the advice of the Prime Minister of India. As in the case of the Centre, the Governor does not directly exercise the powers that are vested in him. They are exercised through the Council of Ministers headed by the Chief Minister. The advice of the Council of Ministers is binding on the Governor. The Council of Ministers works through the secretariat that is headed by a secretary. The main functions of the secretariat relate to assisting the ministers in policy making and in discharging their legislative responsibilities, co-ordination of policies and programmes, supervision and control of expenditure, efficient running of administration, etc. The Council of Ministers has a number of departments functioning under it which can be broadly classified into three categories:

(a) Development departments (having the departments of agriculture and animal husbandry, rural development, public

works and industries).

(b) Social welfare departments (having the departments of education, health and social welfare).

(c) Coordinating departments (having home, revenue, finance and planning departments).

Since the Indian Union is a federation of States, the framers of the Constitution have provided for division of powers between the Centre and the States, so that the chances of confrontation between the two levels of government may be minimised. The division of powers between the Centre and the States is according to the seventh schedule and Article 246 of the Constitution. Article 246 gives three lists of subjects: 1. *Union List* (containing 97 items), 2. *State List* (containing 66 items), and 3. *Concurrent List* (containing 49 items).

The Central Government has the power to legislate on the subjects given in the Union List while the State governments have powers to legislate on the subjects given in the State List. As far as subjects contained in the Concurrent List are concerned, both Central and State governments have powers to legislate on them, but in case of conflict, the Central law prevails. Organised activities such as industries, minerals, railways and telecommunications come under the Centre's responsibilities, while agriculture, collection of land revenue, irrigation, power, public health, education, local self government, and several other important subjects come under the control of states.

3. The District-Level Planning

There are generally three levels below the State level, namely, district, block, and village. The district administration is under the overall charge of the collector, also known as deputy commissioner in some states). He or she is responsible for maintaining law and order in the district. He or she also works as a coordinator among various departments. Important departments located in the district include the agriculture, minor, medium and major irrigation, animal husbandry, dairying, forestry, industries, public works, cooperation, education, medical and publc health, social welfare and panchayati raj, etc.

Besides, there are autonomous agencies with offices in the district such as electricity board, state transport and special agencies set up for special programmes such as SFDA, DPAP, etc. Though the collector generally works as a coordinator, his or her role is not defined properly and differs in different states. This introduces complexities and difficulties in the task of administration.

4. The Block-Level Planning

At levels below the district, there are blocks and villages. The block level planning was started during the First Five-Year Plan. Each district was divided into a number of blocks and each block comprised about 100 villages with a population of about 60,000. The programme visualised (i) mobilisation of local resources, and (ii) participation of people in the decision making and implementation of the development scheme. Hence, a new unit of planning was created at block level under the leadership of a block development officer and a team of various specialists and village level workers. The general supervision of blocks was made by the *Block Samitis* under the chairmanship of the *Block-Pramukh* and elected representatives.

The set-up basically consists of a block development officer who is assisted by five extension officers, one each in the field of agriculture, animal husbandry, cooperation, *panchayats*, and rural industries. The other staff consists of an overseer, a social education organiser, a progress assistant and village level workers. In the block, there are also veterinary stockmen, a medical officer, a sanitary inspector, and a lady health visitor; some of them with necessary supporting staff. In some states, there is also an extension officer for programmes relating to women and children.

The block-level planning focuses more on local problems. At the micro level it becomes easier to identify target groups which enables the optimum utilisation of local resources. The entire strategy of the block-level planning is based on employment planning, growth-centre planning, and credit planning. The following activities are palnned at the block level:

(i) Agriculture and allied activities;

(ii) Minor irrigation;
(iii) Soil conservation and water management;
(iv) Animal husbandry and poultry;
(v) Fisheries;
(vi) Forestry;
(vii) Processing of agricultural produce;
(viii) Organising input supply, credit, and marketing;
(ix) Cottage and small industries;
(x) Local infrastructure; and
(xi) Social Services.
    (a) Drinking water supply;
    (b) Health and nutrition;
    (c) Education;
    (d) Housing;
    (e) Sanitation;
    (f) Local transport;
    (g) Welfare programme; and
(xii) Training of local youth and updating of skills of local population.

5. The Panchayat-Level Planning

A Panchayat is an elected body at village level. The *Panchayat Raj System* involeves a three-tier structure:

(i) village level,
(ii) block level, and
(iii) district level.

The first tier at the village level is commonly known as *Gram Panchayat or Gram Sabha* (Village Assembly), the second tier at block level as *Panchayat Samiti*, and the third tier at district level as *Zila Parishad.* According to the provisions of *Panchayat* Acts 1996, the election to the *Village Panchayat* is held at an interval of five years where there is proportionate seat reservation for Scheduled Castes and Scheduled Tribes, and not less than one third seats for

women. There are about 2.20 lakh *Gram Panchayats*, 5500 *Panchayat Samitis*, and over 600 *Zila Prishads* in the country.

Through the Constitution Amendment Act 1992, the *Panchayat*, also called *Gram Sabha*, has been authorised to look after the preparation and implementation of plans for economic development and social justice on an illustrative list of 29 subjects. The respective states have been given discretionary powers to prescribe powers and functions of the *Gram Sabha* to act as an institution of self government. It has also been advised to constitute a District Planning Committee to consolidate the plans prepared by the *Panchayats* and Municipalities, and prepare an integrated development plan for the district as a whole. It has also been directed to constitute a State Finance Commission (SFC) to review every five years, the financial position of *Panchayats*, to make recommendations about the principle governing the distribution of revenue between the state and the *Panchayats*, and determination of the grants-in-aid to the *Panchayats* from the Consolidated Funds of the State.

The implementation of the plan at the *Panchayat*-level is the responsibility of the Village Development Officer (VDO) and the secretary, and is supervised by the Gram Sabha which is headed by the *Gram Pradhan*. Under the existing provisions, funds for the *Gram-Sabha* (Village Panchayat) are directly being allocated from the centre to execute rural development programmes like Integrated Rural Development (IRDP) and Jawahar Rojgar Yojna (JRY).

The *Panchayat* has also been entrusted with the responsibility for the (i) promotion of agriculture, (ii) rural industries, (iii) provision of medical facilities, (iv) maternity, women, and child welfare, (v) maintaining common grazing grounds, (vi) village roads, tanks, wells, (vii) sanitation, and execution of other socio-economic development programmes. The *Panchayats* have also been authorised to identify the beneficiaries in anti-poverty programmes.

After the Constitution Amendment of 1992, the new status accorded to the *Panchayats* by the Constitution has raised high hopes and expectations among the elected representatives and the

rural folk at large. But owing to the political complexions of the governments in Indian states, the reluctance of the state-level political and administrative functionaries to part with power and authority, and some genuine financial and economic difficulties, the progress in the implementation has been somewhat slow. It has been found that the elected representatives of *Panchayat Raj* Institution are largely unaware of the political and economic dimensions of development issues and lack of planning and managerial skill. This is particularly true of elected women representatives, who are performing their duties under some severe constraints of different kind.

THE DAMODAR VALLEY CORPORATION

Flowing through the states of Jharkhand and West Bengal, the Damodar river is 541 km (336 miles) long. The river is also known as *Deonadi* in its upper reaches. It rises in western Jharkhand in the hills of Chotanagpur and drains areas in the Ranchi, Hazaribagh, Dhanbad, and Santhal Parganas districts of Jharkhand, and Bankura and Burdwan districts of West Bengal. After its confluence with the Barakar river, it enters West Bengal. Flowing through Bankura and Burdwan, it finally merges into the Hugli River opposite to Falta, approximately 50 km north of Kolkata. The river has a total catchment area of 25,820 sq km **(Fig. 15.2).**

The upper reaches of the Damodar presents a highly denuded and desolate topography characterised with gorges and barren rocks, while the lower part in West Bengal is a flat, fertile stretch of thick alluvial soil. It is known for lush green fields of paddy crop. The catchment area of Damodar receives about 125 cm average annual rainfall which occurs mainly in the months of July, August, and September. In its upper reaches it runs rapidly, eroding land and collecting silt on large scale. In its lower reaches it runs too sluggishly, discharging silt all along its banks through flooding.

The river basin had great natural endowment in the form of fertile soils, huge water resources, forest wealth, great variety of good reserves of minerals. It is rich in coal, iron ore, bauxite, mica, fire-clay,

limestone, lead, silver, antimony, quartz, chromite, kyanite, etc. **(Fig. 15.3).** Moreover, the upper part of the basin was the home of several aboriginal tribes including Oraons, Hos, Santhals, and Mundas.

The Damodar River, often called The '*Sorrow of Bengal'*, is notorious and is known for its erratic character. During the last one hundred years it recorded serious floods in 1901, 1905, 1907, 1913, 1916, 1923, 1935, and 1943. During these years the countryside was inundated, crops and cattle were washed away, and transportation and communications seriously disturbed. Between 1862 and 1872 floods in the Damodar River caused a malaria epidemic and almost one third of the population on the left bank of the river died of what came to be known as *Burdwan Fever.* The 1943 flood was even more devastating in which the countryside south of Burdwan was submerged and inundated in some areas to a depth of six or seven feet, villages were swept away, rice fields were devastated, rail traffic was seriously disrupted since railway bridges were washed away, the grand trunk road was breached, and for a time, even the cities of Burdwan and Calcutta were in danger.

In order to overcome the problems of floods, inundation, and to utilise the water of the river for the generation of hydro-electricity, irrigation, and industries and to develop the region agriculturally and industrially, the Central Government, in consultation with the state governments of Bihar and West Bengal, conceived the Damodar Valley Project, later on called the Damodar Valley Corporation in 1945 on the lines of Tennessee Valley Authority (TVA) of USA. One of the leading engineers of the Tennessee Valley Corporation, W.L. Voorduin prepared the Preliminary Memorandum that presented a plan of flood control, irrigation, power generation, and navigation in the Damodar Valley. Finally the Damodar Valley Corporation was instituted through an Act of parliament in 1948 and was assigned INR 131.71 crore for implementation of the project.

The project was designed to construct seven dams, namely, (i) *Maithon*, (ii) *Tilaiya*, (iii) *Balpahari (all on Barakar river)*, (iv) *Panchet Hill*, (v) *Aiyar (on Damodar river)*, *and one across* (vi) *Konar*, *and one*

on (vii) *Bokaro* **(Fig. 15.4).** The total storage capacity of these dams was 4.7 million acre feet. The total storage capacity of 4.7 million acre feet thus created was considered sufficient to supply regular water for irrigation during the entire non-monsoonal period of 8 months. The Damodar, a seasonal river was thus converted into a perennial river to ensure irrigation throughout the year.

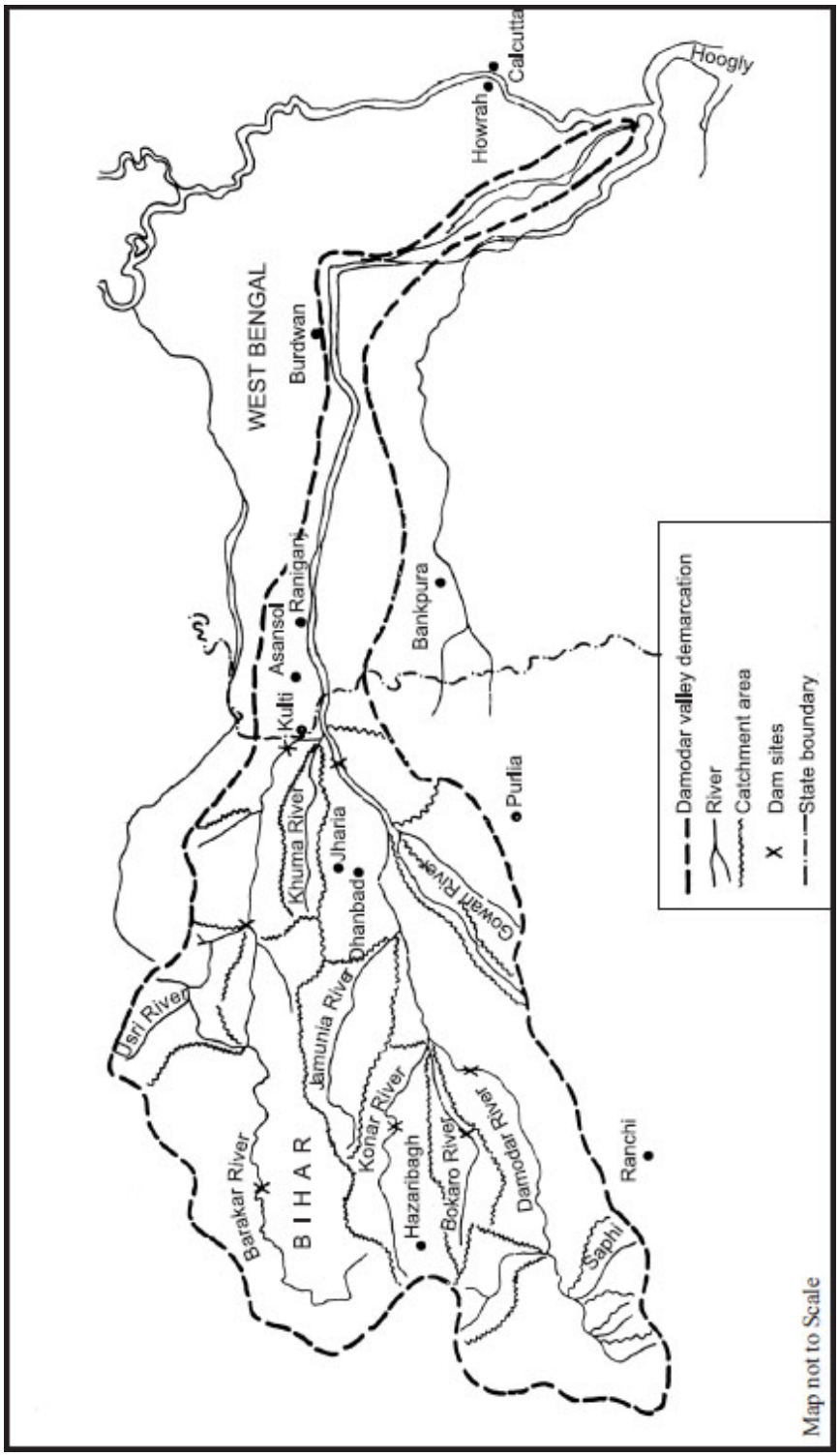


Fig. 15.2 Damodar Valley—Catchment Area (after R.C. Chandna)

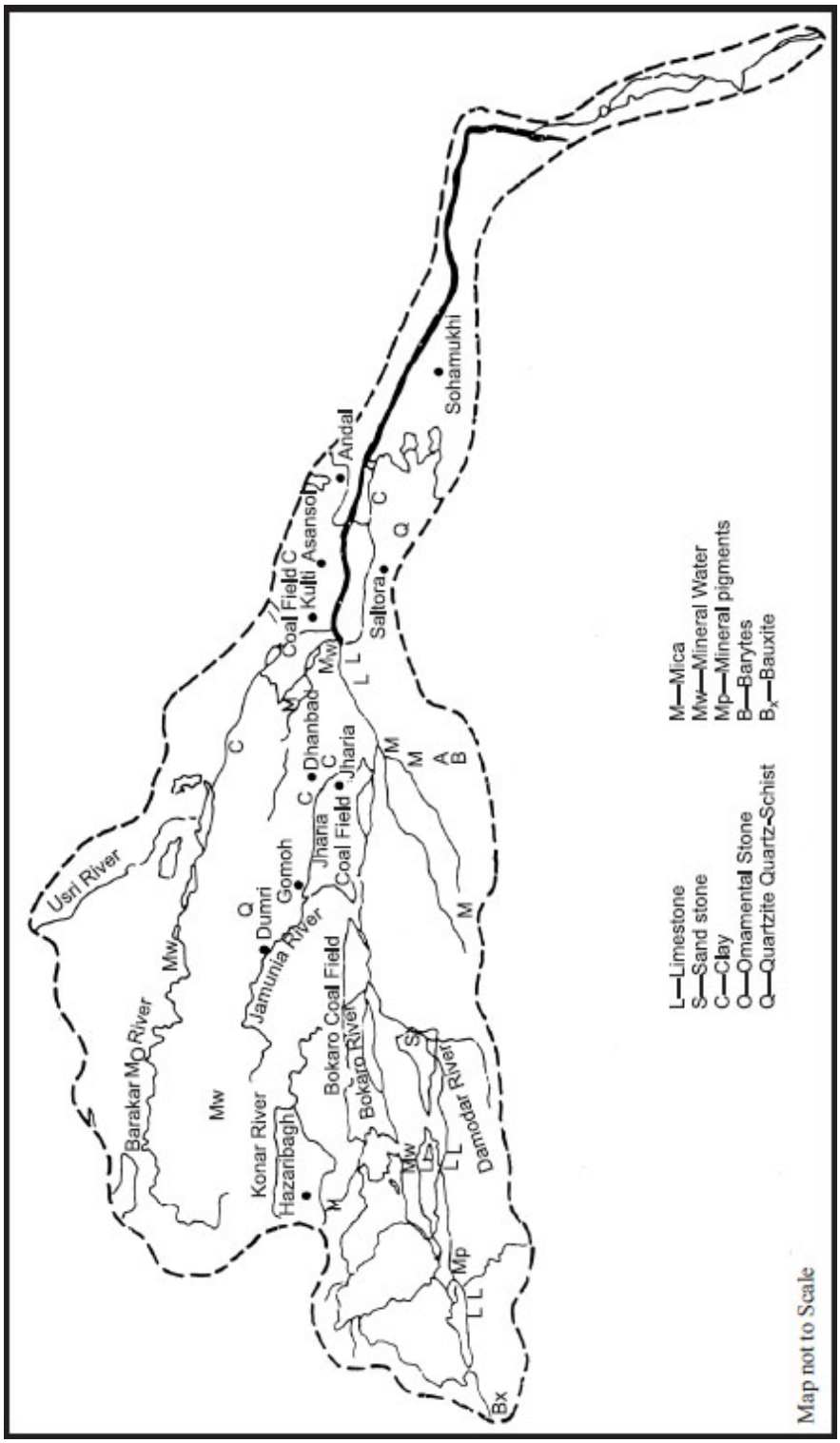


Fig. 15.3 Damodar Valley—Selected Mineral Wealth (after R.C. Chandna)

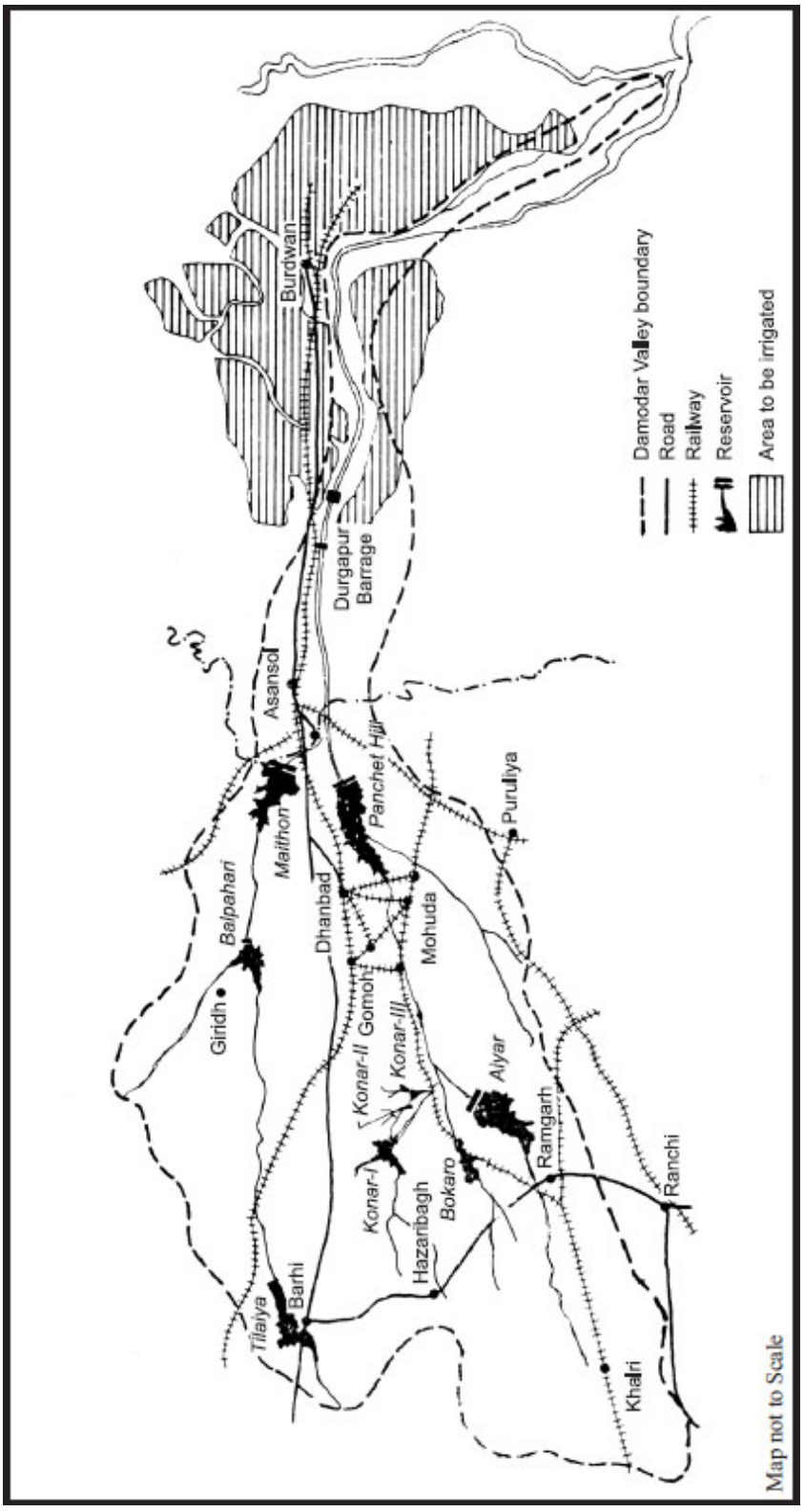


Fig. 15.4 Damodar Valley—Development Plan (after R.C. Chandna)

With regard to power generation, three major hydro-power plants were planned; one each at Tilaiya, Maithon, and Panchet Hill, with a total installed capacity of 104,000 kw of hydropower. A thermal power plant was planned at Bokaro with an installed capacity of 150,000 kw of thermal power.

Benefits from the Project

The Damodar Valley Corporation is a major landmark in the regional planning of India. Some of the important benefits arising out of the project have been briefly presented in the following section:

1. Flood control in the flood prone areas of Jharkhand, Bihar, and West Bengal.
2. Generation of about 260,000 kw of hydro-power at various dam sites.
3. Electricity and Irrigation facilities for about 5.5 lakh hectares**.**
4. Controlling of soil erosion through regulated flow of water.
5. Additional land reclamation for agriculture and pastures.
6. Navigation in the Damodar river and its tributaries.
7. Promotion of public health through control of malaria and other diseases.
8. Encouragement of fishing in the reservoirs and other water-bodies.
9. Promotion of industrial development.
10. Promotion of tourism.

Problems of Damodar Valley Corporation:

1. The main objective of the Damodar Valley Corporation was to control floods. This objective, has, however, not been fully realised.
2. The DVC has constructed only four dams instead of seven suggested by Voorduin. Heavy floods did occur in 1959, 1970, 1971, and 1978.
3. The large scale deforestation in its catchment area and lack of new afforestation have aggravated the problem of siltation

in the dams. The power crisis is threatening the closure of industries.

4. Hydroelectricity is to be supplemented by thermal electricity because the flood control dams cannot be taken as power generation dams in view of the creation of flood cushion before monsoon every year.
5. Industrialisation and mining along the Damodar has made it a river of slurry with minerals, mine rejects and toxic effluents polluting the river.

Despite all these odds, the DVC is an important example of river basin management and integrated regional development.

THE NATIONAL CAPITAL REGION (NATIONAL CAPITAL REGION)

After Independence, Delhi, the capital of India recorded one of the highest population growth rates in the country. The total population of Delhi city was only about two lakhs in 1901 which grew to 168 million in 2011 and 1.6 crore in 2011 and 2.5 crore in 2014 (UN Report). According to the census of 2011 the sex ratio in Delhi has gone up to 868 females per thousand males which is the highest since 1901. The child sex ratio (below 6 years) is 871 given for every 1000 boys. The stupendous growth of population and the influx of immigrants from the surrounding as well as from distant areas have put great pressure on the limited infrastructural facilities and civil amenities of the city.

In order to overcome the problems of rapid growth of population, the hinterland of Delhi was included in its Master Plan (1962) which is known as the '*National Capital Region'* (National Capital Region). The National Capital Board was constituted in 1973 to co-ordinate the urban and rural development of the National Capital Region The National Capital Region (National Capital Region), though a city region, cuts across state boundaries. It stretches over the entire state of the national capital territory of Delhi, parts of adjoining Haryana, Uttar Pradesh, and Rajasthan. The districts included in the National

Capital Region are Faridabad, Gurgaon, Bhiwani, Rohtak, Sonepat, Panipat, Rewari, Narnaul and Mahendergarh; part of Uttar Pradesh, covering the districts of Meerut, Ghaziabad, Bhagpat, Gautam-Budh-Nagar (NOIDA), and Bulandshahar; and parts of Rajasthan covering the districts of Alwar, Ramgarh, Behror, Mandawar, and Tijara Tahsils **(Fig. 15.5)**. The total area of the National Capital Region is 30,242 sq km.

Main Objectives of the National Capital Plan

The main objectives of the regional plan of the National Capital Region are:

1. To contain Delhi's fast growing population size within manageable limits through integrated development of all the components of the National Capital Region.
2. To divert the population to ring towns like Ghaziabad, Faridabad, Gurgaon, Ballabhgarh, Bahadurgarh, Loni, Narela, Sonepat, Panipat, Noida, Moradnagar, Modinagar, Hapur, Dadri, Sikandrabad, Khurja, and Dankaur.
3. To decentralise the economic activities away from the metropolis of Delhi and generate economic activities in the ring or surrounding towns.
4. To check the location of large scale industries in Delhi and encourage the dispersal of industries and other job opportunities elsewhere in the National Capital Region.
5. To decentralise the selected government offices which have been concentrated in Delhi. This can only be achieved by developing integrated infrastructure like transport network, water supply, power supply, housing, education, and medical facilities, etc., in the ring towns.
6. To accomplish a balanced and harmonious growth of urban and rural components of the National Capital Region.
7. To make Delhi a world class city.

Major Problems of the National Capital Region

The National Capital Region faced innumerable problems of serious nature. The problems got accentuated during the last 60 years because of unplanned development of the city. Some of the important problems of Delhi are:

1. In-Migration

The large scale in-migration from both within and without the National Capital Region (more than two lakh in-migrants move into the National Capital Region per year from outside).

Fig. 15.5 National Capital Region (National Capital Region) (after R.C. Chandna)

2. Inadequacy of Essential Services

The gap between the demand and supply of the essential services like water supply, power supply, transport, waste disposal, waste treatment, and management of solid waste is steadily increasing. Ideally, 1500 vehicles should use a single lane during one hour. In Delhi 5000–10000 vehicle occupy the space. WHO recommends 5 beds per 1000-Delhi has only 2.55, and most of these are in the private sector. Delhi has the highest infant mortality rate (IMR) which is 25 per 1000 of births per year.

3. Congestion

The city is becoming congested and there is mushrooming of urban slums.

4. Increasing Employment Opportunities

The job opportunities in Delhi are increasing which is working as a pull-factor and attracting more people from the surrounding areas in to the capital.

5. Heavy Stress on Infrastructure

There is almost complete collapse of its service infrastructure due to perennial inflow of population into Delhi (the population of Delhi at each census has outgrown all its projections).

6. Deteriorating Quality of Life

Rapidly deteriorating quality of life in the national capital is a cause of concern for all the planners. Delhi is one of the most polluted cities of the world.

7. Increase in Crime Rate

Crime against property and crimes against females are on the increase in the capital. According to UN experts, Delhi has become a hub for traffickers. The traffickers in Delhi supply domestic labour, provide child labour, bonded labour to brick-kilns in Haryana, UP, and Punjab, and run flesh trade racket. Tribes like Bedia, Nat and Kanjar are active in trafficking.

8. Life is not enjoyable in Delhi

Planning Strategy

The regional plan for the National Capital Region listed the following strategies and proposals to meet the challenges in this area of rapid population growth and urban congestion:

(i) To decelerate the growth of Delhi City.

(ii) To control the growth rate of Delhi Metropolitan Area.

(iii) To develop regional centres with infrastructure and adequate network of service **(Fig. 15.6).**

(iv) To bring down the population growth rate of National Capital Region to 1.2% per annum from the existing 3 to 4% per annum.

(v) To bring down the volume of in-migration to 84,000 per annum as against the existing in-flow of two lakh persons per year.

(vi) To diversify the functional character of other towns of the National Capital Region.

(vii) To revitalise the economy of stagnating urban centres of the National Capital Region.

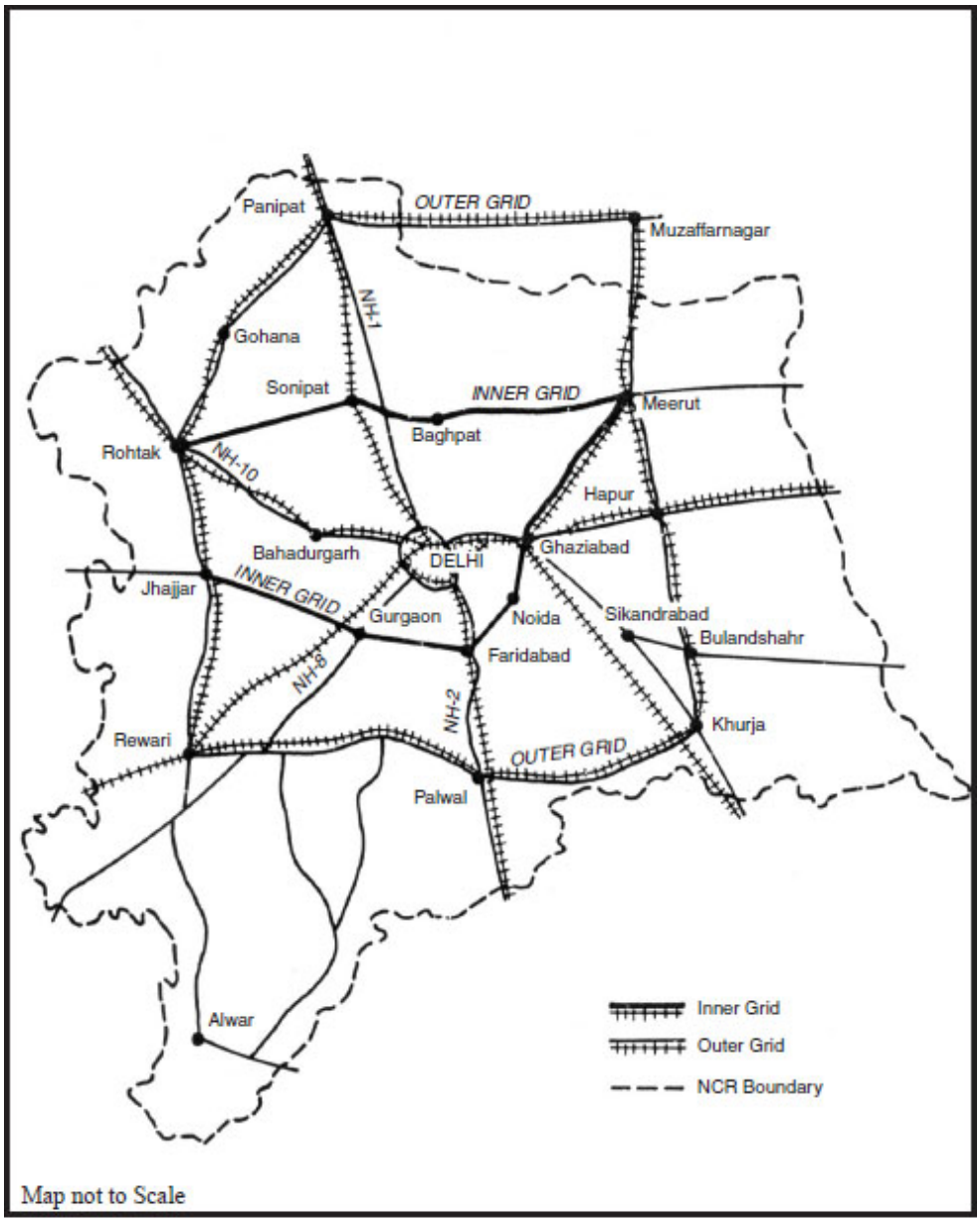


Fig. 15.6 National Capital Region—The Transport System (after R.C. Chandna)

(viii) To select priority towns and develop them on priority basis for a balanced development of the National Capital Region **(Fig. 15.7).**

(ix) To develop selected regional centres namely, Meerut, Bulandshahar, Panipat, Rewari, Alwar, Bhiwadi complex,

Khurja, Dharuhera, Rohtak, Palwal **(Fig. 15.7)** to accommodate Delhi-bound potential migrants by creating employment opportunities in the secondary and tertiary sectors in these regional centres.

(x) To develop a number of sub-regional centres at focal points as sub-divisional headquarters, agricultural markets, service centres, and industrial centres.

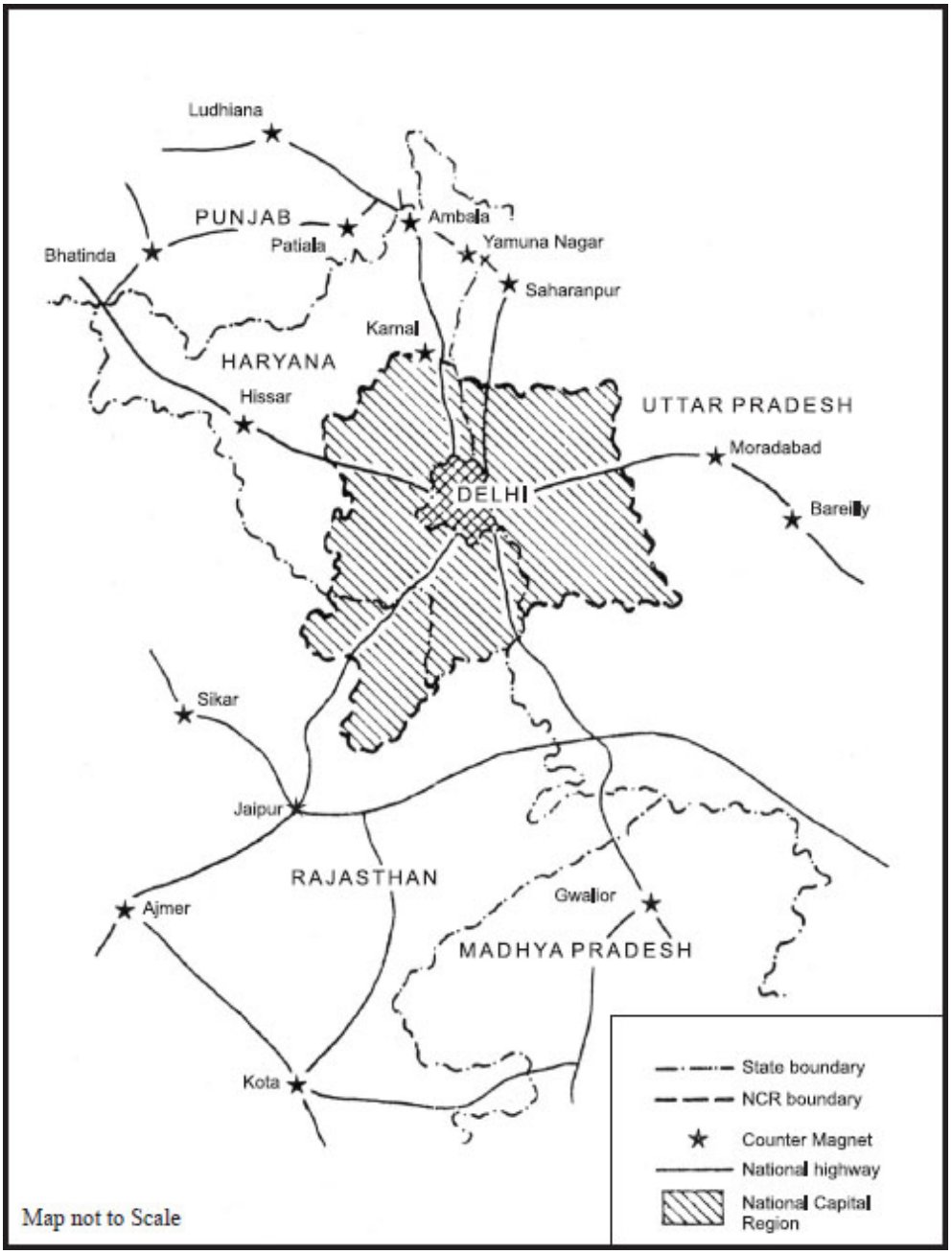


Fig. 15.7 National Capital Region—Proposed Counter Magnets

The following steps can go a long way in making the city of Delhi clean and enjoyable, and overcoming many of its problems:

1. Distribution of Population

The National Capital Region Plan envisages the development of some nodal points in the National Capital Region to arrest the growth of Delhi in terms of its population size. Some such nodal points are Meerut, Hapur, Bulandshahar, Khurja, Palwal, Ballabgarh, Rewari-Bhiwadi-Dharuhera Complex, Rohtak, Alwar, and Panipat**.**

2. Dispersal of Economic Activities

The National Capital Region plan envisages the dispersal of economic activities to ring towns to reduce the pressure of population on Delhi. These activities are location of industries, dispersal of Central Government and public sector offices and dispersal of wholesale trade and commerce. Medium and small industries may only be allowed in Delhi.

3. Transport Sector

Urban transport is one of the key elements of urban infrastructure. An effective urban transport system enhances productivity and growth in the economy. Realising the fast deteriorating condition of transport sector, the regional plan for the National Capital Region came out with many specific proposals for transport sector. An integrated road and rail network is essential to regulate the flow of traffic and goods within the National Capital Region. It is proposed to have an inner grid of roads linking Sonepat-Baghpat, Meerut-Hapur-Bulandshahar-Sikandarabad-Faridabad-Gurgaon-Bahadurgarh-Rohtak-Jhajjar-Gohana and Sonepat. There is also a proposal to develop regional rail bypass connecting Meerut-Hapur-Bulandshahar-Khurja-Palwal-Sohna-Rewari-Jhajjar and Rohtak.

To ease the pressure of passengers on the Transport System, Delhi has introduced Metro Rail System. The Delhi Mass Rapid Transport System (MRTS), a joint venture between the Government of India

and the Government of the National Capital Territory of Delhi is being implemented by the Delhi Metro Rail Corporation.

The Delhi Metro Rail System, technically known as Delhi Mass Rapid Transport System (MRTS) and popularly called Delhi Metro, has two phases:

Delhi MRTS Phase I consists of the following:

Line 1: Delhi–Rithala

Line 2: Vishwavidyalaya–Central Secretariat

Line 3: Barakhamba Road–Dwarka, and extension of Line 3 to Dwarka sub-city and Barakhamba Road–Indraprastha.

Delhi MRTS Project Phase II (50.07 km) at an estimated cost of Rs. 8118 crore, consists of the following alignments:

1. Vishwavidyalaya–Jahangirpuri,
2. Central Secretariat–All India Institute of Medical Sciences (AIIMS),
3. AIIMS–Qutab Minar,
4. Shahadara–Dilshad Garden,
5. Indraprastha–New Ashok Nagar,
6. Yamuna Bank–Anand Vihar Inter–State Bus Terminus (ISBT), and
7. Kirti Nagar–Mundka.

The phase III projects of MRTS are given below:

Phase-III Projects

1. Majlis Park–Shiv Vihar
2. Janakpuri West–Kalindi Kunj
3. Central Secretariat–Kashmere Gate
4. Jahangirpuri–Samaypur Badli
5. Badarpur–Escorts Mujesar
6. Dwarka–Najafgarh
7. Mundka–Bahadurgarh
8. Dilshad Garden–Shaheed Sthal (New Bus Adda)
9. Extension of Metro to Ballabhgarh
10. Kalindikunj to Botanical Garden

11. Najafgarh to Dhansa Bus Stand
12. Extension of Metro from Noida 32–Noida 62

(Source: http://www.delhimetrorail.com/Corridors.aspx)

4. Development of Urban Infrastructure

Urban infrastructure consists of drinking water, sanitation, sewage systems, electricity and gas distribution, urban transport, primary health services, and environment regulation. The process of growth and urbanisation has gathered considerable momentum in the National Capital Region during the last five decades and this has put urban infrastructure and services under severe strain. There is a need to provide uninterrupted supply of clean drinking water, electricity, transport and efficient sewage system. These problems along with sanitation, sewerage disposal and rain water drainage need to be tackled on a priority basis. The National Capital Region Plan needs speedy implementation so as to avoid haphazard development of the city.

5. Rural Development

Various strategies and proposals on rural development aimed: (i) to provide lower order services in each village and to provide higher order facilities in each service centre, (ii) to provide more specialised infrastructure and services such as banks, markets, etc. even in isolated villages,

(iii) to provide a transport system linking all villages with higher order settlements, (iv) to provide marketing facilities for agricultural, dairy and poultry goods in villages, (v) to create job opportunities in non-agricultural sector in the countryside. (vi) to arrest out-migration from rural areas, (vii) to upgrade skills of rural workers in the non-agricultural sector, (viii) to provide adequate housing in rural areas through liberal financial support in the form of housing loans and subsidies, (ix) to provide infrastructure for vocational education to rural youth, (x) to provide adult- education facilities, (xi) to promote emergence of community organisation for participating in the decision making process of regional development programmes of the National Capital Region.

6. Public Health and Environment

The existing conditions of public health and environmental degradation were and are extremely critical. Only one in every five towns in the National Capital Region had sewerage system. Moreover, in most of the towns raw sewerage was let off into rivers through drains with total disregard to various environmental protection laws of the country. The regional plan for the National Capital Region recommended the institutionalisation of a co-ordination committee to look after the environmental degradational aspect. Such a committee should have supervisory authority for the following purposes: (i) location of industry, (ii) afforestation programmes,
(iii) treatment of sewage, and (iv) scientific disposal of the solid waste in the National Capital Region.

7. Employment Opportunities

On the economy front, the major problem identified is that of creating employment opportunities both in rural and urban areas. Three major areas of creating job opportunities identified for the National Capital Region included: (i) government and public sector offices, (ii) wholesale trade, and (iii) industries.

8. Housing

The regional plan for the National Capital Region grants top priority to the problem of housing and proposes: (i) to identify areas for housing development with proper planning of the dwellings,
(ii) to make more land available for urban expansion, and (iii) to constitute land use zoning authority through land use zoning regulation for the National Capital Region as a legislative measure with a view to check future unplanned haphazard growth.

9. Institutional Structure

Regional plans do not exist in vacuum. These remain utopian ideas unless executed and implemented on time with commitment. Keeping this factor in mind and the inter-state nature of the National Capital Region, the National Capital RegionPB has been proposed in institutional structure to ensure that the regional plan really gets

implemented. Apart from the regional authority, it also proposes: (i) the creation of special planning cell for each of the sub-regions of National Capital Region, (ii) the creation of monitoring units within the planning bodies, and (iii) quarterly review of the progress made in each segment of the plan and in each sub-region of the National Capital Region.

POVERTY IN INDIA

Poverty is easy to perceive but difficult to be precise about. About 17% of the world population lives in India, but its share in the total income of the world is only about 2.6% (2012). Poverty and unemployments are the biggest problems of India. More than 5 lakh children (10,000 per week) die because of hunger and malnutrition. The educated unemployed youth are becoming criminals. These problems are the main causes of slow socio-economic transformation. There is a high degree of discrepancy in the income and expenditure of the rich and the poor. While there is no upper limit for spending for the top 5% segment, the maximum spending by those falling at the bottom of the ladder is INR 927 a month (INR 30.50 a day) in rural areas and Rs. 1407 a month (INR 46.25 a day) in urban areas. This implies a monthly consumption expenditure of INR 4,840 in rural areas or INR 7,035 in urban areas for a family of five at 2011-12 prices. As far as states are concerned consumption expenditure in villages of Odisha and Jharkhand were the lowest, followed by Bihar, MP and Uttar Pradesh. In urban areas Bihar had the lowest expenditure (National-Sample Survey Organization–NSSO, July 2011–June 2012). Thus, the Indian growth story is marred by the country's poor performance in human development (Amartya Sen, et. al. India Today, 15.7.2013, p.20).

According to the UNO data 1/3rd of world's extreme poor live in India, i.e., about 400 million (40 crore) out of 1.2 billion are extreme poor (Source: Millenium Development Goals Report, 17.07.2014).

The Planning Commission, as the Government's nodal agency, estimates the incidence of poverty at the national and state levels on

the basis of National Sample Surveys (NSS) on household consumer expenditure conducted every year. About 29% of the total population of India is living below the poverty line. According to the International Food Policy Research Institute (IFPRI), India is among the world's most hunger-ridden countries. India is above only to Congo, Chad, Burundi and Ethiopia. India has been ranked 130 among 187 countries evaluated for human development index (2015). On Gender Inequality Index, India has been ranked 132nd among 148 countries. According to the UN 2014 report, as many as 597 million 50% people in India are practising open defecation, which is the largest number of people defecating in open in the world. Even Bangladesh and Vietnam have lower percentage of population defeacting in the open. Poor sanitation and contaminated water are linked to transmission of diseases such as cholera, diarrhoea, dysentry, typhoide, and hepatitis A.

According to the NSS (2013), income disparity between rich and poor is growing rapidly, spending and consumption by the richest 5% zoomed up by over 60% between 2000 and 2012 in rural areas, while the poorest 5% saw an increase of just 30%. In urban areas, the richest segment's spending increased by 63% while the poorest saw an increase of 33%.

The main causes of poverty are fast population growth, illiteracy, superstition, and protectionist policies pursued since 1947 to 2012 which prevented large amount of foreign investment in the country. The growth rate of unemployment in India according to the 61 round of survey of the NSS is 3.06% which is alarmingly high. Kerala had the highest unemployment rate of about 10% among the larger states. Among the smaller states, Nagaland had a staggering jobless rate of 27%, followed by Tripura of over 15% (NSSO, 2011–12). The employment in the organised sector, public and private combined, declined during the nineties. The Eleventh Plan targets generation of additional employment opportunities in services and manufacturing, in particular, labour intensive manufacturing sectors such as tourism and construction.

The spatial patterns of incidences of poverty in India in 2011–12 have been plotted in
(Fig. 15.8), while the percentages of poverty in different states are given in **(Table 15.1)**.

It may be seen from **(Table 15.1)** that acute poverty prevails in Odisha (57.0%), Bihar (54.4%), and Jharkhand (45.3%). The percentage of poverty is 5.4% in Jammu & Kashmir, Punjab 8.4%, Himachal Pradesh 10.0% Mizoram 12.6%, Goa 13.8%, Haryana 14% and Kerala 15%. The low incidence of poverty in the given states are mainly because of high literacy rate, agricultural and industrial development. The incidence of poverty is higher than the national average (27.5%) in the states of Odisha, Bihar Jharkhand, Chhattisgarh, Madhya Pradesh, Maharashtra,Uttarakhand, and Uttar Pradesh.

The Government of India is trying to strengthen the Public Distribution System (PDS) to reduce the percentage of poverty in the different states of the country. In fact, PDS is a major state intervention to ensure food security to people. In some places, the PDS are not working honestly and efficiently. The main deficiencies of the PDS are leakage and failure in price stabilisation and corruption.

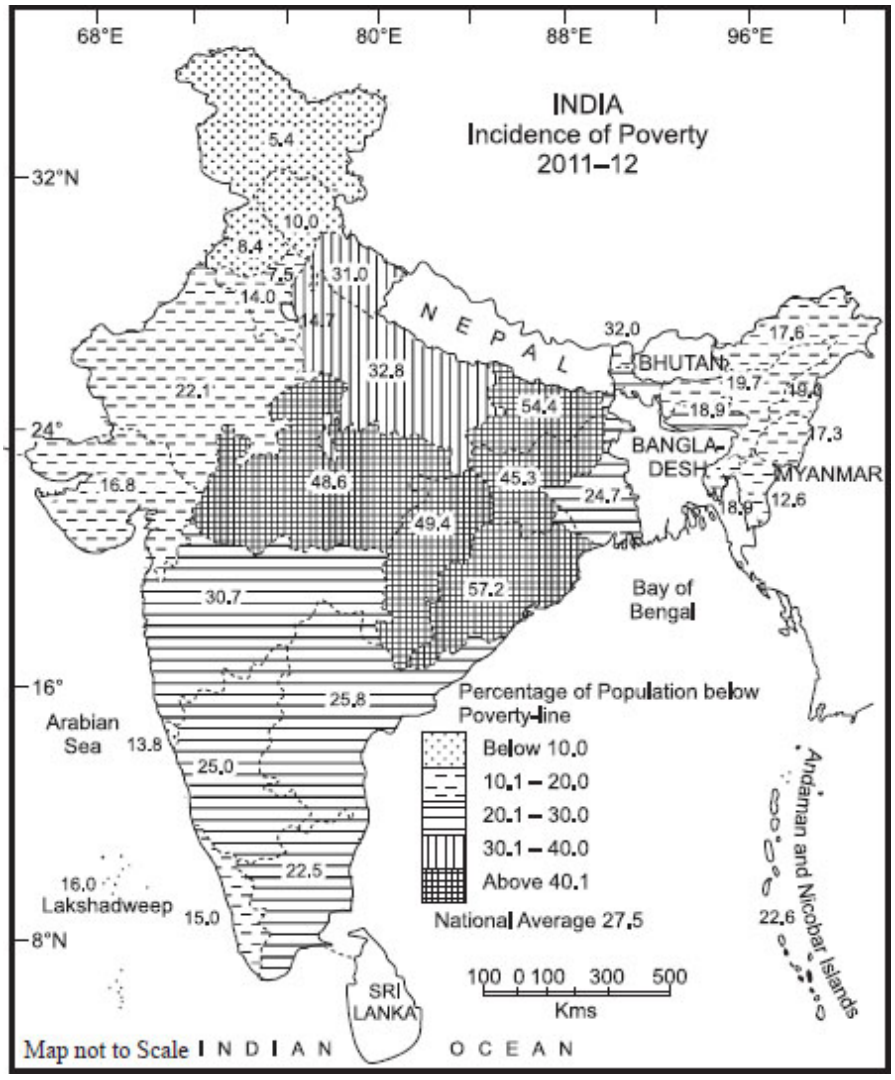


Fig. 15.8 India: Incidence of Poverty

Table 15.1

Incidences of Poverty in Major States

State	Poverty Percent of Popuation (2011–12)
1. Andhra Pradesh	39.7
2. Arunachal Pradesh	35.0
3. Assam	39.7
4. Bihar	41.4
5. Chhattisgarh	54.4

State	Poverty Percent of Popuation (2011–12)
6. Goa	8.0
7. Gujarat	16.8
8. Haryana	14.0
9. Himachal Pradesh	10.0
10. Jammu & Kashmir (inclusive of Ladakh)	9.0
11. Jharkhand	45.3
12. Karnataka	25.0
13. Kerala	15.0
14. Madhya Pradesh	38.3
15. Maharashtra	30.7
16. Manipur	23.0
17. Meghalaya	22.0
18. Mizoram	16.0
19. Nagaland	32.0
20. Odisha	57.0
21. Punjab	8.4
22. Rajasthan	22.1
23. Sikkim	32.0
24. Tamil Nadu	22.5
25. Tripura	33.0
26. Uttarakhand	31.0
27. Uttar Pradesh	32.8
28. West Bengal	24.7
All India	27.5

Source: Economic Survey 2012–13.

Poverty Alleviation and Employment Generation Programmes

To accelerate the reduction in poverty and increase in employment, a number of poverty alleviation, employment generation and basic services programmes are being implemented. Some of the important programmes to eradicate poverty and to generate employment are as under:

1. Pradhan Mantri Gram Sadak Yojna (PMGSY—2006): The basic objective of this programme is to provide all-weather connectivity to all the villages;
2. Indira Awaas Yojna (IAY);
3. Swarnjayanti Gram Swarozgar Yojna (SGSY—1999);
4. Sampoorn Grameen Rozgar Yojna (SGR—2001);
5. Drought Prone Areas Programme (DPAP—1973–74), Drought Development Programme (DDP—1977–78), and Integrated Wasteland Development Development Programme (IWDP—1989–90);
6. Swarn Jayanti Shahri Rozgar Yojna (SJSRY—2006);
7. Valmiki Amdedkar Awaas Yojna (VAMBAY—2001); and
8. Food Security Act (2013).
9. Integrated Rural Development Programme (IRDP)—1980.
10. Food for Work Pragramme.
11. Mahatma Gandhi National Rural Employment Gurantee Act—2005.
12. National Social Assistance Pragramme (NSAP).

GEOGRAPHY AND REGIONAL PLANNING

Regional planning has been recognised as a multi-disciplinary subject both in theory and practice. In the words of Patrick Geddes, the action space of a regional planner, the region, consists of *place* (Geography), *folk* (Sociology and Anthropology), and *work* (Economics and economy).The regional planning, through its actions, seeks the unification of these three into a living whole.

Regional planning is neither economic planning alone, nor physical planning alone. It is such a form of planning whose core area of interest lies in the synthesis of physical, social, economic, political and cultural interests of the concerned region. It implies that regional planning has to deal with the concerned region not in isolation, but in integration with other regions constituting the regional space whole.

In the opinion of Gilkson, ‘Regional Planning’ is a specific type of planning, based on a specific planning structure (regional system), for inducing public action aimed at societal well being. It implies that regional planning is concerned fundamentally with the society in the context of space.

Regional planning is a complex and tedious job as it involves the experts of Physical Sciences, Economics, Sociology, Political Science, Anthropology, Psychology, Engineers, Architects, Lawyers, Planners, and above all, Geographers.

Thus, a regional planner must combine within himself the qualities of all the specialists listed in the foregoing paragraphs. He or she should have an economist in or her, who is now more conscious of the regional problems and is prepared to dig into these problems much more deeply than the economist’s tools are usually capable of. The regional planner shall have to have a sociologist in or her, who shall have to go much beyond the narrow traditional confines of group analysis, and shall have to bring into his or her thinking important economic factors as they interplay in rural-urban regional complexes. The regional planner shall have to combine within himself or herself the qualities of political scientist, who shall have to go beyond the analysis of political behaviour of the system. The regional planner shall have to have an anthropologist in him or her who finds it relevant to introduce in his oe her thinking, the abstract theorising about the cultures prevelant in different territorial areas of both the developed and developing realms. The regional planner shall have to have the qualities of a psychologist in or her who will help him in understanding both individual and mass behaviour, aspirations and potentialities within his region of study. The engineer in or her shall help the regional planner in designing projects and systems of facilities and amenities more rationally and in line with the needs of people and institutions of the region. The lawyer in or her would make or her more concerned with social justice, particularly for the poor. The planner in or her will help in encouraging the shortcomings of

many hastily constructed plans and actions, and thus, shall equip or her to root or her proposals in more solid ground.

Contrary to this, a geographer is in an advantageous position to understand the regional resource base, its problems and can prepare a more balanced model of sustainable regional development. The geographer as a regional planner holds the key position and provides the master key for opening the doors to regional planning, because it is the geographer who knows the rational way of balancing the interaction between natural and human factors in a region, and is able to construct a living bridge between nature and the emerging human societies. Moreover, it is the geographer alone who knows that the equilibrium between natural and human factors is not identical in all regions and varies both in space and time.

The following observations establish beyond doubt that the philosophical foundations of regional planning are deeply rooted in geography:

1. **Delineation of regions is the main area of interest of Geographers:** The concept of region is the main area of specialisation of geographers. The concept of regional space is as old as the discipline of geography itself. The regional concept is extensively used by geographers. In fact, regional geography has been considered as the core area of the discipline of Geography (Hartshorne).
2. **Geography provides a regional synthesis:** Geographers can provide a synthesis of the physical and human phenomena. The delineation of regions done by the geographers are more logical, convincing and suitable for socio-economic and cultural planning.
3. **Geography is essentially an environmental science:** Geography recognises the environment as a system of mutually geo-ecological and socio-economic sub-systems. Thus, the discipline of Geography provides the most comprehensive conceptualisation of environment. Geography, with better understanding of the environment, is

in a position to help the regional planners in their task of environmental construction. After all, no plan is made in vacuum. Instead, every plan is related to a certain regional context in terms of totality of the environment of the concerned region.

4. **Geographers are well trained in the regional issues:** The diffusion of new innovations of the developed countries of Europe may not be suitable for the developing countries of the tropical and sub-tropical regions. Using the new techniques of agriculture prevalent in the environment of developing countries may be counter productive for tropical or sub-tropical countries. This is known better to the experts of geography than that the specialists of any other discipline.
5. **Horizontal unity of regional phenomena is the task of geographers:** Geography believes in the horizontal and the vertical unity of all phenomena operating in a specific regional space. It means that all the phenomena, both natural and human, of a region are intimately integrated with each other as a whole. The whole is not merely a sum of the total of all the phenomena but is something more than the sum total.
6. **Geography believes strongly in horizontal unity of areas:** It implies that though each region is unique in its characteristics, yet it does not exist in isolation from the adjacent areas of other regions. This is known as the comprehensiveness of space. The philosophy of the comprehensiveness of space means that each part of space is integrated with all other parts of the space in the same way as each system of the human body is integrated with the body whole. Thus, each region constitutes a sub-system of the regional systems whole. This philosophy of horizontal unity of area is best united to the philosophy of regional planning as well because no region can be planned independent of the other regions. That is, drainage-basin

based regional planning proved to be most successful as it planned for the entire drainage basin whole.

7. **Geography describes and interprets the variable nature of the Earth's surface:** It believes that the character of area varies both in space and time. The character of two areas can never remain the same through time. It changes in space and time. Each region is a living and continuously evolving organism. Regional planning also believes in this philosophy that no two regions are identical and no region could be static. It is always an ever evolving entity. Therefore, there can be no standard regional plans, which could be applicable to all similar regions (say drainage basin region or city-region). Each region shall have its own plan, and each plan has a life span. As the region evolves, the plan needs modification and revision.
8. **Database:** The discipline of geography provides a strong data-base. In fact, the main concern of a geographer lies in the variable character of areas (regions) and the process of spatial organisation associated with the same. In the words of Geddes, Geography was mainly concerned with what the region is. The geographers, thus, through their regional analysis, know each region, its resource potential, prospects and problems.

   Thus, there is a close relation between Geography, especially regional Geography and Regional Planning. The role of geographers in regional planning is vital and needs recognition and appreciation. This is how Geography and geographers provide the theoretical applied support to the discipline of regional planning.
9. **Geographers are well trained in Cartography:** Maps and diagrams are an indispensable tool of regional planning. A geographer by training is a good cartographer as well.
10. **Geographers are well trained in Surveying:** In the generation of data in the form of structured questionnaires, geographers are well

trained which is so essentially required in regional planning.

Thus Geography offers practical help to the regional planners in many ways such as: (i) providing the data-base, (ii) the demarcation of regional boundaries, (iii) the conduct of regional surveys, (iv) the preparation of maps and diagrams, and (v) the interpretation of the relationship of the physical and cultural phenomena.

Thus, regional planning has become one of geography's allied professions. As stated above, the regional planning process is complex. Geographers, with their cartographic, locational, regional, and analytical skills, are more competent to be involved in regional planning. Moreover, working in a planning project can be a very rewarding experience because it involves the solving of social and economic problems, the conservation and protection of the environment, and much interaction with workers trained in other fields. Planning is a superb learning experience.

Undoubtedly, career in regional planning can be much enhanced by a background in geography, but one will have to adjust and must get training in courses such as public administration, public finance, and other related fields. In many of the universities, the geography department is closely associated with the planning department.

In the above paragraphs, the significance of geography in regional planning has been emphasised. The geographer, by his or her training, however, does not automatically become a good planner. In Geography, he or she has barely learnt enough skills that qualify him or her to enter into the field of a planning. Once in the field of planning, the geographer must acquire the professional skills of planner. His or her training in geography can at best help him in becoming a better planner vis-à-vis a sociologist, an economist, an anthropologist, a geologist, an engineer or architect.

REGIONAL PLANNING AND DEVELOPMENT OF ISLAND TERRITORIES

India has more than one thousand islands in the Bay of Bengal and the Arabian Sea. Most of these islands uninhabited, and lie in the

Union Territories of Andaman and Nicobar, and Lakshadweep. Out of the 247 inhabited islands, 204 lie in the Bay of Bengal and 43 in the Arabian Sea. Being Surrounded by water on all sides, these island territories have unique ecosystems. The islands are occupied by the tribal ethnic groups, having their unique primitive cultures. Islands and their beaches are attracting a large number of tourists. The islands therefore, need special strategy for their socio-economic sustainable development. A brief description of the island territories and a suitable strategy for their socioeconomic development have been presented below:

The Union Territory of Andaman and Nicobar (Population: 380,581–2011)

There are 556 islands in the Union Territory of Andaman and Nicobar Islands. Out of these 36 are permanently inhabited. Most islands are in the Andaman group, 26 of which are inhabited. The Nicobar islands comprises 22 main islands (10 inhabited). The Andaman and Nicobar Islands are separated by the *Ten Degree Channel*, about 150 km wide. The total area of Andaman Islands is about 6408 sq km, and that of Nicobar Islands approximately 1841 sq km.

The Andaman and Nicobar Islands are endowed with tropical rainforest. These forests consists of a mixed flora with elements from Indian, Malaysian, and Myanmarese floral strains. So far, about 2200 varieties of plants have been recorded, out of which 200 are endemic and 1300 do not occur in the mainland of India. Andaman forest is rich in timbers species like **Gurjan** and **Padauk.** About 50 varieties of mammals, 270 varieties of birds, and 225 species of butterflies are found in Andaman and Nicobar. Mount Harriet National Park is one of the richest areas of butterfly and moth diversity on these islands.

The indigenous people of Andamans are: (i) the Jarawa, (ii) the Jangil (or Rutland Jarawa), (iii) the Onge, and (iv) the Sentinelese (most isolated of all groups). The indigenous people of Nicobar are: (i) Nicobarese, and (ii) Shompen—restricted to the interior of Great Nicobar.

A total of 48,700 hectares of land is used for agricultural purposes. Paddy, the main food crop, is mostly cultivated in Andaman group of islands. Coconut and arecanut are the cash crops. Pulses, oilseeds, and vegetables are also grown.

There are 1375 small-scale, village, and handicraft units. Two units are export oriented in the line of fish processing activity. Moreover, there are shell and wood based handicraft units. There are four medium-sized industrial units. The SSI units are engaged in the production of polythene bags, PVC conduit pipes and fittings, paints and varnished, fibre glass and mini flour mills, and soft drinks and beverages, Small-scale and handicraft units are also engaged in shell crafts, bakery products, rice milling, and furniture making. The Andaman and Nicobar Islands Integrated Development Corporation is actively busy in the promotion of tourism, fisheries, and industrial financing and functions as the authorised agency for Alliance Air/Jet Airways.

Lakshadweep

Consisting of 12 atolls, 3 reefs, and 5 submerged banks, with a total of 43 islands, the Lakshadweep is located about 200 to 300 km off the coast of Kerala in the Arabian Sea. The total land area of the territory is 32 sq km. Eleven of the islands are inhabited. The main islands are : (i) Agatti, (ii) Pitti, (iii) Androth, (iv) Kavaratti, (v) Kalpeni, (vi) Suheli Par, (vii) Investigator Bank, and (viii) Minicoy Island. Kavaratti is the capital of Lakshadweep.

The total population of the territory was 64,473–(2011). The islanders are ethnically similar to coastal Kerala's Malayali people, and were influenced by the Arab traders. Inhabitants of Minicoy, the Southernmost and largest island, closely resemble Maldivians. Most of the indigenous population (about 99%) is Muslim. The locals call themselves the *Div-i* or *Aminidivi* (from the mother island). The people are largely dependent on fisheries, as soils on these coral islands are poor.

The coral ecosystems of the islands and the scenic appeal of Lakshadweep are attracting the domestic and international tourists in

a big way. Since such a small region cannot support industries, the government is actively promoting tourism, aquaculture and fisheries. Tourism is one of the important eonomic activities in these islands. These islands have been recognised as an important eco-tourist place of India as a source of income.

The Agatti Aerodrome on Agatti Island is currently the only airport in the Lakshadweep. Also, from April 2007, a private carrier, Kingfisher Airlines, has commenced flights to and from Agatti. Kingfisher connects Kochi and Bangalore to Agatti. The other islands are connected by a helicopter or boat service.

Sagar Island

Sagar Island, also known as Gangasagar, lies on the continental shelf of Bay of Bengal, about 150 km south of Kolkata. The island stretches over 300 sq km and has 43 villages with a population of over 160,000.

The island is the home to the endangered Royal Bengal Tiger. It is also rich in mangrove swamps, waterways, and small rivers. This land is a famous Hindu pilgrim place. Every year on the day of **Makar Sankranti** (mid of January), millions of Hindus gather to take a holy dip in the confluence of Ganga and offer **Puja** in Kapil Muni Temple. The Kolkata Port Trust has a pilot station and a lighthouse on this island **(Fig. 15.9)**.

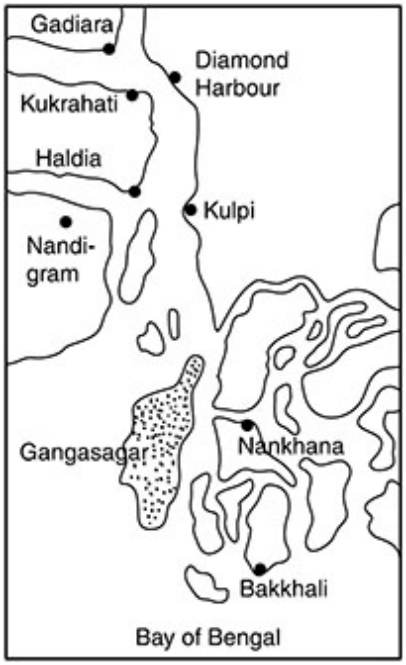


Fig. 15.9 Sagar Island

The Government of West Bengal is planning to connect Sagar Island with the mainland with a 3.3 km bridge costing around INR 6, 000 million.

In addition to these, there are a number of islands around Kathiawad, Mumbai, Kochi, Tuticorin, and along the coasts of Tamil Nadu, Andhra Pradesh, Odisha, and West Bengal.

Strategy for Development

The basic problems of the indigenous people living in the islands of Andaman, Nicobar, Lakshdweep, etc., is that of poverty, low standard of living, hunger, starvation, malnutrition, illiteracy, unemployment, and inadequate housing facilities. Moreover, they have been subjected to exploitation for centuries by the non-tribals.

Keeping all these considerations in view, the Government of India intiated special schemes for the development of Indian island territories. Some of the steps which may go a long way in improving the income, health, education, and quality of life of the inhabitants of the islands territories are as under:

1. The economy of most of islanders, specially that of the

Andaman and Nicobar Islands is forest based. In order to enhance the resilient characteristics of the evergreen and deciduous forests, there is a need to bring more areas under national parks and biosphere reserves.

2. The creeks and inland waters of the island territories are characterised with mangrove, swamps, and wetlands. These ecosystems should be conserved and judiciously utilised.
3. Many of the island territories are made of coral formations; special efforts are to be made to conserve the corals ecosystems.
4. In the hot and humid climate of the islands, the role of soil erosion is significantly high. Afforestation in the degraded lands can help to overcome this problem.
5. Forest-based cottage industries and handicrafts can provide employment at the village level. Processing of arecanut, coconut products, and fisheries, and coir-making industries can generate appreciable employment and income to the people.
6. The lush-green forests, isolation, aesthetic beauty, and scenic appeal of the islands are great tourist attractions. Thus, there is a great potential for the development of eco-tourism.
7. The islands like Andaman and Nicobar lie in the high seismic zone being very close to the subduction plates of Myanmar and Sumatra. The buildings and structures constructed for the tourists should be seismic-proof.
8. There is enough potential for the development of aquaculture and fisheries in the coastal creeks, attols, and bays for which special strategies need to be adopted.
9. The Indian islands are known for some of the primitive cultures. The population of the Jarawa, Jangil, Onge, and Sentinelese tribes is declining fast. These tribes and their cultures need to be preserved.
10. Plantation of coconut, arecanut, spices, cincona, tea, and pineapple, etc., can enhance the income of the inhabitants of

islands. If all these steps are taken in combination with each other, a balanced regional development will be possible, making the standard and quality of life of the people appreciably better.

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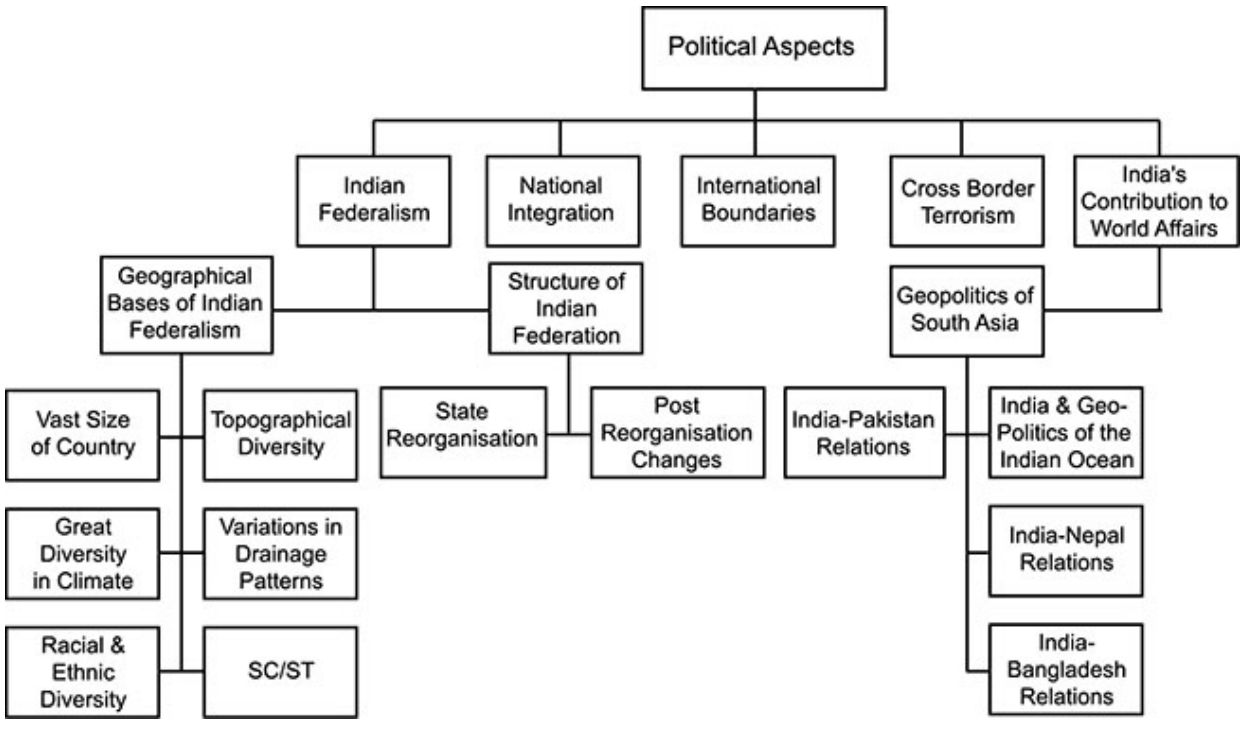
16

India–Political Aspects



POLITICAL GEOGRAPHY

Human Geography is often divided into three sub-disciplines dealing with (i) *economic* , (ii) *social* , and (iii) *political* events respectively. At present, 'political geography deals with the analysis of the systems of class/group conflict over time and space'. In other words political geography studies such topics as spatial organisation of geo-political units, emergence of new states, inter-state issues, international power relationships, nationalism, boundary and frontiers, military conflicts, sub-nationalism, terrorism, and regional separatism within states.

Political geographers are naturally interested in the ways in which states are organised in spatial political terms. Such organisation of political space is the result of lengthy processes of experimentation and modification processes that have not ceased. State systems are continuously being altered, sometimes through deliberation and consultation and at other times because the system cannot withstand certain centrifugal pressures or forces. In the democratic countries of the world, there are two types of government (i) *unitary*, and (ii) *federal*.

Unitary State

The word '*unitary*' derives from the Latin *unitas* (unity) which in turn comes from *unus* (one). It, thus, emphasises the oneness of the state and implies a high degree of internal homogeneity and cohesiveness. The effective governance in countries with relatively homogeneous population, having common ethnicity, language, religion, traditions, customs, and history is possible with the unitary form of government. In a unitary form of government, the central government ordinarily retains most of the administrative powers. The Central Government is the sole custodian of police power, it is the sole taxing authority. The infrastructural facilities as well as the civic amenities like health, safety and welfare of the citizens are provided by the central government. The unitary state, therefore, theoretically has but one strong focus, and its internal differences are few. In the case of the unitary state, the central authority controls the state and local governments and determines how much power they shall have. Thus,

in a unitary form of government the overall administrative authority is vested in the Central Government; the local and regional governments function at the discretion of the central authority.

Federal States

The term *federal* has its origin in the Latin *foederis* meaning league. In a federal state, the government authority is *layered* at different levels. Its implication is one of alliance, contract and coexistence of the state's internal diverse regions and peoples. Under the federal system, the rights and responsibilities of local governments are protected by constitution. Here the central authority cannot simply assume the functions of local governments temporarily to relinquish them when the demands have been met. The individual regions of the state possess a certain degree of autonomy which is protected, and there is a constant watch against over-centralisation. This is what permits diverse peoples and even cultures to join a single state. Their first allegiance may be to their own region, but their ultimate loyalty is towards the state.

A federal government has a written constitution that cannot be unilaterally altered. There is division of power between the central government and the component states/provinces. The federal government is generally formed in the countries where the people belong to different racial, ethnic, cultural groups; speak different languages; follow different religions; and have different customs and traditions.

In a federal set-up, local governments derive their authority from the regional governments, which themselves possess constitutionally guaranteed autonomy in specified matters of importance. Thus, the regional or state governments do not depend on the discretion of the centre.

A federation differs from a unitary government in the sense that in the unitary polity 'states', if any, exist at the mercy of the central government. As opposed to this, in a federation each level of government is, in theory, autonomous within its allocated sphere of competence, and is free from any non-agreed intervention from the

other, except in emergency, if the constitution so provides. Thus, what distinguishes federalism from a unitary government is *guaranteed constitutional autonomy*, not the formal division of powers.

Federalism is, thus, essentially a compromise between centripetal and centrifugal forces that are operative at the same time. A federal system of government is one in which there is a division of powers between one general and several regional authorities, each of which, in its own sphere, is coordinate with the others, and each of which acts directly on the people through its own administrative agencies (Birch, 1955).

India is the largest democracy in the world. It has adopted the democratic form of government for its administration. Its political structure is federal in concept but unitary in functioning. The Indian Constitution provides a federal system of government with a clear division of power between the union and the states.

Indian Federalism

India, a Union of States, is a *Sovereign Socialist Secular Democratic Republic* with a *parliamentary system* of government. The republic is governed in terms of the Constitution, which was adopted by the Constituent Assembly on November 26, 1949 and came into force on January 26, 1950.

The Constitution which envisages parliamentary form of government is federal in structure with unitary features. The President of India is the constitutional head of executive of the Union. Article 74(1) of the Constitution provides that there shall be a Council of Ministers with the Prime Minister as head to aid and advise the President who shall in exercise of his functions, act in accordance with such advice. The real executive power, thus, vests in Council of Ministers with Prime Minister as the head, Council of Ministers is collectively responsible to the House of the People (*Lok Sabha*). Similarly, in states, Governor is head of the executive, but it is the Council of Ministers with Chief Minister as head in whom real executive power vests. The Council of Ministers of a state is collectively responsible to the Legislative Assembly (*Vidhan Sabha*).

The constitution distributes legislative power between Parliament and state legislature and provides for vesting of residual powers in Parliament. Power to amend the Constitution also vests in Parliament. The Constitution has provision for independence of judiciary, Comptroller and Auditor-General, Public Service Commission, and Chief Election Commissioner.

In a federal form of government, a number of separate or autonomous political units mutually agree to merge together to form a nation-state with a single central or federal government. The legislative and executive powers in a federation are divided in systematic manner between the central (federal) and the state governments through a written constitution. In a federation, the central and the state governments exist on the basis of equality. There is a kind of compromise between the centre and states, based on mutual co-operation.

The formation of the Indian federation has a long history. Some kind of loose federation existed through out the history. India was, however, not fully integrated in a sovereign nation even during the period of Ashoka and Akbar. It was the British who unified India and established their domination from the Indus in the west to the Brahmaputra in the east, and from the Himalayas in the north to Kanniyakunari (Cape Camorin) in the south. To make the federal (central) government strong, the British introduced:

(i) All India Administrative Services,
(ii) unified transport and communication,
(iii) uniform educational system through English medium, and
(iv) universal law under modern jurisprudence.

These steps helped in the unification of the country.

The development of the federal structure started in India with the Indian Councils Act, 1909, known as the Minto-Morley Reforms, where the central and the provincial councils were given more powers. The Montagu-Chelmsford Report, submitted by the Indian Constitution Reform Committee in 1918 became the basis of the Government of India Act, 1919, which was the beginning of the

federal system in India. The Act of 1919 provided provision for a system of dual government (central and provincial); relaxation of central control over the provinces through a set of Devolution Rules; provision to prepare budget and levy taxes; and include elected members in the Upper House and Lower House. The genesis of the federal government in India lies in the Simon Commission Report, 1930, which supported the concept of federalism. Between 1930 and 1933, three Round Table Conferences were held in which the federal structure of India was discussed. In 1930, the British Government officially accepted the principle that the form of the new government will be an all-India federation. The recommendations of the Joint Committee became the Government of India Act of 1935. Under the Act, the centre had powers to pass laws on ninety seven subjects set forth in the Central List, and the provincial governments were permitted to pass laws on sixty six subjects, while forty seven subjects were in the Concurrent List. The act set forth the major outline of the federal system of government as finally evolved by the Constituent Assembly which framed the present Constitution of the Republic of India in 1950.

The framers of the Indian Constitution realised that a real political danger to sub-nationalism (regionalism) might develop in India because of the variety of linguistic, religious, cultural, economic, and ideological differences that existed in the country. To unite the Indian nation in the presence of the fissiparous tendencies, a strong central authority was considered necessary. The Constituent Assembly endorsed the principle of federalism as the structure of new India. The constitution of India is the longest constitution in the world with 339 articles and 9 supporting schedules covering 272 pages.

Geographical Bases of India Federalism

The geographical bases of the Indian federalism are as under:

1. **Vast Size of the Country:** India is the seventh largest country in the world (after Russia, Canada, China, USA, Brazil, and Australia) having an area of 32,87, 263 sq km which is about 2.4% of the total land area of the world. It is

supporting 17% of the total population of the world. The vast size and large extent of the country have fostered variegated characteristics in the physical as well as cultural landscape of India.

2. **Topographical Diversity:** The topographical landscape of India has great variations. It has rugged mountains with sharp, pinnacled, snow-covered peaks and steep-sided valleys in the north, flat plateaus with senile topography in the peninsula and highly productive plains of the Satluj-Ganga-Brahmaputra rivers in between the Himalayas and the Peninsular India. Geologically, these three physiographic divisions represent three different structural units. Each one of these has different types of vegetation and soils resulting into different types of land use and lifestyle of the people. India has the second highest peak of the world K^2 (8611 m) as well as the Sunda Trench (depth about 7000m). The Vindhyan Mountains act as a formidable barrier between the northern plains and the peninsular India. Culturally the two regions are separate from each other.
3. **Great Diversity in Climate:** Essentially India has tropical monsoon climate but there are significant variations in the temperature and precipitation distribution. The large size, great latitudinal extension, diverse topographical features, and the presence of the Indian Ocean on three sides of the Peninsula have closely influenced the climatic conditions of the subcontinent. In Dras (Kargil) (Ladakh), the lowest temperature of–45°C was recorded on 28 December, 1910 is recorded during winter season, while the highest temperature 56°C was recorded on 5 June 1991 at Briyawali, Bikaner District of Rajasthan (New Modern–School Atlas, 2009, p.2). In general, in northern India, the amount of rainfall decreases from east to west, while an opposite trend is observed in the Peninsular India. For example, Mausynram (near Cherrapunji) records more than 1150 cm of rainfall annually,

while Phalodi (Rajasthan) records less than 20 cm of rainfall annually. The heterogeneity of relief and variability of rainfall have their bearing on the occupations and lifestyle of the people. The problems of the desert areas are different from that of the areas of heavy rainfall. Consequently, there may be drought in one part of the country, while floods in the other regions. A federal government is ideally suited to overcome the problems of the people living in such diverse type of geo-climatic conditions.

4. **Great Variations in the Drainage Patterns:** There is a marked contrast between the perennial rivers of the Himalayas and the seasonal rivers of the Peninsular India. Problems, like floods and drought, can be better handled in a federal rather than in a unitary form of government.
5. **Racial and Ethnic Diversity:** Ethnic diversity is an important basis of federalism. The presence of sizable groups of people other than that of the majority may pose serious problems for state's internal political coherence and unity. To overcome such a problem a federal structure is more convenient. India is a country of heterogeneous racial and ethnic groups. There are areas of dominance of the Dravidian, Indo-Aryan, Mongoloids, Negrito, and Proto-Australoids racial and ethnic groups. The various ethnic groups exhibit a highly uneven distribution. Each of these ethnic groups is a distinct category of population in a large society whose culture is usually different from that of the society. The members of the ethnic groups are bound together by common ties of race, or culture and traditions. Their problems differ from each other. A proper development of all the ethnic and cultural groups is possible only if more autonomy is given to the states in which they are residing. A federal form of government is, therefore, imperative for the development of all the racial, ethnic, cultural, and social groups.
6. **Linguistic Diversity:** India is a multi-lingual country. There

are 22 major languages in India, written in 13 different scripts with over 720 dialects. Some of the important linguistic groups are: (i) Dravidian, (ii) Indo-Aryan, (iii) Austric, and (iv) Sino-Tibetan. Hindi is the national language (with over 420 million speakers), but all the languages have great significance in the areas of their concentration. In order to promote and develop each of the languages of India, a federal structure is more appropriate.

7. **Religious Diversity:** India is a secular democracy. Religion is an important characteristic of the Indian population. Essentially, the Indians are religious people. Different religions followed by different people living in different parts of the country, have created great diversity of culture. Hinduism is the dominant religion of the country, which accounts for about 82% of the total population of the country. Their concentration is, however, not uniform. There are some districts in Jammu and Kashmir, Assam, Kerala, and West Bengal where Muslims are in majority. Same is the case with Sikhs in Punjab, and Christians in the states of north-east India. For the protection and promotion of the religious and cultural values of all the people a federal form of government is more suitable.
8. **Concentration of Scheduled Castes:** Scheduled castes comprise about 16.20% of the total population of India. These are the socially and economically deprived groups who have been exploited throughout the history. Their concentration is, however, most unequal. For example, in Punjab and Uttar Pradesh about 29 and over 21% of the total population consists of scheduled castes while they are almost absent in Mizoram, Meghalaya, Goa, and Manipur. Special programmes for their upliftment are necessary which can be better formulated and implemented at the state level. The federal form of government can solve such problems in a better way.

9. **Scheduled Tribes:** The Scheduled Tribes of India are living in the areas of isolation and relative isolation. They belong to different ethnic, linguistic and religious groups and have some special social, cultural, and economic characteristics. Their mode of life is very much linked with the environmental conditions of their locales. Most of the tribes have their distinct socio-economic norms. The philosophy of their life is '*from everybody according to his capacity and to every one according to his needs*'. The institution of village headman is very important in their society. Any person who violates the social norms is penalised by the village council, headed by the village headman. In order to maintain their socio-cultural norms, a federal form of government is required.
10. **Demographic Variations:** The structure of population in terms of density, age, sex, literacy, and occupational structure is also an important base of federalism. The sex ratio in the different parts and cultural groups varies from each other. For example, the sex ratio in Kerala is 1058 females and in Haryana only 861, per one thousand of males. Owing to the distorted sex ratio both the states are facing serious population problems which can be tackled in a better way at the grass root level. To overcome such demographic issues, a federal form of government is more appropriate.
11. **Regional Economic Disparities:** Despite an overall sense of belonging together, each of the regional unit has an identity of its own. In large-sized countries having diverse geo-climatic and socio-cultural settings, the economic development may not be at the same level. Some of the areas will be more developed agriculturally and industrially, while the others may be lagging behind. The under-development and economic disparities can be reduced substantially in a federal structure in which the respective governments of the backward areas can formulate and implement their own regional plans.
12. **Historical Factors:** The small states of long historical past are by

and large ethnically based. Such states generally have unitary form of governments. As opposed to this the countries with large area and large population with numerous, ethnic, cultural, and linguistic groups can be governed in a better way by a federal government.

The above description of the physico-cultural, social and historical factors leads us to conclude that geography has played a very vital role in the development of regional units with distinct identity and thus helped the cause of Indian federalism. Thus, for a large country like India with complex physical setting and cultural milieu, federalism is the only practical solution of the political and constitutional problems.

The constitution of India is basically federal in character but incorporates certain unitary features arising out of the special conditions under which it was born. The constitution satisfies all the essential conditions of a federal form of polity, such as: (i) dual government, (ii) distribution of powers between the Centre and the states, (iii) supremacy of the Constitution, and (iv) authority of the courts as the final interpreter of the Constitution. In fact, Indian federalism is unique in the sense that here federation has been created by granting guaranteed constitutional authority to the previously sub-ordinate regional governments in the Provinces. The Constitution was framed by the 'people of India' assembled in the Constituent Assembly and her federation is not the outcome of any bargain between the Centre and the (supposedly sovereign and independent) states. The Constitution does not provide for double citizenship between the state and the Union (unlike the USA) and there is no division of public services or the judiciary at the two levels.

While the Constitution has prescribed cooperative federalism for normal times, it has empowered the Central Government to acquire the strength of a unitary government in times of emergencies. When a proclamation of Emergency is made, the legislative power of the Union extends to state subjects as well (in terms of Articles 353 and 357). Even in normal times the federal government is given sufficient authority to assume legislative power over any subject included in the state list in 'national interest' with the approval of the *Lok Sabha* by

two-third votes (Article 249). Similarly, the Centre has overwhelming powers under Articles 256, 257, and 356.

In a country like India which has great physical diversity and socio-economic disparity, a federal form of government is more suitable to govern the country.

STRUCTURE OF INDIAN FEDERATION

The federal character of the Indian Constitution involves a distribution of sovereignty between the Central Government and the states. The distribution of powers in general follows the Government of India Act of 1935. The Indian Constitution provides for a three-fold distribution of legislative powers between the Union and the States. The Central Government is given exclusive powers to make laws concerning 97 items including defence, national security, foreign affairs, banking, currency and coinage, transportation and communication, commerce, atomic energy, the general framework of criminal and civil laws, revenue collection, the ability to dissolve state governments during crisis, citizenship, immigration, and monetary policies.

The State List comprises 66 subjects including agricultural development, water, forests, fisheries, wildlife, internal trade, public order and police, local government, public health and sanitation, and state taxes and duties.

The Third List, called the Concurrent List, includes 47 items including criminal law and procedure, civil procedure, marriage, contracts, torts and easement, trust, welfare of labour, insurance, economic and social planning, and education. In case of overlapping of any matter between the three lists, the Central Government has been given the paramount power. Likewise all residuary powers are given to the Centre.

State Reorganisation

The evolution of the Indian state has a long history of over 5000 years. The status of the nation was always changing, but it was never a democratic republic where the electorate elected the rulers till its

Independence in 1947. Thus during its long history of over 5000 years, India was never identified as a single state in the modern sense of the term. Interestingly enough, “neither the Mauryas, nor the Guptas, nor the Mughals, nor the British at the height of their power, exercised sovereignty over the whole of India’s territory” (Pannikar, 1964). The British, after extending the political limits of their possession of ‘India’ to the traditional limits of the subcontinent, tried to bring about political stability in the area, and thus brought a ‘*definiteness about the idea of India*’ (*Spate and Learmonth—1967*). The boundaries, except for parts of the Greater Himalayas in the north were fixed. The British also in their turn did not exercise their power of administration uniformly over the entire territory of the sub-continent of India. Politically, there were two India: (i) British India, governed by the crown according to the statutes of the Parliament and enactments of the Indian legislature, and (ii) the Indian states under the personal rule of their rulers/princes.

Before Independence, India was divided into 14 British provinces and 600 princely states. The British Government had full control over the external affairs, defence and communications of the princely states. About half of these native states were in Kathiawad Peninsula and most of the remaining in Rajputana (Rajasthan) and Central India. These were classified in different categories according to the size, revenue, agreements, dignities, and titles of the ruler, etc.

In 1947, the creation of Pakistan was the outcome of the British colonial policy of the 20^{th} century which contributed to Hindu-Muslim schism. Pakistan was carved out from the portions of the British provinces and native states having Muslim majority in the population according to the Census of 1941. India was given possession of 9 British Provinces and 562 princely states of different shape and size. The princely states were legally empowered to accede either to India or Pakistan.

After Independence, some of the princely states were merged with provinces geographically contiguous; some were formed into viable unions of states, while others were converted into centrally

administered areas (Union Territories). Integration, in most of the cases, was achieved swiftly and peacefully. The boundaries formed after integration of princely states and the former British provinces were economically, administratively, linguistically, and culturally illogical. The newly created 27 states were unwieldy, expensive and inequitable. These states were divided into four categories, A, B, C, and D based on the principle of federalism. Category 'A' states (numbering 9) were governed by the governor, while category 'B' states (numbering 7) were governed by the *Rajpramukh.* Category 'C' states were administered by the Centre on unitary basis and were called Commissioner's states. Category 'D' states included island territories of Andaman and Nicobar Islands under the full authority of the Central Government. The integration was a unique type of bloodless revolution unparalleled in the history of the world.

In addition to the British provinces and the princely states, there were five French (Chandernagar, Pondicherry, Karaikal, Mahe and Yanaon) and five Portuguese (Dadar, Nagar-Haveli, Goa, Daman and Diu) possessions in India, majority of which were located along or near the coasts. These possessions were also merged into the Union of India either with negotiations or by applying some force. The protectorate of Sikkim was integrated within the Union of India on April 26, 1975.

Reorganisation of Indian States

The integration of princely states was smooth and swift. But the political units devised in 1950 were not economically viable and politically sustainable. Hence, a need was felt to reorganise the states.

In the reorganisation of the states, language was made the basis as provided by the Congress Party Resolution of 1920 in Nagpur Session and its subsequent reaffirmation in 1927, 1937, 1938, and 1945–46. In December 1953, the Government of India appointed the State Reorganisation Commission to examine 'objectively and dispassionately' the problem of reorganisation of states of the Indian Union.

The States Reorganisation Commission adopted a rational and balanced approach for the reorganisation of states. The bases of reorganisation of states were:

1. ‘to recognise linguistic homogeneity as an important factor … but not to consider it as an exclusive and binding principle. The linguistic basis ensures larger participation of people in the administration. Moreover, the linguistic states helped to pacify the people of different parts of the country whose languages were ignored during the British period;
2. ‘to ensure that communicational, educational and cultural needs of different language groups … are adequately met;
3. ‘where satisfactory conditions exist, and the balance of economic, political and administrative considerations favour composite states, to continue them with the necessary safeguards to ensure that all sections enjoy equal rights and opportunities;
4. ‘to repudiate the ‘home land’ concept … by upholding equal opportunities and equal rights for all citizens throughout the length and breadth of the Union;
5. ‘to reject the theory of ‘one language one state’ …; and
6. ‘finally, to the extent that realisation of Unilingualism at state level would tend to breed a particularist feeling, to counterbalance that feeling by positive measures calculated to give a deeper inter play of different regional cultures, and inter-state cooperation and accord; and to reinforce the links between the Centre and the States in order to secure a greater coordinated working of national policies and programmes’ (Report of the States Reorganisation Commission, 1955).

There are some negative consequences of the re-organisation of states on the linguistic basis.

1. It increases a feeling of regionalism/subnationalism.
2. It hinders economic co-operation.
3. It promotes an antagonistic attitude towards neighbouring

states.

The Commission submitted its report on 30 September, 1955, recommending reorganisation of India into 16 States and 3 Territories. The Government of India examined the report in detail and proposed the reorganisation of India into 15 States and 7 Territories. Finally, the Parliament passed the States Reorganisation Act, 1956 reorganising India into 14 States and 6 Union Territories from November 1, 1956.

As a result of reorganisation, a very large area and population was brought under a similar administrative pattern. All the states, covering 98% of area and population of India, now had a similar legislative, executive and judicial structure as chief components of the Republic. Only a small portion as the Union Territories was left under the Central guidance. The anomaly of the institution of *Rajpramukh* was removed, ending the last hereditary and feudal association with the administration as head of the state. The territorial contiguity was another major achievement. Various blocks of territories, existing as enclaves and exclaves, even after the integration of states were merged with the contiguous units; excepting the one anomaly in respect of Himachal Pradesh which was still left in two major blocks. Most of the reorganised states were larger in area and population than before. Madras and Bihar lost some areas, while Assam, Odisha, Uttar Pradesh, and Jammu and Kashmir underwent no change. Certain former states like Hyderabad, Coorg, Bhopal, Saurashtra, Kutch, Madhya Bharat, Vindhya Pradesh, Ajmer, Mewar, and PEPSU lost their identities as such. There were little boundary changes, old boundaries of different levels had been maintained. The number of states, as reorganised, was less than recommended by the Commission or proposed by the government.

Post-Reorganisation Changes

The reorganisation of states could not satisfy all opinions. It did not mark the end of the process. On linguistic, ethnic, or cultural grounds, pressing demands have been made for readjustment of politico-territorial units. In the years that followed, the territorial changes have

been more towards division and fragmentation. Consequently, there was an increase in the number of units and also certain variations in the administrative set-up.

Although language was made an open issue in the reorganisation, it is evident that 'communalism' played a vital role in the partitioning of Bombay into Maharashtra and Gujarat (May 1, 1960), and Punjab (November 1, 1966) into Punjab and Haryana. The Sikhs felt themselves to be a different community from the Hindus in the Punjab and so did the Marathas from Gujaratis in Bombay (Mumbai). Communalism, thus, in the pre-independent India was responsible for the partition of the subcontinent and in the post-Independence India was responsible for creating further territorial divisions (Rai, Satya, 1965, and Singh, G.S. 1966). The States Reorganisation Commission had recommended one greater Punjab comprising the then East Punjab and Himachal Pradesh including PEPSU and Bilaspur. Various forces have succeeded in getting this region divided into four units as at present: the states of Punjab, Haryana, Himachal Pradesh, and the Union Territory of Chandigarh.

The other region of large territorial changes lies in the North-East. The internal administration of this area was organised and based on traditional tribal ways of life and social customs. Nagaland (1 December, 1963), and Meghalaya (2 April,1970) were carved out from Assam, while Manipur (1971), Tripura (1971), Mizoram (8 December, 1986), and Arunachal Pradesh (8 December, 1986) were upgraded from the status of union territories to the status of full-fledged states. In fact, the geographical and administrative isolation of various hill areas in the North-East was responsible for their economic backwardness. This feeling of political isolation and economic backwardness among the hill people led to their demands for separate political units. These hill people did not wish to be dominated by the people of the plains of the Brahmaputra Valley in Assam.

Table 16.1

Smaller States of India

S. No.	State	Year of Formation	Prior Status
1	Andhra Pradesh	1953	Part of the state of Madras
2	Kerala	1956	State of Travancore and Cochin
3	Gujarat	1960	Part of the state Bombay
4	Maharashtra	1960	Part of the state Bombay
5	Nagaland	1963	Union Territory
6	Haryana	1966	Part of Punjab
7	Karnataka	1956	State of Mysore formed in 1953 and renamed in 1973
8	Himachal Pradesh	1971	Union Territory
9	Manipur, Tripura	1972	Union Territories
10	Meghalaya	1972	Part of State of Assam
11	Sikkim	1975	Union Territory from 1972 to 1987
12	Mizoram	1987	Union Territory from 1972 to 1987
13	Arunachal Pradesh	1987	Union Territory
14	Goa	1987	Union Territory
15	Uttarakhand	2000	Part of Uttar Pradesh
16	Chattisgarh	2000	Part of Madhya Pradesh
17	Jharkhand	2000	Part of Bihar
18	Telangana	2014	Part of Andhra Pradesh

The reorganisation of states left many regions, which are clamouring for new states, dissatisfied. The Gorkhas of the Hill areas of West Bengal are demanding Gorkhaland; the Bodo tribe of Assam

wants a separate Bodoland; the tribes of the North Cachar Hill District also feel antagonistic to the introduction of Assamese as the medium of education; the tribes of Mikir Hills also feel the same way; Ahoms have been demanding a separate Ahom State comprising the districts of Upper Assam; the Garos, inhabiting about 316 villages in South Central Assam, have sought their merger with the adjoining state of Meghalaya.

After the creation of Chhattisgarh, Jharkhand, and Uttarakhand, the demands are being raised for further sub-division of Uttar Pradesh into Bundelkhand, Harit-Pradesh, Purvanchal, and central Uttar Pradesh. There are demands for separate states from Madhya Pradesh in the form of Mahakaushal, Baghelkhand, and Malwanchal, and from Bihar in the form of Bhojpur. Similarly there have been demands for sepatate states of Vidarbha and Konkan to be carved out from Maharashtra.

Demand for Smaller States

Demands for smaller states have been made in the name of better administration and all-round development but such demands ignore the fact that the smaller states of the north-eastern region have not performed so well in the socio-economic development.

The reasons behind the demand for smaller states are being crucially analysed and some of the reasons are listed below:

1. Emotional considerations such as culture, language, religion, sense of economic and regional deprivation.
2. With different languages being spoken in the country a sense of better governance with increased participation and administrative ease prevails in the sub-regions of the bigger states.
3. The smaller states were deprived of industrial development and their mineral resources were being exploited to benefit the other developed regions of the state. In order to enhance the development, smaller states were proposed.
4. The creation of smaller states had also the motive of the betterment of the Indian economy, the growth of GDP and

better governance.

5. The prejudices amongst the people decrease when there is a participation of the people of smaller states in the state affairs.

The tendency towards more and more fragmentation by dividing bigger states on one pretext or the other has been marked in the post-Reorganisation period in the country. The tragic irony of recent history is that peace has been brought at the cost of a strong integrated India by carving out new units and new administrative provisions (Srivastava, R.P. 1996). It has created danger to the national unity since it has encouraged the feeling of regionalism, hindered economic co-operation between the states and developed antagonistic attitude among them. With the beginning of multi-party coalition rule the power of central authority has weakened, and centrifugal forces are gaining strength. Backed by our unfriendly neighbours and foreign powers, demands are being raised in some regions of the country for sovereign states.

REGIONAL CONSCIOUSNESS AND NATIONAL INTEGRATION

Regionalism is a move to foster or protect an indigenous culture in a particular region. This may be a formal move, made by the state as it creates administrative or planning region, or an informal move for some degree of independence arising from a gut feeling, based on territory, of a minority group.

In Political Geography, regionalism is a political ideology that focuses on the interest of a particular region or group of regions, whether traditional or formal (administrative divisions, country, state, political sub-divisions, regional and sub-national units). Regionalism centres on increasing the region's influence and political power, either through movements for limited form of autonomy (Devolution, states' rights, decentralisation) or through stronger measures for a greater degree of autonomy (sovereignty, separatism, independence).

Regionalism in India

India is a highly diverse pluralistic society. At the time of Independence, its population was only 340 million which has risen to 1,350,438,098 in 2019. After Independence the country was consolidated by the integration of princely states. Subsequently, some states, like Maharashtra and Gujarat in 1960, and Punjab and Haryana in 1966, were created on the linguistic basis to accommodate the regional aspirations. The number of states increased and at present, there are 28 provinces and 7 Union Territories.

There have been arguments both for and against regionalism. In the opinion of some of the experts, there is nothing wrong if the people of a region have a strong feeling of regionalism or sub-nationalism because: (i) it fulfills democratic urges, (ii) it facilitates political management, (iii) it smoothens regional and over-all development, (iv) it provides for greater access to and participation in the decision making process, (v) it enhances the local accountability, and (vi) it increases national unity. These experts also refer to recent events in Russia and Pakistan which are becoming more and more unstable for not accommodating the regional demands of the people. Pakistan broke up because it refused to accept and respect the Bangla identity. The mighty Soviet Union also could not sustain itself under the pressure of regionalism and sub-nationalism.

The other school of thought describes regionalism as a process of Balkanisation or disintegration of the country which divides people raising slogans like ‘Maharashtra for Maharashtrians’ or ‘Assam for Assamese only’ and creates agitation and violence. Some people are worried about the demand for greater autonomy for states like Jammu and Kashmir, Manipur and Nagaland which may catalyse such demands for more autonomous regions. Each act of fragmentation is, thus, seen as encouraging further fragmentation.

In fact, at times feelings of regionalism become so strong that states do not want to share even the river water (like Kaveri water by Karnataka and Tamil Nadu) or electricity (like Punjab and Delhi) with their neighbouring states. States like Assam claim that in spite of their

great contribution to national economic growth through oil, tea, jute and timber, they are totally neglected in development schemes, while states like Jammu and Kashmir hold that being a Muslim-dominated state, it is discriminated in industrial development schemes. While economic backwardness of certain regions is at the root of the unrest there (i.e. Gorkhaland in West Bengal, Bodoland in Assam, Ahom state in Upper Assam, Brij, Bundelkhand and Rohilkhand in Uttar Pradesh, Maha-Kaushal in Madhya Pradesh, etc.), some demands for new states are purely politically motivated (like Harit Pradesh and Purvanchal in Uttar Pradesh). With increasing social and political awareness among all levels of people, it is not difficult to mobilise masses for demanding a separate state. If people's protests are ignored, law and order problems are bound to escalate. Such problems cannot be supported for long or put down with a firm hand. It will compel people to take to violence. Their aspirations need to be filled through accommodation and compromises.

The growing population in many states will be difficult to manage. Uttar Pradesh, Madhya Pradesh, Bihar, and Maharashtra are the states where this problem would be seen. While in 1947, there were only 300 districts in India, today there are over 640 districts in the country. Several districts are larger in area and population than many members of the United Nations. Hence, the rationale of smaller states need not necessarily be perceived as disintegration or growth of regionalism. If the number of states increase from 28 to 50 and the number of districts 650 to 700, administrative management may be more efficient, accountability may improve, and democratic participation may increase, provided the new states are economically viable.

Regionalism and sub-nationalism are unavoidable in a country as vast as India. Nothing is more basic to the very concept of federalism than regionalism and sub-regionalism. Once the federal nation-state comes into being and national freedom becomes a reality, the regional sentiments and demands also manifest themselves. This has been the lesson of history.

In India, there is strong case not only for the existence but also for the growth of healthy regionalism perceived from a democratic perspective and political angle. The regions in India may be identified on the basis of following traits of the people: (i) *language/dialect*, (ii) *social composition*, (iii) *ethnicity*, (iv) *demographic features*, (v) *area (geographic contiguity)*, (vi) *economy and economic life*, (vii) *historical antecedents*, (viii) *political background*, and (ix) *psychological make-up and felt-consciousness of group identity*.

The process of national integration itself involves the integration of these viable regions as a precondition for the development of national identity. For instance, 'to be Indian' does not necessarily mean that one cannot be an Assami, Bengali, Gujarati, Kashmiri, Maharashtrian, Naga, Tamilian, Telgu, etc. National unity can genuinely flower out of a healthy reconciliation between regionalism and nationalism. However, it is evident that like chauvinism, unhealthy regional patriotism is equally cancerous and disruptive.

Thus, national integrity will not be imperiled only by creating new political or administrative units. Regionalism, if sensibly handled, will not disintegrate India. Ethnic sentiments and linguistic homogeneity, however, may not be desirable as considerations calling for the creation of new states. There is serious resource crunch. There is also the problem created by militancy in certain parts of the country. Dangers both internal and external, loom large. As such, the demand for new states has to be effectively scotched and conceded only after careful scrutiny of each case on the basis of economic viability and administrative convenience.

One of the most significant aspects of the Indian polity is that it is an aggregation of regions and sub-regions. These regions and sub-regions have a distinct socio-cultural personality within the framework of the natural regions of India.

A harmonious federal polity of Indian can be built on a stable foundation in the measures in which it coincides with the socio-cultural regions of India. Indeed, the history of ancient and Medieval India reveals the existence and, what is more, the recognition of

these socio-cultural units by many enlightened rulers over the centuries. One of the major disasters of the British colonial rule has been the destruction in some cases and distortion in others of these vital socio-cultural segments of our polity. The British rule had disrupted the natural evolution of India into an authentic federal polity.

NATIONAL INTEGRATION IN INDIA

India is a multilingual, multi-religious, and multicultural country. The ethnic, cultural, and religious diversity is bound to create some social and communal tension at the macro and micro levels.

Immediately after the partition of the subcontinent into India and Pakistan, both the countries were confronted with the problems of merging of states, demarcation of boundaries, allocation of assets, sharing of water resources, shortage of food, internal disturbances, regionalism, and communalism.

After successfully overcoming many of these problems, India faced the problem of linguistic communalism. In the face of widespread agitation for linguistic separation of states, beginning with the Telangana Movement in 1953, the reorganisation of states was made. The reorganisation of states on the basis of languages encouraged regionalism and the government had to create more number of provinces to accommodate the political aspirations of many regions. At the top of these problems, communalism raised its head and created serious challenge before the government to develop a secular outlook among the people.

Essence of National Integration

National integration is necessary for modernisation. It involves readjustment of loyalties of the people. The values of tribal, feudal and a parochial ethos give way to conflicting regional ideas and end a democratic, egalitarian and an evolving national society. Integration is thus a movement away from traditionalist allegiances and towards a modernist allegiance—a movement for the establishment of a new national identity.

In other words, '*national integration*' *means*, *and ought to mean cohesion*; *reconciliation but not merger*; *accommodation but not elimination*; *assimilation but not extinction*; *synthesis but not non-existence*; *solidarity but not regimentation of the many segments of the people in a territorial sovereignty*.

National integration may be summed up to signify a condition of *unity in diversity* in which components and the whole are equally valid and mutually interdependent. National integration presumes the existence of unity and diversity because, if there is only unity then integration is not necessary, and if there is only diversity then integration is not possible. Obviously, then integration is not a process of conversion of diversities into uniformities but a congruence of diversities leading to a higher level of unity in which both the varieties and similarities are maintained. It may also be remembered that all diversities are not and need not be constructed as divisive in their operation. Viable plural societies anyhow presume that diversities are reconcilable.

The finest elements in the India National Movement subscribed to the value of humanism and human fraternity, pluralism of cultures and unity of political action for freedom, democratic construction, secular polity, social justice and egalitarianism, and nationalism as part of interdependent internationalism.

Communal tension is often the expression of aspiration of the people who feel they are being discriminated on the basis of religion. In order to overcome the problems of communal riots, discrimination, under-development, poverty, deprivation and regionalism, the Government of India took several steps given below.

A National Integration Council was established in June 1962, which set up two committees to deal with the problems of regionalism and communalism. But this effort was not pursued in the wake of Chinese aggression which created national solidarity. However, this was a shortlived phenomenon and communal violence erupted in a menacing form in the sixties which led to the reconstruction of another National Integration Council in 1968. This time, three

committees were set up to deal with (i) communalism, (ii) regionalism, and (iii) educational problems. All these committees produced worthy recommendations focusing on guidelines, legislation and administrative action. By 1970, all this lapsed into inaction. The Steering Committee was sought to be revived in 1973 but this did not work. In 1980, the second incarnation of the National Integration Council appeared and it met in 1980 and again in 1984. This time also three sub-committees were set up, but they too failed to do anything substantial. The next incarnation took place in 1986 which focused on Punjab. The September 1986 meeting set up a sub-group of five persons to suggest action for the defence of the interest of the minorities and to foster national integration and communal harmony. The report of this group was adopted by the Council which, however, did not meet for the next three years. In February 1990 came yet another incarnation of the National Integration Council (NIC) which took up for its agenda the Punjab, Kashmir and Ayodhya issues. This time too, three committees were set up but they were later boycotted by members of the Bhartiya Janta Party (BJP). In November 1991 came the next reincarnation and the Ayodhya issue was discussed in its meeting. A second meeting was called on December 31, 1991 in which the focus was shifted to Jammu and Kashmir and Punjab. After this, a meeting of the National Integration Council was held on July 18, 1993 but no solution has evolved till date on the burning problem of national integration to create communal harmony.

NATIONALISM

Nationalism refers to the political movement based on the belief that a nation, usually defined in terms of ethnicity or culture, has the right to constitute an independent or autonomous community based on a shared history and common destiny. Most nationalists believe the borders of a state should be congruent with the borders of the nation. Extreme forms of nationalism, such as those propagated by fascist movements in the twentieth century, hold that nationality is the most

important aspect of one's identity and attempt to define the nation in terms of race and ethnicity.

Nationalists see nations as an inclusive categorisation of human beings, assigning every individual to one specific nation. In fact, nationalism sees most human activity as national in character. Nations have national symbols, national culture, national music, national literature, national folklore, national mythology, and—in some cases—national religion. Individuals share national values and national identity, admire the national hero, eat the national dish, and play the national sport.

Nationalists define individual nations on the basis of certain criteria, which distinguish one nation from another and determine who is a member of each nation. These criteria typically include a shared language, culture, and/or shared values which are predominantly presented within a specific ethnic group. National identity refers both to these defining criteria, and to the shared heritage of each group. Membership in a nation is usually involuntary and determined by birth. Individual nationalism vary in their degree of internal uniformity: some are monolithic, and tolerate little variance from national norms. Academic nationalism theory emphasises that national identity is contested, reflecting differences in region, class, gender, and language or dialect. A recent development is the idea of a national core culture—as in Germany the *Leitkultur*—which emphasises a minimal set of non-negotiable values; this is primarily a strategy of cultural assimilation in response to immigration.

Nationalism has strong territorial component, with an inclusive categorisation of territory corresponding to the categorisation of individuals. For each nation, there is a *territory* which is uniquely associated with it, the *national homeland*. This is reflected in the geopolitical claims of nationalism, which seeks to order the world as a series of nation-states, each based on the national homeland of its respective nation. Territorial claims characterise the politics of nationalist movements. Established nation-states make an implicit territorial claim, to secure their own continued existence: sometimes it

is specified in the national constitution. In the nationalist view, each nation has a moral entitlement to sovereign state.

The nation-state is intended to guarantee the existence of a nation, to preserve its distinct identity, and to provide a territory where the national culture and ethos are dominant. Nationalism is also a philosophy of state. It sees a nation-state as a necessity for each nation: secessionist national movements often complain about their second class status as a minority within another nation. This specific view of duties of the state influenced the introduction of national education systems, often teaching a standard curriculum, and national cultural policy. In turn, nation-states appeal to a national cultural-historical mythos to justify their existence, and to confer political legitimacy, acquiescence of the population, in the authority of the government.

Nationalists recognise that ‘non-national’ states exist and existed, but do not see them as legitimate form of state. The struggles of early nationalist movements were often directed against such non-national states, specifically multi-ethnic empires such as the Ottoman Empire and the Austria-Hungary Empire. Most of the multi-ethnic empires have disappeared, but some secessionist movements see Russia and China as comparable to national, imperial states. At least one modern state is clearly not a nation-state: the Vatican City exists solely to provide a sovereign territorial unit for the Roman Catholic Church.

Nationalism as an ideology includes ethical principles: the moral duties of individuals to fellow members of the nation override their moral duties to non-members. Nationalism claims that national loyalty, in case of conflict, overrides local loyalties, and all other loyalties to family, friends, profession, class, caste, or religion.

According to Anthony D. Smith, the precondition for the formation of a nation are as follows:

1. A fixed homeland (current or historical)
2. High autonomy
3. Hostile surroundings

4. Memories of battles
5. Sacred centres
6. Languages and scripts
7. Special customs, traditions and festivals
8. Historical records and thinking

Those preconditions may create powerful common mythology. Therefore, mythic homeland is in reality more important for the national identity than the actual territory occupied by the nation. Smith also stated that nations are formed through the inclusion of the whole populace (not just elites), constitution of legal and political institutions, nationalist ideology, international recognition, and drawing up of borders.

INTERNATIONAL BOUNDARIES OF INDIA

India lies entirely in the Northern Hemisphere. The mainland extends between latitude 8° 4’ and 37° 6’ North, longitudes 68° 7’ to 97° 25’ East, and measures about 3214 km from north to south and about 2933 km from east to west. It has a land frontier of about 15,200 km. the total length of the coastline of the mainland, Lakshadweep Islands, and Andaman and Nicobar Islands is about 7517 km.

Countries having common border with India are Afghanistan and Pakistan in the north-west, China, Bhutan, and Nepal in the north, Myanmar in the east and Bangladesh to the east of West Bengal. Sri Lanka is separated from India by a narrow channel of sea formed by the Palk Strait and the Gulf of Mannar. The longest border of India is with Bangladesh (4096 km), followed by China, Pakistan, Nepal, Myanmar, Bhutan, and Afghanistan. The shortest border of India is with Afghanistan being only 80 km **(Table 16.2)**.

Table 16.2

Boundaries of India with Adjacent Countries

	Country		Length of Border in km.	Percentage of Total Length of Border

1.	Bangladesh	Dawki-Tamabil	4096	26.95
2.	China	Mc-Mahon	3917	25.77
3.	Pakistan	i) Wagah Border ii) Longewala iii) Rann of Kutch	3310	21.78
4.	Nepal	Sonauli	1752	11.53
5.	Myanmar	Moreh	1458	9.59
6.	Bhutan	Jaigaon	587	3.86
7.	Afghanistan	–	80	0.52
	Total		15,200	100.00

The Sino-Indian Border Dispute

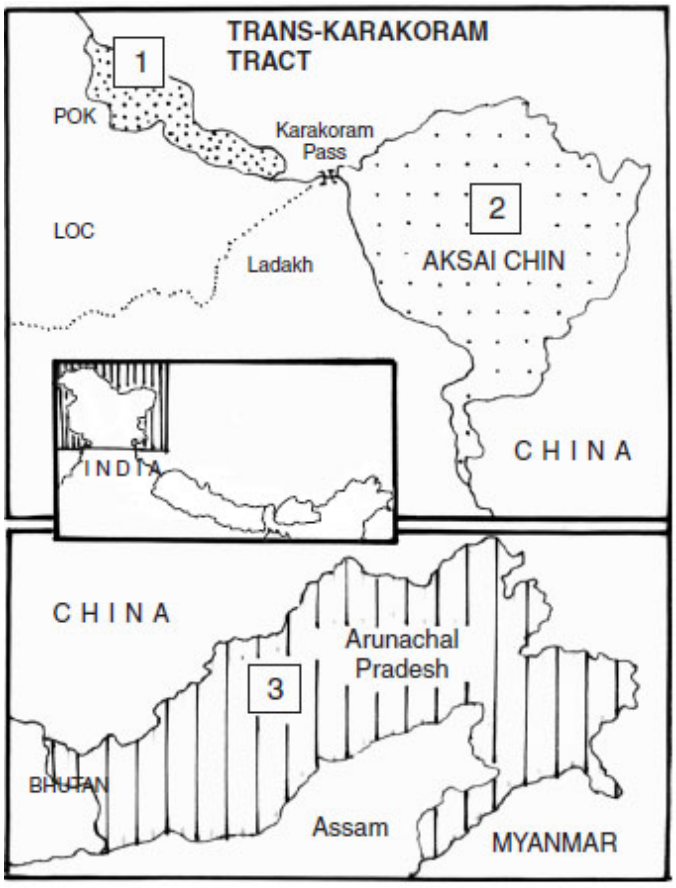
India's border with China is 3917 km. which is about 26% of the total land border of the country. Five Indian states, namely, Jammu and Kashmir, Himachal Pradesh, Uttarakhand, Sikkim and Arunachal Pradesh share the border with China **(Fig. 16.2).** This boundary is the product of Manchu policy, Chinese Republican policy, and the British policy.

Since the topographical features are rugged folded mountains, and the climate harsh, it was not possible to demarcate the boundary on the ground. The boundary was, however, demarcated on the maps. The boundary remained peaceful till India achieved Independence in 1947, and the Communist takeover of China on October 1, 1949.

In 1954, India gave up its extra-territorial rights on Tibet, exercised by the British Raj on the basis of the Anglo-Tibetan Trade Agreement of July 1914. India formally recognised the Chinese Sovereignty over Tibet and, as a consequence, for the first time in several centuries Tibet came under the direct control of a strong military power with an active and aggressive policy. The geo-strategic value of the entire

Himalayan front was drastically changed. The buffer zone of Tibet disappeared and sharp boundary between India and China was established.

Under the changed geo-political situation, the Chinese took advantage of their superior military strength, vast population and communistic administrative machinery to degrade Indian prestige and to teach it a lesson for its interference in Tibetan affairs.



1. Trans-Karakoram Tract
Pakistan handed over this area to China as part of an agreement in 1963. But India said the Trans-Karakoram Tract was part of J&K.

2. Aksai Chin
India accuses China of occupying 8000 Sq. Km. in this region, North-east of Ladakh. Aksai Chin is almost uninhabited.

3. Arunachal Pradesh
China claims 90,000 Sq. Km. of what they call "South Tibet" as their territory, that is almost the whole of Arunachal Pradesh.

Map not to Scale

Fig. 16.1 Three Unresolved Territorial Disputes between India and China

‘The Sino-Indian border is a Line of Actual Control (LAC). Both sides have claims in Aksai Chin. Moreover, in the east China does not recognize MacMahon Line. The MacMahon Line is named after Sir Henry MacMahon who was foreign secretary of the British ***turn*** administered government of India. He was the prominent British negotiator at the 1914 Shimla Conference that decided to break Tibet into Outer and Inner Tibet. China rejected this accord and its representative withdrew from talk. Only in 1937, the British began using the MacMahon Line in Survey of India maps. The Shimla agreement was published officially in 1938. China’s contention was that Tibet was never an independent state and did not have the right to sign a treaty on behalf of Beijing to delineate an international frontier.

In May 1951, China established full control over Tibet. On March 2, 1955, India objected to the inclusion of a portion on the Northeast in the official map of China. In March-April 1959 Dalia-Lama escaped from Lhasa and India gave him asylum which soured ties between the two countries.

On 20th September, 1962 China forces crossed MacMahon Line in the North Eastern Frontier Agency (NEFA). On 20th October, 1962 China launched a massive attack along border from NEFA to Ladakh. Consequently on 21st November China captured Bomdila in NEFA and declared unilateral ceasefire, and withdrew 20 km behind Line of Actual Control (LAC).

India was militarily ill-prepared in Ladakh and NEFA(Present Arunachal Pradesh), while China was much better placed. Indian posts with barely ten soldiers each were swamped. The debacle was military ill-prepared in Ladakh & NEFA **(Fig. 16.2).**

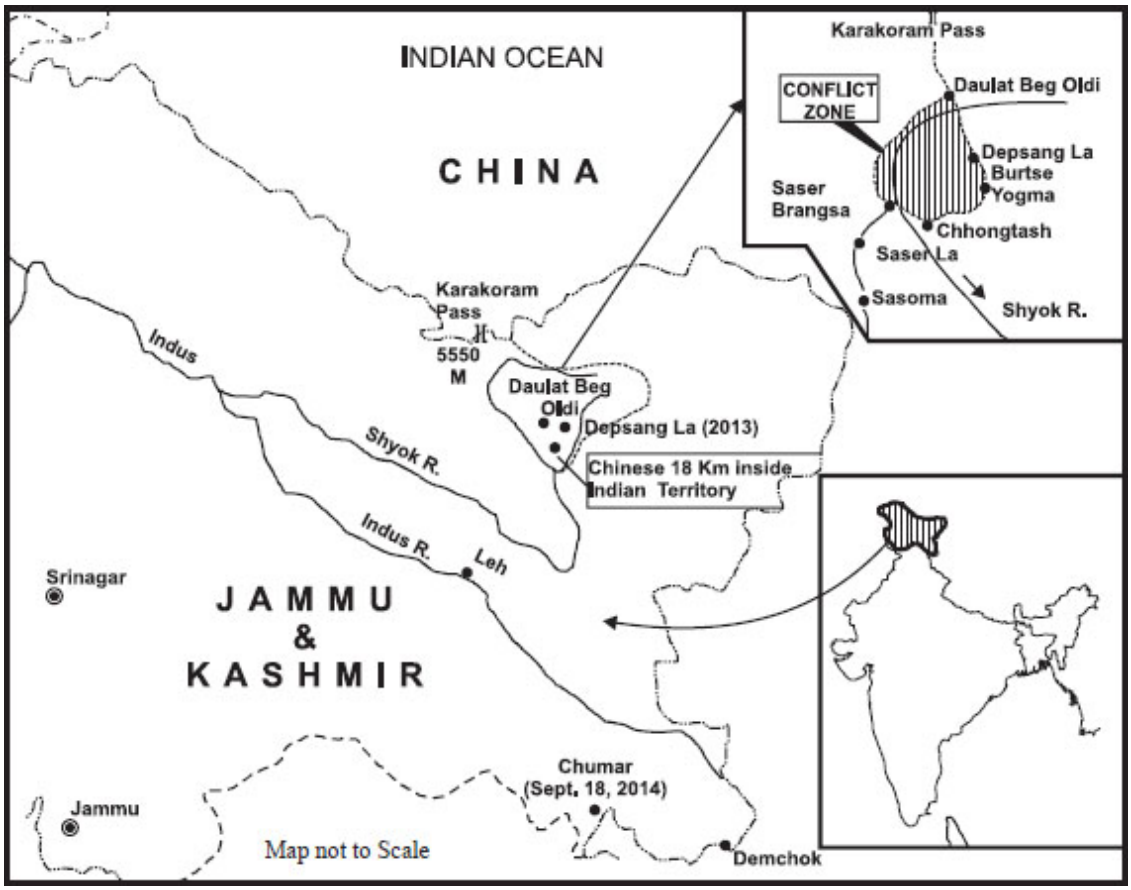


Fig. 16.2 China Incursion Sites Near Barahoti (Uttarakhand) (April 15, 2013)

On Nov.20,1962, after Bomdila fell to Chinese soldiers, Pandit Jawhar Lai Nehru spoke on AIR. He said, 'Huge Chinese armies have been marching in the northern part of NEFA. We have had reverses at Walong, Se La and Bomdila... We shall not rest till the invader goes out of India or is pushed out! I want to make that clear to all of you, and especially our countrymen in Assam, to whom our heart goes out at this moment. 'Many in Assam say the speech rankles to this day. They believe Delhi was not really concerned about the Brahmaputra Valley. Nehru's defenders insist he almost wept as he spoke. The speech was followed by virtual hysteria in Tezpur and many began fleeing.

For a systematic study, the Sino-Indian border may be divided into three sectors: (i) the Western sector, (ii) the Middle sector, and (iii) the

Eastern sector.

The Western Sector

The Indian boundary with China in the western sector is about 2152 km long. It is between Jammu and Kashmir State of India and the Xinjiang (Sinkiang) Province of China. The frontier between Xinjiang and Pakistan-occupied-Kashmir (Azad Kashmir) is about 480 km long and involves a disputed area of about 13,000 to 15,500 square kilometers. The rest of the border is between Ladakh and Tibet. In the western sector, the boundary runs along the Mustagh Range and the Aghil Range, across the Korakoram Pass via Qara-Tagh Pass; and along the main Kun Lun range to a point east of longitude 80° E and 40 km north of Hajit Langar. Further south-east, the boundary runs along the watershed across Lanak La, Kone La, the Pangong Lake and the Kailash Range. Here the boundary constitutes the watershed between the Indus System in India and the Khotan System in China. In this area, China claims the Aksai Chin District, the Changmo Valley, Pangong Tso (lake), and the Spanggar Tso (lake) areas of Ladakh, as well as a strip of about 5000 sq km down the entire length of eastern Ladakh (Bains, 1962).

Since 1954, the Chinese have repeatedly intruded deeper and deeper into the western sector. By 1962, the Chinese had penetrated far deeper and the Chinese occupation line now runs from 16 to 240 km west of the international line The Chinese army penetrated in this region repeatedly in September and October, 2009. In this sector China is in actual possession of about 54,000 sq km of the Indian territory. Of which 38,000 sq km lie in the Ladakh area alone.

The Middle Sector

The middle sector boundary between India and China is about 625 km long which runs along the watershed from Ladakh to Nepal. The states of Himachal Pradesh and Uttarakhand touch this border with Tibet (China). The boundary of Himachal Pradesh follows the water parting between the Spiti and Para Chu rivers and continues along the watershed between the eastern and the western tributaries of the Satluj. The Uttarakhand boundary is demarcated by the watershed

between the Satluj on the one hand and the Kali, the Alaknanda, and the Bhagirathi on the other. There after it runs along the watershed passes of Mana, Niti, Kungri-Bingri, Dharma and Lipu Lekh. It finally joins the tri-junction of China, India and Nepal. This part of the border was approved by the Tibetan and British Governments under the treaties of 1890 and 1919. The Chinese lay claim on nearly 2000 sq km in this area. In the Chimoli District of Uttarakhand China claims the Barahoti area. The Barahoti lies to the south of watershed in India territory. China often intrudes in this area and creates pressure on Indian Government. Intruding Chinese troops spotted on July 22, 2016 by a team of ITBP officials **(Fig. 16.2–A)**.

China is also claiming the Chumbi Valley and Toe-point in Sikkim. Toe-point along the border of Sikkim and Bhutan is important strategically. China is negotiating with Bhutan on this valley and hill, which is not acceptable to India.

The Eastern Sector

The boundary of India with China in the Eastern Sector is 1140 km long which runs from the eastern limit of Bhutan to the point near the Talu Pass at the tri-junction of India, Myanmar and Tibet (China). This line is called the McMahon Line after Henry McMahon, a British representative, who signed the Shimla Convention of 1914. The boundary was established along the Himalayan crest of the northern watershed of the Brahmaputra, except where the Lohit, Dehang, Subansiri, and Kemang rivers break through that watershed.

In the Eastern Sector China claims about 90,000 sq km of the Indian territory. China considers the MacMahon Line illegal and unacceptable claiming that Tibet had no right to sign the 1914 Convention held in Shimla in 1914.

The Chinese government never formally questioned the validity of the agreement between India and Tibet until 1959. China launched a massive attack on India in late 1962, and occupied the Thangla (Thagla) Ridge, Dhole, Tse Dong, Twang, Bomdila, Walong, Longju, and Sela outposts in Arunachal Pradesh.

The China war proved that the Himalayas were not an effective defence barrier. India's confidence in China's friendship was broken. India also realised that the posture of 'non-alignment' was no substitute for defence preparedness.

China's aggression (1962) was motivated not by law, custom, treaties, the facts of geography or administration, but rather by pure ideological, political and strategic considerations. At present, the leaders of both the countries are working towards improving trade and cultural relations at the initial stage and the delicate issue of border settlement later on.

In January 2013, China cleared three dams on the Brahmaputra River namely Dagu-640 MW, Zangum 510 MW and Jiacha-320 MW. The dams and power projects are a point of great concern. The complation of these dams may dry up the Brahmaputra, adversely affecting the eco system of its drainage system.

In April, *2013 China invaded Daulat Beg Oldi in Ladakh*, capturing about 750 sq km of India
(Fig. 16.1A). Morevoer chinese authorities issued a new map in the 3rd weak of June, 2014 showing Arunachal Pradesh as a part of southern Tibet which Beijing claims as its own. On 18th September, 2014 at the time of China's President Xi Jinping, Chinese army entered inside India at Chumar and Demchok **(Fig. 16.1-B)**.

The Indo-Pak Boundary

The Indo-Pakistan boundary is the product of the process of partition of the Sub-continent in 1947 under '*the Radcliffe Award*' decided by Sir Cyril Radcliff who acted as the Chairman of Border Commissions. This boundary runs through varied geographical features. It starts from the marshy Rann of Kutch in Gujarat, and traverses through the sandy desert of Rajasthan, fertile plains of Punjab, hills and mountains of Jammu and Kashmir, and the high mountain ranges of the Himalayas and Karakoram Range **(Fig. 16.3)**.

The boundary with Pakistan may be examined under the following two issues (i) The Rann of Kutch Dispute, and (ii) The Kashmir Problem.

Fig. 16.3 India–Pakistan Border

The Rann of Kutch Dispute

The boundary of India and Pakistan in the Rann of Kutch was well defined according to the Radcliffe award. Pakistan, however, advocated that it was not demarcated on the ground. According to Pakistan, the Rann was not a marsh but a land-locked sea, and as such it should be equally divided between India and Pakistan. India denied the claim and presented proof that the Rann was a marsh and always remained as a part of the Kutch State which after independence had merged with India.

Stretching over an area of 20,720 sq km, the Rann of Kutch consists of salt flats, brackish ponds, and marsh interrupted by a few rocky elevations. Pakistan claimed that 9065 sq km of the Rann of Kutch, and in 1965 its troops invaded the Indian territory. On the intervention of the Prime Minister of Britain, the dispute was referred to the boundary tribunal constituted by the International Court of Justice. With Gunnar Lagergren of Switzerland as the President, the tribunal awarded 9065 sq km

of the Kutch area to Pakistan. Both the sides accepted the award and the demarcation of the boundary was completed in June 1969.

The Kashmir Problem

The state of Jammu and Kashmir has been separated in to two U.Ts- Jammu & Kashmir and Ladakh after article 370 sprawls over an area of about 2,22,236 sq km. The U.Ts has common boundary with Afghanistan and China. Earlier state of Jammu and Kashmir consists of six distinctive regions, which except Hunza, Gilgit and Nagar, were brought as a united political unit by Maharaja Gulab Singh, who entered into a subsidiary alliance with the British within the Indian Empire in 1849. Among the various regions of the Jammu and Kashmir, the most significant is the Valley of Kashmir, the summer capital (Srinagar) and the centre of tourist attraction. The Valley of Kashmir is a Muslim majority region. Kashmiri Pandits (Brahmins) have traditionally held positions of economic, social and political power, whereas the Muslim peasantry has remained very poor and highly oppressed.

The second region Jammu lies in the southern part of Jammu and Kashmir and contains only one-seventh of the total area of the state. Its population is predominantly Hindu. The city of Jammu is the winter capital of the state and home of the state's former rulers (the Dogra Rajputs). The third region is Ladakh (also called little Tibet) which covers over one-third of the total area of the state. Ladakh is a vast barren plateau characterised by desolate rocks. It resembles Tibet in culture and the people are largely the followers of Buddhism. Racially, the Ladakhis are Mongoloids. The fourth region, Gilgit, is the northern part of the state which is mountainous and almost Muslim. It used to be reached from the Srinagar by crossing high mountains and glaciers. But now it is well linked with Pakistan through the Karakoram Highway. The fifth region is Baltistan. It lies in the extreme northern part of the state, which is largely mountainous, dominated by the Muslim population and accessible by a road along the Indus river. The sixth region, Punch lies to the north-west of the Jammu region and south-west of the Kashmir Valley. It is mostly Muslim and easily

accessible from Pakistan. The two important rivers flowing through this region are Chenab and Jhelum. Thus, the region of Jammu and Kashmir has great cultural heterogeneity.

Pakistan claims to Kashmir is largely based on religious and economic considerations. It is a Muslim-dominated state, contiguous and connected by several roads with Pakistan. Moreover, the four rivers—Ravi, Chenab, Jhelum and Indus, vital to the agriculture of Pakistan flow through the state of Jammu and Kashmir. Pakistan has also maintained that the plebiscite, as recommended by the United Nations, is the only solution of this problem.

India, on the other hand has held the position that Kashmir's accession had given India sovereignty over Kashmir. It had at first agreed to a plebiscite on condition that the Pakistani invading forces were withdrawn, a condition which was stipulated by the United Nations' resolution, but never been realised. India, therefore, stresses the illegality of Pakistani support of the raiders and the Azad Kashmir government. Apart from the legalistic view of Kashmir's accession to India, there are other consideration also. India is a secular state and surrender of Kashmir on religious grounds would be a denial of the nation's basic principles. The strategic location of Kashmir to China is believed to be of critical importance to India. China's aggressive policy in the late 1950s, its occupation of Tibet, later a part of Ladakh (1965), and construction of a road through the Aksai-Chin in Ladakh, as well as its threatening posture towards India, have all lent urgency to the strategic aspect of Kashmir's location.

The instrument of accession was signed by Maharaja Hari Singh on October 26, 1947 and the accession was accepted by Governor General of India on October 27, 1947. Thus Kashmir became legally and constitutionally a part of Indian Union. Pakistan described Maharaja's accession to India as based upon 'fraud, deceit, and violence,' and maintained that it was totally against the wishes of its long oppressed Muslim subjects.

Since 1947, there had been wars between India and Pakistan in 1948, 1965, 1971, and 1999 (Kargil) on the issue of Kashmir. Unable

to beat India in regular warfare, its forces are launching proxy war in Kashmir and sending foreign mercenaries and militants to disturb law and order in the state. Even after over six decades of Kashmir's accession to India, there seems no sign of resolving the Kashmir problem as both sides stick to their respective positions. India is taking no chance to safeguard the integrity and sovereignty of the country, particularly of Jammu and Kashmir. There are over five lakh Indian troops ranged along 188 km International border, the 788 km Line of Control (LoC) and 150 km Actual Ground Position Line (AGPL). Under the present circumstances, it is difficult to compel China and Pakistan to vacate these areas. It is, therefore, necessary for the contesting parties to realise the ground realities and resolve the crisis amicably.

The Siachin/Aksai chin Glacier Dispute

Situated near Karakoram, the Siachin glacier is about 75 km long and 2 to 8 km wide. This glacier is about 5800 m above the sea-level with an average temperature of–50°C. The glacier lies to the north-west of K^2 in Pakistan-occupied-Kashmir and pierced through the Saltoro Range. The Karakoram Highway between China and Pakistan is very close to the glacier. Hence, the Siachin glacier has great strategic importance.

India occupies about two-third area of the glacier in its south eastern part. Here, Nubra river emerges from the Karakoram glaciers and meet the Shyok river which is a tributary of the Indus river. Indian troops use the Nubra valley route KhardungLa (Pass) to reach the Aksai chin glacier. The glacier has four passes. Of these, Gesar, Broom, Saltoro, and Vilafondala are under the possession of India and Gyong under Pakistan.

The Siachin glacier attracted the attention of India when Pakistan permitted some Japanese tourists for skiing in this area and scaling some peaks under India possession. Moreover, Pakistan established an observation post on the Saltoro Range. To counter this move, India launched Operation Meghdoot on April 3, 1984. Since then both the countries are maintaining troops in this inhospitable environment

in the position of high alert. India alone spends over Rupees 20 million per day to maintain its force in the Siachin glacier area.

Sir Creek

Sir Creek forms the boundary between Gujarat State of India and the Sind Province of Pakistan. This creek is rich in marine life and both the countries desire to include it in their respective territories. While India claims that the middle point of Sir Creek should be demarcated as the international boundary, Pakistan wants the eastern bank of the Creek as the border between India and Pakistan **(Fig. 16.4)**. As such Sir Creek has become a contentious issue between the two countries as Pakistan has been claiming a 40 km region inside India's territory.

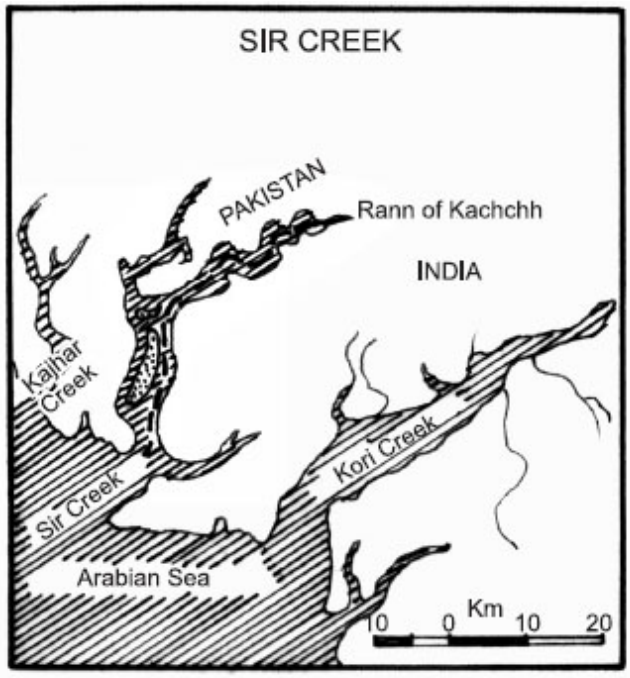


Fig. 16.4 India–Pakistan Boundary in Sir Creek

Pakistan has responded positively to the goodwill gesture initiated by India since early 2004 to ease tension on the border. This can lead to friendly relations between the two countries if the concerned countries take necessary positive steps for normalisation of relation. Some of the steps taken/proposed are:

1. Delhi-Lahore bus and rail service.
2. Srinagar-Muzaffrabad bus service.
3. Kargil-Skardu bus service.
4. Rail-link between Munnabao (Rajasthan) and Khokrapar (Sind).
5. Link between Ferozpur and Sahiwal.
6. Meeting points for divided families across LoC at Poonch, Mendhar, Suchetgarh, Uri, Tangdar, and Kargil.
7. Trade across the LoC border.
8. Mechanism to permit two-way religious pilgrims.
9. Promotion of cultural interaction and cooperation.
10. Joint efforts to promote tourism.

These steps, if taken together can go a long way in improving the relationship between India and Pakistan and can finally help in resolving the disputes like that of Kashmir and Siachin.

Violation of Ceasefire

Violation of ceasefire is a frequent feature along the Line of Control between India and Pakistan. The border hostilities have, however, increased in 2015–16. The main sectors of border hostilities have been shown in **(Fig. 16.5)**. These sectors include Kargil, Drass, Nowgam, Uri, Poonch, Krishnaghati, Bimber-Gali, Jammu and Samba.

Fig. 16.5 India–Pakistan: Main Hostility Sectors

Indo-Bangladesh Border

India's boundary with Bangladesh is 4096 km long **(Fig. 16.6)** out of which 2,450 km has been demarcated on the ground. This boundary, demarcated under the Radcliffe Award, divided the erstwhile province of Bengal into two parts; eastern part as East Pakistan (present Bangladesh) and the western part remaining with India as West Bengal. In the Koch-Bihar region, the boundary award left a number of small enclaves of India and Pakistan (Bangladesh) territory on either side of the border. A number of disputes were referred to a tribunal in which Justice A. Bagge of Sweden was selected by both the parties for arbitration.

Bangladesh was liberated in 1971 with the help of the Indian forces. Along the India-Bangladesh border, there were four areas of overlapping claims. The first was between Rajshahi (Bangladesh) and Murshidabad (India) where the shifting channel of the Ganga fluctuates frequently and the boundary shifted several times. The award here was in favour of India and the northern portion of the

district boundary was confirmed as the international boundary. Another dispute area was along the West Bengal-Bangladesh border between Karimpur (India) and Daulatpur (Bangladesh). In this instance, the Radcliffe Award was amended, and a half way line in the Matabhanga River was designated to form the boundary. As a result of this change, Bangladesh gained nearly 13 sq km from this award. The third disputed area was in the Sylhet District of Bangladesh and the Garo, Khasi and Jaintia Hill districts of India. This is a densely-forested area, and hence demarcation of the boundary on land was difficult. Here the Radcliffe Award was confirmed, and India's claim was sustained. A fourth dispute existed in the boundary zone between Barisari and Gobindpur. In April 2001, Bangladesh Rifles personnel forcefully occupied Peerdivah village (Meghalaya) and brutally killed 18 Border Security Force personnel. This generated a lot of tension between India and Bangladesh.

Exchange of India and Bangladesh Enclaves

An enclave is a piece of land encircled by another country's territory. The enclaves exchanged by India and Bangladesh on the midnight of 31st July, 2015 were created in India's Cooch Behar District after the treaties signed between the local rulers and the Mughal Empire in 1713.

India had 111 such enclaves within Bangladesh and Bangladesh 51 its enclaves within India. The total area covered by the 111 Indian enclaves in Bangladesh was about 17160 acres and the Bangladesh enclaves in India about 7110 acres. The total population of the enclaves was about 51,549 (37,334 in Indian enclaves and 14,215 in the enclaves of Bangladesh). These enclaves lacked even basic infrastructure facilities such as electricity, schools, and health services. In fact, some of them became as a hot bed of criminal activities **(Fig. 16.6)**.

At the time of Independence 1947, the demarcation of India and Pakistan borders by the British left the boundary issue unresolved in the eastern province of Bengal. In 1958 India and Pakistan agreed to

‘exchange of enclaves on the basis of enclaves for enclaves’ without consideration for the territorial loss or gain.

In 1974, Bangladesh Prime Minister, Sheikh Mujibur Rahman and the Prime Minister of India, Indira Gandhi signed ‘Land Boundary Pact’ under which the two countries agreed to transfer the enclaves to each other’s sovereignty. The legislature of Bangladesh ratified the pact, but India’s Parliament refused to do so. The pact implementation was put on hold after the assassination of Sheikh Mujeebur Rahman.

India, however, in 1974, agreed to grant Bangladesh the right to use a 178 meter long and 85-meter wide corridor, known as-Teen Bigha Corridor. This gave Bangladesh access to its South Berubari Enclabe in Cooch Behar.

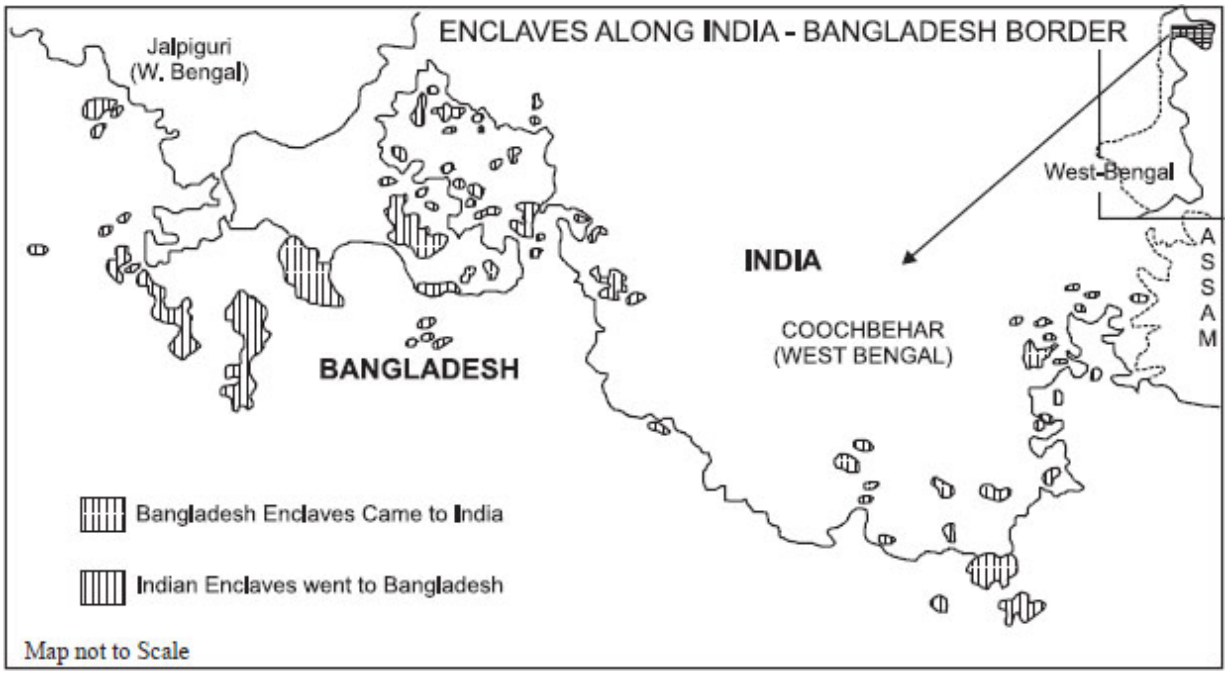


Fig. 16.6 Enclaves along India–Bangladesh International Border (W. Bengal Sector)
(After: Times of India, New Delhi, 1 August, 2015)

In 1992, Narasimha Government granted Dhaka Government the right to use Teen Bigha Corridor. Initially, access was given for limited

hours. From 2011, Bangla got full access. In September, 2011, the Prime Minister of India, Manmohan Sigh visited Dhaka and the leaders agreed to implement the pact. As per the protocol, enclave residents were allowed to accept nationality of either of the two countries and stay on 'as-is-where-is' basis. The agreed protocol also addressed the outstanding issues including undemarcated land boundary in the three sectors, enclaves, and adverse possessions.

In November 2014, all the Indian political parties okayed the draft bill. In May 2015, the political parties approved the deal in Parliament. In June 2015, Prime Minister Narendra Modi ratified the agreement during a visit to Dhaka. Finally in the midnight of July 31, the border deal regarding enclaves became effective.

Before the transfer of enclaves, the two governments recorded the wishes of all the people of enclaves involved. The people were asked whether they would want to become Indian or Bangladeshi citizens. As a result of exchange of enclaves between India and Bangladesh the following transfers have been made:

1. The 51 Bangladesh Enclaves in India with a territory of area of 7110 acres became of part of Indian territory.
2. The residents of Bangladesh 14,215 in number became the citizens of India.
3. Indian enclaves 111 in number in Bangladesh to be swapped with a total area of about 17,160 acres.
4. In the exchange of population, 979 opted to return to India, while 36,355 residents remained in Bangladesh.
5. The transfer of population started on 1st August and was to be completed on 30th November, 2015.

It is a commendable exchange of territory and population in the subcontinent, as it resolved the complex boundary problem left over from history. It is for the first time since 1947 that India can draw a firm, uncontested boundary with Bangladesh.

People coming to India from Bangladesh enclaves, however, would not be given citizenship immediately. They must wait till the end of November, 2015 for the formalities to be completed and are likely to

face problems arising out of lack of state-backed residency proof and identity documents.

Now onwards the residents of enclaves on both sides of the border will enjoy the benefits of nationality of India or Bangladesh, and thus access to civic services, education, healthcare and other facilities provided by the two governments to their respective nationals.

India-Nepal Boundary

India's border with Nepal is 1752 km. Five states of India, namely, Uttarakhand, Uttar Pradesh, Bihar, West Bengal and Sikkim touch the border of Nepal **(Fig. 16.7).** India-Nepal border, mainly runs in a west-east direction almost along the foothills of the Siwalik Range. A treaty between British and Nepal government was signed in 1814–15. This treaty was ratified in March 1916. Under the terms of the Treaty, the Himalayan districts of Nainital, Almora, Garhwal, Dehra Dun, and Shimla were annexed by the British. (Karan, et.al. 1963). The eastern tracts were given to Sikkim. The present boundary between India and Nepal is the boundary demarcated in 1858. Both the governments also agreed upon the preservation of a strip of uncultivated land of 10 yards in width on both sides of the British-Nepalese border. The principle on which the boundary between British India and Nepal was decided, is given in the resolution of June 1882. The resolution directed that 'except where natural obstacle intervenes the line from pillar to pillar may be regarded as straight.' The present Indo-Nepal boundary is peaceful and there is no boundary dispute between the two countries.

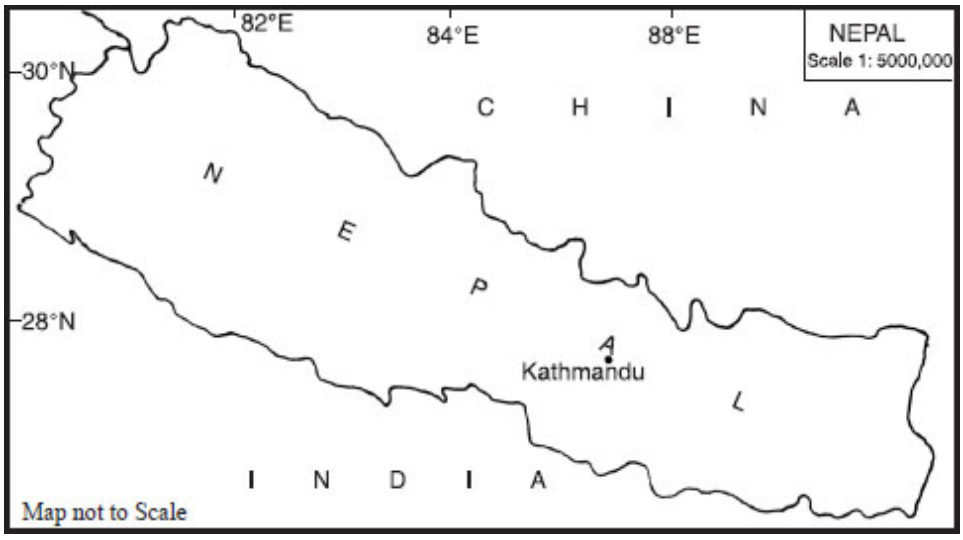


Fig. 16.7 India–Nepal Border

India-Bhutan Boundary

India Bhutan boundary is the outcome of long history of border conflict between the British and the Bhutanese since 1775. A treaty was signed at Sanchula in 1865 between Bhutan and Great Britain. The British annexed Bhutanese territory along Bengal, Koch-Bihar, and Assam border. Through this treaty, Bhutan was awarded Rs. 50,000 per annum as a British subsidy to Bhutan which was raised to Rupees one lakh in June 1911.

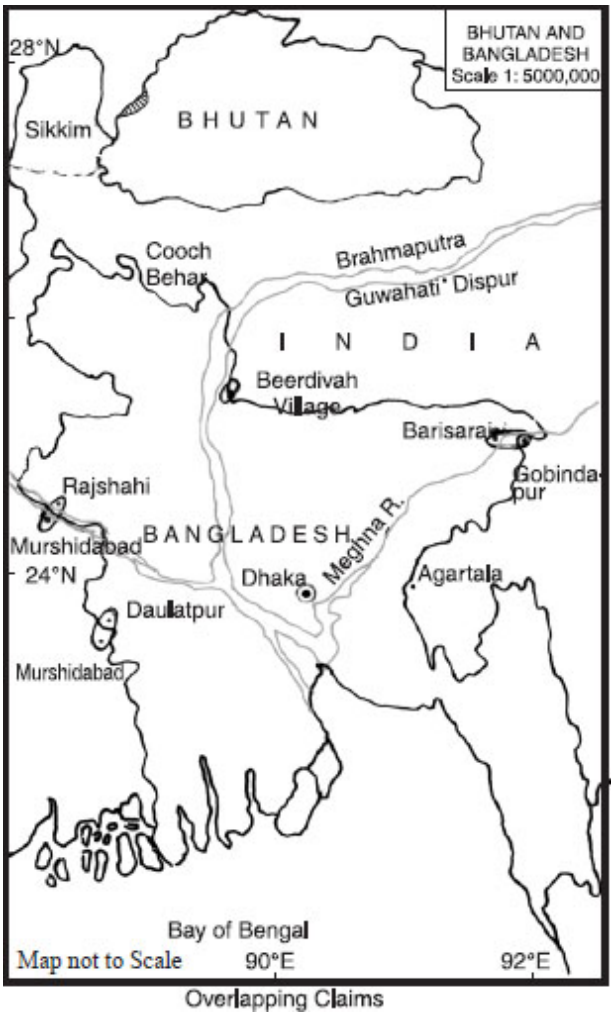


Fig. 16.8 India–Bangladesh Border

India's friendly relations with Bhutan are based on the Treaty of 1949 which provides a sound framework to ensure 'perpetual peace and friendship'. Under the treaty, India has got the rights to protect Bhutan's sovereignty and defend its borders. Indian army units are permanently stationed all along the Bhutan-Tibet border.

India-Myanmar Boundary

The boundary between India and Myanmar is 1458 km. long which runs from India-China-Myanmar tri-junction in the north to the southern tip of Mizoram **(Fig. 16.9).** This boundary runs roughly along the watershed between the Brahmaputra and Ayeyarwady (Irrawaddy). This border passes through the thickly forested hills

along the borders of Mizoram, Manipur, Nagaland and Arunachal Pradesh. The boundary was delimited precisely by a bilateral treaty signed on March 10, 1967. Some difficulty came near Diphu Pass, the tri-junction between India, Myanmar, and China. India's claim has been that the Dipuh Pass is not the tri-junction but rather a few kilometers south of it.

Fig. 16.9 India–Myanmar Border

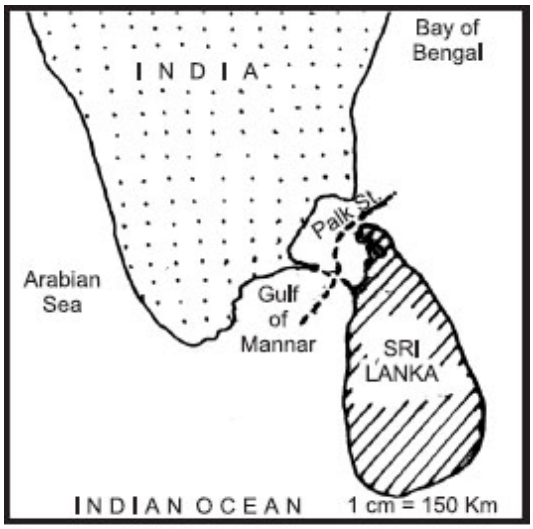
Insurgency and smuggling are the main problems along the India-Myanmar border. On the Myanmar side, the Communist supported rebels are playing a significant role in instigating Karnes, Kachins and

Shans to fight for their independence from Myanamar. Similarly, on the Indian side, the Nagas and Mizos, etc. are receiving aid and encouragement from the Chinese and Myanmar (Burmese) Communists. Moreover, a lot of smuggling of drugs and narcotics is going on along this border. Barring a few minor incidents, the Indo-Myanmar border has remained peaceful.

India-Sri Lanka Boundary

India Sri-Lanka boundary is a maritime boundary. It is separated by a 30 km wide shallow sea called the Palk Strait **(Fig. 16.10)**. The nearest points of the two countries are Dhanushkodi in Tamil Nadu (India) and Talai Mannar in Jaffna (Sri Lanka).

Historically, the India-Sri Lanka boundary has remained generally peaceful. Some bitterness was, however, created over the ownership of Kachchitevu Island (area 1.92 sq km) in the Palk Strait which was given to Sri Lanka by India in 1974. The maritime boundary has become lively with the insurgent activity of LTTE which is demanding a separate homeland for Sri Lanka Tamils.



Map not to Scale

Fig. 16.10 India–Sri Lanka Border

CROSS BORDER TERRORISM

Terrorism, especially the cross border terrorism is a matter of great concen for the internal security of the country. Terrorism from across the border is creating a lot of problems for the government and the people of India. Pakistan, in keeping with its anti-India stance over the years, has been using state-sponsored terrorism as an important instrument of its foreign policy. The threat from ISI is real and grave and its primary focus is to undermine India's internal security.

Terrorism in India can be attributed to India's low intensity conflicts within its borders. If terrorism can be defined as '*peacetime equivalent of war crime*', then these sites of low intensity conflicts are prime spots for terrorism in India. The region with long-term terrorist activities today are Jammu and Kashmir, Central India (Naxalism) and the Seven Sister States (independence and autonomy movements). In the past, the Punjab insurgency led to militant activities in the Indian state of Punjab as well as the national capital Delhi (Delhi serial blast, and anti-Sikh riots).

Northern and Northwestern India

Terrorism in Jammu and Kashmir

Violence has existed in various forms in Jammu and Kashmir. Thousands of lives have been lost since 1989 due to intensified insurgency. Casualities include civilians, Indian security forces, and Kashmiri and non-Kashmiri militants who are supported by Pakistan.

Though there had been instances of sporadic conflict in many regions for many years, intensified attacks occurred in the late 1980s, when Mujahideen fighters from Afghanistan slowly infiltrated the region, allegedly with Pakistan's help, following the end of Soviet-Afghan War in 1989. Since then, violence has increased significantly in strength. Many separatists have carried out attacks on Indian civilians and Indian army installations in response to what they see as Indian army occupation.

India claims that most of the separatist terrorist groups are based in Pakistan and Pakistan-administered Kashmir (also known as Azad Kashmir). Some like the All Parties Hurriyat Conference and the Jammu and Kashmir Liberation Front, demand an independent

Kashmir. Other terrorist groups such as Lashkar-e-Taiba and Jaish-e Mohammad favour a Pakistani-Kashmir. These terror groups have contacts with Taliban and Bin-Laden. Both the organisations no longer operate under these names after they were banned by the Indian and Pakistani government, and by other countries including the USA and UK. Of the larger militant groups, the Hizbul Mujahadeen, a militant organisation based in Indian-administered Kashmir, unlike other groups, has only kept its name. Despite casualties, the militants are still believed to number in thousands rather than hundreds. Several new separatist organisations have also emerged. According to US Intelligence, Al-Qaeda also has a base in Pakistani Kashmir and is helping to foment terrorism in Jammu and Kashmir.

It is hard to determine the total number of casualties. According to a report by the Government of India in the year 2000, 31,000 Indian civilians had lost their lives due to insurgency. Human rights groups and Local NGOs put the total figure at more than 84,000 (2005 figure). Kashmir, however, continues to remain as the most volatile region in the world with an average of 2,500 incidences every year. Not all Kashmiri separatists and militant organisations share the same ideology. Some fight in the name of religion, some are pro-Pakistan and some favour an independent Kashmir.

India claims it is the presence of these numerous anti-India insurgent groups that has compelled New Delhi to deploy massive number of troops in Jammu and Kashmir for the task of counter insurgency.

Involvement of Pakistan: A 1994 report by Human Rights Watch group lends support to both India and Pakistan charges. In support of India's claims, it states that "there are compelling evidence that elements of the Pakkistani government have sponsored a significant flow of arms to Kashmiri militants as well as arranged for them an extensive training programme." While in support of Pakistan's claims, it states that "the human rights record of the Indian government in

Punjab and Kashmir is appalling. Abuse in Kashmir are clearly on the rise."

The insurgents who initially started a pro-Kashmiri independence movement, have gone through a lot of change in their ideology. Most of the insurgents portray their struggle as a religious one.

The main terrorist groups active along the Line of Actual Control are:

(i) *Lashkar-e-Toiba* (LeT): The group is blamed for some of the high profile terrorist attacks in India.

(ii) *Jaish-e-Mohammad* (JeM): Founded in 2000, it tries to drive India out of Jammu and Kashmir.

(iii) *Harkat-ul-Mujahedeen* (HuM): Founded in 1985 as an anti-Soviet group to fight in Afghanistan.

(iv) *Harkat-ul-Jihad-e-Islami* (HjI): Founded in 1980 to fight Soviet soldiers in Afghanistan.

(v) *The United Liberation Front of* Founded in 1979 to establish independent state of Assam.

Assam (ULFA):

(vi) *Peoples War Group* (PWG) Founded in Andhra Pradesh, Chhattisgarh, Jharkhand, West Bengal, etc.

Indian analysts allege that by supporting these insurgents, Pakistan is trying to wage a proxy war against India while Pakistan claims that it regards most of these insurgent groups as 'freedom fighters' rather than militants. On 5th August, 2015, a suspected Pak militant-45mah Naved alias Qasim is the first militant to be captured alive in India after Ajmal Kasab in 2008.

Internationally known to be the most deadly theatre of conflict, nearly 10 million people, including Muslims, Hindus, and Buddhists have been fighting a daily battle for survival. The cross border firing between India and Pakistan, and the terrorist attacks combined have mostly taken its toll on the Muslim Kashmiris, who have suffered poor living standards and erosion of human rights.

Punjab

During the 1970s, the Indian Green Revolution brought increased economic prosperity for the Sikh community in Punjab. The prosperity kindled an age old fear in the Sikh community—that of being absorbed into the Hindu fold and led to the rise of Sikh militants. The insurgency was intensified during the 1980s when the movement turned violent and the name Khalistan resurfaced and sought independence from the Indian Union. This movement was led by Sant Jarnail Singh Bhidarwalw. Soon things turned bloody with India alleging that Pakistan supported these militants.

In 1984, Operation Blue Star was constructed by the Indian government to stem the movement. It involved an assault on the Golden Temple Complex. In this operation militant casualties were as follows: 493 killed and 68 injured. Human rights observers assert that the number of civilian casualties may have been as high as 500.

During the same year, the assassination of Indira Gandhi by two Sikh bodyguards, believed to be driven by the Golden Temple affair, resulted in widespread anti-Sikh riots, especially in New Delhi. The riots claimed the lives of over 2000 Sikhs.

The ending of the overt Sikh militancy in 1993 led to a period of relevant calm, punctuated by militant acts attributed to half a dozen or so operating Sikh militant organisations. These organisations include Babbar Khalsa International, Khalistan Commondo Force, Khalistan Liberation Force, and Khalistan Zindabad Force.

Bihar

Terrorism is not considered a major isuue in the Bihar state. There is, however, existence of certain groups like the *CPI-ML, Peoples War Group*, *MCC*, *Ranvir Sena*, and *Balbir militias*. The *Ranvir Sena* is a militia of forward caste land owners which is taking on the powerful Naxalite in the area. The state has witnessed many massacares and mass killings. Many a times politicians use these groups for their advantage.

New Delhi

The explosions went off in New Delhi on October 29, 2005 (on the eve of the Hindu festival Diwali) which killed more than 60 people and

injured at least 200 others.

Uttar Pradesh

Ayodhya Crisis: The long simmering Ayodhya crisis finally culminated in a terrorist attack on the site of the 16th century Babri Masjid-Ram Janambhoomi Hindu temple in Ayodhya on 5 July, 2005. Following the two hour gunfight between Lashkar-e-Taiba terrorists based in Pakistan and Indian police, in which six terrorists were killed, opposition parties called for nationwide strike with the country's leaders condemning the attack, believed to have been masterminded by Dawood Ibrahim.

Varanasi Blasts: A series of blasts occurred across the Hindu holy city of Varanasi on 7 March, 2006. Fifteen people were killed and 101 others were injured. It is believed that the attack was made by the Lashkr-e-Taiba group.

Terrorism in North-eastern India

North Eastern India consists of seven states (also known as the *seven sisters*): Arunachal Pradesh, Assam, Manipur, Mizoram, Nagaland, Meghalaya, and Tripura. Tension exists between these states and the Central Government, as well as amongst the tribal people, who are natives of these states, and migrant peoples from other parts of India. The states have accused New Delhi of ignoring the issues concerning them. It is this feeling which has led to the greater participation of the natives of these states in self governance. There are existing territorial disputes between Manipur and Nagaland. There is a rise of insurgent activities and regional movements in the northeast, especially in the states of Assam, Mizoram, Nagaland, and Tripura. Most of the organisations demand independent state status or increased autonomy and sovereignty.

The first, perhaps the most significant, insurgency was in Nagaland from the early 1950s until it was finally quelled in the early 1980s through a mixture of repression and cooperation. The *National Socialist Council of Nagaland-Isak-Muivah* (NSCN-IM), demands an independent Nagaland and has carried out several attacks on Indian

Military installations in the region. According to government officials, 599 civilians, 235 security forces, and 862 terrorists have lost their lives between 1992 and 2000.

Assam

Beginning in 1979, the indigenous people of Assam demanded that the illegal immigrants who had emigrated from Bangladesh to Assam be detected and deported. The movement led by *All Assam Student Union* began non-violently with satyagrah, boycotts, picketing, and courting arrests. Those protesting frequently came under police action. The Bodos demanded for separate Bodoland which led to a clash between the Bengalis, the Bodos, and the Indian military resulting in hundreds of deaths.

There are several organisations which advocate the independence of Assam. The most prominent of which is the ULFA (*United Liberation Front of Asom*). Formed in 1971, the ULFA has two main goals, the independence of Assam and the establishment of socialist government. The ULFA has carried out several terrorist attacks in the region targeting the Indian Military and noncombatants. The group assassinated poltical opponents, attacked police and other security forces, blasted railroad tracks and attacked other infrastructure facilities. The ULFA is believed to have strong links with Nationalist Socialist Council of Nagaland (NSCN), Maoist, and the Naxalites.

Assam remains the only state in the northeast where terrorism is still a major issue. The Indian Military was successful in dismantling terrorist outfits in other areas, but have been criticised by human right groups for allegedly using harsh methods when dealing with terrorists.

Tripura

Tripura witnessed a surge in terrorist activities in 1990s. New Delhi blamed Bangladesh for providing a safe heaven to insurgents from its territory.

Manipur

In Manipur, militants formed an organisation known as *People's Liberation Army.* Their main goal was to unite the Meitei tribe of

Myanmar (Burma) and establish an independent state of Manipur. However, the movement was thought to have been suppressed after fierce clash with the Indian army in the mid 1990s.

Mizoram

The *Mizo National Front* fought for over two decades with the Indian military in an effort to gain independence. As in neighbouring states, the insurgency was quelled by force.

South India

Andhra Pradesh

The terrorism in Andhra Pradesh stems from the *People's War Group* or *PWG*, popularly known as Naxalites. This group has been operating for more than two decades with most operations in Telangana region of Andhra Pradesh. The group is also active in Chattisgarh, Jharkhand, and Odisha. The ideological extremists aim to create equality in the society by attacking the rich and powerful landlords. The group targets Indian police, multinational companies, and other influential institutions in the name of rights of landless labour. It is believed they have links with the Maoists in Nepal and LTTE of Sri Lanka.

Chhattisgarh

The state of Chhattisgarh is infested with Maoist. The people's Liberation Guerrilla Army (PLGA), is active in the state and creating serious problems for the civilians, CRPF and other security forces. Between 2013 and 2014 the casualty was significantly high. In July 2010, 76 CRPF men were killed. In May, 2013, the Congress leadership was almost wiped out before the Assembly election. In 2014, the casualty of CRPF was 28 in Naraynpur, 54 Dantewada, 49 in Sukma, 40 in Bijapur, 60 in Kankar and 92 in Kondagaon. These places are the strongholds of Maoist. According to Rangarajan Committee (7.7.2014) almost 40% of the total population of the Chhattisgarh is below the poverty line. The physical infrastructure needs to be strengthened to improve the quality of life. More jobs also

need to be created to overcome the problem of Naxalism in the region.

INDIA'S ROLE IN WORLD AFFAIRS

The Republic of India, the second most populous country and one of the fastest growing economies in the world, is considered as a major power and a potential superpower. It is India's growing international influence that gives it a prominent voice in global affairs.

India has a long history of collaboration with several countries and is considered as a leader of the developing world. India was one of the founding members of the United Nations and the Non-Aligned Movement and is an active member of several international organisations, most notably the WTO, ADB, SAARC, G8+5, East Asian Summit, and G20.

During the Cold War, India adopted a foreign policy of not aligning itself with any major power bloc. However, India developed close ties with the Soviet Union and received extensive military support from the USSR. The end of the Cold War significantly affected Indian foreign policy, as it did for much of the world. The country now seeks to strengthen its diplomatic and economic ties with the United States, the People's Republic of China, European Union, Japan, Israel, Latin America, and Association of Southeast Asian Nations. India also has ties with the African Union, the Commonwealth states and the Arab World. India continues to have a strong military relationship with Russia.

India has actively participated in several UN peacekeeping missions. India is among the largest troop contributors to the UN and currently seeking a permanent seat in the UN Security Council.

India's growing economy, strategic location, friendly foreign policy and large and vibrant diaspora has won it more allies than enemies. India has friendly relations with several countries in the developing world. Though India is not a part of any major military alliance, it has close strategic and military relationship with most of the major powers.

Countries considered India's closest allies include the Russian Federation, Israel, Bhutan, Nepal, and Tajikistan. After the collapse of the Soviet Union and economic liberalisation in 1992, India has fostered a close relationship with the United States, the United Kingdom, Germany, France, and Japan. India's military and economic collaboration with the United States, Japan, and Israel has grown significantly in the past few years. India has also forged relationships with developing countries, especially South Africa, Brazil, Mexico, and the Republic of China. India, along with Mexico, Brazil, and South Africa, often represents the interests of the developing countries through economic forums such as the G8+5, IBSA, G4, and WTO. India's decade old '*Look East*' policy has helped it to develop greater economic and strategic partnership with South East Asian countries, South Korea, China, Japan, and Taiwan. India also enjoys friendly relations with Persian Gulf countries and the African Union.

On June 2-3, 2017, the Prime Minister, of India and the President of France discussed the 'The Paris Climate Accord' and the Climate issue, in an attempt to save the environment and global terrorism. The two countries jointly launched the 'International Solar Alliance' whose headquarter is in New Delhi. France and India have also developed their bilateral cooperation on energy storage, e-mobility and sustainable development. In India, attention is being given to climate modeling for a better understanding of climate.

The first trilateral meeting between India, Iran and Afghanistan of the Coordination Council of Chabahar Agreement was held in Tehran in 2016. It entailed the transit and transport corridor among the three nations.

India and U.S. signed Communications Compatibility and Security agreement in order to strengthen the bilateral military partnership in the defence sector. High end secured communication equipment shall be installed on the military platforms (*C-130 J, C-17, P-81, Apache and Chinook*).

India and China signed an agreement on May 9, 2019, with the purpose of MoU being SPS protocol for export of Indian Chilli meal to

China.

Sri Lanka, Japan and India jointly signed an agreement on May 28, 2019, to develop the East Container Terminal at the Colombo Port.

An agreement has been signed between India and Maldives on May 30, 2019 to develop socio-economic infrastructure in the neighbouring countries in the field of health and sanitation.

Konkan Railways with Nepal signed an agreement on May 10, 2019 to supply two 1600 HP DEMU trains sets.

The Ministry of External Affairs with the National Bank for Agriculture and Rural Development Consultancy Service (NABCONS) on April 2, 2019 shall establish India-Africa Institute of Agriculture and Rural Development in Malawi.

GEOPOLITICS OF SOUTH ASIA

Some of the aspects of India's relations within the subcontinent are conducted through the *South Asian Association for Regional Cooperation* (SAARC). Its members are Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan, and Sri Lanka. Established in 1985, SAARC encourages cooperation in agriculture, rural development, science and technology, culture, health, population control, narcotics control, and anti-terrorism.

SAARC has internationally stressed these 'core issues' and avoided more divisive issues, although political dialogue is often conducted on the margins of SAARC meeting. In 1993, India and its SAARC partners signed an agreement gradually to lower tariffs within the region. Forward movement in SAARC has come to a standstill because of the tension between India and Pakistan, and the SAARC summit originally scheduled for, but not held, November 1999 has not been rescheduled.

In November 1988, at the behest of Maldivian government, Indian paratroopers and naval forces crushed a coup attempt by mercenaries. India's action, viewed by some critics as an indication of Indian ambition to be a regional hegemon, were regarded by the United States, the Soviet Union, Britain, Nepal, and Bangladesh as

legitimate assistance to a friendly government, and to be fully in keeping with India's strategic role in South Asia.

India-Pakistan Relations

The principal source of contention between India and Pakistan has been Kashmir, the Hindu Maharaja of this state, Hari Singh of Dogra, and its Muslim Prime Minister, Sheikh Abdullah, chose in 1947 to join India conditionally and provisionally. A Standstill Agreement with Pakistan was in place, so the matter is still much disputed, since the decision was hastened by the loss of territory to invading irregulars from Pakistan, backed by Pakistan Army. India maintains that this decision was the norm for every other princely state at Independence and that subsequent elections in Kashmir for over 40 years have made it an integral part of India. This dispute triggered wars between the two countries in 1947 and 1965 and a limited conflict in 1999. The state remains divided between the two countries by the Line of Control (LoC), which demarcates the ceasefire line agreed upon in the 1947 conflict. Pakistan refers to its part of the state as Azad Kashmir, India terms it Pakistan Occupied Kashmir (PoK).

India-Nepal Relations

Relations between India and Nepal are close yet fraught with difficulties stemming from geography, economics, the problems inherent in big-power-small power relations, and common ethic and linguistic identities that overlap the two countries' borders. In 1950, New Delhi and Kathmandu initiated their intertwined relationship with the Treaty of Peace and Friendship and accompanying letters that defined security relations between the two countries, and an agreement governing both bilateral trade and trade transiting Indian soil. The 1950 treaty and letters stated that "neither government shall tolerate any threat to the security of the other by a foreign aggressor" and obligated both sides "to inform each other of any serious friction or misunderstanding with any neighouring state likely to cause any breach in the friendly relations subsiting between the two governments." These accords cemented a 'special relationship'

between India and Nepal that granted Nepal preferential economic treatment and provided Nepalese in India the same economic and educational opportunities as Indian citizens.

India-Bangladesh Relations

Since Bangladesh is almost entirely encircled by India via a land border which stretches over 4096 kilometers, its relations with India are vital.These relations furthermore have been subject to the traditional Pakistan-India polarity of the subcontinent.This effect was most marked during the administration of Mujibur Rahman, after whose assassination Pakistan immediately recognised Bangladesh and announced an aid package while India's attitude cooled considerably.

The border of India and Bangladesh was one of a number of disputes that arose as a result of the partition of India. The partion of India had left a poorly demarcated international border between the two neighbours. The ownership of several villages on both sides of the border are disputed and claimed by both the countries.

According to the Bangladesh government, ever since the liberation of Bangladesh (erstwhile East Pakistan) by Indian forces in 1971. Indian forces have occupied a small silver of land along their border near the village of Beerdivah. On April 18, troops of Bangladesh Rifles (border troops) occupied the disputed village. In the fighting that followed, Bangladesh claims that 15 members of India's border forces. In Border Security Forces (BSF) were killed, along with several of the attacking soldiers. However, India claims a far lower death toll.

In recent years India has increasingly complained that Bangladesh does not maintain its border properly. It fears an increasing flow of economically depressed Bangladeshis and it accuses Bangladesh for harbouring Indian separatist group like ULFA and alleged terrorist groups.

But recent relations are not without bright spots. In May 2007, they announced that for the first time since the 1965 Indo-Pak War, rail service between Kolkata and Dhaka will be restored, after a lapse 42

years. Moreover, in 2007 some cooperation was announced on the border issues mentioned above. These signs of cooperation coincide for the moment at least with a change in Bangladeshi leadership that is expected to last until the end of 2008.

INDIA AND THE GEO-POLITICS OF THE INDIAN OCEAN

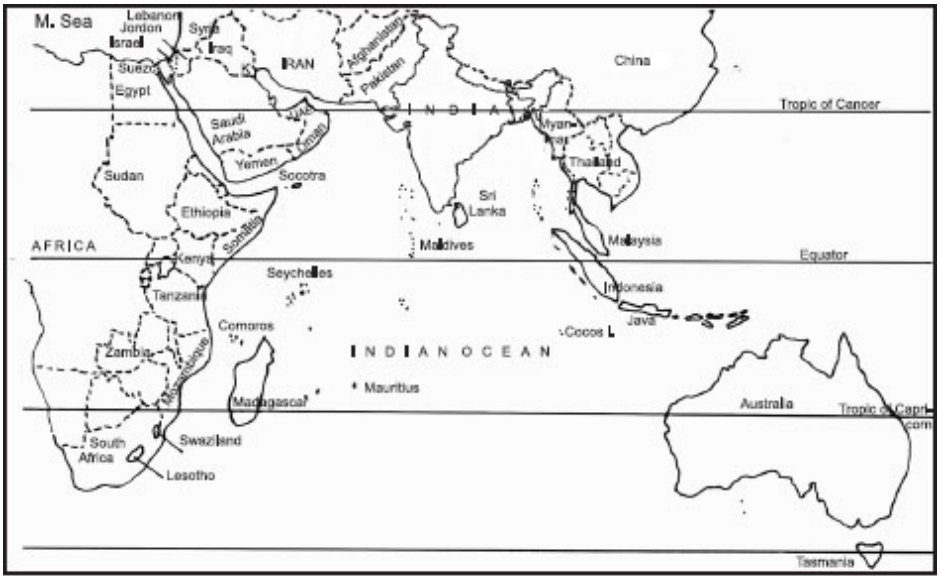
The Indian Ocean is the third largest ocean of the world after the Pacific and the Atlantic oceans. Stretching over an area of 73,556,000 sq km it coves 20% of the earth's water surface. It is bounded on the north by Asia (including the Indian Sub-continent, after which it is named); on the west by Africa; on the east by Indonesia, the Sunda Islands, and Australia and on the south by Southern Ocean. The Indian Ocean is delineated from the Atlantic Ocean by 20° east longitude running south from Cape Agulhas, and from the Pacific Ocean by the 147° east longitude. The ocean is nearly 10,000 kilometres (6200 miles) wide at the southern tips of Africa and Australia.

The International Hydrographic Organisation delimited a fifth world ocean, stripping the southern portions of the Indian Ocean. The new ocean extends from the coast of Antarctica north to 60° south latitude which coincides with the Antarctic Treaty Limit. Thus, the 60° south has been taken as its southern most limit of the Indian Ocean, separating it from the Antarctic Ocean
(Fig. 16.10). The Indian Ocean remains the third largest of the world's five oceans. Pack ice and icebergs are found throughout the year south of 65° south latitude. The average northern limit of icebergs is 45° south latitude.

Being located to the south of India, it has great geo-political, socio-cultural and economic importance. This is the only ocean in the world to be named after the name of a country, i.e. Indian Ocean after India. The Indian Ocean has immense strategic importance for India. The landlocked nature of the Indian Ocean has given India a commanding position. The strategic importance of this ocean is further enhanced

by the fact that it is accessible from the west and the east through the narrow straits only. The Red Sea is the narrow outlet in the west, while the Strait of Malacca and Timor and Arafura seas are the narrow outlets in the east. Through the Suez Canal, the Indian Ocean is connected with the Mediterranean Sea and the Europe. Before the opening of the Suez Canal in 1869, India was connected with Europe and America via the Cape of Good Hope (South Africa). The Indian Ocean can be choked any time by controlling these outlets. The major choke points include Bab-al-Mandeb, Strait of Hormuz, Strait of Malacca, southern access to the Suez Canal, and the Lombok Strait.

The ocean is encircled by 47 littoral countries, 7 island countries (Madagascar-the world's fourth largest island, Comoros, Seychelles, Maldives, Mauritius, Singapore, and Sri Lanka) and 13 landlocked countries (Afghanistan, Bhutan, Burundi, Ethiopia, Laos, Lesotho, Malawi, Nepal, Rwanda, Swaziland, Uganda, Zambia, and Zimbabwe) who are the members of the UNO. All these countries are traditionally and economically associated with the Indian Ocean. These countries have about 33% (2 billion) population of the world. **(Fig. 16.11).**



Map not to Scale

Fig. 16.11 Indian Ocean and Adjacent Countries

The Indian Ocean consists of marginal seas of Andaman Sea, Arabian Sea, Arafura Sea, Lakshadweep Sea, Malagassi Sea, Red Sea, Sawu Sea, and Timor Sea. The main gulfs of the Indian Ocean are: Bay of Bengal, Great Australian Bight, Gulf of Aden, Gulf of Carpentaria, Gulf of Oman, Gulf of Martaban, Persian Gulf, and Gulf of Spencer. The important straits are: Bass Strait, Malacca Strait, Mozambique Channel, Palk Strait, Singapore Strait, Selat-Sunda Strait, and Tore Strait. In comparison to other oceans, it has the largest number of marginal seas-touching warm water.

The Oceanic ridges in the Indian ocean are Socotra Ridge, Chagos Ridge, Gavssberg Ridge, Madagascar Ridge, Seychelles Ridge, St. Paul Ridge, 90° Ridge, Kerguelen Plateau, Prince Edward Island (Ridge), and Seychelles-Mauritius Ridge.

Economic Significance

The economic importance of the Indian Ocean may be appreciated from the following facts:

1. Agricultural Products

The littoral countries of the Indian Ocean are the leading producers of some of the important cereal and cash crops. The Indian Ocean countries produce 77% of rubber, 76% of tea, 60% of dates, 55% of cashew-nuts, 45% of wool, 27% of cotton, and 20% of coffee of the world. There is much demand of these cash crops in Europe, America, and Japan. Moreover, Rice, wheat, maize, millets, pulses, dates, coconut, arecanut, oilseeds, sugarcane, are the other important crops grown in the littoral countries of the Indian Ocean **(Fig. 16.12).** Countries like Indonesia, Malaysia, Myanmar, and Thailand are among the leading exporters of heavy wood.

INDIAN OCEAN: AGRICULTURAL RESOURCES

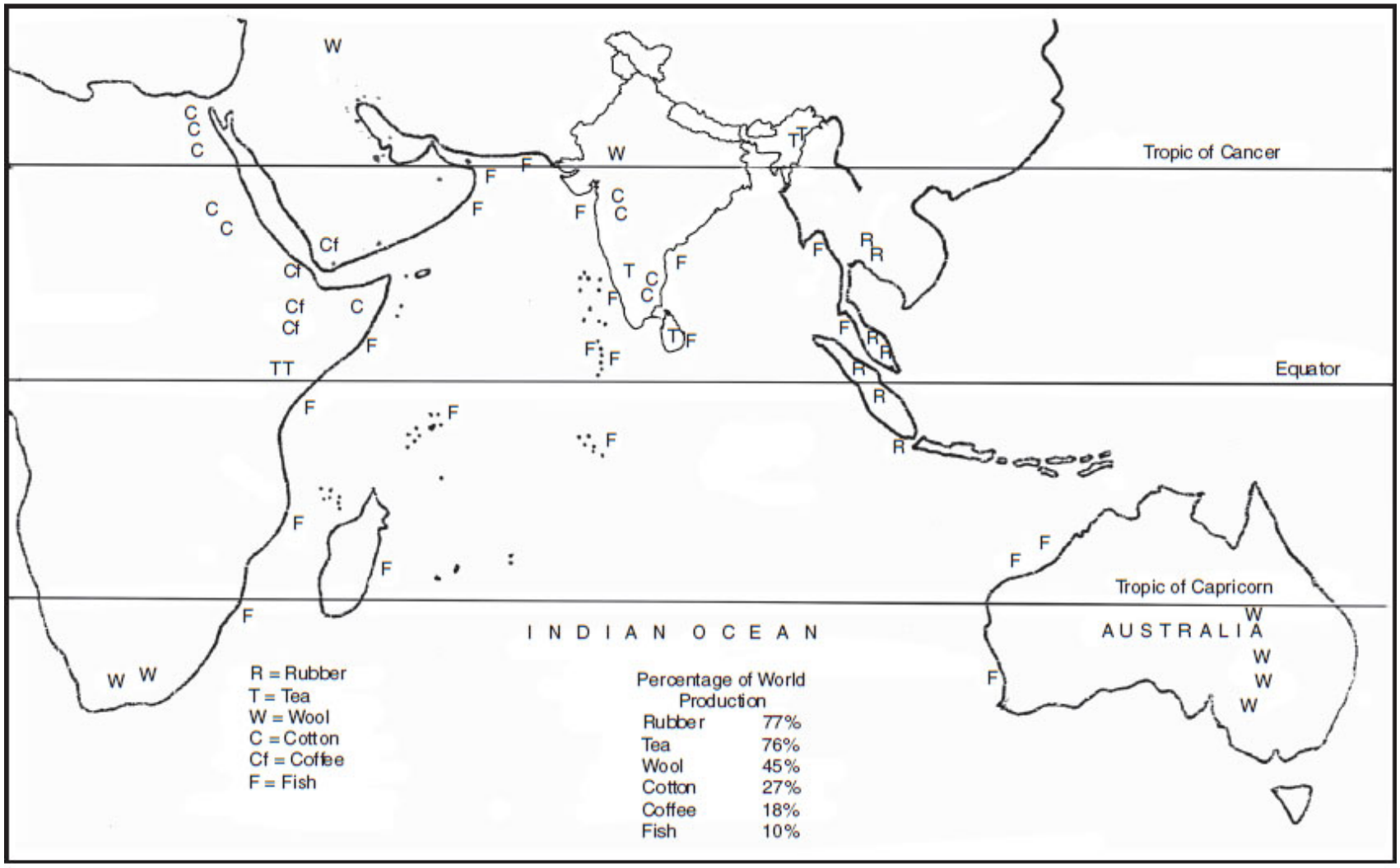


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Fig. 16.12 Indian Ocean and Agricultural Products in Adjacent Countries

2. Fisheries

The continental shelf of the Indian Ocean is rich in sea-food, fisheries, especially in shrimps and tuna fish which have great regional and international demand. Its fish are of great and growing importance to the bordering countries for domestic consumption and export. Fishing fleets from Japan, Russia, South Korea, and Taiwan also exploit the Indian Ocean, mainly for shrimp and tuna.

3. Minerals

The Indian Ocean is quite rich in mineral wealth. Oil and natural gas are the most valuable minerals extracted from the continental shelf of the Indian Ocean. An estimated 40% of the world's offshore oil production comes from the Indian Ocean. The offshore production of oil and natural gas is from the Persian Gulf, the continental shelf of Indonesia, Malaysia, Australia, India and Myanmar. Huge gas deposits are also found in the Persian Gulf along the coasts of Saudi Arabia, Kuwait, Bahrain, United Arab Emirates, Oman, Iran, Iraq, Bombay High, Indonesia, Malaysia, and Australia **(Fig. 16.13)**.

Beach sand rich in heavy minerals, and offshore placer deposits are actively exploited by the bordering countries, particularly, India, South Africa, Indonesia, Sri Lanka, and Thailand.

4. Other Minerals

The littoral countries of the Indian Ocean are rich in some of the important minerals. For example, 82% of gold, 60% of tin, 40% of antimony, 35% of copper, 30% of iron ore, 30% of manganese, 25% of nickel, 22% of mica, 20% of bauxite, and 18% of lead of the world are produced in the littoral countries of the Indian Ocean.

5. Heavy Minerals

The Indian Ocean is rich in beach sand. The sand is rich in heavy-minerals, and offshore placer deposits which are being actively exploited by the bordering countries, particularly India, Indonesia, Malaysia, South Africa, Sri Lanka, and Thailand. In addition to these

minerals, there are enormous deposits of placers, polymetallic nodules, manganese nodules, and metalliferous nodules.

6. Non-Conventional Source of Energy

There is a great scope for harnessing the non-conventional sources of energy in the various parts of the Indian Ocean.

7. Trade Routes

The Indian Ocean provides major sea routes connecting the Middle East, Africa, and East Asia with Europe and Americas. It carries particularly heavy traffic of petroleum and petroleum products from the oil-fields of the Persian Gulf and Indonesia. The Indian Ocean is far calmer and was thus, opened to trade earlier than the Atlantic Ocean and the Pacific Ocean.

8. Tourism

The sea beaches of the Indian Ocean are the attractions for the European, American, Australian and Japanese tourists.

9. Cradle of Ancient Civilisations

The world’s earliest civilisations in Egypt, Mesopotamia, and Indus had all developed around the Indian Ocean.

INDIAN OCEAN MINERAL WEALTH

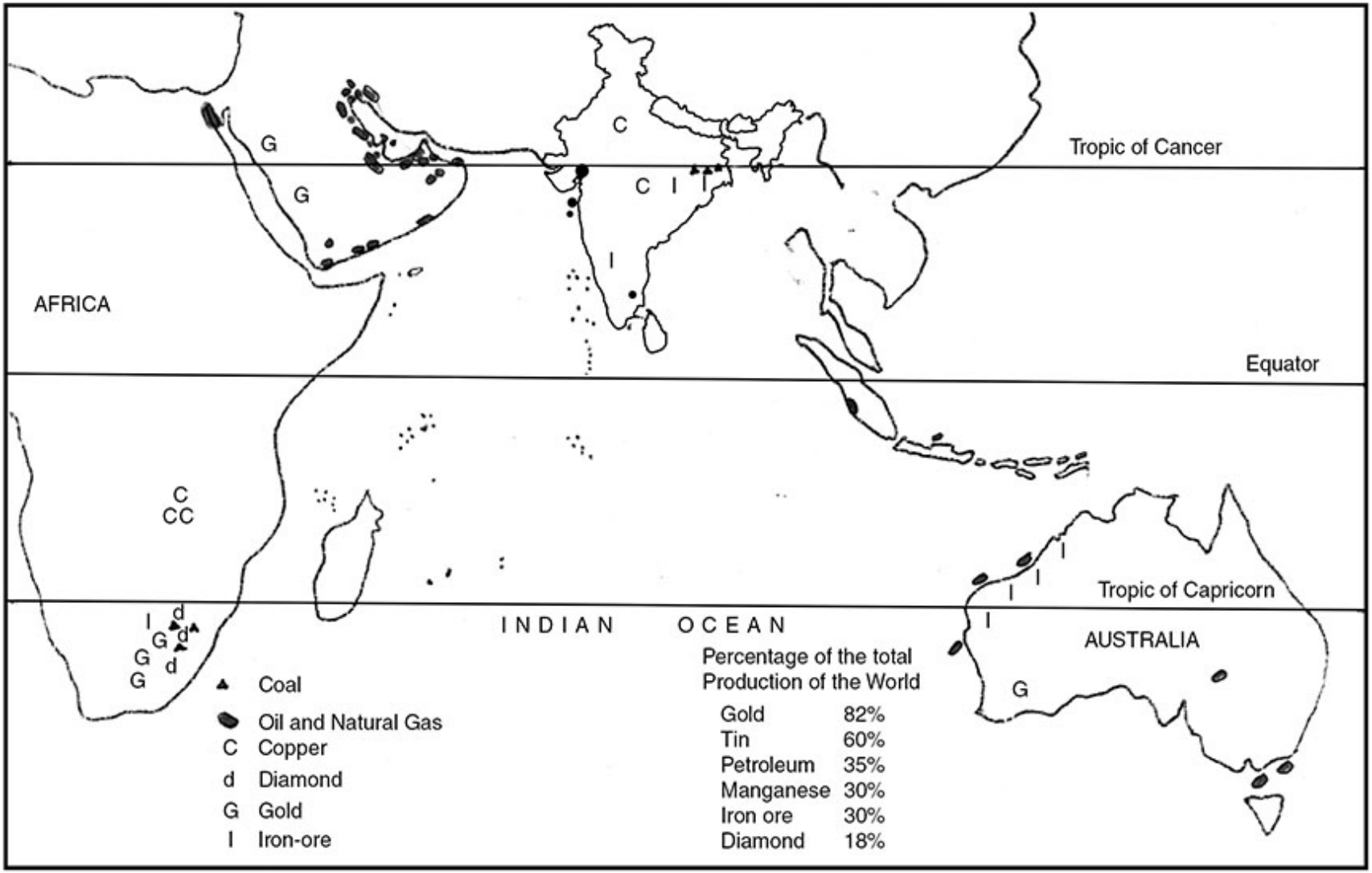


Fig. 16.13 Indian Ocean—Mineral Resources in Littoral Countries

History

The earliest civilisations of the world like Mesopotamia (beginning with Sumer), ancient Egypt, and the Indian subcontinent (beginning with the Indus Valley Civilisation), which began along the valleys of the Tigris-Euphrates, Nile, and Indus rivers respectively, developed around the Indian Ocean. Civilisations soon arose in Persia (beginning with Elm) and later in South-east Asia (beginning with Funan).

In the second or first BC, Eudoxus was the first Greek to cross the Indian Ocean. Hippalus is said to have discovered the direct route from Arabia to India around this time. During the first and second centuries, intensive trade relations developed between Roman Egypt and Tamil kingdoms of Cheras, Cholas, and Pandyas in Southern India. Like the Indonesian people above, the western sailors used the monsoon to cross the ocean.

From 1405 to 1433, Admiral Zheng led large fleets of the Ming Dynasty on several voyages to the Western Ocean (Chinese name of the Indian Ocean) and reached the coastal country of East Africa.

In 1498, Vasco da Gama rounded the Cape of Good Hope and became the first European to sail to India. The European ships, armed with heavy cannon, quickly dominated trade. Portugal at first attempted to achieve pre-eminence by setting up forts at the important straits and ports, but in the 17^{th} century they were replaced by the other European powers. The Dutch East India Company (1602–1798) sought control of trade with the East across the Indian Ocean. Eventually, Britain became the principal power and by 1815 dominated the area.

The opening of the Suez Canal in 1869 revived European interest in the East, but no nation was successful in establishing trade dominance. The commanding position of the British over the region made the Indian Ocean as the *British Lake*. K.M. Panikkar's geo-

political axiom that 'one who rules the sea, rules the adjoining land' became true.

After the Second World War, the littoral states of the Indian Ocean started becoming independent and the British power started declining. British practically lost all their colonies by mid-1960s. On January 16, 1968, Prime Minister Harold Wilson announced the withdrawal of its forces from the east of the Suez to be completed by the end of 1971. British withdrawal from the India Ocean created '*power vacuum'* in the area and made room for super power rivalry in the region. The United States stepped in to fill up this power vacuum and used the opportunity to contain Soviet Union and to safeguard its economic and political interests in the region.

In order to fulfil its ambitions, the United States purchased Diego Garcia Island (Chagos Archipelago) from Great Britain to build a strong military base. The aims of U.S.A. to enter into the Indian Ocean were:

(i) to keep the drilling and transport of oil by pipe-lines operative.
(ii) to check the spread of Communism in Asia and Africa.
(iii) to control trade in the Indian Ocean.
(iv) to influence the politics of the littoral countries.
(v) to pressurise the states around Indian Ocean on various International issues.

The '*Diego Garcia Island*' is about 730 km south of Maldives, about 2,225 km from Kanniyakumari (India) and about 2,300 km north of Mauritius. It is a shallow, horseshoe shaped atoll. This base is fully equipped with nuclear weapons and provides decisive advantage to the United States to command the areas of South-west Asia, Central Asia, South Asia, South-east Asia, China, and Russia.

In addition to Diego Garcia, the United States of America has established its strategic bases at Asmara (Eritrea), Woomera and Harsld E. Holt (Australia), Bahrain (UAE), Doha (Qatar), Dammam, Jubail (Saudi Arabia), Mahe (Seychelles). Similarly Vacaaos (Mauritius), Gan Island, Masirah Island, Simonstown (South Africa), and Port Louis (Seychelles) military bases are jointly owned by USA

and UK. The French defence installations are at Djibouti, Reunion, Diego Suareg in and around the Indian Ocean **(Fig. 16.14).**

The tremendous increase of American naval presence in the Indian Ocean led to the counter presence of the Russian fleet. Soviet Union wanted to have control over some of the littoral states for defence purposes. In March 1968, just two months after the announcement to withdraw its forces from east of the Suez, it sent a naval squadron of four warships and two auxiliaries from the Pacific Fleet at Valadivostok. Since then it is maintaining a few surface warships and support ships in the region. Recent reports say that there are 35 Russian warships in the region. Russia has made diplomatic moves and has entered into bilateral treaties with several countries of the region, like Mauritius, Somalia, Seychelles, Sri Lanka, India, Bangladesh, and Egypt. It maintains military bases at Berbera (Somalia), Masirah (Oman), Umakas, and Dahalak (Red Sea) Islands.

After the disintegration of Soviet Union, China has also started taking keen. interest in the Indian Ocean. With the disintegration of the Soviet Union, China is trying to fill in the vacuum and assume leadership of the Afro-Asian world. In recent years China has built naval bases in Kyaukpyu and Cacos Islands (Myanmar), Hambantota and Colombo (Sri Lanka) and Gwadar (Pakistan). These naval bases are known as string of Pearls **(Fig. 16.15)**.

Response of the Bordering Countries

As stated at the outset, there are 47 littoral countries, 7 island countries and 13 landlocked countries in the Indian Ocean. All these countries are, historically, economically and culturally, associated with the Indian Ocean. Moreover, there are 13 landlocked countries which are also included into this ocean as their trade is carried on through the Indian Ocean.

With the growing military build up and big power rivalries the littoral states are greatly concerned about their political stability and economic development. All the littoral states want the Indian Ocean to remain a '*zone of peace'*. This is imperative considering the socio-

economic milieu of these littoral countries as they are in the process of economic development, social transformation and building the democratic system. All of them, except Australia, belong to the non-aligned group of the world and therefore, they do not wish to see any power to interfere in their domestic affairs. There is an emerging opinion among these states that the socio-economic prosperity in the region will only usher in when there is an atmosphere of economic cooperation and coordination among the littoral units of the richly endowed region of the Indian Ocean, which could be feasible when the external powers do not exist there.

The first conference of the Heads of the littoral states of the Indian Ocean was held at Cairo in 1964. Later, the Third Conference of the Non-aligned countries held at Lusaka (Zambia) in 1970 called upon all states to identify the Indian Ocean as a zone of peace. Subsequently, in September 1971, the foreign ministers of 54 non-aligned nations resolved that "concrete steps should be taken at the 26th session of the General Assembly to implement the decision relating to the declaration of the Indian Ocean as a *'zone of peace'."* It was further strengthened by the UN Resolution (2832, 16 December, 1971) of the 26th General Assembly Session, stating that "the Indian Ocean within limits to be determined, together with the air-space above and the ocean floor subjacent thereto … for all time as a zone of peace." Such resolutions were successfully passed and adopted in the UN General Assembly.

INDIAN OCEAN MINERAL WEALTH

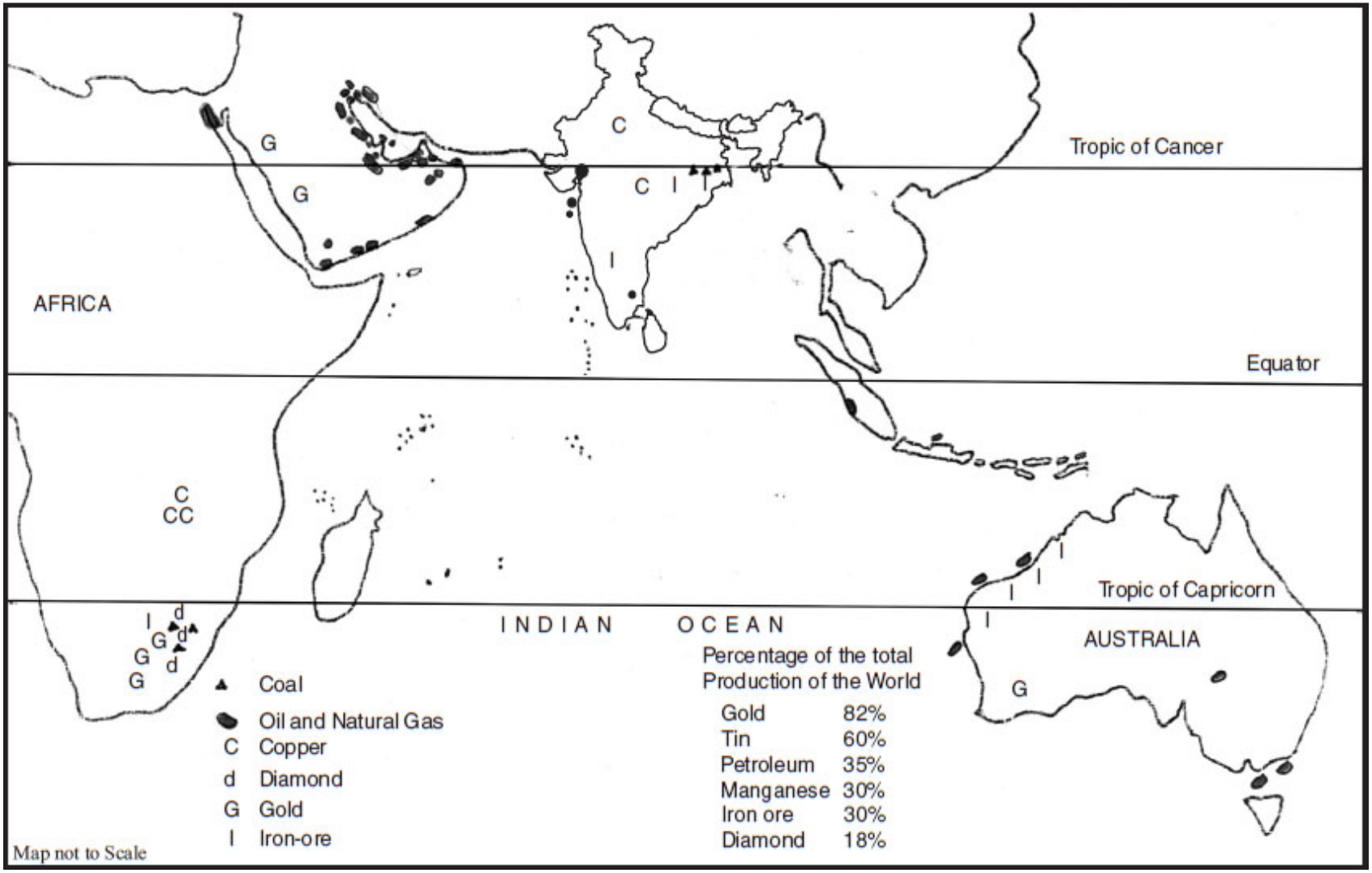
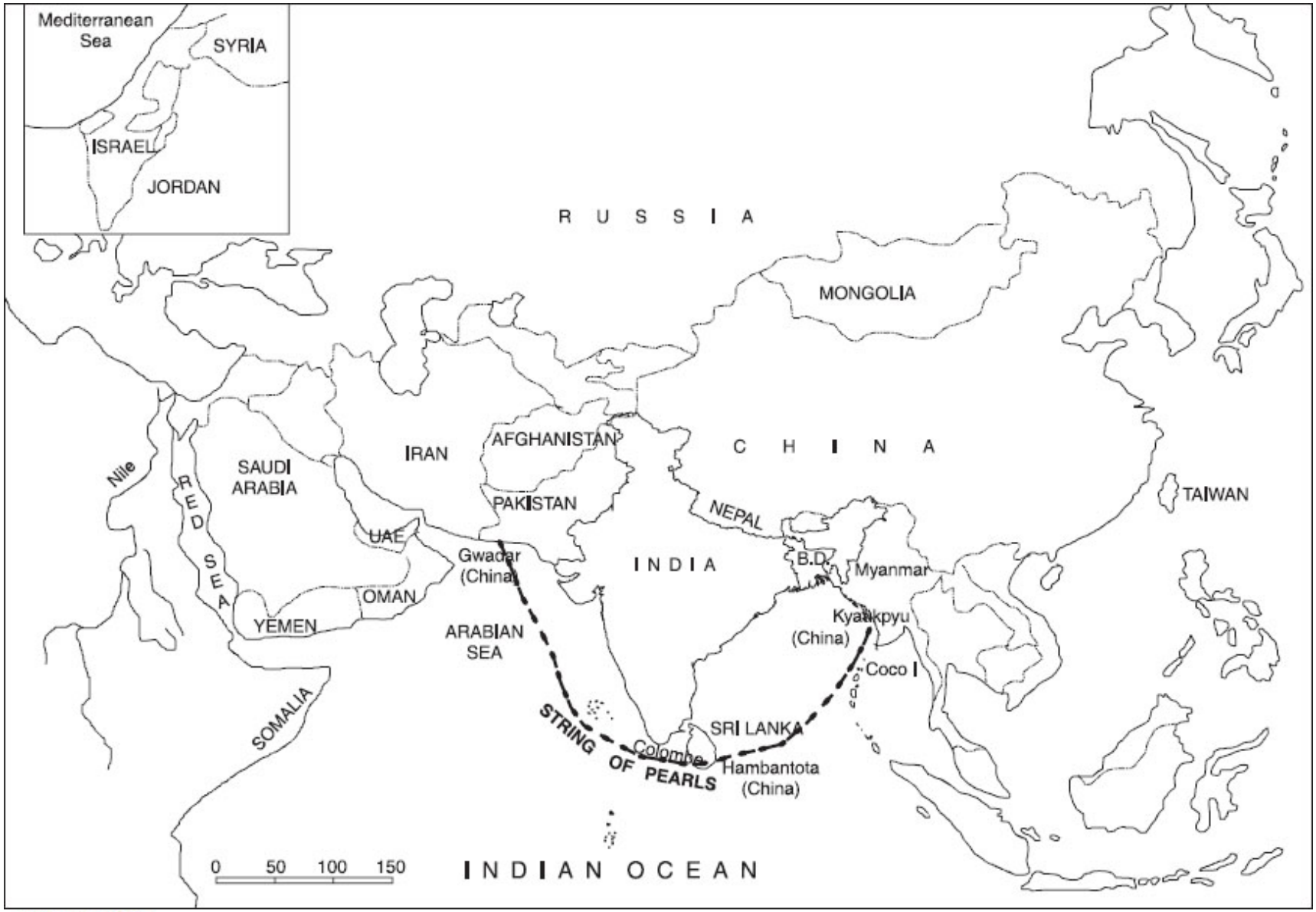


Fig. 16.14 Indian Ocean—Strategic Locations



Map not to Scale

Fig. 16.15 China's Ring of Pearls in Indian Ocean

The littoral states not only desire to stop the escalation and expansion of military presence, termination of all manifestations of super-power rivalry in the region but to impose voluntary restrictions of various kinds for the littoral powers of the region also.

There are, however, many obstacles in achieving the objective of making the Indian Ocean free from the power rivalry and international conflict. Majority of the littoral states are either small or economically poor and politically in the youth stage. There a few littoral regional powers like Australia, India, Indonesia, Iran, and South Africa which are capable of influencing the events at the regional as well as international levels. There is too much diversity among the littoral states and many of them do not have peaceful cordial relations with their neighbours. Even large littoral states have own economic interests. With the growing activity of China and Japan in the Indian Ocean, the complexity has been further enhanced. China wants to establish itself as a major power in the Afro-Asian region. Similarly, Japan has economic interests in the region because 60% of its trade is carried through the Indian Ocean. After the American war in Iraq, the Indian Ocean has assumed paramount importance in the geo-politics of the world.

Thus, there are external as well regional forces which have been working in the Indian Ocean region and are responsible for bringing this region in the focus of world politics. There is a need to bring more economic cooperation among the littoral states so as to avoid their exploitation by the erstwhile colonial masters and the developed countries. A regional organisation like the Indian Ocean Community is the need of the day. The sooner it is done the better.

India and the Geopolitics of the Indian Ocean

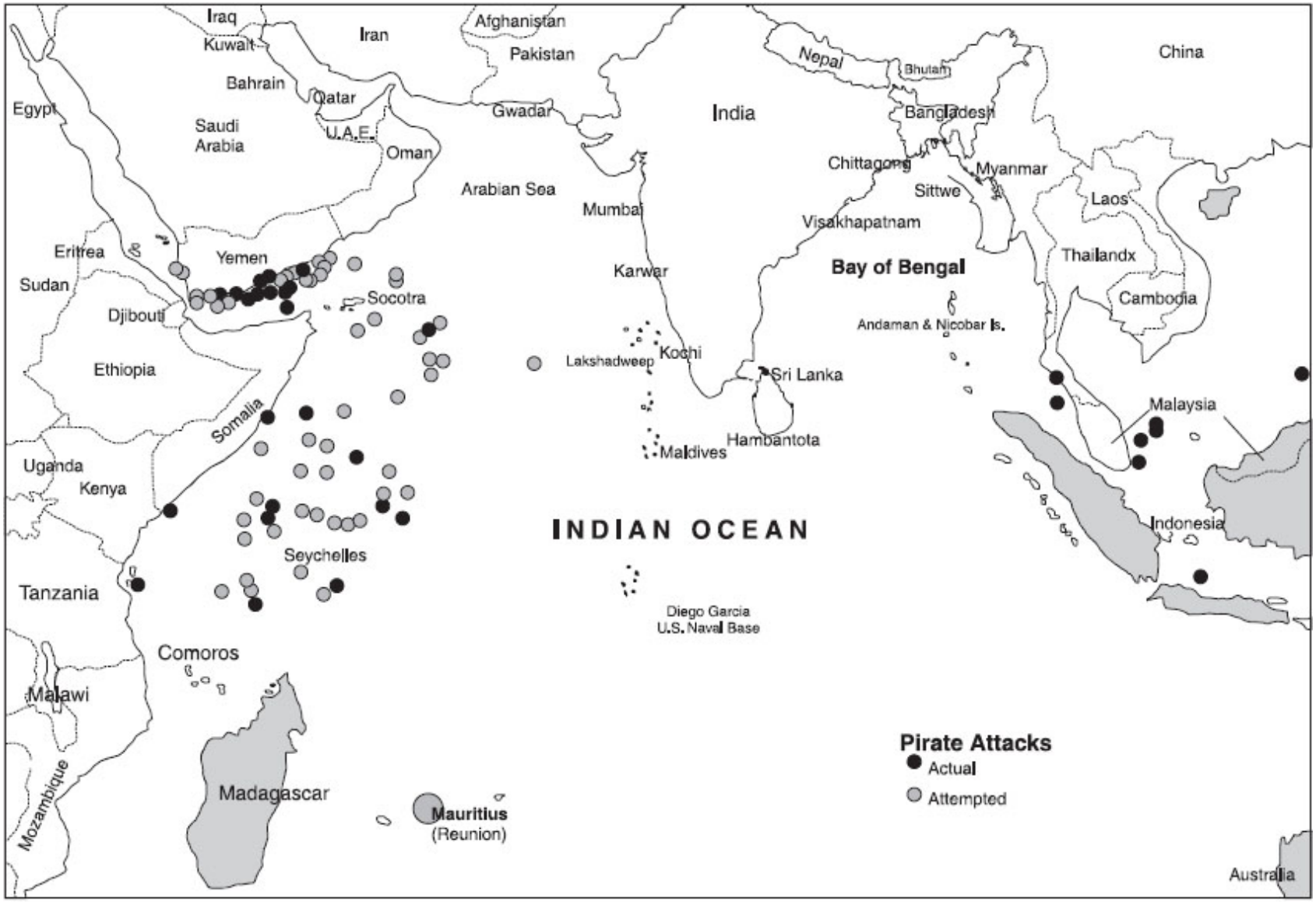
- The Indian Ocean is named after a country. It is because in the past, during the ancient and medieval periods, India had a dominant position in the region.
- The Indian culture spread to Southeast Asia and African

countries through the Indian Ocean.

- Islands like Mauritius, Chagos, Seychelles, etc. have a dominance of population of the Indian origin.
- The continental Shelf of Mumbai High, Gulf of Khambat, and Godawari-Krishna deltas are rich in petroleum, natural gas, fisheries, and biodiversity.
- The coral formations along the coast of India have great ecological and economic significance.
- Over 98% of the international trade of India is carried through the Indian Ocean.
- About 60% of India's petroleum requirements are imported through the Indian Ocean. Thus Indian Ocean is termed "*Oil-line of India*." Moreover, the bulk of Indian salt is also obtained from the Indian Ocean.
- The potential tidal energy, especially along the Gulf of Khambat, is enormous.
- India occupies an important geographical position is the central arc of the Indian Ocean.
- India has significant economic and technical co-operation programmes in the littoral states of the Indian ocean. India has been providing technical training facilities, Indian expertise, and consultancy services to the countries around the Indian Ocean.

Issue of Piracy

In recent years, piracy in Indian Ocean has given a new significance to the geopolitics of the Indian Ocean. The attacks made by the pirates in the last two decades in Indian Ocean have been shown in **(Fig. 16.16).**



Map not to Scale

Fig. 16.16 Pirate Attacks in Indian Ocean

It may be observed from this figure that of attacks by the pirates were mainly made in the Arabian sea, especially in the Gulf of Aden around Somalia, and the triangle between Seychelles, Zanzibar (Tanzania) and Somalia. In February, 2012, two Indian fishermen fell to firing opened by Italian oil-tanker Enrica Lexie, sailing through or near our territorial waters. Italian marines Massimiliano Latorre and Salvatore Girone have been held in India since allegedly shooting to deah Kerala fisherman Valentine and Ajesh Binke.

These developments show that a myriad challenges facing the Indian Ocean Region (IOR) are becoming more complex. Among the challenges piracy deserves priority, being a live issue with wide spread repercussions. The experts agree that piracy can be lynched *"not on the sea but on land*" i.e. by attacking all the vested interests and a long web of players involved in piracy industry as well as rebuilding the state of Somalia.

The Association of South East Asian Nations, the SAARC countries and Indian Ocean Rim Association for Regional Co-operation can foster regional economic and technical cooperation. Moreover, India needs to strengthen its relation with the island countries like Seychelles, Comors, Zanzibar, as well as the east African countries and Yemen. Finally, India needs to deploy the full range of weapons to strengthen its Indian Ocean Region (IOR) identity.

Kolkata-Kunming Corridor-A Proposed Highway

The Kolkata-Kunming project connects China, Myanmar, India and Bangladesh. This is an ambitious project that hopes to connect Kolkata with Kunming, capital of Yunnan Province of China. The project was announced by the Chinese President XI Jinping-led regime in 2013. It is also called the ' New Silk Road Economic Belt.

The main objective of this corridor is to form a thriving economic belt with focus on cross border transport, energy, tourism and communication network. It was also visualised that this corridor could play an important role in 'global economic recovery'. The project has

acquired a fresh momentum under the theme of sub-regional co-operation.

The route of this corridor passes through nodal points, such as Mandaly, Lashio and Kalewa in Myanmar. It heads towards Kolkata after passing through Imphal (Manipur) and Silchar (Assam), before crossing Bangladesh via Sylhet and Dhaka, with branches extending to the ports of Cox's-Bazar and Chittagong **(Fig. 16.17)**.

Dr. Ren pointed out that ethnic insurgencies, terrorism, communal violence (involving Buddhists and Muslims in Myanmar) drug-trafficking and the accompanying spread of HIV infections, antiques smuggling as well as cross-border human trafficking, threatened to derail the project.

The Leading Chinese scholars have proposed setting up a security mechanism and accelerating a legal dialogue among Bangladesh, China, India and Myanmar. Ren Jia, President of the Yunnan Academy of Social Sciences, advocated a four-country institutional tie-up to address have non-traditional security threats to the Bangladesh, China, India, Myanmar project.

Chinese experts in Yunnan say that except for a 200 km stretch between Silchar (Assam) and Manipur, and the similar length between Kalewa and Monywah in Myanmar, the central artery of the corridor is nearly functional. Some experts in India are of the opinion that it is a major strategic initiative by China. It is a project to encircle India and constituting a challenge of a geopolitical type. These strategic concerns seem overblown and derived from the fact that there are significant political issues, the boundary question in particular, that are unresolved between India and China. Despite all these concerns and apprehensions, Dr. Ren pointed out that ethnic insurgencies, terrorism, communal violence (involving Buddhists and Muslims in Myanmar) drug-trafficking and the accompanying spread of HIV infections, antiques smuggling, as well as cross-border human trafficking, threatened to derail the project. China is planning a summit of various countries to take the project forward both bilaterally and multilaterally in 2016.

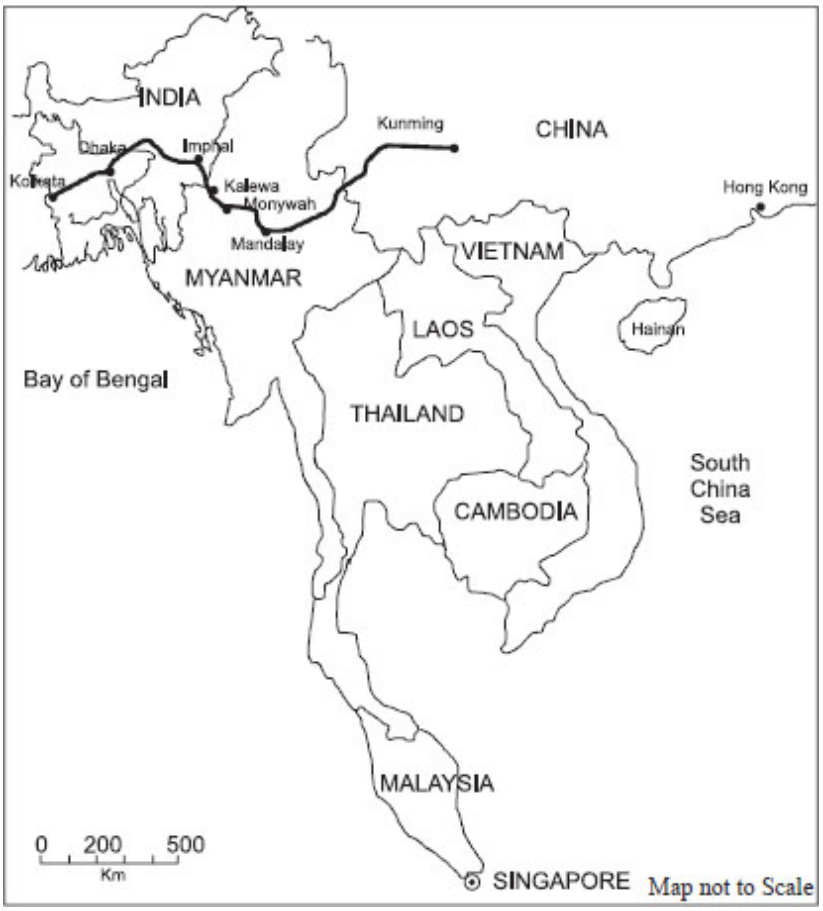


Fig. 16.17 Kolkata, Kunning Corridor–Project Connects China, Myanmar, India and Bangladesh

China Pakistan Economic Corridor

China Pakistan Economic Corridor connects the Kashgar city of Xingiang (Sinkiang) Province (China) with the Gwadar Seaport-Bilochistan (Pakistan). The corrridor passes through the part of Kashmir occupied by Pakistan, Islamabad, Dera Ismail Khan, Quetta, Besima and finally terminates at Gwadar–a seaport along the coast of Bilochistan. (Fig. 16.18).

The Karakoram Highway road will connect China with Pakistan. The 3,218 km long route is likely to cost over $75 billion and expected to be completed by 2020. There will be placement of fibre-optic line to ensure better communication between the two countries. The China-Pakistan Economic Corridor is said to be the biggest overseas

investment of China. By 2017, completion of Gwadar International Airport and major development of Gwadar Seaport is expected.

Gains to Pakistan: The Pakistani government and the country's press are calling China Pakistan Economic Corridor (CPEC) a 'game changer', with many analysts saying $46 billion Chinese investment will make Pakistan the next Asian Tiger. The main gains likely to Pakistan are briefly given below:

(i) Islamabad believes that if all projects are implemented, value will exceed all FDI in the country since 1970. (ii) Pakistan hopes to create over 7 lakh jobs between 2015 and 2030 and add up to 2.5 percentage points to its growth rate. (iii) As China's vital energy supply route will pass through Pakistan, China will be compelled to support Pakistan's security. (iv) Chinese naval assets at Gwadar will check India's aspirations to dominate the Arabian Sea. After the lease period, it would be Pakistan's second naval base.

Gains to China: The expectations of China from the China Pakistan Economic Corridor are as under: (i) The corridor gives Beijing a locational advantage to compete with major Middle Eastern ports. (ii) The corridor will help Beijing to reach destinations in Europe via West Asia in 10 days instead of the 45 days it takes now via the Strait of Malacca. (iii) Actual sea routes from the Persian Gulf: 12,900 km to Beijing will be reduced to only 2,000 km. (iv) The alternate trade route will help to counter US's purported 'Contain China Policy'. (v) The corridor will help in upliftment of Xinjiang economy-inhabited by about ten million Uighur (Muslims).

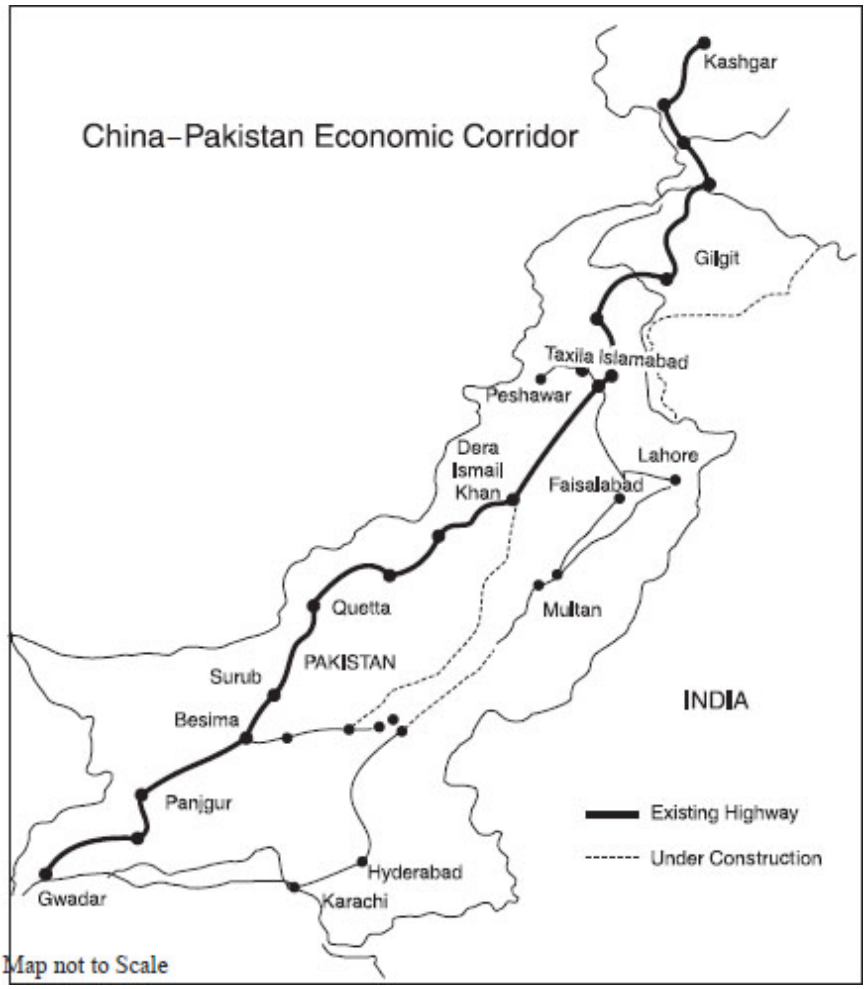


Fig. 16.18 China–Pakistan Economic Corridor

Despite all these merits, the increasing cost of security of the corridor is becoming a big problem in efficiently pushing forward and implementing the project. The economic corridor passes through some turbulent regions including Kashmir. There are many challenges such as a complex environment. It is because of these problems that China wanted the establishment of a separate ministry or authority that would focus exclusively on the China-Pakistan Economic Corridor.

India's Stance: New Delhi has objected to China Pakistan Economic Corridor (CPEC) on point of principle, Saying its projects are located and/or pass through Pakistan Occupied Kashmir (Pok) which belongs to India.

One Belt-One Road (The New Silk Route)

The ambitious project based on $5 trillion, of 'One Belt One Road (OBOR)' has been designed by the Chinese government under the Chinese President Xi Jinpings. It is an 'umbrella of trade initiatives' some of which are already under way. Bringing together ongoing and planned and future infra projects, it includes the 'China-Pakistan Economic Corridor (CPEC), and Bangladesh, China, India-Myanmar Corridor, which India opposes. India is strongly opposed to the China Pakistan Economic Corridor (CPEC) on the ground that Occupied Kashmir, a key chunk of OBOR is an integral part of India. China however, organized a meeting in Beijing on May 14–15, 2017, about the One Belt-One Road (OBOR). The main objectives of the One Belt One Road project are:

1. To develop a land route (old Silk Road) to connect 65 countries of Asia, Africa and Europe (Fig. 16.19).
2. To connect the 4.4 billion of the world's population.
3. To cover about 33% of the world economy.
4. To provide a land route from Inner China to southern Europe via the Netherlands.
5. To provide a sea route connecting Shanghai Port with the endpoint of the land-based route in Venice, via Indonesia, Malaysia, India and Africa.
6. To develop ports, railways, airways, airports, power plants, oil and natural gas pipe-lines and refineries.
7. To develop free trade zones, and in infra for IT and telecom and fintech.
8. The implementation could take at least 30 to 40 years.

The key Belt and Road Projects: Following are the main roads and railways of the projects.

1. Moscow-Kazan High Speed Railway.
2. Khorgos Gateway: Cargo-hub on China-Kazakhstan border to handle one million containers per year .
3. Tehran Rail line: First freight line China to Tehran.
4. China Pakistan Econmic Corridor: Under construction at a cost

of about $46 billion (Economic Corridor) between China and Pakistan which shall link China to port of Gwadar (Baluchistan-Pakistan) on the Arabian Sea.

As stated above, India did not participated in the ‘One Belt One Road’ forum meeting in Beijing held on May 14–15, 2017. India is strongly opposed to the China Pakistan Economic Corridor (CPEC), a key chunk of One Belt One Road (OBOR) that passes through the Occupied Kashmir under Pakistan. India’s main objection is that occupied Kashmir is an integral part of India and a question of sovereignty and territorial integrity. This road cuts down transport time from factories in eastern China to West Asia 12 days to 36 hours.

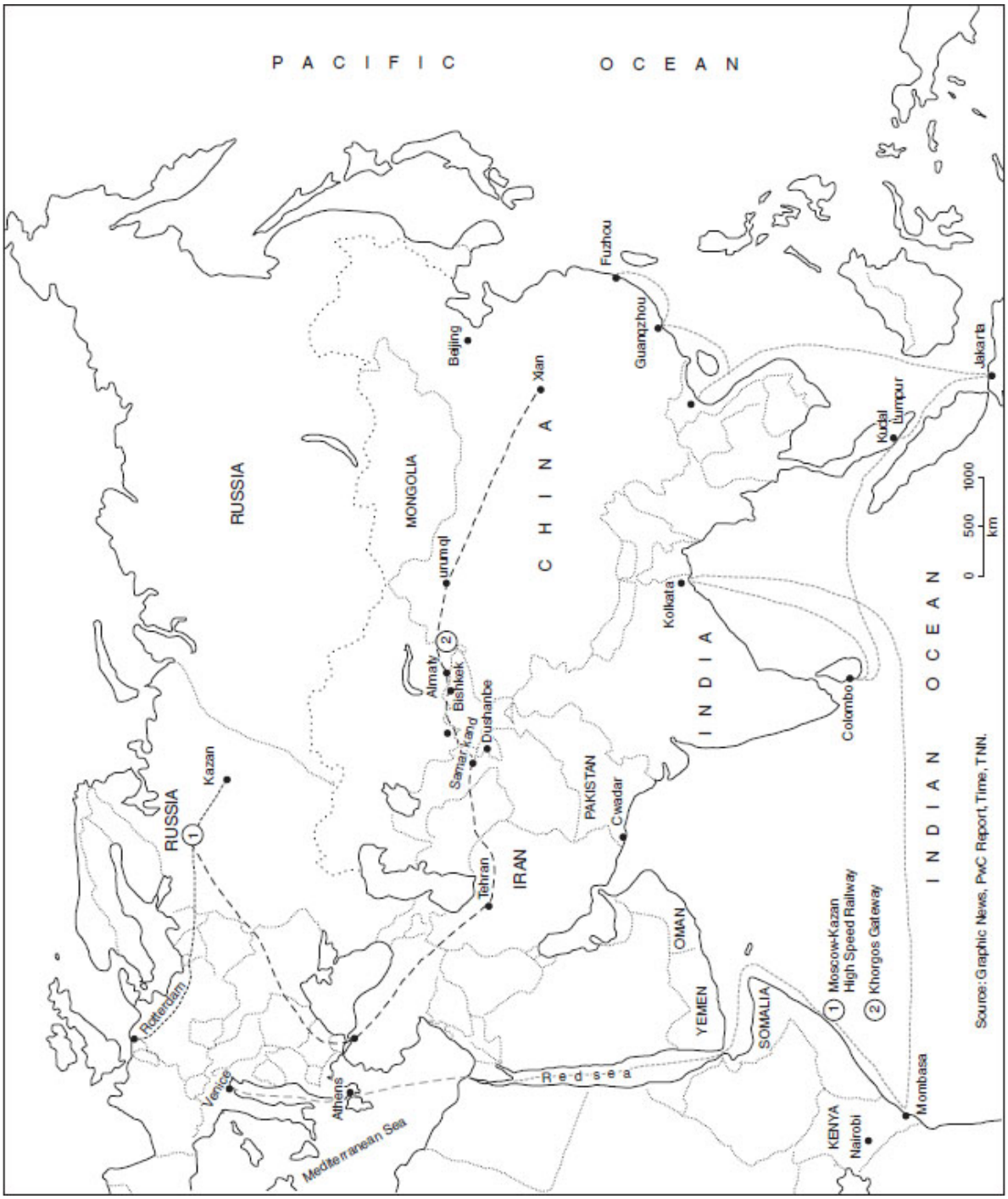
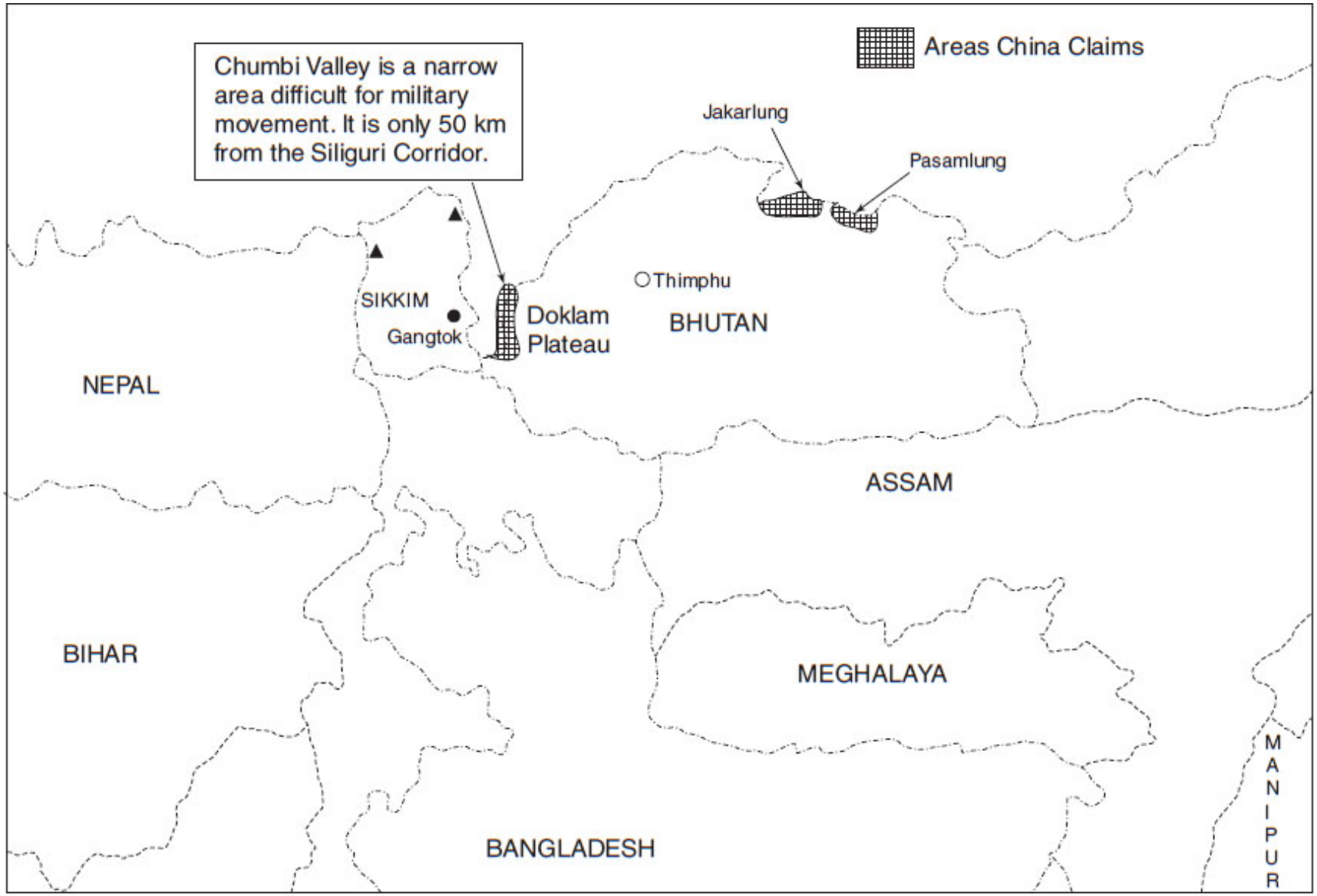


Fig. 16.19 One Belt one Road (The New Sikl Route)



Map not to Scale

Fig. 16.20 Sino-Indian border in Sikkim (after Rajat Pandit, Times of India, 30.6.17).

The sea route is a distance away in the near future, because China is facing opposition from several nations over its activities in the south China Sea. Moreover, questions are being raised about the security of the CPEC too, given it passes through Taliban territory.

Source: Graphic News, PwC Report, Time, TNN.

China's Plan in Bhutan: In the third week of June, 2017, China started the widening its claim in the Chumbi Valley, especially at the Doklam Plateau, which lies at the tri-junction of Sikkim, Bhutan and Tibet. China stressed that Doklam Plateau belongs to China and it is indisputable. According to the Chinese Government, the area where they are constructing road, totally belongs to chinese territory, as this plateau has been the traditional pasture for Tibetan residents.

Though India and China have no dispute along the Sikkkim boundary, China's bid to increase access to India's border via Bhutan has resulted into diplomatic standoff. The eastern parts of the Chumbi Valley, east of the Nathu La (Pass) along the border of Bhutan has emerged as a new flashpoint (Fig. 16.20). In addition to parts of Chumbi Valley, China is claiming the Jakarlung and Pasamlung areas of Bhutan as parts of Tibet-China (Fig. 16.20).

India and China have a 2012 agreement that the boundary points of the tri-junction would be finalized in consultation with all concerned parties (Bhutan, India and China). Thus, this is an unjustified claim of China and not acceptable to Bhutan and India. India has a Friendship Treaty with Bhutan as old as 1865. Bhutan is a sovereign country, but India has to defend its territorial integrity and sovereignty. In opposition to this Bhutan is China's only neighbour with whom it has no diplomatic tries. It maintains contact with China via the Delhi embassy. Unfortunately, ever since entering Tibet in the 1950s, China has been eyeing Nepal, Sikkim and Bhutan. A bilateral talk between India and China is imperative to overcome this boundary dispute problem.

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17

Contemporary Issues

ENVIRONMENTAL HAZARDS

Most of the environmental hazards are natural hazards such as earthquakes, volcanoes, landslides, hurricanes, cyclones, typhoons, tornadoes, lightning, tsunami, floods, droughts, and epidemics. It has been noted that human beings consistently underrate natural hazards; the growth of Bhuj did not halt after the earthquake of 2001, nor, for that matter, after the shocks of 2005 and 2007. Some geographers see this response—or lack of response—as an example of cognitive dissonance. Other hazards are manmade, for example, air, water, soil and noise pollution, oil spills, pesticides, wars and nuclear disasters.

NATURAL DISASTERS

Humans have always dealt with the fury of nature such as earthquakes, volcanic eruptions, cyclones, droughts, floods, and

epidemics. These phenomena affect human habitation over large areas causing loss of lives and property. However, technologically advanced we think of ourselves, the forces of nature, at times, remind us exactly how fragile our existence is on the earth. We often use terms like natural hazards and natural disasters to describe these terrifying forces of nature. Is there any difference between these terms? Let us consider what do they mean.

We use the word hazard to describe the *latent danger* which may be posed by material, event or phenomenon, e.g., a hazardous substance or a hazardous act. Natural hazards, therefore, literally mean those natural happenings, which are potentially dangerous to humans—causing injury, loss of life and property, and disrupting social and economic life. In our context, they include a wide variety of phenomena, e.g. earthquakes, volcanic eruptions, landslides, cyclones, droughts, floods, tsunami, and epidemics. While some of these occurrences are exclusively of natural origin, e.g., volcanic eruptions, folding and faulting of rocks, others are due to a combination of both, natural and human factors, e.g., landslides, floods, and droughts.

The earth surface is unstable and changes with space and time. While some changes on the earth are slow and invisible, others are fast and perceptible. The drift of the continents, and the formation of mountains are slow processes, but a volcanic eruption and an earthquake are sudden events, which bring rapid changes on the earth surface. Any natural occurrence or event that is infrequent and fast enough to threaten life is referred to as a natural hazard. Most of these hazards are periodic, and can be predicted. But some are difficult to predict. Natural hazards may be grouped under the following two major categories:

1. **Geological Hazards**: Earthquakes, volcanoes, tsunami, landslides, and avalanches.
2. **Weather associated hazards**: Cyclones, thunderstorms, droughts, floods, and epidemics.

It has been estimated that the world endures on an average about 100,000 thunderstorms, thousands of floods, hundreds of landslides and earthquakes, and scores of cyclones and volcanic eruptions every year **(Fig. 17.1** and **17.2)**.

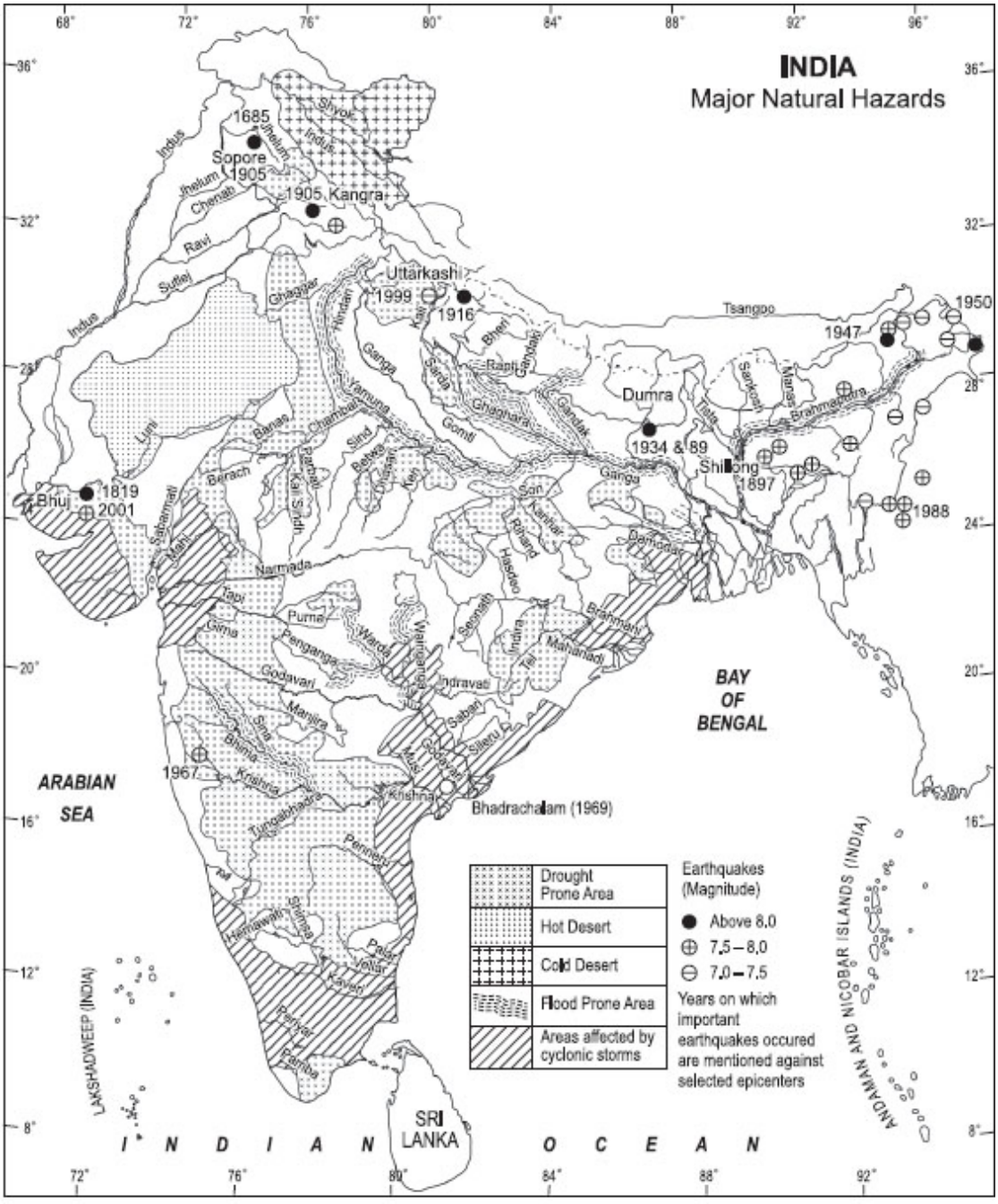


Fig. 17.1 Major Natural Hazards

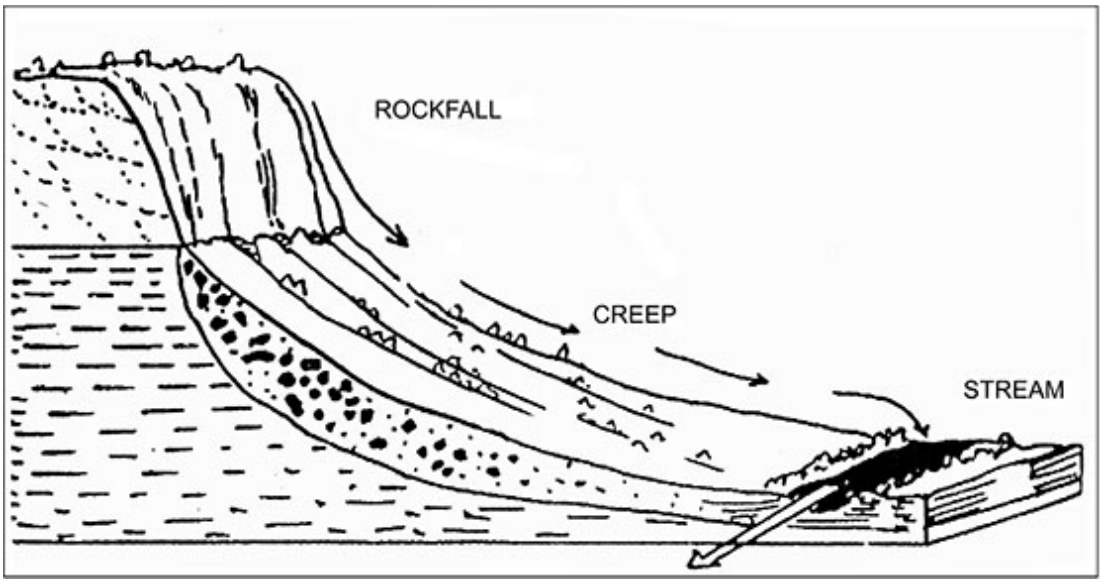


Fig. 17.2 Landslide

All of them are obviously not equally harmful. Depending upon the location of such occurrences and the intensity, the natural hazards become *natural disasters*. For example, in the 20th century, 30 major natural disasters took place in the world. Of these, there were 17 earthquakes, 10 cyclones, two floods, and one volcanic eruption. It is, thus, clear that of many natural hazards, a few create widespread havoc and are, therefore, called disasters.

Humans take risk in life, they also adapt to varying environments. For example, in many developing countries, people live in river-flood-prone plains, abandon it at annual high flow of the river water, and come back after the flood recedes. Normally, the event is cyclic and predictable and hence, adaptation is possible. However, any change in normal pattern, i.e., in the timing of flood or increase in the volume of water, may have a devastating impact on people. Thus this event, which otherwise is a normal annual feature, turns into a disaster.

Most natural disasters are infrequent and unpredictable. Besides, most people are unaware of their causes as well as their potential dangers. Therefore, it is difficult for them to change their behaviour to reduce risk from a hazard that few have ever witnessed. People cannot be expected to modify their habits and behaviour to avoid

building on sites where powerful cyclonic storms may occur once every fifty years, or where geological hazards occur once in every 1000 years? As compared to weather associated disasters, geological disasters such as earthquakes and volcanoes do occur rarely. But they are dramatic in their impact. Damage caused by them may be severe, especially in an area where people think themselves to be at little risk, before the event. For example, the peninsular part of India was always considered stable, i.e., having little risk of earthquakes. But the earthquakes of Koyna (1967) and in Latur, Maharashtra (1993) proved this belief to be wrong. These earthquakes caused extensive damage to human lives and property as people were not prepared for such an eventuality.

Although many acts of nature cannot be predicted or controlled, better knowledge about these hazards can at least reduce the extent of damage. The Government of India has issued certain dos and don'ts for the people living in areas affected by earthquakes, cyclones, and floods.

Landslides (Landslips)

The down-slope gravitational movement of a body of rock or earth as a unit owing to failure of the material is known as landslide. It may be induced by natural agencies, e.g., heavy rain, earthquake, or it may be caused by human over-interference with the slope-stability. Earth flow, mass movement, mud flow, rotational slip, and avalanches are all examples of landslides.

Landslides are rarely on a scale comparable to seismic or volcanic events. Moreover, many of the larger landslides are slow moving so while destructive to propert— they present little threat to life. The intensity and magnitude of landslide, however, depends on the geological structure, angle of dip of slope, nature of sedimentary rocks, and the human interaction with the slope **(Fig. 17.1(a)).**

The major causes of landslides are:

1. Earthquakes

Earthquakes are the most important cause of landslides in the folded mountainous areas (Himalayas, etc.). In India, landslides are more

frequent in the folded mountains of the Tertiary Period **(Fig. 17.2).** In the Kashmir Valley, earthquake of 1905 resulted into landslides in the Lesser and the Greater Himalayas in which several thousand people lost their lives.

2. Rainfall and Snowfall

Occurrence of heavy or continuous rainfall may lead to heavy landslides in the areas of steep slopes where National Highways and roads have been constructed. The Nashri area between Batote-Ramban-Ramsu and Banihal (Jammu and Kashmir State) is frequently subjected to landslides. The landslides in this region are particularly severe during the rainy and winter seasons when the vehicular traffic is disturbed for several days **(Fig. 17.3 and 17.5)**. In Malin village, Pune district, 44 houses were destroyed killing nearly a hundred people and over 200 feared trapped on 30 July 2014. Envionmentalists say large scale deforestation has made the are a vulnerable (Times of India, 31.7, 2014).

3. Mining, Quarrying, and Road-cutting

The continuous extraction of coal, minerals, and stones from the mines and quarries and development of roads by cutting the steep slopes in the folded mountains create a conducive condition for the occurrence of landslides. Such landslides may be observed throughout the Himalayas and in the Eastern and Western Ghats.

4. Loading by Construction of Houses

Unplanned growth of towns and cities in the hilly areas without testing soil and rocks is also an important cause of landslides. The eastern slope of Nainital (Uttarakhand) is sinking because of heavy load of hotels and residential structures.

5. Deforestation

Deforestation and other human activities also induce landslides. Most of the landslides are small involving some blocks up to a few metres across. But some are large enough to cause catastrophe. They may bury roads, buildings, and other structures. The adverse effect of landslides can be reduced by checking deforestation on mountain

slopes, following building codes for such areas, and by avoiding construction of buildings on steep slopes (Fig. 17.4).

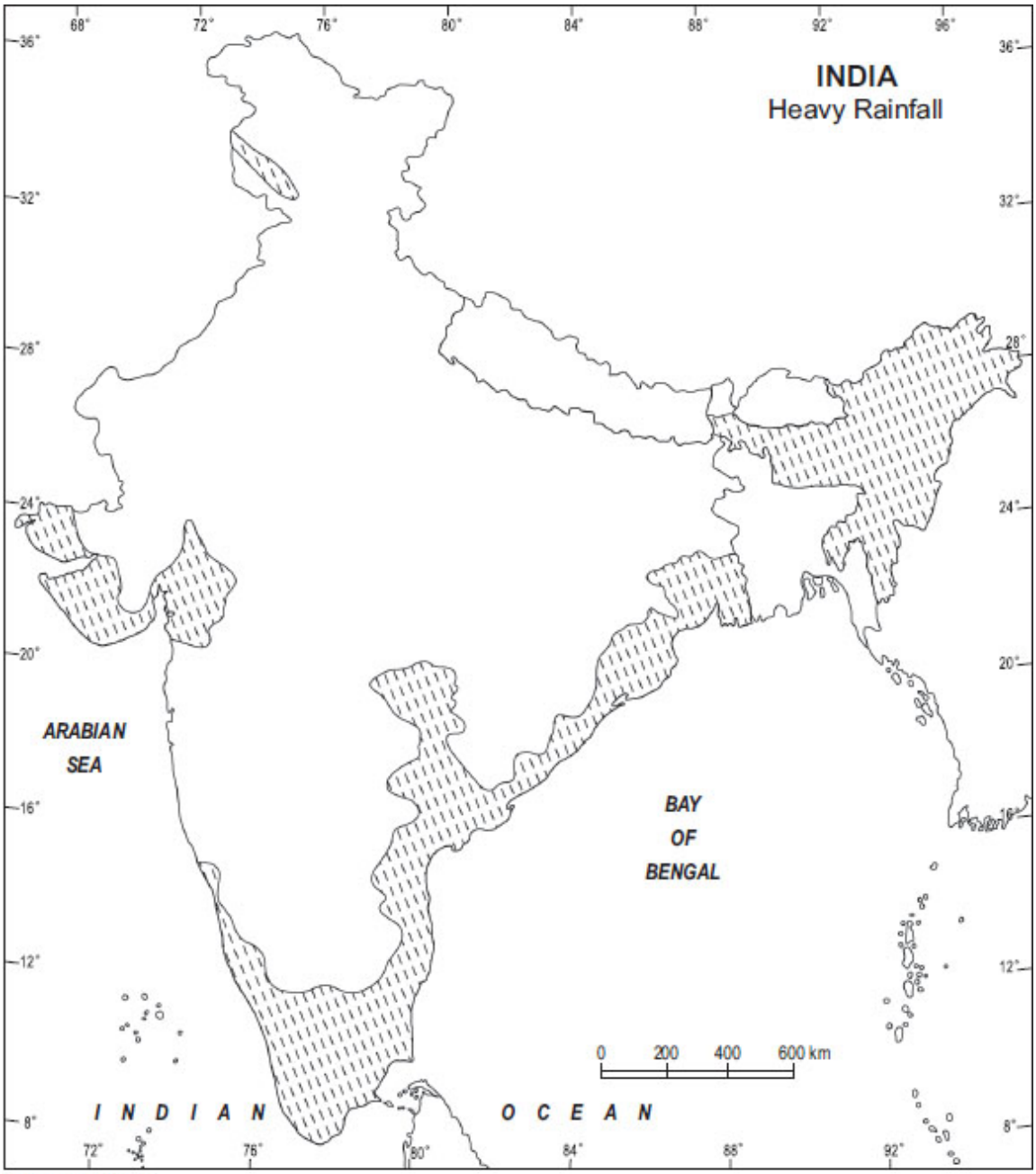


Fig. 17.3 Heavy Rainfall Affected Areas

Earthquakes

Earthquakes are vibrations of earth caused by ruptures and sudden movements of rocks that have been strained beyond their elastic limits. In other words, an earthquake is a motion of ground surface ranging from faint tremor to a wild motion capable of shaking buildings apart.

The main causes of earthquakes are: (i) movement of the plates of the earth crust (plate tectonics); (ii) volcanic eruptions; (iii) folding and faulting of rocks; (iv) landslides; and (v) anthropogenic factors.

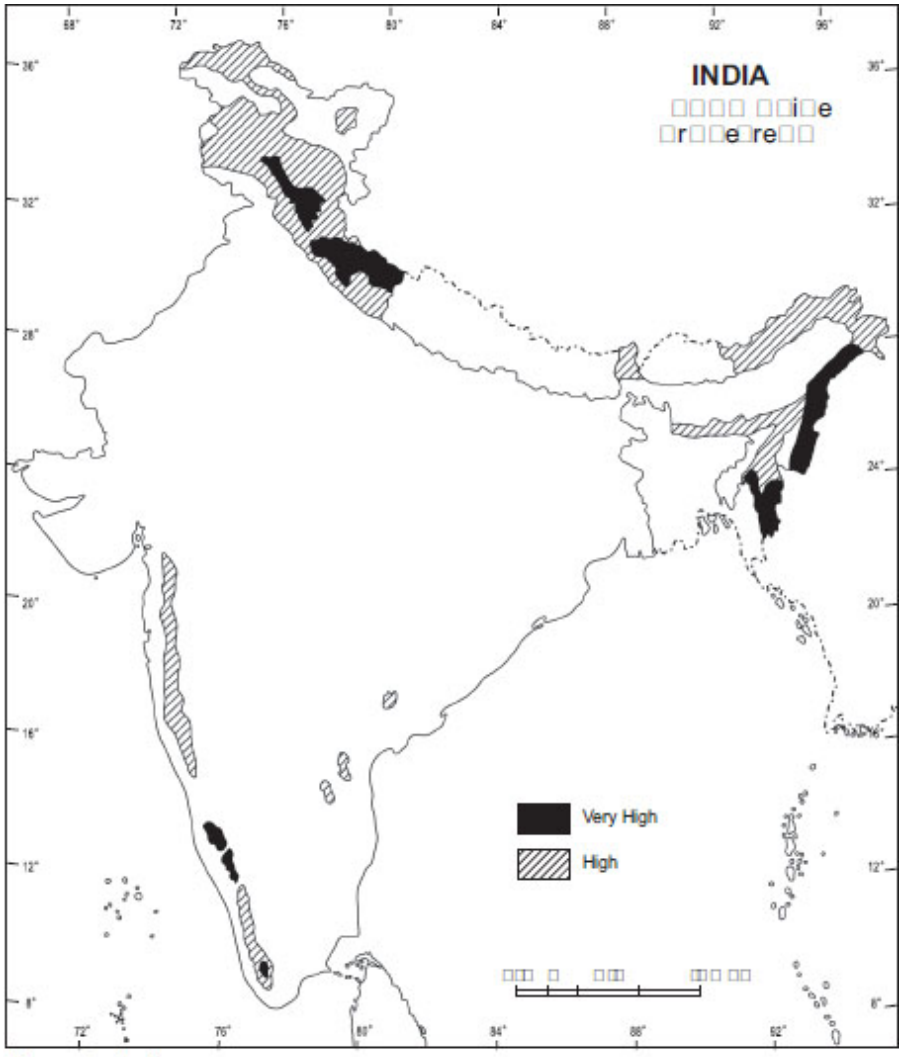
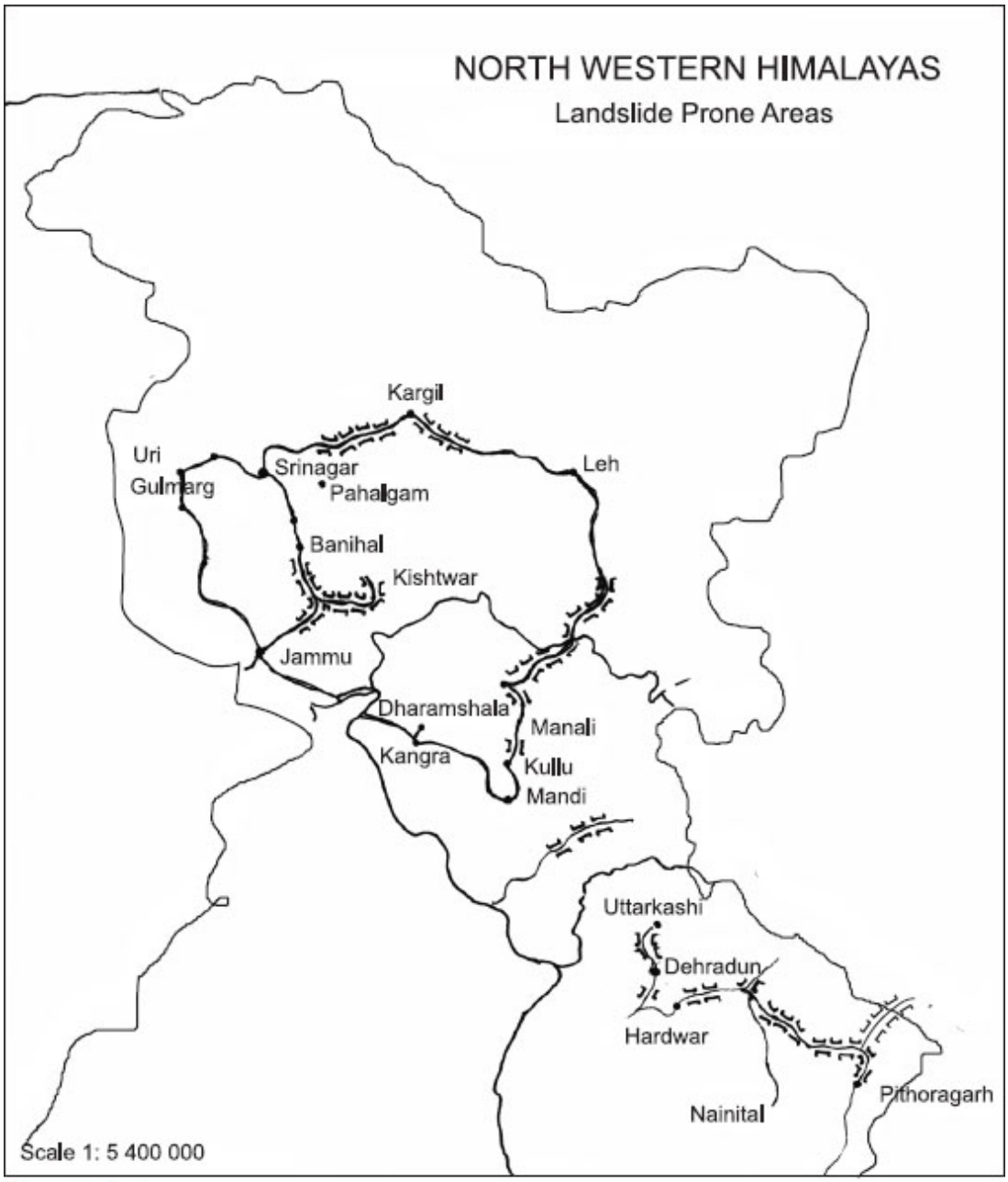


Fig. 17.4 Landslide Prone Areas

The sudden displacement of the earth's crust releases energy stored within the earth's interior (crust and mantle). It causes tremors or waves which move in different directions from the centre of disturbance. These waves are similar to ripples in water or light waves. The centre from which the earthquake waves originate is called the *seismic focus.* The point vertically above on the earth's surface is called *epicentre*. Most of the earthquakes have focus at depths of less than 60 km. The focus of deepest earthquake ever recorded was at a depth of 70 km in the mantle. The intensity of the earthquake is maximum at the epicenter and decreases with distance from the epicenter. Earthquake waves are recorded by an instrument called *seismograph*. The magnitude of an earthquake is recorded by using *Richter Scale.* It measures the amount of energy released by an earthquake.

Earthquakes of mild intensity take place daily. Strong tremors causing large scale destruction are, however, less frequent. Earthquakes are more frequent in the areas of plate boundaries, especially along the convergent boundaries. In India, the region of convergence of the Indian Plate and the Eurasian Plate is more vulnerable to earthquakes.



Map not to Scale

Fig. 17.5 National Highways and Landslide Areas

The peninsular part of India is considered to be a stable block. Occasionally, however, some earthquakes are felt along the margins of minor plates. The Koyna earthquake of 1967 and the Latur earthquake of 1993 are the examples of earthquakes which occurred in Peninsular India. The Gujarat earthquake on 26 January, 2001 was

recorded 8.1 at the Richter Scale. Its epicenter was at 23°8' east—about 20 km north-east of Bhuj **(Fig. 17.6).**

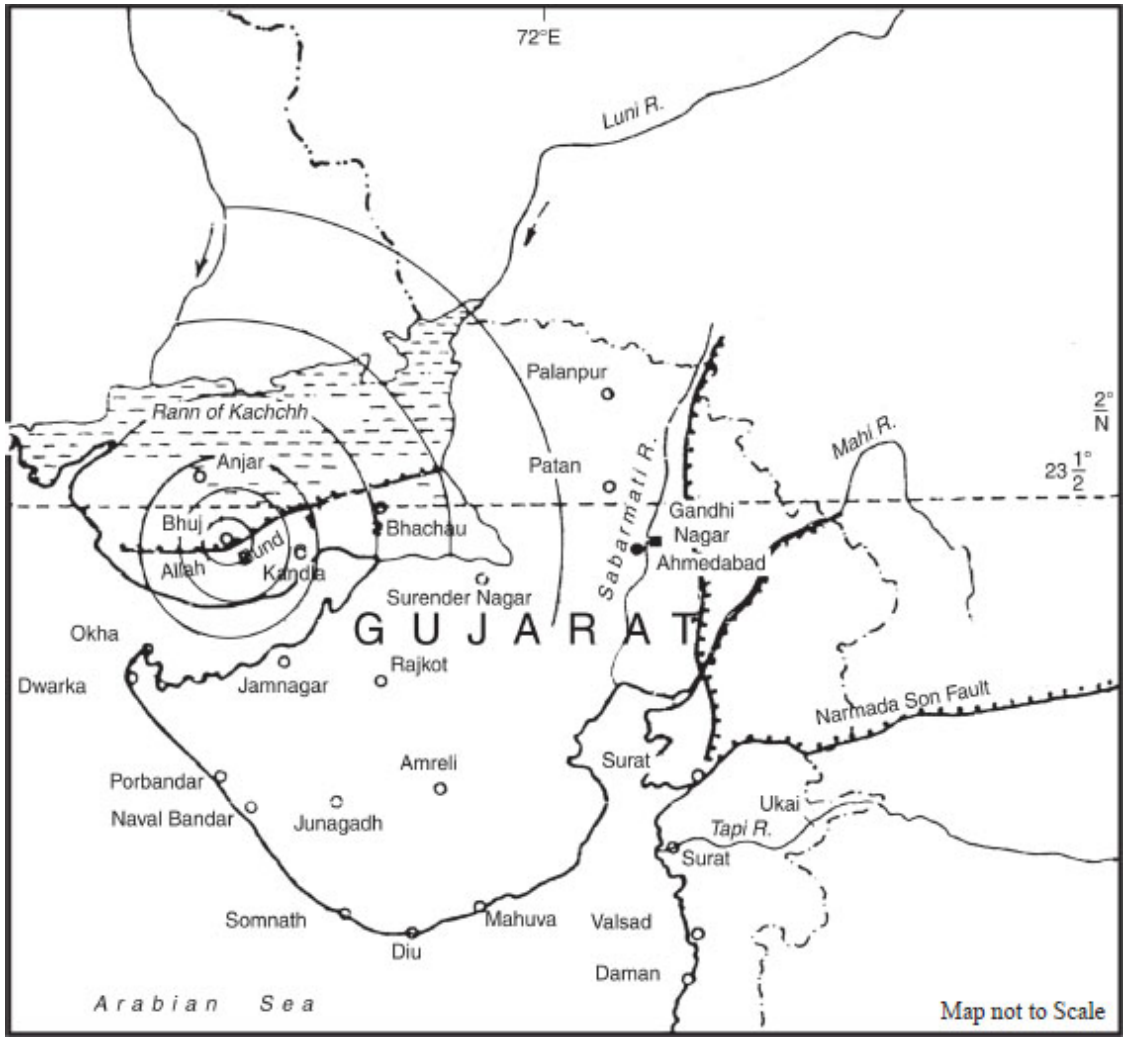


Fig. 17.6 Allah Bund and Bhuj Earthquake— 26 January, 2001

The experts of Indian seismology have divided India into five seismic zones given in **(Fig. 17.7)**.

It may be observed from **(Fig. 17.7)** that the entire Himalayan region, the states of North-East India, western and northern Punjab, Haryana, Uttar Pradesh, Delhi, and parts of Gujarat (Katch) belong to the highest and the high risk categories (Zone V and IV). The remaining parts of the northern plains and western coastal areas fall

in moderate risk zone. A large part of the Peninsular India lies in the low risk zone.

Consequences

The major consequences of earthquakes are:

(i) Deformation of ground surface because of the vertical and horizontal movement of the earth's crust.

(ii) Loss of human and cattle lives.

(iii) Damage and destruction of human establishments and structures, such as buildings, railway lines, roads, bridges, dams, towns, and cities.

(iv) Devastating fires, flash floods, and disturbances in the underground water-tables.

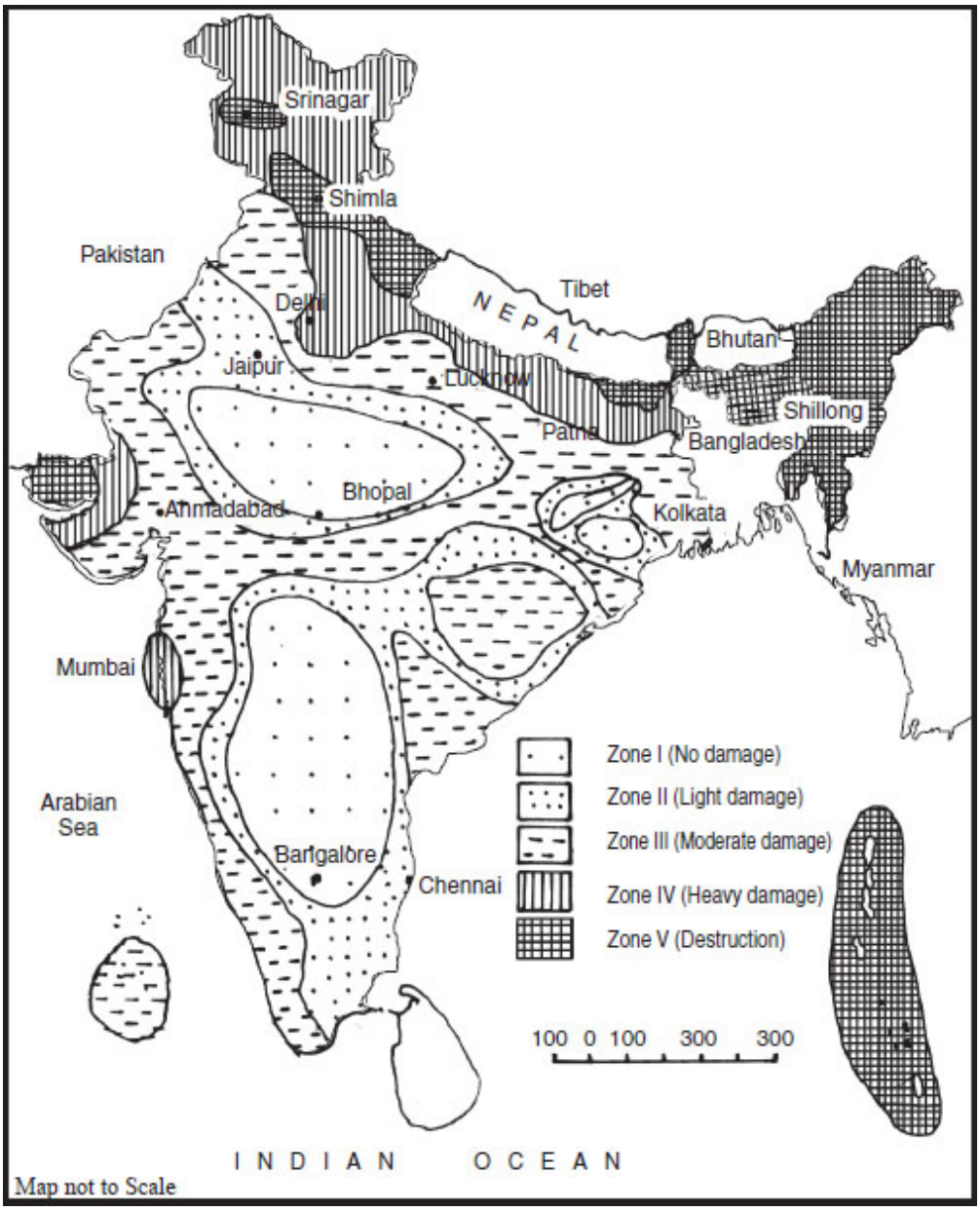


Fig. 17.7 Seismic Zones

(v) Landslides and avalanches which may block the river courses, leading to floods.

(vi) *Tsunami:* The tsunami of 26th December 2004 earthquake killed about 2.4 lakh people in the countries in and around the Indian Ocean. The massive Tohoku earthquake of Japan on 11 March 2011 resulted into Tsunami waves of 10 m. The tsunami destroyed the civil structures killing over 28,000

people. The radioactive fallout from the Fukushima Daiichi caused world-wide concern.

The Gujarat earthquake that struck the region at 8.46 a.m. on 26 January, 2001 was of the magnitude of 8.1 on the Richter Scale. Its epicenter was 20 km north-east of Bhuj town. In the opinion of seismologists, the Bhuj earthquake was a reawakening of the *Allah-Bund Fault*, which came into existence in 1819. The main cause of Bhuj earthquake is attributed to the movement of Indian Plate northward. The Indian Plate is moving in north-east direction at the rate of about 5 cm per annum against the Eurasian Plate along the Himalayas. The Sikkim earthquake of 18th September 2011 caused landslides and serious damage to the life and property, especially the Singik and Upper Teesta hydel projects. Some of the important earthquakes that occurred in the subcontinent during the last 100 years have been given in **(Table 17.1).**

Table 17.1

India—Major Earthquakes and their Magnitudes

Year	Region	Magnitude	Death Toll
1819	Bhuj (Kachchh)	8.0	2000
1885	Sopore (Kashmir)	7.0	2000
1897	Shillong	8.7	1542
1905	Kangra (HP)	8.0	19,500
1934	Bihar-Nepal border	8.3	10,700
1950	Arunachal Pradesh-Assam border	8.5	1526
1967	Koyna (Maharashtra)	6.5	177
1991	Uttar Kashi	6.6	2000
2001	Bhuj (Gujarat)	8.1	30,000
2004	Andaman-Nicobar (Banda-Ache-Sumatra)	9.0	India = 20,000 Total = 240,000
2011	11th March, Tohoku (Japan)	9.25	= 15,854 (Missing 3274)

Source: Meteorological Department of India, New Delhi.

Earthquake Forecasting

Making a precise forecast about the occurrence of an earthquake in a region and time is still a difficult proposition. The seismologists are increasingly concentrating on the aspect of earthquake forecasting. One approach for making prediction is to examine the history of each plate boundary and determine the frequency of earthquakes in the past. Seismologists then construct maps that provide an estimate of expected earthquake activity. Areas that are quite and overdue for an earthquake are termed as *seismic gaps*, such an area forms a gap in the earthquake occurrence record and is, therefore, a place that possesses accumulated strain.

The Chinese, on the basis of seismographic studies on animal behaviour, made fairly accurate predictions in the 1970s. The Chinese experts provided a list of indicative animal behaviour. Before the earthquake, cattle refused to enter corals, and ducks refond to enter water, snakes come out of hibernation, and fish jumped out of water. The Haicheng earthquake of China (1975) was accurately predicted by the Chinese experts on the basis of animal behaviour and the loss due to this earthquake of life and property was minimum.

The Tang-Shan earthquake of China (1976) in which 750,000 people lost their lives could not be predicted. People interviewed after the event talked about unusual activity that might have been related to the quake. Water level in wells changed abruptly; in a nearby village one well spouted a gas jet; another spouted water several metres into the air; and chicken, mules, goldfish, mice, and dogs were all said to have shown unusual behaviour on the night preceding the earthquake. Just before the earthquake occurred, residents were awakened by a brilliant red light in the early morning sky, which was present in the area spreading hundreds of kilometers around.

Despite animals' behaviour and other unusual phenomena, effective earthquake prediction, which could save many lives and property damage, is, however, proving an elusive goal.

An earthquake of magnitude 8.5 or higher is likely to hit the Central Himalayas in India. Such an earthquake could be a cause of unseen destruction in Uttarakhand and Himachal Pradesh. The earthquake in 2001 of measure 7.7 in Gujarat killed more than 13,000 people.

Tsunami

A tsunami is a series of waves created when a body of water, such as ocean, is rapidly displaced. Earthquakes, mass movements above or below water, volcanic eruptions and other underwater explosions, landslides, underwater earthquakes, large meteorides or asteroid impacts, and testing with nuclear weapons at sea all have the potential to generate a tsunami. The effects of a tsunami can range from being unnoticeable to devastating.

The term tsunami comes from the Japanese words *tsu* meaning harbour and *nami* meaning wave. Tsunamis are common throughout Japanese history; approximately 195 events in Japan have been recorded.

A tsunami has much smaller amplitude (wave height) offshore, and a very strong wavelength (often hundreds of kilometres along), which is why they generally pass unnoticed at sea, forming only a passing ‘hump’ in the ocean. Tsunamis have been historically referred to as *tidal waves* because as they approach land, they take on the characteristics of a violent onrushing tide rather than the sort of cresting waves that are formed by wind action upon the ocean (with which people are more familiar). Since they are not actually related to tides the term is considered misleading and its usage is discouraged by oceanographers.

Causes

A tsunami can be generated when the plate boundaries abruptly deform and vertically displace the overlying water. Such large vertical movements of Earth’s crust occur at plate boundaries. Subduction earthquakes are particularly effective in generating tsunami. Also, one tsunami in the 1940s in Hilo, Hawaii, was actually caused by an earthquake on one of the Aluetian Islands in Alaska. That earthquake measured was 7.8 on the Richter Scale. Tsunami takes place when a

huge earthquake occurs causing the plates below the water to push up causing the water to create a huge wave. The Japan earthquake, 11-9-0 resulted into devastating. tsunami in which 15,854 people lost their lives, and 3274 went missing. It caused the Fukushima Daiichi Nuclear crisis. Today only 2 of 54 reactions remain in operation. It was Japan's largest earthquake on records as a result of which 320,000 people are etill living in temporary homes.

In the 1950s, it was discovered that a larger tsunami, than previously believed possible, could be caused by landslides, explosive volcanic action, and impact events when they contact water. These phenomena rapidly displace large volumes of water, as energy from falling debris or expansion is transferred to water into which the debris falls. Tsunami caused by these mechanisms, unlike the ocean-wide tsunami caused by some earthquakes, generally dissipate quickly and rarely affect coastlines distant from the source due to the small area of sea affected. These events can give rise to much larger local shock waves (solitons), such as the landslide at the head of Lituya Bay which produced a water wave of the size estimated at 50–150 m and reached 524 m up the local mountain. However, an extremely large landslide could generate a 'megatsunami' that might have ocean-wide impacts.

Signs of an Approaching Tsunami

Often no, there is indication of an approaching tsunami. However, since earthquakes are often a cause of tsunami, an earthquake felt near a body of water may be considered an indication that a tsunami will shortly follow.

When the first part of a tsunami to reach land is a trough rather than a crest of the wave, the water along the shoreline may recede dramatically, exposing areas that are normally always submerged. This can serve as an advance warning of the approaching crest of the tsunami, although the warning arrives only a very short time before the crest, which typically arrives seconds to minutes later.

In the 2004 tsunami that occurred in the Indian Ocean, the sea receding was not reported on the African coast or any other western

coasts it hit, when the tsunami approached from the east.

Tsunamis occur most frequently in the Pacific Ocean, but are a global phenomenon; they are possible wherever large bodies of water are found, including inland lakes,where they can be caused by landslides.Very small tsunamis, non-destructive and undetectable without specialised equipment, occur frequently as a result of minor earthquakes and other events.

Tsunami in History: In history, tsunamis are not rare, with at least 25 tsunamis occurring in the last century. Of these, many were recorded in the Asia-Pacific region—particularly Japan. The Boxing Day Tsunami 2004, caused appropximatly 200,000 deaths and many more injuries **(Fig. 17.8)**. The destruction of the Egyptian City of Alexandria in AD 365 is now presumed to have been caused by a tsunami.

Warnings and Prevention

A tsunami cannot be prevented or precisely predicted, but there are some warning signs of an impending tsunami, and there are many systems being developed and in use to reduce the damage from tsunami.

In instances where the leading edge of the tsunami wave is its trough, the sea will recede from the coast half of the wave's period before the wave's arrival. If the slope of the coastal seabed is shallow, this recession can exceed many hundreds of metres. People unaware of the danger may remain at the shore due to curiosity, or for collecting fish from the exposed seabed.

Regions with a high risk of tsunami may use tsunami warning systems to detect tsunami and warn the general population before the wave reaches land. In some communities on the west coast of the United States, which is prone to Pacific Ocean tsunami, warning signs advise people where to run in the event of an incoming tsunami. Computer models can roughly predict tsunami arrival and impact based on information about the event that triggered it and the shape of the seafloor (bathmetry) and coastal land (topography).

One of the early warnings comes from animals. Many animals sense danger and flee to higher grounds before the water arrives.The Lisbon quake is the first documented case of such a phenomenon in Europe.The phenomenon was also noted in Sri lanka in the 2004 Indian Ocean earthquake.

While it is not possible to prevent a tsunami, in some particularly tsunami-prone countries some measures have been taken to reduce the damage caused on shore. Japan has implemented an extensive programme of buiding tsunami walls of upto 4.5 m (13.5 ft) high in front of populated coastal areas. Other localities have built floodgates and channels to redirect the water from incoming tsunami. However, their effectiveness has been questioned, as tsunamis are often higher than the barriers.

The effects of a tsunami can be mitigated by natural factors such as tree cover on the shoreline. Some locations in the path of the 2004 India Ocean Tsunami escaped almost unscathed as a result of the tsunami’s energy being sapped by a belt of trees such as coconut, palms, and mangroves. In one striking example, the village of Naluvedapathy in India’s Tamil Nadu region suffered minimal damage and few deaths as waves broke up on a forest of 80,244 trees, planted along the shoreline in 2002, in a bid to enter the Guiness Book of Records.

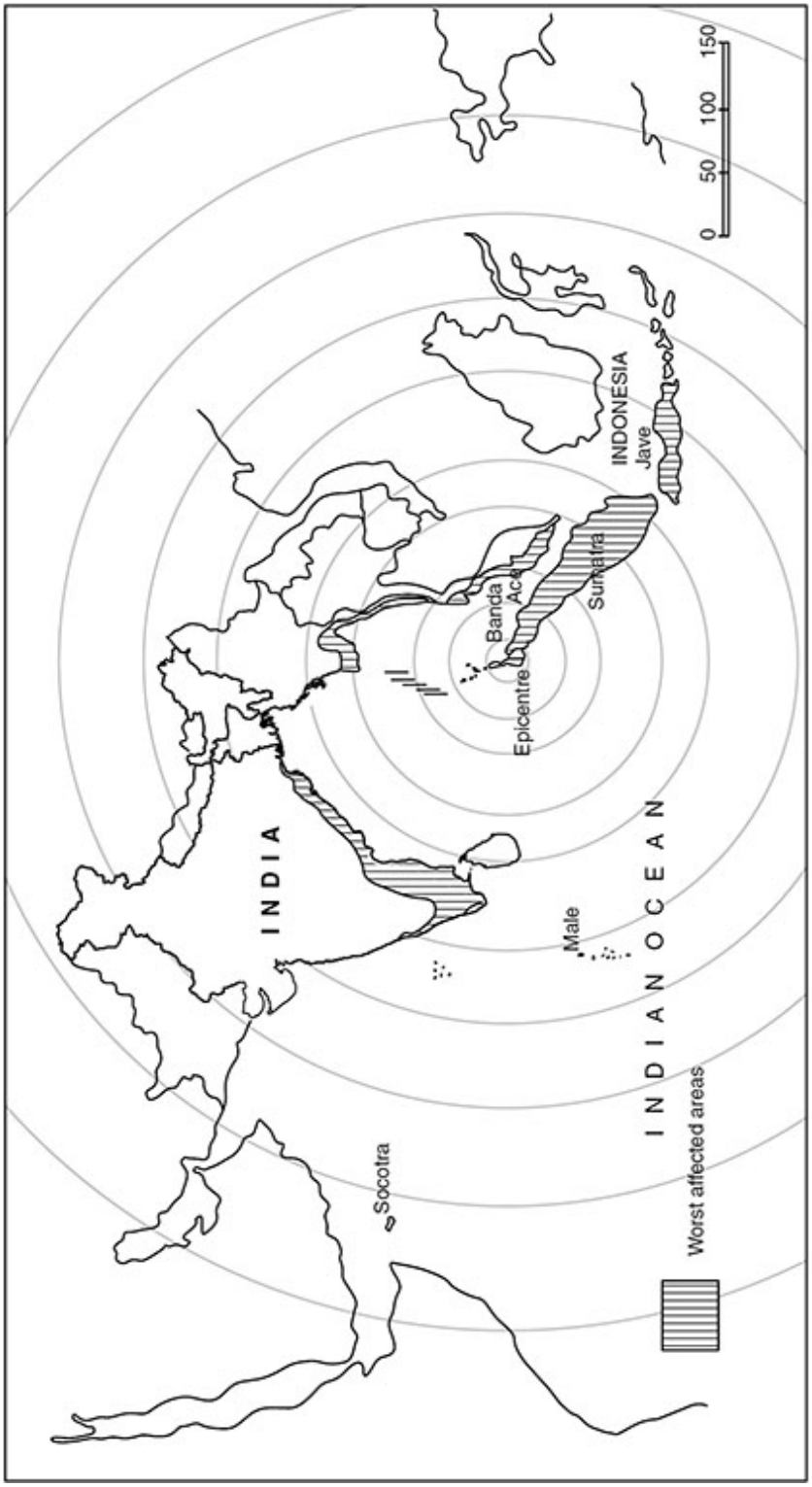


Fig. 17.8 Tsunami—2004

Cylcone '*Fani*' struck India's eastern Coastline of Odisha on May 2019. It cast floods and a storm surge of 1.5 m threatened the low lying areas.

Floods

A high water level that overflows the natural banks along any portion of a stream is called flood. Thus, floods are commonly associated with a stream or river. A stream floods when its discharge is greater than the capacity of its river channel. Excess water flows over the river banks and submerges the adjacent land which is usually dry. When it happens, the channel and the flood plain together allow passage of water. Floods are natural phenomena that occur in all river systems. Some floods are seasonal, for example, those associated with monsoon rains. Sometimes flash floods also occur. There are many causes of floods. Some of the important causes of floods are:

1. Heavy rainfall, cloud burst, and prolonged rainfall;
2. Heavy melting of snow and ice;
3. Large catchment area;
4. Changes in the river courses;
5. Silting of river beds;
6. Collapse of dams;
7. Landslides;
8. Deforestation;
9. Unscientific use of land utilisation;
10. Transgression of sea at the occasion of tropical cyclones; and
11. Tsunami.

Consequences of Floods

The frequency, intensity, and magnitude of floods are increasing day by day. Floods cause more damage than any other single disaster in India. It creates numerous problems for the society and environment. Floods in India are taking thousands of lives every year. The crops are adversely affected, there is loss of livestock, the lines of communication are disturbed creating great problems for the

movement of people and goods. Unfortunately, floods are becoming more frequent and devastating.

Flood Affected Areas in India

Floods are universal in India. Except the hilly and mountainous regions, they occur in almost all the parts states of the country. The frequency and intensity of floods is, however, more serious in the states of Andhra Pradesh, Assam, Bihar, Gujarat, Haryana, Jammu and Kashmir, Manipur, Odisha, Punjab, Rajasthan, Tamil Nadu, Uttar Pradesh and West Bengal. The flood-prone areas have been shown in **(Fig. 17.8).**

It may be observed from **(Fig. 17.8)** that the middle and lower Ganga-Brahmaputra plains have the largest flood affected areas. Floods are a common phenomenon in the deltas of Mahanadi, Krishna, Kaveri, Mahi, Sabarmati, and in the Sutluj plains of Punjab.

Brahmaputra, Ganga, Yamuna and their tributaries—Gandak, Kosi, Mahanadi, Godavari, Jhelum, Krishna, Kaveri, Mahi, and Satluj record frequent floods, especially during the rainy season (July to September).

Worst floods of the century affected the state of Kerala in August 2018, due to unusually high rainfall during the monsoon season. The *Wayanad* and *Idukki* areas suffered from landslides. Thirteen out of 14 districts were affected. 2346.6 mm of rainfall was recorded from June 1, 2018 till August 19, 2018. The expected rainfall was 1649.5 mm. Floods occurred in Mananthavady, Vythiri in Wayanad district.

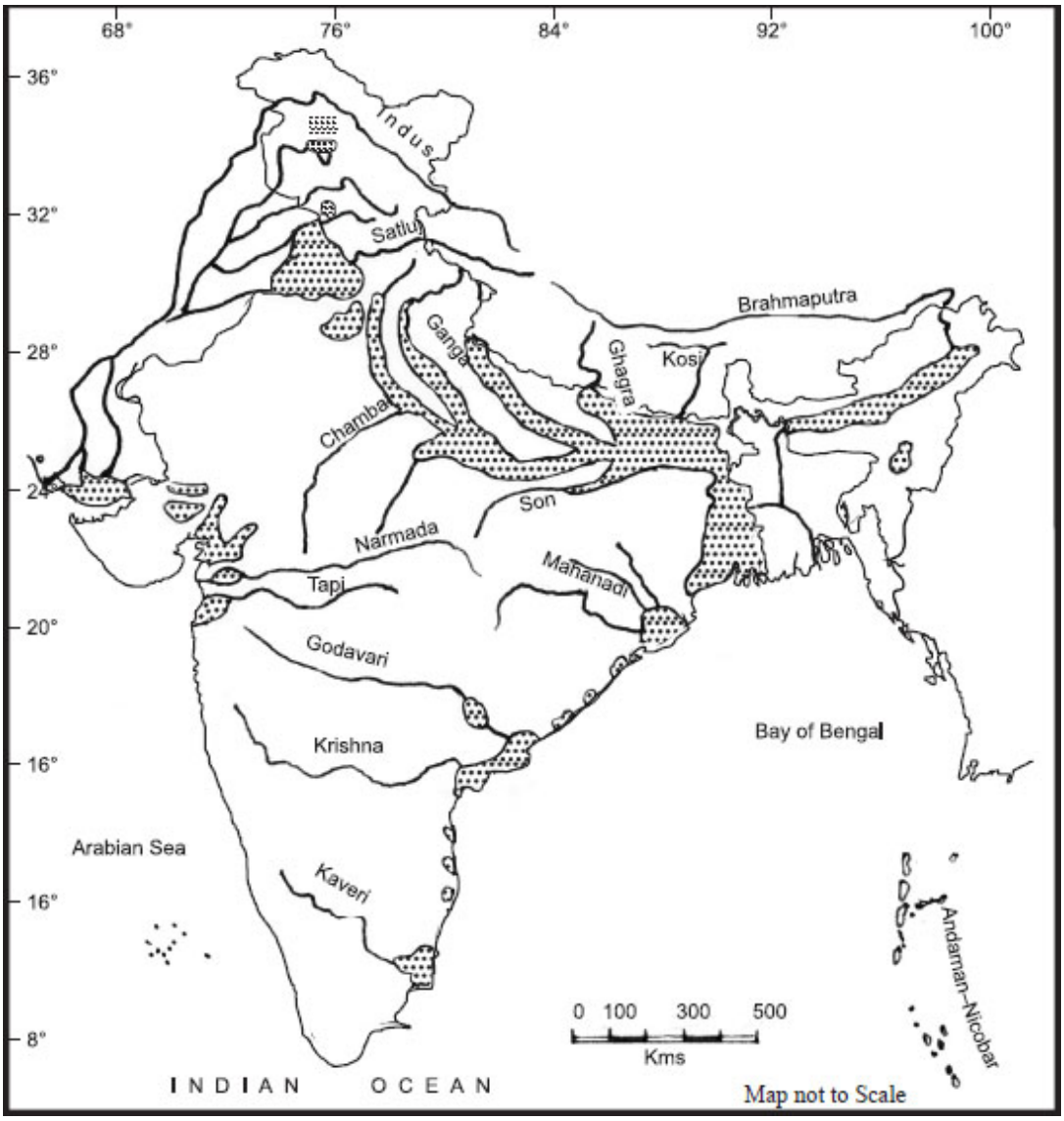


Fig. 17.9 Flood Prone Areas

Flood Control

Looking at the devastating floods of 1949 and 1954, the Government of India took planned steps to control the floods. In the initial phase of flood control planning, the main thrust was towards the development and construction of multi-purpose projects. The Damodar Valley, Kosi, Bhakhra, Hirakud, Rihand projects were some of the important multi-purpose projects designed to control the floods and to generate electricity. In 1976, the *Rashtraya Barh Ayog* (National Flood Commission) was constituted. The Commission identified the areas in

different states which are more prone and vulnerable to floods. According to one estimate about 50 million hectares of land of India is flood prone. The Central and State Governments have taken the following steps to reduce the menace of floods:

1. **Flood Forecasting:** Flood forecasting involves giving prior information regarding the occurrence of floods. It is of immense help in taking timely action to reduce the loss of human lives, livestock, and movable property. The Central Water Commission started flood forecasting in November 1958 when the first flood forecasting station was established near the old railway bridge of Delhi. At present there are 175 flood forecasting stations on various rivers in the country. The Flood Forecasting Network covers 15 states and two Union Territories. The flood forecasting stations issue daily-flood forecasts and warnings throughout the flood season from May to October.
2. **Reduction in Run-off:** Reduction of run-off is one of the very effective methods of flood control. Run-off can be reduced by inducing and increasing infiltration of the surface water into the ground in the catchment area. This can be done by large scale afforestation, especially in the upper catchments of the rivers.
3. **Construction of Dams:** Dams and multi-purpose projects are being constructed across the rivers to store the surplus water in the reservoirs. A number of such reservoirs were constructed during the First Five Year Plan. In the subsequent plans also, many dams were constructed to reduce the run-off of the rivers.
4. **Channel Improvements:** The channels of the rivers more prone to flood are improved by deepening and widening. The water of such rivers is also diverted into canals.
5. **Construction of Embankments:** The central and state governments have constructed a number of embankments along the rivers to reduce the menace of floods. Such

embankments have been constructed along the Brahmaputra, Beas, Gandak, Ganga, Ghagra, Godavari, Kaveri, Kosi, Krishna, Mahananda, Narmada, Ravi, Sabarmati, Son, Satluj, Tapi, and Tista rivers and their tributaries.

6. **Flood Plain Zoning:** Flood plain zoning is also an important step to control the floods. It is based on information regarding flood plains, particularly the identification of floodways in relation to land use. Detailed maps of the flood-prone areas are prepared after a detailed study of flood cycles.

7. Legal Measures: In addition to the steps given above, legislative measures are taken to restrict the construction of industrial and residential units in the flood-prone areas.

Droughts

Occurrence of substantially lower than average precipitation in a season that normally has ample precipitation in a season for the support of cereal and non-cereal crops, is known as a drought. In India, the erratic nature of summer monsoon with long dry spells and high temperature are responsible for the drought conditions. On an average, one in every five years is a drought year; in Rajasthan, one in every three years is a drought year.

Amphan Cyclone

Cyclones are low pressure systems that form over warm tropical waters with gale winds near the centre. The winds can extend upto kilometres from the eye of the storm. They bring about torrential rains and flooding resulting in loss of human life and damage to property. Tropical cyclones, also known as Hurricanes are the fiery and destructive in nature.

The north coast of Bay of Bengal is more prone to catastrophic surges as compared to anywhere on the earth since it has high surface temperatures that favours strong cyclones.

Amphan was a super tropical cyclone that caused extreme damage to West Bengal and Odisha on the eastern states of India and also to

Bangladesh. It struck the coastline of India on May 20, 2020. It was the strongest storm that has struck the Bay of Bengal after 1999.

'Amphan' name was given by Thailand in 2004. The name of any cyclone is finalized from a list by WMO/ESCAP. It is pronounced as UM-PUN. The cyclone hit the India–Bangladesh border with winds upto the speeds of 185 km/hr. The meterological department at Alipur recorded exceptionally high rainfall of 236 mm. The cyclone caused 72 reported deaths across Bengal. 88,000 hectares of paddy and one lakh hectares each of vegetable and sesame crops have been destroyed in the state of Bengal.

About 500 trees got uprooted and 14 lakhs consumers were left without power. In the state of Odisha, the standing crops got damaged, 45 lakh people across 6745 villages have been affected.

The super cyclone lost its intensity and weekend into a deep depression with a reduced speed of wind of about 60km/hr.

Transmission

Epidemic diseases are transmitted in many ways. Some of the ways of their transmission are as under:

1. By direct contact, for example, droplets sprayed about when a patient coughs or sneezes
2. By contaminated food and water.
3. By arthropods—filth associated flies—of various types that may serve as mechanical carriers of disease germs as in dysentery and cholera. Blood-sucking arthropods like mosquitoes are more effective transmitters.

Factors Affecting Incidence

Apart from weather and climate, there are many factors which influence the spread and intensity of epidemics. Some of the factors are:

(i) ethnicity,
(ii) age and sex composition,
(iii) literacy and education,
(iv) occupation and lifestyle,

(v) standard of living,
(vi) size of family and degree of crowding,
(vii) residential locality,
(viii) food habits,
(ix) smoking and consumption of liquer, and
(x) association with animals and birds.

Incidence Curve

Incidence curve of different epidemic diseases are similar in that the rise in number of cases in an epidemic is usually more rapid than its fall.

Corona Pandemic

The corona virus COVID-19 pandemic is the greatest challenge being faced by the world after the World War-II. It was started as an outbreak in the city of Wuhan in China in December 2019. It is a highly contagious disease caused by the virus Corona. Its symptoms include mild to moderate respiratory illness. Older people with history of cardio vascular disease, diabetes, chronic respiratory diseases and cancer along with the children of ages less than ten years are more to prone to corona fatality.

The mode of transmission of the virus is through saliva, or discharge from the nose when an infected person sneezes or coughs. The best way of protection against the infection is wearing mask, frequent washing of hands, sanitizing and maintaining a social distance of more than 1 metres.

Till May 23, 2020, the number of active cases in India have been reported as 69597, the cured/
discharged patients being 51783, number of deaths being 3720 (as available from the website of MOHFW, Government of India). Maharashtra, Tamil Nadu, Gujarat, Delhi and Rajasthan are the top states with maximum reported cases in the country. The statistics across the globe as reported till date is 5,320,865 cases, with 2,170,472 having recovered and deaths of 3,40,261.

In order to control the exponentially growing cases in the country India went for the world's largest total lockdown on March 25, 2020 that restricted 1.3 billion people.

Locust Infestation

Locust is a herbivore and its scientific name is 'Acrididae' and it belongs to a group of invertebrate. Its size is usually from 0.5 to 3 inches. Plagues of locusts have devastated societies since the Pharoahs led ancient Egypt.

During the dry spells, individual locusts collect in patchy areas with little vegetation. Serotonin a chemical is secreted in their nervous system and it makes them social and increases their movement and apetite. Locust swarms can cover large distances upto 81 miles/day. In 1954, locust swarm flew from north west Africa to Great Britain while in 1988 from West Africa to Carribean. One swarm may cover an area of one square kilometer and contain upto 80 million locusts.

The desert Locust swarms (migratory pests) invaded the northern part of the country (Rajasthan, Gujarat, Madhya Pradesh, Uttar Pradesh, Haryana) via Pakistan amidst the Corona crisis in the month of April, 2020. It is the worst invasion after three decades. It was about two or three kilometres long. The locusts cause a severe damage to the crops and even entered cities.The pest threatens vegetable and pulse crops but the rabi crop produce remained un impacted as it was harvested.

Vizag Gas leak

A styrene gas leakage occurred from the South Korean battery maker LG chemical Ltd., at Gopalapatnam, 14 kms away from Vishakhapatnam in Andhra Pradesh, during the early morning hours on May 7, 2020 .The leak killed at least 11 people and more than 800 people had to be hospitalized. 1500 people were evacuated over a radius of 3 kilometre from the plant. The inhaled gas was poisonous and smelled foul. It induced symptoms of nausea, fatigue, dizziness and fainting amongst the people.

The gas leakage was a consequence of the Corona virus lockdown , when the gas was left there due to sudden closure , that led to a chemical reaction and generation of heat that led to the gas leakage.

Cloudburst

Cloudburst is a sudden rainfall which can be quite unexpected, very abrupt, characterised with flash floods. In other words, cloudburst is an extreme amount of precipitation, sometimes with hail and thunder which normally lasts no longer than a few minutes but is capable of creating havoc condition of flood, landslides, avalanches, mudflow, earth-flow, rock-fall, subsidence, slumps, soil-creep and mass wasting. In some cloudbursts up to 13 cm (5 inche) of rain can fall in one hour, often in the form of extremely large droplets.

The rain characteristic of a cloudburst is caused by a phenomenon known as *Langmuir Precipitation,* in which drops of rain fuse together to create large drops as they fall, falling quickly as they grow. The *Langmuir Precipitation* process is applicable only to those clouds which do not extend beyond the freezing point. The temperature in the uppermost part of the cloudburst seldom falls below 5°C. The occurrence of precipitation from such cumulus-nimbus clouds involve the coalescence of cloud droplets of different sizes. Since the rate of fall of these unequal rain-drops is different, they collide with each other within the cloud, and the larger drops grow at the expense of smaller ones. The size of the rain drops become sufficiently large which fall faster, and results into cloudburst. Sometimes, the rain in cloudburst falls so fast and is large that it is actually scary, fearful and frightening.

According to the meteorologists, the rainfall rate equals to greater than 10 cm (about 4 inche) per hour in cloudburst. There are cases when more than 20 mm (2 cm) of rainfall fall in a few minutes in a cloudburst. The extremes of rainfall in cloudbursts have been given in **(Table 17.2).**

Table 17.2

Extremes of Rainfall in Cloudburst

Duration	Rainfall	Location	Date
1 minute	38.10 mm (1.5 inches)	Basse-Terre, Guadeloupe (West Indies	26 November, 1970
15 minutes	198.12 mm (7.9 inches)	Plumb-Point, Jamaica	12 May, 1916
1 hour	250 mm (9.84 inches)	Leh (ladakh) India	6 August, 2010
10 hours	940 mm (37 inches)	Mumbai, India	26 July, 2005
20 hours	2329 mm (91.69 inches)	Ganges Delta (India)	January 1966
2 hours	220 mm	Uttrakhand	June 16, 2015
	100 mm	Singhals area & Chanali districts	July 1, 2016

In the Indian subcontinent, a cloudburst usually occurs when moisture laden monsoon clouds drift northward from the Bay of Bengal or the Arabian Sea across the Northern Plains, then on to the Himalaya and burst, bringing as high as 10 cm rainfall per hour. Some of the devastating cloudburst of India in recent years have been given in the **(Table 17.3):**

Table 17.3

Devastating Cloudbursts in India

Date of Cloudburst	Place of Cloudburst	Deaths
August 1998	Golpara	400
July 1970	Alaknanda Basin (Uttarakahand)	500
15 August, 1997	Chirgaon, Shimla District (H.P)	115
17 August, 1998	Milpa Village, Kumaun Division (Uttarakhand)	250
26 July, 2004	Mumbai cloudburst	>5000
6 August, 2010	Leh-Ladakh	>1000
16 June, 2013	Kedarnath cloudburst (Uttarakhand)	>10,000*

**** Unofficial estimated figures.***

Cloudburst or Himalayan Tsunami in Uttarakhand

Uttarakhand the *abode of gods* with its magnificent snow-covered peaks, pristine forests and awe-inspiring rivers is a sacred place in Indian culture. Million of pilgrims and tourists visit the state of Uttarakhand every year, especially during the summer season. Despite all these attractions, the state of Uttarakhand has a highly vulnerable ecosphere. There are sudden cloudburst, flashfloods, landslides, avalanches, rock fall, soil creep and mass wasting.

The cloudburst of 16^{t} June, 2013 has been termed by the experts as the *Himalayan Tsunami*
(Fig. 17.10(a)). Although the rain falling would have been fairly cold, it would not have frozen as it moved down through the snowpack. The rain released small quantity of heat that caused snow to melt. The high intensity of rains aotuelly accelerated the process. The heat released was sufficient to raise the heat of the snowpack from a few degrees below zero to the melting point, a substantial change in temperatures in the Himalayan heights. The rain on snow rather than glacial melts aggravated the floods which were caused by phenomenal 34 cm of precipitation in less than 20 hours.

Fig. 17.10(a) Bhagirathi, Mandakni, Alaknanda Basins, the worst affected
Region of 'Himalayan Tsunami' 16 June, 2013

The pace at which melted snow melted moved affected the volume of rain and ground conditions. As the turbulent rivers of Uttarakhand moved downwards on steep slopes, they carried more weathered material, and the water gained speed and volume, rapidly increasing the levels downstream. It led to flash flood and became an unparalleled disaster in the modern history of the state. In the words of a poet *"Tabahi ka Manzar Dukhon ki Kahani: Pahdon pe Barsa Pahadon ka Pani"*

This cloudburst resulted into enormous loss of life, property, and infrastructure. With many National Highways damaged, bridges washed away, electricity and phones networks down, hotels and motels destroyed, towns and rural settlements buried and numerous ravaged places marooned. The Uttarakhand cloudburst affected about 48,000 sq km of Bhagirathi, Alaknanda rivers and their tributaries. The worst affected was the Mandakni Basin, in which the Kedarnath -Shrine, Rambara, Gaurikund, and Guptkashi towns were devastated. According to official figures, the number of villages affected by the flash flood was 3978, roads damaged–1636 km, worth INR 500 crore; cost of restoring drinking water network Rs.100 crore; damage to hydropower projects about INR 150-200 crore **(Fig. 17.10(b))**.

According to the experts of environment, ecology, glaciers and himalayan geology, the disaster might have been waiting to happen. The illegal construction on the hills, the exponential growth of rickety structures that serve as hotels which are sordidly maintained, the lackadaisical attitude towards warnings of natural catastrophes are largely responsible for the disaster of *Mandakni*, *Bhilangna*, *Bhagirathi*, *Alaknanda*, *Pindar*, *Nandakini*, *and Dhauli Ganga basins* **(Fig.17.11)**.

Apart from natural disaster the cloudburst of June 2013, according to experts, is largely a man-made disaster. The main anthropogenic

causes of the disaster are given below:

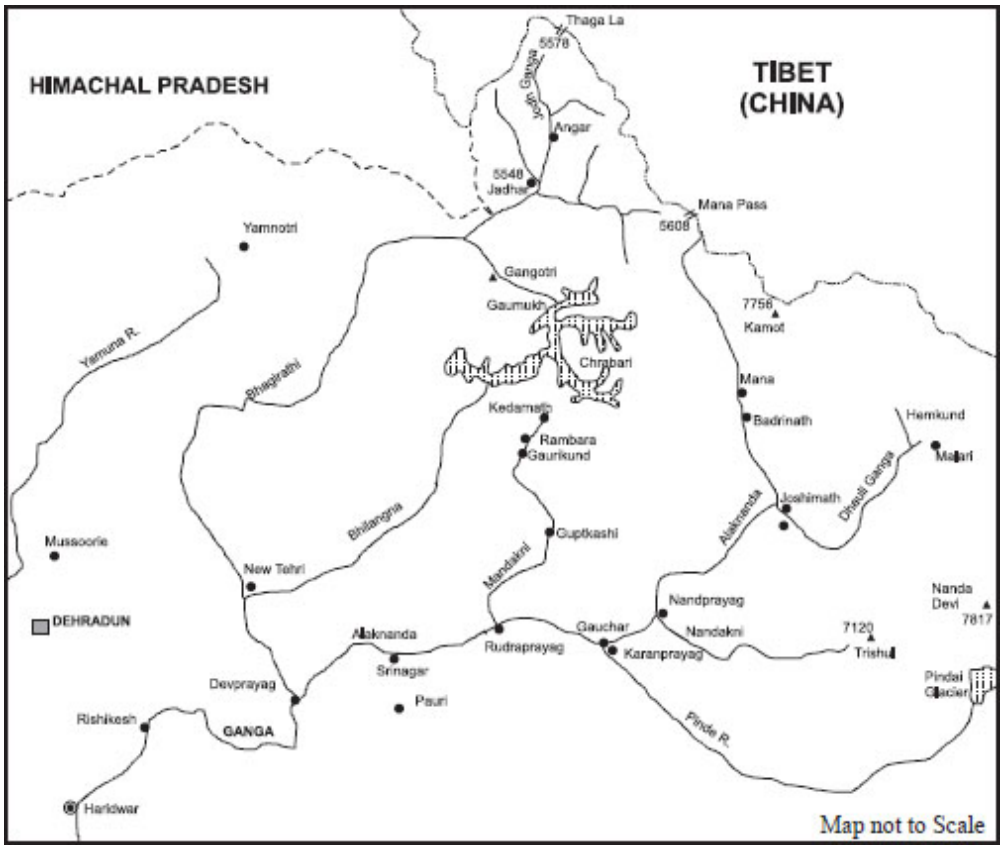


Fig. 17.10(b) Cloudburst Region of Uttarakhand (16 June 2013)

1. The development and extension of roads and National Highways NH-58 -Mana Pass, NH-108 -Gangotri-Angar, NH-109-Guptkashi-Kedarnath, etc., have disturbed the ecology and the highly fragile ecosystems of the region.
2. Under the pressure of rapid growth of population and religious and cultural tourism, there is tremendous growth of urban settlements, hotels, business-houses, hill resorts, camping sites, *Chattis* (night-halts) in the hills. Unfortunately, the pilgrimage today has become the ugly face of tourism. The tourism lobby has destroyed the sanctity of the shrines of *Deobhumi.* Hotels and *Dhabas* play the latest film music at the top volume. All these developments are damaging and polluting the natural beauty and ecology of the Himalayas.

Regulate the limited number of healthy cultural tourists who can walk on foot up to the shrines.

3. Restrictions like that of biosphere reserves, strict rules should be framed that each shrine will take so many pilgrims in a day and in a year. There should not be any VIP quotas. Hotels and *Dharamshalas* should not be allowed near the ehriars. The basic principle of religious tourism should be *"Have a **darshan** and proceed back".*
4. The business of ponies and *dandi-rides* should be abolished or minimised.
5. The uncontrolled cultural tourism at religious places like *Char Dham-Yatra* (Gangotri, Yamontri, Kedanarnath, and Badrinath), and other places like Hemkund Sahib, Rudranath, Hanuman Chatti, Govind Pahuvihar, etc., also need to be regulated.
6. The Himalayas are unstable that are still in the process of rising. Blasting them to make tunnels, dams, roads and other such projects destabilises the entire mountain range. The earth and rocks become loose, and when there is torrential rains, they come tumbling down with a humongous amount of debris and slush, which destroys everything in their way.
7. Construction of multistoried buildings, hotels, motels, *Sarais, Dharamshalas* (inns), rest-houses, factories, picnic spots, tourists and pilgrims camping grounds have damaged the fragile ecosystems of the Himalayas. They should be developed according to the principles of ecology and environment after Environmental Impact Assessment.
8. Several major multipurpose projects (Tehri Dam, Koteshwar, Maneri-Bhali Hydel Project, Loharinga-Pala Hydel Project, Srinagar Project, Asi-Ganga Project and Vishnu-Prayag Project) and numerous other big and small dams across the tributaries of the Bhagirathi, Alaknanda, Ganga and Yamuna rivers and their reservoirs have disturbed the isostatic-equilibrium in the young folded mountains of the Himalayas.

The Supreme Court on 13 August, 2013 ordered fresh environment impact assessment of the 24 major hydro-electric projects on Alaknanda and Bhaghirathi rivers. Small dams and projects should be preferred to big and multipurpose projects.

9. The ambitious target of the decision makers and planners to convert the State of Uttarakhand into the ***Urja Pradesh*** (Energy Province) and the Tourist Capital of India have launched many projects against the principles of ecology and geomorphology.
10. Indiscriminate deforestation of oak trees for commercial purposes have changed the forest ecosystems of the Himalayas. The oak forests have been replaced by pine trees. But what is profitable to man, may not be profitable to Himalayas. Pine is not a good substitute to oak trees. We may have little or no control over cloudburst and very heavy rainfall, but we can mitigate the devastation of the Himalayas by foresting it with trees that can hold the mountains together; the trusty old oak. Oak trees help create through natural masonry a fortification of soil against erosion caused even by heavy rainfall.

It was because of the above factors, the havoc that followed cloudburst on 16 June, 2013 in Uttarakhand had been in the making for many years. The devastating floods in Uttarakhand have proved that taming natural forces through human intervention is all but a myth. The plans now made should work with nature and not in opposition to it.

Cloudburst Management in Uttarakhand

Cloudburst is a natural disaster that causes great damage to life and property. Some of the steps that can help in reducing the damage to life and property in the affected areas are given below:

1. Based on scientific data, there should be delineation of eco-sensitive zones.

2. The maximum and minimum discharge of water of all the rivers and their tributaries should be ascertained and the people should be made aware about the possible discharge of the rivers. Unfortunately, the state of Uttarakhand has not defined the minimum discharge of water in the major rivers. Such laxity is bound to result into large scale destruction and disasters.
3. Better and effective system of weather forecasting and dissemination of weather related information are imperative to mitigate the cloudburst disaster. For this purpose Doppler-Radars should be installed by the Meteorological Department of India in the upper reaches of the Himalayan rivers to make advance weather forecasting for events like cloudburst.
4. There should be proper site selection and planning of the rural and urban settlements.
5. There is an urgent need for a comprehensive renewal and relook at construction techniques of roads, houses, factories and other infrastructures.
6. Blasting which is done for the construction of dams and roads, weaken the hills and shakes the roots of the trees. Such blasting should be stopped or minimised.
7. Encroachment of river beds by buildings, and blasting of mountains to build National Highways are making hilly regions more susceptible to disasters. These activities should be done keeping in mind the ecological principles.
8. There should be strict regulation of religious, cultural and aesthetic tourism. The tourism industry need to be converted into eco-tourism to avert the disasters and catastrophe.
9. There is urgent need to establish the Bhagirathi-Alaknanda Development Authority on the pattern of the Damodar Valley Corporation.
10. Uncontrolled traffic-jams and increased environmental pollution play havoc in the fragile ecosystems of the Himalayas. They should be checked and mitigated.

11. There should be state-level, and district level Disaster Management and Mitigation Centres to combat the humongous natural calamities, like that of the 16 June, 2013.
12. Human resource required to tackle the natural disasters should be strengthened.
13. There should be no discrimination while providing relief and rehabilitation measures. Calls from influential people and politicians should not be entertained by the persons engaged in rescue operation.
14. The post cloudburst operations should include quick action of rescue of stranded people buried under thick cover of debris to evacuate them to safer places. Immediate medical help should be provided to survivors.
15. The state government should take strict and effective action against the encroachment around the shrines in the form of make-shift shops, hotels, *dhabas* and temporary shelters.
16. The *Sant-Samaj* (religious community) should work as a pressure group to check the construction of multi-story buildings, hotels, motels, *Sarais* and *Dharamshalas.*
17. People should be educated about the small size of families to reduce the pressure of population on the limited resource base of the mountainous ecosystems.

If these steps are taken together, the consequent damage to life, property, infrastructure, ecology and environment may be reduced substantially.

Epidemics

The term *epidemic* originally denotes a sudden excessive prevalence of disease in a population. Epidemic means *upon the people*. Historically, epidemic was applied to infectious disease, but in more recent times, it also includes major non-infectious diseases, such as AIDS, and cancer, too. The main epidemics in India are malaria, small-pox, cholera, typhoid, dengue, swine flue, cancer, AIDS, etc. AIDs according to a UN report caused 1,30,000 deaths in 2013.

Ebola fever in West Africa (Guinea, Sierra-Leone, Libara, etc.) has emerged as a serious epidemics during the last decade.

In 2015, the swine flu outbreak occurred in the states of Rajasthan and Gujarat. More than 2000 people died in the epidemic. The '*Nipah*' virus in 2018 resulted in a total of 19 deaths of which 17 were from Kerala. The outbreak was localised in the districts of Kozhikode and Mallapuram and three deaths due to '*Zika*' virus infection were also reported from Kerala.

The *Encephalitis syndrome* has resulted in death of over 153 children in the state of Bihar in June 2019.

ENVIRONMENTAL POLLUTION

A substance which causes an undesirable change in the physical, chemical, or biological characteristics of natural environment is known as pollution. Although there are some natural pollutants such as volcanoes, pollution generally occurs because of human activity. Biodegradable pollutants, like sewage, cause no permanent damage if they are adequately dispersed, but non-biodegradable pollutants, such as lead, may be concentrated as they move up the food chain. At present, air pollution—associated with basic industries such as oil refining, chemicals, iron and steel, and coal, as well as with internal combustion engine—is probably the principal offender, followed by water, and land pollution. Other forms of environmental pollution include noise and the emission of heat into waterways, which may damage aquatic life. Present day problems of pollution include acid rain and the burning of fossil fuels to produce excessive carbon dioxide.

There are four types of environmental pollution which have been described briefly in the following:

1. Air pollution
2. Water pollution
3. Soil and land pollution
4. Noise pollution

1. Air Pollution

The presence in the Earth's atmosphere of man-caused or man-made contaminants which may adversely affect property, or the lives of plants, animals, or humans. Commonly air pollution include carbon dioxide, carbon monoxide, lead, nitrogen oxide, ozone, smoke, and sulphur dioxide.

The tremendous increase of vehicles during the last three decades in the country has increased the air pollution, especially in the large cities. Consequently, the urban population is suffering more by cough, nausea, irritation of eyes, and various bronchial and visibility problems.

Because of the emission of carbon-dioxide, carbon monoxide, nitrogen oxide, and suspended particles of lead and heavy metals, the urban environment is more polluted than the rural environment. The air pollution is the main cause of many of the diseases and ailments. Air pollution has led to the increase in the incidence of some of the diseases like cough, shortness of breath, bronchitis, colds and fatigue, bronchopneumonia, lung infection, infection and anaemia, high blood pressure, nervous system, emphysema, angina, fatal arrhythmia, or myocardial damage, coronary disease, cancer in kidney, testis, brain, pancreas, stomach, lung, respiratory tract bladder and uterus, leukaemia and problems associated with gastro-intestinal and liver damage.

In addition to human beings, air pollution has damaging effect on vegetation, animals, and micro-organism.

Some of the important steps which can be helpful in reducing the air pollution are: (i) reduction in vehicular emission, (ii) improvement in the quality of diesel and petroleum, (iii) use of alternate sources of fuel, e.g., CNG (Compressed Natural Gas) and LPG (Liquid Petroleum Gas), (iv) more use of public transport, (v) improved vehicular technology, and (vi) mass awareness.

2. Water Pollution

Water is one of the most important requirements of human survival. Clean water is necessary for health and human development. Water

pollution has been defined as alteration of the physical, chemical, and biological properties of water, which may cause harmful effect on human and aquatic life. Unfortunately, over 70% of the available water in India is polluted. The main causes of water pollution are:

(i) **Urbanisation**: The unplanned growth of urban centres, especially that of the million cities, inevitably lead to huge quantities of domestic and industrial waste. The high energy consumption in the urban places give rise to large quantities of waste-water, sewage, and domestic trash. The inadequate development of sewerage disposal has polluted most of the rivers, lakes, ponds, and wells. Municipal water treatment facilities in India are not up to the international standard. The quality of water is poor in the urban areas, especially in the unauthorised colonies and slums.

(ii) **Industrialisation**: The industrial waste consists of chemicals, detergents, metal and synthetic compounds, besides the solid waste and garbage. These pollutants are generally discharged in the rivers. Consequently, most of the rivers, lakes, ponds, and wells in India are polluted by industrial wastes.

(iii) **Withdrawal of Water**: The water of most of the rivers is utilised for irrigation, industries, and domestic purposes. In the plain areas, generally the rivers contain very little water during the winter and summer seasons. What flows into the rivers is the water merging from the small streams and drains carrying untreated sewage and effluent. Rivers like Yamuna river, for example, contains very little water near Delhi. Each river must contain a minimum discharge throughout the year to maintain its ecology.

(iv) **Application of Plants Protection Chemicals:** After the introduction of High Yielding Varieties in the agricultural landscape, the use of chemical fertilisers, insecticides, and pesticides has increased substantially. These inputs can be used with success only in the areas where controlled

irrigation is available. In fact, these chemicals have changed the soil chemistry. The use of insecticides have killed the useful bacteria in the soil as well as the fertility enhancing earthworms.

(v) **Religious and Cultural Practices**: Religious practices have accentuated the pollution of problem of the rivers. The carcasses of cattle and humans are disposed off into the rivers. The deadbodies are often cremated on the river banks. These practices pollute the rivers. The Ganga is the worst affected river by such rituals.

Ground Water Quality

The underground water is largely consumed for irrigation, drinking, and industrial purposes. The underground water is being polluted by the heavy application of chemical fertilisers, plant protection chemicals, infiltration of contaminated water, and waste disposal. Once polluted, underground water may remain in hazardous condition for decades or even for centuries.

Consequences of Water Pollution

Consumption of polluted water is a major cause of poor health in India. The polluted water causes diseases like cholera, dysentery, diarrhoea, jaundice, and tuberculosis. About 80% of stomach diseases in India are water-borne. Children, especially in the urban slums and rural areas are the worst affected.

3. Soil and Land Pollution

Soil is the life support system of mankind. Soil becomes polluted either due to misdeeds of man or at times by the environmental hazards. The main factors of soil and land pollution are soil erosion, excessive use of chemical fertilisers and plant protection chemicals. Soil is also polluted by liquid and solid wastes from urban and industrial areas, forest fires, water-logging, related capillary processes, and mining wastes.

The soil pollution can be reduced by a judicious application of chemical fertilisers, insecticides, and pesticides. The urban and industrial effluents can be used for irrigation after proper treatment.

4. Noise Pollution

Noise is one of the important form of atmospheric pollution. Noise pollution may be defined as the state of discomfort and restlessness caused to human by unwanted high intensity sound. Noise pollution has increased in India considerably with the increase in urbanisation and industrialisation. The automobiles, factory machines, and loudspeakers at religious places are the main noise pollutants.

Noise pollution leads to impairment of hearing. It also results in mental tension, blood pressure, heart diseases, irritation, fatigue, and stomach trouble.

Noise pollution can be reduced by:

(i) locating the noise producing industries away from the residential areas,
(ii) replacement of old machinery,
(iii) minimum use of horn,
(iv) improvement in rail tracks, and
(v) to educate the younger generation about the adverse consequence of noise pollution.

ENVIRONMENTAL AWARENESS

The rapid increase in population and economic development have led to severe environmental degradation that undermines the environmental resource base upon which sustainable development depends. The economics of environmental pollution, depletion, and degradation of resources has in fact been neglected as compared to the issues of growth and expansion. India has been no exception to this worldwide phenomenon; rather, the trends in environmental deterioration in India, because of the substantial increase in population, have been far more prominent as compared to other developing economies.

The increasing pressure of population in the developing countries like India and over-consumerism in the developed countries are putting great stress on the resources and environment. A judicious

utilisation of the limited natural resources is imperative for the sustainable development and the very survival of humanity. Unfortunately, human activities are transforming the global environment, and these global changes have many faces: ozone depletion, tropical deforestation, acid depositrion, and increased atmospheric concentration of gases that trap heat and may warm the global climate.

Human activities now match or even surpass nature as an agent of change in the global environment, as evidenced by a growing list of seemingly diverse human-induced environmental changes that have gripped public attention in recent years. These are as follows:

1. Rapid changes in the global atmosphere due to fossil fuel combustion and industrial activities predicted to change global climate.
2. Massive ozone depletion.
3. Large scale destruction of tropical forests for timber, fuel, conversion to agriculture, and economic development, with consequent additions to the greenhouse effect and losses in plant and animal species.
4. Damaged lakes with acid deposition from fossil fuel combustion.

In order to overcome the problems of resource depletion, an environmental awareness is imperative at all levels of the societies. The scientists, politicians, decision-makers, planners, and public have to work collectively to keep the environment in a healthy condition for which a rational use of resources is required. Some of the steps which can go a long way in achieving these objectives are:

1. Population policies of the developing countries need to be oriented to check the uncontrolled growth of population.
2. There should be controlled consumerism in the developed countries.
3. There should be reduction in the emission of greenhouse gases.
4. Development aid policies should focus on improving the

indigenous scientific and technological knowledge base and infrastructure of the developing countries.

5. There should be rigorous environmental impact assessment of the developmental projects all over the world.
6. Environmental studies should be made a part of the syllabi at all the levels of education.
7. There is an urgent need of the participation of the corporate sector through investment in the programmes and projects of mass education and environmental awareness.
8. There should be involvement of government and NGOs to educate the masses on environmental issues.
9. Regular conferences, seminars, workshops and discussions on television and radio need to be organised in different parts of the country to develop environmental awareness.
10. More funds need to be provided for environmental research.
11. The media can play a very vital role in the propagation and education of environmental awareness.
12. Masses need to be educated about the wholeness need of the environment, i.e., the natural, social, economic, political, historical, technological, and aesthetic aspects of environment, and its significance for the survival of mankind.

CHANGES IN PATTERNS OF LAND USE

From the point of view of land use, the area of India was divided into the following five categories in 1950–51.

1. Forest.
2. Land not available for cultivation: (i) land put to non-agricultural uses, and (ii) barren and uncultivated land.
3. Other uncultivated land excluding fallow lands: (i) permanent pastures and other grazing lands, (ii) miscellaneous trees and groves not included in the net area sown.
4. Fallow-land: (i) old fallow, and (ii) current fallow.
5. Net area sown.

There are significant spatial and temporal changes in the patterns of land use in India. The radical changes in land use have occurred all over the country, but the changes in agricultural land use are substantial in the areas of green revolution. The changing patterns of land use during the agricultural years of 1950–51 and 2006–07 have been presented in **(Table 17.4)**.

Table 17.4

Changing Land Use Patterns—1950–51, 2006–07 and 2017–18

(Total Reporting Area 30.48 Crore Hectares)				
	Land Use	**1950–51 (Area in Percent)**	**2006–07 (Area in Percent)**	**2017–18**
1.	Forests	14.23	22.8	24.39
2.	Land not available for cultivation	14.72	14.2	12.63
3.	Other uncultivated land	17.39	8.6	10.27
4.	Fallow-land	9.89	8.2	7.13
5.	Net area sown	43.77	46.2	46.15
	Total	100.00	100.00	

Source: Directorate of Economics and statistics, Govt. of India, 2017–18

It may be seen from **(Table 17.4)** that the net area sown was 43.77% in 1950–51 which increased to 46.17% in 2017–18. There is an increase of about 10.16% in the forest cover. The area under the category of land not available for cultivation has decreased by over 2%, while the fallow land has declined from about 10% in 1950–51 to only 6% in 2006–07 and 7.13% in 2017–18. In fact, the double-cropped and multiple-cropped area has increased, and most of the farmers, especially in the areas of irrigation are not interested in keeping their land fallow. (*For details see Chapter-9*)

The agricultural land use has also recorded some substantial changes in the cropping pattern. The area under wheat and rice has increased at the cost of pulses, millets, and fodder, while the area under oilseeds, sugarcane, vegetables, and orchards has increased.

Unfortunately, about 7% of the good quality agricultural land has gone out of agriculture (urbanisation, industrialisation, and development of roads and railways, etc.) and the former fallow-land, pastures, and degraded forests areas have been brought under agriculture.

ENVIRONMENTAL IMPACT ASSESSMENT (EIA)

Environment Impact Assessment seeks to consider the probable consequences of human intervention in the environment so as to restrict environmental damage. The US National Environmental Policy Act of 1969 required such assessment to be drawn up for all major federal developments, giving information about the technology and location of the development, an appraisal of the likely environmental effects, both positive and negative, an outline of possible alternatives, and an estimate of any irreversible commitment of scarce resources.

Environmental Impact Assessment is necessary before the commencement of any project because the pollutants like the greenhouse gases and CFCs deplete ozone layer, and lead to global warming, acidification, acid-rains, desertification, and overall environmental degradation. The deterioration of environment adversely affects the quality of life and the very survival of mankind.

Environmental Impact Assessment is an imperative requirement before any major or medium project is installed. Projects like multi-purpose projects, thermal and nuclear power plants, defence installations, ordnance factories, irrigation projects, agro-based and mineral-based industries, national highways, railways, airports, and seaports require Environmental Impact Assessment.

In the environmental impact assessment, an inventory of *pollutants* is prepared in which an account is maintained of the inputs and outputs. It is also known as *environmental auditing*. On the basis of standards laid down by the government, the positive and negative consequences of the conceived project on economy, society, and ecology are assessed.The project should be economically viable, socially acceptable, and ecologically sustainable.

ENVIRONMENTAL MANAGEMENT

Our natural resources are limited and wants unlimited. A judicious utilisation of natural resources is imperative to make them sustainable. There is, thus, a need of conservation of resources. Conservation of environment (resources) means to use the resources in such a way so that the present generation may fulfil its needs without compromising the needs of the future generations. The main objectives of conservation are:

(i) To maintain the resilience characteristics of the ecosystem.
(ii) To preserve the biodiversity.
(iii) To enhance the sustainability of the environment.

Some of the measures which may go a long way in achieving these objectives have been given briefly here:

1. **Forest Conservation:** Forest resource is renewable but not inexhaustible. The rate of forest removal and their complete clearance by cutting and burning has been accentuated during the last few decades after Independence. About four thousand years back, the
   Indo-Gangetic Plain was under thick forest cover, but now hardly any forest is found in this physiographic fertile region. The rapid urbanisation and industrialisation during the post-Independence period, the requirements for construction of houses, railway sleepers, paper and timber, fuel-wood, and furniture, have degraded the Indian forests significantly. Moreover, forest fires and shifting cultivation have converted many of the lush green tracts into ecological slums. The following steps are needed to keep the forests in a healthy and regenerative condition:
2. **Afforestation:** The cultivable waste, village community land, fallow-land and degraded forest need to be brought under social forestry. Every tree cut must be replaced by a new tree. For this purpose, incentives should be given to rural females and youths. The forests can be conserved by adopting better

harvesting techniques. Only the mature trees be cut. Simultaneously, there is a need to reduce wastage. Waste paper could be recycled. The forest trees should be pruned for healthy growth.

3. Protection of forests from natural hazards such as large-scale fires, insects, and diseases need immediate attention.
4. Over-grazing should be prevented in the forest areas.
5. There should be more areas marked as reserved forests, national parks, and biosphere reserves.
6. Agri-silviculture and agri-silvi-pastoral systems should be encouraged.
7. Scientific rotation of crops can make the agricultural land more sustainable.
8. Recycling of metallic minerals should be encouraged.
9. Mass awareness is required to keep the environment in a sound condition.

(For more details read Chapters 5 and 6).

POPULATION EXPLOSION AND FOOD SECURITY

India is the second most populous country accounting for over 17% of the total population of the world. The scale at which India's population is increasing is simply mind-boggling. While the total population of the country was 361 million (36.1 crore) in 1951, it rose to 1210 million in 2011. According to one estimate, if the present trend continues the population will be 1400 million (140 crore) in 2025 and in that year India may surpass the population of China, achieving the first rank in the total population of the world. According to the Population Division of UNO, India's population is likely to stabilise in 2060, when the total population of India is likely to be 1700 million.

The rapid growth of population may be appreciated from the fact that a little less than 20 million (2 crore) persons are added every year, or about 17 lakh persons every month, or about 74 thousand persons every day, or about 52 persons every minute. The disaster of

population explosion in India is that every sixth person in the world population is an Indian. Number of people equal to Chandigarh's population are added to India's population in a week, and equal to Australia's population in a year.

At the present rate of growth of population, the life for most Indians would be unbearable—medical facilities would be difficult to provide, expenses on education, housing, etc., would be exorbitant, technical and professional education would become the exclusive prerogative of the elite, and scarcity of food once again may plunge more than half of the nation below the poverty line.

Causes of Rapid Growth of Population

Following are the main causes of rapid growth of population:

(i) High birth rate, low death rate.

(ii) Low age of marriage—over 65% of the girls get married before 18 years of age at which they are not ready for marriage socially and emotionally.

(iii) High illiteracy—The overall female literacy is low. Family planning has a direct link with female education, general status of women, their fertility, and infant mortality.

(iv) Religious attitude towards family planning. The religious orthodox views are against family planning.

(v) Joint family system and lack of responsibility of younger couples.

(vi) Low per capita income and low standard of living.

Effects of Population Explosion

The rapid growth of population has a direct effect on the living standards of the people. This is why, despite our spectacular progress in agriculture and industries, our per capita income has not increased appreciably.

As a result of population explosion about 30 million people are homeless, over 200 million have no access to safe drinking water, over 360 million are illiterate, over 40 million are unemployed, about 60% children below five years are underweight, and country as a

whole has 119th position among 177 countries in the World Human Development Index (2011). The overcrowding of cities has brought about a virtual breakdown of transportation, electricity, medical , educational, and recreational services. It has also led to the rise in crime and an increase in violence in urban and semi-urban areas. The fast growth of population has raised an army of unemployed, hungry and desperate people who are threatening the very foundation of our social, economic, and political systems in the country. It is futile to think of jobs for all or shelter for all or health protection for all.

Food Security

The issue of population explosion and food security was raised for the first time by Robert Malthus in his theory (1798) when he said that the world population grows in geometric progression and the means of subsistence in arithmetic progression **(Fig. 17.11).**

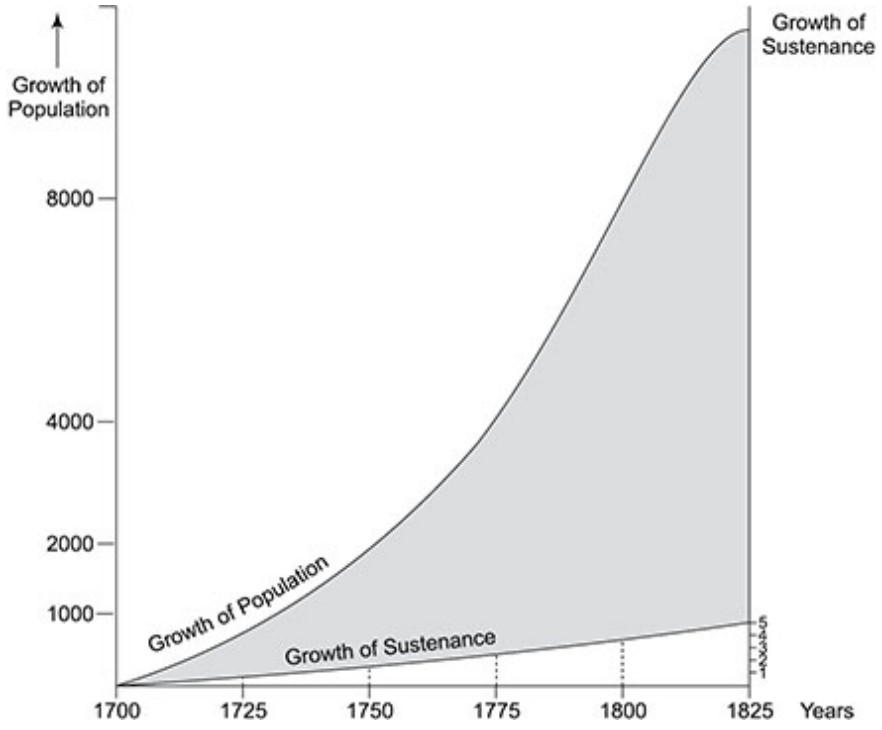


Fig. 17.11 Malthusian Model of Population Growth

The World Development Report, 1986 defined food security as a situation of “access by all people at all times to enough food for an active healthy life”. The Food and Agriculture Organisation (FAO) defined food security in 1986 as “ensuring that all the people at all times have both physical and economic access to basic food they need”.

According to the International Food Policy Research Institutes, Global Hunger Index, 2011, India’s food security situation continues to rank as alarming. India ranks 67 of the 81 countries of the world with the worst food security status. This means that there are only 14 countries in the world whose people have a worse nutritional status. The poorest people bear the heaviest burden when food prices spike or swing unpredictably. The world has crossed the milestone of seven billion people on 31st oct. 2011. It is distressing that 800 million people live in slums and similar number mostly women are not literate. Thus growing population can only have a negative outcome, depleting scarce resource faster. Education and women empowerment can play an important role to check the faster growth of population. Moreover, a rising quality of life can lead to voluntary stabilization of world population, which is projected by the United Nations to touch 9.3 billion by 2050.

In India, the population growth after Independence exceeded the agricultural growth. In order to maintain a balance between food and population, the annual agricultural growth should be over 4.5% and over all economic growth over 7%. At present, the annual agriculture growth is much lower being only around 2% **(Fig. 17.12).**

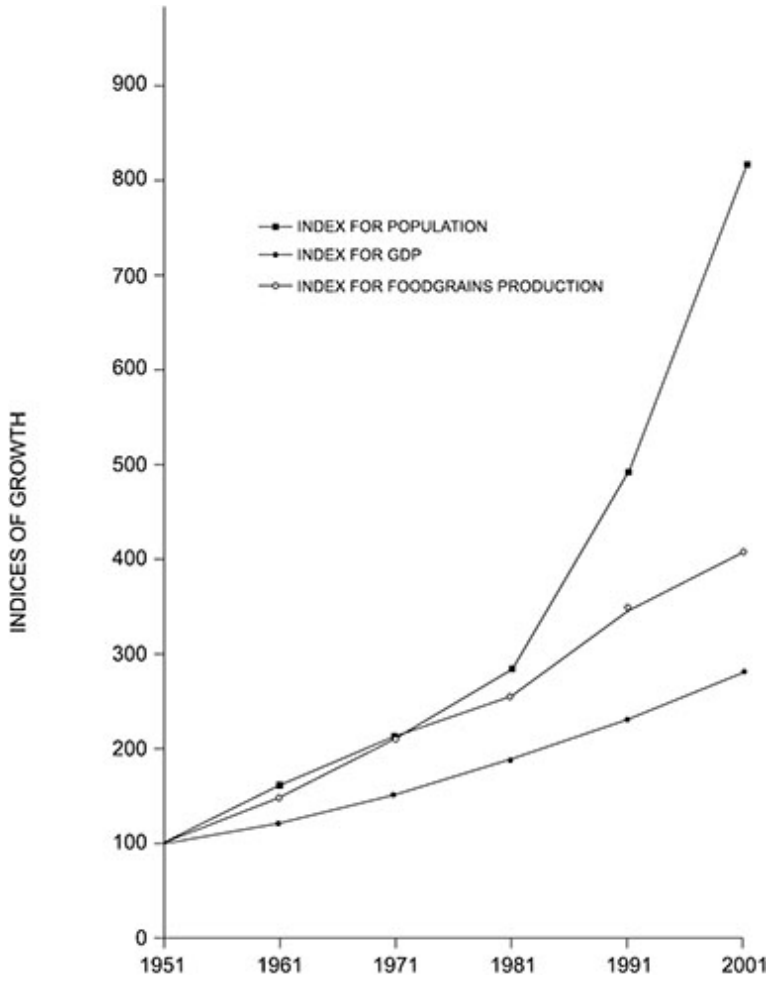


Fig. 17.12 Indices of Growth of Population, GDP and Foodgrains Production 1951 to 2001

Under the impact of rapid growth of population, the average size of holding has decreased to less than 1.5 hectares, and about 80% of the land holdings are less than one hectare, rendering most of the holdings economically unviable. The per capita availability of food has been decreasing for the last ten years.

Population explosion is the most challenging problem, the country is facing. Today, the food security system is near collapse. As against the total requirement of 25 million tonnes of food grains for buffer stock and Public Distribution System (PDS) together, the public stock is over 60 million tonnes. A substantial share of this is not even properly stored and may not be suitable for human consumption. On

the other hand due to large-scale unemployment, hunger, and starvation, deaths are reported from several states. The PDS system introduced during the inter-war period, which was emanated in 1960, and the Targeted Public Distribution System (TPDS) introduced in 1997 were virtually collapsed. Poor people cannot afford the 'so called economic price' of food-grains available in the Public Distribution Shops (PDS).

The population explosion and the food security problems need to be tackled by taking the following steps:

(i) Delay in the marriageable age—21 years for bride and 24 years for bridegroom.
(ii) Implementation of strict two-child policy.
(iii) Implementing an efficient Public Distribution System.
(iv) Ensuring sustainable agriculture.
(v) Generation of more employment in the rural areas.
(vi) Scientific rotation of crops based on mixture of cereals and leguminous crops.
(vii) Water harvesting and moisture conservation.
(viii) Diversification of agriculture.
(ix) Credit facilities to the rural marginal farmers, and unemployed youths.
(x) Use of marine food resources.
(xi) Change in the food habits.
(xii) Check urban and industrial encroachment on good quality of agricultural land.

URBAN DISASTERS

As the cities are growing as the centres of administrative and economic importance, there is an unregulated and uncontrolled growth of the population. The '*Metropolitanisation*' is leading to stagnant growths of the smaller towns. The migrant population settles in marginalized areas like floodplains, unstable slopes, unhygienic slums and squatter settlements. The impacts of the climate changes

are also making the cities susceptible to the environmental risks. The risks from natural hazards arise due to the location of an urban area increase due to problem in the built up forms, spatial planning, networks, complex emergencies and disasters.

By 2025, it is expected that 65% of the population of the world shall live in the cities. The developing countries are at a greater risk since it is expected that 70% of the world's urban population shall be living in poorer countries. In India, 31.2% of the population lives in the cities. Moreover, 85% of the area is vulnerable to one or more hazards. About 59% of the Indian landmass is prone to earthquakes and landslides of different magnitudes and about 12% of its geographical area (about 40 million hectares) is subject to riverine and flash floods of which about 8 million hectares are susceptible to annual flooding.

Indian cities are at a higher risk because of poverty, poor education, limited access to water, sanitation, lack of urban planning and governance contributing to the urban disasters. Mumbai, Chennai, Delhi, Bangalore and Hyderabad are placed at high risk according to Mapplecroft Index.

Disasters like earthquakes, fire outbreaks, chemical leaks are some of the examples of the urban disasters. About 60 cities with population with more than half a million lie within zones III, IV and V. With return periods of earthquakes ranging from 5 years to 50 years, severe earthquakes hitting highly populated cities is expected.

Cities that are located in various climatic or geological zones are prone to hazards of varying nature. The challenge is to create resilient cities, based on the premise that disasters are a reality most cities have to face. A resilient city can bend due to a disaster, but will not break down entirely because the systems, people and functions are integrated in a risk resilience format. Stricter building laws and urban planning, disaster risk reduction and contingency plans for restoring transport and amenities, better education and awareness are required.

(Source: *NIDM, Ministry of Home Affairs, Govt. of India)*

ENVIRONMENTAL DEGRADATION

Environmental change is a continual process that has been in operation since the earth came into existence about 4600 million years ago. Since then, dynamic systems of energy transfer and material transfers have operated on a global scale to bring about gradual and sometimes catastrophic transformations of the lithosphere, atmosphere, hydrosphere, and biosphere. For most of the history of the earth, the agents of change have been the natural forces like volcanoes, earthquakes and natural elements of running water, wind, glaciers, sea-waves, plants and animals. About 4.5 million years ago, however, a new agent of change emerged in the form of *homonids* (ancestor of ape) culminating in the revolution of *homo sapiens* or *modern humans*, that are considered by many to be the most powerful generators of environmental change.

Man is the product of the earth (Miss Semple). He is, however, a creator of the cultural landscape and a transformer of the physical landscape. In fact, human beings have dominated ecology since their emergence on the earth. Man has interacted with his physical environment right from the primeval stage of human development and continues to do so even today in this age of space technology with greater vigour. With the advancement in science and technology, the frontiers of knowledge have expanded beyond the cosmic space and the environment of the earth has transformed beyond recognition. People, in increasing numbers, with their technological skills have placed their imprints upon the natural landscape and environment, transforming the physical surroundings to conform to their needs. In some instances, such as that of mega-cities, conurbations, and modern towns on land and seas, the human imprints may be so complete that the original landscape of nature has been totally wiped away and replaced by man-made cultural environment.

About 10,000 BP (Before Present), the Neolithic Period started. It was during this period that man domesticated plants and animals. The emergence of agriculture has been regarded as revolutionary. Obviously, it was a very important development in the human history.

The start of agriculture provided a more reliable food resource. This might have enabled the development of permanent settlements, and higher population to be supported. The beginning of agriculture played a dominant role in the development of human communities and resulted into significant environmental changes. The natural systems were transformed into agro-ecosystems. With this development, homo sapiens started to control the ecosystems more vigorously.

The domestication of plants and animals brought about substantial changes in the physical environment. The introduction of domesticated species provided a food source that supplemented natural plant and animal resources. In this sense, diversity of species increased, but as clearance of natural vegetation was undertaken to make way for arable fields, diversity of species at least locally, decreased. Geomorphic and hydrological systems were also radically altered as cultivation replaced growth of natural vegetation.

With the passage of time, apart from the expansion and intensification of agriculture, the most conspicuous development during the historic period was the Industrial Revolution in Britain in 1779. The industrial revolution led to industrialisation and urbanisation. This revolution brought about environmental changes not only in Europe and Asia, but also transformed tremendously the environment in America, Africa, and Australia. The Europeans did large-scale deforestation in these continents for the expansion and development of agriculture which brought about further changes in ecology and environment.

The extraction of minerals became imperative for industrial development. The extraction of minerals at a large scale, however, creates environmental changes in the immediate vicinity of the mining centres. Mining also results into a widespread change in soil, aquatic and atmospheric environmental quality at distance which may be related to both extraction methods and mineral processing. Of particular significance is the use of fossil fuels in mineral processing and vehicular traffic that are contributing to enhanced greenhouse

effect. One of the major causes of changes in environmental quality is contamination by heavy metals such as lead, cadmium, and mercury.

The over-interaction of man with environment has resulted into the following adverse consequences**:**

(i) **Increase in Greenhouse Gases:** One of the most important causes of environmental degradation is the widespread use of fossil fuels. The use of fossil fuels is increasing concentration of carbon dioxide in the atmosphere. The increase in greenhouse gases is the main cause of climatic change and rise of sea level. Consequently, the biosphere and hydrosphere are adversely affected.

(ii) **Acidification:** Acidification is damaging the aquatic ecosystems. According to ecologists, acidification has damaged the marine and aquatic ecosystems in lakes and rivers. The fish stock in the rivers of India is declining at a faster pace **(Fig. 17.13)**.

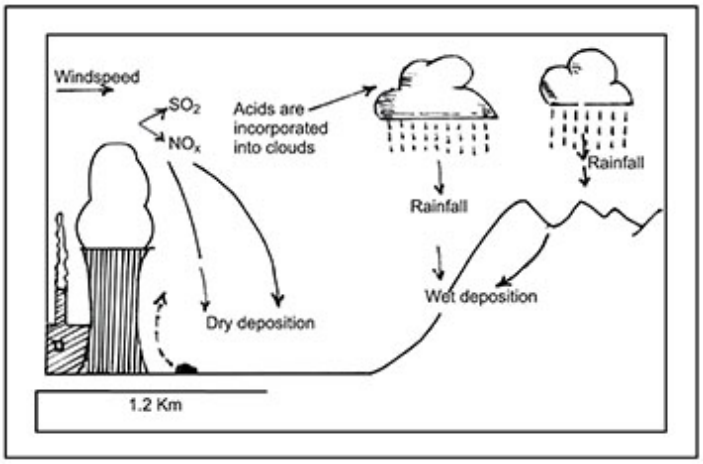


Fig. 17.13 Acidification

(iii) **Ozone Depletion:** Ozone is formed from reaction between gases in the air under the influence of sunlight and high temperature. The presence of ozone in the atmosphere is particularly important because it filters out the incoming ultraviolet (UV) radiation and thus, acts as a screen against

ultraviolet B (UV-B) radiation that can increase the occurrence of some forms of skin cancer, cataracts and other diseases of eye **(Fig. 17.14**).

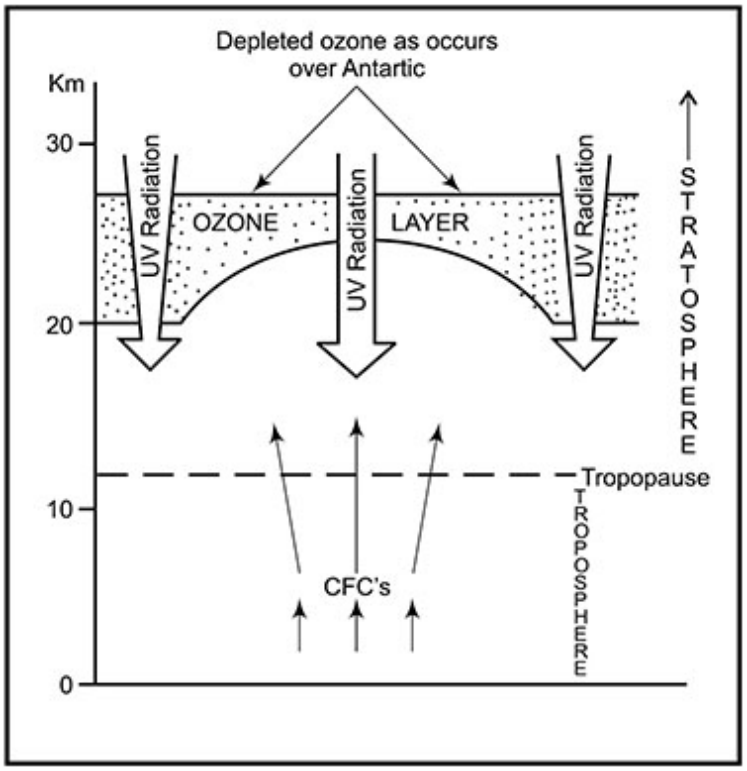


Fig. 17.14 Ozone Depletion [*Source: After Farman, 1985*]

Analysis by Centre for Science and Environment (CSE) reveals that the capital city is experiencing greater ozone pollution. The eight hour standard for ozone exposure is 100 micrograms per cubic meter. The levels of ozone have exceeded by 16% (122 µg/m3) as against 5% (106 µg/m3) in 2018. Delhi's hotspots have been identified as Najafgarh (92%), Sri Aurbindo Marg (87%), Narela (80%), Bawana (78%), Siri Fort (76%). (Source: T.O.I.; June 20, 2019)

(iv) **Air, Water, Soil, and Noise Pollution:** The quality of air, water, and soil is depleting both in the urban and the rural areas. The air and water borne diseases in the country are increasing. The population of the mega and million cities are

the worst affected. In the states of Punjab, Haryana, and western Uttar Pradesh, the quality of underground water has gone substantially below the consumption level. Same is the case in Bhopal, Delhi, Jaipur, Meerut, and Saharanpur.

(v) **Problem of Disposal of Waste Material:** The rapid industrialisation and urbanisation is leading to huge quantities of domestic, industrial, and nuclear wastes.The industrial wastes consist of chemicals, detergents, metals, and synthetic compounds besides the solid waste and garbage. The heavy metal chemical pollute the water system which may be a threat to aquatic life and human health. In addition to urban and industrial wastes, the nuclear fuel is becoming an increased nuisance. The nuclear waste contains radioactive isotopes which generate large quantities of heat. These radio-active elements remain in the atmosphere for many hundreds and thousands of years. The disposal of nuclear waste is, therefore, a serious danger to the biosphere.

(vi) **Soil Erosion:** Soil erosion is a universal problem. It is all the more serious in the tropical and sub-tropical countries like India. In India, the problem of soil erosion is a serious menace as about 6000 million tonnes of soil is being eroded annually. According to one estimate, about 80 million hectares is adversely affected by soil erosion, out of which over four million hectares is affected by ravines and gullies. Soil erosion affects not only the areas from where soil is removed but also the ecosystems and environment of the places where the eroded soils are deposited.

(vii) **Salinisation:** The application of chemical fertilisers in heavy doses in the irrigated parts of the country have salinised the rivers, lakes, ponds, and underground water-bodies. Consequently, many of the water-bodies and wetland habitats have suffered adversely and their fauna and flora have been damaged or disappeared.

(viii) **Desertification:** The causes of desertification are numerous. The over interaction of humans in the form of overgrazing, deforestation, and encroachment of forests for agricultural land have increased the desert area and resulted into more desertification within the Thar Desert.

(ix) **Deforestation:** Deforestation is a cause and effect of environmental degradation.The density of trees in the Indian forests is decreasing at a faster pace. There is only about 22.5% of the total reporting area under forest. Destruction of forests changes surface and air temperatures, moisture content, and reflectivity. Conversion of forests to grasslands, for example, increases surface temperature, raises air temperature above the treeless ground and therefore, increases the water-holding capacity of warmer air. As the wind moves and becomes hotter and drier, it tends to exert a drying effect on adjacent forests and agricultural lands.

Afforestation is the need of the day to keep the forests in a healthy condition. Moreover, social forestry needs to be developed in the community land and the degraded forest areas of the country.

(x) **Reduction in Biodiversity:** The rapid growth of population in India, especially after Independence resulted into cutting of trees and deforestation. A major setback of deforestation is the loss of biological diversity. Many of the medicinal plants and useful insects are already in danger.

PROBLEMS OF AGRARIAN UNREST

Despite over sixty years of planning, there is enough unrest both in the agricultural and the industrial sectors. The agricultural and industrial unrests are the manifestation of collective disillusionment, discontentment, and frustration of the agricultural and industrial workers. In fact, agricultural and industrial unrest brings collective discontentment among all the workers in agriculture and different industries in the country. The main issues may be minimum wages, employment, security of employment, and social amenities. Thus, the

emphasis in the agricultural and industrial unrest is on 'collective frustration and disillusionment on common issues of these groups in the society.' Apart from labour unrest, according to the National Crime Records Bureau (NCRB), the total number of farm suicides since 1995 has reached the figure of 270,940. The state of Maharashtra leads in the list of farmers' suicide (Hindu, 7.7. 12).

The number of the unemployed persons in the country registered in the Employed Exchange was 4.37 lakh in 1952, which rose to over 42 million (5 crore) in 2011. Out of which about 30 million are directly or indirectly dependent on agriculture. There is greater unemployment in agricultural sector than in industrial and tertiary sectors.

During the post-Independence period, the main agrarian movements occurred are:

(i) Telengana Movement (1967), (ii) Bhil Movement (Dhulia 1967–75), (iii) New Farmers Movement, Nasik (1980), (iv) Bhartiya Kisan Union (1986) (U.P., Haryana and Punjab), (v) Bharatiya Kisan Sangh (BKU) in Gujarat, (vi) Tamilaga Vyavasavavigal Sangham (TVS) in Tamil Nadu and

(vii) Karnataka Rajya Ryota Sangha (KRRS) in Karnataka.

Causes of Rural Unemployment

The main reasons for the agricultural and rural unemployment may be:

(i) **Unchecked Growth of Population:** There is surplus labour in most of the rural families because of population explosion in some of the states of the country. In fact, there is more supply of labour than demand.

(ii) **Seasonal Employment**: There is peak labour demand in agriculture at the time of sowing and harvesting, and rest of the time is generally lean period when the agricultural workers are without work. According to one estimate, the Indian cultivators do work for only about 180 days and the dependent labourers even less than six months.

(iii) **Small Size of Holdings**: Agricultural unemployment is also caused due to the small size of holdings which may not

absorb the family workers. In general, a cultivator in India remains unemployed for about four to six months in a year.

(iv) **Technological Unemployment**: The technology of tractors and other agricultural machinery has increased in the agriculturally developed parts of the country. The tractors and harvesters are a displacement of agricultural workers.

(v) **Illiteracy and Inadequate Education**: The uneducated or semi-educated youth cannot finding jobs in the secondary and tertiary sectors.

(vi) **Symbol of Low Social Status:** The semi-educated youth consider agricultural work below dignity.

(vii) **Dishonest Leadership**: Education and awareness has increased in the rural society as a result of which, the rural unskilled and skilled youths, who are unemployed become angry when they notice the widening chasm between what the leaders preach and what they practice, when the leaders call for sacrifice, they themselves loll in luxury; when the leaders talk morality but they themselves maintain liaison with smugglers, criminals and anti-social elements; when they appeal for peace and harmony but they themselves revel in factional squabbles; when they shed crocodile tears for the poor but always live with and support the rich. Disappointed and disillusioned by this, the frustrated rural youth agitate and protest. This is also an important contributory factor of agricultural unrest. The agricultural unrest is mainly because of unemployment and under-employment. The activities of the unemployed are so restricted and their attitude so bitter that in this phase of disillusionment and discouragement, they lose their desire to work and their skill may deteriorate with a resultant loss to the whole community.

In many parts of the country, agricultural unrest has taken shape of a serious menace and shaken the administration and the governments. The Naxalite movement in Andhra Pradesh, Bihar, Chhattisgarh, Jharkhand, Karnataka, Madhya Pradesh, Odisha and

parts of West Bengal has become a serious threat to the administration and governments. In some of the states, *Bhoomi Sena*, and *Ranvir Sena* have been launched. The Government of Uttar Pradesh has organised the *Bhoomi Sainik* force to create work and to help landless labourers to lead a life of economic self-sufficiency. Under this scheme, one *sainik* gets one acre of land for afforestation. The *Adarsh Gram Yojna* also provides employment to the rural unemployed.

The Unemployment Insurance can be a right policy. If unemployment insurance scheme is introduced in India, can this scheme be funded in a sustainable manner? With the increasing unemployment and increased cost of living, how can our government fulfil the rising expectations of the people? The problem needs to be tackled at several fronts. There is a need to adopt a more strict two-child policy of population with increasing education and awareness among the masses in general and Hindi speaking states in particular.

Agricultural Unrest and Remedial Measures

The agricultural unrest and its remedy depends on what is our agricultural policy and what are our goals? Is our goal to accelerate the growth rate of agricultural output, or achieve national food self-sufficiency, or keep enriching the large farmers with periodic hikes in procurement and support prices, or make work available to all cultivators and the agricultural labourers in the villages? So far our emphasis had been to give the highest priority to production of foodgrains. But during the last ten years, there has been a decline in the cultivated area and the foodgrains. The per capita availability of foodgrains now shows a declining trend. And now comes the wheat scam of the Food Corporation of India (FCI), which reveals the functioning of agricultural sector. The Food Corporation of India diverted wheat meant for domestic consumption to export and black-market circuit.

The Government of India and the state governments have taken a number of steps to reduce the unrest of agricultural workers. The first measure adopted to contain rural unemployment was the introduction

of Prime Minister's cash programme in 1972–73. In 1980, the *Kam Ke Badle Anaj* programme was introduced. The IRDP and Jawahar Rozgar Yojna (JRY) in April 1989 were very significant to tackle the growing rural unemployment.

Is the guarantee of the *Right to Work* feasible in the present situation of our country? The feeling is that the slogan 'Right to Work' is nothing but a digressive, escapist, tangential, flippant, and comic proposal. The 'Right to Work' is neither viable nor enforceable without massive inputs. It is a distant fantasy and an empty slogan. Where are the resources and billions and trillions of Rupees to give jobs to over 40 million unemployed persons considering the large amount of money that is needed per year to create even one job? Will this slogan not lead to wholesale frustration and disillusionment all around? Is not 'Right to Work' linked with obtaining a mandate from a court of law to get a job and the right of employer to terminate the employee's services, collective bargaining, and so forth? Is it not contradictory to the right of reservation? Do we, therefore, not find a pathological fallacy in the thinking of politicians who give such populist slogans and promises?

There is a need to create awareness about the self-employment in the rural areas. Self-employment in rural areas can be found in: (i) farming on modern scientific lines, (ii) small scale industries, (iii) agro-industries, (iv) forest-based industries, (v) dairy farming, (vi) poultry farming, (vi) village arts and crafts (like hand-made paper, pottery, leather goods, hand-printing and dyeing of textiles, carpets, rugs, etc.).

INDUSTRIAL UNREST

Industrial unrest means the collective discontentment among all workers in different industries in the country on issues of minimum wages, safety measures, security of employment, and certain intra-mural and extra-mural facilities that is referred to as the problem of 'industrial unrest'. In other words, it is a collective frustration and disillusionment on common issues of the groups in the society. The

industrial unrest, in fact, means, conflict between employees and workers in industries. The industrial workers display their protest in the form of strikes, *gherao*, *bandh*, slow tactics, rallies and demonstrations. The employers (industrialsts) show their might by retrenchment, dismissal, lockouts, etc. Industrial unrest causes industrial recession, decline in production and national income.

Causes of Industrial Unrest

The main causes on industrial unrest are:

(i) The upsurge in the birth rate or rapid growth of population and the consequent supply of surplus labour.
(ii) The hire and fire policy of the industrialists and employers.
(iii) The cyclical unemployment in industries, which is caused because of ups and downs in trade and business.
(iv) Large scale migration of the people from rural to urban and industrial areas.
(v) Losses occurred by industries.
(vi) Slow growth of industries.
(vii) Competition with foreign industries.
(viii) Most of the Industrialisation is unplanned.
(ix) Lack of welfare and social security.
(x) Frequent instances of labour strikes or employers' lockouts and rationalisation.
(xi) Rising wages and low productivity.
(xii) Liberalisation and Globalisation.
(xiii) New lifestyle.

In addition to the above factors, the introduction of new technology also leads to technological unemployment. The automation or other technological changes in industry results in reduction of manpower necessary to produce a finished product. Throughout the course of economic development, particularly since the industrial revolution man has been forced to adjust to the process of mechanisation. Industry has diminished an average man's economic security since

every advancement in technology leads to the displacement of human labour.

Thus, improvements in machine technology, over-production, falsely stimulated speculation, social emphasis upon monetary success and the inevitable depression—all these make for crippling disruption in the demand of labour which ultimately leads to industrial labour unrest. The Maruti Suzuki India Limited Plant (MSIL) faced a serious labour problem of unrest in July, 2012. The company accused 147 workers, some of them under Section 302 of the Indian Panel Code (IPC) of killing the HR Managers of the industry.

The central and states governments have taken several remedial measures to redress the grievances of the industrial workers. The solution of the unrest, however, may not be found unless the growth of population and supply of labour are checked. Moreover, there is an urgent need to implement the self-employment schemes more vigorously.

The industrial unrest may be reduced appreciably if great emphasis it put on: (i) creation of opportunities for self employment; (ii) augmentation of productivity and income levels of the working poor; and (iii) shift in emphasis from creation of relief type of employment to building up of durable productive assets.

REGIONAL DISPARITIES IN ECONOMIC DEVELOPMENT

A disparity between the standards of living applying within a nation is known as regional disparity. It is difficult to quantify the prosperity or poverty of a region, but there are two basic indicators. The first indicator is the *per capita income* and the second is *unemployment*. Other indicators of disparity are the type of industry, and its growth or decline, numbers of young people in further education, housing standards, and the quality of the environment. In the opinion of some of the experts, economic development brings about regional inequality.

Regional disparity shows the uneven development. Uneven development is the condition of an economy, which has not benefitted

equally from development either in a regional sense and/or within classes in society. It may also occur between ‘consumer goods’ and ‘capital goods’ industries, and between sectors of economy. It may be seen as a result of capitalism, based on competition and accumulation, but it is not unique to capitalism. State socialism has often led to concentration on one sector at the expense of another.

Regional imbalances and regional inequalities is a universal phenomena. Most of the countries of the world are faced with the problem of regional disparity. The problem is, however, more serious in the developing countries like India where it has assumed such a magnitude that their political and economic stability are threatened.

The reasons for the regional disparities are not only physical but also historical, socio-economic, and political. Some regions of the country are endowed with more natural resources (soil, water, forest, minerals, etc.). The variations in natural resources in the different parts of the country may create regional disparities.

India, during the period of Akbar the Great, was considered as prosperous a country as the best in the world. Though mainly agrarian, India was a leading manufacturing nation, at least at par with pre-industrial Europe. India lost its relative advantage only after Europe achieved a revolution in technology (Industrial Revolution, 1779).

During the Mughal period, the economy was village-based, though under Muslim rule for over 500 years, the society continued to be organised according to Hindu traditions. Caste system was intact. The social disparity often added another dimension to economic exploitation. While the *Jajmani System* ensured social security, the caste system ensured social immobility.

During the medieval period of Indian history, there is historical evidence to indicate that there were food surplus and deficit regions, as trade in food-grains between regions took place. Indian village was highly segmented both socially and economically. There was significant inequality in distribution of farm land. In comparison to rural rich, the urban merchants in coastal towns and cities were much

wealthier. The pattern of life of the nobility and the upper class in Mughal India has become a byword for luxury and ostentation. There is hardly any evidence to show that the puritan style of life set up by Aurangzeb had any marked effect on the lives of the nobility.

Regional Disparities during British Period

Regional disparity in economic development in India is a British legacy. Some of the important causes of regional disparity are given below:

1. Raw Material Based Industries

The mechanised industrial development in India started mainly in the middle of the nineteenth century. In the location of industries the availability of raw material was the main consideration. In such areas once the economic development started, they attracted more industries, especially industries where raw material considerations played an important role in deciding the location. For example, the location of jute, cotton textiles, sugar, tea, and mineral-based industries were located in the areas where their raw materials were available.

2. Historical Factors

The present regional disparity in India has its roots during the British period. In India, the historical factors guided the development of the port cities of Mumbai (Bombay), Kolkata (Calcutta), and Chennai (Madras). These three cities have, in turn, worked as nuclei for the development of Maharashtra, West Bengal and Tamil Nadu respectively, which are, at present, the most industrially advanced states of the country. On the other hand, the areas having natural advantages in the form of mineral resources such as Bihar, Odisha, Madhya Pradesh have lagged far behind.

The interest of the ruling countries was to ‘use’ the dependent countries or colonies both as supplier of raw material to home country’s manufacturing industry and as a market for its products.

3. Migration and Investment of Capital

The dynamic, healthy, skilled, and young labour that migrated to the industrial centres of Mumbai (Bombay), Kolkata (Calcutta), and Chennai (Madras), affected the age structure. The age structure of the place of origin of migration became lopsided where the dependency ratio increased resulting into an excessive burden of children and old people. The economy of such areas could not flourish and lagged behind. Moreover, because of increasing investment opportunities and expectations of higher profits, capital also moved to the rich region. This outflow of capital further depressed economy in backward regions. Consequently, a circular process of more investment and more income was initiated in the rich regions, while an opposite circular process operated in the poor regions.

4. Transport Development

During the pre-Independence period, the processes of industrialisation and urbanisation were closely linked with the development of port-cities of Bombay, Calcutta and Madras in India. From its beginning in 1853, India's railway system expanded rapidly to become, by 1910, the fourth largest in the world. The development of transport (and especially railways) also increased the regional disparities as all parts of the country were not well connected by railway network.

5. Development of Canal Irrigation

In order to overcome the problem of food shortage and famine, the British Government in India developed canal irrigation by digging the Upper-Ganga Canal (1854), the Lower-Ganga Canal (1878), the Agra Canal (1878), the Bari Doab Canal (1878), etc. The benefit of irrigation development were mainly concentrated in northern, western, and southern provinces during the British Period. Central and Eastern India were relatively neglected.

The historical growth processes described above explain the creation of regional imbalances in the early stage of development and modernisation in India.

Regional Disparities after Independence

Disparities in economic and social development across the regions and intra-regional disparities among different segments of the society have been the major plank for adopting planning process in India since Independence. Apart from massive investment in backward regions, encouragement to private investors in such regions was given. While efforts to reduce regional disparities were not lacking, achievements were not often commensurate with these efforts. Considerable level of regional disparities remained at the end of the 1970s. The accelerated economic growth since the early eighties appears to have aggravated regional disparities.

The fifteen large states of the country accounting for over 96% of the total population of the country may be grouped into two categories:

(i) the forward group which consists of Andhra Pradesh, Goa, Gujarat, Haryana, Karnataka, Kerala, Maharashtra, Punjab, and Tamil Nadu; and

(ii) the backward group which comprises Assam, Bihar, Chhattisgarh, J&K, Ladakh, Madhya Pradesh, N.E.Hill states, Odisha, Rajasthan, Uttarakhand, Uttar Pradesh, and West Bengal.

Geographically, the forward group of states fall in the Western and Southern parts of the country except Punjab and Haryana, while the group of backward states are in the Eastern and Northern parts of the country making a contiguous region **(Fig. 17.15).**

The population growth in the forward states is below the national average, while in the backward states it is significantly high. Moreover, the literacy rate in the forward states is much higher to that of the backward states. The per capita income of the forward states has gone up, while in the backward states it is declining.

An important aspect of regional disparities in India is the significant regional, inter-district, and intra-district disparities in economic and social development. An important cause of regional tensions which lead to popular agitation and at times militant activities and in some

cases the creation of some of the states in the past. The broad basic trends that emerged in regional inequality after independence are given here.

1. Disparities in Per Capita Income

In respect of per capita income, the states of Punjab, Maharashtra, and Haryana, have continuously maintained a considerable lead over other states. Many states have continuously remained backward in terms of per capita income. The position of some of the states like Bihar, Chhattisgarh, Jharkhand, Jammu & Kashmir, Madhya Pradesh, Odisha, Rajasthan, Uttarakhand and Uttar Pradesh appeared to be particularly bad.

2. Disparities in Literacy Rate

The literacy rate according to the Census of 2011 was over 73%; it means that about 27% or about 327 million (32.7 crore) of the total population was illiterate. In 1951, the total population of India was 36.1 crore. Most of the illiterates are living in the backward states of Assam, Bihar, Chhattisgarh, J&K, Jharkhand, Madhya Pradesh, Odisha, Rajasthan, and Uttar Pradesh. The absolute number of illiterates is bound to increase in these states. Unfortunately, there is more illiteracy in the Hindi belt in comparison to other regions of the country. Ashish Bose has termed the four states in the Hindi belt—Bihar, Madhya Pradesh, Rajasthan, and Uttar Pradesh—as '*Bimaru*' (sick) states.

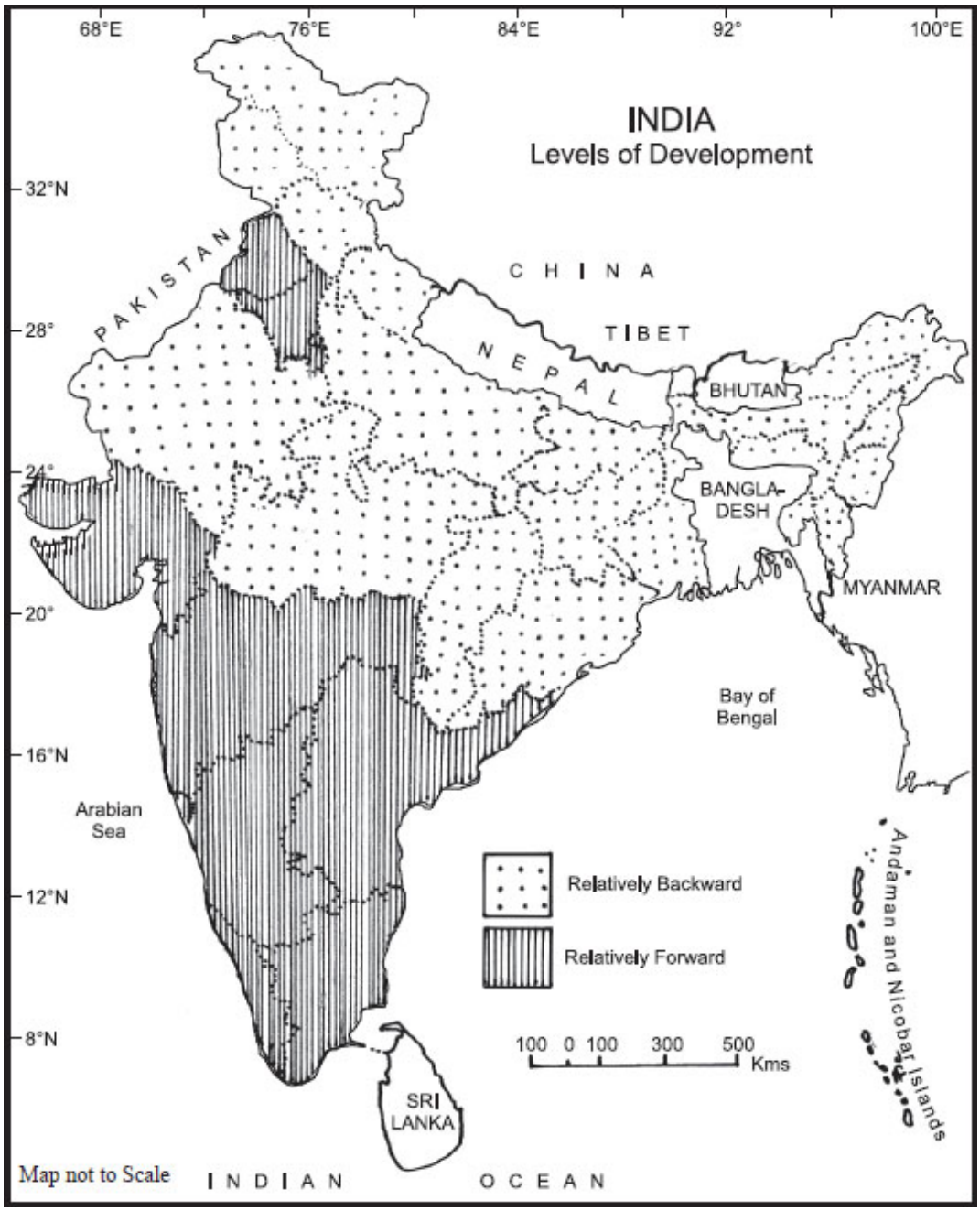


Fig. 17.15 Relatively Backward and Forward Areas

3. Disparities in Longevity

The longevity in India has gone up from 37 years in 1951 to about 66.4 years in 2013. Both Pakistan and Bangladesh are better when compared to India in average age. The life expectancy is high (around 70 years) in Punjab, Haryana, Gujarat, Maharashtra,

Karnataka, and Tamil Nadu, while it is below 55 years in Assam, Bihar, Madhya Pradesh, Odisha, Rajasthan, and Uttar Pradesh. The average longevity of whold world in 70.8 years, in China 75.3 years, while in developed countries 80.2 years.

4. Disparity in Industrial Growth

As stated above, the initial distribution of industries was determined by the historical processes reflected in the development of transport facilities and the interest of the British rulers.

Thus, at the time of Independence, we inherited a lopsided pattern of industrial development with most of the industries concentrated at a few centres. After Independence, most of the industries remained concentrated in Gujarat and Maharashtra.

The states of Punjab and Haryana made tremendous progress in the agro-based industries after Independence. The per capita consumption of electricity was highest in Punjab, followed by Maharashtra and Gujarat. The under-developed states of Bihar, Madhya Pradesh, Rajasthan, and Uttar Pradesh remained way behind these states in industrial consumption of electricity.

5. Disparities in Agricultural Growth and Development

After the introduction of High Yielding Varieties of wheat and rice, the production and productivity has substantially increased in Punjab, Haryana, western Uttar Pradesh, Gujarat, and Tamil Nadu. The performance of agriculture is, however, much below in Bihar, Chhattisgarh, Jharkhand, Madhya Pradesh, Odisha, and Rajasthan. The erratic and low performance of agriculture is affecting the standard of living of the people adversely.

6. Disparities in Private Investments

The flow of private investment, both domestic and foreign, has been extremely biased in favour of the more developed regions/states of the country because these regions were having better infrastructural facilities. The backward regions of the country, which were unable to attract any significant private investment flows, experienced decelerated economic growth during the post-Independence period.

The poor infrastructure and the poor law and order in the backward states discouraged the private investors.

7. Illiteracy, Corruption and Lack of Political Vision

The above mentioned factors are also responsible to a great extent for regional disparities in India.

If the post-Independence trend of socioeconomic development continues, India will be highly uneven nation in terms of quality of life. Income and standard of living will vary considerably across the country. People in most of the southern and western parts of the country will be enjoying fairly high per capita incomes, which may be comparable to those of middle income developed countries today. The literacy rate will be high and the gender difference will disappear. The situation will be just the reverse in the backward states.

Profile of Regional Disparities for Different Growth Scenarios—2025

The historical and the post Independence trends show that disparities are bound to increase in the coming decades. The backward regions have very weak growth impulses. The backward states like Bihar, Jharkhand, Madhya Pradesh, Chhattisgarh, Odisha, Rajasthan, and Uttar Pradesh have the demographic disadvantage. Because of fast growth of population in the backward states, employment opportunities, social sector investments, and the overall development can be very slow.

RELATIONSHIP BETWEEN POPULATION AND DEVELOPMENT

Population and development are closely interrelated. There are a number of population traits that influence developmental issues. In the same way, the developmental issues also affect the different traits of population. The ways by which population issues are affected also vary from region to region; for example, after the Industrial Revolution, there was a need in the West (Europe) to have a large population to use the manufactured goods.

In contrast, it is alleged that the increase in population in countries of the East became an obstacle to development. It puts pressure on the finite resources and their utilisation. Since there is no industrialisation on a large scale, the demands of the increasing population could not be fulfilled. A major section of the population depends on agriculture, which also leads to under-utilisation of human resources.

When a country develops, there is general improvement in the standard of living. It is reflected in a number of ways—people get health facilities, safe drinking water, and sufficient amount of food. This leads to decline in death rates. Many people including women go to work in factories and offices. Since the survival of children is ensured, people tend to have small families. Moreover, in order to maintain a better standard of living, small families are preferred. Over the years, all these factors have led to reduction in birth rates. Reduction in both birth rate and death rate results in low natural growth rate. Thus, a gradual and smooth population growth is established.

When population increases in a country, it has to be provided with a corresponding increase in basic amenities, such as food, clothing, shelter, educational institutions, hospitals, drinking water, transport, and electricity. Government also have to ensure availability of food-grains. A rise in population may lead to difficulties for government to provide the required facilities for all. It also raises the demand for goods and services and more employment opportunities.

The different aspects of the population also affect the development. For instance, age structure of the population directly affects the distribution of resources available for developmental activities in a country. If a country has a greater proportion of young people (aged 0–14), they are to be provided with education and health-care facilities. In the same way elderly people (aged 60–65) are also to be provided with health-care and retirement or old age benefits. This will divert the resources mobilised by the government programmes in such a way that the requirements of these age groups are met.

Sustainable Growth and Development

Substainability is the capacity to endure. In ecology, the word describes how biological systems remain diverse and productive over time. For humans, sustainability is the potential for long-term maintenance of well-being, which has environmental, economic, and social dimensions.

The concept of sustainable development, first articulated by the World Commission on Environment and Development, 1987, through the now well-known Brundtland Report entitled 'Our Common Future' recognised that the natural resources are not inexhaustible and the development process should be *aimed to meet the needs of the present generation without compromising the ability of future generations to meet their own needs.*

Global change (involving climate change, biological invasion, land use/cover changes, and biodiversity depletion) has triggered this shift in paradigm on economic development linked to sustainable natural resource management.

The environmental issues to be resolved are:

(i) global warming because of greenhouse gas emissions,
(ii) acidification of soil and water,
(iii) drastic decline in the quality of air,
(iv) reckless exploitation of mineral resources through mining activities,
(v) drastic changes in land use and land cover through over-exploitation of natural resources leading to desertification of landscape, and
(vi) rapid depletion of biodiversity, etc. These issues need to be addressed immediately.

In the developing world, resource exploitation is largely geared to meet the very basic needs of food, fodder, fuel-wood, clothing, and shelter for a large section of the deprived societies. On the other hand, in the developed countries maintenance and acceleration of the already very high levels of resource consumption are the primary concern, blessed as they are with early initiative taken and advantage

gained through industrialisation. Natural resources are often viewed as unlimited by the industrialised world.

The Guiding Principles of Sustainability

The guiding principles of sustainability cut across ecological, economic, social, and cultural dimensions. The basic principles of sustainability are as under:

(i) *Inter-generational equity and Intra generating equity*: providing for today while retaining resources for tomorrow.

(ii) Conservation of cultural and biological diversity and ecological integrity.

(iii) There should be sustained rise in real per capita income.

(iv) Anticipatory and precautionary policy approach to resource use.

(v) Resource use in a manner that contributes to equity and social justice while avoiding social disruptions.

(vi) Limits on natural resource use within the capacity of the environment to supply renewable resources and assimilate wastes.

(vii) Qualitative rather than quantitative development of human well-being.

(viii) Pricing of environmental values and natural resources to cover full environmental and social costs.

(ix) Controlled growth of population.

(x) Global rather than national or regional perspective of environmental issues.

(xi) Efficiency of resource use by all societies.

(xii) Strong community participation in policy and practice, during the process of transition to an ecologically sustainable policy.

Indicators of Ecological Evaluation

There are ecological, economic, and social indicators for the measurement of sustainability. The main indicators of sustainability are:

(i) Land use and land cover changes.
(ii) Biomass quality and quantity.
(iii) Water quality and quantity.
(iv) Soil fertility.
(v) Energy management.

Sustainable Agriculture

Agriculture, forestry, and fisheries are the traditional activities in the rural environment of India. During the last few decades, India chose the *green revolution* as the path for self-sufficiency. The High Yielding Varieties of rice and wheat were diffused in the 1960s which performed superbly. However, this was achieved at a great cost, both economic, social and ecological, to the nation. On the bio-physical level, the negative impacts have led to organic carbon depletion, increased soil salinity, drastic changes in underground water-table, and chemical pollution due to the heavy application of chemical fertilisers, insecticides, and pesticides. In the heartland of green revolution, in the north-western parts of the Satluj-Ganga plains, agricultural production has already started showing tendencies of decline—a matter of concern for agricultural scientists; this is in spite of increased energy inputs. Warmer soil temperatures in the tropical countries, compounded by highly seasonal monsoonic rainfall, as in India and many parts of Asia, accelerate soil degradation.

On the social side, the damage is even more visible. The exploding population pressure is becoming more and more difficult to cope up with. Agriculture still remains an important economic activity for a large section of the Indian population; yet a substantial section of this farming community still follows traditional agricultural practices. What green revolution has done is the marginalisation of a very large segment of the farming community, through lesser access to both resources and market. Much of the tree cover that once sustained the traditional farming system has disappeared from our rural landscape, without any alternate way to sustain the minimal productivity levels of these traditional systems. With food self-sufficiency at the national level, the marginalised farmer often has not even the purchasing

power to buy food to fulfil his minimal needs. Further, major governmental agricultural policies, indeed, are to a large extent influenced by the more affluent farmers, who contribute to national food self-sufficiency. Resorting to industrialisation or some form of gainful employment is unthinkable as an alternative solution to the problem in near future. This social situation may become even more serious with the fast growth of rural population, especially that of the small and marginal farmers and the landless workers.

Localised famine conditions among more vulnerable sections of our society, in the drought-prone areas of the country are not uncommon. This is paradoxical in the context of overall self-sufficiency in food. Famine in India is related to problems of distribution, lack of purchasing power due to extreme poverty and competition from those who can afford it. In essence, it is a consequence of marginalisation of the already vulnerable sections of the society. Within a given household, there is overwhelming evidence to suggest that women as a group are more vulnerable than men to poverty and its consequences.

In the areas where green revolution is a success, cropping patterns are based on monoculture or two-crop combination. A substantial section of the rural population of about 60%, living outside the green revolution region, struggle with their traditional production systems. There is another 15% of the total population living in the hilly and mountainous areas. These hill dwellers are also involved in a whole variety of low input, multi-species, and more complex agro-ecosystems.

Sustainable agriculture demands efficient use of water and nutrients, largely based on locally available resources and regulated cropping done in a manner that would contribute towards soil fertility. Application of wrong technology and/or over-exploitation of natural resources may give short-term gains but could often lead to ecological degradation no longer sustainable by man.

India has great diversity in its agricultural systems and cropping patterns. Separate strategies need to be evolved for the different

agro-climatic agricultural systems.

1. Shifting Cultivation or Jhuming

Shifting cultivation is largely confined to those hilly and mountainous areas where the average annual rainfall is more than 100 cm. The basic principle of this extensive form of agriculture is the alternation of short crop phases (usually one or two years of cropping) with long phase of vegetation- fallow. It is a system in which 16 to 60 crops are mixed and sown in the same field. The crop diversity alone does not express the total biodiversity in the system. The species- diversity in a field may go up to sixty or more, if weeds are considered; when the fallow phase is included, the number may run into more than a hundred.

In the recent past, as a result of increasing population pressure, there has been a shift from extensive to more intensive system of land use in *Jhuming* many parts of the country. Consequently, the shifting cultivation cycle has fallen to only five to six years, and thus, the fallow system of about 25 years has been replaced by shorter period of fallowing.

In order to make the shifting cultivation more productive and sustainable, the gentle slopes (less than 15°) may be cut into terraces for sedentary cultivation; the slopes up to 30° may be brought under orchards and perennial cash crops; while the slopes more than 30° should be left as reserved forests to be utilised for forest products **(Fig. 17.16).**

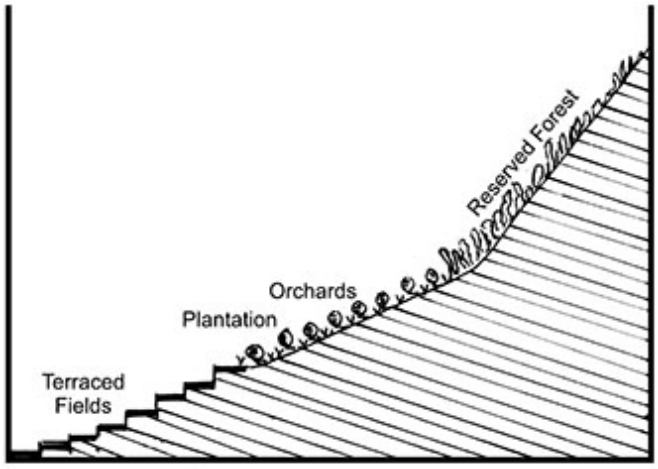


Fig. 17.16 Land Use Model in Jhum-Region

The system may be made more effective and productive by: (i) mixed arable-livestock farming, (ii) traditional agro-forestry systems, (iii) agri-silviculture, (iv) silvi-pastoralism, (v) agri-silvi-pastrolism, (vi) specialised cash crop systems (ginger, pineapple, yams, bamboo, tea, coffee, etc.), and (vii) poultry farming and bee-keeping. (*See chapter 10 for details*)

2. Sedentary Agriculture

In the sedentary agriculture of reasonable good rainfall areas, intercropping and alley farming can enhance the income of the cultivators and the fertility of the soil.

(i) **Intercropping System:** In the sedentary agriculture, mixed cropping, and inter-cropping with the inclusion of leguminous crops in cereal rotation is well recognised for fertility improvement.

(ii) **Alley Cropping:** In the alley cropping, trees are planted closely in rows to form hedgerows and the crops are grown in the 'alleys' between them. The trees are regularly pruned to provide mulch or fodder and to reduce above ground competition effects with crops. The major function of alley cropping is to improve soil fertility without necessitating a break in cultivation for fallowing. On sloping lands, planting hedgerows along the contours greatly reduces soil erosion, while the pruning of the trees generate more rural employment, firewood, and fodder. (*For details see Chapter 10*)

3. Modern Monocropping Agriculture

In the areas where green revolution is a success, the farmers are interested in the cultivation of wheat and rice (Punjab, Haryana, western Uttar Pradesh, and southern Uttarakhand) or only rice (Kaveri-delta). Thus, modern agriculture in these areas is based on the use of High Yielding Varieties of a particular wheat and rice replacing bio diversity in agriculture. It is monotonous not only

because of uniformity being imposed at a landscape level, but also because of intensive use of external energy based on chemical fertilisers, insecticides, and pesticides, which leads to reduction in above ground and below ground bio diversity. Intensively managed fruit plantations and orchards, vegetable crops, and intensive cereal production systems are some of the examples.

The diversified crop combinations in which several soil-exhausting crops are mixed with the soil enriching crops, like legumes, are more stable and less risky compared to those with fewer crops in combination. Mixed cropping systems are also receiving attention from modern agricultural scientists for their role in biological pest suppression.

Environmental Awareness

The rapid increase in population in the developing countries and consumerism in the developed nations have created numerous environmental problems. It is imperative to spread environmental awareness to make our resources sustainable.

All over the world, various media and means are used to spread environmental awareness among the people. The electronic media and print media are the major mediums of spreading information among the masses. Environmental issues like global warming, climate change, air, water and noise pollution, over use of chemical fertilisers, insecticides, pesticides, adverse effects of plastics and polythene, etc., are regularly debated in the media to create environmental awareness.

Schools, colleges, and universities play a vital role in generating environmental awareness. Textbooks reveal an increasing concern with environmental problems and their solutions. Several national and international conferences have been organised with a focal theme on environmental sustainability. The United Nations conference on Human Environment in 1972 played a key role in the emergence of the United Nations' Environmental Programme (UNEP). The Earth Summit was held in 1992 at Rio-de-Jenerio. The Kyoto Protocol and the Copenhagen conference were also held to address the issue of

environmental degradation and awareness. The main objectives of environmental education are as under:

(i) To develop awareness of the whole environment;
(ii) To develop skill among groups and individuals to identify and to solve the environmental problems;
(iii) To help people to gain experiences and basic understanding of the environment;
(iv) To involve people in the problems of environment;
(v) To educate the people about the real causes of environmental degradation;
(vi) To introduce environmental education at the school and university level;
(vii) To make masses aware about the existing and potential environmental situation;
(viii) To design all plans for socio-economic development to incorporate the issue of environmental degradation, conservation and sustainability.

GLOBALISATION AND INDIAN ECONOMY

The concept of globalisation was introduced in the Indian economy in July 1991. It was a part of the structural adjustment package during the period of the government of Prime Minister Narasimha Rao. The government soon applied for a loan of $2.3 billion from the International Monetory Fund (IMF) and began fulfilling the anticipatory conditionalities, including two successive devaluations and withdrawal of export subsidies, to smoothen the passage of the application (*Agarwal (eds), 1995, p. 182*). The annual budget of July 1991 fully reflected the structural adjustment package, which came to be known as the New Economic Policy (NEP).

Liberalisation, *Privatisation*, and *Globalisation* were the essential components of the New Economic Policy of 1991.

Liberalisation

Liberalization of economy means freedom of the producing units from direct or physical controls by the government.

Measures Taken for Liberalization

1. With the exception of five industries, industrial licensing has been abolished for all other industries for which licences are still necessary are: (i) liquor, (ii) cigratte, (iii) defence equipments, (iv) industrial explosive, and (v) dangerous chemicals.
2. Under the policy of liberalization, industries are free to expand and produce. They need no prior official approval. (3) Investment limit of the small industries has been raised to Rs 5 crore so as to enable them to introduce modernization. Investment limit of tiny industries has been increased to Rs 25 lakhs.

The essence of the liberalisation is that economic management should be left to market. The prices determined by the interaction of demand and supply forces, whether they are for commodities, labour power, capital, land, or foreign exchange, should be flexible in either direction and should be capable of clearing the market.

The resulting allocation of resources, commodities, labour power, foreign currency, etc., would be optimal and efficient, while any deviation from it would entail avoidable social cost. In order to ensure that markets are allowed to undertake their jobs, all controls and regulations, as also measures that constitute barriers to entry, should be done away with.

It follows from this that the state should take a backseat in economic matters. Any intervention by the state—in the form of controls, subsidies, selective protection, etc.—would distort prices and make the resulting allocation inefficient, thus hindering economic growth.

Privatisation

Privatization is the general process of involving the private sector in the ownership or operation of a state owned enterprise. It may

happen in two ways: (i) outright sale of the government enterprises, or (ii) withdrawl of the government ownership and management from the mixed enterprises—jointly owned and managed by the government and the private enterprises.

It follows from this philosophy of New Economic Policy that all *public sector economic activities* that are in operation should be closed down, phased out, trimmed or passed on to private sector. Public ownership should be allowed only in case of natural monopolies and strategic industries, such as defence and research establishments, and should not be allowed to become a drain on the state resources. The privatisation of such enterprises would improve their efficiency and bring in much-needed funds for reducing the fiscal deficit.

Globalisation

The third major objective of the New Economic Policy is globalisation. Globalisation implies removal of barriers to international trade and investment. *The core idea of globalisation is that more trade is better for all the parties concerned—some may gain more from it than others, but all would eventually gain.* Any action that interferes with the free flow of capital, goods, and services, would produce sub-optimal results. Import substitution and consequent protective measures, such as tariffs, controls, and restrictions, raise the cost of domestic production. Such a policy discriminates against export items that could be traded in the global market on the basis of country's comparative advantage, by reducing their competitiveness. In this way, import substitution hinders exports and makes the economy inward looking and increasingly backward, and isolates it from technical and other changes taking place worldwide.

According to the policy of Liberalisation, a country should specialise in production and export only in those items wherein it enjoys a comparative advantage, that is, wherein given its factor endowment (resources), it can produce best. Earning from such exports should be able to pay for the necessary imports. In most poor countries, such a comparative advantage lies in labour-intensive production

including that of *food products* and *textiles*, and not in large-scale capital and knowledge-intensive industries involving long gestation periods such as *steel, oil refining, perochemicals, or ship-building*. Accordingly, those who take this view recommend only those technologies which involve lots of labour.

Consequences

The consequences of Globalisation may be summarised as under:

1. **A New Trade Policy:** Two successive devaluation in two days were followed by a policy of phased Rupee convertibility (except for capital account), the withdrawal of export subsidies, liberalisation of imports, tariffication of quantitative restrictions, and a substantial reduction in tariffs in line with the levels in East and South-East Asian countries, and ultimately the signing of the Marakesh GATT Treaty by India in 1994 and its joining of the WTO in January 1995 as a founding member.
2. **A New Industrial Policy:** The share of foreign equity ownership was raised from 40% to 51% (and upto 100% in some cases) while the approval of licence applications was reduced to a mere formality in the cases of 34 major areas, including electronics, minerals, transport, and chemicals. For all but a few areas, industrial licences were no longer necessary and virtually the entire economy was now opened upto the private sector, both Indian and foreign.
3. **A New Banking and Finance Policy:** The new policy package moved away from directed credit and investment, avoided the *crowding out* of the private sector by curtailing lending to the public sector and brought about a reduction in statutory liquidity ratios that favoured bank investment in the securities floated by public sector units. A number of share market reforms were undertaken, including the legal setting up of the SEBI (Securities and Exchange Board of India), depositories of shares, and permitting foreign participation.
4. **Fiscal Reform:** Following the recommendations of the Raja

Cheilah Committee, the emphasis on customs and excise duties was reduced and a simplified tax structure with reduced rates and stricter enforcement, fewer slabs, and better coverage and compliance was recommended.

5. **Public Sector Reform:** MoUs (Memorandum of Understanding) were signed with public sector units delineating respective financial and other obligations and emphasising PSU autonomy in raising capital, deployment of resources, recruitment of personnel, and other management divisions.
6. **Agricultural Reform:** Subsidies on fertiliser, irrigation, power, and interest were reduced, a trimmed and better-targeted distribution system (PDS) was proposed, procurement prices for agricultural crops were raised, export orientation was encouraged, and there was a clear move away from the policy of self sufficiency in food and agricultural inputs including seeds and fertilisers. Poverty was to be tackled by way of growth and better agricultural prices, while agriculture was to be modernised and industrialised. Extensive poverty alleviation programmes were not considered necessary any longer, as growth was expected to take care of that in due course. Foreign agribusiness firms were invited to invest while the rhetoric of land reform was conspicuous by its absence.
7. **Labour Market Reform:** A new policy on the labour market that allowed for downward flexibility of wages and the right of an industrial owner to *hire and fire* his personnel was proposed. In India, accelerating output growth in recent years has not been accompanied by faster rate of job growth. Indian economy needs to create a large number of jobs for the unskilled workers who are currently unemployed or employed in the informal sector and for those who will be entering the labour force in the next few years. The high rate

of job creation in the IT (Information Technology) sector will be for the educated (skilled) workers with specific skill sets.

8. **A Better Macro-Economic Management:** This was sought by reducing fiscal deficit below 4% of GDP within 3–4 years and by pushing inflation below 5%, mainly by reducing government expenditure, restraining demand (at least in the initial years) and raising money by selling shares in PSUs. This was to be the key objective and a test of success for the new policy package.The second key objective was to create an incentive structure, through the reforms indicated above, that favoured production and growth, mainly under private management.

AGRICULTURE AND GLOBALISATION

It is being seen that the pressure on forests and forest-based natural resources will get further accentuated in the context of liberalisation and globalisation of economies, currently occurring in many Third World countries. It is in this context that the following conclusions became significant. Some of the important changes in the cropping patterns of India are given here.

1. **Emphasis on Cash Crops**: Energy-intensive agriculture triggered either by market forces or by governmental policies have already led to cash crop system in southern India—coffee in the Kodagu region in the Western Ghats, rubber and coconut in the Kerala state, etc.
2. **Emphasis on few Crop Combinations**: In the north-western parts of the Northern Plains of India, where green revolution is a success, the cereals like wheat and rice have taken the status of cash crops as they are largely grown for the market. Thus, the agriculture in these areas is becoming market-oriented.
3. **Increase in Income Disparity**: The gulf between the incomes of the big and small and marginal farmers is increasing. It is a cause of great concern for the planners. In

fact, the rural society is getting polarised between the poors and the rich.

4. **Deforestation and Land Degradation:** Large-scale deforestation for fuelwood has reduced the size of trees. Moreover, there is weed invasion particularly by exotic varieties leading to biodiversity depletion and soil erosion.

Biodiversity Depletion:

5. Natural ecosystems have been altered and/or degraded in a variety of different ways resulting in rapid loss of natural biodiversity.

6. **Climate Change:** The climate change is emerging as the biggest developmental challenge for the planet earth. It is the long-term alteration of temperature and normal weather patterns of a place. The substantial changes in land use, deforestation and green house effect is the major cause of the climatic changes occurring across the globe. '*Global warming*' is a consequence of the burning of fossil fuels that release gases (carbon dioxide) into the atmosphere. This gas forms a cover in the layers of the atmosphere, thereby trapping the solar heat. This results in the increase in the earth's temperature.

A warmer atmosphere also pushes the hydrologic cycle that alters the rainfall because of the greater moisture content. More evaporation occurs that leads to drying up of lakes and water bodies.

The effects of the climate changes make the temperatures and weather patterns more unpredictable. For example, in a temperate region lot of rain might be experienced in one year with very little rain the following year. Such changes affect the growth of the crops adversely. There are predictions that there might be an increase in the rainfall in some areas up to 50%. The climate changes also result in intense hurricanes, floods, downpours and winter storms.

In polar regions, the warming global temperatures are resulting into melting of the ice sheets and the glaciers. As a result an increase in the sea level is being noticed. The coastlines are being damaged and resulting into increased floods.

A drop in the production of wheat yields of up to 5–10% has been predicted with an increase of 1° C .This accounts to total of 30% decrease in India. Assam due to its geographical location and poor socio-economic conditions is an example where the economic burden is on the increase due to climate change. According to a report of the State Action Plan for Climate Change, the annual temperature of Assam has increased by 0.59° C over the last 60 years. It is likely to increase by 1.7–2.2° C by 2050. The extreme rainfall events are likely to increase by 30% in the state.

Utilisation of Resources

The resource depletion may be checked appreciably, if every citizen takes some steps in the utilisation of resources. Some of these steps are as under:

(i) People must change their worldviews and value systems. They should recognise that the natural resources are finite and limited.

(ii) People should recognise that development and environment are compatible. In other words, development is the best guarantor of future economic well-being.

(iii) People all over the world should consider the needs of future generations more than we do now.

(iv) People should strive for self reliance through the use of appropriate technologies. Villagers, for example, should rely increasingly on solar power of electricity.

(v) People of the developing countries need to limit their population growth. Advances in status of women, and improvements in education, health, family planning and social services are desired.

(vi) In the developing countries, steps should be taken to ensure a more equitable distribution of land.

(vii) In the developed countries, steps should be taken to reduce over-consumerism.

(viii) Wealth should be redistributed among the developed and the

developing countries, because poverty is the fundamental cause of environmental degradation. Proponents of sustainable development argue that this does not mean that rich countries should give cash outright to the poor. Instead, the lending institutions of the developed countries could forgive some existing debts owed by developing countries, or use much innovations as *'debt-for-nature swaps'* in which a certain portion of debt is forgiven in return for the borrower's pledge to invest that amount in national parks, biosphere reserves, and other conservation programmes.

Environmental Protection

We cannot expect science to solve all the problem for us. Most of the decisions and necessary actions fall outside pure science in the areas of values, ethics, morality, and philosophy. The solution of environmental problems, if one exists, lies in *education and action*.

Each of us can make daily decisions that collectively make a difference. Something we can do for environmental protection and sustainable development are listed below:

Energy

(i) Drive fewer miles. Ride your bike or walk. Use public transport system.

(ii) Replace lamps with smaller wattage light bulbs. Replace incandescent bulbs with compact
18-watt fluorescent ones. Turn off lights not being used. Switch to low-voltage exterior lightning.

(iii) Purchase energy efficient appliances and cars. Government-mandated stickers on cars and appliances allow you to compare the amounts of energy each will use. Spend a bit more now to save a lot later.

(iv) Check insulation and weather stripping around your house, apartment, or room. Keep thermostat relatively high in summer, and low in winter.

(v) Check into alternative energy sources such as solar and wind

power units.

(vi) Buy and use rechargeable batteries. The disposable ones contain heavy metals that can pollute.

Food

(i) Try to avoid foods from areas where pesticides have been used. Never eat fish caught in developed harbours or bays.

(ii) Consider how your food choices affect the environment. Eat foods low on the food chain and avoid foods raised in places where endangered environments (such as rain forests) were destroyed.

(iii) Raise a garden.

(iv) Buy bulk or unpackaged goods, or buy food in reusable containers.

Water

(i) Use water-saving showerheads, sinks, faucet aerators, and toilets.

(ii) Sweep walks and driveways.

(iii) Water lawns infrequently but thoroughly. Better yet, replace them with drought-tolerant ground covers.

(iv) Take shorter showers.

(v) Run only fully loaded dish-washer and clothes-washer.

Toxic Materials and Pollutants

(i) Check for toxic materials (old cans, pesticides, engine oil, pool chemicals and so on) you may have around the house. Take them to an approved toxic waste receiving station.

(ii) Read labels. Buy the least toxic materials available.

(iii) Swat flies; do not spray them.

(iv) Buy clothing that does not require a cleaning. Dry clean only when necessary.

Recycling and Waste Reduction

(i) Recycle newspapers, cans, glass, and plastics.

(ii) Buy products that are refillable and recyclable.

(iii) Use cloth rags and diapers instead of paper diapers and paper towels.
(iv) Take your grocery bags back to the market for another trip home.
(v) Volunteer your time in a beach, pond or water-body cleaning campaign.

Preservation of Life and the Environment

(i) Plant trees.
(ii) Preserve open space.
(iii) Avoid CFCs (Check air-conditioners to see if they leak).
(iv) Boycott organisations that violate sound environmental practices.
(v) Don't litter. Pack your trash out of campsites.

Other Things

(i) Write to your councilors, and legislators. Better yet visit them.
(ii) Get involved in an environmental group such as Greenpeace, *Chipko Movement*, etc.
(iii) Talk to your friends about environmental matters.
(iv) Buy quality. Purchase a well built item, maintain it, repair it, keep it forever.
(v) Sit quietly on beach at sunset or sunrise.
(vi) Read and listen.
(vii) Be positive about results.

Social Media

(i) Of the total population of India, about 462 million people use the internet facilities and go online. India has the second largest internet users in the world (34.8% of the total population). There has been a significant growth of 30.5% since 2015. In 2017, 1.06 billion people had mobile subscriptions. The increase can be attributed to affordable, ubiquitous mobile data plans.
(ii) There are over 326.1 million active social media users in 2018. Facebook is the biggest social network in India with

over 281 million users in 2018, LinkedIn has around 53 million users, and Twitter has 30.4 million users. It forms the second largest base in Asia Pacific region after Japan.

(iii) Some of the new social media sites are Tumblr, Snapchat, WeChat, Google+, Flickr, Viber, Orkut.

Cyber Crime

(i) The cyber crime includes the criminal activities carried out by means of computers or the internet. The cyber crime may threaten a person's or the nation's security and finances. The cyber criminals use the computer technology to access personal information, business trade secrets and harm the target. Such criminals are referred to as 'hackers'.

(ii) Cyber bullying, Cyber stalking, Cyber terrorism, Cyber warfare, Troll, Cyberveillance are few common crimes in which the criminals get indulged through the computers.

(iii) The National Crime Records Bureau in India was set up in 1986. It is the Central Nodal Agency that controls and manages the functions of Online 'Cyber Crime Reporting Portal'.

(iv) It initiated the programme Crime and Criminal Tracking System (CCTS) in 2009. 15,000 police stations and 6000 higher offices of the police are connected in the country.

(v) Bengaluru city (Silicon Valley) recorded the most number of cyber cases in 2018 with 5035 FIRs. Delhi registered 84 cases as in 2017. Other metro cities with high record of cybercrime cases are Hyderabad, Kolkata and Chennai. The increasing use of laptops and mobile phones have caused a negative impact on the youth. The cyber crimes against the children and the women account for about 30%.

(vi) Technology evolution and innovations make it challenging to detect the cyber crimes.

CAA

The Citizenship Amendment Act, 2019 was passed in the parliament on December 11, 2019. It amended the Citizenship Act of 1955 allowing the Indian citizenship to Hindu, Sikh, Buddhist, Jain, Parsi, and Christian religious minorities who migrated from the neighbouring Muslim majority countries of Pakistan, Bangladesh and Afghanistan before December, 2014 due to religious persecution or fear of religious persecution in their country of origin. These category of migrants would be granted the Indian citizenship in six years. (Source: E-Times of India, Jan 9, 2020)

NPR

NPR is the usual register of usual residents of India. It is being prepared at the local (village/sub town), sub district, District, State and the National level under the provisions of Citizenship Act 1955 and the citizenship (Registration of Citizens and issue of National Identity cards) Rules, 2003.Its objective is to create a comprehensive identity database of every usual resident in the country. It shall include demographic and biometric particulars. The updation of the national population register alongwith the house listing phase of Census 2021 during April to September 2020 would be completed in all the states/UT except Assam. (Source: https://censusindia.gov.in/2011-Common/IntroductionToNpr.html)

NRC

The National Register of Citizens is a register maintained by the Government of India. It is an official record of those who are legal Indian citizens. It includes the demographic information of the Indian citizens as per the Citizenship Act 1955. The register was first prepared in 1951, Census of India. So far the database has been maintained for the state of Assam.

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Multiple Choice Questions

[This set of carefully prepared questions will be of valuable help in preparation for General Studies Paper-I of the civil services preliminary examination.]

1. The north coast of Bay of Bengal is prone to catastrophic surges as compared to any part of the Earth. The reason being:
   (a) it has high surface temperatures that favours strong cyclones
   (b) because it lies between latitudes that favour cyclones
   (c) Strong winds blow from north east to south west
   (d) All of the above

2. Tropical cyclones are also known as
   (a) Tornados
   (b) Typedos
   (c) Hurricanes
   (d) Blizzards

3. Amphan was a super cyclone that caused severe damages to

(a) West Bengal and Bangladesh
(b) West Bengal and Odisha
(c) Odisha and Bihar
(d) Odisha and Andhra Pradesh

4. The city of Wuhan is located on the river:
(a) Huang Ho
(b) Tarim
(c) Si Kiang
(d) Yangtze

5. The state associated with the styrene gas leakage that occurred from South Korean battery maker LG chemical Ltd on May 7, 2020 is:
(a) Karnataka
(b) Chattisgarh
(c) Andhra Pradesh
(d) Tamil Nadu

6. Consider the following statements about the earthquakes and select the correct answer using the code given below:
1. In India, the region of convergence of the Indian Plate and the Eurasian Plate is more vulnerable to earthquakes
2. An earthquake of magnitude 8.5 or higher is likely to hit Deccan Plateau in India

Code:

(a) Only 1 is correct
(b) Both 1 and 2 are correct
(c) None of the two is correct
(d) Only 2 is correct

7. Match List I with List II and select the correct answer using the code given below

List I	List II
A. Black revolution	1. Cotton

B. Brown Revolution	2. Apiculture
C. Golden revolution	3. Horticulture
D. Silver Fibre revolution	4. Petroleum

Code:

A	B	C	D	
(a)	2	1	4	3
(b)	4	2	3	1
(c)	3	1	2	4
(d)	1	3	4	2

8. The Metro line that connects Mukundpur Depot to Shiv Nagar is:
(a) Red Line
(b) Green Line
(c) Blue Line
(d) Pink Line

9. The governmental body that has replace the Planning Commission in 2015 is:
(a) Niti Ayog
(b) National Development Council
(c) Regional Planning Commission
(d) Regional Planning Commission

10. Match the tribes of List I with the states of List II and select the correct answer using the code given below

List I	List II
A. Gond	1. Chattisgarh
B. Lushai	2. Ladakh
C.Todas	3. Tamil Nadu
D. Bota	4. Mizoram

Code:

	A	B	C	D
(a)	4	2	1	3
(b)	3	1	4	2
(c)	1	4	3	2
(d)	1	2	4	3

11. The silkworms thrive on the leaves of:
(a) Mulberry, mahua, sal, ber and kusum
(b) Mulberry, Neem, Peepal, Jamun
(c) Mulberry, Sal, Jamun, Rosemerry
(d) Mango, Mahua, Ber, Jamun

12. The hydro electric power plant Hirakud is located on the river
(a) Sutlej
(b) Gomti
(c) Kaveri
(d) Mahanadi

13. The Bakra Nangal Project is a joint venture of the following states:
(a) Uttrakhand, Punjab, Uttar Pradesh
(b) Punjab, Haryana, Rajasthan
(c) Uttrakhand, Punjab, Haryana
(d) Uttar Pradesh, Punjab, Haryana

14. The largest slum of Asia that is located in Mumbai is:
(a) Lokandwala
(b) Worli
(c) Dharavi
(d) Kankaria

15. The Kumaun Himalaya are located between the following rivers:
(a) Satluj and Jhelum
(b) Satluj and Kali
(c) Jhelum and Ganga
(d) Ganga and Yamuna

Chapters 1 and 2 – Structure of India and Physiography

16. Arrange the following geological formations in a chronological order and select the correct answer using the code given below:

1. Cuddapah Formation
2. Deccan Lava Trap
3. Gondwana Formation
3. Vindhyan Formation

Code:

(a) 3, 1, 4 and 2
(b) 1, 2, 4 and 3
(c) 1, 4, 3 and 2
(d) 3, 1, 2 and 4

17. The Great Boundary Fault lies in the state of:

(a) Gujarat and Rajasthan
(b) Himachal Pradesh and Uttarakhand
(c) Jammu & Kashmir and Himachal Pradesh
(d) Jammu & Kashmir, Himachal Pradesh and Uttarakhand

18. Arrange the following mountain ranges from north to south and select the correct answer using the code given below:

1. Karakoram Range
2. Ladakh Range
3. Pir-Panjal Range
4. Zaskar Range

Code:

(a) 1, 2, 3 and 4
(b) 2, 1, 4 and 3
(c) 2, 1, 3 and 4
(d) 1, 2, 4 and 3

19. The Main Boundary Fault separates:

(a) The Karakoram from the Greater Himalayas
(b) The Greater Himalayas from the Lesser Himalayas
(c) The Lesser Himalayas from the Shiwalik
(d) The Shiwalik from the Great Plains of India

20. Which one of the following mountain peaks is not correctly matched?

(a) Amarkantak - Madhya Pradesh
(b) Dunagiri - Uttarakahand
(c) Kylas - Meghalaya
(d) Nimgiri - Orissa

21. Arrange the following passes from west to east and select the correct answer using the code given below:

1. Balcha Dhura
2. Mana La
3. Niti La
4. Thaga La

Code:

(a) 4, 3, 2 and 1
(b) 4, 2, 3 and 1
(c) 2, 3, 1 and 4
(d) 2, 1, 3 and 4

22. Consider the following statements and select the correct answer using the code given below:

1. Mikir Hills of Assam are a part of the Peninsula of India.
2. Kanchenjunga is the second highest peak in the world.

Code:

(a) Only 1 is correct
(b) Only 2 is correct
(b) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

23. Which one of the following statements is not correct?

(a) K2 is the highest peak in India
(b) Jog waterfall is the highest waterfall in India
(c) Cherrapunji is the wettest place in India
(d) Deccan Plateau is the largest plateau of India

24. Which one of the following statements is not correct?

(a) Gujarat has the longest coastline in India.
(b) Indira Gandhi Canal is the longest canal in India.

(c) Kaharbude is the longest tunnel in India.
(d) Kovalam is the longest sea-beach in India.

25. Arrange the following peaks of the Peninsular India in a descending order of height and select the correct answer using the code given below:

1. Aroyakonda
2. Dewodimunda
3. Mahendragiri
4. Nimgiri

Code:
(a) 1, 2, 3 and 4
(b) 1, 2, 4 and 3
(c) 2, 3, 4 and 1
(d) 2, 3, 1 and 4

26. Consider the following statements and select the correct answer using the code given below:

1. The Northern Plain of India is a 'foredeep' formed in front of the high crust waves of the Himalayas.
2. The concept of 'foredeep' about the origin of the Northern Plains of India was given by Sir S. G. Burrard.

Code:
(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

27. Consider the following statements and select the correct answer using the code given below:

1. The Gujarat Plain was formed by the depositional work of parallel consequent rivers.
2. The Gujarat Plain contains the Gondwana and Cretaceous lava formations.

Code:
(a) Only 1 is correct
(b) Only two is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

28. Match List I with List II and select the correct answer using the code given below:

List I (Vulcanacity) (Landform)	**List II**
A. Cretaceous	1. Bijapur and Gwalior series
B. Cuddapah	2. Champion Series
C. Dharwar	3. Deccan Trap
D. Vindhyan	4. Malani-series

Code:

	A	**B**	**C**	**D**
(a) 3	1	4	2	
(b) 3	1	2	4	
(c) 2	3	1	4	
(d) 2	3	4	1	

29. Which one of the following is not correctly matched?

(a) Kangra earthquake - 1934
(b) Koyna earthquake - 1967
(c) Latur earthquake - 1993
(d) Shillong earthquake - 1897

30. Match List I with List II and select the correct answer using the code given below:

List I (Mountain Pass)	**List II (State)**
A. Bum La	1. Sikkim
B. Khardungla	2. Jammu & Kashmir
C. Nathu La	3. Himachal Pradesh
D. Shipki La	4. Arunachal Pradesh

Code:

A	**B**	**C**	**D**

(a)	2	1	4	3
(b)	2	4	3	1
(c)	4	2	3	1
(d)	4	2	1	3

Chapter 3 – Drainage

31. Which one of the following is an example of epigenetic (discordant) drainage?
(a) Chenab
(b) Chambal
(c) Narmada
(d) Sabarmati

32. Consider the following statements and select the correct answer using the code given below:
1. In India, the drainage basin of Brahmaputra is bigger than that of the Indus basin.
2. The Mahanadi drainage basin is bigger than that of the Krishna River basin.

Code:
(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

33. Which one of the following is not correctly matched?
(a) Jodh Ganga - Alaknanda
(b) Penganga - Godavari
(c) Parbati - Chambal
(d) Purna - Tapi

34. Which one of the following rivers is shortest in length?
(a) Chambal
(b) Damodar
(c) Gandak
(d) Kaveri

35. Consider the following statements and select the correct answer using the code given below:

1. Tso Moriri is a tectonic lake.
2. The Gohana Lake of Garhwal was created by the dissolution work of rivers.

Code:

(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

36. Consider the following statements and select the correct answer using the code given below:

1. Dudhsagar waterfall lies in the Jabalpur District of Madhya Pradesh on the Narmada River.
2. Gokak falls lie in the Belgaum District of Karnataka on the Ghatabarbha River.

Code:

(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

37. Which one of the following is not correctly matched?

(a) Barkana waterfall lies in the Shimoga District of Karnataka.
(b) Dudama waterfall lies in the Koraput District of Odisha.
(c) Kiliyur waterfalls lie in the Nilgiri (Tamil Nadu).
(d) Lodh Waterfalls lies in Jharkhand.

38. Consider the following statements and select the correct answer using the code given below:

1. Dhansiri is right hand tributary of the Brahmaputra River.
2. Ringit is a left hand tributary of the Tista River.
3. Banas is a left hand tributary of the Chambal.
4. Amravati is a right hand tributary of the Kaveri River.

Code:
(a) 1, 2 and 3 are correct
(b) 2, 3 and 4 are correct
(c) 1, 2, and 4 are correct
(d) 1, 3 and 4 are correct

39. Which one of the following is not correctly matched?
(a) Hathmati – Mahi River
(b) Matatila – Betwa River
(c) Nagarjuna-Sagar – Krishna River
(d) Ukai Project – Tapi River

40. Match List I with List II and select the correct answer using the code given below:

List I (River)	**List II (Source of the River)**
A. Chenab	1. Bara Lacha
B. Jhelum	2. Rakas Lake
C. Ravi	3. Rohtang Pass
D. Satluj	4. Verinag

Code:

	A	**B**	**C**	**D**
(a)	3	1	4	2
(b)	3	1	2	4
(c)	1	4	2	3
(d)	1	4	3	2

Chapter 4 – Climate

41. "The Indian Monsoon is extended land breeze and sea breeze."
This statement is related to:
(a) Flohn
(b) Halley
(c) Koteshwaram
(d) Walker

42. Consider the following statements and select the correct answer using the code given below:

1. The variability of rainfall at Chennai is more than that of Mumbai.
2. Chennai records rainfall from the retreating monsoon.

(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

43. Consider the following statements and select the correct answer using the code given below:

1. The Western - disturbances originate from the Mediterranean Sea.
2. The origin of the Western - disturbances is associated with the Polar Front Theory.

Code:

(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

44. Consider the following statements and select the correct answer using the code given below:

1. The Greater Himalayas receive more rainfall than that of the Lesser Himalayas.
2. The rate of environmental degradation is more in the Lesser Himalayas than that of the Greater Himalayas.

Code:

(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

45. Who among the following scholars did not delineate 'the climatic regions of India*'?*

(a) Blandford, H.E
(b) Coppack, J

(c) Spate, O.H.K
(d) Singh, R.L

Chapter 5 – Natural Vegetation and National Parks

46. The Floristic Regions of India were delineated by:
(a) Calder, C.C
(b) Chatterji, S.P
(c) Johnson, B.L.C
(d) Subramanyam V.P

47. Which set of the following biosphere reserves is included in the World Network of Biosphere Reserves?
(a) Gulf of Mannar, Kanchenjunga, Nokrek and Seshachalam
(b) Nilgiri, Nokrek, Panchmarhi and Seshachalam
(c) Nilgiri, Nokrek, Panchmarhi and Panna
(d) Gulf of Mannar, Nokrek, Panchmarhi and Simlipal

48. Which one of the following National parks is not correctly matched?
(a) Balpakhran – Meghalaya
(b) Bhagwan Mahavir – Bihar
(c) Guindy – Tamil Nadu
(d) Molem – Goa

49. The largest percentage of forest area in India is in which of the following regions?
(a) Eastern Ghats
(b) Himalayan Region
(c) Peninsular Plateau
(d) Western Ghats

50. Consider the following statements and select the correct answer using the code given below:
1. Teak forest is deciduous in nature.
2. The largest area of teak forest is in the state of Assam.

Code:
(a) Only 1 is correct

(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct.

51. The Western Ghats was included in the UNESCO World Heritage List in:
(a) 2010
(b) 2011
(c) 2012
(d) 2013

52. Which one of the following states has the highest percentage of the forest area?
(a) Arunachal Pradesh
(b) Mizoram
(c) Nagaland
(d) Tripura

53. Which one of the following states has the largest area under forest?
(a) Arunachal Pradesh
(b) Chhattisgarh
(c) Karnataka
(d) Madhya Pradesh

54. Consider the following statements and select the correct answer using the code given below:
1. Rosewood forest is mainly found in the states of North East India.
2. Ebony forests are largely found in Peninsular India.

Code:
(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

55. Consider the following statements and select the correct answer using the code given below:

1. Biosphere reserves are rich in biological and cultural diversity.
2. The Biosphere Reserves Programme in India was initiated in 1986.
3. A Wildlife Sanctuary is dedicated to protect wildlife and particular species.
4. The total number of biosphere reserves in India is 18.

Code:

(a) 1, 2 and 3 are correct
(b) 1, 3 and 4 are correct
(c) 2, 3 and 4 are correct
(d) 1, 2, 3 and 4 are correct

Chapter 6 – Soils

56. Arrange the following soils in descending order of area and select the correct answer using the code given below:

1. Alluvial soil 2. Black (*Regur*) soil
3. Desert soil 4. Red soil

Code:

(a) 1, 2, 3 and 4
(b) 1, 4, 2 and 3
(c) 1, 2, 4 and 3
(d) 1, 4, 3 and 2

57. Arrange the following particles of soils in a descending order and select the correct answer using the code given below:

1. Clay 2. Gravel
3. Sand 4. Silt

Code:

(a) 3, 1, 4, 2
(b) 2, 3, 1, 4
(c) 2, 3, 4, 1
(d) 3, 1, 2, 4

58. Grey-brown soil is found in:

(a) Haryana and Rajasthan
(b) Haryana and Gujarat

(c) Gujarat and Madhya Pradesh
(d) Gujarat and Rajasthan

59. Consider the following statements and select the correct answer using the code given below:
1. The All India Soil and Land Use Survey attempted the soil classification of India in 1956.
2. The Indian Council of Agricultural Research published the first map of soils of India in 1963.

(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

60. Consider the following statements about the Land-capability classification and select the correct answer using the code given below:
1. Land capability classification system is designated primarily for soil conservation.
2. Land capability classification is intended for determining maximum intensity of land use.

Code:
(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

Chapter 7 – Resources

61. Which one of the following is not correctly matched?
(a) Hazaribagh – Mica
(b) Hutti – Gold
(c) Jaduguda – Uranium
(d) Zawar – Copper

62. Consider the following statements and select the correct answer using the code given below:
1. Maharashtra is the leading producer of manganese.

2. Odisha is the leading producer of iron-ore.

Code:

(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

63. Arrange the following states in a descending order of iron-ore production and select the correct answer using the code given below:

1. Chhattisgarh 2. Goa
3. Karnataka 4. Odisha

Code:

(a) 3, 4, 1, 2
(b) 3, 4, 2, 1
(c) 4, 3, 2, 1
(d) 4, 1, 3, 2

64. Arrange the following iron-ores in descending order of metal content and select the correct answer using the code given below:

1. Haematite 2. Limonite
3. Magnetite 4. Siderite

Code:

(a) 1, 3, 4, 2
(b) 1, 3, 2, 4
(c) 3, 1 , 2, 4
(d) 3, 1, 4, 2

65. Which one of the following has the richest deposits of Mica?

1. Andhra Pradesh 2. Bihar
3. Jharkhand 4. Rajasthan

Chapter 8 – Energy Resources

66. Which one of the following mines has the best quality of coking coal?

(a) Bailadila

(b) Jharia
(c) Raniganj
(d) Singrauli

67. Which one of the following states has the highest coal reserves?
(a) Andhra Pradesh
(b) Chhattisgarh
(c) Jharkhand
(d) Madhya Pradesh

68. Consider the following statements and select the correct answer using the code given below:
1. The Tertiary coal deposit came into existence during the Eocene Perio(d)
2. Tertiary coal deposits are found in Assam and Arunachal Pradesh.

Code:
(a) Only 1 is correct
(b) Only 2 is correct
(c) 1 and 2 are correct
(d) Neither 1 nor 2 is correct

69. Which one of the following statements is not correct?
(a) Crude oil is found in the metamorphic rocks.
(b) Crude oil is found at the place of its formation.
(c) Crude oil overlain with gases.
(d) The origin of crude oil is considered to be organic.

70. Which one of the following states is the largest producer of crude-oil?
(a) Assam
(b) Andhra Pradesh
(c) Gujarat
(d) Tamil Nadu

71. Out of the following which region has the largest reserve of natural gas?

(a) Bassein basin
(b) Gulf of Khambat
(c) Godavari & Krishna basin
(d) Barmer (Rajasthan)

72. Consider the following statements and select the correct answer using the code given below:
1. Electricity is a clean source of energy.
2. Electricity is relatively cheap, pollution free and renewable source of energy.

Code:
(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

73. Which one of the following is not correctly matched?
(a) Dhuvaram – Gujarat
(b) Kahalgaon – Bihar
(c) Manuguru – Andhra Pradesh
(d) Unchahar – West Bengal

74. The maximum potential of wind-energy lies in the states of:
(a) Gujarat and Maharashtra
(b) Gujarat and Tamil Nadu
(c) Andhra Pradesh and Tamil Nadu
(d) Andhra Pradesh and Rajasthan

75. Which one of the following nuclear plants is not correctly matched?
(a) Bagri-Chutka, Chhattisgarh
(b) Kawada, Andhra Pradesh
(c) Kumharia, Haryana
(d) Mithi Virdi, Gujarat

Chapters 9 and 10 – Agriculture and Spatial Organisation of Agriculture

76. The net sown area of India is about:

(a) 42%
(b) 46%
(c) 50%
(d) 52%

77. The land affected by salinity and alkalinity in India is about what percentage of the degraded land?
(a) 1%
(b) 2%
(c) 3%
(d) 4%

78. Arrange the following states of India in respect of net cultivated area and select the correct answer using the code given below:

1. Haryana
2. Punjab
3. Uttar Pradesh
4. West Bengal

Code:
(a) Haryana, Punjab, West Bengal, Uttar Pradesh
(b) Haryana, Punjab, Uttar Pradesh, West Bengal
(c) Punjab, Haryana, Uttar Pradesh, West Bengal
(d) Punjab, Haryana, West Bengal, Uttar Pradesh

79. The marginal farmers constitute about what percentage of the total holdings in India?
(a) 40%
(b) 45%
(c) about 50%
(d) about 55%

80. Consider the following statements and select the correct answer using the code given below:
1. Canals used to be the main source irrigation in India during the pre-Independence period.
2. The percentage share of canal irrigated area is decreasing with the passage of time.

Code:

(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

81. Arrange the area of the following staple crops in descending order and select the correct answer using the code given below:

1. Bajra
2. Jowar
3. Rice
4. Wheat

Code:
(a) 3, 4, 1, 2
(b) 3, 4, 2, 1
(c) 4, 3, 1, 2
(d) 4, 3, 2, 1

82. Which one of the following states of India is the leading producer of pulses?
(a) Bihar
(b) Karnataka
(c) Madhya Pradesh
(d) Uttar Pradesh

83. Operation Flood is associated with:
(a) Blue Revolution
(b) Brown Revolution
(c) Green Revolution
(d) White Revolution

84. Which one of the following is known as the *Shrimp Capital of India*?
(a) Chennai
(b) Machilipatnam
(c) Nagapattinam
(d) Nellore

85. Consider the following statements about rice cultivation and select the correct answer using the code given below:

1. China is the largest producer of rice in the world.
2. China has the largest area under rice cultivation in the world.

Code:
(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

86. Consider the following statements and select the correct answer using the code given below:
1. Rice is grown successfully from 39° S (Australia) to 50° N latitude (China).
2. In India, rice is grown between 8° N to 34°N.
3. In India rice is grown in areas of Kuttanad (Kerala) which is below the sea level.
4. Rice is grown above 2000 m in parts of Jammu and Kashmir.

Code:
(a) 1, 2 and 3 are correct
(b) 1, 2 and 4 are correct
(c) 1, 3 and 4 are correct
(d) 1, 2, 3 and 4 are correct

87. Consider the following statements and select the correct answer using the code given:
1. The optimum climatic requirements for rice is 20°-35°C temperature throughout the period of crop duration.
2. Rice is grown in all types of soils.

Code:
(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

88. Consider the following statements about wheat cultivation in India and select the correct answer using the code given below:

1. The cycle of wheat crop is about 100 days south of the Vindhyan Range and 145 days in Punjab.
2. High temperatures in October permit early seeding of the wheat crop.

Code:

(a) Only 1 is correct
(b) Both 1 and 2 are correct
(c) Only 2 is correct
(d) Neither 1 nor 2 is correct

89. Which one of the following statements is not correct?

(a) Sugarcane cultivation requires over 20°C and below 50°C temperature.
(b) Temperature below 10°C is injurious to sugarcane cultivation.
(c) Sugarcane needs 75 cm to 150 cm of rainfall.
(d) Sugarcane performs better in temperate conditions.

90. Which one of the following statements is not correct?

(a) Cotton is a semi-xerophyte crop.
(b) Cotton requires about 85 cm of rainfall.
(c) The optimum temperature for vegetative growth of cotton is 21°-27°C.
(d) Temperature below 21°C is detrimental to rice crop.

91. Consider the following statements and select the correct answer using the code given below:

1. China is the leading producer of cotton in the world.
2. India has the largest area under cotton in the world.

Code:

(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(c) Neither 1 nor 2 is correct

92. Consider the following statements and select the correct answer using the code given below:

1. The ideal temperature for tea is 20°-30°C.

2. Temperature below 5°C is harmful to tea-bush.
3. Heavy fog and dew favour the rapid development of young leaves of tea.
4. Tea bush develops vigorously when planted along with shady trees.

Code:

(a) 1, 2 and 3 are correct
(b) 1, 3 and 4 are correct
(c) 2, 3 and 4 are correct
(d) 1, 2, 3 and 4 are correct

93. Which one of the following is the leading producer of coffee in India?

(a) Andhra Pradesh
(b) Karnataka
(c) Kerala
(d) Tamil Nadu

94. Which one of the following is known as the *king of spices*?

(a) Clove
(b) Cardamom
(c) Pepper
(d) Turmeric

95. Consider the following statements and select the correct answer using the code given below:

1. India is the leading producer and exporter of cardamom.
2. Kerala is the leading producer of spices in India.

Code:

(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

Chapter 11 – Industries

96. In the production of cotton textiles, arrange the following

states in a descending order and select the correct answer using the code given below:

1. Gujarat
2. Maharashtra
3. Punjab
4. Tamil Nadu

Code:

(a) 1, 2, 4, 3
(b) 1, 2, 3, 4
(c) 2, 1, 3, 4
(d) 2, 1, 4, 3

97. In India, the first jute factory was established in 1854 at:
(a) Agarpara
(b) Budge Budge
(c) Serampore
(d) Titagarh

98. Which one of the following iron and steel industries is not correctly matched?
(a) Burnpur – West Bengal
(b) Dolvi – Jharkhand
(c) Diatari – Odisha
(d) Vijayanagar – Karnataka

99. Consider the following statements and select the correct answer using the code given below:
1. The Bhilai Iron and Steel Plant was established with the technical co-operation of the Government.
2. The Durgapur Iron and Steel Plant was established with the technical co-operation of German firm, Krupps and Demang.

Code:
(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

100. Which one of the following statements is not correct about

the Vijayanagar Steel Plant?

(a) It is located in the Bellary District of Karnataka.
(b) It obtains coking coal from Jharia.
(c) It obtains hydroelectricity from the Tungbhadra Project.
(d) It makes use of non-cocking coal known as Corex process.

101. In which of the following cities the Pohang Steel Company of South Korea collaborated to establish an iron and steel plant?

(a) Daitari
(b) Kalinganagar
(c) Koraput
(d) Paradwip

102. Consider the following statements and select the correct answer using the code given below:

1. Aluminum is the second most important metallurgical industry of India.
2. Its location is determined on the availability of bauxite and hydroelectricity.

(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

103. Which one of the following aluminum plants is not correct?

(a) BALCO – Korba
(b) HINDALCO – Renukoot
(c) INDAL – Hirakud
(d) MALCO – Madras

104. The first state-owned fertiliser unit was set up in 1951 at:

(a) Durgapur
(b) Korba
(c) Ramagundam
(d) Sindri

105. Consider the following statements and select the correct answer using the code given below:

1. The first successful modern paper mill was located at Ballyganj (Kolkata) in 1870.
2. The paper industry was delicensed and decontrolled in 1997.

Code:

(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

Chapter 12 – Transport, Communication and Trade

106. Which one of the following is the longest National Highway (NH) of India?

(a) NH-5
(b) NH-6
(c) NH-7 (old)
(d) NH-8

107. The four segments of the Golden Quadrilateral join the following cities:

1. Delhi–Mumbai 2. Mumbai–Chennai
3. Chennai–Kolkata 4. Kolkata–Delhi

Consider the lengths of the above segments in descending order and select the correct answer using the code given below:

Code:

(a) 3, 4, 1, 2
(b) 3, 4, 2, 1
(c) 4, 3, 2, 1
(d) 4, 3, 1, 2

108. The East-West Corridor joins:

(a) Aizawl with Kanyakumari
(b) Aizawl with Mumbai
(c) Porbandar with Guwahati

(d) Porbandar with Silchar

109. The Headquarters of the South-Central Railway is at:
(a) Bangalore
(b) Hubli
(c) Hyderabad
(d) Secundrabad

110. The Headquarters of the Central Water Transport Corporation is located at:
(a) Kolkata
(b) New Delhi
(c) NOIDA
(d) Guwahati

111. The length of metallic roads per 100 sq km is longest in the state of:
(a) Gujarat
(b) Kerala
(c) Maharashtra
(d) Punjab

112. The length of railways per 1000 sq km is longest in the state of:
(a) Bihar
(b) Haryana
(c) Punjab
(d) West Bengal

113. Oil and natural gas is not found in the:
(a) Andaman and Nicobar Basin
(b) Ganga Basin
(c) Lakshadweep Basin
(d) Saurashtra Kachh Basin

114. Which one of the following seaports of India has the best natural harbour?
(a) Kochi

(b) Mangalore
(c) Mumbai
(d) Vishakhapatnam

115. Consider the following statements and select the correct answer using the code given below:
1. The Adam's Bridge is made up of coral rocks.
2. The length of Adam's Bridge is about 52 km.

Code:
(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 is correct
(d) Neither 1 nor 2 is correct

Chapter 13 – Cultural Setting

116. The highest percentage of scheduled tribe population is in the state of:
(a) Lakshadweep
(b) Meghalaya
(c) Mizoram
(d) Nagaland

117. The highest percentage of scheduled castes population is in the state of:
(a) Haryana
(b) Himachal Pradesh
(c) Punjab
(d) West Bengal

118. The highest percentage of Christian population is in the state of:
(a) Arunachal Pradesh
(b) Meghalaya
(c) Mizoram
(d) Nagaland

119. The highest decadal growth of population between the years 2001-2011 was in the state of:
(a) Arunachal Pradesh

(b) Bihar
(c) Jammu & Kashmir
(d) Meghalaya

120. The lowest decadal growth of population between 2001 and 2011 was in the state of:
(a) Andhra Pradesh
(b) Kerala
(c) Nagaland
(d) West Bengal

121. The highest density of population is in the state of:
(a) Bihar
(b) Kerala
(c) Uttar Pradesh
(d) West Bengal

122. The lowest sex ratio is in the Union Territory of:
(a) Andaman & Nicobar
(b) Daman & Diu
(c) Dadar & Nagar Haveli
(d) Lakshadweep

123. The frizzy-hair, bulbous forehead, flat nose are the physical traits of:
(a) Bhils
(b) Bodos
(c) Chakmas
(d) Kadars

124. The lowest female literacy rate is in the state of:
(a) Bihar
(b) Jharkhand
(c) Rajasthan
(d) Uttar Pradesh

125. Consider the following statements and select the correct answer using the code given below:

1. Population of India has increased by more than 181 million during the decade 2001-2011.
2. India constitutes about 17.5% of the total population of the worl(d)

Code:

(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

126. Censider the following statement and select the correct answer using the code given below:

1. The decadal growth rate during 2001-2011 has registered the sharpest decline since Independence of India .
2. The decadal growth of population was 17.64 during the decade of 2001-2011.

Code:

(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

127. Consider the following statements and select the correct answer using the code below:

1. The total number of children in the age-group 0-6 years according to the population of 2011 is about 159 million.
2. Bihar has the largest number of children population.

Code:

(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

128. Which one of the following is not correct?

(a) The density of population of India is 380 persons per sq km.

(b) The lowest sex ratio in 2011 has been recorded in Punjab.
(c) Among the largest states, the highest agricultural density is found in Kerala.
(d) The female children sex ratio is lowest in the state of Haryana.

129. Out of the following which state has the lowest literacy rate?
(a) Andhra Pradesh
(b) Karnataka
(c) Odisha
(d) Tamil Nadu

130. Out of the following which state has the lowest sex ratio?
(a) Gujarat
(b) Madhya Pradesh
(c) Maharashtra
(d) Rajasthan

Chapter 14 – Settlement

131. Consider the following statements and select the correct answer using the code given below:
1. A compact settlement has an ill-defined form.
2. Compact settlements are not found in the Himalayan region.

Code:
(a) Only 1 is correct
(b) Only 2 is correct
(b) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

132. The 'Socio-Spatial Model of rural settlements' morphology was advocated by:
(a) J.P. Singh
(b) M.P Singh
(c) K. N. Singh
(d) R.L. Singh

133. Which one of the following is the most urbanized state of India?
(a) Karnataka
(b) Kerala
(c) Maharashtra
(d) Tamil Nadu

134. Which one of the following is the least urbanised state?
(a) Chhattisgarh
(b) Jharkhand
(c) Odisha
(d) Rajasthan

135. The concept of conurbation was given by:
(a) Geddes, P.
(b) Jonston, R.J.
(b) Kendall, M.G.
(d) Weaver, J. C.

136. According to the Census of India 2011, the total number of million cities in India is:
(a) 35
(b) 43
(c) 53
(d) 63

137. The number of mega cities in India is:
(a) 3
(b) 4
(c) 5
(d) 6

138. The percentage of urban population in India according to the Census of 2011, is about:
(a) 28
(b) 30
(c) 32
(d) 34

139. The percentage of slum population is highest in the state of:
(a) Andhra Pradesh
(b) Madhya Pradesh
(c) Maharashtra
(d) Uttar Pradesh

140. The highest percentage of slum population is in the city of:
(a) Delhi
(b) Kolkata
(c) Mumbai
(d) Nagpur

Chapter 15 – Regional Development and Planning

141. Which one of the following is not a component of Planning in India?
(a) The National Development Council
(b) The Panchayat Level Planning Committee
(c) The State Level Planning Board
(d) The Planning Commission of India

142. About the functions of the Planning Commission of India consider the following statements and select the correct answer using the code given below:
1. To formulate Five Year/Annual Plan
2. To supervise the work of the National Planning
3. To encourage peoples participation in the process of planning
4. To monitor the plan from time to time

Code:
(a) 1, 2 and 3 are correct
(b) 1, 3 and 4 are correct
(c) 2, 3 and 4 are correct
(d) 1, 2, 3, and 4 are correct

143. Which one of the following is considered as successful regional plan in India?
(a) The Damodar Valley Corporation

(b) The Hugli Valley Corporation
(c) The National Capital Region
(d) The Mumbai Corporation Region

144. Consider the following statements and select the correct answer using the code given below:

1. The Command Area Development Programme was launched in 1974.
2. The Command Area Development was restructured and named as 'Command Area Development and Water Management Programme' in 2004.

Code:

(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

145. Which one of the following is not the objective of the Integrated Rural Development?

(a) To promote employment among the graduates.
(b) To provide assistance in self employment opportunities.
(c) To give assistance to a target group of rural poor.
(d) The target group includes marginal and small farmers.

146. Arrange the following dams of the Damodar River from west to east and select the correct answer using the code given below:

1. Balpari 2. Maithon 3. Tilaiya

Code:

(a) 1, 2, 3
(b) 2, 3, 1
(c) 3, 2, 1
(d) 3, 1, 2

147. The incidence of poverty is highest in which of the following states?

(a) Bihar

(b) Chhattisgarh
(c) Jharkhand
(d) Odisha

148. The incidence is poverty is lowest in the state of:
(a) Himachal Pradesh
(b) Haryana
(c) Jammu & Kashmir
(d) Kerala

149. In the North Eastern states of India the incidence of poverty is lowest in which of the following states:
(a) Arunachal Pradesh
(b) Manipur
(c) Meghalaya
(d) Mizoram

150. Consider the following statements and select the correct answer using the code given below:
1. Regional Planning is a multi-disciplinary subject.
2. Regional planning is a synthesis of physical, social, economic and cultural planning.

Code:
(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

Chapter 16 – India: Political Aspects

151. Consider the following statements and select the correct answer using the code given below:
1. The total population of India at the time of Independence was about 340 million.
2. The states of Maharashtra and Gujarat were created in 1960.
3. The basis of delineation of Gujarat and Maharashtra was language.

4. The states of Punjab and Haryana were created in 1966.

Code:

(a) 1, 2, 3 and 4 are correct
(b) 1, 2 and 3 are correct
(c) 1, 3 and 4 are correct
(d) 2, 3 and 4 are correct

152. The longest boundary of India is with:

(a) Bangladesh
(b) China
(c) Myanmar
(d) Pakistan

153. The Sir Creek in the Rann of Kachchh lies:

(a) To the west of Kajhar Creek
(b) To the east of Kori Creek
(c) To the west of Kori Creek
(d) To the west of Kori and Kajhar Creeks

154. Most of the pirate attacks in the Arabian Sea were along the coast of:

(a) Kenya
(b) Somalia
(c) Tanzania
(d) Yemen

155. The China's Ring of Pearls joins:

(a) Gwadar-Male-Hambantota
(b) Gwadar-Hambantota-Kyaukpyu
(c) Gwadar-Colombo-Yangon
(d) Gwadar- Male-Kyaukpyu

Chapter 17 – Contemporary Issues

156. Which one of the following is more vulnerable to landslides?

(a) Greater Himalayas
(b) Lesser Himalayas
(c) Shiwaliks

(d) Western Ghats

157. Which one of the following states never recorded an earthquake of serious magnitude?

(a) Arunachal Pradesh
(b) Meghalaya
(c) Manipur
(d) Tripura

158. The highest casualty occurred in the earthquake of:

(a) Assam-1950
(b) Bhuj-2001
(c) Bihar-1934
(d) Kangra-1905

159. Which one of the following states has the largest percentage of the forest cover in relation to geographical area?

(a) Gujarat
(b) Karnataka
(c) Mizoram
(d) Uttar Pradesh

160. Match List I with List II and select the correct answer using the code given below:

List I (Mountain Pass)	**List II (State)**
A. Bum La	1. Sikkim
B. Khardungla	2. Jammu & Kashmir
C. Nathu La	3. Himachal Pradesh
D. Shipki La	4. Arunachal Pradesh

Code:

	A	**B**	**C**	**D**
(a)	2	1	4	3
(b)	2	4	3	1
(c)	4	2	3	1
(d)	4	2	1	3

161. Consider the following statements and select the correct answer using the code given below:

1. The forest area in India is increasing.
2. The density of trees in the forests is decreasing.

Code:

(a) Only 1 is correct
(b) Only 2 is correct
(c) 1 and 2 are correct
(d) Neither 1 nor 2 is correct

162. Consider the following statements about the soil degradation and select the correct answer using the code given below:

The main causes of soil degradation are:

1. Removal of natural vegetation
2. Overgrazing
3. Agricultural activities
4. Bio-industrial activities

Code:

(a) 1, 2, and 3 are correct
(b) 1, 3, and 4 are correct
(c) 2, 3 and 4 are correct
(d) 1, 2, 3 and 4 are correct

163. The rank of India in groundnut production in the world is:

(a) 1
(b) 2
(c) 3
(d) 4

164. Consider the following statements and select the correct answer using the code given below:

1. About 50% of total oilseeds area is under soybean in the world.
2. Soybean provides about 60% of the vegetable protein and 30% of the oil in the world.
3. Soybean ranks second in vegetable oil economy of

India after groundnut.

Code:

(a) 1 and 2 are correct
(b) 1 and 3 are correct
(c) 2 and 3 are correct
(d) 1, 2 and 3 are correct

165. Consider the following statements and select the correct answer using the code given below:

1. *Barseem* is a prominent legume fodder crop of *rabi* in entire North Zone of India.
2. *Barseem* makes most digestible green fodder to the milch–cattle.
3. It increases the fertility of soil.
4. *Barseem* performs well in light sandy soil.

Code:

(a) 1, 2, and 3 are correct
(b) 1, 2 and 4 are correct
(c) 2, 3, and 4 are correct
(d) 1, 2, 3, and 4 are correct

166. Arrange the operational size of holdings of the following states of India in a descending order and select the correct answer using the code given below:

1. Gujarat 2. Madhya Pradesh
3. Punjab 4. Rajasthan

Code:

(a) Punjab, Rajasthan, Madhya Pradesh, Gujarat
(b) Rajasthan, Punjab, Gujarat, Madhya, Pradesh
(c) Punjab, Madhya Pradesh, Rajasthan, Gujarat
(d) Rajasthan, Punjab, Madhya Pradesh, Gujarat

167. Arrange the following states of India, in respect of percentages of irrigated area and select the correct answer using the code given below:

1. Haryana 2. Punjab
3. Tamil Nadu 4.Uttar Pradesh

Code:
(a) Haryana, Punjab, Uttar Pradesh, Tamil Nadu
(b) Haryana, Punjab, Tamil Nadu, Uttar Pradesh
(c) Punjab, Haryana, Tamil Nadu, Uttar Pradesh
(d) Punjab, Haryana, Uttar Pradesh, Tamil Nadu

168. Consider the following statements and select the correct answer using the code given below:
1. A minimum temperature of 10°C is required for the germination of cotton.
2. Cotton can tolerate a temperature up to 42°C.
3. Cotton cultivation needs 120 frost free days.

Code:
(a) 1 and 2 are correct
(b) 2 and 3 are correct
(c) 1 and 3 are correct
(d) 1, 2 and 3 are correct

169. Consider the following statements about the cultivation of sugarcane and select the correct answer using the code given below:
1. Sugarcane performs well at a temperature between 21°C and 25° C.
2. Sugarcane grows well in well-drained soil.
3. In India, north–west Uttar Pradesh is the largest producer of sugarcane.
4. North-west Uttar Pradesh has the ideal geo-climatic conditions for the cultivation of sugarcane.

Code:
(a) 1, 2 and 3 are correct
(b) 2, 3 and 4 are correct
(c) 1, 3 and 4 are correct
(d) 1, 2, 3 and 4 are correct

170. Consider the following statements and select the correct answer using the code given below:
1. Jute cultivation requires a temperature between 25°C to 35°C.

2. Jute cultivation requires over 100 cm of rainfall.
3. Water should remain standing in the Jute field for a good harvest.

Code:
(a) 1, 2 and 3 are correct
(b) 1 and 2 are correct
(c) 2 and 3 are correct
(d) 1 and 3 are correct

171. Arrange the following states of India in a descending order in respect of thermal power production and select the correct answer using the code given below:
1. Gujarat 2. Maharashtra
3. Uttar Pradesh 4. West Bengal

(a) Uttar Pradesh, Maharashtra, Gujarat, West Bengal
(b) Uttar Pradesh, Maharashtra, West Bengal, Gujarat
(c) Maharashtra, Uttar Pradesh, Gujarat, West Bengal
(d) Maharashtra, Gujarat, Uttar Pradesh, West Bengal

172. Consider the following statements and select the correct answer using the code given below:
1. The younger population adopts the language of the power in the world.
2. There are 192 languages in India spoken by more than 10,000 people each.
3. More than 96 per cent of the total population of India speak the 22 scheduled languages.

Code:
(a) 1 and 2 are correct
(b) 2 and 3 are correct
(c) 1 and 2 are correct
(d) 1, 2 and 3 are correct

173. Match list I with list II and select the correct answer using the code given below:

List I	**List II**
(Languages at the brink of	**(State)**

Extinction)

A. Bharwad	1. Gujarat
B. Chinali	2. Himachal Pradesh
C. Hakki-Pikki	3.Karnataka
D. Khash	4. Himachal Pradesh

Code:

	A	**B**	**C**	**D**
(a)	1	2	3	4
(b)	4	3	2	1
(c)	2	3	4	1
(d)	2	1	3	4

174. Arrange the urban population (2011) of the following states of India in a descending order and select the correct answer using the code given below:

1. Gujarat 2. Maharashtra
3. Gujarat 4. Punjab

Code:

(a) Tamil Nadu, Maharashtra, Punjab, Gujarat
(b) Tamil Nadu, Maharashtra, Gujarat, Punjab
(c) Maharashtra, Tamil Nadu, Gujarat and Punjab
(d) Maharashtra, Tamil Nadu, Punjab and Gujarat

175. Consider the following statements and select the correct answer using the code given below:

1. bihar has lower percentage of urban population to that of Himachal Pradesh
2. odisha has more urban population than that of Assam.

Code:

(a) only 1 is correct
(b) only 2 is correct
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

176. Consider the following statements and select the correct answer using the code given below:

1. The child sex ratio in Kerala is higher to that of Megahalya.
2. the child sex ratio in Chhattisgarh is higher to that of Nagaland.

code:
(a) only 1 is correct
(b) only 2 is correct
(c) both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

177. Consider the following statements and select the correct answer using the code given below:

1. The literacy rate in Uttarakhand is higher to that of Himachal Pradesh
2. The literacy rate in Maharashtra is higher to that to Tamil Nadu.

Code:
(a) only 1 is correct
(b) only 2 is correct
(c) both 1 and 2 are correct
(d) neither 1 nor 2 is correct

178. Consider the follows statements and select the correct answer using the code given below:

1. Coffee grows well when the mean temperature varies between 15°C and 28°C.
2. Coffee estates are located below 600 meters above the sea level.
3. Karnataka is the largest producer of coffee in India.

Code:
(a) 1 and 2 are correct

(b) 1 and 3 are correct
(c) 2 and 3 are correct
(d) 1, 2 and 3 are correct

179. Which one of the following power projects is not correctly matched?
(a) Matatila project – Betwa
(b) Mayurakshi project – Damodar river
(c) Tawa project – Narmada river
(d) Ukai power project – Tapi river

180. Consider the following statements and select the correct answer using the code given below:
1. Barakar is right hand tributary of the Damodar river.
2. Panchat dam is located on the Damodar river.

Code:
(a) only 1 is correct
(b) only 2 is correct
(c) both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

181. Consider the following statements and select the correct answer using the code given below:
1. The sex ratio of the tribal population of India is higher to that of the national average.
2. The literacy rate of the females of scheduled tribes population is below 50 per cent.

Code:
(a) only 1 is correct
(b) only 2 is correct
(c) both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

182. which one of the following is/are not the tributaries of the Tapi river?
1. Girna
2. kundi
3. Panjhra
4. purna

(a) 1 and 2
(b) 1, 2 and 3
(c) 2 and 3
(d) 1, 3 and 4

183. Consider the following river and identify the tributaries of the Kaveri river:
1. Amravati 2. bhavanisagar
3. Dudhganga 4. hemavati

code:
(a) 1, 2 and 3
(b) 1, 2 and 4
(c) 2, 3 and 4
(d) 1, 2, 3 and 4

184. Consider the following statements select the correct answer using the code given below:
1. Machh- kund project lies on the Sabari river.
2. Dudama waterfall lies on the Sileru river.

(a) only 1 is correct
(b) only 2 is correct
(c) both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

185. which one of the following statements is not correct?
1. Aroya Konda is the highest peak of the eastern ghats.
2. Aroya Konda lies in the state of Odisha.
3. Deodi Munda is the second highest peak of the eastern ghats.
4. Deodi Munda lies in the state of Andhra Pradesh.

Code:
(a) 1 and 2 are correct
(b) 1 and 3 are correct
(c) 1 and 4 are correct
(d) 1, 2, 3 and 4 are correct

186. which one of the following is not correctly matched?
(a) Mahadev Hills – Satpura Mt.

(b) Meghasani peak – Odisha
(c) Deograh peak – Madhya Pradesh
(d) Satmala Hills – Vindhyan Range

187. Consider the following statements and select the correct answer using the code given below:
1. Parbhati is a right hand tributary of the Chambal River.
2. Banas is a left hand tributary of the Chambal River.
3. Kali Sind is a right hand tributary of Betwa.

Code:
(a) 1 and 2 are correct
(b) 2 and 3 are correct
(c) 1 and 3 are correct
(d) 1, 2 and 3 are correct

188. Consider the following statements and select the correct answer using the code given below:
1. Mikir hills lie in the state of Arunachal Pradesh.
2. Mikir hills are an extension of the peninsular India.

Code:
(a) only 1 is correct
(b) only 2 is correct
(c) both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

189. Which one of the following is not correctly matched?
(a) Abor hills – Arunachal Pradesh
(b) Blue mountain – Mizoram
(c) Kylas peak – Meghalaya
(d) Saramati peak – Tripura

190. Which one of the following is not correctly Matched?
(a) Dafla hills – Arunachal Pradesh
(b) Parasnath – Jharkhand
(c) Pauhunri peak – Sikkim
(d) Tiger hill – Tripura

ANSWER KEY

1. (a)	2. (c)	3. (b)	4. (d)
5. (c)	6. (a)	7. (b)	8. (d)
9. (a)	10. (c)	11. (a)	12. (d)
13. (b)	14. (c)	15. (b)	16. (c)
17. (a)	18. (d)	19. (c)	
20. (a)	21. (b)	22. (a)	23. (c)
24. (d)	25. (b)	26. (a)	
27. (c)	28. (b)	29. (a)	30. (d)
31. (b)	32. (b)	33. (a)	
34. (c)	35. (a)	36. (b)	37. (c)
38. (b)	39. (a)	40. (d)	
41. (b)	42. (c)	43. (a)	44. (b)
45. (b)	46. (a)	47. (d)	
48. (b)	49. (c)	50. (a)	51. (c)
52. (b)	53. (d)	54. (b)	
55. (d)	56. (b)	57. (c)	58. (d)
59. (c)	60. (c)	61. (d)	
62. (c)	63. (a)	64. (b)	65. (a)
66. (b)	67. (c)	68. (c)	
69. (a)	70. (c)	71. (a)	72. (c)
73. (d)	74. (b)	75. (a)	
76. (b)	77. (b)	78. (d)	79. (c)
80. (c)	81. (b)	82. (c)	
83. (d)	84. (d)	85. (a)	86. (d)
87. (c)	88. (c)	89. (d)	
90. (b)	91. (c)	92. (d)	93. (b)
94. (c)	95. (b)	96. (d)	
97. (c)	98. (b)	99. (a)	100. (b)
101. (d)	102. (c)	103. (d)	
104. (d)	105. (c)	106. (c)	107. (a)
108. (d)	109. (d)	110. (a)	
111. (b)	112. (c)	113. (c)	114. (d)
115. (a)	116. (a)	117. (c)	
118. (d)	119. (a)	120. (c)	121. (a)
122. (b)	123. (d)	124. (c)	
125. (b)	126. (c)	127. (a)	128. (b)

129. (a)	130. (a)	131. (d)	
132. (c)	133. (d)	134. (c)	135. (a)
136. (c)	137. (a)	138. (b)	
139. (a)	140. (c)	141. (b)	142. (d)
143. (a)	144. (c)	145. (a)	
146. (d)	147. (d)	148. (c)	149. (d)
150. (c)	151. (a)	152. (a)	
153. (c)	154. (d)	155. (b)	156. (b)
157. (d)	158. (b)	159. (c)	
160. (d)	161. (c)	162. (d)	163. (b)
164. (d)	165. (a)	166. (b)	
167. (d)	168. (b)	169. (a)	170. (b)
171. (c)	172. (d)	173. (a)	
174. (b)	175. (b)	176. (d)	177. (b)
178. (b)	179. (b)	180. (b)	
181. (c)	182. (d)	183. (b)	184. (b)
185. (b)	186. (c)	187. (a)	
188. (b)	189. (d)	190.(d)	